Analysis Chart of Important Computer Questions asked in Previous Year Examination

S.L.	EXAM NAME	EXAM DATE	No. of Questions
LICENTIATES TEACHER GRADE			
1.	LT Grade C.S.	2018	120
KENDRIYA VIDYALAYA SANGATHAN			
2.			
3.	KVS PGT C.S.	2017	100
4.	KVS PGT C.S.	23.03.2023 (Shift-II)	80
5.	KVS PGT C.S.	2013	100
6.	KVS PGT C.S.	2018	80
NAVODAYA VIDYALAYA SAMITI			
7.	NVS PGT C.S.	15-.12.2022 (Shift-II)	80
	NVS PGT C.S.	16.12.2022 (Shift-II)	80
DELHI SUBORDINATE SERVICES SELECTION BOARD			
8.	DSSSB TGT C.S.	07.02.2024 (Shift-II)	100
9.	DSSSB TGT C.S.	01.08.2021 (Shift-II)	100
10.	DSSSB TGT C.S.	01.08.2021 (Shift-I)	100
11.	DSSSB TGT C.S.	07.08.2021 (Shift-I)	100
12.	DSSSB TGT C.S.	07.08.2021 (Shift-II)	100
13.	DSSSB TGT C.S.	08.08.2021 (Shift-I)	100
14.	DSSSB PGT C.S. (Male/Female)	11.07.2021	200
15.	DSSSB PGT C.S. (Male)	27.07.2018 (Shift-I)	200
BIHAR PUBLIC SERVICE COMMISSION			
16.	Bihar STET C.S.	12.09.2023 (Shift-I)	100
17.	Bihar STET C.S.	12.09.2023 (Shift-II)	100
18.	Bihar STET C.S.	18.09.2020 (Shift-I)	100
19.	Bihar STET C.S.	2020	100
20.	Bihar STET C.S.	2019 (Shift-II)	100
21.	Bihar TRE 3.0	22.07.2024 (Shift-I)	80
22.	Bihar TRE 3.0	22.07.2024 (Shift-II)	80
23.	Bihar TRE 2.0	08.12.2023	80
24.	Bihar TRE 2.0	15.12.2023	80
25.	Bihar TRE 1.0	26.08.2023	80
HARYANA TEACHER ELIGIBILITY TEST			
26.	HTET PGT C.S.	30.07.2025	60
27.	Haryana PGT C.S. Asist. Prof.	2024	100
28.	HTET PGT C.S.	2023	60
29.	HTET PGT C.S.	2022	60
30.	HTET PGT C.S.	2021	60

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31.	HTET PGT C.S.	2020	60
32.	HTET PGT C.S.	2019	60
33.	HTET PGT C.S.	2018	60
UTTRAKHAND PUBLIC SERVICE COMMISSION			
34.	UKPSC PGT C.S. Asist. Prof.	2017	100
CHHATTISGARH PUBLIC SERVICE COMMISSION			
35.	CGPSC C.S. Set-A	2020	100
RAJASTHAN PUBLIC SERVICE COMMISSION			
36.	RPSC C.S.	2016 (Paper-I)	150
37.	RPSC C.S.	2016 (Paper-II)	150
EKLAVYA MODEL RESIDENTIAL SCHOOL			
38.	EMRS PGT C.S.	16.12.2024	80
UNIVERSITY GRANTS COMMISSION (NATIONAL ELIGIBILITY TEST)			
39.	UGC NET C.S.	June-2014 (Paper-II)	50
40.	UGC NET C.S.	June-2014 (Paper-III)	75
41.	UGC NET C.S.	December-2014 (Paper-II)	50
42.	UGC NET C.S.	December-2014 (Paper-III)	75
43.	UGC NET C.S.	June-2015 (Paper-II)	50
44.	UGC NET C.S.	June-2015 (Paper-III)	75
45.	UGC NET C.S.	December-2015 (Paper-II)	50
46.	UGC NET C.S.	December-2015 (Paper-III)	75
47.	UGC NET C.S.	June-2016 (Paper-II)	50
48.	UGC NET C.S.	June-2016 (Paper-III)	75
49.	UGC NET C.S.	Augest-2016 (Paper-II)	50
50.	UGC NET C.S.	August-2016 (Paper-III)	75
51.	UGC NET C.S.	June-2017 (Paper-II)	50
52.	UGC NET C.S.	June-2017 (Paper-III)	75
53.	UGC NET C.S.	November-2017 (Paper-II)	50
54.	UGC NET C.S.	November-2017 (Paper-III)	75
55.	UGC NET C.S.	June-2018 (Paper-II)	100
56.	UGC NET C.S.	December-2018 (Paper-II)	100
57.	UGC NET C.S.	24 June-2019 (Paper-II)	100
58.	UGC NET C.S.	4 December 2019 (Paper-II)	100
59.	UGC NET C.S.	June-2020 (Paper-II)	100
60	UGC NET C.S.	December 2020	100
61.	UGC NET C.S.	December 2021	100
62.	UGC NET C.S.	December-2022 (11.0.2023)	100
63.	UGC NET C.S.	December-2022 (15.03.2023)	100
64.	UGC NET C.S.	June-2023	100
65.	UGC NET C.S.	December-2023	100
66.	UGC NET C.S.	Re-Exam- June 2024	100
67.	UGC NET C.S.	December-2024	100
	Total		**5940**

Note: After detailed analysis of above question papers of different exams related to **Computer Science** total **5940** questions have been presented chapter wise. Questions of repeated and similar nature have been included so that technique of asking question can benefit the competitors.

Sarkari Resource

01. Digital Logic Circuits and Discrete Mathematical Structures

(i) Number Systems

1. Write the binary code for EXAM in ASCII-7. (decimal code of A is 65)
ASCII-7 में EXAM लिए बाइनरी कोड लिखें । (A का दशमलव कोड 65 है)

(a) 1001101101100010000011001101
(b) 1000101101100010000011001101
(c) 1011000100000110110001000001
(d) 1000001101100010011011011000

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : First let's find ASCII decimal codes
E = 69
X = 88
A = 65
M = 77
Now convert each to 7 bit binary-
E(69) = 1000101
X(88) = 1011000
A(65) = 1000001
M(77) = 1001101
Now concatenate

E	X	A	M
1000101	1011000	1000001	1001101

Final binary string-
1000101101100010000011001101

2. 1000 of 8421 binary coding system is equal to 1011 in ..... binary coding.
बाइनरी कोडिंग सिस्टम 8421 का 1000, बाइनरी कोडिंग सिस्टम ..... में 1011 के बराबर है।

(a) 5211 (b) Excess-3
(c) 2421 (d) 4221

HTET PGT C.S. 2022

Ans. (b) :
- "1000" in 8421 (Standard BCD) represents the decimal value 8.
- We need a coding system where "1011" also represent 8.
- **In Excess - 3,** the code for decimal 'd' is the binary representation of d + 3
- For d = 8, 8 + 3 = 11, which is binary 1011
- Thus, excess - 3 matches even then both differences are empty)

Therefore:- (R - S) n (S - R) = ϕ is always true

3. The BCD number for decimal 473 is
डेसीमल 473 के लिए बीसीडी संख्या क्या है?

(a) 111011010 (b) 110001110011
(c) 010001110011 (d) 10011110011

HTET PGT C.S. 2021

Ans. (c) : BCD means: Convert each decimal digit into its 4 bit binary form.
Break 473 into digits:
- 4 → 0100
- 7 → 0111
- 3 → 0011
- Now just combine them:

BCD of 473 = 010001110011

4. What is representation of $(539)_{10}$ in hexadecimal ?
$(539)_{10}$ का हेक्साडेसिमल में निरूपण क्या है?

(a) ABE (b) CDB
(c) DES (d) 21B

HTET PGT C.S. 2021

Ans. (d) : $(539)_{10}$ to Hexadecimal
Divide by 16 repeatedly and write remainders in reverse:
- 539 ÷ 16 = 33 remainder 11 → B
- 33 ÷ 16 = 2 remainder 1 → 1
- 2 ÷ 16 = 0 remainder 2 → 2

So $(539)_{10} = (21B)_{16}$
= 21B

5. Full form of ASCII is:
ASCII का फुल फॉर्म है–

(a) American Simulation Code for Information Interchange/अमेरिकन सिम्युलेशन कोड फॉर इन्फॉर्मेशन इंटरचेंज
(b) African Simulation Code for Information अफ्रीकन सिम्युलेशन कोड फॉर इन्फॉर्मेशन इंटरचेंज
(c) African Standard Code for Information Interchange/अफ्रीकन स्टैंडर्ड कोड फॉर इन्फॉर्मेशन
(d) American Standard Code for Information Interchange/अमेरिकन स्टैंडर्ड कोड फॉर इन्फॉर्मेशन इंटरचेंज

HTET PGT C.S. 2023

Ans. (d) : ASCII:- American Standard Code for Information Interchange.

6. ASCII (American Standard Code for Information Interchange) can only handle _____or at most _____ languages and that shows its limitation to handle all world languages.
ASCII (अमेरिकन स्टैंडर्ड कोड फॉर इन्फॉर्मेशन इंटरचेंज) केवल यह ____ या अधिक ____ भाषाओं को भी संभाल सकता है। और यह सभी विश्व भाषाओं को संभालने की इसकी सीमा को दर्शाता है।

(a) 1.2,5 (b) 2.1,2
(c) 3.1,5 (d) 4.3,4

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : ASCII (American Standard Code for Information Interchange) uses 7 bits (128 characters) and was originally designed only for English and at most to few western language with very limited symbols.
so, ASCII can only handle 1 language or at most 2 languages (English + another with limited overlap).

7. Which type of computer uses the 8-byte code called EBCDIC?
किस प्रकार के कम्प्यूटर में 8 बिट कोड का उपयोग किया जाता है, जिसे EBCDIC कहा जाता है?
(a) Minicomputers
(b) Microcomputers
(c) Mainframe computers
(d) Supercomputer

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : EBCDIC stands for Extended Binary Coded Decimal Interchange Code. It is an 8-bit character coding scheme that was developed by IBM. It is mainly used in mainframe of computers, especially in older IBM systems.

8. $(929.19)_{10}$ = ?
(a) $(1640.41)_8$ (b) $(1641.141)_8$
(c) $(1632.141)_8$ (d) $(1543.34)_8$

HTET PGT C.S. 2020

Ans. (b) : Step 1 :- Integer part (929 → octal) use direct division
:- 929 ÷ 8 = 116R1
116 ÷ 8 = 14R4
14 ÷ 8 = 1R6
1 ÷ 8 = 0R1
→ Reverse the remainders $929_{10} = 1641_8$
Step 2 :- Fractional part (0.19→octal) multiply by 8 repeatedly
0.19 × 8 = 1.52 → 1
0.52 × 8 = 4.16 → 4
0.16 × 8 = 1.28 → 1
So, $0.19_{10} \approx 0.141_8$
total = $(1641 \cdot 141)_8$

9. EBCDIC stands for :
EBCDIC का अर्थ है:
(a) Extended Binary Coded Decimal Interchange Code/एक्सटेंडेड बाइनरी कोडेड डेसीमल इंटरचेंज कोड
(b) Extended Bit Code Decimal Interconversion Code/एक्सटेंडेड बिट कोड डेसीमल इंटरकन्वर्जन कोड
(c) Extended Bit Case Decimal Interconversion Code/एक्सटेंडेड बिट केस डेसीमल इंटरकन्वर्जन कोड
(d) Extended Binary Case Decimal Interchange Code/एक्सटेंडेड बाइनरी केस डेसीमल इंटरफेस कोड

HTET PGT C.S. 2020

Ans. (a) : Step 1 :- Recall that EBCDIC is a well know IBM character encoding
Step 2:- Break the acronym :
- E → Extended
- BCD → Binary Coded Decimal
- IC → Interchange Code

Match with options → Option (a) fits perfectly

10. In Binary to Octal Number System Conversion, what is the octal equivalent for binary number 1101010 ?
बाइनरी से ऑक्टल नम्बर सिस्टम रूपांतरण में बाइनरी नम्बर 1101010 के लिए ऑक्टल समकक्ष क्या है?
(a) 125 (b) 137
(c) 152 (d) 155

HTET PGT C.S. 2020

Ans. (c) : Binary : 1101010
Group into 3 bits (right to left): 001 101 010 convert each group to octal
- 001→1
- 101 → 5
- 010 → 2

11. Value of $(34.36)_{10}$ in binary is :
बाइनरी (द्विचर) में $(34.36)_{10}$ का मान है:
(a) $(10010.0101)_2$ (b) $(10010.10111)_2$
(c) $(1010.10111)_2$ (d) $(100010.010111)_2$

HTET PGT C.S. 2019

Ans. (d) : Steps to convert from decimal to Binary
Divide the number by 2.
Get the integer quotient for the next iteration
Get the remainder for the binary digit
Repeat the steps until the quotient is equal to 0.

Division by 2	Quotient	Remainder
34/2	17	0
17/2	8	1
8/2	4	0
4/2	2	0
2/2	1	0
2/1	2	0

$(34)_{10} = (100010)_2$
Now convert 0.36 into binary
0.36 × 2 = 0.72 (0)
0.72 × 2 = 1.44 (1)
0.44 × 2 = 0.88 (0)
0.88 × 2 = 1.76 (1)
0.76 × 2 = 1.52 (1)
0.52 × 2 = 1.04 (1)
$0.36 = (010111)_2$
Therefore $(34,36)_{10} = (100010.010111)_2$

12. Which of the following value is not equal to $(7214)_8$?
निम्न में से कौन-सा मान $(7214)_8$ के बराबर नही है?
(a) $(111010001100)_2$ (b) $(EAC)_{16}$
(c) $(3724)_{10}$ (d) $(322030)_4$

HTET PGT C.S. 2018

Ans. (b) : Convert $(7214)_8$ into Decimal
$= 7 \times 8^3 + 2 \times 8^2 + 1 \times 8^1 + 4 \times 8^0$
$= 7 \times 512 + 2 \times 64 + 8+4$
$= 3584 + 128 + 8 + 4$
$= (3724)_{10}$
So, option (c) is correct
From option (a), $(111\ 01\ 000\ 11\ 00)_2$
Convert this binary number into octal
$(\underline{111}\ \underline{010}\ \underline{001}\ \underline{1\ 00})_2 = (7214)_8$
So, option (a) is correct

Sarkari Resource

Decimal number to $(7214)_8$ is $(3724)_{10}$
Convert decimal number $(3724)_{10}$ equivalent to base 4
Divide 3724 by 4 repeatedly.
3724 ÷ 4⟶ Quotient 931 and remainder 0
931 ÷ 4⟶ Quotient 232 and remainder 3
232 ÷ 4⟶ Quotient 58 and remainder 0
58 ÷ 4⟶ Quotient 14 and remainder 2
14 ÷ 4⟶ Quotient 3 and remainder 2
3 ÷ 4⟶ Quotient 0 and remainder 3 (finally, last bit of MSB)
Hence, $(3724)_{10} = (322030)_4$
So, option (d) is correct.
Binary equivalent to octal number $(7214)_8$ is $(111010001100)_2$
Convert this binary into Hexadecimal
$(\underline{1110}\ \underline{1000}\ \underline{1100})_2 = (E8C)_{16}$
So, option (b) is not equal to $(7214)_8$

13. Which of the following value is not equal to $(6103)_8$?
निम्न में से कौन-सा मान $(6103)_8$ के बराबर नहीं है?

(a) $(110001000011)_2$ (b) $(C43)_{16}$
(c) $(3139)_{10}$ (d) $(30103)_4$

HTET PGT C.S. 2019

Ans. (d) : Convert $(6103)_8$ into Decimal
$= 6 \times 8^3 + 1 \times 8^2 + 0 \times 8^1 + 3 \times 8^0$
$= 6 \times 512 + 1 \times 64 + 0 + 3$
$= 3072 + 64 + 0 + 3$
$= (3139)_{10}$
So, option (c) is correct
From option (a), $(110001000011)_2$
Convert this binary number into octal
$(\underline{110}\ \underline{001}\ \underline{000}\ \underline{011})_2 = (6103)_8$
So, option (a) is correct
Decimal number to $(6103)_8$ is $(3139)_{10}$
Convert decimal number $(3139)_{10}$ equivalent to base 4
Divide 3139 by 4 repeatedly.
3139 ÷ 4⟶ Quotient 784 and remainder 3
784 ÷ 4⟶ Quotient 196 and remainder 0
196 ÷ 4⟶ Quotient 49 and remainder 0
49 ÷ 4⟶ Quotient 12 and remainder 1
12 ÷ 4⟶ Quotient 3 and remainder 0
3 ÷ 4⟶ Quotient 0 and remainder 3 (finally, last bit of MSB)
Hence, $(3139)_{10} = (301003)_4$
So, option (d) is not correct.
Binary equivalent to octal number $(6103)_8$ is $(110001000011)_2$
Convert this binary into Hexadecimal
$(\underline{1100}\ \underline{0100}\ \underline{0011})_2 = (C43)_{16}$
So, option (d) is not eqal to $(6103)_8$

14. On dividing $(101011)_2$ by $(11)_2$ the quotient and remainder respectively are :
$(101011)_2$ में $(11)_2$ का भाग देने पर भागफल तथा शेषफल क्रमशः होगे:

(a) $(1110)_2$ and $(1)_2$ (b) $(1101)_2$ and $(0)_2$
(c) $(1110)_2$ and $(0)_2$ (d) $(1011)_2$ and $(1)_2$

HTET PGT C.S. 2018

Ans. (a) : Convert $(101011)_2$ into decimal
$= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
$= 32 + 0 + 8 + 0 + 2 + 1 = 43$
Convert $(11)_2$ into decimal
$= 1 \times 2^1 + 1 \times 2^0$
$2 + 1 = 3$
After dividing 43 by 3, we find 14 as quotient and run as remainder
Binary conversion of 14 and 3 is $(1110)_2$ and $(1)_2$ repectively

15. Sum $(89D)_{16} + (259)_{10}$ is :
$(89D)_{16} + (259)_{10}$ का योग है:

(a) $(9EA)_{16}$ (b) $(3472)_8$
(c) $(2464)_{10}$ (d) $(63215)_8$

HTET PGT C.S. 2018

Ans. (c) : Convert Hexadecimal number $(89D)_{16}$ into decimal number
$= 8 \times 16^2 + 9 \times 16^1 + 13 \times 16^0$ {D represents as 13}
$= 8 \times 256 + 9 \times 16 + 13$
$= 2048 + 144 + 13 = (2205)_{10}$
Now $(2205)_{10} + (259)_{10} = (2464)_{10}$

16. Find the octal equivalent of 952_{10}:
$(952)_{10}$ का ऑक्टल समतुल्य ज्ञात कीजिए–

(a) $(1670)_8$ (b) $(167)_8$
(c) $(1607)_8$ (d) $(1660)_8$

HTET PGT C.S. 2023

Ans.(a): convert decimal $(952)_{10}$ into octal number divide 952 by 8 repeatedly to get octal digit.
952 ÷ 8 = quotient 119 & remainder 0
119 ÷ 8 = quotient 14 & remainder 7
14 ÷ 8 = quotient 1 & remainder 6
1 ÷ 8 = quotient 0 & remainder 1 (Finally, the lost remainder is 1 → MSB)
So,octal number is $(1670)_8$.

17. Find the decimal equivalent of octal number $(127.54)_8$:
ऑक्टल संख्या $(127.54)_8$ का डेसिमल समतुल्य ज्ञात कीजिए–

(a) $(78.6875)_{10}$ (b) $(48.6875)_{10}$
(c) $(87.6875)_{10}$ (d) $(67.6875)_{10}$

HTET PGT C.S. 2023

Ans.(c): • To convert the octal number $(127{\cdot}54)_8$ to decimal.
Separate the integer part $(127)_8$ and the fractional part $(0.54)_8$.
• Covert the integer part $(127)_8$ to decimal:-
$1 \times 8^2 + 2 \times 8^1 + 7 \times 8^\circ$
$64 + 16 + 7 = 87$
• Convert the fractional part $(0.54)_8$ to decimal :-
$= 5 \times 8^{-1} + 4 \times 8^{-2}$
$= 5 \times \frac{1}{8} + 4 \times \frac{1}{64}$
$= 0.625 + 0.0625$
$= 0.6875$
Combine the integer and fractional parts.
$= 87 + 0.6875$
$= (87.6875)_{10}$

Sarkari Resource

18. What is the hexadecimal equivalent number of the octal number 726?

(a) EBO (b) E4C
(c) 1E3 (d) 1D6
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : Convert octal number $(726)_8$ into decimal number
$= 7 \times 8^2 + 2 \times 8^1 + 6 \times 8^0$
$= 7 \times 64 + 16 + 6$
$= 448 + 16 + 6$
$= (470)_{10}$
Convert decimal number $(470)_{10}$ into Hexadecimal
Now, divide 470 by 16 repeatedly to get hexadecimal.
$470 \div 16 \rightarrow$ quotient 29 & remainder 6
$29 \div 16 \rightarrow$ quotient 1 & remainder 13 (D)
$1 \div 16 \rightarrow$ quotient 0 & remainder 1 (finally, the last quotient is 1→ most significant
So, hexadecimal number = $(1D6)_{16}$

19. The sum of two octal numbers 12 and 17 would be represented in octal as

(a) 21 (b) 23
(c) 31 (d) 29
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : Step 1: Convert to decimal
$12_8 = 1 \times 8 + 2 = 10_{10}$
$17_8 = 1 \times 8 + 7 = 15_{10}$
So,
$10 + 15 = 25_{10}$
Step 2: Convert 25_{10} to octal
$25 \div 8 =$ 3remainder 1
So,
$25_{10} = 31_8$

20. Which is the binary exponent part corresponding to the IEEE single-precision representation C1C00000 (hex) of a floating-point number?

(a) 1000 0011 (b) 11000011
(c) 00000011 (d) 1100 0000
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : C1C00000→ Binary =1100 0001 1100 0000 0000 0000 0000 0000
IEEE 754 single precision = 1 sign bit + 8 exponent bits + 23 mantissa bits
sign bit = 1
Exponent bits = 10000011

21. The translation from symbolic program into binary is done in_____.
सिम्बोलिक प्रोग्राम को बाइनरी में _____ में बदला जाता है।

(a) Two passes/दो पास (b) Directly/सीधे
(c) Three passes/तीन पास (d) Four passes/चार पास

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : Symbolic program = Assembly language
Binary= machine language
Translation is done by an Assembler
Assemblers are of two types:
1- One-pass assembler-translate line by line
2- Two-pass assembler-commonly used, because in the first pass it builds the symbol table, and in the second pass it generates the machine code.

22. What will be binary equivalent of 562_8?
562_8 का बाइनरी समतुल्य क्या होगा?

(a) 101110010_2 (b) 101100010_2
(c) 111110010_2 (d) 101110011_2

HTET PGT C.S. 30.07.2025

Ans. (a) : Step 1: Recall
Each octet digit= 3 binary bits
Step 2: convert each digit of 562_8
$5_8 = 101_2$
$6_8 = 110_2$
$2_8 = 010_2$
Step 3: combine
$562_8 = 10110010_2$
That is
$562_8 = 101110010_2$

23. What will be binary equivalent of $2APB_{16}$?
$2APB_{16}$ का बाइनरी समतुल्य क्या होगा?

(a) 1011101011_2 (b) 1110101011_2
(c) 1000101011_2 (d) 1010101011_2

HTET PGT C.S. 30.07.2025

Ans. (d) : Step 1: Assume it is $2AB_{16}$
Convert each hex digit into 4-bit binary
2 = 0010
A = 1010
B = 1011
Step 2: Combine
$2AB_{16} = 001010101011_2 = 1010101011_2$

24. To convert a Gray Code to its binary equivalent, which technique is commonly used?
ग्रे कोड को उसके बाइनरी समकक्ष में परिवर्तित करने के लिए कौन सी तकनीक आमतौर पर उपयोग की जाती है?

(a) Subtraction method/घटाव विधि
(b) Addition method/जोड़ विधि
(c) Exclusive OR (XOR) operation
विशिष्ट या (XOR) क्रिया
(d) Division method/ भाग विधि

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans, (c)
- To convert Gray code in binary, the most significant bit (MSB) of binary code is same as MSB of Gray code.
- Each subsequent binary bit is obtained by XORing the previous binary bit with the next Gray code bit.

Example:-
Gray: 1011 in Binary: 1110 using XOR operations.

25. The octal equivalent of 111010 is?
111010 का Octal संख्या है?

(a) 81 (b) 72
(c) 81 (d) None of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : To convert binary into octal we group each 3 bits from left to right and write octal number.
$(\underline{111}\ \underline{010})_2 = (72)_8$
So, the correct option is (b)72.

Sarkari Resource

26. ASCII and EBCDIC are the popular character coding systems. What does EBCDIC stand for?
ASCII और EBCDIC लोकप्रिय Character Coding System हैं। EBCDIC का पूरा नाम है

(a) Extended Binary Coded Decimal Interchange Code
(b) Extended Bit Code Decimal Interchange Code
(c) Extended Bit Case Decimal Interchange Code
(d) Extended Binary Case Decimal Interchange Code

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : EBCDIC (Extended Binary Coded Decimal Interchange Code).
It is a character encoding system. It tells the computer how to represent letters (A-Z), numbers (0-9), punctuation marks and control characters using 8 bits (1 byte).
ASCII (American Standard Code for Information Interchange) is a character encoding standard that assigns a unique number (from 0 to 127) to each letter, digit, punctuation mark or control character.
So, the correct option is (a). Extended Binary Coded Decimal Interchange Code

27. The hexadecimal number F4 is equivalent to which binary number?
हेक्साडेसीमल संख्या F4 किस बाइनरी संख्या के समकक्ष है?

(a) 11111001 (b) 11110100
(c) 11111100 (d) 11111010

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : In Hexadecimal number system 'F' represents 15
Binary of 15 : 1111
Binary of 4 : 0100
After combined both we get 11110100
i.e. $(F4)_{16} = (11110100)_2$
So, the correct option is (b) 11110100

28. Which of the following is exact equivalent of given *binary* number in Octal Number ?
निम्नलिखित में से कौन सी संख्या अष्टाधारी संख्या दी गई द्विआधारी संख्या के बिल्कुल समतुल्य है ?
$(101110001011)_2$

(a) $(232023)_8$ (b) $(A8A)_8$
(c) $(5613)_8$ (d) $(3165)_8$

KVS PGT 23.03.2023 (Shift-II)

Ans, (c) : Convering binary number $(101110001011)_2$
To octal by grouping 3 bits. From LSB.

101	110	001	011
↓	↓	↓	↓
5	6	1	3

So, $(101110001011)_2 = (5613)_8$

29. What is full form of ISCll encoding scheme ?
आई एस सी आई आई एनकोडिंग स्कीम की विस्तृत परिभाषा क्या है ?

(a) Indian Standard Code for Interchange of Information
(b) Indian Standard Code for Information interchange.
(c) Indian Standard Coding for Information interchange
(d) Indian Scheme of Coding for information Interchange

KVS PGT 23.03.2023 (Shift-II)

Ans, (b) : Full form of ISCII encoding scheme is "Indian standard code for Information interchange". It is a coding scheme for representing various writing systems of India. It encodes the main Indic scripts and a roman transliteration.

30. The accuracy of the floating-point numbers represent able in two 16-bit words of a computer is approximately
कंप्यूटर के दो 16 बिट शब्दों में फ्लोटिंग पॉइंट नंबरों की सटीकता तालिका का प्रतिनिधित्व करती है लगभग

(a) 16 digits
(b) 6 digits
(c) 9 digits
(d) All of these/इनमें से सभी

Bihar STET C.S. 2020

Ans. (b) : The floating point representation using two 16-bit words (total 32-bits) corresponds to single precision floating point format.
The mantissa has 23 bits (with an implicit leading bit, effectively 24-bits of precision).
The number of significant decimal digits is calculated as:
$\log_{10} 2^{24} \approx 7.22$, but due to rounding errors and practical limitations, the effective accuracy is about 6 decimal digits.
So, the correct option is (b) 6 digits.

31. A binary-to-gray code converter is an example of:/एक बाइनरी-टू-ग्रे कोड कनवर्टर एनकोडर का एक उदाहरण है–

(a) Encoder (b) Decoder
(c) Comparator (d) Multiplexer

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : A binary to gray code converter is a type of Encoder. Encoder converts information from one code to another (like decimal to binary, or binary to gray).
Other options:
Decoder: converts a coved input to activate one of many outputs (e.g., 3 to 8 line decoder).
Comparator: compares two binary numbers to indicate equality/greater.
Multiplexer: selects one of many inputs to pass to output using select lines.
So, the correct options (a) Encoder

32. Which of the following is a self-complementing binary code, where adjacent codes differ in only one bit position?
निम्नलिखित में से कौन सा एक स्वयं पूरक (Self-Complementing) बाइनरी कोड है, जहाँ सन्निकट कोड केवल एक बिट पोजीशन में भिन्न होते है?

(a) BCD (b) ASCII
(c) Gray Code (d) EBCDIC

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : Gray code is a binary numeral system where two successive values differ in only one bit (binary digit). It is also known as a reflected binary code.
Gray code is self complementing.
So, the correct option is (c) Gray code

33. The number of bits require to represent decimal number 4096 in binary from is_____.
दशमलव संख्या 4096 को द्विआधारी संख्या में प्रस्तुत करने में _____बिट्स की आवश्यकता है।
(a) 16 (b) 10
(c) 12 (d) 13

KVS PGT C.S. 2017

Ans, (d) : Decimal number = 4096.
convert this decimal number in to binary.
$4096 = (1000000000000)_2$
Total number of bits in this binary form is 13. So, The number of bits require to represent decimal number 4096 binary form is 13.

34. What is the octal equivalent of the hexadecimal number 132A?
दशमलव (हेक्साडेसीमल) संख्या 132A का अष्टक (ऑक्टल) समतुल्य क्या है ?
(a) 46252 (b) 11450
(c) 11452 (d) 46250

KVS PGT C.S. 2017

Ans, (c) : Given Hexadecimal Number = 132 A
This hexadecimal number is converted in to binary number :
$(132\ A)_{16} = (0001001100101010)_2$
Now group this binary number into 3 bits starting from right to left.
i.e 0 001 001 100 101 010
This 3 binary bits converted in to octal number.
= 011452
Hence,
$(132\ A)_{16} = (11452)_8$.

35. Consider an arbitrary number system with the independent digits as 0, 1 and X. What is the radix of this number system ?
किसी एक यादृच्छिक संख्या पद्धति की कल्पना कीजिए । जिसमें स्वतंत्र अंक जैसे 0, 1 और X हों । इस संख्या पद्धति का रेडिक्स क्या है ?
(a) 1 (b) 2
(c) 3 (d) 4

KVS PGT C.S. 2013

Ans, (c) : The radix of the proposed number system is 3. The first 10 numbers in this number system would be 0, 1, x, 10, 11, 1x, x0, x1, xx and 100.

36. Identify the Hexadecimal equivalent of the decimal number :
$(3564)_{10} = (?)_{16}$
निम्नलिखित दशमलव संख्या के लिए षोडश आधारी समतुल्य की पहचान कीजिए:
$(3564)_{10} = (?)_{16}$
(a) DAD (b) DEF
(c) DEC (d) DAC

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans, (c) : $(35\ 64)_{10} \rightarrow (?)_{16}$

16	3564	
16	222	12
16	13	14
	0	13

13 14 12
↓ ↓ ↓
D E C

Note- 10-A
11-B
12-C
13-D
14-E
15-F

37. Binary equivalent of the octal number $(173)_8$ is:
अष्टाधारी संख्या $(173)_8$ का द्विआधारी समतुल्य बताइए:
(a) $(1110111)_2$ (b) $(1111011)_2$
(c) $(1101111)_2$ (d) $(1101011)_2$

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans, (b) : Octal number $(173)_8$
⇒ Binary Equivalent
⇒ $(173)_8 = ?$
⇒ Look up Each octal digit to obtain the equivalent group of three binary digits.
oct = 1, 3, 7
Bin = 1, 011, 111
⇒ $(173)_8 = (1111011)_2$ is the binary number

38. Binary equivalent of the decimal number $(11)_{10}$ is:
दशमलव संख्या (decimal number) $(11)_{10}$ का द्वि-आधारी (Binary) समतुल्य (equivalent) है:
(a) 1101 (b) 1001
(c) 1111 (d) 1011

EMRS PGT C.S. 16.12.2023

Ans, (d) : To convert the decimal number 11 to binary. you can use the method of successive division by 2.

2	11	
2	5	1
2	2	1
	1	0

$(11)10 = (1011)_2$

39. The decimal representation for the character 'a' (lower A) in ASCII is _____.
ASCII में करेक्टर 'a' (lower A) के लिए दशमलव निरूपण (decimal representation) है:
(a) 90 (b) 97
(c) 122 (d) 65

EMRS PGT C.S. 16.12.2023

Ans, (b) : The decimal representation for the characters 'a' (lower a) in ASCII is 97.

40. Decimal equivalent of the hexadecimal number $(ADD)_{16}$ is:
षोडशआधारी संख्या (hexadecimal number) $(ADD)_{16}$ का दशमलव समतुल्य (Decimal equivalent) है:

Sarkari Resource

(a) 2871 (b) 2178
(c) 2187 (d) 2781

EMRS PGT C.S. 16.12.2023

Ans, (d) : To find the decimal equivalent of the hexadecimal number $(ADD)_{16}$. Hexadecimal is a base 16 number system meaning it uses 16 symbols to represent numbers from 0 to 9 and them A to F.
The hexadecimal number $(ADD)_{16}$ can be expanded as follows.

$(ADD)16 = (A*16^2) + (D*16^1) + (D*16^0)$
$= (10*16^2) + (13*16^1) + (13*16^0)$
$= (10\times256) + (13\times16^1) + (13\times16^0)$
$= 2560 + 208 + 13$

So, the decimal equivalent of the hexadecimal number $(ADD)_{16}$ is 2781.

41. A and B are two 8-bit numbers such that $A + B \le 2^8$. The number of possible combination of A and B is
A और B दो 8 बिट की ऐसी संख्याएँ है, जहाँ $A + B \le 2^8$. A और B के सम्भव कॉम्बिनेशन की संख्या होगी

(a) 2^9 (b) 2^8
(c) 2^{16} (d) $2^4 - 1$

LT. Grade C.S. 2018

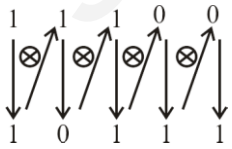
Ans, (c) : A and B are both 8-bit numbers, so each has 2^8 possible values.
To satisfy the condition $A + B \le 2^8$, we need to count all possible combinations of A and B. Since A and B are both 8-bit numbers, the total number of combinations of A and B is $2^8 \times 2^8 = 2^{16}$.
Therefore, the correct answer is 2^{16}.

42. The binary equivalent of the Gray code 11100 is /ग्रे कोड 11100 का बाइनरी समकक्ष है

(a) 10111 (b) 00111
(c) 01011 (d) 10101

LT. Grade C.S. 2018

Ans, (a) : To convert the gray code 11100 to binary.
1. The most significant bit (MSB) of the binary code is the same as the MSB of the gray code.
2. Each subsequent binary bit is obtained by XORing the previous binary bit with the corresponding gray code bit.



Thus, the binary equivalent of gray code 11100 is 10111.

43. Convert the binary number 1011.1101 to its decimal equivalent. Which of the following is the correct decimal value?/बाइनरी संख्या 1011.1101 को उसके दशमलव समकक्ष में बदलें। निम्नलिखित में से कौन-सा सही दशमलव मान है?

(a) 11.75
(b) 12.125
(c) 13.5
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (e) : Converting binary number (1011·1101) into decimal:

1	0	1	1	·	1	1	0	1
↓	↓	↓	↓		↓	↓	↓	↓
2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	2^{-4}

Integer part $= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
$= (8 + 0 + 2 + 1)$
$= 11$

Fractional part $= 1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$
$= \frac{1}{2} + \frac{1}{4} + 0 + \frac{1}{16}$
$= \frac{1}{2} + \frac{1}{4} + \frac{1}{16}$
$= 0{\cdot}50 + 0{\cdot}25 + 0{\cdot}0625$
$= 0{\cdot}8125$

Combine integer and fractional part, we get (11·8125)
So, the correct option is (e) None of the above

44. Convert the decimal number 1234 to binary, octal, and hexadecimal. Which of the following options correctly represents these conversions?/दशमलव संख्या 1234 को बाइनरी, ऑक्टल और हेक्साडेसिमल में बदलें।
निम्नलिखित में से कौन-सा विकल्प इन रूपांतरणों को सही ढंग से दर्शाता है?

(a) Binary: 10011010010, Octal: 2322, Hexadecimal: 4D2/बाइनरी : 10011010010, ऑक्टल : 2322, हेक्साडेसिमल : 4D2
(b) Binary: 11001110010, Octal: 1712, Hexadecimal: 72A /बाइनरी : 11001110010, ऑक्टल : 1712, हेक्साडेसिमल : 72A
(c) Binary: 11100110010, Octal: 1642, Hexadecimal: 3CD/बाइनरी : 11100110010, ऑक्टल : 1642, हेक्साडेसिमल : 3CD
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans, (a) :

$(1234)_{10} = ()_2$

2	1234	
2	617	0
2	308	1
2	154	0
2	77	0
2	38	1
2	19	0
2	9	1
2	4	1
2	2	0
	1	0

$(10011010010)2 = (1234)_{10}$

$(1234)_{10} = ()_8$

8	1234	
8	154	2
8	19	2
	2	3

$(1234)_{10} = (2322)_8$

$(1234)_{10} = ()_{16}$

16	1234	
16	77	2
16	4	13
	0	4

$(1234)_{10} = (4D_2)_{16}$

Sarkari Resource

45. The octal number $(651.124)_8$ is equivalent to _____.
ऑक्टल संख्या $(651.124)_8$ किसके बराबर है _____

(a) $(1A9.2A)_{16}$
(b) $(1B0.l0)_{16}$
(c) $(1A8.A3)_{16}$
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : Converting octal number $(651.124)_8$ into binary.
$(651.124)_8 = (110\ 101\ 001 \cdot 001\ 010\ 100)_2$
Now, convert this binary number into hexadecimal by grouping 4.
$(\underline{0001}\ \underline{1010}\ \underline{1001}.\ \underline{0010}\ \underline{1010}\ \underline{0000})_2$
$= (1A9 \cdot 2A)_{16}$
{ ∵ 10 is written by A in hexadecimal}

46. An n bit binary number, represented in sign-magnitude representation has a range?
साइन मैग्निट्यूड रूप में प्रदर्शित एक n बिट के बाइनरी नंबर की रेंज होती है?

(a) $-(2^n - 1)$ to $+(2^n - 1)$
(b) $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
(c) -2^n to $+(2^n - 1)$
(d) -2^{n-1} to $+(2^n - 1)$

RPSC C.S.-I 2016

Ans, (b) :
1. Sign - magnitude representation–
• In sign magnitude, the first bit (most significant bit) is the sign bit
• 0 = Positive number
• 1 = Negative number
• The remaining (n–1) bits represent the magnitude (the absolute value)
2. • Total number of different bit patterns 2^n
• But both +0 and –0 exist (unlike 2's complement), so zero is duplicated.
3. The range–
For an n–bit sign - magnitude number.
• Positive range
• From 0 (all zeros) to $+(2^{n-1} - 1)$
• Negative range
• From –0 to $-(2^{n-1} - 1)$
Ex- (if n = n – bits):
• Positive : 0000 (0) to 0111 (+7)
• Negative : 1000 (–0) to 1111 (–7)
Range of an n–bit sign magnitude binary number is:
$-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$

47. In 4 bit two's (2's) complement notation, addition of + 3 and –4 result into:
4 बिट 2's कॉम्प्लीमेंट रूप में +3 एवं –4 का योग करने का परिणाम प्राप्त होता है?

(a) 1001 (b) 1010
(c) 1100 (d) 1111

RPSC C.S.-I 2016

Ans, (d) : Step-1– Represent +3 in 1–bit 2's complement.
• +3 to binary (normal) = 0011
Step-2– Represent –4 in 4–bit two's complement
• +4 in binary = 0100
• To get –4, take the two's complement
1. invert the bits - 0100 → 1011
2. Add1: 1011 + 1 = 1100
–4 = 1100
Step-3– Add them

```
  0011  (+3)
+ 1100  (-4)
------
  1111
```

Step-4– Interpret the result
• The result is 1111
• In 4–bit two's complement, 1111 = –1
(Because it's negative : leading bit is 1)
1111

48. What is the value of IEEE 754 single precision floating point number represented by binary bit string given below: 0100 0000 0110 0000 0000 0000 0000 0000
दी गयी बाइनरी बिट श्रृंखला द्वारा प्रदर्शित IEEE 754 एकल प्रिसिजन फ्लोटिंग प्वाइंट संख्या का मान है: 0100 0000 0110 0000 0000 0000 0000 0000

(a) 3.5 (b) –3.5
(c) 4.8 (d) –4.8

RPSC C.S.-I 2016

Ans, (a) :
Step-1– Break into IEEE 754 parts for single precision (32- bit IEEE 754)
• 1 bit : Sign bit (S)
• 8 bits : Exponent (E)
• 23 bits : Fraction / Mantissa (M)
Let's split–

Part	Bits
Sign (S)	0
Exponent (E)	10000000 (binary)
Mantissa (M)	11000000000000000000000 (23bits)

Interpret each part.
1. Sign bit (S)–
• 0 → positive number
2. Exponent (E)–
• 10000000 (binary) = 128 (decimal)
• IEEE 754 uses a bias or 127 for single precision
• So, the actual exponent = 128 – 127 = 1
3. Mantissa (M)–
• 11000000000000000000000
• This is function part
• IEEE 754 assumes a leading 1 before the binary point (normalized form) 1.11000000000000000000000
• This = 1 + 1 × (1/2) + 1 × (1/4)
1 + 0.5 + 0.25 = 1.75
Step-3– Assemble the value
$= (-1)^5 \times (1.m) \times 2^{(E\text{-bias})}$
Plug in the numbers:
$= (-1)^0 \times (1.75) \times 2^1 = 1 \times 1.75 \times 2 =$ **3.5**

49. Given the following equation : $(142)_b + (112)_{b-2} = (75)_8$, find base b.

(a) 3 (b) 6
(c) 7 (d) 5

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (d) : $(142)_b + (112)_{b-2} = (75)_8$
$b^2 + 4 \times b + 2 + 1 \times (b-2)^2 + 1 \times (b-2) + 2 = 7 \times 8 + 5$
$2b^2 + b + 6 = 61$
$2b^2 + b - 55 = 0$
$2b^2 + 11b - 10b - 55 = 0$
$b(2b + 11) - 5(2b + 11) = 0$
$(b - 5)(2b + 11) = 0$
$\therefore b = 5$

Alternate :

Option 1 : b = 3
It has base 3 which is not possible because (142) having a digit 4 cannot exist in base 3

Option 2 : b = 6
$(142)_6 = (62)_{10}$
$(112)_4 = (22)_{10}$
$62 + 22 = (84)_{10}$
$(75)_8 = (61)_{10}$
Therefore, 6 is not base

Option 3 : b = 7
$(142)_7 = (79)_{10}$
$(112)_5 = (32)_{10}$
$79 + 32 = (111)_{10}$
$(75)_8 = (61)_{10}$
Therefore, 7 is not base

Option 4 – b = 5
$(142)_5 = (47)_{10}$
$(112)_3 = (14)_{10}$
$47 + 14 = (61)_{10}$
$(75)_8 = (61)_{10}$
Therefore, 5 is the base.
Hence, $(142)_5 + (112)_3 = (75)_8$

50. Which of the following binary codes for decimal digits are self complementing?
(A) 8421 code
(B) 2421 code
(C) excess-3 code
(D) excess-3 gray code
Choose the correct option:
(a) (A) and (B) (b) (B) and (C)
(c) (C) and (D) (d) (D) and (A)

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) : A binary coded decimal code in which of the combination for the complement of a digit is the complement of the combination for that digit is called a Self-complementing codes are : excess-3 code and 2421 code.

(1) 2421 code- It is positive weighted code values in binary terms for each digit from 0 to 9 are given as:

Decimal Digit	2421 code
0	0000
1	0001
2	0010
3	0011
4	0100
5	1011
6	1100
7	1101
8	1110
9	1111

(2) Excess-3 code: It is a non-weighted code and self complementary code. It is derived from 8421 code by adding 3 to each digit.

Decimal Digit	Excess – 3 code
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

51. What is the radix of the numbers if the solution to the quadratic equation x^2 - 10x + 26 = 0 is x = 4 and x = 7 ?
(a) 8 (b) 9
(c) 10 (d) 11

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Ans. (d) : If one solve the quadratic equation, $ax^2 + bx + c$, then the sum of root of quadratic equation $= -b/a$
Quadratic equation $x^2 - 10x + 26 = 0$
Since, 4 and 7 are the root of the equation

$$-b/a = \frac{(10)r}{(1)r} = (4)r + (7)r$$

$r = 4 + 7 = 11$

52. The Octal equivalent of hexadecimal $(D.C)_{16}$ is:
(a) $(15.6)_8$ (b) $(61.6)_8$
(c) $(15.3)_8$ (d) $(61.3)_8$

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Ans. (a) : The Octal equivalent of hexadecimal $(D.C)_{16}$ is $(15.6)_8$.

53. In computers, subtraction is generally carried out by
(a) 1's complement (b) 10's complement
(c) 2's complement (d) 9's complement

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Ans. (c) : In computers, subtraction is generally carried out by 2's complement.

54. Perform the following operation for the binary equivalent of the decimal numbers $(-14)_{10} + (-15)_{10}$
The solution in 8 bit representation is :
(a) 11100011 (b) 00011101
(c) 10011101 (d) 11110011

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Ans. (a) : $(-14)_{10} + (-15)_{10}$
Now, 29 in binary (8 bit) is 00011101. But the number is negative. Take its one's compliements 11100010 and then add 1 to it Resulting in
i.e. – 11100011.

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55. The hexadecimal equivalent of the binary integer number 110101101 is :

(a) D24 (b) 1 B D
(c) 1 A E (d) 1 A D

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Ans. (d) : Group the given number in subgroup of 4 from right and assign alphabet when it exceeds 1001:

1010 – A 1110 – E
1011 – B 1111 – F
1100 – C
1101 – D

Given number is 110101101

1101 – D
1010 – A
1 – 1

Hexadecimal equivalent is – 1AD.

56. Which of the following is a non-positional number system?
निम्नलिखित में से कौन सी एक नॉन-पोजिशन संख्या प्रणाली है?

(a) Decimal/डेसिमल (b) Roman/रोमन
(c) Octal/ऑक्टल (d) Binary/बाइनरी

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Ans. (b) : A non-positional system is one in which a value of a symbol does not change with its position.
e.g. Roman numerals
Decimal, Binary and Octal are positional number system where value depends on its place.

57. How many bits are required to represent $(1000)_{10}$ in BCD code?
BCD कोड में $(1000)_{10}$ को दर्शाने के लिए कितने बिट्स की आवश्यकता होती है?

(a) 16 bits (b) 10 bits
(c) 32 bits (d) 4 bits

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Ans. (a) : BCD (Binary Coded Decimal): Each decimal digit is written separately using 4 binary bits.
For example:- the given number is $(1000)_{10}$
It has 4 deciaml digits: 1,0,0,0
Each digit represents 4 binary bits in BCD.
So, total number of bits required in BCD= 4×4=16

58. What will be the quotient when $(100001)_2$ is divided by $(110)_2$?
भागफल क्या होगा जब $(100001)_2$ को $(110)_2$ से विभाजित किया जाता है?

(a) $(101)_2$ (b) $(110)_2$
(c) $(001)_2$ (d) $(100)_2$

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Ans. (a) : Number $(100001)_2$ converted into decimal system
$(100001)_2 = 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = (33)_{10}$
Number $(110)_2$ converted into decimal system $(110)_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = (6)_{10}$
After dividing
$= \frac{33}{6} = 5.5 \approx 5$
converting $(5)_{10}$ into binary
$\frac{5}{2} = 2$, remainder = 1
$\frac{2}{2} = 1$, remainder = 0
$\frac{1}{2} = 0$, remainder = 1
$= (101)_2$

59. If $(X567)_8 + (2YX5)_8 = (71YX)_8$ then the value of X and Y are
अगर $(X567)_8 + (2YX5)_8 = (71YX)_8$ तो X और Y के मान (वैल्यू) हैं

(a) 4, 3 (b) 3, 3
(c) 4, 4
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : converting these octal numbers into decimal number
$(X567)_8 = (X \times 8^3 + 5 \times 8^2 + 6 \times 8^1 + 7 \times 8^0)_{10}$
$= (512X + 320 + 48 + 7)_{10}$
$= (512X + 375)_{10}$
$(2YX5)_8 = (2 \times 8^3 + Y \times 8^2 + X \times 8^1 + 5 \times 8^0)_{10}$
$= (1024 + 64Y + 8X + 5)_{10}$
$= (8X + 64Y + 1029)_{10}$
$(71YX)_8 = (7 \times 8^3 + 1 \times 8^2 + Y \times 8^1 + X \times 8^0)_{10}$
$= (3584 + 64 + 8Y + X)_{10}$
$= (X + 8Y + 3648)_{10}$
Substituting these values in equation
$(512X + 375)_{10} + (8X + 64Y + 1029)_{10} = (X + 8Y + 3648)_{10}$
$(520X + 64Y + 1404)_{10} = (X + 8Y + 3648)_{10}$
Or $520X + 64Y + 1404 = (X + 8Y + 3648)$
$519X + 56Y = 2244$
From option (a) X = 4 , Y = 3
$519 \times 4 + 56 \times 3 = 2076 + 168 = 2244$
So, option (a) is correct.
From option (b) X = 3, Y = 3
$519 \times 3 + 56 \times 3 = 1557 + 168 = 1725$, not satisfies
So, option (b) is not correct.
From option (c) X = 4, Y = 4
$519 \times 4 + 56 \times 4 = 2076 + 224 = 2300$, not satisfies
So, option (c) is not correct.
So, correct value of X = 4 and Y = 3

60. The range of the number that can be represented in 8-bit using 2's complement representation is
2's कॉम्प्लिमेन्ट (Complement) रिप्रजेन्टेशन का प्रयोग करके 8 बिट में रिप्र जेन्ट होने वाली संख्याओं का रेन्ज है

(a) –128 to +128/ –128 से +128
(b) –127 to +127/ –127 से +127
(c) –128 to +127/ –128 से +127
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : The range of bit word in 2's complement representation is,
(-2^{n-1}) to $(2^{n-1} - 1)$
On putting n = 8
Lower bound: $-2^{8-1} = 2^7 = -128$
Upper bound: $2^{8-1} - 1 = 2^7 - 1 = 128 - 1 = 127$
Range: (-2^{8-1}) to $(2^{8-1}-1)$
So the range is: –128 to 127.

61. If $(11X1Y)_8 = (12C9)_{16}$, then the values of X and Y are
अगर $(11X1Y)_8 = (12C9)_{16}$, हो, तो X और Y के मान (वैल्यू) हैं
(a) 5, 1 (b) 5, 7
(c) 3, 1
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
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Ans. (c): Converting $(12C9)_{16}$ into decimal number

$$(12C9)_{16} = (1\times16^3 + 2\times16^2 + 12\times16^1 + 9\times16^0)_{10}$$
$$= (1\times4096 + 2\times256 + 12\times16 + 9)_{10}$$
$$= (4096 + 512 + 192 + 9)_{10}$$
$$= (4809)_{10}$$

Converting $(11X1Y)_8$ into decimal number

$$(11X1Y)_8 = (1\times8^4 + 1\times8^3 + X\times8^2 + 1\times8^1 + Y\times8^0)_{10}$$
$$= (1\times4096 + 1\times512 + X\times64 + 8 + Y)_{10}$$
$$= (4096 + 512 + 64X + 8 + Y)_{10}$$
$$= (4616 + 64X + Y)_{10}$$

Equating both
$(4616 + 64X + Y)_{10} = (4809)_{10}$
We obtain
$4616 + 64X + Y = 4809$
$64X + Y = 193$
From option (a) x = 5, y = 1
64 × 5 + 1 = 320 + 1 = 321 no satisfies
So, option (a) is not correct
From option (b) x = 5, y = 7
64 × 5 + 7 = 320 + 7 = 327 not satisfies
So, option (b) is not correct.
From option (c) . x = 3, y = 1
64 × 3 + 1 = 192 + 1 = 193 satisfies
So, option (c) x = 3, y = 1 is correct

62. Which of the following 4-bit numbers equal its 2's complement?
निम्नांकित 4 - बिट नम्बरों में से कौन-सा अपने 2's कॉम्प्लिमेन्ट के बराबर है?
(a) 1010
(b) 0101
(c) 1000
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans. (c) : 2's Complement: 2's complement is a method to represent negative number in binary. It allows easy addition and subtraction of signed number in digital circuits. To find 2's complement of a binary number, first find, 1's complement (invert all bits: 0 → 1, 1 → 0) and then add 1 to the result.
Option (a): 1010
1's complement = 1010
$\underline{\quad\quad +1}$
2's complement = 1011 ; so option (a) is not correct.
Option (b) : 0101
1's complement = 1010
$\underline{\quad\quad +1}$
2's complement = 1011 ; so option (a) is not correct.
Option (c) : 1000
1's complement = 0111
$\underline{\quad\quad +1}$
2's complement = 1000 ; so option (c) 1000 is equal to it 2's complement.

63. The number of 1's in the binary representation of the value of the decimal expression of $16^3 \times 9 + 16^2 \times 7 + 16 \times 5 + 3$ is
डेसिमल एक्सप्रेशन $16^3 \times 9 + 16^2 \times 7 + 16 \times 5 + 3$ के मान (वैल्यू) को बाइनरी रिप्रजेन्टेशन में बदलने पर 1 की संख्या कितनी होगी?
(a) 15
(b) 9
(c) 12
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
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Ans. (b) : Decimal expression = $16^3 \times 9 + 16^2 \times 7 + 16 \times 5 + 3$
$= (2^4)^3 (8 + 1) + (2^4)^2 (4 + 2 + 1) + 2^4 (4 + 1) + (2 + 1)$
$= 2^{12} (2^3 + 2^0) + 2^8 (2^2 + 2^1 + 2^0) + 2^4 (2^2 + 2^0) + (2^1 + 2^0)$
$= 2^{15} + 2^{12} + 2^{10} + 2^9 + 2^8 + 2^6 + 2^4 + 2^1 + 2^0$
In binary representation, any number which is represented by power of 2, the number of 1 in that number is 1.
Here total number of term = 9
Hence, total number of 1 in this decimal expression = 9.

64. Match the List-I with List-II

	List-I		List-II
A.	$(100110011)_2$	I.	Error detection code
B.	Complements	II.	CISC
C.	Parity Bits	III.	Negative number
D.	Variable length Instruction	IV.	$(133)_{16}$

Choose the correct answer from the options given below :
(a) A-IV, B-I, C-III, D-II
(b) A-IV, B-III, C-I, D-II
(c) A-IV, B-II, C-III, D-I
(d) A-I, B-IV, C-II, D-III
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Ans. (b) : The correct match is-

List-I			List-II
A.	$(100110011)_2$	–	$(133)_{16}$
B.	Complements	–	Negative number
C.	Parity Bits	–	Error detection code
D.	Variable length Instruction	–	CISC

65. The hamming distance between 10101 and 11110 is.

(a) 2 (b) 3
(c) 4 (d) 5

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Ans. (b) : The Hamming distance is calculated by comparing each corresponding bit of two binary numbers and counting the position where they differ.

Given,

First number – 10101
Second number – 11110

Comparing bit by bit-

1 vs 1 → same
0 vs 1 → Different (Count = 1)
1 vs 1 → same
0 vs 1 → Different (count = 2)
1 vs 0 → Different (count = 3)

Therefore, the Hamming distance is 3.

66. A code in which only one bit changes between successive numbers is known as _____ code.
वह कोड, जिसमें क्रमागत संख्याओं के बीच केवल एक बिट परिवर्तित होती है, _____ कोड के रूप में जाना जाता है।

(a) ASCII (b) HAMMING
(c) GRAY (d) BCD

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Ans. (c) : GRAY Code– the binary numbers system can be arranged in the reflected binary code, also known as the gray code, so that two subsequent values only differ in one bit (binary digit). In the typical sequence of binary numbers produced by the hardware that could provide an error or ambiguity during the change from one number to the next, gray codes are highly helpful. because the first (n/2) values compare with the last (n/2) values in reverse order, gray code is also known as reflected binary code.

Conversion of Gray Code–

Gray code and binary conversion are essential for understanding and resolving issues in digital systems.

Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100

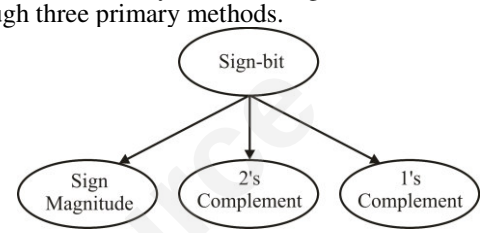
67. Which of the following pair regarding sign bit is correct?
निम्नलिखित में से कौन सा/से कथन युग्म चिन्ह(साइन) बिट के संदर्भ में सही है/हैं?

I. 0 – positive/पॉजिटिव
II. 1 – negative/नेगेटिव

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

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Ans. (c) : Sign bit– Sign bit represens both positive and negative values in computing by allocating one bit (Typically the MSB) as a sign indicator. In binary representation, they enable negative value storage through three primary methods.



1. Sign Magnitude– This method uses the left most bit as a sign flag (0 for Positive, 1 for negative) while remaining bits store the absolute value. It creates two zero representations (+0 and –0) and complicates arithmetic operations due to separate sign handling.

- 00000101 = (Sign bit 0, magnitude 5)
- 10000101 = –5 (sing bit 1, magnitude 5)

2. 1's Complement– In 1's complement negative numbers are created by flipping all bits of the positive counterpart the leftmost bit still indicates sign (0 for positive, 1 for negative) this method also suffers from dual zero representations (0000 as to and 1111 as –0) but it simplifies subtractions through bit inversion.

- 0000101 = +5
- 11111010 = –5 (invert all bits of +5)

3. 2's Complement– 2's complements forms negative by inverting all bits of the positive value and adding 1. The leftmost bit serves as the sign indicator while enabling single zero representation. It simplifies hardware design by allowing identical addition and substraction operation

2's Complement
= 1' Complement +1
= (Flipping of bits forget) 1's Complement) +1
00000101 = +5
11111011 = –5 Convert 00000101.
→ 11111010, add 1 → 11111011

68. If $(21121)_3 = (C)_4$, then what is the value of C?
यदि $(21121)_3 = (C)_4$, तो C का मान क्या है?

(a) $(2023)_4$ (b) $(3032)_4$
(c) $(2112)_4$ (d) $(3031)_4$

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Ans. (d) :
Step-1
Convert $(21121)_3$ to Base 10
$(21121)3 = 2 \times 3^4 + 1 \times 3^3 + 1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0$
Calculate each term–
$\Rightarrow 3^4 = 81$, So $2 \times 81 = 162$
$\Rightarrow 3^3 = 27$, So $1 \times 27 = 27$

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$\Rightarrow 3^2 = 9$, So $1 \times 9 = 9$
$\Rightarrow 3^1 = 3$, So $2 \times 3 = 6$
$\Rightarrow 3^0 = 1$, So $1 \times 1 = 1$
Sum the values–
$162 + 27 + 9 + 6 + 1 = 205$
$(21121)_3 = (205)_{10}$
Step-2
Convert $(205)_{10}$ to Base 4
$\Rightarrow 205 \div 4 = 51$ remainder 1($51 \times 4 = 204$, and $205 - 204 = 1$)
$\Rightarrow 51 \div 4 = 12$ remainder 3 (since $12 \times 4 = 48$ and $51 - 48 = 3$)
$\Rightarrow 12 \div 4 = 3$ remainder 0 (since $3 \times 4 = 12$ and $12 - 12 = 0$)
$\Rightarrow 3 \div 4 = 0$ remainder 3 (since $0 \times 4 = 0$ and $3 - 0 = 3$
Verify the Conversion–
$(3031)_4$ back to base 10
$(3031)_4 = 3 \times 4^3 + 0 \times 4^2 + 3 \times 4^1 + 1 \times 4^0$
$\Rightarrow 4^3 = 64$, So $3.64 = 192$
$\Rightarrow 4^2 = 16$, So $0 \times 16 = 0$
$\Rightarrow 4^1 = 4$, So $3 \times 4 = 12$
$\Rightarrow 4^0 = 1$, So $1 \times 1 = 1$
Sum : $192 + 0 + 12 + 1 = 205$
Confirm :- $(21121)_3 = (3031)_4$

69. 1 picoseconds = _____:
1 पिकोसेकंड = _____:
(a) 10^{-9} seconds/10^{-9} सेकंड
(b) 10^{-12} seconds/10^{-12} सेकंड
(c) 10^{-15} seconds/10^{-15} सेकंड
(d) 10^{-18} seconds/10^{-18} सेकंड

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Ans. (b) : A picosecond is a unit of time in the International system of units (SI) equal to one trillionth of a second, that is 10^{-12} seconds.

70. Which of the following statement (s) is/are correct?
निम्नलिखित में से कौन सा/से कथन सही है/हैं ?
I. $(4052)_7 = (1409)_{10}$
I. $(1715)_{10} = (BAB)_{12}$
(a) Only I /केवल I
(b) Only II / केवल II
(c) Both I and II / I तथा दोनो II
(d) Neither I nor II /ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Statement I: $(4052)_7 = (1409)_{10}$ converting $(4052)_7$ into decimal:
$(4052)_7 = (4 \times 7^3 + 0 \times 7^2 + 5 \times 7^1 + 2 \times 7^0)_{10}$
$= (4 \times 343 + 0 + 35 + 2)_{10}$
$= (1372 + 37)_{10}$
$= (1409)_{10}$
So, Statement I is correct
Statement II: $(1715)_{10} = (BAB)_{12}$
In $(BAB)_{12}$, A and B represents 10 and 11 respectively.
Convert $(BAB)_{12}$ into decimal:
$(BAB)_{12} = (B \times 12^2 + A \times 12^1 + B \times 12^0)_{10}$
$= (11 \times 144 + 10 \times 12 + 11)_{10}$
$= (1584 + 120 + 11)_{10}$
$= (1715)_{10}$
So, statement II is also correct.
So, both the statements are correct.

71. What is the full form of EBCDIC?
ई.बी.सी.डी.आई.सी का असंक्षिप्त रुप क्या है ?
(a) External Binary Coded Digit Interchange Code
एक्सटर्नल बाइनरी कोडेड डिजिट इंटरचेंज कोड
(b) External Bit Code Digit Internal Code
एक्सटर्नल बीट कोड डिजिट इंटर्नल कोड
(c) External Bit Code Decimal Interchange Code
एक्सटर्नल बीट कोड डिमिसल इंटर्नल कोड
(d) Extended Binary Coded Decimal Interchange Code
एक्स्टेंडेड बाइनरी कोडेड डेसिमल इंटरचेंज कोड

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Ans. (d) : EBCDIC stands for Extended Binary Coded Decimal Interchange Code.
EBCDIC is a 8-bit character encoding developed by IBM in the early 1960s.

72. What is the value of $(1715)_{10}$ in base 12?
बेस 12 में $(1715)_{10}$ का मान क्या है
(a) $(BAA)_{12}$ (b) $(ABA)_{12}$
(c) $(BCB)_{12}$ (d) $(BAB)_{12}$

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Ans. (d) : Step-by step conversion confirmed:
- $1715 \div 12 = 142$ rem 11 (B)
- $142 \div 12 = 11$ rem 10 (A)
- $11 \div 12 = 0$ rem 11 (B)

So, $1715_{10} = (BAB)_{12}$.
Thus, the value is $(BAB)_{12}$, which is option (d)

73. If b is the base of the positional number system. What is the range of symbols it consists of?
यदि b स्थितीय संख्या प्रणाली का बेस है। इसके सिबंल्स की रेंज क्या है?
(a) 0 to b (b) 1 to b
(c) 0 to (b – 1) (d) 1 to (b – 1)

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : For a number system with base b, the valid digits begin at 0 and end at (b –1).
So, the symbols range from 0 through (b – 1).

74. How many bits are required to represent any decimal number in the range from 0 to 2^n–1 in base2 ?
बेस 2 में 0 से 2^n–1 के बीच किसी भी दशमलव संख्या को दर्शाने के लिए कितने बिट्स की आवश्यकता होती है?
(a) n (b) n –1
(c) 2^n –1 (d) n+1

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Ans. (a) : To cover all numbers from 0 up to 2^n –1 in base 2, we need n binary digits (bits), because n bits give 2^n unique patterns.
So, the correct is option (a) n

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75. The binary equivalent of $(25.625)_{10}$ is _____.
$(25.625)_{10}$ का बाईनरी समकक्ष _____ है।

(a) $(10001.11)_2$ (b) $(11101.001)_2$
(c) $(11011.100)_2$ (d) $(11001.101)_2$

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Ans. (d) : To convert the decimal number 25.625 to binary we can split it into its integer and fractional parts and convert each separately.
(1)

2	25	
2	12	1
2	6	0
2	3	0
	1	1

25 in binary is 11001.
(2) Convert the fractional part (0.625) to binary Integer part
$0.625 \times 2 = 1.250$ 1
$0.25 \times 2 = 0.50$ 0
$0.5 \times 2 = 1.00$ 1
Reading the integer parts from top to bottom, 0.625 in binary is 0.101 combine the integer and fractional parts: $(25.625)_{10}$ is $(11001.101)_2$

76. What is minuend value in the following?
subtract $(011100)_2$ from $(1011100)_2$?
निम्नलिखित में मिन्यूएन्ड मान क्या है?
$(011100)_2$ से $(1011100)_2$ घटाना?

(a) $(111000)_2$ (b) $(1011100)_2$
(c) $(100100)_2$ (d) $(100101)_2$

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Ans. (b) : The minuend is the number from which another number is subtracted in binary subtraction.
Minuend value is $(1011100)_2$.

77. What is the value of product of $(10110)_2$ and $(10011)_2$?
$(10110)_2$ तथा $(10011)_2$ के गुणनफल का मान क्या है?

(a) $(110110101)_2$ (b) $(110100010)_2$
(c) $(110100001)_2$ (d) $(110100100)_2$

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Ans. (b) :

$$\begin{array}{r} 10110 \\ \times\ 10011 \\ \hline 10110 \\ 10110 \\ 1011000 \\ \hline 110100010 \end{array}$$

78. What is the value of $(3564)_8 + (323232)_4$?
$(3564)_8 + (323232)_4$ का मान क्या है?

(a) $(IE62)_{16}$ (b) $(1662)_{16}$
(c) $(1762)_{16}$ (d) $(1562)_{16}$

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Ans. (b) :
$= (323232)_4$
$= 3 \times 4^5 + 2 \times 4^4 + 3 \times 4^3 + 2 \times 4^2 + 3 \times 4^1 + 2 \times 4^0$
$= 3072 + 512 + 192 + 32 + 12 + 2$
$= (3822)_{10}$
$= (3564)_8$
$= 3 \times 8^3 + 5 \times 8^2 + 6 \times 8^1 + 4 \times 8^0$
$= 1536 + 320 + 48 + 4$
$= (1908)_{10}$
$= (1908)_{10} + (3822)_{10}$
$= (5730)_{10}$

16	5730	
16	358	2
16	22	6
	1	6

$= (1662)_{16}$

79. What is the 2's complement of $(-28)_{10}$ in 16 bit representation?
16 बिट रिप्रजेंटेशन में $(-28)_{10}$ का 2's कॉम्प्लीमेंट क्या है?

(a) 1000000000011100
(b) 1111111111100011
(c) 1111111111100100
(d) 1111111111100101

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Ans. (c) : To find the 2's complement of –28 in a 16-bit representation–
(i) Convert the absolute value to binary
$28_{10} = 11100_2$
(ii) Pad the binary number to 16 bits
$28_{10} = 0000000000011100_2$
(iii) Find the 1's complement (Invert all the bits)
(change 0s to 1s and 1s to 0s)
1111111111100011
(iv) Add 1 to the first complement to get the 2's complement

$$\begin{array}{r} 1111111111100011 \\ +\qquad\qquad 1 \\ \hline 1111111111100100 \end{array}$$

Therefore, the 2's complement of –28 in 16-bit representation is 1111111111100100.

80. What is 6's complement of $(1543)_7$ of the computer?
कंप्यूटर के $(1543)_7$ का 6's कॉम्प्लीमेंट क्या है ?

(a) $(6132)_7$ (b) $(5132)_7$
(c) $(5123)_7$ (d) $(5432)_7$

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Ans. (c) : The 6's complement of a number in base 7
(1) Determine the largest digit in base 7:
In base 7, digit range from 0 to 6.
(2) Subtract each digits from 6:
Given the number $(1543)_7$, we subtract each digit from 6:

$$\begin{array}{r} 6666 \\ -\ 1543 \\ \hline 5123 \end{array}$$

So, the 6's complement of $(1543)_7$ is $(5123)_7$.

81. What is the Hamming distance between 1101 and 110?
1101 और 110 के बीच हैमिंग दूरी क्या है?

(a) 2 (b) 3
(c) 4 (d) 0

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Ans. (b) : Hamming distance is a metric for comparing two binary strings of equal length, hamming distance is the number of bit positions in which the two bits are different.
Hamming distance can bed calculated by taking XOR of two code words.
XOR operation:

$$\begin{array}{r} 1\ 1\ 0\ 1 \\ +\ 0\ 1\ 1\ 0 \\ \hline 1\ 0\ 1\ 1 \end{array}$$

The number of 1's in XOR result is 3.
So, Hamming distance is 3.

82

सामान्यीकृत साइंटिफिक (वैज्ञानिक) नोटेशन का उपयोग करके $(0.10101)_2$ को कैसे निरूपित किया जाएगा?

(a) 0.010101×2^{1} (b) 0.10101×2^{0}
(c) 1.0101×2^{-1} (d) 10.101×2^{-2}

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Normalized scientific notation requires the radix point to be placed after the first non-zero digit. For a binary number, there is the first '1'
For $(010101)_2$, move the binary point one place to the right to get 1.0101. Since the point move right, the exponent is negative, and its magnitude is the number of places moved which is '1'

83. Which of the following is the base-4 representation of the decimal number 213?
दशमलव संख्या 213 का बेस-4 का रिप्रेजेंटेशन (प्रतिनिधित्व) निम्नलिखित में से कौन सा है?

(a) $(1111)_4$ (b) $(3113)_4$
(c) $(3133)_4$ (d) $(3111)_4$

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Ans. (d) : To find the base-4 representation of the decimal number 213, we can use repeated division by 4;
1. Divide 213 by 4 : 213÷4 = 53 with a remainder of '1'
2. Divide 53 by 4 : 53÷4 = 13 with a remainder of '1'
3. Divide 13 by 4 : 13÷ 4=3 with a remainder of '1'
4. Divide 3 by 4 : 3÷4 =0 with a remainder of 3.
Reading the remainders from bottom to top, the base -4 representation of 213 is $(3111)_4$

84. What is the 15's complement of the hexadecimal number $(C00D0)_{16}$?
हेक्साडेसिमल संख्या $(C00D0)_{16}$ में 15 के पूरक क्या होगा?

(a) $(2FF3F)_{16}$ (b) $(3FF2F)_{16}$
(c) $(3EE2E)_{16}$ (d) $(2EE3E)_{16}$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : 15s complement Explained: the 15s complement of a hexadecimal number is found by substracting each digit from 15(F in hexadecimal)
• Solution : Substract each digit of $(C00D0)_{16}$ from $(FFFFF)_{16}$

$$\begin{array}{r} F\ F\ F\ F\ F \\ -\ C\ 0\ 0\ D\ 0 \\ \hline 3\ F\ F\ 2\ F \end{array}$$

A = 10
B = 11
C = 12
D = 13
E = 14
F = 15

Therefore, answer will be $(3FF2F)_{16}$

85. What is the 10's complement of the decimal number 20100?
दशमलव संख्या 20100 में 10 का पूरक क्या होगा?

(a) 79899 (b) 89999
(c) 79900 (d) 79800

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : 1. Find the 9's complement
The 9's complement of a number is found by substracting each digit from 9. For 20100, the 9's complement is : 99999 – 20100 = 79899
2. Find the 10's complement of 20100 is 79900.

86. If $(1101)_x = (241)_{16}$ then what is the value of x?
यदि $(1101)_x = (241)_{16}$ तो x का मान क्या होगा?

(a) 2 (b) 4
(c) 8 (d) 10

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Ans. (c) : $(1101)_x = (241)_{16}$
Step 1: Convert RHS to decimal
$(241)_{16} = 2\times16^2 + 4\times16^1 + 1\times16^0$
$= 2\times256 + 4\times16 + 1$
$= 512 + 64 + 1 = 577$
तो, $(241)_{16} = (577)_{10}$
Step 2: Convert LHS to decimal
$(1101)_x = 1\times x^3 + 1\times x^2 + 0\times x + 1$
$= x^3 + x^2 + 1$
Step 3: Make equation
$x^3 + x^2 + 1 = 577$
$x^3 + x^2 = 576$
Step 4: Check the value
On putting x = 8
$8^3 + 8^2 = 512 + 64 = 576$

87. If $(12?07)_8 = (1010110000111)_2$, then what is the missing digit in $(12?07)_8$?
यदि $(12?07)_8 = (1010110000111)_2$, तो $(12?07)_8$ में गुम अंक क्या है?

(a) 5 (b) 6
(c) 7 (d) 8

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Ans. (b) : We are given:
$(12\ ?\ 07)_8 = (1010110000111)_2$
Step 1:- Convert the binary to decimal
$(1010110000111)_2 = 4096 + 1024 + 256 + 128 + 4 + 2 + 1$
$= (5511)_{10}$
Step 2:- Write the octal number in decimal form
$(12\ ?\ 07)_8 = 1\cdot8^4 + 2\cdot8^3 + ?\cdot8^2 + 0\cdot8 + 7$
$= 4096 + 1024 + 64? + 7$
$= 5127 + 64?$

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Step 3:- Equate both sides

$5127 + 64? = 5511$

$64? = 384$

$? = \frac{384}{64} = 6$

88. Considering 2's complement representation for negative number, what is the decimal equivalent of an 8-bit register content 10000000?

ऋणात्मक संख्याओं के लिए 2 के कांप्लीमेंट रिप्रेजेंटेशन (पूरक प्रतिनिधित्व) कॉन्टेंट 10000000 का दशमलव समतुल्य क्या है?

(a) –0 (b) +0

(c) +128 (d) –128

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : The decimal equivalent of 8-bit register content $(10000000)_2$ in 2's complement representation is (d) - 128.

89. Identify the missing digits in the following expression so that left hand side is equal to the right hand side.

$(1010?)_2 + (?7)8 = (24)_{16}$

निम्नलिखित व्यंजक में गुम अंकों की पहचान करें ताकि बाएं पक्ष दाएं पक्ष के बराबर हो।

$(1010?)_2 + (?7)8 = (24)_{16}$

(a) 1.2 (b) 1, 1

(c) 1, 6 (d) 0, 1

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Ans. (b) : Converting binary number $(1010x)_2$ into decimal

$1010x = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + x \times 2^0$

$= (16 + 4 + x)_{10}$

$= (20 + x)_{10}$

Converting octal number $(y7)_8$ into decimal

$(y\ 7)_8 = y \times 8^1 + 7 \times 8^0$

$= (8y + 7)_{10}$

Converting Hexadecimal number $(24)_{16}$ into decimal

$(24)_{16} = 2 \times 16 + 4 \times 16^0$

$= 32 + 4$

$= (36)_{10}$

Equating–

$(20 + x)_{10} + (8y + 7)_{10} = (36)_{10}$

$20 + x + 8y + 7 = 36$

$x + 8y = 9$

By hit & trail method

Option (b) (1,1) is correct

90. Which of the following is the 2's complement of the binary number $(101100)_2$?

निम्नलिखित में से कौन सा बाइनरी नंबर $(101100)_2$ का 2 का पूरक है?

(a) $(010011)_2$ (b) $(101111)_2$

(c) $(010100)_2$ (d) $(111100)_2$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : We are asked : 2's complement of $(101100)_2$

Step 1 : Recall definition The 2's complement of a binary number = 1's complement + 1

Step 2 : Find 1's complement

$(101100)_2$

$101100 \rightarrow 010011$

Step 3: Add 1

$010011 + 1 = 010100$

Step 4: Match with option

2's complement = $(010100)_2$

91. The Excess-3 decimal code is self-complementing code because

(1) The binary sum of a code and its 9's complement is equal to 9

(2) It is a weighted code

(3) Complement can be generated by inverting each bit pattern

(4) The binary sum of a code and its 10's complement is equal to 9

(a) (1) (b) (2) and (3)

(c) (1) and (3) (d) All are correct.

UGC NET C.S. December-2014 (Paper-II)

Ans : (c) The Excess-3 decimal code is a self-complementing code because the binary sum of a code and its 9's complement is equal to 9 and complement can be generated by inverting each bit pattern.

92. For the 8-bit word 00111001, the check bits stored with it would be 0111. Suppose when the word is read from memory, the check bits are calculated to be 1101. What is the data word that was read from memory?

(a) 10011001 (b) 00011001

(c) 00111000 (d) 11000110

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Given 8 bit word 00111001. encoded in $d_8 = 0, d_7 = 0, d_6 = 1\ d_5 = 1, d_4 = 1\ d_3 = 0, d_2 = 0, d_1 = 1$

parity is calculated as 0111 which are written as $p_8p_4p_2p_1$

$p_1 = 1 \quad p_4 = 1 \quad p_2 = 1 \quad p_8 = 0$

encoded the string as

$d_8d_7d_6d_5p_8d_4d_3d_2p_4d_1p_2p_1$

001101001111

check bit at other end is 1101

$C_1 = 1 \quad C_4 = 1$

$C_2 = 0 \quad C_8 = 1$

XOR(1101, 0111) = 1010. Its decimal conversion is 10, so 10th is changed. 10th digit is d6. so we flip & 4x get 00011001

93. The equivalent hexadecimal notation for octal number 2550276 is :

(a) FADED (b) AE0BE

(c) AD0BE (d) ACABE

UGC NET C.S. June-2015 (Paper-III)

Ans. (c) : Given octal No. is

$(2550276)_8 = (010\ 101\ 101\ 000\ 010\ 111\ 110)_2$

$\Rightarrow (01\ 010\ 1101\ 0000\ 1011\ 1110)_2$

$\Rightarrow (0AD0BE)_{16}$

$(AD0BE)_{16}$

94. Convert the octal number 0.4051 into its equivalent decimal number.

(a) 0.5100098 (b) 0.2096

(c) 0.52 (d) 0.4192

UGC NET C.S. June-2017 (Paper-II)

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Ans. (a): Enter the number = $(0.4051)_8$
decimal equivalent of $(0.4051)_8 = 0.8^0 + 4.8^{-1} + 0.8^{-2} + 5.8^{-3} + 1.8^{-4} = 0 + 0.5 + 0 + 0.000244140625$
$= 0.510009765625_{10} = 0.5100098$

95. The hexadecimal equivalent of the octal number 2357 is :
(a) 2EE (b) 2FF
(c) 4EF (d) 4FE

UGC NET C.S. June-2017 (Paper-II)

Ans. (c) : $(2357)_8 = (010\ 011\ 101\ 111)_2$
now grouping them of 4 from right to left
0100 1110 1111 – so it will be 4EF

96. The Octal equivalent of the binary number 1011101011 is :
(a) 7353 (b) 1353
(c) 5651 (d) 5657

UGC NET C.S. November-2017 (Paper-II)

Ans. (b)**:** To solve this (1011101011) question first we make pair of three digit from right side
i.e. 001 011 101 011
011 = 3, 101 = 5, 011= 3, 001 = 1
So, (001 011 101 011) = (1353) in octal

97. Let m = $(313)_4$ and n = $(322)_4$. Find three base 4 expansion of m + n.
(a) $(635)_4$ (b) $(32312)_4$
(c) $(21323)_4$ (d) $(1301)_4$

UGC NET C.S. November-2017 (Paper-II)

Ans.(d) **:** This question can also be solved by first converting both the number in decimal system then per forming addition and then finally converting addition result into base 4.
$(313)_4 = 3*4^2 + 4^1*1+3*4^0 = 55$;
$(322)_4 = 3*4^2 + 4^1* 2 + 2 * 4^0 = 58$;
addition is 55 + 58 = 113
now 113 in base 4 can be represented as 1301 to the base 4.

98. Identify the code sequence :
1010 1011 1001 1000
(a) BCD (b) Excess-3
(c) Gray (d) Excess-3 gray

UGC NET C.S. December-2023

Ans. (c) : The Gray code is a sequence of binary number systems, which is also known as reflected binary code. The reason for calling this code as reflected binary code is the first N/2 values compared with those of the last N/2 values in reverse order. In this code, two consecutive value are differed by one bit of binary digits.
The gray code is a very light weighted code because it doesn't depend on the value of the digit specified by the position. This code is also called a cyclic variable code as the transition of one value to its successive value carries a change of one bit only. Example-

Decimal Number	Binary code	Gray code
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

99. What is the (4+4) fit binary fixed point equivalent of $-(3.72)_{10}$?
(a) 0011.1100 (b) 0011.1010
(c) 1100.0100 (d) 0011.1011

UGC NET C.S. June-2023

Ans. (c) : Convert $-(3.72)_{10}$ in binary fixed point-
Integer part (–3)
- The binary representation of 3 is 001.
- 2's complement representation of -3 gives 1101.
- We need to consider only 4 bits which is 1100 offer discarding overflow bits, So, the binary representation of –3 is 1100.

Fraction part (0.720)-
$0.72 \times 2 = 1.44$
the first digit after decimal point is 1
$0.44 \times 2 = 0.88$
the second digit after decimal point is 0
$0.88 \times 2 = 1.76$
the third digit after decimal point is1
$0.76 \times 2 = 1.52$
the fourth digit after decimal point is 1
So, the binary equivalent of 0.72 is .1011.
Therefore, after 4 bit binary fixed point 0.72 is .0100.
So, the (4 + 4) bit binary fixed point equivalent of $-(3.72)_{10}$ is 1100.0100.

100. In 16-bit 2's Complementation representation, the decimal number 28 is
(a) 1111 1111 0001 1100
(b) 1111 1111 1111 0100
(c) 1111 1111 1110 0100
(d) 1111 1111 1110 0011

UGC NET C.S. December-2022

Ans. (c) : Given decimal number
$(28)_{10} = (11100)_2 = (0000\ 0000\ 0001\ 1100)_2$
2's complement of 0000 0000 0001 1100
= 1111111111100100
Note : Start reading the bits from Least significant bits (right hand side) and write it unless first 1 is encounter leave the first 1 as it is and complement the remaining bits.

101. The octal number 326.4 is equivalent to :
(a) $(214.2)_{10}$ and $(D6.8)_{16}$
(b) $(212.5)_{10}$ and $(D6.8)_{16}$
(c) $(214.5)_{10}$ and $(D6.8)_{16}$
(d) $(214.5)_{10}$ and $(D6.4)_{16}$

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Ans : (c) $(326.4)_8 = (3 \times 8^2 + 2 \times 8^1 + 6 \times 8^0 + 4 \times 8^{-1})_{10} = (214.5)_{10}$
$(326.4)_8 = (011010110.100)_2$
$= (011010110.1000)_2$
$= (OD6.8)_{16}$
$= (D6.8)_{16}$

102. Which of the following is the most efficient to perform arithmetic operations on the numbers?
(a) Sign–magnitude (b) 1's complement
(c) 2's complement (d) 9's complement

UGC NET C.S. Augest-2016 (Paper-II)

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Ans. (c) **:** In 1's complement there are two values of zero. A – 0 (11111111) and a + (00000000). On the other hand in 2's complement, there is only one value for 0 (00000000). 2's complement is efficient to perform arithmetic operation on the numbers.

103. What shall be the 2's complement representation of 24 in a 16 bit computer?
किसी 16 बिट कम्प्यूटर में 24 का 2's पूरक निरूपण होगा?

(a) 1111 1111 1110 1011
(b) 1111 1111 1110 1001
(c) 1111 1111 1110 0111
(d) 1111 1111 1110 1000

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Ans, (d) : Convert decimal number 24 into binary number in 16 bit computer.
$(24)_{10} = (0000000000011000)_2$
1's complement of 24 is-
$= 1111111111100111$
2's complement of 24 is
$= 111111111100111$
$+ 1$
1111111111101000
So, the 2's complement representation of 24 in 16 bit is 1111111111101000.

104. Guard bits are used to:

(a) Avoid unnecessary loss of MSB
(b) Avoid unnecessary loss of LSB
(c) The loss of MSB
(d) The loss of LSB

CGPSC C.S. 2020 Set-A

Ans.(b): Avoid unnecessary loss of LSB: –
When we do floating point arithmetic (like addition, subtraction, multiplication, division) numbers are stored in a limited number of bits. Because of this, some digits can get cut off.
Hence, guard bits are used in floating point arithmetic (especially during division or subtraction) to improve precision and avoid loss of significant low order bits (LSB).

(ii) Boolean algebra and Logic Gates

105. Based on the current technology, _____ is the fastest logic family.
वर्तमान प्रौद्योगिकी के आधार पर,_____एक तीव्रतम लौजिक फैमिली है।

(a) CMOS (b) TTL
(c) MOS (d) ECL

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Ans. (d) : Based on the current technology, ECL (Emitter-coupled Logic) is the fastest logic family. ECL is a high-speed integrated circuit bipolar transistor logic family. ECL uses an overdriven bipolar junction transistor (BJT) differential amplifier with single-ended input and limited emitter current to avoid the saturated region of operation and its slow turn-of behavior.

106. How many output signals can a gate have ?
एक गेट में कितने आउटपुट सिग्नल हो सकते हैं?

(a) Zero/शून्य
(b) Two/दो
(c) More than two/दो से अधिक
(d) One/एक

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Ans. (d) : • A logic gate (AND , OR, NOT, etc) processes input (s) and gives only one output
• Even if it has multiple inputs, the result is always a single output signal.

107. If all the inputs of a NAND gate are connected together then the consequential circuit is …… .
यदि NAND गेट के सभी इनपुट एक साथ जुड़े हुए हैं, तो परिणामी सर्किट …….. है।

(a) NOR gate/नॉर गेट
(b) AND gate/एंड गेट
(c) NOT gate/नॉट गेट
(d) None of the three/तीनों में से कोई भी नहीं

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Ans. (c) : • A NAND gate output 'O' only when all inputs are 1, otherwise it gives 1
• If all inputs are tied together, say input = A_1 then:
output = NOT (A and A) = NOT A

108. The solitary purpose of NOT gate is which of the following ?
नॉट गेट का एकमात्र कार्य निम्न में से कौन-सा है?

(a) To stop signal/सिगनल बंद करना
(b) Invert input signal/इनपुट संकेत को उल्टा करना
(c) Act as a universal gate
एक सार्वभौमिक द्वार के रूप में कार्य करना
(d) None of the three/तीनों में से कोई भी नहीं

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Ans. (b) : A NOT gate, also called an inverter, takes a single input and reverses its logic.
• If input is 1. output is 0
If input is 0, output is 1
It's the simplest logic gate and its sole purpose is to invert the input signal.

109. Which of the following is not a valid rule of Boolean Algebra ?
निम्न में से कौन-सा बूलियन ऐलजेब्रा का एक उचित नियम नहीं है?

(a) A + A' = A (b) A . A = A
(c) A . A' = 0 (d) A + A'= 1

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Ans. (a) : From booleam Algebra we know that
A + A' = 1 (Complement law)
A . A = A (Idempotent law)
A . A' = 0 (Complement law)
But A + A' ≠ A
Hence A + A' = A is not a valid rule of Boolean Algebra

Sarkari Resource

110. Logistic regression can be bionomial, _____or multinomial.
लॉजिस्टिक रिग्रेशन बायोनॉमियल, _____या मल्टीनॉमियल हो सकता है।
(a) Ordinal/ऑर्डिनल (b) Numbered/नम्बर्ड
(c) Unimial/यूनिमियल (d) Binary/बाइनरी
HTET PGT C.S. 2023

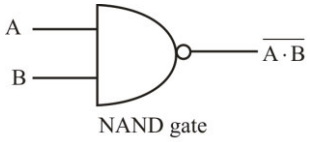
Ans.(a): Logistic regression can be bionomial ordinal or multinomial. When the dependent variable has more than two ordered categories (e.g., low/medium/high) is called ordinal.
When the dependent variable has more than two unordered categories (e.g., Red/Blue/Green) is called Multinomial.
When the dependent variable has two possible categories (e.g., Yes/No, 0/1) is called Binomial/Binary.

111. Consider an XOR gate with 4 inputs X, Y, Z, and W. For which of the following combinations of the inputs, the output of this XOR will be 1?
(a) X = 1, Y = 1, Z = 0, W = 0
(b) X = 1, Y = 1, Z = 0, W = 0
(c) X = 1, Y = 0, Z = 1, W = 1
(d) X = 1, Y = 1, Z = 1, W = 1
(e) Not attempted
Haryana PGT C.S. Asstt. Prof. 2024

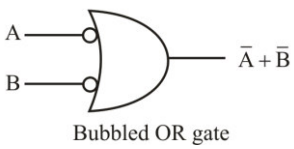
Ans. (c) : For an XOR gate with 4 inputs, the output is 1 only when an odd number of the inputs are 1.
From option (a) X = 1, Y = 1, Z = 0, W = 0 → 1+1+0+0=2→ 1+1+0+0=2(even)→output 0
From option(b) X = 1, Y = 1, Z = 0, W = 0, 0+1+1+0=2(even) →output 0
From option (c) X = 1, Y = 0, Z = 1, W = 1, 1+0+1+1=3(odd)→output 1
From option (d) X = 1, Y = 1, Z = 1, W = 1→1+1+1+1=4(even)→ output 0
Hence the correct option is (c) X=1, Y= 0, Z=1, W= 1

112. A NAND gate is equivalent to a
(a) Bubbled OR gate
(b) Bubbled XOR gate
(c) NOR gate followed by an AND gate
(d) Bubbled AND gate
(e) Not attempted
Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : A NAND gate is equivalent to Bubbled OR gate.



NAND gate



Bubbled OR gate

From De-morgan's theorem , we know that $\overline{A \cdot B} = \overline{A} + \overline{B}$

113. In binary multiplication using the simple method, what is the first step after aligning the digits and preparing for multiplication?
सरल विधि से बाइनरी गुणा करते समय, अंकों को संरेखित करने और गुणन के लिए तैयार करने के बाद पहला कदम क्या होता है?
(a) Multiply the least significant bit (LSB) by the multiplier/सबसे कम महत्वपूर्ण बिट (LSB) को गुणक से गुणा करें
(b) Multiply the most significant bit (MSB) by the multiplier/सबसे महत्वपूर्ण बिट (MSB) को गुणक से गुणा करें
(c) Multiply the least significant bit (LSB) by the multiplicand/सबसे कम महत्वपूर्ण बिट (LSB) को गुणक से गुणा करें
(d) Multiply the most significant bit (MSB) by the multiplicand/सबसे महत्वपूर्ण बिट (MSB) को गुणक से गुणा करें
Bihar STET C.S. 12.09.2023 (Shift-I)

Ans, (a) : Correctly stated in simple binary multiplication this process proceeds from right to left (from LSB to MSB) in this the multiplicand is multiplied by the right most bit (LSB) of the multiplier, and then the process is repeated for all the bits moving left, just as in decimal multiplication.

114. In Boolean algebra A.A is equal to
Boolean बीजगणित में A. A बराबर है
(a) A (b) 2A
(c) A^2 (d) 1
Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : In Boolean algebra,
$A \cdot A = A$ {Idempotent law}

115. In Boolean algebra the function of AND operator is
Boolean बीजगणित में AND operator का कार्य है
(a) + (b) –
(c) / (d) *
Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : In Boolean algebra, the AND operator is represented by the multiplication sign(*).
AND means both input must be 1 to get 1.
Truth table of AND:

A	B	A*B
0	0	0
0	1	0
1	0	0
1	1	1

116. Which of the following logical operations is represented by the + sign in Boolean algebra?
निम्नलिखित Logical कार्यों में से कौन सा Boolean बीजगणित में + संकेत द्वारा दर्शाया गया है?
(a) OR (b) NOT
(c) NOR (d) AND
Bihar STET C.S. 2019 (Shift-II)

Sarkari Resource

Ans. (a) : In Boolean algebra, the OR operator is represented by plus sign (+).
OR operator gives output if and only if at least one input is 1.
Truth table of OR:

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

117. Which one is used to express logic gate
लॉजिक गेट किसके द्वारा प्रदर्शित होता है।
(a) Set theory
(b) Numerical Technique
(c) Artificial Intelligence
(d) Boolean Algebra

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : Logic gates work with 0s and 1s and the rules for calculating with 0 and 1 come from Boolean Algebra. This helps us understand AND, OR, NOT gates and write their equations.
So, the correct option is (d) Boolean Algebra.

118. According to Boolean law : A + 1 = ?
बूलियन लॉ के अनुसार : A + 1 = ?
(a) 1 (b) A
(c) 0 (d) A'

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : According to Boolean law:
A+1= 1
This rule states that for any Boolean variable A, when it is ORed with 1 (true), the result is always 1 (true).
So, the correct option is (a) 1

119. A (A + B) = ?
(a) AB (b) 1
(c) (1 + AB) (d) A

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : Boolean Expression
=A(A+B)
= A·A +A·B {Distributive property}
= A +A·B {Idempotent law A·A=A}
=A {Absorption law}
So, the correct option is (d) A

120. The logic gate that provides high output for same inputs.....
.....तर्क गेट जो समान इनपुट के लिए उच्च आउटपुट प्रदान करता है।
(a) NOT (b) X-NOR
(c) AND (d) XOR

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : The logic gate that provides high output (1) for same inputs is the X-NOR gate.
For inputs A and B.
If A = 0 and B = 0, output is 1.
If A = 1 and B = 1, output is 1.
Otherwise, output is 0.
Truth table of X-NOR gate:

A	B	A ⊙ B
0	0	1
0	1	0
1	0	0
1	1	1

So, the correct option (b) X-NOR

121. Boolean algebra can be used.....
बूलियन बीजगणित का उपयोग किया जा सकता है.....?
(a) For designing of the digital computers
(b) In building logic symbols
(c) Circuit theory
(d) Building algebraic functions

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a): Boolean algebra is the mathematical foundation for digital logic circuits, which are the building block of all digital computers. It is used to design, analyze and optimize circuits like processors, memory and other hardware components.
So, the correct option is (a) for designing of the digital computers.

122. A..... value is represented by a Boolean expression.
एक..... मूल्य एक बुलियन अभिव्यक्ति द्वारा दर्शाया गया है।
(a) Positive (b) Recursive
(c) Negative (d) Boolean

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : A Boolean value is represented by a Boolean expression. Boolean expressions are used to represent true or false values, often denoted as 1 and 0.
So, the correct option is (d) Boolean

123. Which of the following expression results in zero?
निम्नलिखित में से कौन से व्यंजक का उत्तर शून्य होगा।
(a) (0+0+1)(0+0+1) (b) (0+0+0)(0+1+1)
(c) (1+0+0)(1+1+1) (d) (0+1+0)(1+0+1)

KVS PGT C.S. 2018

Ans, (b) : Given expression
(a) (0 + 0 + 1) (0 + 0 + 1) = 1.1 = 1
(b) (0 + 0 + 0) (0 + 1 + 1) = 0.1 = 0
(c) (1+ 0 + 0) (1 + 1 + 1) = 1.1 = 1
(d) (0 + 1 + 0) (1 + 0 + 1) = 1.1= 1
Hence, option (b) expression (0 + 0 + 0) (0 + 1 + 1) results is zero.

124. Exclusive OR (XOR) is a special gate whose output is 1 only if:
एक्सक्लूसिव ऑर (XOR) एक विशेष गेट है जिसका आउटपुट 1 तभी होगा यदि:
(a) all inputs are 0/सभी इनपुट 0 हों।
(b) all inputs are 1/सभी इनपुट 1 हों।
(c) odd number of inputs are 1
विषम संख्यक इनपुट 1 हों।
(d) even number of inputs are 1
सम संख्यक इनपुट 1 हों।

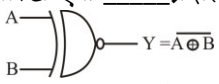
KVS PGT C.S. 2018

Sarkari Resource

Ans, (c) : The exclusive OR (XOR) gate is a digital logic gate with two or more inputs and one output that performs exclusive disjunction. If an XOR gate has more than two inputs, then its behavior depends on its implementation. An XOR gate will output 1 if an odd number of its inputs is 1. The following truth table for an XOR gate with two inputs.

Input A	Input B	Output Y
0	0	0
0	1	1
1	0	1
1	1	0

125. The following diagram shows a
निम्नांकित आरेख एक _____प्रदर्शित करता है ।



(a) Exclusive-Nor gate (b) NAND gate
(c) AND gate (d) OR gate

KVS PGT C.S. 2013

Ans, (a) : The XNOR gate is a digital logic gate whose function is the logical complement of the Exclusive OR (XOR) gate. It is equivalent to the logical connective (↔) from mathematical logic also known as the material Biconditional.
The Boolean expression for the XNOR gate is given by

$$y = \overline{(A \oplus B)} = \overline{A}.B + A.\overline{B}$$

126. Identify the Boolean Gate, which represents the following truth table:
उस बूलियन गेट (Boolean Gate) की पहचान कीजिए, जो निम्नलिखित सत्य सारणी (truth table) को निरूपित करता है:

A	B	Output
0	**0**	**1**
0	**1**	**1**
1	**0**	**1**
1	**1**	**0**

(a) XOR (b) NOR
(c) NAND (d) OR

EMRS PGT C.S. 16.12.2023

Ans, (c) : The NAND gate is special type of logic gate in the digital logic circuit. The NAND gate is the universal gate. It means all the basic gates such as AND, OR and NOT gate can be constructed using a NAND gate. The NAND gate is the combination of the NOT - AND gate.
This table belongs to NAND gate truth table.

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

127. If ⊕ represents XOR Gate in (Boolean algebra) and A ⊕ B = C, then B ⊕ C equals to:
यदि बूलियन बीजगणित Boolean algebra में ⊕, XOR गेट को निरूपित करता है और A ⊕ B = C, तो B ⊕ C किसके बराबर है?

(a) 1 (b) A
(c) A' (d) 0

EMRS PGT C.S. 16.12.2023

Ans, (b) : Given that $A \oplus B = C$
Then, $B \oplus C$

$= B \oplus (A \oplus B)$ $\quad \because C = A \oplus B$
$= B \oplus (B \oplus A)$
$= (B \oplus B) \oplus A$ $\quad \because B \oplus B = 0$
$= 0 \oplus A$
$= A$

128. What is the maximum number of different functions involving n Boolean variables?
n बूलियन वैरिएबलों से अधिकतम कितने भिन्न फलन बनाए जा सकते हैं?

(a) n^2 (b) 2^n
(c) 2^{n^2} (d) 2^{2^n}

LT. Grade C.S. 2018

Ans, (d) : In Boolean algebra, the number of different Boolean functions that can be define involving n Boolean variables is 2^{2^n}. This is because each Boolean function can have 2^n possible input combinations (for n variables), and for each combination there can be two possible outputs (0 or 1).

129. Which of the following is a universal gate?
निम्न में से कौन-सा एक यूनिवर्सल गेट है?

(a) NOT (b) NAND
(c) OR (d) AND

LT. Grade C.S. 2018

Ans, (b) : A universal gate is a type of logic gate that can be used to implement any Boolean function without the need to use any other type of gate. The NAND gate is considered a universal gate because any Boolean function can be implemented using a combination of NAND gates.

130. What is the result of the operation 1010 AND 1100 in binary?/बाइनरी में ऑपरेशन 1010 AND 1100 का परिणाम क्या है?

(a) 1000
(b) 1110
(c) 100
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : AND (x) operation of 1011 and 1100

$$\begin{array}{r} 1011 \\ \times 1100 \\ \hline 1000 \end{array}$$

Truth table of AND gate:

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

So, the correct option is (a) 1000.

Sarkari Resource

131. An OR gate has 4 inputs. The number of words in truth table will be _____./OR गेट में 4 इनपुट होते हैं। सत्य तालिका में शब्दों की संख्या होगी _____

(a) 4
(b) 8
(c) 16
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : The 'number of words' in a truth table means the number of rows, it has. Each row shows a different possible combination of inputs (0s and 1s).
The rule is:
Number of rows = 2^n; where n = number of inputs
On putting n = 4
Number of row = $2^4 = 16$
So, the correct option is (c) 16

132. What are the basic gates in MOS logic family?
एम. ओ. एस. लॉजिक परिवार में बुनियादी द्वार क्या हैं?

(a) NAND and NOR
(b) AND and OR
(c) NAND and OR
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : In the MOS (Metal-Oxide-Semi conductor) logic family, the basic gates are NAND and NOR. These gates are fundamental building block for digital circuits and are widely used in MOS technology due to their simplicity and efficiency in implementation.
So, the correct option is (a) NAND and NOR.

133. The logic gate that provides high output for same input is:
लॉजिक गेट, जो समान इनपुट के लिए उच्च आउटपुट प्रदान करता है,

(a) NOT
(b) X-NOR
(c) XOR
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): The X-NOR gate (Exclusive-NOR gate) provides a high output when both inputs are the same.
If both inputs are 0, the output is 1 (high).
If both input are 1, the output is 1 (high).
Truth table of X-NOR gate.

A	B	output A ⊙ B
0	0	0
0	1	0
1	0	0
1	1	1

134. Boolean algebra can be used:
बूलियन बीजगणित का उपयोग किया जा सकता है–

(a) For designing the digital computers/designing the digital computers में
(b) In building logic symbols/building logic symbols में
(c) In circuit theory/circuit theory में
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d): Boolean algebra can be used:
(a) Boolean algebra is fundamental in designing digital circuits and systems, including computers.
(b) Boolean algebra is used to represent and manipulate logical expressions and symbols.
(c) Boolean algebra is applied in the analysis and design of digital circuits.
So, the correct option is (d) More than one of the above.

135. The dual of a Boolean expression is obtained by:
बूलियन अभिव्यक्ति का द्वैत किसके द्वारा प्राप्त किया जाता है?

(a) Interchanging all 0s and 1s
(b) Interchanging all 0s and 1s, all + and '.' signs
(c) Interchanging all 0s and 1s, all + and '.' signs and complementing all the variables
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(b): The dual of a Boolean expression is obtained by applying the following transforms:
1. Interchanging all 0s and 1s and vice versa.
2. Interchanging all '+' (OR) operations with '·' (AND) operations and vice versa.
This means that if we have a Boolean expression, we can find its dual by swapping the AND and OR operators and swapping the 0s and 1s
For example, if we have the expression A + 0 = A, its dual would be A·1 = A
So, the correct option is (b) Interchanging all 0s and 1s, all + and '·' signs

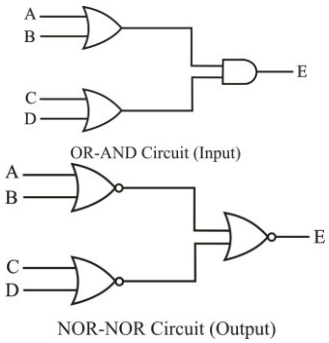
136. Which logical gate is used to represent product of sum expression?
सम एक्सप्रेशन के प्रोडक्ट को दर्शाने के लिए किस लॉजिकल गेट का उपयोग किया जाता है?

(a) EX-OR gate/ EX-OR गेट
(b) AND gate/ AND गेट
(c) OR gate/ OR गेट
(d) NOR gate/ NOR गेट

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : The product of sums form can be implemented using a two-level gate network NOR-NOR logic. NOR realization is the same as OR-AND. A Boolean function is expressed algebraically from a given truth table by forming a maxterm.

Sarkari Resource



Input= (A+B).)(C+D)
Output=[(A+B)' + (C+D)']'
= [(A+B)']' . [(C+D)']'
= (A+B).(C+D)

137. Which gate is represented by the following truth table?
निम्नलिखित ट्रूथ टेबल द्वारा कौन सा गेट दर्शाया जाता है?

Input/इनपुट		Output/आउटपुट
A	B	
0	0	0
0	1	1
1	0	1
1	1	1

(a) OR (b) AND
(c) XOR (d) NOT

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : An OR gate is a basic digital logic gate. It gives an output of 1 (true) if at least one input is 1 otherwise out is 0.
So, in the question, given truth table represents the OR gate.

138. Boolean algebra can be used
बूलियन बीजगणित का उपयोग किया जा सकता है
(a) For designing the digital computers
डिजिटल कम्प्यूटर की डिजाइन के लिए
(b) In building logic symbols/तर्क प्रतीकों के निर्माण में
(c) In circuit theory/सर्किट सिद्धांत में
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (d) : Boolean algebra is a branch of algebra that deals with variables having only two values: 1(true) and 0(false) and uses logical operations like AND, OR, NOT. It is used to design and analyze digital circuits, create logic symbols and simplify circuit operation.
So, the correct option is (d) more than one of the above

139. In two-valued Boolean algebra, the maximum number of Boolean function for two variables will be
दो मान वाले बूलियन अलजेब्रा में 2 वैरिएबल्स के लिए अधिकतम बूलियन फंक्शन (Boolean function) की संख्या होगी
(a) 8
(b) 12
(c) 16
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : For n Boolean variables, the number of possible Boolean function is: 2^{2^n}
Applying for 2 variables.
Maximum number of Boolean function = $2^{2^2} = 2^4 = 16$

140. Which of the following logic gates is called universal gate?
निम्नांकित में से कौन-सा लॉजिक गेट, यूनिवर्सल गेट कहलाता है?
(a) AND
(b) OR
(c) NAND
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : A universal Gate is a logic gate that can be used to implement any other logic gates (AND, OR, NOT) using only that gate.
Both NAND and NOR gates are universal gates
So, correct option is (c) NAND

141. Match the List-I with List-II

List-I (Boolean Algebra Law)		**List-II (Axioms)**	
A.	**Absorption Law**	**I.**	**a + 1 = 1**
B.	**Bounded Law**	**II.**	**a + 0 = a**
C.	**Identity Law**	**III.**	**a*(b + c) = (a*b) + (a*c)**
D.	**Distributive Law**	**IV.**	**a + (a*b) = a**

Choose the correct answer from the options given below :
(a) A-IV, B-I, C-II, D-III
(b) A-IV, B-III, C-I, D-II
(c) A-III, B-IV, C-II, D-I
(d) A-II, B-III, C-IV, D-I

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Ans. (a) : Correct match-

List-I (Boolean Algebra Law)		**List-II (Expression)**	
A.	Absorption Law	–	a +(a * b) = a
B.	Bounded Law	–	a + 1 = 1
C.	Identity Law	–	a + 0 = a
D.	Distributive Law	–	a * (b + c) = (a * b) + (a * c)

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142. A 2 way switch works as follows:
एक 2 वे (way) स्विच निम्नानुसार काम करता है:
Both switches ON = lights off
दोनों स्विच ऑन = लाइट ऑफ
Both switches OFF = light off
दोनों स्विच ऑफ = लाइट ऑफ
One switch ON and other OFF = light on
एक स्विच ऑन तथा दूसरा ऑफ = लाइट ऑन
Which gate can be used to implement this?
इसे लागू करने के लिए किस गेट का उपयोग किया जा सकता है?

(a) OR (b) AND
(c) NAND (d) XOR

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) : To implement the logic of a 2-ways Switch s described, we head a logic gate that produces on output where the light is on (1) when exactly one of the two switches is on, and the light is off. This behavior matches the truth table of an XOR gate.
Truth Table for 2-way switch–

Switch A	Switch B	Light (Output)
0	0	0
0	1	1
1	0	1
1	1	0

This is identical to the truth table of an XOR gate, where the output is 1 only when the inputs are different (one is 1 and the other is 0)
The XOR gate can be used to Implement 2*way switch.

143. The input-output relationship of the binary variable for each gate can be represented in tabular form by a _____.
प्रत्येक गेट के लिए बाइनरी वेरिएबल के इनपुट-आउटपुट संबंध को _____ द्वारा सारणीबद्ध रूप में दर्शाया जा सकता है।

(a) Symbol table/सिंबल टेबल
(b) Decision table/डिसिजन टेबल
(c) Truth table/ट्रुथ टेबल
(d) Characteristic table/कैरेक्टरस्टिक टेबल

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : In digital logic and Boolean algebra, the truth table is a fundamental tool used to describe the input-output relationship of logic gates.
Truth table– A truth lists all possible combinations of inputs for a logic gate and the corresponding outputs. It is specifically designed for binary (0 or 1) variables used for gates like AND, OR, NOT, NAND, NOR, XOR, XNOR.
Truth table for AND gate–

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

This shows the output for all input combinations.

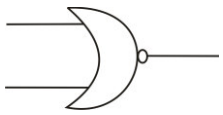
144. Which statement(s) is/are correct regarding the Boolean algebra?
बूलियन बीजगणित के संदर्भ में कौन सा/से कथन सही है/हैं?
I. It facilitate the analysis and design of digital circuits./यह डिजिटल सर्किटस के विश्लेषण और डिजाइन की सुविधा प्रदान करता है।
II. Expresses in algebraic form the input-output relationship of logic diagram.
तर्क आरेख के इनपुट-आउटपुट संबंध को बीजीय रूप में व्यक्त करता है।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

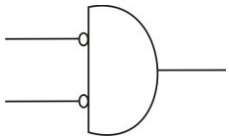
DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Statement-I – Boolean algebra is a fundamental tool in the analysis and design of digital circuits , as it simplifies complex logical expressions and helps in implementing logic gates efficiently.
Statement-II – Boolean algebra expresses the input output relationship of a logic circuit in algebraic form using variables and logical operations (Like AND, OR, NOT). Which is essential for circuit analysis and synthesis.
Hence statement-I and II are correct regarding the Boolean algebra.

145. Which of the following symbol represent the NOR gate?
निम्नलिखित में से कौन प्रतीक NOR गेट का प्रतिनिधित्व करता है?
I.



II.



(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Both symbols I and II represent a NOR gate.
Symbol I is the standard logic Symbol for a NOR gate. It's an OR gate with a bubble at the output indicating negation.
Symbol II is an alternative symbol for a NOR gate, sometimes referred to as a "distinctive shape" Symbol. In also represents the OR operation followed by negation.

146. Which gate is represented by the following truth table?

Input		**Output**
A	**B**	
0	**0**	**0**
0	**1**	**0**
1	**0**	**0**
1	**1**	**1**

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निम्नलिखित ट्रूथ टेबल द्वारा किस गेट का प्रतिनिधित्व किया जाता है

इनपुट		आउटपुट
A	B	
0	0	0
0	1	0
1	0	0
1	1	1

(a) OR (b) AND
(c) NAND (d) NOR

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : The given truth table represents the AND gate because AND gate given output 1 only when both inputs are 1, if either of one input is 0 output will be 0.

147. Both the inputs of XOR gate are connected to B. What will be the output ?
XOR गेट के दोनो इनपुट B से जुडे.है। आउटपुट क्या होगा ?

(a) 0 (b) 1
(c) B (d) B

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : Truth Table of XOR gate

Input A	Input B	output A + B
0	0	0
0	1	1
1	0	1
1	1	0

From the truth table of XOR gate, it is clear that XOR gate gives output 1 when both inputs are different. In the case of same inputs, output will be 0.
So, if both the inputs of XOR gate are connected to B, then its output will be 0.

148. Which gate is represented by the following truth table?

Input			Output
A	B	C	
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

निम्नलिखित ट्रूथ तालिका द्वारा किस गेट को दर्शाया जाता है?इनपुट			आउटपुट
A	B	C	
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

(a) NOR (b) NAND
(c) XOR (d) XNOR

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) **:**
1. Analyze the inputs (A, B, C) and the outputs
2. Match the resulting outputs to a known logic gate function or composite logic.

इनपुट			आउटपुट
A	B	C	
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step-by Step Analysis:
We have 3 inputs (A, B, C) → so 8 possible combinations.
The output are:

Input	→	**Output**
000	→	1
001	→	1
010	→	1
011	→	1
100	→	1
101	→	1
110	→	1
111	→	0

So the only time the output is 0 is when A = 1, B = 1, C = 1.
This behavior mataches the logic of a NAND gate with three inputs:
A NAND gate gives output 0 only when all inputs are 1.
The gate represented by the truth table is a 3-input NAND gate.

149. Which of the following is correct regarding figure of merit of an IC?
IC की मेरिट के फिगर के संबंध में निम्नलिखित में से कौन सा सही है?

(a) Propagation delay time – power
प्रोपेगेशन डीले टाइम– पॉवर

(b) Propagation delay time ÷ power/प्रोपेगेशन डीले टाइम÷पॉवर

(c) Propagation delay time × power / प्रोपेगेशन डीले टाइम ×पॉवर

(d) Propagation delay time + power / प्रोपेगेशन डीले टाइम+पॉवर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : The figure of merit (FOM) of an Integrated circuit (IC) or a logic family is a measure of its performance, and it is defind as the prouduct of the

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propagation delay time and the power dissipation. A lower figure of merit indicates better performance.

150. Which of the following represents XNOR gate?
निम्नलिखित में से कौन सा XNOR गेट को दर्शाता है?

(a) $Y = A \oplus B$
(b) $Y = A \odot B$
(c) $Y = \overline{A + B}$
(d) $Y = \overline{A.B}$

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : The XNOR (Exclusive NOR) gate is a digital logic gate that output a high (1) only when both input are the same (both 0 or both 1).
XNOR Boolean Expression,
$y = AB + A'B' = A \odot B$

151. What is the correct answer of the given statements in the given order?
दिए गए क्रम में दिए गए कथनों का सही उत्तर क्या है?

I. Gate in which output of logical gate is high when all inputs are high is _____
गेट जिसमें जब सभी इनपुट हाई होते हैं तो गेट का आउटपुट हाई होता _____ है

II. Half of byte is known as _____
आधा बाइट _____ केस रूप में जाना जाता है

III. A flip-flop whose output follows the input when triggered is _____
एक फ्लिप-फ्लॉप जिसके ट्रिगर होने पर आउटपुट, इनपुट का अनुसरण करता है, _____ है

(a) NAND gate, bit, T – flip-flop
नेंड गेट, बिट, टी - फ्लिप-फ्लॉप
(b) OR gate, nibble, T – flip-flop
ओर गेट, निब्बल, टी - फ्लिप-फ्लॉप
(c) AND gate, nibble, D – flip-flop
एंड गेट, निब्बल, डी - फ्लिप-फ्लॉप
(d) NOR gate, byte, S-R flip-flop
नोर गेट, बाइट, S-R फ्लिप-फ्लॉप

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) :
(i) The AND Gate is a logic gate that produces a high output (1) when all of its inputs are high (1). If any of the inputs are low (0), the output will also be low (0).
(ii) Half of a byte is known as a "nibble". A nibble consists of 4 bits.
(iii) A flip-flop whose output follows the input when triggered is called a D flip-flop or data or delay (Flip-flop). When the clock signal is active, the output of the D flip-flop takes the value of the input and holds it until the next triggering event.

152. The XOR gate gives _____ as output when both inputs are different and _____ when both inputs are same.
XOR गेट आउटपुट के रूप में _____ देता है, जब दोनों इनपुट असमान होते हैं और _____ जब दोनों इनपुट समान होते हैं।

(a) 0, 0 (b) 0, 1
(c) 1, 0 (d) 1, 1

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Let's check carefully:
An XOR gate truth table is :
- Input (A = 0, B = 0) → Output = 0
- Input (A = 0, B = 1) → Output = 1
- Input (A =1, B = 0) → Output = 1
- Input (A = 1, B = 1) → Output = 0

So,
- When both inputs are\ same → Output = 0
- When both inputs are different → Output = 1

153. Which of the following represents the Idempotent law in Boolean Algebra?
निम्नलिखित में से कौन सा विकल्प बूलियन बीजगणित में वर्गसम नियम को दर्शाता है?

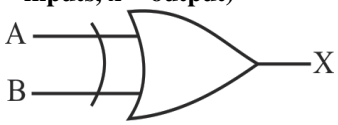
(a) A + 0 = A (b) A + A = A
(c) A + A' = 1 (d) (A')' = A

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : In Boolean Algebra, the Idempotent Law states:
- $A + A = A$
- $A \cdot A = A$

Among the given options:
(a) $A + 0 = A \rightarrow$ Identity law
(b) $A + A = A \rightarrow$ Idempotent law
(c) $A + A' = 1 \rightarrow$ Complement law
(d) $(A')' = A \rightarrow$ Double Negation law

154. Which of the following logic gate is shown in the given Graphic Symbol? (where, A and B = inputs, x = output)



(a) OR gate
(b) NAND gate
(c) NOR gate
(d) Exclusive − OR (XOR) gate

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (d) : The graphic symbol shown in the image represents an Exclusive -OR (XOR) gate.
Explanation:-
The Symbol depicted in the image is characterized by a curved input side with an shape of an OR gate, and a single output line. This distinct graphic representation is universally recognized for an XOR gate in digital electronics.
- **OR gate :** A NAND gate symbol is an AND gate symbol with a small circle (inversion bubble) at the output.
- **NOR gate:-** A NOR gate symbol is an OR gate symbol with a small circle (inversion bubble) at the output.
- **Exclusive - OR (XOR) gate:-**

The XOR gate produces a high output (i) if and only if its inputs are different. Its symbol is a distinctive OR gate shape with an extra curved line on the input side.

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155. Consider the following statements :

S_1: Boolean expressions and logic networks correspond to labeled acrylic digraphs.

S_2: Optimal Boolean expressions may not correspond to simplest networks.

S_3: Choosing essential blocks first in a Karnaugh map and then greedily choosing the largest remaining blocks to cover may not give an optimal expression.

Which of these statements(s) is/are **correct?**

(a) S_1 only (b) S_2 only

(c) S_1 and S_2 (d) S_1, S_2 and S_3

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) :

- **S_1:** Boolean expressions – acylic diagraphs.
- **S_2:** Minimal expressions ≠ always simplest network
- **S_3:** Greedy K-map grouping may not give optimal expression.

So, S_1, S_2 and S_3 all are correct.

156. Which of the following 2 input Boolean logic functions is linearly inseparable?

(i) AND (ii) OR

(iii) NOR (iv) XOR

(v) NOTXOR

(a) (i) and (ii) (b) (ii) and (iii)

(c) (iii), (iv) and (v) (d) (iv) and (v)

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) XOR is linearly inseparable, if XOR is, then swapping the labels on the points will not alter the structure of the space, and NOTXOR is just XOR with the 1's swapped with 0's and vice-versa.

157. Let $\oplus$ denote XOR operation. Let 1 and 0 denote the binary constants and F is the Boolean expression over two variables P and Q

$$F(P,Q) = ((1\oplus P))\oplus((P\oplus Q))\oplus((P\oplus Q))\oplus((Q\oplus 0))$$

Which of the following is equivalent expression to F?

(a) $(P\oplus Q)$ (b) $(P+Q)$

(c) $\overline{(P+Q)}$ (d) $\overline{(P\oplus Q)}$

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Ans. (d) : Given,

Boolean expression-

$$F(P,Q) = ((1\oplus P)\oplus(P\oplus Q))\oplus((P\oplus Q)\oplus(Q\oplus 0))$$

$$= (\bar{P}\oplus P\oplus Q)\oplus(P\oplus Q\oplus Q) \quad [\because 1\oplus P = \bar{P}]$$

$$[Q\oplus 0 = Q]$$

$$= (1\oplus Q)\oplus(P\oplus 0) \quad [\because \bar{P}\oplus \ = P]$$

$$[Q\oplus Q = 0]$$

$$= \bar{Q}\oplus P \quad [\because P\oplus Q = \bar{P}Q + P\bar{Q}]$$

$$[P\oplus\bar{Q} = PQ + \bar{P}\bar{Q}]$$

$$= PQ\oplus\overline{PQ}$$

Now, we need to check for the options, if we simplify expression $\overline{P\oplus Q}$.

$$\overline{P\oplus Q} = \overline{P\bar{Q} + \bar{P}Q}$$

$$= (\bar{P}+Q).(P+\bar{Q}) \quad \text{by Demorgan's Law}$$

$$[\because \bar{P}P = 0,\ Q\bar{Q} = 0]$$

$$= \bar{P}P + \bar{P}\bar{Q} + PQ + Q\bar{Q}$$

$$= PQ + \bar{P}\bar{Q}$$

158. If we convert a decision tree to a set of logical rules then :

(a) The internal nodes in a branch are connected by AND and branches by AND

(b) The internal nodes in a branch are connected by OR and branches by OR

(c) The internal nodes in a branch are connected by AND and branches by OR

(d) The internal nodes in a branch are connected by OR and branches by AND

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Ans. (c) : The correct statement is : The internal nodes in a branch are connected by "AND" and branches by "OR".

When converting a decision tree into a set of logical rules, each path from the root to a leaf in the decision tree represents a conjunction of conditions, and the different paths branches are combined using the logical OR operation. Each internal node in a branch represents a condition connected by logical AND. By following this approach you can express the decision tree as a set of logical rules that capture the same decision - making process.

159. Which Boolean operation on two variables can be represented by a single perception layer?

A. X1 AND X2 B. X1 OR X2

C. X1 NOR X2 D. X1 XOR X2

Choose the most appropriate answer from the options given below :

(a) A and B Only (b) B and C Only

(c) A, B and C Only (d) A, B, C and D Only

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Ans. (c) : A single perception can represent any linearly separable function. AND, OR and NOR are linearly separable in 2D plane and hence a single perception (single perception layer) can represent them.

So option (c) is correct.

160. Consider the following statements:

A. There exists a Boolean algebra with '5' elements.

B. Every element of Boolean algebra has unique complement.

C. If a Lattice 'L' is a Boolean algebra then 'L' is not relatively complemented.

D. The direct product of two Boolean Algebras is also a Boolean algebra.

Choose the correct answer about the four statements given above.

(a) Only A and D are correct

(b) Only B and D are correct

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(c) All statements are NOT correct
(d) All statements are correct

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Ans. (b) : Since each Boolean Algebra must have 2n element for some integer n.
So first statement in incorrect.
- Every element of Boolean algebra has unique complement.
- If a Lattice 'L' is a Boolean algebra then 'L' is relatively complemented.
- The direct product of two Boolean Algebras is also a Boolean algebra.

So option (b) is correct.

161. Which of the following statements are true?
(i) Every logic network is equivalent to one using just NAND gates or just NOR gates.
(ii) Boolean expressions and logic networks correspond to labelled acyclic digraphs.
(iii) No two Boolean algebras with n atoms are isomorphic.
(iv) Non-zero elements of finite Boolean algebras are not uniquely expressible as joins of atoms.
Choose the correct answer from the code given below:
Code :
(a) (ii), (iii) and (iv) only
(b) (i), (ii) and (iii) only
(c) (i) and (iv) only
(d) (i) and (ii) only

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Ans. (d) : NOR gates can be combined to generate any other logical function. They shares this property with the NAND gate; that's why every logic network is equivalent to one using just NAND gates or just NOR gates. Boolean expressions and logic networks correspond to labelled acyclic digraphs is correct.
So, statement (i) & (ii) are true.

(iii) Simplification of Boolean Functions

162. Which of the following is the simplified form of the given function.
f(A, B, C) = $\sum$M(0,1, 3, 7)
(a) A'B'C' + A'B'C + A'BC + ABC
(b) A'B' + A'BC + ABC
(c) A'B' + BC
(d) A'B' + BC'

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Ans. (c) : F(A, B, C) = Σm (0, 1, 3, 7)
This means the function is 1 for the minterms corresponding to the binary representations
- m(0) = 000
- m(1) = 001
- m(3) = 011
- m(7) = 111

Step1 : write the sum of minterms form F (A, B, C) = A'B'C' + A'B'C + A'BC + ABC
Step 2 : Simplify using Boolen Algebra
Factor A'B' From the first two terms
A' B' C' + A' B' C + A' B' (C' + C) = A' B'
(Since C' + C = 1)
Factor BC from the last two terms:
A' B C + A B C = BC (A' + A) = BC
(Since A' + A = 1)
Thus, the simplipied form is:
F (A, B, C) = A' B' + B C
Step 3: Find the correct Answer
Looking at the given options is A' × B' + BC

163. The Boolean expression F = (x.y' + x'.y + x'.y') is equivalent to :
बूलियन व्यंजक F = (x.y' + x'.y + x'.y') के समान है:
(a) x + y'
(b) x' + y'
(c) x' + y
(d) x + y

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Ans. (b) : 1. Group terms with x:
F xy' + x' (y + y')
Since y + y' = 1
F = xy' + x'·1 = xy' + x'
2 . Apply the Boolean identity
A' + AB = A' + B
Let A = x , B = y
x' + xy' = x' + y'
3. Final simplified form
F = x' + y'

164. Simplify the following expression.
निम्नलिखित व्यंजक को सरलीकृत कीजिए।
$(A+B+1).(\overline{A}+\overline{B}+1).(A+B+C).(\overline{A}+\overline{B}+\overline{C})$
(a) 1
(b) $\left[(A+C)\overline{B}\right]+\left[(B+C)\overline{A}\right]+\left[(A+B)\overline{C}\right]$
(c) $\left[(A+C)\overline{B}\right]+\left[(B+C)\overline{A}\right]$
(d) A + B + C

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : To Simplify the given expression-
$(A+B+1).(\overline{A}+\overline{B}+1).(A+B+C).(\overline{A}+\overline{B}+\overline{C})$
Simplify the first two terms-
The term (A + B + 1) simplifies to 1 because 1 in an OR operation makes the entire expression 1.
$(\overline{A}+\overline{B}+1)$ also simplifies 1.
So, the expression now reduces to
$1.1.(A+B+C).(\overline{A}+\overline{B}+\overline{C})$
now, expand the remaining product.
$(A+B+C).(\overline{A}+\overline{B}+\overline{C})$
$= A\overline{A}+A\overline{B}+A\overline{C}+B\overline{A}+B\overline{B}+B\overline{C}+C\overline{A}+C\overline{B}+C\overline{C}$
$= 0+A\overline{B}+A\overline{C}+B\overline{A}+0+B\overline{C}+C\overline{A}+C\overline{B}+0$
$= A\overline{B}+A\overline{C}+B\overline{A}+B\overline{C}+C\overline{A}+C\overline{B}$

$= \left(A\overline{B} + C\overline{B}\right) + \left(B\overline{A} + C\overline{A}\right) + \left(A\overline{C} + B\overline{C}\right)$

$= \left[(A+C)\overline{B}\right] + \left[(B+C)\overline{A}\right] + \left[(A+B)\overline{C}\right]$

165. Which of the following is not a valid expression of Boolean Algebra ?
निम्न में से कोन-सा बूलियन अलजेब्रा का एक उचित व्यंजक नहीं है?

(a) x + x.y = x (b) x'.y + x.y = y
(c) x'.y' + x.y = x (d) (x + y + z)' = x'.y'.z'

HTET PGT C.S. 2019

Ans. (c) : x'·y' +x·y ≠ it's actually the XNOR of x any % not equal to just x.

166. Boolean expression y.z + z is equal to which of the following ?
निम्न में से कौन-सा बूलियन अलजेब्रा का एक उचित व्यंजक नहीं है?

(a) y + y. z (b) y + z
(c) z (d) y.z

HTET PGT C.S. 2019

Ans. (c) : Let f = y.z + z
= z (y + 1)
= z (1) {∵ 1 + y = 1}
= z

167. Consider the following Boolean expressions :
निम्न बूलियन व्यंजकों पर विचार करें:

I. x.y + x'y'
II. XOR(x', y')
III. XOR(x', y)

Which of the above expressions represents exclusive NOR operation ?
उपरोक्त व्यंजकों में से कौन-से एक्सक्लूसिव NOR ऑपरेशन को निरूपित करते है?

(a) I and II/ I एवं II
(b) I and III/ I एवं III
(c) II and III/ II एवं III
(d) I, II and III/ I, II एवं III

HTET PGT C.S. 2019

Ans. (b) : XNOR (x,y) = x ⊙ y = x.y + x'.y'
XOR (x,y) = x ⊕ y = x.y'+x'y
Statement I: x.y + x'y'
reperesents exclusive NOR operation
Statement II: XOR (x',y')
It does not reperesents exclusive NOR operation
XOR (x', y') = x' ⊕ y' = x'(y') +(x')(y')' = x'.y + x.y' = x ⊕ y Which is not exclusive NOR operation.
Statement III: XOR (x'y) represents exclusive NOR operation.
XOR (x',y) = x' ⊕ y = x'.y' + (x').y = x'.y' + xy = x ⊙ y

168. Consider the Boolean function f = (a + bc) . (pq + r). Complement f ' of function f is :
बूलियन फलन f = (a + bc) . (pq + r) पर विचार करें। फलन f का कॉम्प्लिमेंट f' होगा।

(a) (a' + b'c') . (p'q' + r')
(b) a'(b' + c') + (p' + q')r'
(c) (a' + b'c') + (p'q' + r')
(d) (a'b'c') + (p'q'r')

HTET PGT C.S. 2018

Ans. (b) : By using Demorgan's law:
(A + B)' = A'.B' and (A.B)' = A' + B'
Boolean function f = (a + bc). (qp + r)
f = ((a + bc). (pq + r))
Complement of f is,
f' = ((a + bc). (pq + r))'
f' = (a + bc)' + (pq + r)'
f' = a'. (bc)' + (pq)'.r'
f' = a'. (b' + c') + (p' + q').r'

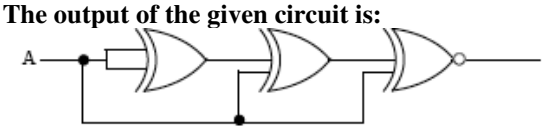
169. The number of NAND gates required for the implementation of the function f (A, B, C) = A + $\overline{BC}$ is:

(a) 3 (b) 4
(c) 5 (d) 6

CGPSC C.S. 2020 Set-A

Ans. (b) : The Boolean function:
f (A, B, C) = A + $\overline{BC}$
The function is:
f = A + $\overline{B \cdot C}$
This means we need:
• The AND of B and C
NAND gate
A NAND B = (A . B)
Total NAND Gates Used:
1. B NAND C → 1 gate
2. A NAND A → A → 1 gate
3. T1 NAND T1 → T1 → 1 gate
4. Final NAND for OR → combine A' and T1' → 1 gate
Total = 1 + 1 + 1 + 1 = 4 NAND gates

170. The output of the given circuit is:



(a) 0 (b) 1
(c) A (d) $\overline{A}$

CGPSC C.S. 2020 Set-A

Ans.(b):
• A ⊕ A
0
• 0 ⊕ A
A
• A ⊕ A
0
• $\overline{0}$
⇒ 1

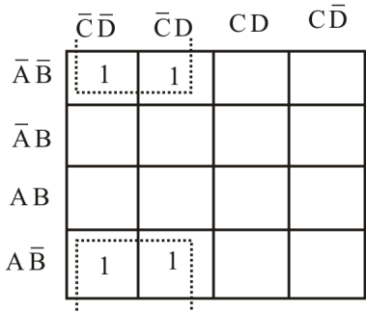
171. Simplified form of $\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + A\overline{B}\overline{C}\overline{D} + A\overline{B}\overline{C}D$ is:

(a) B + C (b) A B + CD
(c) $\overline{B}\overline{C}$ (d) None of the above

CGPSC C.S. 2020 Set-A

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Ans. (c) :
Let $f = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$
K– map:



$f = \bar{B}\cdot\bar{C}$

172. Simplified form of $\overline{(AB+\bar{C})}(\overline{\bar{A}+\bar{B}}+C)$ is:

(a) $(\bar{A}+\bar{B}+\bar{C})\cdot(A+B+C)$
(b) $(\bar{A}+B+C)\cdot(A+\bar{B}+\bar{C})$
(c) $(\bar{A}+\bar{B})\cdot(A+B+C)$
(d) None of the above

CGPSC C.S. 2020 Set-A

Ans. (d) : We need to apply the distributive law:
$(AB+\bar{C})(\bar{A}+B+C) = AB(\bar{A}+B+C)+\bar{C}(\bar{A}+B+C)$
First term:
$AB(\bar{A}+B+C)$
Distribute AB over each term:
- AB . B = ABB = AB (since B . B = B)
- AB . C = ABC

Second term:
$\bar{C}(\bar{A}+B+C)$
Distribute:
- $\bar{C}.\bar{A} = \overline{AC}$
- $\bar{C}.C = \bar{C}C = 0$

So,
$\bar{C}(\bar{A}+B+C) = \overline{AC}+B\bar{C}$
Step 3:
From above:
$AB+\overline{AC}+B\bar{C}$

173. The binary number 110011 is to be converted to Gray code. The number of gates and types required are:

(a) 6, AND (b) 6, X-NOR
(c) 6, X-OR (d) 5, X-OR

CGPSC C.S. 2020 Set-A

Ans.(d): Binary to Gray code conversion:
- MSB of Gray = MSB of Binary
- Each next Gray bit = XOR of current and previous Binary bits
- Example: Binary 110011 → 101010
- Implementation: 5 XOR gates (MSB passes directly)

174. The Boolean expression AB (A + B) can be reduced to

(a) ABA (b) AB
(c) $A^2B + AB^2$ (d) None of these
(e) Not attemtped

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : Step 1: Apply distributive law
AB (A + B) = ABA + ABB
Step 2 : Simplify
ABA = A.B.A = A^2B = AB (Since A^2 = A)
ABB = A.B.B = AB^2 = AB (since B^2 = B)
So,
AB (A + B) = AB + AB = AB

175. The expression for Absorption law is given by.....
अवशासन लॉ की अभीव्यक्ति .....द्वारा दी गई है।

(a) A + AB = A (b) A + AB = B
(c) AB + AA' = A (d) A + B = B + A

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : Absorption Law is a fundamental law in Boolean algebra that states the following two identities:
1. $A + A\cdot B = A$
2. $A\cdot(A+B) = A$

The first form A + A·B= A means that if A is true, then adding AB does not change the result.
The second form A·(A+B) = A means if A is true, then the entire expression is true regardless of B.

176. What are the canonical forms of Boolean Expression?
बूलियन एक्सप्रेशंस के विहित रूप क्या है?

(a) OR and XOR (b) NOR and XNOR
(c) MAX and MIN (d) SOM and POM

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : The canonical forms of Boolean expressions are SOM (Sum of Minterms) and POM (Product of Maxterms). These canonical are used to standardize and simplify Boolean expressions.

177. When converting a binary number to Gray Code, what is the result of XORing the most significant bit (MSB) of the binary number with the next bit?
जब किसी बाइनरी संख्या को ग्रे कोड में बदला जाता है, तो सबसे महत्त्वपूर्ण बिट (MSB) को अगली बिट के साथ XOR करने का क्या परिणाम होता है?

(a) The MSB of the Gray Code/ग्रे कोड की MSB
(b) The LSB of the Gray Code/ग्रे कोड की LSB
(c) The XOR of all the bits in the binary number बाइनरी संख्या की सभी बिट्स का XOR
(d) The XOR of all the bits in the Gray Code ग्रे कोड की सभी बिट्स का XOR

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : When changing a binary number into Gray code, the first Gray bit is identical to the first (most significant bit -MSB) bit of the binary number. The later Gray bits are found by taking an XOR of each binary bit with the bit just before it.

178. De Morgan's First Theorem states that the complement of a _____ of variables is equal to the _____ of their individual

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complements./डी मॉर्गन का पहला प्रमेय क्या कहता है: चर के _____का पूरक उनके व्यक्तिगत पूरकों के _____ के बराबर होता है।

(a) AND, OR (b) OR, AND
(c) NAND, NOR (d) NOR, NAND

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : De Morgan's first Theorem states that the complement of the OR of variables is equal to the AND of their individual complements.
In Boolean algebra from:
$(A + B)' = A' \cdot B'$

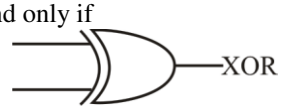
179. Which Boolean expression is correct interpretation of the following logic circuit ?
निम्नलिखित तर्क परिपथ (लॉजिक सर्किट) की सही व्याख्या वाला बूलियन व्यंजक कौन सा है ?



(a) $(X + Y)^1$ (b) $(X,Y)^1$
(c) $X \oplus Y$ (d) $(X \oplus Y)^1$

KVS PGT 23.03.2023 (Shift-II)

Ans, (c) : For the given logic circuit XOR represents its symbol +.
For 2 inputs, an XOR gate
Outputs 1 if and only if



One of the inputs is 1 and the other is O. If both inputs are the same, the output is O.
Hence, for XOR logic circuit, the correct interpretation is XOY.

180. Which of the following represents *Absorption Law* correctly ?
निम्नलिखित में से कौन सा अवशोषण नियम (एब्जॉर्प्शन लॉ) को सही रूप से निरूपित करता है?

(a) $X+Y.Z = (X+Y).(X+Z)$
(b) $X+X.Y = X$
(c) $(X+Y)^1 = X^1Y^1$
(d) $(X.X)^1 = X^1$

KVS PGT 23.03.2023 (Shift-II)

Ans, (b) : Absorption law enables a reduction in a complicated expression to a simpler one by absorbing like terms.
The expression for absorption laws is given by,
$X + X.Y = X$ or $X.(X + Y) = X$ is absorption law.

181. The term sum-of-product in Boolean algebra means :
बूलियन के बीजगणित में गुणनफल का जोड़ (सम-ऑफ-प्रोडक्ट) शब्द का अर्थ है :

(a) The AND function of several AND functions.
(b) The AND function of several OR functions.
(c) The OR function of several AND functions.
(d) The OR function of several OR functions.

KVS PGT C.S. 2017

Ans, (c) : The term Sum of product (SOP) is one kind of Boolean algebra expression. In this, the different product inputs are being added together. The product of input is Boolean logical AND whereas the sum or addition is Boolean logical OR. example of sum of product (SOP) is.
$F = X'Y + XY' + XY'$
Therefore the term sum-of-product in Boolean algebra means the OR function of several AND functions.

182. How many gate (s) would be required to implement the following boolean expression after simplification?
Expression : AC+ABC
सरलीकरण के पश्चात् निम्नलिखित बुलिअन व्यंजक को लागू करने के लिए कितने द्वारों (गेट) की आवश्यकता होगी?
व्यंजक : AC + ABC

(a) 4 (b) 1
(c) 2 (d) 3

KVS PGT C.S. 2017

Ans, (b) : Given Expression
$= AC + ABC$
$= AC(1+B)$
$= AC$



Therefore one 2- input AND gate is required to implement this simplified Boolean-expression.

183. The Boolean expression
बूलियन व्यंजक

$(A+C)(A\bar{B}+AC)(\overline{AC}+\bar{B})$

Can be simplified as :/को किस रूप में सरल किया जा सकता है?

(a) $\bar{A}B + BC$ (b) $A\bar{B}$
(c) $AB + BC$ (d) $AB + \bar{A}C$

KVS PGT C.S. 2017

Ans, (b) : Given Boolean Expression

$$(A+C)(A\bar{B}+AC)(\overline{AC}+\bar{B})$$
$$= (A+C)(A\bar{B}(\overline{AC}+\bar{B})+AC(\overline{AC}+\bar{B}))$$
$$= (A+C)(A\bar{B}\overline{AC}+A\bar{B}\bar{B}+AC\overline{AC}+AC\bar{B})$$
$$= (A+C)(A\bar{B}+AC\bar{B})$$
$$= (A+C)A\bar{B}(1+C)$$
$$= AA\bar{B}+A\bar{B}C$$
$$= A\bar{B}(1+C)$$
$$= A\bar{B}$$

184. How many gate(s) would be required to implement the following Boolean expression after simplification?
सरलीकरण के पश्चात् निम्नलिखित बुलयन व्यंजक को क्रियान्वित करने के लिए कितने गेट की आवश्यकता होगी?

Expression:

$$\bar{A}(\bar{C}\bar{D}+\bar{C}D)+AB(\bar{C}\bar{D}+\bar{C}D)+A\bar{B}\bar{C}$$

(a) 1 (b) 2
(c) 3 (d) 4

KVS PGT C.S. 2018

Sarkari Resource

Ans, (a) : Given Boolean expression-

$$= \bar{A}(\bar{C}\bar{D}+\bar{C}D)+AB(\bar{C}\bar{D}+\bar{C}D)+A\bar{B}\bar{C}$$

$$= \bar{A}\bar{C}(\bar{D}+D)+AB\bar{C}(\bar{D}+D)+A\bar{B}\bar{C} \qquad \left(\because \ \bar{D}+D=1\right)$$

$$= \bar{A}\bar{C}+AB\bar{C}+A\bar{B}\bar{C}$$

$$= \bar{A}\bar{C}+A\bar{C}\left(B+\bar{B}\right)$$

$$= \bar{C}\left(A+\bar{A}\right)$$

$$= \bar{C}$$

Therefore, only one NOT gate is enough to represent the given Boolean expression.

185. The simplified form of the Boolean expression; $\left(A\bar{B}(C+BD)+\bar{A}\bar{B}\right)C$ is
दिए गए बूलियन व्यंजक का सरल रूप क्या होगा?
$\left(A\bar{B}(C+BD)+\bar{A}\bar{B}\right)C$

(a) $\bar{B}C$ (b) $\bar{A}\bar{B}C$
(c) $A\bar{B}C$ (d) $\bar{A}BC$

KVS PGT C.S. 2018

Ans, (a) : Given Boolean expression

$$= \left(A\bar{B}(C+BD)+\bar{A}\bar{B}\right)C$$

$$= \left(A\bar{B}C+A\bar{B}BD+\bar{A}\bar{B}\right)C$$

$$= \left(A\bar{B}C+\bar{A}\bar{B}\right)C \qquad \because \bar{B}B=0$$

$$= \bar{B}C\left(AC+\bar{A}\right) \qquad \because \bar{A}+AC=\bar{A}+C$$

$$= \bar{B}C\left(\bar{A}+C\right)$$

$$= \bar{B}C\bar{A}+\bar{B}C.C$$

$$= \bar{B}C\left(\bar{A}+1\right) \ = \ \bar{B}C$$

186. The reduced expression for the following expression using a Karnaugh Map F (W,X,Y,Z) =∑(0,4,8,12) is :
एक कारनॉफ मानचित्र का प्रयोग करके निम्नलिखित व्यंजक के लिए लघुकरण व्यंजक है : F (W,X,Y,Z) =∑(0,4,8,12) :

(a) YZ (b) Y'Z'
(c) Y+Z (d) Y'+Z'

KVS PGT C.S. 2013

Ans, (b) : A karnaugh map is a visual method used to simplify the algebraic expressions in Boolean function without having to resort to complex theorems or equation manipulations.

Mapping the given function in a K-map we get 1 quad i.e.

WX \ YZ	[00] Y'Z'	[01]Y'Z	[11]YZ	[10]YZ'
[00]W'X'	1 (0)	0 (1)	0 (3)	0 (2)
[01]W'X	1 (4)	0 (5)	0 (7)	0 (6)
[11]WX	1 (12)	0 (13)	0 (15)	0 (14)
[10]WX'	1 (8)	0 (9)	0 (11)	0 (10)

$M_0+M_4+M_8+M_{12}$
$= W'X'Y'Z'+ W'XY'Z' + WX'Y'Z' + WX'Y'Z'$
$= W'Y'Z' (X'+X) + WY'Z' (X+X')$
$= W'Y'Z' + WY'Z'$
$= Y'Z' (W'+W)$
$= Y'Z'$

Simplified Boolean expression forgiven k- map is.
F (W,X, Y, Z) = Y'Z'

187. The Boolean expression :
बूलियन व्यंजक
a' · b + a · b'
Where a, b are Boolean operands, can equivalently represented by :
जहाँ a, b बूलियन संकार्य हैं, को समान रूप से किससे निरूपित किया जा सकता हैं:

(a) NAND (b) NOR
(c) XOR (d) OR

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans, (c) : Boolean Expression:-
a'.b + a.b'
⇒ (a' + ab') (b + ab')
⇒ (a' + a) (a' + b') (b + b') (b + a)
⇒ (a' + b') (b + a) → this is represented by – XOR

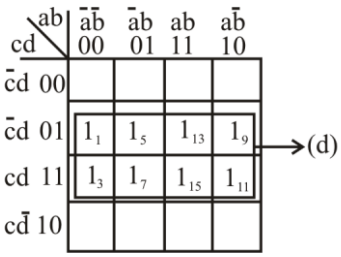
188. Identify the simplified and equivalent Boolean expression for the K-Map :
के-मैप के लिए सरल तथा समतुल्य बूलियन व्यंजक की पहचान कीजिए:

cd \ ab	00	01	11	10
00	0	4	12	8
01	1_1	1_5	1_{13}	1_9
11	1_3	1_7	1_{15}	1_{11}
10	2	6	14	10

(a) a (b) b
(c) c (d) d

NVS PGT C.S. 15.12.2022 (Shift-II)

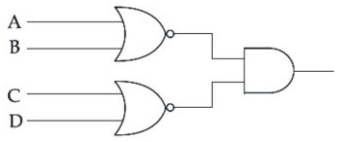
Ans, (d) :



Simplified and equivalent Boolean Expression
$= \left(\bar{c}d\right)+\left(cd\right)$
= (d)

189. Which of the following Boolean expression correctly represents the logic circuit diagram ?
लॉजिक परिपथ आरेख को सही प्रकार निरूपित करने वाला बूलियन व्यंजक कौन सा है?

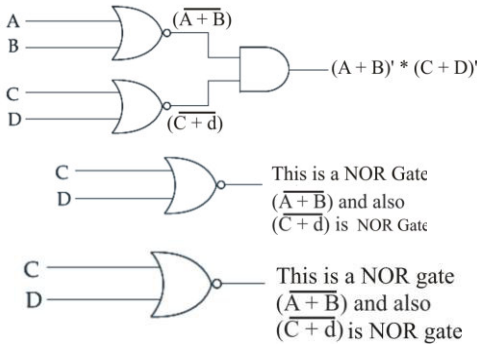
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(a) (A+B) . (C+D) (b) (A . B)' + (C . D)'
(c) A . B + C . D (d) (A+B)' . (C+D)'

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans, (d) :



⇒ Boolean Expression represents logic circuit diagram = two NOR gate * 1 AND gate (A.B)

⇒ (A + B)' . (C + D)'

190. In Boolean Algebra, the expression a + a . b is equivalent to :
बूलियन बीजगणित में व्यंजक a + a . b किसके समतुल्य है:

(a) a (b) b
(c) a . b (d) a + b

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans, (a) : Boolean Algebra Expression = a + a.b

⇒ according Absorption Law and Theorems of Boolean Algebra-
a + a.b

⇒ a(1 + b) ∵{1 + b = 1}

⇒ a(1) = (a)

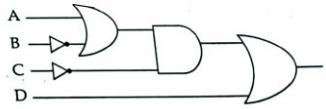
191. In Boolean algebra, A.(A+B) is logically equivalent to:
बूलियन बीजगणित (Boolean algebra) में A.(A+B) तार्किक रूप से किसके समतुल्य (logically equivalent) है?

(a) B (b) A.B
(c) A+B (d) A

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Ans, (d) : The expression A.(A + B)
= A.A + AB (∵ A.A = A)
= A + AB
= A (1 + B) (∵ 1 + B = 1)
= A

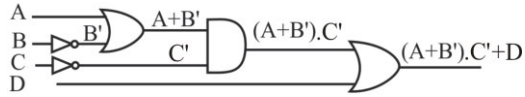
192. Identify the Boolean expression from the given options, which is equivalent to the Logic Circuit Diagram.
दिए गए विकल्पों में से उस बूलियन व्यंजक (Boolean expression) की पहचान कीजिए, जो लॉजिक सर्किट डायग्राम (Logic Circuit Diagram) के समतुल्य हो:



(a) (A + B').C' + D (b) A.B' + C'.D
(c) A+B'.C'+D (d) (A.B' + C').D

EMRS PGT C.S. 16.12.2023

Ans, (a) :



The output of this logic circuit Diagram is (A+B').C'+D.

193. The Boolean expression (A.B)' is equivalent to:
बूलियन व्यंजक (Boolean expression) (A.B)' किसके समतुल्य है?

(a) A'.B' (b) A+B'
(c) A.B' (d) A'+B'

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Ans, (d) : Given expression (A.B)'
A.B represents the AND operation between A and B. (A.B), represents the complement of the AND operation between A and B.
From the truth table of the AND operation we can see that the only case where A.B is true (1) is when both A and B are true (1). Therefore the complement of A.B will be false (0) is this case and true (1) for all other cases. So, the expression (A.B) is equivalent to A'+B' , where + represents the OR operation. This is known as De Morgan's Law.

194. The Boolean expression X + X'Y equals
बूलियन अभिव्यक्ति X + X'Y किसके समान है?

(a) X + Y (b) X + XY
(c) X + YX (d) X' Y + Y' X

LT. Grade C.S. 2018

Ans, (a) : According to Distributive law -
X + X'Y ⇒ (X + X') (X + Y)
= 1. (X + Y) ∵ X + X' = 1
= X + Y.

195. What is the purpose of the "don't care" condition in digital logic?/डिजिटल तर्क में ''डोन्ट केयर'' स्थिति का उद्देश्य क्या है?

(a) To indicate that the value of a variable does not affect the output
यह इंगित करने के लिए कि किसी चर का मान आउटपुट को प्रभावित नहीं करता है

(b) To prioritize certain inputs over others/कुछ इनपुट को दूसरों पर प्राथमिकता देना

(c) To ensure that all possible input combinations are covered in truth tables
यह सुनिश्चित करना कि सभी संभावित इनपुट संयोजन सत्य तालिकाओं में शामिल हैं

(d) More than one of the above
उपर्युक्त में से एक से अधिक

(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

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Ans. (a) : The don't care condition (often denoted as 'x') used in digital logic to represent input combinations that are irrelevant or impossible for the circuit. This means the output for those inputs can be either 0 or 1 without affecting the intended functionality.
So, the correct option is (a) To indicate that the value of a variable does not affect the output.

196. What is De Morgan's theorem used for in digital logic?/ डिजिटल तर्क में डी मॉर्गन प्रमेय का उपयोग किस लिए किया जाता है?

(a) Simplifying Boolean expressions
बूलियन अभिव्यक्तियों को सरल बनाना
(b) Implementing NAND gates
NAND गेट्स का कार्यान्वयन
(c) Performing binary addition
बाइनरी जोड़ करना
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d): De-Morgan's theorem is a fundamental concept in digital logic that provides rules for transforming logical expressions. It states that:
1. The complement of a conjunction (AND) is the disjunction (OR) of the complements.
$\overline{A \cdot B} = \overline{A} + \overline{B}$
2. The complement of a disjunction (OR) is the conjunction (AND) of the complements.
$\overline{A + B} = \overline{A} \cdot \overline{B}$
This theorem is used for:
Simplifying Boolean expressions: It helps in reducing the complexity of expression by transforming ANDs into ORs and vice versa.
Implementing logic gates: It allows for the implementation of any logic function using only NAND Gate or only NOR Gate.
So, the correct option is (d) More than one of the above.

197. The simplified form of the Boolean expression (X + Y + XY) (X + Z) is:
बूलियन अभिव्यक्ति (X + Y + XY) (X + Z) का सरलीकृत रूप है–

(a) XY + YZ
(b) X + YZ
(c) XZ + Y
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(b): Boolean expression (X + Y + XY) (X + Z)
= [X + Y (1 + X)] (X + Z)
= (X + Y) (X + Z) $\because \{1 + X = 1\}$
= X. X + X.Z + Y.X + Y.Z
= X + XZ + YX + YZ
= X (1 + Z) + YX + YZ $\because \{1 + Z = 1\}$
= X + YX + YZ
= X (1 + Y) + YZ $\because \{1 + Y = 1\}$
= X + YZ

198. Let $F_1 = (\overline{A + B}).(\overline{A' + B'})$ and $F_2 = A + \overline{A}B + \overline{AB}$ Choose a correct statement:
माना $F_1 = (\overline{A + B}).(\overline{A' + B'})$ तथा $F_2 = A + \overline{A}B + \overline{AB}$ सही कथन चुनें:

(a) $F_1 = 0$ and $F_2 = 0/F_1 = 0$ एवम $F_2 = 0$
(b) $F_1 = 0$ and $F_2 = 1/F_1 = 0$ एवम $F_2 = 1$
(c) $F_1 = 1$ and $F_2 = 0/F_1 = 1$ एवम $F_2 = 0$
(d) $F_1 = 1$ and $F_2 = 1/F_1 = 1$ एवम $F_2 = 1$

RPSC C.S.-I 2016

Ans, (b) : $F_1 = (\overline{A + B})(\overline{A' + B'})$
$= (\overline{A} \cdot \overline{B})(\overline{\overline{A}} \cdot \overline{\overline{B}})$
$= \overline{A} \cdot \overline{B} \cdot A \cdot B$
$= 0$
$F_2 = A + \overline{A}B + \overline{A}\overline{B}$
$= (A + \overline{A})(A + B) + \overline{A} + \overline{B}$
$= A + B + \overline{A} + \overline{B}$
$= (A + \overline{A}) + (B + \overline{B})$
$= 1 + 1$
$= 1$
Hence, $F_1 = 0$ and $F_2 = 1$

199. Simplified sum of product form of Boolean expression $(P + \overline{Q} + \overline{R}).(P + \overline{Q} + R).(P + Q + \overline{R})$ is:
बूलियन सूत्र $(P + \overline{Q} + \overline{R}).(P + \overline{Q} + R).(P + Q + \overline{R})$ का सरल सम ऑफ प्रोडक्ट रूप है:

(a) $(\overline{P}.\overline{Q} + \overline{R})$ (b) $(\overline{P}.Q + R)$
(c) $(P + Q.\overline{R})$ (d) $(P + \overline{Q}.\overline{R})$

RPSC C.S.-I 2016

Ans, (d) : Boolean Expression–
$F = (P + \overline{Q} + \overline{R}).(P + \overline{Q} + R).(P + Q + \overline{R})$
Step-1– use distributive property
The first two terms first–
$(P + \overline{Q} + \overline{R})$
$(P + \overline{Q} + R)$
both have P+ \ overline {Q} in common, so we apply (A + B) (A + C) = A + (B - C)
• $A = P + \overline{Q}$
• $B = \overline{R}$
• $C = R$
$(P + \overline{Q} + \overline{R})(P + \overline{Q} + R) = (P + \overline{Q}) + (\overline{R} \cdot R)$
$\overline{R} \cdot R = 0$
$\Rightarrow P = \overline{Q}(P + Q + \overline{R})$

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$F = (P + \bar{Q}) \cdot (P + Q + \bar{R})$

Step-2– remaining expression–

$(P + \bar{Q}) - (P + Q + \bar{R})$

$= (P \cdot (P + Q + \bar{R}) + (\bar{Q} (P + Q + \bar{R}))$

First term–

$(P \cdot (P + Q + \bar{R}) = P$

Second term–

$\bar{Q} \cdot (P + Q + \bar{R})$

We can expand–

$= (\bar{Q} \cdot P) + (\bar{Q} \cdot Q) + (\bar{Q} \cdot \bar{R})$

- $\bar{Q} \cdot P = P\bar{Q}$
- $\bar{Q} \cdot Q = 0$
- $\bar{Q} \cdot \bar{R}\ \overline{QR}$

$= P\bar{Q} + \overline{QR}$

Combine everything

$F = P + P\bar{Q} + \overline{QR}$

Now simplify–

- $P + P\bar{Q} = P(1 + \bar{Q}) = P(1) = P$

So, final expression:

$F = P + \overline{QR}$

Simplified sum of products–

$F = P + \bar{Q}.\bar{R}$

200. The Boolean expression $AB + A\bar{B} + \bar{A}C + AC$ is unaffected by the value of the Boolean variable_____.

(a) A (b) B
(c) C (d) A,B and C

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) : Let $F(A, B, C) = AB + A\bar{B} + \bar{A}C + AC$

$F(A, B, C) = A(B + \bar{B}) + (\bar{A} + A).C$

$F(A, B, C) = A + C$ ..... $(\because A + \bar{A} = 1$ and $1.A = A)$

Since the minimized Boolean expression does not contain Boolean variable B. Therefore, F(A, B, C) is unaffected by the value of the Boolean variable B.

201. The boolean expression $\bar{A}.B + A.\bar{B} + A.B$ is equivalent to

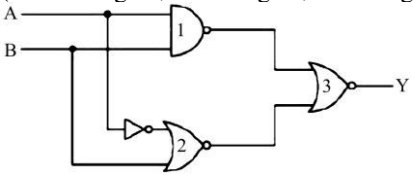
(a) A.B (b) $\overline{A + B}$
(c) A + B (d) $\bar{A}.B$

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : $A' . B + A . B' + A . B$
$= A' . B + A(B' + B)$
$= A' . B + A$
$= A + AB = (A + A)(A + B)$
$= A + B$

202. Find the boolean expression for the logic circuit shown below :

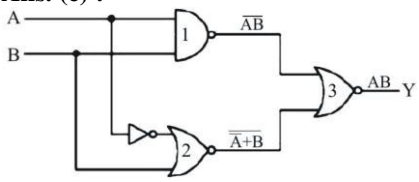
(1-NAND gate, 2-NOR gate, 3-NOR gate)



(a) $\bar{A}\bar{B}$ (b) $\bar{A}B$
(c) AB (d) $A\bar{B}$

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) :



So, option (c) is correct.

203. Simplify the following using K-map :

$F(A, B, C, D) = \Sigma (0, 1, 2, 8, 9, 12, 13)$

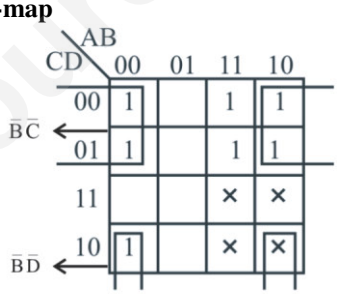
$d(A, B, C, D) = \Sigma (10, 11, 14, 15)$

d stands for don't care condition.

(a) $A + \bar{B}\bar{D} + BC$ (b) $A + \bar{B}\bar{D} + \bar{B}\bar{C}$
(c) $\bar{A} + \bar{B}\bar{D}$ (d) $\bar{A} + \bar{B}\bar{C} + \bar{B}\bar{D}$

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : K-map



$F(A,B,C,D) = A + \bar{B}\bar{D} + \bar{B}\bar{C}$

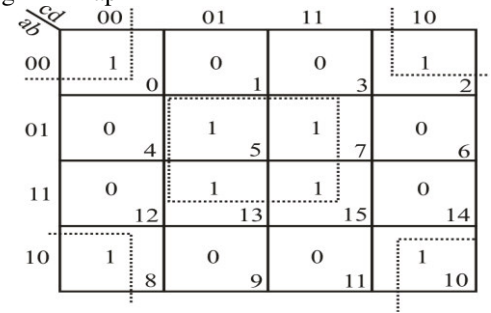
204. Simplify the following using K-Map

$F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$

(a) BD + B'D' (b) AC + A'C'
(c) BC + B'C' (d) AD + A'D'

UGC NET C.S. December-2022

Ans. (a) : Minterm = 0, 2, 5, 7, 8, 10, 13, 15
Variable = A, B, C, D
Using K = Map



Positions = 0, 2, 8, 10
Simplified B' D'
Positions = 5, 7, 13, 15
Simplified = B.D.
So Σ (0, 2, 5, 7, 8, 10, 13, 15) = BD + B'D'

205. The expression Y=AB+BC+AC shows the _____ operation.

अभिव्यक्ति Y=AB+BC+AC_____ ऑपरेशन को दर्शाती है।

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(a) EX-OR
(b) SOP
(c) POS
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : Expression Y=AB+BC+CA represents a set of AND terms that are ORed together. This is the Sum of Products (SOP) form. Each product term is also called a minterm.

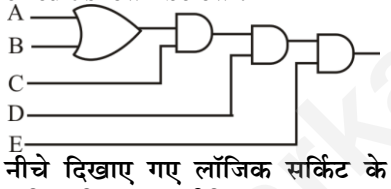
206. What are the canonical forms of Boolean expressions?
बूलियन एक्सप्रेशन के कैनॉनिकल रूप क्या हैं?

(a) OR and XOR/ OR और XOR
(b) NOR and XNOR/ NOR और XNOR
(c) SOM and POM/ SOM और POM
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : Canonical form: A canonical form of a Boolean expression is a standardized way of writing it so that every term contains all the variables of the function exactly once, either in true or complementary form.
Two types of canonical form-
(1). Sum of minterms (SOM/SOP canonical form)
(2).Product of maxterm (POM/POS canonical form)

207. Derive the Boolean expression for the logic circuit shown below :

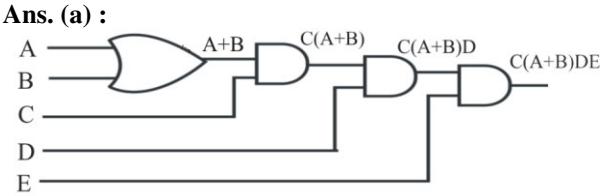


नीचे दिखाए गए लॉजिक सर्किट के लिए, बूलियन अभिव्यक्ति प्राप्त कीजिए :

(a) C (A + B)DE
(b) [C (A + B)D + E']
(c) [[C(A + B)D]E']
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) :



So, the Boolean expression for the logic circuit is : C(A + B)DE

208. The Boolean expression x + x'y equals
बूलियन एक्सप्रेशन x + x'y बराबर है

(a) x + y
(b) x + xy
(c) y + yx
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a) : Boolean expression = x + x'y
= x.1 + x'y
= x (1 + y) + x'y
{∵ 1 + y = 1 Null law}
= x + xy + x'y
= x + y (x + x')
= x + y
{∵ x + x' = 1 complement law}

209. The Boolean expression AB + AB' + A'C + AC is independent of Boolean variable
बूलियन एक्सप्रेशन AB + AB' + A'C + AC बूलियन वैरिएबल _____ पर निर्भर नहीं है।

(a) A
(b) B
(c) C
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): Boolean expression
⇒ AB + AB' + A'C + AC
⇒ A(B + B') + A'C + AC
⇒ A(1) + A'C + AC {(B + B' = 1) Inverse Law}
⇒ A + AC + A'C {(A+B = B+A) Commutative Law}
⇒ A + C(A + A')
⇒ A + C(1) { (A + A' = 1) Inverse Law}
⇒ A + C
Which is independent of B ..

210. The Boolean expression A + BC equals
बूलियन एक्सप्रेशन A + BC _____ के बराबर है।

(a) (A' + B) (A' + C)
(b) (A + B) (A' + C)
(c) (A + B) (A + C)
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : Boolean expression A + BC equals (A + B) (A + C)
This is obtained from Absorption law which states that A + AB = (A + A) (A + B)

211. The Boolean expression x + xy will be equal to
बूलियन एक्सप्रेशन x + xy के बराबर होगा

(a) x
(b) y
(c) x + y
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Sarkari Resource

Ans. (a) : Boolean expression = x + xy
= x (1 + y)
= x.1 {∵ 1 + y = 1 Null law}
= x

212. The Boolean expression x(x + y) is equal to
बूलियन एक्सप्रेशन x(x + y) बराबर है
(a) x
(b) y
(c) xy
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a) : Boolean expression = x (x + y)
= x.x + x.y
= x + x.y {∵ x.x = x Idempotent law}
= x (1 + y)
= x.1 {∵ 1 + y = 1 Null law}
= x

213. If x is a Boolean variable, then (((x')')')' will be
अगर x बूलियन वैरिएबल है, तो (((x')')')' होगा
(a) x′
(b) x
(c) x + x′
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : Taking the complement of twice, we bring back to the original number
Hence (((x')')')' = x

214. Simplified Boolean function of Boolean function F = A'C + A'B + AB'C + BC is
बूलियन फंक्शन (Boolean function)
F = A'C + A'B + AB'C + BC का सरलीकृत बूलियन फंक्शन है
(a) F = C + A'B
(b) F = A'B + B'C
(c) F = BC +A' B
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a): Simplification of Boolean function
F = A'C + A'B + AB'C + BC
= A'C + A'B + C (AB' +B)
= A'C + A'B + C [(A+B) (B+B')] {∵ B+B' = 1}
= A'C + A'B + C (A+B)
= A'C + A'B + AC + BC
= A'B + C (A+A') + BC {∵ A+A' = 1}
= A'B + C + BC
= A'B + C (1+B) {∵ 1 + B = 1}
= A'B + C

215. The minimum number of NAND gates required to implement Boolean function f = x'y + xy' is
बूलियन फंक्शन को इम्प्लीमेन्ट करने के लिए न्यूनतम कितने NAND गेट चाहिए?
(a) 3
(b) 4
(c) 6
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : f = x'y + xy'
This is already the standard XOR form because.
x ⊕ y = x'y + xy'
NAND only implementation of XOR:
A well know minimal NAND-only realization of XOR requires 4 NAND gates.
Steps:
1. P = NAND (x, y)
2. Q = NAND (x, p) (givens x ∧ ¬ y)
3. R = NAND (y, p) (given ¬ x ∧y)
4. F = NAND (Q, R) (Final XOR)

This outputs F = x'y + xy'
Hence, minimum number of NAND gates required = 4

216. Which of the following logic expressions is incorrect
(a) 1 ⊕ 0 = 1 (b) 1 ⊕ 1 ⊕ 1 = 1
(c) 1 ⊕ 1 ⊕ 0 = 1 (d) 1 ⊕ 1 = 0

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) With EX-OR operation, odd number of 1's returns 1 and even number of 1's return 0.
Here's the truth table of XOR (Exclusive-OR) gate:

Input A	Input B	Output (A⊕B)
0	0	0
0	1	1
1	0	1
1	1	0

• XOR gives 1 only when the inputs are different
• Expression: A⊕B = A'B + AB'
Therefore the correct option is (c) 1 ⊕ 1 ⊕ 0 = 1

217. Simplified Boolean equation for the following truth table is:

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Sarkari Resource

(a) $F = y\bar{z} + \bar{y}z$ (b) $F = x\bar{y} + \bar{x}y$
(c) $F = \bar{x}z + x\bar{z}$ (d) $F = \bar{x}z + x\bar{z} + xyz$

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Function f will be:

$$f = x'y'z + x'yz + xy'z' + xyz'$$
$$= x'z(y' + y) + xz'(y' + y)$$
$$= x'z + xz'$$

218. The simplified form of a Boolean equation $(A\bar{B} + A\bar{B}C + AC)(\bar{A}\bar{C} + \bar{B})$ is:

(a) $A\bar{B}$ (b) $A\bar{B}C$
(c) $\bar{A}B$ (d) ABC

UGC NET C.S. June-2016 (Paper-II)

Ans : (a) Given Boolean expression is:

$$= (A\bar{B} + A\bar{B}C + AC)(\bar{A}\bar{C} + \bar{B})$$
$$= (A\bar{B}(1 + C) + AC)(\bar{A}\bar{C} + \bar{B})$$
$$= (A\bar{B} + AC)(\bar{A}\bar{C} + \bar{B}) \quad \{\because 1 + c = 1\}$$
$$= A\bar{A}\bar{B}\bar{C} + A\bar{B} + A\bar{A}C\bar{C} + A\bar{B}C \quad \{\bar{B} \cdot \bar{B} = \bar{B}\}$$
$$= A\bar{B} + A\bar{B}C \quad \{A \cdot \bar{A} = 0\}$$
$$= A\bar{B}(1 + \bar{C}) \quad \{1 + \bar{C} = 1\}$$
$$= A\bar{B}$$

219. The Karnaugh map for a Boolean function is given as

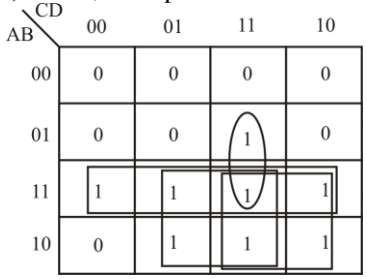
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	0	0	1	0
AB	1	1	1	1
$A\bar{B}$	0	1	1	1

The simplified Boolean equation for the above Karnaugh Map is

(a) AB + CD + $A\bar{B}$ + AD
(b) AB + AC + AD + BCD
(c) AB + AD + BC + ACD
(d) AB + AC + BC + BCD

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) Given, K map is



Therefore, boolean function is-
f = AB + AD + AC + BCD

220. Which of the following is the complement of the Boolean function.
A'B + CD' + A'B + CD'

(a) A'B + CD' (b) (A' + B) (C + D')
(c) (A + B') (C' + D) (d) AB' + CD'

UGC NET C.S. December-2024

Ans. (c) : The original Boolean function is given as A'B + CD' + A'B + CD'.
Simplifying this using the idempotent law, we get.
F = A'B + CD'
To find the complement F', we apply De Morgan's laws,
F' = (A'B + CD')'
F' = (A'B)'. (CD')'
Breaking down each part using De Morgan's law's
The complement of A'B is–
(A' B)' = (A')' + B' = A + B'
The complement of CD' is–
(CD')' = C' + (D')' = C' + D
Combining these result.
F' = (A + B'). (C' + D)
So, the correct answer is (c).

221. What is equivalent of the following Boolean expression?
निम्नलिखित बूलियन अभिव्यक्ति के समान क्या है?

$\overline{(x + y)}$

(a) $\bar{x} + \bar{y}$ (b) xy
(c) $\bar{x}\bar{y}$ (d) x

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Boolean expression-

$\overline{(x + y)}$

De Morgan's law states that

$$\overline{x + y} = \bar{x} \cdot \bar{y}$$

222. Simplify the following Boolean Expression?
निम्नलिखित बूलियन व्यंजक को सरल कीजिए?

$(1 + A).(A + B).(\bar{A} + B)$

(a) $\bar{B}$ (b) AB
(c) $\bar{A}\bar{B}$ (d) A + B

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Ans. (a) : Given Boolean expression-

$(1 + A) \cdot (A + B) \cdot (\bar{A} + B)$
$1 \cdot (A + B) \cdot (\bar{A} + B)$ ($\because 1 + A = 1$)
$(A + B) \cdot (\bar{A} + B)$

Now apply the distributive law.

$= A \cdot \bar{A} + A \cdot B + B \cdot \bar{A} + B \cdot B$
$= 0 + A \cdot B + B \cdot \bar{A} + B$ $\quad \because A \cdot \bar{A} = 0$
$= A \cdot B + B \cdot \bar{A} + B$ $\quad B \cdot B = B$

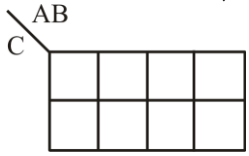
Now factor-

$= B \cdot A + B \cdot \bar{A} + B$

Take B Common–

$= B(A + \bar{A}) + B = B \cdot 1 + B$ $\quad \because (A + \bar{A} = 1)$
$= B + B$
$= B$

223. Minimize the three variable logic function using the following K-Map.
निम्नलिखित K-Map का उपयोग करके तीन वेरिएबल वाले लॉजिक फ़ंक्शन को मिनीमाइज करें।



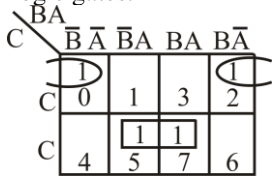
f(A, B, C) = Sm(0, 2, 5, 7)

(a) $\overline{A}\overline{C}+AB$ (b) $\overline{A}\overline{C}+AC$

(c) $AC+B$ (d) $\overline{A}C\overline{B}$

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : A Karnaugh-Map (k map) is a diagram used in Boolean algebra and digital logic design to simplify Boolean expressions. It helps reduce complex logic functions into simpler expressions, which are easier to implement with logic gates.



$=\overline{A}\overline{C}+AC$

224. What is the value of $\overline{F}$?

$F=AB+\overline{C}\,\overline{D}+\overline{B}D$

$\overline{F}$ **का मान क्या है**

$F=AB+\overline{C}\,\overline{D}+\overline{B}D$

(a) $\overline{A}\overline{B}+\overline{C}\overline{D}+\overline{B}D$

(b) $(\overline{A}+\overline{B})(C+D)(B+\overline{D})$

(c) $(A+B)(\overline{C}+\overline{D})(\overline{B}+D)$

(d) $(A+\overline{B})(C+\overline{D})(B+\overline{D})$

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : $F=AB+\overline{C}\overline{D}+\overline{B}D$

$\overline{F}=\overline{AB+\overline{C}\overline{D}+\overline{B}D}$

$=\overline{AB}\cdot\overline{\overline{C}\overline{D}}\cdot\overline{\overline{B}D}$ $\{\because \overline{A+B}=\overline{A}.\overline{B}$ De-Morgan's law$\}$

$=(\overline{A}+\overline{B}).(\overline{\overline{C}}+\overline{D}).(\overline{\overline{B}}+\overline{D})$

$=(\overline{A}+\overline{B})(C+D)(B+\overline{D})$ $\{\because \overline{\overline{A}}=A\}$

Hence correct option is (b) $(\overline{A}+\overline{B})(C+D)(B+\overline{D})$

225. Which of the following K-Map represent the function F?

F(A, B, C, D) = S(0, 1, 2, 6, 8, 9, 10)

निम्नलिखित में से कौन सा K-Map फलन का प्रतिनिधित्व करता है ?

F(A, B, C, D) = S(0, 1, 2, 6, 8, 9, 10)

(a)

AB \ CD	00	01	11	10
00	1	1		1
01				1
11				
10	1	1		1

(b)

AB \ CD	00	01	11	10
00		1	1	1
01				1
11				
10	1	1		1

(c)

AB \ CD	00	01	11	10
00	1			1
01	1			1
11				
10	1	1	1	

(d)

AB \ CD	00	01	11	10
00	1	1		1
01				1
11	1	1		
10				1

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : We are asked; which k-map represents
F (A, B, C, D) = $\sum$m (0, 1, 2, 6, 8, 9, 10)

Step 1. Recall minterm → cell mapping in 4-variable k-map

- Rows = AB in Gray code order: 00, 01, 11, 10
- Columns = CD in Gray code order: 00, 01, 11, 10
- Cell (AB, CD) corresponds to minterm m = ABCD (binary).

Step 2. Locate given minterms

- m(0) = 0000 ⇒ (AB = 00, CD = 00)
- m (1) = 0001 ⇒ (00, 01)
- m(2) = 0010 ⇒ (00,10)
- m (6) = 0110 ⇒ (10,10)
- m (8) = 1000 ⇒ (10,00)
- m (9) = 1001 ⇒ (10,01)
- m (10) = 1010 ⇒ (10, 10)

Step 3. Mark 1s in the k-map

- Row AB = 00 → cols 00,01,10 → 1s at (0,1,2)
- Row AB = 01 → col 10 → 1 at (6)
- Row AB = 10 → cols 00,01, 10 → 1s at (8,9,10)

226. Simplify the following Boolean expression.

E(E + F) + DE + D(E + F)

निम्नलिखित बूलियन व्यंजन का सरलीकरण ।

E(E + F) + DE + D(E + F)

(a) E + DF (b) F + DE

(c) D + EF (d) D + E+ F

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : Boolean expression = E (E + F) + DE + D (E + F)

= EE + EF + DE + DE + DF

= E + EF + (DE + DE) + DF $\{\because$ E.E=E$\}$

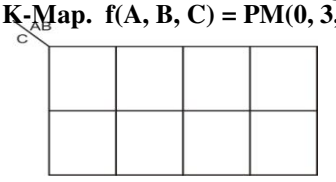
= E (1 + F) + DE + DF $\{\because$ DE + DE = DE$\}$

= E + DE + DF $\{\because$ 1 + F = 1$\}$

= E (1 + D) + DF

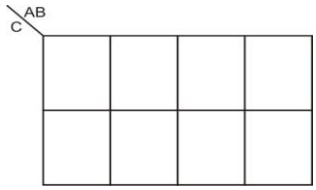
= E + DF $\{\because$ 1 + D = 1$\}$

227. Minimize the three variable logic function using K-Map. f(A, B, C) = PM(0, 3, 4, 7)



Sarkari Resource

K-Map. का उपयोग करके तीन वेरिएबल लॉजिक फंक्शन को मिनीमाइज करें
f(A, B, C) = PM(0, 3, 4, 7)



(a) $(\bar{B}+C)(B+\bar{C})$
(b) $(B+\bar{C})(B+C)$
(c) $(\bar{B}+C)(B+C)$
(d) $(\bar{B}+\bar{C})(B+C)$

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : Step-1– Understand the notation
f (A, B, C) = IIM (0,3,4,7) means the function is 0 at minterms 0, 3, 4, 7,
So, f = 1 at minterms 1,2,5,6
Step-2– Fill K-map

A	BC = 00	01	11	10
0	0	1	0	1
1	0	1	0	1

Step-3– Group the 1's
• Column 01 (minterms 1 and 5) → $\bar{B}C$
• Column 10 (minterms 2 and 6) → $B\bar{C}$
Step-4– Write minimal SOP
$$f = \bar{B}C + B\bar{C}$$
This is B ⊕ C.

228. The Boolean expression
$AB + \bar{A}C + BC =$ _____
बूलियन व्यंजक
$AB + \bar{A}C + BC =$ _____
(a) AB (b) AB + $\bar{A}B$
(c) AB + $\bar{A}C$ (d) $\bar{A}B$ + $\bar{A}C$

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Using the consensus theorem, the expression simplifies to: $AB + \bar{A}C + BC = AB + \bar{A}C$

229. A product of sum (POS) form of logic function consist of only _____ while a sum of product (SOP) form of logic function consist of only _____.
तर्क व्यंजक के प्रोडक्ट ऑफ सम (POS) फॉर्म में केवल होते हैं जबकि सम ऑफ प्रोडक्ट (POS) फॉर्म में केवल _____ होते हैं।
(a) maxterms; maxterms/मैक्सटर्म्स; मैक्सटर्म्स
(b) miniterms; minitersm/मिनीटर्म्स; मिनीटर्म्स
(c) miniterms; maxterms/मिनीटर्म्स; मैक्सटर्म्स
(d) maxterms; miniterms/मैक्सटर्म्स; मिनीटर्म्स

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : A Sum of Products (SOP) form of a logic function consist of only miniterms.
Product of Sums (POS)– this form is a product (and operation) of sums (OR operations) and consists of maxterms.
Sum of products (SOP)– This form is a sum (OR operation) of products (AND operations) and consists of miniterms.

230. Which of the following options is equivalent to the boolean expression A'C+A'BC'+A'B'C'?
निम्नलिखित में से कौन सा विकल्प बूलियन व्यंजक A'C+A'BC'+A'B'C'?
(a) A (b) A'
(c) AC (d) BC'

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Which of the following options is equivalent to the Boolean expression:
$$F = A'C + A'BC' + A'B'C'?$$
Step-1 : Factorize the terms
$$F = A'C + A'BC' + A'B'C'$$
$$F = A'(C + BC' + B'C')$$
Step-2 : Simplify inside the bracket
Inside: $C + BC' + B'$
Notice
$$BC' + B'C' = C'(B + B') = C' \cdot 1$$
So,
$$C + BC' + B'C' = C + C' = 1$$
Step-3: Final simplification
$$F = A'(1) = A'$$

231. Which of the following is the simplified from of the Boolean expression (AB)' (A'+B)(B'+B)?
निम्नलिखित में से कौन सा विकल्प बूलियन व्यंजक (AB)' (A'+B)(B'+B) का सरलीकृत रूप है?
(a) A (b) A'
(c) B (d) B'

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : We are asked to simplify:
$$F = (AB)' \cdot (A' + B') \cdot (B' + B)'$$
• (A + B)' (A' + B) (B' + B)
• (A + B)' (A' + B) (B + B' = 1)
• A'·A' + A' B + A' B' + B' B'
A'·A' = A'
• A' + A' (B + B')
• A' + A'
• A' Ans.

232. Using K-map, simplify the Boolean function: F(x, y, z) = S (0, 2, 4, 5, 6)
Which of the following is simplified function?
(a) F = z + x'y' (b) F = z' + xy'
(c) F = z + x'y (d) F = z' + xy

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : Reason: K-map (with columns yz = 00,01, 11, 10 and raws x = 0, 1) has 1s at m(0,2,4,5,6). All minterms with z = 0 (m0, m2, m4, m6 form a 4-group → gives z'.
The remaining 1 is m5 (x = 1, y = 0, z = 1) →
Gives, x·y'. Combine: F = z' + xy'.

Sarkari Resource

233. **A Boolean function F is called self-dual if and only if**
$F(x_1, x_2, ...x_n) = F(\bar{x}_1, \bar{x}_2, ...\bar{x}_n)$
How many Boolean function of degree n are self-dual ?

(a) 2^n (b) $(2)^{2^n}$
(c) $(2)^{n^2}$ (d) $(2)^{2^{n-1}}$

UGC NET C.S. June-2014 (Paper-II)

Ans : (d) A function F is called self dual if it has equal number of minterms and maxterms, also mutually exclusive terms should not be included the number of mutually exclusive term is (pair wise) is $2^n/2^{n-1} = 2$ number of functions possible by taking any of the one term from above mentioned mutually exclusive pair is $(2)^{2^{n-1}}$.

234. **Consider the following statements:**
(i) If primal (dual) problem has a finite optimal solution, then its dual (primal) problem has a finite optimal solution.
(ii) If primal (dual) problem has an unbounded optimum solution, then its dual (primal) has no feasible solution at all.
(iii) Both primal and dual problems may be infeasible.
Which of the following is correct?
(a) (i) and (ii) only (b) (i) and (iii) only
(c) (ii) and (iii) only (d) (i), (ii) and (iii)

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) These are fundamental duality theorem. All options are correct.

235. **If X is a binary number which is power of 2, then the value of X & (X – 1) is :**
(a) 11.....11 (b) 00.....00
(c) 100.....0 (d) 000.....1

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) Let $x = 2 \wedge 3 = 8 = 1000$
then $x - 1 = 7 = 0111$
now $x \,\&\, (x - 1) = 0000$
(here & is bitwise and) = if both in the compared position of the bits patterns are 1, the bit in the resulting bit pattern is 1, otherwise 0)

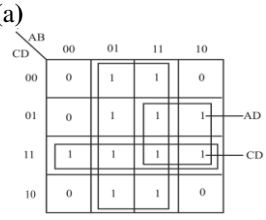
236. **The Boolean function with the Karnaugh map**

CD \ AB	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	0

(a) (A + C) . D + B (b) (A + B). C + D
(c) (A + D). C + B (d) (A + C). B + D

UGC NET C.S. November-2017 (Paper-II)

Ans : (a)



So, SoP from = AD + CD + B
= (A + C). D + B

237. **Let** $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **and** $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ **Find the Boolean product A ⊙ B of the two matrices.**

(a) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

UGC NET C.S. November-2017 (Paper-II)

Ans : (a)

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

we have to perform logical AND logical or in place of multiplication and addition for ex. consider following matrix Boolean product.
Now,

$$A \odot B = \begin{bmatrix} 1\vee0\vee0 & 0\vee1\vee0 & 0\vee1\vee0 & 0\vee0\vee0 \\ 0\vee0\vee0 & 0\vee1\vee0 & 0\vee1\vee0 & 0\vee0\vee0 \\ 1\vee0\vee0 & 0\vee1\vee0 & 0\vee1\vee0 & 0\vee0\vee0 \\ 0\vee0\vee1 & 0\vee0\vee0 & 0\vee0\vee1 & 0\vee0\vee1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

So, option (a) is correct.

238. **Out the following steps in the proper sequence for simplifying a Boolean function using a Karnaugh map (K-map).**
A. Identify and group the largest possible cluster of 1' s.
B. Draw the K-map for the given Boolean function.
C. Write the simplified Boolean expression from the grouped clusters.
D. Transfer the truth table values to the K-map.
Choose the correct answer from the options given below :
(a) (B), (D), (A), (C) (b) (D), (B), (A), (C)
(c) (B), (A), (D), (C) (d) (A), (B), (C), (D)

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Ans. (a) : The correct sequence for simplifying Boolean function using a Karnaugh map (k-map) is-
(B) Draw the k-map for the given Boolean function- Start by creating the k-map layout based on the number of variables.

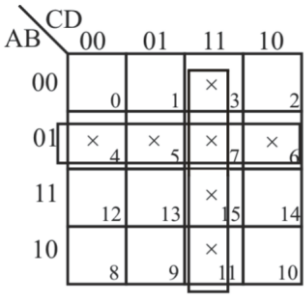
(D) Transfer the truth table values to the k-map-place the values from the truth table into the corresponding cell of the k-map.
(A) Identify and group the largest possible cluster of I's - Look for the largest groups of 1's in the k-map to simplify the function.
(C) write the simplified Boolean expression from the grouped clusters, Derive the simplified Boolean expression based on the Indented groups.

239. Simplify the following using K-Map F(A, B, C, D) = ∑(3, 4, 5, 6, 7, 11, 15)

(a) AB + C'D (b) A'B + CD
(c) AC + BD (d) A'B + B'C

UGC NET C.S. December-2022

Ans. (b) : Given K-map
F(A, B, C, D) = ∑(3, 4, 5, 6, 7, 11, 15)



So simplified Expression = A'B + CD

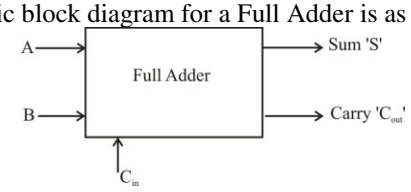
(iv) Combinational Circuits

240. A 2-bit full adder circuit may be designed with the help of the following 2-input logic gates :
एक 2-बिट फुल ऐडर सर्किट निम्न 2-इनपुट लॉजिक गेट्स की सहायता से डिजाइन किया जा सकता है:

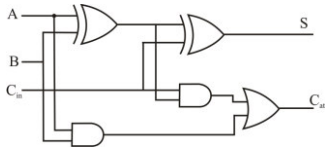
(a) 2-XOR, 2-AND, 1-OR
(b) 2-NOR, 2-NAND
(c) 1-XOR, 1-AND, 1-OR
(d) 2-AND, 2-OR

HTET PGT C.S. 2018

Ans. (a) : Add two bit:
A= 10 (A_1 A_0)
B = 11 (B_1 B_0)
$A_0 + B_0$ will give carry C_{in} → using half adder
Now using full adding Add $C_{in} + A_1 + B_1$
The basic block diagram for a Full Adder is as shown:

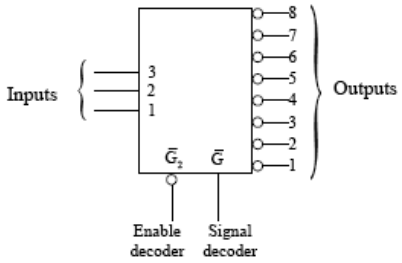


A full adder can be implemented using 2 XOR, 2 AND, 1 OR



Therefore option 1 is the most optimal answer.

241. In a 3-to-8 decoder given below:



all the output lines of the chip, except pin number 8 will be high when all the inputs 1, 2 and 3:

(a) are high; and G, G_2 are low
(b) are high; and G is low and G_2 is high
(c) are high; and G, G_2 are high
(d) are high; and G is high and G_2 is low

CGPSC C.S. 2020 Set-A

Ans.(d): Are high; and G is high and G2 is low: This combination correctly enables a decoder with an active-high G and active-low G2. When inputs 1, 2, and 3 are also high output pin 8 will be active (LOW), and all other output pins will be inactive (HIGH). This perfectly matches the condition given in the question.

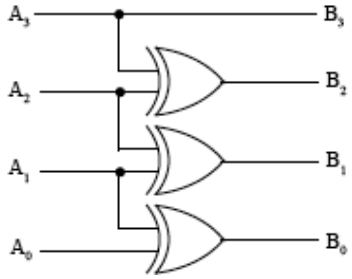
242. A 4-bit parallel Full Adder without input carry requires:

(a) 8 HA, 4 OR gates (b) 8 HA, 3 OR gates
(c) 7 HA, 4 OR gates (d) 7 HA, 3 OR gates

CGPSC C.S. 2020 Set-A

Ans. (d) : Step 1: Recall
A Full Adder = 2 Half Adders (HA) + 1OR gate
4-bit parallel adder normally needs 4FAs→ 8HA + 4OR
Without input carry, first stage can user 1HA instead of FA.
Remaining 3FAs → 6HA + 3OR
Total = 7 HA, 3 OR gates

243. The given circuit is used to convert one code to another. Identity if:



(a) Binary to Gray (b) Gray to binary
(c) Gray to XS-3 (d) Gray to 8421

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Ans. (a) : A binary→ Gray converter outputs the MSB unchanged and each next Gray bit = XOR of the current binary bit with the previous binary bit. The given signal names (A_1 B_1A_2, B_2,A_1..) Match that pattern (MSB forwarded, other outputs are XORs), so it implements binary to Gray conversion.

244. A full subtractor can be implemented by using:

(a) 3 to 8 line decoder only
(b) 3 to 8 line decoder and 1 OR gate

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(c) 3 to 8 line decoder and 2 OR gate
(d) None of the above

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Ans. (c) : A full subtractor has 3 inputs (A,B, Borrow-in) and 2 outputs (Difference, Borrow-out)
Using a 3-to-8 line decoder, all 8 minterms can be generated.
To get the outputs (Difference and Borrow), some minterms need to 0 red.
Difference = 1OR gate (OR selected minterms).
Borrow may need another OR gate.

245. In a PLA, what components are used to implement the combinational logic functions?
PLA मे संयोजनात्मक तर्क फंक्शन्स को क्रियान्वित करने के लिए कौन से कम्पोनेन्ट्स का उपयोग किया जाता है?
(a) AND gates and OR gates
(b) NAND gates and XOR gates
(c) NOR gates and XNOR gates
(d) NOT gates and multiplexers

Bihar STET C.S. 12.09.2023 (Shift-I)

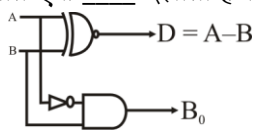
Ans. (a) : In a PLA (Programmable Logic Array), AND gates and OR gates are used to implement combinational logic functions.
AND array generates product terms based on input signals. These product terms are essentially min-terms of the input variables.
OR array combines the product terms to generate the final output functions. Each output is a sum of selected product terms.
So, the correct option is (a) AND gates and OR gates.

246. The logic circuit of binary adder which is used to add two 4-bits binary numbers, requires _____half adder (s) and _____full adder (s).
द्विआधारी योजक का तर्क परिपथ जिसका प्रयोग दो 4-बिट्स द्विआधारी संख्या को जोड़ने के लिए किया जाता है, इसमें _____अर्द्ध योजक और _____ पूर्ण योजक की आवश्यकता होती है।
(a) 4 , 0 (b) 1 , 3
(c) 2 , 2 (d) 3 , 1

KVS PGT C.S. 2017

Ans, (b) : Half adder is combinational logic circuit which is designed by connecting one EX-OR gate and one AND gate. The half adder circuit has two inputs, which add two input digits and generates a carry and a sum, Full adder is the circuit that consists of two EX-OR gates, two AND gates, and one OR gate, Full adder is the adder that adds three inputs and produces two outputs which consist of two EX-OR gates, two AND gates, and one OR gate,
If we want to add two n-bit binary adders then it requires 1 half adder and n-1 full adder to complete the circuit. So, add 4-bit binary numbers, requires 1 half adder and 3 full adders.

247. The diagram below shows a
निम्न आकृति एक ____ दर्शाती है ।



(a) Half adder/अर्द्ध योजक
(b) Half subtractor/अर्द्ध व्यकलक
(c) Full adder/पूर्ण योजक
(d) Full subtractor/पूर्ण व्यकलक

KVS PGT C.S. 2013

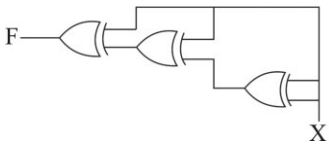
Ans, (b) : Half subtractor is a combinational logic circuit intended to perform the subtractor of two single bits and generate the output. A subtractor circuit with two input variable as A and B displays two output i.e. D = A–B and B_o
The logical expression for halt-subtractor
Difference (D) = $\overline{A}.B + A\overline{B} = A \oplus B$
Borrow = $\overline{A}B$

248. Determine the number of programmable inter-connections in the following programmable logic device- PAL device with eight input variables, 16 AND gates and four OR gates.
निम्नलिखित प्रोग्रामेबल लॉजिक डिवाइस के लिए प्रोग्रामेबल परस्पर संबंधों की संख्या निर्धारित कीजिए । जिसमें आठ इनपुट वेरीयेबल्स, 16 AND गेट्स और चार OR गेट्स के साथ PAL डिवाइस हैं
(a) 384 (b) 512
(c) 256 (d) 128

KVS PGT C.S. 2013

Ans (c) : In a PAL the AND plane is programmable (the OR plane is fixed). Each input variable provides two literal lins (True and complemented) so for 8 inputs there are 2 × 8 = 16 Posible literal conections per AND gate with 16 AND gate the total programmable interconnections = 16 × 16 = 256
PAL ⇒ Programmable Array Logic.

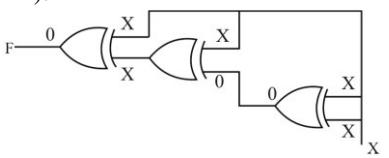
249. The output F of the circuit is given by-
परिपथ का निर्गत (output) F निम्न में से क्या होगा?



(a) 1 (b) 0
(c) X (d) X'

LT. Grade C.S. 2018

Ans, (b) : The circuit involves XOR gates (denoted by ⊕).



- In first XOR gate both inputs are X. So,
  $X \oplus X = 0$
- In second XOR gate one input is the output of the first XOR gate (0), and the other input is X. So,
  $0 \oplus X = X$
- In third XOR gate both inputs are X. So,
  $X \oplus X = 0$

Thus, the final output is 0.

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250. In a half-adder circuit, what are the outputs?/अर्द्ध-योजक सर्किट में आउटपुट क्या है?

(a) Sum and Carry
(b) Difference and Borrow
(c) Sum and Difference
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : A half adder is a simple digital circuit that adds two single bit binary numbers.
It gives Sum (S) output → result of $A \oplus B$ (XOR gate)
It gives Carry (C) output → result of $A \cdot B$ (AND gate)
Truth table for a half A adder:

Input A	Input B	Sum $A \oplus B$	Carry $A \cdot B$
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

251. What is a PLA?/पीएलए (PLA) क्या है?

(a) Programmable Logic Application
प्रोग्रामेबल लॉजिक एप्लीकेशन
(b) Programmable Logic Array
प्रोग्रामेबल लॉजिक ऐरे
(c) Programmable Logic Architecture
प्रोग्रामेबल लॉजिक आर्किटेक्चर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : PLA stands for Programmable Logic Array. A PLA is a digital circuit used to implement combinational logic. It contains a programmable AND array followed by a programmable OR array.
So, the correct option is (b) Programmable Logic Array

252. Digital circuit can be made by the repeated use of the _____./डिजिटल सर्किट _____ के बार-बार उपयोग से बनाया जा सकता है।

(a) NAND gate
(b) NOR gate
(c) AND gate
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) **:** Both NAND and NOR gates are universal gates. Any digital circuit can be built using only NAND gates or only NOR gates.
So, the correct option is (d) More than one of the above.

253. Which of the following is correct for Digital Circuits?/डिजिटल सर्किट के लिए निम्नलिखित में से कौन-सा सही है?

(a) They use analog signals for communication
वे एनालॉग सिग्नल का संचार में उपयोग करते हैं
(b) They process information using continuous voltage levels/निरंतर वोल्टेज स्तर, वे सूचना का उपयोग कर प्रक्रिया करते हैं
(c) They are not suitable for high speed operations
गति संचालन में वे उच्च के लिए उपयुक्त नहीं है
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (e) : Digital circuits use digital signals (discrete 0 and 1 levels) for communication, not analog signals (continuous).
So option (a) is incorrect.
Digital circuits process information using discrete voltage levels (e.g. 0V for logic 0 and 5V for logic 1), not continuous.
So, option (b) is incorrect.
Digital circuits are highly suitable for high-speed operations due to their noise immunity and precise switching characteristics.
So, option (c) is incorrect.
So, the correct option is (e) None of the above

254. How many AND gates are required to construct a 4 - bit parallel multiplier if four 4 - bit parallel binary adders are given?
यदि चार 4 बिट समांतर द्विआधारी योजक दिए गए हैं, तो 4 बिट समांतर गुणक के निर्माण के लिए कितने AND गेटों की आवश्यकता होगी?

(a) Four 2 - input AND gates
(b) Eight 2 - input AND gates
(c) Sixteen 2 - input AND gates
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : To construct a 4-bit parallel multiplier, we need to perform multiplication of two 4-bit numbers. let two numbers are A (A3A2A1A0) and B (B3B2B1B0). For each bit of A, we need to multiply it with each bit of B. This squires $4 \times 4 = 16$ AND operations, each of which can be performed by a 2-input AND gate.
So, the correct option is (c) sixteen 2-input AND gates.

255. If x, y, z are input variables of full adder-output sum (S) and carry (C) are given by:
यदि x, y, z फुल ऐडर के इनपुट चर है तो आउटपुट सम (S) तथा कैरी (C) निम्नानुसार होंगे:

(a) $S = xy + (x \oplus y)z$ and $C = x \oplus y \oplus z$
(b) $S = (x + y) \oplus z$ and $C = xy + y \oplus z$
(c) $S = x \oplus y \oplus z$ and $C = xy + (x \oplus y)z$
(d) $S = x \oplus y + z$ and $C = x \oplus y \oplus z$

RPSC C.S.-I 2016

Ans, (c) : The standard full adder logic equation–
• Sum(S) = $S = x \oplus y \oplus z$
• Carry (C) = $C = xy + (x \oplus y)z$
Now checking the options–
A. $S = xy + (x \oplus y)z$ and $C = x \oplus y \oplus z$
This is incorrect for both S and C
B. $S = (x + y) \oplus z$ and $C = xy + y \oplus z$
The sum part is not the standard full adder sum
C. $S = x \oplus y \oplus z$ and $C = xy + (x \oplus y)z$
This matches the standard full adder logic.
D. $S = x \oplus y + z$ and $C = x \oplus y \oplus z$
The sum and carry do not fit the full adder
Correct and → (c).

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256. **Consider the following properties for combinational digital circuit (C) and sequential digital circuit (S) :**
कॉम्बिनेशनल डिजिटल सर्किट (C) तथा सिक्वेन्शिअल डिजिटल सर्किट्स (S) के निम्न गुणधर्मो पर विचार करे:

I. Output is entirely dependent on current input./आउटपुट पूर्णतः वर्तमान इनपुट पर आधारित है।

II. Implemented using flip-flop./फ्लिप फ्लॉप के प्रयोग से बनाये जाते है।

III. Output is determined by sequence of inputs./आउटपुट इनपुट की श्रृंखला पर आधारित है।

Which of the following is valid match.
निम्न में से कौन सी सही जोड़ी है?

(a) C - I, II and S - III/ C - I, II एवम् S - III
(b) C - III and S - I/ C - III एवम् S - I
(c) C - II, III and S - I/ C - II, III एवम् S - I
(d) C - I and S - II, III/ C - I एवम् S - II, III

RPSC C.S.-I 2016

Ans, (d) : The properties listed are:
- I: Output is entirely dependent on current
- II: Implemented using flip-flop
- III: Output is determined by the sequence of inputs.

Correct mapping are–
- Combinational Circuits (C):
- I (output depends only on current input)
- Sequential Circuits (S):
- II (implemented using flip-flops)
- III (Output depend on sequence / history of input)

So, the valid match-
- C – I
- S – II, III

257. **Consider the following statements related to AND-OR Search algorithm.**

S1 A solution is a subtree that has a goal node at every leaf.
S2 OR nodes are analogous to the branching in a deterministic environment.
S3 AND nodes are analogous to the branching in a non-deterministic environment.

Which one of the following is true referencing the above statements?
Choose the correct answer from the code given below :
Code :

(a) S1 – False, S2 – True, S3 – True
(b) S1 – True, S2 – True, S3 – False
(c) S1 – True, S2 – True, S3 – True
(d) S1 – False, S2 – True, S3 – False

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : A solution for an AND-OR search problem is a subtree that
(1) has a goal node at every leaf
(2) specifies one action at each of its OR nodes
(3) includes every outcome branch at each of its AND nodes
(4) OR nodes are analogous to the branching in a deterministic environment.
(5) AND nodes are analogous to the branching in a non-deterministic environment.

258. **Match the following :**

A.	**Line coding**	**i.**	**A technique to change analog signal to digital data**
B.	**Block coding**	**ii.**	**Provides synchronization without increasing number of bits**
C.	**Scrambling**	**iii.**	**Process of converting digital data to digital signal**
D.	**Pulse code modulation**	**iv.**	**Provides redundancy to ensure synchronization and inherits error detection**

Code :

	A	B	C	D
(a)	(iv)	(iii)	(ii)	(i)
(b)	(iii)	(iv)	(ii)	(i)
(c)	(i)	(iii)	(ii)	(iv)
(d)	(ii)	(i)	(iv)	(iii)

UGC NET C.S. June-2016 (Paper-III)

Ans : (b)
(A) Line coding is a process of converting digital data into digital signal (III)
(B) Block coding Provides redundancy to ensure synchronization and inherits error detection (IV)
(C) Scrambling is Provides synchronization without increasing number of bits (II)
(D) Pulse code modulation (PCM) is a technique to change analog signal to digital data (I)

259. **Both the inputs of NAND gate are connected to A. What will be the the output?**
NAND गेट के दोनों इनपुट A से जुड़े हुए हैं। आउटपुट क्या होगा?

(a) 0 (b) 1
(c) A (d) $\overline{A}$

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : A NAND gate is a two input and single output device. It is a combinational logic circuit and a NAND gate if formed by combining an AND gate and followed by a NOT gate. If any of the one input is 0 then the output will be 1. If the both inputs of NAND gate is A then its out will be $\overline{A}$



Sarkari Resource

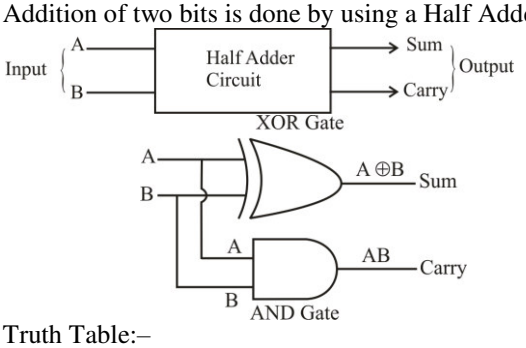
Truth table:			
A	A	A.A	$y = \overline{A \cdot A} = \overline{A}$
0	0	0	1
1	1	1	0

260. Which of the following represents carry in half adder?
निम्नलिखित मे से कौन सा हाफ एडर में कैरी को दर्शाता है?

(a) $A\overline{B}$ (b) $\overline{A}B$
(c) $\overline{AB}$ (d) $\overline{A}\,\overline{B}$

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : Half Adder: It is a combinational logic circuit of two inputs and two outputs. The first output gives the sum and the second output gives the carry. Addition of two bits is done by using a Half Adder.



Truth Table:–

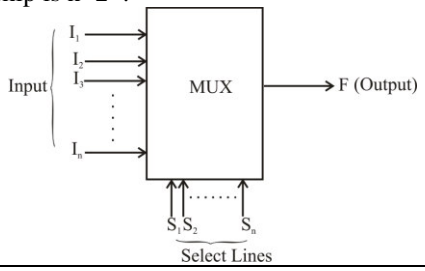
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

261. In a multiplexer, the number of input is 'n' and the number of select lines is 'm'. Which of the following is correct?
एक मल्टीप्लेक्सर में इनपुट लाइनों की संख्या 'n' है और सेलेक्ट लाइनों की संख्या 'm' है। निम्नलिखित में से कौन सा सही है?

(a) $2^m = n$ (b) $2^n = m$
(c) $2m = n$ (d) $2n = m$

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : Multiplexer: A multiplexer (MUX) is a digital circuit that selects one input line from many input lines and sends it to a single output line, based on the combination of control signals called select lines.
It there are 'm' select lines, they can produce 2^m different combinations, that means the multiplexer can choose from 2^m input lines.
So, if the number of input lines is n, the correct relationship is $n=2^m$.



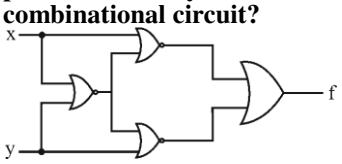
262. A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. What is the bit rate?

(a) 100 kbps (b) 200 kbps
(c) 400 kbps (d) 1000 kbps

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Ans : (c) bit rate = no. of channels × Input bit rate
= 4 × 100 kbps = 400 kbps

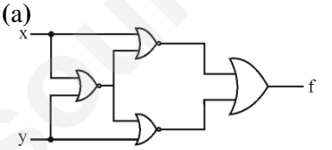
263. Which of the following logic operations is performed by the following given combinational circuit?



(a) EXCLUSIVE-OR
(b) EXCLUSIVE-NOR
(c) NAND
(d) NOR

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Ans : (a)



$$f = \overline{\overline{((x+y)+y)}+\overline{((x+y)+x)}}$$

$$f = \overline{\overline{((x+y)\cdot\bar{y})}+\overline{((x+y)\cdot\bar{x})}}$$

$$f = ((x+y)\cdot\bar{y})+((x+y)\cdot\bar{x}) \quad \because y.y' \text{ or } x.x' = 1$$

$$f = (x\bar{y})+(\bar{x}y)$$

$$f = x \oplus y = x(\text{EXOR})_y$$

264. Match the following :

List - I		List - II	
A.	**Controlled Inverter**	**i.**	**a circuit that can add 3 bits**
B.	**Full adder**	**ii.**	**a circuit that can add two binary numbers**
C.	**Half adder**	**iii.**	**a circuit that transmits a binary word or its 1's complement**
D.	**Binary adder**	**iv.**	**a logic circuit that adds 2 bits**

Codes :

	A	B	C	D
(a)	iii	ii	iv	i
(b)	ii	iv	i	iii
(c)	iii	iv	i	ii
(d)	iii	i	iv	ii

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Ans. (d) **:**
- Half adder used to add 2 bits
- Full adder used to add 3 bits
- Controlled inverter is a circuit that transmits a binary word or its 1's complement
- Binary adder used to add 2 binary number.

Hence, answer is (d).

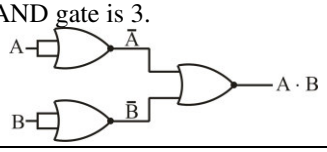
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265. How many NOR gates are used to get AND gate form them?
AND गेट बनाने के लिए कितने NOR गेट का उपयोग किया जाता है?

(a) 1 (b) 3
(c) 2 (d) 4

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : The minimum number of NOR gates needed to create an AND gate is 3.



266. A railway crossing has 4 railway tracks. It a train comes on at least 1 of the tracks, the crossing has to closed. Which gate can be used to implement this?
एक रेलवे क्रॉसिंग में 4 रेलवे ट्रैक होते हैं, यदि कम से कम 1 ट्रैक पर कोई ट्रेन आती है, तो क्रॉसिंग को बंद करना पड़ता है। इसे लागू करने के लिए किस गेट का इस्तेमाल किया जा सकता है?

(a) NAND (b) NOR
(c) OR (d) AND

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Ans. (c) : The railway crossing must be closed it a train comes on at least one of the tracks.
This means the output (Close gate) should be 1 (true) if any one or more inputs (tracks) are 1 (true).
This is exactly the behavior of an OR gate. OR Gate Truth Table (for 2 inputs, generalizes to more)-

0 OR 0 = 0
0 OR 1 = 1
1 OR 0 = 1
1 OR 1 = 1

So for 4 inputs, the OR gate will output 1 if at least one input is 1, which is the required behavior.

267. Which of the following is an advantage of use of multiplexers?
निम्नलिखित में से कौन सा मल्टीप्लेक्सर के उपयोग का एक लाभ है?

I. Logic design is simplified
लॉजिक डिजाइन सरलीकृत होता है।

II. Minimizes IC package count
IC पैकेज संख्या को कम करता है।

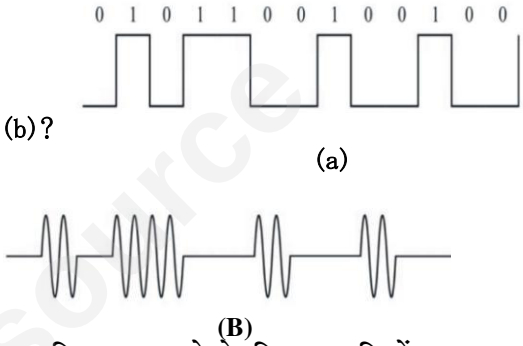
III. Simplification of logic expression is not required/लॉजिक अभिव्यक्ति के सरलीकरण की आवश्यकता नही होती है।

(a) I and II/ I तथा II
(b) II and III/ तथा III
(c) I and III/ I तथा III
(d) I, II and III/ I, II तथा III

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Multiplexers offer several advantages in digital circuit design.
1. Logic design is simplified– Multiplexers allow complex logic functions to be implemented easily by selecting appropriate inputs, reducing the need for extensive combinational logic design.
2. Minimizes IC package count– Since a single multiplexer can replace multiple logic gates, it reduces the number of integrated circuits (ICs) needed, saving board space and cost.
3. Simplification of logic expression is not required– When using multiplexers Boolean algebra simplification is often unnecessary because the MUX inherently implements the logic function based on its input selection.
Frequency– Division Multiplexing is a very simple and easy modulation that allows a large number of signals to be sent simultaneously.

268. Which modulation technique is applied to a binary signal in figure (a) to achieve figure (b)?



आकृति प्राप्त करने के लिए आकृति में एक बाइनरी सिग्नल पर कौन सी मॉड्यूलेशन तकनीक लागू की जाती है ?

(a) Amplitude modulation /एम्प्लीट्यूड मॉड्यूलेशन
(b) Frequency modulation /फ्रीक्वेंसी मॉड्यूलेशन
(c) Phase modulation /फेज मॉड्यूलेशन
(d) Quadrature phase shift keying /क्वडरेचर फेज़ शिफ्ट कियिंग

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : Frequency modulation (FM) is a technique that is used to encode information in an analog signal by modulating the frequency of the signal. FM is commonly used in radio communication systems to transmit audio signals, can also be used to transmit other types of information, such as images.
To encode an image using FM, the image would first need to be converted into digital representation, such as a series of pixel values. The pixel values could then be used to modulate the frequency of a carrier signal, such as a radio frequency (RF) signal, in a way that reflects the information contained in the image.

269. Which of the following represents the borrow in a half subtraction ?
निम्नलिखित में से कौन सा हॉल्फ सब्ट्रैक्शन में उधार का प्रतिनिधित्व करता है ?

(a) AB (b) $A\bar{B}$
(c) $\bar{A}B$ (d) $\bar{A}\bar{B}$

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Half Subtractor Circuit:–
A half Subtractor is a combinational logic circuit used for binary subtraction. It subtracts two binary digits and generates two outputs. The difference and the borrow-out. It is called a "Half Subtractor" because it does not

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take into account any borrow input from previous stages, unlike the Full Subtractor.

Turth Table:

Input A	Input B	Difference (A ⊕B)	Borrow-out $(\bar{A}B)$
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

So, Borrow is represented by $(\bar{A}B)$

270. In a demultiplexer, the number of output lines is n and the number of select lines is m. Which of the following is correct?
एक डीमल्टीप्लेक्सर में, आउटपुट लाइनों की संख्या n है और सेलेक्ट लाइनों की संख्या m है। निम्नलिखित में से कौन सा सही है ?

(a) $n = 2^m$ (b) $m = 2^n$
(c) $n = 2m$ (d) $m = 2n$

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : Demultiplexer:- A Demultiplexer (DEMUX) is a combinational logic circuit that takes one input and routes it to one of many output lines based on the values of the select (control) lines. It performs the reversal operation of multiplexer. In a Demultiplexer, if the number of output lines is n and the number of select lines is m then $n = 2^m$.

271. What is the relation between N select inputs and M inputs lines in a multiplexer?
एक मल्टीप्लेक्सेर में, N चुनिंदा इनपुट और M इनपुट लाइनों के बीच क्या संबंध है?

(a) $2^M = N$ (b) $2^N = M$
(c) $2M = N$ (d) $2N = M$

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : In a multiplexer (MUX), the relationship between the number of select inputs N and the number of input lines M is given by the formula N.

$M = 2$

This means that a multiplexer with N select inputs can choose between 2^N different input lines.

272. What is the full form of DCTL in logic family?
लॉजिक फेमिली में DCTL का संक्षिप्त रूप क्या है?

(a) Direct – coupled transistor logic
डायरेक्ट – कपल्ड ट्रांजिस्टर लॉजिक
(b) Diode – coupled transistor logic
डायोड – कपल्ड ट्रांजिस्टर लॉजिक
(c) Direct – circuit transistor logic
डायरेक्ट – सर्किट ट्रांजिस्टर लॉजिक
(d) Direct – circuit threshold logic
डायरेक्ट – सर्किट थ्रेशहोल्ड लॉजिक

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : The full form of DCTL in logic family is Direct Coupled Transistor Logic.

273. A decimal to Binary Coded Encoder have _____ inputs and _____ outputs.
एक डेसीमल बाइनरी कोडेड एनकोडर में _____ इनपुट और _____ आउटपुट होता है।

(a) 4 : 4 (b) 10 : 4
(c) 4 : 10 (d) 2 : 10

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : A decimal-to-binary encoder is defined as a digital circuit that converts a decimal number into its binary equivalent. It is also known as a decimal to BCD (binary-coded decimal) encoder, it represent each decimal digit with a 4-bit binary code.
The encoder has 10 input lines for the decimal digits and 4 output lines for the binary code, converting one active input at a time.

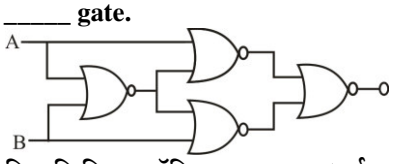
274. A logic circuit that accepts one input and distributes it over several output is known as _____.
एक लॉजिक सर्किट जो एक इनपुट को स्वीकार करता है और इसे कई आउटपुट में वितरित करता है, _____ के रूप में जाना जाता है।

(a) multiplexer/मल्टीप्लेक्सेर
(b) encoder/एनकोडर
(c) flip flop/फ्लिप फ्लॉप
(d) demultiplexer/डीमल्टीप्लेक्सेर

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : A demultiplexer (also known as a demux or data distributor) is defined as a circuit that can distribute or deliver multiple outputs from a single input. A demultiplexer can perform as a single input with many output & switch.

275. The following logic diagram is equivalent to _____ gate.



निम्नलिखित लॉजिक डायग्राम (तर्क आरेख) _____ गेट के बराबर है।

(a) XOR (b) NAND
(c) NOR (d) XNOR

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Step 1: Analyze the first stage of the circuit
The inputs A and B are fed into a combination of NOT gates and NAND gates
1. Input A is inverted to produce A'
2. Input B is inverted to produce B'
3. A NAND gate receives inputs A' and B.
Its output 01 is given by the Boolean expression:
$01 = \overline{A' \cdot B}$.
4. Another NAND gate receives inputs A and B'. Its output 02 is given by the Boolean expression:
$02 = \overline{A \cdot B'}$
Step 2 : Analyze the second and third stages of the circuit
The outputs 01 and 02 are then fed into another NAND gate, whose output is then inverted by a NOT gate to produce the final output Y.
1. The output of the third NAND gate (03) is :
$03 = \overline{01.02}$.

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2. The final output Y is the inversion of 03 $Y = \overline{03.} = \overline{01} \cdot \overline{02}$.

Step 3: Simplify the Boolean expression for Y

Using the property that $\overline{\overline{X}} = X$, the expression for Y simplifies to

$Y = 01 \cdot 02$.

Now substitute the expression for 01 and 02 :

$Y = \overline{(A' \cdot B)} \cdot \overline{(A \cdot B')} \cdot$

Apply De Morgan's Theorem, which states that $\overline{X \cdot Y} = \overline{X} + \overline{Y}$ and $\overline{X + Y} = \overline{X} \cdot \overline{Y}$.

Also, $\overline{\overline{X}} = X$.

Applying De Morgan's to $\overline{A' \cdot B}$:

$\overline{A \cdot B} = \overline{A'} + \overline{B'} = A + B$.

Applying De Morgan's to $\overline{A \cdot B'}$:

$\overline{A \cdot B'} = \overline{A'} + \overline{B'} = \overline{A} + B$.

Substitute these simplified expressions back into the equation for Y.

$\mathbf{Y} = \left(A + \overline{B}\right) \cdot \left(\overline{A} + B\right)$.

Expand the expression:

$Y = A \cdot \overline{A} + A \cdot B + \overline{A} \cdot \overline{B} + \overline{B} \cdot B$

Using Boolean algebra identities, $A \cdot \overline{A}. = 0$

and $B \cdot \overline{B} = 0$.

So, the expression becomes

$Y = 0 + A \cdot B + \overline{A} \cdot \overline{B} + 0$

$Y = A \cdot B + \overline{A} \cdot \overline{B}$.

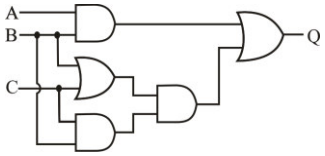
Step 4 identify the equivalent gate

The Boolean expression $Y = A \cdot B + \overline{A} \cdot \overline{B}$

Corresponds to the Exclusive NOR (XNOR) gate.

The XNOR gate outputs a '1' when both input are the same (both '0' or both '1'), and a '0' when the inputs are different.

276. For A = 1, B = 0 and C = 1, what will be the value of Q in the following logic diagram?
A = 1, B = 0 और C = 1 के लिए, निम्नलिखित लॉजिक डायग्राम (तर्क आरेख) में Q का मान क्या होगा?



(a) 0
(b) 1
(c) Cannot be determined/निर्धारित नहीं किया जा सकता है।
(d) Both 0 and 1/ 0 और 1 दोनों

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : The problem asks us to determine the Value of Q in the given logic diagram when A = 1, B = 0, and C = 1

Step 1 :- Analyze the first level of gates.

• The first gate is an AND gate with inputs A and B.

⇒ Given A = 1 and B = 0 the output of this AND gate is A.B = 1.0 = 0.

• The second gate (connected to c) is a NOT gate.

⇒ Given c =1, the output of this NOT gate is $\overline{c} = \overline{1} = 0$

Step 2:- Analyze the second level of gates.

• The third gate is an OR gate with inputs from the output of the first AND gate (0) and the output of this OR gate is 0 + 0 = 0

Step 3:- Analyze the final gate-

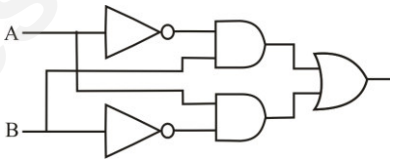
• The last gate is an AND gate with inputs A and the output of the previous OR gate (0).

⇒ Given A = 1 and the output of the OR gate is O, the final output Q is

A. (output of or) = 1.0 =0.

The value of Q in the given logic diagram for A = 1, B = 0, and c = 1 is o.

277. What is the output of the following logic diagram? In the given options, ⊕ represents XOR operation, whereas ⊗ represents XNOR operation.
निम्नलिखित लॉजिक डायग्राम (तर्क आरेख) का आउटपुट क्या है? दिए गए विकल्पों में, ⊕ XOR ऑपरेशन को दर्शाता है, जबकि XNOR ऑपरेशन को दर्शाता है।



(a) A⊕B
(b) A⊗B
(c) AB
(d) A'B'

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : To find output of the given logic diagram we need to analyze the circuit step by step.

Step-1 : Identify the gates and their operations

The diagram contains the following gates.

• Two Not gates (inverters), indicated by the circles on the inputs of the first tow gates.

• Two AND gates, with inverted inputs (Making then effectively NAND gates with one input inverted or a combination that creates an XNOR gate function).

• One OR gates at the final Stage.

The problem states that "⊕ represents XOR operation whereas represents XNOR operation"

Step 2: Determine the output of the first stage (after the Not gates)

The inputs are A and B.

• The upper path has input A going through a NOT gate, so its output is $\overline{A}$.

• The lower path has input B going through a NOT gate, So its output is $\overline{B}$.

Step 3: Determine the output of the intermediate gates.

• The upper intermediate gate takes $\overline{A}$ and B as inputs to an AND gate. The output of this gate is $\overline{A}$ .B. to an AND gate. The ouput of this gate is A. $\overline{B}$.

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Step 4: Determine the final output
The final gate an OR gate, taking the outputs of the two intermediate AND gates as its inputs. SO the final output Y is:

$$Y = (\bar{A}B) + (A.\bar{B}.)$$

Step 5: Recognize the Boolean expression
The Boolean expression $Y = \bar{A}.B + A.\bar{B}.$ is the definition of an XOR (Exclusive OR) gate. Therefore, the output of the entire logic diagram is $A \oplus B$.

278. Consider a full - adder with the following input values :
(a) x = 1, y = 0 and C_1 (carry input) = 0
(b) x = 0, y = 1 and C_1 = 1
Compute the values of S(sum) and C_0 (carry output) for the above input values.
(1) S = 1, C_0 = 0 and S = 0, C_0 = 1
(2) S = 0, C_0 = 0 and S = 1, C_0 = 1
(3) S = 1, C_0 = 1 and S = 0, C_0 = 1
(4) S = 0, C_0 = 1 and S = 1, C_0 = 0

UGC NET C.S. June-2015 (Paper-II)

Ans.(a) : A full adder can be implemented in many different ways such as with a custom transistor level circuit or composed of other gates.
One implemented is with $S = A \oplus B \oplus C_{in}$ and
$C_{out} = (A \cdot B) + (C_{in}(A \oplus B))$.
Given,
S_1: x = A = 1, y = B = 0 and $C_{in} = 0$
Then, $S = 1\ S = 1 \oplus 0 \oplus 0 = 1 \oplus 0 = 1$
$C = (1.0) + (0(1 \oplus 0)) = 0$
S_2: x = A = 0, y = B = 1 and $C_{in} = 1$
Then, $S = 0 \oplus 1 \oplus 1 = 0 \oplus 0 = 0$
$C = (0.1) + (1(0 \oplus 1)) = 0 + 1 = 1$

279. The BCD adder to add two decimal digits needs minimum of
(a) 6 full adders and 2 half adders
(b) 5 full adders and 3 half adders
(c) 4 full adders and 3 half adders
(d) 5 full adders and 2 half adders

UGC NET C.S. December-2014 (Paper-II)

Ans : (c) To add two BCD decimal digits we need two 4-bit address. First one for adding the two BCD number and second one the make the result a valid BCD number. The first simple 4-bit adder can be constructed using 3 full adders and 1 half adder (let it be A_3 A_2 A_1 A_0) In second step to make the result a valid BCD 0110 is added to A_3 A_2 A_1 A_0

	C_3	C_2	C_1	C_0	
Result from 1st adder		A_3	A_2	A_1	A_0
Adding b		0	1	1	0

Since A_0 is always added to 0, no need for any adder. A_1 is added to only '1' and carry C_0 is zero, so half adder A_2 is added to '1' and non-zero cary C, so full adder is usedC_3
A_3 is added to C_2 only hence a half-adder is used C_3 and resulting bits from the BCD of resulting sum of the decimal digits.
Total 3+1 = 4 full adders and 1+2=3 half adders is used.

280. Consider the following statements regarding combinational and sequential circuits.
A. Output of combinational circuits depends on the only current input.
B. Output of combinational circuit depends on the both current input and previous output.
C. Output of sequential circuit depends on the current input.
D. Output of sequential circuit depends on both current input and previous output.
Choose the correct answer from the options given below :
(a) (A) and (C) Only (b) (A) and (D) Only
(c) (B) and (D) Only (d) (B) and (C) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : Combinational circuits produces outputs based slowly on the current inputs. They do not have memory, So the previous outputs do not affect the current output.
Sequential circuits produces outputs based on both the current inputs and the previous outputs as they includes memory elements to store information about past inputs. Thus, combinational circuits are simpler and faster, while sequential circuits are more complex, involving feedback and memory.
So, the given statements (A) and (D) are correct.

281. To display time in railway stations which digital circuit is used?
रेलवे स्टेशनो में समय प्रदर्शित करने के लिए किस डिजिडल सर्किट का उपयोग किया जाता है?
(a) Seven-segment decoder/सात -सेगमेंट डिकोडर
(b) Eight-segment encoder/आठ-सेगमेंट एनकोडर
(c) 8 : 3 multiplexer/8 : 3 मल्टीप्लेक्सर
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : A seven segment Decoder is a digital circuit that converts binary coded into the signals required to drive a seven-segment display.
• Input : usually 4 bits (eg. 8421 BCD)
• Output : 7 control signal , one for each segment (a - g)
• logic : The decodes uses logic gates (AND, OR, NOT, NAND, etc) to generate correct output.

282. How many full adders are required for n-bit binary adder?
n-बिट बाइनरी एडेर के लिए कितने फुल एडेर्स की आवश्यकता है?
(a) 2× n (b) $\frac{n}{2}$
(c) n (d) n^2

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : An-n-bit binary adder requires n full adders. Each full adder is responsible for adding one pair of bits (one from each of the n-bit numbers) and a carry-in-bit from the previous stage. The carry out from each full adder is then passed on as the carry-in to the next stage, forming what is known as a ripple – carry adder.

Sarkari Resource

283. By including which logical gate binary adder circuit is used for both binary addition and subtraction ?
किस लॉजिकल गेट को शामिल करके बाइनरी एडेर को बाइनरी जोड़ और घटाव दोनों के लिए उपयोग किया जाता है?

(a) Ex – AND gate/ Ex – AND गेट
(b) AND gate/ AND गेट
(c) Ex– OR gate/ Ex– OR गेट
(d) OR gate/ OR गेट

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : A binary added circuit can be converted into a binary added-subtract by using an Ex-OR (Exclusive-OR) gate. This gate is crucial because it can be used as a controllable inverter.

284. The size of the ROM required to build an 8-bit adder/subtracter with mode control, carry input, carry output and two's complement overflow output is given as :

(a) $2^{16} \times 8$ (b) $2^{18} \times 10$
(c) $2^{16} \times 10$ (d) $2^{18} \times 8$

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because, total input to the ROM decoder will be (8 + 8(two 8 bit number) + 1(mode) + 1(carry in))
So total number of words out of decoder will be 2^{18}.
Result will be 8 bit so 8 vertical lines + (1 for carry) + 1 (for saying under flow) means $2^{18} \times 10$
So the answer is (b)

285. A 4 bit combinational circuit "shifter" to implement left and right shift operations on 4 bit number may be designed using:
एक 4 बिट नंबर पर लेफ्ट व राईट शिफ्ट ऑपरेशन करने वाला 4 बिट कॉम्बिनेशनल सर्किट ''शिफ्टर'' निम्न के प्रयोग से बनाया जाता है?

(a) Four 2 - to - 4 decoders/चार 2 - टू - 4 डिकोडरस्
(b) Two 3 - to - 8 decoders/दो 3 - टू - 8 डिकोडरस्
(c) Four 2 - to - 1 multiplexers
चार 2 - टू - 1 मल्टीप्लेक्सरस्
(d) Two 4 - to 1 multiplexers
दो 4 - टू - 1 मल्टीप्लेक्सरस्

RPSC C.S. 2016 (PAPER-I)

Ans, (c) :
(a) Four 2 - to - 4 decoders–
- Decoders convert binary inputs into one hot outputs, they are not useful for routing data paths directly which is needed in shifting.
- Not suitable

(b) Two 3 - to - 8 decoders–
- Again, there are for decoding binary inputs into one-hot format not typically used for shifting bits.
- Not suitable

(c) Four 2 - to - 1 multiplexers–
- Each 2 - to - 1 MUX can select between two inputs based on one control bit.
- For shifting you need to select which bit goes into each position with careful design, this could work for either left or right shift but not both.
- Only enough for a single shift direction (Left or right), not both.
- Not sufficient for a full shifter

(d) Tow 4 - to - 1 multiplexers–
- Each 4 - to - 1 MUX can select one of 4 inputs based on 2 control lines.
- You would need one 4 - to - 1 MUX per output bit to select among all possible bits for left/right shift. So for a 4 - bit shifter, you had need four 4 - to - 1 muxes, not two.
- Two are not enough.

286. Determine the size of PROM required for implementing the 16-to-1 multiplexer.
16 से 1 मल्टीप्लेक्सर के क्रियान्वयन के लिए PROM की साईज निर्धारित कीजिए ।

(a) 1M x 1 (b) 2M x 1
(c) 8M x 1 (d) 32M x 1

KVS PGT C.S. 2013

Ans, (a) : A PROM in general has n input and m output lines and is designated as $2^n \times$ m PROM.
Combinational circuit we need $2^n \times$ m size PROM
Here inputs are 16+ (4 selection lines in 16×1 mux) =20
So, n = 20 and m= 1
Hence PROM size = 2^{20} = 1m
So answer will be 1m × 1

(v) Sequential Circuits

287. A 2-bit synchronous counter uses flip-flops with propagation delay times of 25 nsec each. The maximum possible time required for change of state will be:

(a) 25 nsec (b) 50 nsec
(c) 75 nsec (d) 100 nsec

CGPSC C.S. 2020 Set-A

Ans. (a) : In a 2-bit synchronous counter, all flip-flops are triggered simultaneously by the clock.
Each flip-flop has a propagation delay of 25 nsec.
Since, it is synchronous the maximum delay is just the propagation delay of a single flip-flop, not cumulative.

288. 12 MHz clock frequency is applied to a cascaded counter of MOD 3 counter, MOD 4 counter and MOD 5 counter. The lowest output frequency is:

(a) 200 kHz (b) 1 MHz
(c) 3 MHz (d) 4 MHz

CGPSC C.S. 2020 Set-A

Ans. (a) : Frequency division formula-
For cascaded counters, output frequency $F_{out} = \frac{F_{in}}{MOD1 \times MOD2 \times MOD3}$
MOD-3 → divides by 3
MOD-4 → divides by 4
MOD-5→ divides by 5

$$F_{out} = \frac{12MHz}{3 \times 4 \times 5} = \frac{12MHz}{60} = 0.2MHz = 200kHz$$

Sarkari Resource

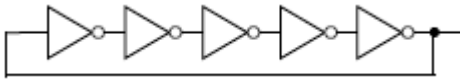
289. N flip-flops can be used to divide the input clock frequency by:

(a) N (b) 2N
(c) 2^N (d) 2^{N-1}

CGPSC C.S. 2020 Set-A

Ans. (c) : A counter made of N flip-flops can count from 0 to 2^N -1.
So, it divides the input clock frequency by 2^N

290. For the oscillator, find the fundamental frequency if the propagation delay of each inverter is 1000 psec.



(a) 100 MHz (b) 10 MHz
(c) 1 GHz (d) 10 GHz

CGPSC C.S. 2020 Set-A

Ans. (a) : Ring oscillator frequency; $f = \frac{1}{2Nt_p}$
Propagation delay, $t_p = 1000$ psec. $= 10^{-9}$ sec
With 3 inverters, $f \approx 100$ MHz.

291. For a NOR latch, A and B are made first (0, 1) and after a few seconds (1. 1). The corresponding output Q1 and $\overline{Q1}$ are:

(a) first (1, 0) and then (0, 0)
(b) first (1, 0) and then (1, 0)
(c) first (1, 0) and then (1, 1)
(d) first (1, 0) and then (0, 1)

CGPSC C.S. 2020 Set-A

Ans. (a) • First inputs (A,B) = (0,1):-
A = 0, B = 1
When, B = 1 and A = 0 $\rightarrow$ $\overline{Q}$ = NOR (1, Q) $\rightarrow \overline{Q} = 0$
(because NOR of any input being 1 (gives 0)
Then, Q = NOR (0, $\overline{Q}$) $\rightarrow$ Q = NOR (0, 0) $\rightarrow$ Q = 1
So, Q = 1, $\overline{Q}$ = 0
• After few seconds inputs (A,B) = (1, 1) :-
A = 1, B = 1
Both inputs are 1 $\rightarrow$ this is the invalid or forbidden state for NOR latch, but typically in such case both outputs go to 0 (Q = 0 and $\overline{Q}$ = 0).
Hence, first (1, 0) and then (0, 0).

292. Which of the following flip-flops does not have a problem of race condition?

(a) JK (b) Clocked D
(c) Clocked RS (d) Unclocked RS
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

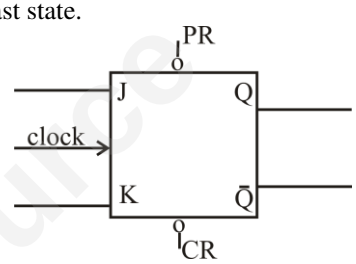
Ans. (b) : Race condition in flip-flops
JK flip-flop$\rightarrow$ has race around condition when J = K = 1 and clock is high.
Clocked RS flip-flop$\rightarrow$can have invalid state (R=S=1) also race-like issue.
Unlocked RS (latch)$\rightarrow$ Sensitive to input changes$\rightarrow$ race problems.
D flip- flop$\rightarrow$designed to remove the ambiguity of RS and race problem.
D flip-flop does not have race condition.

293. In a positive edge triggered Jk flip-flop, a low J and a low K produces :
किसी पाजिटिव एज प्रवर्तित (ट्रिगर्ड) जे.के. फ्लिपफ्लॉप में, निम्न 'जे' और निम्न 'के' से प्राप्त होता है:

(a) No change/कोई परिवर्तन नहीं
(b) Low state/निम्न स्थिति
(c) High state/उच्च स्थिति
(d) Toggle state/टोगल स्थिति

KVS PGT C.S. 2017

Ans, (a) : JK flip-flop is improvement over RS flip-flop in which forbidden condition is defined. when J=0, K=0 irrespective of the clock pulse both the AND gate are disabled resulting S=0, R=0. Therefore output Q- remains in the last state.



So, In positive edge triggered JK flip-flop with a low J and a low K will have no change in the output.

294. When does a negative level triggered flip-flop in Digital Electronics changes its state?/डिजिटल इलेक्ट्रॉनिक्स में नकारात्मक स्तर ट्रिगर फ्लिप-फ्लॉप अपनी स्थिति कब बदलता है?

(a) When the clock is Negative
जब घड़ी नेगेटिव हो
(b) When the clock is Positive
जब घड़ी सकारात्मक हो
(c) When the inputs are all zero
जब सभी इनपुट शून्य हो
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans (a) : A negative level-triggered flip-flop changes its state only when the clock signal is at low level (0 or negative). It does not change when the clock is high (positive). The inputs (like, J, K, D etc) are only read and the state is updated during this negative clock level.

295. Which of the following statement is not true about J-K flip-flop?
J-K फ्लिप फ्लॉप के बारे में कौन-सा कथन सत्य नहीं है?

(a) It behaves like S-R flip flop when both J & K are not 1 simultaneously
यह S-R फ्लिप फ्लॉप की तरह कार्य करता है जब J व K दोनों साथ 1 ना हो
(b) It behaves like T Flip flop when J = K = 1
यह T फ्लिप फ्लॉप की तरह कार्य करता है जब J = K = 1 है
(c) It behaves like D flip flop when J = K = 1
यह D फ्लिप फ्लॉप की तरह कार्य करता है जब J = K = 1 है

Sarkari Resource

(d) When J = K = 0, the state remain unchanged on a clock signal/जब J = K = 0 हो तो क्लॉक सिग्नल पर इसकी स्टेट अपरिवर्तित रहती है

RPSC C.S.-I 2016

Ans, (c) :
(a) It behaves like S-R flip-flop when both J& K are not 1 simultaneously
→ This is true
When J and K are not both 1, The J-K flip-flop works like an S-R flip-flop.
- J = 1, K = 0 → set
- J = 0, K = 1 → Reset
- J = 0, K = 0 → No change

(b) It behaves like T flip-flop when J = K = 1
→ This is true
When both inputs are 1, the J-K flip-flop toggles (changes its state to the opposite), just like a T flip-flop.
(c) It behaves like D flip-flop when J = K = 1
→ This is false
AD flip-flop transfers the D input to the Q output directly on the clock edge but in a J-K flip-flop, when J = K = 1, The output toggles it does not just copy a single input value like a D flip-flop.
(d) When J = K = 0, the state remains unchanged on a clock signal.
→ This is true
With both inputs at 0, the J-K flip-flop holds (latches) it's current state, regardless of the clock.
Option (c) is not true.

296. The minimum number of D flip flop needed to design a mod-264 counter is:
एक मोड़-264 काउंटर डिजाइन करने के लिए कितने D फ्लिप फ्लॉप की आवश्यकता होगी
(a) 8 (b) 9
(c) 264 (d) 265

RPSC C.S.-I 2016

Ans, (b) : Step-1– A mod - 264 counter counts from 0 to 263 this gives 264 unique states.
So, we need a counter that can represent at least 264 different states.
Step-2– Each flip-flop can store 1 bit (0 to 1) with n flip-flops, we can represent.
2^n different states
$2^n \geq 264$
Such that–
- $2^7 = 128 < 264$
- $2^8 = 256$ → not enough (only counts up to 255)
- $2^9 = 512$ → enough (can count up to 511)

$2^9 = 512$ → Satisfied
⇒ **9 Ans**

297. In RS flip-flop, the output of the flip-flop at time (t + 1) is same as the output at time t, after the occurrence of a clock pulse if :
(a) S = R = 1 (b) S = 0, R = 1
(c) S = 1, R = 0 (d) S = R = 0

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : S = R = 0 is the normal resting condition of the flip-flop. It has no. effect on the output state of the flip-flop. Both QQ and Q^-Q^- output remain in the logic state they were in prior to this input condition.

S = R = 0

R	S	Q_{t+1}
0	0	No change
0	1	0
1	0	1
1	1	* race

298. In D flip-flop, if D = 1, then the output of the D flip-flop is _____, but if D = 0, then the output of D flip-flop goes to _____ state.
D फ्लिप-फ्लॉप में यदि D = 1 है, तो D फ्लिप-फ्लॉप का आउटपुट _____ है, लेकिन यदि D = 0 है, तो D फ्लिप-फ्लॉप का आउटपुट _____ अवस्था में जाता है।
(a) 0; 1 (b) 1; 0
(c) 0; 0 (d) 1; 1

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : D flip-flop (Data or Delay flip flop): A D flip flop is a type of sequential digital circuit that stores one bit of data. It is edge-triggered, meaning it changes its output only on a specific transition of the clock signal (rising edge or falling edge).
Truth table of D flip-flop:

Clock	Input	Output	
	D	Q	Q'
Low	x	0	1
High	0	0	1
High	1	1	0

From the truth table of D flip flop, if D=1, then the output of the D flip flop is 1 but if D=0, then the output of D flip flop goes to 0 state.

299. In a JK flip flop, if J = K, then resulting flip flop is _____.
JK फ्लिप फ्लॉप में, यदि J = K है, तो परिणामी फ्लिप फ्लॉप _____ है।
(a) SR
(b) Clocked SR/क्लौक्ड SR
(c) T
(d) D

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : J-K flip-flop :- A JK flip flop is a bistable (two state) memory device. It has two inputs J and K, a clock input and two outputs Q and $\bar{Q}$.
Working of JK flip flop:
If J=0, K=0→ output stays the same
If J=0, K=1→ output reset to 0
If J=1, K=0→ output sets to 1
If J=1, K=1→ output toggles (changes to the opposite state) on every clock pulse.
T flip flop: A T flip flop is a basically a special case of a JK flip flop where both J and K inputs are connected together and act as a single T input.
T flip flop= JK flip flop with J=K=1

Sarkari Resource

300. In a positive-edge-triggered JK flip-flop, if *J* and *K* both are high then the output will be _____ on the rising edge of the clock.

(a) No change (b) Set
(c) Reset (d) Toggle

UGC NET C.S. June-2016 (Paper-II)

Ans : (d) Racing is problem of toggling more than once when the inputs are $J = 1$ and $K = 1$. The proper combination of propagation delay and edge triggering prevents the racing problem.

301. Which counter is a shift register with its complement output of the last stage connected to the D-input of the first stage?
कौन सा काउंटर पहले स्टेज के D-इनपुट से जुड़े अंतिम स्टेज के अपने कॉम्प्लिमेंट आउटपुट के साथ एक शिफ्ट रजिस्टर है?

(a) Synchronous/सिन्क्रोनस
(b) Twisted ring/ट्विस्टेड रिंग
(c) Up/अप
(d) Down/डाउन

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) :
(i) Synchronous– A synchronous counter is one where all flip-flops are clocked simultaneously by a common clock signal while a Johnson counter is typically synchronous (all flip-flops share the same clock), " synchronous" describes the timing.
(ii) Twisted ring– This directly refers to the Johnson counter, where the complement output of the lost stage is fed back to the first stage's D-input, this matches the question's description exactly.
(iii) UP– An up counter increments it's count (000, 001, 010, 011,….in binary). The sequence of Johnson Computer (000, 100, 110, 111, 011, 001) does not follow.
(iv) Down– A down counter decrements its count (111, 110, 101, 100….). The Johnson counter's sequence does not follow.

302. Which of the following pair is/are correct?
निम्नलिखित में से कौन सा/से युग्म सही है/है?

I. Astable multi vibrator – Flip Flop
एस्टेबल मल्टी वाईब्रेटर – फ्लिप फ्लॉप

II. Bistable multi vibrator – Free running
बाईस्टेबल मल्टी वाईब्रेटर – फ्री रनिंग

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Astable Multivibrator– This is a type of multivibrator that continuously oscillates between two states, meaning it has no stable states. It is also known as a "free running" oscillator. However it is not a flip-flop.
Bistable multivibrator– This is a type of multivibrator with two stable states. It is also known as a flip-flop. The key difference is that an astable multi vibrator constantly switches between states, while a bistable multi vibrator stays in one stable until triggered to switch.

303. A flip-flop is capable of storing _____ information.
एक फ्लिप –फ्लॉप _____ जानकारी संग्रहीत करने में सक्षम होता है ।

(a) 1byte /1 बाइट
(b) 1 bit /1 बिट
(c) 1byte word /1 बाइट वर्ड
(d) 1 nibble /1 निब्बल

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b): A flip-flop is the smallest sequential logic circuit used to store a single binary stats (either 0 or 1). It has two stable states, so, it is called a bi-stable multivibrator A flip flop is capable of storing 1 bit information.

304. Astable multivibrator has_____quasi-stable states.
एस्टेबल मल्टीवाईब्रेटर में _____ क्वासि स्टेबल अवस्थाएँ होती हैं।

(a) 0 (b) 1
(c) 2 (d) 3

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : An astable multivibrator is a type of multivibrator that has no stable states and continuously switches between two unstable (quasi-stable) states , generating a continuous square wave output .
Therefore, an astable multivibrator has two quasi-stable states

305. Which of the following is correct regarding T type flip flop ?
I. If T = 1, changes state for lining clock pulse
II. If T = 0, does not changes state
T टाइप फ्लिप फ्लाप के संदर्भ में निम्नलिखित में से कौन सा सही है?
I. If T = 1, लाइनिंग क्लॉक पल्स की स्टेट बदलता है।
II. If T = 0,स्टेट नहीं बदलता है

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Therefore , both statements accurately describe the behavior of a T type flip flop under the specified input condition.

306. What is the correct answer of the given statements in the given order?
दिए गए क्रम में दिए गए कथनों का सही उत्तर क्या है?

I. Gate in which output of logical gate is high when all inputs are high is _____
गेट जिसमें जब सभी इनपुट हाई होते हैं तो गेट का आउटपुट हाई होता _____ है

II. Half of byte is known as _____
आधा बाइट _____ किस रूप में जाना जाता है

III. A flip-flop whose output follows the input when triggered is _____

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एक फ्लिप-फ्लॉप जिसके ट्रिगर होने पर आउटपुट, इनपुट का अनुसरण करता है, _____ है

(a) NAND gate, bit, T – flip-flop
नेंड गेट, बिट, टी - फ्लिप-फ्लॉप

(b) OR gate, nibble, T – flip-flop
ऑर गेट, निब्बल, टी - फ्लिप-फ्लॉप

(c) AND gate, nibble, D – flip-flop
एंड गेट, निब्बल, डी - फ्लिप-फ्लॉप

(d) NOR gate, byte, S-R flip-flop
नॉर गेट, बाइट, S-R फ्लिप-फ्लॉप

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) :
(i) The AND Gate is a logic gate that produces a high output (1) when all of its inputs are high (1). If any of the inputs are low (0), the output will also be low (0).
(ii) Half of a byte is known as a "nibble". A nibble consists of 4 bits.
(iii) A flip-flop whose output follows the input when triggered is called a D flip-flop or data or delay (Flip-flop). When the clock signal is active, the output of the D flip-flop takes the value of the input and holds it until the next triggering event.

307. A ripple counter is a(n) _____ counter where only the _____ flip-flop is clocked by an external clock.
रिप्पल काउंटर एक _____ काउंटर है, जहां केवल _____ फ्लिप-फ्लॉप को बाहरी घड़ी द्वारा दिखाया जाता है।

(a) Synchronous, First
सिंक्रोनस, फर्स्ट (तुल्यकालिक, पहला)

(b) Asynchronous, Second
एसिंक्रोनस, सेकेण्ड (असीमित, दूसरा)

(c) Asynchronous, First
एसिंक्रोनस, फर्स्ट असीमित, पहला

(d) Synchronous, Second
सिंक्रोनस, सेकेण्ड तुल्यकालिक,दूसरा

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Let's carefully analyze the question:
A ripple counter is a counter in which only the first flip-flop is clocked by the external clock. The output of one flip-flop acts as the clock input for the next flip-flop.
• This makes the counter asynchronous (since not all flip-flops are clocked simultaneously).
• And it is specifically the first flip-flop that is directly connected to the external clock.

308. If n flip-flops are used in a ripple counter, then it will count from _____ to _____.
यदि n फ्लिप-फ्लॉप का उपयोग रिप्पल काउंटर में किया जाता है, तो इसे _____ से _____ तक गिना जाएगा।

(a) $0, 2^n$ (b) $0, 2^n - 1$
(c) $1, 2^n$ (d) $1, 2^n - 1$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : We are asked:
If n flip-flops are used in a ripple counter,
Then it will count from ___ to ___ ?
Step 1: Recall property of ripple counter
• A ripple counter with n flip-flops can represent 2^n distinct states.
• It counts starting from 0 up to $2^n - 1$.
Step 2 : Match with options
• Correct range = 0 to $2^n - 1$.

309. State whether the following statements are true or false.
Statements:
(i) The JK flip-flop is a basic building block of combinational logic circuits.
(ii) In a ripple counter, if the uncomplemented output counts up, then complemented output counts down.
यह बताएं कि निम्नलिखित कथन सत्य है या असत्य है।
कथन:
(i) JK फ्लिप-फ्लॉप संयोजन तर्क परिपथ (कम्बिनेशनल लॉजिक सर्किट्स) का मूल निर्माण (बेसिक बिल्डिंग) ब्लॉक है।
(ii) रिप्पल काउंटर में, यदि असम्बद्ध आउटपुट संख्या बढ़ती है, तो फिर सम्बद्ध आउटपुट संख्या घट जाती है।

(a) (i) True, (ii) True/ (i) सत्य, (ii) सत्य
(b) (i) True, (ii) False/ (i) सत्य, (ii) असत्य
(c) (i) False, (ii) True/ (i) असत्य, (ii) सत्य
(d) (i) False, (ii) False/ (i) असत्य, (ii) असत्य

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : (i) The JK flip-flop is a basic building block of combinational logic circuits.
• Flip- flops (including JK flip-flops) are sequential logic elements, not combinational.
• Combinational circuits use only logic gates (AND, OR, NOT, etc.), no memory.
•A flip-flop stores state, so it belongs to sequential logic.
Therefore, False.
(ii) In a ripple counter, if the uncomplemented output counts up, then complemented output counts down.
• In counters, the complemented output (Q') always toggles opposite to the normal output (Q).
So if Q produces an up-count sequence, then Q' naturally produces the down-count sequence.
Therefore, True.

310. Which of the following flip-flops is free from race condition?

(a) T flip-flop
(b) SR flip-flop
(c) Master-slave JK flip-flop
(d) None of the above

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, In the case of JK FF when
J = 1 K = 1 the Q toggle to 0 and 1.
It happens continuously fill the clock is enable ..... and prediction of Q is uncertain (race condition)..... Master-slave JK flip-flop is used to resolve this problem.

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312. The output of a sequential circuit depends on

(a) present input only
(b) past input only
(c) both present and past input
(d) past output only

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, There are 2 types of circuit
(1) Combinational, (2) Sequential
Combinational circuit are those whose outputs depends upon correct input like ADDER, SUBTRACTOR, MUX, DEMUX etc.
Sequential circuits are those whose output depends on both current and previous input.
Hence the answer is (c).

313. The number of flip-flops required to design a modulo – 272 counter is :

(a) 8 (b) 9
(c) 27 (d) 11

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Remember with 'n' number of flip-flop you can get upto maximum modulo - (2^n) counter.
Since $2^9 = 512 > 272$ it can satisfy the condition.

314. A ripple counter is a (*n*):

(a) Synchronous Counter
(b) Asnchronous counter
(c) Parallel counter
(d) None of the above

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) A **ripple counter** is an Asnchronous**:** counter where only the first flip-flop is clocked by an external clock. All subsequent flip-flops are clocked by the output of the preceding flip-flop. Asnchronous **counters** are also called ripple-counters because of the way the clock pulse ripples its way through the flip-flops.

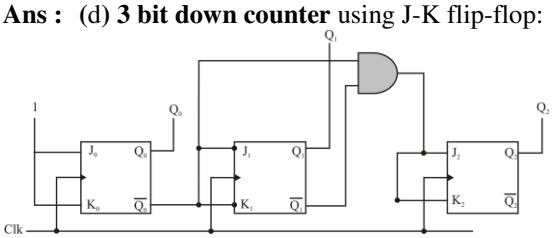
315. A binary -bit down counter uses J-K flip-flops, FF_i with inputs J_i, K_i and outputs Q_i, i = 0, 1, 2 respectively. The minimized expression for the input from following, is

I. $J_0 = K_0 = 0$
II. $J_0 = K_0 = 1$
III. $J_1 = K_1 = Q_0$
IV. $J_1 = K_1 = \overline{Q}_0$
V. $J_2 = K_2 = Q_1\, Q_0$
VI. $J_2 = K_2 = \overline{Q}_1\, \overline{Q}_0$

(a) I, III, V (b) I, IV, VI
(c) II, III, V (d) II, IV, VI

UGC NET C.S. June-2017 (Paper-II)

Ans : (d) **3 bit down counter** using J-K flip-flop:



So, $J_0 = K_0 = 1$, $J_1 = K_1 = \overline{Q}_0$ and $J_2 = K_2 = \overline{Q}_1.\overline{Q}_0$

316. Consider the following statements regarding combinational and sequential circuits.

A. Output of combinational circuits depends on the only current input.
B. Output of combinational circuit depends on the both current input and previous output.
C. Output of sequential circuit depends on the current input.
D. Output of sequential circuit depends on both current input and previous output.

Choose the correct answer from the options given below :

(a) (A) and (C) Only (b) (A) and (D) Only
(c) (B) and (D) Only (d) (B) and (C) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : Combinational circuits produces outputs based slowly on the current inputs. They do not have memory, So the previous outputs do not affect the current output.
Sequential circuits produces outputs based on both the current inputs and the previous outputs as they includes memory elements to store information about past inputs. Thus, combinational circuits are simpler and faster, while sequential circuits are more complex, involving feedback and memory.
So, the given statements (A) and (D) are correct.

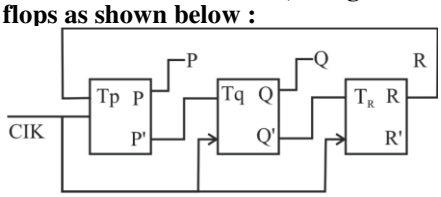
317. Which of the following circuit is used to store one bit of data?

(a) Encoder (b) Decoder
(c) Flip-Flop (d) Register

UGC NET C.S. December-2023

Ans. (c) : A flip-flop is a fundamental digital electronic component used in digital circuits to store binary data. It is capable of storing one bit of data, which can be either 0 or 1. Filp - flops are widely used in various digital systems such as computers, microcontrollers and communication devices to store and manipulate data.

318. Consider a 3-bit counter, designed using T flip flops as shown below :



Assume the initial state of the counter given by PQR as 000, what are the next three states?

(a) 011, 101, 000 (b) 001, 010, 111
(c) 011, 101, 111 (d) 001, 010, 000

UGC NET C.S. December-2022

Ans. (a) : From the above counter-
$T_P = R$, $T_Q = \overline{P}$, $T_R = \overline{Q}$ and PQR = 000

Present Input			Next state		
$T_P = R$	$T_Q = \overline{P}$	$T_R = \overline{Q}$	P^+	Q^+	R^+
0	1	1	0	1	1
1	1	0	1	0	1
1	0	1	0	0	0

Where $P^+ = P \oplus P$, $Q^+ = \overline{P} \oplus Q$, $R^+ = \overline{Q} \oplus R$
So, next three states are- 011, 101, 000

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319. Match List - I with List - II.

List - I (Counter)	List - II (uses/working)
(A) N-bit Ring Counter	(I) Uses universal clock
(B) Synchronous Counter	(II) Counts exactly N states
(C) Asynchronous Counter	(III) Counts 0 to 9
(D) Decimal Counter	(IV) Main clock is applied to first flip-flop only

Choose the correct answer from the options given below :

(a) (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
(b) (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
(c) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
(d) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

UGC NET C.S. Re-Exam- June 2024

Ans. (c) : Correct match-

List-I (Counter)		List -II (Uses/working)	
A.	N-bit Ring counter	–	Counts exactly N states
B.	Synchronous Counter	–	Uses universal clock
C.	Asynchronous Counter	–	Main clock is applied to first flip-flop only
D.	Decimal counter	–	Counts 0 to 9

320. In order to read multiple bytes of a row at the same time, we make use of _____.
एक ही समय में एक पंक्ति के कई बाइट्स पढ़ने के लिए, हम इसका उपयोग करते हैं_____

(a) Memory extension
(b) Latch
(c) Shift register
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) : A latch is a basic digital memory device used in electronics and computer systems. It stores one bit of information (O or 1) and can maintain its state until is changed by input signal. Essentially, it "Latches onto" a value and hold it.

321. The number of columns in a state table for a sequential circuit with m flip-flops and n inputs is
एक सिक्वेंशियल सर्किट की स्टेट टेबल में, जिसमें की m flip-flops और n इनपुट है, कितने कॉलम होंगे?

(a) m + n (b) m +2n
(c) 2m + n (d) 2m + 2n

LT. Grade C.S. 2018

Ans. (c) : In a state table for a sequential circuit, each row represents a unique combination of the current state and input values. The state table typically includes the following columns:

(i) **Current state :** There are m filp- flops, so there are m columns for the current state.
(ii) **Inputs :** There are n input variables, so there are n columns for the inputs.
(iii) **Next state :** Since there are m flip- flops, there are m colums for the next state.

Thus, the total number of columns in the state table:
m (current state) + n (inputs) + m (next state) = 2m + n.

322. Which of the following is a memory element?
निम्न में से कौन-सा एक memory element है?

(a) NAND gate/NAND गेट
(b) SR Flip-Flop/SR flip-flop
(c) NOT gate/NOT गेट
(d) Universal gate/यूनिवर्सल गेट

LT. Grade C.S. 2018

Ans. (b) : An SR flip-flop (Set - Reset flip - flop) is a memory element because it can store one bit of information. It is a bistable circuit that maintains its state until an input causes it to change.

323. Which of the following flipflops does not have a problem of race condition?
निम्नलिखित में से कौन से फ्लिपफ्लॉप में रेस कंडीशन समस्या नहीं है?

(a) T flipflop/T फ्लिपफ्लॉप
(b) Master JK flipflop/JK फ्लिपफ्लॉप
(c) Clocked RS flipflop/क्लॉक्ड- RS फ्लिपफ्लॉप
(d) Clocked D flipflop/क्लॉक्ड- D फ्लिपफ्लॉप

KVS PGT C.S. 2018

Ans. (b) : JK flip plop true table:–

J	K	Qn+1
0	0	Qn (No change)
1	0	1 (Set)
0	1	0 (Reset)
1	1	Qn Toggles

A race condition occurs in JK flipflop when it is implemented, hence, to avoid a race condition, Master-Slave JK Flip Flop is implemented.
Master-Slave JK flipflop has two JK flipflop connected together in inverter which gives inverted clock pulse to slave Flip Flop.
Race conditions: J = K = 1 and CLK = 1 for a long period of time then the output Q will toggle till CLK is high.
This is avoided in the Master-Slave JK flipflop by ensuring the clock input is at "logic 1" for a very short time.

(vi) Sets

324. The difference (–) operation removes common tuples from the first relation. This operation is
अंतर (–) ऑपरेशन पहले सम्बन्ध में से आम tuples हटा देता है। यह ऑपरेशन होता है

(a) Commutative and associative
कॉम्यूटेटिव और सहयोगी
(b) Non-commutative and associative
गैर-कॉम्यूटेटिव और सहयोगी
(c) Commutative and non-associative
कॉम्यूटेटिव और गैर-सहयोगी
(d) Non-commutative and non-associative
गैर-कॉम्यूटेटिव और गैर-सहयोगी

LT. Grade C.S. 2018

Sarkari Resource

Ans. (b) : The Difference operation (–) in rational algebra works as follows :

• It removes all tuples from the first relation that are also present in the second relation.

• That is,

R – S gives all tuples in relation R that are not in S.

1. Non-Commutative property:

• In general :

$R - S \neq S - R$

Example :

Let

R = 1, 2, 3 S = 2, 3, 4

Then,

R – S = 1

But,

S – R = 4

So clearly,

R – S ≠ S – R is Non-commutative.

2. Associative property:

• The difference operation is associative in a special form :

$(R - S) - T = R - (S \cup T)$

• This mean that removing S first and then T is equivalent to removing all elements that are in either S or T from R.

Example :

Let,

R = 1,2,3,4,5

S = 2, 3 T = 4, 5

Then,

$(R - S) - T = (1, 4, 5) - 4, 5 = 1$

And,

$R - (S \cup T) = 1,2,3,4,5 - 2,3,4,5 = 1$

Both give the same.

The difference (–) operation is non-commutative but associative.

325. Let A and B be sets in a finite universal set U. Given the following :

$|A-B|, |A \oplus B|, |A| + |B|$ **and** $|A \cup B|$

Which of the following is in order of increasing size?

(a) $|A-B| \le |A \oplus B| \le |A| + |B| \le |A \cup B|$

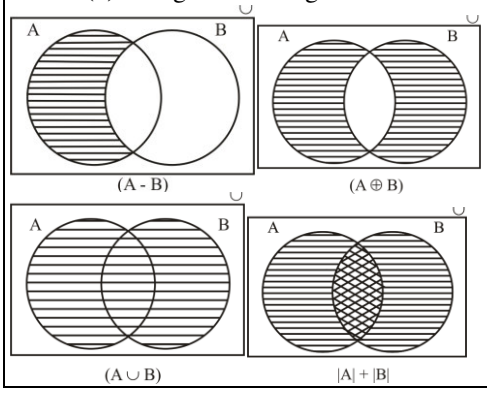
(b) $|A \oplus B| \le |A - B| \le |A \cup B| \le |A| + |B|$

(c) $|A \oplus B| \le |A| + |B| \le |A - B| \le |A \cup B|$

(d) $|A-B| \le |A \oplus B| \le |A \cup B| \le |A| + |B|$

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (d) Using Venn's diagram:



Therefore, $|A - B| \le |A \oplus B| \le |A \cup B| \le |A| + |B|$

326. Let $R = \{x : x \in N, x \text{ is multiple of } 3 \text{ and } x \le 100\}$ **and** $S = \{x : x \in N, x \text{ is multiple of } 5 \text{ and } x < 100\}$. **What is the number of elements in** $(R \cap S) \times (S \cap R)$**?**

(a) 36 (b) 33

(c) 20 (d) 6

UGC NET C.S. June-2023

Ans. (a) : Given that-

$R = \{x : x \in N, x \text{ is multiple of } 3 \text{ and } x \le 100\}$

$R = \{3, 6, 9, ...99\}$

$S = \{x : x \in N, x \text{ is multiple of } 5 \text{ and } x < 100\}$

$S = \{5, 10, 15, ...95\}$

The intersection of these two sets, R ∩ S, would be multiples of both 3 and 5, or multiples of 15, that are less than 100.

$R \cap S = \{15, 30, 45, 60, 75, 90\}$

So, there are 6 elements in the intersection of these sets. Since the sets $R \cap S$ and $S \cap R$ are the same, which has 6 elements.

The number of elements in $(R \cap S) \times (S \cap R)$

$= 6 \times 6 = 36$

327. Match the following :

List-I		List-II	
A.	**Absurd**	**i.**	**Clearly impossible being contrary to some evident truth**
B.	**Ambiguous**	**ii.**	**Capable of more than one interpretation or meaning**
C.	**Axiom**	**iii.**	**An assertion that is accepted and used without a proof**
D.	**Conjecture**	**iv.**	**An opinion preferably based on some experience or wisdom**

Codes :

	A	B	C	D
(a)	i	ii	iii	iv
(b)	i	iii	iv	ii
(c)	ii	iii	iv	i
(d)	ii	i	iii	iv

UGC NET C.S. June-2017 (Paper-II)

Ans. (a) :

(i) Absurd is an adjective, which means utterly or obviously senseless, illogical or untrue i.e., contrary to all reason or common sense or evident truth.

Sarkari Resource

(ii) **Ambiguous** is an adjective, which means open to or having several possible meaning or interpretations or structure.

(iii) **Axiom** is a noun, which means a self evident truth that requires no proof or universally accepted principle or rule.

(iv) **Conjecture** is a noun, which means the formation or expression of an opinion or theory without sufficient evidence for proof i.e., an opinion base on some past experience.

(vii) Relations and Functions

328. In context of databases consider set operations on relation R and S, and select a true statement:
डाटाबेस के संदर्भ में रिलेशन R तथा S पर सेट ऑपरेशनों पर विचार करें तथा सत्य कथन का चुनाव कीजिए:

(a) $(R - S) \cup (S - R) = \phi$
(b) $(R - S) \cup (S - R) = R \cap S$
(c) $(R - S) \cap (S - R) = \phi$
(d) $(R - S) \cup (S - R) = R \cup S$

HTET PGT C.S. 2022

Ans. (C) :
• R - S : Elements in R but not in S.
• S - R : Elements in S but not in R.
These two sets have no common elements because an element cannot be both in R but not in and S but not in R at the same time (unless R=S, but
Step 2 :- Check BC → DEF
• BC → D (given)
• B → E ⇒ BC – E
• BC → DEF holds
step - 3 Check A → EF
• A → C ⇒ C → E ⇒ A → F
• A → B ⇒ B → E ⇒ A → E
A → EF holds
step-4 Check BD → EF
• BD ⇒ B ⇒ E
• But there's no way to derive F from B or D (F needs C which we can't get from BD) BD → EF does NOT hold

329. A function F(x) is invertible, if it is:–
एक फंक्शन व्युत्क्रमणीय होता है यदि वह-

(a) One - One
(b) Onto
(c) Both one - one and onto
(d) Neither one - one nor onto

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : For a function F(x) to be invertible, it must satisfy both of the following conditions-
One-one (Injective)– No two different inputs given the same output. This ensures that the inverse function can uniquely map each output back to its original input.
Onto (Surjective)– Every possible output is covered by the function. This ensures that the inverse function is defined for every element in the code main.
A function that is both one-one and onto is called a injection, and only such functions are invertible.

330. Let X = (1, 2, 3, …….} be ordered by divisibility, which of the following subset is totally ordered?
मान लीजिए X = (1, 2, 3, …….} विभाज्यता द्वारा क्रमित किया जाता है, निम्नलिखित में से कौन सा सबसेट पूर्णतया क्रमित है?

(a) (2, 9, 16)
(b) (3, 5, 15)
(c) (2, 6, 24)
(d) (4, 15, 30)

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : A totally ordered set is a set, where every pair of elements can be compared. When the ordering is based on divisibility, it means for any two elements one must divide the other.
(c) (2,6,24) 2 divides 6, and 6 divides 24. Since every element divides the next, this is a totally ordered set (also known as a chain)

331. In the relation R = {(1,2), (2,3)}, the minimum number of ordered pairs that must be added to this set, so that the enlarged relation is reflexive, symmetric and transitive, is
दिए गए रिलेशन R = {(1,2), (2,3)} में निम्नतम कितने ऑर्डर पेयर जोड़े जाए ताकि एन्लार्ज्ड रिलेशन रिफ्लेक्सिव, सिमेट्रिक और ट्रान्जिटिव हो जाए?

(a) 4
(b) 5
(c) 6
(d) 7

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Ans. (d) : Make the relation R = {(1, 2), (2, 3)} reflexive, symmetric and transitive, we need to systematically add pairs to satisfy these properties.
• Reflexive pairs : (1, 1), (2, 2), (3, 3)
• Symmetric pairs : (2, 1), (3, 2)
• Transitive pairs : (1, 3), (3, 1)
Adding all the required pairs:
{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2), (1, 3), (3, 1)}
This results in 7 new pairs.

332. The range of the function $f(x) = x^2/(1+x^2)$ is
फलन $f(x) = x^2/(1+x^2)$ की रेन्ज है

(a) $(-\infty, +\infty)$
(b) $(0, \infty)$
(c) $(-\infty, 0]$
(d) $[0, 1)$

LT. Grade C.S. 2018

Ans. (d) : Given function $f(x) = \frac{x^2}{1+x^2}$
x^2 is always non-negative, means $x^2 \geq 0$
The denominator $1 + x^2$ is always positive, as $1 + x^2 \geq 1$.
As x varies from $-\infty$ to ∞, the value of f(x) moves from 0 up to but never reaching 1. Therefore, the range of f(x) is $0 \leq f(x) < 1$.

The range of the function $f(x) = \frac{x^2}{1+x^2}$ is [0, 1)

333. The number of equivalence relations of the set {1, 2, 3, 4} is
सेट {1, 2, 3, 4} के समकक्ष सम्बन्धों की संख्या है।

(a) 4
(b) 24
(c) 25
(d) 15

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Ans. (d) : The number of equivalence relations on a set is equal to the number of partitions of the set. This is given by the Bell number, denoted as B_n, where n is the size of the set.
For the set {1, 2, 3, 4,}; n = 4
The Bell numbers for n are, $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$.
Thus, the number of equivalence relations for the set {1, 2, 3, 4} is 15.

334. What is the range of the following function?
निम्न फलन की रेंज क्या है?

{(4, 8), (9, 2), (–3, –4)}

(a) {4, 9, –3}
(b) {8}
(c) {(4, 8), (9, 2), (–3, –4)}
(d) {8, 2, –4}

LT. Grade C.S. 2018

Ans. (d) : The range of a function is the set of all possible output values (the second components of the ordered pairs).
Given the function:
{(4, 8), (9, 2), (–3, –4)}
The output are 8, 2 and –4.
Thus, the range is {8, 2, –4}.

335. What is the possible number of reflexive relations on a set of 5 elements?
5 अवयवों के सेट पर रिफ्लेक्सिव सम्बन्धों की संभावित संख्या क्या है?

(a) 2^{10} (b) 2^{15}
(c) 2^{20} (d) 2^{25}

LT. Grade C.S. 2018

Ans. (c) : A reflexive relation on a set requires that every element is related to itself. If the set has n elements, the relation is a subset of the cartesian product A × A, where A is the set. Since reflexivity requires that all diagonal elements (a_i, a_i) must be included, only the off -diagonal elements are optional.
For a set A with 5 elements, the number of pairs in A×A is 5^2. Reflexivity fixes 5 diagonal pairs, leaving $5^2 - 5$ off diagonal pairs that can either be included or excluded independently. Thus, the number of reflexive relations is $2^{25-5} = 2^{20}$.

336. A relation 'R' is defined on ordered pairs of integers as :
(x, y) R (u, v) if x<u and y>v. Then R is

(a) Neither a partial order nor an equivalence relation
(b) A partial order but not a total order
(c) A total order
(d) An equivalence relation

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Ans. (a) : An equivalence relation on set x is a subset of x*x, i.e., a collection R of ordered pairs of elements of x, satisfying certain properties. Write "xRy" to mean (x, y) is an element of R and We say "x is related to y. Then the properties are Reflexive, Symmetric, Transitive.
An partial order relation on a set x is a subset of x*x, satisfying certain properties. The properties are Reflexive, Anti-Symmetric, Transitive.
An total order relation a set x is a subset of x*x, satisfying certain properties. The properties are Reflexive, Anti-Symmetric, Transitive, Comparability.
As given in question, a relation "R" is defined on ordered pairs of integers as : (x, y) R (u, v) if x < u and y > v, Reflexive property is not satisfied here, because there is > or = v (or x = v and y = v) kind of relation among elements of sets then reflexive property could have been satisfied. Since reflexive property in not satisfied here, so given relation cannot be equivalence, partial order or total order relation.
So, option (a) is correct.

337. How many different equivalence relations with exactly three different equivalence classes are there on a set with five elements?

(a) 10 (b) 15
(c) 25 (d) 30

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) The set of equivalence relations on a set are in direct bijection with the set of partitions on a set. We know that the only partition types for a 5 element set are of the form (1 + 1 + 3) or (1 + 2 + 2) of the first type, pick the three in the triple, $^5C_3 = 10$ in case of the second type, first pick which is the singleton and then pick which is paired with smallest remaining, i.e., $5 \times {}^3C_1 = 15$ of this for a total of 10 + 15 = 25.

338. Suppose that R_1 and R_2 are reflexive relations on a set A.
Which of the following statements is correct?

(a) $R_1 \cap R_2$ is reflexive and $R_1 \cup R_2$ is irreflexive.
(b) $R_1 \cap R_2$ is irreflexive and $R_1 \cup R_2$ is reflexive.
(c) Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are reflexive.
(d) Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are irreflexive.

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) If R_1 and R_2 are reflexive relations on a set A.
$R_1 \cup R_2$ is reflexive because.
$\Delta \subseteq R_1 \subseteq R_1 \cup R_2$ $(or\ \Delta \subseteq R_2 \subseteq R_1 \cup R_2)$
$R_1 \cap R_2$ is also reflexive, because
$\Delta \subseteq R_1 \wedge \Delta \subseteq R_2 \Rightarrow \Delta \subseteq R_1 \cap R_2$

339. Write the name of the relational algebra operation as per the given order of their notation.
π, X , s
उनके संकेतन (नोटेशन) के दिए गए क्रम के अनुसार रिलेशनल अलजेब्रा ऑपरेशन का नाम लिखिए।
π, X , s

(a) Projection, Cartesian product, division
प्रोजेक्शन, कार्टिसिअन गुणनफल, विभाजन
(b) Projection, Cartesian product, select
प्रोजेक्शन, कार्टिसिअन गुणनफल, सेलेक्ट

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(c) Intersection join, outer join, select
इंटरसेक्शन ज्वाइन, आउटर ज्वाइन, सेलेक्ट
(d) Projection, outer join, select/
प्रोजेक्शन, आउटर ज्वाइन, सेलेक्ट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) :
(i) Projection (π)
(ii) Cartesian product (X)
(iii) Select (s)

340. What is the cardinality and degree of relation having 50 Tuples and 20 Attributes?
50 टुपल्स तथा 20 एट्रीब्यूट्स वाले रिलेशन की कार्डिनैलिटी तथा डिग्री क्या है?

(a) Cardinality – 70 Degree – 30
कार्डिनैलिटी – 70 डिग्री – 30
(b) Cardinality – 20 Degree – 50
कार्डिनैलिटी – 20 डिग्री – 50
(c) Cardinality – 50 Degree – 20
कार्डिनैलिटी – 50 डिग्री – 20
(d) Cardinality – 50 Degree – 50
कार्डिनैलिटी – 50 डिग्री – 50

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Cardinality– Number of tuples (rows) in a relation (table).
Degree– Number of attributes (columns) in a relation
Given
The relation has 50 tuples.
The relation has 20 attributes.
Cardinality– 50
Degree– 20

341. Given U = {1, 2, 3, 4, 5, 6, 7}
A = {(3, 0.7), (5, 1), (6, 0.8)}
then A will be : (where ~ → complement)

(a) {(4, 0.7), (2, 1), (1, 0.8)}
(b) {(4, 0.3), (5, 0), (6, 0.2)}
(c) {(1, 1), (2, 1), (3, 0.3), (4, 1), (6, 0.2), & (7, 1)}
(d) {(3, 0.3), (6, 0.2)}

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because,
$\mu A'(x) = ((\mu A(x)) = 1 - \mu A(x)$ for all $x \in U$
The given set A will be seen as A = {(1, 0), (2, 0), (3, 0.7), (4, 0), (5, 2) (6, 0.8), (7, 0)}
So the compliment of A will be {(1, 1), (2, 1), (3, 0.3) (4, 1), (6, 0.2) (7, 1)}
Hence the answer is (c).

342. Let *R* and *S* be two fuzzy relations defined as

$$R = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{matrix} \text{ and } S = \begin{matrix} \\ y_1 \\ y_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{matrix}$$

Then, the resulting relation, *T*, which relates elements of universe of *X* to elements of universe of *Z* using max-product composition is given by

(a) $T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.68 & 0.89 & 0.39 \\ 0.76 & 0.72 & 0.32 \end{bmatrix} \end{matrix}$

(b) $T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.68 & 0.89 & 0.39 \\ 0.72 & 0.76 & 0.32 \end{bmatrix} \end{matrix}$

(c) $T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.63 & 0.42 & 0.35 \\ 0.72 & 0.48 & 0.20 \end{bmatrix} \end{matrix}$

(d) $T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.05 & 0.35 & 0.14 \\ 0.04 & 0.28 & 0.16 \end{bmatrix} \end{matrix}$

UGC NET C.S. August-2016 (Paper-III)

Ans. (c) : By **max-min composition:**
$\mu_T(x_1, z_1) = \max(\min(0.7, 0.9), \min(0.5, 0.1))$
$= 0.7$
and the rest

$$T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix} \end{matrix}$$

and by max-product composition:
$\mu_T(x_2, z_2) = \max((0.8, 0.6), (0.4, 0.7)) = 0.48$

$$T = \begin{matrix} \\ x_1 \\ x_2 \end{matrix}\begin{matrix} \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{bmatrix} 0.63 & 0.42 & 0.35 \\ 0.72 & 0.48 & 0.20 \end{bmatrix} \end{matrix}$$

343. Given a vector with Cartesian components (x, y), if scaling is done with matrix $\begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$, which of the following are true.

(A) Decrease the vertical by three halves
(B) Increases the vertical by three halves
(C) Doubles the horizontal
(D) Halves the horizontal

Choose the correct answer from the options given below :

(a) A and C only (b) A and D only
(c) B and C only (d) B and D only

UGC NET C.S. December-2022

Ans. (d) : To determine the effect of scaling on the given vector with Cartesian components (x, y) using the matrix $\begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$, we can perform matrix multiplication

$$\begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0.5x + 0 \\ 0 + 1.5y \end{bmatrix} = \begin{bmatrix} 0.5x \\ 1.5y \end{bmatrix}$$

The resulting vector is (0.5x, 1.5y) comparing this to the original vector (x, y) we can observe the following changes.

- The horizontal component has been halved (0.5x)
- The vertical component has been increased by three halves (1.5y)

So option (d) is correct.

Sarkari Resource

(viii) Mathematical Logic

344. Boolean expression ~ (A * B) * (~A + B) * (~B + B) is equal to which of the following ?
बूलियन व्यंजक ~ (A * B) * (~A + B) * (~B + B) निम्न में से किसके बराबर है?

(a) ~A (b) ~B
(c) ~A + ~B (d) A * B

HTET PGT C.S. 2022

Ans. (a) : Boolean expression:-
~ (A * B)* (~A + B)* (~B + B)
step 1:- Notice that ~ B + B =1 (always true)
so the expression reduces to:
~(A * B)* (~A + B)
staep-2:- Apply De Morgan's law ~ (A * B) = ~ A + ~ B Now the expression is
(~ A + ~B) * (~ A + B)
Step 3 :- Factor out ~ A (common term)
~A + (~ B * B)
Step 4:- ~ B * B = 0 (always false)
So we get
~ A + 0 = ~ A

345. Which of the following is not a valid expression of Boolean algebra ?
निम्न में से कौन सा बूलियन एल्जेब्रा का एक उचित व्यंजक नहीं है?

(a) ~ x + (y * ~ z) + x + ~ y = 1
(b) x * ~ y + ~ x * y + ~ x * ~ y = 1
(c) ~ (x + y + z) = ~ x * ~ y * ~ z
(d) x * ~ y + ~ x * y + x * y = x + y

HTET PGT C.S. 2022

Ans. (b) : x = 1
y = 1
x* ~ y + ~ x* y + ~ x* ~ y = 1.0 + 0.1 + 0.0 = 0 + 0 + 0 = 0 ≠ 1 (This fails, so it is invalid)

346. Which of the following are tautology?
उपरोक्त में किसमें पुनरुक्ति है?

(A) $(P \rightarrow (P \wedge Q)) \rightarrow (P \rightarrow Q)$
(B) $((P \rightarrow Q) \rightarrow Q) \rightarrow (P \vee Q)$
(C) $((P \vee \neg P) \rightarrow Q) \rightarrow ((P \vee \neg P) \rightarrow R)$
(D) $(Q \rightarrow (P \wedge \neg P)) \rightarrow (R \rightarrow (P \wedge \neg P))$
Choose the correct answer from the options given below :

(a) (A) Only (b) (B) Only
(c) (A) and (B) Only (d) (C) and (D) Only

UGC NET C.S. December-2023

Ans. (c) : A tautology is a statement that is always true regardless of the truth values of its of its components. for example-
$(P \rightarrow (P \wedge Q)) \rightarrow (P \rightarrow Q)$
This statement is a tautology because it's always true.
$((P \rightarrow Q) \rightarrow Q) \rightarrow (P \vee Q)$
This statement is a tautology because it's always true.

347. Consider the set of all possible five-card poker hands dealt fairly from a standard deck of fifty-two cards. How many atomic events are there in the joint probability distribution ?
बावन पत्तों के एक मानक डेक से निष्पक्ष रूप से बांटे गये सभी सम्भावित पाँच-पत्तों वाले पोकर हाथों के समुच्चय पर विचार करें। संयुक्त प्रायिकता वितरण में कितनी परमाणु घटनाएँ हैं?

(a) 2, 598, 960 (b) 3, 468, 960
(c) 3, 958, 590 (d) 2, 645, 590

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : Joint probability distribution that gives the probability that each of x, y, ...
falls in any particular range or discrete set of values specified for that variable.
Pb = 52C5 ⇒ 2598960

348. Algebra of logic is termed as?
तर्क का बीजगणित कहा जाता है?

(a) Numerical logic (b) Boolean algebra
(c) Arithmetic logic (d) Boolean number

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : The algebra of logic is termed as Boolean algebra. It is a mathematical structure used in logic and set theory and it is also widely used in digital electronics and computer science.
In Boolean algebra, there are two fundamental values: true and false represented as 1 and 0. It includes various operations such as AND, OR and NOT which help in creating and simplifying logical expressions.
So, the correct option is (b) Boolean algebra
Note: But as per commission option (c) Arithmetic logic is correct.

349. What will be the result of the following expression ?
निम्नलिखित व्यंजक का परिणाम क्या होगा?

3−(−3−3)−3

(a) 6 (b) 0
(c) –3 (d) 3

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans. (a) : Expression are–
= 3 – (–3 – 3) – 3
= 3 – (–6) – 3
= 9 – 3 = 6

350. Which of the following expression will evaluate to true ?
निम्नलिखित में से कौन सा व्यंजक सही है?

Assuming the values of A, B, C as follows :
मान लीजिए A, B, C का मान इस प्रकार है:

A = −10
B = 30
C = −15

(a) A > C and B < =C/A > C तथा B < =C
(b) A <=B and B < =C/A <=B तथा B < =C
(c) B > =C and A > C/B > =C तथा A > C
(d) A < C and C < =B/A < C तथा C < =B

NVS PGT C.S. 15.12.2022 (Shift-II)

Sarkari Resource

Ans. (c) : Assuming the valus of ABC follow
A = –10
B = 30
C = –15
opation will be (c) correct B > = C and A > C put the values of ABC
30 > = – 15 or d – 10 > – 15 is correct opation is (c)

351. Which of the following is tautology?
निम्न में से कौन - सा टॉटोलॉजी है?

(a) $a \vee b \rightarrow b \wedge c$
(b) $a \wedge b \rightarrow b \vee c$
(c) $a \vee b \rightarrow (b \rightarrow c)$
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (b) : A tautology is a statement that is always true, irrespective of the truth values of its components.
If $a \wedge b$ is true, then $b \vee c$ is always true.

352. With respect to relational algebra, which of the following operations are included from mathematical set theory?
(A) Join
(B) Intersection
(C) Cartesian product
(D) Project

(a) (A) and (D) (b) (B) and (C)
(c) (C) and (D) (d) (B) and (D)

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) : Relational Algebra :→ It is a procedure oriented language. In this, we use relation as values instead of numbers. It works on relational model. Main work is to retrieve data, perform operations such as insert, delete, update etc.

* Basic operation in relational algebra are select, Project, union, set difference, Cartesian product, rename.
* other operations are natural join, intersection division.
* from the above options, operator that are included from mathematical set theory are intersection and Cartesian product.
* In mathematical set theory, intersection is used to find the common elements between two sets. It works same in relational algebra.
* Cartesian product is used to find the multiplication of two sets by including common elements only once.

353. Which of the following has same expressive power with regard to relational query language?
(A) Relational algebra and domain relational calculus
(B) Relational algebra and tuples relational calculus
(C) Relational algebra and domain relational calculus restricted to safe expression
(D) Relational algebra and tuples relational calculus restricted to safe expression

(a) (A) and (B) only
(b) (C) and (D) only
(c) (A) and (C) only
(d) (B) and (D) only

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) :

* Relational algebra is a procedural query language, which takes instances of relations as input and yields instances of relations as output. It uses operation (selection, projection, union etc) to perform queries
* Relational calculus consists of two calculi, the tuple relational calculus and the domain relational calculus, that are part of the relational model for databases and provide a declarative way to specify database queries.
* Relational algebra, Domain relational calculus restricted to safe expression and tuple relational calculus restricted to safe expression have the same expressive power.
  Relation algebra's expressive power is not equal to Domain relational calculus or Tuple relational calculus unless they are restricted to safe operation.

354. Which of the following pairs of propositions are not logically equivalent?
(a) $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$
(b) $p \leftrightarrow q$ and $(-p \leftrightarrow -q)$
(c) $((p \wedge q) \vee (-p \wedge -q))$ and $p \leftrightarrow q$
(d) $((p \wedge q \rightarrow r))$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) :
Options A:- $(p \rightarrow r) \wedge (q \rightarrow r)$ and $((p \vee q) \rightarrow r)$
Both are equal and it gives the same truth table.
Hence it is logically equivalent.
Options B:- $p \leftrightarrow q$ and $(-p \leftrightarrow -q)$
$(p \rightarrow q)$ $(q \rightarrow p)$ and $(> p \rightarrow q)$
$(\rightharpoonup q \rightarrow p)$ $(p + \bar{q})$ $(q + \bar{p})$ and $(p + \bar{q})$ $(q + \bar{p})$
Both the equal and it gives the same truth table.
Options C:- $((p \wedge q) \vee (\neg p \rightarrow q))$ and $p \leftrightarrow q$
hence logically equivalent.
Options D:- $((p \wedge q \rightarrow r))$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

P	Q	R	¬ P	¬ Q	P⇒Q	Q⇒R	(P⇒R)∩(Q ⇒R)
T	T	T	F	F	T	T	T
T	T	F	F	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	F	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

The above truth table is not equivalent.
Hence the above statement is true, logically not equivalent.
Hence, the correct answer option d.

Sarkari Resource

355. The negation of "Some students like hockey" is :

(a) Some students dislike hockey
(b) Every student dislike hockey
(c) Every student like hockey
(d) All students like hockey

UGC NET C.S. December-2022

Ans. (b) : Negation is an operation that takes a proposition P to another proposition 'not P', standing for 'P is not true', written $\neg P$, ~ P, or $\overline{P}$. The Statement "Some students like hockey" says that 'atleast' one student like hockey. So, the negation of this statement is "Every student not like (dislike) hockey".

356. The Boolean function [~(~p∧q)∧~(~p∧~q)] ∨(p∧r) is equal to Boolean function :

(a) q (b) $p \wedge r$
(c) $p \vee r$ (d) p

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (d) : Given,
Boolean function is equivalent

$\Rightarrow [(\overline{p} \cdot q) \cdot (\overline{p} \cdot \overline{q})] + p.r$

$\Rightarrow [(p \cdot \overline{q}) \cdot (p+q)] + p.r$

$\Rightarrow [(p+\overline{q}q)] + p.r$

$\Rightarrow p(1+r)$

$= p$

357. The mathematical notation to describe logical entailment of a sentence "a entails another sentence β" is

(a) $a \models \beta$ (b) $a \subseteq \beta$
(c) $\beta \models \alpha$ (d) $\beta \subseteq \alpha$

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Ans. (a) : $\alpha = \beta$ mean that α semantically entails β, meaning that in every model where is true β must also be true.

- The entailment symbol |= represent semantic entailment.
- It Indicates that β is necessary consequence of α.
- Example: If α is "it is raining" and β is "the ground is wet," then if the first statement is always true, the second one must also be true.

358. The kind of symbols for basic syntactic elements of first order logic are

A. Constant **B. Domain**
C. Predicate **D. Temporal**
E. Functions

Choose the correct answer from the options given below :

(a) B, D only (b) A, B, C only
(c) A, C, E only (d) C, D, only

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Ans. (c) : The basic syntactic elements of first order logic include.
A. constant, Represents specific objects in the domain.
C. predicate, Represents relations or properties of objects.
E. Function, Represents operations that map objects to other objects.
The other options are not basic syntactic elements of first order logic.
B. Domain, The domain is the set of objects over which the logic is interpreted, but it is not a syntactic element.
D. Temporal, Temporal elements are related to temporal logic not basic first order logic.

359. "If may computations are correct and I pay the electric bill, then I will run out of money. If 1 don't pay the electric bill, the power will be turned off. Therefore, if I don't run out of money and the power is still on, then my computations are incorrect."
Convert this argument into logical notations using the variables c, b, r, p for propositions of computations, electric bills, out of money and the power respectively. (Where – means NOT)

(a) if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$
(b) if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(r \wedge p) \rightarrow c$
(c) if $(c \wedge b) \rightarrow r$ and $\neg\ p \rightarrow \neg b$, then $(\neg r \wedge p) \rightarrow \neg c$
(d) if $(c \vee b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$

UGC NET C.S. June-2015 (Paper-II)

Ans.(a) : S_1: $(c \wedge b) \rightarrow r$
S_2: $(\neg b \rightarrow \neg p)$

Conclusion = $(\neg r \wedge p) \rightarrow \neg c$

Same as if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$,

Then, $(\neg r \wedge p) \rightarrow \neg c$

360. Match the following :

	List–I	List–II
a.	$(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$	i. Contra positive
b.	$[(p \wedge q) \rightarrow r] \Leftrightarrow [p \rightarrow (q \rightarrow r)]$	ii. Exportation law
c.	$(p \rightarrow q) \Leftrightarrow [(p \wedge -q) \rightarrow 0)$	iii. Reduction as absurdum
d.	$(p \leftrightarrow q) \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$	iv. Equivalence

Codes :

	a	b	c	d
(a)	(i)	(ii)	(iii)	(iv)
(b)	(ii)	(iii)	(i)	(iv)
(c)	(iii)	(ii)	(iv)	(i)
(d)	(iv)	(ii)	(iii)	(i)

UGC NET C.S. June-2015 (Paper-II)

Ans.(a) :
(a) The contra positive of a conditional statement is formed by negating both the hypothesis & the conclusion, and then interchanging the resulting negations.
ex. $(P \rightarrow Q) \Leftrightarrow (7\ Q \rightarrow 7\ P)$
(b) Exportation is a valid rule of replacement in propositional logic
$((P \wedge Q) \rightarrow R) \Leftrightarrow (P \rightarrow (Q \rightarrow R))$
(c) Reduction and absurdum ⇒

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$(P \to Q) \Leftrightarrow [(P \wedge Q) \to 0]$
(d) Logical equality is an operation on two logical values, typically the values of two proposition that produces a value of true & false
$(P \to Q) \wedge (Q \to P)$

361. Consider a proposition given as :
" $x \geq 6$, if $x^2 \geq 25$ and its proof as :
If $x \geq 6$, then $x^2 = x \,.\, x \geq 6 \,.\, 6 = 36 \geq 25$
Which of the following is correct w.r.to the given proposition and its proof?
(1) The proof shows the converse of what is to be proved
(2) The proof starts by assuming what is to be shown
(3) The proof is correct and there is nothing wrong
(a) (1) only (b) (3) only
(c) (1) and (2) (d) (2) only

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : "$X \geq 6$, if $x^2 \geq 5$ and its proof as: I f $x \geq 6$, then $x^2 = X.X \geq 6.6 = 36 \geq 25$ it is clear from the above statement that proof starts by assuming g what is to be shown and it is clear that the proof shows the converse of what is to be proved hence the proof is not correct. Statement (a) and (b) are correct but (3) statement is not correct. So, option (C) is correct.

362. Consider the following identities for regular expressions:
(i) $(r + s)^* = (s + r)^*$
(ii) $(r^*) = r^*$
(iii) $(r^* s^*)^* = (r + s)^*$
Which of the above identities are true?
(a) (i) and (ii) only (b) (ii) and (iii) only
(c) (iii) and (i) only (d) (i), (ii) and (iii)

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) $\Rightarrow (r + s)^*$ will generate any string containing r or s or both. We can draw DFA for $(r + s)^*$ and it is same as $(s + r)^*$. It is a regular expression.
$\Rightarrow (r^*)^*$ will generate any string containing r and its DFA can be draw easily and it is same as r^*. It is also a regular expression.
$\Rightarrow (r^* s^*)^*$ will generate any string containing r or s or both. We can draw DFA for $(r^* s^*)^*$ and it is same as $(r + s)^*$. It is a regular expression. All option are true.

363. In propositional logic if $(P \to Q) \wedge (R \to S)$ and $(P \wedge R)$ are two premises such that
$(P \to Q) \wedge (R \to S)$
$$\frac{P \vee R}{Y}$$
Y is the premise :
(a) $P \vee R$ (b) $R \vee S$
(c) $Q \vee R$ (d) $Q \vee S$

UGC NET C.S. June-2017 (Paper-II)

Ans : (d) Given that premises are
$(P \to Q) \wedge (R \to S)$
$(P \vee R)$
$(P \to Q) = \sim P \vee Q$
$(R \to Q) = \sim R \vee S$
$(P \vee R)$
$Q \vee S$:- There will be resolution (rule of inference) b/w there premises to give conclusion.
~ P & P, R & R' will resolve out and then we contract are disjunction of remaining clauses to given $S \vee Q$ is correct option.

364. Let P and Q be two propositions, $\neg (P \leftrightarrow Q)$ is equivalent to :
(a) $P \leftrightarrow \neg Q$ (b) $\neg P \leftrightarrow Q$
(c) $\neg P \leftrightarrow \neg Q$ (d) $Q \to P$

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Ans : (a, b)

P	Q	~ P	~ Q	Q→ P	~P↔~ Q	~P↔ Q	~P↔ Q	~P ↔~ Q
0	0	1	1	1	0	0	0	1
0	1	1	0	0	1	1	1	0
1	0	0	1	1	1	1	1	0
1	1	0	0	1	0	0	0	1

Given statement: $\neg(P \leftrightarrow Q)$
We know $A \leftrightarrow B = A'B' + AB$
So, $\neg(P'Q' + PQ)$
$(P'Q + PQ') \cong P \leftrightarrow \neg Q \cong \neg P \leftrightarrow Q$

365. Negation of the proposition $\exists$ x H(x) is :
(a) $\exists x \neg H(x)$ (b) $\forall x \neg H(x)$
(c) $\forall x\, H(x)$ (d) $\neg x\, H(x)$

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Ans : (b) Proposition $\Rightarrow \exists x\, H(x)$
Negation $\Rightarrow \neg[\exists x\, H(x)]$
$\Rightarrow \forall x \neg H(x)$

366. If the time is now 4 O'clock, what will be the time after 101 hours from now?
(a) 9 O'clock (b) 8 O'clock
(c) 5 O'clock (d) 4 O'clock

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Ans : (a) Current time = 4 O'clock
Time to come back at same position = 12 hour
So clock will be completed cycles in 101 hour and remaining hour is $\to$ 101 mod 12
= 5 hour
So, time after 101 hours = 4+5 = 9 O'clock

367. Consider universe positive integer $X = \{1 \leq n \leq 8\}$, proposition P = "n is an even integers", $Q = "(3 \leq n \leq 7) \wedge (n \neq 6)"$ .Then truth set of $P \leftrightarrow Q$ is
(a) {1, 4} (b) {2, 6}
(c) {3, 4, 5} (d) {1}

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Ans. (a) : Given that,
$X = \{1 \leq n \leq 8\}$
Propositions-
P = 'n is an even integer' for n in the set X.

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$Q = \{3 \le n \le 7\} \wedge (n \ne 6)$ for n in the set x.

Now, evaluate each proposition.

P = The even integers in the set X are {2, 4, 6, 8}.

Q = The integers satisfying $3 \le n \le 7$ and $n \ne 6$ are $\{3,4,5,7\}$.

For the bidirectional implication P ↔ Q, it means both proposition have the same truth values for all elements in the set X.

P ↔ Q is true for n = 4, as both propositions are true for these values.

P ↔ Q is false for n = 2, 3, 5, 6, 7, 8 as at least one of the propositions is false for each of these values.

So, , the truth set of P ↔ Q is {1, 4}.

368. The precedence of regular expression operators from lowest to highest precedence is :

(a) Star, Concatenation, Union
(b) Union, Star, Concatenation
(c) Union, Concatenation, Star
(d) Star, Union, Concatenation

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Ans. (c) : The precedence of regular expression operators from lowest to highest precedence is -

Star is highest then concatenation and union is lowest precedence.

So option (c) is correct.

(ix) Combinatorics and Recurrence Relations

369. Find the sum of all four digit numbers formed using the digits 1,2,4 and 6.
अंक 1, 2, 4 और 6 का उपयोग करके बनाई गई सभी चार अंकों की संख्याओं का योग ज्ञात कीजिए?

(a) 86,658 (b) 88,8858
(c) 91,958 (d) 93,358

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Ans. (a) : To find the sum of all four-digit numbers formed using the digits 1, 2, 4 and 6 we can calculate the sum of each digit in each position and then multiply it by the number of times each digit appears in that position.

The sum of digits in that place = 1 + 2 + 4 + 6 = 13

Each digit appears in that place = 6 times

So, the sum contributed by-

Thousands place = 13 × 6 × 1000 = 78000

Hundreds place = 13 × 6 × 100 = 7800

Tens place = 13 × 6 × 10 = 780

Unit place = 13 × 6 × 1 = 78

Adding these sums to gather = 78000 + 7800 + 780 + 78 = 86658

Hence, the sum of all four digit number is 86,658.

370. In how many ways can 8 Indians, 4 Americans and 4 Englishmen be seated in a row so that all persons of same nationality sit together?
एक पंक्ति में 8 भारतीय, 4 अमेरिकन और 4 इंग्लिश आदमी कितनी तरह से बैठ सकते हैं, ताकि एक ही राष्ट्र के सभी आदमी एक साथ हों?

(a) 3! 4! 8! 4! (b) 3! 8!
(c) 4! 4! (d) 8! 4! 4!

LT. Grade C.S. 2018

Ans, (a) : Given, 8 Indians, 4 Americans and 4 Englishmen such that all persons of the same nationality sit together. This involves treating each nationality as a single 'block', then arranging the people within each block.

- The 8 Indians can be arranged among themselves in 8! ways.
- The 4 Americans can be arranged among themselves in 4! ways.
- The 4 Englishmen can be arranged among themselves in 4! ways.
- There are 3 blocks (Indians, Americans, Englishmen). The block can be arranged among themselves in 3! ways.

Multiply the arrangements of the blocks and the arrangements within each block.

Total ways = 3! × 8! × 4! × 4!

371. From the given digits
दिए गए अंकों
2, 2, 3, 3, 3, 4, 4, 4, 4
how many 4-digit numbers greater than 3000 can be formed?
से 3000 से अधिक 4 अंकों की कितनी संख्या बनाई जा सकती है?

(a) 50 (b) 51 (c) 52 (d) 54

LT. Grade C.S. 2018

Ans, (b) : The first digit must be 3 or 4, as the number needs to be greater than 3000.

- If first digit is 3 then the rest of the numbers must come from the list 2, 2, 3, 3, 4, 4, 4, 4. Therefore we may choose any 3-digit sequence except 222 and 333 for the rest of the digits. This shows there are 1 × 3 × 3 × 3 – 2 = 25 numbers in this case.
- If first digit is 4 then the rest of the number must come from the list 2, 2, 3, 3, 3, 4, 4, 4. Therefore we may choose any 3-digit sequence except 222 for the rest of the digits. This shows there are 1×3×3×3 – 1 = 26 number in this case.

Now, the total numbers is just 25 + 26 = 51.

372. Consider a sequence F_{00} defined as :

$$F_{00}(0) = 1, F_{00}(1) = 1$$

$$F_{00}(n) \frac{10 * F_{00}(n-1) + 100}{F_{00}(n-2)} \text{ for } n^3 > 2$$

Then what shall be the set of values of the sequence F_{00} ?

(a) (1, 110, 1200)
(b) (1, 110, 600, 1200)
(c) (1, 2, 55, 110, 600, 1200)
(d) (1, 55, 110, 600, 1200)

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) $F_{00}(0) = 1, F_{00}(1) = 1$

$F_{00}(n) = 10*F_{00}(n-1) + 100$

$F_{00}(2) = (10*F_{00}(1) + 100)*F_{00}(0)$

$= (10*1 + 100)*1 = 110$

$F_{00}(3) = 10*F_{00}(2) + 100)*F_{00}(1)$

$= (10*110 + 100)*1 = 1200.$

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373. How many distinguishable permutations of the letters in the word BANANA are there?

(a) 720 (b) 120
(c) 60 (d) 360

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Number of permutation of n objects with n_1 identical object of type 1, n_2 identical object of type 2 & nk identical objects of type ki $\frac{n!}{n_1!n_2!}....nk!$

Permutation of latter BANANA are $\frac{6!}{3!2!1!} = 60$

(x) Graph Theory

374. Sum of degrees of all nodes for a graph G(V, E) can be given by ..... .
एक ग्राफ G(V, E) के लिए सभी नोड्स की डिग्रीज का योग ..... के द्वारा दिया जा सकता है।

(a) |E| (b) 2 × |E|
(c) 2 × |V| (d) |V|

HTET PGT C.S. 2022

Ans. (b)
1. for any simple undirected graph, the well know handshaking lemma States
$\sum_{\upsilon\varepsilon V} \deg(\upsilon) = 2|E|$
2. Reason:- Each edge contributes 2 to the sum of degrees (one for each of its two endpoints)
3. Even for a directed graph, if you count in degree add out degree, the total also equals 2 × |E|

375. In directed graph G(V, E) where V = {P, Q, R, S, T} and E = {PQ, RS, TP, PS, QR, PT, ST, QP, RQ, TS}, the vertex with maximum indegree is ..... and vertex with maximum out-degree is ..... .
V = {P, Q, R, S, T} तथा E = {PQ, RS, TP, PS, QR, PT, ST, QP, RQ, TS}वाले एक ग्राफ G(V, E) में अधिकतम इन-डिग्री का वर्टेक्स .....है तथा अधिकतम आउट डिग्री का वर्टेक्स .....है।

(a) P, S (b) T, Q
(c) Q, R (d) S, P

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Ans. (d) : graph G (V, ε) with
- V {P, Q, R, S, T}
- E = { PQ, RS, TP, QP, PS, QR, ST, PT, RQ, TS}
- [edges pointing to vertex]
- S gets edges from Rs (R→S), PS(P→S), TS (T→S) → in - degree = 3 (max)
- Others: P (from TP, QP → 2), Q (from PQ, RQ→2), R (from QR→1), T (from PT, ST →2)

[edges starting from vertex]
- P has edges PQ, PS, PT → Out degree = 3 (max)
- Others: Q (QP, QR→2), R (RQ, RS →2), S (ST→1), T(TP, TS→2)

376. Consider a graph G of 8 vertices containing no self loop. If there is no more than one edge between any pair of vertices, maximum possible number of edges in G are :
8 वर्टेक्स वाले एक ग्राफ G पर विचार करें। यदि किसी शीर्ष (वर्टेक्स) युग्म के बीच एक से अधिक कोर (एज) नहीं हो, तो G में अधिकतम संभव कोरों (एजेज) की संख्या होगी:

(a) 8 (b) 16
(c) 28 (d) 64

HTET PGT C.S. 2019

Ans. (c) : Maximum edges = $= \frac{n(n-1)}{2}$

Where, (n) = 8

$\frac{8(8-1)}{2} = \frac{8\times7}{2} = 28$

377. Consider a directed graph G = {V, E} with vertex set V = {A, B, C, D} and edge set E = {AB, AC, BC, BD, DC}. Which of the following is a valid topological ordering of graph G = {V, E} ?
शीर्ष समुच्चय (विर्टेक्स सेट) V = {A, B, C, D} तथा कोर समुच्चय (एज सेट) E = {AB, AC, BC, BD, DC} वाले एक डायरेक्टेड ग्राफ G = {V, E} पर विचार करें।
निम्न में से कौन-सा ग्राफ G का एक उचित टोपोलॉजिकल कम है?

(a) ABCD (b) BACD
(c) BADC (d) ABDC

HTET PGT C.S. 2019

Ans. (d) : Edges: A → B, A → C, B → C, B → D, D→C
- A must come before B and C
- B must come before C and D
- D must come before C

378. Number of edges in an undirected complete graph of 7 vertices is:
7 नोक के एक अनडायरेक्टड कम्पलीट ग्राफ में कुल कितनी एजेज होगी?

(a) 20 (b) 21
(c) 14 (d) 28

RPSC C.S.-I 2016

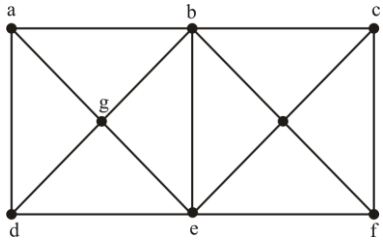
Ans, (b) : To find the number of edges in an undirected complete graph with 7 vertices, we use the formula:

Number of edges = $\frac{n(n-1)}{2}$

Where n = 7

Edges $= \frac{7(7-1)}{2} = \frac{7\times6}{2} = \frac{42}{2} \Rightarrow 21$

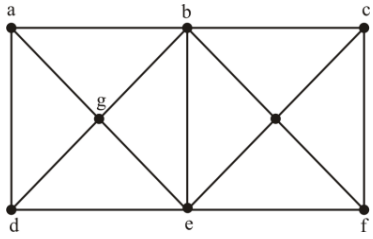
379. A clique in a simple undirected graph is a complete subgraph that is not contained in any larger complete subgraph. How many cliques are there in the graph shown below?



(a) 2 (b) 4
(c) 5 (d) 6

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Ans : (c) Given graph is



Complete subgraph means "each vertex should be connected with all other vertices in the subgraph". Such subgraph are:

(i) {b, c, e, f} (ii) {a, d, g}
(iii) {a, b, g} (iv) {b, e, g}
(v) {d, e, g}

There are only 5 subgraphs.

380. Consider a complete bipartite graph $k_{m,n}$. For which values of m and n does this, complete graph have a Hamilton circuit

(a) m = 3, n = 2 (b) m = 2, n = 3
(c) m = n ≥ 2 (d) m = n ≥ 3

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, $K_{m,n}$ is Hamilton circuit if and only if m = n ≥ 2, (Circuit is a path similar to a cycle that starts and ends at the same vertex) and $K_{m,n}$ is a Hamilton circuit if m = n ≥ 2, so the correct answer is (c).

381. Consider a Hamiltonian Graph G with loops or parallel edges and with $|V(G)| = n \geq 3$. Then which of the following is true?

(a) $\deg(v) \geq \frac{n}{2}$ for each vertex v.

(b) $|E(G)| \geq \frac{1}{2}(n-1)(n-2)+2$

(c) $\deg(v)+\deg(w) \geq n$ whenever v and w are not connected by an edge.

(d) All of the above

UGC NET C.S. June-2017 (Paper-II)

Ans : (d) For Hamiltonian Graph 'G' with are loops or parallel edges with |v (G)| = n ≥ 3 then:

- According to ore's theorem sum of degree of every non-adjacent pair of vertices i.e., deg (v) +d (w) must be greater than equal to n(total number of vertices).
- According to Dirac's theorem degree of every vertex must be greater than n/2.
- Since according to Dirac's theorem

$\deg(v)+\deg(w) \geq n$ .....(i)

Hence, $|E| \leq |E(G-\{u,v\})| + \deg G(v) + \deg G(w)$

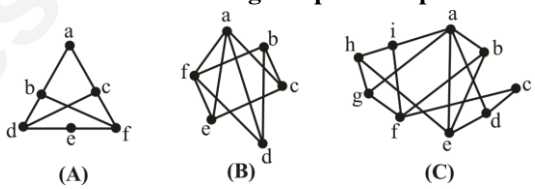
$\leq \binom{n-2}{2} + n$ using equation (i)

$\leq \frac{(n-2)(n-3)}{2} + n$

$\leq \frac{n^2-3n-2n+6+2n}{2}$

$\leq \frac{n^2-3n-6}{2}$

382. Which of the following Graph is/are planer?



Choose the most appropriate answer from the options given below :

(a) A and C only (b) B only
(c) A only (d) A and B only

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Ans. (d) : Graph planer- A planer graph is a graph that can be drawn on a plane (a flat surface without any of its edges crossing. In other words, it's posssible to arrange the vertices and edges in a way that they to not intersect except at the vertices.

Graph A:- is planer you can redraw it without any edges crossing here's one way to do it imagine the rectangle as the outer boundary, and then draw one diagonal inside the rectangle and the other diagonal outside.

Graph B :- is planer you can redraw it without any edges crossing. Think of the house shape as the outer boundary, and then draw the extra edge as a line connecting the roof's peak to the bottom side of the house.

Graph C:- is non-planer, This one is a bit trickier to visualize, but it's a classic example of a non-planer graph called the "complete bipartite graph k. {2, 3}". No matter how you try to redraw it, you all always end up with at least one pair of edges crossing.

Sarkari Resource

02. Computer Organization and Architecture

(i) Stored Program Concept

1. The speed of supercomputer is measured in _____.
सुपर कम्प्यूटर की गति को _____ में मापा जाता है।

(a) Floating – point operations per hour
फ्लोटिंग – पॉइंट ऑपरेशन्स प्रति घंटा

(b) Floating – point operations per day
फ्लोटिंग – पॉइंट ऑपरेशन्स प्रति दिन

(c) Floating – point operations per second
फ्लोटिंग – पॉइंट ऑपरेशन्स प्रति सेकंड

(d) Floating – point operations per minute
फ्लोटिंग – पॉइंट ऑपरेशन्स प्रति मिनट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The speed of supercomputer is measured in Floating-Point Operations Per Second. (FLOPS). These measurements help determine the computational power and performance of a supercomputer.

2. A computer based information system is needed :

(I) as it is difficult for administrative staff to process data
(II) due to rapid growth of information and communication technology
(III) due to growing size of organizations which need to process large volume of data
(IV) as timely and accurate decisions are to be taken

Which of the above statement (s) is/are true?

(a) I and II (b) III and IV
(c) II and III (d) II and IV

UGC NET C.S. December-2014 (Paper-III)

Ans : (b) A computer based information system is needed because of growing size of organizations which need to process large volume of data & so large data need timely & accurate decisions.
So, the correct choice is (b).

3. Choose a false statement about Von Neumann's stored program computer architecture:
वॉन-न्यूमैन के ''स्टोर्ड प्रोग्राम आर्किटेक्चर'' के बारे में असत्य कथन चुनिए:

(a) Program and data are stored in same address space in main memory/प्रोग्राम व डेटा मेन मेमोरी के एक ही समान एड्रेस स्पेस में स्टोर किये जाते है

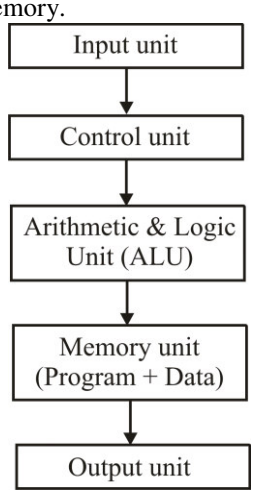
(b) Allow self modifying program
सेल्फ मॉडिफाइंग प्रोग्राम अनुमत है

(c) Programs and data are stored in secondary memory
प्रोग्राम व डाटा सेकेंडरी मेमोरी में स्टोर किये जाते है

(d) Set of control signal is same for instruction and data fetch./इंस्ट्रक्शन व डाटा फैच के लिए समान कंट्रोल सिग्नल का सेट होता है

RPSC C.S. 2016 (PAPER-I)

Ans: (c) :
(a). Program and data are stored in same address space in main memory is a core feature of Von Neumann's architecture.
(b). Allow self modifying programs because both data and Instructions are stored in the same memory, self modifying code is possible.
(c). Programs and data are not stored in secondary memory in Von Neumann's architecture, programs and data are stored in main memory (RAM) during execution secondary memory (like a hard disk) is used for long-term storage, but the Von Neumann's model specifically refers to programs being loaded in to main memory for processing.
(d). Set of control signal is same for instruction and data fetch since both use the same memory and buses, the control signals are similar when fetching data or instructions.
The false statement is (c) Programs and data are stored in secondary memory.



4. The term ____ is used for the larger, more powerful computers other than supercomputers. It supercomputer a large database, has elaborate I/O hardware, and is used in a central data processing facility.
सुपर कम्प्यूटर्स के अलावा बड़े, अधिक शक्तिशाली कम्प्यूटरों के लिए _____ पद प्रयुक्त होता है यह विशाल डाटाबेस का अवलम्बन करता है , जो I/O हार्डवेयर को विस्तृत करता है और सेन्ट्रल डाटा प्रोसेसिंग फैसिलिटी में प्रयुक्त होता है ।

(a) minicomputer/मिनीकम्प्यूटर
(b) supercomputer/सुपरकम्प्यूटर

Sarkari Resource

(c) mainframe/मेनफ्रेम
(d) microcomputer/माइक्रोकम्प्यूटर

KVS PGT C.S. 2013

Ans, (c) : A mainframe computer informally called a mainframe or big iron, is a computer used primarily by large organization for critical applications like bulk data processing for tasks such as censuses. Enterprise resource planning and large scale transaction processing.

5. Von Neumann computer architecture is _____.
वॉन न्यूमैन कम्प्यूटर आर्किटेक्चर किस प्रकार का है।
(a) SISD (b) SIMD
(c) MIMD (d) MISD

KVS PGT C.S. 2018

Ans, (a) : Von-Neumann proposed his computer architecture design in 1945 which was later known as Von-Neumann Architecture. It consisted of a Control Unit, Arithmetic Logical Memory Unit (ALU), Registers and Inputs/Outputs, Von-Neumann architecture is based on the stored program computer concept, where instruction data and program data are stored in the same memory. Von-Neumann computer architecture is SISD. SISD stands for Single Instruction, Single Data.

6. Which of the following is not considered in Booth's algorithm for multiplication?
निम्न में से किसका गुणन के लिए बूथ एल्गोरिथ्म में ध्यान नहीं रखते हैं?
(a) Positive multipliers and positive multiplicands
धनात्मक गुणक तथा धनात्मक गुण्य
(b) Negative multipliers and negative multiplicands /ऋणात्मक गुणक तथा ऋणात्मक गुण्य
(c) Positive multipliers and negative multiplicands /धनात्मक गुणक तथा ऋणात्मक गुण्य
(d) All of the above /उपरोक्त सभी

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : Booth's algorithm efficiently handles. Both positive and negative multipliers and multiplicands, so all cases are considered.

(ii) Components of a Computer System

7. Computers works only on 1s and 0s pattern, instructions made up with it is called?
कम्प्यूटर में 1 और 0 द्वारा निर्मित निर्देश कहलाता है?
(a) Programming language
(b) Machine code
(c) Instructions
(d) Data

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : The instructions made up of 1s and 0s that computer works with are called Machine Code. Machine code is the native language of computers, which they can directly understand and execute. It is written in binary code, consisting of only 0s and 1s.
So, the correct option is (b) Machine code.

8. Which of the following is not true regarding registers?
निम्न में से कौन-सा रजिस्टर के बारे में सही नही है?
(a) Internal storage of CPU/CPU का आन्तरिक संचय
(b) Can hold either data or instruction
डेटा अथवा इंस्ट्रक्शन को होल्ड कर सकता है
(c) Made up of flip-flop/Flip-flop से बना हुआ है
(d) Cannot store intermediate results
मध्यवर्ती रिजल्ट को स्टोर नहीं करता है

LT. Grade C.S. 2018

Ans, (d) : Registers are small high speed memory locations with in the Central Processing Unit that store data temporarily while it is being processed.
So, the option (d) false regarding to registers.

9. _____ is used to transfer the data directly to main memory without using CPU.
CPU का उपयोग किए बिना डेटा को सीधे मुख्य मेमोरी में स्थानांतरित करने लिए _____ का उपयोग किया जाता है।
(a) Device drive/डिवाइस ड्राइव
(b) Input/output interface/इनपुट/आउपुट इंटरफेस
(c) DMA controller/डी.एम.ए कंट्रोलर
(d) Memory address Register/मेमोरी एड्रेस रजिस्टर

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : DMA controller (Direct memory Access controller) is a special hardware component in a computer system.
That manages data transfer directly between I/O devices and main memory without involving the CPU for every byte /word of data.

10. Which of the following statement(s) is/are true control unit?
क़ंट्रोल यूनिट के बारे में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?
I. It manages and coordinate the entirecomputer system./यह संपूर्ण कंप्यूटर सिस्टम का प्रबंधन और समन्वित करता है
II. It does not perform any actual processing on data/यह डेटा पर कोई वास्तविक प्रक्रमण नहीं करता है
(a) Only I (b) Only II
(c) Both I and II (d) Neither I and II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : The control unit is a part of the CPU (Central Processing Unit) that direct and coordinates all the operation of the computer.
In simple term the control unit acts like the "manager" of "traffic controller" of the CPU.
The control unit does not perform calculation or store data it self-it simply control, how data and instruction more Through the CPU

11. Which of the following natural elements is the primary element in computer chips?
निम्नलिखित में से कौन-सा प्राकृतिक तत्व, कम्प्यूटर चिप्स में प्राथमिक तत्त्व है?
(a) Carbon (b) Silicon
(c) Iron
(d) More than one of the above
उपर्युक्त में से एक अधिक

Sarkari Resource

(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : Silicon is a semiconductor material abundant in nature (found as silicon dioxide), and is used to manufacture integrated circuits (ICs) and microchips.

12. Secondary memory is the long term store for programs and data while main memory holds program and data currently in use. What kind of an organization is this?
माध्यमिक मेमोरी कार्यक्रमों और डेटा के लिए दीर्घकालिक स्टोर है जबकि मुख्य मेमोरी वर्तमान में उपयोग में आने वाले प्रोग्राम और डेटा रखती है। यह किस तरह का संगठन है?

(a) Physical/भौतिक
(b) Logical/तार्किक
(c) Structural/संरचनात्मक
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : Physical organization deals with how the hardware components are physically arranged and connected to perform computing tasks.
1. Memory organization
* How main memory (RAM) and secondary memory (hard drives, SSD) are physically structured.
2. CPU Organization
* Internal layout of CPU:ALU (Arithmetic logic unit)
3. Input /output organization
4. Bus organization
5. Storage organization.

13. _____ is used to store software that does not update.
_____ का प्रयोग ऐसे सॉफ्टवेयर को स्टोर करने के लिए किया जाता है, जिसको अपडेट करना नहीं पड़ता है।

(a) SRAM (b) DRAM
(c) ROM
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c): ROM is a type of non-volatile memory in a computer, meaning its contents cannot be lost when the power is turned off. Data stored in ROM is usually permanent and preinstalled by the manufacturer.

14. A storage device used to compensate the difference in flow of data is known as
डेटा के प्रवाह में अंतर की भरपाई के लिए उपयोग की जाने वाली स्टोरेज डिवाइस को क्या कहा जाता है?

(a) Main memory/मुख्य मेमोरी
(b) Auxiliary memory/सहायक मेमोरी
(c) Buffer/बफर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : A Buffer in computer organization is a temporary storage area used to hold data while it is being transferred between two devices or processes that operate at different speeds.
In CPU context, the memory buffer register (MBR) (also called memory data register MDR) is a type of buffer that hold data being transferred to or from memory.

15. Memory in which any location can be reached in a fixed amount of time after specifying its address is called
वह मेमोरी, जिसके किसी लोकेशन पर उस लोकेशन को बताने के बाद एक निश्चित समय में पहुँचा जा सकता है, कहलाती है

(a) Sequential Access Memory
सिक्वेन्शियल एक्सेस मेमोरी
(b) Random Access Memory
रैंडम एक्सेस मेमोरी
(c) Secondary Memory/सेकेंडरी मेमोरी
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): RAM is temporary, fast and volatile memory used by the CPU to store and access data quickly. RAM is a type of computer memory that is used to store data and instruction temporarily which a computer is running. It is called "random access" because any memory cell can be accessed directly it.

The row and column are know, unlike sequential access (like in magnetic tapes).

16. The brain of computer is
कम्प्यूटर का दिमाग है

(a) CPU/सीपीयू
(b) Memory/ मेमोरी
(c) I/O device/आई/ओ डिवाइस
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a): The CPU (Central Processing Unit) is called the "brain" of the computer because it performs all calculations, processes instruction, and controls the functioning of other unit such as memory and input/output devices.

17. The cost of storing a bit is minimum in
एक बिट को स्टोर करने का न्यूनतम खर्च _____ में होता है।

(a) Cache/कैश
(b) Register/रजिस्टर
(c) Magnetic Tape/मैग्नेटिक टेप
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Sarkari Resource

Ans. (c): Magnetic tape offers the lowest cost per bit for storing data compared to other memory types like cache, registers or magnetic disks.

18. Tera means
Tera का अर्थ/मतलब होता है
(a) 2^{20}
(b) 2^{40}
(c) 2^{30}
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): Memory size in power,

1 KB	1024 bytes	2^{10}
1 MB	1024 KB	2^{20}
1 GB	1024 MB	2^{30}
1 TB	1024 GB	2^{40}
1 PB	1024 TB	2^{50}
1 EB	1024 PB	2^{60}
1 ZB	1024 EB	2^{70}
1 YB	1024 ZB	2^{80}

19. Arrange that following in the increasing order with respect to access time.
A. Solid State Drive **B. Optical Discs**
C. DRAMs **D. SRAMs**
E. Registers
Choose the correct answer from the options given below :
(a) D, C, A, B, E (b) B, D, C, A, E
(c) A, E, B, D, C (d) E, D, C, A, B

UGC NET C.S. December-2024

Ans. (d) :
(E) Registers (Fastest, located inside the CPU)
(D) SRAM (Static RAM) (Faster than DRAM, used for CPU cache)
(C) DRAM (Dynamic RAM) (Main memory; slower than SRAM)
(A) Solid state Drive (SSD) (Non-Volatile storage much slower than RAM)
(B) Optical Discs (Slowest, used for long term storage like CDS/DVDs)

20. Which input device is used for reading answer to question in a test or survey?
एक परीक्षण या सर्वे में प्रश्न के उत्तर को पढ़ने के लिए इनपुट उपकरण का उपयोग किया जाता है?
(a) OCR
(b) OMR
(c) Barcode reader/बारकोड रीडर
(d) MICR

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : An Optical Mark Reader (OMR) is a specified input device used to read and process answers on pre-printed forms, such as those used in tests, surveys, or questionnaires. Explain of how it works and its role.

1. Form Design– OMR relies on standardized forms with predefined areas (bubbles, circles, boxes) where respondents mark their answers using a pen or pencil. These forms are designed with specific layouts that the OMR device can interpret.

2. Mark Detection– The OMR device uses optical sensors to scan the form and detect the presence of marks in designated areas. It identifies whether a bubble is filled based on the contrast between the mark (Usually dark, like pencil graphite or ink) and the unmarked paper.

3. Data Conversion– Once the marks are detected the OMR converts the information into digital data for a filled bubble for option "A" on Question 1 might be recorded as "A" in the system.

4. Processing– The collected data is sent to a computer or software for analysis, scoring or reporting. For tests, this could mean calculating scores; for surveys, it could involve aggregating responses.

Application–

- **Standardized Testing –** OMR is widely used for multiple-choice tests (SAT, ACT) where students fill in bubbles corresponding to their answers.
- **Surveys and Polls–** Organizations use OMR to collect responses from customer feedback forms or opinion surveys.
- **Elections–** some voting system use OMR to read marked ballots.

21. Which of the following is a impact printer?
निम्नलिखित में से कौन सा एक इम्पैक्ट प्रिंटर है?
(a) Inkjet printer/इंकजेट प्रिंटर
(b) Dot-matrix printer/डॉट-मैट्रिक्स प्रिंटर
(c) Laser printer/लेज़र प्रिंटर
(d) Thermal Printer/थर्मल प्रिंटर

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) :

- **Dot Matrix Printer–** Dot matrix printers contain print heads of thin metal pins that strike an ink ribbon and transfer dots onto paper.Varied combinations of pins are activated to produce complete characters and graphics to compose printable output augmented with carriage movement.
- **Print Head–** Houses a vertical column of metal pins that are individually triggered to extend out and print or retract back when not needed for a specific dot position. High quality heads use jeweled tips.
- **Inked Ribbon Cartridge–** An inked ribbon cartridge is typically a fabric ribbon saturated with quick drying ink that sits between the print head and paper allowing ink transfer upon pin impact.
- **Stepper Motor–** Controls horizontal print head motion to facilitate printing full page width rather than Just printable column under fixed head this allows composing the dot matrix output.
- **Paper Feed Roller–** Feeds continuous stationery paper vertically with perforations that engage the sprocket tractor. Ensure paper alignment.
- **Electronic Interface–** Inputs digital signals from computer I/O and converts them into electrical commands for print head pin firing order and hammer retractions based on dot pattern requirements.

Sarkari Resource

22. Which of the following is an input device?
निम्नलिखित में से कौन सा एक इनपुट डिवाईस है?
(a) Plotter/'प्लॉटर' (b) Printer/प्रिंटर
(c) Joystick/'जॉयस्टिक' (d) Speaker/स्पीकर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : An input device is a piece of hardware that sends data and instructions to a computer.
Joystick:- Joystick is an input device It is used to control movement or direction, typically in video games, and sends signals based on its position to the computer.

23. Arrange the following memory units in descending order
Gigabyte, Megabyte, Exabyte, Zettabyte, Petabyte, Yottabyte
निम्नलिखित मेमोरी यूनिट को अवरोही क्रम में व्यवस्थित करें।
गीगाबाइट, मेगाबाइट, एक्साबाइट, जेटाबाइट, पेटाबाइट, योटाबाइट
(a) Yottabyte, Zettabyte, Exabyte, Petabyte, Gigabyte, Megabyte/योटाबाइट, जेटाबाइट, एक्साबाइट, पेटाबाइट, गीगाबाइट, मेगाबाइट
(b) Zettabyte, Yottabyte, Petabyte, Exabyte, Megabyte, Gigabyte/जेटाबाइट, योटाबाइट, पेटाबाइट, एक्साबाइट, मेगाबाइट, गीगाबाइट
(c) Yottabyte, Exabyte, Petabyte, Zettabyte, Gigabyte, Megabyte/योटाबाइट, एक्साबाइट, पेटाबाइट, जेटाबाइट, गीगाबाइट, मेगाबाइट
(d) Zettabyte, Petabyte, Yottabyte, Exabyte, Gigabyte, Megabyte/जेटाबाइट, पेटाबाइट, योटाबाइट, एक्साबाइट, गीगाबाइट, मेगाबाइट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Memory units in descending order–
Yottabyte (YB)
Zettabyte (ZB)
Exabyte (EB)
Petabyte (PB)
Terabyte (TB)
Gigabyte (GB)
Megabyte (MB)
Kilobyte (KB)

24. With respect to size, which of the following represents the correct relationships between different categories of cache memory?
आकार के संबंध में निम्नलिखित में से कौन कैश मेमोरी की विभिन्न श्रेणियों के बीच सही संबंधों को दर्शाता है?
(a) L1 < L2 < L3 (b) L1 < L3 < L2
(c) L2 < L1 < L3 (d) L3 < L2 < L1

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : • Cache memory is divided into levels depending on speed and size
• L1 cache → smallest in size (usually 32 KB-128 KB per core) but fastest
• L2 cache → Larger than L1 (typically 256 KB - 1 MB per core), a bit slower
• L3 cache → Largest (several MB, shared across cores), but slower compared to L1 and L2
Therefore, in terms of size relationship L1 < L2 < L3.

25. Identify the valid cache organisation techniques:
मान्य कैश संगठन तकनीकों की पहचान करें:
(i) Direct / डायरेक्ट (प्रत्यक्ष)
(ii) Associative / एसोसिएटिव
(iii) Set-associative / सेट-एसोसिएटिव
(a) (i) and (ii) only / केवल (i) और (ii)
(b) (ii) and (iii) only / केवल (ii) और (iii)
(c) (i) and (iii) only / केवल (i) और (iii)
(d) (i), (ii) and (iii) / (i), (ii) और (iii)

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Cache memory can be organized in three standard ways.
(1) Direct mapping
(2) Associative mapping
(3) Set associative mapping
All three are valid techniques

26. Which of the following refers to a memory access mechanism in which the time to access a given location is constant and independent of the sequence of prior accesses?
निम्नलिखित में से कौन सा मेमोरी एक्सेस मैकेनिज्म से संबंधित होता है जिसमें किसी दिए गए स्थान तक पहुंचने का समय निरंतर और पूर्व-एक्सेस अनुक्रम से स्वतंत्र होता है?
(a) Random access/रैंडम एक्सेस
(b) Direct access/डाइरेक्ट एक्सेस
(c) Sequential access/सिक्वेंशियल एक्सेस
(d) Associative/एसोसिएटिव

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Random access is a memory access method where any location in memory can be accessed directly and in a constant amount of time regardless of the physical location of the data or the sequence of prior accesses. This is defining characteristic of Random Access Memory **(RAM)** where the time required to read or write data to any memory address is theoretically the same.
Therefore, option (a) is correct.

27. State whether the following statements are true of false.
यह बताएं कि निम्नलिखित में से कौन सा कथन सत्य या असत्य है।
Statements:/कथन:
(i) For main memory, the unit of transfer is the number of bytes read out of or written into memory at a time.
मेन मेमोरी (मुख्य स्मृति) के लिए, स्थानांतरण की इकाई एक समय में मेमोरी (स्मृति) में रीड किए गए या लिखित बाइट्स की संख्या होती है।
(ii) The unit of transfer need not equal a word an addressable unit.
स्थानांतरण की इकाई किसी शब्द या एड्रेसेबल यूनिट के बराबर नहीं होनी चाहिए।
(a) (i) Ture, (ii) True/ सत्य, (ii) सत्य
(b) (i) Ture, (ii) False/ सत्य, (ii) असत्य

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(c) (i) False, (ii) True/ असत्य, (ii) सत्य
(d) (i) False, (ii) False/ असत्य, (ii) असत्य

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Ans. (c) : statement I.
(For main memory the unit of transfer is the number of bytes read out of or written into memory at time)
⇒ In most computer – organization reference (e.g. William Stallings, computer Organization and Architecture), the unit of transfer for main memory is normally a word.(For example, 32 bits or 64 bits)
⇒ Saying that it is specifically "the number of bytes read or written" makes it sound like bytes are always the transfer unit which is not necessarily true.
Verdict : False
Statement II.
The unit of transfer need not equal a word or an addressable unit.
This is correct : A machine might have an addressable unit of 1 byte but transfer data in 4 byte words or even larger blocks for cache.
Correct option ….(i) False (ii) true ⇒ options – c

28. Which of the following statements is correct?
निम्नलिखित कथनों में से कौन सा सही है?

(i) A hard disk is also used to provide an extension to the main memory known as virtual memory/ वर्चुअल मेमोरी के रूप में जाने वाली मेन मेमोरी (मुख्य स्मृति) को एक्सटेंशन प्रदान करने के लिए हार्ड डिस्क का उपयोग भी किया जाता है।

(ii) DRAM is faster than SRAM.
एस.आर.ए.एम (SRAM) की तुलना में डी.आर.ए.एम (DRAM) तेज है।

(iii) External, non-volatile memory is also referred to as 'secondary memory' or 'auxiliary memory'
बाहरी, नॉन वोलेटाइल (गैर-अस्थिर) मेमोरी को 'सेकंडरी मेमोरी' या 'ऑग्जिलरी मेमोरी' के रूप में भी जाना जाता है।

(a) Statements (i) and (ii) are correct
कथन (i) और (ii) सत्य हैं।
(b) Statements (ii) and (iii) are correct
कथन (ii) और (iii) सत्य हैं।
(c) Statements (i) and (iii) are correct
कथन (i) और (iii) सत्य हैं।
(d) Statements (i), (ii) and (iii) are correct
कथन (i), (ii) और (iii) सत्य हैं।

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Floppy Disk→ Very small storage capacity (1.44 MB).
- Compact Disk (CD) → Around 700 MB
- Magneto-optic Disk→ Typically a few GB, (less common)
- Hard Disk (HDD) → Can store hundreds of GB to several TBs, making it the Largest storage capacity among the given option.

29. Which of the following part of the CPU has the necessary electronic circuitry to perform binary addition and shift operations in computer system?
(a) Arithmetic Logic Unit (ALU)
(b) Memory Buses
(c) Registers
(d) Control Unit

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : The part of the CPU that contains the necessary electronic circuitry to perform binary addition, subtraction, and shift operations in a computer system is the **Arthmetic Logic Unit (ALU).**

30. Various storage devices used by an operating system can be arranged as follows in increasing order of accessing speed :
(a) Magnetic tapes → magnetic disks → optical disks → electronic disks → main memory → cache → registers
(b) Magnetic tapes → magnetic disks → electronic disks → optical disks → main memory → cache → registers
(c) Magnetic tapes → electronic disks → magnetic disks → optical disks → main memory → cache → registers
(d) Magnetic tapes → optical disks → magnetic disks → electronic disks → main memory → cache → registers

UGC NET C.S. December-2014 (Paper-III)

Ans. (d) : The storage devices used by a comuter can be arranged in order of increasing speed like this.
Magnetic tapes < Optical disks < Hard disks < Flash/SSD (electronic disks) < Main memory < cache < Registers.
So the correct answer is (D).

31. Which of the following (s) are main memory?
(A) Virtual memory
(B) Cache memory
(C) RAM
(D) SSD
Choose the correct answer from the options given below :
(a) (A) and (C) Only
(b) (B) and (C) Only
(c) (C) and (D) Only
(d) (A), (B) and (C) Only

UGC NET C.S. December-2023

Ans. (b) : The main memory is the fundamental storage unit in a computer system. It is associatively large and quick memory and saves programs and information during computer operations. The technology that make the main memory work is base on semiconductor integrated circuits . It is also known as primary memory. RAM and cache memory is the main memory.

32. Which of the following is/are true for Dynamic RAM (DRAM)?
A. It is slower than static RAM (SRAM)
B. Packing Density is higher than Static RAM (SRAM)

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C. It is faster than static RAM(SRAM)
D. It requires data refreshing
Choose the correct answer from the options given below:
(a) A, B and D Only (b) A, B and C Only
(c) C and D Only (d) B and D Only

UGC NET C.S. December-2022

Ans. (a) : Difference Between static RAM and Dynamic RAM

Basic of comparison	SRAM	DRAM
Speed	Faster	Slower
Size	Small	Large
Cost	Expensive	Cheap
Used in	Cache Memory	Main Memory
Destruction	Less Dense	Highly Dense
Power Consumption	Low	High

So option (a) is correct.

33. CMOS is a Computer Chip on the motherboard, which is :
(a) RAM (b) ROM
(c) EPROM (d) Auxillary storage

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : CMOS stands for "Complementary Metal-Oxide-Semiconductor". It's the manufacturing process used to create, processors, RAM, and digital logic circuits, and is also the name for chips created using that process.

34. Choose a true statement:
सत्य कथन चुनें:
(a) SRAM bit cell uses a capacitor and a transistor
SRAM बिट सेल में एक कैपासिटर व एक ट्रांजिस्टर का उपयोग होता है
(b) DRAM bit cell uses two inverter and two transistors/ DRAM बिट सेल में दो इन्वर्टरस् तथा दो ट्रांजिस्टर्स का उपयोग होता है
(c) SRAMs are faster than DRAMs
SRAMs, DRAMs की तुलना में तेज होती है
(d) DRAM bit cell uses two capacitors and one transistor./ DRAM बिट सेल में दो कैपासिटर एवं एक ट्रांजिस्टर का उपयोग होता है।

RPSC C.S. 2016 (PAPER-I)

Ans, (c) :
(a) SRAM bit cell uses a capacitor and a transistor–
- SRAM (Static RAM) uses 6 transistors (usually) per bit and does not use capacitors.
- Capacitors are used in DRAM not SRAM

(b) DRAM bit cell uses two inverters and two transistors–
- DRAM (Dynamic RAM) uses 1 capacitors and 1 transistors per bit.
- Two inverters are used in SRAM, not DRAM.

(c) SRAMs are faster than DRAMs–
- SRAM is faster than DRAM because it does not need to be refreshed and has simpler access.
- That's why it's used for CPU cache.

(d) DRAM bit cell uses two capacitor and one transistors–
- DRAM uses one capacitor and one transistor per bit.

Options (c) correct answer.

35. In context of computer system architecture a stack is:
कंप्यूटर सिस्टम आर्किटेक्चर के सन्दर्भ में स्टैक है?
(a) Reserved RAM address space
रैम (RAM) एड्रेस स्पेस में आरक्षित स्थान
(b) Reserved ROM address space
रोम (ROM) एड्रेस स्पेस में आरक्षित स्थान
(c) Reserved I/O address space
I/O एड्रेस स्पेस में आरक्षित स्थान
(d) A list of instruction/इंस्ट्रक्शन की एक लिस्ट

RPSC C.S. 2016 (PAPER-I)

Ans, (a) :
(a) Reserved RAM address space–
- The stack resides in RAM (Random access memory), which is volatile memory that allows both for temporary storage of data, typically used for managing function calls, local variables, and returns addresses it operates on a Last-in-First-Out (LIFO) principle, where data is pushed onto the stack and popped off in reverse order.

(b) Reserved ROM address space–
- ROM (Read-only memory) is used for permanent storage of firmware or instructions, not for dynamic data structures like a stack.

(c) Reserved I/O address space-
- I/O address space is used for communication with peripheral devices, not for stack operations.

(d) A list of instruction–
A stack is not a list of instructions it is a data structure for storing temporary data, not executable code.

36. The brain of any computer system is-
किसी भी कंप्यूटर सिस्टम का मस्तिष्क_____होता है।
(a) ALU
(b) Memory
(c) CPU
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c) : The CPU (Central processing unit) is after called the brain of the computer because it perform all the processing, decision-making, and controls the operation of other component.
- ALU is part of the CPU that handles arithmetic and logical operation
- Memory store data and instruction but does not processes them.

37. The memory devices which are similar to EEPROM but differ in the cost effectiveness is _____.
मेमोरी डिवाइस जो EEPROM के समान हैं लेकिन लागत प्रभावशीलता में भिन्न है_____
(a) CMOS
(b) Memory sticks

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(c) Flash memory
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : Flash memory is derived from EEPROM technology, work similarly (can be electrically erased and reprogrammed), but is more cost-effective for larger memory sizes.
• EEPROM (Electrically Erasable Programmable Read Only Memory)

38. Which of the following is the type of Read Only Memory (ROM)?/निम्नलिखित में से कौन-सा रीड ओनली मेमोरी (ROM) का प्रकार है?

(a) PROM
(b) EPROM
(c) MROM
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : A cording to question all three are types of ROM.
• PROM → This is a type of ROM that can be programmed once after manufacturing
• EPROM → This types of ROM that can be erased by UV light and reprogrammed.
• MROM → This is a pre programmed ROM at the time of manufacturing.

39. Which of the following memory is non-volatile? निम्नलिखित में से कौन सी मेमोरी क्षणभंगुर नहीं है?

(a) ROM/रोम
(b) Cache/कैश
(c) ROM and Cache/रोम और कैश
(d) RAM/रैम

EMRS PGT C.S. 16.12.2024

Ans, (a) : Non Volatile memory is memory that retains its values even when power is removed. Earlier forms of volatile memory included various forms of read only memory (ROM).
ROM, EPROM, EEPROM and Flash memory are non-volatile memory.

40. Which of the following options states the correct storage - device hierarchy in terms of decreasing access time ?
निम्नलिखित में कौन सा विकल्प अवरोही एक्सेस समय की दृष्टि से सही स्टोरेज उपकरण पदानुक्रम को व्यक्त करता है?

(a) Magnetic tape, magnetic disk, cache, main memory
मैग्नेटिक टेप, मैग्नेटिक डिस्क, कैश, मेन मेमोरी
(b) Magnetic disk, optical disk, main memory, register
मैग्नेटिक डिस्क, ऑप्टिकल डिस्क, मेन मेमोरी, रजिस्टर
(c) Magnetic tape, magnetic disk, main memory, register
मैग्नेटिक टेप, मैग्नेटिक डिस्क, मेन मेमोरी, रजिस्टर
(d) Register, main memory, magnetic disk, magnetic tape
रजिस्टर, मेन मेमोरी, मैग्नेटिक डिस्क, मैग्नेटिक टेप

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans, (d) : In computer architecture, memory hierarchy separates computer storage into a hierarchy based on response time. As we move from the bottom level to the top Level in the Hierarchy, the access time decreasing.
⇒ magnetic tape < magnetic disk < main memory < register.

41. A program is a/an__ entity.
एक प्रोग्राम ____ सत्ता (इन्टिटी) है ।

(a) active/क्रियाशील (एक्टिव)
(b) passive/निश्चेष्ट (पेसिव)
(c) dormant/प्रसुप्त (डोर्मन्ट)
(d) hyperactive/अतिक्रियाशील (हाइपरएक्टिव)

KVS PGT C.S. 2013

Ans, (b) : A program is a/an passive entity as it resides in the secondary memory such as the contents of a file stored on disk.

42. _____ refers to those attributes of a system visible to a programmer or, put another way, those attributes that have a direct impact on the logical execution of a program.
_____प्रोग्रामर को दिखने वाले पद्धति के उन लक्षणों को या अन्य तरीके से कहें तो उन लक्षणों को जो प्रोग्राम के तार्किक क्रियान्वयन पर सीधे प्रभाव डालते हैं, की ओर इंगित करता है ।

(a) Computer organization/कम्प्यूटर संगठन
(b) Computer architecture/कम्प्यूटर आर्किटेक्चर
(c) Microprocessor/माइक्रोप्रोसेसर
(d) Bus/बस

KVS PGT C.S. 2013

Ans, (b) : Computer architecture refers to those attributes of a system visible to a programmer or, put another way, those attributes that have a direct impact on the logical execution of a program.

43. Which of the following is/are wrong?
निम्नलिखित में से कौन-सा/कौन-से कथन गलत है।

(A) RAM and ROM are volatile memories,/RAM तथा ROM वोलेटाइल मेमोरी हैं।
(B) ROMs, PROMs and EPROMs are non-volatile memories./ROMs, PROMs तथा EPROMs नॉन-वोलेटाइल मेमोरी हैं।
(C) RAM and Dynamic RAM are same./RAM तथा डायनैमिक RAM समान हैं।
(D) A Random Access Memory (RAM) is a read write memory./रैंडम एक्सेस मेमोरी (RAM) रीड राइट मेमोरी
(a) (A) and (B)/(A) तथा (B)
(b) (A) and (C)/(A) तथा (C)
(c) (A) and (D)/(A) तथा (D)
(d) (C) and (D)/(C) तथा (D)

KVS PGT C.S. 2018

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Ans, (b) : RAM:- RAM is also called read write memory or the primary memory. It is a volatile memory as the data is lost when the power is turned off. RAM is further classified into two types, static RAM and Dynamic RAM.
ROM:- ROM is non-volatile memory ROM is further classified into four types- M ROM, PROM EPROM and EEPROM, which are also non-volatile memory.

44. Which of the following memory improves the speed of execution of a program?
निम्नलिखित में से कौन सी स्मृति समय जटिलता में सुधार लाती है?
(a) Virtual memory/आभासी स्मृति (वर्चुअल मेमोरी)
(b) Primary memory/प्राथमिक स्मृति
(c) Secondary memory/द्वितीयक स्मृति
(d) Cache memory/द्रुणिका स्मृति (कैशे मेमोरी)

KVS PGT C.S. 2017

Ans, (d) : Cache memory is a special very high-speed memory. It is used to speed up and Synchronize with high-speed CPU.
Cache memory improves the speed of execution of a program. cache memory is costlier than main memory or disk memory but more economical than CPU registers. Cache memory is an extremely fast memory type that acts as a buffer between RAM an the CPU.

45. On which of the following storage media, storage of information is organized as a single continuous spiral groove ?
निम्नलिखित में से किस भण्डारण माध्यम में, सूचना का भण्डारण एकल सतत घुमावदार खांचे (सिंगल कंटिन्यूअस स्पाइरल ग्रुव) के रूप में व्यवस्थित होती है?
(a) CD-ROM/सी डी - रोम
(b) RAM/रैम
(c) Hard disk/हार्ड डिस्क
(d) Floppy disk/फ्लॉपी डिस्क

KVS PGT C.S. 2017

Ans, (a) : A CD-ROM, (compact disc read-only memory) is a type of read-only memory consisting of a pre-pressed optical compact disc that contains data. In CD-Rom storage of information is organized as a single continuous spiral groove.

46. The memory which does not loose its content on failure of power supply is known as _____ memory.
वे स्मृतियाँ विद्युत आपूर्ति के बंद हो जाने पर भी अन्तर्निहित विषयवस्तु नष्ट नहीं होती है, को _____ स्मृतियाँ कहा जाता है।
(a) Main memory (b) Volatile
(c) Non-volatile (d) RAM

KVS PGT C.S. 2017

Ans, (c) : Non-volatile memory is a type of computer memory, that can retain stored information even after power is removed. In contrast, volatile memory needs constant power in order to retain data.

47. Which out of the following is the least capacity of memory unit compared to other options ?
निम्नलिखित मे से किसकी मेमोरी यूनिट की क्षमता अन्य विकल्पों की तुलना में सबसे कम है?
(a) MB (b) KB
(c) GB (d) TB

KVS PGT 23.03.2023 (Shift-II)

Ans, (b) : Tabular representation of various memory size is:-

Name	Equal to	Size (In byte)
Bit	1 bit	1/8
Nibble	4 bits	1/2 (rare)
Prate	8 bits	1
Kilobyte	1024 bytes	$1024 = 2^{10}$
Megabyte	1024 KB	$1048576 = 2^{20}$
Gigabyte	1024 MB	$1079741924 = 2^{30}$
Terabyte	1024 GB	$1099\ 511627776 = 2^{40}$
Peta byte	1024 TB	$11,25,899,906,842,624 = 2^{50}$

Hence, out of the above table, KB is the least capacity of memory unit compared to other options.

48. A collection of 4 bits is known as ----------.
4 बिट्स के समूह को क्या कहते है?
(a) BYTE (b) WORD
(c) NIBBLE (d) K. BYTE

KVS PGT 23.03.2023 (Shift-II)

Ans, (c) : From the above question explanation, a collection of 4 bits is known as Nibble.

49. Which of the following can work as Input as well as an Output device ?
निम्नलिखित में से क्या एक इनपुट के साथ-साथ आउटपुट डिवाइस के रूप काम कर सकता है?
(a) Scanner/स्कैनर
(b) Interactive Panel/इंटरेक्टिव पैनल
(c) LED Screen/एल ई डी स्क्रीन
(d) Touch Pad/टच पैड

NVS PGT- 16.12.2022 (Shift-II)

Ans. (b) : An interactive panel can function as both an input and output device. so it effectively (output) and touch sensitive input device in one.

50. Resolution of printer is measured in :
प्रिंटर के रेज्योलूशन को किसमें मापा जाता है?
(a) dpi/डी पी आई (b) kbps/के बी पी एस
(c) MHz/मेगाहर्ट्ज (d) MB/एम बी

NVS PGT- 16.12.2022 (Shift-II)

Ans. (a) : The resolution of a printer indicates the clarity and detail of the printer output. It is measured in DPI (dots per inch), which tell how many individual dots
The printer can place in one inch.

51. A CPU contains
एक CPU रखता है
(a) A card reader and a printing device
(b) An analytical engine and a control unit
(c) A control unit and an arithmetic logic unit
(d) An arithmetic logic unit and a card reader

Bihar STET C.S. 2020

Ans. (c) : A CPU is the "brain" of a computer.
• Control Unit → Direct and coordinates other component what to do
• Arithmetic logic unit→ Perform arithmetic operation (like addition, subtraction) and logical operations (like comparisons)

Sarkari Resource

52. In most IBM PCs, the CPU, the device drives, memory expansion slots and active components are mounted on a single board. What is the name of this board?
अधिकांश IBM पीसी में, CPU, डिवाइस ड्राइव्स, मेमोरी विस्तार स्लॉट और सक्रिय घटकों को एक ही बोर्ड पर लगाया जाता है। इस बोर्ड का नाम क्या है?

(a) Motherboard (b) Breadboard
(c) Daughter board (d) Grandmother board

Bihar STET C.S. 2020

Ans. (a) : Motherboard is the main circuit board of a computer, on which the key component of the system is mounted. It acts as the central hub connecting all parts of the computer.

53. Regarding a VDU, which statement is more correct?
एक VDU के बारे में, कौन सा कथन अधिक सही है?

(a) It is an output device
(b) It is an input device
(c) It is a peripheral device
(d) It is hardware item

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Ans. (a) : VDU (Visual Display Unit) is basically a monitor/screen that display output to the user.
It work with the graphics card (Video adapter) to show processed data.

54. A modern electronic computer is a machine that is meant for
एक आधुनिक इलेक्ट्रॉनिक कंप्यूटर एक ऐसी मशीन है जिसका मतलब है

(a) Doing quick mathematical calculations
(b) Input, storage, manipulation and outputting of data
(c) Electronic data processing
(d) Performing repetitive tasks accurately

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Ans. (b) : A modern electronic computer is a machine that is meant for input storage, manipulation of outputting of data. The processor, more formally known as the central processing unit (CPU), has the electronic data into the information people want.

55. Which of the following storage devices can store maximum amount of data?
निम्नलिखित में से कौन सा स्टोरेज डिवाइस अधिकतम मात्रा में डेटा स्टोर कर सकता है

(a) Floppy Disk
(b) Hard Disk
(c) Compact Disk
(d) Magneto-optic Disk

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Ans. (b) : The storage device that can typically store the maximum amount of data among the options listed is the Hard Disk.

56. Computer memory consists of?
Computer memory में होता है?

(a) RAM (b) ROM
(c) PROM (d) All of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : • RAM (Random Access Memory): Volatile memory, temporary storage .
• ROM (Read Only Memory): Non Volatile, stores permanent instructions (like BIOS).
• PROM (Programmable ROM): A type of ROM that can be programmed once by the user.

57. Control unit is called?
Control Unit कहलाता है?

(a) Nerve centre (b) Clock
(c) Master dispatcher (d) All of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : In computer science, the nerve center of a computer is the CPU (central processing unit)
• Controls all operation of the system.
• Interprets and executes instructions.
• Coordinates input, processing, storage, and output
• Contains the control unit (CU) and Arithmetic logic unit (ALU).

58. Which function of a computer is responsible for permanent storage of data?
इनमें कौन Storage function के लिए जिम्मेदार है?

(a) Input (b) Output
(c) CPU (d) Memory

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : • Memory → Stores data (temporary in RAM and permanent in ROM, Hard Disk, SSD, etc).
For permanent storage, memory (non volatile memory) is responsible.

59. Main memory is called?
Main memory कहलाता है?

(a) Cells (b) RAM
(c) Hard Disk (d) Floppy Disk

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : Main memory is called RAM (Random Access Memory)
• It is volatile memory, which means data is lost when the power is switched off.
• It stores program and data that the CPU needs for immediate use.

60. Bubble memory is a?
Bubble memory है एक?

(a) Sequential access device only
(b) Direct access device only
(c) Combination of sequential and direct access device
(d) All of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : • Bubble Memory is a type of non-volatile memory that store data using tiny magnetized areas called magnetic bubble.
• It can access data Both.
• Sequentially (like a tape)
• Directly (like a disk)
• Because of this dual nature it is considered a combination of sequential and direct access device.

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61. Data entry can be performed with all of the following except.....
.....को छोड़कर Data Entry सभी से होता है।
(a) OCR (b) OMR
(c) COM
(d) Voice-recognition systems
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : COM (computer output to microfilm micro fiche) → This is not used for data entry. It is used to store computer output on microfilm for archiving not used for data entry.
Where as OCR, OMR, Voice-recognition system used for data entry.

62. Computer is a/an .....
कम्प्यूटर एक .....
(a) Electrical device
(b) Electronic device
(c) Mechanical device
(d) Electro/Mechanical device
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : A Computer is primarily an electronic device because it operates using the controlled flow of electrons through circuits, specifically using components like transistors and microchips to process and store data.

63. .....is the brain of computer.
.....कंप्यूटर का मस्तिष्क है।
(a) CPU (b) Memory
(c) VDU (d) Keyboard
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : The CPU (Central processing Unit) is the "Brain" of computer.
It is the main unit that control, processor and executes instructions.

64. VGA stand for?
VGA का अर्थ है?
(a) Video Guard Ample
(b) Visual Gate Adapter
(c) Video Graphics Array
(d) Visual Game Adapter
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : VGA (Video Graphic Array) is a display standard developed by IBM in 1987. It became one of the most widely used video display system for computer.

65. Computer cannot function without a?
कंप्यूटर.....के बिना कुछ नहीं कर सकता है।
(a) Paper (b) Cable
(c) Battery (d) Program
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : A computer cannot function without a program because programs (set of instruction) tell the computer, what task to perform. Without programs, the hardware is just an idle machine.

66. The most common input device used today is?
आज के समय का सर्वाधिक उपयोगी Input Device है।
(a) Mouse (b) ALU
(c) Keyboard (d) Printer
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : The keyboard is the most commonly used input device even today, as it allows users to enter text, numbers, and commands into the computer.

67. Which of the following is a part of the Central Processing unit?
इनमें से कौन CPU का भाग है?
(a) Mouse (b) ALU
(c) Keyboard (d) Printer
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : CPU (Central Processing Unit) is made up of ALU (Arithmetic logic unit), CU (Control unit) and register So, ALU is part of CPU.

68. Which of the following is an example of non-volatile memory?
इनमें से कौन का Non-volatile memory उदाहरण है?
(a) RAM (b) LSI
(c) ROM (d) VLSI
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : • ROM (Read Only Memory) is non-volatile memory, meaning it retains data even when the power is turned off.
• ROM-Stores firmware and system level programs permanently.

69. Large computer system typically use.....
बड़ा कंप्यूटर सिस्टम .....उपयोग करता है।
(a) Line Printer
(b) Ink jet printer
(c) Dot-matrix printers
(d) Daisy wheel printers
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : Line printer: Used in large computer system (like main frames) for high speed, bulk printing. It prints one full line at a time

70. CD-ROM is a
CD-ROM है एक
(a) Semiconductor memory
(b) Memory register
(c) Magnetic Memory
(d) None of these/इनमें से कोई नहीं
Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : A CD-ROM (Compact Disc Read Only Memory) is an optical storage medium.
It is read-only meaning data cannot be modified after it is written during manufacturing. Where as magnetic memory, semiconductor memory register it is store the data. using pits and land on the disc surface that read by a laser.

71. The ALU of a computer responds in the commands coming from
Computer का ALU ..... से आने वाले आदेश का जवाब देता है।
(a) Primary memory (b) Control section
(c) External memory (d) Cache memory
Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : The ALU (Arithmetic logic unit) perform arithmetic and logical operations in a computer. However, it does not act on its own- it reposed to commands sent by the control unit (control section).

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The control unit fetches instructions from memory decodes them, and sent the appropriate signals to the ALU and other part of the computer to excute those instructions.

72. The basic operations performed by a computer are
एक Computer द्वारा किए गए मूल संचालन है
(a) Arithmetic operation
(b) Logical operation
(c) Storage and related
(d) All of these/इनमें से सभी

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) :
- Arithmetic operation - addition, subtraction, multiplication, division,.
- Logical operation → Comparisons like, greater than is than, equal to, AND, OR, NOT.
- Storage and related operation → Storing data in memory.

Where as all the option are fundamental operation of a computer.

73. Storage capacity of magnetic disk depends on
चुम्बकीय डिस्क की भंडारण क्षमता निर्भर करती है
(a) Tracks per inch of surface
(b) Bits per inch of tracks
(c) Disk pack in disk surface
(d) All of these/इनमें से सभी

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : The storage capacity of a magnetic disk depends on multiple factors.
- Tracks per inch of surface → More tracks per inch means more concentric circle on the disk so more data can be stored .
- Bits per inch of track→ more bits per inch along a track increases data density also increases storage
- Disk pack in disk surface→ using multiple disk in pack increses total storage.

74. The two kinds of main memory are
दो प्रकार के Main Memory हैं
(a) Primary and Secondary
(b) Random and Sequential
(c) ROM and RAM
(d) All of these/इनमें से सभी

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : ROM and RAM → There are two types of main memory
Since main memory include volatile RAM and non-volatile (ROM) memory.
Main memory is the memory that the CPU directly accesses to read and write data. it is fast, volatile and essential for running programs.

75. What is the primary difference between ROM and RAM?
ROM और RAM के बीच मुख्य अंतर क्या है?
(a) ROM is faster than RAM
ROM, RAM की तुलना में तेज है।
(b) ROM is volatile, while RAM is non-volatile
ROM वोलाटाइल है, जबकि RAM नॉन-वोलाटाइल है।
(c) ROM is used for data storage, while RAM is used for program execution/ROM डाटा स्टोरेज के लिए उपयोग होती है, जबकि RAM प्रोग्राम निष्पादन के लिए।
(d) ROM stores permanent data that cannot be changed, while RAM stores temporary data that can be read and written/ ROM स्थायी डाटा को संग्रहित करता है जिसे बदला नहीं जा सकता, जबकि RAM अस्थायी डाटा को संग्रहित करता है जिसे पढ़ा और लिखा जा सकता है।

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : RAM (Random Access Memory: volatile memory, used for temporary storage during program execution. Data is lost when power is off-
ROM (Read Only Memory) : Non volatile memory, stores permanent instructions (like firmware /BIOS). Data remains even after power off.

76. EPROM stands for:
EPROM का आशय है–
(a) Erased Programmable Read Only Memory
इरेज्ड प्रोग्रामेबल रीड ओनली मेमोरी
(b) Erasable Program Read Only Memory
इरेजेबल प्रोग्राम रीड ओनली मेमोरी
(c) Erasable Programmable Read Only Memory
इरेजेबल प्रोग्रामेबल रीड ओनली मेमोरी
(d) Extensive Program Read Only Memory
एक्सटेन्सिव प्रोग्राम रीड ओनली मेमोरी

HTET PGT C.S. 30.07.2025

Ans (c) : EPROM Stands for Erasable Programmable Read Only Memory. It is a non-volatile memory and it can erased by UV radiation and programmed again.

77. Access to which of the following memories is fastest in a computer system?
(a) RAM (b) Hard disk
(c) Cache (d) CPU registers
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : CPU. Register is the fastest memory of the computer system. It present in CPU. And provide fastest data to the computer. CPU. Register > Cache Memory>Magnetic Disk

78. Full form of DRAM is:
DRAM का पूर्ण रूप है–
(a) Dynamic Random Access Memory
(b) Download Random Access Memory
(c) Dynamic Ransom Access Memory
(d) Different Random Access Memory

HTET PGT C.S. 2023

Ans.(a): Dynamic Random Access Memory
It is a type of volatile memory that stores each bit of data in a separate capacitor with an integrated circuit.

79. All the calculations and decision making operations are performed in which unit?
सभी गणनाएँ एवं निर्णय लेने की प्रक्रियाएँ_____इकाई में की जाती हैं।
(a) Control Unit/कंट्रोल यूनिट
(b) Algorithm Unit/एल्गोरिथ्म यूनिट

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(c) Arithmetic Logical Unit
अर्थमेटिक लॉजिकल यूनिट
(d) Memory Unit/मेमोरी यूनिट

HTET PGT C.S. 2023

Ans. (c) : All calculation and decision making operation within a computer are performed in the Arithmetic Logic Unit ALU is fundamental unit of CPU.

80. Fast and small memory between CPU and main memory whose access time is closer to the processing speed of CPU is:
सीपीयू और मुख्य मेमोरी के बीच तेज और छोटी मेमोरी जिसका एक्सेस समय सीपीयू की प्रोसेसिंग स्पीड के करीब होता है, है–

(a) Erasable Read Only Memory
इरेजेबल रीड ओनली मेमोरी
(b) Random Access Memory/रैंडम एक्सेस मेमोरी
(c) Read Only Memory/रीड ओनली मेमोरी
(d) Cache Memory/कैश मेमोरी

HTET PGT C.S. 2023

Ans.(d): Cache memory is a small fast memory located between CPU and main memory its purpose is to store frequently accessed data and instructions reducing the time the C.P.U spends waiting for data from the slower main memory improving processing speed

81. It is required to refresh _____after a certain time interval.
_____ को निश्चित समय अन्तराल के बाद रिफ्रेश करने की आवश्यकता पड़ती है।

(a) Static RAM/स्टेटिक रैम
(b) Dynamic RAM/डायनेमिक रैम
(c) Magnetic Memory/मैग्नेटिक मेमोरी
(d) Optical Memory/ऑप्टिकल मेमोरी

HTET PGT C.S. 2018

Ans. (b) : Dynamic RAM :– Dynamic volatile memory in computer, storing data in tiny capacitors that need to constant their charge and Prevent data loss.

82. EEPROM stands for :
EEPROM का तात्पर्य है:

(a) Electrically Erasable Programmable Read Only Memory/इलेक्ट्रिकली इरेजेबल प्रोग्रामेबल रीड ओनली मेमोरी
(b) Entry Erasable Programmable Reference Only Memory/एंट्री इरेजेबल प्रोग्रामेबल रेफरेंस ओनली मेमोरी
(c) Extra Internet Participation Read On Memory/एक्स्ट्रा इंटरनेट पार्टिसिपेशन रीड ऑन मेमोरी
(d) Electrical Examination Portable Reference On Medium/इलेक्ट्रिकल एग्जामिनेशन पोर्टेबल रेफरेंस ऑन मीडियम

HTET PGT C.S. 2018

Ans. (a) : EEPROM – Electrically Erasable Programmable Read Only Memory is a type of non – volatile (retains data even when the power is turned off) memory that allows data to be erased and rewritten repeatedly.

83. Which of the following is responsible for extending commands to Arithmetic Logic Unit of a computer ?
निम्नलिखित में से कौन कंप्यूटर की अंकगणितीय लॉजिक यूनिट को कमांड देने के लिए जिम्मेदार है?

(a) Primary memory/प्राथमिक मेमोरी
(b) Control Unit/नियंत्रण इकाई
(c) External Memory/बाह्य मेमोरी
(d) Cache Memory/कैश मेमोरी

HTET PGT C.S. 2020

Ans. (b) : Control Unit :– The part of the (CPU) that controls all the functions of the computer and the functionality of other components is the control unit, often refused to as CU.

84. Which one of the following is an impact printer?
निम्नलिखित में से कौन सा एक इम्पेक्ट प्रिंटर है?

(a) Inkjet printer/इन्कजेट प्रिंटर
(b) Dot-matrix printer/डॉट-मैट्रिक्स प्रिंटर
(c) Laser printer/लेज़र प्रिंटर
(d) Thermal printer/थर्मल प्रिंटर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : Impact printers :- This printer works by physically striking an ink ribbon against the paper to create an image where as non impact printers like inkjet, laser, and thermal printers use heat or other methods to transfer ink without direct contact.
Inkjet printer :- This printer sprays tiny droplets of ink onto the paper, making it a non-impact printer.
Laser printer :- This printer uses a laser beam to create an image on a photosensitive drum, which is then transferred to the paper, making it a non-impact printer.
Thermal Printer :- This printer uses heat to activate color-changing paper, making it a non impact printer.

85. ..... is used by banking industry for pre printing of characters on the cheque.
..... का उपयोग बैंकिंग उद्योग द्वारा चेक पर वर्णों की पूर्व छपाई के लिए किया जाता है।

(a) MICR
(b) OCR
(c) Bar Code Reader/बार कोड रीडर
(d) None of these/इनमें से कोई नहीं

HTET PGT C.S. 2021

Ans. (a) : Magnetic ink character recognition :- It is a technology used in the banking industry to process cheques and other documents quickly and accurately.

86. The part of the CPU that coordinates all the functionality of the computer and other components :
CPU का वह भाग जो कम्प्यूटर और अन्य घटकों की सभी कार्यक्षमता का समन्वय करता है:

(a) Motherboard/मदरबोर्ड
(b) ALU/ए.एल.यू
(c) Control Unit/नियंत्रण इकाई
(d) (a) and (c)/ (1) और (3)

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Ans. (c) : Control unit : The part of the CPU that interprets instruction and coordinates the activities of all other component (ALU, memory, I/O devices).

Main function of control unit
- Instruction decoding.
- Control of data flow
- Generating control singnals.
- Maintains sequence

Type of control unit
- Hardwired control unit
- Microprogrammed control unit

87. ……. transform the information from binary form to human acceptable form.
……. सूचना को द्विआधारी रूप से मानव स्वीकार्य रूप में बदलता है।
(a) Output interfaces/आउटपुट इंटरफेस
(b) Processor/प्रोसेसर
(c) Input interfaces/इनपुट इंटरफेस
(d) None of these/इनमें से कोई नहीं

HTET PGT C.S. 2021

Ans. (a) : An output interface is system that connects a computer's central processing unit (CPU) to external device such as a monitor, printer, or speakers, and central the exchange of data between them it acts as on intermediary between hardware and software managing the flow of data from the CPU to output devices information to be displayed, printed, or listened to.

88. Part of computer which does addition, subtraction, multiplication, division and comparison is:
कम्प्यूटर का वह भाग जो जोड़, घटाव, गुणा, भाग और तुलना करता है:
(a) ALU/ए.एल.यू. (b) Memory/मेमोरी
(c) CPU/सी.पी.यू. (d) Control/नियंत्रण

HTET PGT C.S. 2021

Ans. (a) : An arithmetic logic unit (ALU) is a fundamental digital circuit within a computer central processing unit that performs both arithmetic and logical opration an binary numbers. It's functions as the processor's calculator.

89. 2^{30} bytes and 2^{50} bytes are respectively equal to ..... and ..... .
2^{30} बाइट तथा 2^{50} बाइट क्रमशः.....तथा.....के समान हैं।
(a) 1 Megabyte, 1 Terabyte
(b) 1 Gigabyte, 1 Petabyte
(c) 1 Exabyte, 1 Zettabyte
(d) 1 Terabyte, 1 Yottabyte

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Ans. (b) : 1 Gigabyte, 1 Petabyte

90. VGA stands for :
VGA का तात्पर्य है:
(a) Video Graphics Array/वीडियो ग्राफिक्स ऐरे
(b) Visual Graphical Array/विजुअल ग्राफिकल ऐरे
(c) Versatile Graph Adapter/वर्सेटाइल ग्राफ एडाप्टर
(d) Vast Graphics Adapter/वास्ट ग्राफिक्स एडाप्टर

HTET PGT C.S. 2022

Ans. (a) : Video Graphics Array is an analog interface and connector standard for sending video signals from a computer to a display device such as a monitor or projector.

(iii) Machine Instruction

91. What is the meaning of this arithmetic micro-operation?
$R3 \leftarrow R1 + \overline{R2} + 1$
इस अर्थमेटिक माइक्रो ऑपरेशन का अर्थ क्या है?
$R3 \leftarrow R1 + \overline{R2} + 1$
(a) Content of R1 plus R2 transferred R3/ R1 प्लस R2 की अंतर्वस्तुयें R3 में स्थानांतरित की जाती हैं
(b) Content of R1 plus the 2's complement of R2 transferred to R3/ R1 प्लस R2 के 2's कॉम्प्लिमेंट की अंतर्वस्तुयें R3 में स्थानांतरित की जाती है
(c) Content of R1 plus the 1's complement of R2 transferred to R3/ R1 प्लस R2 के 1's के कॉम्प्लिमेंट की अंतर्वस्तुयें R3 में स्थानांतरित की जाती है
(d) Content of R1 plus R2 after increments transferred to R3/ R1 प्लस R2 की अंतर्वस्तुयें वृद्धि के बाद R3 में स्थानांतरित की जाती है

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Micro operations are the elementary operations carried out on the data stored in the registers. Arithmetic micro-operations is a part of micro-operations.
(1). Add micro operation
$R3 \leftarrow R1 + R2$
The content of R1 is added to R2 and the result is stored in R3.
(2). Subtract micro operation
$R3 \leftarrow R1 - R2$
The content of R2 is subtracted from R1 and the result is stored in R3 but in Binary Number System R1 – R2 is also represented in the form of 2's complement.
$R3 \leftarrow R1 + \overline{R2} + 1$
So, the meaning of this micro operation is: content of R1 plus 2's complement of R2 is transferred to R3.

92. By how many different ways micro-operation encountered in digital computers are classified?
डिजिटल कंप्यूटर में आने वाले माइक्रो-ऑपरेशन को कितने अलग-अलग तरीकों से वर्गीकृत किया जाता है?
(a) One/एक (b) Two/दो
(c) Three/तीन (d) Four/चार

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Ans. (d) In digital computers, micro-operations are the very small steps that the CPU performs on the data stored in registers. They are grouped into 4 main types:
1. **Arithmetic micro-operations:** math actions like add, subtract, increment, or decrement.
2. **Logic micro-operations-:** logical actions like AND, OR, NOT, XOR.
3. **Shift micro-operations:** moving bits left, right, or in a circular way.
4. **Control (or Register-Transfer):** micro-operations - moving data from one register to another or controlling transfers.

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93. **The 4 bit registers R1 and R2 initially have the following values.**
4 बिट रजिस्टर R1 और R2 में शुरू में निम्नलिखित मान होते है।
R1 : 1010
R2 : 1100
What is the content of R1 and R2 after execution of this micro-operation?
इस माइक्रो-ऑपरेशन के निष्पादन के बाद R1 और R2 की अंतर्वस्तुएं क्या है?
$R1 \leftarrow R1 \oplus R2$
$\oplus$ is symbol of Ex-OR.
$\oplus$, Ex-OR का एक प्रतीक है
(a) R1 : 0110, R2 : 1100
(b) R1 : .0110, R2 : 0110
(c) R1: 0110, R2 : 0110
(d) R1 : 1010, R2 : 1100
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Ans. (a) : The XOR Operation compares corresponding bits of the two registers. The result for each bit position is '1' if the bits are different, and '0' if the bits are the same.
Performing the XOR operation:
Bit 3 : R_1 (1) XOR R_2 (1) = 0
Bit 2 : R_1 (0) XOR R_2 (1) = 1
Bit 1 : R_1 (1) XOR R_2 (0) = 1
Bit 0 : R_1 (0) XOR R_2 (0) = 0
Therefore, after the execution of the micro-operation: The content of 'R_1' will be 0110. The content of 'R_2' will remain 1100 as it was not modified by the operation.

94. **The content of the accumulator after the execution of the following 8085 assembly language program, is :**
MVI A, 35H
MOV B, A
STC
CMC
RAR
XRA B
(a) 00H (b) 35H
(c) EFH (d) 2FH
UGC NET C.S. Augest-2016 (Paper-II)

Ans. (d) :
MUI A, 35 H || Content of accumulator = 35 H
MOV B, A || Content of register B = 35 H
STC || set carry flag = 1
CMC || Complement carry flag = hot (1) =0
RAR
XRA B
(00110101) XOR (0001 1010)
= 0010 1111= 2FH

95. **The number of addressable memory units for a 10-bit size address bus is:**
10-बिट आकार वाले एड्रेस बस के संबंध में एड्रेसेबल मेमोरी इकाइयों की संख्या कितनी होती है?
(a) 512 (b) 256
(c) 1023 (d) 1024
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : The number of addressable memory locations is:
Number of memory units = $2^{\text{number of address lines}}$
Hence,
2^{10} = 1024 memory units

96. **Which of the following sequences of one-address instructions can be used to evaluate Y – (A – B)/(C + D ×E)?**
निम्नलिखित में से किन वन-एड्रेस इंस्ट्रक्शन्स के अनुक्रमों का उपयोग Y – (A – B)/(C + D ×E) का मूल्यांकन करने के लिए किया जा सकता है?
(a) LOAD D
MPY E
ADD C
STORE Y
LOAD A
SUB B
DIV Y
STORE Y
(b) LOAD D
MPY E
ADD C
LOAD A
STORE Y
SUB B
DIV Y
STORE Y
(c) LOAD D
MPY E
ADD C
LOAD A
STORE Y
SUB B
DIV Y
STORE Y
(d) LOAD D
MPY E
ADD C
STORE Y
LOAD A
DIV Y
SUB B
STORE Y
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Ans. (a) : LOAD D
MPY E
ADD C
STORE Y
LOAD A
SUB B
DIV Y
STORE Y

97. **Which of the following sequences of three-address instructions can be used to evaluate Y = (A – B)/(C + D × E)?**
निम्नलिखित में से किन थ्री-एड्रेस इंस्ट्रक्शन्स के अनुक्रमों का उपयोग Y = (A – B)/(C + D × E) का मूल्यांकन करने के लिए किया जा सकता है?

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(a) SUB Y, A, B
MPY T, D, E
ADD T, T, C
DIV T, Y, T

(b) SUB Y, A, B
MPY T, D, E
ADD T, T, C
DIV Y, T, T

(c) SUB Y, A, B
ADD T, T, C
MPY T, D, E
DIV Y, Y, T

(d) SUB Y, A, B
MPY T, D, E
ADD T, T, C
DIV Y, Y, T

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Ans. (d) : SUB Y, A, B
MPY T, D, E
ADD T, T, C
DIV Y, Y, T

(iv) Op codes and Operands

98. The time required to wait read/Write head under a desired sector is known as _____ .
रीड/राइट शीर्ष के वांछित सेक्टर के अंतर्गत आने तक प्रतीक्षा करने के लिए आवश्यक समय को _____ के रूप में जाना जाता है।

(a) Seek time/सिक टाईम
(b) Execution time/एक्सीक्यूजन टाईम
(c) Wait time/वेट टाईम
(d) Rotational delay/रोटेशनल डिले

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Ans. (d) : It is the time taken by a desired and come under the read/write head after the head has already reached.
The correct track. The average waiting time for disk is called Rotational delay.

99. Which of the following do not represent memory reference instruction.

A. | 1 | opcode | | **mode = 1, opcode = 010**

B. | 0 | opcode | | **mode = 0, opcode = 111**

C. | 1 | opcode | | **mode = 1, opcode = 111**

D. | 1 | opcode | | **mode = 1, opcode = 110**

E. | 1 | opcode | | **mode = 1, opcode = 001**

Choose the correct answer from the options given below :

(a) B & C only (b) B & D only
(c) A & D only (d) A & E only

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Ans. (b) : Memory Reference Instructions- It is typically involve load, store or other operations that access memory and opcodes define the type of operation being performed.
Step-1. Identify Non-memory Reference Instructions opcodes like 111, 110 and 001 are often used for register-based or immediate operations. If an instruction does not access memory, it should be considered a non-memory reference instructions.
Step-2. Find Incorrect memory references- B(mode = 0, opcode = 111) → likely a register based instructions.
C (mode = 1, opcode = 111) → also likely a non memory instruction.
D (mode = 1, opcode = 110) → could be a non-memory instruction.
Step-3. Choose the correct Answer-
B, C and D seem to be non-memory reference instructions. The correct answer corresponds to option (b) B & D only.

100. How many memory references are to fetch an operand in indirect addressing?
ऑपरेंड को इन्डायरेक्ट एड्रेसिंग में लाने के लिए कितने मेमोरी रेफरेंस की आवश्यकता होती है?

(a) 1 (b) 2
(c) 3 (d) 4

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : In indirect addressing, the address field of the instruction contains the address of a memory location where the effective address (EA) of the operand is stored.
So, two memory references are needed.
(i) First memory reference– To fetch the effective address from memory.
(ii) Second memory reference– To fetch the operand using that effective address.
Example– Suppose the instruction is.
LOAD M
And this uses indirect addressing, where:
- M = 500 (means the address field in instruction is 500)
- Memory location 500 contains value 700 → this is the effective address.
- Memory location 700 contains the operand, say value 25.

Steps:
(A) First Memory Reference– Access memory location 500 to get value 700 (effective address).
(B) Second Memory Reference– Access memory location 700 to get operand 25.
Thus, 2 memory references are required in indirect addressing:
- One to get the effective address.
- One to get the actual operand.

101. When n bits are used to represent an operation (op) code, then how many different operations a computer can perform?
जब एक ऑपरेशन (op) कोड का प्रतिनिधित्व करने के लिए n बिट्स का उपयोग किया जाता है, तो एक कंप्यूटर कितने अलग-अलग ऑपरेशन कर सकता है?

(a) n operations/ n ऑपरेशन
(b) 2 × n operations/ 2 × n ऑपरेशन

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(c) 2^n operations/ 2^n ऑपरेशन
(d) 2/n operations/ 2/n ऑपरेशन

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The number n different operations a computer can perform is determined by the number of bits used to represent the operation (op) code. It follows the formula: 2^n, where 'n' is the number of bits. So if 'n' bits are used the computer can perform 2^n different operations.

102. Instruction format of machine level language consists of OPCODE and_____
मशीन लेवल लैंग्वेज के निर्देश प्रारूप में OPCODE और_____शामिल है।
(a) Operators (b) Operand
(c) Variable (d) Function

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Ans.(b): The instruction format machine level language primarily consists of OPCODE (Operation code) and Operand. The OPCODE specifies the operation to be performed, while the operand provides the data or the address of the data on which the operation is performed.

(v) Instruction Cycle

103. Pipelining improves performance by:
(a) decreasing instruction latency
(b) eliminating data hazards
(c) exploiting instruction level parallelism
(d) decreasing the cache miss rate

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Instruction pipelining is a technique that implements a form of parallelism called instruction level parallelism with a single processor. It therefore allows faster CPU throughput that would otherwise be possible at a given clock rate.

104. Correct the order of instruction cycle:
A. Read the effective address
B. Fetch the information
C. Execute the instruction
D. Decode the instruction
Choose the correct answer from the options given below :
(a) A, B, C, D (b) B, D, A, C
(c) B, A, D, C (d) A, B, D, C

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Ans. (b) : The correct order of the instruction cycle is -
Fetch the information - Retrieve the instruction from memory.
Decode the instruction- The fetched instruction is decoded to understand the operation to be performed.
Read the effective address- Fetch operands if needed (Only for indirect addressing)
Execute the instruction-perform the operation.

105. Move the Read/Write head of disk in or out to position on a correct track is known as _____.
डिस्क के रीड/राइट को एक सही ट्रैक पर पॉजिशन करने के लिए अंदर या बाहर ले जाने को _____ के रुप में जाना जाता है ।
(a) Waiting time/वेटिंग टाइम
(b) Access time /ऐक्सेस टाइम
(c) Seek time /सीक टाइम
(d) Latency time /लैटेनसी टाइम

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Seek time refers to the time it takes for the read/write head of a hard disk drive to physically move from its current position to the specific track where the desired data is located.

106. Consider a 32-bit microprocessor, with a 16 - bit external data bus, driven by an 8 MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate for this microprocessor?
(a) 8×10^6 bytes/sec (b) 4×10^6 bytes/sec
(c) 16×10^6 bytes/sec (d) 4×10^9 bytes/sec

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Since minimum bus cycle duration = 4 clock cycle bus clock = 8 MHZ
then, maximum bus cycle rate = 8 m/4sec = 2m/sec
Data transferred per bus cycle = 16 bit = 2 B
Data transfer rate per second = bus cycle rate × data per bus cycle = 2 M × 2 = 4×10^6 bytes/sec
So, option (b) is correct.

107. The CPU a system having 1 MIPS execution needs 4 machine cycles on an average for executing an instruction. The fifty percent of the cycles use memory bus. A memory read/write employs one machine cycle. For execution of the programs, the system utilizes 90 percent of the CPU time. For block data transfer, an IO device in attached to the system while CPU executes the background programs continuously. What is the maximum I/O data transfer rate if programmed I/O data transfer technique is used?
(a) 500 Kbytes/sec (b) 2.2 Mbytes/sec
(c) 125Kbytes/sec (d) 250 Kbytes/sec

UGC NET C.S. June-2015 (Paper-III)

Ans : (d) CPU speed 1 MPS = 1096 instruction per sec
1 CPU instruction = 4 machine cycle (AVG)
1 memory read/access = 1 machine cycle
CPU utilization = 90%
Programmed I/0
each byte transfer requires 4 cycle (instruction) in status check status
Branch
Read/write data
max data transfer rate

$$\frac{\text{CPU speed}}{4} = \frac{10^6}{4} = \frac{1000000}{4} = \frac{250000}{1000} = 250$$

Kbytes/sec

108. In most general case, the computer needs to process each instruction with the following sequence of steps:
(A) Calculate the effective address
(B) Execute the instruction
(C) Fetch the instruction from memory

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(D) Fetch the operand from memory
(E) Decode the instruction
Choose the correct answer from the options given below :
(a) (A), (B), (C), (D), (E)
(b) (A), (B), (C), (E), (D)
(c) (C), (E), (A), (D), (B)
(d) (C), (E), (D), (A), (B)

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Ans. (c) : The correct sequence of steps in the most general case of processing an instruction by a computer is:
(3) fetch the instruction from memory
(5) Decode the instruction
(1) Calculate the effective address
(4) Fetch the operand from memory
(2) Execute the instruction
Therefore, the correct answer is
(c) - (C), (E), (A), (D), (B).

109. The sequence of events that happen during a typical fetch operation is/are :
A. Memory B. MBR C. MAR
D. PC E. IR
Choose the correct order from the options given below :
(a) A, B, C, D, E (b) D, C, A, B, E
(c) E, D, C, B, A (d) D, C, A, E, B

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Ans. (b) : During a typical fetch operation the sequence of events is a follows.
- The program counter (PC) holds the address of the next instruction to fetched.
- The Memory address register (MAR) receives the address from the PC, indicating the location of the in memory.
- The memory unit retrieves the instruction from the memory location specified by the MAR.
- The fetched instruction is stored in the memory Buffer register.
- The fetched instruction is also loaded into the instruction register (IR) allowing the processor to decode and execute the instruction.

So, option (b) is correct.

110. The correct sequence in fetch-execute cycle is _____
A. Decode
B. Fetch
C. Execute
Chose the correct answer from the following
(a) A-B-C (b) B-C-A
(c) C-B-A (d) B-A-C

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Ans. (d) : The fetch execute cycle, which shows how the CPU can conduct calculations. The stages of the fetch execute cycle are broken into three parts :
1. Fetch 2. Decode 3. Execute
So, option (d) is correct answer.

111. Which of the following statements are correct?
A. The first three-address instruction in the intermediate code is a leader.
B. The instruction which is exactly at the middle of the given three address code is a leader.
C. Any instruction that is the target of a conditional or unconditional jump is a leader.
D. Any instruction that immediately follows a conditional or unconditional jump is a leader.
Choose the correct answer from the options given below :
(a) A, B and C (b) B, C and D
(c) A, C and D (d) A, B, C and D

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Ans. (c) : Basic Block Process– The process of forming a basic block determines the instructions from the intermediate code that are leaders.
Following are the rules used for finding a leader.
- The first three-address instruction of the intermediate code is a leader.
- Instructions that are targets of unconditional or conditional jump/go to statements are leaders.
- Instructions that immediately follow unconditional or conditional jump/go to statements are considered leaders.

So, A, C, D is correct.

112. Which of the following is not a characteristics of reduced instruction set (RISC)?
निम्न में से कौन-सा ''रिड्युस्ड इंस्ट्रक्शन सेट कंप्यूटर (RISC)'' का गुणधर्म नहीं है?
(a) Hardwired control/हार्डवायर्ड कण्ट्रोल
(b) Relatively few addressing modes
तुलनात्मक रूप से एड्रेसिंग मोड्स की कम संख्या
(c) Memory access limited to load and store
केवल लोड व स्टोर तक सिमित मेमोरी एक्सेस
(d) Variable length instruction formats
चर लम्बाई के इंस्ट्रक्शन फॉर्मेट

RPSC C.S. 2016 (PAPER-I)

Ans, (d) :
A. Hardwired control
- RISC architecture used hardwired control units because they are faster and more efficient for simple instructions compared to micro programmed control.
- This is characteristic of RICS.

B. Relatively few addressing modes–
- RISC architecture have a small number of addressing modes (register, immediate, displacement) to simplify the instructions set and improve performance.
- This is characteristic of RISC.

C. Memory access limited to load and store–
- In RISC, memory access is typically restricted to load and store instruction computational instructions (ADD SUB) operate on registers, not directly on memory.
- This is a characteristic of RISC.

D. Variable Length Instruction Formats–
- RISC architectures use fixed-length instruction fetching, decoding, and pipelining variable length instruction formats are a characteristic of CESC architectures not RISC.
- This is not a characteristic of RISC.

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113. Fetch Operands (FO), Fetch Instruction (FI), Execute (E) calculated effective address (EA), Decode Instruction (DI) and Store Result (SR) are steps taken by CPU control to execute an instruction. Which of the following specifies a valid sequence of these steps:
CPU कण्ट्रोल द्वारा एक इंस्ट्रक्शन को एग्जीक्यूट करने के लिए निम्न कार्य किया जाते है, फेच ऑपरेंड (FO), फेच इंस्ट्रक्शन (FI), एग्जीक्यूट (E), इफेक्टिव एड्रेस की गणना (EA), डिकोड इंस्ट्रक्शन (DI) तथा स्टोर रिजल्ट (SR)। निम्न में से कौन सा इन कार्यो के उचित क्रम को दर्शाता है?

(a) EA → FI → DI → FO → E → SR
(b) EA → FO → FI → DI → E → SR
(c) FI → DI → EA → FO → E → SR
(d) DI → EA → FO → FI→ E → SR

RPSC C.S. 2016 (PAPER-I)

Ans, (c) : Evaluating the Options–
A. EA → FI → DI → FO → E → SR
- EA (Calculate Effective Address) because the CPU does not know which instruction or operands to address until it fetches and decodes the instruction. EA typically follows FI and DI.
- The sequence is invalid.

B. EA → FO → FI → DI → E → SR
- EA, which as notes, is premature without fetching and decoding the instruction additionally, FO (Fetch operands) before FI (Fetch Instruction) is illogical since the CPU needs the instruction to know which operands to fetch.
- This sequence is invalid.

C. FI → DI → EA → FO → E → SR
- FI⇒ Fetch the instruction.
- DI : Decode it to understand the operations.
- EA : Calculate the effective address for memory operands (if applicable).
- FO : Fetch the operands.
- E : Execute the operation.
- SR : Store the result.
- This sequence is valid.

4. DI → EA → FO → FI→ E → SR
- DI (Decode Instruction) because the CPU cannot decode an instruction before fetching it (FI). The instruction must be retried from memory first.
- This sequence is invalid.

The valid sequence–
(c) FI → DI → EA → FO → E → SR

114. Considering equal processing time for each segment, speed - up S achieved by a K segment instruction pipeline operating on a straight sequence of N instruction is given by:
एक K सेगमेंट इंस्ट्रक्शन पाइपलाइन एक N इंस्ट्रक्शन की सीधी श्रृंखला पर कार्यरत है। इसके द्वारा प्राप्त स्पीड अप S यदि प्रत्येक सेगमेंट का प्रोसेसिंग टाइम समान हो तो, दिया जा सकता है:

(a) $S = \frac{K(N-K)}{(K+N-1)}$ (b) $S = \frac{K+N}{KN}$
(c) $S = \frac{K(N+1)}{(K+1)N}$ (d) $S = \frac{KN}{(K+N-1)}$

RPSC C.S. 2016 (PAPER-I)

Ans, (d) : Evaluating the given options–

A. $S = \frac{K(N-K)}{(K+N-1)}$

- This suggests the numerator is K (N-K), which implies a subtraction of K2. This does not align with the derived N-K, as N-K accounts for the total non-pipelined time. the numerator K(N-K) would understanding time, and this, formula is incorrect.

B. $S = \frac{K+N}{KN}$

- This formula has a numerator of K + N and a denominator of K-N this does not match the structure of the speedup formula, where the numerator should reflect the non pipelined time (N-K) and the denominator the pipelined time (K+N-1). Additionally for large N, this gives a very small s, which contradicts the expectation that speedup increases with N.
- This is Incorrect.

C. $S = \frac{K(N+1)}{(K+1)N}$

- The numerator K(N+1) and denominator (K+1), N does not simplify to the derived formula testing with values (K = 2, N = 10)

$$S = \frac{2\times(10+1)}{(2+1)\times 10} = \frac{22}{30} = 0.733$$

Speed up should be grater then 1 (since pipelining reduce time).
- This option is incorrect.

D. $S = \frac{KN}{(K+N-1)}$

- $S = \frac{N\times K}{K+N-1}$

It correctly represents the speed up with the numerator N.K (Non-Pipelined time) and denominator K+N-1 (Pipelined times in units of t). Let's verify with
K = 2, N = 10
This is resonal, as pipelining reduces execution time, and speed up approaches K for large N.

Correct Ans = $S = \frac{K.N}{K+N-1}$

115. Which of the following type of Data Hazard in pipelined processor is harmless hazard?
निम्न में से कौन सा पाइप लाइन्ड प्रोसेसर में डाटा हजार्ड का एक रूप है जो नुकसानदेय नहीं है?

(a) Read after Write (RAW)
रीड आफ्टर राईट (RAW)
(b) Read after Read (RAR)
रीड आफ्टर रीड (RAR)
(c) Write after Write (WAW)
राईट आफ्टर राईट (WAW)

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(d) Write after Read (WAR)
राईट आफ्टर रीड (WAR)

RPSC C.S. 2016 (PAPER-I)

Ans, (b) : Data Hazard In Pipeline Processor Method–

Data Hazards– Occur when an instruction depends on the Result of previous instruction and that result of instruction has not yet been computed, whenever two different instructions use the same storage, the location must appear as if it is executed in sequential order.

There are four types of data–

(a) Read after Write (RAW)– it is also known as true dependency or flow dependency it occurs when the value produced by an instructions is required by a subsequent instruction.

ADD R_1……..;
SUB…..R_1…..;
Stalls are required to handle these hazards

(b) Write after Read (WAR)– It is also known anti dependency these hazards occur when the output register of an instruction is used right after read by a previous instruction.

ADD….,R_1….;
SUB,….,….,.....;

(c) Write after Write (WAW)– It is also known as output dependency these hazards occur when the output register of an instruction is used for write after written by previous instruction.

ADD R_1,….,….;
SUB R_1,….,….;

(d) Read after Read (RAR)– It occurs when the instruction both read from the same register.

ADD …,R_1,…;
SUB…,R_1,…;
Since reading a register value does not change the register value, these Read after Read (RAR)hazards do not cause of a problem for the processor.

116. If FD denotes hardware that fetches and decode instructions then, which of the following best defines data path of processor?
यदि वह हार्डवेयर जो इंस्ट्रक्शन को फेच व डिकोड करता है, FD द्वारा दर्शाया जाये तो निम्न में से कौनसी प्रोसेसर के डाटा पाथ की सर्वोत्तम परिभाषा है?

(a) FD + Secondary Memory + Primary Memory
FD + सेकंड्री मेमोरी + प्राइमरी मेमोरी
(b) FD + ALU + Register File
FD + ALU + रजिस्टर फाइल
(c) FD + ALU + CU
(d) FD + ALU + CU + Register File
FD + ALU + CU + रजिस्टर फाइल

RPSC C.S. 2016 (PAPER-I)

Ans, (b) :

(a) FD + Secondary memory + Primary memory– as secondary and primary memory are not part of the processor's data path; they are external storage components.

(b) FD + ALU + Register file– Partially correct, as ALU and Register file are part of the data path, but FD (Fetch and Decode) is not.

(c) FD + ALU + CU– as CU is part of the control path, not the data path, and FD is also not included.

(d) FD + ALU + CU + Register file– as it includes both FD and CU, which are not part of the data path.

117. _____ is not included in an instruction cycle of computer processor.
_____ कंप्यूटर प्रोसेसर के इंस्ट्रक्शन साइकिल का भाग नहीं है?

(a) Fetch and decode instruction
इंस्ट्रक्शन को फेच व डिकोड करना
(b) Handle an interrupt/इंटरप्ट को हैंडल करना
(c) Calculate effective address and fetch data
इफेक्टिव एड्रेस की गणना करना व डाटा फेच करना
(d) Execute instruction and store results/इंस्ट्रक्शन को एक्सीक्यूट करना व परिणामों को स्टोर करना

RPSC C.S. 2016 (PAPER-I)

Ans, (b) :

(a) Fetch and decode instruction– This is a fundamental part of the instruction cycle, as the processor must retrieve and interpret the instruction included.

(b) Handle an interrupt– Handle an interrupt occurs outside the normal instruction cycle. Interrupts are asynchronous events that temporally pause the instruction cycle to execute an interrupt service routine this is not a standard step in the instruction cycle not included.

(c) Calculate effective address and fetch– This is part of the instruction cycle, particularly for instructions requiring memory access, where the processor calculates the memory address and retrieves data included.

(d) Execute instruction and store results– Executing the instruction and storing results (if needed) are core parts of the instructions cycle included.

118. Which of the following is the full form of CISC?
निम्नलिखित में से कौन-सी सी. आई. एस. सी. का पूर्ण रूप है ?

(a) Complex Instruction Sequential Compilation
(b) Complex Instruction Set Computer
(c) Computer Integrated Sequential Compiler
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) : CISC is a type of computer architecture that has complex instructions, where each instruction can perform multiple machine-level operations. The main goal is to make programming easier and reduce memory usage.

119. The technology that stores only the essential instructions on a microprocessors chip and thus enhances its speed is referred to as :
वह प्रौद्योगिकी जो माइक्रोप्रोसेसर चिप पर केवल आवश्यक अनुदेशों का भण्डार करता है और इस प्रकार से गति में वृद्धि करता है, को कहा जाता है:

(a) MIMD (b) CISC
(c) RISC (d) SIMD

KVS PGT C.S. 2017

Sarkari Resource

Ans, (c) : RISC (Reduced Instruction set computer.) is a microprocessor that is designed to perform a smaller number of types of computer instructions so that it can operate at a higher speed (MIPS). Since each instruction type that a computer must perform requires additional transistors and circuitry, a larger list or set of computer instructions tends to make the microprocessor more complicated and slower in operation.

120. In a well-tuned pipeline machine with five-stages, after how many clock cycles one-one instruction rolls out every clock cycle assuming that all the stages take the same amount of time?

(a) 1 (b) 4
(c) 5 (d) 6
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : In a 5-stage pipeline, the first instruction takes 5 cycles to complete. After the pipeline is filled, each subsequent instruction completes every clock cycle. The pipeline fills in number of stages 1 = 5 – 1 = 4 cycles. Hence, after 4 clock cycles, the machine reaches steady state.

121. After how many clock cycles, the second instruction will complete execution in a seven-stage pipeline machine?

(a) 1 (b) 6
(c) 7 (d) 8
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : • 1st instruction completes in 7 cycles
• Each next instruction finishes 1cycle later
• So, 2nd instruction completes in 7 + 1 = 8 cycles

122. The instruction cycle time in a generic microprocessor is:

(a) Longer than the machine cycle time
(b) Shorter than the machine cycle time
(c) Same as the machine cycle time
(d) Double the machine cycle time

CGPSC C.S. 2020 Set-A

Ans.(c): (1) Instruction cycle = the time taken to fetch, decode and execute one instruction
• Machine cycle = the basic operation cycle of the CPU (memory read, memory write, op code fetch, etc)
Normally, an instruction cycle consists of one or more machine cycle
(2) However, in some simplified definitions (especially in competitive exams), the terms instruction cycle and machine cycle are treated as equivalent because
For a simple instruction (e.g. NOP), execution may require only one machine cycle.
[Instruction Cycle Time = Machine Cycle Time]

123. Which table for a dynamic pipeline becomes more interesting when a nonlinear pattern is follows?
एक गतिशील पाइपलाइन के लिए कौन सी तालिका अधिक इन्टेरेस्टिंग हो जाती है, जब एक गैर-रेखीय पैटर्न का अनुसरण किया जाता है?

(a) Reservation Table/ रिजर्वेशन तालिका
(b) Confusion Table/कन्फ्यूजन तालिका
(c) Inverted Table/इनवर्टेड तालिका
(d) Greedy Table/ग्रिडी तालिका

UGC NET C.S. December-2024

Ans. (a) : Non-linear Pipeline pattern:- Non-linear pipeline is a pipeline which is made of different pipelines that are present at different stages. The different pipelines are connected to perform multiple functions. It also has feedback and feed-forward connections. It is made such that it performs various function at different time intervals in non-linear pipeline the functions are dynamically assigned.
Key-points:- In Non-linear pipeline different pipelines are present at different stages.
• Non-linear pipelines is also called as dynamic pipeline as it performs different functions.
• The output is not necessarily produced from the last black.
• Non-Linear pipeline has feedback and feed forward connections.
• It can generate more than one **Reservation table.**
• Functional portioning is difficult in non-linear pipeline.

124. State whether the following statements are true of false.
यह बताएं कि निम्नलिखित में से कौन सा कथन सत्य या असत्य है।
Statemens:
कथन:

(i) SCSI is itself a type of bus used to support local disk drives and other peripherals.
एस.सी.एस.आई. (SCSI) स्वतः एक तरह की बस है, जिसका उपयोग लोकल डिस्क ड्राइव और अन्य पेरिफेरल्स (परिधीय) को सपोर्ट करने के लिए किया जाता है।

(ii) A serial port could be used to support a printer or scanner.
प्रिंटर या स्कैनर सपोर्ट के लिए सीरियल पोर्ट का उपयोग किया जा सकता है।

(a) (i) True, (ii) True/ सत्य, (ii) सत्य
(b) (i) True, (ii) False/ सत्य, (ii) असत्य
(c) (i) False, (ii) True/ असत्य, (ii) सत्य
(d) (i) False, (ii) False/ असत्य, (ii) असत्य

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : • SCSI- Stands for Small Computer System interface it is one such bus standard bus is a system that transfers data between computer components it connects and communicates with local storage devices (like hard drives, CD/DVD drives) and other peripheral.
• Serial port is an interface that sends data one bit at a time over a communication line it is also knows RS-232 port in older computers.
Although printers were commonly connected through parallel ports, some models did use serial ports.

Sarkari Resource

(vi) Organization of Central Processing

125. The mask logical micro operation is equivalent to which logical gate?
मास्क लॉजिकल माइक्रो ऑपरेशन किस लॉजिकल गेट के समतुल्य है?
(a) OR gate/ OR गेट
(b) NOR gate/ NOR गेट
(c) AND gate/ AND गेट
(d) NAND gate/ NAND गेट

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : In the context of computer architecture and micro-operations, "masking" refers to a bitwise operation used to selectively isolate or modify specific bits within a data word (like the content of a register). This is achieved by using a "mask", which is another bit pattern that interacts with the original data through a logical operation.
The 'AND' logical operation is the most common gate used for masking because of its behavior. Therefore, the correct answer is 3 'AND' Gate.

126. In ..... method, the word is written to the block in both the cache and main memory, in parallel.
(a) Write through (b) Write back
(c) Write protected (d) Direct mapping

UGC NET C.S. June-2016 (Paper-III)

Ans : (a) In write through method, data is written into the cache and the corresponding main memory location at the same time.

127. What does the control unit generate to control other units?
अन्य Units को नियन्त्रित करने के लिए Control Unit क्या उत्पन्न करती है?
(a) Timing signals
(b) Command signals
(c) Control signals
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : A computer is an electronic device because it processes, stores, and , manipulates data using electronic circuits and semi conductor devices [like transistors and IC_s)

128. Which component of the computer is used to resolve the difference between CPU and the peripheral device?
सीपीयू तथा पेरिफेरल के बीच विभिन्नता को हल करने के लिए कंप्यूटर के किस कॉम्पोनेन्ट का उपयोग किया जाता है?
(a) Memory unit/मेमोरी यूनिट
(b) Arithmetic Logic Unit/अरिथमेटिक लॉजिक यूनिट
(c) Interface unit/इंटरफेस यूनिट
(d) Control unit/कंट्रोल यूनिट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : An Interface unit is a component of a computer system that converts signal so that computer can communicate with its environment. It acts as an interface between the processor bus and peripheral devices, and supervises and synchronizes all input and output transfers.

129. Which one of the following types of memory is fastest
(a) Cache Memory (b) Register Memory
(c) Main Memory (d) Secondary Memory

UGC NET C.S. December-2022

Ans. (b) : Register memory is the smallest and fastest memory in a computer. It is not a part of the main memory and is located in the CPU in the form of registers, which are the smallest data holding elements. A register temporarily holds frequently used data, instructions and memory addresses that are to be used by the CPU.

130. Consider an unpipelined machine with 10nsec clock cycles which uses four cycles for ALU operations and branches where as five cycles for memory operation. Assume that the relative frequencies of these operations are: 40%, 20% and 40% respectively. Due to clock skew and setup pipeline let us consider that the machine adds one nsec overhead to the clock. How much speedup is observed in the instruction execution rate when a pipelined machine is considered.
(a) 2 times (b) 4 times (c) 6 times (d) 8 times

UGC NET C.S. December-2022

Ans. (b) : Consider the unpipelined machine with 10 ns clock cycles. It

Operation	Frequency	Clock cycles
ALU	40%	4
Branch	20%	4
Memory	40%	5

Total clock cycle = $0.40 \times 4 + 0.20 \times 4 + 0.40 \times 5$
$= 1.6 + 0.8 + 2$
$= 4.4$ clock cycle
Under unpipeline time = $4.4 \times (10) = 44$
under pipeline time
$= 10 + \text{overhead} = 10 + 1 = 11$
So, speedup = $\frac{44}{11} = 4$ times.

131. In a memory hierarchy access time, hit ratio pairs for cache, main and virtual memory are given by (5ns, 80%), (100ns, 99.5%) and (10ms, 100%) respectively. The closest value of average access time of hierarchy is :
एक मेमोरी हैरेरकी में कैश, मेन तथा वर्चुअल मेमोरी के लिए एक्सेस टाइम एवं हिट रेशियों का युग्म क्रमानुसार (5ns, 80%), (100ns, 99.5%) एवं (10ms, 100%) है। मेमोरी हैरेरकी की एवरेज एक्सेस टाइम के समीपतम मान है?
(a) 10032 ns (b) 10024 ns
(c) 10008 ns (d) 10064 ns

RPSC C.S. 2016 (PAPER-I)

Sarkari Resource

Ans, (b) : 3-Level Memory Hierarchy–
• Cache-
Access time = 5ns
Hit ratio = 80% = 0.80
• Main memory (If cache miss)
Access time = 100ns
Hit ratio = 99.5% = 0.995 (of those not found in cache)
• Virtual memory (if both cache and main memory miss)
Access time = 10ms = 10,000,000ns
Hit ratio = 100% (everything is found here if missed in others)
Step-by-Step Calculation of Average Access Time (CAAT)–
Let's denote–
• H_1 = Cache hit ratio = 0.80
• H_2 = Main memory hit ratio = 0.995
• T_1 = Cache access times = 5ns
• T_2 = Main memory access time = 100ns
• T_3 = Virtual memory access time = 10,000,000ns
Formula for Average Access Time–
$AAT = H_1 \cdot T_1 + (1-H_1) \cdot (H_2 \cdot (T_1+T_2) + (1-H_2) \cdot (T_1+T_2+T_3)$
Plug in the values–
AAT = 0.80.5+0.20×[0.995×(5+100)+0.005×(5+100+10,000,000)]
(1) 0.80.5 = 4
(2) 5+100 = 105
(3) 5+100+10,000,000 = 10,000,105
(4) 0.995×105 = 104.475
(5) 0.005×10,000,105 = 50,000.525
(6) Sum inside brackets :
104.475+50,000.525 = 50,105
(7) 0.20×50,105 = 10021
(8) Final AAT = 4+10021 = 10025
Closest answer ⇒ 10024ns

132. How is the TLB typically organized?/TLB का आयोजन सामान्यतः किस प्रकार किया जाता है?
(a) As a direct-mapped cache
प्रत्यक्ष-मैप किए गए कैश के रूप में
(b) As an associative cache
एक सहयोगी कैश के रूप में
(c) As a set-associative cache
सेट-एसोसिएटिव कैश के रूप में
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : "TLB is organized as an associative cache"
The Translation look aside Buffer (TLB) can store virtual-to-physical address mapping in any entry.

133. ____ memory is intended to give memory speed approaching that of the fastest memories available, and at the same time provide a large memory size at the price of less expensive types of semiconductor memories.
____ मेमोरी को उस उपलब्ध शीघ्रातिशीघ्र-मेमोरी गति तक पहुँचने के लिये अभिलक्षित किया जाता है और साथ ही साथ यह कम खर्चीले प्रकार की अर्धचालक मेमोरी की कीमत पर बड़ी मेमोरी साईज प्रदान करता है।
(a) Register/रजिस्टर
(b) Counter/काउन्टर
(c) Flip flop/फ्लिप-फ्लॉप
(d) Cache/कैचे

KVS PGT C.S. 2013

Ans, (d) : Cache memory is intended to give memory speed approaching that of the fastest memories available , and at the Same time provide a large memory size at the price of less expensive types of Semi conductor memories.

134. A Series of statement explaining how the data is to be processed is called.
Data Processing के लिए Statement का समूह को कहते है?
(a) Editing
(b) Compiler
(c) Interpreter
(d) Program

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d): A program is a series of statement or instruction written to perform a specific task or to processes data. It tell the computer how to process the data step by step.

135. If the average access time of CPU is 20 ns, access time of main memory is 110 ns and the cache access time is 10 ns, what is the hit ratio?
(a) 100%
(b) 90%
(c) 80%
(d) 70%

CGPSC C.S. 2020 Set-A

Ans.(b): $AMAT = (h \times T_c) + (1-h) \times T_m$
Where:
• h = hit ratio
• T_c = cache access time = 10 ns
• T_m = main memory access time = 110 ns
• AMAT = average access time = 20 ns
$20 = h \times 10 + (1-h) \times 110$
$20 = 10h + 110 - 110h$
$20 = 110 - 100h$
$100h = 110 - 20 = 90$
$h = \frac{90}{100} = 0.9 = 0.9 \times 100\% = 90\%$

136. In a three level memory hierarchy, the access time of cache, main and virtual memory is 5 nano-seconds, 100 nanoseconds and 10 milli-seconds respectively. If the hit ratio is 80% for the cache and 99.5% for the main memory, then the closest average access time of memory hierarchy in nano-seconds is :
एक तीन स्तरीय मेमोरी अनुक्रम में कैश, मैन तथा वर्चुअल मेमोरी का अभिगम समय (एक्सेस टाइम) क्रमशः 5 नैनो-सेकण्ड, 100 नैनो-सेकण्ड तथा 10 मिली-सेकण्ड है। यदि हिट रेशियो कैश मेमोरी के लिए 80% तथा मेन मेमोरी के लिए 99.5% है, तो मेमोरी अनुक्रम का सर्वाधिक नजदीकी औसत अभिगम समय है (नैनो सेकण्ड में)
(a) 512
(b) 124
(c) 10024
(d) 128

HTET PGT C.S. 2019

Sarkari Resource

Ans. (c) : Cache
Cache = T_1 = 5ns
Hit ratio = P = 0.8
Miss ratio = 1 – p
= 1 – 0.8 = 0.2
Main memory = T_2 = 100ns
Miss ratio = 1 – 0.995 = 0.005
Secondary memory
Hit ratio = 100% = 1
$T_3 = 10 \times 10^{-3} = 10 \times 10^6$ns
$= 10^7$ ns

$$T_{avg} = T_1 + (1-P)T_2 + (1-P)(1-q)T_3$$

$= 5 + (0.2) \times 100 + ((0.005) \times (0.2)) \times 10^7$
$= 5 + 20 + 10000$
$T_{avg} = 10025$
The closest average access time of the memory Hierarchy is 10025 nano-seconds

(vii) ALU, Hardwired and Micro programmed Control Unit

137. The microoperation which divides a signed binary number by 2 is :
(a) Circular shift (b) Logical shift
(c) Arithmetic shift right (d) Arithmetic shift left

UGC NET C.S. December-2023

Ans. (c) : The microoperation that divides a singned binary number by 2 is
Arithmetic shift right.
It shifts all bits to the right.
Presser the sign bit (MSB)
Effectively performs division by 2 for signed numbers.

138. The three main components of a digital computer system are
डिजिटल कम्प्यूटर सिस्टम के तीन मुख्य घटक हैं
(a) Memory/मेमोरी, I/O, DMA
(b) ALU, CPU, Memory/मेमोरी
(c) Memory/मेमोरी, CPU, I/O
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c): CPU (Central Processing Unit) → The brain of the computer, performs processing and control.
• Memory → Store instruction and data.
• I/O (Input/output devices) Used for communication between the computer and the outside world.

139. Which component of computer act as nervous system?
कंप्यूटर का कौन सा कॉम्पोनेन्ट नर्वस सिस्टम के रूप में कार्य करता है?
(a) Arithmetic Logic Unit/अरिथमेटिक लॉजिक यूनिट
(b) Input unit/इनपुट यूनिट
(c) Control unit/कंट्रोल यूनिट
(d) Storage unit/स्टोरेज यूनिट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c): In a computer system the CPU (central processing unit) acts as nervous system. It process instructions. controls the operations of other hardware components, and coordinates the overall functioning of the system, much like how the nervous system process the information and controls various functions in living organisms.

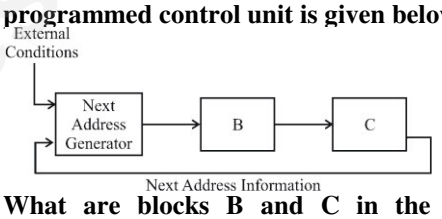
140. Which of the following is not valid with reference to message passing interface (MPI)?
(a) MPI can run on any hardware platform
(b) The programming model is a distributed memory model
(c) All parallelism is implicit
(d) MPI - Comm. - size returns the total number of MPI processes in specified communication

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Message passing interface (MPI) is a standardized and portable message passing system.
- It can run on any hardware platform. It is portable, efficient, flexible.
- It has become a de facto standard for communication among memory system.
- MPI – Comm. – Size returns the total number of MPI processes in specified communication.

So answer is (c)

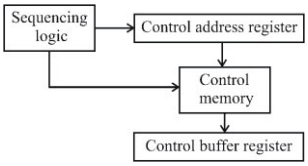
141. The general configuration of the micro programmed control unit is given below :



What are blocks B and C in the diagram respectively?
(a) Block address register and cache memory
(b) Control address register and control memory
(c) Branch register and cache memory
(d) Control address register and random access memory

UGC NET C.S. June-2017 (Paper-III)

Ans : (b)



So Block B is control address register and block C is control memory.

142. Match List-I with List-II:

List-I		List-II	
A.	**Micro operation**	**1.**	**Specify micro operations**
B.	**Micro programmed control unit**	**2.**	**Improve CPU utilization**
C.	**Interrupts**	**3.**	**Control Memory**
D.	**Micro instruction**	**4.**	**Elementary operation performed on data stored in registers**

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Choose the correct option from those given below:

(a) A-4, B-3, C-2, D-1
(b) A-4, B-3, C-1, D-2
(c) A-3, B-4, C-1, D-2
(d) A-3, B-4, C-2, D-1

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (a) :

List-I		List-II	
A.	Micro operation	4.	Elementary operation performed on data stored in registers
B.	Micro programmed control unit	3.	Control Memory
C.	Interrupts	2.	Improve CPU utilization
D.	Micro instruction	1.	Specify micro operations

143. Microprogramming is a technique for:

(a) Programming input or output routines
(b) Progamming the microprocessor
(c) Writing small programs
(d) Programming the control steps of a computer

CGPSC C.S. 2020 Set-A

Ans.(d): Programming the control steps of a computer:– Microprogramming is a method used to implement the control unit of a CPU. It uses a microprogram (stored in control memory) to generate control signals for executing instructions. Each machine instruction is broken down into a sequence of microinstructions that define the low-level steps (e.g., register transfers, ALU operation), effectively "programming" the control flow of the computer.

(viii) General Purpose and Special Purpose Registers Memory Organization

144. While implementing a stack on a register stack, the stack pointer register is
रजिस्टर स्टैक पर किसी स्टैक के कार्यान्वयन में स्टैक पॉइंटर रजिस्टर

(a) Incremented first during push operation
पुश ऑपरेशन के दौरान पहले अंक ह्रास (decrement) करता है
(b) Decremented first during push operation
पुश ऑपरेशन के दौरान पहले अंक ह्रास (decrement) करता है
(c) Incremented first during pop operation
पॉप ऑपरेशन के दौरान पहले अंक वृद्धि (increment) करता है
(d) Decremented first during pop operation
पॉप ऑपरेशन के दौरान पहले अंक ह्रास (decrement) करता है।

LT. Grade C.S. 2018

Ans, (b) : In a register based stack the stack pointer (SP) typically points to the top of the stack. During the push operation, you first decrement the stack pointer (since the stack grows towards lower memory addresses), and then you store the value at the memory location pointed to by the stack pointer. Similarly, for the pop operation you first access the value at the location pointed to by the stack pointer and then increment the stack pointer.

145. In IA-32 architecture along with the general flags, which of the following conditional flags is provided?
IA-32 वास्तुकला में, सामान्य झंडे के साथ, निम्नलिखित में से कौन-सा सशर्त झंडा प्रदान किया जाता है?

(a) TF
(b) IOPL
(c) IF
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (d) : IA – 32 (Intel architecture 32-bit), also known as X86 Architecture. It is a widely used computer architecture for 32-bit processors. It consist of a set of general registers and flags. Some common flags used in the IA– 32 architecture are:
i. Zero Flag (ZF)
ii. Carry Flag (CF)
iii. Trap Flag (TF)
iv. Interrupt Flag (IF)
v. Overflow Flag (OF)

146. In various devices, _____ are used to overcome the difference in data transfer speed.
विभिन्न उपकरणों में, डेटा ट्रांसफर गति में अन्तर को दूर करने के लिए _____ का उपयोग किया जाता है।

(a) Multiple buses
(b) Buffer registers
(c) Speed-enhancing circuitries
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : A buffer Register (often called an I/O buffer register or just buffer) is a temporary storage area inside the CPU or I/O module used to hold data being transfer between two devices that work at different speeds (for examples CPU and memory, or CPU and I/O devices)

147. The register that contains the address of the next instruction to be executed is called
वह रजिस्टर, जिसमें अगला इन्सट्रक्शन जो एक्जिक्यूट होने वाला होता है उसका ऐड्रेस होता है, कहलाता है

(a) Program counter/प्रोग्राम काउन्टर
(b) Instruction register/इन्सट्रक्शन रजिस्टर
(c) Accumulator/एक्यूमुलेटर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a): The program counter is a register that keeps track of the address of the next instruction to be executed by the CPU. It ensures that instruction are executed in correct sequence.

Sarkari Resource

148. The register that contains the instruction that is being executed is called
एक्जिक्यूट हो रहे इन्सट्रक्शन को रखने वाले रजिस्टर को कहा जाता है
(a) Accumulator/एक्यूमुलेटर
(b) Program counter/प्रोग्राम काउन्टर
(c) Instruction register/इन्सट्रक्शन रजिस्टर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c): The instruction register is the CPU register that hold the current instruction being executed, which is then decoded and carried out by the control unit The instruction is first stored in the Instruction Register (IR). The Control Unit Then decodes it do understand that the CPU must add the content of R_1 and R_2

149. The register which holds the address of location of memory to and from which data are to be transferred is known as
वह रजिस्टर, जिसमें मेमोरी के स्थान का पता होता है जहाँ से या जहाँ तक डेटा स्थानांतरित किया जाना होता है, कहलाता है
(a) Index register/इन्डेक्स रजिस्टर
(b) Memory address register
मेमोरी एड्रेस रजिस्टर
(c) Memory buffer register/मेमोरी बफर रजिस्टर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): The register which hold the address of the memory location from or to which data is to be transferred is called the memory address Register (MAR)
- Index Register → used for modifying address during indexed addressing
- Memory Buffer Register → Hold the actual data being transferred to/from memory
- MAR → Hold the address of the required memory location.

150. The 8 bit register A and B initially have the following values.
A = 01110110
B = 10000101
What is the 8–bit value in each register after execution of the following micro–operation?
A ← A + B
B ← B + 1
8–बिट रजिस्टर A तथा B के शुरू में निम्नलिखित मान हैं।
A = 01110110
B = 10000101
दिए गए माइक्रो-ऑपरेशन के निष्पादन के बाद प्रत्येक रजिस्टर में 8 बिट के मान क्या हैं?
A ← A + B
B ← B + 1
(a) A = 11111110, B = 10000101
(b) A = 11111011, B = 10001110
(c) A = 11111011, B = 10000110
(d) A = 11111101, B = 10000101

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Initial values A = 01110110
B = 10000101
Operation A = A + B this operation performs a left array Binary addition of the value A & B = 01110110(A) + 10000101 (B) = 11111 (New A)
After this operation value of A 11111011.
The value of register B remains unchanged.
Operation B = B + 1 operation increments the original value of register B by one = 10000101 (B) + 00000001 = 10000110 NewB)
After this operation value of B 10000110

151. What is the function of the program counter register?
प्रोग्राम काउंटर रजिस्टर का क्या कार्य होता है?
(a) It holds the operand address
यह संकार्य (आपरेंड) का पता (एड्रेस) रखता है
(b) It holds the address of current instruction/ यह वर्तमान अनुदेश (इंस्ट्रक्शन) का पता (एड्रेस) रखता है।
(c) It holds the address of previous instruction
यह पिछले अनुदेश (इंस्ट्रक्शन) का पता (एड्रेस) रखता है।
(d) It holds the address of next instruction/यह अगले अनुदेश (इंस्ट्रक्शन) का पता (एड्रेस) रखता है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : The program counter (PC) register is a crucial component in a processor. It keeps track of the memory address of the next instruction to be executed from memory.

152. How many 16 K × 1 bit RAM chips are needed to provide a memory capacity of 128 K × 1 byte?
128 K × 1 बाइट्स की मेमोरी क्षमता प्रदान करने के लिए कितने 16 K × 1 बिट RAM चिप्स की आवश्यकता है?
(a) 8 (b) 12
(c) 64 (d) 128

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) :
RAM chips capacity = 16 K × 1 =16 × 1024 × 1 bits
Memory capacity = 128 × 1024 × 8 bits

$$\text{No of chips needed} = \frac{\text{Memory capacity}}{\text{Ram chips capacity}}$$

$$\text{No of chips needed} = \frac{128\times1024\times8}{16\times1024\times1}$$
$$= 64$$

153. In CPU, which of the following type of register is required by the control unit and contains the instruction currently being executed?
(a) Instruction register
(b) Accumulator
(c) Program register
(d) Memory data register

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Sarkari Resource

Ans. (a) : The Instruction Register (IR) is a crucial component within the CPU's control unit that holds the instruction currently being executed or decoded. After an instruction is fetched from memory, it is loaded into the IR for processing by the control unit.

154. A byte addressable computer has a memory capacity of 2^m K bytes and can perform 2^n operations. An instruction involving 3 operands and one operator needs a maximum of

(a) 3m bits (b) m + n bits
(c) 3m + n bits (d) 3m + n + 30 bits

UGC NET C.S. June-2014 (Paper-III)

Ans. (d) : Because, Memory capacity is 2^m Kbytes = 2^{m+10} bytes. Whole memory can be represented by M + 10 bits.2^n operations can be represented by n bits.
Now instruction of from
Instruction OP_1, OP_2, OP_3 will take (3m + 30 + n) bits.
So the answer is (d).

155. In the indexed scheme of blocks to a file, the maximum possible size of the file depends on :

(a) The number of blocks used for index and the size of index
(b) Size of Block and size of Address
(c) Size of index
(d) Size of Block

UGC NET C.S. June-2015 (Paper-III)

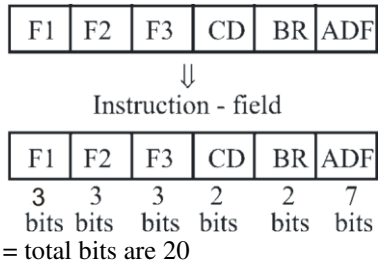
Ans:(b) Number of pointers that can be stored in the index block – which depends on the size of the index block and the size of each address. Size of each data block – because each pointer points to a block, so the larger the block size, the more data each pointer represents.

156. A micro-instruction format has micro-ops field which is divided into three subfields F1, F2, F3 each having seven distinct micro-operations, condition field CD for four status bits, branch field BR having four options used in conjunction with address field ADF. The address space is of 128 memory locations. The size of micro-instruction is :

(a) 17 bits (b) 20 bits
(c) 24 bits (d) 32 bits

UGC NET C.S. November-2017 (Paper-III)

Ans : (b) Microprocessor instruction format, which is divided into three subfields F_1, F_2, F_3 each having seven distinct micro-operation condition field CD for four status bits, branch fields BR having four options used in conjunction with address field ADF. The address space is of 128 memory locations i.e.



= total bits are 20

157. The register that stores the bits required to mask the interrupts is

(a) Status register
(b) Interrupt service register
(c) Interrupt mask register
(d) Interrupt request register

UGC NET C.S. August-2016 (Paper-III)

Ans : (c) Interrupt Mask Register (IMR): An Interrupt mask register is a specialized register in a computer's CPU or a dedicated peripheral used to enable or disable (mask) individual interrupt lines.

158. Consider the following assembly language instructions :
mov al, 15
mov ah, 15
xor al, al
mov cl, 3
shr ax, cl
add al, 90H
adc ah, 0
What is the value in ax register after execution of above instructions?

(a) 0270H (b) 01070H
(c) 01E0H (d) 0370H

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) We know that Ax is a register containing al, ah.
MOV al, 15 → AX = al ah 0F H;]
MOV ah, 15 → AX = 0F0FH;
XOR al, al → AX = 0F00H;
MOV cl, 3 → AX = 0F00H; cl = 03
SHR ax, c1 → AX = 01E0H; shift right 3 times
Add al, q ◦ h → AX = 0170H;
ADC ah, 0 → AX = 0270H; cy = 1 (carry)
ADC = add carry
So output will be AX = 0270H

159. The contents of register (BL) and register (AL) of 8085 microprocessor are 49H and 3AH respectively. The contents of Al, the status of carry flag (CF) and sign flag (SF) after executing 'SUB AL, BL' assembly language instruction, are

(a) AL = 0FH; CF = 1; SF = 1
(b) AL = F0H; CF = 0; SF = 0
(c) AL = F1H; CF = 1; SF = 1
(d) AL = 1FH; CF = 1; SF = 1

UGC NET C.S. June-2017 (Paper-II)

Ans : (c)
(AL) ⇒ 3AH ⇒ 00111010
(BL) ⇒ 49H ⇒ 01001001
11110001
(AL) ⇒ F1H, (CF) ⇒ 1, (SF) ⇒ 1
The carry flag is set since the first operand is less than the second operand. Since the result produce the negative result sign flag is set.

Sarkari Resource

160. Consider the following program fragment in assembly language :

```
mov ax, 0h
mov cx, 0A, h
do loop :
dec ax
loop doloop
```

What is the value of ax and cx registers after the completion of the doloop?

(a) ax = FFF5 h and cx = 0 h
(b) ax = FFF6 h and cx = 0 h
(c) ax = FFF7 h and cx = 0A h
(d) ax = FFF5 and cx = 0A h

UGC NET C.S. November-2017 (Paper-II)

Ans : (b)

dec ax =

```
  1 1 1 1 1 1 1 1
  Ø Ø Ø Ø Ø Ø Ø Ø
               -1
  ---------------
  1 1 1 1 1 1 1 1
```

now, loop do loop

```
cx =  0 0 0 0 1 0 1 0     ax =  1 1 1 1 1 1 1 1
                 -1                        -1
      ---------------           ---------------
(1)   0 0 0 0 1 0 1 0           1 1 1 1 1 1 1 0
cx =  0 0 0 0 1 0 0 1     ax =  1 1 1 1 1 1 1 0
                 -1                        -1
      ---------------           ---------------
(2)   0 0 0 0 1 0 0 0           1 1 1 1 1 1 0 1
```

now, do as same for 10 time & in the last step is.

```
cx =  0 0 0 0 0 0 0 1
                 -1
      ---------------
      0 0 0 0 0 0 0 0
```

cx = 0h so the loop stops here
ax = 11110110
i.e. F6h
Which could be written as FFF6 h

161. Consider the following assembly program fragment :

```
stc
mov al, 110101110b
mov cl, 2
rcl al, 3
rol al, 4
shr al, cl
mul cl
```

The contents of the destination register ax (in hexadecimal) and the status of Carry Flag (CF) after the execution of above instructions, are :

(a) ax = 003CH; CF = 0
(b) ax = 001EH; CF = 0
(c) ax = 007BH; CF = 1
(d) ax = 00B7H; CF = 1

UGC NET C.S. November-2017 (Paper-II)

Ans : (a) If you multiply any binary number by 2, Simply add a 0 bit to its L.S.B. (Right End).
Here, d = 2, so we are multiplying the binary number is al by 2.
Hence, al = 000111100
Now, it becomes 0000000000111100 = 003CH
CF = 0
CF = 0, if high byte/word/double word = 0
CF = 1, if high byte/word/double word < > 0.
Hence, the correct answer is ax = 003 CH; CF = 0.

162. Consider the following statement regarding special purpose registers.

A. Program Counter (PC) keeps the track of next instruction executed.
B. Instruction register holds the address of first instruction to be executed.
C. Accumulator holds the output of ALU.
D. Program Counter (PC) keeps the track of only first instruction of the program.

Choose the correct answer from the options given below :

(a) (A) and (C) Only (b) (A) and (D) Only
(c) (B) and (D) Only (d) (A), (B), and (C) Only

UGC NET C.S. Re-Exam- June 2024

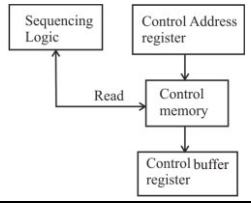
Ans. (a) : (A) Program Counter (PC) keeps the track of the next instruction executed. This is correct. The program Counter holds the address of the Next instruction that is to be executed.
(B) Instruction register holds the address of the first instruction to be executed. This is incorrect. The Instruction Register holds the actual instruction that is currently being executed, not the address.
(C) Accumulator holds the output of ALU. This is correct. The Accumulator is a register that typically holds the result of arithmetic and logical operations performed by the Arithmetic logical Unit.
(D) Program Counter keeps the track of only the first instruction of the program, This is incorrect.
The program counter continuously updates to point to the next instruction of the program, not just the first instruction.

163. The fault can be easily diagnosed in the micro-program control unit using diagnostic tools by maintaining the contents of

(a) Flags and counters
(b) Registers and counters
(c) Flags and registers
(d) Flags, registers and counters

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : Flag registers and counters: Within the computer, a control unit manages the computer's resources and orchestrated the performance of its functional parts in response to the instructions.
Microprogrammed control unit operates by executing instructions that define the functionality of the control unit. In this, set of micro instructions is stored in the control memory. The control address register contains the address of the next micro instruction to be read. When a micro instruction is read from the control memory, it is transferred to a control buffer register. Next element is sequencing unit that loads the control address register and issues a read command.



Sarkari Resource

164. Consider single accumulator (SA), general register (GR) and stack (S) is type of CPU organization. Which of the following specifies valid number of address fields in a computational type of instruction of each organization?
CPU आर्गेनाइजेशन के विभिन्न प्रकारों जैसे सिंगल अक्यूमुलेटर (SA), जनरल रजिस्टर (GR) तथा स्टैक (S) पर विचार करे। निम्न में से कौन प्रत्येक आर्गेनाइजेशन के एक गणना सम्बन्धी इंस्ट्रक्शन में एड्रेस फील्ड की उचित संख्या दर्शाता है?

(a) SA - 1, GR - 2 or 3, S - 0
(b) SA - 2 or 3, GR - 1, S - 0
(c) SA - 2 or 3, GR - 1, S - 1
(d) SA - 0, GR - 1, S - 2 or 3

RPSC C.S. 2016 (PAPER-I)

Ans, (a) : Single Accumulator (SA) Organization–
• In a single accumulator CPU, there is one dedicated register (the accumulator) used to most arithmetic and logical operations.
• A computational instruction (ADD, SUB) typically operators requires one address field to specify the memory location of the operand. The accumulator is implied as the second operand and destination.
• Number of Address fields: 1
2. General Register (GR) Organization–
• In GR, instructions typically involve registers for operands, and sometimes a memory location.
• A common computational instruction (ADD) may specify two registers (source and destination) and possibly a memory address or another register.
• For a typical instruction like ADD R_1, R_2, R_3 ($R_1 = R_2 + R_3$)
• Number of Address fields : 2 (or 3 in some cases, but 2 is more standard for two operand instructions).
3. Stack (S) Organization–
• In a stack based CPU, operands are implicitly taken from the top of the stack, and results are pushed back on to the stack.
• Computational instructions ADD do not required address field because operands are the top fields.
• Number of address fields = 0
SA = 1
GR = 2
S = 0

165. Which one of the following is used to read the control words sequentially from the control memory?
निम्न में से किसका कन्ट्रोल मेमोरी से कन्ट्रोल वर्ड को क्रमबद्ध पढ़ने के लिए प्रयोग किया जाता है?

(a) Microprogram address register
माइक्रोप्रोग्राम एड्रेस रजिस्टर
(b) Microprogram counter/माइक्रोप्रोग्राम काउन्टर
(c) Control memory address register
कन्ट्रोल मेमोरी एड्रेस रजिस्टर
(d) Program counter/प्रोग्राम काउन्टर

LT. Grade C.S. 2018

Ans, (b) : A micro program counter (MPC) is a register that keeps track of the current microinstruction being executed in a micro programmed control unit. The MPC is used to read control words sequentially from the control memory.

166. Shifting a register content to left by one bit is equivalent to _____.
रजिस्टर कन्टैन्ट को एक बिट बाएँ खिसकाने का अर्थ है:

(a) Davison by 2 /2 से भाग करना
(b) Addition by 2/2 जोड़ना
(c) Multiplication by 2/2 से गुणा करना
(d) Subtraction by 2/2 घटाना

KVS PGT C.S. 2018

Ans, (c) : Shifting a register content to left by one bit is equivalent to multiplication by 2. An shifting register content to right by one bit is equivalent to division by 2.

167. The operation executed on data stored in registers is called _____.
रजिस्टरों में स्टोर किए गए डाटा पर ऑपरेशन निष्पादित करना क्या कहलाता है?

(a) Macro-operation/मेक्रो ऑपरेशन
(b) Micro operation/माइक्रो ऑपरेशन
(c) Bit-operation/बिट ऑपरेशन
(d) Byte operation/बाइट ऑपरेशन

KVS PGT C.S. 2018

Ans, (b) : The operation executed on data stored in registers in called Micro operation. Micro operations are the functional or atomic, operations of a processor, These are low level instructions used in some designs to implement complex machine instructions.

168. A register in the microprocessor that keeps track of the answer or result of any arithmetic or logic operation is the _____.
माइक्रोप्रोसेसर में वह रजिस्टर जो किसी अंकगणितीय या तार्किक प्रचालन के उत्तर या परिणाम के पथ की रखता है उसे _____कहा जाता है।

(a) Accumulator/एक्युमुलेटर
(b) Stack pointer/स्टैक प्वाइंटर
(c) Program counter/प्रोग्राम काउण्टर
(d) Instruction pointer/अनुदेश प्वाइंटर

KVS PGT C.S. 2017

Ans, (a) : An accumulator is a type of register for short-term intermediate storage of arithmetic and logic data in a computer's central processing unit (CPU). However, the term is rarely used in reference to contemporary CPUs. having been replaced around the turn of the millennium by the term register. It keeps track of result of any arithmetic or logic operation.

169. Which type of register is primarily used to temporarily hold data during arithmetic and logical operations in a microprocessor?
माइक्रोप्रोसेसर में अंकगणितीय और तार्किक क्रियाओं के दौरान डेटा को अस्थायी रूप से रखने के लिए किस प्रकार के रजिस्टर का मुख्य रूप से उपयोग किया जाता है?

(a) Shift register (b) Parallel register
(c) Accumulator register (d) Counter register

Bihar STET C.S. 12.09.2023 (Shift-I)

Sarkari Resource

Ans. (c) : In a microprocessor, the Accumulator register is mainly used to temporarily hold data during arithmetic and logical operation.
The ALU (Arithmetic logic unit) perform operation, and the intermediated or final result is store in the Accumulator.

170. A stack pointer is:
(a) A 16-bit register in the microprocessor that indicates the beginning of the stack memory
(b) A register that decodes and executes 16-bit arithmetic operation
(c) The first memory address where a subroutine address is stored
(d) A register in which flag bits are stored

CGPSC C.S. 2020 Set-A

Ans.(a): A stack pointer is a special purpose register (after 16 bit in microprocessors like the 8085) that holds the memory address of the top the stack, indicating where the next data will be pushed or popped from.

171. The minimum time delay that is required between the initiation of two successive memory operations is called:
(a) Memory access time (b) Transmission time
(c) Seek time (d) Memory cycle

CGPSC C.S. 2020 Set-A

Ans.(d): Memory cycle time is defined as the minimum time interval required between the ignition of two successive memory operation in same memory unit. It includes the memory access time and additional recovery time needed before the next operation can begin.

172. If an error is detected, a part of the memory can be erased in:
(a) PROM (b) EPROM
(c) EAROM (d) EROM

CGPSC C.S. 2020 Set-A

Ans.(c): EAROM (Electrically Alterable Read Only Memory) is a type of non-volatile memory that individual words or bytes to electrically Erased and rewritten. EAROM offers more granular, selective electrical erasure and reprogramming.

(ix) I/O Organization

173. The method of accessing the I/O devices by repeatedly checking the status flags is
बार-बार स्थिति झंडे की जाँच करके I/O उपकरणों तक पहुँचने की विधि _____ है।
(a) Program-controlled I/O/प्रोग्राम-नियंत्रित I/O
(b) Memory-mapped I/O/मेमोरी-मैप किया गया I/O
(c) I/O-mapped I/O/ I/O मैप किया गया
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : The method of accessing I/O devices by repeatedly checking status flags is called Program-controlled I/O (Polling).
In this method, the CPU continuously monitors the status register of the I/O device. If the device is ready, then the CPU performs data transfer, otherwise, it keeps checking again and again.
This wastes CPU time because it stays busy in checking instead of doing other tasks. Hence, the correct answer is (a).
Program-controlled I/O.

174. In memory mapped I/O
मेमोरी-मैप्ड में किए गए I/O
(a) The I/O devices and the memory share the same address space/ I/O डिवाइस और मेमोरी एक ही पते को साझा करते हैं
(b) The I/O devices have a separate address space/ I/O उपकरणों का एक अलग पता होता है
(c) The memory and I/O devices\ have an associated address space/मेमोरी और I/O उपकरणों में एक सम्बद्ध पता स्थान होता है
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : • In Memory Mapped I/O, the I/O devices and memory share the same address apace
• If I/O devices had a seprate address pace that would be called Isolated I/o (or port Mapped I/O).
Hence, standard instruction like LOAD and STORE can directly access I/O devices.

175. What is the function of data input command in I/O command?
I/O कमांड में डेटा इनपुट कमांड का क्या कार्य करता है?
(a) Activate the peripheral and to inform it what to do/पेरिफेरल को सक्रिय करना तथा उसे यह सूचित करना कि क्या करना है
(b) To check status conditions in the interface and the peripheral/इंटरफेस तथा पेरिफेरल में स्टेटस परिस्थितियों की जाँच करना
(c) It cause the interface to respond by transferring data from bus into one of its registers/इंटरफेस को बस से डेटा को उसके किसी एक रजिस्टर में स्थानांतरित करके प्रतिक्रिया (रिस्पॉन्ड) देने के लिए कहना
(d) It cause the interface to respond by transferring data from one of its register to bus/इंटरफेस को अपने किसी रजिस्टर से बस में डेटा स्थानांतरित करके प्रतिक्रिया (रिस्पॉन्ड) देने के लिए कहना

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In Computer science I/O command the interface to respond by transferring data from one of its registor to bus.
• Control command.
• States command.
• Data input command.
• Data output command.

176. When Bus Grant (BG) = 1 in direct memory access (DMA) transfer, then what function read line and write lines perform?
जब डायरेक्ट मेमोरी एक्सेस (DMA) ट्रांसफर में बस ग्रांट (BG) = 1 होता है, तो रीड लाइन तथा राइट लाइन फंक्शन क्या कार्य करते हैं?

Sarkari Resource

(a) Read line and write line act as input line allowing CPU to communicate with internal DMA register/रीड लाइन तथा राइट लाइन इनपुट लाइन के रूप में कार्य करते हुए सीपीयू को आंतरिक डीएमए रजिस्टर के साथ संचार करने की अनुमति देते हैं

(b) Read line act as input line and write line act as output line from DMA/डीएमए से रीड लाइन इनपुट लाइन के रूप में कार्य करती है तथा राइट लाइन आउटपुट लाइन के रूप में कार्य करती है

(c) Read line act as output line and write line act as input line from DMA controller/डीएमए कंट्रोलर से रीड लाइन आउटपुट लाइन के रूप में कार्य करती है तथा राइट लाइन इनपुट लाइन के रूप में कार्य करती है

(d) Read line and write line act as output lines from DMA controller/रीड लाइन तथा राइट लाइन डीएमए कंट्रोलर से आउटपुट लाइट के रूप में कार्य करती है

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : When Bus Grant (BG) = 1
in direct memory access (DMA) transfer, the read line and write lines act as output lines from DMA controller.

177. Which of the following data transfer schemes requires constant monitoring by the CPU of the peripheral devices?
निम्नलिखित डेटा स्थानांतरण योजनाओं में से किसे परिधीय उपकरणों के सी.पी.यू. (CPU) द्वारा निरंतर निगरानी की आवश्यकता होती है?

(a) Programmed IO/ प्रोग्राम्ड (क्रमादेशित) आई.ओ.

(b) Interrupt-driven IO इंटरप्ट-ड्राइवेन आई.ओ.

(c) Direct Memory Access (DMA)
डायरेक्ट मेमोरी एक्सेस (DMA)

(d) Both, Interrupt-driven IO and (DMA)
दोनों, इंटरप्ट-ड्राइवेन आई.ओ. (IO) और डी.एम.ए. (DMA)

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Programmed IO:- The CPU Continuously monitors (polls) the peripheral device to check if it is ready to send or receive data. This requires constant CPU attention, which can be inefficient.

178. In Direct Memory Access (DMA) _____, a block sequence consisting of a number of memory words is transferred in a continuous burst.

(a) Bus granting (b) Burst transfer
(c) Cycle stealing (d) Controller count

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : In Direct Memory Access (DMA), "burst transfer" is a mode where a block of data, consisting of a number of memory words, is transferred in a continuous burst. Once the DMA controller gains control of the system bus, it transfers the entire block of data before releasing control back to the CPU, thus optimizing the transfer of large data blocks.

179. A DMA controller transfers 32-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much will the CPU be slowed down because of the DMA Transfer?

(a) 0.06% (b) 0.12%
(c) 1.2% (d) 2.5%

UGC NET C.S. December-2015 (Paper-III)

Ans : (b) 1 Character = 1 byte
32 bit word = 4 byte
4800 bytes in 1 sec
so for 1 byte it takes 1/4800 sec
so for 4 byte it takes 1/1200 sec
so 1200 words are transferred in 1 sec through cycle stealing & gives 1 million instructions are fetching & executing in 1 sec out of which 1200 words are executing through cycle stealing so
% of slow down or cycle wasted in dma transfer = (1200/1000000) * 100 = 0.12%

180. Which of the following is correct statement?

(a) In memory-mapped I/O, the CPU can manipulate I/O data residing in interface registers that are not used to manipulate memory words

(b) The isolated I/O method isolates memory and I/O addresses so that memory address range is not affected by interface address assignment

(c) In asynchronous serial transmission of data the two units share a common clock

(d) In synchronous serial transmission of data two units have different clocks

UGC NET C.S. November-2017 (Paper-III)

Ans : (b) The Isolated I/O method isolates memory & I/O addresses so that memory address range is not affected by interface address assignment.

- Memory based I/O uses same address space for memory & I/O devices
- In asynchronous serial transfer of data the two units do not share a common clock.
- In synchronous serial transfer of data the two units share a common clock.

181. Arrange the given steps required for a Direct memory Access (DMA) transfer in the correct order.

(A) Initiate DMA transfer request.

(B) Transfer data directly between peripheral and memory.

(C) Processor grants DMA Control over the system bus.

(D) DMA controller completes data transfer and signals completion.

Choose the correct answer from the options given below :

(a) (C), (A), (B), (D) (b) (A), (C), (B), (D)
(c) (A), (B), (C), (D) (d) (C), (B), (A), (D)

UGC NET C.S. Re-Exam- June 2024

Sarkari Resource

Ans. (b) : The correct order of steps for a direct memory Access (DMA) transfer is:

(A) Initiate DMA transfer request: The processor initiates a DMA transfer request by sending a signal to the DMA controller.

(C) Processor grants DMA control over the system bus: The processor grants the DMA controller control over the system bus, allowing it to manage data transfer.

(B) Transfer data directly between the peripheral to memory: The DMA controller transfer data directly between the peripheral device and memory, without involving the processor.

(D) DMA controller completes data transfer and signals completion: The DMA controller completes the data transfer and signals the processor that the transfer is complete.

This order makes sense because the processor needs to initiate the transfer request, them grant the DMA controller control over the system bus. The DMA controller can then perform the data transfer, and finally signal completion to the processor.

182. The interface (S) that provide (S) I/O transfer of data directly to and from the memory unit peripheral is/are termed as:

(A) DMA (Direct Memory Access)
(B) IOP (Input- Output Processor)
(C) Serial Interface
(D) Parallel Interface

Choose the correct answer from the options given below :

(a) (A) Only (b) (B) Only
(c) (A) and (B) Only (d) (C) and (D) Only

UGC NET C.S. December-2023

Ans. (c) : DMA (Direct Memory Access) is an interface that provides I/0 transfer of data directly to and from the memory unit and the peripheral. 'IOP (Input-Output Process) is a specialized processed designed to manage input and output operations in a Computer system. It handles tasks such as data transfer between peripheral devices and the CPU or memory.

So, the interface that provide I/0 transfer of data directly to and from the memory unit peripheral are termed as. DMA and IOP.

183. Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O ?

(a) I/O protection is ensured by operating system routines.
(b) I/O protection is ensured by a hardware trap.
(c) I/O protection is ensured during system configuration.
(d) I/O protection is not possible.

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : User applications are not allowed to perform I/O in user mode – All I/O requests are handled through system calls that must be performed in kernel mode.

I/O protection is ensured by operating system routines.

184. Match list I with List II

	List I		List II
A.	**Handshaking**	**I.**	**I/O interface inform the CPU that device is ready for transfer**
B.	**Programmed I/O**	**II.**	**requires two control signals working in opposite directions**
C	**Interrupt initiated I/O**	**III.**	**has local memory & control large set of I/O devices**
D	**I/O processor**	**IV.**	**require CPU to check the I/O flag & perform transfer**

Choose the correct answer from the options given below:

(a) A-I, B-II, C-III, D-IV
(b) A-II, B-IV, C-III, D-I
(c) A-II, B-IV, C-I, D-III
(d) A-IV, B-III, C-II, D-I

UGC NET C.S. June-2020 (Paper-II)

Ans. (c) : Hand shaking:- When one computer sends a message to another indicating that it wants to create a contact channel, it is called Handshaking. It requires two control signals working in opposite directions.

Programmed I/O:- The output of the I/O Instructions written in the computer programme is the trigger. It requires the CPU to check the I/O flag & perform the transfer.

Interrupt - Initiated I/O:- Interrupt powered I/O is a different way to deal with I/O. Interrupt I/O is a method of monitoring Input/output operation by sending a signal from a peripheral or terminal that needs to send or receive data. I/O interface informs the CPU that the device is ready for transfer.

I/O processor:- A processor with direct memory access is known as an input output processor (IOP). The computer system is divided into memory units and processors in this way. I/O processor has local memory & control a large set of I/O devices.

185. Match List-I and List-II

List-I		List-II	
A.	**Isolated I/O**	**1.**	**same set of control signal for I/O and memory communication**
B.	**Memory mapped I/O**	**2.**	**separate instructions for I/O and memory communication**

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C.	**I/O interface**	**3.**	**requires control signals to be transmitted between the communication units**
D.	**Asynchronous data transfer**	**4.**	**resolve the differences in central computer and peripherals**

Choose the correct option from those given below:

(a) A-2, B-3, C-4, D-1
(b) A-1, B-2, C-3, D-4
(c) A-2, B-1, C-4, D-3
(d) A-1, B-2, C-4, D-3

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) :

	List-I		List-II
A.	Isolated I/O	2.	separate instructions for I/O and memory communication
B.	Memory mapped I/O	1.	same set of control signal for I/O and memory communication
C.	I/O interface	4.	resolve the differences in central computer and peripherals
D.	Asynchronous data transfer	3.	requires control signals to be transmitted between the communication units

186. Choose a true statement for memory-mapped I/O system:
मेमोरी मैप्ड I/O व्यवस्था के लिए सत्य कथन चुनें:

(a) It uses two separate buses for memory and I/O.
इसमें मेमोरी तथा I/O के लिए दो अलग अलग बसों का प्रयोग होता है

(b) It uses one common bus for memory and I/O but has separate control line of each./इसमें मेमोरी तथा I/O के लिए एक ही बस परन्तु अलग अलग कण्ट्रोल लाइन का प्रयोग होता है

(c) It uses common bus for memory and I/O with common control lines./इसमें मेमोरी तथा I/O के लिए एक ही बस एवं एक समान कण्ट्रोल लाइन का प्रयोग होता है

(d) It uses distinct instruction for I/O transfer and memory transfer operations./इसमें I/O ट्रान्सफर तथा मेमोरी ट्रान्सफर के लिए पृथक इंस्ट्रक्शन का प्रयोग होता है

RPSC C.S. 2016 (PAPER-I)

Ans, (c) :
(1) **Memory–** mapped I/O does not use separate buses for memory and I/O it uses a single, shared bus.
• It is incorrect value.
(2) While a common bus is used, there are no separate control lines for memory and I/O in memory mapped I/O the same control lines are used for both.
• It is incorrect value.
(3) In a memory– mapped I/O system, I/O devices and memory share the same address space and use a common lines (read/write signals) are used for both memory and I/O operations. The CPU treats I/O devices as if they were memory locations, and no separate instructions are required for I/O transfers.
• It's correct value.
(4) In memory– mapped I/O, the same instructions such as (load/store) are used for both memory and I/O operations not distinct instructions.

187. Which of the following is not a valid bus arbitration policy used to decide which device may use the bus at a given point of time?
निम्न में से कौन सी, एक दिए हुए समय पर कौन सी डिवाइस बस का प्रयोग करेगी यह निर्धारित करने वाली बस आर्बिट्रेशन पालिसी नहीं है?

(a) Daisy chain method/डेजी चैन मेथड
(b) Independent request and grant method
इंडिपैडेंट रिक्वेस्ट तथा ग्रांट मेथड
(c) Polling Method/पोलिंग मेथड
(d) DMA control method/डी.एम.ए. कण्ट्रोल मेथड

RPSC C.S. 2016 (PAPER-I)

Ans, (d) : Valid Bus Arbitration Methods–
A. Daisy Chain Method–
• Devices are connected in series
• Priority is fixed the devices closer to the controller gets preference.
• Oldest and simplest method.
B. Independent Request and Grant–
• Each device sends its own request signal to a bus controller.
• Controller grants access based on a priority logic.
• More flexible than daisy chaining.
C. Polling Method–
• The controller asks each device one by one if it needs the bus.
• Devices are checked one by one in a specific order.
• Simple to implement but can be slower.
D. DMA Control Method–
• Not a bus arbitration policy
• DMA (Direct memory Access) is a data transfer mechanism not an arbitration method.
• It allows peripherals (like a hard drive or sound card) to transfer data directly to from memory without CPU involvement.
• PMA control method is not it's a way of transferring data, not arbitrating bus access.
Correct answer (d).

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188. The major functions or requirements for an I/O module fall into which of the following categories?
किसी I/O मॉड्यूल के लिए मुख्य कार्य फंक्शन या आवश्यकताएँ निम्न में से किस कोटि के अन्तर्गत आती हैं ?
(a) Control and timing./नियंत्रण और समय
(b) Processor communication./प्रोसेसर संचार
(c) Data buffering./डेटा बफरिंग
(d) All of these/यह सभी

KVS PGT C.S. 2013

Ans, (d) : major requirements for an I/O module
1. Processor communication
2. Control and timing
3. Data buffering
4. Device communication
5. Error detection

189. An interface that provides I/O transfer of data directly to and from the memory unit and the peripheral, is termed as _____.
वह इंटरफेस, जो कि मेमोरी यूनिट तथा पेरिफेरल को सीधे तौर पर डाटा का इनपुट/आउटपुट ट्रांसफर करने की सुविधा प्रदान करता है, क्या कहलाता है?
(a) DDA/डी.डी.ए.
(b) Serial Interface/सीरियल इंटरफेस
(c) BR/बी.आर.
(d) DMA/डी.एम.ए.

KVS PGT C.S. 2018

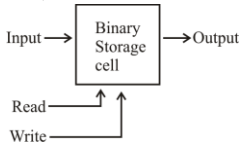
Ans, (d) : DMA (Direct Memory Access) controller is a hardware device that allows I/O devices to directly access memory with less participation of the processor. DMA controller needs the same old circuits of an interface to communicate with the CPU and Input/output devices.

190. The I/O subsystem in an operating system is responsible for:
ऑपरेटिंग सिस्टम में I/O सबसिस्टम का कार्य है:
(a) Allocating memory for processes
प्रोसेसेस के लिए मेमोरी अलॉट करना
(b) Managing the file system
फाइल सिस्टम का प्रबंधन करना
(c) Managing input/output operations between processes and input/output devices
प्रोसेसेस और इनपुट/आउटपुट डिवाइसेस के बीच इनपुट/आउटपुट ऑपरेशन्स को प्रबंधित करना
(d) Allocating CPU time to process
प्रोसेसेस को CPU समय आवंटित करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : • Handing input and output requests
• Acting as an interface between processes and I/O devices (Key board, mouse, printer, disk, etc).
• Providing device driver and buffering
• Managing device communication and error handing.

191. The following diagram depicts a __ cell.
निम्न आकृति एक __सेल को दर्शाती है ।



(a) storage/संचायक
(b) mobile/मोबाइल
(c) memory/मेमोरी
(d) register/रजिस्टर

KVS PGT C.S. 2013

Ans, (c) : The following diagram depicts a memory cell.
Computer memory is the storage space in the Computer. and we perform read and write operations.

192. Which of the following devices is used to connect a peripheral to a bus?
(a) Control register
(b) Interface
(c) Communication protocol
(d) None of the there

CGPSC C.S. 2020 Set-A

Ans.(b): An interface is device or circuit that allows a peripheral device to communicate with the computer's bus it handles the necessary signal conversions and protocols. to ensure proper data transfer between the peripheral and the main system.

193. SATA stands for :
SATA का तात्पर्य है:
(a) Serial Active Technology Association
सीरिअल एक्टिव टेक्नोलॉजी ऐसोसिएशन
(b) Simple Algorithm Technique Association
सिपंल एल्गोरिथम टेक्निक ऐसोसिएशन
(c) Serial Advanced Technology Attachment
सिरिअल एडवांस्ड टेक्नोलॉजी अटैचमेंट
(d) Simple Active Technique Attachment
सिंपल एक्टिव टेक्निक अटैचमेंट

HTET PGT C.S. 2019

Ans. (c) : SATA - serial advanced technology Attachment.
It is computer bus interface used to connect storage devices like hard drives and solid state drives to a computer motherboard.

(x) Functioning of CPU

194. What do we use to extend the connectivity of the processor bus?
प्रोसेसर बस की कनेक्टिविटी बढ़ाने के लिए हम किसका उपयोग करते हैं?
(a) PCI bus
(b) Multiple bus
(c) SCSI bus
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : In computer science PCI Bus is define as Peripheral component Inter connect Bus.
The PCI bus is a high speed parlled computer bus used to connect peripheral devices (like network cards, sound cards, modems, storage controller graphics card, etc.)

195. The number of lines used in a data bus is referred to as the _____ of the data bus.
डेटा बस में उपयोग की जाने वाली लाइनों की संख्या को डेटा बस के/की _____ के रूप में जाना जाता है।

Sarkari Resource

(a) length/लंबाई (b) density/घनत्व
(c) width/चौड़ाई (d) volatility/अस्थिरता

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : In computer architecture, the data bus width indicates how many bits can be transferred simultaneously. For example, a 32 bit bus can transfer 32 bits of data in one operation. The "width" of the data bus determines the speed and capacity of data transfer.103.With respect to system buses, SCST stands for:

196. सिस्टम बस के संबंध में, एस.सी.एस.आई. (SCSI) का अर्थ क्या है?

(a) Small Computer Side Interface
स्मॉल कंप्यूटर साइड इंटरफेस
(b) Small Computer System Interface
स्मॉल कंप्यूटर सिस्टम इंटरफेस
(c) Solid Computer System Interface
सॉलिड कंप्यूटर सिस्टम इंटरफेस
(d) Synchronous Computer System Interface
सिंक्रोनस कंप्यूटर सिस्टम इंटरफेस

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : SCSI- Small computer system Interface-
• It is used to boost performance, transfer received data faster and provide extensive expansion for machines such as CD-ROM drivers DVD drives and CD Writers.
• It is the controller responsible for transmitting data over the small computer system Interface bus and between computers.

197. With respect to system bus. _____ are use to control the access to and the use of the data and address lines.
सिस्टम बस के संबंध में, _____ का उपयोग डेटा और एड्रेस लाइन्स का एक्सेस करने और उपयोग को नियंत्रित करने के लिए किया जाता है।

(a) Address lines/एड्रेस लाइन्स
(b) Decoders/डिकोडर्स
(c) Encoders/इनकोर्डस
(d) Control lines/कंट्रोल लाइन्स

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Control Bus or (Control line) – Carries Control signals that manage the access and use of the data and address lines.
These signals ensure proper communication and synchronization between different components.
Therefore control lines are used to control the access to and the use of the and address lines.

198. Which of the following determines the time it takes for devices to coordinate the use of the bus?
निम्नलिखित में से कौन सा बस के उपयोग को समन्वयित करने वाले डिवाइस के लिए लगने वाले समय को निर्धारित करता है?

(a) Propagation delay/प्रोपेगेशन डिले
(b) Clock delay/क्लॉक डिले
(c) Cycle delay/साइकल डिले
(d) Data transfer delay/डेटा ट्रांसफर डिले

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Propagation delay is the time it takes for a signal to travel form one point to another within a system. Including the time for the signal to travel through wires and electronic components.

199. With respect to cache organisation, direct mapping maps each block of main memory into _____ possible cache line.
कैश ऑर्गनाइजेशन के संबंध में, डायरेक्ट मैपिंग _____ संभावित कैश लाइन में मेन मेमोरी (मुख्य स्मृति) के प्रत्येक ब्लॉक को मैप करता है।

(a) Only one/केवल एक (b) Two/दो
(c) Four/चार (d) Five/पाँच

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In direct-mapped caches, each block of main memory is mapped to exactly one possible cache.

200. Consider a uniprocessor system where new processes arrive at an average of five processes per minute and each process needs an average of 6 seconds of service time. What will be the CPU utilization?

(a) 80% (b) 50%
(c) 60% (d) 30%

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because, Given in system there are on average 5 new process arrive pertime.
So, arrival rate = 5 process / min
In every 12 second a $\left(\frac{60}{5}\right)$ new process arrive is system or we can say every process stays 12 second with the CPU.
Give service time = 6 seconds

$$\text{CPU utilization} = \frac{\text{Total Service Time Required}}{\text{Total Available Time}} \times 100$$

$$= \frac{6}{12} \times 100 = 50\%$$

Hence the right option is (b).

201. Assume that a program will experience 200 failures in infinite time. It has now experienced 100 failures. The initial failure intensity was 20 failures/CPU hr. Then the current failure intensity will be

(a) 5 failures/CPU hr.
(b) 10 failures/CPU hr.
(c) 20 failures/CPU hr.
(d) 40 failures/CPU hr.

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because, $V_0 = 200$ failures
$\mu = 100$ failures
$\lambda_0 = 20$ failures / CPU hr.
Current failures intensity $= \lambda_0 (1 - \mu/V_0)$
$= 20 (1 - 100/200) = 20 (1 - 0.5)$
$\Rightarrow 10$ / failures / CPU hr
and the answer is (b).

202. For switching from a CPU user mode to the supervisor mode following type of interrupt is most appropriate :

Sarkari Resource

(a) Internal interrupts (b) External interrupts
(c) Software interrupts (d) None of the above

UGC NET C.S. December-2014 (Paper-III)

Ans : (c) Because, internal interrupts occur due to some internal error (like division of zero)
External interrupts occur due H/w problem
Software interrupts are the most appropriate method for switching user mode to kernel mode.

203. A CPU handles interrupt by executing interrupt service subroutine .....

(a) By checking interrupt register after execution of each instruction
(b) By checking interrupt register at the end of the fetch cycle
(c) Whenever an interrupt is registered
(d) By checking interrupt register at regular time interval

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) A CPU handles interrupt by executing interrupt service subroutine by checking the interrupt register after finishing the executing of the current instruction.

204. Which of the following statement is/are True?

A. The processor has direct access to both primary and secondary memory.
B. Primary memory stores the active instructions and data for the program being executed on the process
C. Secondary memory is used as a backup memory
D. Memory system is implemented on a single level memory

Choose the correct answer from the options given below :

(a) A and B only (b) B and C only
(c) A, B and C only (d) A and D only

UGC NET C.S. December-2022

Ans. (b) : The processor has direct access to both primary and secondary memory is not entirely accurate. While the processor can directly access primary memory (RAM), it typically does not have direct access to secondary memory.

- Memory system is implemented on a single level memory is incorrect. Memory systems are typically implemented with multiple levels of memory hierarchy.

So correct Answer is option (b)

205. Which of the following statement is/are false?

(A) The processor has direct access to both primary and secondary memory
(B) Primary memory stores the active instructions and data for the program being executed on the process
(C) Secondary memory is used as a backup memory
(D) Memory system is implemented on a single level memory

Choose the correct answer from the options given below :

(a) A and B only (b) B and C only
(c) A, B and C only (d) A and D Only

UGC NET C.S. December-2022

Ans. (d) : Primary memory can be accessed by the data bus where as secondary memory is accessed by I/O channels.
Primary memory data is directly accessed by the processing unit. Secondary memory data cannot be directly accessed by the processing unit.
In memory system
Level O : CPU registers
Level 1 : Cache memory
Level 2 : Main memory or primary memory
Level 3 : Magnetic disks or secondary memory
Level 4 : Optical disks or magnetic tape tertiary memory.
So option (d) is correct.

206. Which of the following is NOT a type of interrupt?
निम्नलिखित में से कौन-सा इंटरप्ट का प्रकार नहीं है?

(a) Hardware interrupt/हार्डवेयर इंटरप्ट
(b) Memory interrupt/मेमोरी इंटरप्ट
(c) Software interrupt/सॉफ्टवेयर इंटरप्ट
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : A memory interrupt is not a standard type of interrupt in computer system where are two type interrupt in computer system
- Hardware interrupts
- Software interrupts

207. Switching between a supervisor mode and a user mode for a processor is usually done by a
प्रोसेसर में सुपरवाइजर मोड तथा यूजर मोड के बीच स्विचिंग निम्न में से किसके द्वारा होती है?

(a) Subroutine call/सबरूटिन कॉल
(b) Branch instruction/ब्रांच इंस्ट्रक्शन
(c) Software interrupt/सॉफ्टवेयर इन्टरप्ट
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans, (c) : Switching between supervisor mode (also known as kernel mode) and user mode is typically done using a software interrupt, also known as a trap or exception. Software interrupts can be used to trigger a context switch, which involves saving the current state of the processor and switching to a different mode.

208. Interrupts are sent by _____ and received by _____.
इंटरप्ट्स _____ भेजे जाते और _____ द्वारा प्राप्त किए जाते हैं।

(a) CPU, I/O Devices
सीपीयू (CPU), आई/ओ (I/O) उपकरण
(b) I/O Devices, CPU
आई/ओ (I/O) उपकरण, सीपीयू (CPU)
(c) Timer, I/O Devices
टाइमर, आई/ओ (I/O) उपकरण
(d) CPU, ALU/सीपीयू (CPU), ए एल यू (ALU)

NVS PGT C.S. 15-.12.2022 (Shift-II)

Sarkari Resource

Ans: (b) : Interrupts are signals sent by hardware devices (like keyboards, mouse, network cards etc.) to the CPU to get its attention when they need processing. These signals are sent to interrupt controller and interrupt controller then sends the interrupt signal to the CPU. The CPU receives the interrupt and pauses its current tasks to handle the interrupt via an interrupt handler or interrupt service routine(ISR).

209. Hardware mechanism that enables a device to notify the CPU is called:
वह हार्डवेयर मैकेनिज्म, जो कि किसी यंत्र को CPU को सूचित करने में सक्षम बनाती है, _____ है।

(a) busy-waiting (b) interrupt
(c) polling (d) DMA

KVS PGT C.S. 2018

Ans, (b) : Hardware mechanism that enables a device to notify the CPU is called interrupt The CPU hardware has a wire called the interrupt request line that the CPU senses after executing every instruction.

210. Which of the following is a type of interrupt?
निम्न में से कौन सा इन्टरप्ट का प्रकार है?

(a) External interrupt/बाह्य इन्टरप्ट
(b) Internal interrupt/आंतरिक इन्टरप्ट
(c) Software interrupt/सॉफ्टवेयर इन्टरप्ट
(d) All of the above/उपरोक्त सभी

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : Interrupts can be classified as external (from input/output devices), internal (from CPU) e.g. divide by zero or software (triggered by instruction).

211. An exceptional condition in a computer system caused by an event external to the CPU is known as
कम्प्यूटर सिस्टम में एक विशेष दशा, जो कि सी.पी.यू. के बाहर से घटित होने के कारण आती है, उसे कहते हैं

(a) Halt /हाल्ट
(b) Process /प्रक्रिया
(c) Interrupt / इन्टरप्ट
(d) None of these /उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : An Interrupt is a signal from a external source from Input /Output device that requests. The CPU to pause its current task and handle the event.

212. When a subroutine is called, the address of the instruction following the CALL is stored in the:

(a) Stack (b) Accumulator
(c) Program counter (d) Stack pointer

CGPSC C.S. 2020 Set-A

Ans.(a): When a subroutine is called the return address (the address of the instruction immediately following the CALL instruction) is saved on the stack to ensure that program execution can return to the correct point after the subroutine completes

213. Data path consists of:

(a) Register (b) ALU
(c) BUS (d) All of the above

CGPSC C.S. 2020 Set-A

Ans.(d): All of path in a computer's CPU is responsible for performing data processing operation. It typically indulges the ALU for Arithmetic logical operations, registers for temporary storage of data and internal buses to facilitate the movement of data between these components.

214. Processor needs software interrupt to:

(a) Return from subroutine
(b) Implement as routines
(c) Test the interrupt system of the processor
(d) Obtain system services which need execution of privileged instructions

CGPSC C.S. 2020 Set-A

Ans.(d): Obtain system services which need execution of privileged instructions.
Software interrupts (also known as traps or exceptions) are mechanism wed by user-level programs to request services from the operating kernel.

215. Software interrupt is:

(a) Used to stimulate an external device
(b) Generated by an external device
(c) Both (A) and (B)
(d) None of there

CGPSC C.S. 2020 Set-A

Ans.(a): Software interrupt is generated by a program instruction (like INT in X86). It is used to simulate the effect of a hardware interrupt without requiring an external device.

216. Which speed up could be achieved according to Amdahl's law of infinite number of processes if 5% of a program is sequential and the remaining part is ideally parallel?
यदि किसी प्रोगाम का 5% अनुक्रमिक है और शेष भाग आदर्शतः समानांतर है, तो एम्डल के अनंत प्रक्रियाओं के नियम के अनुसार कौन सी गति वृद्धि प्राप्त की जा सकती है?

(a) Infinite/अनंत (b) 5
(c) 20 (d) 50

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Sequential part – 5%
Parallel part (P) = 1 – Sequential part
= 1 – (5 ÷ 100)
= 1 – 0.05
= 0.95 or 95%
According to
Amdahl's Law,

$$S = \frac{1}{1-0.95} = 20$$

Hence, the value of speed up is 20.

(xi) Instruction Formats

217. _____ is CPU independent machine language style instruction set into which NET Framework program are compiled.
_____ एक CPU इंडिपेंडेंट मशीन भाषा शैली इंस्ट्रक्शन सेट है जिसमें NET फ्रेमवर्क प्रोग्राम कमपाईल किया जाता है?

(a) Machine language/मशीन लैंग्वेज़
(b) High level language/हाई लेवल लैंग्वेज़

Sarkari Resource

(c) Intermediate language/इंटरमेडिएट लैंग्वेज़
(d) Assembly language/ऐसैम्बली लैंग्वेज़

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : A program is a series of statement or instruction written to perform a specific talks or to processes data. It ell the computer how to process the data step by step.

218. Pseudo instruction is given to_____
स्यूडो इंस्ट्रक्शन _____ को जाता है।
(a) compiler/कंपाइलर
(b) linker/लिंकर
(c) assembler/एशेम्बलर
(d) loader/लोडर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : A pseudo- instruction, also know as an assembler directive, is not a real machine instruction instead, it is a compound or instruction specifically for the assembler program.

219. State whether the following statements are true or false.
यह बताएं कि निम्नलिखित में से कौन सा कथन सत्य या असत्य है।
(i) With one-address instructions, the programmer generally has available only one general-purpose register, the accumulator./ वन-एड्रेस इंस्ट्रक्शन्स के लिए प्रोग्रामर के पास आम तौर पर केवल एक जनरल परपज रजिस्टर, एक्यूम्यूलेटर होता है।
(ii) Zero-address instructions are applicable to a special memory organization called a stack.
जीरो-एड्रेस इंस्ट्रक्शन्स एक स्पेशल मेमोरी ऑर्गनाइजेशन (विशेष स्मृति संगठन) के लिए लागू होते हैं, जिसे स्टैक कहा जाता है।
(a) (i) True, (ii) True/ (i) सत्य (ii) सत्य
(b) (i) True, (ii) False/ (i) सत्य (ii) असत्य
(c) (i) False, (ii) True/ (i) असत्य (ii) सत्य
(d) (i) False, (ii) False/ (i) असत्य (ii) असत्य

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : With one-address instructions, the programmer generally has available only one general-purpose register, the accumulator
Zero-address instructions are applicable to a special memory organization called a stack.
So,
Both statements are true.

220. Which of the following statements is/are NOT CORRECT about NUMA?
(A) LOAD and STORE instructions are used to access remote memory.
(B) There is a single address space visible to all CPU.
(C) Access to local memory is slower than access to remote memory.
(D) When the access time to remote memory is hidden, the system is called NC - NUMA.
(E) In CC - NUMA, Coherent caches are present.
Choose the correct answer from the options given below :
(a) (A) and (C) Only (b) (B) and (D) Only
(c) (A) and (E) Only (d) (C) and (D) Only

UGC NET C.S. December-2023

Ans. (d) : Non-Uniform Memory Access (NUMA) is a computer memory design used in multiprocessing. where the memory access time depends or the memory location relative to the processor. under NUMA, a processor can access its own local memory faster than non-local memory. In NC-NUMA systems, remote memory access times are not hidden by cache mechanisms. So, if the access time to remote memory is not hidden, the system is called NC- NUMA.
Thus, the statements (c) and (d) are not correct.

(xii) Instruction Types

221. Which of the following is characteristic of a RISC (Reduced Instruction Set Computer) processor?
(a) Microprogrammed control
(b) Instruction that manipulate operands in memory
(c) Fixed-length, easily decoded instruction format
(d) A large number (typically form 100 to 250) of instructions

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : A characteristic of a RISC (Reduced Instruction Set Computer) processor is its use of a fixed-length, easily decoded instruction format. This simplifies instruction decoding and allows for efficient pipelining.

222. A CPU has a 5-stage pipeline with the following stages Fetch (F), Decode (D), Execute (E), Memory (M) and Write-back (W). Each stage takes one clock cycle to complete. Assume there are no pipeline stalls and the pipeline is initially empty. How many clock cycles are required to complete the execution of 10 instructions?
(a) 10 (b) 14
(c) 15 (d) 19

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : In a 5 stage pipeline (Fetch, Decode, Execute, Memory, Write back), each instruction takes 5 clock cycles to complete once it enters the pipeline, multiple instructions are in different stages at the same time after the initial delay.
Steps to calculate–1. Initial Filling of the pipeline, the first instruction completes after 5 cycles, as each stage takes 1 clock cycle and there are 5 stages.
2. Subsequent Instructions, Each subsequent instruction completes after 1 additional clock cycle once the pipeline is filled because a new instruction enters the pipeline in every cycle.
Total clock cycles–
Total cycles = 5 (for the first instruction) + 9 (for the remaining instructions) = 14
Therefore, 14 clock cycles are required to complete the execution of 10 instructions.

Sarkari Resource

(xiii) Addressing Modes

223. The ..... addressing mode is similar to register indirect addressing mode, except that an offset is added to the contents of the register. The offset and register are specified in the instruction.

(a) Base indexed
(b) Base indexed plus displacement
(c) Indexed
(d) Displacement

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) The displacement addressing mode is similar to register indirect addressing mode, except that an offset is added to the contents of the register. The offset and register are specified in the instruction.
Based Indexed Addressing: The operand's offset is sum of the content of a base register BX or BP and an index register SI or DI.
Based Indexed plus displacement addressing mode: In this mode of addressing the operand's offset is given by offset=[BX or BP]+[SI or DI]+8 bit or 16 bit displacement. Indexed addressing mode: The operand's offset is the sum of the content of an index register SI or DI and an 8 bit or 16 bit displacement.
So, option (D) is correct.

224. How many bits are required to distinguish between a direct and indirect address?
प्रत्यक्ष और अप्रत्यक्ष के बीच अंतर करने के लिए कितने विट्स की आवश्यकता होती है?

(a) 1 (b) 2
(c) 3 (d) 4

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : To distinguish between direct and Indirect addressing, we only need 1 bit (called the indirect bit/addressing mode bit).
- If the bit = 0 → direct addressing
- If the bit = 1 → indirect addressing

225. The most common addressing technique employed by a CPU is
CPU द्वारा नियोजित सबसे आम ऐड्रेसिंग तकनीक है

(a) Direct
(b) Indirect
(c) Immediate
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (d) : CPU Addressing Techniques CPU, employ various addressing techniques to access data. some common techniques include.
- Direct Addressing: The instruction contains the direct memory address.
- Indirect Addressing: The instruction contain an address that points to the actual data address.
- Immediate Addressing: The data itself is part of the instruction.

226. What is the addressing called that is used in an instruction of the form ADD X Y?
उस संबोधन को क्या कहा जाता है, जिसका उपयोग ADD X Y फॉर्म के निर्देश में किया जाता है?

(a) Indirect
(b) Index
(c) Absolute
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : Absolute addressing mode: In this addressing mode, The address of location of
The operand is given explicitly as a part of the instruction. This type of addressing is also knows as direct addressing mode. Hence, ADD X,Y represents direct addressing mode.

227. FFFF will be the last memory location in a memory of size
कितने साइज के मेमोरी में आखिरी मेमोरी लोकेशन FFFF होगा?

(a) 16K
(b) 32K
(c) 64 K
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c): The last memory location given a FFFF This is in Hexa decimal (base 16)
In decimal, FFFF = $(16^4 - 1) = 65535$
So, the addresses range is from OOOOH to FFFFH
That means the total number of addressable location = 65536 = 64K
(since 1K = 1024 location).

228. In which of the following type of addressing mode, the instruction specifies a register in the CPU whose contents give the address of the operand in memory?

(a) Register indirect mode
(b) Indirect address mode
(c) Implied mode
(d) Immediate mode

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : In register indirect addressing mode, the instruction specifies a register whose contents hold the actual memory address of the operand. This means the CPU first reads the content of the specified register, and that content is then used as the memory address to fetch the operand.

229. The advantage of ——— is that it can reference memory without paying the price of having a full memory address in the instruction.

(a) Direct addressing
(b) Indexed addressing
(c) Register addressing
(d) Register Indirect addressing

UGC NET C.S. June-2014 (Paper-III)

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Ans. (d) : Because, Register indirect addressing means that the location of an operand is held in a register. It is also called indexed addressing Register indirect addressing modes requires three read operation to access an operand. Hence the answer is (d).

230. In _____ addressing mode, the operands are stored in the memory. The address of the corresponding memory location is given in a register which is specified in the instruction.

(a) Register direct (b) Register indirect
(c) Base indexed (d) Displacement

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Register indirect addressing means that the location of an operand is held in a register. It is also called indexed addressing or base addressing.

231. An Address in main memory is called:

(a) Virtual address (b) Memory address
(c) Logical address (d) Physical address

UGC NET C.S. December-2023

Ans. (d) : An address in main memory is called a location or physical address. The set of such locations is called the memory space, which consists of the actual main memory locations directly addressable for processing. As an example, consider a computer with main memory capacity of 32M words. 25 bits are needed to specify a physical address in memory since 32M = 225.

232. A system bus in which each data item is transferred during a time slice known in advance to both units source and destination is called:

(a) MIMD (b) DMA
(c) Asynchronous bus (d) Synchronous bus

UGC NET C.S. December-2023

Ans. (d) : Data transfers over the system bus may be synchronous or asynchronous. In a synchronous bus, each data item is transferred during a time slice known in advance to both source and destination units. Synchronization is achieved by driving both units from a common clock source. An alternative procedure is to have separate clocks of approximately the same frequency in each unit. Synchronization signals are transmitted periodically in order to keep all clocks in the system in step with each other.

233. Match List I with List II

List-I		List-II	
A.	**Direct Addressing**	**i.**	**MVI A, FFH**
B.	**Indirect Addressing**	**ii.**	**MOV A, B**
C.	**Register Addressing**	**iii.**	**STAFFFFH**
D.	**Immediate Addressing**	**iv.**	**MOV A, M**

Choose the correct answer from the options given below:

(a) A-I, B-II, C-III, D-IV
(b) A-II, B-III, C-IV, D-I
(c) A-III, B-IV, C-II, D-I
(d) A-IV, B-III, C-I, D-II

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Ans. (c) : Direct Addressing Mode
In direct addressing mode, the data to be operated on is available inside a memory location and that memory location is directly specified as an operand. The operand is directly available in the instruction itself. Example; STAFFFFH
(A) (iii) Direct addressing → STAFFFFH
(B) Indirect Addressing–In register indirect addressing mode, the data to be operated on is available inside a memory location and that memory location is indirectly specified by register pair.
Example
MOV A, M (Move the contents of the memory location pointed by the H–L pair to the accumulator.
(B) (iv) Indirect Addressing mode
→ MOV A,M
(C) Register Addressing–In register addressing mode the data to be operated on is available inside the register (S), and the register (S) are the operands.
Therefore, the operation is performed within various registers of the microprocessor.
Example :
MOV A, B (move the contents of register B to register A)
(C) (ii) Register Addressing → Move A, B
(D) Immediate Addressing– In immediate addressing mode the source operand is always data. If the data is 8 bit, then the instruction will be 2 bytes, if the data is 16 bit then the instruction will be 3 bytes.
(D) (I) MVI A, FFH →Immediate Addressing
So option (c) is correct.

234. In_____ addressing mode the operands are specified within the definition of instruction such as" complement accumulator" while in _____ addressing mode operands are explicitly specified within the address field of instruction itself.
_____एड्रेसिंग मोड में ऑपरेंड्स, इंस्ट्रक्शन की परिभाषा जैसे की ''कॉम्प्लीमेंट अक्यूमुलेटर'' में निहित रहते है जबकि ____ एड्रेसिंग मोड में ऑपरेंड्स, सीधे ही इंस्ट्रक्शन के एड्रेस भाग में ही लिख दिए जाते है।

(a) Implied, Immediate/इम्प्लाइड,इमीडियेट
(b) Direct, Immediate/डायरेक्ट, इमीडियेट
(c) Immediate, Direct/इमीडियेट, डायरेक्ट
(d) Direct, Indirect/डायरेक्ट, इनडायरेक्ट

RPSC C.S. 2016 (PAPER-I)

Ans, (a) : Implied Addressing mode– In Implied addressing mode, the operands are specified implicitly in the definition of the instruction. All the instructions which reference resisters that use an accumulator are Implied mode instruction zero address instructions zero address instructions in a stack organized computer are also implied mode Instructions Thus, it is also known as stack addressing mode.
Advantage of Implied addressing mode–
• Advantages it in that it simplifies given instructions because the operand is not included.
• Delivery of faster results because the execution of the instruction does not involve an extra memory reference to acquire the operand.
Disadvantages of implied addressing mode–
• Relatively rigid as the operand cannot be changed other than in the physical sense by the programmer

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• Appropriate only for particular codes to be executed where the operands is suggested by the operation.

Immediate Addressing Mode–

In immediate addressing mode, the operand is specified in the instruction itself in this mode, the operand field contains the actual operand to be used in conduction with operation specified in the instruction.

Advantages of Immediate Addressing Mode–

- Since operand is easily accessible from the instruction then it executes fast.
- It helps to free the machine from the necessity of obtaining operands from the memory.

Disadvantages of Immediate Addressing–

- Restricted to minute operand specifications since the operand must fit in instruction area.
- Inapplicable to the operands preferment for static data as the operands.

235. "Effective Address = address part of instruction + content of CPU register" is applicable for which pair of addressing modes given below:
"इफेक्टिव एड्रेस = इंस्ट्रक्शन का एड्रेस भाग + सी.पी.यू. रजिस्टर का मान" निम्न में से किस एड्रेसिंग मोड़ युग्म पर लागू होता है?

(a) Register Indirect, Relative
रजिस्टर इनडायरेक्ट, रिलेटिव
(b) Direct, Relative/डायरेक्ट, रिलेटिव
(c) Indirect, Indexed/इनडायरेक्ट, इंडेक्स्ड
(d) Relative, Indexed/रिलेटिव, इंडेक्स्ड

RPSC C.S. 2016 (PAPER-I)

Ans, (d) : • **Relative Addressing–** the effective address is calculated by adding the address part of the instruction (often and offset) to the content of a CPU register, typically the Program Counter (PC).

• **Indexed Addressing–** the effective address is computed by adding the address part of the instruction to the content of an index register.

236. Which of the following addressing modes are suitable for program relocation at run time?
(i) Direct addressing (ii) Based Register addressing (iii) PC relative addressing (iv) Indexed addressing.

(a) (i) and (ii) (b) (ii) and (iii)
(c) (iii) and (iv) (d) (ii), (iii) and (iv)

CGPSC C.S. 2020 Set-A

Ans.(b): Based Register Addressing:- This mode uses a base register to determine the effective address. The operand address is calculated by adding the register value of the off set specified in the instruction

PC Relative Addressing :- In this mode, the operand address is calculated by adding on offset to current program counter (PC). Since the offset is relative to the PC, the instruction can be relocated anywhere in memory without needing to change the address within the instruction itself.

237. Which of the following addressing modes requires more number of memory access?

(a) Direct (b) Immediate
(c) Indirect (d) Implied

CGPSC C.S. 2020 Set-A

Ans.(c): Indirect Addressing mode requires more memory access because the address field in the instruction points to a memory location that contain the effective address of the operand meaning two memory access are needed to retrieve the data.

238. Stack addressing is same as:

(a) Direct addressing (b) Indirect addressing
(c) Zero addressing (d) Relative addressing

CGPSC C.S. 2020 Set-A

Ans.(c): Stack addressing is also known as zero-address addressing because the operand are implicitly located at the top of the stack and the instruction themselves do not explicitly specify memory address for operands.

(xiv) Common Microprocessor Instructions

239. Which of the following does not have 8 data lines?
निम्न में से किसमें 8 डेटा लाइनें नहीं होती हैं?

(a) 8085 (b) 8086
(c) 8088 (d) Z80

LT. Grade C.S. 2018

Ans, (b) : The 8086 microprocessor has a 16 bit data bus, meaning it has 16 data lines, not 8. The other processors listed, 8085, 8088 and Z80 all have an 8-bit data bus with 8 data lines.

240. To achieve parallelism, one needs a minimum of/समांतरता के लिए न्यूनतम कितने प्रोसेसर की आवश्यकता होती है?

(a) 2 processors/2 प्रोसेसर
(b) 3 processors/3 प्रोसेसर
(c) 4 processors/4 प्रोसेसर
(d) None of the above/उपर्युक्त में से कोई नही

LT. Grade C.S. 2018

Ans, (a) : Parallelism is achieved when multiple processors, cores or processing units work together simultaneously to perform multiple tasks or operations. The minimum number of processors required to achieve parallelism is 2, as this allows for the simultaneous execution of at least two tasks or threads.

If there is a single processor, only one task can be executed at a time, and parallelism is not possible. So, having at least two processors is essential to achieve parallelism.

241. Intel 8085 microprocessor is of ____ generation.
इन्टेल 8085 माइक्रोप्रोसेसर _____ जनरेशन का है।

(a) First/प्रथम
(b) Second/द्वितीय
(c) Third/तृतीय
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): INTEL 8085 is a second generation microprocessor. Intel 8085 microprocessor is an 8-bit microprocessor developed by the Intel in 1976.

It is one of the most popular microprocessor used for learning and small embedded system application.

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242. The maximum memory size of third generation microprocessor is
तृतीय जनरेशन माइक्रोप्रोसेसर का अधिकतम मेमोरी साइज है
(a) 4 GB (b) 16 MB
(c) 16 GB
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans. (b): The maximum memory size of third generation microprocessor is 16MB.
- Third generation microprossar generally used a 16 bit address bus.
- A 16 bit address bus can access 2^{16} = 65536 memory locations, which equals 64KB.
- However, using segmented or bank- Switching techniques, the addressable memory was extended up to 16MB.
- A good example is the Intel 80286, which is a third generation microprocessor, and it supports up to 16 MB of memory.

So, the correct answer (b).

243. CPU on a single chip is called
एकल चिप पर सीपीयू (CPU) को कहा जाता है
(a) Microprocessor/माइक्रोप्रोसेसर
(b) Microcontroller/माइक्रोकन्ट्रोलर
(c) Solid-state device/सॉलिड-स्टेट युक्ति
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans. (a): • When the entire CPU (Central Processing Unit) is integrated on a single-chip (IC) it is called a microprocessor.
• A Microcontroller has the CPU along with memory and I/O devices on the same chip.
• Therefore, a single chip CPU is specifically called a Microprocessor.

244. Which of the following 8085 microprocessor hardware interrupt has the lowest priority?
(a) RST 6.5 (b) RST 7.5
(c) TRAP (d) INTR
UGC NET C.S. December-2015 (Paper-III)

Ans : (d) Hardware interrupts in 8085.
TRAP RST 4.5> RST 7.5> RST 6.5> RST 5.5> INTR.
INTR is level triggered and maskable interrupt. It has lowest priority.

245. In 8085 microprocessor which of the following flag(s) is (are) affected by an arithmetic operation?
(a) AC flag only (b) CY flag only
(c) Z flag only (d) AC, CY, Z flags
UGC NET C.S. November-2017 (Paper-III)

Ans : (d) AC is auxiliary – carry flag, CY is carry flag & Z is zero flag. All these flags will be affected during arithmetic operation.

246. In 8085 microprocessor the address bus is of ..... bits
(a) 4 (b) 8 (c) 16 (d) 32
UGC NET C.S. November-2017 (Paper-III)

Ans : (c) In 8085 microprocessor 16 bits are used for address bus & 65,536 (2^{16} = 65,536) different memory location are possible.

247. In the architecture of 8085 microprocessor match the following :

A.	**Processing unit**	**i.**	**Interrupt**
B.	**Instruction unit**	**ii.**	**General purpose Register**
C.	**Storage and Interface unit**	**iii.**	**ALU**
D.		**iv.**	**Timing and control**

Code :

	A	B	C
(a)	iv	i	ii
(b)	iii	iv	ii
(c)	ii	iii	i
(d)	i	ii	iv

UGC NET C.S. November-2017 (Paper-III)

Ans : (b)
- ALU is the Arithmetic Logic Unit and it involves processing of input desired output.
- Timing and control instruction are covered in instruction unit of microprocessor.
- There are some general-purpose register in storage & interface unit
- While an interrupt is signal to the processor which required attention from processor, an interrupt is serviced on the basis of priority & need

248. 8085 microprocessor has _____ bit ALU.
(a) 32 (b) 16
(c) 8 (d) 4
UGC NET C.S. August-2016 (Paper-III)

Ans : (c) ALU is designed to operate on 8-bit data therefore the 8085 microprocessor has 8-bit ALU.

249. Which of the following in 8085 microprocessor performs
HL = HL + HL ?
(a) DAD D (b) DAD H
(c) DAD B (d) DAD SP
UGC NET C.S. August-2016 (Paper-III)

Ans:(b) DAD H: double add HL to HL⇒HL=HL+HL

250. In 8085 microprocessor, the digit 5 indicates that the microprocessor needs
(a) –5 volts, +5 volts supply
(b) +5 volts supply only
(c) –5 volts supply only
(d) 5 MHz clock
UGC NET C.S. June-2017 (Paper-III)

Ans : (b) The "5" in the part number highlighted the fact that the 8085 uses a single +5 volt (V). Power supply by using depletion mode transistors, rather than requiring the +5V, –5V and +12V supplies need by the 8085.
It uses a single +5 volt D.C. (Direct Current) supply for its operation. It clock speeds is 3 MHz.

251. In 8085, which of the following performs : load register pair immediate operation?
(a) LDAS rp (b) LHLD addr
(c) LXI rp, data (d) INS rp
UGC NET C.S. June-2017 (Paper-III)

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Ans : (c) LXI is load register pair immediate which take one register as input and 2 bytes immediate data. There are seven integral 8-bit register.
This is generally used to load SP, memory pointer. Generally H–L, D–E, B–C register pairs are used. B, D, H hold the MSB and L, E, C hold the LSB

252. In 8085 microprocessor, what is the output of following program ?
LDA 8000H
MVI B, 30H
ADD B
STA 8001H
(a) Read a number from input port and store it in memory
(b) Read a number from input device with address 8000H and store it in memory at location 8001H
(c) Read a number from memory at location 8000H and store it in memory location 8001H
(d) Load A with data from input device with address 8000H and display it on the output device with address 8001H

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : LDA 8000H (Memory location)
MVI B, 30H
ADD B
STA 8001H (Stored in memory location)
Read a number from memory at location 8000H and store it in memory location 8001H.

253. ____ chips are high speed processors that are known for their small die size and low power requirements. They are widely used in PDAs and other hand held devices, including games and phones as well as a large variety of consumer products. It is probably the most widely used embedded processor architecture and indeed the most widely used processor architecture of any kind in the world.
____ चिपें उच्च गति प्रोसेसर हैं जो अपने छोटे डाई आकार (die size) तथा निम्न शक्ति आवश्यकता के लिए जानी जाती है । इनका बड़े पैमाने पर PDAs तथा अन्य हस्त-पकड़ युक्तियों में उपयोग होता है, जिसमें खेल और फोन के साथ ही साथ उपभोक्ता उत्पादों की अनेक किस्में आती हैं । यह संभवतः सबसे बड़े पैमाने पर उपयोग की जाने वाली एम्बेडेड प्रोसेसर आर्किटेक्चर तथा निश्चित तौर पर किसी भी तरह का विश्व का अत्याधिक बड़े पैमाने पर प्रयुक्त प्रोसेसर आर्किटेक्चर है ।
(a) HAND (b) LEG
(c) ARM (d) SUN

KVS PGT C.S. 2013

Ans, (c) : ARM stands for Advanced RISC Machine. An Arm processor is one of a family of central processing units based on the reduced instruction set computer architecture for compute processors.

254. In which of the following machines, a single control unit broadcasts instructions, which are carried out simultaneously by multiple processor/memory elements?
(a) Multiprocessor (b) Array processors
(c) Multicomputers (d) Pipeline machines
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : An array processors is a type of parallel computer where a single control unit sends instructions to multiple processing elements (PEs) which all execute the instruction simultaneously on their own data.

255. Microprocessor unit consists of:
(a) Control and timing circuit
(b) ALU
(c) Memory
(d) All of the above

CGPSC C.S. 2020 Set-A

Ans.(d): A microprocessor unit (C.P.U.) fundamentally consist of an Arithmetic Logic Unit for calculations, a control and timing unit to manage operations, and memory to temporarily store data and instruction.

256. Microprocessor memory and other parts of the micro computer are physically connected through which of the following ?
माइक्रोप्रोसेसर मेमोरी और माइक्रो कम्प्यूटर के अन्य हिस्सों के बीच संबंध को क्या नाम दिया गया है?
(a) Route/रूट
(b) Cache memory/कैश मेमोरी
(c) Path/पाथ
(d) Address bus/एड्रेस बस

HTET PGT C.S. 2020

Ans. (d) : Address bus is a set of wires that carry physical addresses from the CPU to main memory or input/output devices. The function of the address bus is to the CPU which memory location address or devices to access.

257. _____ Have a micro-chip embedded on it:-
जिसमें एक माइक्रो-चिप लगी हो:-
(a) Smart card/स्मार्ट कार्ड
(b) Credit card/क्रेडिट कार्ड
(c) Debit card/डेबिट कार्ड
(d) All of these/ये सभी

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : A smart card has a micro-chip embedded on it. specifically a smart card is a plastic card with an embedded microchip that can store and process data. This chip allows the card to be used for various applications, such as payments, identification and transportation ticketing.

(xv) Multi-core Architecture

258. Computer on a single chip is called
एकल चिप पर कम्प्यूटर (Computer) को कहा जाता है
(a) Microcontroller/माइक्रोकन्ट्रोलर
(b) Microprocessor/माइक्रोप्रोसेसर
(c) Assembler/एसेम्बलर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a): A microcontroller (MCU) is a compact integrated circuit (IC) designed to perform specific control function in embedded systems.
A microcontroller = CPU + memory + I/O
devices on a single chip, used for dedicated control task.

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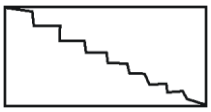
(xvi) Multiprocessor and Multicomputer

259. A multi computer with 256 CUPs is organized as 16 × 16 grid. What is the worst case delay (in hops) that a message might have to take?

(a) 16 (b) 15
(c) 32 (d) 30

UGC NET C.S. June-2016 (Paper-III)

Ans : (d)



In the 16 × 16 CPU case, the worst case delay happens when message passes through longest of the upper right corner to lower left corner or upper left corner to lower right corner.
Longest path is indicated in red color
It passes through 2(N –1) = 2(16 – 1) = 30 hops

260. The translator which performs macro calls expansion is called :

(a) Macro processor
(b) Micro pre-processor
(c) Macro pre-processor
(d) Dynamic Linker

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Translator which performs macro calls expansion is called macro pre-processors

261. One of the main features that distinguish microprocessor from micro-computers is

(a) words are usually larger in microprocessors.
(b) words are shorter in microprocessors.
(c) microprocessor does not contain I/O devices.
(d) None of the above.

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : A microprocessor is a computer processor which incorporates the functions of a computer's control processing unit (CPU) on a single integrated circuit (IC), or at most a few integrated circuits.
A microcomputer is small, relatively, inexpensive computer with a microprocessor as its control processing unit (CPU). It includes a microprocessor, memory, and minimal input/output (I/O) circuitry mounted on a single printed circuit board.
That means, microprocessor does not contain input/output devices.

262. For multiprocessor system, interconnection network - cross bar switch is an example of

(a) Non blocking network
(b) Blocking network
(c) That varies from connection to connection
(d) Recurrent network

UGC NET C.S. December 2021

Ans. (a) : For a multiprocessor system, interconnection network- crossbar switch is an example of non-blocking network.
A non-blocking network is a type of inter connection network in which concurrent communication can occur between multiple nodes without blocking. The crossbar switch is a non-blocking network because it can connect any input to any output without any blocking. This allows for efficient communication and data transfer between multiple processors memory modules and input/output devices in a multiprocessor system.

263. Which of the following is correct?
निम्नलिखित में से कौन-सा कथन सत्य है?

(a) In asymmetric multiprocessing, the processors are peers./असिमिट्रिक मल्टी-प्रोसेसिंग में प्रोसेसर पीयर्स हैं।
(b) In symmetric multiprocessing, the processors are placed symmetrically on the mother board./सिमिट्रिक मल्टी-प्रोससिंग में, प्रोसेसर मदरबोर्ड पर सिमिट्रिकली रखे जाते हैं।
(c) Clustered systems are used for high performance computing./उच्च निष्पादन कम्प्यूटिंग के लिए क्लस्टर्ड सिस्टम का उपयोग किया जाता है।
(d) All multiprocessor systems are multicore systems./सभी मल्टी-प्रोसेसर सिस्टम मल्टी-कोर सिस्टम होते हैं।

KVS PGT C.S. 2018

Ans, (c) : The clustered systems are a combination of hardware clusters and software clusters. The hardware clusters help in sharing of high performance disks between the systems. The software clusters makes all the systems work together. So, clustered systems are used for high performance computing.

264. Which component of an ideal microcomputer is responsible for temporarily holding data and instructions during processing?
एक आदर्श माइक्रोकंप्यूटर का कौन सा घटक डेटा और निर्देशों को प्रसंस्करण के दौरान अस्थायी रूप से धारण करने के लिए जिम्मेदार है?

(a) CPU
(b) ALU
(c) Memory/मेमोरी
(d) Output devices/आउटपुट उपकरण

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Memory (RAM) is used to temporarily hold data and instruction while the CPU processes them.
The temporary storage during processing is the job of memory (RAM).

265. How is the width of the data bus typically measured in a microcomputer system?
एक माइक्रोकंप्यूटर सिस्टम में डेटा बस की चौड़ाई को आमतौर पर किस प्रकार से मापी जाती है?

(a) In kilobytes (KB)/किलोबाइट्स (KB) म
(b) In megahertz (MHz)/मेगाहर्ट्ज़ (MHz) में
(c) In bits/बिट्स में
(d) In address lines/पता रेखाओं में

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : • Data bus carries actual data between CPU, memory, and I/O devices.
• Its width (e.g., 8 - bit, 16 bit, 32 bit, 64 bit) determines how many bits can be transferred simultaneously
• Wider data bus = Higher data a transfer capacity.

Sarkari Resource

03. Data Structures and Algorithm

(i) Definition and types

1. In data structure, data may of …… types.
डेटा संरचना में, डेटा ..... प्रकार के हो सकते हैं।
(a) 1 (b) 2
(c) 3 (d) 4

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : There are mainly two types of data in a data structure which.
1. Primitive Data structure- Chart float ivt, and Boolean etc
2. Non primitive Data structure-Array list stack queue, Tree, Graph etc.

2. Representation of data structure in memory is known as
मेमोरी में डेटा संरचना का प्रतिनिधित्व जाना जाता है
(a) recursive./रिकर्सिव
(b) abstract data type./एबस्ट्रेक्ट डेटा टाईप
(c) storage structure./संचायक संरचना
(d) file structure./फाईल संरचना

KVS PGT C.S. 2013

Ans.(c) : The represention of a data structure in memory is known as its storage structure. This refers to how the data is actually stored in the computer's memory including the allocation of memory for elements and the way they are linked or organized.

3. Which of the following is NOT a basicoperation performed on a data structure?/निम्नलिखित में से कौन-सा डेटा संरचना पर किया जाने वाला मूल ऑपरेशन नहीं है?
(a) Encryption/एन्क्रिप्शन
(b) Deletion/विलोपन
(c) Insertion/सम्मिलन
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : The basic operations that can be performed an a data structure are insertion, deletion, searching, and traversal. Insertion means adding a new element. Searching and traversal are used to access and process elements stored in the data structure.
On the other hand, encryption is not a basic operation of a data structure, as it deals with securing data rather than organizing or managing it.

(ii) Linear Structures

4. The following postfix expression with single-digit operands is evaluated using a stack:
सिंगल-डिजिट ऑपरेन्ड्स के साथ निम्न पोस्टफिक्स व्यंजक का स्टैक के प्रयोग से मूल्यांकन हुआ है :
823^/23*+51*–
(Note that ^ is the exponentiation operator.)
The top two elements of the stack after the first * is evaluated, are
(ध्यान दें ^ एक एक्सपोनेन्शियेशन ऑपरेटर है।) पहले * के मूल्यांकन के बाद स्टैक के दो शीर्ष एलिमेन्ट्स होंगे
(a) 6, 1 (b) 5, 7
(c) 3, 2 (d) 1, 5

LT. Grade C.S. 2018

Ans, (a) : Given postfix expression-
823^/23*+51*–
To evaluate the given postfix expression step by step using a stack.
Push 8, Stack : [8]
Push 2, Stack : [8, 2]
Push 3, Stack : [8, 2, 3]
Operator ^ (exponentiation)
compute $2^3 = 8$, Stack: [8, 8]
Operator/ (Division),
compute 8/8 = 1, Stack : [1]
Push 2, Stack : [1, 2]
Push 3, Stack : [1, 2, 3]
Operator * (Multiplication)
compute 2×3 = 6, Stack : [1, 6]
At this point, the first * is evaluated. The top two elements are 6, 1.

5. Which one of the following is prefix representation of the infix expression A*(B+C)/D?
इनफिक्स व्यंजक A*(B+C)/D का प्रिफिक्स व्यंजक निम्न में से कौन - सा है?
(a) /*A+BCD
(b) A+B/CD
(c) +*AB/CD
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans, (a) : Given infix expression
A*(B+C)/D
To convert the infix expression in to prefix notation, follow these steps
((A*(B+C))/D)
((A*(+BC))/D)
((*A(+BC))/D)
/(*A(+BC))D
/*A+BCD
So, the prefix expression is /*A+BCD.

6. What is the polish notation of the given infix expression?
(A * B + C)/(E – F * G)
दिये गये इंफिक्स एक्सप्रेशन का पॉलिश नोटेशन क्या है?
(A*B+C)/(E–F*G)
(a) +*/ABCE*FG (b) /+* ABC–E*FG
(c) /ABC+*–E*FG (d) +/ABC*–E*FG

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Sarkari Resource

Ans. (b) Correct the given infix expression (A*B+C)/(E–F*G) in to polish notation (also known as prefix notation)

1. **Understand Polish Notation–** In polish (prefix) notation, operators are placed before their operands, eliminating the need for Parentheses. The expression is processed such that for any operator its operands follow it.
2. **Operator Precedence–** Identify the order of operations based on precedence

⇒ Parentheses have the highest precedence

⇒ Multiplication (*) and division (/) have higher precedence that addition (+) and subtraction (–).

⇒ Process the expression from left to right for operators of the same precedence

3. Convert the Expression–

⇒ The given expression is (A*B+C)/(E–F*G)

⇒ Break it into sub-expressions based on parentheses and operator precedence.

Numerator : A*B+C

⇒ First, handle A*B in prefix, this becomes *AB.

⇒ Then, add C So, A*B+C becomes +(*DB)C or in prefix : +*ABC.

Denominator : E–F*G

⇒ First, handle F*G in prefix, this becomes *FG

⇒ Then, subtract from E. So, E–F*G becomes –E(*FG), or in prefix : –E*FG

Combine with Division–

⇒ The full expression is the numerator divided by the denominator : (A*B+C)*/(E–F*G)

⇒ In Prefix, division is written as / numerator denominator. So, we combine : /(+*ABC) (–E*FG)

The Polish notation (Prefix notation) of the given infix expression.

(A*B+C)/(E–F*G): /+*ABC–E*FG.

7. What is the total number of elements in a array E[–4 : 1, 6 : 10]?
एक ऐरे E[–4 : 1, 6 : 10] में अवयवों की कुल संख्या क्या है?

(a) 30 (b) 20
(c) 9 (d) 11

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : Given,
Array E[–4: 1, 6 : 10]
Step-1: Understand the array dimensions
•First dimension (rows) = –4 to 1
•Second dimension (columns) = 6 to 10
Step -2: Count element in each dimension
1- Rows:- –4, –3,–2,–1, 0,1 = 6 elements.
2- Columns :- 6, 7, 8, 9, 10 = 5 elements.
Step-3: Total number of elements
Total elements = Rows × Columns
= 6 × 5 = 30

8. What is the equivalent infix expression of the following postfix expression?
M, N, O, +, *, P, /, Q, R, S, T, /, +, *, –
निम्नलिखित पोस्टफिक्स व्यंजक का समतुल्य इंफिक्स व्यंजक क्या है
M, N, O, +, *, P, /, Q, R, S, T, /, +, *, –

(a) O * (M + N)/P-Q * (R + S/T)
(b) M * (N + O)/Q - P * (R/S + T)
(c) N * (M + O)/Q-P * (S+ R/T)
(d) (((M*(N + O))/P)- (Q*(R + (S/T))))

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (d) : Postfix : M N O + * p / Q R S T / + * -
Convert with a stack (combine the last two operands with each operator):
Final infix (fully parenthesized):
(((M * (N + O)) / P) – (Q * (R + (S/T)))

9. Convert the following arithmetic expression into Postfix form:
A – B – D * E / F + B * C
निम्नलिखित अंकगणितीय व्यंजक को पोस्टफिक्स फ्रॉर्म में रूपांतरित करें:

(a) AB–DE / F*–BC*+
(b) AB–DE* / F–BC*+
(c) AB–DE*F / –BC+*
(d) AB–DE*F / –BC*+

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Convert each part into postfix
D * E → DE*
(DE*)/ F → DE*F/
(A – B) → AB –
(AB–) – (DE*F/) → AB – DE* F/–
B * C → BC*
(AB – DE* F/–) + (BC*) → AB – DE* F/– BC*+

10. Given an expression
(A + B * D)/(E – F) + G

A. The prefix notation is +/+A *DB-EFG
B. The prefix expression is the reverse of the postfix expression
C. The order of operands in infix expression and postfix expression are the same.
D. The order of operands in infix expression, prefix expression and postfix expression are the same.

Choose the most appropriate answer from the options given below :

(a) A, C and D Only (b) C and D Only
(c) A and B Only (d) A and D Only

UGC NET C.S. December-2022

Ans. (b) : Given expression
(A+B*D) / (E – F) + G
Prefix Notation
+ / + A* BD – EFG
Postfix Notation
ABD* + EF – / G +
So option (b) is correct.

11. 2 Dimensional array (Matrices) with relatively high proportion of zero entries called _____ while with low proportion of zero entries are called _____ .
शून्य प्रविष्टियों के अपेक्षाकृत उच्च अनुपात केसाथ 2 आयामी एरे (मैट्रिसिस) _____ कहलाती हैं जबकि शून्य प्रविष्टियों के कम अनुपात के साथ _____ कहलाती हैं।

(a) Regular matrix; Non-Regular matrix/रेगुलर मैट्रिक्स; नॉन रेगुलर मैट्रिक्स

Sarkari Resource

(b) Sparse Matrix; Dense matrix/स्पार्स मैट्रिक्स; डेंस मैटरिक्स

(c) Transpose Matrix; Inverse Matrix /ट्रांसपोज मैट्रिक्स; इंवर्स मैट्रिक्स

(d) Square Matrix Dense Matrix/स्कवायर मैट्रिक्स; डेंस मैट्रिक्स

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) :A sparse matrix is a type of matrix in which most of the element are zero. In contrast, a dense matrix has mostly non-zero elements. sparse matrix save memory space and reduces computation time in algorithms.
Where as dense matrix requires more memory than sparse representation because all elements are explicity stored.

12. Consider an array A[2 : 8, –4 : 1, 6 : 10]. Base address of array A is = 400 and size of each element = 2 word per memory cell. What is theaddress of A[5, –1, 8] in row major form?
एक एरे A[2 : 8, –4 : 1, 6 : 10]पर विचार करें। एरे Aका बेस एड्रेस = 400 है और प्रत्येक एलिमेंट का साईज = 2 शब्द प्रति मेमोरी सेल है। रो मेजर रूप में A[5, –1, 8] का एड्रेस क्या है?

(a) 628 (b) 614
(c) 630 (d) 626

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : Array : A[2 : 8, – 4 : 1, 6 : 10]
Base address = 400
Size of each element = 2 words per memory cell
Required address = A[5, – 1, 8] according to row major order.
Size of dimension :
1st dimension (i) : From 2 to 8
Size = (8 – 2 + 1) = 7
2nd dimension (j) : From – 4 to 1
Size = [1 – (–4) + 1] = 6
3rd dimension (k) : From 6 to 10
Size = (10 – 6 + 1) = 5
Row major formula
$A[i, j, k] = Base + (((i - L_1) \times (N_2 \times N_3)) + ((j - L_2) \times N_3)) + (k - L_3)) \times w$
Where,
L_1, L_2, L_3 = lower bounds of dimension
N_2, N_3 = Sizes of 2nd and 3rd dimensions
w = Size of each element
Substitute values,
$L_1 = 2, L_2 = -4, L_3 = 6$
$N_2 = 6, N_3 = 5$
w = 2
Base = 400
$A[5, -1, 8] = 400 + (((5 - 2) \times (6 \times 5)) + ((-1 - (-4)) \times 5) + (8 - 6)) \times 2$
= 614
Hence, option (b) is correct answer.
Note :- As per commission option (a) 628 is correct.

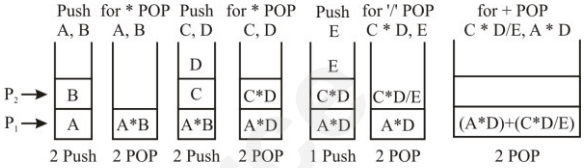
13. How many PUSH and POP operations will be needed to evaluate the following expression by reverse polish notation in a stack machine (A ×B) + (C × D/E)?

(a) 4 PUSH and 3 POP instructions
(b) 5 PUSH and 4 POP instructions
(c) 6 PUSH and 2 POP instructions
(d) 5 PUSH and 3 POP instructions

UGC NET C.S. December-2014 (Paper-II)

Ans :(b)To evaluate postfix expression we need operand stack given expression is infix, so convert it into postfix. So, resulting postfix expression:

AB * CD * E/ +



So, total 5 push and 9 POP. We can reduce POP since keep one operand in separate register, so need only 1 POP operation instead of 2 POP. So, can be answer but if 5 push and 8 POP but the question format counts only "push of operands" and "POP for operators" (ignoring result pushes). So it's 5 PUSH and 4 POP.

14. Which ordered pair correctly represents the data structures used in implementing graph traversing algorithms BFS & DFS?

(a) (Queue, Stack) (b) (Stack, Queue)
(c) (Stack, Stack) (d) (Queue, Queue)

RPSC C.S. 2016 (Paper-II)

Ans.(a) : The correct ordered pair representing the data structures used in implementing graph traversal algorithms BFS (Breadth-First Search) and DFS (Depth-First Search)is Queue, and Stack.
BFS (Breadth-First Search)– Uses a Queue, it explores all neighbors at the current depth before moving to the next. The queue ensures nodes are processed in the order they were discovered.
DFS (Depth-First Search) uses a stack (or recursion which uses the call stack)– It goes as deep as possible along a branch before backtracking.

15. Which traversal algorithm is typically implemented using a stack data structure?
कौन-सा ट्रैवर्सल एल्गोरिदम आमतौर पर स्टैक डेटा संरचना का उपयोग करके लागू किया जाता है?

(a) DFS
(b) BFS
(c) Both DFS and BFS/दोनों DFS और BFS
(d) Neither DFS nor BFS/न DFS और न ही BFS

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : Among traversal algorithms, Depth first (DFS) is generally implemented using a stack because DFS explores nodes deeply and requires backtracking, for which stack is most suitable. On the other hand, Breadth First search (BFS) is implemented using a queue since it processes nodes level by level. Therefore, the traversal algorithm typically implemented using a stacks is DFS.

Sarkari Resource

16. Stack, Queue and Linked List can be implemented using
Stack, Queue और Linked List..... का उपयोग कर लागू किया जा सकता है।

(a) Numeric variable (b) String variable
(c) Logical function (d) array

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : Data types like Stocks queues and linked lists can be implemented using arrays.

17. Which of the following is not a common example of an abstract data type?
निम्नलिखित में से कौन अमूर्त डेटा प्रकार (Abstract Data Type) का समान्य उदाहरण नहीं है?

(a) Stack/स्टैक (b) Integer/पूर्णांक
(c) Queue/कतार (d) List/सूची

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : Abstract data Types (ADTs) are data structures defined by their operations (like, push, pop, enqueue, dequeue) rather than their implementation.

- Stack, Queue and list are ADTs because they describe how data can be used.
- Integer is just a basic data type, not an abstract data type.

18. Which of the following data structures is most suitable for evaluating postfix expressions?
पोस्टफिक्स व्यंजकों के मूल्यांकन के लिए निम्नलिखित में से कौन सी डेटा संरचना सबसे अधिक उपयुक्त है?

(a) Tree/ट्री (b) Stack/स्टैक
(c) Linked list/लिंक्ड लिस्ट (d) Queue/क्यू

KVS PGT C.S. 2017

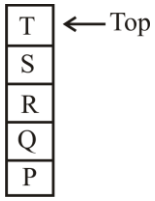
Ans.(b) : Stack is a linear data structure in which elements are inserted and deleted from one end only (top of the stack). It follows a order to insert the elements into stack which is known as LIFO (Last in First out). stack can be used for evaluating arithmetic expression. So, Stack data structure is most suitable for evaluating postfix expressions.

19. The five items P,Q,R,S and T are pushed in a stack, one after the other starting from P. The stack is popped four times and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack, The popped item is :
पाँच मदें P,Q, R, S और T को एक स्टैक पर P से शुरू करते हुए एक के बाद एक धकेला जाता है। स्टैक चार बार उछला (पॉपअप) और प्रत्येक घटक को पंक्ति में प्रविष्ट किया गया तत्पश्चात पंक्ति (क्यू) में से दो घटकों का विलोपन किया गया और चट्टे (स्टैक) पर दुबारा धकेल दिया गया। अब मद चट्टे से उछल (पाप्ड) गई है। उछलने वाली मद (पाप्ड आइटम) है:

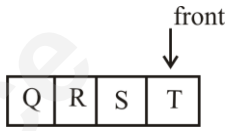
(a) S (b) P
(c) Q (d) R

KVS PGT C.S. 2017

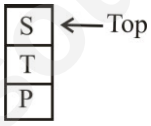
Ans.(a) : Given,
Five items P,Q,R,S, and T.
These are pushed into the stack as-



Now, stack is popped four times, and each item popped is inserted into the queue. So, items that are popped from the stack are
T, S, R, Q,
After insertioninto queue, -



Now, Two items are deleted from the Queue which is T, S
These are inserted back into the stack.



Now, one item is popped from the stack which is S.

20. When a circular queue is implemented in an array, then which of the following condition holds when there is only one element in the queue?
यदि किसी ऐरे में सर्कुलर क्यू क्रियान्वित की जाती है, तो क्यू में केवल अवयव के होने की स्थिति में निम्नलिखित में कौन सी शर्त बनी रहेगी?

(a) Front = Rear = null
(b) Front = Rear ≠ null
(c) Front = Rear + 1
(d) Front = Rear – 1

KVS PGT C.S. 2018

Ans.(b) :A circular queue is similar to a linear queue as it is also based on the FIFO (first in first out) principle except that the last position is connected to the first position in a circular queue that forms a circle.
Circular queue is implemented in an array a circular queue will be full when front = – 1 and Rear = max – 1,and when there is only one element in the queue then the condition is front = Rear ≠ Null.

21. Which of the following applications may use a stack?
निम्नलिखित में से कौन-सा एप्लीकेशन में स्टैक का प्रयोग किया जा सकता है?

(A) parenthesis balancing program/पैरेंथेसिस बैलेंसिंग प्रोग्राम
(B) process scheduling in operating system/ऑपरेटिंग सिस्टम में प्रोसेस शेड्यूलिंग
(C) conversion of infix arithmetic expression to postfix form/इनफिक्स अंकगणितीय व्यंजक को पोस्ट फिक्स में परिवर्तित करना

Sarkari Resource

(a) (A) and (B)/(A) तथा (B)
(b) (B) and (C)/(B) तथा (C)
(c) (A) and (C)/(A) तथा (C)
(d) (A), (B) and (C)/(A), (B) तथा (C)

KVS PGT C.S. 2018

Ans.(c) : Astack is a linear data structure that holds a linear, ordered sequence of elements. A stack works on the LIFO process. Some applications of stack are given below–
- A stack can be used for evaluating expression consisting of operands and operators.
- Stacks can be used for Backtracking i.e. to check parenthesis matching in an expression
- It can also be used to convert one form of expression to another form.

22. The largest element of an array index is called its
किसी ऐरे सूचकांक का सबसे बड़ा तत्व कहलाता है
(a) lower bound./इसका निम्नतर बाऊँड
(b) range./इसका परास (रेंज)
(c) upper bound/इसका उच्चतर बाऊँड
(d) All of these/ये सभी

KVS PGT C.S. 2013

Ans.(c) : Largest element of an array index is called upper bound and smallest array index is called lower bound.

23. The number of pointer changes required for inserting a node between two given nodes in thelinked implementation of a doubly linked list is _____.
किसी डबल लिंक्ड लिस्ट के लिंक्ड इम्प्लीमेंटेशन में दिए गए दो नोड्स के मध्य एक नोड प्रविष्ट करने के लिए कितने पॉइंटर परिवर्तन होंगे:
(a) One/एक (b) Two/दो
(c) Three/तीन (d) Four/चार

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans.(d) : The number of pointer changes required for inserting a node b/w two given nodes in the linked implementation of a doubly Linked List is four

24. Zero address instructions are necessarily used in _____ architecture.
_____ आर्किटेक्चर में जीरो एड्रेस इंस्ट्रक्शंस का अनिवार्य रूप से उपयोग किया जाता है।
(a) Stack/स्टैक (b) CISC/सीआईएससी
(c) RISC/आरआईएससी (d) Parallel/पैरेलल

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans.(a) : Zero address instruction are necessarily used in stack architecture.
⇒ In zero address instruction stack is used. Also called stuck based organization.

25. Which of the following options is the correct code for traversing a link list staring at pointer P, uptoelement X ?
पॉइंटर P से प्रारंभ करते हुए एलीमेंट X तक किसी लिंब लिस्ट को तिर्यक (ट्रावर्स) करने के लिए निम्नलिखित में से कौन सा कोड सही विकल्प है?
(a) While (P == NULL ?? P → Element ! = X)
P = P → Next ;
(b) While (P == NULL && P → Element ! = X)
P = P → Next ;
(c) While (P ! = NULL ?? P → Element ! = X)
P = P → Next ;
(d) While (P ! = NULL && P → Element ! = X)
P = P → Next ;

NVS PGT C.S. 15.12.2022 (Shift-II)

Ans.(d) : Linked List is a Linear data structure Like arrays. In a Linked list Each Element is actually a separate object Linked to the next Element using a reference stored in a pointer.
Correct the (d) options
While (P! = NULL && P → Element ! = X)
P = P → Next.

26. The concatenation of two lists is to be performed in O(1) time. Which of the following implementations of lists could be used?
दो लिस्ट का कॉनकैटिनेशन O(1) समय में होना है। निम्न में से किस लिस्ट-कार्यान्वयन का उपयोग होना चाहिए?
(a) Singly linked list/सिंगली लिंक्ड लिस्ट
(b) Doubly linked list/डबली लिंक्ड लिस्ट
(c) Circular doubly linked list
सर्कुलर डबली लिंक्ड लिस्ट
(d) Array implementation of list
लिस्ट का ऐरे कार्यान्वयन

LT. Grade C.S. 2018

Ans.(c) : As list concatenation requires traversing at least one list to the end. So singly linked list and doubly linked list requires O(n) time complexity whereas circular doubly linked list required O (1) time.

27. Zero address instructions are implemented with the help of
ज़ीरो ऐड्रेस इंस्ट्रक्शन का निम्न में किसकी सहायता से उपयोग होता है?
(a) Queue/क्यू
(b) Stack/स्टैक
(c) Register/रजिस्टर
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans.(b) : Zero address to instructions is a stack based architecture, where operations implicity employ the top items of the stack, zero address instruction are commonly used.
In a zero address instruction architecture, the stack is used to store operands and instructions operate on the top elements of the stack.

28. Which of the following is useful in traversing a given graph by breadth-first search (BFS)?
निम्न में से कौन-सा, एक ग्राफ को ब्रेड्थ-फर्स्ट सर्च(BFS) द्वारा ट्रैवर्स करने में उपयोगी है?
(a) Stack/स्टैक (b) Set/सेट
(c) List/लिस्ट (d) Queue/क्यू

LT. Grade C.S. 2018

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Ans.(d) :A queue data structure is useful in traversing a given graph by breadth first search because it maintains the order in which the vertices are visited it easy to visit all the vertices at the same level before moving to the next level.

29. A queue has configuration a, b, c, d. If youwant to get the configuration d, c, b, a, youneed a minimum of _____./कतार में विन्यास a, b, c, d है। यदि आप विन्यास d, c, b, a प्राप्त करना चाहते हैं, तो आपको कम से कम _____

(a) 2 deletions and 3 additions
2 विलोपन और 3 जोड़
(b) 3 deletions and 3 additions
3 विलोपन और 3 जोड़
(c) 4 deletions and 4 additions
4 विलोपन और 4 जोड़
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b) : Initial queue: a, b, c, d
Goal: d, c, b, a
Steps:
1. Delete a (1 deletion) → Queue: b, c, d
2. Delete b (2 deletions) → Queue: c, d
3. Delete c (3 deletions) → Queue: d
4. Add c (1 addition) → Queue: d, c
5. Add b (2 additions) → Queue: d, c, b
6. Add a (3 additions) → Queue: d, c, b, a
Total: 3 deletions and 3 additios.

30. Which of the following application makes useof a circular linked list?
निम्नलिखित में से कौन-सा अनुप्रयोगवृत्ताकार लिंक्ड सूची का उपयोग करता है?

(a) Recursive function calls
(b) Allocating CPU to resources
(c) Implement Hash Tables
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(b): Circular linked lists are often used in applications like allocating CPU to resources, particularly in scheduling algorithms such as round-robin. This data structure allows the system to cycle through tasks efficiently, allocating time slices in a circular manner.

31. Which data structures are considered linear?
कौन-सी डेटा स्ट्रक्चर्स लीनियर मानी जाती हैं?

(a) Binary tree
(b) Graph
(c) Stack
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c):A stack is considered a linear data structure because elements are arranged sequentially and follow a specific order, such as Last-In-First-Out (LIFIO). In contrast, binary trees and graphs are non-linear due to their branching and complex relationships.

32. Linked lists are not suitable for theimplementation of _____.
लिंक की गई सूचियां _____ के लिए उपयुक्त नहीं है ?

(a) Insertion sort
(b) Radix sort
(c) Binary search
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c): Linked lists are not suitable for implementing binary search because they do not allow random access to elements. Binary search requires direct access to the middle element of a list, which is inefficient in linked lists due to their sequential access nature. In contrast, linked lists are suitable for operations like insertion and deletion at arbitrary positions, making them a good fit for algorithms like insertion sort.

33. Which of the following is not the applicationof stack?
निम्नलिखित में से कौन-सा स्टैक का अनुप्रयोग नहीं है?

(a) Data Transfer between twoasynchronous process
(b) Compiler Syntax Analyzer
(c) Tracking of local variables at run time
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(a):• Stack is used for things like tracking function calls, local variables, recursion, and complier syntax checking.
• But transferring data between two processes is not done with a stack. It usually uses a queue.

34. What is the need for a circular queue?
वृत्ताकार कतार की क्या आवश्यकता है ?

(a) Easier computations
(b) Implement LIFO principle in queues
(c) Effective usage of memory
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c): The main need for a circular queue is to ensure effective usage of memory. In a simple linear queue, when elements are deleted from the fronts the empty spaces created cannot be reused, and after some time, the queue may appear full even though free space is available. This leads to wastage of memory. To overcome this problem, the circular queue connects the last position back to the first position, so that the freed space can be reused.

35. What are the characteristics of stack basedIDS?
स्टैक आधारित आई.डी.एस. की विशेषताएं क्या हैं?

(a) It is programmed to interpret a certainseries of packets
(b) It models the normal usage of thenetwork as a noise characterization
(c) They are integrated closely with theTCP/IP stack and watch packets
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

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Ans.(c): Stack based IDS integrate directly with the TCP/IP stack allowed them to monitor all incoming and outgoing packets in real-time. This allows them to detect malicious activity by analyzing data passing through the networks stack.

36. In the deque implementation, using singly linked list, what would be the time complexity of deleting an element from the rear end?
एकल लिंक्ड सूची का उपयोग करके डेक कार्यान्वयन में, किसी तत्व को पीछे के छोर से हटाने की समय जटिलता क्या होगी?

(a) O(1)
(b) O(N log N)
(c) O(N)
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): In a dequeue (double- ended queue) implemented using a singly linked list deleting an element from the rear end is not efficient. This is because, in a single linked list we only have a refrence to the next node, not the previous one. To reach the last node's predecessor (so we can remove the rear node) we must traverse the entire list from the head to the second - last node.This travensal takes O (N) time in the worst case. Therefore, The time complexity of deleting an element from the rear end in this implementation is O(N).

37. Which one of the following is the overflow condition if linear queue is implemented using an array with a size MAX_SIZE?
यदि MAX_SIZE आकार वाले ऐरे का उपयोग करके रैखिक क्यू लागू की जाती है, तो निम्नलिखित में से कौन-सी अतिप्रवाह स्थिति है?

(a) rear = MAX_SIZE–1
(b) rear = MAX_SIZE
(c) rear = front + 1
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Rear = MAX_SIZE - 1, Since the size of the array is MAX_SIZE, We can insert element up to MAX_SIZE 1. If we try if insert elements of size greater than MAX_SIZE or MAX_SIZE into a queue, it leads to an overflow condition.
Note - BPSC has considered (d) as the answer to this question, which is incorrect.

38. What is another name for the circular queue among the following options?
निम्नलिखित विकल्पों में से वृत्ताकार क्यू का दूसरा नाम क्या है?

(a) Rectangle buffer
(b) Square buffer
(c) Ring buffer
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Another name for circular queue is ring buffer ring buffer is a data protection technique that stores data in a circular area. In this data is placed in a circular area which has a wide circular area around it.

39. Which of the following determines the needs for the circular queue?
निम्नलिखित में से कौन-सा सर्कुलर क्यू की आवश्यकता निर्धारित करता है?

(a) Access the queue using priority
(b) Avoid wastage of memory
(c) Follow the FIFO principle
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(b): The circular queue is needed to avoid wastage of memory.
In a normal queue, empty spaces are created after items are removed, and those spaces can not be used again. A circular queue fixex this by connecting the end back to the start, reusing the empty spaces and saving memory.

40. Which data structure is the best for implementing a priority queue?
प्राथमिक क्यू को लागू करने के लिए कौन-सी डेटा संरचना सबसे अच्छी है?

(a) Linked list
(b) Array
(c) Binary heap
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): The best data structure for implementing a priority queue is the binary heap. Although arrays and linked lists can also be used they are not efficient because either insertion or deletion becomes slow. In contrast a binary heap allows insertion and detetion operation in O (logn) time and finding the highest or lowest priority element in O(1) time. makes it the most suitable and widely used structure for priority queues hence the correct answer is binary heap.

41. Which of the following data structures finds its use in recursion?
निम्नलिखित में से किस डेटा संरचना का उपयोग रिकर्शन में किया जाता है?

(a) Stack
(b) Linked list
(c) Array
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Recursion works by storing the function call one after another. Each time a function calls itself, the system keeps track of the return address, parameters, and local variables. This is managed using a stack (called the function call stack). when the function finishes, its data is popped from the stack, and control goes back to the previous call.
Therefore, recursion uses the stack data structure.

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42. Which one of the following is not the application of the stack data structure?
निम्नलिखित में से कौन-सा स्टैक डेटा संरचना का अनुप्रयोग नहीं है?
(a) Asynchronous data transfer
(b) String reversal
(c) Backtracking
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Asynchronous data transfer is less directly related to the typical applications of stacks compared to string reversal and backtracking. Stacks are commonly used for string reversal and backtracking due to their Last-In- First (1IFO) nature but asysnchronous data transfer often involves other data structures like queues.

43. Which of the following is the prefix form of A + B*C?
निम्नलिखित में से कौन-सा A+B*C का उपसर्ग रूप है?
(a) + A*BC
(b) ABC+*
(c) +AB*C
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Perfix notation means all operators that come before the operands. To convert an infix expression to a perfix expression we will move the operator to the left of the brackets.

44. Which data structure is required to covert the infix to prefix notation?
इनफिक्स को प्रीफ्रिक्स नोटेशन में बदलने के लिए किस डेटा संरचना की आवश्यकता है?
(a) Queue
(b) Stack
(c) Linked list
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): To convert infix to perfix notation, data structure called stack is required. With this , you can convert a person given in infix notation to perfix notation.

45. Which of the following highly uses the concept of an array?
निम्नलिखित में से कौन-सा ऐरे की अवधारणा का अत्यधिक उपयोग करता है?
(a) Binary search tree
(b) Scheduling of processes
(c) Spatial locality
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Spatial locality makes heavy use of the concept of arrays . This concept is such that a program continuously operator near an array and retrievers the data stored in that array. This takes advantage of cache memory and other spatial data structures, thereby increasing the efficiency of the program.

46. The necessary condition to be checked before deletion from the queue is:
क्यू से हटाने के पहले जाँच की जाने वाली आवश्यक शर्त है–
(a) Front value
(b) Underflow
(c) Overflow
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): The necessary condition to be checked before deleting from the queue is underflow, if the queue is empty and we try to delete an item from it, will underflow. This condition is necessary to check so there is no error in the code.

47. Which one of the following is not the application of the queue data structure?
निम्नलिखित में से कौन-सा क्यू डेटा स्ट्रक्चर का अनुप्रयोग नहीं है?
(a) Data is transferred asynchronously
(b) Resource is shared between various systems
(c) Balancing of symbols
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Balancing of symbols is not an application of queue data structure. A queue data structure transfers and stores items in sequence, the system uses shaming of resources and helps in transferring data asynchronously.

48. A double - ended queue (dequeue) supports adding and removing items from both the ends of the queue. The operations supported by dequeue are AddFront(adding item to front of the queue), AddRear(adding item to the rear of the queue), RemoveFront(removing item from the front of the queue), and RemoveRear(removing item from the rear of the queue). You aregiven only stacks to implement this data structure. You can implement only push and pop operations. What' s the time complexity of performing AddFront() and AddRear() assuming *m* is the size of the stack and *n* is the number of elements?
(a) O(m) and O(n)
(b) O(1) and O(n)
(c) O(n) and O(1)
(d) O(n) and O(m)

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Ans. (b) : A double-ended queue (deque) allows insertion and deletion elements from both ends. When implementing a deque using only stacks, which support

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only push and pop operations, we must consider the limitations of the stack's LIFO (last-in-first-out) nature. To add an element to the front of the deque, we can simply push it onto the top of one of the stacks, which takes O(1) time because it involves only a single push operation. However, adding an element to the rear of the deque is more involved. Since the stack only allows access to the top element, inserting at the bottom requires moving all existing elements (say, n elements) to another temporary stack, pushing the new element, and then moving the element back. This process takes O(n) time. Therefore, in a stack-based implementation of a deque, the time complexity for AddFront ( ) is O(1), while the time complexity for AddRear ( ) is O(n).

49. Suppose a circular queue of capacity (n–1) elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variable respectively. Initially, REAR = FRONT=0. The conditions to detect queue empty and queue full are

(a) EMPTY : REAR ==FRONT
FULL : (REAR +1) mod n == FRONT

(b) EMPTY : (FRONT +1) mod n ==REAR
FULL: (REAR +1) mod n == FRONT

(c) EMPTY (REAR +1) mod n == FRONT
FULL: REAR == FRONT

(d) EMPTY : REAR == FRONT
FULL:(FRONT +1) mod n == REAR

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Ans. (a) : For a circular queue implemented with an array of (n-1) elements, the conditions to detect queue empty and queue full are as follows:
1. Queue Empty: when REAR = = FRONT
2. Queue full : when (REAR + 1) mod n = = FRONT.
These conditions ensure that the circular nature of the queue is properly maintained, allowing elements to wrap around the array when necessary.

50. Which of the following addressing mode is best suited to access elements of an array of contiguous memory locations?

(a) Indexed addressing mode
(b) Base Register addressing mode
(c) Relative address mode
(d) Displacement mode

UGC NET C.S. November-2017 (Paper-III)

Ans :(a) Indexed addressing mode is best suited for accessing an array in contiguous memory location.

51. The reverse polish notation equivalent to the infix expression
((A + B) * C + D)/(E + F + G)

(a) A B + C * D + EF + G + /
(b) A B + C D * + E F + G + /
(c) A B + C * D + E F G + +/
(d) A B + C* D + E + F G +/

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : Because, ((A + B) * C + D) / (E + F + G)
(AB + *C + D) / (E + F + G)
(AB + C * + D) / (E + F + G)
AB + C * D + / (E + F + G)
AB + C * D) + / (EF + G)
AB + C * D + / EF + G +
AB + C * D + EF + G + /
Hence, the answer is (a).

52. Consider an array A[20, 10], assume 4 words per memory cell and the base address of array A is 100. What is the address of A [11, 5]? Assume row major storage.

(a) 560 (b) 565
(c) 570 (d) 575

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Ans :(a)Because, Each cell needs 4 words
- A [0][0], A[0] [1],..... A[0][9] needs 4 × 10 = 40 words
- Since array starts at 100, A [0] [9] will be at 100 + 40 = 140th location
- Similarly from A[0][0] to A[10][9] there are 11 × 10 = 110 elements
- From A[11][0] to A[11][5] there are 5 elements
- Total 115 elements are present before A[11][15]
- Hence A[11][15] will be 100 + 4 × 115 = 560

So, the answer is (a)

53. Convert the following infix expression into its equivalent post fix expression
(A + B^D)/(E – F) + G

(a) ABD^ + EF –/G+ (b) ABD+ ^EF –/G+
(c) ABD+ ^EF/ –G+ (d) ABD^ + EF /–G+

UGC NET C.S. December-2014 (Paper-II)

Ans : (a) Given infix notation is
$=(A+B\hat{}D)/(E-F)+G$
= Convert it into postix expression
$=((((A+(B\hat{}D))/(E-F))+G)$
$=((((ABD\hat{}+)/(EF-))+G)$
$=ABD\hat{}+EF-/G+$

54. _____ is a two-dimensional labelled data structure with columns of potentially different type, in data handling with pandas.

(a) Series (b) Missing data
(c) Iteration (d) Data frames

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans.(d) :• Data frame is a table-like data structure. Which is used to organize and analysis big data.
• Dataframe is a two dimensional (rows and columns) data structure.
• It is like a table where there are rows and columns.

55. Which of the following data structures is used in implementing recursive calls?
निम्नलिखित में से कोन सी डेटा स्ट्रक्चर्स(संरचनाओं) का उपयोग रिकर्सिव कॉल को लागू करने के लिए किया जाता है?

(a) Queue/क्यू
(b) Binary Tree/बाइनरी ट्री
(c) Stack/स्टैक
(d) Hash Table/हैश टेबल

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Ans. (c) : Stack is a linear data structure in which data can be added (Insert 1 push) and deleted (delete/pop) from only one end (TOP).
It is called a data structure working on LTFO principle. That means the element which is inserted last is removed first.

56. State whether the following statements are true of false.
Statements:
(i) A queue can be implemented using two stacks.
(ii) A stack can be implemented using two queues.
यह बताएं कि निम्नलिखित में से कौन सा कथन सत्य या असत्य है।
कथन:
(i) एक क्यू को दो स्टैक्स का उपयोग करके कार्यान्वित किया जा सकता है।
(ii) दो क्यूज का उपयोग करके एक स्टैक को कार्यान्वित किया जा सकता है।
(a) (i) True, (ii) True/ (i) सत्य, (ii) सत्य
(b) (i) True, (ii) False/ (i) सत्य, (ii) असत्य
(c) (i) False, (ii) True/ (i) असत्य, (ii) सत्य
(d) (i) False, (ii) False/ (i) असत्य, (ii) असत्य
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) :• A queue is implemented using two stacks that's right, by taking two stacks, that's right, by taking two stacks, an element is pushed into one and when needed, de-queue operation can be performed by transferring to its stack.
• A stack can be implemented using two queue correct.
• Push operation is easy, but pop operation requires rotating the queue.

57. A queue is___data structure in which elements can be inserted only at one end called _____ and deletd only at the other end called _____.
क्यू एक _____ डेटा है जिसमें तत्वों को केवल _____ नामक एक छो पर इंसर्ट(सम्मिलित) किया जा सकता है और केवल दूसरे छोर पर डिलीट कर (हटा) दिया जाता है जिसे _____ कहा जाता है।
(a) Linear, Front, Rear/लाइनर, फ्रंट, रियर
(b) Non-linear, Rear, Front/नॉन-लाइनर, रियर, फ्रंट
(c) Non-linear, Front, Rear/नॉन-लाइनर, फ्रंट, रियर
(d) Linear. Rear. Front/लाइनर, रियर, फ्रंट
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Queue is linear data structure in which
• Insertion always occurs at the rear end.
• Deletion always occurs from the front end.

58. Which of the following data structures can be used to efficiently implement a priority queue?
प्राइयोरिटी क्यू (प्राथमिकता कतार) को कुशलतापूर्वक कार्यान्वित करने के लिए निम्नलिखित में से किन डेटा स्ट्रक्चर्स का उपयोग किया जा सकता है?
(a) Array/अरै
(b) Single Linked List/एकल लिंक्ड लिस्ट
(c) Circular Linked List/सर्कुलर लिंक्ड लिस्ट
(d) Heap Tree/हीप ट्री
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Ans. (d) : Heap tree is a special type of binary tree which is called complete binary tree.
• All level of complete binary tree are completely filled.
• The last level is filled frem left to right. Heap tree is used in priority queue application.
Heap tree is of two types
1. Max-heap
2. Min-heap

59. Which of the following statements is NOT true?
निम्नलिखित कथनों में से कौन स सत्य नहीं है?
(a) The size of an array is generally fixed whereas the size of a linked list is dynamic./एक अरै का आकार आम तौर पर निश्चित किया जाता है, जबकि एक लिंक की गई सूची का आकार गतिशील हो
(b) Inserting a new element in an array is expensive compared to inserting a new element in a linked list./अरै में एक नया एलीमेंट डालने की तुलना में लिंक सूची में नया एलीमेंट डालना महंगा होता है
(c) Both, random access and direct access are allowed in a linked list./रैंडम ऐक्सेस (यादृच्छिक पहुंच) और डाइरेक्ट ऐक्सेस (प्रत्यक्ष पहुंच) दोनों, को लिंक की गई सूची में रखे जाने की अनुमति होती है
(d) Deleting an element from array is expensive compared to deleting an element from a linked list./अरै से एलीमेंट को हटाने की तुलना में लिंक सूची से एलीमेंट को हटाना महंगा होता है
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Arrays allow for both random access (accessing any elements directly by its index) and direct access because elements are stored in contiguous memory locations. Linked lists, however, do not allow for random or direct access to lements by index. To access an element in a linked list, you must traverse the list sequentially from the head, making it a sequential access data structure.

60. Linked List is a _____ data structure.
लिंक्ड लिस्ट एक _____ डेटा स्ट्रक्चर है
(a) Linear/लाइनर (रेखीय)
(b) Non-linear/नॉन-लाइनर (गैर रेखीय)
(c) Virtual/वर्चुअल (वास्तविक)
(d) Imaginary/इमैजिनरी (काल्पनिक)
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Linked list is a linear data structure because its elements (called nodes) are arranged in a sequence, with each node confairing a pointer to the next node in the chain, forming a linear order rather than being stored in contiguous memory Locations.

61. Consider a n × n tri-diagonal sparse matrix A. Suppose we want to store sparse A in a one-dimensional array. What is the index of the (i.j)thelement of A in the one-dimensional array.
एक n × nट्राई-डायगोनल स्पार्स मेट्रिक्स Aहै। मान लीजिए हमे स्पार्सAको एक वन डायमेंशनल ऐरे में स्टोर करना है। इस वन डायमेंशनल ऐरे में Aके (i.j)thएलेमेंट का इंडेक्स क्या होगा?

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(a) 2i + 2j (b) 2i + j – 2
(c) i + j (d) i + j – 1

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Ans. (b) : The index of the (i, j)th element of a tridiagonal sparse matrix is stored in a one dimensional array is given by the formula 2i + j–2. this formula is used for efficient storage by mapping the non-zero elements of the matrix to a consecutive series of indices in the array, preventing memory waste.

62. Which statement is true about condition of overflow in the circular Queue?
सर्कुलर क्यु के ओवरफ्लो की स्थिति के संदर्भ में निम्नलिखित में से कौन सा/से सही है/हैं?

I. Front = 1 and Rear = N
फ्रंट = 1 और रियर = N

II. Front = Rear + 1
फ्रंट = रियर+ 1

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ Iतथा IIदोनों
(d) Neither I nor II/ ना ही Iना ही II

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Ans. (c) : In a circular queue, overflow occurs when the queue, is full and a new element is attempted to be inserted. This happens when the position where the new element is to be inserted is the same as the start position, which is the position of the first element pointed to by the front end. Overflow happen when the Position/Index where the new element is to be inserted [(Rear + 1) +h Position] is the start position.

63. A 2D array in which there are different number of elements in each row is known as____.
एक 2Dऐरे जिसमें प्रत्येक पंक्ति में तत्वों की भिन्न-भिन्न संख्या होती है, वो_____के रूप में जानी हाती है।

(a) regular/रेगुलर (b) irregular/इर्रेगुलर
(c) sparse/स्पार्स (d) dense/डेंस

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) :A 2D array with different number of elements in each row is called a jagged or Irregular Array.
(a). Regular Array:-Each row has the same number of elements.
(c). Sparse Array:-Most of the elements are zero (or empty), but structure remains Rectangular.
(d). Dense array:- Most of the elements are filled, but still rectangular (opposite to sparse).

64. What is the length of an array when its first index is denoted by i and last index is denoted by j?
एक ऐरे की लंबाई क्या है जब इसके पहले इंडेक्स कोi से निरुपित जाता है और अंतिम इंडेक्स कोj द्वारा विरुपित किया जाता है ?

(a) length of an array = i+j/एक ऐरे की लंबाई=i+j
(b) length of an array = j-i /एक ऐरे की लंबाई=j-i
(c) length of an array = j-i+1
एक ऐरे की लंबाई =j-i+1
(d) length of an array = j-i-1
एक ऐरे की लंबाई=j-i-1

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Calculating the length of an Aray –
To calculate the length of an array. We need to find the difference between the last index and first index, and then add 1. If the first index is 'i' and the last index is j. the length of the array
Would be,
Length = j – i + 1
Hence, the correct answer is (c).

65. Which data structure is used to indicate that the order of the processing of data must be postponed until other conditions are fulfilled?
किस डेटा संरचना का उपयोग यह बताने के लिए किया जाता है कि डेटा की प्रोसेसिंग के क्रम व अन्य शर्तो के पूरा होने तक स्थगित किया जाना चाहिए?

(a) Array/ऐरे (b) Linked list/लिंक्ड लिस्ट
(c) Trees/ट्रीज़ (d) Stack/स्टैक

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) :
(a) **Stack data Structure**– A stack is a linear data structure that follows a particular order in which the operations are performed the order may be LiFO (Last - in, First Out) or FiLO (First - in, Last Out).
(b) LiFO Implies–The element that is inserted first, comes out last it behaves like a stack of plates, where the last plate added is the first one to be removed. Think of it this way (It's value is indeterminate until assigned).
(c) Expression Evaluation–
• i = k / ++v;
• The ++v is a pre-increment operator meaning v is incremented by 1 before its value is used in the expression.
• Initially; v = 8 after ++v, v becomes 9.
• Now, the expression is k/v, which is 8/9.
• Since both k and v are integers, this is integer division, which discard the fractional part.
• 8/9 = 0.888…, but in integer division the result is truncated to 0.
• So, i is assigned the value 0.
(d) Output–
• This prints the value of i as a decimal integer, followed by a newline (\n).
• Since i = 0 the output will be 0 followed by a newline.

66. Which of the following uses only increment operations for adding and removing element at either end?

(a) Queues (b) Stacks
(c) Priority Queues (d) Deques

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Ans. (d) :A deque (Double ended queue) allows insertion and deletion from both ends. It can use only increment operations (like push-front, push-back, pop-front, and pop-back) to add and remove elements efficiently from either end.

67. The entire set of parameters, including return address that is stored for a procedure invocations is referred to

(a) Stack frame (b) Stack base
(c) stack limit (d) stack record

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Ans. (a) : Stack frame:- A stack frame (also known as an activation record) is a structure in the call stack that stores parameters, local variables, returns addresses, and saved registers for a procedure invocation. It helps manage function calls and returns in programs.
Stack base:- The button (usually the highest memory address) of the stack.
Stack record:- While the term "record" might be used loosely, "stack frame" is the started and precise term for the entire set of Information function call on stack.
Therefore, is the correct and stack frame.

68. Consider the following statements :

S_1 : A queue can be implemented using two stacks.

S_2 : A stack can be implemented using two queues.

Which of the following is correct?

(a) S_1 is correct and S_2 is not correct
(b) S_1 is not correct and S_2 is correct
(c) Both S_1 and S_2 are correct
(d) Both S_1 and S_2 are not correct

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Ans :(c) **Explanation of statement 1:** Let us how 2 stack S1& S2 for our purpose. For enqueue operation, we have 2 cases.
(a) S2 is not empty : In that case, we simply pop from this stack
(b) S2 is empty : In that case first we pop all elements from S1 and push into S2. After pushing all the elements so S1 is correct and as same, simply do for S2. So S2 is also correct.
Hence, the answer is option (c).

69. Given the following prefix expression :

*** + 3 + ↑3 + 333**

What is the value of the prefix expression?

(a) 2178 (b) 2199
(c) 2205 (d) 2232

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Ans :(c)Prefix expression = *+3+↑3 + 333
⇒*+3+3∧3(3+3)3
⇒*+3+3∧3(6)3
⇒*+3+3(3∧6)3
⇒*+3+3(729)3
⇒*+3(3+729)3
⇒*+3(732)3
⇒*(3+732)3
⇒*735 3
⇒(735)*3
⇒2205 (ans c)

70. Consider an implementation of unsorted single linked list. Suppose it has its representation with a head and a tail pointer (i.e. pointers to the first and last nodes of the linked list). Given the representation, which of the following operation can not be implemented in O(1) time?

(a) Insertion at the front of the linked list
(b) Insertion at the end of the linked list
(c) Deletion of the front node of the linked list
(d) Deletion of the last node of the linked list

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Ans :(d) Here although we have tail pointer but if will be of no use because these is no previous pointer. We need to here seemed last pointer so that we make it the last pointer & can free the last pointer i.e. deleting the last node. So we need to traverse the whole linked list making it O(n).

71. Which of the following is not an inherent application of stack?

(a) Implementation of recursion
(b) Evaluation of a postfix expression
(c) Job scheduling
(d) Reverse a string

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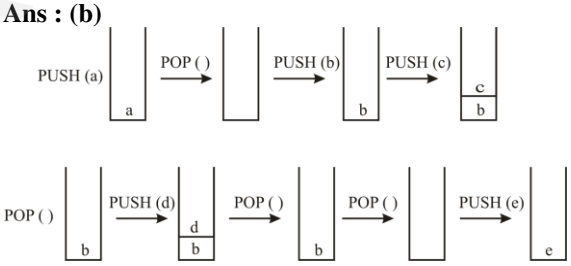
Ans : (c) Application of stack(s) are:
(i) Implementation of recursion.
(ii) Evaluation of a postfix expression.
(iii) Reverse a string.
Job scheduling is commonly done via queue data structure.

72. Consider the following operations performed on a stack of size 5:
Push (a); Pop ( ); Push (b); Push (c); Pop ( ); Push (d); Pop ( ); Pop ( ); Push (e)
Which of the following statements is correct?

(a) Underflow occurs
(b) Stack operations are performed smoothly
(c) Overflow occurs
(d) None of the above

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Ans : (b)



Since, given stack size is 5, so, there is no overflow or underflow.

73. Which of the following statementsis not true about thedoubly linked list?
निम्नलिखित में से कौन-सा कथन Doubly Linked List के बारे में सत्य नहीं है?

(a) We can traverse in both the directions.
(b) It requires extra space.
(c) Implementation of doublylinked list is easier than thesingly linked list.
(d) More than one of the above
उपर्युक्त में से एकअधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : Doubly Linked List is a data structure in which every node is accompanied by two pointers, the first pointer represents the previous node of the list and the second pointer represents the next node of the list. It can be traversed in both direction. It is a combined data structure that allows data to be stored in a large manner it require more space- Implementation of singly linked List is easier them Doubly linked list.

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74. The postfix expression equivalent of the prefix expression * + ab – cd is
प्रिफिक्स एक्सप्रेशन (prefix expression)* + ab – cd का समतुल्य पोस्टफिक्स एक्सप्रेशन (postfix expression)
(a) ab + cd – *
(b) ab + cd + –*
(c) ab + cd * –
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans.(a): The equivalent of prefix expression *+ab-cd will be ab+cd–* in the postfix expression.

75. Which data structure is used to convert infix notation to postfix notation?
इनफिक्स नोटेशन (Infix Notation) को पोस्टफिक्स नोटेशन (Postfix Notation)में बदलने के लिए किस data structure का प्रयोग किया जाता है?
(a) Stack/स्टैक
(b) Queue/क्यू
(c) Tree/ट्री
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans.(a): • To convert Infix → postfix, we use a stack.
• Operators are pushed on to the stack, and based on their precedence and associativity, they are either popped or pushed.
• The process is explained by the shunting yard algorithm developed by Edsger Dijkstra.

76. The data structure in which data are added at the rear and deleted from front is called
वह Data Structure जिसमें Data पीछे (Rear) से जोड़ा जाता है औरआगे(Front)से हटाया जाता है, कहलाता है
(a) Stack
(b) Queue
(c) Linked List
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
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Ans.(b): • In a Queue, insertion happens at the rear (back).
• Deletion happens at the front.
• That's way it follows the the rule FIFO (First In, First Out).

77. Queue data structure uses
Queue data structureप्रयोग करता है
(a) FIFO
(b) LIFO
(c) LILO
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans.(a): Queue data structure follows FIFO (First in First out) principle i.e. first element is inserted in the list and last element is detected.

78. Stack data structure uses
Stack data structure प्रयोग करता है
(a) FIFO
(b) LIFO
(c) LILO
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 2023 (11-12)

Ans. (b): A Stack is a linear data structure that follows the principle of Last In, First Out (LIFO). This means that the element which is inserted last into the stack is the one that will be removed first, just like a pile of plates where the last plate placed on the top is the first to be removed. In other words, insertion and deletion in a stack take place only at one end, called the top of the stack. Hence, the stack data structure uses the LIFO approach.

79. Which of the following is theadvantage of the array datastructure?
निम्नलिखित में से कौन-सा Array Data Structure का लाभ है?
(a) Elements of mixed data types can be stored/ मिश्रित डेटा प्रकार के तत्वों को संग्रहीत किया जा सकता है
(b) Easier to access the elements in an array/ किसी सारणी में तत्वों तक पहुँच आसान हो जाती है
(c) Index of the first element starts from 1/ पहले तत्व का सूचकांक 1 से शुरू होता है
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 1.0 2023 (11-12)

Ans. (b) : Array Data Structure has many advantages such as, it is easy to access the element in an array, array allows to keep similar data together which can save time and space.

80. Which data structure is mainlyused for implementing therecursive algorithm?
Recursive Algorithmको लागू करने के लिए मुख्य रूप से किस Data Structure का उपयोग किया जाता है?
(a) Queue
(b) Stack
(c) Binary tree
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 1.0 2023 (11-12)

Ans. (b) :stack data structure is mainly used to implement Recursive Algorithm. Stack is a data structure that keeps data in LIFO order and is useful for pushing recursion calls.

Sarkari Resource

81. What is the postfix expressionfor the corresponding infixexpression?
सम्बन्धित infixअभिव्यक्ति के लिए postfix अभिव्यक्ति क्या है?
a+ b* c + (d * e)
(a) abc * + de * +
(b) abc + * de * +
(c) a+ bc * de +*
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : Converting Infix Expression to Postfix Expression-
Given infix = a + b* c + (d*e)

Input String	Output Stack	Operator Stack
a	a	
+	a	+
b	ab	+
*	ab	+*
c	abc*	+
+	abc*+	+
(	abc*+	+ (
d	abc*+d	+(
*	abc*+d	+(*
e	abc*+de	+(*
)	abc*+de*	+
	abc*+de*+	

So, for the infix expression a + b * c + (d*e), the postfix expression will be abc* + de* +.

82. Which one of the following is not the type of queue?
निम्नलिखित में से कौन-सा queue का प्रकार नहीं है?
(a) Linear queue
(b) Circular queue
(c) Double-ended queue
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (e) : Queue is a non-primitive and linear data structure. It works on the principle of FIFO (First - in, First Out). There are four types of queue –
1. Liner queue
2. Circular queue
3.Priority queue
4.Double-ended queue (Deque)

83. The necessary condition to bechecked before deletion from thequeue is
Queueसे Deletionसे पहले जाँच की जाने वाली आवश्यक शर्त है
(a) Overflow
(b) Underflow
(c) Rear value
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : The necessary condition to be checked before deletion form the queue is that the queue should not be empty. Delete operation can be done only if there is an item in the queue. Delete operation is not possible if there is no item in the queue attempting to do usually results in an error called underflow.

84. After performing following set ofoperations, what does the final list look contain?
संचालन के निम्नलिखित सेटों को करने के बाद, अन्तिम सूची में क्या शामिल है?
InsertFront(10);
InsertFront(20);
InsertRear(30);
DeleteFront( );
InsertRear(40);
InsertRear(10);
DeleteRear( );
InsertRear(15);
display( );
(a) 10 30 10 15
(b) 20 30 40 15
(c) 10 30 40 15
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : On operating the sets

Insert front (10):	List = [10]
Insert front (20):	List = [20, 10]
Insert front (30):	List = [20, 10, 30]
Delete front ( ) :	List = [10, 30]
Insert Rear (40):	List = [10, 30, 40]
Insert Rear (10):	List = [10, 30, 40, 10]
Delete Rear ( ):	List = [10, 30, 40]
Insert Rear (15):	List = [10, 30, 40, 15]

Hence the element of the final list are [10,30,40,15].

85. A linear data structure in whichinsertion and deletion operationscan be performed fromboth the ends is
एक Linear Data Structure, जिसमें दोनों सिरों से सम्मिलन और विलोपन संचालन किया जा सकता है, है
(a) Queue
(b) Circular queue
(c) Deque
(d) More than one of the above
उपर्युक्त में से एकअधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : Deque (Double-ended queue is a linear data structure in which elements can be added and removed from both the front and rear ends.

86. Let A = 4, B = 3, C = 6 and D = 2, What is value of postfix expression ABCD/*+?
यदि A = 4, B = 3, C = 6 and D = 2 ही तो पोस्टफिक्स एक्सप्रेशन ABCD/*+ का मान होगा?

Sarkari Resource

(a) 21 (b) 13
(c) 22 (d) 10

RPSC C.S. 2016 (Paper-I)

Ans.(b) : Step-1– Replace variables with values expression becomes
4362/*+
Step-2– Evaluate postfix expression using a stage
Postfix means–
• Push operands onto stack
• When operator comes, pop required number of operands, apply operator, push result.
Postfix means–
Expression : 4362/*+
1. Push 4 → stack [4]
2. Push 3 → stack [4, 3]
3. Push 6 → stack [4, 3, 6]
4. Push 2 → stack [4, 2, 6, 2]
5. / → pop2 and 6 → 6/2 = 3 → push → stack [4, 3, 3]
6. * → pop3 and 3 → 3*3 = 9 → push → stack [4, 9]
7. + → pop 9 and 4 → 4 + 9 **= 13**

87. Choose a valid statement to insert a node PTR into a queue pointed by FRONT and REAR pointers?
प्वॉइंटरFRONT व REAR से प्वॉइंटेड एक क्यू में नया नोड PTR इन्सर्ट करने के लिए उपयुक्त स्टेटमेंट है?

(a) REAR → next = PTR ; REAR = PTR;
(b) REAR → next = PTR ; PTR= REAR;
(c) FRONT → next = PTR ; PTR = FRONT;
(d) FRONT → next = PTR ; FRONT= PTR;

RPSC C.S. 2016 (Paper-I)

Ans.(a) :
(a)REAR → next = PTR ; REAR = PTR;
- First, adds PTR to the end of the queue
- Then, update REAR to point to the new last node.
- This is the correct way to insert a node of the rear.

(b) REAR → next = PTR ; PTR= REAR;
- The first part (REAR → next = PTR) is correct as it links the new node.
- But PTR = REAR is incorrect because it overwrites PTR with REAR, losing the reference to the new node and breaking the queue structure.

(c) FRONT → next = PTR ; PTR = FRONT;
- This modifies the FRONT pointer's next which is incorrect for enquire (insertion should happen at REAR, not FRONT)
- Also, PTR = FRONT overwrites PTR, which is wrong.

(d) FRONT → next = PTR ; FRONT= PTR;
- This also incorrectly modifies FRONTnext instead of REAR.
- Updating FRONT = PTR is wrong for enqueue since FRONT should only change if they queue was initially empty.

88. Let a number X denote a Push (X) operation and P denotes POP() operation on stack. What will be sum of elements of stack after applying following sequence of operations on empty stack: 2, 22, 12, 15, P, P, 17, 18, P, P, P, 23
माना एक संख्या X स्टैक पर Push (X) तथा POP() ऑपरेशन को दर्शाते है। एक खाली स्टैक पर ऑपरेशन की निम्न शृंखला लगाने के उपरांत स्टैक के अवयवों का योग क्या होगा: 2, 22, 12, 15, P, P, 17, 18 , P, P, P, 23

(a) 44 (b) 2
(c) 25 (d) 24

RPSC C.S. 2016 (Paper-I)

Ans.(c) : Given
• A number like X means Push (X)
• P means POP()
Sequence– 2, 22, 12, 15, P, P, 17, 18, P, P, P, 23 Let's simulate:
(1) Push (2) → stack = [2]
(2) Push (22) → stack = [2, 22]
(3) Push (12) → stack = [2, 22, 12]
(4) Push (15) → stack = [2, 22, 12, 15]
(5) POP() → removes 15 → stack = [2, 22, 12]
(6) POP() → removes 12 → stack = [2, 22]
(7) Push (17) → stack = [2, 22, 17]
(8) Push (18) → stack = [2, 22, 17, 18]
(9) POP() → removes 18 → stack = [2, 22, 17]
(10) POP() → removes 17 → stack = [2, 22]
(11) POP() → removes 22 → stack = [2]
(12) Push (23) → stack = [2, 23]
final stack : [2, 23]
sum = 2 + 23 ⇒ 25

89. A singly circular linked list without header mode is maintaining pointer to first made (START) and last node (END). If first node is to be deleted valid statements are:
बिना हैडर नोड की एक सिंगली सर्कुलर लिंक्ड लिस्ट अपने प्रथम नोड (START) तथा अंतिम नोड (END) का पॉइंटर मेन्टेन करती है, यदि प्रथम नोड डिलीट किया जाना है तो उपयुक्त स्टेटमेंट होगें

(a) START → NEXT = START ; END = START;
(b) START = START → NEXT ; END = END → NEXT;
(c) END = END → NEXT ; START = END;
(d) START = START → NEXT ; END → NEXT = START;

RPSC C.S. 2016 (Paper-I)

Ans.(d) : In a singly circular linked list (without a header node), where START points to the first node and END points to the Last node, and we want to delete the first node, the correct approach is:
⇒ Move START to the next node:
START = START → NEXT
⇒ Update END → NEXT to point to the new START
This maintains the circular structure.
• START = START → NEXT;
This moves the START pointer to the second node (i.e, deletes the first logically)
• END → NEXT = START;
Ensures the circular link is maintained from last to new first

Sarkari Resource

90. If circular queue is implementedusing array having sizeMAX_SIZE in which array indexstarts with 0, front points to the first element in the queue, and rear points to the last element inthe queue. Which one of thefollowing conditions is used to specify that the circular queue isempty?
यदि परिपत्र कतार को MAX_SIZE आकार वाले सारणी का उपयोग करके कार्यान्वित किया जाता है, जिसमें सारणी सूचकांक 0 से शुरू होता है, समाने कतार में पहले तत्त्व को इंगित करता है और पीछे कतार में अन्तिम तत्त्व को इंगित करता है। निम्नलिखित में से किस शर्त का उपयोग यह निर्दिष्ट करने के लिए किया जाता है कि Circular Queue खाली है?

(a) Front=rear= –1
(b) Front=rear=0
(c) Front=rear +1
(d) More than one of the above
उपर्युक्त में से एकअधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : If circular queue is implemented using array of size max-SIZE where in array index stats from 0, front points to the first element in queue then using front = Rear = –1 specifies that circular queue is empty, here both front and rear are set to 1 which means there is no element in circular queue.

91. Given the list of tasks in Col-A and list of data structure in Col-B. Identify the best match.
कॉलम-A में कार्यों की सूची दी गई है तथा कॉलम-B में डाटा स्ट्रक्चर की सूचना दी गई है। सही मिलान कीजिए:

A		B	
A.	Recursion रिकर्शन	(i)	Binary tree बाइनरी ट्री
B.	Scheduling शेड्यूलिंग	(ii)	Stack/स्टैक
C.	Sorting/सॉर्टिंग	(iii)	Queue /क्यू

	A	B	C
(a)	(i)	(iii)	(ii)
(b)	(ii)	(i)	(iii)
(c)	(iii)	(ii)	(i)
(d)	(ii)	(iii)	(i)

KVS PGT C.S. 2018

Ans.(d) : Best match of Col-A and Col-B
Recursion – Stack
Scheduling – Queue
Sorting – Binary tree.

92. Recursive procedures is implement using:
रिकर्सिव प्रक्रिया को किसकी सहायता से लागू किया जाता है?

(a) String/स्ट्रिंग (b) Queue/क्यू
(c) Stack/स्टैक (d) Linked List/लिंक्ड लिस्ट

KVS PGT C.S. 2017

Ans.(c) : When a function calls itself in the program then it is known as recursion It is implemented by using a stack data structure when a function calls another function or itself, then control is transferred to the called function and the data of the caller function is stored on the stack so that of after processing of called function it can resume its execution where it put itself on hold. hence, the Recursive procedure is implemented using Stack.

93. In pointer implementation of linked lists, which operation can be performed in constant time (O(1)) when inserting or deleting nodes?
लिंक्ड लिस्ट के पॉइंटर इम्प्लीमेंटेशन में, जब नोड्स को जोड़ने या हटाने की बात आती है, तो कौन-सा ऑपरेशन स्थिर समय (O(1)) में किया जा सकता है?

(a) Inserting at the beginning/शुरुआत में जोड़ना
(b) Inserting at the end/अंत में जोड़ना
(c) Inserting at a specific position
किसी विशिष्ट स्थिति पर जोड़ना
(d) Deleting a specific node
किसी विशिष्ट नोड को हटाना

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : In linked lists, inserting at the beginning can be perfumed in constant time, 0 (1), because it only requires updating the head pointer to the new node and setting the new node's next pointer to the current head. This operation does not depend on the size of the list, making it efficient for adding elements at the start.

94. A..... is a container that implements Last-In-First-Out (LIFO) protocol.
यह एक कंटेनर है जो लास्ट-इन-फर्स्ट-आउट (LIFO) प्रोटोकॉल लागू करता है–

(a) Queue (b) Stack
(c) Object (d) Interface

HTET PGT C.S. 30.07.2025

Ans.(b): Stack, implements LIFO (last -In- First - out).

95. Toll Plaza simulation is an application of ..... .
टोल प्लाजा सिमुलेशन ..... का अनुप्रयोग है।

(a) Queue (b) Stack
(c) Object (d) Class

HTET PGT C.S. 30.07.2025

Ans.(a): Queue : follows FIFO (first -in-First-out). Which aligns with the processing order in a toll plaza.

96. Which data structure is mainly used for implementing the recursive algorithm?
कौन सी डेटा संरचना मुख्य रूप से पुनरावर्ती एल्गोरिथ्म को लागू करने के लिए उपयोग की जाती है?

(a) Stack (b) Queue
(c) Tree (d) Array
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Stack, when a recursive function is called, the system needs to keep track of the state of each function call, including local variables, parameters and the return address to resume execution after the current call completes. This information is stored in an activation record or stack frame which is then pushed on to the call stack.

Sarkari Resource

97. Which of the following isa postfix expression?
निम्नलिखित में से कौन-सा पोस्टफिक्स एक्सप्रेशन है?

(a) +an-c (b) a–b*c
(c) abc*+de-+ (d) (a+b)*(a-c)
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) :A postfix expression is a mathematical notation where operators follow their operands. In this format, there is no need for parentheses to define the order of operations, as the order is determined by the position of the operators and operands. So the correct answer is (c) = abc* + de – +.

98. Initially a stack is empty. What shall be the content of the stack after the following operations?
PUSH (A)
PUSH (B)
PUSH (C)
POP
PUSH (D); POP; POP;
PUSH (E)
PUSH (F)
POP

(a) ABE (b) AE
(c) A (d) ABCE

CGPSC C.S. 2020 Set-A

Ans.(b): Step 1: Initial state and first push operations
Initially, the stack is empty.
PUSH (A) : stack : [A]
PUSH (B) : stack : [A,B]
PUSH (C) : stack : [A,B,C]
Step 2 : first pop operation
POP : removes c. stack : [A, B]
Step 3: subsequent PUSH and POP operations
PUSH (D) : stack : [A, B, D]
POP : Removes D. stack : [A, B]
So that correct answer- AE

99. Stack is useful for implementing:

(a) Radix sort (b) Breadth first search
(c) Quick sort (d) Recursion

CGPSC C.S. 2020 Set-A

Ans.(d):Recursion:- stacks are fundamentally used in implementing recursion as they manage the function call frames and local variables during recursive calls.

100. Which of the following is used for the evaluation of a prefix expression?

(a) Queue (b) Heap
(c) Stack (d) Hash

CGPSC C.S. 2020 Set-A

Ans. (c) :stack, A stack is the Primary data structure used for the evaluation of Prefix expressions (also known as Polish notation).
Operands are pushed onto the stack and when an operator is encountered, the necessary number of operands are popped, the operation is performed and the result is pushed back onto stack.

101. A linear list in which elements can be added or removed at either end but not in the middle is:

(a) Queue (b) Dequeue
(c) Array (d) Tree

CGPSC C.S. 2020 Set-A

Ans. (b) : Dequeue (or Deque) short for Double –ended queue elements can be added or removed at both ends so that correct answer – dequeue.

102. TOLL Plaza simulation is an application of_____.
टोल प्लाजा सिम्युलेशन____के अनुप्रयोग के रूप में है।

(a) Stack/स्टैक (b) Queue/क्यू
(c) Linked List/लिंक्ड लिस्ट (d) Graph/ग्राफ

HTET PGT C.S. 2023

Ans.(b): A TOLL Plaza simulation typically models the behavior of vehicles lining up to pass through toll booths. In such a scenario:
•Vehicles arrive and wait their turn to be served (i.e., pay toll and pass through)
•The First Come, First Served (FCFS) principle is applied– the first vehicle to arrive is the first to be processed.
This behavior matches the Queue data structure, which follows FIFO (First-In-First-out) ordering.
Other options:
• **(a) Stack** –Follows LIFO (Last-In-First-out); not suitable for toll simulations.
• **(c) Linked List** –A data structure used to implement queues or stacks but doesn't directly represent the toll simulation behaviour.
• **(d) Graph** –Used for network representations, routing, etc., not ideal for modeling queue, scenarios.

103. Let the postfix expression 3, 2, +, 7, *,6, 5, +, 4, *, – is being evaluated using astack. What are the values on top andbottom of stack respectively, just beforesecond + operator is evaluated ?
माना पोस्टफिक्स एक्सप्रेशन 3, 2, +, 7, *,6, 5, +, 4, *, – का मान एक स्टेक की सहायता से ज्ञात किया जा रहा है। दूसरें + ऑपरेटर को हल करने से ठीक पहले, स्टेक में सबसे ऊपर तथा सबसे नीचे क्रमशः क्या मान होंगे?

(a) 27, 5 (b) 11, 35
(c) 11, 27 (d) 5, 35

HTET PGT C.S. 2018

Ans. (d) : We need stack contents just before the second + operator is evaluated.
1. Process expression step by step
• 3 → Push → stack:[3]
• 2 → Push → stack:[3,2]
• + → Pop 2 & 3 →2+3 = 5 → push → stack [5] (This is the first +)
• 7 → push → stack: [5,7]
• * → Pop 7&5→5×7 = 35→push →stack [35]
• 6 → push → stack: [35,6]
• 6 → push →stack : [35, 6, 5]
Now the next symbol is + → this is the second +operator
2. Stack state just before second + is evaluated
stack: [35, 6, 5]
• Top = 5
• Bottom = 35

Sarkari Resource

104. A linear list of elements in whichdeletion can be done from one end(front) and insertion can take placeonly at the other end (rear), is knownas :
तत्त्वों की एक रैखिक सूची जिसमें एक छोर (सामने) से एलिमेंट को हटाया या विलोपन किया जा सकता है और दूसरे छोर (पीछे) पर ही निवेशन हो सकता है, किस रूप में जाना जाता है:

(a) Queue/क्यू (b) Stack/स्टैक
(c) Tree/ट्री (d) Linked list/लिंक्ड सूची

HTET PGT C.S. 2021

Ans. (a) :A queue is linear data structure where elements are inserted at the rear end and deleted from the front end. This follows the First-In-First-Out (FIFO) principle, meaning the first element inserted is the first one to be removed.
- Insertion (enqueue) occurs only at the rear.
- Deletion (dequeue) occurs only at the front.

105. In a stack, if a user tries to removean element from an empty stack, it iscalled :
एक स्टैक में, यदि कोई उपयोगकर्ता (यूजर) किसी खाली स्टैक से किसी एलिमेंट (तत्त्व) को हटाने का प्रयास करता है, तो उसे क्या कहा जाता है?

(a) Underflow/अंडरफ्लो
(b) Empty collection/खाली संग्रह
(c) Overflow/ओवरफ्लो
(d) Garbage collection/गार्बेज संग्रह

HTET PGT C.S. 2021

Ans. (a) : In a stack, if a user tries to remove an element (pop operation) from an empty stack, this situation is known as "underflow". It is an error condition that occurs when there are no elements in the stack to be removed.

106. In the given declaration of an array :float height[50];
एक ऐरे के निम्न डिक्लेरेशन में: float height[50];

(a) height[0] refers to first elementof array.
height[0]ऐरे के पहले तत्त्वको संदर्भित करता है।
(b) height refers to first element ofarray.
height ऐरे के पहले तत्त्व को संदर्भित करता है।
(c) height[0 + 1] refers to first elementof array.
height[0 + 1]ऐरे के पहले तत्त्व को संदर्भित करती है।
(d) height[1] refers to first elementof array.
height[1]ऐरे के पहले तत्त्व को संदर्भित करता है।

HTET PGT C.S. 2021

Ans. (a) :float height [50] means an array named height with 50 elements of type float.
In C language, array indexing starts from 0, not 1.
So,
height [0] refers to first element of array.
height [1] refers to second element of array.
height [4] refers to fifth element of array.

107. A queue follows ……. .
एक क्यू ……..का अनुसरण करती है।

(a) FIFO (First In First Out) principle
फीफो (फर्स्ट इन फर्स्ट आउट) सिद्धान्त
(b) LIFO (Last In First Out) principle
लीफो (लास्ट इन फर्स्ट आउट) सिद्धान्त
(c) Ordered array/आदेशित ऐरे
(d) Linear tree/रैखिक ट्री

HTET PGT C.S. 2021

Ans. (a) :A queue is a fundamental data structure that operates on the FIFO (First In First Out) principle. This means the first element added to the queue will be the first one to be removed.

108. Process of inserting an element instack is called:
स्टैक में एलिमेंट डालने की प्रक्रिया क्या कहलाती है?

(a) Create/क्रिएट (b) Push/पुश
(c) Evaluation/मूल्यांकन (d) Pop/पॉप

HTET PGT C.S. 2021

Ans. (b) : Push → Means to put (insert) a new element on the top of the stack.
Pop⟶Means to remove the top element from the stack.
Create → Means to initialize or make a new stack.
Evaluation → Means to calculate or process an expression (not related to insertion)
So,
Push is the right answer.

109. If the elements 'A', 'B', 'C' and 'D' areplaced in a queue and are deleted oneat a time, in what order will theseelements be removed ?
यदि तत्त्वों 'A', 'B', 'C' तथा 'D'को एक कतार (क्यू) में रखा जाता है और उन्हें एक-एक कर हटाया जाता है, तो इन तत्त्वों को किस क्रम में हटाया जायेगा?

(a) ABCD (b) DCBA
(c) DCAB (d) ABDC

HTET PGT C.S. 2021

Ans. (a) :A queue works on the FIFO principle .
This means the first element added to the queue will be the first one to be removed.
So,
Order of insertion → A B C D
Order of deletion → A B C D

(iii) Non-Linear Data Structures

110. A threaded binary tree is a binary tree in which:
एक थ्रेडेड बाइनरी ट्री वह होता है जिसमें:

(a) Each node has two children
प्रत्येक नोड के दो बच्चे होते है
(b) Each node has at most one child
प्रत्येक नोड के अधिकतम एक बच्चा होता है
(c) Each node is connected to its parent
प्रत्येक नोड अपने पैरेंट से जुड़ा होता है
(d) Each node has a thread connecting it to its predecessor or successor/प्रत्येक नोड अपने पूर्ववर्ती या उत्तरवर्ती से जुड़ी एक थ्रेड रखता है

Bihar STET C.S. 12.09.2023 (Shift-I)

Sarkari Resource

Ans. (d) : Threaded Binary Tree: A Binary tree in which some of the null pointers are replaced by threads that point to the in-order predecessor or in-order success of the node. This allows for more efficient traversal without using recursion or stack.
In a threaded binary tree:
If a node has a left child, the left pointer points to the left child, otherwise, it points to the in-order predecessor.
If a node has a right child, the right pointer point to the right child, otherwise, it points to the in-order successor.
So, the correct option is (d).

111. The order of the binary search algorithm is_____:
बाइनरी सर्च एल्गोरिथम का क्रम है _____
(a) N
(b) N log n
(c) N^2
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans.(e) : The binary search algorithm works efficiently on a sorted list by repeatedly dividing the search range in half. First, it checks the middle element of the list. If the target value matches this element, the search ends immediately. If the target is smaller, the search continues in the left half; if it is larger, the search continues in the right half. Each step cuts the remaining elements to half of their previous size. Starting with n items, after one step the size is about n/2, after two steps about n/4 and so on, until only one element remains. The number of steps needed is therefore about $\log_2 n$, because 2 raised to this number of steps equals n.
In the best case, when the first middle element is the target, the time is O(1). In the worst and average cases, the time grows only as $O(\log_2 n)$.
Thus the order of binary search algorithm is $O(\log_2 n)$.
So, the correct option is (e) None of the above.

112. A multimedia project is said to be _____and user-interactive when users are give navigational control.
किसी मल्टीमीडिया परियोजना को _____ और यूजर-इंटरेक्टिव कहा जाता है जब प्रयोक्ता को नेविगेट करने के लिए कंट्रोल दिया जाता है।
(a) Secure/सुरक्षित
(b) Hypertext/हाइपरटेक्स्ट
(c) Linear/रैखिक (लिनियर)
(d) Non-linear/अरैखिक

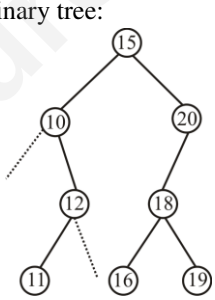
KVS PGT C.S. 2017

Ans, (d) : A multimedia project is said to be Non-linear and user-interactive when users are given navigational control. Multimedia project are the multimedia materials which are presented on computer screen. There are two categories in multimedia ; linear and non-linear.

113. The preorder traversal of a binary search treeis 15, 10, 12, 11,20, 18, 16, 19. Which oneof the following is the postorder traversal ofthe tree?
बाइनरी सर्च ट्री का प्रीऑर्डर ट्रैवर्सल 15, 10, 12, 11, 20, 18, 16, 19 है।
निम्नलिखित में से कौन-सा ट्री का पोस्टऑर्डर ट्रैवर्सल है?
(a) 20, 19, 18, 16, 15, 12, 11, 10
(b) 11, 12, 10, 16, 19, 18, 20, 15
(c) 19, 16, 18, 20, 11, 12, 10, 15
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans.(b) : Preorder traversal (Root→Left→Right)
15, 10 , 12, 11, 20, 18, 16, 19
Root
In order traversal (Left→Root→Right) gives numbers in ascending order.
In order traversal
10,11,12, 15 ,16,18,19,20
Left Root Right
Construction of Binary tree:



Post-order traversal: Left→Right→Root
11, 12, 10, 16, 19, 18, 20, 15

114. How many distinct binary search trees can be created out of 4 distinct keys?
4 अलग-अलग कुंजियों से कितने अलग-अलग बाइनरी सर्च ट्री बनाए जा सकते हैं?
(a) 8 (b) 24
(c) 14
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans.(c) : The number of binary search tree with n distinct keys = $\frac{2n!}{(n+1)!n!}$
On putting n = 4
Total number of binary search tree = $\frac{(2\times4)!}{(4+1)!4!}$

$= \frac{8!}{5!4!} = \frac{8\times7\times6\times5!}{5!4\times3\times2\times1} = 14$

So, the correct option is (c) 14

115. Which type of best first searchalgorithm was used to predict thecloseness of the end of path and itssolution ?
पथ के अंत और उसके समाधान की निकटता की भविष्यवाणी करने के लिए किस प्रकार का सबसे अच्छा पहला खोज एल्गोरिथम का उपयोग किया गया था?

Sarkari Resource

(a) Greedy BFS (b) Divide and Conquer
(c) Heuristic BFS (d) Combinatorial

HTET PGT C.S. 2020

Ans. (a) : The algorithm used to predict the closeness of the end of the path and its solution is the **Greedy Best-First Search (Greedy BFS).**
Greedy BFS uses a heuristic function to estimate the cost from the current node to the goal. It prioritizes nodes that appear to be closest to the goal based on this heuristic (e.g., Euclidean or Manhattan distance in pathfinding problesm). This makes it efficient for predicting the closeness to the solution, though it does not guarantee the shortest path.

116. Which of the following is a supervised learning algorithm?/निम्नलिखित में से कौन-सा पर्यवेक्षित (suprevised) शिक्षण एल्गोरिथम है?

(a) Decision Tree
(b) K-means Clustering
(c) Reinforcement Learning
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : A Decision Tree is a supervised learning algorithm because it is trained using labeled data, where both inputs and their correct output are provided.
K-mans clustering is an unsupervised learning method, as it groups data without using labels. Similarly.
Reinforcement Learning is not supervised; instead, it learns by interacting with an environment and receiving rewards or penalties.

117. Let QSPTRU and PQSRTU are inordertraversal and pre-order traversalrespectively for a given binary tree T.What will be the post-order traversal T ?
माना QSPTRU तथा PQSRTU क्रमशः एक दिए हुए बाइनरी ट्री T के इन-आर्डर तथा प्री- आर्डर ट्रैवर्सल है। T का पोस्ट-आर्डर ट्रैवर्सल क्या होगा?

(a) SQUTRP (b) QSUTRP
(c) SQTURP (d) QSUTPR

HTET PGT C.S. 2018

Ans. (c) : Given - Inorder (LNR): QSPTRU
Preoder (NLR): PQSRTU
Step 1: Indentifying the root
The first node of the preorder = P ⇒ Root is P
In Inorder (QSPTRU), the left - subtree of P = QS (left subtree), right = TRU (right subtree)
Step-2: Left subtree (QS)
The left-subtree node after the root P in the preorder = QS
Root = Q (because Q comes first in the preorder)
In inorder QS,Q is left, S is right
⇒ Left subtree postorder = SQ
Step - 3: Right Subtree (TRU)
The remaining nodes in the preorder = RTU
Root = R.
Inorder TRU. left = T, Right = U
⇒ Right subtree postorder = TUR
Step-4: Postorder of complete Tree
Postorder = (Left subtree)+(Right subtree)+(Root)
= SQ+TUR+P = SQTURP

118. A binary tree T has n leaf nodes. The number of nodes of degree two in T is:

(a) n (b) n – 1
(c) log n (d) n + 1

CGPSC C.S. 2020 Set-A

Ans.(b) : In a binary tree if **T**has**n** leaf nodes, the number of nodes with degree two (internal nodes with two children) is (n-1). This is a property of binary trees where the number of internal nodes with two children is always one less than the number of leaf nodes.

119. A full binary tree with a non-leaf nodes contains:

(a) 2n nodes (b) $\log_2 n$ nodes
(c) n + 1 nodes (d) 2n+ 1 nodes

CGPSC C.S. 2020 Set-A

Ans.(d) : For a full binary tree where **n** represents the number of nodes is 2n-1. In a full binary tree, every node has either zero or two children. If **n** is the number of internal nodes the number of leaves is **n+1** and the total number of nodes is n+(n+1) = 2n+1.

120. A complete binary tree of level 5 has:

(a) 20 nodes (b) 63 nodes
(c) 30 nodes (d) 73 nodes

CGPSC C.S. 2020 Set-A

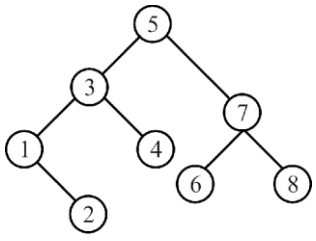
Ans.(b) : The number of nodes in a complete binary tree is determined by the formula $2^{L+1}-1$, where L represents the level of the tree.
For a complete binary tree of level 5, the value of L is 5.
Substituting L is 5.
Substituting L = 5 in to the formula, the calculation becomes $2^{5+1}=1$
This simplifies to 2^6-1
The value of 2^6 is 6^4
Therefore, the total number of nodes is 64 – 1 = 63
The correct Answer = 63 nodes.

121. A binary search tree contains the values 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in preorder and the value are printed out. Which of the following is a valid output?

(a) 53124786 (b) 53126487
(c) 53241678 (d) 53124768

CGPSC C.S. 2020 Set-A

Ans. (d):



Option verify and try to draw each preorder and in order of a binary search tree. Only option will correct the binary tree.

122. The number of null pointers of a binary tree of n nodes is:

(a) n + 1 (b) n (n + 1)
(c) n^2 (d) 2n

CGPSC C.S. 2020 Set-A

Sarkari Resource

Ans.(a): (n + 1), in a binary tree with 'n' nodes the number of null pointers (or empty child pointers) is always n + 1. This is because each node in a binary tree can have at most two children, and the 'extra'

123. Which data structure cannot have cycles?

(a) Graph (b) Linked lists
(c) Tree (d) Circular queue
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : If there is only one Path between any two vertices of a graph, then that graph is a tree. A tree is a connected graph that has no cycles, and because of this property there is a unique one way path between every of vertices.

124. Which of the following processes is faster for threaded trees compared with their unthreaded counterparts?

(a) Insertion
(b) Deletion
(c) Finding predecessor or successor
(d) Update
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : Finding Predecessor or successor. Threaded binary trees are designed to make traversal easier and more efficient. Particularly for finding the in order Predecessor and successor of a node without using a stack or recursion. In a standard binary tree. Finding the predecessor or successor might require traversing up the tree or using a stack to keep track of visited nodes.

125. Level of any node of a tree is
किसी भी नोड का स्तर एक ट्री में है?

(a) Its distance from the root.
(b) Height of its left subtree minus height of its right subtree.
(c) Height of its right subtree minus height of its left subtree
(d) Height of its right subtree plus height of its left subtree.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Its distance from the root. The correct answer. The level of a node in a tree is its depth, defined as the number of edges from the root to that node. In tree data structures, the local of a node is determined by how for it is from the root node. The root node is at level (00.1) depending on the convention, but it's always a fixed distance from itself.

126. Which of the following is NOT a valid operation on a binary search tree?
निम्न में से कौन-सा बाइनरी सर्च ट्री पर वैध ऑपरेशन नहीं है?

(a) Sorting elements
(b) Searching an element
(c) Inserting a new element
(d) Deleting an existing element
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Sorting elements, as a Binary search Tree (BST) is not designed for sorting elements directly. While an in-order traversal of a (BST) yields traversal of a BST yields sorted elements, it's the traversal method itself that performs the sort, not a fundamental operation of the BTS data structure for the purpose of sorting.

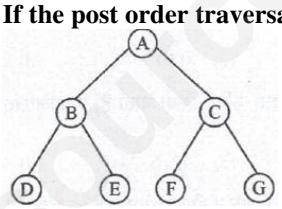
127. Which of the following is a non-linear data structure?
निम्न में से कौन नान-लिनियर डाटा स्ट्रक्चर है?

(a) Linked list /लिंक्ड लिस्ट (b) Array/ऐरे
(c) Graph/ग्राफ (d) Queue/क्यू

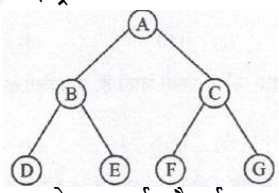
UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : Graph. Graphs are non-linear data structures because there elements can be interconnected in a non-sequential, or complex manner unlike linear structures such as linked lists, Arrays, and Queues which arrange elements in a simple sequence.

128. If the post order traversal of tree



gives ab — cd * +, then the label of the nodes A, B, C....., G will be
यदि ट्री



का पोस्ट आर्डर ट्रैवर्सल ab — cd * +,है तो नोड A, B, C....., G का लेबल होगा-

(a) a,-.,b,+,c,*,d
(b) +, —, *, a, b, c, d
(c) —, a, +, b, c, d, *
(d) a,b,c;d,—,*,+

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : Post order expression ab – cd* +.
Post order expression tree can be convert
ab – means (a – b)
cd* means (c*d)
(a – b) (c * d) + means
$\left[(a-b)+(c*d)\right]$
Post order traversal – a, b, c, d leaf noads.
-noad a and b parent.
* noad c and d parent.
+ noad – and * parent.
noads A, B, C,… G level. where noads tree different so correct answer.(+, –, *, a, b, c, d) is correct.

129. Pruning a decision tree always

(a) Increase the error rate
(b) Reduce the size of the tree
(c) Provide partitions with lower entropy
(d) Provide partitions with higher entropy

RPSC C.S. 2016 (Paper-II)

Sarkari Resource

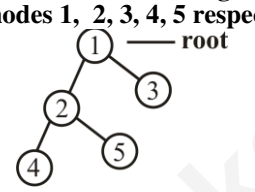
Ans. (b) : Pruning a decision tree is the process of removing parts of the tree that do not provide significant power in predicting target values. It is done to.
- Avoid overfitting.
- Improve generalization to unseen data.
- Simplify the model.

130. Which of the following statement is not valid about binary search?
(a) It perform search in O(log2n) time
(b) It constructs a Binary search tree
(c) It is applicable only on a sorted list
(d) It uses divide and conquer strategy

RPSC C.S. 2016 (Paper-II)

Ans.(b) : Examine the all statements about binary search tree.
1. **Search Time Complexity–** Binary search indeed performs search operations in O(log2n) time, making it efficient for large datasets.
2. **Constructing a Binary Search Tree–** Binary Search itself does not construct a binary search tree, its an algorithm for finding on element in a sorted array.
3. **Applicability–** Binary search is applicable only on sorted lists, which is a fundamental requirements for its operation.
4. **Strategy–** Binary search uses the divide and conquer strategy dividing the search space in half with each comparison.

131. Consider the tree T given below.
If post order traversal of T gives BCEAD, the labels of nodes 1, 2, 3, 4, 5 respectively are:–



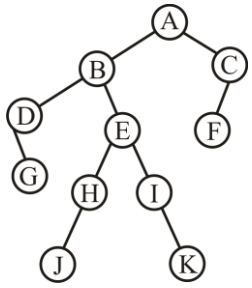
Tree - T

(a) DEBAC (b) DBEAC
(c) EDABC (d) DEABC

RPSC C.S. 2016 (Paper-II)

Ans.(d) : Find the labels of the nodes of a tree given its post-order traversal.
Step-1–The last element in the post-order traversal is the root, A.
B and C are in the left subtree and E is in the right subtree. Since B and C are together in the post-order C is the right child of B
The tree structure is- A, B, C, E
Step-2– Node 1 (root) A
Node 2 B
Node 3 E
Node 4 C
Node 5 D
Step-3– The labels of nodes 1, 2, 3, 4, 5 are D, E, A, B, C respectively.

132. A node of rooted binary tree is said to have preorder number l, if the node occurs in the lth position in the preorder traversal sequence. The notions of postorder number and inorder number are defined in a similar manner. We use the notations PRE(X), POST(X) and IN(X)to denote the preorder number, postorder number and inorder number of a node X respectively. Consider the following tree:



For the tree given above which of the following is TRUE.
(a) (PRE (A), POST (H), IN(F)) = (1, 4, 9)
(b) (PRE (B), POST (I), IN(D)) = (2, 6, 1)
(c) (PRE (C), POST (D), IN(E)) = (10, 3, 6)
(d) (PRE (J), POST (K), IN(G)) = (7, 5, 3)

RPSC C.S. 2016 (Paper-II)

Ans.(b) :Step -1 Understand the Traversal Types
- **Preoder (PRE):** Visit the root, then traversal the left subtree, then the right subtree.
- **Inorder (IN) :** Traverse the left subtree, visit the root. Then traverse the right subtree.
- **Postorder (POST):** Traverse the left subtree, traverse the right subtree, then visit the root.

Step -2 Perform traversals on the given Tree
- **Preocrder (PRE) :** A = 1, B = 2, D = 3, G = 4, E = 5, H = 6, J = 7 , I = 8 , K = 9, C = 10, F = 11
- **Inorder (IN) :** D = 1, G = 2, B = 3, H = 4, J = 5, E = 6, I = 7, K = 8, A = 9, C = 10, F = 11
- **Postorder (POST) :** G = 1, D = 2, J = 3, H = 4, K = 5, I = 6, E = 7, B = 8, F = 9, C = 10, A = 11

Step - 3 Comparing the calculated value with the options.
Option (b)matches: PRE (B) = 2, POST (I) = 6, IN (D) = 1

133. Which one is a non-linear data structure?
निम्न में से कौन non-linear डेटा संरचना है?
(a) Stack (b) Queue
(c) Linked list (d) Tree

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : In Linear Data structure data is stored in sequential manner Each element has a next and a previous element such as stack, Queue, Linked list.
In non-Linear data structure data is stored in hierarchical or non-sequential manner. Such as Tree. Graph etc.

134. Which traversal technique is used to visit each node of a linked list exactly once?
लिंक्ड लिस्ट के प्रत्येक नोड को केवल एक बार विज़िट करने के लिए कौन-सी ट्रैवर्सल तकनीक का उपयोग किया जाता है?
(a) Preorder traversal/प्रीऑर्डर ट्रैवर्सल
(b) Inorder traversal/इनऑर्डरट्रैवर्सल
(c) Postorder traversal/पोस्टऑर्डरट्रैवर्सल
(d) Linear traversal/रैखिकट्रैवर्सल

Bihar STET C.S. 12.09.2023 (Shift-II)

Sarkari Resource

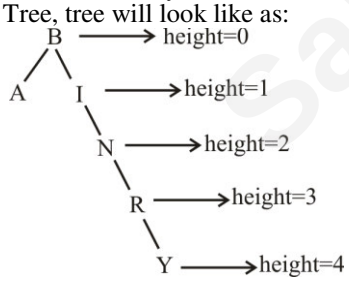
Ans. (d) : In a linked list, the nodes are connected one after another in a linear sequence. To visit each node exactly once, We start from the head node and move step by step to the next node until we reach the end of the list. This method of going through all elements one by one in order is called linear traversal. Unlike trees, where we use preorder, inorder, or postorder traversals, a linked list always requires a linear traversal technique. Therefore, the correct answer is linear traversal.

135. For a tree with just one node, the root node is the height of a binary tree is defined to zero, if there are 2 levels of nodes, the height is 1 and so on.
A Binary Search Tree is built according to the usual rules with the following six key inserted one at a time as given :
B, I, N, A, R, Y. What is the height of the tree?
केवल एक नोड के वृक्ष (ट्री) के लिए, जड़ नोड, द्वयी वृक्ष की ऊँचाई शून्य परिभाषित की गई है, यदि इसमें नोड्स के दो स्तर है, ऊँचा 1 और इसी प्रकार आगे होगी ।
एक द्वयी खोज वृक्ष निम्नलिखित छह कुंजियों, जिन्हें यहाँ दिए गए अनुसार एक समय पर एक अंतस्थापित किया जाता है, के अनुसार सामान्य नियमों से बनाया जाता है :
B, I, N, A, R, Y । वृक्ष की ऊँचाई क्या है?

(a) 5 (b) 2
(c) 3 (d) 4

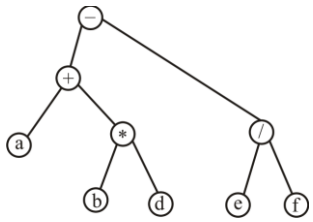
KVS PGT C.S. 2017

Ans. (d) : Binary Search Tree is a tree in which all nodes in the left subtree of root are smaller than the root and nodes in the right subtree are greater or equal to the root node.
Given keys are:
B, I, N, A, R, Y
Sorted keys: A, B, I, N, R, Y
When these keys are inserted to form a Binary Search Tree, tree will look like as:



So, height of this Binary Search Tree will be 4.

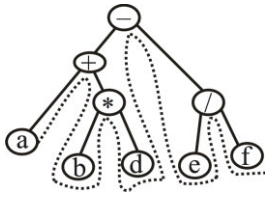
136. For the following binary tree inorder traversal yields the expression :
निम्नलिखित द्वयी वक्ष (बायनरी ट्री) के लिए ठीक क्रम में (इनआर्डर) अनुप्रस्थ (ट्रैवर्सल) से प्राप्त होने वाला व्यंजक है :



(a) – + * / abdef (b) a + bd * – ef /
(c) abdef * / + – (d) a + b * d – e / f

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Ans.(d) : Given binary tree-
In the in-order traversal method, the left subtree is visited first, then the root and then the right subtree. So, in-order traversal is.
A + b * d – e / f



137. For which of the following tasks, stack is not a suitable data structure?
निम्नलिखित में से कौन से कार्यों के लिए स्टैक डाटा संरचना नहीं है?

(A) Binary search in an array/किसी ऐरे में बाइनरी सर्च
(B) Breadth first search/ब्रिद्थ फर्स्ट सर्च
(C) Implementing function calls/इम्प्लीमेंटिंग फंक्शन कॉल्स
(D) Process scheduling/प्रॉसेस शेड्यूलिंग

(a) (B) and (D)/(B) तथा (D)
(b) (B) and (C)/(B) तथा (C)
(c) (A) and (C)/(A) तथा (C)
(d) (C) and (D)/(C) तथा (D)

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Ans.(a) : Breadth first search (BFS) is an algorithm for searching a tree data structure for a node that satisfies a given property. It starts at the tree root and explores all nodes at the present depth prior to moving on to the nodes at the next depth level. Extra memory, usually a queue, is needed to keep track of the child nodes that were encountered but not yet explored.
Process scheduling is implemented using the queue data structure. Ready queue is maintained for the processes which are ready for the execution.

138. Following numbers (items) are inserted in order into binary search tree.
40, 60, 50, 33, 55, 11
Then the number of items in left and right sub trees of the root are _____.
निम्नलिखित संख्याएँ (मदें) बाइनरी सर्च ट्री में इस क्रम से डाली गई हैं।
40, 60, 50, 33, 55, 11
तो रूट के बाएँ तथा दाएँ सब-ट्री में मदों की संख्या _____ होगी।

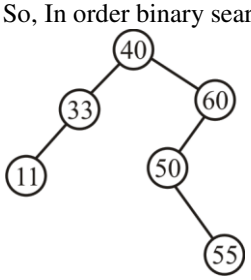
(a) (3, 3) (b) (2, 3)
(c) (3, 2) (d) (2, 4)

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Ans.(b) : Binary search Tree (BST) is a rooted binary tree data structure with the key of each internal node being greater than all the keys in the respective node's left subtree and less than the ones in its right subtree

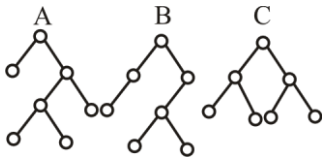
Sarkari Resource

Given items:- 40, 60, 50, 33, 55, 11
So, In order binary search tree-



So, 40 is the root of this binary search tree the number of items in left subtree is 2 (11, 33) and number of items in Right subtree is 3 (60, 50, 55)

139. Consider the following three binary trees, each with 7 nodes.
निम्नलिखित तीन बाइनरी ट्री पर विचार कीजिए, प्रत्येक में 7 नोड है।



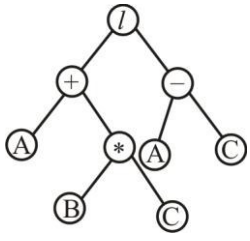
Then A is _____ binary tree, B is _____ binary tree and C is _____ binary tree.
तो A _____ बाइनरी ट्री है, B _____ बाइनरी ट्री है तथा C _____ बाइनरी ट्री है।

(a) strictly, strictly, complete
(b) complete, not strictly, strictly
(c) strictly, not strictly, strictly
(d) strictly, not strictly, complete

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Ans.(d) :A binary tree is identified as a strictly binary tree if each parent node contains either no or two children. All nodes contain in a strictly binary tree except the leaf nodes which have a children.
A complete binary tree is a special type of binary tree where all the levels of the tree are filled completely except the lowest level nodes which are filled from as left as possible.
Hence, the given A is strictly binary tree, B is not strictly binary tree and C is complete binary tree.

140. Given the following expression tree, identify the correct arithmetic expression:
नीचे एक एक्सप्रेशन ट्री दिया गया है। सही अंकगणितीय व्यंजक की पहचान कीजिए:

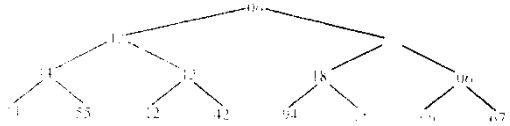


(a) (A+B)*/A–C
(b) A+(B*C)/(A–C)
(c) A+B*C/A–C
(d) (A+B*C)/(A–C)

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Ans.(b) : The expression tree is a binary tree in which each external or leaf node corresponds to the operand and each internal or parent node corresponds to the operators so, the used infix expression evaluation, Hence the correct arithmetic
expression is A + (B * C)/ (A – C).

141. The figure below represents a ___ sort.
निम्न आकृति एक _____छाँट (sort)को प्रदर्शित करती है ।

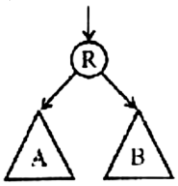


(a) Bubble/बबल
(b) Shake/शेक
(c) Tree/ट्री
(d) Insertion/अंतरवेशन

KVS PGT C.S. 2013

Ans.(c) : The above figure represent the tree sort. A tree sort is a sort algorithm that builds a binary search tree from the elements to be sorted, and then traverses the tree so that the elements come out in sorted order.

142. The figure below represents a
निम्न आकृति एक _____को प्रदर्शित करती है ।



(a) Binary tree/बाइनरी ट्री
(b) Recursive tree/रीकर्सिव ट्री
(c) Insert sort/इन्सर्ट-सोर्ट
(d) Unitary tree/यूनिटरी-ट्री

KVS PGT C.S. 2013

Ans.(a) : Binary tree is defined as a tree data structure where each node has at most 2 children.
The recursion tree method is a way of solving recurrence relations. A recurrence relation is converted into recursive trees.
Insertion sort is a simple sorting algorithm that builds the final sorted array one item at a time by comparisons.

143. An important quantitative measure of the complexity of a binary tree is its _____It also provides a measure of the average depth of all nodes in the tree.
किसी बाइनरी ट्री के जटिलता के महत्वपूर्ण परिमाणात्मक मापन में उसको _____हैं । यह ट्री में सभी नोड्स की औसत गहराई का माप भी प्रदान करता है ।

(a) average path length/औसत पथ लम्बाई
(b) median path length/माध्यिका पथ लम्बाई
(c) mode path length/बहुलक पथ लम्बाई
(d) simple deviation path length/सरल विचलन पथ लम्बाई

KVS PGT C.S. 2013

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Ans.(a) : Average path length or average shortest path length is a concept in network topology that is defined as the average number of steps alone the shortest paths for all possible pairs of network nodes. It is a measure of the efficiency of information or mass transport on a network.

144. Which of the following search algorithm requires a sorted array ?
निम्नलिखित में से कौन से सर्चएल्गोरिथम में सोर्टेड ऐरे की आवश्यकता है ?

(a) Linear search /लिनियर सर्च
(b) Hash search/हेश सर्च
(c) Binary search/बाइनरी सर्च
(d) All of these/ये सभी

KVS PGT C.S. 2013

Ans.(c) : Binary search works on sorted arrays. Binary begins by comparing an element in the middle of the array with the target value. If the target value matches the element , its position in the array is returned. If the target value is less than the element , the search continues in the lower half of a the array.

145. A binary search tree with N integers may have height _____ in the best case, and _____in the worst case.
N इंटीजर वाले किसी बाइनरी सर्च ट्री में सर्वोत्तम स्थिति में _____ हाइट और सबसे बुरी स्थिति में _____ हाइट होगी?

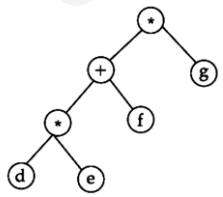
(a) O(N), O(N) (b) $O(N^2)$, N
(c) $O(\log_2 N)$, $O(N^2)$ (d) $O(\log_2 N)$, N−1

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans.(d) :A Binary search tree with N integers may have height $O(\log_2 N)$ in the bestcase, and (N-1) in the worst case.

146. Consider the following expression tree.
निम्नलिखित 'एक्सप्रेशन ट्री' पर विचार कीजिए।

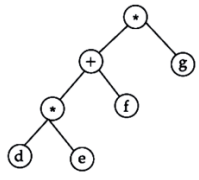
Preorder and post-order traversal of the tree will result into _____ and _____ expressions,respectively.
'ट्री' के प्रीऑर्डर तथा पोस्ट-ऑर्डर तिर्यक (ट्रैवर्सल) के परिणामस्वरूप क्रमशः _____ तथा _____ व्यंजक प्राप्त होंगे?



(a) d * e + f * g , d e * f + g *
(b) * + * d e f g, d e * f + g *
(c) * + * d e f g, d * e + f * g
(d) * + * d e f g, d e f g * + *

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Ans.(b) : Expression tree



- **Preorder** :- the root node is visited first, then the left subtree and finally the right subtree.
- We start from * and following preorder traversal. the out put of pre-order traversal of this tree will be.
- (*) → (+) → (*) → (d) → (e) → (f) → (g)

Post-order :- The root node is visited last hence the name, first we traverse the left subtree, then the right subtree and finally the root node.

- We first visit the left subtree (d) the output of post-order traversal of this tree willbe-

(d) → (e) → (*) → (f) → (+) → (g) → (*)

147. What is common in three different types of traversals (Inorder, Preorder and Postorder)?
निम्न में से कौन-सा सभी तीन ट्रैवर्सलों (इन-आर्डर, प्री-ऑर्डर और पोस्ट-ऑर्डर) में आम होता है?

(a) Root is visited before right subtree
रूट को राइट सब-ट्री से पहले विजिट करते हैं।
(b) Left subtree is always visited before right subtree/बाएँ सब-ट्री को हमेशा दाएँ सब-ट्री से पहले विजिट करते हैं।
(c) Root is visited after left subtree
रूट को बाएँ सब-ट्री) के बाद विजिट करते हैं।
(d) All of the above/उपर्युक्त सभी

LT. Grade C.S. 2018

Ans.(b) : In all three types of binary tree traversals (inorder, preorder and post order) the left subtree is always visited before the right subtree, irrespective of when the root node is visited.

(i) Preorder traversal - Root → Left subtree → Right subtree
(ii) Inorder traversal - Left subtree → Root → Right subtree
(iii) Post order traversal - Left subtree → Right subtree → Root

In each case, the left subtree is consistently visited before the right subtree.

148. In a binary search tree, which subtree of anode contains elements that are greater thanthe node's value?/बाइनरी सर्च ट्री में, कौन-सा उपवृक्ष एक नोड में ऐसे तत्व होते हैं जो नोड के मान से अधिक?

(a) Left subtree/बायां उपवृक्ष
(b) Right subtree दायां उपवृक्ष
(c) Both subtrees/दोनों उपवृक्ष
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : In a Binary Search Tree (BST), the left subtree of a node contains all the elements that are smaller than the node's value, while the right subtree contains all the elements that are greater than the node's value. This property of BST helps in efficient searching, insertion and deletion operations. Therefore, the elements greater than a node's value are always found in its right subtree.

Sarkari Resource

149. Which of the following data structures storeselements in a non-linear relationship?
निम्नलिखित में से कौन-सी डेटा संरचना तत्वों को गैर-रैखिक रूप में संबंध संग्रहीत करता है?

(a) Stack
(b) Queue
(c) Array
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

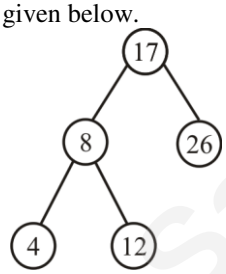
Ans. (e) :A stack, queue, and array are all examples of linear data structures because they store elements in a sequence, one after another, like in a straight line. In contrast, non-linear data structures, such as trees and graphs, store element in a hierarchical or interconnected way, not in a straight line.

150. Consider the original array 17 8 12 4 26.How many comparisons are needed toconstruct the BST on the original array?
मूल सारणी 17 8 12 4 26 पर विचार करें। मूल सरणी पर बी. एस. टी. के निर्माण के लिए कितनी तुलनाओं की आवश्यकता है?

(a) 5
(b) 10
(c) 7
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(e): The BST formed from the given array 17, 8, 12, 4, 26 is given below.



Sequence : 17, 8 , 12, 4, 26
• Insert 17: root – 0 comparisons
• Insert 8 : compare with 17 → goes left – 1 comparison
• Insert 12: compare with 17 (1), then with 8(2) → goes right of 8 – 2 comparisions
• Insert 4: compare with 17 (1), then with 8 (2) → goes left of 8 – 2 comparisions
• Insert 26: compare with 17 → goes right – 1 comparison
Total comparisons = 1 + 2 + 2 +1 = 6

151. Which search method takes less memory?
कौन-सी खोज विधि कम मेमोरी लेती है?

(a) Depth-First Search
(b) Breadth-First search
(c) Linear Search
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(a) Depth -First Search (DFS) consumes less memory a only the nodes at present are stored, but in Breadth first search (BFS), all the nodes generated in the tree are stored.

152. Which one of the following techniques is not used in the binary tree?
बाइनरी ट्री में निम्नलिखित में से किस तकनीक का उपयोग नहीं किया जाता है?

(a) Preorder traversal
(b) Inorder tranversal
(c) Postorder traversal
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

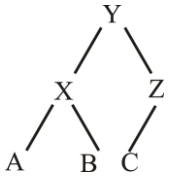
Ans. (e): In binary tree all the technique like preorder traversal, inorder traversal and postorder traversal can be used.

153. Let A, B, C, X, Y, Z are positive integer such that A < X < B < Y < C < Z. If preorder traversal of a binary search tree T is given by Y X A B Z C then its post order traversal will be:
माना A, B, C, X, Y, Z प्राकृतिक संख्याये है तथा A < X < B < Y < C < Z है। यदि एक बाइनरी ट्री T का प्री आर्डर ट्रैवर्सलY X A B Z C है तो उसका पोस्ट आर्डर ट्रैवर्सलहोगा:

(a) A B C X Z Y (b) A B X C Z Y
(c) A B Z Y C X (d) A B Y Z C X

RPSC C.S. 2016 (Paper-I)

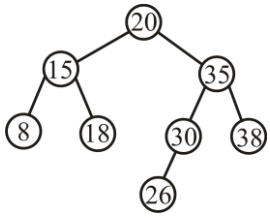
Ans.(b) : Step-1– understand preorder in preorder traversal
• First element is root
• Then comes the left subtree
•Then the right subtree
So, from Y X A B Z C
• Root = Y
Step-2– Identify left and right subtrees in a BST, left subtree < root and right subtree > root
From the inequality : A < X < B < Y < C < Z so values less than Y go to the left
• X, A, B → left subtree values greater than Y.
• Z, C → right subtree



Step-3– Post-order traversal–
Post order = Left → Right → Root
1. Left Subtree (X A B):
• Root = X
• Left = A, Right = B ⇒ Post order = A B X
2. Right subtree (Z C):
• Root = Z
• Left = C ⇒ Post order = C Z
3. Full Post order = Left (A B X) + Right (C Z) + Root (Y)
Post-order = A B X C Z Y

Sarkari Resource

154. Consider the binary search tree T given below:
नीचे दिए गए बाइनरी सर्च ट्री T पर विचार करें



If root of T is deleted it may be replaced by any one of:
यदि ट्री T के रुट को डिलीट कर देते है तो इसे दिए गए युग्म में किसी भी एक से बदला जा सकता है:

(a) 30, 26 (b) 15, 38
(c) 18, 26 (d) 8, 38

RPSC C.S. 2016 (Paper-I)

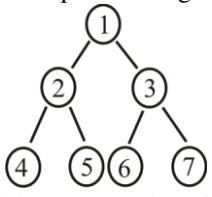
Ans. (c) : The root of the tree is 20.
When we delete a node like this (with two children), we replace it with:
The biggest number in the left side (called in-order predecessor), or
The smallest number in the right side (called in-order successor).
Left side of 20 has: 15,8,18→ biggest is 18
Right side of 20 has: 35,30,26,38→ smallest is 26.
So, we can replace 20 with either 18 or 26.

155. Which of the following is application of depth-first search?

(a) Only topological sort
(b) Only strongly connected components
(c) Both topological sort and strongly connected components
(d) Neither topological sort nor strongly connected components

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : Depth first search: Depth first search algorithm is used to traverse the vertices of a graph. It uses the concept of back tracking.
Depth first search uses a data structure for its implementation. Which is known as stack.
Application of depth first search :
1) It is used to find the strongly connected components.
2) It helps in topological sorting
3) To check the bipartiteness of a graph.
4) It helps in detecting cycles in a graph.
5) To find articulation points in a graph.



DFS traversal for this – 1, 2, 4, 5, 3, 6, 7

156. A complete n-ary tree is a tree in which each node has n children or no children. Let I be the number of Internal nodes and L be the number of leaves in a complete n-ray tree. If L = 41 and I = 10. What is the value of n ?

(a) 3 (b) 4
(c) 5 (d) 6

UGC NET C.S. June-2020 (Paper-II)

Ans. (c) : If the tree is 3-ary and 'l' is an internal node the number of leaves $2l + 1$.
• If the tree is 5-ary and 'l ' an internal node, the number of leaves is $4l + 1$.
• If the tree is n-ary and 'l ' is an internal node the number of leaves is $(n - 1)l + 1$.
given that leaves L = 41, internal nodes I = 10

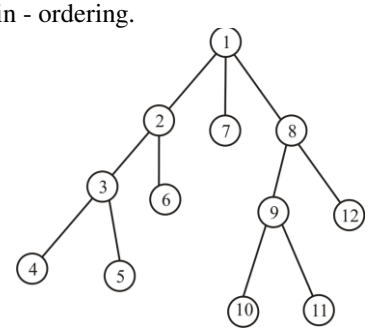
$L = (n - 1)l + 1$
$41 = 10(n - 1) + 1$
$10n = 50$
$n = 5$

157. Which of the following concepts can be used to identify loops?
A. Depth first ordering
B. Dominators
C. Reducible graphs
Choose the correctanswer from the options given below:

(a) A and B only
(b) A and C only
(c) B and C only
(d) A, B and C

UGC NET C.S. December 2020

Ans. (d) :A Depth first ordering :- Depth First Search (DFS) is an algorithm for traversing or searching tree or graph data structures. The algorithm starts at the root node (selecting some orbitrary node as the root node in the case of a graph) and explores as for as possible along each branch before back tracking.
Vertex Orderings :-
i) pre ordering
ii) post ordering
iii) reverse preordering
iv) reverse postordering
for binary trees there is additionally-
e) in ordering
f) reverse in - ordering.



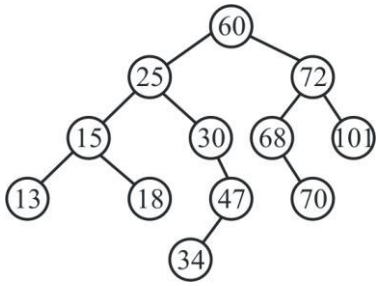
Dominators :- Dominators are defined in a directed graph with respect to a source, vertex S. formally, a node a is said to dominate a node w art source vertex S if all the paths from S to w in the graph must pass through node u. Note that only the vertices that are reachable from source vertex in the directed graph are considered here.
Reducible Graph :- A control how graph G is said to be reducible if the removal of its back edges leads to an acyclic graph where each node can be reached from the initial node of G.
So All above concepts can be used to identify loops.

Sarkari Resource

158. A binary search tree is constructed by inserting the following numbers in order :
60, 25, 72, 15, 30, 68, 101, 13, 18, 47, 70, 34
The number of nodes is the left subtree is
(a) 5 (b) 6
(c) 3 (d) 7

UGC NET C.S. December-2018 (Paper-II)

Ans. (d) :Binary Search Tree–The left subtree of a contains only < ROOT
The right subtree of a node contains only nodes > ROOT
The left and right subtree each must also be a binary search tree.



So, option is 7 on left subtree.

159. A 5-ary tree is tree in which every internal node has exactly 5 children. The number of leftnodes in such a tree with 8 internal nodes will be :
(a) 30 (b) 33 (c) 45 (d) 125

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : In complete n-ary tree,
$L = (n - 1) * I + 1$
Where L = No. of leaves, I = no. of internal nodes
Here n = 5, I = 8
Therefore, $L = (5 - 1)*8 + 1 = 4 * 8 + 1 = 33$

160. A binary search tree in which every non-leaf node has non-empty left and right subtrees iscalled a strictly binary tree. Such a tree with 19 leaves :
(a) cannot have more than 37 nodes
(b) has exactly 37 nodes
(c) has exactly 35 nodes
(d) cannot have more than 35 nodes

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : 2(n) –1, where n is the for the leaves nodes. so by that way we have exactly 37 nodes.
$2(19) - 1$
$= 38 - 1 = 37$

161. How many different trees are possible with 'n' nodes?
(a) n_{-1} (b) $2^n - 1$
(c) 2^n (d) 2^n-n

UGC NET C.S. December-2022

Ans. (*) :Different tree are possible with n nodes, there exist (2n)!/(n+1)! n!

162. Where does the values of alpha-beta search get updated?
(a) Along the path of search.
(b) Initial state itself.
(c) At the end.
(d) None of these.

UGC NET C.S. December-2022

Ans. (a) : Alpha-Beta search updates the value of Alpha and Beta gets along and prunes the remaining branches at node.

163. Consider the following statements:
A. The number of labelled binary trees possible with 'n' nodes is (2n)!/(n+1)!
B. The number of binary search trees possible with 'n' nodes is (2n)!/((n+1)!n!)
C. The number of unlabeled binary trees possible with 'n' nodes is C(2n, n)!/(n+1)!
D. The number of simple, undirected and labelled graphs possible with 'n' vertices is $2^{n(n-1)/2}$
Choose the correct answer from the options given below.
(a) A, B and C only correct
(b) A, B and D only correct
(c) B, C and D only correct
(d) A, C and D only correct

UGC NET C.S. December-2022

Ans. (b) :Binary Trees - A tree whose element have 0 or 1 or 2 children is called a binary tree. Since each element in a binary tree can have only 2 children, we typically name them the left and right child.
- The number of binary search trees possible with 'n' nodes is (2n)!/((n+1)!n!)
- The number labeled binary search trees possible with 'n' nodes is (2n)!(n+1)!
- The number of simple, undirected and labeled graphs possible with n nodes can be calculated using the catalan number formula.

Cn = (2n)!/[n+1!*n!]
So statement C is false.
So option (b) is correct.

164. Which of the following is/are NOT CORRECT statement?
(A) The first record in each block of the data file is known as actor record.
(B) Dense index has index entries for every search key value in the data file.
(C) Searching is harder in the B^+ tree than B^- tree as the all external nodes linked to each other.
(D) In extendible hashing the size of directory is just an array of 2^{d-1}, where d is global depth.
Choose the correct answer from the options given below:
(a) (A), (B) and (C) Only
(b) (A), (C) and (D) Only
(c) (A), (B) and (D) Only
(d) (A), (B), (C) and (C) Only

UGC NET C.S. December-2023

Ans. (b) :(A)- In the data file, the first record of any of the block is called anchor record. This is not correct because the term actor record is not standard in data base systems.
(C) In B^+ tree, only the leaf nodes are linked together, not all external nodes. This is incorrect. Searching is easier in a B^+ tree because all external nodes are linked together, which allows efficient range queries and sequential access. In a B^- tree nodes (leaves) are not typically linked. B^+ tree is not harder than in a B^- tree.

Sarkari Resource

Also note that B^- tree is not a common data structure. In stardard B^- trees, leaves are not linked, while in B^+ tree they are. Searching in B^+ trees is efficient due to the linked leaves.
(D) In extendible hashing the size of directory is just an array of 2^{d-1}, where d is global depth. This is not correct. The size of the directory in extendible hashing is 2^d. It is not 2^{d-1}
So, the statements (A),(C) and (D) are not correct in the question.

165. Post order traversal of a given binary search tree T produces following sequence of keys : Which one of the following sequences of keys can be the result of an in-order traversal of the tree T?
(a) 3, 4, 5, 7, 9, 14, 20, 18, 17, 16, 15
(b) 20, 18, 17, 16, 15, 14, 3, 4, 5, 7, 9
(c) 20, 18, 17, 16, 15, 14, 9, 7, 5, 4, 3
(d) 3, 4, 5, 7, 9, 14, 15, 16, 17, 18, 20

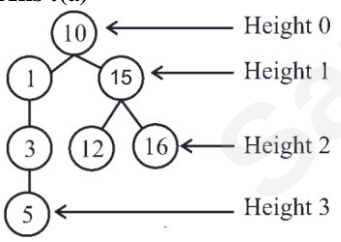
UGC NET C.S. November-2017 (Paper-II)

Ans :(d)In order traversal always gives sorted list in BST. 3, 4, 5, 7, 9, 14, 15, 16, 17, 18, 20 .
• An in-order traversal of the Binary Search Tree (BST) always given elements in ascending or increasing order.
So, In order traversal of the sequence of keys given in post- order traversal in sorted ascending order of that keys.

166. The following numbers are inserted into an empty binary search tree in the given order : 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree?
(a) 3 (b) 4
(c) 5 (d) 6

UGC NET C.S. November-2017 (Paper-II)

Ans :(a)



167. Which of the following statements is false?
(A) Optimal binary search tree construction can be performed efficiently using dynamic programming.
(B) Breadth-first search cannot be used to find connected components of a graph.
(C) Given the prefix and postfix walks of a binary tree, the tree cannot be re-constructed uniquely.
(D) Depth-first-search can be used to find the connected components of a graph.
(a) A (b) B
(c) C (d) D

UGC NET C.S. June-2017 (Paper-II)

Ans :(b) Optimal binary search tree construction can be performed efficiently using dynamic programming. This is well-known application of dynamic programming. Where the optimal substructure and over lapping subproblems allow for efficient computation of the optimal binary search tree.
Given the prefix and postfix walks of binary tree, the tree can not be re-constructed uniguely. Depth-first-search can be used to find the connected components of a graph.

168. Which of the following statements is false?
(A) Optimal binary search tree construction can be performed efficiently using dynamic programming
(B) Breadth-first search cannot be used to find connected components of a graph
(C) Given the prefix and postfix walks of a binary tree, the tree cannot be re-constructed uniquely
(D) Depth-first-search can be used to find the connected components of a graph
(a) A (b) B
(c) C (d) D

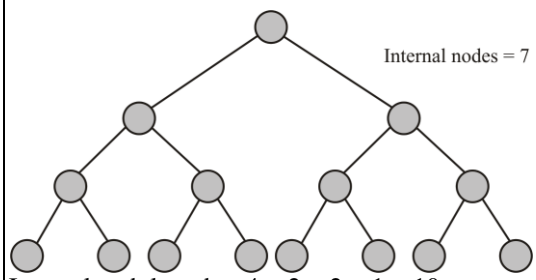
UGC NET C.S. June-2017 (Paper-II)

Ans :(b)BFS and DFS both algorithm are used to find connected component. In BFS, a search that begins at some particular 'N' will find the entire connected component V before returning.

169. Consider a full binary tree with n internal nodes, internal path length i, and external path length e. The internal path length of a full binary tree is the sum, taken over all nodes of the tree, of the depth of each node. Similarly, the external path length is the sum, taken over all leaves of the tree, of the depth of each leaf. Which of the following is correct for the full binary tree?
(a) $e = i + n$ (b) $e = i + 2n$
(c) $e = 2i + n$ (d) $e = 2^n + i$

UGC NET C.S. November-2017 (Paper-III)

Ans :(b) Consider full binary tree:



Internal path length = $4 \times 2 + 2 \times 1 = 10$
External path length $=3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 = 24$
So, $e = i + 2n$
$24 = 10 + 2 \times 7$
$24 = 10 + 14$
$24 = 24$

170. ———— is used in game trees to reduce the number of branches of the search tree to be traversed without affecting the solution.
(a) Best first search
(b) Goal stack planning
(c) Alpha-beta pruning procedure
(d) Min-max search

UGC NET C.S. June-2014 (Paper-III)

Sarkari Resource

Ans. (c) : Because, It is a search algorithm that seeks to decrease the number nodes that are evaluated by the Min-max algorithm in its search tree. It is an adversial search algorithm used commonly for machine playing of two-player games.

171. A full binary tree with n leaves contains

(a) n nodes (b) $\log_2$ nodes
(c) 2n – 1 nodes (d) 2^n nodes

UGC NET C.S. December-2014 (Paper-II)

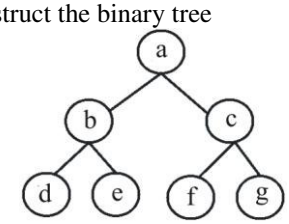
Ans :(c)Because, full binary tree with n leaves contains n–1 internal nodes
So total nodes = internal nodes + leaves = n – 1 + n
= 2n–1
So the answer is (c)

172. The in order and preorder Traversal of binary tree are dbeafcg and abdecfg respectively. The post-order Traversal is .....

(a) dbefacg (b) debfagc
(c) dbefcga (d) debfgca

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : Take the first node in preorder traversal a will be the root.
Now we construct the binary tree



Therefore, postorder is - debfgca

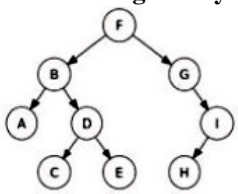
173. Level order Traversal of a rooted Tree can be done by starting from root and performing :

(a) Breadth first search (b) Depth first search
(c) Root search (d) Deep search

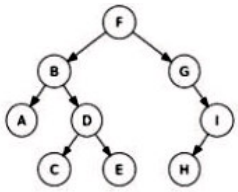
UGC NET C.S. June-2015 (Paper-II)

Ans.(a) : Level order traversal is similar to breadth first search whereas preorder traversal is similar to depth first search.
So answer is (a)

174. Which of the following options present the correct postorder and preorder traversals of the following binary tree?



निम्नलिखित में से कौन सा विकल्प निम्नलिखित बाइनरी ट्री के सही पोस्टऑर्डर और प्रीऑर्डर ट्रैवर्सल को दर्शाता है?



(a) Postorder: A, C, E, D, B, H, I, G, F
Preorder: F, B, A, D, C, E, I, G, H
पोस्ट ऑर्डर: A, C, E, D, B, H, I, G, F
प्रीऑर्डर: F, B, A, D, C, E, I, G, H

(b) Postorder: A, C, E, D, B, H, I, G, F
Preorder: F, B, A, D, C, E, G, I, H
पोस्ट ऑर्डर: A, C, E, D, B, H, I, G, F
प्रीऑर्डर: F, B, A, D, C, E, G, I, H

(c) Postorder: A, C, E, D, B, H, I, G, F
Preorder: F, B, A, D, E, C, G, I, H
पोस्ट ऑर्डर: A, C, E, D, B, H, I, G, F
प्रीऑर्डर: F, B, A, D, E, C, G, I, H

(d) Postorder: A, C, D, E, B, H, I, G, F
Preorder: F, B, A, D, C, E, G, I, H
पोस्ट ऑर्डर: A, C, D, E, B, H, I, G, F
प्रीऑर्डर: F, B, A, D, C, E, G, I, H

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Let us assume the general structure of the tree as follows root ps F, its left sub tree is B, right subtree is G. whose subordinate is I and child of I is H.
• Pre order = Root, left, right → F →
(B sub tree → B, A, D, C, E) →
(G SUB TREE → G, I, H,) ⇒ F, B, A, D, C, E, G, I, H
• post order = left right, root →
(B sub tree → A, C, E, D, B) →
(F sub tree → H, I, G) → f ⇒ A, C, E, D, B, H, I, G, F

175. Considering the height of a binary tree as the number of edges between the tree's root and its furthest leaf, what is the height of the binary search tree created using the following key values (in order)?
19, 14, 18, 15, 16, 13, 12
ट्री रूट और उसके फर्थस्ट लीफ के बीच किनारों की संख्या के रूप में एक बाइनरी ट्री की ऊंचाई को ध्यान में रखते हुए, निम्नलिखित की वैल्यूज (क्रम में) का उपयोग करके बनाई गई बाइनरी सर्च ट्री की ऊंचाई क्या है?
19, 14, 18, 15, 16, 13, 12

(a) 3 (b) 4
(c) 5 (d) 6

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

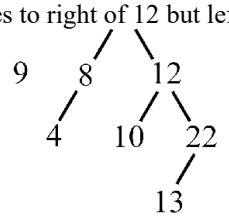
Ans.(b) : Insert the given key value (19, 14, 18, 15, 16, 13, 12) into a BST in the given order
- Start with 19 as the root
- insert 14 : 14 < 19, so 14 becomes the left child of 14.
- insert 18 : 18 < 19, 18 > 14, so 18 becomes the right child of 14.
- insert 15 : 15 < 19, 15 > 14, 15 < 18, so 15 becomes the left child of 18.
- insert 16 : 16 < 19, 13 < 14, so 13 becomes the left child of 14
- insert 13 : 12 < 19, 12 < 14, 12 < 13 > 12 (length : 3 edges)

The longest path has 4 edges.

Sarkari Resource

176. What is the pre-order traversal of the binary search tree corresponding to the following key values (in order)?
9, 8, 12, 22, 4, 10, 13
निम्न की वैल्यूज (क्रम में) के अनुरूप बाइनरी सर्च-ट्री का प्री-ऑर्डर ट्रैवर्सल क्या है?
9, 8, 12, 22, 4, 10, 13
(a) 8, 9, 4, 12, 10, 22, 13
(b) 9, 8, 4, 12, 22, 10, 13
(c) 9, 8, 4, 12, 10, 22, 13
(d) 13, 22, 10, 12, 4, 8. 9
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : We are given the insertion order for the BST
9, 8, 12, 22, 4, 10, 13
• insert 9 → root = 9
• insert 8 → goes to left of 9
• insert 12 → goes to right of 9
• insert 22 → goes to right of 12
• insert 4 → goes to left of 8
• insert 10 → goes to left of 12
• insert 13 → goes to right of 12 but left of 22



Pre order Traversal
• Start at root : 9
• left sub tree of 9 → 12 → 10→ 22→ 13
So pre order = 9, 8, 4, 12, 10, 22, 13

177. Identify the non-linear data structures from among the following.
निम्नलिखित में से नॉन-लाइनर (गैर-रेखीय) डेटा स्ट्रक्चर्स की पहचान करें।
(i) Linked List/लिंक्ड लिस्ट
(ii) Stack/स्टैक
(iii) Tree/ट्री
(a) Both (i) and (ii)/ दोनों (i) और (ii)
(b) Only (iii)/ केवल (iii)
(c) Both (i) and (iii)/ दोनों (i) और (iii)
(d) (i), (ii) and (iii)/ (i), (ii) और (iii)
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Linked list:
• Linked list is a linear data structure, its elements are connected sequentially.
• **Stack** ⇒ Stact Ps-also a linear data structure it is based on LIFO (Last-In, first-Out) principle.
• **Tree** ⇒ Tree is a non-Linear structure because its elements are connected by hierarchical structures.

178. If _____ traversal of a binary tree is sorted (in increasing order). Then the binary tree is a Binary Search Tree (BST).
यदि बाइनरी ट्री के _____ ट्रैवर्सल का क्रमबद्ध किया जाता है (बढ़ते क्रम में), तो बाइनरी ट्री एक बाइनरी सर्च ट्री (बीएसटी) हो जाती है।
(a) Preorder/प्री-ऑर्डर
(b) Postorder/पोस्ट ऑर्डर
(c) Inorder/इनऑर्डर
(d) Level-order/लेवल-ऑर्डर
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Ans. (c) : In a Binary search tree (BST), the in order traversal visit nodes in ascending order. Therefore, if the inorder traversal of a binary tree is sorted in increasing order, it confirms that the tree. is a BST
Other traversals (Preorder, postorder, level-order) being sorted do not guarantee that the tree is a BST, as they may produce sorted sequences even in non-BST structures due to specific arrangements.

179. The number of subtrees of a node is called the _____ of the node.
नोड के सबट्री की संख्या को नोड का _____ कहा जाता है।
(a) Size/साइल (आकार) (b) Rank/रैंक (श्रेणी)
(c) Degree/डिग्री (d) Order/ऑर्डर (क्रम)
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Ans. (c) : The number of subtrees of a node in a tree data structure is called its degree.

180. The degree of tree is the _____ degree of a node in the tree.
ट्री की डिग्री ट्री में एक नोड की _____ डिग्री है।
(a) Maximum/अधिकतम
(b) Minimum/न्यूनतम
(c) Second minimum/दूसरा न्यूनतम
(d) Second maximum/दूसरा अधिकतम
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Ans. (a) : The degree of a node in a tree is the number of its children. The degree of a tree is the maximum degree among all nodes in the tree.

181. In a tree, a node of degree _____ is called a terminal node or leaf node.
एक ट्री मे डिग्री _____ को एक नोड टर्मिनल नोड या लीफ नोड कहा जाता है।
(a) 0 (b) 1 (c) 2 (d) 3
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Ans. (a) : In a tree data structure, node with a degree of zero is referred to as a terminal node or a leaf node. This indicates that the node has no children or subtrees branching from it.

182. Inorder and postorder traversal of a binary tree are given.
In: E, I, C, F, B, G, D, J, H, K
Post: I, E, F, C, G, J, K, H, D, B
What is the preorder traversal of the binary tree?
बाइनरी ट्री का इन ऑर्डर और पोस्ट ऑर्डर ट्रैवर्सल दिया गया है।
इनः E, I, C, F, B, G, D, J, H, K
पोस्टः I, E, F, C, G, J, K, H, D, B
बाइनरी ट्री का प्रीऑर्डर ट्रैवर्सल क्या है।
(a) B, C, E, I, F, D, G, H, J, K
(b) B, F, I, E, C, D, G, H, J, K
(c) B, C, E, I, F, D, G, H, K, J
(d) B, C, E, F, G, H, K, D, J, I
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Ans. (a) : Given,
In order (In) : E, I, C, F, B, G, D, J, H, K
Post order (post): I, E, F, C, G, J, K, H, D, B
(i) Root of the tree
- In post order, the last elements is the root

Root = B
(ii) Split in order around root
- In order sequence : E, I, C, F, B, G, D, J, H, K
- Root B splits it into:
- Left subtree : E, I, C, F
- Right Subtree : G, D, J, H, K

(iii) Left Subtree
- Left in order = E, I, C, F
- Corresponding post order = I, E, F, C
- Root of left subtree = last elements of left post order = C
- Left of C = E , I → Left post order = I, E

→ Preorder = E, I
- Right of C = F → Preorder = F

(iv) Right subtree
- Right in order = G, D, J , H, K
- Corresponding post order = G, J, K, H, D
- Root of right subtree = last element of right post order = D
- Split right inorder around D :
- Left = G → preorder = G
- Right = J, H, K → post order = J , K, H

→ Root H
→ Left of H = J → preorder = J
- Right of H = K → preorder = K

Preorder of right subtree = D, G, H, J , K
(v) Combined
- Root = B
- preorder = Root + left subtree + right subtree

preorder = B, C, E, I, F, D , G , H, J, K
So the correct answer (a)
B, C, E, I, F, D, G, H, J, K

183. How many different binary search trees can be constructed with 6 different nodes?
6 विभिन्न नोड्स वाले कितने विभिन्न बाइनरी सर्च ट्री बन सकते हैं?
(a) 64 (b) 128
(c) 32 (d) 132

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Ans. (d) : The number of distinct Binary search trees that can be made with n distinct nodes = catalan number Cn

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

For n = 6

$$C_6 = \frac{1}{6+1}\binom{12}{6}$$

1. $\binom{12}{6} = \frac{12\times11\times10\times9\times8\times7}{6\times5\times4\times3\times2\times1}$

$= \frac{665280}{720} = 924$

2. Divide by 7

$$C_6 = \frac{924}{7} = 132$$

184. Consider a sorted array of 10,00,000 elements. What is the maximum number of comparison required to find the location an item using binarysearch?
10,00,000 तत्वों का एक सॉर्टेड ऐरे है। बाइनरी सर्च का उपयोग करके किसी आइटम के स्थान को खोजने के लिए आवश्यक अधिकतम तुलनाएँ कितनी होंगी?
(a) 20 (b) 6
(c) 31 (d) 7

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Ans. (a) : Binary search redress the search space by half each time.
So the maximum number of comparisons = $[\log_2 (n)]$
Apply for n = 1000000
We need
$\log_2 (1000,000)$
2^{19} = 524,288 (less than 1000000)
2^{20} = 1,048,576 (just greater than 1000,000)
So $\log_2 (1000,000) \approx 19{\cdot}93 = 20$

185. The numbers 70, 50, 10, 80, 30, 60, 100, 90, 40, 20 are inserted in the given order in to a binary search tree. What is the in-order traversalsequence of the resultant binary search tree?
बाइनरी सर्च ट्री में दिए गए क्रम में संख्याओं 70, 50, 10, 80, 30, 60, 100, 90, 40, 20को डाला जाता है। परिणामी बाइनरी सर्च ट्री का इन-ऑर्डर ट्रैवर्सल अनुक्रम क्या है?
(a) 70, 50, 10, 80, 30, 60, 100, 90, 40, 20
(b) 100, 90, 80, 70, 60, 50, 40, 30, 20, 10
(c) 10, 30, 50, 70, 90, 20, 40, 60, 80, 100
(d) 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

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Ans. (d) : In-order traversal of a BST =
Nodes always in ascending order
Given numbers : 70, 50, 10, 80, 30, 60, 100, 90, 40, 20
When these are inserted in BST, the In- order sequence will be:
10, 20, 30, 40, 50, 60, 70, 80, 90, 100

186. A complete binary tree T has n leaf nodes. What is the number of nodes with degree of 2 in tree T?
एक कम्पलीट बाइनरी ट्री T में n लीफ नोड्स हैं। ट्री T में डिग्री 2 के नोड्स की संख्या क्या है?
(a) n (b) n/2
(c) n – 1 (d) 2n

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In a complete binary tree– A node with degree 2 is a node that has exactly two children. If the tree has n leaf nodes the total number of nodes in the tree is 2n–1. Now let's find the number of nodes with a degree of 2.
(1) In a complete binary tree, about half of the non-leaf nodes are internal nodes with exactly 2 children.
(2) A complete binary tree has n–1 internal nodes (total nodes minus the leaf nodes).
(3) The number of nodes with degree 2 (which are internal nodes with 2 children) is n–1.
Thus, the number of nodes with a degree of 2 in the tree T is n–1.

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187. Which traversal of binary search tree is in ascending order?
I. Preorder traversal
II. Postorder traversal
III. Inorder traversal
बाइनरी सर्च ट्री का कौन सा ट्रैवर्सल आरोही क्रम में है?
I. प्री ऑर्डर ट्रैवर्सल
II. पोस्ट ऑर्डर ट्रैवर्सल
III. इन ऑर्डर ट्रैवर्सल
(a) Ony II/केवल II
(b) I and III// I तथा III
(c) Only III/केवल /III
(d) II and III// II तथा III

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans.(c) :A binary search three (BST) is a three data structure where for node.
•All values in the left sub tree are less than the node's value.
•All values in the right subtree are greater than the node's value.
•Both the left and right subtrees are also BsTs.
Pre-order:– The root is visited first. This does n't guarantee ascending order because the root's value might be anywhere within the range of values in the tree. For example, in a simple tree root 10, lift child 5, right child 15, pre order is 10, 5, 15 – not ascending.
Past order:– The root is visited last. The left and right subtrees are visited first again, this does not naturally produce ascending order because the large values in the right subtree might be visited before the smaller values in the root node. For the example tree (10, 5, 15) post order might be 5, 15, 10 – nor ascending.
In order:–The lift subtree is visited first then the root node, then the right subtree (all larger values), Due to the BST property, visiting the lift subtree nodes, then the root, then the right subtree nodes naturally sorts the elements in ascending order. For the example tree (10, 5, 15), in order is 5, 10, 15 ascending. The correct answer is only III.

188. What is the condition for binary search of be unsuccessful where Beg is the beginning and End is ending location of sorted array ?
बाइनरी सर्च के असफल होने की परिस्थिति क्या है जहां Beg सोरटेड ऐरे का शुरुआती और End अंतिम स्थान है ?
(a) End > Beg (b) End = = Beg
(c) End < Beg (d) End !=Beg

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : End < Beg
If end is les than beg, it means the beginning index has moved post the ending index. The search interval is empty or inverted. The condition Beg ≤ end for continuing the search is false.
The binary seatch loop terminates. If the elements has not been found by this point, the search is unsuccessfull.

Binary search conditions –

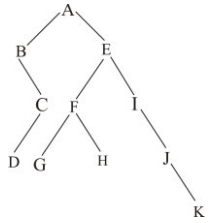
Condition	Interpretation	Search state
Beg ≤ end	Valid interval (0 or more elements)	Search continuls (potent iol success)
Beg > end	Invalid interval (empty)	Search terminates
end < Beg	Invalid interval (empty)	Search terminates (Unsuccessful if element not found)
Beg = = end	Interval with 1 elements	Search continuls (valid state)
Beg < end	Interval with > 1 element	Search continues (Valid state)

189. Preorder traversal of binary search tree is 38, 14, 8, 23, 18, 20, 56, 45, 82, 70.What is the postorder traversal of binary search tree?
बाइनरी सर्च ट्री का प्री ऑर्डर ट्रैवर्सल है। बाइनरी सर्च ट्री का पोस्ट ऑर्डर ट्रैवर्सल क्या है?
(a) 23, 20, 18, 14, 8, 38, 45, 56, 70, 82
(b) 8, 18, 20, 23, 14, 45, 70, 82, 56, 38
(c) 8, 20, 18, 23, 14, 45, 70, 82, 56, 38
(d) 82, 70, 56, 45, 38, 23, 20, 18, 14, 8

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) :
Preorder list: 38, 14, 8, 23, 18, 20, 56, 45, 82, 70
First number = root→38
Split number:
Number < 38→ left side of tree → 14, 8, 23, 18, 20
Number > 38 → right side of tree → 56, 45, 82, 70
Post order rule: left → right → root
Left side post order = 8, 20, 18, 23, 14
Right side post order = 45, 70, 82, 56
Add root 38 at the end.
Post result: 8, 20, 18, 23, 14, 45, 70, 82, 56, 38
Binary tree:

190. Consider the following New-order strategy for traversing a binary tree.
बाइनरी ट्री को ट्रैवर्स करने के लिए निम्नलिखित न्यू-ऑर्डर रणनीति पर विचार करें।
I. Visit the right subtree of node using new-order/नोड के न्यू-ऑर्डर का उपयोग करते हुए दाएँ सबट्री पर जाएँ
II. Visti the left subtree of node using new-order/नोड के न्यू-ऑर्डर का उपयोग करके बाएं सबट्री पर जाएं
III. Visit the node/नोड पर जाएँ
What is the new-order traversal of the given tree?/दिए गए का न्यू-ऑर्डर ट्रैवर्सल क्या है ?



Sarkari Resource

(a) K, J, I, H, G, F, E, D, C, B, A
(b) A, B, C, D, E, F, G, H, I, J, K
(c) B, D, C, A, G, F, H, E, I, J, K
(d) A, B, C, D, K, J, I, H, G, F, E

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Ans. (a) : New-order is a custom tree traversal strategy defined specifically for a problem. It is not a standard traversal like inorder, preorder, or postorder, but it follows a clear pattern.
This is essentially a post order traversal but with the order of subtrees reversed, instead of left-right-root it is right-left-root.
Step by step traversal–
Start at A:
Traverse E,
Visit K, J, I
Traverse F:
Visit H,G
Visit F, E
Traverse B:
Visit D, C
Visit B
Visit A
Traversal order is - K, J, I, H, G, F, E, D, C, B, A

191. Which statement(s) is/are correct regarding node N deletion in binary search tree?
बाइनरी सर्च ट्री में नोड N डीलेशन के संदर्भ में कौन सा/से कथन सही है/हैं?

I. If node N has no children then just replace location of N in the parent node by the null pointer./यदि नोड N की कोई चिल्ड्रेन नहीं है, तो पैरेंट नो में N के स्थान को नल पॉइंटर से बदलें।

II. If node N has two children, then replace it by preorder successor./यदि नोड N के दो चिल्ड्रेन हैं तो फिर इसे प्रीऑर्डर सक्सेसर द्वारा प्रतिस्थापित करें।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

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Ans. (a) : Statement-I – If node N has no children then just replace the location of N in the parent node by the null pointer. This is correct, when a node has no children (i.e., it is a leaf node), you can simply remove it by setting the corresponding child point in its parent to null. This does not violate any BST properties.
Statement-II – If node N has two children, then replace it by preorder successor. It's incorrect, when a node has two children, the standard approach is to replace it with either its.
In order successor (the smallest node in its right subtree).
In order predecessor (the largest node in its left subtree).

192. Which statement is correct regarding binary search tree?
बाइनरी सर्च ट्री के संदर्भ में कौन सा कथन सही है?

I. The value at node N is less than every value in the left subtree of node./नोड N पर मान नोड के बाएं सबट्री में प्रत्येक मान से कम है।

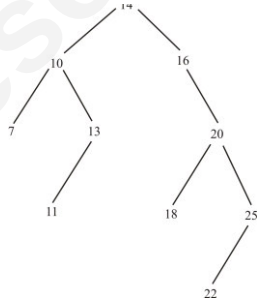
II. The value at node N is greater than or equal to every value in the right subtree of N./नोड N का मान N के दाएँ सबट्री में प्रत्येक मान से अधिक या उसके बराबर है।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : In Binary Search Tree–
For given node N- All nodes in the left Subtree of N have values less than the value at N. All nodes in the right subtree of N have values greater than the value at N.
Statement-I – The value at node N is less than every value in the left subtree of node. This statement is incorrect. In a BST, the value at node N is greater than every value in its left subtree.
Statement-II – The value at node N is greater than or equal to every value in the right subtree of N. This statement is also incorrect. In a BST, the value at node N is less than every value in its right subtree. Some definitions of BST allow for equal values to be in the right subtree, but the core principle is greater than.
Therefore, neither statement I nor statement II is correct.

193. Consider a binary search tree given below.
नीचे दिए गए बाइनरी सर्च-ट्री विचार करें।

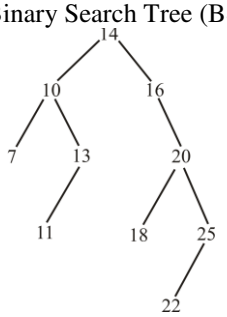


What is the per-order traversal of binary search after 14 is deleted?
नोड 14को हटाने के बाद बाइनरी सर्च-ट्री का प्री-ऑर्डर ट्रैवर्सल क्या है?

(a) 7, 10, 11, 13, 16, 18, 20, 22, 25
(b) 16, 10, 7, 13, 11, 20, 18, 25, 22
(c) 16, 7, 10, 13, 11, 18, 22, 20, 25
(d) 25, 20, 22, 18, 11, 13, 10, 16, 7

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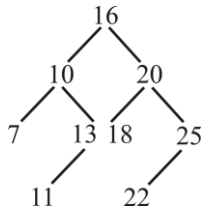
Ans. (b) : Given Binary Search Tree (BST)



Step-1– Delete node 14
• Node 14 is the root
• It has two children 10 and 16
• In a BST, when deleting a node with two children, we replace it with its in-order successor (Smallest made in right's subtree)

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• In order successor of 14 is 16 (it's the Immediate right child and the smallest in that subtree)
So 16 because the new root updated tree after deleting 14



Step-2 – Pre-order traversal (Node →Left→ Right)
Pre-order traversal of this tree
1. Visit 16
2. Left Subtree of 16
   - Visit 10
   - Left → 7
   - Right → 13
   - Left →11
3. Right Subtree of 16
   - Visit →20
   - Left →18
   - Right →25
   - Left →22

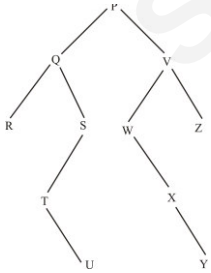
(16, 10, 7, 13, 11, 20, 18, 25, 22)

194. Consider the following new-order strategy for traversing a given tree.
दिए गए ट्री ट्रैवर्स करने के लिए निम्नलिखित न्यू-ऑर्डर रणनीति पर विचार करें।

I. Visit the right subtree of a Node using new-order/न्यू-ऑर्डर का उपयोग करते हुए नोड के दाएँ सब-ट्री पर जाएँ

II. Visit the Node/नोड पर जाएँ

III. Visit the left subtree of a Node using new-order/न्यू-ऑर्डर का उपयोग करते हुए नोड के बाएं सब-ट्री पर जाएं



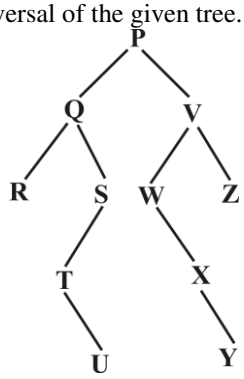
What is the new-order traversal of the given tree?/दिए गए का न्यू-ऑर्डर ट्रैवर्सल क्या है?

(a) Z, V, Y, X, W, P, S, U, T, Q, R
(b) R, Q, T, U, S, P, W, X, Y, V, Z
(c) P, W, X, Y, V, Z, R, Q, T, U, S
(d) S, U, T, Q, R, Z, V, Y, X, W, P

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : New order Strategy–
1. Visit the right subtree (Recursively using new order)
2. Visit the current node
3. Visit the left subtree (Recursively using new-order)

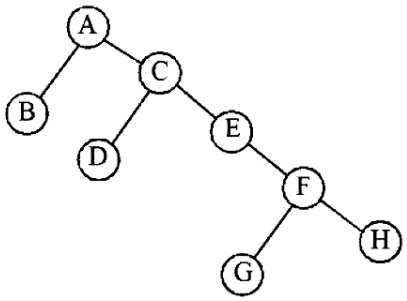
Reverse in order traversal
Right→Node→Left
Step-by-step traversal of the given tree.



Let's traverse it step by step using reverse in order (Right→Node→Left)
Starting at root:P
i. traverse right subtree (ii)
• Traverse right to V→Z (No Children)→Z
• Visit V
• Traverse left of V→W
• Right of W→X
• Right of X→Y→visit Y
visit X
• No left
• Visit W
• No left
So V's traversal = Z, V, Y, X, W
ii. Visit P
iii. Traverse left subtree (O)
• Right of O→S
• Right → Null
• Visit S
• Left → T
• Right → Null
• Visit T
• Left → U → visit U
• Visit → Q
Left → R

Z, V, Y, X, W, P, S, U, T, Q, R

195.



Considering above binary tree, what will be the in order traversal

(a) B A D C E G F H (b) G H F E D C B A
(c) B A C D E G F H (d) G H F D E B C A

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Ans. (a) : The in-order traversal of a binary tree involves visiting the left sub-tree, the root node, and then the right sub-tree let's determine the in order traversal of the given binary tree in the image.

1. Start from the leftmost node: B(B is the left child of A, and D is the left child of B, E is the left child of D)
2. Visit the root node: A
3. Move to the right sub-tree: C(C is the right child of A)

Following this process, the in-order traversal for the given binary tree is:
BADCEGFH
So, the correct answer is : A

196. Arrange the following steps of the Inorder Traversal of Binary Tree in the correct order.
A. Visit the Left subtree
B. Visit the Root node
C. Visit the Right sub tree
D. Start traversing by visiting the nodes in rooted tree
E. Repeat the above three steps.
Choose the correct answer from the options given below :
(a) A, B, C, D, E (b) E, A, B, C, D
(c) D, A, B, C, E (d) D, A, E, B, C

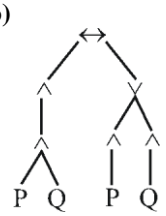
UGC NET C.S. December-2024

Ans. (c) : The correct sequence in order traversal of Binary tree is :
(D) Start traversing by visiting the nodes in the rooted tree.
(A) Visit the left sub-tree.
(B) Visit the Root node.
(C) Visit the right sub-tree
(E) Repeat the above three steps

197. Let us assume that you construct ordered tree to represent the compound proposition (~(p ∧ q)) ↔ (~p ∧ ~ q).
Then, the prefix expression and post-fix expression determined using this ordered tree are given as ..... and ..... respectively.
(a) ↔~∧pq∨~~pq, pq∧~p~q~ ∨↔
(b) ↔~∧pq∧~p~q, pq∧~p~q~∧↔
(c) ↔∧pq∨~~ pq, pq∧~p~~q∧↔
(d) ↔~∧pq∧~p~q, pq∧~p~~q∧↔

UGC NET C.S. August-2016 (Paper-II)

Ans :(b)



Prefix : ↔~∧pq∧~p~q
Postfix : pq∧~p~q~∧↔
option (b) is answer

198. In how many ways can the string
A ∩ B – A ∩ B – A
Be fully parenthesized to yield an infix expression?
(a) 15 (b) 14
(c) 13 (d) 12

UGC NET C.S. June-2016 (Paper-II)

Ans : (b) Number of ways we can parenthesize$n$ operands is catalun number.

$$= \frac{^{2n}C_n}{n+1}$$

Given number of total operators ($n = 4$).

So, $$= \frac{^{8}C_4}{5}$$

$$^{n}C_r = \frac{n!}{r!(n-1)!}$$

$$\frac{^{8}C_4}{5} = \frac{\left[\frac{8!}{4!4!}\right]}{5}$$

$$\frac{\left[\frac{8\times7\times6\times5\times4!}{4!\,4!}\right]}{5}$$

$$\frac{\left[\frac{8\times7\times6\times5}{4\times3\times2\times1}\right]}{5}$$

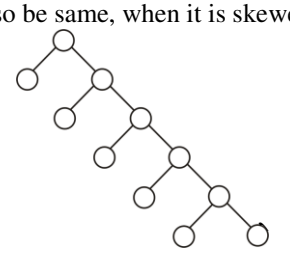
$$\frac{2\times7\times5}{5} = 14$$

Note: All operators work as nodes of binary search tree.

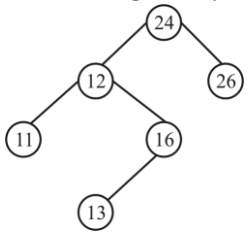
199. Suppose you are given a binary tree with n nodes, such that each node has exactly either zero or two children. The maximum height of the tree will be
(a) $\frac{n}{2}-1$ (b) $\frac{n}{2}+1$
(c) $\frac{(n-1)}{2}$ (d) $\frac{(n+1)}{2}$

UGC NET C.S. June-2016 Paper-II

Ans : (c) In full binary tree, if T has a total of n nodes, the number of internal nodes is I = (N – 1)/2 and height will also be same, when it is skewed full binary tree.



200. Consider the following binary search tree:

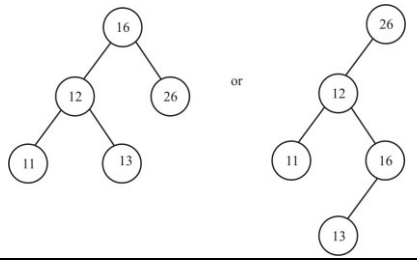


If we remove the root node, which of the node from the left subtree will be the new root?
(a) 11 (b) 12
(c) 13 (d) 16

UGC NET C.S. June-2016 Paper-II

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Ans : (d) If we delete root node of a binary search tree, then it can be replaced by inorder successor or inorder predecessor of the node.
So, either 16 or 26 can be root of the tree.



201. Which one of the followingtechniques is not used in thebinary tree?
Binary Tree में, निम्नलिखित में से किस तकनीक का उपयोग नहीं किया जाता है?
(a) Randomized traversal
(b) Preorder traversal
(c) Postorder traversal
(d) More than one of the above
(e) None of the above

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : Binary Tree is a type of data structure in which every node refers to two children. The main purpose of binary tree is to store data in an organized manner, which facilitates search, sort and other operations of data. There are different types of traversal techniques of binary tree –
1. In order Traversal
2. Preorder Traversal
3. Post order Traversal

202. Post-order Binary tree traversal is_____ .
पोस्ट-ऑर्डर बाइनरी ट्री ट्रैवर्सल _____ है।
(a) Left, Node, Right/लेफ्ट, नोड, राइट
(b) Left, Right, Node/लेफ्ट, राइट, नोड
(c) Node, Left, Node/नोड, लेफ्ट, राइट
(d) Right, Left, Node/राइट, लेफ्ट, नोड

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Post order traversal means:
1. Traverse the Left subtree first
2. Then traverse the Right subtree
3. Finally, visit the Node itself.
So, the order is Left→ Right→ Node
Other traversals:
Inorder traversal- Left → Node → Right
Pre-order traversal- Node→ Left → Right

203. The height of a tree is the length of the longest path from root to leaf. What is the maximum and minimum number of nodes in a binary tree of height 'n'?
एक ट्री की ऊंचाई जड़ से पत्ती तक के सबसे लंबे रास्ते की लंबाई है। ऊंचाई'n' के एक बाइनरी ट्री में नोड्स की अधिकतम और न्यूनतम संख्या क्या है?
(a) $2^{n+1}-1$, n +1 (b) 2^{n-1}, n – 1
(c) 2^n, n + 1 (d) $2^{n-1}-1$

DSSSB TGT C.S. 07.08.2021 Shift-I

Ans. (a) : The height of a binary tree is the length of the longest path from the root to a leaf.
For binary tree of height n
When the binary tree is complete (fully filled at all levels except possible the last), the maximum number of nodes us $2^{n+1}-1$.
And the minimum number of nodes is n +1.

204. Which of the following statement(s) is/are true about binary search algorithm?
बाइनरी सर्च एल्गोरिथम के बारे में निम्नलिखित में कौन सा/से कथन सत्य है/हैं?
I. It is implemented on sorted array.
इसे क्रमबद्ध सारणी पर लागू किया गया है।
II. One must have direct access to the middle element in any sub-list./किसी भी सब-लिस्ट में मध्य एलिमेंट तक डायरेक्ट एक्सेस होना चाहिए।
III. It is implemented on sorted linked list.
इसे क्रमबद्ध लिंक्ड सूची किया गया है।
(a) II and III (b) I and III
(c) I and II (d) I, II and III

DSSSB TGT C.S. 07.08.2021 Shift-I

Ans. (c) : Binary search tree works on sorted arrays.
We need direct (random) access to the middle element for binary search to be efficient.
Binary search tree on linked lists is insufficient because linked lists don't allow direct access to the middle element.

205. Which of the following options isnot true about the binarysearch tree?
BinarySearch Treeके बारे में, निम्नलिखित में से कौन-सा विकल्प सत्य नहीं है?
(a) The value of the left child should be less than the rootnode.
(b) The value of the right childshould be greater than theroot node.
(c) The left and right subtreesshould also be a binarysearch tree.
(d) More than one of the above
(e) None of the above

BPSC TRE 1.0 2023 (11-12)

Ans. (e) : Binary search tree (BST) is a data structure that stores data. In this every node has two child nodes – one left child and the other right. BST has some main rules like-
1. All left child nodes have a value less than their root node.
2.All right child nodes have a value greater than their root node.
3.Every sub-tree of a binary search tree is also a binary search tree.

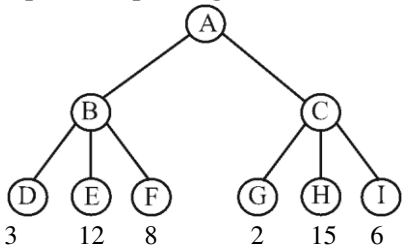
206. Suppose that we have numbers between 1 and 1,000 in a binary search tree and want to search for the number 364. Which of the following sequences could not be the sequence of nodes examined?
(a) 925, 221, 912, 245, 899, 259, 363, 364
(b) 3, 400, 388, 220, 267, 383, 382, 279, 364
(c) 926, 203, 912, 241, 913, 246, 364
(d) 3, 253, 402, 399, 331, 345, 398, 364

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Sarkari Resource

Ans.(c) : 913 can not be searched after 912, after 912 any number must be less than 912 since 912>364 So we will move towards 912 to 1.
So, answer is (c)

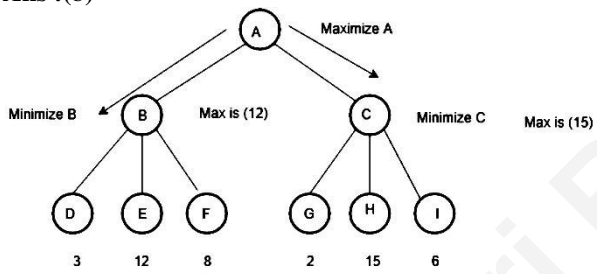
207. Consider the following game tree in which root is a maximizing node and children are visited left to right. What nodes will be pruned by the alpha-beta pruning?



(a) I (b) HI
(c) CHI (d) GHI

UGC NET C.S. June-2016 (Paper-III)

Ans :(b)



We have to maximize A and minimize B and C. Minimum of B is 3 and C is 2 But max from B is 12 and max from C is 15. So, there is no need of further expanding H and I. Because whatever value their successor will produce will be immaterial. Hence H I are pruned. For more information on Game theory Refer:Minimum Algorithm in Game Theory | Set 4 (Alpha-Beta Pruning) Option (B) is correct.

208. Which of the following traversal techniques lists the nodes of a binary search tree in ascending order?
निम्न ट्रैवर्सल तकनीकों में से कौन-सा, बाइनरी सर्च ट्री के नोडों को आरोही क्रम में सूचीबद्ध करता है?

(a) Postorder/पोस्ट-ऑर्डर
(b) Preorder/प्री-ऑर्डर
(c) Inorder/इन-ऑर्डर
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans.(c) : Inorder traversal lists the nodes of a binary search tree in ascending order. This traversal technique visits the nodes in the following order.
(i) Left subtree
(ii) Root
(iii) Right subtree
As a result, inorder traversal lists the nodes of a binary search tree in increasing order.

(iv) Hashing and Collision Resolution Techniques. Searching and Sorting

209. Which of the following is not collision resolution technique of hashing?
निम्नलिखित में से कौन सी हैशिंग की कोलिजन रिजोल्युशन टेक्निक नहीं हैं?

(a) Double hashing/डबल हैशिंग
(b) Quadratic probing/क्वाड्रैटिक प्रोबिंग
(c) Chaining/चैनिंग
(d) Folding/फोल्डिंग

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) :Folding: Folding is actually a method used in generating hash codes, not a collision resoluting technique. It involves dividing they key into parts and combining them to produce a hash code.
(a) Double Hashing: This is a technique of open addressing where a second hash function is used to determine the probe sequence.
(b) Quadratic probing: This is another open addressing technique where the interval between probes increases quadratically.
(c) Chaining: As Mentioned, this involves storing colliding elements in a linked list a the hashed index.

210. Which of the following recurrencerelation is related to worst case timecomplexity of quick sort ?
निम्न में से कौन सा रिकरेंस रिलेशन क्विक सॉर्ट एल्गोरिथ्म की वर्स्ट केस टाइम कॉम्प्लेक्सिटी से संबंधित है?

(a) $T(n) = T(n/2) + C$
(b) $T(n) = 2 . T(n/2) + Cn$
(c) $T(n) = T(n–1) + Cn$
(d) $T(n) = 7 . T(n/2) + Cn^2$

HTET PGT C.S. 2022

Ans. (c) : The worst case of Quick sort happens when the pivot choosen is always the smallest or largest element.
That means at each recursive call, one side has (n–1) elements and other has 0 elements.
So,
The recurrence relation becomes –
$T_{(n)} = T_{(n-1)} + C_n$ ($C_n \rightarrow$ partitioning cost)
Now →
(a) $T(n) = T(n/2) + C \Rightarrow$ This looks like binary search, not quicksort.
(b) $T(n) = 2 . T(n/2) + Cn \Rightarrow$ This is merge short, not quicksort.
(c) $T(n) = T(n–1) + Cn \Rightarrow$ Worst case of quiksort.
(d) $T(n) = 7 . T(n/2) + Cn^2 \Rightarrow$ It is similar to Strassen's matrix multiplication, not quicksort.
Thus Option (c) $T_{(n)} = T_{(n-1)} + C_n$ is correct.

211. Hashing refers to the process of
हैशिंग उस प्रक्रिया को कहते है जिसमें?

(a) Deriving a record key from storage address.
(b) Deriving storage address from a record key

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(c) Deriving an encoded address from a record key
(d) Deriving a record key from an encoded address.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : Deriving storage address, from a record key. Hashing is a process that takes an input and converts it into a fixed-size value, which often represents a storage address or index in a data structure like a hash table. This allows for efficient retrieval of data based on its key.

212. Which of the following table implementation is based on the property of the locality ofreference?
निम्न में से कौन सी टेबल का क्रियान्वयन लोकेलिटी ऑफ रिफरेन्स की प्रकृति पर निर्भर है?
(a) Hash table/हैश टेबल
(b) Search table/सर्च टेबल
(c) Linear list/लिनियर लिस्ट
(d) Self organizing list/सेल्फ आर्गनाइजिंग लिस्ट

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Ans.(d) : Self organizinglists are a table implementation that leverages locality of reference for efficient access by moving frequently accessed items closer to the front. Self organizing lists improve access times by dynamically adjusting their structure based on access patterns a concept closely tied to temporal locality.

213. Which of the following search method takes less memory?
निम्न में से कौन सा सर्च विधि कम मेमोरी लेता है?
(a) Depth-first search/डेप्थ-फर्स्ट सर्च
(b) Breadth-First search/ब्रिड्थ-फर्स्ट सर्च
(c) Linear search/लिनीयर सर्च
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans.(a) : Generally, Depth - First search (DFS) tends to useless memory compared to breadth- First search (BFS) in many scenarios. This is because DFS explores one both completely before backtracking, storing only the nodes on the current path in its stack.

214. Which of the following is not used for hash function?
निम्न में से कौन हैश फलन हेतु प्रयुक्त नहीं किया जाता है?
(a) Mid-square method /मिडस्क्वायर विधि
(b) Division method/डिविजन विधि
(c) Folding method/फोल्डिंग विधि
(d) Probe method/प्रोब विधि

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Ans. (d) : The Probe method on the other hand is a technique used in collision resolution within hash table, specifically during the process of finding an empty slot or locating and existing key when a collision occurs.

215. Which of the following is a pair of stable sorting algorithm
(a) Quick, Selection (b) Bubble, Insertion
(c) Heap, Bubble (d) Selection, Insertion

RPSC C.S. 2016 (Paper-II)

Ans.(b) : A stable sorting algorithm maintains the relative order of equal elements Both Bubble Sort and Insertion Sort are stable sorting algorithms. They preserve the original order of elements with the same value when sorting.
Quick, Selection– Quick sort is an unstable sorting algorithm. It does not guarantee the relative order of equal elements selection sort is also unstable.
Heap Bubble– Heap sort is an unstable sorting algorithm. While Bubble sort is stable, the combination of an unstable and stable sort does not make it stable.
Selection Insertion– Selection sort is an unstable sorting algorithm. While insertion sort is stable, again, the combination of an unstable and stable sort is not stable.

216. The number of swapping needed to sort the number 8, 22, 7, 9, 31 in ascending order using bubble sort is:-
(a) 3 (b) 5
(c) 7 (d) 10

RPSC C.S. 2016 (Paper-II)

Ans.(a) : Bubble Sort– To determine the number of swaps needed to sort the numbers 8, 22, 7, 9, 31 in ascending order using bubble sort, lets go through the process step by step.
Given– 8, 22, 7, 9, 31
Bubble Sort Steps–
1. First pass–
Compare 8 and 22- No swap (8 < 22)
Compare 22 and 7- Swap (22 > 7)⇒ 8, 7, 22, 9, 31 (1 swap)
Compare 22 and 9- Swap (22 > 9) ⇒ 8, 7, 9, 22, 31 (1 swap)
Compare 22 and 31= No swap (22 < 31)
Total swap in first pass = 2
2. Second pass–
Compare 8 and 7- Swap (8 > 7) ⇒ 7, 8, 9, 22, 31 (1 swap)
Compare 8 and 9- No swap
Compare 9 and 22- No swap
Compare 22 and 31- No swap
Total swaps in second pass =1
Total swaps = 2 (First pass) + 1 (Second pass) = 3

217. Which of the following is not a collision resolution technique in hash table using linear open addressing?
(a) Rehashing (b) Clustering
(c) Quadratic probing (d) Linear probing

RPSC C.S. 2016 (Paper-II)

Ans.(b) : Clustering is not a collision resolution technique. It's a problem that can arise with certain collision resolution techniques like linear probing.
Rehashing– Rehashing is a technique where the hash table is resized to accommodate more data, effectively creating a new hash function. This is a way to mitigate collisions, not a technique for resolving them when they occur.
Quadratic probing– It is a collision resolution technique that uses a quadratic formula to probe the hash table for an open slot.
Linear probing– Linear probing is a collision resolution technique the searches for an open slot sequentially.

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218. Separate chaining involves storing collided elements in separate data structures, typically in:
सेपरेट चेनिंग में टकराव वाले एलिमेंट्स को अलग-अलग डेटा संरचनाओं में संग्रहित किया जाता है, आमतौर पर:
(a) Linked lists/लिंक्ड लिस्ट्स
(b) Arrays/एरेज़
(c) Stacks/स्टैक्स
(d) Queues/क्यूज़

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : In hashing, collisions occur when two or more keys map to the same hash tuble index. separate chaining is a technique used to resolve such collisions.
In this method, each slot of the hash table holds a pointer to a linked list that stores all the elements hashing to that index. Thus, collided elements are kept in separate date structures, usually linked lists.

219. Which collision resolution technique involves placing collided elements in the next available empty slot in the hash table?
कौन-सी कोलिज़न रेज़ोल्यूशन तकनीक हेश टेबल में टकराव वाले एलिमेंट्स को अगले उपलब्ध खाली स्लॉट में रखने में शामिल होती है?
(a) Linear probing/लीनियर प्रोबिंग
(b) Quadratic probing/क्वाड्रेटिक प्रोबिंग
(c) Separate chaining/सेपरेट चेनिंग
(d) Double hashing/डबल हैशिंग

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : Collision resolution in hashing can be handled in several ways. Linear probing is on such technique where it the desired slot is already occupied, the algorithm sequentially searches for the next empty slot in the hash table and places the element there.
This makes it simple to implement but can cause flustering problems. Other method like quadratic probing or double hashing use different strategies to find the next available slot, while separate chaining uses liked list.

220. Which data structure often results in a time-space tradeoff by using extra memory to speed up operations?
कौन सा डेटा संरचना अतिरिक्त मेमोरी का उपयोग करके संचालन को तेज़ करके समय-स्पेस ट्रेडऑफ का कारण बनता है?
(a) Arrays/एरेज़
(b) Linked lists/लिंक्ड लिस्ट्स
(c) Hash tables/हैश टेबल्स
(d) Stacks/स्टैक्स

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Hash tables extra memory (hashing mechanism) to achieve faster search insert and delete operations which is a classic example of time-space tradeoff.

221. Which of the following sorting algorithms uses recursion?
निम्नलिखित में से किस वर्गीकरण एल्गोरिथ्म (सार्टिंग एल्गोरिथ्म) में आवर्तन (रिकर्सन) का प्रयोग किया जाता है?
(a) Insertion Sort (b) Heap Sort
(c) Merge sort (d) Bubble Sort

KVS PGT C.S. 2017

Ans.(c) : The Merge sort algorithm is a sorting algorithm that is based on the Divide and conquer paradigm. In this algorithm, the array is initially divided into two equal halves and then they are combined in a sorted manner. Merge sort as a recursive algorithm continuously splits the array in half until it cannot be further divided. It uses recursion to sort the elements of the array.

222. The maximum number of comparisons for a particular record among 32 sorted records through Binary Search method will be :
बाइनरी सर्च पद्धति के माध्यम से 32 सॉर्ट किए गए रिकॉर्ड के बीच किसी विशेष रिकार्ड के लिए तुलना की अधिकतम संख्या होगी:
(a) 2 (b) 10 (c) 8 (d) 5

KVS PGT C.S. 2017

Ans.(d) : Binary search, also known as half-interval search is a search algorithm that finds the position of a target value within a sorted array. Binary search runs in logarithmic time in the worst case, making O (log n) comparisons, where n is the number of elements in the array.
Here, we have to apply the binary search on 32 elements. so, it will take $\log_2 32 = 5$ comparisons to search for the element.

223. Which of the following algorithms use recursion for sorting an array of integers?
निम्नलिखित में से कौन सा एल्गोरिथम पूर्णांकों की एक सरणी को छाँटने के लिए पुनरावर्तन का उपयोग करता है?
(a) Bubble sort and Insertion sort
(b) Bubble sort and Quick sort
(c) Bubble sort and merge sort
(d) Quick sort and merge sort

KVS PGT C.S. 2018

Ans.(d) : Quick sort is a Divide and Conquer algorithm. It picks an element as a pivot and partitions the given array around the picked pivot, The merge sort algorithm is a sorting algorithm that is based on the Divide and Conquer paradigm. In this algorithm, the array is initially divided into two equal halves and then so, quick sort and merge sort algorithms are based on the divide and conquer algorithm which works in the recursive manner.

224. Sorting is
छँटाई (Sorting) है
(a) a process of re-arranging a given set of objects in a specific order./किसी विशिष्ट क्रम में दिए गए लक्ष्य समुच्चयों को पुनः व्यवस्थित करने को एक प्रक्रिया
(b) to facilitate the later search for members of the sorted set./छंटे हुए समुच्चय के सदस्यों को बाद में ढूँढने की सुविधा
(c) is a relevant and essential activity, particularly in data processing./प्रासंगिक तथा अनिवार्य क्रियाकलाप है, विशेष रूप से डेटा प्रोसेसिंग में।
(d) All of these/यह सभी

KVS PGT C.S. 2013

Sarkari Resource

Ans.(d) : All three statements are accurate and complete descriptions of the purpose and importance of sorting. Therefore, the correct choice is "All of these".

225. Consider the following statements about searching algorithms and choose the correct option.
सर्चिंग एल्गोरिथम के बारे में निम्नलिखित कथनों पर विचार कीजिए और सही विकल्प का चयन कीजिए:

(a) Sequential search requires the list to be ordered, while there is no such constraint for Binarysearch./सीक्वेंशियल सर्च के लिए लिस्ट का क्रमबद्ध होना आवश्यक है जबकि बाइनरी सर्च के लिए ऐसी कोई बाधा नहीं है।

(b) Binary search has time complexity of $O(\log_2 n)$, while sequential search has time complexityof O(n). Here, n is the number of elements in the list./बाइनरी सर्च में $O(\log_2 n)$ की टाइम कॉम्प्लेक्सिटी है, जबकि सीक्वेंशियल सर्च में O(n) की टाइम कॉम्प्लेक्सिटी है। यहाँ n सूची में अवयवों की संख्या है।

(c) Link list is a suitable data structure for storing a list, and performing Binary search.
लिंक लिस्ट, लिस्ट को स्टोर करने तथा बाइनरी सर्च करने के लिए एक उपयुक्त डाटा स्ट्रक्चर है।

(d) The best case in Binary search is, when the key element exists in the begining of the list./बाइनरी सर्च में सर्वोत्तम मामलो वह है जब ''की एलीमेंट'' लिस्ट के प्रारंभ में हो।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans.(b) : Binary search has time complexity of $O(\log_2 n)$, while sequential search has time complexityof O(n). Here, n is the number of elements in the list.

226. A hash table of length 10 uses open addressing with hash function h(k) = kmod 10 and linear probing. After inserting 6 values into an empty hash table, the table is shown below:
लम्बाई10की एक हैश तालिका, हैश फलन h(k) = kmod 10 और लीनियर प्रोबिंग के साथ ओपन ऐड्रेसिंग का उपयोग करती है। एक खाली हैश तालिका में 6 मान डालने के बाद तालिका निम्नानुसार है :

0	
1	
2	**42**
3	**23**
4	**34**
5	**52**
6	**46**
7	**33**
8	
9	

Which of the following choices gives a possible order in which the key values could have been inserted in the table?
निम्न में से कौन-सा विकल्प एक सम्भावित क्रम है जिसमें से तालिका में महत्वपूर्ण मान डाले जा सकते है?

(a) 46, 42, 34, 52, 23, 33
(b) 34, 42, 23, 52, 33, 46
(c) 46, 34, 42, 23, 52, 33
(d) 42, 46, 33, 23, 34, 52

LT. Grade C.S. 2018

Ans.(c) : The question involves a has table of length 10 using open addressing and linear probing with the hash function h(k) = kmod 10. We need to determine the correct sequence of insertion based on the given table.
Option (c) : 46, 34, 42, 23, 52, 33
• Insert 46 → 46 mod 10 = 6
placed at index 6.
• Insert 34 → 34 mod 10 = 4
placed at index 4.
• Insert 42 → 42 mod 10 = 2
placed at index 2.
• Insert 23 → 23 mod 10 = 3
placed at index 3.
• Insert 52 → 52 mod 10 = 2
collision → next available index is 5.
• Insert 33 → 33 mod 10 = 3
collision → next available index is 7.
Matches the given table, So the possible orders of insertion is 46, 34, 42, 23, 52, 33.

227. A hash function defined as f(key) = key mod 7 with linear probing is used to insert the keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0 to 6. What will be the location of 11?
एक हैश फंक्शन f(key) = key mod 7 के रूप में लीनियर प्रोबिंग के साथ, परिभाषित है। इसका प्रयोग कुंजियों 37, 38, 72, 48, 98, 11, 56 को 0 से 6 तक की तालिका में शामिल करने के लिए किया गया है। कुंजी 11 का स्थान क्या होगा?

(a) 3 (b) 4
(c) 5 (d) 6

LT. Grade C.S. 2018

Ans.(c) : To determine the location of the key 11 in the hash table, apply the given hash function and resolve collisions using linear probing.
Given hash function
f(key) = key mod 7
Step-by-step insertion of keys–

Index	Key
0	98
1	56
2	37
3	38
4	72
5	11
6	48

Final Hash Table-

Insert 37:
37 mod 7 = 2
key 37 is placed at index 2
Insert 38:
38 mod 7 = 3
key 38 is placed at index 3

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Insert 72:

72 mod 7 = 2
Index 2 is occupied (collision),
So we use linear probing to find the next available index:
check index 3 → Occupied
check index 4 → Free
key 72 is placed at index 4

Insert 48:

48 mod 7 = 6
key 48 is placed at index 6

Insert 98:

98 mod 7 = 0
key 98 is placed at index 0.

Insert 11:

11 mod 7 = 4
Index 4 is occupied, so we use linear probing.
Check index 5 → free
key 11 is placed at index 5.

Insert 56:

56 mod 7 = 0
index 0 is occupied, so we use linear probing check index 1 → free
key 56 is placed at index 1.

So, the location of key 11 is index 5.

228. Which of the following algorithm design techniques is used in the quicksort algorithm?
निम्न में से किस ऐल्गॉरिद्म डिजाइन तकनीक का उपयोग quicksort ऐल्गॉरिद्म में किया जाता है?

(a) Dynamic programing
डायनामिक प्रोग्रामिंग
(b) Backtracking/बैकट्रैकिंग
(c) Divide and conquer/डिवाइड एवं कॉन्क्वेर
(d) Greedy method/ग्रिडी मेथड

LT. Grade C.S. 2018

Ans.(c) : Divide and conquer quicksort algorithm uses divide and conquer technique, because it solves the problem by recursively dividing the problem into parts, until problem becomes trival.

229. Which data structure is used for efficientsearching, insertion, and deletion of elements?/किस डेटा संरचना का उपयोग किया जाता है, कुशल खोज, सम्मिलन और तत्वों का विलोपन?

(a) Stack
(b) Queue
(c) Hash table
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : The correct data structure for efficient searching, insertion, and deletion is the hash table. A stack follows the Last-in, First-out (FIFO) rules 80 it is useful only for specific tasks like function calls or expression evaluation, not for fast searching.

- A queue follows the First-In-First-Out (FIFO) rule, mainly used in scheduling and buffering, and is also not good for searching.
- On the other hand, a hash table uses a hash function to quickly locate where data should be stored or found.

230. Merge Sort is an example of which algorithm design paradigm?/Merge sort किस एल्गोरिथम डिजाइन प्रतिमान का एक उदाहरण है?

(a) Greedy
(b) Divide and Conquer
(c) Dynamic Programming
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : Marge sort is a sorting algorithm that follows the divide and conquer design paradigm. In this method, the array is first divided into two halves until only single elements remain. Then, these smaller parts are solved by sorting them individually. Finally, the sorted parts are combined together to form the completely sorted array. Because it uses this process of dividing, solving and combining, merge sort is a perfect example of the Divide and conquer approach.

231. Which of the following is NOT a step in theDivide and Conquer algorithm?/निम्नलिखित में से कौन-सा डिवाइड एंड कॉनकर एल्गोरिथम का चरण नहीं है?

(a) Combine/संयोजित
(b) Conquer/जीतना
(c) Divide/विभाजित करें
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (e) : Divide and Conquer algonthm works in three main steps. First, we Divide the big problem into smaller problems. Second, we Conquer by solving these smaller problems, often using recursin.
Third, we Combine the solutions of the smaller problems to get the final answer Since all three steps are part of Divide and Conquer, none of them is outside the process.
Correct ansawer:(e) None of the above

232. Which of the following is the biggestadvantage of selection sort?
निम्नलिखित में से कौन-सा चयनप्रकार का सबसे बड़ा लाभ है ?

(a) it has low time complexity
(b) it has low space complexity
(c) it requires only n swaps under any condition
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) :• Selection sort's biggest plus point is that it only swaps 'n' times.
• This is good because swapping can be slow or expensive (like swapping big items).
• It is not fast for large lists (It's slow).
• It does use less memory, but that's not its special feature.

233. Which of the following is a divide-and-conquer algorithm?
निम्नलिखित में से कौन-सा फूट डालो और जीतो एल्गोरिथ्म है?

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(a) Merge sort
(b) Heap sert
(c) Bubble sort
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

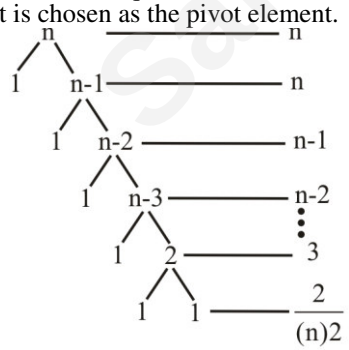
BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Merge sort is a divide and conquer algorithm. It splits the input array into two ports, calls itself for the two parts and then merges the two sorted parts. Merry sort and quick sort are both based on the divide and conquer method.

234. There are many sorting algorithms based on comparison. The running time of heapsot algorithm is O(Nlgn). Like P, but unlike Q, heapsort sorts in place where (P, Q) is equal to
(a) Merge sort, Quick sort
(b) Quick, sort, insertion sort
(c) Insertion sort, Quick sort
(d) Insertion sort, Merge sort

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : The question asks specifically for inplace sorting algorithms, which means the algorithms in which only O(1) space is used.
Insertion sort :
In insertion sort, the worst case takes $O(n_2)$ time, worst case of insertion sort is when elements are sorted in reverse order. It is in place sorting algorithms.
In that case the number of comparison and be like.
Merge Sort –
In merge sort, the worst case takes O(inlogin) time. Merge sort is based on the divide and conquer approach. It is out of place sorting algorithms. Recurrence relation for merge sort will become
$T(n) = 2T(n/2) + O(n)$
$T(n) = n + O(n)$
$T(n) = n \times \text{Log } n$
Quicksort : In Quicksort, the worst case takes $O(n_2)$ time the worst case of quicksort is when the first or the last element is chosen as the pivot element.



Recurrence relation for quick sort $T(n) = T(n-1) + (n)$
This will give the worst case time Complexity as $O(n^2)$.
It is in place sorting algorithms.
Conclusion :P needs to be an in-place sorting algorithm, and Q needs to be out of place sort in algorithm, Hence option d is correct

235. A hash function h defined h(key)=key mod 7, with linear probing, is used to insert the keys44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?
(a) 3 (b) 4
(c) 5 (d) 6

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) :

0	1	2	3	4	5	6

Keys = 44, 45, 79, 55, 91, 18, 63, insert one by one, start with 44, 44% 7 = 2

0	1	2	3	4	5	6
		44				

Insert 41, 45% 7 = 3

0	1	2	3	4	5	6
		44	45			

Insert 79,

0	1	2	3	4	5	6
		44	45	79		

79% 7 = 27 collision → check
Next cell; collision ⇒ check next cell ⇒ fit at 4
Insert – 55, 55%7 = 6

0	1	2	3	4	5	6
		44	45	79		55

Insert 18, 18%7 = 4 .....collision ⇒ check next cell ⇒ fit at 5

0	1	2	3	4	5	6
91		44	45	79	18	55

Therefore 18 fits at location 5.

236. Which of the following algorithm design approach is used in quick sort algorithm?
(a) Dynamic programming
(b) Back Tracking
(c) Divide and Conquer
(d) Greedy approach

UGC NET C.S. December 2021

Ans. (c) : Merge sort, quicksort uses divide and conquer, so it's a recursive algorithm.

237. Arrange the following search and sort algorithms in ascending order of their worst-case complexities.
A. Merge sort **B. Linear Search**
C. Bubble Sort **D. Binary Search**
Choose the correct answer from the options given below :
(a) A, B, D, C (b) B, D, A , C
(c) B, D, C, A (d) B, A, D, C

UGC NET C.S. December-2022

Ans. (b) : Here are the search and sort algorithms arranged in ascending order of their worst-case complexities.
B. Linear search - worst-case complexity : O(n)
D. Binary search - worst - case complexity : O(log n)
A. Merge sort - worst - case complexity : O (nlog n)
C. Bubble sort - worst - case complexity : O (n^2)
So option (b) is correct.

238. Which of the following symbol table implementation is best suited if access time is to be minimum?
(a) Linear list (b) Search tree
(c) Hash Table (d) Self organisation list

UGC NET C.S. December-2023

Sarkari Resource

Ans. (c) : Hash table implementations are best suited for minimizing access time, especially for lookups and inserts, as they provide constant time average case performance for these operations. Its access time is 0(1). However, their performance can degrade if there are many collisions or if the hash function is poorly designed.

239. Which collision resolution technique involves maintaining a linked list of collided keys?

(a) Linear probing (b) Quadratic probing
(c) Chaining (d) Double hashing

UGC NET C.S. December-2023

Ans. (c) : Chaining is a collision resolution technique used in hash tables. In chaining, when a collision occurs (two keys hash to the same index), the collided keys are stored in linked lists at that index. This allows multiple keys to be stored at the same location in the hash table. when searching for a key, the hash function is used to find the correct index, and then the linked list at that index is traversed to find the desired key.

240. If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m, where n ≤ m, the expected number of collisions involving a particular key x is less than .....

(a) 1 (b) 1/n
(c) 1/m (d) n/m

UGC NET C.S. June-2017 (Paper-II)

Ans :(a) Let X_1 be random variable that defines as follows :–
X is particular key and X is the random variable
$X_1 = 1$ if X collide with another key
$X_1 = 0$ if it does not collide
Now single pair collide with probability 1/m as 'm' is the number of slots
Now $E(X_1) = 1 * 1/m + 0 * (m - 1)/m$
$E(X_1) = 1/m$
Now, X_2, X_3, X_4 …….. X_n be the random variable define same as
$X_i = 1$ if X collide with key C[i]
= 0 otherwise
$E(X_1 + X_2 \ldots\ldots X_n) = E(X_1) + E(X_2) + \ldots\ldots E(X_n)$
[as all event are independent]
= 1/m + 1/m + ….1/m
[(n – 1) time as x already inserted one place]
= (n – 1)/m
Now n < = m
So E (X) < = 1
(So option A is correct answer)

241. Which of the following is true for computation time in insertion, deletion and finding maximum and minimum element in a sorted array?

(a) Insertion – 0(1), Deletion – 0(1), Maximum – 0(1), Minimum – 0(1)
(b) Insertion – 0(1), Deletion – 0(1), Maximum – 0(n), Minimum – 0(n)
(c) Insertion – 0(n), Deletion – 0(n), Maximum – 0(1), Minimum – 0(1)
(d) Insertion – 0(n), Deletion – 0(n), Maximum – 0(n), Minimum – 0(n)

UGC NET C.S. June-2017 (Paper-II)

Ans : (c) Array is sorted, so in worst case(last item insert) Insertion = O(n), in worst case, last item delete deletion O(n) maximum and minimum is always at end.
maximum →O(1), minimum →O(1)

242. Suppose that the splits at every level of quicksort are in the proportion (1 –α) to α. where 0 <α≤ 1/2 is a constant. The minimum depth of a leaf in the recursion tree is approximately given by

(a) $-\frac{lgn}{lg(1-\alpha)}$ (b) $-\frac{lg(1-\alpha)}{lgn}$
(c) $-\frac{lgn}{lg\alpha}$ (d) $-\frac{lg\alpha}{lgn}$

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : In Quicksort, the depth of a leaf depends on how the input size reduces at each level.
If the split is always in the ratio (1–α) to α, the minimum depth corresponds to the path where the recursion always follows the smaller partition, i.e., of size αn.
At each level, the size becomes:
n, αn, α^2n, α^3n, …
We stop when the size becomes 1:
So,
We have –
$\alpha^k n = 1$
$\log(\alpha^k n) = \log(1)$
$\log(\alpha^k) + \log(n) = 0$
$k \log(\alpha) = -\log(n)$
$k = -\frac{\log(n)}{\log(\alpha)}$

243. If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m, where n ≤ m, the expected number of collisions involving a particular key K is

(a) less than 1 (b) less than *l*g n
(c) greater than 1 (d) greater than *l*g n

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : Because, This is a theorem in universal hashing :
"if h is chosen from a universal collection and is used for hash n keys in table of size m, where n < = m, the expected number of collision involving a particular key is less than 1")
So the answer is (a).

244. Mergesort makes two recursive calls. Which statement is true after these two recursive calls finish, but before the merge step?

(a) The array elements form a heap.
(b) Elements in each half of the array are sorted amongst themselves.
(c) Elements in the first half of the array are less than or equal to elements in second half of the array.
(d) All of the above

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Ans : (b) After the two recursive calls in Mergesort complete, but before the merge step, elements in each half of the array are sorted amongst themselves because, Mergesort is possible when two arrays are sorted. If two unsorted arrays are submitted to Mergesort algorithm. It will sort each of them first before merging them, because Mergesort is a divide and conquer algorithm, Hence the answer is option (b).

245. You have to sort a list L. consisting of a sorted list followed by a few 'random' elements. Which of the following sorting method would be most suitable for such a task?

(a) Bubble sort (b) Selection sort
(c) Quick sort (d) Insertion sort

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Ans :(d)For a sorted list with few random numbers:
(i) Bubble sort will take $O(n^2)$ time in best case.
(ii) Selection sort will take $O(n^2)$ time in best case.
(iii) Quick sort will take $O(n^2)$ time in this case because it is the worst case for quick sort.
(iv) Insertion sort will take O(n)

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246. Two way merge sort algorithm is used to sort the following elements in ascending order.
200, 470, 150, 80, 90, 40, 400, 300, 120, 70.
What is the order of these elements after second pass of the merge sort algorithm?
टू वे मर्ज सॉर्ट एल्गोरिथ्म का उपयोग निम्नलिखित तत्वों को आरोही क्रम में क्रमबद्ध करने के लिए किया जाता है।
200, 470, 150, 80, 90, 40, 400, 300, 120, 70.
मर्ज सॉर्ट एल्गोरिथ्म के दूसरे पास के बाद इन तत्वों का क्रम क्या है?

(a) 40, 70, 80, 90, 120, 150, 200, 300, 400, 470
(b) 200, 470, 80, 150, 40, 90, 300, 400, 70, 120
(c) 80, 150, 200, 470, 40, 90, 300, 400, 70, 120
(d) 40, 80, 90, 150, 200, 300, 400, 470, 70, 120

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Ans. (c) :
(1) First Pass–
Split into individual elements and then merge into pairs.
(200, 470) (150, 80), (90, 40), (400, 300), (120, 70)
Sort each pair–
(200, 470), (80, 150), (40, 90), (300, 400), (70, 120).
(2) Second Pass–
Merge and sort adjacent pairs from the first pass:
(200, 470) and (80, 150)→ (80, 150, 200, 470) → 40, 90, 300, 400.
The last pair (70, 120) remains as is after the second pass, the elements are grouped into
(80, 150, 200, 470)
(40, 90, 300, 400)
(70, 120)
Final order after second pass:
The elements, after the second pass of the merge sort algorithm, are:
(80, 150, 200, 470), (40, 90, 300, 400) (70, 120)
This order represents the partially merged and sorted sub arrays after the second pass.

247. Using division method [n(k) = k mod m)], at which position 177 and 197 key values are stored in hash table when the size of hash table is 57?
विभाजन विधि [n(k) = k mod m)]का उपयोग करते हुए, 177 तथा 197 की मान जो हैश तालिका में संचय है, जब हैश तालिका का आकार 57 है किस पद पर है?

(a) 6, 26 (b) 7, 27
(c) 26, 6 (d) 27, 7

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Ans. (a) : We are asked to find the positions of key 177 and 197 in a hash table of size 57 using the division method, where -:
$h(K) = K \bmod m$
Here, $m = 57$
(i) Compute positions for 177
177 mod 57
57×3=171
177 – 171 = 6
(ii) Compute positions for 197
197 mod 57
57×3 = 171
197–171 = 26
So, the correct answer 6, 26

248. What is the average number of comparisons required in sequential search to find an element?
किसी तत्व को खोजने के लिए सिक्युएन्शिअल सर्च में आवश्यक तुलनाओं की औसत संख्या क्या है?

(a) n^2 (b) n
(c) $\frac{n(n+1)}{2}$ (d) $\frac{(n+1)}{2}$

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Ans. (d) : •Bestcase → element at first position
⇒comparison
Worst case element at last position
(also absent) ⇒n
Comparisons
Average Case
Assume that the first element is at any position with equal probability.
Finding average comparisons

$$\text{Average} = \frac{1+2+3---+n}{n}$$

$$= \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$$

249. Quicksort algorithm is based on which algorithm design technique?
क्विक सॉर्ट एल्गोरिथम किस एल्गोरिथम डिजाइन तकनीक पर आधारित है?

(a) Greedy Algorithm/ग्रीडी अल्गोरिथम
(b) Brute – Force/ब्रूट – फोर्स
(c) Divide and Conquer/डिवाइड एंड कॉन्कर
(d) Dynamic programming/डायनामिक प्रोग्रामिंग

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Ans. (c) : Quicksort algorithm is based on Divide and conquer algorithm design technique that picks an element as a pivot and partitions the given array around the picked pivot by placing the pivot in its correct position in the sorted array.

250. Sort the following elements using Bubble sort in assending order. What is the intermediate sequence of 37, 54, 21, 85, 68, 12, 9, 57 after second pass ?
बबल सॉर्ट का उपयोग करके निम्नलिखित तत्वों को आरोही क्रम में क्रमबद्ध करे। सेकेंड पास के बाद 37, 54, 21, 85, 68, 12, 9, 57 का इंटरमीडिएट सिक्वेंस क्या है?

(a) 21, 37, 12, 9, 54, 57, 68, 85
(b) 37, 21, 54, 68, 12, 9, 57, 85
(c) 21, 37, 54, 12, 9, 57, 68, 85
(d) 21, 12, 9, 37, 54, 57, 68, 85

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Ans. (c) : In the first pass, we compare and swap adjacent element and swaps them if they are in to its correct position at the end of the at the end of the unsorted portion of the array.
Compare (37 and 54) -37 is not > 54.
Sequence remains: 37, 54, 21, 85, 68 12, 9, 57
Compare (54 and 85)- 54 is not > 85
Sequence remains – 37, 21, 54, 85, 68, 12, 9, 57
Compare (85 and 68) – 85 > 68. Swap.
Sequence becomes: 37, 21, 54, 68, 85, 12, 9, 57
Compare (85 and 12) -85 > 12, swap.
Sequence becomes – 37, 21, 54, 68, 12, 85, 9, 57
Compare (85 and 9) – 85 > 9. Swap.
Sequence becomes – 37, 21, 54, 68, 12, 9, 85, 57
Compare (85 and 57)- 85 > 57 , swap.
Sequence becomes: 37, 21, 54, 68, 12, 9, 57, 85
At the first pass, the sequence is: 37, 21, 54, 68, 12, 9, 57, 85
In the second pass, we repeat the process but this time we only need to compare up to the second already sorted, the second largest unsorted element will bubble up' to its correct position (the second to last)
Sequence after pass 1: 37, 21 54, 68, 12, 9, 57 85
Compare 37 and 21-37> 21. Swap.
Sequence becomes: 21- 37 , 54, 68, 12, 9, 57 85
Compare 37 and 54 - 37 is not > 54.
Sequence remains – 21, 37, 54, 68, 12, 9 57, 85
Compare 68 and 12 – 68 > 12. Swap
Sequence becomes – 21, 37, 54, 12, 68, 9, 57, 85
Compare 68 and 9 – 68 > 9. Swap.
Sequence becomes – 21, 37, 54, 12, 9, 68, 57, 85
Compare 68 and 57 – 68 > 57 . swap.
Sequence becomes : 21, 37, 54 12, 9, 57, 68, 85
After the second pass, the sequence is: 21, 37, 54, 21, 85, 68, 12, 9 and 57 after the second pass of Bubble sort in ascending order is 21, 37, 54, 12, 9, 57, 68, 85
Hence the correct answer is option (c).

251. The following keys 22, 28, 23, 12 and 13 are inserted into initially empty hash table of length 9 using open addressing with hash function h(k)=k mode 9 and linear probing. At wht index key 13 is inserted?
निम्नलिखित कुंजियाँ 22, 28, 23, 12 और 13 को ओपन एड्रेसिंग के साथ हैश फंक्शन h(k)=k मोड 9 और लीनियर प्रोबिंग का उपयोग करके 9 लम्बाई की प्रारंभिक खाली हैश टेबल में डाला गया है। किस इंडेक्स पर की 13 को डाला गया है?

(a) 4 (b) 5
(c) 6 (d) 7

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Ans. (c) : Hash Table :– A hash table is a data structure that stores key- value pairs. It uses a hash function to compute an index, also called a hash code into an array of buckets or slots, from which the desired value can be found. Let's derail each insertion step.
1.Insert 22:– calculate hash index
h(22) = 22 (mod 9) = 4. Index 4 is empty.
Insert 22 at index 4.
2. Insert 28: Calculate hash index
h (28) = 28 (mod 9) = 1. Index 1 is empty. Insert 28 at index 1.
3. Insert 23: Calculate hash index
h(23) = 23 (mod 9) = 5. Index 5 is empty. Insert 12 at index 5.
5. Insert 13: calculate has index
h(13) = 13 (mod 9) = 4
The initial has index is 4. Index 4 is occupied by key 22.
• According to linear probing, we check the next index: (4 + 1)
(mod 9) = 5. Index 5 is occupied by key 23.
• We check the next index : (5 +1)
(mod 9) = b. index 6 is empty.
• Insert key 13 at index 6.
After all the keys (22, 28, 23, 12 and 13) are inserted using the specified hash function and linear probing, the key 13 is located at index 6.

252. Sort the following list using Radix sort algorithm.
329, 457, 839, 436, 720, 355, 657
What is the output of the algorithm after 2nd pass ?
रेडिक्स सॉर्ट एल्गोरिदम का उपयोग करके निम्न सूची को क्रमबद्ध करें।
329, 457, 839, 436, 720, 355, 657
दूसरी पास के बाद एल्गोरिदम का आउटपुट क्या है?

(a) 720, 355, 436, 457, 657, 329, 839
(b) 720, 329, 436, 839, 355, 457, 657
(c) 329, 355, 436, 457, 657, 720, 839
(d) 355, 329, 457, 436, 720, 657, 839

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Ans. (b) : Radix sort algorithm – Radix sort is a non-comparative sorting algorithm that sorts integers by sorting on the individual digits. It starts from the least significant digit (ones place) and moves to the most significant digit.
Passes of Radix soot for : given listg : 329, 457, 839, 436, 720, 355, 657
1. First pass (ones place) : sort based on the ones digit.
Digits : 9(39), 7 (457), 9(839), 6(436), 0(720), 5(355), 7(657) output after first pass.
Second pass (tens place) – sort based on the tens digit using the output of the first pass

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(720, 355, 436, 657, 457, 329, 839),
Digits : 2(720), 5 (355), 3(436), 5(657), 5(457), 2(329), 3(939)
The output after second pass - 329, 720, 436, 839, 355, 457, 657
Hence the correct answer is : 720, 329, 436, 839, 355, 457, 657.

253. Sort the following list using selection sort algorithm
H, V, A, T, L, N, K
What is the output of the algorithm after 3rd pass ?
सिलेक्शन सॉर्ट एल्गोरिदम का उपयोग करके निम्न सूची को क्रमबद्ध करें।
H, V, A, T, L, N, K
तीसरी पास के बाद एल्गोरिदम का आउटपुट क्या है?

(a) A, V, H, T, L, N, K
(b) A, H, K, T, L, N, V
(c) A, V, T, H, L, N, K
(d) A, T, H, K, L, N, V

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Ans. (b) : Sortingusing selection sort algorithm –
In the selection sort algorithm we find the smallest element in each pass and place it in the position.
Passes Description. –
Initial list : H, V, A, T, L, N, K
1.first pass the smallest element is A. A is placed in the first position.
Output: A, V, H, T, L, N, K
2.Second pass - The smallest element in the remaining list is H. H is placed in the second position.
Output: A, H, V, T, L, N, K
3. third pass - The smallest element in the remaining list is k. k is placed in the third position.
Output: A, G, K, T, L, N, V
The output of the algorithm after the 3rd pass.
is: A, H, K, T, L, N, V
The correct answer is (b)

254. Arrange the following in ascending order of their average case complexity.
Selection-sort, Quick-sort, Linear search, Binary search
निम्नलिखित को उनकी औसत केस कंप्लैक्सिटी के आरोही क्रम में व्यवस्थित करें ।
सेलेक्शन –सॉर्ट,क्विक-सॉर्ट, लिनियर सर्च, बाइनरी सर्च

(a) Quick-Sort, Selection-sort, Binary search, Linear search /क्विक-सॉर्ट, सेलेक्शन-सॉर्ट,बाइनरी सर्च, लिनियर सर्च
(b) Linear search, Binary search, selection-sort, Quick-sort /लिनियर सर्च, बाइनरी सर्च, क्विक-सॉर्ट, सेलेक्शन-सॉर्ट
(c) Binary search, Linear search, Quick-sort Selection-sort /बाइनरी सर्च, लिनियर सर्च, क्विक-सॉर्ट, सेलेक्शन-सॉर्ट
(d) Selection-sort, Quick-sort, Linear search, Binary search /सेलेक्शन-सॉर्ट, क्विक-सॉर्ट, लिनियर सर्च, बाइनरी सर्च

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Ans. (c) : Ascending order of average case complexity–
Arranging these in ascending order of their average case complexity–
1. Binary search (O(log n))
2. Linear search (O (n))
3. Quick sort (O (n log n))
4. Selection sort (O (n^2))
The correct order is : Binary search, Linear search, Quick-sort, selection sort.

255. While inserting the elements 71, 56, 48, 96, 76, 82, 98 in an empty binary search tree in the sequence shown, which element will come in the lowest level?
दिखाए गए क्रम में एक खाली बाइनरी सर्च ट्री में 71, 56, 48, 96, 76, 82, 98 अवयवों को सम्मिलित करते हुए, निम्नतम स्तर पर कौन सा अवयव आयेगा

(a) 96 (b) 98
(c) 71 (d) 82

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Ans. (d) : The elements are inserted into an empty binary search tree in the sequence : 71, 56, 48, 96, 76, 82, 98.
After insertion, the level of the elements are:
Level 0 : 71 (root)
Level 1 : 56 : 96
Level 2 : 48, 76, 98
Level 3 : 82
The lowest level is 3 and the element at this level is 82.
Indeed 82.

256. How many interchanges will happen when the sequence of letters of "SORTED" is sorted using bubble sort?
बबल सॉर्ट का उपयोग करते हुए "SORTED" के अक्षरों के क्रम को क्रमबद्ध करने पर कितने इंटरचेंज होंगे?

(a) 36 (b) 11
(c) 18 (d) 6

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Ans. (b) : Babble sort is a simple sorting algorithm that repeatedly steps through the list, compares adjacent element and swaps them if they are in the wrong order. The pass through the list is repeated until the list is sorted.
The sequence is - S, O, R, T, E, D
Position 1 : S
Position 2 : O
Position 3 : R
Position 4 : T
Position 5 : E
Position 6 : D
ASCII values-
To compare the letters, it's helpful to know their ASCII values.
D : 68
E : 69
O : 79
R : 82
S : 83
T : 84
So, the order is D < E < O < R < S < T
Initial Sequence: S, O, R, T, E, D

Sarkari Resource

First Pass–
1. Compare S (83) and O (79) : S > O
→ swap
swap : O, S, R, T, E, D
Interchange count : 1
2. Compare S (83) and R (82) : S > R → Swap
swap : O, R, S, T, E, D
interchange count : 2
3. Compare S (83) and T (84) : S < T → no swap
4. Compare T (84) and E (69) : T > E → swap
swap : O, R, S, E, T, D
interchange count : 3
5. Compare T (84) and D (68) : T > D → swap
swap : O, R, S, E, D, T
interchange count : 4
After first pass : O, R, S, E, D, T

Second pass–
1. Compare O (79) and R (82) : O < R → no swap
2. Compare R (82) and S (83) : R < S → no swap
3. Compare S (83) and E (69) : S > E → swap
swap : O, R, E, S, D, T
interchange count : 5
4. Compare S (83) and D (68) : S > O → swap
swap : O, R, E, D, S, T
interchange count : 6
After second pass : O, R, E, D, S, T

Third pass–
1. Compare O (79) and R (82) : O < R → no swap
2. Compare R (32) and E (69) : R > E → swap
swap : O, E, R, D, S, T
interchange count : 7
3. Compare R (82) and D (68) : R > D → swap
swap : O, E, D, R, S, T
interchange count : 8
After Third pass : O, E, D, R, S, T

Fourth Pass–
1. Compare O (79) and E (69) : O > E → swap
swap : E, O, D, R, S, T
interchange count : 9
2. Compare O (79) and D (68) : O > D → swap
swap : E, D, O, R, S, T
interchange count : 10
After fourth pass : E, D, O, R, S, T

Fifth pass–
1. Compare E (69) and D (68) : E > D → swap
swap : D, E, O, R, S, T
interchange count : 11
After fifth pass : D, E, O, R, S, T

Total interchanges–
Counting all the swaps we made-
1. First pass - 4 swaps
2. Second pass - 2 swaps
3. Third pass - 2 swaps
4. Fourth pass - 2 swaps
5. Fifth pass - 1 swaps

Total interchanges = 4 + 2 + 2 + 2 + 1 = **11**

257. Which of the following is not used to calculate Hash functions?
निम्नलिखित में से किसका उपयोग हैश फ़ंक्शन की गणना के लिए नहीं किया जाता है?
(a) Folding method/फोल्डिंग विधि
(b) Modular method/मॉडल विधि
(c) Division method/डिवीजन विधि
(d) Rehashing/रीहैशिंग

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Ans. (d) : Rehashing is not a method to calculate a hash function. Instead, it is a collision resolution technique where a new has function is applied when the first one leads to a collision.

258. Match the List-I with List-II

	List-I		List-II
A.	**Breadth First Search**	**I.**	**LISP**
B.	**Depth First Search**	**II.**	**Syntax tree**
C.	**Prefix**	**III.**	**Stack**
D.	**Infix**	**IV.**	**Queue**

Choose the correct answer from the options given below :
(a) A-III, B-IV, C-I, D-II
(b) A-III, B-IV, C-II, D-I
(c) A-IV, B-II, C-III, D-I
(d) A-IV, B-III, C-I, D-II

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Ans. (d) : The correct match is-

	List-I		List-II
A.	Breadth First Search	–	Queue
B.	Depth First Search	–	Stack
C.	Prefix	–	LISP
D.	Infix	–	Syntax tree

259. Match the List-I with List-II

	List-I Algorithm		List-II Complexity
A.	**Insertion Sort**	**I.**	**O(log n)**
B.	**Binary Search**	**II.**	**$O(n^2)$**
C.	**Quick Sort**	**III.**	**O(n–1)**
D.	**Selection Sort**	**IV.**	**O(n log n)**

Choose the correct answer from the options given below :
(a) A-III, B-I, C-IV, D-II
(b) A-II, B-III, C-I, D-IV
(c) A-I, B-II, C-IV, D-III
(d) A-II, B-III, C-IV, D-I

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Ans. (a) :

	List-I Algorithm		List-II Complexity
A.	Insertion Sort	–	O(n–1)
B.	Binary Search	–	O(log n)
C.	Quick Sort	–	O(n log n)
D.	Selection Sort	–	$O(n^2)$

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260. Which of the following is stable sort?
निम्नलिखित में से कौन सा स्टेबल सॉर्ट है?
(a) Selection sort/सेलेक्शन सॉर्ट
(b) Quick sort/क्विक सॉर्ट
(c) Heap sort/हीप सॉर्ट
(d) Merge sort/मर्ज सॉर्ट

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Ans. (d) : A stable sorting algorithm maintains the relative order of records with equal keys.
Merge sort is a stable sort.

(v) Algorithms

261. Pictorial representation of an algorithm is called as:
किसी एल्गोरिथ्म का सचित्र निरूपण कहलाता है–
(a) Image/इमेज
(b) Picture/पिक्चर
(c) Flowchart/फ्लोचार्ट
(d) Pseudocode/स्युडोकोड

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Ans.(c):
- Flowcharts are diagrams that use symbols and lines to visually represent the step-by-step process of an algorithm or program.
- They help in understanding the sequence of operations, decisions, and the flow of control within an algorithm.
- The visual representation makes complex processes easier to comprehend and communicate.

262. Step by step description of how to arrive at a solution to a given problem is known as:
किसी दी गई समस्या के समाधान तक पहुँचने के स्टेप-बाइ-स्टेप विवरण को कहा जाता है–
(a) Algorithm/एल्गोरिथ्म
(b) Plan/प्लान
(c) ASCII Code/एएससीआईआई कोड
(d) UNICODE/यूनीकोड

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Ans.(a): Step by step description of how to arrive at a solution to a given problem is known as algorithm.
An algorithm is a clear, finite, step by step procedure to solve a problem or perform a task.

263. How may distinct stage are there in DES algorithm, which is parameterized by a 56-bit key?
(a) 16 (b) 17
(c) 18 (d) 19

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Ans :(d)The DES algorithm works with a 64-bit block size, employs 16 processing rounds, and is driven by a 56-bit key (with 8 additional parity bits), making up a total of 19 distinct stages.

264. In the Tower of Hanoi problem with "n" disks, how many moves are required to solve the problem?
'n' डिस्क के साथ टॉवर ऑफ हनोई समस्या को हल करने के लिए कितने मूव्स की आवश्यकता होती है?
(a) n (b) 2n
(c) 2^n-1 (d) 2^n

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : It the tower of Hanoi problem if there are n disks. The minimum of moves required is
$2^n - 1$
- For n = 1→ $2^1 - 1 = 1$ move
- For n = 2 → $2^2 - 1 = 3$ moves
- For n = 3 → $2^3 - 1 = 7$ moves

265. Which of the following problem cannot be solved without recursion?
निम्न में से किस प्राब्लम को रिकर्जन के बिना नहीं हल कर सकते हैं?
(a) Tower of Hanoi/टावर ऑफ हनोई
(b) Fibonacci series/फिबोनासी सीरीज
(c) Tree traversal/ट्री ट्रैवर्सल
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : While recursion is a natural and often elegant way to solute problems like the tower of Hanoi Fibonacci series and Tree Traversal, none of them can not solved without recursion.

266. The term "DFD" stands for?
'DFD' का पूर्ण रूप क्या है?
(a) Data file diagram/डेटा फाइल डायग्राम
(b) Data flow document/डेटा फ्लो डॉक्यूमेंट
(c) Data flow diagram/डेटा फ्लो डॉयग्राम
(d) None of the above/इनमें से कोई नहीं

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The term DFD stands for data flow diagram. In simple terms, a DFD is a diagram that shows hour data moves through a system. It illustrate where the data comes from, hour it is processed, and where it goes. This makes it easier to understand, analyze, and design a system especially in software development or information system. It is a widly used tool in system analysis because it clearly represents the flow of information.

267. The algorithm used in the Game tree to make decisions of Win/Lose is ……?
गेम ट्री में जीत/हार के निर्णय के लिए कौन-सा एल्गोरिदम उपयोग होता है?
(a) Heuristic Search Algorithm
हयूरिस्टिक सर्च एल्गोरिदम
(b) DFS/BFL algorithm/DFS/BFS एल्गोरिदम
(c) Greedy Search algorithm/ग्रीडी सर्च एल्गोरिदम
(d) Min/Max algorithm/मिन/मैक्स एल्गोरिदम

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : In game trees the minimax Algorithm is used to decide the best move for a player by assuming that the opponent also plays optimally. It helps determine win/loss or draw out comes.

Sarkari Resource

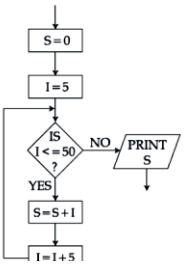
268. .....is a pictorial representation of a program.
.....Program का pictorial प्रस्तुती है।

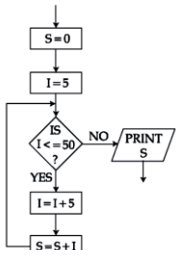
(a) Icon (b) Folder
(c) Flowchart (d) Command

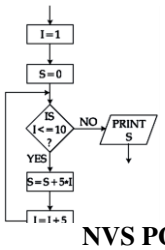
Bihar STET C.S. 18.09.2020 (Shift-I)

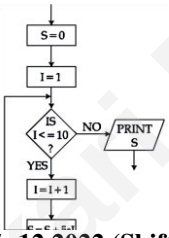
Ans. (c) :A flowchart is a pictorial method used to represent program's process or logic in the form of a step-by step diagram. It represents actions and decisions using various symbols (such as a rectangle diamond or arrow).

269. Which of the following flow charts will be most appropriate for finding sum of first 10 multiples of5?
5 के प्रथम 10 गुणज का योग प्राप्त करने के लिए निम्नलिखित में से कौन सा प्रवाह संचित्र सर्वाधिक उपयुक्त है?

(a)


(b)


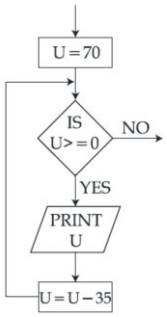
(c)


(d)


NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans.(a) : So correct opation is (a)
⇒ (i) S = 0, I = 5
→ 5 < = 50 → True
→ S = 0 + 5
⇒ I = I + S = 5 + 5 = 10
(ii) 10 < = 50
⇒ S = S + I = S = 5 + 10 = 15
I = I + 5 = 10 + 15 = 15 → True
it continues pro(es) – 10 times and print the code.

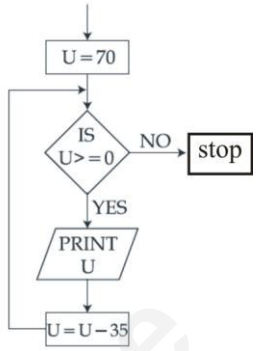
270. How many times loop will be executed in the following flow chart ?
निम्नलिखित प्रवाह संचित्र में लूप कितनी बार क्रियान्वित होगा?



(a) 1 (b) 2
(c) 3 (d) 70

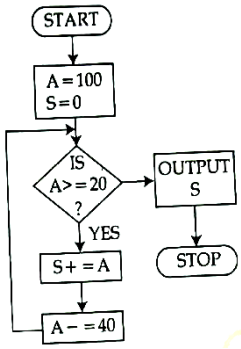
NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans.(c) :Flow chart: flow chart is a graphical representation of the algorithm.



⇒ Condition (1)
⇒ U is > = 0, then now print U = 70 then, U = U – 35, U = 70 – 35 = 35 and one time Exccuted
Condition (2) U = 35 is > = 0 then nowprint U = 35 then process to next U = 35 – 35
U = 0
0 ≥ 0, then condction is also true,
⇒ 3 time Loop will be Executed in the following flow chart.

271. Observe the following flowchart and find the output.
निम्नलिखित फ्लोचार्ट का अवलोकन करें और उसका आउटपुट बताएं।



(a) 180 (b) 120
(c) 100 (d) 160

EMRS PGT C.S. 16.12.2024

Ans.(a) : The flowchart provided starts with initializing variables A to 100 and S to 0.
1. Since A is initially 100 and it's greater than 20. We don't execute the block inside the No branch.
2. Instead, we add the value of A to S, making S 100.
3. Then we subtract 40 from A, making it 60.
4. We go back to the condition check is A greater than or equal to 20? Yes, it is (60 is greater than 20).
5. Again, we don't execute the block inside the No branch, but we add the value of A to S, making S 160 now.
6. Then we subtract 40 from A again, making if 20.
7. We go back the condition check - is A greater than or equal to 20? yes, it is (20 is equal to 20).

Sarkari Resource

8. This time, we don't execute the block inside the No branch, but we add the value of A to S, making S 180 now.
9. Then we subtract 40 from A again, making it -20.
10. Now, A is no longer greater than or equal to 20, So we processed to the No branch.
11. We output the value of S, Which is 180, and the program stops.

So, The output of the flowchart is 180.

272. What is Booth's algorithm used for?/बूथ एल्गोरिथम का उपयोग किस लिए किया जाता है?

(a) Binary to decimal conversion
बाइनरी से दशमलव रूपांतरण
(b) Decimal to binary conversion
दशमलव से बाइनरी रूपांतरण
(c) Binary multiplication/बाइनरी गुणन
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : Booth's algorithm is a method used for performing binary multiplication, especially when dealing with signed numbers. It reduces the number of addition and subtraction operations required by encoding runs of 1s in the multiplier efficiently. This makes multiplication faster and more efficient compared to the simple repeated addition method. Therefore, booth's algorithm is specifically used for binary multiplication, not for number system conversions.

273. Match the following with respect to algorithm paradigms :

	List - I		**List - II**
a.	The 8-Queen's problem	i.	Dynamic programming
b.	Single-Source shortest paths	ii.	Divide and conquer
c.	STRASSEN's Matrix multiplication	iii.	Greedy approach
d.	Optimal binary search trees	iv.	Backtracking

Code :

	(a)	**(b)**	**(c)**	**(d)**
(a)	(iv)	(i)	(iii)	(ii)
(b)	(iv)	(iii)	(i)	(ii)
(c)	(iii)	(iv)	(ii)	(i)
(d)	(iv)	(iii)	(ii)	(i)

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : The 8-Queen's problem is solved using back tracking since we try placements step by step and backtrack when conflicts arise; The single-source shortest path problem (like Dijkstra's) fallows the greedy approach; Strassen's matrix multiplication is based on the divide and conquer paradigm, where, the large multiplication is broken into smaller ones;
Optimal binary search trees are constructed using dynamic programming, as the problem involves optimal substructure and overlapping subproblems.
Thus the correct order is
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i).

274. The A* algorithm is optimal when,

(a) It always finds the solution with the lowest total cost if the heuristic 'h' is admissible.
(b) Always finds the solution with the highest total cost it the heuristic 'h' is admissible.
(c) Find the solution with the lowest total cost if the heuristic 'h' is not admissible.
(d) It always finds the solution with the highest total cost if the heuristic 'h' is not admissible.

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Ans. (a) : Optimality of the A* algorithm
A search algorithm is optimal if no other search algorithm uses less time or space or expands fewer nodes both with a guarantee of solution quality.
Optimality of A*–Among search algorithms that only use are costs and a heuristic estimate of the cost and a heuristicestimate of the cost from a node to a goal no algorithm expands fewer nodes. Then A* and guarantees to find a lowest-cost path.
So option (a) is correct.

275. Match List I with List II

	List-I		**List-II**
A.	**Dijkstra's Algorithm**	**i.**	**Calculates path matrix**
B.	**Prim's Algorithm**	**ii.**	**Stores minimum cost edge**
C.	**Warshall's Algorithm**	**iii.**	**Stores the total cost from a source node to the current node**
D.	**Kruskal's algorithm**	**iv.**	**Finds Minimum Spanning Tree**

Choose the correct answer from the options given below :

(a) A-I, B-II, C-III, D-IV
(b) A-III, B-II, C-I, D-IV
(c) A-II, B-I, C-IV, D-III
(d) A-III, B-IV, C-II, D-I

UGC NET C.S. December-2022

Ans. (b) : Dijkstra's Algorithm–
Once the algorithm has found the shortest path between the source node and another node that node is marked as "visited" and added to the path.
A (iii) Dijkstra's stores the total cost from a source node to the current node.
Prim's Algorithm→
Prim's algorithm is a minimum spanning tree algorithm that takes a graph is input and finds the subset of the edges of that graph.
B (ii) Prim's Algorithm→stores minimum cost edge.
C. Warshall's Algorithm→is used to determine the transitive closure of a directed graph or all paths in a directed graph by using the adjacency matrix.
C. (1) Warshall's Algorithm→Calculate path matrix.
D. Kruskal's Algorithm–Kruskal's Algorithm is used to find the minimum spanning tree for a connected weighted graph.
D. (iv) Kruskal's algorithm→Findminimum spaning tree.
So option (b) is correct answer.

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276. Match List I with List II

List-I		List-II	
A.	**8-Queens problem**	**I.**	**Divide and conquer**
B.	**Closest pair of Points**	**II.**	**Dynamic Programming**
C.	**String Edit problem**	**III.**	**Greedy Method**
D.	**Knapsack problem**	**IV.**	**Backtracking**

Choose the correct answer from the options given below :

(a) A-I; B-II; C-III; D-IV
(b) A-I; B-IV; C-II; D-III
(c) A-IV; B-I; C-II; D-III
(d) A-II; B-III; C-IV; D-I

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Ans. (c) : Correct Match are -

A.	8-Queens problem	IV.	Backtracking
B.	Closest pair of Points	I.	Divide and conquer
C.	String Edit problem	II.	Dynamic Programming
D.	Knapsack problem	III.	Greedy Method

277. Standard planning algorithms assume environment to be.....

(a) Both deterministic and fully observable
(b) Neither deterministic nor fully observable
(c) Deterministic but not fully observable
(d) Not deterministic but fully observable

UGC NET C.S. November-2017 (Paper-III)

Ans. (a):Standard planning algorithms assume environment to be both deterministic and fully observable.

278. Code blocks allow many algorithms to be implemented with the following parameters :

(a) clarity, elegancy, performance
(b) clarity, elegance, efficiency
(c) elegance, performance, execution
(d) execution, clarity, performance

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because, Code blocks not only allow many algorithm to be implemented with clarity elegance and efficiency. More over, they help the programmer better conceptualize the true nature of algorithm being implement.

279. Consider f (N) = g(N) + h(N)
Where function g is a measure of the cost of getting from the start node to the current node N and h is an estimate of additional cost of getting from the current node N to the goal node. Then f(N) = h(N) is used in which one of the following algorithms?

(a) A* algorithm
(b) AO* algorithm
(c) Greedy best first search algorithm
(d) Iterative A* algorithm

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, the greedy algorithm work in a way that it has same estimate (called a heuristic) of how for from the goal any vertex is. Instead of selecting the vertex closet to the starting point, it selects the vertex closet to the goal, so the answer is (c).

280. Match the following cryptographic algorithms with their design issues :

	List–I		List–II
a.	DES	i.	Message Digest
b.	AES	ii.	Public Key
c.	RSA	iii.	56-bit key
d.	SHA-1	iv.	128-bit key

Codes :

	a	**b**	**c**	**d**
(a)	ii	i	iv	iii
(b)	iii	i	iv	ii
(c)	iii	iv	ii	i
(d)	iv	i	ii	iii

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because,
(a) DES : DES is an implementation of a fiestel cipher. It uses 16 round fiestel structure. DES has an effective key length of 56 bits.
(b) AES : The more popular and widely adopted symmetric encryption algorithm is AES. The features of AES is symmetric key symmetric block cipher and it has 128 bit key.
(c) RSA : RSA algorithm works on two different keys i.e public key & private key. As the name public key given to everyone & private key is kept private.
(d) SHA-1 : It is message digest.

281. Which of the following properties ensures that an algorithm must always terminate after a finite number of steps?
निम्नलिखित में से कौन सी प्रॉपर्टी यह सुनिश्चित करती है कि एक एल्गोरिदम हमेशा चरणों के सीमित संख्या के बाद समाप्त होना चाहिए?

(a) Effectiveness/प्रभावशीलता
(b) Definiteness/निश्चितता
(c) Finiteness/नियत्ता
(d) Output/आउटपुट

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Finiteness (in the content of algorithms) means that an algorithan must always computer after a finite number of steps It should not run forwer or go into an intinite loop
- The algorithm must terminate
- for any valid input, if should produce an out put within a limited number of step.

282. Which symbol is used to represent an input/output operation in the flow chart?
फ्लो चार्ट में इनपुट/आउटपुट ऑपरेशन को दर्शाने के लिए किस प्रतीक का उपयोग किया जाता है?

(a) Oval/अंडाकार

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(b) Circle/वृत्त
(c) Rectangle/आयत
(d) Parallelogram/समान्तर चतुर्भुज

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) :A flow chart is a visual representation of a process, showing the sequence of steps using various symbols.
(1) Oval (Start/End)– Represents the start or end of a process.
(2) Rectangle (Process)– Represents a step or action in the process.
(3) Diamond (Decision)– Represent a decision point, where the flow can branch based on a condition.
(4) Parallelogram (Input / Output)– Represents data input or output.
(5) Circle (Connector)– Used to connect flow chart elements on different pages or sections.

283. Which of the following statement(s) is/are true regarding cryptography?
क्रिप्टोग्राफी के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. Cipher text is encrypted text transformed from plain text using an encryption algorithm./सिफर टेक्स्ट एन्क्रिप्टेड टेक्स्ट है जो एक एन्क्रिप्शन एल्गोरिथम का उपयोग करके सादे टेक्स्ट से रूपांतरित होता है।

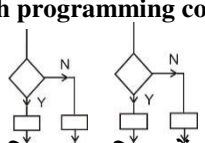
II. According to Kerckhoff's principle, all algorithms must be private and only the keys are secret./केरचॉफ के सिद्धांत के अनुसार, सभी एल्गोरिदम निजी होने चाहिए और केवल कुंजियाँ गुप्त होनी चाहिए।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ IऔरIIदोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Statement I is true.
"Cipher text is encrypted text transformed from plain text using an encryption algorithm. This is a correct definition of cipher text.
Statement II is false
Kerchoffs's principle states that the security of a cryptographic system should rely only on the secrecy of the key, and the algorithm should be public.
Hence, only statement I is true.

284. Flowchart given in below figure represents which programming construct?



ऊपर दिए गए चित्र में दिया गया फ्लोचार्ट किस प्रोग्रामिंग कन्स्ट्रक्ट को दर्शाता है?

(a) Sequence structure/सिक्वेंस स्ट्रक्चर
(b) Decision structure/डिशिज़न स्ट्रक्चर
(c) Loop structure /लूप स्ट्रक्चर
(d) Case structure /केस स्ट्रक्चर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : The flowchart given in the figure represents a decision structure. A decision structure, also known as selection structure, is a programming construct that allow a program to execute different blocks of code based on a condition the diamond shape in the flowchart represents a decision point where a condition is evaluated.
(e.g. 'Y' or 'N' for yes or no).
The program flow then proceeds along one of two or more paths depending on the outcome of the decision. In the given flowchart, the flow splits into two paths based on the condition, illustrating a decision structure.

285. Pictorial representation of a program is known is_____
किसी प्रोग्राम के पिक्टोरियल निरूपण को _____कहा जाता है।

(a) flowchart/फ्लोचार्ट
(b) pseudocode/स्वीडोकोड
(c) code/कोड
(d) algorithm/ऐल्गोरिदम

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) :A pictorial representation of a program is known as a flowchart. A flowchart is a graphical representation that illustrates the lagic an flow of a program.

286. Which of the following algorithms are based on the Breadth First Search (BFS)?
A. Prism's algorithms
B. Kruskal algorithms
C. Dijkstra algorithms
D. Greedy algorithms
E. Dynamic Programming
Choose the correct answer from the options given below :

(a) A & B only (b) A, C & D only
(c) D & E only (d) A & C only

UGC NET C.S. December-2024

Ans. (d) : The algorithms that are based on Breadth first search (BFS) from the given option:-
A. Prism's algorithm : Prism's algorithm is a greedy algorithm used to find a minimum spanning tree for a weighted, undirected graph. While it can be implemented using a priority queue (which can be seen as a variation of BFS), the core logic of prism's algorithms is not directly based on BFS.
C. Dijkstra's algorithms : Dijkstra's algorithm is used to find the shortest paths between nodes in a graph. It can be seen as a generalization of BFs, where instead of exploring all neighbors at the same level, it explores the neighbor with the smallest distance.

287. Which of the following statement(s) are true about Algorithm?
एल्गोरिथम के बारे में निम्नलिखित मे से कौन-सा कथन सत्य है।

I. It is step by step solution of problem
यह समस्या का चरणबद्ध समाधान है

II. It is difficult to debug
इसको डीबग करना कठिन है

III. It is not dependent on the programming language/यह प्रोग्रामिंग लैंग्वेज पर निर्भर नहीं है

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(a) I and II (b) II and III
(c) I and III (d) I, II and III

DSSSB TGT C.S. 07.08.2021 Shift-I

Ans. (c) : Algorithm is a step-by-step solution of a program.
Dubugging applies to code, not to algorithms in conceptual form.
Algorithms are written in pseudocode or natural language, making them independent of any programming syntax.
Hence, only statement I and III are correct.

288. Efficiency of an algorithm is measured in terms of _____.
एल्गोरिथम की दक्षता को _____ के संदर्भ में मापा जाता है।

(a) Memory and processor speed
मेमोरी तथा प्रोसेसर स्पीड
(b) Time and space/टाइम तथा स्पेस
(c) Response time and completion time
रिस्पॉंस टाइम तथा कम्पलीशन टाइम
(d) Data and space/डेटा तथा स्पेस

DSSSB TGT C.S. 07.08.2021 Shift-I

Ans. (b) : Efficiency of an algorithm is measured in terms of Time and Space.
Time: measures how long the algorithm takes to run typically based on input size (e.g., O(n), O(logn))
Space: measures how much memory the algorithm uses during execution.

289. Match the following :

A.	**Huffman codes**	**i.**	**$O(n^2)$**
B.	**Optimal polygon triangulation**	**ii.**	**$\theta(n^3)$**
C.	**Activity selection problem**	**iii.**	**O(nlogn)**
D.	**Quicksort**	**iv.**	**$\theta(n)$**

Code :

	A	B	C	D
(a)	(i)	(ii)	(iv)	(iii)
(b)	(i)	(iv)	(ii)	(iii)
(c)	(iii)	(ii)	(iv)	(i)
(d)	(iii)	(iv)	(ii)	(i)

UGC NET C.S. June-2016 Paper-III

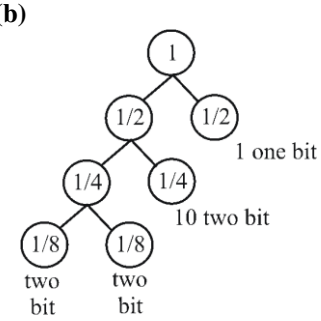
Ans :(c) **correct matching is :**
Huffman codes – O(nlogn)
Optimal polygon triangulation – $\theta(n^3)$
Activity selection problem –$\theta(n)$
Quicksort – $O(n^2)$

290. Consider a source with symbols A, B, C, D with probabilities 1/2, 1/4, 1/8, and 1/8 respectively. What is the average number of bits per symbol for the Huffman code generated from above information?

(a) 2 bits per symbol
(b) 1.75 bits per symbol
(c) 1.50 bits per symbol
(d) 1.25 bits per symbol

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Ans : (b)



Length of bits 11

$$\frac{3\times1}{8}+\frac{3\times1}{8}+\frac{2\times1}{4}\times\frac{1\times1}{2}=\frac{14}{8}=1.75 \text{ bits per symbol}$$

(vi) Analyzing. Algorithms

291. Which of the following is the correct order of problem solving steps ?
समस्या समाधान चरणों का निम्नलिखित में से कौन सा क्रम सही है?

(a) Testing and Debugging, Coding, Analysing the problem, Developing an Algorithm/टेस्टिंग व डीबगिंग, कोडिंग, समस्या का विश्लेषण, एल्गोरिथ्म का विकास
(b) Developing an Algorithm, Coding, Testing and Debugging, Analysing the problem/एल्गोरिथ्म का विकास, कोडिंग, जाँच व डीबगिंग, समस्या का विश्लेषण
(c) Analysing the problem, Developing an Algorithm, Coding, Testing and Debugging/समस्या का विश्लेषण, एल्गोरिथ्म का विकास, कोडिंग, जाँच व डीबगिंग
(d) Developing an Algorithm, Analysing the problem, Coding, Testing and Debugging/एल्गोरिथ्म का विकास, समस्या का विश्लेषण, कोडिंग, जाँच व डीबगिंग

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans.(c) : The standard sequence for solving a programming problem always follows
analysis → algorithm → coding → testing / debugging
- Testing & Debugging must always be last, because you can only test a program after it's written.
- Coding comes after planning (algorithm development)
- Analyzing the problem is always the first step-without understanding the problem, you can't design a solution

292. The solution of the recurrence relation of T(n) $= 3T\left(\text{floor}\left(\frac{n}{4}\right)\right)+n$ is

(a) $O(n^2)$ (b) O(n/g n)
(c) O(n) (d) O(/ g n)

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Given recurrence relation is:

$$T(n)=3T\left(\left\lceil\frac{n}{4}\right\rceil\right)+n$$

Sarkari Resource

Using master theorem case 3 is:

$(n^{\log_b(a)}) \le f(n)$ then $T(n) = \Theta(n)$

Given a = 3, b = 4 and f(n) = n then

$(n^{\log_4(3)}) \le n$, then $T(n) = \Theta(n)$

So, $T(n) = O(n) = O(n^2)$

Both option (a) and (c) are true, but option (c) is best choice.

293. Which of the following asymptotic notation is used to provide lower boundary constraints ?
निम्नलिखित में से कौन सा एसिम्पटोटिक नोटेशन लोअर बाउंड्री कोन्सट्रैंट्स को प्रदान करने के लिए उपयोग किया जाता है?

(a) O (b) θ
(c) Ω (d) ×

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Asymptotic Notation for Lower Bound. _____
The asymptotic notation used to provide Lower boundary constrains is Ω.
Ω notation gives a lower bound on the time or space complexity of an algorithm, indicating the best – case scenario or the minimum amount of time space required.

(vii) Complexity of Algorithms

294. Consider a singly linked list. What is the worst case time complexity of the best-known algorithm to delete the node a, pointer to this node is q, from the list?
एकल लिंक्ड सूची विचार करे।
सूची से नोड a, इस नोड का सूचक q, है, को हटाने के लिए सबसे प्रसिद्ध एल्गोरिथम की सबसे खराब स्थिति समय जटिलता क्या है?

(a) O(1) (b) O(n)
(c) O(lg n) (d) O(n lg n)

UGC NET C.S. December-2018 (Paper-II)

Ans. (b) :A **simple solution** is to traverse the linked list until you find the node you want to delete. But this solution requires pointer to the head node which contradicts the problem statement.
Fast solution is to copy the data from the next node to the node to be deleted and delete the next node. Something like following.

```
// Find next node using next pointer
    struct node *temp  = node_ptr->next;
    // Copy data of next node to this node
    node_ptr->data  = temp->data;
    // Unlink next node
    node_ptr->next  = temp->next;
    // Delete next node
    free(temp);
```

Time complexity of this approach is O(n).

295. Time complexity of Merge SortAlgorithm and Binary SearchAlgorithm are respectively _____.
मर्ज-सॉर्टएल्गोरिथम तथा बाइनरी-सर्चएल्गोरिथम की टाइम कॉम्प्लेक्सिटी क्रमशः _____ है।

(a) $O(\log_2 n)$ and $O(n\log_2 n)$
(b) $O(n\log_2 n)$ and $O(\log_2 n)$
(c) $O(n^2)$ and $O(\log_2 n)$
(d) $O(2^n)$ and $O(n^2)$

HTET PGT C.S. 2018

Ans. (b) : Merge Sort → It's a Divide & conquer algorithm.
- The array is divided into halves → about $\log_2^n$ levels.
- At each level. merging takes o(n) time.
- So total time Complexity = o (n $\log_2^n$)
- Binary Search→ It repeatedly halves the search space.
- Each step reduces the problem size by half.

So time complexity = o($\log_2^n$)

296. Time complexity of $f(x) = 4x^2 - 5x + 3$ is:

(a) O(x) (b) $O(x^2)$
(c) $O(x^{3/2})$ (d) $O(x^{0.5})$

CGPSC C.S. 2020 Set-A

Ans.(b) : The time complexity of a polynomial function is determined by the term with the highest power. In the function $f(x)=4x^2-5x+3$, the term with the highest power is $4x^2$. Therefore, the time complexity in big (o) notation is $o(x^2)$.

297. Which of the following statements is true?

(a) Both time and space efficiencies are measured as the functions of algorithm input size
(b) Only time efficiencies are measured as the function of the algorithm input size
(c) Only space efficiencies are measured as the function of the algorithm input size
(d) Neither time nor space efficiencies are measured as the function of the algorithm input size

CGPSC C.S. 2020 Set-A

Ans.(a):1. Time complexity → How time grows with input size (n).
2. Space complexity → How memory grows with input size (n)
3. Algorithm analysis always measures both as a function of input size.
Both time and space efficiencies are measured as functions of algorithm in input size therefore, (a) statement is correct.

298. What is the worst case analysis for the given code?

```
int search (int a [ ], int x, int n)
{int i;
for (i = 0; i < n; + i + + )
if (a [i] = = x)
return i;
return – 1;}
```

(a) O (n) (b) On (n log n)
(c) O (log n) (d) $O(n^2)$

CGPSC C.S. 2020 Set-A

Ans.(a): • The loop runs from i = 0 to I < n, checking each element of the array a [] to see if it matches x.
• In the worst case, the element x is not present in the array, so the loop runs 'n' times without returning early

Sarkari Resource

• Each iteration performs a constant-time comparison: a [i] = x
• It checks each element one by one. If it finds x, it returns the index if not returns - 1,
• If x is not in the array, it checks all (n) elements so, worst case = O(n)

299. What is the worst case complexity of sorting n numbers using quicksort?

(a) $\theta(n)$ (b) $\theta(n \log n)$
(c) $\theta(n^2)$ (d) $\theta(n!)$

CGPSC C.S. 2020 Set-A

Ans.(c): The worst-case time complexity of quicksort occurs when the pivot selection is poor - e.g., always picking the smallest or largest element in an already sorted or reverse-sorted list. In that case, the partitioning is extremely unbalanced, and the recursion essentially becomes:

$\mathbf{T(n) = T(n-1)+T(0) +\theta(n) =T(n-1)+\theta(n)}$

Solving this recurrence:

$\mathbf{T(n) = T(n-1)+n =T(n-2) +(n-1)+(n) =\ldots.. \theta(n^2)}$

300. Time complexity of strassen's matrix multiplication algorithm is given by

(a) $O(n^{2.81})$ (b) $O(n^3)$
(c) $O(n^{8.21})$ (d) $O(n^2)$

RPSC C.S. 2016 (Paper-II)

Ans. (a) : The time complexity of strassen's matrix multiplication algorithm is given by $O\left(n^{\log 27}\right)$. Which is approximately $O(n^{2.81})$.

Strassen's algorithm improves upon the standard matrix multiplication (Which has a time complexity of $O(n^3)$).

It reduces the number of recursive multiplications from 8 to 7 when dividing matrices into sub matrices. This gives a recurrence relation.

$$T(n) = 7T(n/2) + 18(n^2)$$

Solving this recurrence using the master Theorem gives.

$$T(n) = O\left(n^{\log 27}\right) \approx O(n^{2.81})$$

301. Using dynamic programming approach, longest common subsequence (LCS) of two sequences of size n each, may be computed in:-

(a) $O(n^2)$ (b) $O(2^n)$
(c) $O(n^3)$ (d) $O(n \log_2 n)$

RPSC C.S. 2016 (Paper-II)

Ans. (a) : The Longest Common Subsequence Problem can be solved using dynamic programming with a time complexity of $O(n^2)$, where n is the length of each sequence.

Dynamic Programming Approach–

1. We construct a DP table of size $(n + 1) \times (n + 1)$ (Since we include an extra row and column for empty subsequences).

2. We fill this table by comparing characters of the two sequences.

If characters match - dp[i][j] =

dp [i – 1] [j – 1] + 1

If they don't match- dp [i] [j] = max (dp [i – 1] [j], dp [i] [j – 1])

The final answer (length of LCS) is found in dp [n] [n].

Since we fill an $n \times n$ table with each cell taking O(1) time, the total time complexity is $O(n^2)$.

302. Choose correct increasing order of growth rates of asymptotic notations:-

(a) $O(1), O(n), O(\log n), O(n \log n), O(n^2), O(2^n)$
(b) $O(1), O(\log n), O(n), O(n \log n), O(2^n), O(n^2)$
(c) $O(1), O(\log n), O(n), O(n \log n), O(n^2), O(2^n)$
(d) $O(\log n), O(1), O(n \log n), O(n), O(n^2), O(2^n)$

RPSC C.S. 2016 (Paper-II)

Ans. (c) : The common asymptotic notations from slowest to fastest growth, are generally.

O(1) (Constant Time)– The time taken is independent of the input size.

O(log n) (Logarithmic time)– The time grows very slowly with the input size.

O(n) (Linear time)– The time grows proportionally to the input size.

O(n log n) (Linearithmic time)– A bit slower than linear, but faster than polynomial.

$O(n^2)$ (Quadratic time)– The time grows with the square of the input size.

$O(2^n)$ (Exponential time)– The time grows extremely rapidly with the input size.

Now examine each option–

(a) O(1), O(n), O(log n), O(n log n), $O(n^2)$, $O(2^n)$–
This is incorrect because O(n) comes before O(log n), which is wrong. Also, O(2) is just O(1), and its placement is inconsistent.

(b) O(1), O(log n), O(n), O(n log n), $O(2^n)$, $O(n^2)$–
This is incorrect because $O(2^n)$ grows much faster than $O(n^2)$, $O(n^2)$ should come before $O(2^n)$.

(c) O(1), O(log n), O(n), O(n log n), $O(n^2)$, $O(2^n)$–
This order accurately reflects the increasing growth rates.
Constant (O(1))
Logarithmic (O(log n))
Linear (O(n))
Linearithmic (O(n log n))
Polynomial ($O(n^2)$)
Exponential ($O(2^n)$)

(d) O(log n), O(1), O(n log n), O(n), $O(n^2)$, $O(2^n)$–
This is incorrect because O(log n) comes before O(1), which is wrong O(1) is the slowest growth.
Therefore, the correct option is (c).

303. Find the running time of the fastest algorithm to calculate the shortest path between any two vertices of a graph where all edges have equal weights'.

(a) $O(V \log_2 V - E)$ (b) $O(E + V)$
(c) $O(V \log_2 V + E)$ (d) $O(V + E)\log_2 V$

RPSC C.S. 2016 (Paper-II)

Ans. (b) : To determine the fastest algorithm for computing the shortest path between any two vertices in a graph where all edges have equal weights.

BFS can compute the shortest path from a single source to all vertices in O(V + E) time.

To compute shortest paths between any two vertices, one can run BFS from each vertex a total of V times. So the total running time is $O(V \times (V \times E))$ for all pairs shortest paths using BFS.

For finding the shortest path between any two given vertices, use BFS from the source node.

Running Time - O (V + E).

Sarkari Resource

304. An algorithm performs $(\log N)^{1/2}$ find operations, N insert operations, $(\log N)^{1/2}$ delete operations, and $(\log N)^{1/2}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set. For a delete operation, a point is provided to the record that must be deleted. For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for by the algorithm to use, it the goal is to achieve the best total asymptotic complexity considering all the operations?

(a) Unsorted array
(b) Min-heap
(c) Sorted array
(d) Sorted doubly linked list

RPSC C.S. 2016 (Paper-II)

Ans. (a) :Unsorted Array–

$$T(\text{Find}) \simeq \frac{N}{2}$$

$$T(\text{Insert}) \simeq 0(1)$$

$$T(\text{Delete}) \simeq \frac{N}{2}$$

$$T(\text{Decrease Key}) \simeq \frac{N}{2}$$

$$\therefore \quad T(\text{Total} \simeq \sqrt{\log N}\left[\frac{3}{2}N\right] + N)$$

(ii) Min-Heap–

$$T(\text{Find}) \simeq \frac{N}{2}$$

$$T(\text{Insert}) \simeq \log N$$

$$T(\text{Delete}) \simeq \frac{N}{2} + \log N$$

$$T(\text{Decrease Key}) \simeq \frac{N}{2} + \log N$$

$$\therefore \quad T(\text{Total}) \simeq \sqrt{\log N}\left[\frac{3N}{2} + 2\log N\right] + N \log N$$

(iii) Sorter Array–

$$T(\text{Find}) \simeq \log N$$

$$T(\text{Insert}) \simeq \log N + \frac{N}{2}$$

$$T(\text{Delete}) \simeq \log N + \frac{N}{2}$$

$$T(\text{Total}) \simeq \sqrt{\log N}\left[3\log N + \frac{3}{2}N\right] + \frac{N^2}{2} + N \log N$$

Sorted doubly linked list will be even worst choice.

305. The worst case time complexity of AVL tree is better in comparison to binary search tree for

(a) Search and Insert Operations
(b) Search and Delete Operations
(c) Insert and Delete Operations
(d) Search, Insert and Delete Operations

RPSC C.S. 2016 (Paper-II)

Ans. (d) : In a Binary Search Tree (BST), the worst case time complexity for operations like Search, Insert and Delete can degrade to O(n) when the tree becomes skewed (like a linked list).
However an AVL Tree is a self-balancing binary search tree, where the height difference (balance factor) between left and right subtrees is at most 1. This strict balancing ensures that the height of the AVL tree remains O(log n) time complexity for-

Search
Insert
Delete

Thus, AVL trees provides better worst case time complexity for all three operations compared to unbalanced BST.

306. Let M1 is a single tape Turing machine simulating a multi-tape turning machine M (k-tapes). Time taken by M1 to simulate n moves of M, considering 'k' as constant, is given by:-

(a) $O(n^2)$ (b) $O(n)$
(c) $O(n^k)$ (d) $O(k^n)$

RPSC C.S. 2016 (Paper-II)

Ans.(a) : When a single-tape Turing machine M1 simulates a multi-tape Turing machine M (with k tapes), the time taken for M1 to simulate 'n' moves of M, considering 'k' as a constant is given by $O(n^2)$.

307. Which one of the following is the tightest bound that represents the time complexity of inserting an object into a binary search tree of n nodes?

(a) O(1) (b) O(log n)
(c) O(n) (d) O(n, log n)

RPSC C.S. 2016 (Paper-II)

Ans.(*) : Suppose that we need to insert a node z such that k = key[z]. Using binary search we find a nil such that replacing it by z does not break the BST-property.

BST–
Insert (x, z, k)
if x = nil then return "Error"
y × x
while true do{
if key [y] < k
then z ← left [y]
else z ←right [y]
if z = nil break
}
if key [y] > k then left [y] ← z
else right [p[y]]← z

Time Complexity Analysis–

1. Best case = O(1), when it is smallest/greatest element and BST contains only all greater/smaller element than inserting element respectively.
2. Avg case = O(log n), when it belongs between some elements.
3. Worst case = O(n), when it is smallest/greatest element and BST contains only all smaller/greater element than inserting element respectively

Note– This question is dropped by RPSC.

308. What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

(a) $\ominus(n^2)$ (b) $\ominus(n^2 \log n)$
(c) $\ominus(n^3)$ (d) $\ominus(n^3 \log n)$

RPSC C.S. 2016 (Paper-II)

Sarkari Resource

Ans.(c) : Time complexity of Bellman-Ford algorithm is $\ominus(|V||E|)$ where $|V|$ is number of vertices and $|E|$ is number of edges. If the graph is complete the value of $|E|$ becomes $\ominus(|V|^2)$. So overall time complexity becomes $\ominus(|V|^3)$. And given here is n vertices. So, the answer ends up to be $\ominus(n^3)$.

309. The average number of comparisons performed in sequential search on a list of n elements is:-

(a) n^2 (b) n(n+1)/2
(c) n(n–1)/2 (d) (n+1)/2

RPSC C.S. 2016 (Paper-II)

Ans.(d): Expected number of comparisons
= 1 × probability of first element being x + 2x probability of second element being x+……+ nx probability of last element being x.

$$= \frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \ldots\ldots + \frac{n}{n}$$

$$\frac{\left(\frac{n \times (n+1)}{2}\right)}{n} = \frac{n+1}{2}$$

310. For positive n and f(n), choose time complexity of f(n), if
$f(n) = a_0 + a_1n + a_2n^2$ …..+ a_mn^m, and $a_m > 0$.

(a) $\theta(n^m)$ (b) $\theta(m^n)$
(c) $\theta(a^m)$ (d) $\theta(n^a)$

RPSC C.S. 2016 (Paper-II)

Ans.(a) : Time Complexity–
Given the function $f(n) = a_0 + a_1n + a_2n^2$ …..+ a_mn^m, where $a_m > 0$, the time complexity is determined by the highest degree term in the polynomial, which dominates as n approaches infinity.
The time complexity of f(n) is $\theta(n^m)$, where m is the degree of the polynomial. This is because the term with the highest power of $n(a_mn^m)$ dominates the growth of the function for large values of n.
So, the best answer is $\theta(n^m)$.

311. If an algorithm's behavior is bounded by "O(f(n)) if g(n)", what notation does it become when the condition is removed?
यदि किसी एल्गोरिदम का व्यवहार 'O(f(n)) यदि g(n)' द्वारा सीमित है, तो जब यह शर्त हटा दी जाती है, तो यह कौन-सी नोटेशन बन जाती है?

(a) O(f(n)) (b) O(g(n))
(c) O(f(n) + g(n)) (d) O(f(n) * g(n))

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : If an algorithm's running time is written as "O(f(n)) if g (n)", it means the algorithm is faster than f(n), but only when the condition g(n) is true.
This is called conditional asymptotic notation, because it depends on a specific situation. If we remove the condition g(n), then it becomes just O(f(n)), which means the algorithm is always faster than f(n), without depending on any condition. In short conditional notation reduces to normal asymptotic notation when the condition is removed.

312. The recurrence equation T(n) = T(n/2) + 1 represents the time complexity of which algorithmic paradigm?
पुनरावृत्ति समीकरण T(n) = T(n/2)+1 किस एल्गोरिदमिक प्रतिमान की समय जटिलता को दर्शाता है?

(a) Divide and Conquer/डिवाइड एंड कॉन्कर
(b) Greedy Algorithms/ग्रीडी एल्गोरिदम
(c) Dynamic Programming/डायनामिक प्रोग्रामिंग
(d) Brute Force/ब्रूट फोर्स

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : The recurrence relation $T(n) = T(n/2) + 1$ shows that a problem of size n is being divided into a smaller sub problem of size n/2, and at each step only a constant amount of extra work is done. This continuous unit the input becomes very small (base case). Such a process, where the problem is broken into smaller parts and solved step by step is the main idea of the divide and conquer paradigm. A common example is Binary search, where the array is divided into half in each step, and only one half is processed further. Thus this recurrence represents the time complexity of the divide and conquer approach.

313. Conditional asymptotic notation allows for a more nuanced analysis of algorithm behavior under specific conditions. This is particularly useful when:
कंडीशनल एसिम्प्टोटिक नोटेशन एल्गोरिदम के व्यवहार का अधिक सूक्ष्म विश्लेषण करने की अनुमति देता है। यह विशेष रूप से उपयोगी होता है जब:

(a) Analyzing algorithms with constant time complexity/एल्गोरिदम का विश्लेषण स्थिर समय जटिलता के साथ किया जा रहा हो
(b) The input size is fixed
इनपुट आकार स्थिर हो
(c) The input data is random
इनपुट डेटा यादृच्छिक हो
(d) Algorithm behaviour varies based on certain input characteristics/एल्गोरिदम का व्यवहार कुछ इनपुट विशेषताओं के आधार पर बदलता हो

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : • Conditional asymptotic notation is used when or algorithm does not always take the same time.
• Its running time changes depending on the type of input.
• Example: quick sort works fast if the input is random, but slow if the input is already sorted (depending on pivot choice).
So, it is useful when the algorithm's speed depends on the input's nature.

314. What is the time complexity of the Divide and Conquer algorithm for finding the maximum and minimum elements in an array of size "n"?
'n' आकार की ऐरे में अधिकतम और न्यूनतम तत्वों को खोजने के लिए डिवाइड एंड कॉन्कर एल्गोरिदम की समय जटिलता क्या होती है?

(a) O(n) (b) O(log n)
(c) O(n log n) (d) O(n^2)

Bihar STET C.S. 12.09.2023 (Shift-II)

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Ans. (a) : The divide and conquer algorithm for finding the maximum and minimum element in an array of size n divides the array into parts, solves recursively, and then combines the results.

- Even though the array is divided, every element has to be checked at least once.
- Hence the number of comparisons is still proportional to n.
- Therefore, the complexity is O(n).

315. What is the worst-case time complexity of Binary Search for an array of size "n"?
'n' आकार के ऐरे के लिए बाइनरी सर्च की सबसे खराब स्थिति में समय जटिलता क्या होती है?

(a) O(log n) (b) O(n)
(c) O(n log n) (d) O(1)

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : The Worst case time complexity of Binary search for an array ofsize n is O(log n). Binary search works on a sorted array and finds the target elements by repeatedly dividing the array into two havles. In each step, it compares the middle element with the target. If the target is not found, it continues the search in the half where the target may exist. This halving process continues until the element is found or the array cannot be divided further. Because the array size is reduced by half in each step, the number of steps grows logarithmically with the size of the array, making the worst-case time complexity O(log n).

316. The complexity of linear search algorithm is :
रैखिक खोज एल्गोरिथ्म की जटिलता है :

(a) O (n log n) (b) O (n)
(c) O (log n) (d) O (n*n)

KVS PGT C.S. 2017

Ans.(b) : Linear Search defined as a sequential search algorithm that starts at one end and goes through each element of a list until the desired element is found, otherwise the search continues till the end of the data set. It is the easiest searching algorithm. Complexity in the worst case of linear search algorithm is O (n).

317. The worst case complexity for searching an element in binary search tree is:
किसी बाइनरी सर्च ट्री में कोई एलीमेंट खोजने के लिए वर्स्ट केस कॉम्प्लेक्सिटी क्या है?

(a) O(lgn) (b) O(nlgn)
(c) $O(n^2)$ (d) O(n)

KVS PGT C.S. 2018

Ans.(d) : Binary search Tree (BST) is a special type of binary tree in which left child of a node has value less than the parent and right child has value greater than parent. for searching element, we have to transverse all elements. Therefore, searching in binary search tree has worst case complexity of O (n).

318. Which of the following is correct recurrence relation for worst case of binary search?
बाइनरी सर्च के वर्स्ट केस के लिए निम्न में से कौन-सा सही रिकरेन्स रिलेशन है?

(a) T(n) = 2T(n/2) + O(1)
T(1) = T(0) = O(1)
(b) T(n) = T(n/2) + O(n)
T(1) = T(0) = O(1)
(c) T(n) = T(n/2) + O(1)
T(1) = T(0) = O(1)
(d) T(n) = T(n/2) + O(log n)
T(1) = T(0) = O(1)

LT. Grade C.S. 2018

Ans.(c) : In Binary Search, we first compare the given element with middle of the array. If given element matches with middle element, then we return middle index otherwise, we either recur for left half of array or right half of array. So recurrence is T (n) = T (n/2) + O (1) and T (1) = T (0) = O (1).

319. Which of the following sorting algorithms has the worst time complexity of nlog(n)?
निम्न में से किस सॉर्टिंग ऐल्गॉरिद्म की वर्स्ट टाइम कॉम्प्लेक्सिटी nlog(n) होती है?

(a) Heapsort/हीपसॉर्ट
(b) Quicksort/क्विकसॉर्ट
(c) Insertion sort/इन्सर्शन सॉर्ट
(d) Selection sort/सिलेक्शन सॉर्ट

LT. Grade C.S. 2018

Ans.(a) : Heapsort is a comparison - based sorting algorithm which can be thought of as implementation of selection sort using the right data structure. In heap sort, we use binary heap so that we can quickly find and move the max element in O (log n) instead of O (n) and hence achieve the O (n log n) time complexity.

320. Match List-I with List-II and select the correct answer using the codes given below the Lists:
सूची-I को सूची-II से सुमेलित कीजिए तथा सूचियों के नीचे दिए गए कूट का प्रयोग कर सही उत्तर चुनिए:

List-I/सूची-I		List-II/सूची-II	
A.	All-pairs shortest path/ऑल-पेयर्स शॉर्टेस्ट पाथ	1.	Greedy/ग्रिडी
B.	Quicksort/ क्विकसॉर्ट	2.	Depth-first search डेप्थ-फर्स्ट सर्च
C.	Minimum weight spanning tree मिनिमम वेट स्पैनिंग ट्री	3.	Dynamic programming डायनामिक प्रोग्रामिंग
D.	Connected components कनेक्टेड कॉम्पोनेन्ट	4.	Divide and conquer/डिवाइड एवं कॉन्क्वेर

(a) A B C D
2 4 1 3
(b) A B C D
3 4 1 2
(c) A B C D
3 4 2 1
(d) A B C D
4 1 2 3

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Ans.(b) : Correct Match-

List -I		List-II
A. All - Pairs shortest path	3.	Dynamic programming
B. Quicksort	4.	Divide and conquer
C. Minimum weight spanning	1.	Greedy
D. Connected components	2.	Depth - first search

321. For merging two sorted lists of sizes m and n into a sorted list of size m+n, how many comparisons are required?
mएवं n अमाप की दो सॉर्टेड लिस्ट के m + n अमाप की एक सॉर्टेड लिस्ट में विलय के लिए कितनी तुलनाओं की आवश्यकता होगी?

(a) O(m) (b) O(n)
(c) O(log m + log n) (d) O(m+n)

LT. Grade C.S. 2018

Ans.(d) : To merge two sorted lists of size m and n you need to compare elements from both lists and place them in the merged list. In the worst case, every element from both lists will need to be compared once, leading to a total of m + n comparisons. Hence the time complexity is O(m + n).

322. Which of the following statements is true aboutBig-O notation?/बिग-ओ नोटेशन के बारे में निम्नलिखित में से कौन-सा कथन सत्य है?

(a) It represents the lower bound of analgorithm's runtime./यह एल्गोरिथम के रनटाइम की निचली सीमा का प्रतिनिधित्व करता है।
(b) It represents the upper bound of analgorithm's runtime/यह एल्गोरिथम के रनटाइम की ऊपरी सीमा का प्रतिनिधित्व करता है।
(c) It represents the average runtime of an algorithm./यह एक एल्गोरिथम के औसत रनटाइम का प्रतिनिधित्व करता है।
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : Big-O notation is used to describe how the runtime of an algorithm grows as the input size increases. It specifically represents the upper bound of the algorithm's runtime, showing the worst-case scenario. It does not indicate the lower bound (Big-omega) or the average runtime. By using Big-O, programmers can compare algorithms and predict performance for large input.

323. What is the computational complexity of thehalting problem?/हॉल्टिंग समस्या की कम्प्यूटेशनल जटिलता क्या है?

(a) O(I)
(b) O(n)
(c) Not computable
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : The Halting problem is a famous problem in computer science that asks whether a given program will eventually stop or keep running forever. Alan turing proved that there is no algorithm that can solve this problem for all possible programs and inputs. This means the problem is undecidable and its computational complexity can not be measured in terms of time like O(n) or O(I).

324. What is the worst case time complexity ofthe binary tree sort?
बाइनरी ट्री सॉर्ट की सबसे खराब स्थिति समय जटिलता क्या है?

(a) O(n)
(b) O(nlogn)
(c) $O(n^2)$
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c): The worst-case time complexity of binary tree sort is $O(n^2)$. This happens when the binary search tree becomes skewed, which can occur if sorted in ascending or descending order. In such a case, each element has to be inserted at the end of the tree, effectively making the tree behave like a linked list. As a result, inserting each of the n elements takes O(n) time, leading to a total time of $O(n^2)$ for the sorting process. While binary tree sort can be efficient with a balanced tree, in the worst case it degenerates and becomes much slower.

325. Which of the following sorting algorithms canbe used to sort a random linked list withminimum time complexity?
निम्नलिखित में से किस सॉर्टिंग एल्गोरिदम का उपयोग न्यूनतम समय जटिलता के साथ यादृच्छिक लिंक्ड सूची को सॉर्ट करने के लिए किया जा सकता है?

(a) Insertion Sort
(b) Quick Sort
(c) Merge Sort
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c): For sorting a random linked list with minimum time complexity, merge sort is the most suitable algorithm. Insertion sort, although simple, has a time complexity of $O(n^2)$ for random lists, making it inefficient. Quick sort works well for arrays because it relies on random access to partition elements, but linked lists do not allow efficient random access, which makes quick sort less suitable. On the other hand, merge sort works perfectly with linked lists because splitting a list into two halves can be done easily using the slow and fast pointer method, and merging two sorted list can be done efficiently in
O(n) time without extra space. Overall, merge sort achieves a time complexity of O(n log n) for linked lists, making it the optimal choice.

326. What is the time complexity of inserting at theend in dynamic arrays?
गतिशील सारणियों में अंत में डालने की समय जटिलता क्या है?

(a) O(1)
(b) O(n)

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(c) Either O(1) or O(n) /या तो O(1) या तो O(n)
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c): In a dynamic array, inserting an element at the end normally takes O(1) time, because we just place the element in the next free index. However, when the array becomes full, a new larger array must be created, and all elements are copied into it before inserting the new one. This resizing operation takes O(n) time in that specific case. Therefore, the insertion at the end in a dynamic array can be O(1) in most cases and O(n) in the worst case.

327. Match List-I with List-II

	List-I		List-II
A.	Prim's algorithm	(i)	$O(V^3 \log V)$
B.	Dijkstra's algorithm	(ii)	$O(VE^2)$
C.	Faster all-paris shortest path	(iii)	O(ElogV)
D.	Edmonds-Karp algorithm	(iv)	$O(V^2)$

Choose the correct option from those given below :
(a) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
(b) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
(c) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)
(d) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) :Prim's Algorithm:-It is used to find the minimum spanning tree for a weighted undirected graph. It uses greedy approach to find the MST for a graph. Unlike the Kruskal's algorithm, we add vertex to growing spanning tree in this. It there are V vertices and E edges in the graph. Then time complexity to for MST is O(E log V)
Dijkstra's Algorithm :-
It is used to find the shortest path from source vertex to all other vertices in a graph. Initially, distance from source to source is considered as O and to all other vertices is infinity. It uses the condition "current vertex distance edge weight < next vertex distance" to find shortest distance. Time complexity for this is $O(V^2)$
Faster all pairs shortest path : It finds shortest the distance from a vertex to all other vertices in a graph. It there are E edges and V vertices in a graph, then it takes $O(V^3 \log V)$ time to find the shortest path.
Edmonds-Karp algorithm :- It is used to find the maximum flow in a network. It uses BFS in ford fulkerson method. It time complexity is $O(V.E^2)$

328. What is the worst case running time of Insert and Extract-min, in an implementation of a priority queue using an unsorted array? Assume that all insertions can be accommodated.
(a) $\theta(1)$, $\theta(n)$ (b) $\theta(n)$, $\theta(1)$
(c) $\theta(1)$, $\theta(1)$ (d) $\theta(n)$, $\theta(n)$

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (a) :A priority queue is a queue in which each element is assigned a priority and they are served according to the priority. If element having same priority, they are processes according to their order in queue.
Insert, extract – min is performed on unsorted array.
Insert : It means insertion of element in unsorted array. In the worst case, element is inserted at the end of the queue. Simply inserting the element at the end of queue takes O (1) complexity.
Extract-min : To do extract-min, use linear search approach for worst case to find the minimum element. For this, we need to traverse all the elements of the queue in worst case. So it will take 0(n) time complexity to extract - min in priority queue.

329. Consider the following statements:
(A) The running time of dynamic programming algorithm is always θ(ρ) where ρ is number of sub problems.
(B) When a recurrence relation has cyclic dependency. It is impossible to use that recurrence relation (unmodified) in a correct dynamic program.
(C) For a dynamic programming algorithm computing all values in a bottom-up fashion is asymptotically faster than using recursion and memorization.
(D) If a problem X can be reduced to a known NP-hard problem then X must be NP-hard.
Which of the statement(s) is (are) true?
(a) Only (B) and (A)
(b) Only (B)
(c) Only (B) and (C)
(d) Only (B) and (D)

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) :
(a) The running time of dynamic programming algorithm is always θ (ρ) where P is number of sub problems.
This statement is incorrect running time for dynamic algorithms is the number of sub-problems multiplies with time for each sub-problem. It means running time of dynamic programming algorithms is O(1).
(b) When a recurrence relation has cyclic dependency it is impossible to use that recurrence relation (unmodified) in a correct dynamic program.
Cyclic dependency is relation between two or more modules which depends on each other indirectly or directly, these are mutually recursive. So, given statement is correct.
(c) Fora dynamic programming algorithms, computing all values in a bottom up fashion is asymptotically faster than using recursion and memorization.
This statement is incorrect for a dynamic programming algorithm, computing all values using recursion and memorization takes less time than computing in bottom up fashion.
(d) If a problem X can be reduced to known NP-hard problem, then X must be NP-hard.
If a problem X can be reduced to a known NP-hard problem, then X must be NP complete not NP-hard.
So, given statement is incorrect.

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330. Match List-I with List-II

List-I	List-II
(A) Topological sort of DAG	(I) O(V +E)
(B) Kruskal's MST algorithm	(II) O(VE)
(C) Bellman Ford's single source shortest path algorithm	(III) θ(V + E)
(D) Floyd-Warshsall's all pair shortest path algorithm	(IV) θ(V^3)

Choose the correct from the options below:

(a) A-I, B-III, C-IV, D-II
(b) A-III, B-I, C-IV, D-II
(c) A-III, B-I, C-II, D-IV
(d) A-I, B-III, C-II, D-IV

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) : Topological sort of DAG = O (V + E)
Bellman - Ford's single - source shortest path algorithm = O (VE)
Kruskal's MST algorithm = θ (V + E) Due to sorting the edges the dominat term in kruskal time complexity is O (E log E).
Floyd-Warshall's all-pairs shortest path algorithm = θ (V^3).

331. Two balanced binary trees are given with m and n elements, respectively. They can be merged into a balanced binary search tree in ____ time.

(a) O(m*n) (b) O(m+n)
(c) O(m*log n) (d) O(m*log(m+n))

UGC NET C.S. December 2020

Ans. (b) : First we store the inorder traversals of both the trees in two separate arrays and then we can merge these sorted sequences in O(m+n) time. And then we construct the balanced tree from this final sorted array.

332. A data structure is required for storing a set of integers such that each of the following operations can be done in O(log n) time, where *n* is the number of elements in the set.

- **Deletion of the smallest element**
- **Insertion of an element if it is not already present in the set**

Which of the following data structures can be used for this purpose?

(a) A heap can be used but not a balanced binary search tree.
(b) A balanced binary search tree can be used but not a heap.
(c) Both balanced binary search tree and heap can be used.
(d) Neither balanced binary search tree nor heap can be used.

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Ans. (b) : A balanced binary search three is a binary search tree in which the difference between the height of left sub tree and the right sub tree is not more than
Now, to perform deletion of the smallest element, the complexity is : O(log n)

333. In the following table, the left column contains the names of standard graph algorithms and the right column contains the time complexities of the algorithms. Here, *n* and *m* are number of vertices and edges, respectively. Match each algorithm with its time complexity.

	List-I Standard graph algorithms		List-II Time complexities
A.	Bellman - Ford algorithm	I.	O(m*log n)
B.	Kruskal's algorithm	II.	O(n^3)
C.	Floyd - Warshall algorithm	III.	O(n*m)
D.	Topological sorting	IV.	O(n+m)

Choose the correct answer from the options given below :

(a) A - III, B - I, C - II, D - IV
(b) A - II, B - IV, C - III, D - I
(c) A - III, B - IV, C - I, D - II
(d) A - II, B - I, C - III, D - IV

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Ans. (a):

List - I Standard graph algorithms		List - II Time complexities	
A.	Bellman - Ford algorithm	III.	O(n*m)
B.	Kruskal's algorithm	I.	O(m*log n)
C.	Floyd - Warshall algorithm	II.	O(n^3)
D.	Topological sorting	IV.	O(n+m)

334. Given the FFT we can have ___ time procedure for multiplying two polynomials A(x) and B(x) of degree bound n where input and output representations are in coefficient form, assuming n is a power of 2.

(a) O(n^2) (b) O(n.$\log_2$n)
(c) O(2^n) (d) O($\log_2$n)

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Ans. (b) : 1.Double degree-bound : Create coefficient representations of A(x) and B(x) as degree-bounded polynomial by adding n high-order zero coefficients to each.

2. Evaluate : Compute point-value representation of A(x) and B(x) of length 2n through two applications of the FFT of order 2n. These represents contain the values of the two polynomials at the (2n)th roots of unity.

3. Pointwise multiply : Compute a point value representation for the polynomial C(x) = A(x) B(x) by multiplying these values together pointwise. This representation contains the value of c(x) and each (2n)th root unity.

4. Interpolate : Create the coefficient representation of the polynomial c(x) through a single application of an FFT on 2n point value pairs to compute the inverse DFT.

Step (1) and (3) take time O(n) and Step (2) and (4) take time O (nlgn). Thus, once we show how to use the FFT, we will have proven the following.
So the product of two polynomials of degree-bound n can be computed in time O (nlgn), with both the input and output representations in coefficient form.
So option (b) is correct.

Sarkari Resource

335. Match List I with List II

LIST-I		LIST-II	
A.	**Parallel FFT**	**I.**	$\Theta(n^2)$
B.	**Iterative FFT**	**II.**	$\Theta(n)$
C.	**Evaluation of polynomial at n points by Horner's method**	**III.**	$\Theta(\lg n)$
D.	**Product of two polynomials that are represented in point value form**	**IV.**	$\Theta(n \lg n)$

Choose the correct answer from the options given below.

(a) A-III B-I C-II D-III
(b) A-II B-I C-III D-IV
(c) A-III B-IV C-I D-II
(d) A-II B-III C-IV D-I

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Ans. (c) : Correct match-

LIST-I		LIST-II	
A.	Parallel FFT	III.	$\theta(\lg n)$
B.	Iterative FFT	IV.	$\theta(n \lg n)$
C.	Evaluation of polynomial at n points by Horner's method	I.	$\theta(n^2)$
D.	Product of two polynomials that are represented in point value form	II.	$\theta(n)$

336. Match List I with List II

LIST-I		LIST-II	
A.	**The running time of straight forward recursive method to compute nth Fibonacci number Fn**	**I.**	**O (n^2)**
B.	**The running time to compute Fn using memorization**	**II.**	**O(lg n)**
C.	**The running time to compute Fibonacci number Fn using only integer addition and multiplication**	**III.**	**O (n)**
D.	**The running time to determine an optimal bitonic tour**	**IV.**	**$\Theta(\phi^n)$**

Choose the correct answer from the option given below :

(a) A-I B-III C-IV D-II
(b) A-IV B-III C-II D-I
(c) A-I B-II C-IV D-III
(d) A-IV B-II C-III D-I

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Ans. (b) Correct match-

LIST-I		LIST-II	
A.	The running time of straight forward recursive method to compute nth Fibonacci number Fn	IV.	$\Theta(\phi^n)$
B.	The running time to compute Fn using memorization	III.	O (n)
C.	The running time to compute Fibonacci number Fn using only integer addition and multiplication	II.	O(lg n)
D.	The running time to determine an optimal bitonic tour	I.	O (n^2)

337. A list of n strings, each of length n, is sorted into lexicographic order using merge - sort algorithm. The worst case running time of this computation is :

(a) O(n log n) (b) O(n^2 log n)
(c) O(n^2 + log n) (d) O(n^3)

UGC NET C.S. November-2017 (Paper-II)

Ans :(b)Since list contain 'n' string, each of length 'n'. Take 1st character of each string and sort it and place sorted strings into output list, doing this for n times will sort the all the strings into one list, so time complexity = n·(n log n) = O (n^2 log n)
Note: First sort each string individually in list will take O(n log n) time more.

338. Any decision tree that sorts n elements has height .....

(a) Ω(lg n) (b) Ω(n)
(c) Ω(n lg n) (d) Ω (n^2)

UGC NET C.S. June-2017 (Paper-III)

Ans :(c) Any decision tree that sorts n elements has height omega (n lg n) proof :
A binary tree of height h has <=2∧ h leaves
At least n : leaves are required for sorting
Thus n! < = 2∧h
h > = lg (n!) = sum – {i = 2} ∧n lg (i) > sum – {i = n⌋2} ∧n lg (n⌋2)
>(n⌋2) lg (n⌋2) = omega (n lg n)

339. If there are n integers to sort, each integer has d digits, and each digit is in the set {1, 2, ... k}, radix sort can sort the numbers in:

(a) O(k(n + d)) (b) O(d(n + k))
(c) O((n + k)|gd) (d) O((n + d)|gk)

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Radix sort complexity is O(wn) for n keys which are integers of words size w. If there are n integers to sort, each integer has d digits, and each digit is in the set {1, 2,….,k} radix sort can sort the numbers in O(d(n + k)).

340. If b is the branding factor and m is the maximum depth of the search tree, what is the space complexity of greedy search?

(a) O(b + m) (b) O(bm)
(c) O(b^m) (d) O(m^b)

UGC NET C.S. November-2017 (Paper-III)

Sarkari Resource

Ans : (c)
(1) In binary tree branching factor is 2 and space complexity of height n is $O(2^n)$
(2) In ternary tree branching factors is 3& space complexity for height n is $O(3^n)$ similarly
(3) If branching factor is b and height is m for search tree then space complexity for greedy search is $O(b^n)$.

341. If there are n integers to sort, each integer has d digits and each digit is in the set {1, 2, ..., k}, radix sort can sort the numbers in :
(a) O (d n k) (b) O (d n^k)
(c) O((d + n)k) (d) O(d(n + k))
UGC NET C.S. December-2015 (Paper-III)

Ans :(d)Lets consider there are 'k' buckets (as given)
Now, for each digit you repeat the following
(1) Place the digit of each number in the appropriate bin – O(n)
(2) Append all the 'k' bins sequentially thus, d digit, its O (d(n+k))

342. In general, in a recursive and non-recursive implementation of problem (program) :
(a) Both time and space complexities are better in recursive than in non-recursive program
(b) Both time and space complexities are better in non-recursive than in recursive program
(c) Time complexity is better in recursive version but space complexity is better in non-recursive version of the program
(d) Space complexity is better in recursive version but time complexity is better in non-recursive version of the program
UGC NET C.S. December-2015 (Paper-II)

Ans :(b)Recursion may not offer any better space or time complexity it only offers compact & simple to write & understand then non recursion implementation.

343. An ideal sort is an in-place-sort whose additional space requirement is.....
(a) O ($\log_2$ n) (b) O (n$\log_2$ n)
(c) O (1) (d) O (n)
UGC NET C.S. December-2015 (Paper-II)

Ans :(c)In place sort are those sorts which doesn't need extra space it uses own space and in case if space is required then it is O(1).

344. Any decision tree that sorts n elements has height
(a) Ω(n) (b) Ω(lgn)
(c) Ω(nlgn) (d) Ω(n^2)
UGC NET C.S. December-2014 (Paper-III)

Ans :(c)For any comparison based sorting algorithm the decision will have n! leaf nodes, so height will be
log (n!) = Ω(n log n)

345. Searching for an element in the hash table requires O(1) time for the _____ time, whereas for direct addressing it holds for the _____ time.
(a) worst-case, average
(b) worst-case, worst-case
(c) average, worst-case
(d) best, average
UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, In searching for an element in hash table requires O(1) time for the average time, whereas for direct it holds for the worst-case time. So the correct answer is option (c).

346. An algorithm is made up of 2 modules M_1 and M_2. If time complexity of modules M_1 and M_2 are h(n) and g(n) respectively, the time complexity of the algorithm is
(a) min (h(n), g(n)) (b) max (h(n), g(n))
(c) h(n) + g(n) (d) h(n) * g(n)
UGC NET C.S. June-2014 (Paper-II)

Ans : (b) Because, In determining the time complexity of an algorithm, if there is a function $T(n) = n_2 - 11_{n-9}$, n_2 is always considered since it is of highest degree. The lower terms are ignored since they contribute most nothing to the sum when is sufficiently large.

347. Big-O estimates for the factorial function and the logarithm of the factorial function i.e. n! and log n! is given by
(a) O(n!) and O(n log n)
(b) O(n^n) and O(n log n)
(c) O(n!) and O(log n!)
(d) O(n^n) and O(log n!)
UGC NET C.S. June-2014 (Paper-II)

Ans : (b) $n! < n^n$ for all $n \geq 1$
O (n^n)
Logarithm of factorial log (n!)
using stirling's approximation:
log (n!) ≈ n log n – n + O (log n) ≈ O (n log n)

function	Big – O
n!	O (n ^ n)
log(n!)	O (n log n)

Therefore, the correct answer is (b) O (n^n) and O (n* log(n))

348. To determine the efficiency of an algorithm the time factor is measured by:
(a) Counting micro seconds
(b) Counting number of key operations
(c) Counting number of statements
(d) Counting kilobytes of algorithm
UGC NET C.S. June-2015 (Paper-II)

Ans.(b) : The complexity is commonly estimated by counting the number of elementary operations performed by the algorithm, where an elementary operation takes a fixed amount of time to perform. Thus the amount of time taken and the number of elementary operations performed by the algorithm differ by at most a constant factor.

349. The average case occurs in the Linear search Algorithm when
(a) The item to be searched is in some where middle of the array
(b) The item to be searched is not in the array
(c) The item to be searched is in the last of the array
(d) The item to be searcher is either in the last or not in the array
UGC NET C.S. June-2015 (Paper-II)

Sarkari Resource

Ans.(a) : Linear search runs in at worst linear time and make at most n comparisons, where n is the length of the list. If each element is equally likely to be searched then the linear search has an average case of (n/2) comparisons, but the average case can be affected if the search probability for each element vary.

350. Which of the following algorithms sort n integers, having the range 0 to (n^2 – 1), in ascending order in O(n) time?

(a) Selection sort (b) Bubble sort
(c) Radix sort (d) Insertion sort

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Radix sort because it sorts in integers in range 0 … (n^2 – 1) in O(n) time using base-n representation.

351. What is the worst-case time complexity of the Heap sort?
हीप सॉर्ट की बर्स्ट-केस समय जटिलता क्या है?

(a) O(n) (b) $O(n^2)$
(c) O(n log n) (d) $O(n^3)$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) **:**Heap sort is a comparison based sorting algorithm. It uses binary heap data structure Heap sort performs in o (n log n) time even in worst case (fully assorted input).

352. What is the worst-case time complexity of the Bubble sort
बबल शॉर्ट की बर्स्ट-केस समय जटिलता क्या है?

(a) O(n) (b) $O(n^2)$
(c) O(n log n) (d) $O(n^3)$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) **:**Burst case time complexity of Bubble Sort is $o(n^2)$. Bubble sort is a sorting algorithm which starts from the first element is greater than the second, then we merge them.
This process is continued till the end of the array where the largest element is bubbling in according order to woods the end of the array.

353. What is the complexity of selection sort algorithms?
सिलेक्शन सॉर्ट एल्गोरिथम की काम्प्लेक्सिटी क्या है?

(a) O(n) (b) $O(n\log_2 n)$
(c) $O(n^2)$ (d) O(2n)

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The time complexity of the selection sort algorithm is as follows–
(1) Best case $O(n^2)$.
(2) Average case $O(n^2)$.
(3) Worst case $O(n^2)$.
Selection sort works by repeated finding the minimum element by from the unsorted part of the list and placing it at the beginning.

354. What is the maximum running time of binary search algorithm, when n = 12,000?
बाइनरी सर्च एल्गोरिथम का अधिकतम रनिंग टाइम क्या है, जब n = 12,000?

(a) 12 (b) 13
(c) 14 (d) 15

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : The maximum running times of binary search depends on how many times you can divide the array before it becomes size 1. That is the maximum number of comparisons in about $\log_2(n)$ (log base 2 of n), rounded up.
Given– n = 12,000

$$\log_2(12000) = \frac{\log_{10}(12000)}{\log_{10}(2)}$$

(Using change of base formula)
$\log_{10}(12000) = 4.079$
$\log_{10}(2) = 0.301$

$$\text{Thus, } \log_2(12000) = \frac{4.079}{0.301} = 13.55$$

Now, rounding up 13.55 given 14.
So, the maximum running time is 14.

355. What is the worst case time complexity of bubble sort?
बबल सॉर्ट की वर्स्ट केस टाइम कम्प्लेक्सिटी क्या है?

(a) O(x) (b) $O(\log_2 x)$
(c) $O(x^2)$ (d) $O(x\log_2 x)$

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : In bubble sort, in the worst case (When the list is sorted in reverse order). It needs to compare and swap adjacent elements many times, leading to a time complexity of $O(x^2)$.

356. What is the worst case running time of quick sort algorithm?
क्विक सॉर्ट एल्गोरिथम का वर्स्ट केस रनिंग टाइम क्या है?

(a) θ(x) (b) $\theta(x^2)$
(c) $\theta(x\log_2 x)$ (d) $\theta(x\log_2 x^2)$

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) :The Worst– Case running time of the quick sort algorithm is $O(n^2)$, where n is the number of elements in the input array. Below is concise explanation of why this occurs along with relevant details to provide a clear understanding.
Quick Sort Overview– Quick sort is a divide and conquer sorting algorithm.
1. Select a pivot element from the array
2. Partitions the array around the Pivot placing smaller elements before it and larger elements after it.
3. Recursively applies the same process to the sub arrays (elements before and after the Pivot)

Time Complexity Analysis–

$$T(n) = T(n-1) + T(0) + O(n)$$

T(0) is a constant (base case), and the partitioning takes O(n)

$$T(n) = T(n-1) + O(n)$$

Solving the Recurrence–

$$T(n) = T(n-1) + n$$
$$T(n-1) = T(n-2) + (n-1)$$
$$T(n-2) = T(n-3) + (n-2)$$

So, On Until–

$$T(1) = T(0) + 1$$

Summing Up–
$T(n) = T(0) + 1 + 2 + \ldots\ldots\ldots + (n-1) + (n-2)$
the sum of integers from 1 to n is

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

This is $O(n^2)$

Sarkari Resource

357. The run time for traversing all the nodes of a binary search tree with n nodes and printing them in an order is :

(a) O (lg n) (b) O(n lg n)
(c) O(n) (d) $O(n^2)$

UGC NET C.S. August-2016 (Paper-II)

Ans :(c)Printing the elements in an order means the element should be printed in sorted order & we know inorder traversal of BST takes gives the sorted order & takes O(n) time. The recurrence involved is :
T(n) = 2T (n/2) + c for balanced BST which on solving gives O(n) and also for skewed tree T(n) = T(n – 1) + c which also on solving gives O(n) time.

358. What is the time complexity ofan infix to postfix conversionalgorithm?
Postfixरूपान्तरण Algorithmके लिए Infixकी समय जटिलता क्या है

(a) O (n log n) (b) O (n)
(c) $O(n^2)$
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : The average case time complexity of infix to postfix conversion algorithm is O(n) where "n" relates to the size of the infix expression i.e., total number of operators and operands in the infix expression.

(viii) Advanced Data Structures

359. Which of the following descriptions (s) is/are true :

A. Red Black Tree – Guarantees worst case of O(log n) time for search, insert & delete.
B. Trie - Used for efficient prefix - based searches.
C. AVL Tree – Self balancing binary search tree with stricker balance criteria.
D. B- Tree Allows efficient search, inert, delete operation in disk based system.

Choose the correct answer from the options given below :

(a) A & B only (b) C & D only
(c) B only (d) A, B & D only

UGC NET C.S. December-2024

Ans. (*) :The correct answer is (d) A, B & D only.
(A) Red Black Tree- A Red Black tree is a type of self-balancing binary search tree that guarantees O(log n) time complexity for search, insert, and delete operations in the worst case.
(B) Trie-A Trie (Prefix tree) is specifically designed for efficient prefix based searches, making it ideal for tasks like auto complete or directionary implementations.
(C) AVL Tree-An AVL tree is indeed a self balancing binary search tree with stricter balance criteria compared to Red-Black Tree.
(D) B-Tree-B trees are designed for efficient search, insert and delete operations especially in disk based systems as they minimize the number of disk accesses.
Note- This question dropped by UGC.

(ix) Red-Black Trees

360. Which of the following satisfies the property of the red-black tree?
निम्नलिखित में से कौन-सा लाल-काले ट्री के गुण को संतुष्ट करता है?

(a) Black, if the new node is a root node
(b) Red, if the new node is not a root node
(c) Black, if the new node is not a root node
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d):A red-black tree satisfies the following black. If a node has a NULL child, then that child is market black. If a node is red, then both of its children are black.

361. _____ is a Self Balancing binary search tree, where the path from the root to the farthest leaf is no more than twice as long as the path from the root to nearest leaf.

(a) Expression tree (b) Game tree
(c) Red-Black tree (d) Threaded tree

UGC NET C.S. Re-Exam- June 2024

Ans. (c) :A Red Black tree is a type of self balancing binary search tree. It maintains balance by ensuring that the longest path from the root to a leaf is no more than twice as long as the shortest path from the root to any leaf.

362. Red–black trees are one of may search tree schemes that are "balanced" in order to guarantee that that basic dynamic-set operations take ..... time in the worst case.

(a) O(1) (b) O(lg n)
(c) O(n) (d) O(n lg n)

UGC NET C.S. June-2017 (Paper-III)

Ans :(b) Red black trees are one of many search tree schemes that are "balanced". in order to guarantee that basic dynamic set operations O(lg n) time in worst case.

363. Which of the following statement(s) is/are correct about red-black binary tree?
रेड-ब्लैक बाइनरी ट्री के बारे में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. Every node is either red or black.
प्रत्येक नोड या तो लाल या काला होता है।

I. Every leaf node is red.
हर लीफ नोड लाल होता है।

III. When a node is red and both its children are black.
जब एक नोड लाल होता है तो उसके दोनों बच्चे काले होते हैं।

(a) I and II/ Iतथा II
(b) I, II and III/ I, II तथा III
(c) I andIII/ केवल III
(d) III andII/ IIIतथा II

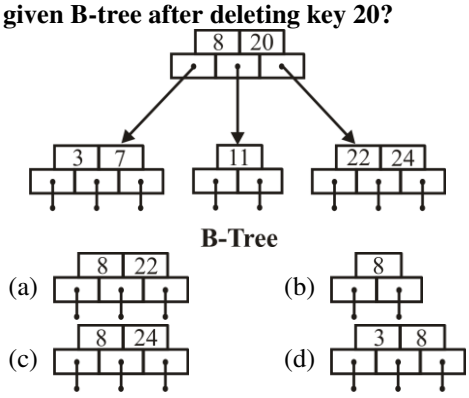
DSSSB PGT C.S. M/F 11.07.2021

Sarkari Resource

Ans. (c) :
(i) Every node is either red or black. This is correct. In a red-black tree, each node must be either red or black.
(ii) Every leaf node is red. This is incorrect. In a red-black tree, the leaf nodes (also called NIL nodes) are typically black.
(iii) When a node is red and both its children must be black. This property helps maintain the balance of the tree.
So the correct statements are I and III.

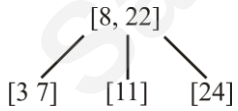
(x) B –Trees

364. Which of the following is a valid root of the given B-tree after deleting key 20?



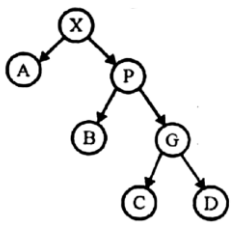
RPSC C.S. 2016 (Paper-II)

Ans. (a) : Step by step deletion of 20.
20 is in the root node [8, 20]
It is an internal node (not a leaf).
We replace 20 with the inorder predecessor or inorder successor.
Use the inorder successor (i.e., the smallest key in the right subtree, which is 22 from the [22, 24] node).
So, replace 20 with 22 → root becomes [8, 22]. Now, delete 22 from the leaf [22, 24] → is becomes [24].
Now the tree looks like-



So, the correct option (a) is the valid root after deleting key 20.

365. The following figure is a _____ tree after Zig-Zig rotations.
निम्न आकृति जिग-जेग घूर्णन के पश्चात एक ____ ट्री है ।



(a) Heap/हीप
(b) Bubble/बदल
(c) Splay/स्प्ले
(d) Binary/बाईनरी

KVS PGT C.S. 2013

Ans.(c) : A splay tree is a self balancing tree , but AVL and Redblack trees are also self balancing trees
There are six types of rotations used for splaying
1. Zig rotation (Right rotation)
2. Zag rotation (left rotation)
3. Zig Zag (Zig followed by zag)
4. Zag Zig (Zag followed by Zig)
5. Zig Zig (two right rotation)
6. Zag Zag (two left rotation)

366. A B-Tree is of order p and consists of n keys. Its maximum height is
एक B-Tree, जिसका क्रम p है एवं जिसमें n कुंजियाँ हैं, की अधिकतम ऊँचाई है
(a) $\log_{(p/2)}(n+1)/2$
(b) $\log_p n$
(c) $\log_{(p/2)}(n+1)$
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans.(a) :B-tree is a self-balancing tree data structure that to maintains sorted data and allows searches, sequential access, insertions and deletions in logarithmic time. A B-tree is of order p and consists of n keys. The maximum of the B-tree is

$$h_{max} = \log_{p/2}\left(\frac{n+1}{2}\right)$$

The minimum height of the B-tree is

$$h_{min} = \left[\log_p (n+1)\right] - 1$$

367. In a balanced binary tree, the height of twosub-trees of every node can not differ by morethan:/एक संतुलित बाइनरी वृक्ष में, प्रत्येक नोड के दो उप-वृक्षों की ऊँचाई में इससे अधिक का अंतर नहीं हो सकता है।
(a) 2
(b) 1
(c) 0
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : In balanced tree (more specifically, an ABL tree), the height of the left and right subtrees of any node can differ by 1. If the difference becomes more than 1, the tree is considered unbalanced, and rotations are needed to restore balance. This property ensures that searching, insertion, and deletion operations remain efficient, keeping the time complexity close to O (log n).
Therefore, the maximum allowed height difference between two subtrees of any node is 1.

368. Which of the following tree data structures isnot a balanced binary tree?
निम्नलिखित में से कौन-सी वृक्ष डेटा संरचना संतुलित बाइनरी ट्री नहीं है?
(a) Red - black tree
(b) B-tree
(c) AVL tree
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Sarkari Resource

Ans.(b): B-tree is not a balanced binary tree; its a self-balancing search tree where each node can have more than two children, making it suitable for disk storage systems. In contrast. AVL trees and Red-black trees are examples of balanced binary trees.

369. Which of the following is the most widely usedexternal memory data structure?
निम्नलिखित में से कौन-सी सबसे व्यापक रूप से इस्तेमाल की जाने वाली बाहरी मेमोरी डेटा संरचना है?

(a) B-tree
(b) Red-black tree
(c) AVL tree
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(a): When working with very large data that cannot for entirely into main memory, external memory data structures are used to minimize disk access. Among all options, the B-tree is the most widely used external memory data structure. It keeps its height low by allowing multiple keys in a single node, which reduces the number of disk accesses required during searching, insertion, or deletion. That is why B-tree are extensively used in databases and file systems. On the other hand, Red-black trees and AVL trees are mostly used for internal memory, not external memory.

370. An AVL tree is called in the unbalanced stage and requires proper rotation for balancing whenever:
एकAVL ट्री अनबैलेंस स्टेट में कहलाता है तथा उसे बैलेंस करने के लिए उपयुक्त रोटेशन की आवश्यकता होती है जब भी:

(a) Any of its node has balance factor -1 or 1
किसी भी नोड का बैलेंस फैक्टर-1 अथवा 1 हो जावे
(b) Root has balance factor -1 or 1
रूट नोड का बैलेंस फैक्टर-1 अथवा 1 हो जावे
(c) Any of its node has balance factor -2 or 2
किसी भी नोड का बैलेंस फैक्टर-2 अथवा 2 हो जावे
(d) Root has balance factor zero
रूट नोड का बैलेंस फैक्टर शून्य हो जावे

RPSC C.S. 2016 (Paper-I)

Ans.(c) : In an AVL Tree, a nodes becomes unbalance when the balance factor is:
Balance factor = Height (Left subtree)- Height (Right subtree)
- Valid balance factors : –1, 0, 1 → Tree is balanced.
- Invalid (unbalanced) : –2 or 2 → Tree need rotation.

So, any node (not just the root) with a balance factor of –2 or 2 causes an imbalance.

371. In a B-Tree, each node represents a disk block. Suppose one block holds 8192 bytes. Each key uses 32 bytes. In a B-tree of order M there are M - 1 keys. Since each branch is on another disk block, we assume a branch is of 4 bytes. The total memory requirement for a non-leaf node is

(a) 32 M - 32 (b) 36 M - 32
(c) 36 M - 36 (d) 32 M - 36

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) :
Data :- Block size = 8192 bytes = 2^{13} bytes
Key size = 32 bytes
Block pointer = 4 bytes
Order of the b-tree = M
Number of keys = M – 1
FORMULA→In a B – Tree, the size of non - leaf node = M x (Block pointer) + (M – I) x (Key + Record pointer)
CALCULATION :- Assuming Record pointer = 0 (since nothing is mentioned about it)
Total size = M(4) + (M – 1) (32) = 36 M – 32

372. Consider a B-tree of height h. minimum degree $t \geq 2$ that contains any n-key, where $n \geq 1$. Which of the following is correct?

(a) $h \geq \log_t \frac{n+1}{2}$ (b) $h \leq \log_t \frac{n+1}{2}$
(c) $h \geq \log_t \frac{n-1}{2}$ (d) $h \leq \log_t \frac{n-1}{2}$

UGC NET C.S. December 2021

Ans. (b) : In the case of B-tree with a minimum no of keys, there is one key in the root, 2 nodes at depth 1, $2t^i$–1 nodes at depth *i*.
Let h be the height of the tree and n be the number of nodes. Then

$$n \geq 1 + (t-1)\sum_{i=1}^{n} 2t^i - 1$$

Which workout to $t^h \leq \frac{n+1}{2}$
So, $\log_t$ taking the both sides

$$h \leq \log_t \frac{(n+1)}{t}$$

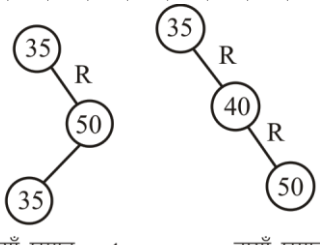
So option (b) is correct.

373. How many rotations are required during the construction of an AVL tree if the following elements are to be added in the given sequence?
35, 50, 40, 25, 30, 60, 78, 20, 28

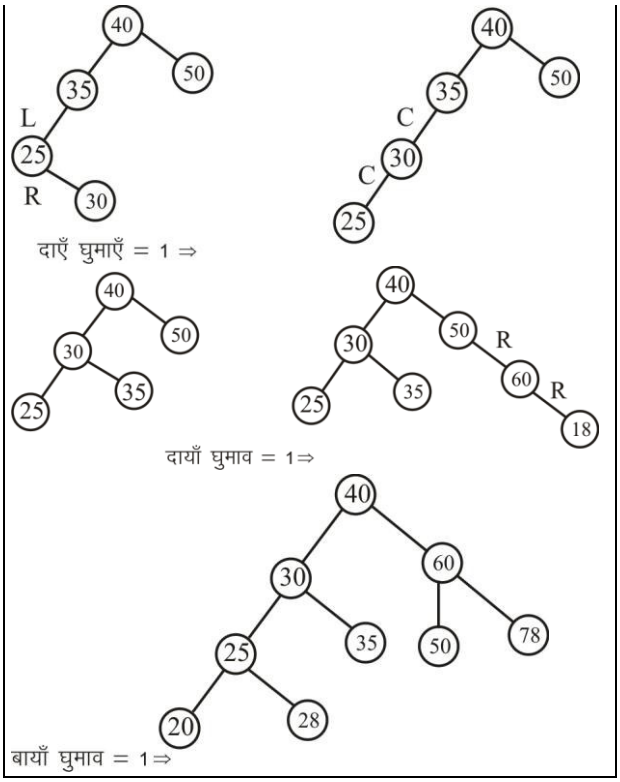
(a) 2 left rotations, 2 right rotations
(b) 2 left rotations, 3 right rotations
(c) 3 left rotations, 2 right rotations
(d) 3 left rotations, 1 right rotations

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Ans. (b) : The first type of self balancing binary search tree to be invented is the AVL tree.
There are usually four cases of rotation in the balancing algorithm of AVL trees : LL, RR, LR, RL.
given sequence
35, 50, 40, 25, 30, 60, 78, 20, 28



Sarkari Resource



374. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The AVL trees are more balanced as compared to Red Black trees, but they may cause more rotations during insertion and deletion

Reason R : A Red Black tree with n nodes has height that is greater than $2\log_2 (n + 1)$ and the AVL tree with n nodes has height less than $\log_{\Phi}\left(\sqrt{5}(n+2)\right) - 2$ (where Φ is golden ratio)

In the light of the above statement choose the correct answer from the option given below.

(a) Both A and R are correct and R is the correct explanation of A
(b) Both A and R are correct and R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

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Ans. (c) : The AVL trees are generally more balanced than Red-Black trees because AVL trees strictly enforce balance by maintaining a height difference of at most 1 between the left and right subtrees of any node. However, this strict enforcement can lead to more rotations during insertion and deletion operations.

A Red Black trees with n nodes has height that is no more than $2 \log_2 (n + 1)$. The height of an AVL trees with n nodes is less than or equal to $1.44 \log_2(n + 2) - 1.328$. There is not a direct relationship between the height of an AVL trees and the golden ratio. So, Assertion A is true but Reason R is false.

375. The total cost of retrieving records in sorted order using an unclustered B^+ tree is (P-Average number of records per data page, N-Data pages
F- Ratio of the size of a data entry to the size of a data record)

(a) (F*N)+P (b) (F+P)*N
(c) F*N*P (d) F+P/N

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Ans. (b) : To calculate the total cost of retrieving records in sorted order using an unclustered B+ tree. We need to consider the cost of accessing each data page. This can be calculated by taking into account the average number of records per data page (P), the number of data page entry (N), and the ratio of the size of a data entry to the size of a data record (F). The formula for the total cost can be expressed as:
Total cost = (F + P) * N.

376. A B-tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then maximum number of nodes that could be newly created in the process is

(a) 5 (b) 4
(c) 3 (d) 2

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Ans. (a) : All nodes are completely full means every node has n-1 keys. Given tree has 4 levels if a new key is inserted then at every level there will be created a new node. In worst case, root node will also be broken into two parts, and we have 4 levels so, answer should be 5 because tree will be increased with one more level.

377. 2 – 3 – 4 trees are B – trees of order 4. They are isometric of _____trees.

(a) AVL (b) AA
(c) 2 – 3 (d) Red-Black

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Ans. (d) :2 – 3 – 4 tree (also called a 2 – 4 tree) is a self-balancing data structure that can be used to implement dictionaries. 2 – 3 – 4 trees are B-trees of order 4, like B-trees in general, they can search, insert and delete in O(log n) time, 2 – 3 – 4 trees are isomorphic to Red-Black trees, meaning that they are equivalent data structures. In other words, for every 2 – 3 – 4 tree, there exists at least one and at most one red-black tree with data elements in the same order.

378. Which of the following is TRUE?

(a) The cost of searching an AVL tree is θ (log n) but that of binary search is O(n)
(b) The cost of searching an AVL tree in θ (log n) but that of complete binary tree is θ (n log n)
(c) The cost of searching a binary tree is O(log n) but that of AVL tree is θ (n)
(d) The cost of searching an AVL tree is θ (n log n) but that of binary search tree is O(n)

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Ans. (a) :AVL tree is a balanced tree. AVL tree's time complexity of searching = θ (log n). But a binary search tree may be a skewed tree, so in the worst case Binary search tree searching time = O(n).

379. The order of a leaf in a B+ tree is the maximum number of children it can have. Suppose that block size is 1 kilobyte, the child pointer takes 7 bytes long and search field value takes 14 bytes long. The order of the leaf node is .....

(a) 16 (b) 63
(c) 64 (d) 65

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Ans :(a)Data pointer = child pointer + search field value = 14 + 7 = 21bytes
So max number of children it can have = 1 KB/21bytes = 1024/21 = 48.76 = 45 (approx)
All other option are greater than 48.

380. Which of the following statement is false about a B-tree of order M?
निम्न में से कौन सा कथन M आर्डर के B-ट्री के बारे में गलत है?

(a) The root has at least two child nodes and at most M child nodes./रूट में कम से कम दो चाइल्ड नोड्स और अधिकतम Mचाइल्ड नोड होते हैं।
(b) The internal nodes except the root have atleast $\left\lceil \frac{M}{2} \right\rceil$ child nodes.
रूट को छोड़कर आंतरिक नोड्स में कम से कम $\left\lceil \frac{M}{2} \right\rceil$ चाइल्ड नोड होते हैं।
(c) Leaf nodes are not at the same level.
लीफ नोड्स समान स्तर पर नहीं होते हैं।
(d) A node is full if it has (M – 1) keys./एक नोड भरा हुआ है यदि उसके पास (M – 1)कुंजियाँ हैं।

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Ans. (c) : Leaf nodes are not at the same level this statement is false about a B- tree of order M. In a B-tree, all leaf nodes are at the same level. This property ensures that the tree remains balanced which is essential for maintain the B- tree's efficiency in terms of search, insertion and deletion operations.

381. Which rotations in an AVL tree is known as double rotation ?
ए.वी.एल. ट्री में कौन सा रोटेशन दोहरा रोटेशन कहलाता है?

(a) LL and RR/LL तथा RR
(b) LR and LL/LR तथा LL
(c) RL and RR/RL तथा RR
(d) LR and RL/LR तथा RL

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Ans. (d) :AVL Rotations- Rotation in AVL tree only in case if balance factor is other than –1,0 and +1. There are basically four types of rotations which are as –
1- LL Rotations - Inserted node is in the left sub tree of left sub tree of A.
2- RR rotations- Inserted node is in the right sub tree of left sub tree of A.
3- LR rotations- Inserted node is in the right sub tree of left sub tree of A.
4- RL rotations- Inserted node is in the left sub tree of right sub tree of A.
Hence the correct answer is option d.

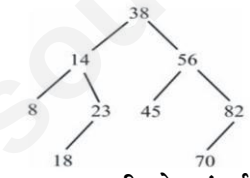
382. Which rotations in AVL trees are known as single rotations ?
AVL ट्री में कौन से घूर्णन एकल घूर्णन के रुप में जाने जाते हैं ?

(a) LL and RR (b) LL and LR
(c) RR AND RL (d) LR and RL

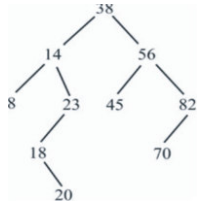
DSSSB TGT C.S. 01.08.2021 (Shift-I)

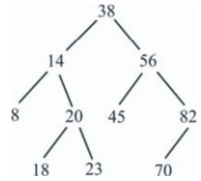
Ans. (a): In AVL trees, rotations are used to balance the tree when the balance factor becomes +2 or –2.
There are two types of rotations.
(1). Single Rotation (one step)
LL (Left Left Rotation)
RR (Right Right Rotation)
(2). Double Rotations (Combine of two single rotations)LR (Left Right Rotation)
RL (Right Left Rotation)
So, the correct answer is option (a) LL and RR

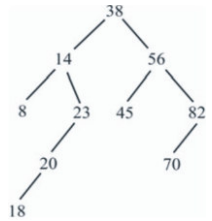
383. Consider the following AVL tree.
What will be the resultant AVL tree after insertion of node with value 20?
निम्नलिखित AVL ट्री पर विचार करें ।

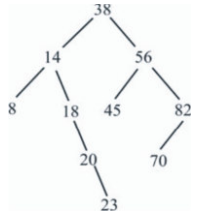


मान 20 वाली नोड इंसर्ट के बाद परिणामी AVL ट्री क्या होगा

(a)


(b)


(c)


(d)


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Ans. (b) : To insert the node with the value 20 into the AVL tree. Starting from the root (38), traverse left (to 14), then right (to 23), then left (to 18). The value 20 is greater than 18, so it is inserted as the right child of the node 18.
After insertion, the tree becomes unbalanced at node 23. The balance factor of node 23 is height (left subtree) – height (right subtree) which becomes 0 – 2= –2. This imbalance is a left right case because the new node (20) was inserted into the right subtree of the left child (18) of the unbalanced node(23).
To correct this imbalance, a double rotation is performed.
1. Left Rotation on node 18. Node 20 becomes the new root of this subtree, with 18 as its left child.
2. Right Rotation on the original unbalanced node 23. Node 20 becomes the new root of the entire subtree, with 14 as its left child and 23 as its right child.
The rebalanced tree after rotation is given in option (b).

(xi) Binomial Heaps

384. In a binary max heap containing a number, the smallest element can be found in _____ time.

(a) O(n) (b) O($\log_2$n)
(c) O(1) (d) O($\log_2\log_2$n)

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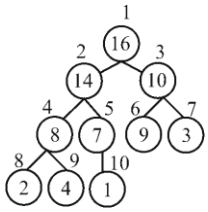
Ans. (a) : The heap's smallest values will always be present on the last level or second last of the heap in a max heap and the time complexity of reaching the last level of the Heap is O(n).
• It will take O(n) time to traverse all of the elements using linear search.

385. Consider an array representation an n element binary heap where the elements are stored from index 1 to index n of the array. For the element stored at index i of the array (i < = n), the index of the parent is :

(a) floor ((i + 1)/2) (b) ceiling ((i + 1)/2)
(c) floor (i/2) (d) ceiling (i/2)

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Ans :(c)Consider a binary heap of n elements in which elements are stored in array from index 1 to n.



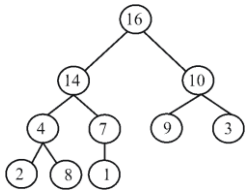
(5)

1	2	3	4	5	6	7	8	9	10
16	14	10	8	7	9	3	2	4	1

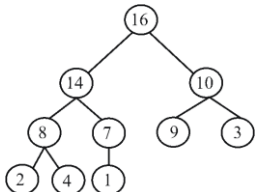
Parent of any element at index i is calculated as $\left\lfloor \frac{i}{2} \right\rfloor$

386. Which of the following is a valid heap?

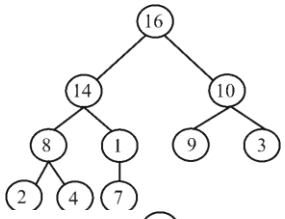
(A)



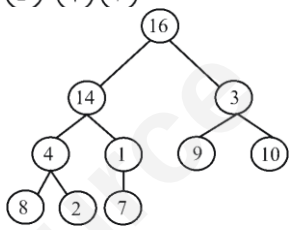
(B)



(C)



(D)

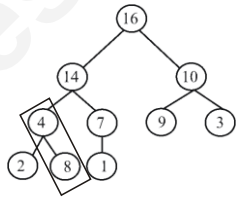


(a) A (b) B
(c) C (d) D

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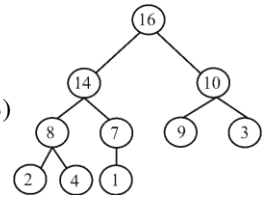
Ans :(b)

(A)



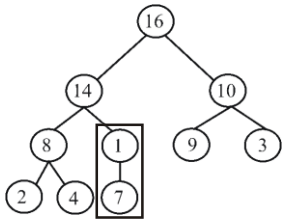
Max heap property failed because parent ≥ child

(B)



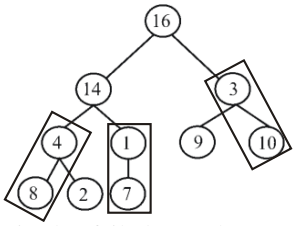
valid max heap because every parent ≥ child

(C)



Here is also failed, max heap property.

(D)



Here is also failed, max heap property.

387. Heap allocation is required for languages that :

(a) use dynamic scope rules
(b) support dynamic data structures
(c) support recursion
(d) support recursion and dynamic data structures

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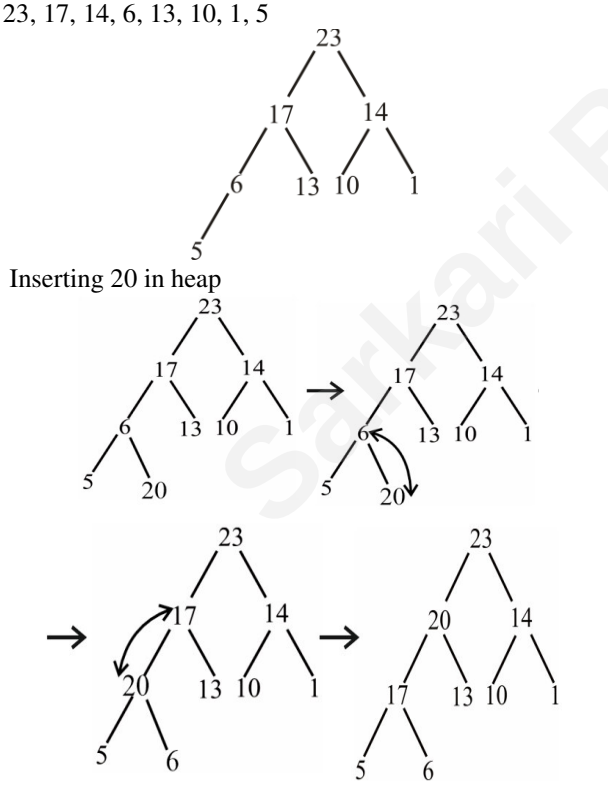
Ans :(b) Heap allocation is required for language which support dynamic data structure.

388. Consider a max heap, represented by the array:
सारणी द्वारा दर्शाए गए मैक्स हीप पर विचार करें।
23, 17, 14, 6, 13, 10, 1, 5
If a value 20 is inserted into this heap, then what will be the new max heap
यदि इस हीप में एक मान 20 डाला जाता है, तो नया मैक्स हीप क्या होगा?

(a) 23, 17, 4, 6, 13, 10, 1, 5, 20
(b) 23, 20, 14, 17, 13, 10, 1, 5, 6
(c) 23, 20, 17, 14, 13, 10, 6, 5, 1
(d) 23, 17, 20, 14, 13, 10, 6, 51

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Ans. (b) : Original Max Heap–
23, 17, 14, 6, 13, 10, 1, 5

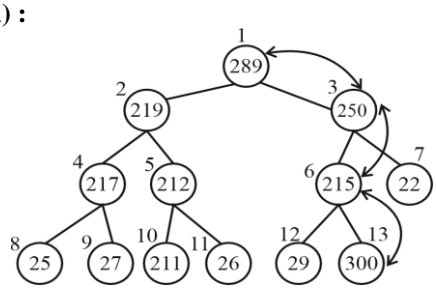


389. Consider the following arrays of elements 289, 219, 250, 217, 212, 215, 22, 25, 27, 211, 26, 29, 300. What is the minimum number of interchanges needed to convert it into max-heap?
289, 219, 250, 217, 212, 215, 22, 25, 27, 211, 26, 29, 300 तत्वों के निम्नलिखित एरे पर विचार करें। इस मैक्स-हीप में बदलने के लिए आवश्यक न्यूनतम इंटरचेंज की संख्या क्या है?

(a) 4 (b) 5
(c) 2 (d) 3

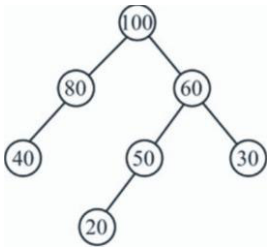
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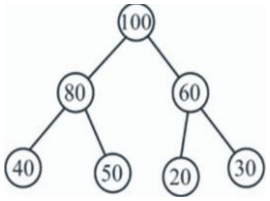
Ans. (d) :

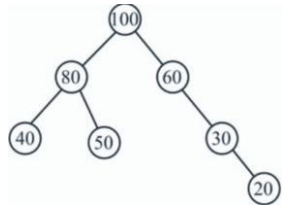


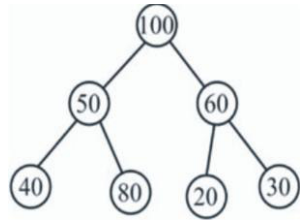
If you build max-heap using the usual bottom–up shift down from the given array you perform 3 swaps:
1. swap index 6 and 13: 215 ↔ 300
2. Swap index 3 and 6 : 250 ↔ 300
3. Swap index 1 and 3 : 289 ↔ 300
After these 3 interchanges the array becomes a valid max-heap [300, 219, 289, 217, 212, 250, 22, 25,. 27, 211, 26, 29, 215]
So,
minimum number of exchanges = 3

390. Which of the following is a max-heap?
निम्नलिखित में से कौन सा एक मैक्स-हीप है ?

(a)


(b)


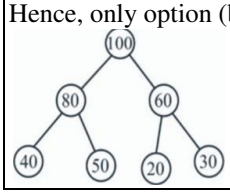
(c)


(d)


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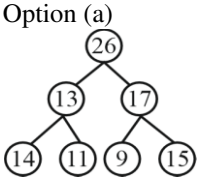
Ans. (b) : Properties of Max Heap:
Must be complete binary tree.
All levels are completely filled except possibly the last level.
The last level is filled from Left to Right.
Each parent node has a value greater than or equal to its children.
Option (a) and (c) do not satisfy the property of Complete Binary tree while in option (d) parent node (50) is less than its right child (80) which does not fulfill the property of Max Heap.
Hence, only option (b) is a Max Heap.



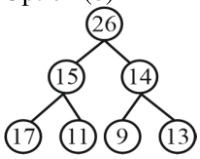
391. Which one of the following array represents a binary max-heap?
(a) [26, 13, 17, 14, 11, 9, 15]
(b) [26, 15, 14, 17, 11, 9, 13]
(c) [26, 15, 17, 14, 11, 9, 13]
(d) [26, 15, 13, 14, 11, 9, 17]

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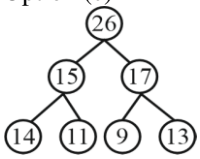
Ans : (c) Properties of Max Heap:
Must be complete binary tree.
All levels are completely filled except possibly the last level.
The last level is filled from Left to Right.
Each parent node has a value greater than or equal to its children.
Option (a)



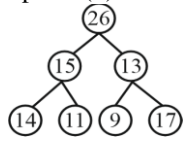
Not fulfill the property of Max Heap.
Option (b)



Not fulfill the property of Max Heap.
Option (c)



Fulfill the property of Max Heap.
Option (d)



Not fulfill the property of Max Heap.
So, the correct option is (c).

(xii) Divide and Conquer

392. Divide and Conquer is an algorithmic paradigm that solves problems by:
डिवाइड एंड कॉन्कर एक एल्गोरिदमिक प्रतिमान है जो समस्याओं को हल करता है:
(a) Iteratively solving subproblems
उपसमस्याओं को पुनरावृत्त रूप से हल करके
(b) Recursively solving subproblems
उपसमस्याओं को पुनरावृत्त (Recursive) रूप से हल करके
(c) Using heuristics to find solutions
समाधान खोजने के लिए हीयुरिस्टिक का उपयोग करके
(d) Greedily combining solutions
समाधानों को ग्रीडिली संयोजित करके

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : Divide and conquer is an algorithmic paradigm in which a large problem is broken down into smaller sub problems. Each of these sub problems is then solved recursively, meaning the same method is applied again and again on smaller inputs. Finally, the results of these sub problems are combined to form the solution to the original big problem. This approach is commonly used in algorithms like merge sort, quiche sort and Binary search. Therefore, Divide and conquer solves problems by recursively solving subproblems.

393. Which of the following is not a divide and conquer method.
(a) Binary Search (b) Merge Sort
(c) Quick Sort (d) Heap Sort

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Ans. (d) : Heap sort is not a divide and conquer algorithm. Instead, it is based on the Heap Data structure and works by first building a max heap and then repeatedly extracting the maximum element.
Binary Search merge sort and quick sort all follow the divide and conquer approach by breaking the problem into smaller sub problems, solving them recursively and combining the results.

394. Which of the following algorithm design technique is used in designing quick sort algorithm:–
(a) Dynamic programming method
(b) Back tracking strategy
(c) Divide and conquer strategy
(d) Greedy strategy.

RPSC C.S. 2016 (Paper-II)

Ans.(c) : Quick sort works by the divide and conquer approach-
Divide– The array is divided into two sub arrays based on a pivot element such that all elements less than the pivot go to the left, and all elements greater go to the right.
Conquer– The sub arrays are then sorted recursively using the same process.
Combine– No actual combination step is needed as the sorting happens in place.

Sarkari Resource

(xiii) Greedy Algorithms

395. Statement I : In greedy method the input stream is divided into two parts.
Statement II: In greedy method, one input is considered at time for decision making.
Which of the following is correct?
कथन : ग्रीडी विधि में इनपुट स्ट्रीम को दो भागों में बांटा जाता है।
कथन : ग्रीडी विधि में एक समय में एक इनपुट को निर्णय करने हेतु कन्सीडर किया जाता है।
निम्न में से कौन सा सही है?

(a) Both Statements are true./दोनों कथन सही हैं।
(b) Both Statements are false./दोनों कथन गलत हैं।
(c) Statement I is true and Statement II is false./कथन 1 सही है तथा कथन 2 गलत है।
(d) Statement I is false and Statement II is true./कथन 1 गलत है तथा कथन 2 सही है।

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : Statement (I) is false and statements (II) is true.
Statement (I) : "in greedy method the input stream is divided in two parts." This statement is false. It does not necessarily involve dividing the input stream into two distinct parts in a general sense.
Statemet (II) : "In greedy method, one input is considered at time for decision making" This statement is true.

396. Choose a false statement:–

(a) Greedy approach is based on local optimal solution.
(b) Greedy approach is faster than dynamic programming.
(c) Greedy approach examines all feasible solutions.
(d) MST prim's algorithm is based on greedy approach.

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Ans.(c) : Greedy approach– This approach makes the best choice at each step, hoping to head to the overall optimal solution, but it does not necessarily examine all possible feasible solutions. It focuses on the immediate benefit and moves forward without looking back.
Dynamic Programming– This approach, on the other hand, systematically explores all feasible solutions, often using memorization to avoid redundant calculations and guarantee an optimal solutions.
So, the correct option is (c).

(xiv) Knapsack

397. Fractional knapsack problem is solved mostefficiently by which of the followingalgorithm?
भिन्नात्मक नैपसेक को निम्नलिखित में से किस एल्गोरिथमद्वारा सबसे कुशलता से हल किया जाता है?

(a) Backtracking
(b) Greedy algorithm
(c) Dynamic programming
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) : The Fractional knapsack problem is most efficiently solved by the Greedy algorithm. In this problem, we are allowed to take fractions of items instead of whole items.
Since this method directly leads to the optimal solution with less time and effort coopered to backtracking or dynamic programming, the Greedy algorithm is the best choice.

398. Consider the fractional knapsack instance n = 4, (p_1, p_2, p_3, p_4) = (10, 10, 12, 18), (w_1, w_2, w_3, w_4) = (2, 4, 6, 9) and M = 15. The maximum profit is given by
(Assume p and w denotes profit and weight of objects respectively)

(a) 40 (b) 38
(c) 32 (d) 30

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because,

p_1 / w_1	5
p_2 / w_2	2.5
p_3 / w_3	2
p_4 / w_4	2

Now select the one which has max (p/w) ratio that is $p_1/w_1 = 5$ So select 10
Next $p_2/w_2 = 2.5$ select 10
Now p_3/w_3 and p_4/w_4 has same ratio but p_4 gives maximum profit 80 select p_4,
therefore total weight = (2 + 4 + 9) = 15
And max profit = 10 + 10 + 18 = 38
So the answer is (b).

(xv) Minimum Spanning trees Prim's and Kruskal's algorithms

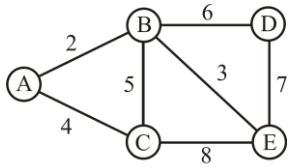
399. Which statement is true about Krushal's algorithm?

(a) It is a greedy algorithm for the minimum spanning tree problem
(b) It constructs spanning tree by selecting edges in increasing order of their weights
(c) It does not accept creation of cycles in spanning tree
(d) All of the above are true

CGPSC C.S. 2020 Set-A

Ans.(d):Greedy → Picks smallest edge each time.
Sorted edges → Chooses edges in increasing weight order
No cycles → Uses union-Find to avoid loops.
So, **all statements are true.**

400. What is weight of minimum cost spanning tree rooted at 'A' extracted from the graph G given below:-



Graph G

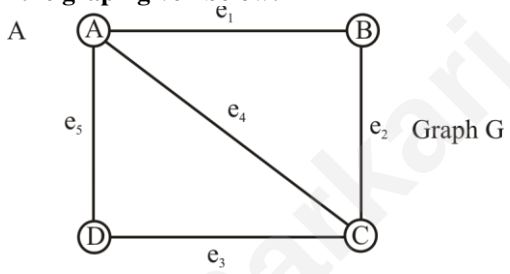
Sarkari Resource

(a) 20 (b) 15
(c) 9 (d) 27

RPSC C.S. 2016 (Paper-II)

Ans. (b) : To find the minimum cost spanning tree of the given graph rooted at A, we can use prim's algorithm, which is especially convenient when a starting node is specified.
Graph G (edges with weights)
A - B = 2
A - C = 4
B - C = 5
B - D = 6
C - E = 8
D - E = 7
B - E = 3 (Note- there are two edges between D and E, choose 3 since it's minimum)
Primes Algorithm starting from A-
1. Start at A
   Add edge A-B (weight 2) → total = 2
2. From A, B → choose edge A-C (4) or B-C (5) → Pick A-C
Add A-C (weight 4) → total = 6
3. From A, B, C → next minimum edge is B-D (6)
   Add B-D (weight 6) → total = 12
4. From A, B, C, D → edge B-E (3)
   Add B-E (weight 3) → total = 15
Total weight = 2 + 4 + 6 + 3 = 15

401. How many spanning trees can be formed from the graph given below:-



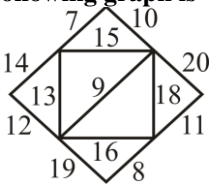
(a) 1 (b) 5
(c) 7 (d) 8

RPSC C.S. 2016 (Paper-II)

Ans. (d) : Step by step process-
The vertices be-
Top - left : A
Top - right : B
Bottom - right ; C
Bottom - left : D
Edges in the graph.
e_1 : A - B
e_2 : B - C
e_3 : C - D
e_4 : A - C (diagonal)
e_5 : A - D
Properties of a spanning Tree– A spanning tree for a graph with n vertices has exactly n-1 edges, and must be connected and acyclic.
Since the graph has 4 vertices, each spanning tree must have exactly 3 edges.
Total number of 3-edge combinations from 5 edges = c (5, 3) = 10
Now we evaluate each combination-
1. $\{e_1, e_2, e_3\}$= A-B, B-C, C-D
   Forms a tree (A-B-C-D)
2. $\{e_1, e_2, e_4\}$ = A-B, B-C, A-C
   Has a cycle (A-B-C-A)
3. $\{e_1, e_2, e_5\}$ = A-B, B-C, A-D
   Tree, A-B-C and A-D
4. $\{e_1, e_3, e_4\}$ = A-B, C-D, A-C
   Tree, A-B, A-C, C-D
5. $\{e_1, e_3, e_5\}$ = A-B, C-D, A-D
   Tree, A-B, A-D, D-C
6. $\{e_1, e_4, e_5\}$ = A-B, A-C, A-D
   Tree, A-B, A-C, A-D
7. $\{e_2, e_3, e_4\}$ = B-C, C-D, A-C
   Tree, A-C, C-B, C-D
8. $\{e_2, e_3, e_5\}$ = B-C, C-D, A-D
   Tree, A-D, D-C, C-B
9. $\{e_2, e_4, e_5\}$ = B-C, A-C, A-D
   Tree, A-C, C-B, A-D
10. $\{e_3, e_4, e_5\}$ = C-D, A-C, A-D
    Has a cycle (A-C-D-A)

Valid spanning trees- combinations 1, 3, 4, 5, 6, 7, 8, 9
That's 8 spanning trees.

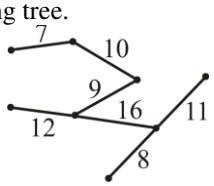
402. The cost of minimum cost spanning tree of the following graph is



(a) 72 (b) 73
(c) 74 (d) 75

RPSC C.S. 2016 (Paper-II)

Ans.(b) : Kruskal algorithm is a minimum spanning tree algorithm which finds an edge of the least possible weight that connects any one two trees in the forest. It is a greedy algorithm in graph theory as it finds a minimum spanning tree for a connected weighted graph adding increasing cost arcs at each step.
Minimum spanning tree.



Cost of minimum cost spanning tree = 7 + 8 + 9 + 10 + 11 + 12 + 16 = 73.

403. Choose an invalid property of a minimum spanning tree T, extracted from a connected graph G of n vertices:–
(a) T has n–1 edges (b) T has n vertices
(c) T is connected (d) T has n/2 edges

RPSC C.S. 2016 (Paper-II)

Ans.(d) : A minimum spanning tree (MST) of a connected graph G with n vertices has the following properties.
1. Number of Edges– An MST has n–1 edges.
2. Number of vertices– An MST has the same number of vertices as the original graph G, which is n.
3. Connectivity– An MST is connected.

Sarkari Resource

Invalid property–
a– T has n–1 edges- valid property of an MST
b– T has n vertices- valid property of an MST
c– T is connected- valid property of an MST
d– T has n/2 edges- This would not be a valid property for an MST of a graph with n vertices, especially considering the number of edges in an MST is determined to be n–1 not n/2.
So, the best answer is (d) T has n/2 edges.

404. The minimum number of edges in a spanning tree of a connected graph with "n" vertices is:
'n' शीर्षों वाले एक संयोजित ग्राफ के स्पैनिंग ट्री में न्यूनतम कितनी धाराएँ (edges) होती हैं?
(a) n (b) n – 1
(c) n + 1 (d) n(n–1)/2

Bihar STET C.S. 12.09.2023 (Shift-II)

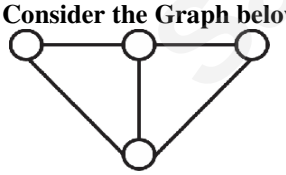
Ans. (b) :A spanning tree is a Subgraph that connects all the vertices of the original graph without any cycles. For any connected graph with n vertices, the spanning tree must have exactly n-1 edges to connect all vertices without forming a cycle.

405. How many edges does a spanning tree of a graph with N vertices have?/N शीर्षों वाले ग्राफ के फैले हुए वृक्ष में कितने किनारे होते हैं?
(a) N
(b) N – 1
(c) N(N –1)/2
(d) More than one of the above
(e) None of the above

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Ans. (b) : A spamming tree is a special type of graph that includes all the vertices of the original graph but has no cycles. for any tree with N vertices, the number of edges is always one less than the number of vertices that is N-1. This is because if we add more edges, it will form a cycle , and if we remove edges the graph will become disconnect . Therefore a spanning tree of a graph with N vertices will always have exactly N-1 edges.

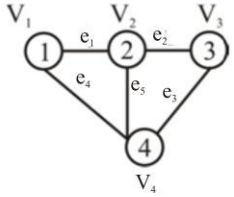
406. Consider the Graph below:



How many spanning trees can be found?
(a) 10 (b) 5
(c) 9 (d) 8

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Ans. (d) : If a graph is a complete graph with n vertices then total number of spanning trees is $n^{(n-2)}$.
Where n is the number of nodes in the graph given graph.



We know that
for a 3×3 matrix (3 rows and 3 column)

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The determinant is

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Matrix representation

	1	2	3	4
1	2	–1	0	–1
2	–1	3	–1	–1
3	0	–1	2	–1
4	–1	–1	–1	3

Then living first row and first columns

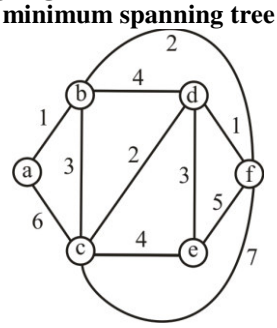
$$A = \det\begin{pmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{pmatrix}$$

Calculate the determinate is

$$|A| = 3\times5 - (-1)(-4) + (-1)\times3$$
$$= 15 - 4 - 3$$
$$= 8$$

So option (d) is correct.

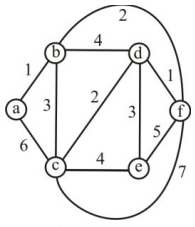
407. Given the graph below which one of the following edges cannot be added in that order to find a minimum spanning tree algorithm.



A. a-b **B. d-f**
C. b-f **D. d-c**
E. d-e
Choose the correct answer from the options given below :
(a) A, B, C, D, E (b) A, B, D, C, E
(c) B, A, C, E, D (d) B, A, D, C, E

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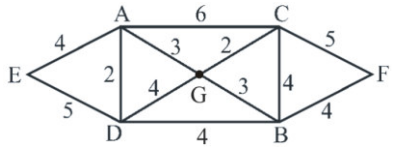
Ans. (c) : Given graph using minimum spanning tree algorithms.



d – f, a – b, b – f, d – e, d – c
So option (c) is correct answer.

Sarkari Resource

408. Consider the graph given below :

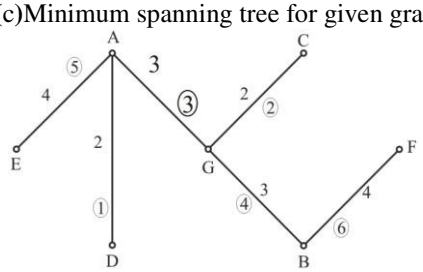


Use Kruskal's algorithm to find a minimal spanning tree for the graph. The list of edges of the tree in the order in which they are choosen it?

(a) AD, AE, AG, GC, GB, BF
(b) GC, GB, BF, GA, AD, AE
(c) GC, AD, GB, GA, BF, AE
(d) AD, AG, GC, AE, GB, BF

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Ans :(c)Minimum spanning tree for given graph:



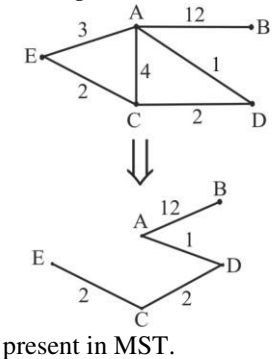
Kruskal's algorithm start with picking edge with smallest edge weight i.e. "AD = 2" or "GC = 2".
After that "AD = 2".
Since for 1st choice we choose "AD = 2".
So, order will GC, AD, GB, GA, BF, EA.
Note: There are more than 1 order possible since edge weights are repeated.

409. Let G be an undirected connected graph with distinct edge weight. Let E_{max} be the edge with maximum weight and E_{min} the edge with minimum weight. Which of the following statements is false?

(a) Every minimum spanning tree of G must contain E_{min}
(b) If E_{max} is in minimum spanning tree, then its removal must disconnect G.
(c) No minimum spanning tree contains E_{max}
(d) G has a unique minimum spanning tree

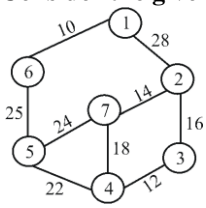
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Ans :(c)There may exist some spanning tree which consist of E_{max} in it & its removal must disconnect of the graph G.
It cannot always be the case that the edge with E_{max} should not be present in MST



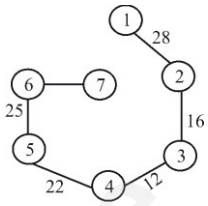
E_{max} is present in MST.

410. Consider the given graph

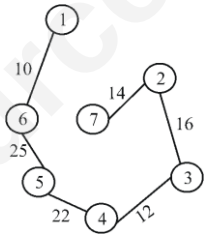


Its Minimum cost spanning tree is .....

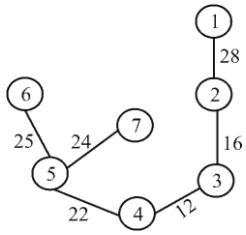
(a)



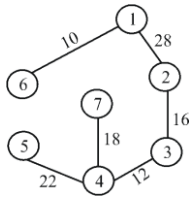
(b)



(c)

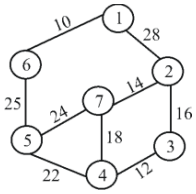


(d)

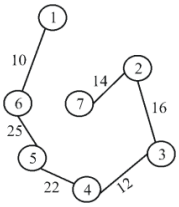


UGC NET C.S. June-2015 (Paper-II)

Ans.(b) : Using



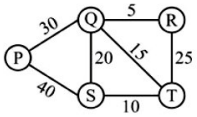
MST :



Option (b) = MST : 10 + 12 + 14 + 16 + 22 + 25 = 99
So the minimum is option (b), hence the answer is (b)

Sarkari Resource

411. Consider the following graph.
निम्नलिखित ग्राफ पर विचार करें।

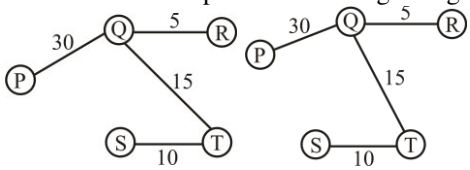


What is the weight of minimum spanning tree?
मिनिमम स्पैनिंग ट्री का वजन क्या होगा?

(a) 50 (b) 60
(c) 70 (d) 80

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : The weight of a Minimum Spanning Tree (MST) is the sum of the weights of the edges in the spanning tree that connects all the vertices of a graph with the minimum possible total edge weight.



The weight of minimum spanning tree is 60.

412. The number of different spanning trees in complete graph, K_4 and bipartite graph, $K_{2,2}$ have _____ and _____ respectively.

(a) 14, 14 (b) 16, 14
(c) 16, 4 (d) 14, 4

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) The number of spanning trees in complete graph with n vertices.
$= n^{n-2} = 4^{4-2} = 16$
A complete bipartite graph $k_{m,n}$ has m^{n-1} n^{m-1} spanning trees $= 2^1 . 2^1 = 4$.

(xvi) Single source shortest paths – Dijkstra's and Bellman Ford algorithms

413. Which of the following is the mostcommonly used data structure forimplementing Dijkstra's Algorithm ?
निम्नलिखित में से कौन-सा सबसे अधिक इस्तेमाल किया जाने वाला डेटा स्ट्रक्चर है जो दिज्कस्ट्रा के एल्गोरिथ्म को लागू करने के लिए है?

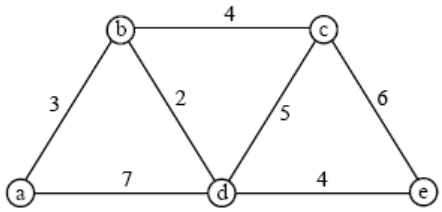
(a) Max priority queue/मैक्स प्रिऑरिटी क्यू
(b) Stack/स्टैक
(c) Circular queue/सर्कुलर क्यू
(d) Min priority queue/मिन प्रिऑरिटी क्यू

HTET PGT C.S. 2020

Ans. (d) : Dijkstra's algorithm is used to find the shortest path from a source node to all other nodes in graph with non–negative edge–weights. The algorithm requires efficienty selecting the node with the smallest tentative distance (current shortest known distance) at each step. A min priority queue is ideal for this because it allows.

- Efficient extraction of the node with the minimum distance (using extract–min)
- Efficient updating of priorities (using decrease – key) when a shorter path to a node is found.

414. Consider the following graph:



Find the shortest path using Dijkstra's algorithm.

(a) a – b – d – e (b) a – b – c – d
(c) a – c – d – e (d) a – b – c – e

CGPSC C.S. 2020 Set-A

Ans.(a): Finding the shortest path using Dijkstra's algorithm from vertex a to d, we follow following steps:

	a	b	c	d	e
	0	∞	∞	∞	∞
a	0	(3)	∞	7	∞
a, b	0	0	4	(2)	∞
a, b, d	0	0	4	0	4

a, b, d, e or a, b, d, c
from given option a - b - d - e is the shortest path.

415. Uniform-cost search expands the node n with the ...../यूनिफॉर्म-कॉस्ट सर्च n नोड को विस्तारित करता है जिसका ..... होता है:

(a) Lowest path cost/सबसे कम पथ लागत
(b) Heuristic cost/अनुमानी लागत
(c) Highest path cost/सबसे अधिक पथ लागत
(d) Average path cost/औसत पथ लागत

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : Uniform-Cost Search (UCS) always expands the node with the lowest path cost (g(n)) it works similarly toDijkstra's Algorithm and guarantees an optional Solution if all step coasts are positive.

416. In multistage graphs, what is the property of optimality?/मल्टीस्टेज ग्राफ्स में, ऑप्टिमलिटी की विशेषता क्या होती है?

(a) The shortest path between any two stages is the one with the fewest edges
किसी दो चरणों के बीच सबसे छोटा रास्ता वह होता है जिसमें सबसे कम धाराएँ होती हैं

(b) The shortest path between any two stages is the one with the lowest total cost
किसी दो चरणों के बीच सबसे छोटा रास्ता वह होता है जिसकी कुललागत सबसे कम होती है

(c) The shortest path between any two stages is the one with the highest total cost
किसी दो चरणों के बीच सबसे छोटा रास्ता वह होता है जिसकी कुललागत सबसे अधिक होती है

(d) The shortest path between any two stages is the one with the most vertices
किसी दो चरणों के बीच सबसे छोटा रास्ता वह होता है जिसमें सबसे अधिक शीर्ष(Vertices) होते हैं

Bihar STET C.S. 12.09.2023 (Shift-II)

Sarkari Resource

Ans. (b) : In a multistage graph, the property of optimality states that the best solution to a problem is made up of the best solutions to its subproblems. This means that when we look for the shortest path between two stages, the path chosen must be the one with the minimum total cast, not necessary the one with the fewest edges or the most vertices. Thus the optimal path is always the one that gives the lowest total cost form the starting stage to the destination.

417. Bellman Ford algorithm is an example for

बेलमैन फोर्ड एल्गोरिथम _____ के लिए एकउदाहरण है

(a) Linear Programming
(b) Greedy Algorithms
(c) Dynamic Programming
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

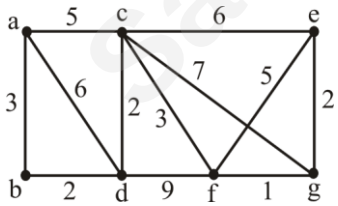
Ans.(c): The Bellman Ford algorithm is an example of Dynamic programming, as it solves the single source shortest path problem by breaking it down into smaller sub problems and iteratively improving the solution. It can handle graphs with negative weight edges and detects negative cycles.

418. When using Dijkstra's algorithm to find shortest path in a graph, which of the following statement is not true?

(a) It can find shortest path within the same graph data structure
(b) Every time a new node in visited. we choose the node with smallest known distance/cost (weight) to visit first
(c) Shortest path always passes through least number of vertices
(d) The graph needs to have a non-negative weight on every edge

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) :



Dijkstra's algorithm is used to find single source shortest path i.e. path between any two vertices in a graph. All edges must have non-negative weights and graph must be connected.

v	a	b	c	d	e	f	g
a	0_a	3_a	5_a	6_a	Inf	Inf	Inf
b	0_a	3_a	5_a	5_b	Inf	Inf	Inf
c	0_a	3_a	5_a	5_b	11_c	8_c	12_c
d	0_a	3_a	5_a	5_b	11_c	8_c	12_c
	0_a	3_a	5_a	5_b	11_c	8_c	9_f

So, we get the shortest distance from a to g. (a → b → c → d → f → g).

419. Which of the following algorithms solves the single-source shortest paths ?

(a) Prim's algorithm
(b) Floyd - Warshall algorithm
(c) Johnson's algorithm
(d) Dijkstra's algorithm

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : Dijkstra's algorithm to find the shortest path from a single source vertex to all other vertices.
1. Create a set spt set (shortest path tree set.
2. Assign Q distance value to all vertices in the input graph.
3. While spt set don't include all vertices.

420. Dijkstra's algorithm is based on

(a) Divide and conquer paradigm
(b) Dynamic programming
(c) Greedy Approach
(d) Backtracking paradigm

UGC NET C.S. June-2017 (Paper-III)

Ans :(c) Dijkstra always choose the closest vertex in V–S to add to set S where V = vertex set & S = set of vertices whose final path shortest path weights from the source have been obtained.
So, Dijkstra algorithm is a greedy algorithm.

421. Dijkstra algorithm, which solves the single-source shortest-paths problem, is a ....., and the Floyd-Warshall algorithm, which finds shortest paths between all pairs of vertices, is a.....

(a) Greedy algorithm, Divide-conquer algorithm
(b) Divide-conquer algorithm, Greedy algorithm
(c) Greedy algorithm, Dynamic programming algorithm
(d) Dynamic programming algorithm, Greedy algorithm

UGC NET C.S. December-2014 (Paper-III)

Ans :(c)Dijkstra algorithm, which solves the single source shortest paths problem, is a Greedy algorithm & the Floyd – Warshall algorithm, which finds shortest paths between all pair of vertices, is a dynamic programming algorithm.

(xvii) Dynamic Programming

422. Dynamic programming is a technique for solving problem with:

(a) Overlapped sub-problems
(b) Huge size sub-problems
(c) Small size sub-problems
(d) None of the above

CGPSC C.S. 2020 Set-A

Ans.(a):DP solves problems by reusing solutions of **overlapped sub-problems** to avoid repeated work.

423. Which property of dynamic programming is particularly useful for solving Optimal Binary Search Tree problems?

डायनेमिक प्रोग्रामिंग की कौन-सी विशेषता ऑप्टिमल बाइनरी सर्च ट्री समस्याओं को हल करने में विशेष रूप से उपयोगी होती है?

Sarkari Resource

(a) Optimal substructure/ऑप्टिमल सबस्ट्रक्चर
(b) Greedy choice property/ग्रीडी चॉइस प्रॉपर्टी
(c) Divide and Conquer/डिवाइड एंड कॉन्कर
(d) Linear programming/रैखिक प्रोग्रामिंग

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : Dynamic programming's optimal substructure property is key to solving optimal binary search tree problems. This property allows the problem to be broken down into smaller sub problems, solving each optimally, and then combining these solutions to form the overall optimal solution. By leveraging optimal substructure, dynamic programming efficiently constructs the most efficient binary search tree for a given set of keys.

424. Consider the following steps :
S_1 : Characterize the structure of an optimal solution
S_2 : Computer the value of an optimal solution in bottom-up fashion
Which of the step(s) is/are common to both dynamic programming and greedy algorithms?
(a) Only S_1
(b) Only S_2
(c) Both S_1 and S_2
(d) Neither S_1 nor S_2

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (a) : Both Dynamic Programming and greedy algorithms find optimal substructure in the problem but only dynamic programming uses the bottom up approach.

(xviii) All Pair Shortest Paths – Warshal's and Floyd's Algorithms

425. Which of the following is an all pair shortest path algorithm of graphs?
(a) Bellman-Ford algorithm
(b) Ford - Fulkerson algorithm
(c) Floyd- Warshall algorithm
(d) Dijkstra's algorithm

RPSC C.S. 2016 (Paper-II)

Ans.(c) : The Floyd-Warshall algorithm is an all pairs shortest path algorithm, meaning it computes the shorts paths between all pairs of vertices in a graph. It works for both directed sand undirected graphs and can handle negative edge weight (but not negative cycles).
Bellman-Ford algorithm– single source shortest path (works for one source vertex).
Ford-Fulkerson algorithm– Solves the maximum flow problem, not shortest paths.
Dijkstra's algorithm– single source shortest path (does not handle negative weights).
Thus, the Floyd-Warshall algorithm is the correct choice for finding all pairs shortest paths.

426. The Floyd-Warshall algorithm is a popular method for solving the All-Pair Shortest Paths problem. What is the time complexity of the Floyd-Warshallalgorithm for a graph with "n" vertices?
'n' शीर्षों वाले ग्राफ के लिए ऑल-पेयर शॉर्टेस्ट पाथ समस्या को हल करने के लिए फ्लॉयड-वॉरशाल एल्गोरिदम की समय जटिलता क्या है?
(a) O(n) (b) O(n log n)
(c) O(n^2) (d) O(n^3)

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : The Floyd-Warshall algorithm is used to find the shortest paths between all pairs of vertices in a graph. If works by considering each vertex as an intermediate point and checking whether going through that vertex gives a shorter path between any two vertices. To do this, the algorithm uses three nested loops over all vertices, comparing and updating the shortest distances. Because it examines all possible combination of starting vertex, eding vertex, and intermediate vertex, the total number of operations grows as n × n × n, where n is the number of vertices. Therefore, the time complexity of the Floyd-Warshall algorithm is 0 (n^3), making it suitable for graphs with a moderate number of vertices.

427. Which of the following algorithms solves the all-pairs shortest path problem?
निम्न में से कौन-सा एल्गोरिदम ऑल-पेयर्स शॉर्टेस्ट पाथ प्रॉब्लम को हल कर सकता है?
(a) Dijkstra's algorithm/डिज्कस्ट्रा एल्गोरिदम
(b) Floyd's algorithm/फ्लॉयड एल्गोरिदम
(c) Prim's algorithm/प्रिम एल्गोरिदम
(d) Warshall's algorithm/वॉरशालएल्गोरिदम

LT. Grade C.S. 2018

Ans.(b) :The floyd– Warshall algorithm also known as floyd's algorithm is an algorithm for finding shortest paths in a directed weighted graph with positive or negative or zero edge weights but with no negative cycles. It is used to solve the all - pairs shortest path problem.

428. Floyd–Warshall algorithm utilizes ..... to solve the all-pairs shortest paths problem on a directed graph in ..... time.
(a) Greedy algorithm, $\Theta(V^3)$
(b) Greedy algorithm, $\Theta \log n(V^2 lgn)$
(c) Dynamic programming, $\Theta(V^3)$
(d) Dynamic programming, $\Theta \log n(V^2 lgn)$

UGC NET C.S. December-2015 (Paper-III)

Ans :(c) The Floyd -Warshall algorithm compares all possible paths through the graph between each pair of vertices. It is able to do this with $\Theta(|V|^3)$ comparisons in a graph.

429. An all-pairs shortest-paths problem is efficiently solved using :
(a) Dijkstra' algorithm
(b) Bellman-Ford algorithm
(c) Kruskal algorithm
(d) Floyd-Warshall algorithm

UGC NET C.S. June-2015 (Paper-III)

Ans :(d)Floyd-Warshallalgorithm is used for all-pairs shortest paths problem efficiently with time complexity $O(n^3)$ using programming.

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(xix) Backtracking

430. Which of the following is not a backtracking algorithm?
निम्न में से कौन-सा बैकट्रैकिंग एल्गोरिदम नहीं है?

(a) Knight tour problem/नाइट टूर प्रॉब्लम
(b) N-queen problem/N-क्वीन प्रॉब्लम
(c) Towers of Hanoi/टावर्स ऑफ हनोई
(d) M-coloring problem/M-क्लरिंग प्रॉब्लम

LT. Grade C.S. 2018

Ans.(c) : The Tower of Hanoi is not a backtracking algorithm, but rather a recursion problem. Backtracking algorithms incrementally build a solution and abandon a path if it won't lead to a solution. The Tower of Hanoi however is a classic problem that's typically using recursive algorithms.

(xx) Branch and Bound with Examples such as Travelling Salesman Problem

431. Match List I With List II

List-I		List-II	
A.	Branch and bound	I.	Keeps track of all partial Paths which can be candidate for further exploration
B.	Steepest accent hill climbing	II.	Detects difference between current state and goal state
C.	Constraint satisfaction	III.	Discover problem state(s) that satisfy a set of constraints
D.	Means end analysis	IV.	Considers all moves from current state and selects best move

Choose the correct answer form the options given below:

(a) A-I, B-IV, C-III, D-II
(b) A-I, B-II, C-III, D-IV
(c) A-II, B-I, C-III, D-IV
(d) A-II, B-IV, C-III, D-I

UGC NET C.S. June-2020 (Paper-II)

Ans. (a): Branch-and-Bound:- Keeps track of all partial paths which can be a candidate for further exploration (Ex: A * search, BFS)
Steepest-accent hill Climbing:- Considers all moves from the current state and selects the best move (Ex. Hill-climbing algorithm)
Constraint satisfaction discovers problem state (s) that satisfy a set of constraints Ex (map coloring problem)
Means-end-analysis:- Detects the difference b/w the current state and the goal state Ex (means-end-analysis).

432. The travelling salesman problem can be solved in :

(a) Polynomial time using dynamic programming algorithm
(b) Polynomial time using branch-and-bound algorithm
(c) Exponential time using dynamic programming algorithm or branch-and-bound algorithm
(d) Polynomial time using backtracking algorithm

UGC NET C.S. June-2015 (Paper-III)

Ans :(c)It will be solved by using branch and bound technique. The travelling salesman problem is NP-complete problem. It can also be solved by- one of the earliest application of dynamic programming is the Held-Karp algorithm to solve TSP in time $O(n^2.2^n)$.

433. Which of the following statements is true for branch - and - bound search?

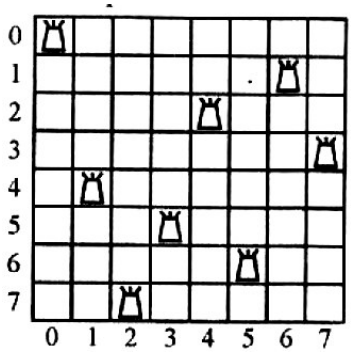
(a) Underestimates of remaining distance may cause deviation from optimal path
(b) Overestimates can't cause right path to be overlooked.
(c) Dynamic programming principle can be used to discard redundant partial paths.
(d) All of the above

UGC NET C.S. June-2015 (Paper-III)

Ans :(c)Branch & bound is an algorithm design paradigm for discrete & combinational optimization problems branch & bound algorithm consist of a systematic enumeration of candidate solution by means of state space search. A branch & bound is a dynamic programming principle can be used to discard radiant partial paths.

(xxi) N-Queen Problem

434. The diagram below represents a solution to the नीचे का आरेख ____ के एक हल को प्रदर्शित करता है ।



(a) Knights Tour.
(b) Eight Queens Problem.
(c) Kings Problem.
(d) Rooks Problem.

KVS PGT C.S. 2013

Sarkari Resource

Ans.(b) : The eight queens puzzle is the problem of placing eight chess queens on an 8×8 chessboard so that no two queens threaten each other thus a solution requires that no two queens share the same row, column or diagonal.
The eight queens puzzle is an example of the more general n queens problem of playing n non attacking queens on an nxn chessboard all natural numbers n with the exception of n = 2 and n = 3

(xxii) Algebraic computation

435. Consider the following Linear programming problem (LPP) :
Maximize $z = x_1 + x_2$
Subject to the constraints :
$x_1 + 2x_2 \leq 2000$
$x_1 + x_2 \leq 1500$
$x_2 \leq 600$
and $x_1, x_2 \geq 0$
The solution of the above LPP is :
(a) $x_1 = 750, x_2 = 750, z = 1500$
(b) $x_1 = 500, x_2 = 1000, z = 1500$
(c) $x_1 = 1000, x_2 = 500, z = 1500$
(d) $x_1 = 900, x_2 = 600, z = 1500$

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) : Graphical method can be used. On solving all the constraints, the peaks of the Common area are : A(0, 0) B(0, 600), C(1500, 0), D(1000, 500) z is maximum on all the points of the constraint $x_1 + x_2 \leq 1500$ (as slope of z is same as the constraint)
Now lets try to eliminate the options option 1 and 2 can be eliminated as they are violating 3rd constraint option 4 can be eliminated as it is violating 1st constraint.

436. Consider the following statements with respect to duality in LPP:
(A) The final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself.
(B) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.
(C) If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all.
Which of the statements is (are) correct?
(a) only (A) and (B) (b) only (A) and (C)
(c) only (B) and (C) (d) (A), (B) and (C)

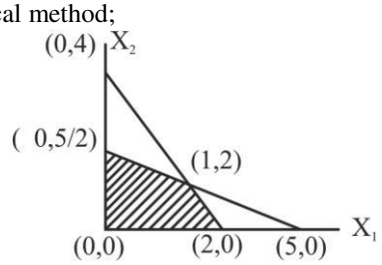
UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (d) : Duality in Linear programming problem (LPP):- It means a linear programming problem has another LPP which is derived from it original LPP which is derived from it.
Original LPP is known as primal and derived LPP is known as DUAL.
Some points about dual LPP.
(1) In final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself.
(2) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.
(3) If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all.
Example-We have a set of equations as:
Maximize $z = 10x_1 + 20x_2$
Subject to:
$3x_1 + 4x_2 <= 100$
$5x_1 + 6x_2 <= 150$
$x_1, x_2 >= 0$
Dual of this will be:
Minimize $z = 100y_1 + 150y_2$
Subject to : $3y_1 + 4y_2 >= 10$
$5y_1 + 6y_2 >= 20$
$y_1, y_2 <= 0$

437. Consider the following linear programming (LP):
Max, $z = 2x_1 + 3x_2$
Such that $2x_1 + x_2 \leq 4$
$x_1 + 2x_2 \leq 5$
$x_1, x_2 \geq 0$
The optimum value of the LP is
(a) 23
(b) 9.5
(c) 13
(d) 8

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) : Max $z = 2x_1 + 3x_2$
$2x_1 + x_2 \leq 4$
$x_1 + 2x_2 \leq 5$ $x_1, x_2 \geq 0$
Graphical method;



Draw lines by removing in equality z at
(1, 2) = 2 + 6 = 8
z at (2, 0) = 4 + 0 = 4
z at (0, 5/2) = 0 + 15/2 = 7.5 = 8 Ans.

438. The following LPP
Maximize $z = 100x_1 + 2x_2 + 5x_3$
Subject to
$14x_1 + x_2 - 6x_3 + 3x_4 = 7$
$32x_1 + x_2 - 12x_3 \leq 10$
$3x_1 - x_2 - x_3 \leq 0$
$x_1, x_2, x_3, x_4 \geq 0$
has
(a) Solution : $x_1 = 100, x_2 = 0, x_3 = 0$
(b) Unbounded solution
(c) No solution
(d) Solution : $x_1 = 50, x_2 = 70, x_3 = 60$

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) :$14x_1 + x_2 - 6x_3 + 3x_4 = 7$
$32x_1 + x_2 - 12x_3 \leq 10$
$3x_1 - x_2 - x_3 \leq 0$

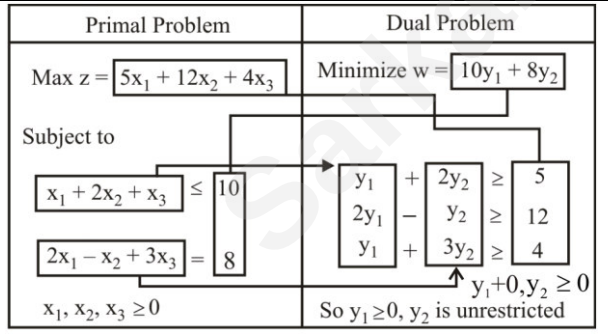
Sarkari Resource

According to this the unbounded feasible solution cannot be determined, instead there are infinite many solutions. It is not possible to solve the problem in which this situation occurs.

439. Consider the primal problem:
maximimze $z = 5x_1 + 12x_2 + 4x_3$
Subject to $x_1 + 2x_2 + x_3 = 10$
$2x_1 - x_2 + 3x_3 = 8$
$x_1, x_2, x_3 \geq 0$
its dual problem is
Minimize $\omega = 10y_1 + 8y_2$
Subject to $y_1 + 2y_2 \geq 5$
$2y_1 - y_2 \geq 12$
$y_1 + 3y_2 \geq 4$
Which of the following is correct?
(a) $y_1 \geq 0$, y_2 unrestricted
(b) $y_1 \geq 0$, $y_2 \geq 0$
(c) y_1 is unrestricted, $y_2 \geq 0$
(d) y_1 is unrestricted, y_2 restricted

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Ans. (a) : Given that
Max $Z = 5x_1 + 12x_2 + 4x_3$
Subject to $x_1 + 2x_2 + x_3 = 10$
$2x_1 - x_2 + 3x_3 = 8$
$x_1 . x_2 . x_3 \geq 0$
Its dual problem is
Minimize $w = 10y_1 + 8y_2$
subject to
$y_1 + 2y_2 \geq 5$
$2y_1 - y_2 \geq 12$
$y_1 + 3y_2 \geq 4$



440. There are M points on one straight line AB and n points on another straight line AC none of them being A. How many triangles can be formed with these points as vertices?

(a) $mn\,(m + n - 2)$ (b) $\frac{1}{2}mn(m+n-2)$

(c) $\frac{1}{2}mn(m+n-1)$ (d) $mn\,(m + n - 1)$

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Ans. (b) : Given,
There are m points on one straight line AB and n points on another straight line AC none of them being A.
So, the total number of triangles that can be formed is

$$= {}^{m}C_1 \times {}^{n}C_2 + {}^{m}C_2 \times {}^{n}C_1$$

$$= \frac{m!}{1!(m-1)!} \times \frac{n!}{2!(n-2)!} + \frac{m!}{2!(m-2)!} \times \frac{n!}{1!(n-1)!}$$

$$= m \times \frac{1}{2}n(n-1) + \frac{1}{2}m(m-1) \times n$$

$$= \frac{1}{2}mn(m+n-2)$$

441. Consider the following linear programming problem:
Max, $z = 0.50x_2 - 0.10x_1$
Subject to the constraints
$2x_1 + 5x_2 \leq 80$
$x_1 + x_2 \leq 20$
and $x_1, x_2 \geq 0$
The total maximum profit (z) for the above problem is:
(a) 6 (b) 8
(c) 10 (d) 12

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Ans : (b) when,
$x_2 = 16$ and $x_2 = 0$
then Max $z = 8$

442. A linear programming problem (LPP) is as follows :
Min z = 30x-18y, subject to the constraints: 3x + 4y ≤ 60, 5x – 3y ≥ 20 and x, y ≥ 0. In this feasible region, the solution of LPP is/are
A. (4, 0) B. (2, 0) C. (7, 5) D. (0, 15) E. (8, 5)
Choose the correct answer from the options given below :
(a) A and C only (b) B only
(c) E only (d) D only

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Ans. (a) : The feasible region is determined by the constraints $3x + 4y \leq 60$, $5x - 3y \geq 20$ and $x, y \geq 0$.
Checking each given point.
A (4, 0) – Satisfies all constraint.
z = 30 (4) –18(0) = 120.
B (2, 0) – Fails 5x – 3y ≥ 20. Not feasible.
C (7, 5) – Satisfies all constraints.
Z = 30(7) –18(5) = 120
D (0, 15) – Fails 5x – 3y ≥ 20. Not feasible.
E. (8, 5) – Fails 5x – 3y ≥ 20. Not feasible.
Point A and C yield the minimum z = 120.
The edge connecting these points lies entirely within the feasible region, making both optional.

443. Which one of the following is not a basic solution of system of linear equation.
$x_1 + 2x_2 + x_3 = 4$
$2x_1 + x_2 + 5x_3 = 5$
(a) $x_1 = -1$, $x_2 = 2$, $x_3 = 1$
(b) $x_1 = 2$, $x_2 = 1$
(c) $x_1 = 5$, $x_3 = -1$
(d) $x_2 = 5/3$, $x_3 = 2/3$

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Ans. (a) : Describe the Step by Step to determine which option is not a basic solution.
Step-1 System of Equations- The given system of linear equations is-

$$x_1 - 2x_2 + x_3 = 4$$
$$2x_1 + x_2 + 5x_3 = 5$$

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Step-2 Basic Solutions- A basic solution is found by setting any two variables to zero and solving for the remaining variable. The number of variables should match the number of equations.
we have three variables (x_1, x_2, x_3) and two equation's, So a basic solution is obtained by choosing two variables are zero and solving for the third.
Step-3 verify the given solution-
option (a)- $x_1 = -1, x_2 = 2, x_3 = 1$
Substituting in the first equation.
$(-1) -2 (2) + (1) = -1- 4 + 1 = -4 \neq 4$
option (b) - for first equation
$x_1 = 2 , x_2 = 1$
The system has three variables, but this options only specifies two values. If $x_3 = 0$ substituting in-
$2 - 2 (1) + 0 = 2 - 2 = 0 \neq 4$
option (c)- $x_1 = 5/3, x_3 = -1$
if $x_2 = 0$, substituting
first equation-
$(5/3) - 2(0) + (-1) = 5/3 - 1 = 2/3 \neq 4$
option (d)- $x_2 = 5/3, x_3 = 2/3$
If $x_1 = 0$, Substituting
first equation
$0 - 2 (5/3) + (2/3) = -10/3 + 2/3 = -8/3 \neq 4$
Step 4 - from our calculations, option (a) is not a basic solution as it does not satisfy the system.

444. What is the probability that a randomly selected bit string of length 10 is a palindrome?

(a) $\frac{1}{64}$ (b) $\frac{1}{32}$
(c) $\frac{1}{8}$ (d) $\frac{1}{4}$

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Ans :(b) For palindrome in even length palindrome half length is timed and rest is repeated. So, in 10 bit palindrome, we have 5 position to be filled with 2 choices each i.e. 2^5 choices for first half and 2^5 choices for second half.

0/1	0/1	0/1	0/1	0/1	Fixed	Fixed	Fixed	Fixed	Fixed

Probability = favorable

$$= \frac{2^5}{2^{10}} \Rightarrow \frac{1}{2^5} \Rightarrow \frac{1}{32}$$

445. A discrete random variable X has the following probability distribution.

X	1	2	3	4	5	6	7
P(X)	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

What is the value of K?
न एक डिस्क्रीट रैंडम वेरिएबल X में निम्नलिखित प्रायिकता वितरण है।

X	1	2	3	4	5	6	7
P(X)	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

K का मान क्या है?

(a) $\frac{1}{30}$ (b) $\frac{1}{40}$
(c) $\frac{1}{10}$ (d) $\frac{1}{20}$

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Probability distribution of given data

X	1	2	3	4	5	6	7
P(X)	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

According to probability sum rule
$K + 2 K + 2K + 3K + K^2 + 2K^2 + (7K^2 + K) = 1$
$9K + 10 K^2 = 1$
Standard Form
$10K^2 + 9K - 1 = 0$
$10K^2 + 10K - K - 1 = 0$
$10K (K + 1) - 1 (K + 1) = 0$
$(10K - 1) (K + 1) = 0$
$10K - 1 = 0 \ \& \ (K+1) = 0$
$10K = 1; \quad K \neq 1$ (invalid)
$K = \frac{1}{10}$

446. What is the value of f(5) by newton's backward difference form of interpolating polynomial for the data?

x	2	4	6	8
f(x)	5	8	12	16

डेटा के लिए इंटरपोलिंग बहुपद के न्यूटन के बैकवर्ड डिफरेंस फॉर्म द्वारा f(5) का मान क्या है?

x	2	4	6	8
f(x)	5	8	12	16

(a) 13.125 (b) 8.0625
(c) 10.0625 (d) 11.125

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Using newton's backward difference

$$P_n(x) = f(x_n) + p\nabla f(x_n) + \frac{p(p+1)}{2!}\nabla^2 f(x_n)....$$

$$f(5) = f(8) + p\nabla f(8) + \frac{p(p+1)}{2}\nabla^2 f(8).....$$

$$f(5) = 16 + (-1.5)(4) + \frac{(-1.5)(-1.5+1)}{2} + \frac{(-1.5)(-0.5)(0.5)}{6} \times (-1)$$

$$f(5) = 16 + (-1.5)(4) + \frac{(-1.5)(-0.5)}{2}(0) + \frac{(-1.5)(-0.5)(0.5)}{6}(-1)$$

$$f(5) = 16 - 6 + 0 + \frac{0.375}{6}(-1)$$

$f(5) = 10 - 0.00625$
$f(5) = 9.9375$
The value of f(5) is 9.9375 The closest option is (c) 10.0625 but the calculated value is 10.0625.

447. A computer program select an integer in the set {k : 1≤ k ≤10,00,000} at random and prints out the result. This process is repeated 1 million times. What is the probability that the value k = 1 appears in the printout at least once?

(a) 0.5 (b) 0.704
(c) 0.632121 (d) 0.68

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Ans :(c) Because, probability of k = 1 is 1/1000000 = 10^{-6}
Probability of k! = 1 is $1 - 10^{-6} = 0.999$
Probability that k = 1 is never printed in all 106 print outs = 0.999 × 0.999 × ..... 0.999 (10^6 time)
= 0.99910^6
Probability that 1 is printed at least once = 1 – probability that 1 is never printed
⇒1 – 0.999^10^6
⇒0.6321

448. If we define the functions f, g and h that map R into R by :

$f(x) = x^4$, $g(x) = \sqrt{x^2+1}$, $h(x) = r^2 + 72$, then the value of the composite functions ho(gof) and (hog) of are given as

(a) $x^8 - 71$ and $x^8 - 71$
(b) $x^8 - 73$ and $x^8 - 73$
(c) $x^8 + 71$ and $x^8 + 71$
(d) $x^8 + 73$ and $x^8 + 73$

UGC NET C.S. December-2014 (Paper-II)

Ans :(d)f = x^4

$$gof = \sqrt{(x^8+1)}$$

ho(gof) = x^8 + 1 + 72 = x^8 + 73 // option d hold this!
ho(gof) = x^8 + 73 = x^8 + 73
So the answer is (d)

449. Find the equation of the circle $x^2 + y^2 = 1$ in terms of x'y' coordinates, assuming that the xy coordinate system results from a scaling of 3 units in the x' direction and 4 units in the y' direction.

(a) $3(x')^2 + 4(y')^2 = 1$
(b) $\left(\frac{x'}{3}\right)^2 + \left(\frac{y'}{4}\right)^2 = 1$
(c) $(3x')^2 + (4y')^2 = 1$
(d) $\frac{1}{3}(x')^2 + \frac{1}{4}(y')^2 = 1$

UGC NET C.S. November-2017 (Paper-III)

Ans :(b)Let f[x, y] = = 0 be the equation for a curve in rectangular coordinates.
To scale the curve by s, the new equation would be
f[x/s, y/s] = = 0
Let f[θ, r] = = 0 be the equation for a curve in polar coordinate. To scale it by s, the new formula is f[r/s, θ] = = 0
Replace x to x′/3 and y to y′/4, and substitute the value in the $x^2 + y^2 = 1$.

450. Find the normalization transformation that maps a window whose lower left corner is at (11) and upper right corner is at (3, 5) onto a view port that is the entire normalized device screen

(a) $\begin{pmatrix} \frac{1}{2} & 0 & \frac{-1}{2} \\ 0 & \frac{1}{4} & \frac{-1}{4} \\ 0 & 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{-1}{4} & \frac{1}{4} \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} \frac{1}{2} & 0 & \frac{-1}{2} \\ 0 & \frac{1}{4} & \frac{1}{4} \\ 1 & 0 & 0 \end{pmatrix}$ (d) $\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{-1}{4} \\ 1 & 0 & 0 \end{pmatrix}$

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Ans :(a) The widow parameters are $w_{xmin} = 1$, $w_{ymin} = 1$, $w_{xmax} = 3$, $w_{ymax} = 5$
view port parameters are $v_{xmin} = 0$, $v_{ymin} = 0$, $v_{xmax} = 1$, $v_{ymax} = 1$
hence $s_x = 1/2$, $s_y = 1/4$
So N using =

$$\begin{pmatrix} 1 & 0 & V_{x\min} \\ 0 & 1 & V_{y\max} \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} s_y & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -w_{x\min} \\ 0 & 1 & -w_{y\min} \\ 0 & 0 & 1 \end{pmatrix}$$

$$N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

First, multiply the second and third matrices

$$\begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1/2\times2+0+0 & 1/2\times0+0+0 & (1/2)\times(-1)+0+0 \\ 0+1/4\times1+0 & 0+1/4\times0+0 & 0+(1/4)\times(-1)+0 \\ 0+0+0 & 0+0+0 & 0+0+1\times1 \end{pmatrix}$$

$$= \begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 1/4 & -1/4 \\ 0 & 0 & 1 \end{pmatrix}$$

Now, multiply this result by the firs matrix

$$N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 1/4 & -1/4 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 1/4 & -1/4 \\ 0 & 0 & 1 \end{pmatrix}$$

The transformation matrix N is

$$N = \begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 1/4 & -1/4 \\ 0 & 0 & 1 \end{pmatrix}$$

451. How many multiples of 6 are there between the following pairs of numbers?
0 and 100 and–6 and 34

(a) 16 and 6 (b) 17 and 6
(c) 17 and 7 (d) 16 and 7

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Ans :(a)
(i) Number of multiples of 6 between 1 to $100 = \left\lfloor \frac{100}{6} \right\rfloor = 16$. Since range start from 0 and 0 is multiple of 6.

(ii) Number of multiples of 6 between 0 to $= \left\lfloor \frac{34}{6} \right\rfloor + 1(\text{for } 0) = 5 + 1 = 6.$

452. What is the probability that a positive integer selected at random from the set of positive integer not exceeding 100 is divisible by either 2 or 5?

(a) 10/5 (b) 3/5
(c) 2/5 (d) 1/5

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Ans. (b) : Number of positive integers divisible by 2 in 100.
Number of positive integers divisible by 5 in 100.
So, 100/5 = 20
Number of positive integers divisible by both 2 and 5 in 100.
So, 100/10 = 10
Therefore, the total number of positive integers divisible by either 2 or 5 in 100.

$$50 + 20 - 10 = 60$$

So, the probability is the number of favorable outcomes (60) divided by the total number of outcomes (100).

So, $\frac{60}{100} = \frac{3}{5}$

453. How many integral solutions are there to $x + y + z + w = 29$, where $x \geq 1, y \geq 2, z \geq 3$ and $w \geq 0$?

(a) 2400 (b) 2600
(c) 2800 (d) 3000

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Ans. (b) : Let's first adjust the restrictions to simplify the problem.
Since, $x \geq 1$, $y \geq 2$ and $z \geq 3$.
we can introduce new variables-

$$x' = x - 1$$
$$y' = y - 2$$
$$z' = z - 3$$

The equation becomes-

$$x' + 1 + y' + 2 + z' + 3 + w = 29$$
$$x' + y' + z' + w = 23$$

Now, to find the number of integral solutions-considering x', y', z' and w as indistinguishable stars and the 3 separating bars as the separation between the variables.
We a have a total of [23 + 4–1] position s to place the bars,
Thus, the number of integral suctioned is

$$= {}^{23+4-1}C_{4-1}$$
$$= {}^{26}C_3 = \frac{26!}{3!(26-3)!}$$
$$= 2600$$

454. The coefficients of $x^2y^3z^2$ in the expansion of $(x+y+z)^7$ is

(a) C(7, 3) * C(5, 2) (b) C(7, 2) * C(5, 3)
(c) C(7, 3) * C(7, 2) (d) C(5, 3) * C(5, 2)

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Ans. (b) :
General term of $(x+y+z)^7$

$$= \frac{7!}{p!q!r!} x^p \times y^q \times z^r$$

So coefficient of $x^2y^3z^2$

$$= \frac{7!}{2!3!2!} \times 1^2.1^3.1^2 = \frac{7!}{2!3!2!} = 210$$

So option (b) is correct

455. How many ways are there to pack six copies of the same book into four identical boxes. Where a box can contain as many as six books?
एक बक्से में छह किताबें तक रखी जा सकती हैं। एक ही किताब की छह प्रतियों को चार समान बक्सों में पैक करने के कितने तरीके हो सकते हैं?

(a) 4 (b) 6
(c) 7 (d) 9

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Ans. (d) : Here there are six copies of the same book into four.
Identical boxes (Same box). We will enumerate all ways to pack the books for each way to pack the books we will list the number of books in the box with the largest numbers of books, followed by the numbers of books in each box containing at least one book, in order to decreasing the number of books in a box.

6, 0, 0, 0
5, 1, 0, 0
4, 2, 0, 0
4, 1, 1, 0
3, 3, 0, 0
3, 2, 1, 0
3, 1, 1, 1
2, 2, 2, 0
2, 2, 1, 1

So, the number of way is 9.

(xxiii) String Matching

456. Which of the following strings would match the regular expression : p+[3 – 5]*[xyz]?

I. p443y **II. p6y**
III. 3xyz **IV. p35z**
V. p353535x **VI. ppp5**

(a) I, III and VI only (b) IV, V and VI only
(c) II, IV and V only (d) I, IV and V only

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Ans :(d)P + [3 – 5]* [xyz] = p + (3 + 4 + 5)*(x+y+z)
Since, P has to be present as the first character of each string
(I) p4434→ matched
(II) p6y→ not matched since 6 cannot used
(III) 3xyz → not matched string must start with p
(IV) p35z → matched
(V) p353535X → matched
(VI) ppp5→ not matched strings must end with x, or y, or z

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(xxiv) Theory of NP-Completeness

457. Circuit satisfiability problem: Given a Boolean combinatorial circuit composed of AND. OR and NOT gates, is it satisfiable? A one output Boolean combinatorial circuit is satisfiable means for the given inputs the output will be 1.

A. The circuit satisfiability problem belongs to class NP
B. The circuit satisfiability problem is at least as hard as any language in NP
C. The circuit satisfiability is NP - Complete
D. The size of the circuit is $\Theta(K^2 + 1)$
E. If P ≠ NP, this situation would contradict the NP - Completeness of the problem.

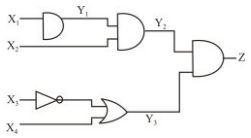
Choose the correct answer from the options given below :

(a) A, C, D Only (b) B, D, E Only
(c) A, B, C Only (d) B, C, D Only

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Ans. (c) :

$Y_1 = X_2 \cap X_3$
$Y_2 = Y_1 \cap X_3$
$Y_3 = X_3 \cup X_4$
$\therefore f = Y_1 \cap Y_2 \cap Y_3 \cap Z$



Therefore the circuit satisfiability is NP – Complete

ex. Circuit satisfiability problem

An instance of the problem is an input specified to the problem. An instance of the problem is a Boolean formula. Since NP- Complete problem is a problem which is both NP and NP-Hard, the proof or statement that a problem is NP-Complete consists of two parts :

1→The problem itself is in NP class.

2→All other problems in NP class can be polynomial-time reducible so that.

If the 2nd conditions is only satisfied then the problem is called NP-Hard. So, option (c) is correct Answer.

458. Choose the correct statement (s)

A. A problem which is NP-Complete will have the property that it can be solved in polynomial time if all other NP-complete problems can also be solved in polynomial time.
B. All NP-complete problem are NP-hard problems.
C. If an NP-hard problem can be solved in polynomial time, then all NP-complete problem can be solved in polynomial time, then all NP-complete problem can be solved in polynomial time
D. All NP-hard-problems are not NP-complete.

Choose the correct answer from the options given below :

(A) एक समस्या जो NP-पूर्ण है, उसमें यह गुण होगा की इसे बहुपद समय में हल किया जा सकता है यदि अन्य सभी NP-पूर्ण समस्याओं को भी बहुपद समय में हल किया जा सकता है।
(B) सभी NP-पूर्ण समस्याएँ NP-हार्ड समस्याएँ है।
(c) यदि एक NP-हार्ड समस्या को बहुपद समय में हल किया जा सकता है, तो सभी NP-पूर्ण समस्या को बहुपद समय मे हल किया जा सकता है, फिर सभी NP-पूर्ण समस्या को बहुपद समय मे हल किया जा सकता है।
(D). सभी NP-हार्ड समस्याएँ NP-पूर्ण नही है।

(a) A, C only (b) B, D only
(c) A, B, C only (d) A, B, C, D

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Ans. (d) :

(A) If any NP-complete problem can be solved in problems can be solved in polynomial time (Since NP-Complete problems are reducible to each other in polynomial time).
(B) NP-Complete problems are both in NP and NP-hard meaning they are at least as hard as any problem in NP.
(C) NP-hard problem (which includes NP-Complete problems) can be solved in polynomial time then all NP-Complete Problems can be solved in polynomial time, implying P = NP
(D) Not all NP-Hard problems are NP- complete some NP-hard problems (like the halting problem) are even harder than NP-Complete problems however, some NP-Hard problems are NP-Complete.

Therefore, is the correct answer: ABCD.

459. Consider a language L for which there exists a Turning Machine (TM) T that accepts every word in L and either rejects of loops for every word that is not in L. The language L is:

(a) NP hard (b) NP-complete
(c) Recursive (d) Recursively enumerable

RPSC C.S. 2016 (Paper-II)

Ans. (d) : We are given a Turing Machine T that-Accept every word in L

For words not in L, it either rejects or loops forever

A language is recursively enumerable if there exists a Turing Machine that accepts all strings in the language and may either reject or loop forever on strings not in the language.

Since the TM accepts all strings in L and may loop on others, L is recursively enumerable not necessarily recursive.

460. Which of the following problems is not an NP-complete class problem?

(a) 3-CNF-SAT Problem
(b) Vertex cover problem
(c) Depth - First - search of graph
(d) Clique problem

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Ans.(c) : NP-complete problems– These are problems where a solution can be verified in polynomial time, but finding the solution itself is believed to be difficult and may take exponentially long.

Depth-First Search (DFS)– This is a graph Traversal algorithm that has a polynomial time complexity. Given a graph, DFS can explore all vertices and edges in a systematic manner. Since the verification of a solution can also be done in polynomial time, Depth-First Search is considered to be in the complexity class P (Polynomial time) and not NP-complete.

Vertex Cover Problem– This problem asks for a set of vertices in a graph such that every edge is incident to at least one vertex in the set. It has been proven to be NP-complete.

3-CNF-SAT Problem– This is a variant of the satisfiability problem (SAT), which is a known NP-complete problem. 3CNF-SAT involves finding a set of truth assignments to variables that make a given Boolean formula true, where each clause consists of at most three literals.

So, the correct option is (c).

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461. Let L1 is NP - complete problem, another problem L2 will also be an NP- complete problem if:-

(a) L2 ∈ NP and L1 is reducible to L2 in polynomial time
(b) L2 ∉ NP and L1 is reducible to L2 in polynomial time
(c) L2 ∈ NP and L2 is reducible to L1 in polynomial time
(d) L2 ∉ NP and L2 is reducible to L1 in polynomial time

RPSC C.S. 2016 (Paper-II)

Ans.(a) :L1 be an NP-complete problem
L2 be another problem.
To prove that L2 is also NP-complete, two conditions must be met.
1. L2∈ NP– That is, L2 is in the class NP (it can be verified in polynomial time).
2. L1≤p L2–That is, L1 is polynomial time reducible to L2
This means that if we can solve L2 efficiently (in polynomial time) then we can also solve L1 efficiently which implies L2 is at least as hard as L1.
Hence, option (a) is correct.

462. Which of the following is NOT an NP Completeproblem?/निम्नलिखित में से कौन-सी NP-पूर्ण समस्या नहीं है?

(a) Traveling Salesman Problem
ट्रैवलिंग सेल्समैन समस्या
(b) Boolean Satisfiability Problem
बूलियन संतुष्टि समस्या
(c) Shortest Path Problem/लघुत्तम पथ समस्या
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : The travelling salesman problem and the Boolean satisfiability problem (SAT) are both well-known NP-complete problems, meaning they are very hard to solve in polynomial time. However, the shortest path problem is different because it can be solved efficiently using algorithms like dijkstara's or Bellman-ford in polynomial time. Therefore, the shortest path problem is not NP-complete, while the others are.

463. A. If some NP-complete problem P is in P that P = NP
B. TSP is in NP C. SAT is in NP
D. Hamilton circuit problem is not NP- complete
Choose the correct answer from the options given below :

(a) A, B and C only (b) B, C and D only
(c) C, D and A only (d) D, A and B only

UGC NET C.S. June-2023

Ans. (a) : If some NP-complete problem P is in P that P = NP. It simply means that this particular NP-complete problem has a polynomial-time algorithm for a single NP-complete problem, then that would mean all NP problems have polynomial-time solutions, i.e. P would equal NP. This is the definition of NP-completeness.
The Traveling salesman problem (TSP) is indeed in NP. This means that given a proposed solution, verify in polynomial time whether it's a valid solution to the problem. The Boolean satisfiability problem (SAT) is in NP. This means that given a proposed solution to a Boolean formula, verify in polynomial time whether is satisfies the formula. However, Similar to TSP whether SAT is in P.
Hamiltonian cycle if graph G contains a Hamiltonian path. Therefore, any instance of the Hamiltonian cycle problem can be reduced to an instance of the Hamiltonian path problem. Thus, the Hamiltonian cycle is NP- Hard. The Hamiltonian cycle is both, a NP. problem and NP-Hard. Therefore, it is a NP-complete problem.
So, the statement A, B and C are correct and statement D is incorrect.

464. A basic feasible solution of a linear programming problem is said to be ..... if at least one of the basic variable is zero.

(a) degenerate (b) non-degenerate
(c) infeasible (d) unbounded

UGC NET C.S. December-2015 (Paper-III)

Ans :(a) Basic feasible solution : A feasible solution to LP problem which is also the basic solution is called the "basic feasible solution". Basic feasible solution are of two types.
Degenerate : A basic feasible solution is called degenerate if value of at least one basic variable is zero.
So answer is option (a)

465. We can show that the clique problem is NP-hard by proving that :

(a) CLIQUE ≤ P 3–CNF_SAT
(b) CLIQUE ≤ P VERTEX_COVER
(c) CLIQUE ≤ P SUBSET_SUM
(d) None of the above

UGC NET C.S. December-2014 (Paper-III)

Ans :(d) **Clique :** A clique in graph G is a complete graph; that is, it is a subset S of the vertices such that every two vertices is S are connected by an edge in G the decision problem is a NP complete.
3 SAT : SAT asks whether the variable of a given Boolean formula can be consistently replaced by the values TRUE or FALSE is such way that the formula evaluates to TRUE SAT is one of the first problems that was proven to be NP-complete & vertex problem & subset sum are NP-complete problem. So option (a, b, c) match with rule (3, 4)
So the correct answer is (d)

466. Which of the following is Not a NP Complete problem?
निम्नलिखित में से कौन एक NPकम्पलीट प्रॉब्लम नहीं है?

(a) The Clique problem/दा क्लिक प्रॉब्लम
(b) Hamiltonian-Cycle problem
हैमिल्टोनियन-साइकिल प्रॉब्लम
(c) The Travelling Salesman Problem
दा ट्रैवेलिंग सेल्समेन प्रॉब्लम
(d) Shortest Path Problem/शॉर्टेस्ट पाथ प्रॉब्लम

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : Shorted Path Problem is not a NP complete problem. The shortest path problem, such as finding the shortest path between two vertices in a graph, can be solved in polynomial time using algorithms, like Dijkstra's or Bellman Ford.

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04. Problem Solving through C-Programming

(i) Basic Programming Concepts

1. An array is
एक ऐरे (सारणी) होता है

(a) Probably the most widely used data structure.
सम्भवतः सर्वाधिक प्रयुक्त डेटा संरचना

(b) A homogeneous structure.
एक समांगी संरचना

(c) A random access structure.
एक यादृच्छिक पहुँच संरचना

(d) All of these/यह सभी

KVS PGT C.S. 2013

Ans. (d) : An array is the most widely used data structure. A homogeneous structure and random access structure homogeneous means they store elements of the Same type.

2. What is the main function of the command interpreter?
कमांड इंटरप्रेटर का मुख्य कार्य क्या है?

(a) To provide the interface between the API and application program

(b) To handle the files in the operating system

(c) To get and execute the next userspecified command

(d) More than one of the above

(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c): Command interpreter acts as an intermediary between the user and the Operating System, allowing users to interact with the system by entering commands. The common interpreter also known as the shell, is a program that reads commands from the user or from a script and execute them. Its main formation is to:
Read the command entered by the user.
Interpret the command and determine what action to take.
Execute the command by interacting with the Operating System.
So, the correct option is (c) to get and execute the next user specified command.

3. The programming of first generation computers was done in_____language.
I. Machine
II. High level
प्रथम पीढ़ी के कंप्यूटरों की प्रोग्रामिंग_____भाषा में की गई थी।
I मशीन
II हाई लेवल

(a) Only I/केवल I

(b) Only II/केवल II

(c) Both I and II/ I तथा II दोनों

(d) Neither I nor II/ना ही I ना ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : First-genration computers (1940-1950) were programmed using machine language (binary code). High-level languages come later, starting with the second generation of computers.

4. Which of the following is false regarding the machine language?
निम्नलिखित में से कौन सा मशीनी भाषा के संदर्भ में गलत है?

(a) It is machine dependent/यह मशीन पर निर्भर है

(b) It easier to modify/इसे संशोधित करना आसान है

(c) It is made up of 1's and 0's
यह 1 और 0 से बनी है

(d) It is difficult to program
इसे प्रोग्राम करना मुश्किल है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : The option "It is easier to modify" is false regarding machine language machine language consist of binary code that is directly executed by the computers CPU. It is the lowest level of programming language and is difficult to read, write and modify compared to higher-level programming languages.

5. What does the following C statement mean?
निम्नलिखित C कथन का क्या अर्थ है?

scanf ("%4s', str);

(a) Read exactly 4 charaters from console
कॉन्सोल से ठीक 4 कैरेक्टर पढ़ेगा

(b) Read maximum 4 characters from console
क़ॉन्सोल से अधिकतम 4 कैरेक्टर पढ़ेगा

(c) Read a string in multiples of 4
4 के गुणज में एक स्ट्रिंग पढ़ेगा

(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans, (b) : The C statement 'Scanf ("%4s",str);' means that the function will read a maximum of 4 characters from the console input and store them in the 'str' array. This includes any spaces and the null terminator, which signifies the end of the string. Therefore, the correct answer is option (b).

6. Which of the following statements (s) regarding a linker software is/are true?

I. A function of a linker is to combine several object modules into a single load module.

II. A function of a linker is to replace absolute references in an object module by symbolic references to locations in other modules

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(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) It is saying that linker replace absolute references in an object which is wrong linker always generate relocatable address.

7. Machine Level Language is a/an
(a) Assembly Language
(b) Low Level Language
(c) High Level Language
(d) Translating language

UGC NET C.S. December-2022

Ans. (b) : Machine Level Language: Machine language is a low-level language made up of binary numbers or bits that a computer can understand. It is also known as machine code or object code and it is extremely tough to comprehend.
The only language that the computer understands is machine language.

8. The structure or format of data is called
डेटा की संरचना या प्रारूप को कहा जाता है
(a) Syntax
(b) Semantics
(c) Struct
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : Syntax refers to the structure and format of data or instructions. It determines how data should be written, organized, and presented so that it can be understood by the system.

9. A source program is the program Written in…… language.
Source program……भाषा में लिखा होता है?
(a) English (b) Machine language
(c) High-Level (d) Symbolic

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : A source program is a program written by a programmer. It is written in a high-level language such as C, C++, Java, Python, etc.
These languages are easily read and understood by humans. A source program is converted into machine language (0s and 1s) using a compiler or interpreter, so that the computer can understand it.

10. A variable must have……
वैरिएबल में ……अवश्य होना चाहिए।
(a) Sign bit (b) Storage location
(c) Data type (d) all of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : In an program, a variable is a named location where data is stored. A variable must have the following sign bit:
This indicates whether the value is positive or negative (only in signed data). Storage Location: Where the variable's data is stored in memory. Data Type: A bit such as int, float char, etc, which determines the type of data the variable will store.
Therefore, these three elements are required in a variable.

11. What is the term for a condition that occurs during program execution that disrupts the normal flow of instructions?
प्रोग्राम निष्पादन के दौरान उत्पन्न होने वाली उस स्थिति को क्या कहते हैं जो निर्देशों के सामान्य प्रवाह को बाधित करती है?
(a) Abstraction (b) Inheritance
(c) Exception (d) Encapsulation

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : A condition that occurs during program execution that disrupts the normal flow of instructions, such as division by zero, incorrect input or missing file, is called an "Exception".

12. Variables which can be accessed inside a module of the program is called
वेरिएब्लस जिसे प्रोग्राम के एक Module के अन्दर एक्सेस किया जा सकता है, उसे कहा जाता है
(a) Global (b) Auto
(c) Static (d) Local

Bihar STET C.S. 2020

Ans. (d) : Local variables are variables that can only be accessed within the same function or module in which they are declared. The remain hidden from other parts of the program.

13. Which of the following Java statements declare and allocate a 2-dimensional array of integers with four rows and five columns.
निम्नलिखित में से कौन सा जावा कथन चार पंक्तिया और पाँच स्तंभों वाले पूर्णांक के दो विमितीय ऐरे (array) को डिक्लेयर और एलोकेट करता है?
(a) int array [][] = new int [5][4];
(b) int array [4][5];
(c) int array [5][4];
(d) int array [][] = new int [4][5];

KVS PGT C.S. 2017

Ans. (d) : Multidimensional array is stored in the form of rows and columns.
Syntax–
int array [][] = new int [row size] [Column Size];
Here, in this question we have allocate 2-d array of integers with 4 rows and 5 columns.
So, int array [][] = new int [4][5];

14. A condition that is caused by run-time error in a computer program is known as :
किसी कम्प्यूटर प्रोग्राम में 'रन टाइम' त्रुटि से होने वाली स्थिति को क्या कहा जाता है ?
(a) Exception/एक्सेप्शन
(b) Syntax error/सिन्टेक्स त्रुटि
(c) Semantic error/सीमेन्टिक त्रुटि
(d) Fault/फ़ाल्ट

KVS PGT C.S. 2017

Ans. (a) : A condition that is caused by run-time error in a computer program is known as exception. An exception is an event, which occurs during the execution of a program, that disrupts the normal flow of the program's instructions.

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15. **_____ variables are not reinitialized each times Visual Basic invokes a procedure and thus retains or preserves value even when a procedure ends.**
प्रत्येक बार जब भी विजुअल बेसिक किसी प्रक्रिया को सक्रिय करता है, _____ चर पुनर्प्रवर्तित नहीं होते और इस प्रकार प्रक्रिया के समाप्त होने पर भी मान को सुरक्षित रखते हैं।
(a) Static/स्टैटिक (b) Dynamic/डायनैमिक
(c) Virtual/वर्चुअल (d) Private/प्राइवेट

KVS PGT C.S. 2018

Ans. (a) : Static variables are not reinitialized each time visual Basic invokes a procedure and thus retains or preserves value even when a procedure ends. A static variables continues to exist and retains its most recent value.

16. **The elements of the triangular array are stored as a vector in the order A[1, 1], A[2, 1], A[2, 2], A[3, 1], A[3, 2], A[3, 3] ..... A[n, n].**
Assuming that A[1, 1] is stored at location 1, addressing function for A[i, j] is given by:
त्रिभुजीय ऐरे को अवयव निम्नलिखित क्रम में वेक्टर के तौर पर स्टोर किए गए है।
A[1, 1], A[2, 1], A[2, 2], A[3, 1], A[3, 2], A[3, 3] ..... A[n, n].
मान लीजिए की A[1, 1] लोकेशन 1 पर स्टोर किया गया है, तो A[i, j] के लिए एड्रेसिंग फंक्शन किससे प्राप्त होगा?
(a) $\frac{(i-1)}{2}+j$ (b) $\frac{(i-1)*i}{2}+j$
(c) $\frac{i\times i}{2}+j$ (d) $\frac{(i*j-1)}{2}$

KVS PGT C.S. 2018

Ans. (b) : Matrix is shown as:

A_{11}	A_{12}	A_{13}	A_{14}
A_{21}	A_{22}	A_{23}	A_{24}
A_{31}	A_{32}	A_{33}	A_{34}
A_{41}	A_{42}	A_{43}	A_{44}

Let, us consider only the lower triangular matrix value from the above matrix.
A [1, 1]
A [2, 1], A[2, 2]
A [3, 1], A[3, 2], A[3, 3]
……..A [n, n]
Assuming that A[1,1] is stored at location 1, addressing function for A [i, j]–
Using the formual–

$$= \frac{(i-1)*i}{2}+j$$

$$= \frac{(1-1)*1}{2}+1 \qquad (\because i = 1, j = 1)$$

= 1
Hence, the memory location of A [1, 1] = 1

17. **Which of the following is not part of programming construct ?**
निम्नलिखित में से कौन सा प्रोग्रामिंग कंस्ट्रक्ट का भाग नहीं है?
(a) Sequencing/सीक्वेंसिंग
(b) Branching/ब्रांचिंग
(c) Overlapping/ओवरलैपिंग
(d) Looping/लूपिंग

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : Programming is usually built from three basic building blocks.
1. **Sequencing-** running statements one after another.
2. **Branching (Selection)** – making decisions, like if/else.
3. **Looping (lteration)-** repeating actions, like for or whilel loops.

The word "overlapping" doesn't describe any of these basic structures, so it is not considered a fundamental programming construct.

18. **Which of the following file stream function will move the file pointer to N^{th} record in a binary file F created with the help of objects of class EMP ?**
निम्नलिखित में से कौन सा फाइल स्ट्रीम फंक्शन क्लास EMP के ऑब्जेक्ट्स की सहायता से बनाई गई बाइनरी फाइल F में फाइल पॉइंटर को N वें रिकॉर्ड पर लेकर जाएगा?
(a) F.tellg(N)
(b) F.seekp(N)
(c) F.seekp((N−1)* size of (EMP))
(d) F.seekp(size of (EMP)/N)

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : Seekp → seekp() is used to move the put pointer to the desired location concerning a reference point.
So, that correct answer is (c) F. Seekp (N – 1) * Size of (EMP).

19. **Which of the following is not a primitive data type?**
निम्न में से कौन-सा डेटा टाइप मौलिक नहीं है?
(a) Boolean/बूलियन (b) Byte/बाइट
(c) String/स्ट्रिंग (d) Double/डबल

LT. Grade C.S. 2018

Ans. (c) : In computer programming language, primitive data types are the basic data types that are built into a programming language. They are typically simple, indivisible values that cannot be broken down into smaller components.
Primitive data type - Integer, float, boolean, char, byte, double.
Non-primitive data type - Arrays, structures classes, strings, pointers.

20. **One of the disadvantages of pass by reference is that the called function may inadvertently corrupt the caller's data. This can be avoided by**
पास बाइ रेफरेन्स की कमियों में से एक यह है कि बुलाया गया फंक्शन अनजाने में कॉलर के डेटा को करप्ट कर सकता है। इससे निम्न में से किसके द्वारा बचा जा सकता है?
(a) Passing pointers/पासिंग पॉइटर्स
(b) Declaring the formal parameters constant फॉर्मल पैरामीटरों को स्थिर घोषित करके

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(c) Declaring the actual parameters constant
ऐक्चुअल पैरामीटरों को स्थिर घोषित करके
(d) All of the above/उपर्युक्त सभी

LT. Grade C.S. 2018

Ans. (b) : Declaring the formal parameters as constants prohibits any change to the contents of the actual parameters. We can also declare the actual parameters constant, which in turn would require formed parameters to be constant, but this is overly restrictive as this would not allow any modification in the caller function also.

21. Among the following, which statement is correct about the modularity?
निम्नलिखित में से मॉड्यूलैरिटी के बारे में कौन-सा कथन सही है?

(a) Modularity means hiding the parts of the program.
(b) Modularity refers to dividing a program into subsequent small modules or independent parts.
(c) It refers to overloading the program's part.
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): Modularity refers to dividing a program into smaller modules that are independent components. This makes development easier and more streamlined and promotes code reusability.

22. What is the scope of the variable declared in the user-defined function?
उपयोगकर्ता परिभाषित फंक्शन में घोषित वेरिएबल का दायरा क्या है?

(a) Whole program
(b) Only inside the { } block
(c) Inside the main function
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(b): The scope of a variable declared in a user defined function is the area within which it can be used. This scope defines the name of the variable along with its data type such as int, float, char, etc. This scope includes the scope and name of the variable, which can be used only within the body of the function. When the function is called, the variable in this scope reaches the body of the function and is available for use. Let us explain with an example

```
int calculate (int x, int y)
{
int result = ++y;
return result;}
```

Here x and y variable are parameters of the calculated function, while result is a declared variable declared within the function itself.

23. Which of the following statements is true about the new and malloc?
न्यू और मॉलोक के बारे में निम्नलिखित में से कौन-सा/से कथन सत्य है/हैं?

I. The 'new' is a type of operator while 'malloc' is a kind of function./ 'न्यू' एक प्रकार का ऑपरेटर है, जबकि 'मॉलोक' एक प्रकार का फंक्शन है।
II. The 'new' invokes a constructor, whereas 'malloc' does not invoke the constructor./ 'न्यू' एक कंस्ट्रक्टर को आमंत्रित करता है, जबकि 'मॉलोक' कंस्ट्रक्टर को आमंत्रित नहीं करता है।
III. The 'malloc' returns void pointer and also needed to typecast whereas 'new' returns required pointer./'मॉलोक' शून्य पॉइंटर लौटाता है और उसे टाइपकास्ट करने की भी आवश्यकता होती है, जबकि 'न्यू' रिटर्न के लिए पॉइंटर की आवश्यकता होती है।

(a) Only I/केवल I
(b) Both I and II/I और II दोनों
(c) Both II and III/II और III दोनों
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d): Both new and malloc are used to allocate dynamic memory in C++ and c programming language. New is a C++ operator that allocates memory for an object and calls its constructor. When using it, you can use the default constructor for the data type specified after new followed by the name of the object's data type. Followed by a colon(:). Malloc is a standard library function in the C programming language that allocates dynamic memory. it only allocates memory but does not call any constructor and return a void pointer that you can typecast to any data type.

24. The M components in MVC are responsible for

(a) user interface
(b) security of the system
(c) business logic and domain objects
(d) translating between user interface actions/events and operations on the domain objects

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : MODEL-VIEW-CONTROLLER (MVC):

* MVC refers to three different realms of responsibility in the design of a graphical user interface.
* when the MVC pattern is applied to a component the model consists of the data that represents the current state of the component.
* The view is simply the visual presentation of the component on the screen.
* The controller is the aspect of the component that carries out actions in response to events generated by the user (or by other source such as timers).

The idea is to assign responsibility for the model, the view, and the controller to different objects.

Sarkari Resource

25. The V components in MVC are responsible for:
(a) User interface.
(b) Security of the system.
(c) Business logic and domain objects.
(d) Translating between user interface actions/events and operations on the domain objects.

UGC NET C.S. December 2020

Ans. (a) : In model-view-controller architectural pattern, there are following parts and there main responsibilities-
Model: Responsible for providing the app's "business logic".
View: Responsible for providing the app's user interface (UI)
Controller: Responsible for translating UI actions to operations on the model.
So V components in MVC are responsible for user interface.

26. Which of the following statement(s) is/are correct about modular programming approach?
मॉड्यूलर प्रोग्रामिंग दृष्टिकोण के बारे में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. Ease of use/उपयोग में आसानी
II. Reusability/पुनर्प्रयोग
(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Both statements I and II are correct about modular programming
(i) Ease of use– Modular programming makes it easier to understand, manage and debug code.
(ii) Reusability– Modular programming promote code reusability. Once a module is created, it can be reused across different parts of the program.

27. In a modular programming approach problem is further divided into _____.
मॉड्यूलर प्रोग्रामिंग दृष्टिकोण में एक समस्या को आगे _____ में विभाजित किया जाता है।
(a) question/प्रश्न (b) solution/समाधान
(c) modules/मॉड्यूल (d) object/ऑब्जेक्ट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Modular programming is a design technique in software development where a program is divided into separate self-contained units or "modules".

28. Match the following types of variables with the corresponding programming language :

A.	**Static variables**	**i.**	**Local variables in Pascal**
B.	**Stack dynamic**	**ii.**	**All variables in APL**
C.	**Explicit heap dynamic**	**iii.**	**Fortran 77**
D.	**implicit heap dynamic**	**iv.**	**All objects in JAVA**

Code :

	A	B	C	D
(a)	(i)	(iii)	(iv)	(ii)
(b)	(iv)	(i)	(iii)	(ii)
(c)	(iii)	(i)	(iv)	(ii)
(d)	(ii)	(i)	(iii)	(iv)

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) FORTRAN I, II & IV contain only static variables. All objects in Java are explicit heap dynamic variable & are destroyed by "Garbage Collector" Languages Such as APL, ALGOL 68, & LISP Use implicit heap dynamic variables.
So, a-iii, b-i, c-iv, d-ii
Hence, correct option is (c).

29. What is the right approach to deal with student errors in computer programming ?
कंप्यूटर प्रोग्रामिंग में छात्र की त्रुटियों से निपटने के लिए सही तरीका क्या है?
(a) There should be zero errors even at stage I.
चरण I पर भी शून्य त्रुटियाँ होनी चाहिए।
(b) Having errors means student does not know programming./त्रुटियाँ होने का अर्थ है कि छात्र प्रोग्रामिंग नहीं जानता है।
(c) Errors are part of a programmer's daily life.
त्रुटियाँ एक प्रोग्रामर के दैनिक जीवन का हिस्सा है।
(d) Should scold the student whenever there is an error./जब भी कोई त्रुटि हो, तो छात्र को डांटना चाहिए।

HTET PGT C.S. 2020

Ans. (c) : Programming involves a lot of trial and error. Making mistakes, debugging and learning from those errors are crucial steps in becoming a proficient programmer. Options (a), (b) and (d) promote unrealistic expectations or helpful disciplinary actions regarding errors, which are detrimental to learning.

30. A variable declared within the block { } has the following scope :
ब्लॉक { } के भीतर घोषित एक चर का निम्नलिखित स्कोप होता है
(a) File Scope/फाइल स्कोप
(b) Local Scope/लोकल स्कोप
(c) Class Scope/क्लास स्कोप
(d) Global Scope/ग्लोबल स्कोप

HTET PGT C.S. 2021

Ans. (b) : In C programming, a variable declared inside a block of code (enclosed within curly braces {}) has a local scope. This means it is only accessible and its lifetime is restricted to that specific block of code where it is declared.

31. If a class contains at least one pure virtual function, it is called as_____class.
यदि किसी क्लास में कम-से-कम एक प्योर वर्चुअल फंक्शन हो, तो उस क्लास को कहा जाता है–
(a) Compiled/कम्पाइल्ड (b) Abstract/एब्सट्रैक्ट
(c) Interpreted/इंटरप्रिटेड (d) Interface/इंटरफेस

HTET PGT C.S. 2023

Sarkari Resource

Ans.(b): In C++ , a pure virtual function is declared in a base class using = 0 and has no implementation in that base class. A class containing atleast one pure virtual function can not be instantiated and is called an abstract class.

32. In C program, variable can not start with :
C प्रोग्राम में चर (Variable) निम्नलिखित में से किसके साथ शुरू नहीं हो सकता है?

(a) An alphabet/एक वर्णमाला
(b) A number/एक संख्या
(c) A special symbol other than underscore
अंडरस्कोर के अलावा एक विशेष प्रतीक
(d) A number and A special symbol other than underscore/एक संख्या और अंडरस्कोर के अलावा एक विशेष प्रतीक

HTET PGT C.S. 2020

Ans. (d) : In C programming a variable name cannot start with a number. Additionally special symbols other than the underscore (_) are not allowed in variable names. Therefore, a variable name can not begin with either a number or a special symbol other than an underscore.

33. Who is known as the father of C Language ?
सी भाषा का जनक किसे कहा जाता है?

(a) Bjarne Stroustrup (b) Dennis Ritchie
(c) James A. Gosling (d) Dr. E. F. Codd

HTET PGT C.S. 2020

Ans. (b) : Dennis Ritchie is widely recognized as the father of the C programming language. He developed C at Bell Labs between 1972 and 1973.

34. An external variable:

(a) is globally accessible by all functions
(b) has a declaration 'exterm' associated with it when declared within a function
(c) will be initialized to 0, if not initialized
(d) all of the above are true

CGPSC C.S. 2020 Set-A

Ans. (d) : (a). External variables (also known as global variables) are declared outside of any function and can be accessed and modified by all functions within the same program file.
(b). When an external variable is declared within a function, the extern keyword is used to indicate that the variable is defined elsewhere (globally) and not being newly created within that function's scope.
(c). If an external variable is not explicitly initialized by the programmer, it is automatically initialized to zero by default.
Therefore, option (d) is correct.

35. When a recursive function is called, all its automatic variable are:

(a) Maintained in stack
(b) Retained from last execution
(c) Initialized during each call of function
(d) None of the above holds true

CGPSC C.S. 2020 Set-A

Ans. (c) : When a recursive function is called, a new stack frame is created for that particular call. This new stack frame includes space for the function's parameters and its local automatic variables. Each time the function is called (including recursive calls), these automatic variables are freshly created and initialized within their respective stack frame.

36. Consider the following two statements:
Statement I: The ternary operator in C requires 4 operands.
Statement II: The operator + in C may be either unary or binary.
In light of the above statements, choose the correct answer from the options given below:

(a) Both statement I and Statement II are true.
(b) Both statement I and statement II are false.
(c) Statement I is correct, but statement II is false
(d) Statement I is incorrect, but statement II is true.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : Statement-I is incorrect because the ternary operator in (c) used 3 operands not 4. Its structure is condition value if true value it false.
Statement-II is true because the + operator can be used in both unary (single operand) and binary (two operands) forms. In unary + x) while in binary form. It performs addition (e.g. a + b).

37. A.....is the specification of legal values for variable of the type.
एक वेरिएबल की मान्य मूल्यों का विनिर्देश है–

(a) Data Types (b) Data Structure
(c) Scope Rules (d) Control Structures

HTET PGT C.S. 30.07.2025

Ans. (a) : A data Type is the specification of legal values for a particular type. It defines the kind of values a variable can hold and the operations that can be performed on it, for example an in data type specific that a variable can store integer values. While a char data type specifies that it can store character values.

38. Consider the following two statements:
Statement I: The while (37) printf ("Message/n"); statement in C will print the message 37 times.
Statement II: The while loop in C is an entry-controlled loop, and if the condition is initially false, the loop body will not execute at all.
In light of the above statements, choose the correct answer from the options given below:

(a) Both statement I and statement II are true.
(b) Both statement I and statement II are false
(c) Statement I is correct, but statement II is false
(d) Statement I is incorrect, but statement II is true.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : Statement-I: The while (37) condition in c evaluates the integer 37. In (any) non-zero integer is considered, true and o is considered false. Therefore while (37) is equivalent to while (true), erecting an infinite loop. The prints statement would. Print message infinitely not just 37 times.
Statement-II : A while loop in c is indeed an entry controlled loop. This means the condition is checked before the loop body is executed for the first time. If the condition is initially false the loop body will not execute even once.

Sarkari Resource

39. Which procedure is characterized by the fact that the procedure calls itself again and again till the base condition is not?
किस प्रक्रिया की यह विशेषता होती है कि वह प्रक्रिया स्वयं को बार-बार कॉल करती है जब तक की आधार शर्त पूरी नहीं हो जाती है?

(a) Nested (b) Functional
(c) Recursive (d) Operational

HTET PGT C.S. 30.07.2025

Ans. (c) : In recursion, a function or procedure keeps calling again and again until a base condition is satisfied. The difference between recursion and iteration (loops) in simple English.

40. In C programming language, which of the following operators has the highest precedence?
C प्रोग्रामिंग भाषा में निम्न ऑपरेटरों में से किसकी सर्वोच्च वरीयता है?

(a) unary +/यूनरी + (b) *
(c) >= (d) ==

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (a) : In programming, the unary operation, including unary + (positive sign)and unary-(negative sign), have a higher precedence than most other operators, including arithmetic, relational, and assignment operators.

- **Unary + and - :** These operators have the highest precedence among he given options.
- **Multiplication :** This has a higher precedence than addition/subtraction, but lower than unary operators.
- Greater than or equal to > = relational operator and has lower precedence than arithmetic and unary operators.
- **Assignment -:** This operator has the lowest precedence among the given options, typically evaluated last in an expression.

41. The conversion characters for displaying data item as a floating point value with an exponent and for displaying data item as an octal integer are
प्रतिपादक के साथ चल-बिन्दु मान तथा ऑक्टल पूर्णांक के डाटा अवयवों को प्रदर्शित करने हेतु रूपान्तरण अक्षर हैं-

(a) e and d respectively/क्रमशः e तथा d
(b) e and o respectively/क्रमशः e तथा o
(c) f and d respectively/क्रमशः f तथा d
(d) e and o respectively/क्रमशः e तथा o

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (b) : In many programming languages, particularly C and C-like languages, the conversion character for displaying a floating-point value with an exponent (scientific notation) is e (or E).
The conversion character for displaying an integer as an octal value is o.

42. In which stage the following code
include < stdio/.h
Gets replaced by the contents of the file stdio.h
किस चरण मे निम्न कोड
include < stdio/.h
फाइल stdio.h के कन्टेन्ट्स से बदलता है-

(a) During linking/लिंकिंग के दौरान
(b) During execution/एक्तिक्यूशन के दौरान
(c) During preprocessing/प्रीपोसेसिंग के दौरान
(d) During compile/कम्पाइल के दौरान

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (c) : The programming is handled by the preprocessor. During the preprocessing stage, the preprocessor replaces the directive with the actual content of the header file before the compilation process begins. This file contains declarations for standard input/output functions like.

43. Which of the following statement about loop is correct?
लूप के लिए निम्न में से कौन सा कथन सही है?

(a) Index value retained outside the loop
लूप से बाहर इंडेक्स मान को बरकरार रखा जाता है।
(b) Index value can be changed from within the loop/लूप में से इंडेक्स मान को परिवर्तित किया जा सकता है।
(c) Goto can be used to jump, out of the loop
लूप से बाहर आने के लिए Goto का उपयोग किया जा सकता है।
(d) All of these/उपरोक्त सभी

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (d) :
(a). In many programming languages, the loop control variable (or index) retains its last value after the loop terminates, making it accessible outside the loop's scope.
(b). The index value of a loop's can indeed be modified within the loop's body, which can affect the loop's iteration and termination.
(c). While generally discouraged in modern unstructured code, the go to statement can be uses in some languages to transfer control to a labeled statement outside the loop, effectively exiting the loop.
The correct statement option (d) is correct option.

44. Variables which can be accessed by all modules of the program is called
Variables जो Program के सभी Modules द्वारा पहुँचा जा सकता है, उसे कहा जाता है

(a) Global (b) Auto
(c) Static (d) Local

Bihar STET C.S. 2019 (Shift-II)

Ans, (a) : Global variable is a variable that can be accessed by all modules and functions of the entire program whereas local variable is valid only inside the function a block where it is declared.

45. Math.h is a
Math.h है एक

(a) Derived data type (b) Function
(c) Header file (d) Built-in data type

Bihar STET C.S. 2019 (Shift-II)

Sarkari Resource

Ans, (c) : Math.h is a header file which is included in languages like C and C++ to use mathematical functions (like sqrt (), Pow (), sin (), cos () etc.)

46. The process of removing recursion involves replacing recursive function calls with:
रिकर्सन को हटाने की प्रक्रिया में रिकर्सिव फंक्शन कॉल्स को किससे बदला जाता है?

(a) More recursive function calls
अधिक रिकर्सिव फंक्शन कॉल्स
(b) Loops/लूप्स
(c) Additional memory allocation
अतिरिक्त मेमोरी अलोकेशन
(d) Non-recursive function calls
गैर-रिकर्सिव फंक्शन कॉल्स

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans, (b): Recursive function calls are replaced with loops (such as for or while loops) to eliminate recursion. This reduces stack memory usage and increases code efficiency.

47. ……. is a basic data type.
.....एक बुनियादी डेटा प्रकार है।

(a) Array (b) Struct
(c) Boolean (d) Int

Bihar STET C.S. 2020

Ans, (d): Int (integer) is a primitive data type, which stores integers (like 1, 5, 100) etc.

48. To store character or text data we need
Character टेक्स्ट डेटा को संग्रहीत करने के लिए हमें जरूरत होती है

(a) Numeric variable (b) String variable
(c) Logical function (d) Array

Bihar STET C.S. 2020

Ans, (b) : A 'string variable' is required to store character or text data. A string is a data type used to store letters, words, or sentence.

49. > = is a/an
> = है एक

(a) Instance (b) Design
(c) Data type (d) Operator

Bihar STET C.S. 2020

Ans, (d) : '> =' is an operator called "greater than or equal to". This operator is used to compare two values.

50. Which of the following is not a storage class specifier in C programming language?
C प्रोग्रामिंग भाषा में निम्न में से कौन-सा एक स्टोरेज क्लास स्पेसिफायर नहीं है?

(a) Register/रजिस्टर (b) Volatile/वोलेटाइल
(c) Extern/एक्सटर्न (d) Typedef/टाइपडेफ

LT. Grade C.S. 2018

Ans, (b) : In C programming language, Register, Extern and typedef are storage class specifiers. However, volatile is not a storage class specifier. It is a type qualifier that tells the compiler that a variable's value may be changed in ways not explicitly specified by the program.

51. In C language
C भाषा में

(a) Parameters are always passed by values
पैरामीटरों को हमेशा भेजा जाता है value द्वारा
(b) Parameter are always passed by reference
पैरामीटरों को हमेशा भेजा जाता है reference द्वारा
(c) Non-pointer variables are passed by value and pointers are passed by reference/non-pointer variables को value द्वारा और pointers को reference द्वारा भेजा जाता है।
(d) Parameters are always passed by value result/पैरामीटरों को हमेशा भेजा जाता है value result द्वारा

LT. Grade C.S. 2018

Ans, (c) : In C language non-pointer variables are passed by value and pointers are passed by reference /non-pointer.

52. C was primarily developed as a
C मुख्यत: विकसित की गई

(a) System programming language
प्रणाली प्रोग्रामिंग भाषा के रूप में
(b) General purpose language
सामान्य उद्देश्य भाषा के रूप में
(c) Data processing language
डेटा प्रसंस्करण भाषा के रूप में
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans, (a) : C was primarily developed as a system programming language, particularly for writing operating systems and low level applications. It was created by Dennis Ritchies in the 1970s for the development of the UNIX operating system.

53. The minimum number of temporary variables needed to swap the contents of two variables is
दो वैरिएबलों के कॉन्टेन्ट को स्वैप करने के लिए न्यूनतम अस्थायी वैरिएबलों की संख्या कितनी होगी?

(a) 1 (b) 2
(c) 3 (d) 0

LT. Grade C.S. 2018

Ans, (d) : The minimum number of temporary variables needed to swap the contents of two variables is 0.

• **Swapping without a temporary variable:** Swapping the contents of two variables can be achieved using arithmetic operations like addition and subtraction, or bitwise XOR.

Here are the steps:

Using arithmetic operations:

a = a + b;
b = a – b;
a = a – b;

Using bitwise XOR:

a = a ∧ b;
b = a ∧ b;
a = a ∧ b;

These methods allow us to swap two variables without requiring an additional temporary variable, making the minimum number of temporary variables needed **0**.

Sarkari Resource

54. The inbuilt C string function "strempi"?
C का इनबिल्ट फंक्शन "strempi"?

(a) Compare two string with case sensitivity
दो स्ट्रिंग की केस सेंसिटिविटी के साथ तुलना करता है
(b) Compare two string without case sensitivity
दो स्ट्रिंग की केस सेंसिटिविटी के बिना तुलना करता है
(c) Return longer string from two input string
दो इनपुट स्ट्रिंग में से बड़ी स्ट्रिंग को रिटर्न करता है
(d) Return one if two input strings are equal
दो इनपुट स्ट्रिंग सामान है तो 1 रिटर्न करता हैं

RPSC C.S. 2016 (Paper-I)

Ans, (b) : The 'strempi' function in C is used to compare two strings, ignoring case. It functions similarly to 'strempi', but it treats uppercase and lowercase letters as the same. The function returns an integer value based on the comparison:
⇒ **0** – If the strings are equal (ignoring case).
⇒ **Negative value** – If the first string is lexicographically less than the second string (ignoring case).
⇒ **Positive value** – If the first string is lexicographically greater than the second string (ignoring case).

55. Which of the following is an advantage of using arrays?/निम्नलिखित में से कौन-सा ऐरे का उपयोग करने का लाभ है?

(a) Constant time insertion and deletion
निरंतर समय प्रविष्टि और विलोपन
(b) Ability to store elements of different data types/विभिन्न डेटा प्रकारों के तत्वों को संग्रहीत करने की क्षमता
(c) Random access to elements using an index
इंडेक्स का उपयोग करके तत्वों तक यादृच्छिक पहुँच
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans, (c) : Arrays store data of a single type. Arrays allow for quick and direct access through indexes, allowing access to an element at a specific index.

56. In context of command line arguments in C program, argv is:
C प्रोग्राम के कमांड लाइन आर्गुमेंट्स के सन्दर्भ में argv है:

(a) Array of characters/ऐरे ऑफ करैक्टरस
(b) Array of character pointers
ऐरे ऑफ करैक्टरस पॉइंटर्स
(c) Pointer to character array/पॉइंटर टू करैक्टर ऐरे
(d) An integer variable/एक इन्टिजर वेरिएबल

RPSC C.S. 2016 (Paper-I)

Ans, (b) : In a C program, when we use command line arguments, the main() function is typically written as:
int main(int argc, char * argv[ ])
- 'argc' is an integer that holds the number of command line arguments, including the name of program itself.
- argv is an array of character pointers, where;

⇒ Each element arg[1] is a pointer to a character i.e., a string (C-style).
⇒ So argv[0] is the name of the program.
⇒ argv[1] to argv [argc –1] are the arguments passed to the program.

57. Which of the following file is not generated by C compiler when a program xyz.c is compiled?
जब एक प्रोग्राम xyz.c को कम्पाइल किया जाता है तब निम्न में से कौन सी फाइल C कम्पाइलर द्वारा नहीं बनायी जाती

(a) xyz.bid (b) xyz.bak
(c) xyz.obj (d) xyz.exe

RPSC C.S. 2016 (Paper-I)

Ans, (a) : xyz.obj– This is an object file, which contains the compiled code of the xyz.c program. It is generated by the compiler before the final executable.
xyz.bak– While not directly generated by the compiler during compilation, a backup file with a "bak" extension might be created by the editor or IDE as a safety measure.
xyz.exe– This is executable file, generated when the object files (like xyz.obj) are linked together with the necessary libraries.
Therefore "xyz.bid" is not a standard file of C programming.

58. Which among the following is language processor?
निम्न में से कौन सा लैंग्वेज प्रोसेसर है?

(a) Assembler/असेम्बलर
(b) Compiler/कम्पाइलर
(c) Interpreter/इंटरप्रेटर
(d) All of these/इनमें से सभी है

RPSC C.S. 2016 (Paper-I)

Ans, (d) : A language processor is a software program that translates source code (written in a programming language) into machine code that the computer can understand.
- **Assembler–** Converts assembly language (a low-level language) into machine code.
- **Compiler–** Translates high-level programming language code into machine code, typically producing an executable file that can be run independently.
- **Interpreter–** Executes code line by line, translating it into machine code as it goes, without generating a separate executable file.

59. Valid C statement to allocate 20 bytes to an integer pointer P is :
एक इन्टिजर प्वॉइंटर P को 20 बाइट एलोकेट करने के लिए उपयुक्त C स्टेटमेंट है?

(a) P = (int*) malloc (20);
(b) *P = (int*) malloc (20);
(c) P = (int* malloc)* 20;
(d) &P = (int*) malloc (20);

RPSC C.S. 2016 (Paper-I)

Ans, (a) : Syntax of malloc–
Pointer = (cast – type*) malloc (size-in-bytes):
• Malloc (size-in-bytes) allocates the specified number of bytes from heap memory

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• It returns a pointer of type void *, which is then cast to the appropriate type (in this case, int*)
• P must be declared as int*
(a) P = (int*) malloc(20);
• This allocates 20 bytes on the heap casts it to int* (because p is an integer pointer) and assigns the resulting memory address to pointer P.
(b) *P = (int*) malloc (20); is wrong because you are trying to assign to the memory pointed by P, but P has not been assigned memory yet.
(c) P = (int* malloc)*20; is invalid C syntax. There's no such type or operation as (int* malloc).
(d) &P = (int*) malloc (20); is invalid because you cannot assign a value to &P it represents the address of pointer p, which is not a variable you can assign to.

60. Which of the following are legal statements in C programming language?
(A) int * P = &44;
(B) int * P = &r;
(C) int P = &a;
(D) int P = a;
Choose the correct option:
(a) (A) and (B) (b) (B) and (C)
(c) (B) and (D) (d) (A) and (D)
UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) :
a – A pointer variable P cannot store the address of a value which is not assigned to a variable.
b – A pointer variable P can store the address of variable r.
c - An integer variable P cannot store the address of variable a.
d - An integer variable P can store the value of variable a.
Hence, int*P = &r; and int P = a; are legal statements in C programming language.

61. Which of the following statements is/are True ?
P : C programming language has a weak type system with static types.
Q : Java programming language has a strong type system with static types.
Code :
(a) P only (b) Q only
(c) Both P and Q (d) Neither P nor Q
UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : C programming language has a weak type system with static types. True
Java programming language has a strong type system with static types. True
Both Statements are True.

62. Which of the following is wrong about the data types?
(a) The number is always positive qualifier 'unsigned' is used
(b) The number can be positive or negative when the qualifier 'signed' is used
(c) The range of values for signed data types is more than that of unsigned data types
(d) The left most bit in unsigned data type is used to represent the value
UGC NET C.S. December-2022

Ans. (c) : In C language, unsigned is also a data type, which is a variable type of int data type can hold zero and positive numbers. There is also a signed into data type, which is a variable type of into data type that can hold negative, zero and positive number. This unsigned into data type cannot represent a negative number.
Unsigned integers range from 0 to 4 billion in 32-bit integers, while the signed version ranges from around –2 billion to +2 billion.
So, option C is correct answer.

63. Of the following, which is NOT a logical error?
(a) Using the '=', instead of ' = =' to determine if two values are equal
(b) Divide by zero
(c) Failing to initialize counter and total variables before the body of loop
(d) Using commas instead of two required semicolon in a for loop header
UGC NET C.S. December 2021

Ans. (b) : Logical error:- A logical error is a mistake a program's source code that results in incorrect or unexpected behaviour. It is a type of runtime error that may simply produce the wrong output or may cause a program to crash while running.
Divide by zero is a runtime error (or arithmetic error), not a logical error. It occurs during program execution when an attempt is made to perform are invalid arithmetic operation, such as dividing a number by zero. This typically causes the program to terminate abnormally or throw an exception, rather than producing incorrect but seemingly valid output.
So, option (b) is correct answer.

64. Which statement is false?
(a) All function call in C pass arguments using call by value
(b) Call by reference enables a called function to modify a variable in calling function
(c) Call by value is always more efficient than call by reference.
(d) Programmers use pointers and indirection operation to simulate call by reference.
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Ans. (c) : Difference between call by value and call by reference.

Call by value	Call by reference
• In this case value of the variable is passed.	• In this case address of the variable is passed.
• Can't change value of actual argument by using formal argument.	• Can change value of actual argument by using formal argument.
• No pointer is used.	• Pointers are used.
• It requires more memory.	• It requires less memory.
• It is less efficient.	• It is more efficient.

So option (c) is correct answer.

65. Pointers cannot be used to
(a) find the address of a variable in memory
(b) reference value directly
(c) simulate call by refrence
(d) manipulate dynamic data structure
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Ans. (b) : A reference value directly cannot be used in pointers.
So directly references a value, where as a pointer indirectly references a value through the memory address it stores.

66. A program that is used by other routines to accomplish a particular task, is called :
(a) Micro program (b) Micro operation
(c) Routine (d) Subroutine

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Ans. (d) : A program that is used by other routine to accomplish a particular task, is called 'subroutine' or 'function'. A subroutine, also known as a method, procedure, is a set of instructions designed to perform a frequently used operation within a program. They can accept arguments, perform, specific operations and often return a result.

67. Which of the following statements about pointers in C are TRUE.
A. Pointers can be used to access array elements.
B. Pointers can store the address of another Pointer.
C. Pointers are automatically differenced in expression.
D. Pointers cannot be used to access structure members.
Choose the correct answer from the options given below :
(a) (A) and (C) Only (b) (A) and (B) Only
(c) (B) and (C) Only (d) (C) and (D) Only

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Ans. (b)
(A) Pointers can be used to access array elements, true. In C, an array name is essentially a pointer to its first element, and you can use pointers to access elements of an array.
(B) Pointers can store the address of another pointer, true. A pointer can store the addresses of another pointer, which is called a pointer to pointer.
(C) Pointer are automatically differenced in expression, False. Pointers are not automatically differenced in expression. You need to explicitly use the differenced operator (*) to differenced a pointer.
(D) Pointers cannot be used to access structure members. False. Pointers can be used to access structure members you can access structure members through pointers using the –> operator.

68. With respect to a loop in the transportation table, which one of the following is not correct?
(a) Every loop has an odd no. of cells and atleast 5.
(b) Closed loops may or may not be square in shape
(c) All the cells in the loop that have a plus or minus sign, except the starting cell, must be occupied cells
(d) Every loop has an even no. of cells and atleast four.

UGC NET C.S. June-2017 (Paper-III)

Ans : (a) In transportation table:
(i) An even number of atleast 4 cells must participate in closed loop and an occupied cell can be considered only once and not more.
(ii) Closed loop may or may not be square or rectangle in shape.
(iii) All cells in the loop that have a plus or minus sign, except the starting cell, must be occupied.

69. Which of the following are not regular?
(A) Strings of even number of a's
(B) Strings of a's, whose length is a prime number
(C) Set of all palindromes made up of a's and b's
(D) Strings of a's whose length is a perfect square
(a) (A) and (B) only
(b) (A), (B) and (C) only
(c) (B), (C) and (D) only
(d) (B) and (D) only

UGC NET C.S. June-2017 (Paper-III)

Ans : (c) (1) Strings of even number of a's is regular because we can draw finite acceptor (FA) for this.
(2) Strings of a's whose length is a prime number. There are infinite no. which are prime and we cannot design FA for infinte language. This is not regular
(3) Set of all palindromes made up of a's and b's. This is not regular because we cannot design FA for infinite language.
(4) Strings of a's whose length is a perfect square. This is not regular because we cannot design FA for infinite language.

70. Converting a primitive type data into its corresponding wrapper class object instance in called
(a) Boxing (b) Wrapping
(c) Instantiation (d) Autoboxing

UGC NET C.S. December-2014 (Paper-III)

Ans : (d) Autoboxing is the automatic conversion that the java compiler makes between the primitive types & their corresponding object wrapper class.

71. Consider the following statements S1, S2 and S3 :
S1 : In call–by–value, anything that is passed into a function call is unchanged in the caller's scope when the function returns.
S2 : In call-by-reference, a function receives implicit reference to a variable used as argument.
S3 : In call-by-reference, caller is unable to see the modified variable used as argument
(a) S3 and S2 are true (b) S3 and S1 are true
(c) S2 and S1 are true (d) S1, S2, S3 are true

UGC NET C.S. December-2014 (Paper-III)

Ans : (c) **S1-** In call-by value, anything passed into a function call is unchanged in the caller's scope when the function returns. **True**
S2- In call-by reference, a function receives an implicit reference to a vaticable used as an argument. **True**
So answer is c, because in call-by reference caller always sees the modified value of the argument.

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72. Which of the following is not a member of class?
(a) Static function
(b) Friend function
(c) Const function
(d) Virtual function

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) A **friend function** that is a "friend" of C given class, is a function that is given the same access than methods to private and protected data. A friend function is declared by the class that is granting access, so friend functions are part of the class interface like methods.

73. When we pass an array as an argument to a function, what actually gets passed?
(a) Address of the array
(b) Values of the elements of the array
(c) Base address of the array
(d) Number of elements of the array

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, Base address of an array is the starting address of an array. One can easily access to all the cells of an array if information about this can be achieved.

74. String is a collection of one or more character which consists of _____ in programming language.
I. Letters
II. Numbers
III. Symbols
(a) I, II and III (b) Only I
(c) Only I and III (d) Only II

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : In programming, a string is a sequence of characters can be:
1. Letters-eg. A, B, Z
2. Numbers-eg, 0, 1, 5
3. Symbols eg, @, #, $, &
So string can contain any combination of letters numbers and symbols.

75. Which of the following statement(s) is/are correct regarding Assembly language?
I. An assembler converts (or assembles) the symbolic program of an Assembly language into its equivalent machine program.
II. A program written in an assembly language is machine dependent.
(a) Both I and II (b) Only II
(c) Only I (d) Neither I nor II

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : Statements I correct
- An assembler translates assembly language code (Symbolic instructions like MOV A,B) into machine language (binary instructions).
  2- Statement II- correct
- Assembly language is machine dependent because its instructions and syntax are designed for a specific CPU architechure (eg, x86, ARM)

76. Unicode in the programming language can be represented using which of the bit format standards in computer science?
(a) 8- bit (b) 17- bit
(c) 65-bit (d) 9-bit

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : Unicode can be stored in formats like UTF-8, UTF-16 and UTF-32. UTF-8 the most common is an 8-bit variable-length encoding using 1-4 bytes per character. While UTF-16 and UTF-32 exist the 8-bit, UTF-8 format is the most fundamental and widely used in computer science.

77. Which of the following is used to access the overridden function of the base class from the derived class?
निम्नलिखित में से किसका उपयोग डिराइव्ड क्लास से बेस क्लास के ओवरराइड फंक्शन तक पहुंचने के लिए किया जाता है?
(a) :: (b) <<
(c) >> (d) :

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In C++ the scope resolution operator (::) is used to access the every then function of the base class from within the derived class.

78. Which of the following is NOT an attribute of good programming languages?
निम्नलिखित में से कौन अच्छी प्रोग्रामिंग भाषाओं की विशेषता नहीं है?
(a) Clarity, simplicity and unity
स्पष्टता, सरलता और एकता
(b) Support for abstraction
एब्सट्रैक्शन के लिए समर्थन
(c) Ease of program verification
प्रोग्राम वेरिफिकेशन में आसानी
(d) Non portability/गैर सुवाह्यता

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : The attribute that is not associated with good programming language is non portability.
Portability of Programs– Transportability of the resulting programs from the computer on which they are developed to other computer system.

79. When the rules of the programming language are not followed it gives _____ errors.
जब प्रोग्रामिंग भाषा के नियमों का पालन नहीं किया जाता है तो यह _____ त्रुटियाँ देता है।
(a) Syntax/सिंटैक्स
(b) Logical/लॉजिकल
(c) run time/रन टाईम
(d) Printing/प्रिंटिंग

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : A syntax error in a programming language occurs when the rules of the programming language are not followed.
This can include missing punctuation, misspelled keyword, incorrect use of operators, and improperly structured statements.

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- A logical error is a type of programming error that occurs when there is a flow in the algorithm or logic of a program.
- A run time error in a program is an error that occurs while the program is running after being successfully compiled.

80. When does the exe file is created in C language?
C भाषा में exe फाइल कब निर्मित होती है?

(a) After compilation/कम्पाईलेशन के बाद
(b) After loading all files/सभी फाईल्स की लोडिंग के बाद
(c) After linking /लिंकिंग के बाद
(d) After writing/लिखने के बाद

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : In C language, the process of creating an executable (.exe) file involves several steps:
1. Writing the source code: After writing, we obtain .c files.
2. Compilation: The compiler converts the source code into object code (.obj files).
3. Linking: The linker combines one or more object files and necessary libraries to create a single executable file (.exe).

Therefore, the .exe file is created after linking phase.

81. Which of the following is a type of subprogram?
निम्नलिखित में से कौन सा सबप्रोग्राम का एक प्रकार है?

I. Functions/फंक्शन
II. Subroutines/सबरूटिन

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Functions and subroutines/procedures are both types of subprograms. Functions return a value while subroutines may not return a value.
A subprogram is a set of instructions designed to perform a specific task, which can be called from different parts of a program.

82. Which statement (s) is are true about formation of identifier in C?
C में इडेंटिफायर्स के गठन के बारे में कौन सा/से कथन सत्य है/हैं ?

I. It consist of alphabets, digits and underscore./इसमें अक्षर, अंक और अंडरस्कोर शामिल हैं।
II. An identifiers can start with an underscore.
एक इडेंटिफायर अंडरस्कोर से शुरु हो सकता है।

(a) Only I /केवल I
(b) Only II /केवल II
(c) Both I and II / I तथा II दोनों
(d) Neither I nor II /ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : In C, a valid identifier can contain uppercase and lowercase letters (A-Z, a-z), Digits (0-9), underscore (–) after the first character.
An identifier may start with a letter or an underscore, but not with a digit. Starting with an underscore is allowed (e.g., –value, –temp), though leading underscore are often reserve for internal or implementation-specific names.
Hence, both the statements are correct.

83. Which of the following statement is/are true about pre-processor in C/C++?
I. To replace each # include directive with the contents of the designated file.
II. To replace each escape sequence with designated character.
III. To process macro definitions and expand macro call.
C/C++में प्री-प्रोसेसर के बारे में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं
I. प्रत्येक # include निर्देश को निर्दिष्ट फाइल की अंतर्वस्तुओं के साथ बदलने के लिए।
II. प्रत्येक एस्केप अनुक्रम को निर्दिष्ट संप्रतीक के साथ बदलने के लिए ।
III. मैक्रो डेफिनिशन को प्रोसेस करने और मैक्रो कॉल का विस्तार करने के लिए ।

(a) I and II / I तथा II
(b) II and III / II तथा III
(c) Only I /केवल I
(d) I,II and III / I, II तथा III

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (d) : The C/C++ preprocessor mainly handles directives before actual compilation.
1. it replaces each *#include* with the contents of the specified file.
2. It processes escape sequences like *\n, \t* by replacing them with their designated characters (as per exam perspective).
3. It handles macro definitions (*#define*) and expands macro calls wherever used.

Thus the preprocessor ensures source code is ready for compilation.
Hence, all three statements are true.
Correct answer: (d) I, II and III.

84. _____ converts a source code to object code.
_____एक सोर्स कोड को ऑब्जेक्ट कोड में परिवर्तित करता है ।

(a) Compiler /कंपाइलर (b) Linker /लिंकर
(c) Editor /एडिटर (d) Loader /लोडर

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : The program that translates source code (human-readable) into object code (machine-readable) is the compiler.
A compiler is a special program that translates source code written in a high-level programming language (like C, C++, Java) into object code or machine code that the computer's processor can execute.

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85. **Which of the following are included in the user documentation of program?**
I. Program formatting
II. Description of program and its purpose
III. Description of program output
निम्नलिखित में से कौन से प्रोग्राम के यूजर डॉक्यूमेंटेशन में शामिल है?
I प्रोग्राम स्वरुप
II प्रोग्राम का विवरण और उसका उद्देश्य
III प्रोग्राम आउटपुट का विवरण
(a) II and III / II तथा III
(b) I and II / I तथा II
(c) I and III / I तथा III
(d) I, II and III / I II तथा III

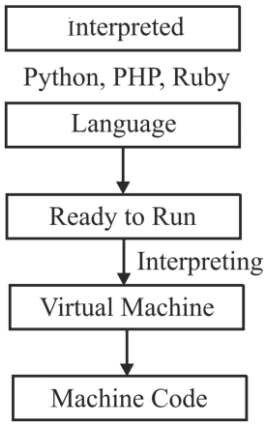
DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : User documentation is the manual or help guided prepared for the people who will actually run or use the software. Its goal is to explain how to install, operate and understand the program and its result, not how the code is written.
User documentation contains only end users need:
- Purpose of the program.
- Explanation of its output.
It does not include the source code.
Hence correct option is (a) II and III.

86. **Which of the following is a typically interpreted language?**
निम्नलिखित में से कौन सी एक विशिष्ट रूप से भाषांतरित लैंग्वेज़ है?
I. C
II. Perl
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Interpreted Language– Interpreted language is a programming language that is generally interpreted, without compiling a program into machine instructions it is one where the instructions are not directly executed by the target machine, but instead, read and executed by some other program interpreted language ranges– JavaScript, Perl, Python, BASIC, etc.



87. **In which debugging approach, the program is loaded with print statements to print the intermediate values with the hope that some of the printed values will help to identify the statement as error?**
किस डिबगिंग अप्रोच में, प्रोग्राम को इस उम्मीद के साथ मध्यवर्ती मानों को प्रिंट करने के लिए प्रिंट स्टेटमेंट के साथ लोड किया जाता है कि कुछ प्रिंटिड मान स्टेटमेंट को त्रुटि के रूप में पहचान करने में मदद करेगे?
(a) Backtracking/बैकट्रैकिंग
(b) Cause Elimination Method
कॉस एलिमिनेशन मेथड
(c) Brute Force Method/ब्रूट फोर्स मेथड
(d) Program Slicing/प्रोग्राम स्लाइसिंग

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : 'Brute force method' is a simple and widely used debugging approach, especially by beginners. In this method, the program is loaded with print statements to display intermediate values of variables, function calls, and program flow. The idea is that by examining these printed outputs, the programmer may be able to identity where the program's behaviour goes wrong.
Some key features of 'Brute Force Method'
→ Involves inserting print () Statement throughout the code.
→ Helps in tracking the flow & spotting incorrect values.
→ Often used when there is no due about the bug's location
→ It's a trial-and-error method.

88. **Which of the following is not true about Global Variables:**
(a) The values of the Global variables which are sent to the called function may be changed inadvertently by the called function.
(b) Functions are supposed to be independent and isolated modules. This Character is lost, if they use global variables.
(c) It is not immediately apparent to the reader which values are being sent to the called function.
(d) A function that uses global variables does not suffer from reusability.

UGC NET C.S. December-2024

Ans. (d) : Statement (d) is false about Global variables – A function using global variables can still be reusable if properly managed. While reliance on global variables can reduce reusability in some cases, it does not necessarily make the function completely Unusable in different contexts.

89. **Which of the following is not a valid variable name in most programming languages?**
अधिकांश प्रोग्रामिंग भाषाओं में, निम्नलिखित में से कौन-सा valid variable नाम नहीं है?
(a) my Variable
(b) 123Variable
(c) _myVariable

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(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : A valid variable name in programming languages must follow the following rules:
(i) Variable names can only consist of letters (a-z,A-Z) digits (0-9), and underscores (-). ii) variable names can begin with any letter and underscore, but cannot begin with a space, special character, or digit.

90. The correct statement for a function that takes pointer to a float, a pointer to a pointer to a char and returns a pointer to a pointer to an integer is:
एक फ़क्शन के लिए कथन, जो एक फ्लोट को पॉइंटर लेता है, एक पॉइंटर को एक चार के लिए एक पॉइंटर देता है और एक पॉइंटर को पूर्णांक पर पॉइंटर देता है, है-

(a) int **fun(float **, char **)
(b) int *fun(float *, char *)
(c) int **fun(float *, char **)
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : int ** fun(float *, char**) is a function that takes a pointer to a float, returns a pointer to a char and returns a pointer to a pointer to an interger.

91. Which of the following is the feature(s) of High level programming languages?
निम्नलिखित में से कौन सी हाई लेवल प्रोग्रामिंग लैंग्वेजेज की विशेषता/विशेषताएं है/हैं?
I. Portability/पोर्टेबिलिटी
II. Easy debugging/आसान डियुगिंग

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : High-level programming languages are characterized by both portability, and ease of debugging.
Portability:- High-level languages are designed to be less dependent on specific hardware, allowing code written in one environment to often run with minimal or no modification on different plateforms.
Easy debugging:- High level languages offer features like clear error massages and integrated development environments (IDE_S) that simplify the process of idenfying and fixing error in, code.

92. If loop does not terminate at all and keeps on executing endlessly then it is known as _____.
यदि लूप बिल्कुल समाप्त नहीं होता है और निरंतर रूप से क्रियान्वित होता रहता है, तो इसे _____ के रूप में जाना जाता है।

(a) For loop/फॉर लूप
(b) Finite loop/फिनाईड लूप
(c) While loop/वाइल लूप
(d) Infinite loop/इन्फायीनाईट लूप

DSSSB TGT C.S. 07.08.2021 Shift-I

Ans. (d) : An infinite loop is sequence of intruction in a computer program that loop endlessly, either became it has no terminating condition, or because the condition in never met, resulting in the loop contiuing forever. This matches the description in the question where a, loop "does not terminate at all and keeps on executing endlessly.

ii. Introduction to: C Programming Language and programming in C

93. The ..... transfers the executable image of a C++ program from hard disk to main memory.

(a) Compiler (b) Linker
(c) Debugger (d) Loader

UGC NET C.S. December-2015 (Paper-II)

Ans : (d) It is the responsibility of loader to transfer executable files from hard disk to main memory.
Debugger is a computer program that is used to test and debug other-programs (target program).
Compiler is a software which converts a program written in high level language (Source Language) to low level language.
Linker takes object files, links them with each other and with any library files that you specify and produces an executable output file.

94. Given that x = 7.5, j = – 1.0. n = 1.0, m = 2.0 the value of – – x + j = = x > n > = m is :

(a) 0 (b) 1
(c) 2 (d) 3

UGC NET C.S. June-2015 (Paper-II)

Ans.(a) : $--x + j == x > n >= m$
Here we need to check LHS = = RHS. So we get answer as 0 or 1 only. By applying values
– – 7.5 + (– 1.0) = = (7.5 > 1.0) = 2.0
(6.5 – 1.0) = = (0 > 2.0)
(5.5 = = 0)
Return false (0) since 5.5 is not equal to 0.

95. In C file handling equivalent to scanf("%d", &val); is :
C फाइल हैंडलिंग में scanf("%d",&val); के तुल्य है?

(a) fscanf(stdin, "%d",&val);
(b) fprintf(stdout, "value = %d\n", val)
(c) scanf(value)
(d) scanf()

HTET PGT C.S. 2021

Ans. (a) : The scan f ( ) function in C is used to read formatted input from the standard input stream, which is typically the keyboard.
The fscanf ( ) function is designed to read formatted input from a file stream. The syntax for fscanf ( ) is int fscanf (file * ptr, const char * format, ….) where ptr is a pointer to the file structure representing the file stream.
In C, stdin is a predefined FILE pointer that refers to the standard input stream.

Sarkari Resource

96. **Consider the following code segment:**
निम्नलिखित कूट खण्ड पर विचार कीजिए-

```
if ( Y<0 )
   {X = –X; Y = – Y;}
Z = 0;
while (Y>0)
            {Z = Z + X; Y = Y –1;}
```

Assume that X, Y and Z are inter variables, and that X and Y have be initialized. Which of the following best describes what this code segment does?
मान लें कि X, Y, और Z पूर्णांक चर है और X और Y आरंभित है। निम्न में से क्या इस कोड खण्ड के कार्य का सर्वोत्तम वर्णन करता है?

(a) Sets Z to be the product X* Y
Z , X* Y का गुणनफल होगा
(b) Sets Z to be the sum X+ Y
Z , X+ Y का योगफल होगा
(c) Sets Z to be the absolute value of X
Z , X का निरपेक्ष मान होगा
(d) Sets Z to be the value of Y
Z,Y का मान होगा

KVS PGT C.S. 2017

Ans. (a) : In this code segment
Consider, X = 4 and Y = 2
CASE 1. If (Y< 0) // 2 < 0 (false)
{ X = –X ; Y = –Y ;}
Z = 0 ;
while (Y > 0) // 2 > 0 (true)
{Z = Z + X ; Y = –1 ;} // Z = 0+4 = 4 ; Y = 2–1=1
CASE 2. When (Z = 4; Y = 1 ; X = 4)
if (Y<0)
{ X = –X ; Y = –Y;}
Z = 0;
while (Y>0) // 1 > 0 (true).
{Z = Z + X; Y = –1;} // Z = 4+4=8, Y = 1–1=0
CASE 3. If (Y<0)
{ X = –X ; Y = –Y ;}
Z =0 ;
while (Y>0) // 0>0 (false)
{Z = Z + X ; Y = –1}
Finally , Z becomes 8. Which is the product of X and Y (4×2=8)

97. **In C++, which of the following statements correctly returns the memory from the dynamic array pointer pp to the free store?**
C++ में, निम्नलिखित में से कौन सा कथन सही रूप से स्मृति को डाइनामिक अरे प्वाइंटर पी पी से मुक्त स्टोर में रिटर्न करता है?

(a) delete pp; (b) delete pp [];
(c) delete [] pp; (d) delete *pp;

KVS PGT C.S. 2017

Ans. (c) : In C++, Dynamic array pointer is created by using the new keyword which returns a pointer to the allocated block of memory.
Syntax for dynamic allocation for array. <data Type>* <pointer name> = new <data Type> [<size>] Now, to remove the memory from dynamic array pointer top syntax is.
delete [] pp;

98. **Assume X and Y are non-zero positive integers. Consider the following pseudo-code fragment:**
मान लीजिए कि X और Y अशून्य धनात्मक पूर्णांक है । निम्नलिखित स्यूडो कोड खण्ड पर विचार कीजिए :

```
while X< >Y do
        if X>Y then
        X < – X–Y
else
        Y < –Y–X
endif
end while
print (X)
```

What is the code doing?
यह कोड क्या कर रहा है ?

(a) It computes the GCD of two numbers.
दो संख्याओं के 'जी.सी.डी.' का परिकलन करता है।
(b) It computes the LCM of two numbers.
यह दो संख्याओं का ल.स.प. की गणना करता है।
(c) It finds the smallest of two numbers.
यह दो संख्याओं में से सबसे छोटी संख्या का पता लगाता है।
(d) It divides the largest number by the smaller.
यह सबसे बड़ी संख्या को कम छोटी संख्या से विभाजित करता है।

KVS PGT C.S. 2017

Ans. (a) : For this pseudo-code fragment
Consider, X = 3 and Y = 6
while X < > Y do
if X > Y then
X < – X – Y // 3 > 6 is false
else
Y < –Y–X // Y = 6 – 3=3
after this X = Y = 3, while condition become false. It prints the value of X is 3. (which is the GCD of 3 and 6.) So, this fragment computs the GCD of two numbers. Greatest Common Divisor (GCD) of two or more integers, which are not all zero, is the largest positive integer that divides each of the integers.

99. **Consider the following code segment:**
निम्नलिखित कोड खंड पर विचार कीजिए:

```
int x = 22, y = 15;
x = (x>y) ? (x+y) : (x–y);
```

What will be the value of x after the code is executed?
कोड के निष्पादित होने के बाद x का मान क्या होगा?

(a) 22
(b) 37
(c) 7
(d) 37 and 7/37 तथा 7

KVS PGT C.S. 2018

Ans. (b) : The conditional operator is a kind of similar to the if-else statement as it does follow the same algorithm as of if-else statement but the conditional operator takes less space and helps to write the if-else statements in the shortest way possible.

Sarkari Resource

Given code segment:
int x = 22, y = 15;
x = (x > y) ? (x + y): (x – y);
// 22 > 15 (condition is true), so output will be 22 + 15
So, output will be x = 37.

100. Which of the following is the correct declaration of linked list?
निम्नलिखित में से कौन सी लिंक्ड लिस्ट की सही घोषणा है?

(a)
```
Struct node*
{
int data;
node * link;
}
```
(b)
```
Struct node
{
int data;
struct node* link;
}
```
(c)
```
Struct node
{
int data;
node link;
}
```
(d)
```
Struct node*
{
int data;
struct node* link;
}
```

KVS PGT C.S. 2018

Ans. (b) : A linked list is a way to store a collection of element. Like an array these can be character or integers each element in a linked list is stored in the form of a node. A linked list can be implemented using structure and pointers.

```
Struct node
{
        int data;
        struct node* link;
}
```

Each node in the linked list is a struct.
So, the option (b) is correct.

101. What is the output of following code?

```
main ()
struct s1
{ char*z;
int i;
Struct s1 * p;
}
static struct s1 a[ ]= {
{"Nagpur", 1, a+1}
{"Raipur', 2, a +2}
{"kanpur' 3.a}
};
struct s1* ptr = a;
Printf (%s%s%s \n", a[0] z.ptr→z,a[2].P→z);
}
```

(a) Nagpur Raipur Kanpur
(b) Nagpur Nagpur Nagpur
(c) Kanpur Kanpur Kanpur
(d) Error

UGC NET C.S. June-2023

Ans. (b) : In given code, each element in the array 'a' is initialized with a pointer to the next element's 'z' member. So,
'a [0], z' refers to the string " Nagpur", which is the 'z' value of the first structure.
'ptr → z' also gives " Nagpur" because 'ptr' was intialized to point to the first structure in the array.
'a[2]. p → z' also gives "Nagpur" because 'p' of the third structure was initialized to point to the first structure in the array.
Therefore, the output of this code would be Nagpur Nagpur Nagpur.

102. Which statement in this code is unreachable switch (x) ?
इस कोड में कौन सा कथन अनरिचेबल (अगम्य) स्विच (x) है ?

```
{
  Case 1:
  a1 = 100;
  Case 2:
   a2 = 1000;
   a3 = 10;
   break ;
   a4 = 10000;
}
```

(a) a1 = 100;
(b) a2 = 1000;
(c) a3 = 10;
(d) a4 = 10000;

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (d) : In a switch statement the break keyword is used to exit the switch block immediately. In the given code, after a3 = 10; there is a break, statement. This means that once case 2 is entered the program will exit the switch block and a4 = 10000; will never be reached or executed.

103. In C program, what is the right syntax to access value of structure variable book {price, page}?
C प्रोग्राम में, संरचना चर पुस्तक (मूल्य पृष्ठ) के मूल्य का उपयोग करने के लिए सही सिंटेक्स क्या है?

(a) printf("%d%d", book.price, book. page);
(b) printf("%d%d", price.book, page. book);
(c) printf("%d%d", price::book, page :: book);
(d) printf("%d%d", price->book, page->book);

HTET PGT C.S. 2020

Ans. (a) : The right syntax to access the value of a structure variable in **C**, like book with members price and page, is using the dot **(.)** operator with the structure variable name.

104. Consider the following C function:

```
int f (int n)
{
static int i = 1;
if (n > = 5) return n;
n = n + i;
i ++;
return f(n);
```

Sarkari Resource

}

The value returned by f(1) is:

(a) 5 (b) 6

(c) 7 (d) 8

CGPSC C.S. 2020 Set-A

Ans.(c): Initially, the static variable i is initialized to 1
f (1) is called :
n is 1, which less then 5.
n is updated to n+i, which becomes 2.
I is incremented to 2.
The function calls itself recursively with the updated value of n, so it becomes f (2).
f (2) is called:
n is 2, which is less than 5.
n is updated to n+ i, which becomes 4.
i is incremented to 3.
The function calls itself recursively with the updated value of n, so it becomes f (4).
f (4) is called:
n is 4, which is less than 5.
n is updated to n+i, which becomes 7.
i is incremented to 4.
The function calls itself recursively with the updated value of n, so it becomes of f (7).
f (7) is called:
n is 7, which is greater than or equal to 5.
The function returns n which is 7.
Now, the recursive calls start to end:
• The return value from f (7) was 7, and this is returned as is to the previous call, f (4).
• The return value from f (4) was 7. And this is returned as is to the previous call, f (2).
• The return value from f (2) was 7, and this is returned as is to the first call, f (1).
So, the value returned by f (1) is 7.
Hence, (c) is the correct answer.

105. What shall be the output of the following C program?

```
main ( )
{
int a = 0;
int b = 20;
char x = 1;
char y = 10;
if (a, b, x, y);
printf (''hello'');
}
```

(a) Logical error (b) Garbage value

(c) Hello (d) 20

CGPSC C.S. 2020 Set-A

Ans.(c): Because the if condition is true, the statement immediately following the if statement, which is print ("hello"); , will be executed. This will print the string "hello" to the console.

106. In C language, x – = y + 1 means:

(a) x = – x – y – 1 (b) x = x – y + 1

(c) x = x – **y** – 1 (d) x = – x + y + 1

CGPSC C.S. 2020 Set-A

Ans. (c) : In x – = y +1 is compound assignment operator the means x = x – (y +1). Due to operator precedence, the addition is performed first, and then the result is subtracted from x. this can be simplified to x = x – y – 1.

107. For the given code

```
for (i = 1; i ≤ 5; i + +)
if (i ! = 3)
printf (''%d'', i);
```

what shall be the output?

(a) 12345 (b) Error

(c) 1245 (d) 0000

CGPSC C.S. 2020 Set-A

Ans.(c): The print ("%d", i); statement will print the value of i if condition is true.
• When i = 1, i ! = 3 is true, so 1 is printed.
• When i = 2, i ! = 3 is true, so 2 is printed.
• When i = 3, i ! = 3 is false, so nothing is printed.
• When i = 4, i ! = 3 is true, so 4 is printed.
Therefore option (c) is correct answer.

108. Find the output of

```
# include < stdio.h>
# define mt char
void main(){inti65;printf('size of(i) %d", size of(i));}
```

आउटपुट बताइए-

```
# include < stdio.h>
# define mt char
void main(){inti65;printf('size of(i) %d", size of(i));}
```

(a) size of(i) 2

(b) size of(i) = 1

(c) compile error/कम्पाइल त्रुटि

(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (c) : The provided code snippet contains several syntax errors that would prevent it from compiling successfully.

109. Find the output of # include <stdio.h> void main () { intk = 5; int*p = &k; mt **m & p; 6; printf ("%d". k); }

आउटपुट बताएँ

```
# include < stdio.h > void main () { intk = 5; int*p = &k; mt **m & p; 6; printf ("%d". k); }
```

(a) 5

(b) 6

(c) Junk value/जंक वैल्यू

(d) Compile time error/कम्पाइल टाइम त्रुटि

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (d) : The line mt **m & p; is syntactically mt is not a valid data type Complete time error It should be a valid data type like.

Sarkari Resource

110. What will be the value of C, in C = x++ + ++x + ++x + x, if x = 10 ?
C = x++ + ++x + ++x + x, में यदि x = 10 है, तो C का मान क्या है ?

(a) 48 (b) 45
(c) 46 (d) 49

KVS PGT C.S. 2013

Ans, (c) : C = X^{++} + ^{++}X + X + X Initial
1. $X^{++} \rightarrow$ returns 10, then X = 11
2. ++X → X = 12 , returns 12
3. X = 12
4. X = 12
How sum
C = 10 + 12 + 12 + 12
C = 46

111. What is the output of the following C program?
निम्न C प्रोग्राम का निर्गत (output) क्या होगा?

```
# include
int main ( )
{
  int index;
  for (index = 1; index < = 5; index ++)
  {
     printf ("%d", index);
     if (index = = 3)
                continue;
  }
}
```

(a) 1245 (b) 12345
(c) 12245 (d) 12354

LT. Grade C.S. 2018

Ans, (b) : In given C program, the continue statement skips the remaining statements in the current iteration of the loop and proceeds to the next iteration. In this program, there are no statements after the continue, so its presence doesn't affect the program's behavior.
The output is simply the numbers from 1 to 5 printed sequentially 12345.

112. What is the meaning of the following declaration in C programming language?
C प्रोग्रामिंग भाषा में निम्न declaration का क्या अर्थ है?

int (*p) [5];

(a) It will result in compile error because there should not be any parenthesis, i.e., "int*p[5]" is valid/यह कॉम्पाइल त्रुटि में परिणत होगी क्योंकि वहाँ कोई पैरेन्थेसिस नहीं होना चाहिए, अर्थात् "int*p[5]" सही है
(b) p is a pointer to 5 integers
p, 5 इंटीजर का पॉइंटर है
(c) p is a pointer to integer array
p, इंटीजर ऐरे का पॉइंटर है
(d) p is a pointer to an array of 5 integers
p, 5 इंटीजर के किसी ऐरे का पॉइंटर है

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Ans, (d) : p is a pointer. The parentheses around *p indicate that p is not just a pointer to an int, but a pointer to an array. [5] indicates that the array has 5 elements. So, the declaration int (*p) [5]; means that p is a pointer to an array of 5 integers.

113. Assume the following C variable declaration:
निम्नलिखित C variable declaration को मान लें:

int * A [10], B [10][10];

Among the following expressions, which will not give compile time errors if used as left-hand side of assignment statements in a C program?
निम्न अभिव्यक्तियों में से कौन-सा/से कम्पाइल टाइम त्रुटि नहीं देगा/देंगे, यदि एक C प्रोग्राम में लेफ्ट-हैंड असाइनमेंट स्टेटमेन्ट की तरह प्रयोग किया जाए?

I. A[2]
II. A[2][3]
III. B[1]
IV. B[2][3]

(a) I, II and IV only/केवल I, II एवं IV
(b) II, III and IV only/केवल II, III एवं IV
(c) II and IV only/केवल II और IV
(d) IV only/केवल IV

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Ans, (a) : Among expression I. A[2], II. A[2] [3], and IV. B[2] [3], only III B [1] will yield a compile time error when used as the left-hand side of assignment statements. Arrays aren't assignable in C. The rest are valid assignments referencing pointers or specific array elements.

114. Consider the following program segment in C programming language :
C प्रोग्रामिंग भाषा के निम्न प्रोग्राम-खण्ड पर विचार करें :

```
i = 6720; j = 4;
while ((i % j) == 0)
{i = i / j; j = j + 1;}
```

On termination, j will have the value
समाप्ति पर j का मान होगा

(a) 4 (b) 8
(c) 9 (d) 6720

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Ans, (c) : Analyze the given C program segment.
- Initially, i = 6720 and j = 4
- The while loop will continue as long as (i%j) = = 0, meaning i is divisible by j.

First iteration (j = 4):
- i % 4 = = 6720 % 4 = = 0 (true),

so the loop executes.
- Now, i = i/j = 6720/4 = 1680
- j = j + 1 = 4 + 1 = 5

Sixth iteration (j = 9):
- i % 9 = = 1 % 9 = = 0 (false)

So the loop terminates.
Hence, the loop terminates when j = 9, so the value of j at termination is 9.

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115. Consider the following segment of C code :
C कोड के निम्न खण्ड पर विचार करें :

```
int j, n;
j = 1
while (j < = n) j = j*2;
```

The number of comparisons made in the execution of the loop for any n > 0 is
किसी भी n > 0 के लिए लूप के निष्पादन में की गई तुलनाओं की संख्या है

(a) $\lceil \log_2 n \rceil + 1$ (b) n
(c) $\lceil \log_2 n \rceil$ (d) $\lfloor \log_2 n \rfloor + 1$

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Ans, (d) : The loop continues while $j \le n$. Each iteration of the loop multiplies j by 2.
j After k iterations:
The value of j after k iterations can be represented as $j = 2^k$.
To find the number of interations k when the loop stops, we need to solve:

$$2^k \le n$$

Taking the logarithm (base2) on both sides

$$k \le \log_2(n)$$

Since k must be an integer, the maximum number of complete iterations is

$$k = \lfloor \log_2 n \rfloor + 1$$

Thus, the number of comparisons made in the execution of the loop is $\lfloor \log_2 n \rfloor + 1$

116. What will be the output of the following?
निम्न का निर्गत (output) क्या होगा?

```
main ( )
{
 int a = 'A';
 printf ("%d", a);
}
```

(a) A
(b) a
(c) 65
(d) Compilation error/कम्पाइलेशन त्रुटि

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Ans, (c) : The variable a is declared as an int and initialized with the value 'A'. In C, characters are represented by their ASCII values. The ASCII value for the character 'A' is 65. The output of the printf function will be 65.

117. What will be the output of the following program?
निम्न प्रोग्राम का निर्गत (output) क्या होगा?

```
int f (int x)
{
 static int y;
y + = x;
return (y);
}
main ( )
{
  int a, i;
for (i = 0; i < 6; i++)
a = f (i);
printf ("%d", a);
}
```

(a) 6 (b) 8
(c) 10 (d) 15

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Ans, (d) : The function f uses a static int y. The static keyword ensures that the variable y retains its value between function calls. The loop in the main function calls f six times with values from 0 to 5, and the return value of f is assigned to a each time.
So, after the last iteration 'a' will be equal to 15. The final output will be 15.

118. The program fragment
प्रोग्राम फ्रैगमेन्ट

```
int a = 5, b = 2;
printf ("%d", a ++ + ++ b);
```

(a) Prints 7/प्रिन्ट करती है 7
(b) Prints 8/ प्रिन्ट करती है 8
(c) Prints 9/ प्रिन्ट करती है 9
(d) None of the above/उपर्युक्त में से कोई नहीं

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Ans, (d) : The expression a +++++ b will cause a syntax error, and the program will fail to compile. So correct answer is none of the above.

119. The number of 3's in the output of the program is
प्रोग्राम के निर्गत (output) में अंक 3 की संख्या होगी

```
main ( )
{
        printf ("1"); fork ( );
        printf ("2"); fork ( );
        fork ( ); printf ("3");
}
```

(a) 16 (b) 8
(c) 4 (d) 2

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Ans, (b) : fork ( ) - Create a new child process which runs concurrently with the parent. Each call to fork ( ) doubles the number of processes.
1. printf (" 1") - This is executed only by the original process, so 1 is printed once.
2. fork ( ) - Two processes now exist, parent P_1 and child C_1.
3. printf ("2") - Both processes (P_1 and C_1) execute this, so 2 is printed twice.
4. fork ( ) - Each of the two processes (P_1 and C_1) creates a child process, resulting in 4 processes.

P_1, C_1, child of P_1 (C_2) and child of C_1(C_3).
5. After the third fork ( ) - Each of the four processes forks again, creating 8 processes in total.
6. printf ("3") this statement is executed by all 8 processes, so 3 is printed 8 times.

120.

```
#include <stdio.h>
int main()
{
int i = 5, j = 10, k = 15;
printf(\"%d\n", sizeof(k/=i+j));
```

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```
printf(\"%d\n", k);
return 0;
}
```

Assume size of an integer as 4 bytes. What is the output of above program?/मान लीजिए कि पूर्णांक का आकार 4 बाइट्स है। उपरोक्त प्रोग्राम का आउटपुट क्या है?

(a) 2 1
(b) 4 1
(c) 4 15
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans, (c) : Size of (k/ = i + j) The expression inside the size of operator returns size of (int) because the result of the expression will be an integer. Since the expression is not evaluated, the value of k will not change.

121. What is the output of the following code snippet?
निम्न संकेत (Code) का निर्गत (Output) क्या है?

```
#include <stdio.h>
#define var 3
  void main ( )
{
  short num [3] [2] = {3, 6, 9, 12, 15, 18};
  printf ("%d %d", *(num+1) [1],
                **(num + 2));
}
```

(a) 12 15 (b) 15 12
(c) 15 15 (d) 12 12

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Ans, (a) : The output of given C language code is 12 15 .
short num [3][2] = {3,6,9,12,15,18}
than this initializes a 2D array with:
num [0] [0] = 3 num [1] [0] = 9 num [2] [0] = 15
num [0] [1] = 6 num [1] [1] = 12 num [2] [1] = 18
Concept 1:
* (num +1) [1] num is a pointer to array of 2 short ints.
* (num +1) points to num [1] → {9,12}
* (num +1) [1] → element at index 1 of num [1] →12
Concept 2:
** (num +2) num +2 → pointer to num [2] → {15, 18}
* (num +2) → num [2]
** (num +2) → num [2][0] = 15

122. Consider following C statement:
int y = 77, x = 10;
x = 5/9*(y – 32);
value of x will be
निम्न C स्टेटमेंट पर विचार करें : int y = 77, x = 10; x = 5/9*(y – 32); x का मान क्या होगा?

(a) 10 (b) 45
(c) 32 (d) 0

RPSC C.S. 2016 (Paper-I)

Ans, (d) :
int y = 77, x = 10;
x = 5/9*(y – 32);
Step by step explanation–
(1) Variable Declaration and Initialization:
int y = 77, x = 10;
This declares two integer variables:
→ y is initialized to 77
→ x is initialized to 10
(2) Expression Evaluation:
x = 5/9*(y – 32);
Let's evaluate this line by line:
• y – 32 = 77 – 32 = 45
• 5/9 = 0, in integer division.
Because both 5 and 9 are integers, and integer division truncates the result to an integer so 5/9 = 0
• Therefore : x = 0 * 45 = 0
Final value of x :
x = 0

123. Consider following declaration in C program : I : void x ; II : void * p ; Choose true statement about I and II:
C प्रोग्राम में निम्न दो डिक्लेरेशन पर विचार करे I : void x ; II : void * p ; I एवं II के लिए सत्य कथन चुनिए:

(a) I - Compilation Error, II – Void
I - कंपाइलेशन एरर, II – वॉयड
(b) I - Void, II - Compilation Error
I - वॉयड, II - कंपाइलेशन एरर
(c) I - Void, II - Void /I - वॉयड, II - वॉयड
(d) I - Compilation Error, II - Compilation Error
I - कंपाइलेशन एरर, II - कंपाइलेशन एरर

RPSC C.S. 2016 (Paper-I)

Ans, (a) : I : void x ;
• Invalid
• In C, void means "no type" or "nothing". You cannot declare a variable of type void, because it doesn't have a size or value.
• So, void x; is a compilation error.
II : void * p;
• Valid
• This declares a pointer to void, i.e. a generic pointer.
• Void * is used to point to any data type.
• So, void * p; is perfectly valid.

124. Consider following array declaration in C : int a1[ ] = {2, 3, 5}; int a2[ ] ; int a3[0] ; int a4[1] ; float a5[3.5] ; The valid declaration are:
C में निम्न एरे डिक्लेरेशन पर विचार करे : int a1[ ] = {2, 3, 5}; int a2[ ] ; int a3[0] ; int a4[1] ; float a5[3.5] ; मान्य डिक्लेरेशन है:

(a) a1, a4 (b) a1, a3, a4
(c) a1, a4, a5 (d) a1, a2, a3

RPSC C.S. 2016 (Paper-I)

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Ans, (a) : int a1[ ] = {2, 3, 5}; // valid
• array size is inferred from initializer
int a2[ ] ; // Invalid
• size or initializer must be provided.
int a3[0] ; // Invalid in standard C
• zero-length arrays are not allowed (except as GCC extension)
int a4[1] ; // valid
• declares an array of 1 integer.
float a5[3.5] ; // Invalid
• array size must be an integer constant.

125. What will be output of : if (printf("%d", printf("Hello"))) printf("India"); else printf("BYE");
निम्न का आउटपुट क्या होगा : if (printf("%d", printf("Hello"))) printf("India"); else printf("BYE");
(a) Hello India (b) Hello5India
(c) Hello BYE (d) BYE
RPSC C.S. 2016 (Paper-I)

Ans, (b) : if (printf("%d", printf("Hello"))) printf("India"); else printf("BYE");
Step by Step evaluation:
(1) printf("Hello");
→ This prints Hello.
→ printf returns the number of characters printed, which is 5.
(2) printf("%d", 5);
→ So now the statement becomes.
printf("%d", 5), which prints 5.
→ This also returns the number of character printed, which is 1.
(3) So, if (1)– this condition is true, hence the else part is skipped.
(4) printf("India") runs and prints India.
Final output = Hello5India

126. Consider the following C code segment :

```
void main () {
int x = 10, y = 10;
F (&x, 10);
printf("%d, %d",x,y);
}
void F(int*x, int y) {
*x = 20;
y = 20;
return;
}
```

Output of program will be:
निम्न C कोड सेगमेंट पर विचार करे : void main () {int x = 10, y = 10 ; F (&x, 10); printf ("%d, %d",x,y);}void F(int*x, int y) {*x = 20; y = 20 ; return ;} प्रोग्राम का आउटपुट क्या होगा:
(a) 20, 20 (b) 20, 10
(c) 10, 20 (d) 10, 10
RPSC C.S. 2016 (Paper-I)

Ans, (b) :

```
void main () {
int x = 10, y = 10;
F (&x, 10);
printf ("%d, %d",x,y);
}
void F(int*x, int y) {
*x = 20;
y = 20;
return;
}
```

Step-by-Step Execution:
• int x = 10, y = 10; → initializes x and y to 10
• F(&x, 10); passes the address of x and the value 10 to function F.
Inside F–
• *x = 20; → This modifies the value at the address pointed by x, i.e., the original x in main() becomes 20.
• y = 20; → This modifies the local copy of y in function F, not the original y in main().
After function call, back in main():
• x has become 20 (changed via pointer)
• y is still 10 (unchanged)
Final output : 20, 10

127. Consider following C code segment : int x = 10, y = 5; while (x > 0){x – ; y – ; if (y > 0) continue ; printf ("India"); if (y == -3) break ;}How many time India will be displayed?
निम्न C कोड सेगमेंट पर विचार करे : int x = 10, y = 5; while (x > 0){x – ; y – ; if (y > 0) continue ; printf ("India"); if (y == -3) break ;} India कितनी बार डिस्प्ले होगा?
(a) 4 (b) 10
(c) 5 (d) 6
RPSC C.S. 2016 (Paper-I)

Ans, (a) : Initial values : x = 10, y = 5
we loop as long as x > 0. In each iteration:
• x-- and y-- happen.
• If y > 0, then 'continue' skips the printf' and the 'break'
• If y < = 0 then printf ("India") is executed
• If y = = -3, then break ends the loop.
Therefore total "India" prints = 4 times.

128. What is the output of following C program?

```
# include <stdio.h>
main( )
{
int i, j, x = 0;
for (i = 0; i < 5; + + i)
for (j = 0; j < i; + + j)
{
x + = (i + j – 1);
break;
}
printf ("%d", x);
}
```

(a) 6 (b) 5
(c) 4 (d) 3
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Ans. (a) : Key Point :-

- In a loop, when a break statement is encountered, the control will come out of that loop immediately, and nothing after the break statement is executed.
- In this question, the J- loop will be executed only once because there is a break statement in it.

So, for is i, the j-loop will execute only for j = 0.

⇒ Round 1 :

Initialization :- Value of i = 0, j = 0 and x = 0

As i = 0, j = 0 and $j \not< i$

So, nothing will be printed there.

Round 2 :

i = 1, j = 0, x = 0

Round 3 :

i = 2, j = 0, x = 1

Round 4 :

i = 3, j = 0, x = 2

Round 5 :

i = 4, j = 0, x = 3

Total: 0+1+2+3 = 6

Hence, x = 6 is the correct answer.

129. Consider the following C-code fragment running on a 32-bit x86 machine :

```
typedef struct {                S B[10];
    union {                     S * p = &B[4];
unsigned char a;                S * q = &B[5];
unsigned short b;               p → U.b = 0x1234;
}U;                             /* structure Stakes 32 – bits */
unsigned char c;
}S :
```

If M is the value of q – p and N is the value of ((int) & (p → c)) – ((int)P), then (M, N) is

(a) (1, 1)
(b) (3, 2)
(c) (1, 2)
(d) (4, 4)

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Ans. (c) : In this, size of short int = 4 bytes

S * p = & B [4]; // p is having the address of B [4]

S * q = & B [5]; // q is having the address of B [5]

Suppose the address B[4] address is 116 [take base address as 100].

So B [5] will come as 120

q – p will be = (address of B[5] – address of B[4])

q – p = (120 – 116)/4 = 4/4 = 1

Also M is equal to q – p (given in the question)

So M = 1

Now, we have to find the value of N =

((int) & (p → c)) – ((int) p),

Here, c is a char and it is type casted to int ((int) & (P → C)) – ((int) p // address difference between p's starting address and c address will be of 2 bytes only.

So N becomes 2

∴ (M, N) = (1, 2)

130. Consider the following method :

```
int f(int m, int n, boolean x, boolean y)
{
    int res = 0;
    if(m < 0) {res = n – m;}
    else if (x || y) {
                    res = –1;
                    if(n = = m) {res = 1;}
            }
            else {res = n;}
        return res;
} /* end of f*/
```

If P is the minimum number of tests to achieve full statement coverage for f(), and Q is the minimum number of tests to achieve full branch coverage for f(), then (P, Q) =

(a) (3, 2)
(b) (3, 4)
(c) (2, 3)
(d) (4, 3)

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Ans. (b) : Let us first consider the full branch coverage. Here there are 4 possible cases of branching which are:-

Case-1:- The first "if" condition is true, which happen when m<0

Case-2:- The else if condition is true which happen when m>0.

Case-3:- The second "if" condition is true, which happen when P=3;

Case-4:- The else if condition is true which happen when Q=4.

131. What does the following function f() in 'C' return?

```
int f(unsigned int N) { unsigned int counter = 0;
while(N > 0) { counter += N & 1;
N = N >> 1;} return counter == 1;}
```

(a) 1 if N is odd, otherwise 0
(b) 1 if N is a power of 2, otherwise 0
(c) 1 if the binary representation of N is all 1's, otherwise 0
(d) 1 if the binary representation of N has any 1's, otherwise 0

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Ans. (b) : 1 if N is a power of 2, otherwise 0

132. What is the output of the following 'C' program ? (Assuming little - endian representation of multi-byte data in which Least Significant Byte (LSB) is stored at the lowest memory address.)

```
#include <stdio.h>
#include <stdlib.h>
/* Assume short int occupies two bytes of storage
*/
int main ( )
{
        union saving
        {
            short int one;
            char two[2];
        };
        union saving m;
```

```
        m.two [0] = 5;
        m.two [1] = 2;
        printf(''%d, %d, %d\n", m.two [0], m.two
        [1], m.one);
}/* end of main */
```

(a) 5, 2, 1282 (b) 5, 2, 52
(c) 5, 2, 25 (d) 5, 2, 517

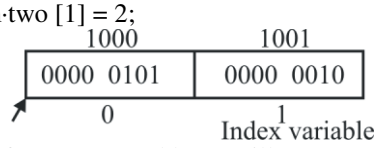
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Ans. (d) : Big-endian and little-endian are terms that describe the order in which a sequence of bytes is stored in computer memory.
Big-endian is an order in which the "big end" (most significant value in the sequence) is stored first (at the lowest storage address). Little-endian is an order in which the "little end" (least significant value in the sequence) is stored first.

Short in tone;
Char two [2]; } – 2 bytes

in union saving m; m is variable
Here, m·two [0] = 5; the 5 is not in single inverted comma, so it is numeric value or number
So, 5 in binary equivalent = 0000 0101
and 2 in binary equivalent = 0000 0010
Now we are using %d for printing the character.

m·two [0] = 5;
m·two [1] = 2;



In case of array "1" address will 1000 and 2nd will 1001.
% d will print decimal value of ·m· two [0] = 5; = 5
Now m· two [1] = 2; = 2
Now m· one



So, according to little endian definition
0000 0101 will be the lowest memory address
0000 0010 will be the most significant byte.
Now evaluate its value–

0000.00 1 0 0 000.0 1 0 1
2^9 2^8 2^7 2^2 2^1 2^0

Total will be = 517

133. What will be the output of the following code?

```
# include < stdio. h>
int main () {
int a, b, c;
a = ox10; b= o10;
c = a + b;
printf ("%d",c);
return 0;
}
```

(a) 20 (b) 24
(c) Garbage (d) error

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Ans. (b) : In C language, if a number starts with '0' it is octal and if it starts with '0x' it is hexadecimal. In the given code '0 × 10' is hexadecimal representation of 16, and '010' is octal representation of 8. Adding them together '16 + 8 = 24'. So the output of given code is 24.

134. What is the output of following code?

```
main ()
{static float a[ ] = {13. 24, 1.5, 4.5, 5.4, 3.5}
float *j, *k;
j = a;
k = a + 4
j = j *2;
k = k/2
printf ("%f%f', *j, *k);
}
```

(a) 13. 25, 4.5
(b) 1.5, 3.5
(c) 13.24, 1.5, 4.5, 5.4, 3.5
(d) Illegal use of pointer in main function

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Ans. (d) : Given code has several syntax errors. The code indeed contains illegal use of pointers in the 'main ()' function. Specifically, the lines j = j * 2; and 'k = k/2; are attempting to perform arithmetic operations directly on pointer variables, which is not allowed in C.

135. How will you free the memory allocated by the following program?

```
# include <stdio.h>
# include <stdib.h>
# define MAXROW 3
# define MAXCOL 4
int main ()
{
int** p,i,j;
P = (int **) malloc (MAXROW * size of (int*));
return ();
}
```

(a) memfree (int p); (b) dealloc (p);
(c) malloc (p, 0); (d) free (p);

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Ans. (d) : This code first allocates memory for an array of pointers 'p' then allocates memory for each row pointed to by p[i]; Finally, it frees the memory for each row and then frees the memory for the array of pointers itself.

136. Which of the following is not a palindromic subsequence of the string "ababcdabba"?

(a) abcba (b) abba
(c) abbbba (d) adba

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Ans. (d) : A palindromic subsequence is a sequence of characters within a larger sequence that reads the same backward as forward. Unlike a palindrome, which must be read in order, a palindromic subsequence can have characters in any order as long as they appear consecutively in the same order as in the original sequence. The given string is "ababcdabba". From this string, 'adba' is not a palindromic subsequence.

137. What is the output of the following program?

```
#include<stdio.h>
int main( )
{ int i = 3;
While (i – –)
```

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```
{ int i = 10;
  i – – ;
printf("%d", i);
  }
printf("%d", i);
  }
```

(a) 990 (b) 9990
(c) 999 – 1 (d) 99 – 1

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Ans. (c) : In this program, there are two variables named 'i'. One is declared in the outer scope of the 'main' function and initialized to 3. The other one is declared inside the 'while' loop and initialized to 10.
During the execution of the while loop, the inner 'i' (initialized to 10) is decremented by 1 and printed as 9, and then the loop condition is checked with the outer 'i'. This loop continues until the outer 'i' becomes 0. After the loop, the outer 'i' is decremented one more time due to the post-decrement in the loop condition, so it becomes –1. Finally, the outer 'i' is printed outside the loop.
So, the output of the program will be 999-1.

138. What is the output of the following program?

```
# include <stdio .h>
# define SQR(x) (x*x)
      int main ( )
{ int a, b = 3;
a = SQR (b + 2);
printf("%d", a);
return 0;
}
```

(a) 25 (b) 11
(c) Garbage value (d) 24

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Ans. (b) : In this program, The macro 'SQR (x)' is defined as 'x * x'. In the ' main () function, 'b' is assigned the value 3. Then 'a' is assigned the result of 'SQR (b + 2)'.
So, 'a' becomes 'b + 2 * b + 2', which is
3 + 2 × 3 + 2
= 3 + 6 + 2 = 11
So, the output of the program will be 11.

139. Consider the following C code :

```
# include <stdio.h>
int temp = 0;
int fun (int x, int y)
  {
    int z; temp + +;
    if (y = 3) return (x*x*x);
    else {
          Z = fun (x, y/3);
          return (z*z*z);
          }
 }
int main ( )
 {
     fun (4, 81);
     printf("%d", temp);
 }
```

What will be the output of the above code ?

(a) 3 (b) 4
(c) 5 (d) 6

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Ans. (b) : Analyze the given C code–
(i) The function fun is called with arguments (4, 81) in the main function.
(ii) In the first call, y is 81, since y is not equal to 3, it goes to the else block, increments temp (temp becomes 1), and makes a recursive call to fun (4, 27).
(iii) In the second call, y is 27. Again, Since y is not equal to 3, it goes to the else block, increments temp (temp becomes 2), and makes a recursive call to fun (4, 9).
(iv) In the third call, y is 9. Again, Since y is not equal to 3, it goes to the else block, increments temp (temp becomes 3), and makes a recursive call to fun (4, 3).
(v) In the fourth call, y is 3. This time, y is equal to 3 so it increments temp (temp becomes 4) and returns 4 * 4 * 4, but we are only interested in the value of temp.
Finally, the value of temp is printed, which is 4.

140. Consider the following code segment:

```
int arr [ ] = {0, 1, 2, 3, 4};
int i = 1, *ptr;
ptr = arr + 2;
```

arrange the following printf statements in the increasing order of their output.

```
(A) printf ("%d", ptr[i]);
(B) printf ("%d", ptr[i + 1]);
(C) printf ("%d", ptr[- i]);
(D) printf ("%d", ptr[- i + 1]);
```

Choose the correct answer from the options given below:

(a) (C), (A), (B), (D)
(b) (C), (D), (A), (B)
(c) (D), (A), (B), (C)
(d) (A), (B), (D), (C)

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Ans. (b) : Let's analyze the code segment.
int arr [] = {0,1,2,3,4}: - This creates an array with 5 elements.
int i = 1, * ptr:- This declare an integer i and a pointer ptr.
Ptr = arr + 2; - This assigns 'ptr' to the address of arr [2] (value 2).
Now let's evaluate the printf statement
(A) printf ("% d", ptr [i]);
prt [i] is equivalent to * (ptr +i), which is * (arr + 2 + 1) = arr [3] = 3
(B) printf ("%d", ptr [i + 1]); - ptr [i + 1] is equivalent to * (ptr + i + 1), which is * (arr + 2 + 1 + 1) = arr [4].
(C) printf ("%d", ptr [–i]); - ptr [–i] is equivalent to * (ptr – i), which is * (arr + 2 – 1) = arr [1] = 1
(D) printf ("%d", ptr [–i + 1]); - ptr [–i + 1] is equivalent to * (ptr – i + 1), which is * (arr + 2 – 1 + 1) = (arr + 2) = arr [2] = 2
So, the correct order of the printf statement in terms of their output values is (C), (D), (A), (B). The correct answer is (b).

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141. What will be the output of the following C code?

```
#include<stdio.h>
void main ()
{
    int arr [5] = {10, 20, 30, 40, 50};
    int *p = (int*) (&arr + 1);
    printf ("%d %d", *(arr + 1), *(p – 1));
}
```

(a) 10 50 (b) 20 50
(c) 30 40 (d) 20 40

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : To determine the output of the given C code, let us analyze it step by step.

1. Declaration of arr– arr is an array of 5 int {10, 20, 30, 40, 50}. arr refers to the base address of the array, and &arr refers to the address of the entire array.
2. Pointer P- &arr points to the whole array, not just the first element. &arr + 1 moves the pointer to one unit beyond the array i.e. after the 5th element. Casting &arr + 1 to (int*) gives a pointer to the memory location just beyond the array but in terms of int.
3. Dereferencing *(arr+1), arr+1 is the address of the second element in the array.
   *(arr+1) fetches the value at the second position, which is 20.
4. Dereferencing *(p –1), p points to the address just beyond the array. p–1 points to the last element of the array (since p was moved beyond the array). *(p–1) fetches the values at the last position, which is 50.

Thus, the print statement will output 20 50.

142. Consider the following program :

```
#include < stdio.h>
main ( )
{
  int i, inp;
  float x, term = 1, sum = 0
  scanff ("% f", & inp, &x);
  for(i = 1, < = inp; i ++)
  {
        term = term * x/t;
        sum = sum + term;
  }
  printf("Result = %f\n", sum);
}
```

The program computes the sum of which of the following series?

(a) $x + x^2/2 + x^3/3 + x^4/4 + \ldots.$
(b) $x + x^2/2! + x^3/3! + x^4/4! + \ldots.$
(c) $1 + x^2/2 + x^3/3 + x^4/4 + \ldots.$
(d) $1 + x^2/2! + x^3/3! + x^4/4! + \ldots.$

UGC NET C.S. December-2015 (Paper-II)

Ans : (b) i = 1, term = 1, term = 1*x/1, sum = 0 + x/1
term = 1, sum = 0
Iteration1 (i = 1):
term = 1 * x/1 = x
sum = 0 + x = x
Interation 2 (i = 2):
term = x* x/2 = x/∧ 2/2
sum = x* x∧ 2/2
Interation 3 (i = 3):
term = (x∧ 2/2)* x/3 = x∧ 3/(2*3) = x∧ 3/6
sum = x + x∧ 2/2 + x∧ 3/6
Interation 4 (i = 4):
term = (x∧ 3/6)* x/4 = x∧ 4/(6*4)
= x∧ 4/24
sum = x + x∧ 2/2 + x∧ 3/6 + x∧ 4/24
which is better shown x/1 ! + x∧ 2/2! + x ∧/3! + ...

143. Assume that the program 'P' is implementing parameter passing with 'call by reference'. What will be printed by following print statements in P?

```
Program P( )
{
    x = 10;
    y = 3;
    funb (y, x, x); print x; print y;
}
funb (x, y, z)
{
    y = y + 4;
    z = x + y + z;
}
```

(a) 10, 7 (b) 31, 3
(c) 10, 3 (d) 31, 7

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Inside funb ( ), *x* will actually refer to *y* of main ( ); and *y* and *z* will refer to *x* of main ( ).
The statement *y* = *y* + 4;
will result in 14 and statement.
z = *x* + *y* + *z* will make
z = 3 + 14 + 14
= 31
(because *y* and *z* point to same variable *x* of main).
Since *z* refers to *x* of main ( ), main will print 31. *z* will be stored back in *x* = 31 and *y* will remain as it is, because there was no change for it. Then y = 3.
Printed output is 31, 3.

144. While (87) printf ("computer");
The above C statement will

(a) print "computer" 87 times
(b) print "computer" 0 times
(c) print "computer" 1 times
(d) print "computer" infinite times

UGC NET C.S. June-2014 (Paper-II)

Ans : (d) Because, Parameter of a while statement in a languages takes two values true in the form of non-zero and false in the form of zero. Here 87 is a non-zero value and hence the while block gets true for infinite times.

145. What will be the output of the following 'C' code?

```
main ( )
{ int x = 128 ;
  pritf("\n%d", 1 + x ++);
}
```

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(a) 128 (b) 129
(c) 130 (d) 131

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) main ( )
{
int x = 128;
printf ("\n%d", 1 + x^{++})}
In the following C program. There is post increment operation: So, printf will print 1 + 128 as output. ++ will increment the x value i.e. x = 129.

146. What is the output of the following program? (Assume that the appropriate directive are included and there is no syntax error)

```
main ()
{ char S[ ] = "ABCDEFGH",
  print f ("%C", * (&S[3]));
  print f ("%s", S + 4);
  print f ("%u", S);
  /* Base address of S is 1000*/ }
```

(a) ABCDEFGH1000
(b) CDEFGH1000
(c) DDEFGHH1000
(d) DEFGH1000

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : S is pointer of character array:

1000	1001	1002	1003	1004	1005	1006	1007	1008
A	B	C	D	E	F	G	H	\0

S[3] is character at index 3 of array S which is 'D' and & (S[3]) is address of 'D' but "%c" printed character at address of 'D' which is 'D' itself.
(S + 4) escapes 4 character of array of S then remaining printed till null which string is "EFGH". "%u" printed address of array S which is given as 1000.
Therefore final printed is "DEFGH1000".

147. What is the output of the code written in C?

```
void main()
{
int k, v;
k = (v = 20, v + 2);
printf("%d/",k);
}
```

C में लिखे कोड का आउटपुट क्या है?

```
void main()
{
int k, v;
k = (v=20, v+2);
printf("%d/",k);
}
```

(a) 2 (b) 20
(c) 22 (d) Error

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Expression k = (v = 20 , v + 2);
Here, , is the comma operator. The comma operator evaluates the expression from left to right. The value of the last expression becomes the result of the whole expression.
v = 20 → assigns 20 to v.
v+2 → adds 2 to v → 20+2 = 22
So, K = 22
printf("%d/", k);
since k is 22, the output will be 22.

148. What is the output of the code written in C?

```
void main()
{
printf("%d",7% –3);
printf("%d", –7%3);
}
```

C में लिखे कोड का आउटपुट क्या है

```
void main()
{
printf("%d",7% –3);
printf("%d", –7% 3);
}
```

(a) 1–1 (b) –1–1
(c) –1 1 (d) 1 1

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a): In C, the result of a % b has the same sign as the sign of numerator.
printf("%d", 7%–3); prints 1 because remainder will come 1 and sign is positive as sign of numerator is positive.
printf("%d", –7%3); prints -1 because sign of numerator is negative.
The printf statements print without spaces.
So the output is 1–1

149. Which inbuilt function in C is used to read string system console?
C में कौन सा इनबिल्ट फंक्शन सिस्टम कोनसोल से स्ट्रिंग को पढ़ने के लिए उपयोग किया जाता है?

(a) Puts() (b) Cgets()
(c) Getche() (d) Getchar()

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : Cgets() is a function used in older versions of C to read a string directly from the console. It stores the input in a buffer and returns a pointer to the string.
Puts()– Used to input a string to the console, not for reading input.
Getche()– Reads a single character from the console and echoes it (not a string).
Getchar()– Read a single character from stdin (not a string).

150. What is the output of the following code written in C?
C में लिखे निम्नलिखित कोड का आउटपुट क्या है?

```
void main()
{
printf("%d%d%d",62,062,0x62);
}
```

(a) 506298 (b) 629850
(c) 625098 (d) 986250

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Ans. (c) : The given code-
void main ()
{
printf("%d%d%d", 62, 062, 0x62);
}
Here's a breakdown of now the printf statement will be evaluated.
62– This is a decimal integer, so its value is 62.
062– This is an octal integer (Indicated by the leading zero) converting it to decimal
$(6 \times 8^1) + (2 \times 8^0) = 48 + 2 = 50$
0x62– This is a hexadecimal integer (indicated by the 0x). Converting it to decimal.
$(6 \times 16^1) + (2 \times 16^0) = 96 + 2 = 98$
The printf statement uses the format string "%d%d%d". This means it will print the three integer values one after the other, without any spaces or other separators in between.
Therefore, the output will be the concatenation of the decimal values- 62, 50, 98 which results in 625098.

151. What is the output of the code written is C?
C में लिखे गए कोड का आउटपुट क्या है?

```
void main()
{
int k = 8, v = 8, i;
i = k/++v;
printf ("%d/n",i);
}
```

(a) 0 (b) 1
(c) 0.8 (d) Error/त्रुटि

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) :

```
Void main()
{
int k = 8, v = 8, i;
i = k/++v;
printf ("%d/n",i);
}
```

Use the order of operations (PEMDAS/BODMAS) to evaluate the expression k/++v and determine the value of I
Step-1
increment v–
• ++v means Pre-increment, so v is incremented to 8 + 1 = 9 before it's used in the division
Step-2
Perform the division–
• i = k/v becomes i = 8/9
• Since both k and v are integers, integers division is performed
• 8/9 results in 0 because the integer part of the quotient is taken.
Step-3
Print the value of i–
• Printf ("%d/n", i) will print the value of I, which is 9, followed by a newline character.
The output code is 0.

152. Correct the following C declaration Assume that objects of data type short, float and long occupy 2 bytes and 8 bytes respectively.
What is memory requirement for variable?
निम्नलिखित C डेक्लरेशन पर विचार करें। मान लें कि शॉर्ट, फ्लोट और लॉन्ग डेटा प्रकार के ऑब्जेक्ट क्रमशः 2 बाइट्स, 4 बाइट्स और 8 बाइट्स ग्रहण करते हैं। t वेरिएबल के लिए कितनी मेमोरी की आवश्यकता होगी?

```
struct {
short a[5];
union{
float v;
long z;}
u;
}t;
```

(a) 22 bytes/22 बाईट्स (b) 18 bytes/18 बाईट्स
(c) 24 bytes/24 बाईट्स (d) 10 bytes/10 बाईट्स

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) :

```
Struct {
Short a [5];
union {
float v;
long z;}
u;
}t;
```

The memory requirement for variable t in bytes
• short occupies 2 bytes
• float occupies 4 bytes
• long occupies 8 bytes
• The structure t contains a short array a[5] and a union u which contains a float v and a long z
Step-1 – Calculate the size of the array a has 5 elements of type short, which occupies 2 bytes each.
The size of a is 5×2=10 bytes
Step-2 – The union u contains a float (4 bytes) and a long (8 bytes)
The size of u is the maximum of these sizes which is 8 bytes.
Step-3 – The structure t contains the array a and the union u.
The size of t is the sum of the sizes of a and u, which is 10+8=18 bytes.
The memory requirement for variable t is 18 bytes.

153. Consider following conditional statement if (m = = 20 – 10 || n > 10). The order of execution of the following operations is
A. == **B. –**
C. || **D. >**
Choose the correct answer from the options given below :
(a) D, B, A, C (b) B, D, C, A
(c) B, D, A, C (d) A, B, D, C

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Ans. (c) : The correct order of execution follows operator precedence and associatively rules in C-like languages:
1- Sub-traction (–) → 20 – 10 is evaluated first.

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2- Greater than (>) → n > 10 is evaluated next.
3- Assignment (=) → Finally, the result of 20–10 is assigned to m
4- Logical OR (||) → The result of m = 10 and n > 10 are combined using ||.

154. Given i = 0, j = 1 k = –1
x = 0.5, y = 0.0
What is the output of given 'C' expression?
x * 3 & & 3 || j I k
(a) –1 (b) 0
(c) 1 (d) 2

UGC NET C.S. Augest-2016 Paper-II

Ans : (c) Given,
C expression with their values-
$= x * y < i + j \| k$
$= 0.5 * 0.0 < 0 + \| 1$
$= 0 < 0 + 1 \| - 1$
$= 0 < 1 \| - 1$
$= 1 \| - 1$
= 1 (returns boolean)

155. Given $i = 0, j = 1, k = -1$
$x = 0.5, y = 0.0$
What is the output of the following expression in *C* language?
x * y < i + j || k
(a) -1 (b) 0
(c) 1 (d) 2

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Given expressions are:
$i = 0, j = 1, k = -1, x = 0.5, y = 0.0$ and
$= x * y < i + j \| k$
$= (((x * y) < (i + j)) \| k)$
$= (((0.5 * 00) < (0 + 1)) \| -1)$
$= ((0 < 1) \| -1)$
= 1

156. The following statement in 'C'
int (*f( ))[ ]:
declares
(a) a function returning a pointer to an array of integers.
(b) a function returning an array of pointers to integers.
(c) array of functions returning pointers to integers.
(d) an illegal statement.

UGC NET C.S. June-2016 (Paper-II)

Ans : (a) The following statement in 'C' int (*f( ))[ ]: a function returning a pointer to an array of integers.

157. What is the value returned by the function *f* given below when *n* = 100?
int *f*(int *n*)
{ if (*n* = = 0) then return *n*;
else
return *n* + *f*(*n* – 2);}
(a) 2550 (b) 2556
(c) 5220 (d) 5520

UGC NET C.S. June-2016 Paper-II

Ans : (a) The given function is a recursive function which takes starting value as 100. It keeps on adding the present value and the value obtained by calling the function $f(n-2)$ where n is current value. It stops at n = 0 where it returns 0. As a result, it forms a arithmetic series, where $a = 100$ and $d = 2$ and $l = 2$. So, sum of series is

$$S_{100} = \frac{n(a_1 + a_n)}{2} = \frac{50(2+100)}{2} = 2550$$

158. What is the output of the code written in C?
C में लिखे कोड का आउटपुट क्या है?
void main()
{
int k, val = 1;
k = ++val* ++ val;
printf ("%d",k);
}
(a) 1 (b) 4
(c) 9 (d) 16

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : 1- Val is initially
2- The first ++ val increments val to 2, and this value (2) is used in the multiplication
3- The second ++ val increments val to 3, and this value (3) is used in the multiplication
4- Thus, k be become 3*3 = 9

159. Which one is the reserved word in C language?
क़ौन सा C लैंग्वेज में एक आरक्षित शब्द हैं?
(a) Automatic/ऑटोमैटिक
(b) Integer/इन्टीजर
(c) Constant/कांस्टेंट
(d) Volatile/वोलेटाइल

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Ans. (d) : In C programming, reserved words (also know as key word) are predefined by the language and can not be used for naming variable, functions, or other identifiers. They represent various language constructs and have a specific meaning in the C language syntax.

160. In context of lexical analyser generator tool 'Lex', choose a valid statement about lex.yy.c:-
(a) It is a document defining tool 'Lex'
(b) It is a Lex language program
(c) It is compiler of tool 'Lex'
(d) It is C language program generated from a Lex program by Lex compiler

RPSC C.S. 2016 (Paper-II)

Ans, (d) : The Lex tool is a lexical analyzer generator that takes a specification file (typically with a extension) and generates a C program that can be used as a lexical analyzer.
Lex.yy.c– When run a Lex specification file through the Lex compiler, it generates a C program named lex.yy.c by default. Thus C program contains the implementation of the lexical analyzer based on the rules specified in the Lex file.

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05. Object Oriented Techniques

(i) Object Orientation

1. Which of the following is NOT an object oriented feature?
निम्नलिखित में से कौन एक ऑब्जेक्ट ओरिएंटेड फीचर नहीं है?

(a) Inheritance/इनहेरिटेंस
(b) Abduction/अबडक्शन
(c) Abstraction/अब्स्ट्रक्शन
(d) Polymorphism/पॉलीमॉर्फिज्म

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : Abduction is not an object oriented feature.
Object Oriented Feature–
(i) Class and Objects
(ii) Encapsulation
(iii) Inheritance
(iv) Polymorphism
(v) Abstraction
(vi) Dynamic Binding
These features make OOP a powerful and flexible approach to software development, enabling code reuse, scalability and maintainability.

2. What are abstract methods?
एब्स्ट्रैक्ट मेथड्स क्या होते हैं?

(a) Methods that are inaccessible to the outside world/ ऐसे मेथड जो बाहरी दुनिया से सुलभ नहीं होते
(b) Methods that have implementation details
ऐसे मेथड जिनमें कार्यान्वयन (implementation) होता है
(c) Methods without a body, defined in an abstract class/ ऐसे मेथड जो बॉडी के बिना होते हैं और एब्स्ट्रैक्ट क्लास में परिभाषित होते हैं
(d) Methods that cannot be inherited
ऐसे मेथड जिन्हें इनहेरिट नहीं किया जा सकता

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : An abstract method is a method that is simply declared but does not contain any body (i.e., code). Its original coding is done in a subclass of that class.

3. How the object interact with each other in object oriented programming?
ऑब्जेक्ट ओरिएंटेड प्रोग्रामिंग में ऑब्जेक्ट एक दूसरे के साथ कैसे इंटरैक्ट करते हैं?

(a) By ringing/रिंगिंग द्वारा
(b) By text/टेक्स्ट द्वारा
(c) By sending message/मैसेज भेजने द्वारा
(d) By transmitting signal/सिग्नल ट्रांसमिट करने द्वारा

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Ans. (c) : In Object-Oriented Programming (OOP), object interact with each-other through method and properties.
(i) Messages sending method
(ii) Properties and state
(iii) Instantiation and relationship.
(iv) Polymorphism
(v) Encapsulation
(vi) Inheritance.

4. Which is an Abstract data type?
निम्न में से कौन एक Abstract डेटा प्रकार है?

(a) Class (b) Double
(c) Int (d) String

Bihar STET C.S. 2020

Ans. (a) : An abstract Data Type (ADT) defines the behavior of a data type in terms of operations, without specifying its underlying implementation details. A class in object - oriented programming serves as a blueprint that encapsulates data (Attributes) and methods (Operations) and Provides an abstract view of an object, hiding the implementation details, making it an example of an ADT.

5. In which case it is mandatory to provide a destructor in a class?
किस मामले में किसी क्लास में डिस्ट्रक्टर का प्रयोग अनिवार्य होता है?

(a) Almost in every class/लगभग सभी क्लास में
(b) Class for which two or more than two objects will be created./क्लास, जिसके लिए दो या दो से अधिक सृजित किए जाएंगे।
(c) Class for which copy constructor is defined./क्लास, जिसके लिए कॉपी कन्स्ट्रक्टर है।
(d) Class whose objects will be created dynamically./क्लास, जिसके ऑब्जेक्ट डायनैमिकली क्रिएट किए जाएंगे।

KVS PGT C.S. 2018

Ans. (d) : A destructor is a member function that is invoked automatically when the object goes out of scope or is explicitly destroyed by a call to delete. A destructor has the same name as the class, preceded by a titled (~). When a class contains dynamic object then it is mandatory to write a destructor function to release memory before the class instance is destroyed this must be done to avoid memory leak.

6. What is the main purpose of using namespaces in programming?
प्रोगामिंग में नेमस्पेस का उपयोग करने का मुख्य उद्देश्य क्या है?

(a) To provide data encapsulation
(b) To organize code into logical groups
(c) To define abstract classes
(d) To handle exceptions

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : Namespaces provide a declarative region that helps organize identifiers (like variables, functions, and classes) into logical groups, preventing naming collisions, especially in larger projects with multiple libraries.

7. There are three main concepts that any language needs to support to be an object-oriented language. Which of the following is one of them?

Sarkari Resource

तीन मुख्य संकल्पनाएँ हैं जिन्हें किसी भी लैंग्वेज को, ऑब्जेक्ट-लैंग्वेज होने के लिए, समर्थन देने की आवश्यकता होती है। निम्नलिखित में से कौनसी उनमें से एक है?

I. Encapsulation/एनकैप्सुलेशन

II. Inheritance/इनहेरिटेंस

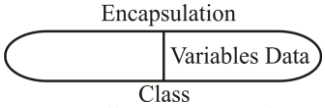
III. Polymorphism/पॉलीमॉर्फिज्म

(a) I and II (b) II and III
(c) I and III (d) I, II and III

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) : Object Oriented Programming– Object Oriented Programming System (OOPS) refers to languages that use objects in programming aims to implement real world entities like inheritance, abstraction (hiding), polymorphism etc in programming the main aim of OOPS is to bind together the data so that no other part of the code can access this data except that function.

(a) Encapsulation– Encapsulation is defined as the wrapping up of data under a single unit. It is the mechanism that binds together code and the data it manipulates in encapsulation, variables or data of a class are hidden.



(b) Inheritance– Allows new classes to inherit properties and methods from existing classes, promoting code reusability.

(c) Polymorphism– Allow objects of different classes to be treated as the same type, enabling single function to work with various objects.

8. Which of the following statements are true regarding C++?

(A) Overloading gives the capability to an existing operator to operate on other data types.

(B) Inheritance in object oriented programming provides support to reusability.

(C) When object of a derived class is defined first the constructor of derived class is executed then constructor of a base class is executed.

(D) Overloading is a type of polymorphism.

Choose the correct option from those given below:

(a) (A) and (B) only
(b) (A), (B) and (C) only
(c) (A), (B) and (D) only
(d) (B), (C) and (D) only

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) : Statement (A) : Overloading gives the capability to an existing operator to operate on other data types.

This statement is correct. Operator overloading is a type of polymorphism in which an operator is overloaded to give a different meaning or user defined meaning. It allows to change the way an operator can work. It gives the capability to an existing operator to operate on other data types.

Statement (B): Inheritance in object oriented programming provides support to reusability.

This statement is correct. When a class acquires the properties of another class, this is known as inheritance. The class which acquire the properties and method is known as derived class and other one is parent or base class. Inheritance another name for reusability.

Statement (C): When object of a derived class is defined, first the constructor of derived class is executed then constructor of a base class is executed.

This statement is incorrect constructor is a special member function of class that is automatically executed or called when objects of that class will be created. As when object or derived class is defined automatically derived class constructor and base class constructor will execute inside them.

Statement (D) : Overloading is a type of polymorphism.

This statement is correct. Operator is a type of polymorphism in which an operator is overloaded to give a different meaning or user defined meaning. It allows to change the way an operator can work.

9. Which of the statement are CORRECT?

(A) Constructors are invoked automatically when the objects are created.

(B) Constructors do not have return types, not even void and therefore they cannot return values.

(C) Constructors cannot be inherited though a derived class can call the base class constructors.

(D) Constructors can be declared as virtual.

Choose the correct answer form the options given below :

(a) (A), (B) and (D) Only
(b) (A), (B) and (C) Only
(c) (B), (C) and (D) Only
(d) (A), (C) and (D) Only

UGC NET C.S. December-2023

Ans. (b) : A constructor in programming is a special type of method that is automatically called when the objects are created constructors do not have return types not even void and therefore they cannot return values. It cannot be inherited though a derived class can call the base class constructors. Constructors can not be declared as virtual.

10. Constructors have _____ return type.

(a) void (b) char
(c) int (d) no

UGC NET C.S. August-2016 (Paper-III)

Ans. (d): A constructor resembles an instance method, but it differs form a method since it has no explicit return type, it is not implicity inherited and it usually has different rules for scope modifiers.

11. Which feature of OOP is indicated by the following code?

OOP की कौन-सी विशेषता निम्नलिखित कोड द्वारा इंगित की गई है?

```
class student{int marks;};
class topper: public student{int age; topper(int age) {this.age=age;}};
```

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(a) Encapsulation and inheritance
इनकैप्सुलेशन और इनहेरिटेंश
(b) Inheritance and polymorphism
वंशानुक्रम और बहुरूपता
(c) Polymorphism/बहुरूपता
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : The given code indicates (Encapsulation) and (Inheritance) two of the features of Object-Oriented Programming (OOP).

12. Which of the following features must be supported by any programming language to become a pure object-oriented programming language?
शुद्ध Object-Oriented Programming भाषा बनने के लिए किसी भी Programming भाषा को निम्नलिखित में से किस विशेषता का समर्थन करना चाहिए?

(a) Encapsulation
(b) Inheritance
(c) Polymorphism
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (d) : Core Features of Object-Oriented Programing (OOP) –

1. Class :-
- A blueprint or template for creating objects
- Example: class Car{ }

2. Object : -
- An instance of a class, representing real-world entities.
- Example: Car my car = new car ();

3. Encapsulation :-
- Wrapping data (variables) and functions (methods) together into a single unit (class)
- It also provides data hiding using access modifiers (Private, Public, etc.)

4. Abstraction :-
- Hiding unnecessary details and showing only the essential features.
- Achieved using abstract classes and interfaces

5. Inheritance :-
- Reusing Properties and methods of an existing class in a new class.
- **Example:** Class Dog extends Animal { }.

(ii) Encapsulation

13. Abstraction and encapsulation are fundamental principles that underlie the object oriented approach to software development. What can you say about the following two statements?

I. Abstraction allows us to focus on what something does without considering the complexities of how it works.
II. Encapsulation allows us to consider complex ideas while ignoring irrelevant detail that would confuse us.

(a) Neither I nor II is correct
(b) Both I and II are correct
(c) Only II is correct
(d) Only I is correct

UGC NET C.S. June-2017 (Paper-III)

Ans. (a) : Abstraction & Encapsulation are fundamental principles that underlie the object oriented approach to s/w development. Encapsulation allows us to focus on what something does without considering the complexities of how it works.
Abstraction allows us to consider complex ideas while ignoring irrelevant details would confuse us.
Here none is correct.

14. Module design is used to maximize cohesion and minimize coupling. Which of the following is the key to implement this rule?

(a) Inheritance (b) Polymorphism
(c) Encapsulation (d) Abstraction

UGC NET C.S. June-2015 (Paper-III)

Ans. (c)**:** We know that the golden rule in module design is used to maximize cohesion and minimize coupling.
- Encapsulation is the key to implement this rule.

(iii) Polymorphism

15. Allowing different derived classes to be used interchangeably though the interface provided by a common base class is called _____

(a) Derived polymorphism
(b) Compile time polymorphism
(c) Static polymorphism
(d) Run time polymorphism

UGC NET C.S. December-2022

Ans. (d) : Run time polymorphism: This is achieved through the use of virtual functions and is a fundamental concept in object oriented programming. At runtime the appropriate derived class's implmentation is called based on the actual type of the object being referred to through a pointer or reference to the base class. This allows for dynamic dispatch and enables the program to behave polymorphically choosing the appropriate method implementation based on the actual runtime type of the object. So option (d) is correct.

16. In C++, polymorphism requires :

(a) Inheritance only
(b) Virtual functions only
(c) References only
(d) Inheritance, Virtual functions and references

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (d) Basically polymorphism means having many forms. In C++, polymorphism requires Inheritance, Virtual functions and references, because polymorphism occurs when there is a hierarchy of classes and they are related by inheritance which include virtual functions and references.

Sarkari Resource

17. Match the following.
निम्नलिखित को सुमेलित करें।

i	**Over riding ओवरराइडिंग**	**1**	**Allows a function of operator to be given more than one definition किसी फंक्शन या ऑपरेटर को एक से अधिक परिभाषाएँ देने की अनुमति देता है**
ii	**Over loading ओवर लोडिंग**	**2**	**A function that is called to deallocate memory एक फंक्शन जिसे मेमोरी डीएलोकेट करने के लिए बुलाया जाता है।**
iii	**Destructor डिस्ट्रक्टर**	**3**	**The ability to change the definition of an inherited method इनहेरिटेड पद्धति की परिभाषा को बदलने की क्षमता**

(a) i - 1, ii - 3, iii - 2 (b) i - 3, ii - 2, iii -1
(c) i - 3, ii - 1, iii - 2 (d) i - 2, ii - 3, iii - 1

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The correct match

i	**Over riding**	**3**	**The ability to change the definition of an inherited method**
ii	**Over loading**	**1**	**Allows a function of operator to be given more than one definition**
iii	**Destructor**	**2**	**A function that is called to deallocate memory**

18. Polymorphism is feature of
Polymorphism .....की एक विशेषता है।
(a) Procedural programming
(b) SQL
(c) Non-procedural programming
(d) Object Oriented Programming

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : Polymorphism is a key feature of Object-Oriented Programming (OOP). This means that functions or operators with the same name can perform different tasks.

19. How can one implement the compile-time polymorphism in the C++ programming language?
कोई C++ प्रोग्रामिंग भाषा में संकलन-समय बहुरूपता को कैसे कार्यान्वित कर सकता है?
(a) By using the template
(b) By using the concepts of inheritance
(c) By using both the virtual functions and inheritance
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Compile time polymorphism is implemented using templates (which can be checked during compile time) to decide which function should be called.

120. Polymorphism is related to?
पॉलीमॉर्फिज्म किससे सम्बन्धित है?
(a) Structured programming
(b) Object Oriented programming
(c) NLP
(d) Machine Learning

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : Polymorphism is an object-oriented programming (OOP) feature in which different functions can have the same name, i.e., a function can take multiple forms.

21. Run-time polymorphism is achieved by _____.
रन-टाइम पॉलीमॉर्फ़िज़्म किससे प्राप्त किया जाता है?
(a) friend function/फ्रेंड फंक्शन
(b) virtual function/वर्चुअल फंक्शन
(c) operator overloading/ऑपरेटर ओवरलोडिंग
(d) function overloading/फंक्शन ओवरलोडिंग

KVS PGT C.S. 2018

Ans. (b) : A virtual function is a member function which is declared within a base class and is re-defined by a derived class. Virtual functions are mainly used to achieve run-time polymorphism.

22. Method over-riding can be prevented by using final as a modifier at
(a) the start of the class.
(b) the start of method declaration.
(c) the start of derived class.
(d) the start of the method declaration in the derived class.

UGC NET C.S. August-2016 (Paper-III)

Ans. (b): To disallow a method from being overridden, specify final as a modifier at the start of its declaration. Methods declared as final cannot be overridden.

23. When a method in a subclass has the same name and type signatures as a method in the superclass, then the method in the subclass _____ the method in the superclass.
(a) Overloads (b) Friendships
(c) Inherits (d) Over rides

UGC NET C.S. June-2016 (Paper-II)

Ans : (d) In a class hierarchy, when a method in a subclass has the name and type signatures as a method in its superclass, then the method in the subclass is said to over ride the method in the superclass.

24. In which UML diagrams, the objects participating in the interaction are shown at the top of the chart as boxes attached to a vertical dashed line?
कौन से UML आरेखों में अन्तः क्रिया में भाग लेने वाली वस्तुओं को चार्ट के शीर्ष पर एक ऊर्ध्वाधर डैश लाइन से जुड़े बॉक्स के रूप मे दिखाया जाता है?
(a) Sequence diagrams/सीक्वेंस डायग्राम
(b) Activity diagrams/एक्टिविटी डायग्राम

Sarkari Resource

(c) Use case diagrams/यूज केस डायग्राम
(d) Class diagrams/क्लास डायग्राम
(e) Not attempted/प्रयास नही किया

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : A sequence Diagram show how objects Interact step-by-step in time order. Objects are shown on the Top as boxes, and their lifeline is a vertical dashed line.

(iv) Object Oriented Design

25. A friend function can be used to
(a) avoid arguments between classes.
(b) allow access to classes whose source code is unavailable.
(c) allow one class to access an unrelated class.
(d) none of the above

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, unrelated class means the classes are not connected via inheritance. But a relation can be achieved between two unrelated classes through friend function.

26. Which of the following statements concerning Object-Oriented databases are FALSE?
(a) Objects in an object-oriented database contain not only data but also methods for processing the data.
(b) Object-oriented databases store computational instructions in the same place as the data.
(c) Object-oriented databases are more adapt at handling structured (analytical) data than relational databases.
(d) Object-oriented databases store more types of data than relational databases and access that data faster.

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) Objects in an object-oriented database contain not only data but also methods for processing the data-**Correct**.
Object-oriented databases store computational instructions in the same place as the data-**Correct**.
Object-oriented databases are more adapt at handling structured (analytical) data than relational databases-**Incorrect**
Object-oriented databases store more types of data than relational databases and access that data faster-**Correct**
So, option (c) is correct.

(v) Object Design

27. Which of the following statements is correct about the class?
Class के बारे में, निम्नलिखित में से कौन-सा कथन सही है?
(a) An object is an instance of its class
एक ऑब्जेक्ट अपने वर्ग का उदाहरण है।
(b) A class is an instance of its object
एक वर्ग अपने ऑब्जेक्ट का उदाहरण है।
(c) An object is the instance of the data type of that class/ऑब्जेक्ट उस क्लास के डेटा का उदाहरण है।
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : A class is a blueprint or template for creating objects. It defines the structure and behavior of objects belonging to that class. An objects is an instance of a class and inherits the attributes and methods defined in the class.

(vi) Object Oriented Programming Style

28. Which of the following is not true in case of public Inheritance in C++ ?
C++ में पब्लिक इन्हेरिटेंस के मामले में निम्नलिखित में से कौन सा कथन सत्य नहीं है?
(a) Each public member in the base class is public in the derived class./बेस क्लास में प्रत्येक पब्लिक मेंबर डेराइव्ड क्लास में पब्लिक होता है।
(b) Each protected member in the base class is protected in the derived class./बेस क्लास में प्रत्येक प्रोटेक्टेड मेंबर डेराइव्ड क्लास में प्रोटेक्टेड होता है।
(c) Each private member in the base class remains private in the base class./बेस क्लास में प्रत्येक प्राइवेट मेंबर बेस क्लास में प्राइवेट रहता है।
(d) Each private member in the base class remains public in the derived class./बेस क्लास में प्रत्येक प्राइवेट मेंबर डेराइव्ड क्लास में पब्लिक रहता है।

KVS PGT C.S. 2018

Ans. (d) : In public inheritance, when deriving a class from a public base class, public members of the base class become public members of the derived class and protected members of the base class become protected members of the derived class. A base class's private members are never accessible directly from a derived class, but can be accessed thorough calls to the public and protected members of the base class.

29. A generic function can be created using the following :
निम्नलिखित का उपयोग करके एक जेनेरिक फंक्शन बनाया जा सकता है:
(a) Inheritance/इनहेरिटेंस
(b) Data hiding/डाटा हाईडिंग
(c) Polymorphism/पॉलीमॉरर्फिज्म
(d) Template/टेम्प्लेट

HTET PGT C.S. 2021

Ans. (d) : Template (in C++) or Generics (in Java) allows us to write one function or class that can work with different data types.
Example:- In C++:

Sarkari Resource

30. In object-oriented programming, the mechanism that allows two related functions to share the same name is called
ऑब्जेक्ट-ओरिएंटेड प्रोग्रामिंग में, वह तंत्र जो दो सम्बन्धित कार्यों को समान नाम साझा करने की अनुमति देता है, कहलाता है

(a) Function over riding/फंक्शन ओवर रीडिंग
(b) Function over writing/फंक्शन ओवर राइटिंग
(c) Function over loading/फंक्शन ओवर लोडिंग
(d) Function over running/फंक्शन ओवर रनिंग
(e) Not attempted/प्रयास नहीं किया

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : Function overloading means having more than one function with the same name but different number or type of parameters.
Example:- Java .

31. What is meant by multiple inheritance ?
मल्टिपल इनहेरिटेंस से क्या तात्पर्य है?

(a) Deriving a base class from derived class
व्युत्पन्न क्लास से आधार क्लास प्राप्त करने
(b) Deriving a derived class from base class
आधार क्लास से व्युत्पन्न क्लास प्राप्त करना
(c) Deriving a derived class from more than one base class/एक से अधिक बेस क्लास से व्युत्पन्न क्लास प्राप्त करना
(d) None of the mentioned/उल्लिखित में से कोई नहीं

HTET PGT C.S. 2021

Ans. (c) : Multiple inheritance means a child class can get features (data and methods) from more than one parent class.

32. ..... refers to the procedure of representing essential features without including the background details.
..... बैकग्राउंड डीटेल्स को शामिल किए बिना आवश्यक विशेषताओं का प्रतिनिधित्व करने की प्रक्रिया को रिप्रेजेंट करता है।

(a) Abstraction/एब्सट्रैक्सन
(b) Encapsulation/इनकैप्सूलेशन
(c) Inheritance/इनहेरिटेन्स
(d) Polymorphism/पॉलिमारफिज्म

HTET PGT C.S. 30.07.2025

Ans.(a): Abstraction means showing only the Important details and hiding unnecessary details.

33. Which programming keys more attention on data rather than functions?
कौन-सी प्रोग्रामिंग कीज फंक्शन्स के बजाय डेटा पर अधिक ध्यान देती है?

(a) Structured Programming/स्ट्रक्चर्ड प्रोग्रामिंग
(b) Conventional Programming/कन्वेंशनल प्रोग्रामिंग
(c) Object Oriented Programming
ऑब्जेक्ट ओरिएंटेड प्रोग्रामिंग
(d) Procedural Programming/प्रोसिजरल प्रोग्रामिंग

HTET PGT C.S. 30.07.2025

Ans.(c): Object Oriented Programming (OOP) is a style of programming where data and the functions that work on that data are kept together in the form of objects.

34. When a derived class inheritance from many base classes, this process is known as
अगर कोई क्लास एक से अधिक बेस क्लास से इनहेरिटेंस अर्जित कर रहा हो तो उसे क्या कहेंगे?

(a) Default inheritance/डिफॉल्ट इनहेरिटेंस
(b) Multiplex inheritance/मल्टीप्लेक्स इनहेरिटेंस
(c) Multilevel inheritance/मल्टीलेवल इनहेरिटेंस
(d) Multiple inheritance/मल्टीपल इनहेरिटेंस

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : Multiple inheritance means a child class can get features (data + functions) from more than one parent class.

35. What is the main purpose of inheritance in OOP?
OOP में इनहेरिटेंस का मुख्य उद्देश्य क्या है?

(a) To create instances of classes
क्लासेस के इंस्टेंस बनाना
(b) To prevent data abstraction
डेटा एब्स्ट्रैक्शन को रोकना
(c) To establish a parent-child relationship between classes
क्लासेस के बीच पैरेंट-चाइल्ड संबंध स्थापित करना
(d) To define exceptions
एक्सेप्शन्स को परिभाषित करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Inheritance is an important principle of object-oriented programming (OOP), the main purpose of which is to allow one class to inherit the properties and methods of another class. This establishes a parent-child relationship between two classes.

36. Which of the following best defines a class?
निम्न में से कौन-सा विकल्प क्लास को सबसे अच्छी तरह परिभाषित करता है?

(a) An instance of an object
एक ऑब्जेक्ट का इंस्टेंस
(b) A blueprint for creating objects
ऑब्जेक्ट्स बनाने के लिए एक ब्लूप्रिंट
(c) A variable holding data
डेटा को रखने वाला एक वेरिएबल
(d) A function with a return value
एक रिटर्न वैल्यू वाली फंक्शन

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : A class in object-oriented programming (OOP) is a blueprint or template that we use to create structure. In a class, we define properties (data/variables) and methods (functions/behavior). When we create an object from a class, that object has all the class's properties.

37. What is an abstract class in OOP?
OOP में एब्स्ट्रैक्ट क्लास क्या होती है?

(a) A class with no methods
एक क्लास जिसमें कोई मेथड नहीं होता
(b) A class that cannot be instantiated
एक क्लास जिसे इंस्टेंसिएट नहीं किया जा सकता

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(c) A class that can only have private members
एक क्लास जिसमें केवल प्राइवेट मेंबर्स हो सकते हैं
(d) A class that cannot have methods with parameters/एक क्लास जिसमें पैरामीटर वाले मेथड नहीं हो सकते

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : An abstract class is a class in object-oriented programming (OOP) that cannot be directly instantiated (created and used and as an object). Its primary purpose is to serve as a base class (parent class) for other classes.

38. Which keyword is used to create an instance of a class in most programming languages?
अधिकांश प्रोग्रामिंग भाषाओं में किसी क्लास का इंस्टेंस बनाने के लिए कौन-सा कीवर्ड उपयोग किया जाता है?
(a) alloc (b) new
(c) create (d) instance

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : In most programming languages (such as Java , C++, C#, etc.), the 'new' keyword is used to create a new instance (object) of a class.
Example: In Java ,
Car my car new Car();

39. The process of removing detail from a given state representation is called…….
किसी स्थिति के निरूपण (representation) से विवरण हटाने की प्रक्रिया को क्या कहा जाता है?
(a) Extraction/एक्सट्रैक्शन
(b) Abstraction/एब्स्ट्रैक्शन
(c) Data Mining/डेटा माइनिंग
(d) Information Retrieval/सूचना पुनःप्राप्ति

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : Abstraction means retaining only the essential information about something and removing unnecessary details. This process makes the system simpler and easier to understand.

40. Objects is/Objects हैं
(a) Design of function (b) Creator of data
(c) Instances of classes (d) Program block5

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) In Object-oriented Programming (OOP), a class is a template or blueprint. When you create an actual entity from that class, it is called an object.
So, objects = instances of classes.

41. Abstraction means
Abstraction का मतलब है
(a) Summarization (b) Elimination
(c) Hiding (d) Casting

Bihar STET C.S. 2019 (Shift-II)

Ans. (c): Abstraction is a key feature of Object-oriented Programming (OOP), which means "hiding complexity and showing only the essential information." This means that users are shown only what is necessary.

42. Inheritance reflects……
इन्हेरिटेन्स प्रदर्शित ……करता है?
(a) Association (b) Difference
(c) Specification (d) Generalization

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : Generalization involves the projection of common properties and behaviors into a super class. Inheritance represents generalization because it involves combining classes that share common characteristics into a single class.

43. What is the primary purpose of an interface in OOP?
OOP में इंटरफेस का प्राथमिक उद्देश्य क्या है?
(a) To provide a blueprint for creating objects
(b) To define a set of methods that must be implemented by implementing classes
(c) To define constructors for classes
(d) To create static methods

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : The main purpose of an interface is to define a set of methods that must be implemented by classes implementing that interface.

44. What is the term for a variable that references an instance of a class?
उस चर के लिए क्या शब्द है जो किसी क्लास के उदाहरण को संदर्भित करता है?
(a) Property (b) Method
(c) Object (d) Constructor

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : When we create an actual instance of a class and reference it through a variable is called an object.

45. What is the term for creating a new class that inherits properties and behaviors from an existing class?
किसी ऐसे नये वर्ग को बनाने के लिए क्या शब्द प्रयोग किया जाता है जो किसी मौजूदा वर्ग से गुण और व्यवहार विरासत में लेता है?
(a) Polymorphism (b) Encapsulation
(c) Abstraction (d) Derivation

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : This process is known as inheritance in OOP, where a new class (derived class or subclass) acquires the characteristics and methods of an existing class (base class or super class). Derivation is a term specifically used for this in the context of creating a derived class.

46. What is the main advantage of using abstract classes over interfaces?
इंटरफेस की तुलना में आमूर्त क्लासों का उपयोग करने का मुख्य लाभ क्या है?
(a) Abstract classes can have multiple inheritance, unlike interfaces/एब्स्ट्रैक्ट क्लासेस में मल्टीपल इन्हेरिटेंस हो सकती है, जबकि इंटरफेस में नहीं।
(b) Interfaces can have constructors, while abstract classes cannot/इंटरफेस में कंस्ट्रक्टर हो सकते है जबकि एब्स्ट्रैक्ट क्लास में नहीं।
(c) Abstract classes can provide default implementations of methods/एब्स्ट्रैक्ट क्लासेस मेथड्स की डिफॉल्ट इम्प्लीमेंटेशन प्रदान कर सकती है।
(d) Interfaces allow for private access modifiers, unlike abstract classes/इंटरफेस में प्राइवेट एक्सेस मॉडिफायर्स की अनुमति होती है, जबकि एब्स्ट्रैक्ट क्लास में नही

Bihar STET C.S. 12.09.2023 (Shift-II)

Sarkari Resource

Ans. (c) : Abstract classes can provide default implementations for some methods while forcing subclasses to implement others, unlike interfaces which can only declare method signatures.

47. What is the term for defining multiple methods in a class with the same name but different parameters?
एक ही नाम लेकिन अलग-अलग पैरामीटर वाले क्लास में कई विधियों को परिभाषित करने के लिये क्या शब्द है?

(a) Overriding (b) Overloading
(c) Inheriting (d) Inheritance

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : Method overloading is a feature of object – oriented programming where multiple methods can be created with the same name provided their parameters different (in number, type, or order). This is determined at compile time.

48. Which is correct for inheritance?
वंशानुक्रम के लिए कौन सा सही है?

(a) Animal to Cat (b) Variable to constant
(c) Lock to key (d) Data to Program

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Ans. (a) : Inheritance in OOP(Oebject-oriented design) signifies an " is – a " relationship where a subclass (child class) derives properties and behaviours from a superclass (Parent class), In the given options, " Cat" is – a " Animal," representing a valid hierarchical relationship type of inheritance in OOP. " Variable to Constant" describes a type of data classification, not an inheritance relationship between entities.

49. Which is encapsulated in a class?
एक Class में कौन सा Encapsulated किया जाता है?

(a) Program & Process
(b) Function and Data
(c) File and Disk
(d) Variable and Constant

Bihar STET C.S. 2020

Ans. (b) : Encapsulation, is a key feature of Oebje-oriented programming (OOP) . It means that both data and the function that operates on that data are kept together hidden (by encapsulating) within a class.

50. The major goal of object – oriented programming is :
अभिलक्ष्य केन्द्रित प्रोग्रामिंग का प्रमुख उद्देश्य है :

(a) Top-down program development
टॉपडाउन प्रोग्राम विकास
(b) Speed/गति
(c) User – interface/प्रयोक्ता-अन्तरपृष्ठ
(d) Re use/पुनः उपयोग

KVS PGT C.S. 2017

Ans. (d) : Languages that support object oriented programming typically use inheritance for code reuse and extensibility in the form of either classes or prototypes. Object oriented programming languages are based on classes and objects. They are used to structure a program into a simple, reusable piece of code.

51. Which one of the following most accurately describes "multiple inheritance" ?
निम्नलिखित में से क्या ''मल्टिपल इन्हेरिट्न्स'' की सबसे अधिक यथार्थ व्याख्या करती है :

(a) When a child class has both an "is a" and a "has a" relationship with its parent class./जब 'चाइल्ड' वर्ग अपने पैरेन्ट वर्ग के साथ ''इज ए '' तथा ''हैज ए'' दोनों संबंध होते हैं।
(b) When two classes inherit from each other./जब दो वर्ग एक दूसरे से इनहेरिट होते हैं।
(c) When a base class has two or more derived classes./जब किसी आधार वर्ग का दो या अधिक व्युत्पन्न वर्ग होता है।
(d) When a child class has two or more parent classes./जब 'चाइल्ड' वर्ग का दो या अधिक 'पैरेन्ट' वर्ग होता है।

KVS PGT C.S. 2017

Ans. (d) : Multiple inheritance is a feature of some Object-oriented computer programming languages in which an object or class can inherit features from more than one parent object or parent class. It is distinct from single inheritance, where an object or class may only inherit from a particular object or class.

52. In object-oriented programming, by wrapping up characteristics and behaviour into one unit, we achieve
ऑब्जेक्ट ओरीएंटेड प्रोग्रामिंग में एक इकाई में विशेषताएँ और व्यवहार को परिवेष्ठित करने पर , प्राप्त होता है

(a) data abstraction./डाटा एबस्ट्रेक्शन
(b) data encapsulation./डाटा एनकेप्स्युलेशन
(c) data hiding./डाटा हाइडिंग
(d) All of these/ये सभी

KVS PGT C.S. 2013

Ans. (b) : In general Encapsulations is the process of wrapping up of data and behaviour of an object into a single unit and the unit here is a class (or interface)

53. Which of the following is a pillar of OOP?/निम्नलिखित में से कौन OOP का एक स्तंभ है?

(a) Inheritance
(b) Encapsulation
(c) Abstraction
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : The main pillars of object-oriented programming are:
Inheritance
Abstraction
Encapsulation and
Polymorphism

54. Which among the following doesn't come under OOP concept?
निम्नलिखित में से कौन ओ.ओ.पी. अवधारणा के अंतर्गत नहीं आता है?

(a) Data hiding
(b) Platform independent

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(c) Message Passing
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) Platform independent is not a concept of Object-oriented programming (OOP).OOP concepts include data hiding (encapsulation), message passing, inheritance, and polymorphism.

55. In which access should a constructor be defined, so that object of the class can be created in any function?
किस एक्सेस में एक कन्स्ट्रक्टर को परिभाषित किया जाना चाहिए, ताकि किसी भी फंक्शन में क्लास का ऑब्जेक्ट बनाया जा सके?

(a) Public
(b) Private
(c) Protected
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) The constructor function is available to all parts of the program where the object is created. Therefore it is recommended to define it publicly so that anyone else can create the object.

56. Which of the following refers to the wrapping of data and its functionality into a single individual entity?
निम्नलिखित में से कौन-सा डेटा और उसकी कार्यक्षमता को एक एकल इकाई में लपेटने को संदर्भित करता है?

(a) Modularity
(b) Abstraction
(c) Encapsulation
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Encapsulation refers to wrapping data and its functionality into a single unit, making it possible to access and use that data seemlessly.

57. Which of the following can be considered as the members that can be inherited but not accessible in any class?
निम्नलिखित में से किसे उन सदस्यों के रूप में माना जा सकता है, जो विरासत में मिल सकते हैं लेकिन किसी भी वर्ग में पहुँच योग्य नहीं है।

(a) Public
(b) Protected
(c) Private
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Memberss that can be inherited but not accessible within any class are called private members. These members can only be accessed from within the classes they are part of but are not publicly available.

58. Which of the following statements is completely true?
निम्नलिखित में से कौन-सा/से कथन पूर्णतः सत्य है/हैं?

I. In an object-oriented programming language, all the function calls are resolved at compile-time./ऑब्जेक्ट-ओरिएंटेड प्रोग्रामिंग भाषा में, सभी फंक्शन कॉल को संकलन-समय पर हल किया जाता है।

II. In a procedure programming language, all the function calls are resolved at compile-time./एक प्रक्रिया प्रोग्रामिंग भाषा में, सभी फंक्शन कॉल को संकलन-समय पर हल किया जाता है।

(a) Only II/केवल II
(b) Only I/केवल I
(c) Both I and II/I और II दोनों
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): Statement-II is correct because all function calls in a programming language are resolved. at compile-time.

59. The programming language that has the ability to create new data types is called:
वह प्रोग्रामिंग language जिसमें नए डेटा प्रकार बनाने की क्षमता होती है, कहलाती है–

(a) Overloaded
(b) Encapsulated
(c) Extensible
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Extensible is a programming language that has the ability to create new data types.

60. Which of the following features must be supported by any programming language to become a pure object-oriented programming language?
शुद्ध ऑब्जेक्ट-ओरिएंटेड प्रोग्रामिंग भाषा बनने के लिए किसी भी प्रोग्रामिंग भाषा द्वारा निम्नलिखित में से किस विशेषता का समर्थन किया जाना चाहिए?

(a) Encapsulation
(b) Inheritance
(c) Polymorphism
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d): To be a pure Object-oriented programming language it should have the following features.
(i) Encapsulation
(ii) Inheritance.
(iii) Polymorphism
(iv) Classes and objects.

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61. The process of removing details from a given state representation is called _____ .

(a) Extraction (b) Mining
(c) Selection (d) Abstraction

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) : The "process of removing detail" from a given state representation is called an abstraction.
• Abstraction is a type artificial Intelligence problem formulation and problem solving process.

62. Match List-I with List-II

	List-I (Programming Paradigm)		List-II (Characteristic)
A.	**Imperative**	**i.**	**Declarative, clausal representation, theorem proving**
B.	**Object-oriented**	**ii.**	**Side-effect free, declarative, expression evaluation**
C.	**Logic**	**iii.**	**Imperative, abstract data type**
D.	**Functional**	**iv.**	**Command-based, procedural**

Choose the correct answer from the options given below:

(a) A - IV, B - III, C - I, D - II
(b) A - III, B - IV, C - I, D - II
(c) A - IV, B - III, C - II, D - I
(d) A - II, B - III, C - I, D - IV

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Ans. (a) :

	List-I (Programming Paradigm)		List-II (Characteristic)
A.	Imperative	iv.	Command-based, procedural
B.	Object-oriented	iii.	Imperative, abstract data type
C.	Logic	i.	Declarative, clausal representation, theorem proving
D.	Functional	ii.	Side-effect free, declarative, expression evaluation

63. Match List-I with List-II :

	List-I		List-II
(A)	**Localization**	**(I)**	**Encapsulation**
(B)	**Packaging or binding of a collection of items**	**(II)**	**Abstraction**
(C)	**Mechanism that enables designer to focus on essential details of a program component**	**(III)**	**Characteristic of software that indicates the manner in which information is concentrated in program**
(D)	**Information hiding**	**(IV)**	**Suppressing the operational details of a program component**

Choose the correct answer from the options given below:

	A	B	C	D
(a)	(I)	(II)	(III)	(IV)
(b)	(II)	(I)	(III)	(IV)
(c)	(III)	(I)	(II)	(IV)
(d)	(III)	(I)	(IV)	(II)

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Ans. (c) : Localization is a characteristic of software that indicates the manner in which information is concentrated in program.
Encapsulation is the packing of data and function into one component and then controlling access to that component to make a ''blackbox'' out of the object. Because of this, a user of that class only needs to know its interface that is, the data and function exposed outside the class, not the hidden implementation.
Abstraction is a mechanism that enables the designer to focus on the essential details of a program component with little concern for lower-level details.
Information hiding is a software design principle, where certain aspects of a program or module are inaccessible to clients. The primary goal is to prevent extensive modification to clients whenever the implementation details of a module or program are changed.
So option (c) is correct answer.

64. Which of the following is a correct statement?

(a) Composition is a strong type of association between two classes with full ownership.
(b) Composition is a strong type of association between two classes with partial ownership.
(c) Composition is a weak type of association between two classes with partial ownership.
(d) Composition is a weak type of association between two classes with strong ownership.

UGC NET C.S. August-2016 (Paper-III)

Ans. (a): Composition is a strong type of association between two classes with full ownership. In this relationship child objects does not have their lifecycle without parent object. If a parent object is deleted, all its child objects will also be deleted.

65. Which of the following is not a correct statement?

(a) Every class containing abstract method must be declared abstract.
(b) Abstract class can directly be initiated with 'new' operator.
(c) Abstract class can be initiated.
(d) Abstract class does not contain any definition of implementation.

UGC NET C.S. August-2016 (Paper-III)

Ans. (b): Abstract class cannot be directly initiated with new operator, since abstract class does not contain any definition of implementation it is not possible to create an abstract object.

66. Which of the following is/are correct with reference to abstract class and interface?

(1) A class can inherit only one Abstract class but may inherit several interfaces
(2) An Abstract class can provide complete and default code but an interface has no code

Codes :
(a) (1) is true
(b) (2) is true
(c) Both (1) and (2) are true
(d) Neither (1) nor (2) is true

UGC NET C.S. December-2015 (Paper-III)

Ans. (c): An interface is an empty shall, there are only the signatures of the methods, which implies that the methods do not have a body. The interface can't do anything. It's just a pattern so b is true.
A child can only extend a single class, where as an interface can extend or a class can implement multiple other interfaces so a is true.

67. Which of the following is used to make an Abstract class?
(a) Making atleast one member function as pure virtual function
(b) Making atleast one member function as virtual function
(c) Declaring as abstract class virtual keyword
(d) Declaring as Abstract class static keyword

UGC NET C.S. December-2015 (Paper-III)

Ans. (a): For making the class as abstract class we should make atleast one member function as pure virtual function.

68. Consider the following two statements :
(1) A publicly derived is a subtype of base class
(2) Inheritance provides for code ruse
Which one of the following statements is correct?
(a) Both the statements (1) and (2) are correct
(b) Neither of the statement (1) and (2) are correct
(c) Statement (1) is correct and (2) is incorrect
(d) Statement (1) is incorrect and (2) is correct

UGC NET C.S. December-2015 (Paper-II)

Ans. (a): Inheritance provide for code reuse
for public inheritance creates sub type
Inheritance only applies to user defined classes.
A publicly derived class is a subtype of its base class.

69. Which methods are utilized to control the access to an object in multi-threaded programming?
(a) Asynchronized methods
(b) Synchronized methods
(c) Serialized methods
(d) None of the above

UGC NET C.S. December-2014 (Paper-III)

Ans. (b): Synchronation is the capability to control the access of multiple threads to shared resources, It is possible for one Java thread to modify a shared variable while another thread is in the process of using or updating same shared variable.

70. Match the following with respect to I/O classes in object oriented programming :

List–I	List–II
A. fopen()	**i. returns end of file**
B. fclose()	**ii. return for any problem report**
C. ferror()	**iii. returns 0**
D. feof()	**iv. returns a file pointer**

Codes :

	A	B	C	D
(a)	iv	i	ii	iii
(b)	iii	i	iv	ii
(c)	ii	iii	iv	i
(d)	iv	iii	i	ii

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : fopen () : Creates a file and open an existing file
Syntax : FILE * fp;
fp = fopen ("<file – name>", "<mode>");
fclose : It closes a file and returns 0 on success and EOF otherwise.
Syntax : fclose (<file – pointer>);
ferror ( ) : Checks the error indicator for the given file stream and returning non-zero if set and 0 otherwise.
feof ( ) : Checks the end of file indicator for the given file stream returning non-zero if set and 0 otherwise.

71. The "part-whole", or "a-part-of", relationship in which objects representing the components of something associated with an object representing the entire assembly is called as
(a) Association (b) Aggregation
(c) Encapsulation (d) Generalisation

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because, Aggregation is defined as a group or mass of distinct or varied things, person, etc. and closets related concept of composition is said to aggregation. Hence the answer is (b).

72. The pure object oriented programming language with extensive metadata available and modifiable at run time is
(a) Small talk (b) C++
(c) Java (d) Eiffel

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : Because, Small talk was one of the earliest OOP's language (with other like simulate and Eiffel) and can be said to be extremely "Pure" in an OOP sense.
In small talk everything is an object and objects are only communicated with via the sending of message. Hence the answer is (a).

73. The syntax of capturing events method for document object is
(a) CaptureEvents()
(b) CaptureEvents(Orgs eventType)
(c) CaptureEvents(eventType)
(d) CaptureEvents(eventVal)

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : captureEvents (EventType) method instructs the window to capture all events of a particular type.

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74. Superficially the term "object-oriented", means that, we organize software as a

(a) collection of continuous objects that incorporates both data structure and behaviour.

(b) collection of discrete objects that incorporates both discrete structure and behaviour.

(c) collection of discrete objects that incorporates both data structure and behaviour.

(d) collection of objects that incorporates both discrete data structure and behaviour.

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, object oriented development makes software closet to abstractions the exists in the real world and the term object oriented means that software is a collection of discrete objects that incorporate both data structure and behaviour.

75. What is the relation between object and class in an object oriented programming?
ऑब्जेक्ट ओरिएंटेड प्रोग्रामिंग में ऑब्जेक्ट और क्लास के बीच क्या संबंध है?

(a) Object is keyword of class
ऑब्जेक्ट क्लास का कीवर्ड है

(b) Object is scope of class
ऑब्जेक्ट क्लास का स्कोप है

(c) Object is instance of class
ऑब्जेक्ट क्लास का इंस्टैंस है

(d) Object is type of class
ऑब्जेक्ट क्लास का प्रकार है

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In object-oriented Programming a class is like a blueprint or template that defines the properties (attributes) and behaviors (method) that an object will have.
An object is a specific instance of a class, created using the class's blueprint.

76. In Object Oriented Programming, when a new class created from an existing class using principle of inheritance, then the new class is known as _____.
ऑब्जेक्ट ओरिएंटेड प्रोग्रामिंग में, जब एक मौजूदा क्लास से इन्हेरिटेंस के सिद्धांत का उपयोग करके एक नई क्लास बनाई जाती है, तो नई क्लास को _____ के रूप में जाना जाता है।

(a) New class/नई क्लास

(b) Child class/चाइल्ड क्लास

(c) Member class/मेंबर क्लास

(d) Derived class/डीराइवड क्लास

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : In Object Oriented Programming, when a new class is created from an existing class using inheritance, it is called a Derived class.
The original class is called Base class.
The new class is called Derived class.

77. Which one of the following is correct regarding Programming Languages?
प्रोग्रामिंग भाषाओं के संबंध में निम्नलिखित में से कौन सा सही है?

(a) Procedural programming – Functions
प्रोसीजरल प्रोग्रामिंग – फंक्शन

(b) Object-oriented programming – Objects
ऑब्जेक्ट ओरिएंटेड प्रोग्रामिंग – ऑब्जेक्ट्स

(c) Functional programming – Logical unit
फंक्शनल प्रोग्रामिंग – लॉजिकल यूनिट

(d) Logical Programming – Function/लॉजिकल प्रोग्रामिंग – फंक्शन

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : The correct option is–
Object-oriented programming – Object
In Object-oriented programming the core concept revolves around objects, which are instances of classes, encapsulating data and behaviors.

(vii) Introduction to Java

78. Which of the following declares an abstract method in an abstract Java class?
निम्नलिखित में से कौन सा एक एब्सट्रेक्ट जावा क्लास में एक एब्सट्रेक्ट विधि घोषित करता है?

(a) Public abstract void method ();
पब्लिक एब्सट्रेक्ट वाइड विधि ()

(b) Public void abstract method ();
पब्लिक वाइड एब्सट्रेक्ट विधि ()

(c) Public void method () { };
पब्लिक वाइड विधि () { }

(d) Public abstract method ();
पब्लिक एब्सट्रेक्ट विधि ()

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : A public Abstract void method is a method that has only a declaration, and no body. It must be written inside an abstract class and implemented by child classes.

79. To override a method in Java , we need to define a method in a sub-class with the
जावा में किसी विधि को प्रत्यादेशित करने के लिए, हमें उपवर्ग में एक विधि को _____के साथ परिभाषित करना आवश्यक है ।

(a) same name, same number of arguments having the same data types as a method in the super class.
समान नाम, सुपर क्लास में एक विधि के रूप में समान डाटा टाइप वाले समान संख्या में दलील

(b) different name, same number of arguments having the same data types as a method in the superclass.
भिन्न नाम, सुपर क्लास में एक विधि के रूप में समान डाटा टाइप वाले समान संख्या में दलील

(c) same name but different number of arguments as a method in the superclass.
समान नाम परंतु सुपर क्लास में एक विधि के रूप में भिन्न संख्या में दलील

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(d) same name, same number of arguments but different data types as a method in the superclass.
समान नाम, समान संख्या में दलील परंतु सुपर क्लास में एक विधि के रूप में भिन्न डाटा टाइप

KVS PGT C.S. 2013

Ans. (a) : Declaring a method in the subclass which already exists there in the parent class is known as method overriding. Given option (a) is true-same name, same number of arguments having the same data types as a method in the super class.

80. Additional information sent when an exception is thrown may be placed in _____.
किसी अपवाद के डाले जाने पर भेजी गई अतिरिक्त सूचना कहाँ रखी जाती है?

(a) The throw keyword/थ्रो कीवर्ड में
(b) The function that caused the error
चूक करने वाले फंक्शन में
(c) The catch block/कैच ब्लॉक में
(d) An object of the exception class
एक्सेप्शन क्लास के ओब्जेक्ट में

KVS PGT C.S. 2018

Ans. (c) : Additional information sent when an exception is thrown may be placed in the catch block. The catch block contains code that is executed if and when the exception handler is invoked. The runtime system invokes the exception handler when the handler is the first one in the call stock whose.
Exception Type matches the type of the exception thrown.

81. Which of the following classes cannot be subclassed in Java ?
निम्नलिखित में से कौन सा वर्ग जावा मे उपवर्गीकृत नहीं किया जा सकता है?

(a) Public/पब्लिक (b) Final/फाइनल
(c) Thread/थ्रेड (d) Object/ऑब्जेक्ट
(e) Not attempted/प्रयास नही किया

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : A Final class cannot be Inherited. No other class can extend a final class.

82. Which of the following statements is TRUE?

(a) Virtual functions do not implement polymorphism
(b) Virtual functions do not permit calling of derived class functions using a base class pointer
(c) We can never build an object from a class containing a pure virtual function
(d) Pure virtual functions can never have a body

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Ans. (c) : A pure virtual function is a function that is declared in a base class but does not have an implementation in that base class. Instead, it must be implemented in any non-abstract derived class, classes containing pure virtual function are considered abstract classes, and you cannot create objects directly from them. You need to derive build from them and implement the pure virtual functions to instantiate objects.

83. Which of the following is a invalid relational operator in Java ?
निम्नलिखित में से कौन सा Java में एक अमान्य रिलेशनल ऑपरेटर है?

(a) < = (b) = >
(c) ! = (d) = =

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : In Java, the full set of relational and equality operators are :

Operator	Meaning
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
= =	Equal to
!=	Not equal to

Since '=>' is not defined, it's invalid in Java .

84. In Java , access to a private member of a class is granted only to
जावा मे, किसी क्लास के प्राइवेट सदस्य तक पहुँच केवल निम्न को दी जाती है -

(a) Other members of that class.
उस वर्ग के अन्य सदस्य
(b) Private members of any other class.
किसी अन्य वर्ग के निजी सदस्य
(c) Public members of that class
उस वर्ग के सार्वजनिक सदस्य
(d) Public members of any other class.
किसी अन्य वर्ग के सार्वजनिक सदस्य
(e) Not attempted/प्रयास नही किया

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Private members are accessible only Inside the same class.
They can not be accessed by outside classes.

85. Which of the following is a character stream in Java ?
निम्नलिखित में से कौन जावा में कैरेक्टर स्ट्रीम है?

(a) Buffered reader/बफर्ड रीडर
(b) Data input stream/डाटा इनपुट स्ट्रीम
(c) Object output stream/ऑब्जेक्ट आउटपुट स्ट्रीम
(d) File output stream/फाइल आउटपुट स्ट्रीम
(e) Not attempted/प्रयास नही किया

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Character stream in Java is used to read and write Text data (characters)
Example:- Buffer Reader, File Reader

86. What is the purpose of the "super" keyword in Java and other similar languages?
Java और अन्य समान भाषाओं में 'super' कीवर्ड का क्या उद्देश्य होता है?

(a) It refers to the superclass in inheritance
इनहेरिटेंस में सुपरक्लास को संदर्भित करता है
(b) It creates a new object instance
एक नया ऑब्जेक्ट इंस्टेंस बनाता है
(c) It defines a new class
एक नई क्लास को परिभाषित करता है
(d) It handles exceptions
अपवादों को संभालता है

Bihar STET C.S. 12.09.2023 (Shift-I)

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Ans. (a) : In Java and other object oriented programming languages, the 'Super' keyword is used to refer to the parent class (super class). When a subclass (child class) extends a super class, we can use the super keyword to call the super class's constructor.
We can access the methods or variables of the superclass, especially when the subclass has overwritten them with the same names.

87. JVM is part of.....
जेवीएम .....का हिस्सा है।

(a) C sharp (b) Oracle
(c) C++ (d) JAVA

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : The JVM (Java Virtual Machine) is a system that runs Java programs. It is a key component of the Java programming language. C++, C#, and Oracle are not part of it.

88. try{ }...catch{ } in JAVA use for?
try{ }...catch{ } का जावा में क्या उपयोग है?

(a) CLASS collaboration
(b) Subclasses inheritance
(c) Exception handling
(d) Debugging

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : In Java , try {} and catch {} blocks are used for exception handing. If an error or exception occurs, it is caught in the try block and handled in the catch block.
Example:-
t r y {
int a = 10/0;} catch (Arithmetic Exception e)
{system.out.print in ("cannot divide by zero");

89. What is the full form of JESS in Expert System Technology?/Expert System Techology में JESS का पूरा नाम क्या है?

(a) Java Expert System Shell
(b) Java script Expert System Shell
(c) Java Expert Sub System
(d) Java script Expert Sub System

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : JESS (Java Expert system shell) is a programming tool used for developing expert systems, based on the Java programming language and used to solve complex problems.

90. What is garbage collection in the context of Java ?
जावा के संदर्भ 'गार्बेज कलेक्शन' क्या होता है?

(a) The Java Virtual Machine (JVM) checks the output of any Java program and deletes anything that does not make sense at all./जावा वर्चुवल मशीन (जे.वी.एम.) किसी जावा प्रोग्राम के आउटपुट की जाँच करता है और उन सभी का विलोपन (डिलीट) करता है जिसका कोई अर्थ नहीं है।
(b) The Operating System periodically deletes all of the Java files available on the system./ऑपरेटिंग सिस्टम, सिस्टम पर उपलब्ध सभी जावा फाइलों की आवधिक रूप से डिलीट करता है।
(c) Any Java package imported in a program and not being used, is automatically deleted./किसी प्रोग्राम में इम्पोर्ट किए गए कोई जावा पैंकेज जिसका उपयोग नही हो रहा है, स्वतः डिलीट हो जाते है।
(d) When all references to an object are gone, then the memory used by the object is automatically reclaimed./जब किसी के सभी रैफरेंस चले जाते है तो ऑब्जेक्ट द्वारा प्रयुक्त स्मृति स्वतः वापस मांग ली जाती (रिक्लैम कर ली जाती) हैं।

KVS PGT C.S. 2017

Ans. (d) : Garbage collection in Java is the process by which Java programs perform automatic memory management. When Java programs run on the JVM, objects are created on the heap, which is a portion of memory dedicated to the program. Eventually, some objects will no longer be needed. The garbage collector finds these unused objects and deletes them to free up memory.

91. What correction is required for following Java code snippet to compile?
निम्नलिखित जावा कोड स्निपेट को संकलित करने के लिए आपको क्या शुद्धि करने की आवश्यकता होगी?

```
int [] X = new int [10];
for (int P = 0; P< = X.length(); P++)
        X [P] = 5;
```

(a) X [P] = 5 should be X (P) = 5
X [P] = 5,X (P) = 5 होना चाहिए।
(b) P< = X.length() should be P< = X.length()
P< = X.length(), P< = X.length() होना चाहिए।
(c) X.length() should be X.length
X.length(), X.length होना चाहिए।
(d) P++ should be P+1
P++, P+1 होना चाहिए।

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Ans. (c) : Given code –

```
int [] X = new int [10];
for (int P = 0; P< = X length (); P++)
  X [P] = 5;
```

In Java , for integer variable X to find the length of the array we should use X length instead of X.length ().

92. Java is a _____language. This means that you must be explicit about what type of data you are working with.
जावा _____ भाषा है। इसका अर्थ यह है कि आपको इस बारे में स्पष्ट होना चाहिए कि आप किस प्रकार के डेटा पर कार्य कर रहे है।

(a) Weakly typed/विकली (weakly) टाइप्ड
(b) Strongly typed/स्ट्रांगली टाइप्ड
(c) Dynamically typed/डायनामिकली टाइप्ड
(d) Loosely typed/लूजली टाइप्ट

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Ans. (b) : Java is a high-level, secure and strongly typed language. This means that you must be explicit about what type of data you are working with, It was developed by sun micro systems.

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93. Parent class of all Java classes is _____.
सभी जावा क्लासों की पैरेंट क्लास क्या है?

(a) Java .lang.system
(b) Java .lang.object
(c) Java .lang.class
(d) Java .lang.reflect.object

KVS PGT C.S. 2018

Ans. (b) : The object class (Java .lang.object) is the parent class of all the classes in Java by default. In other words, it is the topmost class of Java . The Java .lang.object class is the root of the class hierarchy. All objects, including arrays, implement the methods of this class.

94. Usually pure virtual function _____.
आमतौर पर प्योर वर्चुअल फंक्शन _____

(a) has complete function body
संपूर्ण फंक्शन निकाय होता है
(b) will never be called
को कभी कॉल नहीं किया जाएगा
(c) will be called only to delete an object
केवल किसी ऑब्जेक्ट को डिलीट करने के लिए कॉल किया जाएगा।
(d) is defined only in derived class/केवल डेराइव्ड क्लास में परिभाषित किया जाता है।

KVS PGT C.S. 2018

Ans. (d) : Usually pure virtual function is defined only in derived class. Virtual functions ensure that the correct function is called for an object, regardless of the type of reference used for function call.

95. Consider the following Java code segment:
निम्नलिखित जावा कोड खंड पर विचार कीजिए:

```
public class while /* line 1*/
{
  public void loop ( )
  {
  int x = 0;
   While (1)/* line-6*/
    {
system. out. println("x plus one is" + (x+1)):
            }
       }
  }
```

Which of the following is true ?
निम्नलिखित में से कौन-सा कथन सत्य है?

(a) There is a syntax error in line-1
लाइन-1 में सिन्टैक्स एरर है।
(b) There are syntax errors in line- 1 and line- 6
लाइन-1 व लाइन - 6 में सिन्टैक्स एरर हैं।
(c) There is a syntax error in line- 6
लाइन-6 में सिन्टैक्स एरर है।
(d) No syntax error/कोई सिन्टैक्स एरर नहीं है।

KVS PGT C.S. 2018

Ans. (a) : In given Java code segment line-1, a keyword is used as class name i.e. while which is error. So, there is a syntax error in line-1.

96. Use of virtual function implies _____.
वर्चुअल फंक्शन के उपयोग का अर्थ ..... है?

(a) Overloading (b) Overriding
(c) Static binding (d) Dynamic binding

KVS PGT C.S. 2018

Ans. (d) : Use of virtual function implies Dynamic binding. If a body of the method at runtime is bonded to a method call then it is called dynamic binding. Dynamic binding is also referred to as a run-time polymorphism. Overriding a method is a per feet example of dynamic binding in Java .

97. Which of the following statements is true?
निम्न में से कौन सा कथन सही है?

(a) The JAVA compiler produces a code for JVM, which is executed by a JAVA interpreter./JAVA कम्पाईलर JVM के लिए एक कूट उत्पन्न करता है, जिसे JAVA इंटरप्रिटर द्वारा कार्यान्वित किया जाता है ।
(b) The JAVA compiler produces a code directly for JAVA interpreter./JAVA कम्पाईलर JAVA इंटरप्रिटर के लिए सीधे ही एक कूट उत्पन्न करता है ।
(c) The JAVA compiler directly executes code in JVM./JAVA कम्पाईलर सीधे ही JVM में कूट कार्यान्वित करता है ।
(d) The JAVA compiler helps in directly executing a code on different operating systems./JAVA कम्पाईलर विभिन्न आपरेटिग सिस्टम में सीधे ही एक कूट कार्यान्वित करने में मदद करता है ।

KVS PGT C.S. 2013

Ans. (a) : The true statements is, the Java compiler produces a code for JVM. Which is executed by a Java interpreter.
JVM is Java virtual Machine Run/Interprets | translate Byte code into native machine code.

98. From the following Java code determine the attributers of the class student:/ निम्न Java कूट की सहायता से Class Student की गुणों का निर्धारित कीजिए :

```
class student {
  string name;
  int marks;
};
public static void main() {
          student S1 = new student ();
          student S2 = new student();
}
```

(a) Only name./केवल name
(b) Both name and marks./name और marks दोनों
(c) Only S1./केवल S1
(d) Both S1 and S2./S1तथा S2दोनों

KVS PGT C.S. 2013

Ans. (b) : Here we created two objects of the student class S_1 and S_2 Thus S_1 and S_2 are two students having some name and marks.
so attributes of the class student is name and marks.

99. The statements
कथन:-

```
string var1[ ];
var1 = new string[3];
system.out.println (var1[1]);
```

print/प्रिन्ट करेगा

(a) An empty string/एक रिक्त स्ट्रिंग

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(b) Null/नल
(c) 0
(d) a garbage character/एक निरर्थक अक्षर

LT. Grade C.S. 2018

Ans. (b) : Here's a breakdown to each steps -
1. String var1[ ]; declares an array of strings called var1.
2. var1 = new String [3]; initializes the var1 array with a length of 3.
3. Since no value is assigned to the array elements, they are initialized with their default values, which is null for strings in Java .
4. System.out.println(var1[1]), attempts to print the value of the second element of the var1 array.
Since the second element of the array is null, the output will be null.

100. Which keyword is used in Java to implement inheritance?/Java में inheritance को लागू करने के लिए किस कीवर्ड का उपयोग किया जाता है?

(a) extends
(b) inherits
(c) implements
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : The extends keyword is used to implement inheritance in Java .

101. What is the output of the following JAVA program?

```
public class Good {
private int m;
public Good (int m) {this m = m;}
public Boolean equals (Good n) {return n.m = m;}
public static void main (string args []) {
Good m1 = new Good (22);
Good m2 = new Good (22);
Object s1 = new Good (22);
Object s2 = new Good (22);
System . out . println (m1 equals (m2));
System . out . println (s1 equals (s2));
System . out . println (m1 equals (s2));
System . out . println (s1 equals (m2));
}
}
```

(a) True, True, False, False
(b) True, False, True, False
(c) True, True, False, True
(d) True, False, False, False

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) :

System . out . println (m_1.equals (m_2)),	True
System . out . println (s_1 equals (s_2));	False
System . out . println (m_1 equals (s_2));	False
System . out . println (s_1 equals (m_2));	False

102. Consider the following recursive Java function *f* that takes two long arguments and returns a float value :

```
public static float f (long m, long n) {
  float result = (float)m /(float)m;
  if (m < 0 || n < 0) return 0,0f;
  else result – = f (m * 2, n * 3);
return result;
}
```

Which of the following real values best approximates the value of *f* (1, 3)?

(a) 0.2 (b) 0.4
(c) 0.6 (d) 0.8

UGC NET C.S. June-2020 (Paper-II)

Ans. (a) : The function *f*(1, 3) pass the value to m, n then the value will type cast to float and result = 0.3333 (1/3).
Then it enters into else statement the function result as 0.2 (2/9).
Hence, the correct answer is 0.2.

103. Consider the following recursive function F() in Java that takes an integer value and returns a string value:

```
public static String F( int N) {if ( N <= 0) return "–"; return F(N - 3) + N + F(N - 2) + N;}
```

The value of F(5) is:

(a) -2-25-3-135 (b) -2-25-1-3-135
(c) -1-145-2-245 (d) -2-25-3-1-135

UGC NET C.S. December 2020

Ans. (d) : The given Java program that takes an integer value and returns a string value:

```
public static String F (int N)
{
if (N < = 0 ) return "-";
return F (N–3) + F(N–2) + N;
}
```

The value of F (5) is:
it calls a return statement sub-functions F(2) + 5 + F (3) + 5;
it calls a return statement sub- functions F(2) + 5 + F(3) + 5;

F(N)
↓
F(5)
↓
F(2) + 5 + F (3) + 5
↓
F(–1) + 2 + F (0) + 2 F (0) + 3 + F (1) + 3
↓
–2 –2 5 –3 F(–2)+1+F(–1)+1
–1 –1 3 5
–2–25–3–1–135.

104. Consider the following recursive Java function f that takes two long arguments and returns of float value :

```
public static float f{long m, long n}
{
        float result = (float) m / (float) n;
        if (m < 0 | | n < 0)
                return 0.0f;
        else
                result + = f(m*2, n*3);
        return result;
}
```

Which of the following integers best approximates the value of f(2, 3)?

(a) 0 (b) 1
(c) 3 (d) 2

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Ans. (d) : Float class is a wrapper class for the primitive type float which contains several methods to effectively deal with a float value like converting it to a string representation, and vice-versa. An object of Float class can hold a single float value. There are mainly two constructors to initialise a Float object-
Float b: Creates a Float object initialized with the value provided.
float result = (float) m / (float) n;
if (m < 0 | | n < 0)
return 0.0f;
integers best approximates the value of f(2, 3)= f(2,3) = 0.6666667

105. What is the output of the following JAVA program ?

```
class simple
{
    public static void main(String[ ] args)
    {
        simple obj = new simple( );
        obj.start( );
    }
    void start( )
    {
        long [ ] P= {3, 4, 5};
        long [ ] Q= method (P);
        System.out.print (P[0]+P[1]+P[2]+":");
        System.out.print (Q[0] + Q[1] + Q[2]);
    }
    long [ ] method (long [ ] R)
    {
        R [1]=7;
        return R;
    }
} //end of class
```

(a) 12 : 15 (b) 15 : 12
(c) 12 : 12 (d) 15 : 15

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Ans. (d) : Output is 15:15
1. p = 3, 4, 5
2. q = 3, 7, 5

When we have the statement long [ ] Q = method (p). It will call method function and p is passed as parameter and in method function we are changing r [1] = 7.
Actually arrays are passed as parameter and in method function we are changing r[1] = 7.
Actually arrays are passed as reference. So whatever changes you make in r, it actually gets reflected in p. Hence both the arrays will display sum = 3+7+5 = 15.

106. In Java , which of the following statements is/are True ?
S1 : The 'final' keyword applied to a class definition prevents the class from being extended through derivation.
S2 : A class can only inherit one class but can implement multiple interfaces.
S3 : Java permits a class to replace the implementation of a method that it has inherited. It is called method overloading.
Code :
(a) S1 and S2 only
(b) S1 and S3 only
(c) S2 and S3 only
(d) All of S1, S2 and S3

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Ans. (a) : In Java , items with the final modifier cannot be changed.
This includes final classes, final variables, and final methods.
A final class cannot be extended by any other class
A final variable cannot be reassigned another value
A final method cannot be overridden
A final class is simply a class that can't be extended.
(This does not mean that all reference to objects of the class would act if they were declared as final).
⇒ A Java class only extend one parent class. Multiple inheritance (extends) is not allowed. Interfaces are not classes, however, and a class can implement more than one interface.
The parent interfaces are declared in a comma-separated list, after the implements key words. In conclusion, yes, it is possible to do.

107. Which of the following statements(s) with regard to an abstract class in JAVA is/are TRUE?
I. An abstract class is one that is not used to create objects.
II. An abstract class is designed only to act as a base class to be inherited by other classes.
(a) Only I (b) Only II
(c) Neither I nor II (d) Both I and II

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Ans. (d): A class that is declared as abstract is known as abstract class. It needs to be extended and its method implemented. It cannot be instantiated.
An abstract class is designed only to act as a base class to be inherited by other class. It is a design concept in program development & provides a base upon which other classes can be built.

108. Given the following statements :
(A) A class of language that is closed under union and complementation has to be closed under intersection
(B) A class of language that is closed under union and intersection has to be closed under complementation
Which of the following options is correct?
(a) Both (A) and (B) are false
(b) Both (A) and (B) are true
(c) (A) is true, (B) is false
(d) (A) is false, (B) is true

UGC NET C.S. June-2017 (Paper-III)

Ans. (c): Here we can derive this statement with the help of closure properties
S1 : True as it is the case for regular grammar
S2 : False as in RE language it is closed under union, Intersection but not in complementation.

109. Consider the following JAVA program :

```
public class First {
    public static int CBSE (int x) {
        if (x < 100) x = CBSE (x + 10)
        return (x – 1)
    }
  public static void main (String [ ] args) {
```

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system. out. print (First. CBSE (60));
}

What does this program print ?

(a) 59 (b) 95
(c) 69 (d) 99

UGC NET C.S. June-2017 (Paper-III)

Ans. (b):
CBSE (60) 95
↓↑
CBSE (70) 96
↓↑
CBSE (80) 97
↓↑
CBSE (90) 98
↓↑
CBSE (100) 99
Answer is option (b).

110. Given the array of integers 'array' shown below :

13	7	27	2	18	33	9	11	22	8

What is the output of the following Java statements?

```
int [ ] p = new int [10];
int [ ] q = new int [10];
for (ink k = 0; k < 10; k++)
        p[k] = array [k];
q = p;
p[4] = 20;
System out. println (array [4]+ "." + q[4]);
```

(a) 20 : 20 (b) 18 : 18
(c) 18 : 20 (d) 20 : 18

UGC NET C.S. June-2017 (Paper-III)

Ans. (c): For (int k = 0; k < 10; k ++)
p[k] = array [k]; in this loop we are copying all the values from the array to the array P.
Now, Q = P, Q is pointing P, nom, P[4] = 20 but Q contains the old P array where in 4th index i.e. Q[4] is also Q0.
Note : Here pointer is involved, so the location is being update, so Q is getting 20 here but the question is about array [4] which was initial array, so value 18
so answer is 18 : 20.

111. Java uses threads to enable the entire environment to be _____ .

(a) Symmetric (b) Asymmetric
(c) Synchronous (d) Asynchronous

UGC NET C.S. August-2016 (Paper-III)

Ans. (d): Java uses threads to enable the entire environment to be **asynchronous.** This helps in reducing inefficiency by preventing the waste of CPU cycles.

112. In Java , when we implement an interface method, it must be declared as :

(a) Private (b) Protected
(c) Public (d) Friend

UGC NET C.S. June-2015 (Paper-III)

Ans. (c): **Interface:** It is used to provide total abstraction. That means all the methods in interface are declared with empty body and are public and all fields are public, static and final default.

113. What is the output of the following Java code ?
निम्नलिखित Java कोड का आउटपुट क्या है?

String s= "exam" + 9+9+9
System. out. print ln(s);

(a) exam 27
(b) exam : 25
(c) exam : 9
(d) exam : 999

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : In the given code:
String S = "exam" + 9 + 9 + 9;
• First. "exam" is concatenated with 9, resulting in "exam".
• Then, "exam" is concatenated with the next 9, resulting in "exam 99".
Finally, "exam 99" is concatenated with the last 9, resulting in "exam 999".
Therefore, system out.
Prints (s), will print in exam 999.

114. What is function of Java p command?
Java p कमांड का कार्य क्या है?

(a) A command for running the Java compiler/ Java कंपाइलर चलाने के लिए एक कमांड
(b) A command that disassembles Java class files / Java क्लास फाइलों को अलग करने के लिए एक कमांड
(c) A command for running a non-optimized version of Java compiler/ Java कंपाइलर के नॉन ऑप्टिमाइज्ड वर्जन को चलाने क लिए एक कमांड
(d) A command that is used to generate API-style HTML documentation/एक कमांड जिसका उपयोग ए.पी.आई-स्टाइल एच.टी.एम.एल. दस्तावेज उत्पन्न करने के लिए किया जाता है

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : The Java p command is a Java class file disassemble. It inspects compiled Java class files (.class files) and provides information about their structure, including fields, methods, and bytecode instructions. It does not run the Java compiler (which is done by Java p).

115. Which of the following statement is/are correct regarding Java programming?

I. A class marked as final cannot be inherited
II. A method marked final cannot be overridden

Java प्रोग्रामिंग के संदर्भ में निम्नलिखित में से कौन सा/से कथन सत्य हैं?

I. एक क्लास जिसे फाइनल चिह्नित किया गया है को इन्हेरिटेड नहीं किया जा सकता है
II. एक विधि जिसे फाइनल चिह्नित किया गया को ओवरराइड नहीं किया जा सकता है।

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Statement (I) A class marked as final cannot be inherited. This statement is true. (In Java , when a class is declare with the final keyword, it cannot be

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extended or inherited by any other class, there by preventing subclassing.
Statement (II) A method marked as final cannot be overridden.
Both statement are correct option (c) is the correct answer.

116. Java compiler translates source code into _____.
Java कंपाइलर, सोर्स कोड को _____ में ट्रांसलेट करता है।
(a) Symbolic code/सिम्बॉलिक् कोड
(b) Assembly code/असेंबली कोड
(c) Byte code/बाइट कोड
(d) High language code/हाई लैंग्वेज कोड

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : The Java compiler translate Java source code into byte code, not symbolic code, assembly code, or high-level language code.
Byte code can be defined as an intermediate code generated by the compiler after the compilation of source code (Java program). This intermediate code makes Java a platform- independent language.

117. Which of the following statement(s) is/are correct regarding Java programming?
Java प्रोग्रामिंग के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. Place a semicolon immediately after a method header./मेथड हैडर के ठीक बाद सेमीकोलन लगाएं।
II. Define a method outside the braces of a class definition./एक क्लास डेफिनिशन के ब्रेसिज़ के बाहर एक मेथड को डिफाइन करें
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Statement-I–Place a semicolon immediately after a method header. This is incorrect. In Java you don't place a semicolon alter a method header. Instead, you follow it with the method body enclosed in braces{}.
Statement-II – Define a method outside the braces of a class definition. This is also incorrect. In Java , all methods must be defined within the braces of a class definition you can't have methods floating outside of a class.

118. What is the output of the given Java code?
दिए गए Java कोड का आउटपुट क्या है?

```
byte x = 64, y;
  y = (byte) (x << 4);
    system.out.println(y);
```

(a) 0 (b) 128
(c) 256 (d) 32

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Code-

```
byte x = 64, y;
        y = (byte) (x <<4);
        system.out.println(y)
```

1.Initialization– x is declared as a byte with value 64.
2. Bit shifting operation– X << 4 means we are shifting the bits of x left by 4 position. 64 in binary is 01000000 (as a byte).
Shifting left by 4 give 00000000 (because the 1 gets shifted out of the 8 bit byte range)
3. Type casting–
The result of x<<4 is actually 1024 (64×2^4) but since we're casting back to byte, we only keep the least significant 8 bits.
1024 in binary 0000010000000000 (as 16 bits) Taking only the least 8 bits gives 00000000 (which is 0)
4. Final value.
The value stored in y is 0.
The program will print 0.

119. Which one of the following is correct, when a class grants friend status to another class?
(a) The member functions of the class generating friendship can access the members of the friend class.
(b) All member functions of the class granted friendship have unrestricted access to the members of the class granting the friendship.
(c) Class friendship is reciprocal to each other.
(d) There is no such concept.

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Ans : (b) The function of a class which is given friend status by another class can access all its private, protected and public members. Class friendship is not commutative nor transitive unless explicitly specified.

120. A constructor is called whenever
एक Constructor को कॉल (Call) किया जाता है, जब कभी भी
(a) An object is created
एक ऑब्जेक्ट (object) बनाया जाता है
(b) A class is created
एक क्लास (class) बनाया जाता है।
(c) A class is declared
एक क्लास (class) को declare किया जाता है
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a): The main purpose of the constructor is to initialize the object when you create a new object, the constructor is Automatically called.

121. Which of the following is/are correct about Java programming?
निम्नलिखित में से कौन सा/से Java प्रोग्रामिंग के बारे सही है/हैं?
I. All functions in Java must be members of some class/ Java में सभी फलन किसी न किसी क्लास के मेंबर होने चाहिए
II. Member functions are called methods in Java
मेंबर फलन को Java में विधियां कहा जाता है
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : In Java , everything is organized around classes. Java is a pure object-oriented language, and unlike C/C++, there is no standalone functions- all methods must be part of a class.

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In Java , functions defined within a class are called methods. This is standard Java terminology.
Hence, both statements are correct.

122. What is the output of the following Java code?
निम्नलिखित Java कोड का आउटपुट क्या है?

```
int m = 1000;
int k = 3000;
while (+ + m < – – k);
System.out.println(m);
```

(a) Error/त्रुटि (b) 1000
(c) 2000 (d) 4000

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c)
```
int m = 1000;
    int k= 3000;
    while (+ + m < --k);
  System.out.println(m);
```
1. Initialization:
m = 1000
k = 3000
2. Loop condition: + + m < --k
+ + m → pre-increment m before comparing
--k → pre-decrement k before comparing
3. Effect per iteration:
m increases by 1 each loop
k decreases by 1 each loop
condition checks if the new m is still less than the new k
4. When will it stop:
Loop ends when m > = k
They "meet" at the midpoint between original m and k;
midpoint = (1000+3000)/2=2000
5. Printed output: 2000

(viii) Enterprise Java beans (EJB)

123. Every EJB class file has
प्रत्येक EJB वर्ग संचिका के साथ होती है

(a) One accompanying interface and one XML file/एक संलग्न अंतरापृष्ठ एवं एक XML संचिका
(b) One accompanying interface and two XML files/एक संलग्न अंतरापृष्ठ एवं दो XML संचिका
(c) Two accompanying interfaces and one XML file/दो संलग्न अंतरापृष्ठ एवं एक XML संचिका
(d) Two accompanying interfaces and two XML files/दो संलग्न अंतरापृष्ठ एवं दो XML संचिका

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Ans. (c) : Every Enterprise Java Beans (EJB) class file has two accompanying interfaces and one XML file.
Two interfaces –
1. Remote interface - Specifies the business methods that clients of the object can invoke.
2. Home interface - Defines EJB life cycle methods, such as a method to create and retrieve a reference to the bean object.
One XML file - The XML file contains the deployment descriptor, which declares the contents of the module, defines the structure and external dependencies of the beans in the module and describes how the beans are to be used at runtime.

124. Match List-I with List-II

	List-I (Programming Term)		List-II (Meaning)
A.	**JNDI**	**i.**	**Runtime support for running Java programs**
B.	**RMI**	**ii.**	**The API in support of naming and directory services**
C.	**JDK**	**iii.**	**The methods provided by the Java development kit and runtime support for calling remote methods**
D.	**JRE**	**iv.**	**The compiler and class libraries to develop Java applications**

Choose the correct answer from the options given below:
(a) A - II , B - III, C - IV, D - I
(b) A - II, B - IV, C - III, D - I
(c) A - I, B - III, C - IV, D - II
(d) A - III, B - II, C - IV, D - I

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Ans. (a) :

List - I (Programming Term)		List - II (Meaning)	
A.	JNDI	**ii.**	The API in support of naming and directory services
B.	RMI	**iii.**	The methods provided by the Java development kit and runtime support for calling remote methods
C.	JDK	**iv.**	The compiler and class libraries to develop Java applications
D.	JRE	**i.**	Runtime support for running Java programs

(ix) Java as Internet Programming Language

125. Consider the Java Script Code :

```
var y= "12";
function f( ) {
        var y= "6";
        alert (this.y);
        function g( ) {alert (y); }
        g( );
}
f( );
```

If M is the number of alert dialog boxes generated by this Java Script code and D1, D2, ..., DM represents the content displayed in each of the M dialog boxes, then :

(a) M=3; D1 displays "12"; D2 displays "6"; D3 displays "12".

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(b) M=3; D1 displays "6"; D2 displays "12"; D3 displays "6".
(c) M=2; D1 displays "6"; D2 displays "12".
(d) M=2; D1 displays "12"; D2 displays "6".

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Ans. (d) : Alert box will be called 2 times so M = 2
first dialog box alert (this y) is used to access the value of variable which is outside from
f ( ) · So D1 = 12
Second dialog box alert (y) inside nested function g ( ) is used to access the local variable y. So D2 will be 6.

126. A Java program that is embedded in HTML document and runs in the context of Java capable browser is known as _____.
एक Java प्रोग्राम जो HTML डॉक्यूमेंट में अंतर्निहित है और Java सक्षम ब्राउजर के संदर्भ में चलता है, _____के रुप में जान जाता है।

(a) Applet (b) Canvas
(c) Doctags (d) Frame window

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : A Java program that is embedded in HTML (Hyper Text Markup Language) document and runs in the context of Java capable browser is known as Applet.
An Applet is a small Java program that is placed inside an HTML page and runs in a Java -enabled browser or an Applet viewer.

127. Which one of the following is correct?

(a) Java applets can be written in any programming language
(b) An applet is not a small program
(c) An applet can be run on its own
(d) Applets are embedded in another applications

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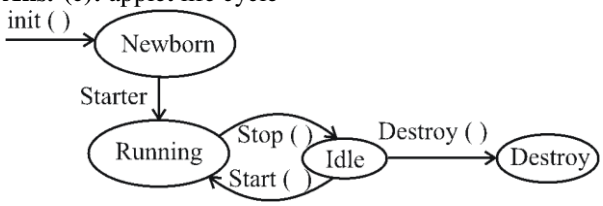
Ans. (d): An applet is a small program that is intended not to be run on its own, but rather to be embedded inside-another application.

128. Which method is called first by an applet program?

(a) start ( ) (b) run ( )
(c) init ( ) (d) begin ( )

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Ans. (c): applet life cycle



So the answer is (c)

129. Which of the following Visual Basic controls is used to accept input from the user as well as to display the output?
निम्नलिखित में से कौन से विजुअल बेसिक कंट्रोल का उपयोग यूजर (उपयोगकर्ता) से इनपुट स्वीकार करने के साथ-साथ आउटपुट प्रदर्शित करने के लिए किया जाता है?

(a) Label/लेबल
(b) Combo Box/कॉम्बो बॉक्स
(c) Text Box/टेक्स्ट बॉक्स
(d) Option Button/ऑप्शन बटन

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Text Box is used to accept input and display output in Visual Basic.
Label→ used only for displaying output, not input
Combo Box→ used for selecting from a list, not ideal for general input / output.
Option Button→used for selection (radio button), not for input/output.

130. Which of the following statements are correct?
निम्नलिखित में से कौन सा कथन सही है?

(i) The 'Label' control of Visual Basic can also be used to display outputs./ विजुअल बेसिक के 'लेबल' कंट्रोल का उपयोग आउटपुट प्रदर्शित करने के लिए भी किया जा सकता है।
(ii) The image in an 'Image Box' control is stretchable./ 'इमेज बॉक्स' कंट्रोल में छवि मापनीय होता है।
(iii) The most common event associated with the 'Command Button' control in Visual Basic is the Click event./विजुअल बेसिक में कमांड बटन कंट्रोल से संबंधित सबसे आम इवेंट क्लिक इवेंट है।

(a) Statements (i) and (ii) are correct
कथन (i) और (ii) सही हैं
(b) Statements (ii) and (iii) are correct
कथन (ii) और (iii) सही हैं
(c) Statements (i) and (iii) are correct
कथन (i) और (iii) सही हैं
(d) Statements (i), (ii) and (iii) are correct
कथन (i), (ii) और (iii) सही हैं

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : All statements are correct.
1. Label control can display.
2. Image Box allows image to be stretched.
3. Click Event is most common for command buttons.

131. Which of the following was the final version of Visual Basic before the release of Visual Basic. NET?
Visual Basic NET के रिलीज होने से पहले निम्नलिखित में से कौन सा विजुअल बेसिक का अंतिम संस्करण था?

(a) Visual Basic 4.0/विजुअल बेसिक 4.0
(b) Visual Basic 5.0/विजुअल बेसिक 5.0
(c) Visual Basic 6.0/विजुअल बेसिक 6.0
(d) Visual Basic 7.0/विजुअल बेसिक 7.0

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Visual Basic 6.0 was the final version before Visual Basic.Net was released. It released in 1998.
Visual Basic.Net came later in 2002 as part of the .NET Framework.

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06. Operating System

(i) Definition, Design Goals, Evolution

1. WINDOWS is popular because of its?
Windows की लोकप्रियता का कारण है?
(a) GUI feature (b) Multi color
(c) Desktop technology (d) Being inexpensive
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : The main reason for the popularity of windows is its GUI (Graphical User Interface). GUI allows the user to work with graphical icons and windows using the mouse, which make it easy and user friendly.

2. Which of the following is not hardware of a computer system ?
निम्नलिखित में से कौन सा एक कम्प्यूटर प्रणाली (सिस्टम) का हार्डवेयर नहीं है?
(a) KEYBOARD
(b) PRINTER
(c) OPERATING SYSTEM
(d) MOUSE
KVS PGT 23.03.2023 (Shift-II)

Ans. (c) : Operating system is not hardware of a computer system where as keyboard, printer and mouse are hardware or peripheral devices because there an be tooth seen and touched. Operating system is system software that manages computer hardware resources and provides common services for computer program.

3. The term "multi-user" in an OS context refers to:
ऑपरेटिंग सिस्टम के संदर्भ में 'मल्टी-यूज़र' शब्द का क्या अर्थ है?
(a) An OS that can execute multiple programs simultaneously/एक ऐसा OS जो एक साथ कई प्रोग्राम चला सकता है
(b) An OS that supports multiple users working on the same machine/एक ऐसा OS जो एक ही मशीन पर कई उपयोगकर्ताओं को समर्थन देता है
(c) An OS that can only run on multiple computers/एक ऐसा OS जो केवल कई कंप्यूटरों पर चल सकता है
(d) An OS with a graphical user interface
एक ऐसा OS जिसमें ग्राफिकल यूज़र इंटरफेस होता है
Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : Multi-User Operating System that allows multiple users to work on the same computer system at the same time. Each user can have his own session or environment and the system manages their tasks.

4. The example of system software is?
System Software का उदाहरण है?
(a) Linux (b) Windows
(c) Both A and B (d) PowerPoint
Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (c) : System software is software that provides an interface between the computer hardware and the user. Linux and Windows are both operating systems that are examples of system software while PowerPoint is an application software.

5. The primary goal of load control is to:
लोड कंट्रोल का मुख्य उद्देश्य है
(a) Prevent processes from running concurrently
प्रोसेसेस को एकसाथ चलने से रोकना
(b) Avoid context switching between processes
प्रोसेसेस के बीच कॉन्टेक्स्ट स्विचिंग से बचना
(c) Ensure that CPU time is evenly distributed among processes/ CPU समय को प्रोसेसेस के बीच समान रूप से वितरित करना
(d) Maximize the usage of virtual memory
वर्चुअल मेमोरी के उपयोग को अधिकतम करना
Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Load control in operating system is primarily concerned with managing how the CPU and system resource are allocated among running processes.
Its mean goal is to balance the CPU load ensuring that no single process hogs. The CPU and that all processes get a fair share of CPU time.

6. Real-Time operating system always runs on
रीयल टाइम ऑपरेटिंग सिस्टम हमेशा किस पर चलता है?
(a) Linux./लिनक्स
(b) Unix./यूनिक्स
(c) Embedded system./ऐम्बैडेड सिस्टम
(d) Apple's Mac OS./ऐपल का मेक ओ.एस.
KVS PGT C.S. 2013

Ans. (c) : Real time operating system always run or embedded system. Embedded System are used in real time environment to communicate with the hardware. Real time operating are designed to handle multiple processes at one time. ensuring that these processes respond to events within a predictable time limit.

7. Which of the following is a type of operating system?
निम्नलिखित में से कौन ऑपरेटिंग सिस्टम का एक प्रकार है?
(a) Real-time operating system
(b) Embeded operating system
(c) Network operating system
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) **:** All of these options are different types of operating system–
Real-Time Operating System– It is designed for sensitive applications where quick response is required.

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Network Operating System– It manages and shares resources between computers through a network.
Embedded Operating System– It is attached to special devices, such as Smartphone's, TVs and other electronic devices.

8. The number of results produced per unit time by machine is called
मशीन द्वारा प्रति यूनिट समय के उत्पादन के परिणामों की संख्या कहा जाता है
(a) Complexity/जटिलता (b) Throughput/थ्रूपुट
(c) Delay/विलंब (d) Stages/स्टेजेज

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : Throughput measures the rate of successful work completion. It measures the efficiency and capacity of the system, indicating how many tasks or operations it can handle in specific time frame such as per unit time.

9. Which of the following is a file system commonly used in operating systems?
निम्नलिखित में से कौन-सा फाइल सिस्टम सामान्यतः ऑपरेटिंग सिस्टम में उपयोग किया जाता है?
(a) NTFS
(b) FAT32
(c) HFS+
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) **:** All the given file systems are commonly used in operating system.
NTFS– It is used in Windows Operating System.
HFS+– It is used in MacOS Operating System.
FAT32– It is used in various operating systems, especially in older systems or when interoperability is required.

10. Which one of the following is not a real time operating system?
निम्नलिखित में से कौन-सा एक वास्तविक समय ऑपरेटिंग सिस्टम नहीं है?
(a) RT Linux
(b) Palm OS
(c) VxWorks
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b): VxWorks, QNX and RT Linux are real-time operating systems. While Palm OS is a mobile operating system. Palm OS is developed for Personal Digital Assistants (PDAs).

11. Consider the following statements:
(A) Windows Azure is a cloud based operating system
(B) Google App Engine is an integrated set of online services for consumers to communicate and share with others.
(C) Amazon Cloud Front is a web service for content delivery.
Which of the statements is (are) correct?
(a) Only (A) and (B) (b) Only (A) and (C)
(c) Only (B) and (C) (d) (A), (B) and (C)

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) : (a) Window Azure is a cloud-based operating system.
This statement is correct. Windows Azure is a cloud based operating system which enables running of business applications, services and work load in the cloud itself. It mainly deals with the foundation of running applications and keeping the data on the cloud.
(b) Google App Engine- is an integrated set of online services for consumers to communicate and share with others. Google app engine is PaaS (Platform as a Service) product. It provides web developing services to developers. Apps must be written in Java or Python. So this statement is incorrect.
(c) Amazon Cloud Front- is a web service content delivery. Amazon cloud front is a fast content delivery network service that securely delivers data, video applications and APIs to customers. Cloud front is integrated with AWS global infrastructure and services. So statement is correct.

12. The extent to which a software tolerates the unexpected problems, is termed as:
(a) Accuracy (b) Reliability
(c) Correctness (d) Robustness

UGC NET C.S. June-2016 Paper-II

Ans : (d) Robustness is the ability of a computer system to cope with errors during execution and cope with erroneous input.
Correctness of an algorithm is asserted when it is said that the algorithm is correct with respect to a specification.

13. If an operating system does not allow a child process to exist when the parent process has been terminated, this phenomenon is called as
(a) Threading (b) Cascading termination
(c) Zombie termination (d) Process killing

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Ans. (b) : Threading– A thread is a separate execution path. It is a lightweight process that the operating system can schedule and run concurrently with other threads.
Cascading termination– Some system, Including VMS, do not allow a child to exist if its parent has terminated. such system, If a process terminates either normally or abnormally, then have to be terminated. This concept is referred to a cascading termination.
Zombie termination– We can say that a zombie process is a terminated or completed process that remains in the process table.
Process killing– terminating a process forcefully is called process killing.

14. Names of some of the Operating systems are given below :
(A) MS–DOS
(B) XENIX
(C) OS/2
In the above list, following operating systems didn't provide multi-user facility.

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(a) (A) only (b) (A) and (B) only
(c) (B) and (C) only (d) (A), (B) and (C)

UGC NET C.S. June-2017 (Paper-III)

Ans : (d)
(i) MS-DOS originally single user, multi-user support provided that did not have present when manufactured.
(ii) XENIX always been multi-user.
(iii) OSI2 does not provide facility of multi-user.
So, none of the O.S. provide facility of multi-user.

15. Which one of the following is NOT an Operating System?
निम्नलिखित में से कौन सा ऑपरेटिंग सिस्टम (प्रचालन तंत्र) नहीं है?

(a) MS-DOS (b) IOS
(c) LDAP (d) LINUX

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : An operating system is system software that acts as an interface between the user and computer hardware. It manages resources like CPU, memory, storage and I/O devices and provides services to run application programs.
Example– Window, Linux, iOS and Android.
LDAP is a protocol, not a software that manages hardware or provides system services.

16. Match the following.
निम्नलिखित को मिलाएं।

I.	**Batch Processing system/ बैच प्रोसेसिंग सिस्टम**	**A.**	**Meeting time constraint/मीटिंग टाइम कॉन्सट्रैन्ट**
II.	**Multi Programming मल्टी प्रोग्रामिंग**	**B.**	**Good response time अच्छा प्रतिक्रिया टाइम**
III.	**Time sharing टाइम शेयरिंग**	**C.**	**Resource utilization संसाधनों के उपयोग**
IV.	**Real time रियल टाइम**	**D.**	**CPU idle time CPU निष्क्रिय टाइम**

(a) I-D, II-C, III-B, IV-A
(b) I-A, II-B, III-C, IV-D
(c) I-D, II-A, III-B, IV-C
(d) I-B, II-C, III-D, IV-A

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Ans. (a) : Correct Matching–

I	Batch Processing system	**D**	CPU idle time
II	Multi Programming	**C**	Resource utilization
III	Time sharing	**B**	Good response time
IV	Real time	**A**	Meeting time constraint

17. Set of one or more programs which controls the operation and/or extends the processing capability of a computer system is:
एक या अधिक प्रोग्रामों का सेट जो संचालन को नियंत्रित करता है और/या कंप्यूटर सिस्टम प्रोसेसिंग क्षमता को बढ़ाता है, हैं–

(a) System Software/सिस्टम सॉफ्टवेयर
(b) Application Software/एप्लीकेशन सॉफ्टवेयर
(c) System Application/सिस्टम एप्लीकेशन
(d) System Hardware/सिस्टम हार्डवेयर

HTET PGT C.S. 2023

Ans.(a): Set of one or more program which control the operation and/or extends the processing capability of computer system is System software.
It acts as a bridge between the hardware and the user/application programs.
Examples of system software are- operating system (Windows,Linux), device drivers, utility programs, language translators (compiler, translators)
Other options:
Application software is for specific user tasks (e.g., MS words , photoshop)
System hardware refers to the physical components.
System application is not a standard category.

18. Which translator is used to convert assembly language to machine language?
असेंबली भाषा को मशीनी भाषा में बदलने के लिए किस ट्रांसलेटर का उपयोग किया जाता है?

(a) Assembler/असेम्बलर
(b) Compiler/कम्पाइलर
(c) Interpreter/इंटरप्रेटर
(d) Linker/लिंकर

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : A programming language translator is a software tool that converts code written in are programming language into another form that can be executed by computer.
There are several types of language translators–
(1) Assembler– An assembler is used to convert assembly language to machine language, which is directly executable by the computers CPU.
(2) Compiler– The compiler convert a program written in high level language (source code) to low level language (object code) in one go.
(3) Interpreter– An Interpreter translates high-level language into machine language line-by-line.

19. Match List – I of firewalls with List–II of their usage. Select the correct answer from the given options.
फायरवॉल की सूची-। को उनके उपयोग की सूची-।। के साथ मिलाएं। दिए गए विकल्पों में से सही उत्तर का चयन करें।

List-I/सूची-।		**List-II/सूची-।।**	
A	**Stateful inspection firewall/स्टेटफुल इंस्पैक्शन फायरवॉल**	**i**	**Telnet टेलनेट**
B	**Packet filter firewalls/पैकेट फिल्टर फायरवॉल**	**ii**	**Operating systems /ऑपरेटिंग सिस्टम**
C	**Proxy server firewalls/प्रोक्सी सर्वर फायरवॉल**	**iii**	**Mail server मेल सर्वर**
D	**Application level firewalls/ऐप्लिकेशन लेवल फायरवॉल**	**iv**	**Inbound network traffic/ इनबाउंड नेटवर्क ट्रैफिक**

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(a) A-iv, B-ii, C-iii, D-i
(b) A-ii, B-i, C-iv, D-iii
(c) A-i, B-iv, C-ii, D-iii
(d) A-iii, B-i, C-iv, D-ii

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : The correct match-

List-I		List-II	
A	Stateful inspection firewall	**iv**	Inbound network traffic
B	Packet filter firewalls	**ii**	Operating systems
C	Proxy server firewalls	**iii**	Mail servers
D	Application level firewalls	**i**	Telnet

20. An operating system that helps in sharing the files across the network is known as _____ .
एक ऑपरेटिंग सिस्टम जो पूरे नेटवर्क में फाइलों को शेयर करने में मदद करता है, _____ के रूप में जाना जाता है।

(a) Network operating system
नेटवर्क ऑपरेटिंग सिस्टम
(b) Real operating system
रियल ऑपरेटिंग सिस्टम
(c) Time operating system
टाइम ऑपरेटिंग सिस्टम
(d) Coupling operating system
कपलिंग ऑपरेटिंग सिस्टम

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : A network operating system specialized software designed to manage and coordinate network resources, services and data or file across the network.

21. Match the List-I with List-II

List-I (Functions)		List-II (Uses of functions)	
A.	**fseek ()**	**I.**	**Gives the current position in the file**
B.	**ftell ()**	**II.**	**Sets the position to the beginning of the file**
C.	**feof ()**	**III.**	**Sets the position to a desired point in the file**
D.	**rewind ()**	**IV.**	**Used to test for an end of file condition.**

Choose the correct answer from the options given below :

(a) A-II, B-III, C-I, D-IV
(b) A-III, B-I, C-IV, D-II
(c) A-III, B-I, C-II, D-IV
(d) A-II, B-IV, C-III, D-I

UGC NET C.S. December-2024

Ans. (b) : Correct match-

List-I (Functions)			List-II (Uses of functions)
A.	fseek ()	–	Sets the position to a desired point in the file
B.	ftell ()	–	Gives the current position in the file
C.	feof ()	–	Used to test for an end of file condition.
D.	rewind ()	–	Sets the position to the beginning of the file

22. Which of the following is not an operating system?
निम्नांकित में से कौन-सा ऑपरेटिंग सिस्टम (Operating System) नहीं है?

(a) Linux/लिनक्स
(b) Windows/विन्डोज
(c) Oracle/ओरेकल
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : Oracle is a database software company that develops database management systems, not an operating system.

23. A software that runs on a computer hardware and serves as a platform for other softwares to run is called
कम्प्यूटर हार्डवेयर पर चलने वाला एक सॉफ्टवेयर जो दूसरे सॉफ्टवेयर को चलाने के लिए प्लेटफार्म का काम करता है, कहलाता है

(a) System software
सिस्टम सॉफ्टवेयर
(b) Operating software
ऑपरेटिंग सिस्टम
(c) Application software
ऐप्लिकेशन सॉफ्टवेयर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : An operating system is software running on computer hardware that serves as a platform for other software to run.

24. Which of the following is a technique that marks the beginning of computer communications?
निम्नलिखित में से कौन-सी तकनीक कम्प्यूटर संचार की शुरूआत को चिह्नित करती है?

(a) User Environment/उपयोगकर्ता पर्यावरण
(b) Batch Environment/बैच पर्यावरण
(c) Time Sharing/समय साझाकरण
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : Time sharing is the process by which multiple users share access to a single computer system or network, allowing for shared use of time and resources. It facilitates the sharing of data and collaboration between multiple users. Time sharing technology marks the beginning of computer communication.

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25. An integrated set of programs that controls the resources of computer system and provides its users an interface that is easy to use:
प्रोग्रामों का एक एकीकृत सेट जो कम्प्यूटर सिस्टम के संसाधनों को नियंत्रित करता है और अपने उपयोगकर्ता को एक ऐसा इंटरफेस प्रदान करता है जिसका उपयोग करना आसान होता है:
(a) Operating system/ऑपरेटिंग सिस्टम
(b) Compiler/कम्पाइलर
(c) High level language program
हाई लेवल लैंग्वेज प्रोग्राम
(d) Interpreter/इन्टरप्रेटर

HTET PGT C.S. 2023

Ans.(a): Operating System (OS) is an integrated set of programs that acts as the manager of computer resources. It controls hardware resources such as the CPU, memory, Input/Output devices and storage. Without the OS, users cannot interact with the hardware directly.
The OS also provides a user-friendly interface so that people can execute programs, manage files and use applications easily, without needing to understand hardware level details.

(ii) Structure and Functions of Operating System

26. The term "FAT" is stands for ……
'FAT' शब्द का पूर्ण रूप क्या है?
(a) File Allocation Tree/फाइल एलोकेशन ट्री
(b) File Allocation Table/ फाइल एलोकेशन टेबल
(c) File Allocation Graph/ फ़ाइल एलोकेशन ग्राफ़
(d) All of the above/उपरोक्त सभी

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : The full form of FAT is File Allocation Table. It is a file system architecture that tracks the position and storage location of files.

27. In which addressing mode, the operand is given explicitly in the instruction (example instruction : ADD R4, #3)?
किस पताभिगमन विधि में संकार्य (ओपरेन्ड) स्पष्ट रूप से अनुदेशों में दिया जाता है (उदाहरण अनुदेश: ADD R4, #3) ?
(a) Based Indexed mode
(b) Absolute mode
(c) Immediate mode
(d) Register indirect mode

KVS PGT C.S. 2017

Ans. (c) : The term addressing modes refers to the way in which the operand of an instruction is specified. The addressing specifies a rule for interpreting or modifying the address field of the instruction before the operand is actually executed. In immediate addressing mode (symbol#), the operand is given explicitly in the instruction. Designed like one address instruction format.

28. Select the correct order of events after power is initialized on a system.
A. Bootstrap loader is loaded from the disk
B. Kernel is loaded onto the memory
C. Firmware ROM loads boot block
Choose the correct answer form the options given below :
(a) B, C, A (b) C, A, B
(c) A, B, C (d) A, C, B

UGC NET C.S. December-2022

Ans. (b) : Power is initialized on system
- Kernel is loaded onto the memory.
- Bootstrap loader is loaded from the disk.
- Firm ware ROM loads boot block.

29. An Operating System (OS) crashes on the average once in 30 days, that is, the Mean Time Between Failures (MTBF) = 30 days. When this happens, it takes 10 minutes to recover the OS, that is, the Mean Time To Repair (MTTR) = 10 minutes. The availability of the OS with these reliability figures is approximately:
(a) 96.97% (b) 97.97%
(c) 99.009% (d) 99.97%

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) System crashes once in 30 days and need 10 minutes to get repaired. Either convert 30 days into minute fraction of time system is not available = (10/(30 × 24 × 60) × 100 = ·023%
Availability = 100 – 0.023 = 99.97%

30. System calls are usually invoked by using :
(a) A privileged instruction
(b) An indirect jump
(c) A software interrupt
(d) Polling

UGC NET C.S. December-2015 (Paper-II)

Ans : (c) Privileged instruction cannot be the answer as system call is done from user mode & privileged instruction cannot be done from user mode.

31. Which of the following command is used in the communication system call of an operating system?
(a) Request device, release device
(b) Attach or detach remote devices
(c) Allocate and free memory
(d) Read, write, reposition

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : Communication system calls are responsible for managing inter process communication (IPC) and network communication.
(b) Attack or detach remote devices– Attack and detaching remote devices in an operating system involves managing how the system interacts with external hardware or virtual resources. This can range from connecting a physical device like a USB drive or printer to establishing a network connection.
(a) Request device, release device– These are typically part of device management system calls, used for allocating and deal locating local hardware resources.

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(c) Allocate and free memory– These are memory management system calls, dealing with the allocation and deal location of memory to processes.
(d) Read, write, reposition– These are primarily file management or device management system calls used for interacting with files or devices for data transfer.

32. An operating system is system software that works as an interface between the computer and an user, it does the following works?
Operating system, user और कंप्यूटर के बीच का माध्यम होता है, और यह निम्न कार्य करता है?

(a) Controls input & output
(b) Control program execution
(c) Manages use of main memory
(d) All of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : An operating system (OS) is system software that acts as in inter face between that acts as in interface between that user and computer hardware. It perform multiple essential tasks:
- Controls input so output
- Control program execution
- Manages us of main memory.

33. Which of the following statements is FALSE?
निम्नलिखित में से कौन सा कथन गलत है?

(a) A trap is a hardware-generated interrupt caused either by an error or a user request.
ट्रैप एक हार्डवेयर-जनरेटेड इंटरप्ट है जो या तो किसी त्रुटि या यूजर (उपयोगकर्ता) रिक्वेस्ट (अनुरोध) के कारण होती है।
(b) An operating system is interrupt-driven.
ऑपरेटिंग सिस्टम इंटरप्ट संचालित होता है।
(c) A bootstrap program is loaded at power-up or reboot./बूटस्ट्रैप प्रोग्राम को पावर-अप या रीबूट पर लोड किया जाता है।
(d) The program running at all times on a computer is the kernel./कंप्यूटर पर हर समय चल रहा प्रोग्राम कर्नेल है।

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) :
(b) An operating system is interrupt-driven– The OS relies on interrupts to handle I/O. CPU scheduling and system calls.
(c) A bootstrap program is loaded at power-up or reboot– The bootstrap program (stored in ROM/firmware) initializes the system and loads the operating system kernel.
(d) **The program running at all times on a computer is the kernel**– The kernel is the core program that always remains in memory to manage resources.
(a) A trap is a hardware generated interrupt caused either by an error or a user request– False A trap is not hardware-generated. It is a software generated interrupt, caused intentionally by a user program (e.g. system call) or unintentionally (e.g. division by zero).

34. Match the following:
निम्नखिखित का मिलान करें:

List-I/सूची-I		List-II/सूची-II	
I	Long-term scheduler लॉन्ग-टर्म शेड्यूलर	A	Swapping स्वैपिंग
II	Medium-term scheduler मीडियम-टर्म शेड्यूलर	B	Job Scheduling जॉब शेड्यूलिंग
III	Short-term scheduler शॉर्ट-टर्म शेड्यूलर	C	CPU Scheduling सी.पी.यू. शेड्यूलिंग

(a) I-C, II-A, III-B (b) I-A, II-B, III-C
(c) I-B, II-A, III-C (d) I-A, II-C, III-B

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Long-term Scheduler (Job Scheduling)– The long-term scheduler decides which jobs should enter the system for execution. These jobs are usually stored in secondary storage (like a disk). Hence long term scheduler is responsible for job scheduling.
Medium-term Scheduler (Swapping)– The medium-term scheduler is responsible for temporarily suspending or resuming processes. Therefore the medium term scheduler is associated with swapping.
Short-term scheduler (CPU Scheduler)– The short-term scheduler decides which process in the ready queue will be executed next by the CPU. Hence the short-term scheduler is responsible for CPU scheduling.

35. Match the following.
निम्नलिखित को मिलाएं।

1.	Disk scheduling डिस्क सेड्यूलिंग	i.	Shortest Job First शॉर्टेस्ट जॉब फर्स्ट
2.	CPU scheduling CPU सेड्यूलिंग	ii.	Banker's Algorithm बैंकर्स अल्गोरिथम
3.	Page replacement पेज रिप्लेसमेंट	iii.	Shortest Seek Time शॉर्टेस्ट सीक टाइम
4.	Deadlock डेडलॉक	iv.	Least Recently Used लीस्ट रिसेंटली यूजड

(a) 1 – iii, 2 – i, 3 – iv, 4 – ii
(b) 1 – i, 2 – iv, 3 – ii, 4 – iii
(c) 1 – iii, 2 – iv, 3 – i, 4 – ii
(d) 1 – iv, 2 – i, 3 – ii, 4 – iii

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Matched List–

1.	Disk scheduling	iii.	Shortest Seek Time
2.	CPU scheduling	i.	Shortest Job First
3.	Page replacement	iv.	Least Recently Used
4.	Deadlock	ii.	Banker's Algorithm

36. Which of the following is a function of an operating system?
निम्नलिखित में से कौन एक ऑपरेटिंग सिस्टम का कार्य है?

I. Memory Allocation/मेमोरी एलोकेशन
II. Resource Utilization/संसाधन उपयोग

(a) Only I/ केवल I
(b) Only II/ केवल II

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(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : An operating system (OS) serves as the intermediary between users and computer hardware. Its primary functions include.
(1) Memory Management– Manages the system's memory, including allocation and de allocation of memory spaces as needed by programs, ensuring optimal use of RAM, and handling memory protection and sharing among process.
(2) Resource Utilization– The operating system optimizes the use of various resources such as CPU, memory, disk space and input/output devices. It ensures that resources are used efficiently and fairly among all processes and users. Thus, both I and II are correct.

37. Which of the following is not a operating system for mobile computing?
निम्नलिखित में से कौन सा मोबाइल कम्प्यूटिंग में एक ओपरेटिंग सिस्टम नहीं है?
(a) Windows Mobile/विंडोज़ मोबाइल
(b) Palm OS/पाम ओ.एस.
(c) Symbian OS/सिम्बियन ओ.एस.
(d) Manet OS/मनेट ओ.एस.

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Windows Mobile– This was a Microsoft-developed mobile operating system. While it's no longer actively supported, it was a legitimate mobile operating system.
Palm OS– Palm OS was a popular mobile operating system used on Palm device and other smart phones. It was later acquired by Palm computing and eventually discontinued.
Symbian OS Symbian was another widely used mobile operating system before being largely superseded by android. It was particularly popular in Europe.
Monet OS– likely refers to a network technology called Mobile Ad-hoc Network (MANET). This is a network formed by portable devices that can communicate directly with each other without the need for a central infrastructure like a router.

38. Which of the following is a type information system?
निम्नलिखित में से कौन सूचना प्रणाली का एक प्रकार है?
(a) Transaction Processing System
ट्रांज़ेक्शन प्रोसैसिंग सिस्टम
(b) Online Processing System
ऑनलाइन प्रोसैसिंग सिस्टम
(c) Real-time Processing System
रीयल-टाइम प्रोसैसिंग सिस्टम
(d) Distributed Processing System
डिस्ट्रीब्युटिड प्रोसैसिंग सिस्टम

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Information systems are classified by organisational levels, mode of data, processing, system objectives and type of support provided.
Types of Information System–
1. Transaction Processing System (TPS)– Transaction Processing System are information system that processes data resulting from the occurrences of business transactions.
Their objectives are to provide transaction in order to update records and general reports i.e. to perform store keeping function. The transaction is performed in two ways, Batching processing and online transaction processing.
2. Management Information System (MIS)
3. Decision Support System (DSS)
3. Export System.

39. Which of the following characteristic(s) of expert systems is/are correct?
एक्सपर्ट सिस्टम्स की निम्नलिखित में से कौन सी विशेषता/विशेषताएं सही है/हैं?
I. The direction of interaction is from machine to the user./इंटरेक्शन की दिशा मशीन से उपयोगकर्ता तक होती है।
II. Its domain of knowledge is narrow.
इसके ज्ञान का क्षेत्र संकीर्ण है।
III. Information is produced by analysis of operational and external data.
सूचना परिचालन और बाहरी डेटा के विश्लेषण द्वारा उत्पादित की जाती है।
(a) I and III (b) Only II
(c) I and II (d) I, II and III

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Statement-I – The direction of interaction is from machine to the user. This statement is correct. Export systems are designed to provide advice, explanations or solutions to users. The system takes user queries or inputs and then processes its knowledge base to generate responses and recommendations back to the user.
Statement-II – Its domain of knowledge is narrow. This statement is correct. Expert systems are typically designed to operate within a very specific and will defined domain. This narrow focus allows them to accumulate deep knowledge and expertise in that particular area.
Statement-III – information is produced by analysis of operational and external data. This is incorrect while some advanced expert systems might incorporate data analysis, the primary way expert systems produce information is by applying rules and reasoning mechanisms to a knowledge base that is typically built by human experts.
Therefore, only statement-I and II are correct characteristics of expert systems.

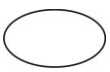
40. Which of the following methods can be used to organize notes for interpreting an information?
किसी सूचना को इन्टरप्रेट करने के लिए नोट्स को व्यवस्थित करने के लिए निम्नलिखित में से किस विधि का उपयोग किया जा सकता है?
I. Flow chart/फ्लो चार्ट
II. Transition devices/ट्रांजिशन डिवाइसेस
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

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Ans. (a) : Flowchart is a diagram representing a process, workflow, or algorithm. It uses various symbols to depict steps, decisions, and the flow of control. Flowchart is excellent tools for visually organising sequential information, relationship between different parts of a process, or decision paths. When interpreting information, especially information related to processes, events, or logical steps, creating a flowchart can be a very effective way to organize your notes and understand the structure and flow of the information.

41. Which symbol in flowchart is used for representing start and end?
फ़्लोचार्ट में कौन सा प्रतीक स्टार्ट और एंड का प्रतिनिधित्व करने प्रयोग किया जाता है?

(a) 
(b) 
(c) 
(d) 

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : In a flowchart the oval (or rounded rectangle) symbol is used to represent the start and end of a process
Start– The first oval in the flowchart marks the beginning of the process.
End– the last oval indicates where the process terminates.

42. The memory resident components of an operating system are commonly known as the - _____ of the operating system.
एक ऑपरेटिंग सिस्टम के मेमोरी रेजिडेंट घटकों को आमतौर पर ऑपरेटिंग सिस्टम के _____ के रूप में जाना जाता है।

(a) Blocks/ब्लॉक्स (b) Kernel/कर्नेल
(c) Partition/पार्टिशन (d) Segments/सेग्मेंट

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Operating System Kernel– The operating System kernel is the core component of an operating system that directly interacts with the hardware, managing memory, disk storage, and low-level networking it provides on abstracted interfaces for the rest of the operating system allowing it to read and write files and communicate on the network without needling to understand the underlying hardware details. The abstract interface presented by the kernel allow other parts of the operating system to read and write files or communicate on the network without knowing or learning what kinds of disks or network adapter installed.

43. Which of the following are monolithic kernel systems?
निम्नलिखित में से कौनसे मोनोलिथिक कर्नेल सिस्टम हैं?

I. Linux **II. windows**
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : A monolithic kernel architecture means the entire operating system runs in kernel space, with all its functionality compiled into a single unit. Linux and windows are both considered monolithic kernels. Examples of operating systems that implement the monolithic kernel operating system architecture are Microsoft Windows, Linux, Solaris, MS-DOS, OpenVMS, etc.

44. Which of the following is not typically a benefit of dynamic linking?
I. Reduction in overall program execution time.
II. Reduction in overall space consumption in memory.
III. Reduction in overall space consumption on disk.
IV. Reduction in the cost of software updates.
(a) I and II (b) I only
(c) II and III (d) IV only

UGC NET C.S. June-2016 (Paper-II)

Ans : (b)
(i) A dynamic linker is the part of an operating system that loads and links the shared libraries needed by an executable when it is executed (at "run time"), by copying the context of libraries from present storage to RAM, and filling jump tables and relocating pointers.
(ii) Static linker has reduction in overall program execution time than dynamic linker.

45. What do a peripheral device generate if it needs immediate attention from the operating system?
यदि एक परिधीय उपकरण को ऑपरेटिंग सिस्टम से तत्काल ध्यान देने की आवश्यकता है, तो यह क्या उत्पन्न करता है?

(a) Interrupt/इंटरप्ट (b) Spool/स्पूल
(c) Stack/स्टैक (d) Page File/पेज फाइल

HTET PGT C.S. 2020

Ans. (a) : Peripheral device– A peripheral device (like a keyboard, mouse, printer, or disk controller) often needs to inform the operating system that it requires immediate attention. For example when a key is pressed or a disk read is completed, the device cannot wait in silence. Instead, it generates an Interrupt signal.
An Interrupt is a hardware or software signal sent to the CPU. It temporarily pauses the current execution, saves the state, and transfers control to an Interrupt Services Routine (ISR) to handle the request. Once the request is serviced, normal execution resumes.

46. Which concept abstracts the hardware of a single computer into several different execution environments enabling execution of multiple operating systems concurrently?
(a) Virtual component
(b) Virtual machine
(c) Virtual compiler
(d) Virtual memory
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Sarkari Resource

Ans. (b) : The concept that abstracts the hardware of a single computer into multiple execution environments is called a virtual machine (VM).
A virtual machine allows a single physical computer to run multiple operating system concurrently.
This is widely used for server virtualization, cloud computing and software testing.

(iii) Process Management and Memory Management

47. In an operating system, a "buffer cache" is used to:
ऑपरेटिंग सिस्टम में 'बफर कैश' का उपयोग किया जाता है:

(a) Store files in memory
फाइलों को मेमोरी में स्टोर करने के लिए
(b) Store copies of frequently used files in memory/अक्सर उपयोग की जाने वाली फाइलों की प्रतियाँ मेमोरी में स्टोर करने के लिए
(c) Store the operating system kernel
ऑपरेटिंग सिस्टम कर्नेल को स्टोर करने के लिए
(d) Store input/output devices
इनपुट/आउटपुट डिवाइस को स्टोर करने के लिए

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : The buffer cache is a high-speed memory area where copies of frequently accessed files or data are kept. This reduce the number of disk accesses and increases system performance.

48. _____ refers to the amount of time required to position the read – write head of a hard disk on appropriate sector.
_____किसी उपयुक्त क्षेत्र में हार्ड डिस्क के रीड-राइट शीर्ष को रखने के लिए अपेक्षित समय की मात्रा को संदर्भित करता है।

(a) Load time (b) Seek time
(c) Access time (d) Rotational Iatency

KVS PGT C.S. 2017

Ans. (b) : Seek Time is the time that is taken by the head of a disc to move from one track to another track on a disk. Seek Time can vary a lot upon where the head is present right now when the read/write request is sent. so, we can say seek Time refers to the amount of time required to position the read-write head of a hard disk on appropriate sector.

49. An operating system contains 3 user processes each requiring 2 units of resource R. The minimum number of units of R such that no deadlock will ever occur is :
किसी ऑपरेटिंग सिस्टम में 3 प्रयोक्ता प्रक्रम है और प्रत्येक को रिसोर्स R की दो इकाइयों की आवश्यकता है 'R' की इकाईयों की न्यूनतम इकाइयाँ क्या होगी जिससे कभी डेडलॉक नहीं होगा :

(a) 5 (b) 2
(c) 3 (d) 4

KVS PGT C.S. 2017

Ans. (d) : Given number of processes = 3 process each process requires 2 units of resource R. Initially, if there is a need for n resources by each process, allocate n-1 resources to each process. Here 3 processes, so each process initially gets 1 resource. Now total resources allocated are 1 + 1 + 1 = 3. But in this situation, the deadlock will occur. So, increase the resource allocated by 1.hence, The total number of units is 4 so that the deadlock will not occur.

50. The Kernel keeps track of the state of each executing program by using a data structure called _____.
करनेल प्रत्येक निष्पादन प्रोग्राम की स्थिति पर ''डेटा स्ट्रक्चर'' की सहायता से संपर्क (ट्रेक) बनाए रखता है जिसे _____कहा जाता है।

(a) Process control block
(b) User control block
(c) File control block
(d) Memory control block

KVS PGT C.S. 2017

Ans. (a) : The kernel keeps track of the state of each executing program by using a data structure called Process Control Block (PCB). PCB is a data structure used by computer operating systems to store all the information about a process It is also known as process descriptor .

51. To load a byte of data parallelly into a shift register with a synchronous load, there must be _____.
सिंक्रोनस लोड के साथ शिफ्ट रजिस्टर में समानांतर रूप से डाटा की एक बाइट लोड करने के लिए निम्नलिखित में से किसकी आवश्यकता होगी?

(a) One clock pulse/एक क्लॉक पल्स
(b) One clock pulse for each 1 in the data
डाटा में प्रत्येक 1 के लिए एक क्लॉक पल्स
(c) Eight clock pulses/आठ क्लॉक पल्स
(d) One clock pulse for each 0 in the data
डाटा में प्रत्येक 0 के लिए एक क्लॉक पल्स

KVS PGT C.S. 2018

Ans. (a) : To load a byte of data parallel into a shift register with synchronous load, there must be one clock pulse. A signal used to synchronize the operations of electronic system. Clock pulses are continuous, precisely spaced changes in voltage.

52. File operations that manipulate the 'open-count' maintained for each entry in open-file table include:
फाइल ऑपरेशंस, जो कि ओपन-फाइल टेबल में प्रत्येक प्रविष्टि के लिए रखे गए ओपन-काउंट को मैनिपुलेट करते हैं, उनमें शामिल है:

(a) Open, write/ओपन, राइट
(b) Read, write/रीड, राइट
(c) Write, close/राइट, क्लोज
(d) Open, close/ओपन, क्लोज

KVS PGT C.S. 2018

Sarkari Resource

Ans. (d) : File operation that manipulate the 'open-count' maintained for each entry in open-file table include open and close. Open file table stores the information about all the files that are open while the OS is running.

53. The process of removing deficiencies and loop holes in the data is called as _____.
डाटा में खामियों तथा लूप होल्स को हटाने की प्रक्रिया _____ कहलाती है।

(a) Data aggregation/डाटा एग्रीगेशन
(b) Extraction of data/एक्स्ट्रेक्शन ऑफ डाटा
(c) Compression of data/कम्प्रेशन ऑफ डाटा
(d) Cleaning of data/क्लीनिंग ऑफ डाटा

KVS PGT C.S. 2018

Ans. (d) : The process of removing deficiencies and loop holes in the data is called as cleaning of data. Data cleaning is the process that removes data that does not belong in your dataset.

54. In a computer system, memory mapped access takes 100 nanoseconds when a page is found in TLB. In case the page is not TLB, it takes 400 nanoseconds to access. Assuming a hit ratio of 80%, the effective access time is:
किसी कम्प्यूटर सिस्टम में, यदि 'TLB' में कोई पेज मिलता है तो मेमोरी मैप्ड एक्सेस 100 नैनोंसेकेंड लगाता है। यदि पेज TLB में नहीं है तो यह एक्सेस में 400 नैनोसेकेंड लगाता है। मान लीजिए हिट अनुपात 80% है तो प्रभावी एक्सेस टाइम होगा:

(a) 120 nanoseconds/120 नैनोसेकेंड
(b) 160 nanoseconds/160 नैनोसेकेंड
(c) 200 nanoseconds/200 नैनोसेकेंड
(d) 500 nanoseconds/500 नैनोसेकेंड

KVS PGT C.S. 2018

Ans. (b) : Given data
Hit Ratio (H) = 80% = 0.8
Miss Ratio (1 – H) = 20% = 0.2
TLB access time = T
Main memory access time = M
Page is found (T + M) = 100 ns
Page is not found (T + 2×M) = 400 ns
So, effective memory access time
= H×(T + M) + (1 – H) (T + 2×M)
= 0.80×100+0.20×400
= 80 + 80
=160 ns

55. Both the first - fit and best - fit strategies for memory allocation suffer from
स्मृति नियतन (मेमोरी एलोकेशन) के लिए फर्स्ट-फिट और बेस्ट-फिट स्ट्रेटेजी किससे हानि पाती है ?

(a) External fragmentation./बाह्य विखंडन
(b) Internal fragmentation./आंतरिक विखंडन
(c) 50-percent rule./50-प्रतिशत नियम
(d) Segmentation./खण्डीकरण

KVS PGT C.S. 2013

Ans. (a) : Both the first fit and best fit strategies for memory allocation suffer from external fragmentation. As the processes are loaded and removed from memory. the free memory space is broken into little pieces.

56. The simplest, but most expensive, approach to introductory redundancy is duplicate to every disk. This technique is called
प्रारंभिक अतिरेकता के लिए सरलतम परंतु सबसे खर्चीला अभिगम प्रत्येक डिस्क की प्रतिकृति है । इस तकनीक को कहते है

(a) Swap space./स्वेप स्पेस
(b) Mirroring./मिररींग
(c) Page slots./पेज स्लोट
(d) None of these/इनमें से कोई नहीं

KVS PGT C.S. 2013

Ans. (b) : The simplest but most expensive approach to introductory redundancy is duplicate to every disk. This technique is called mirroring or shadowing.

57. This begins at the root and follows a path down to the specified file, giving the directory names on the path. This is known as
यह मूल से आरंभ होता है और निश्चित फाइल के नीचे की ओर पथ पर डायरेक्ट्री नामों को देकर चलता है । इसे कहते है

(a) Absolute path name./निरपेक्ष पथ नाम
(b) Relative path name./सापेक्ष पथ नाम
(c) Definite path name./निश्चित पथ नाम
(d) Indefinite path name./अनिश्चित पथ नाम

KVS PGT C.S. 2013

Ans. (a) : An absolute path name begins at the root and follows a path down to the specified file , giving the directory names on the path. A relative path name defines a path from the current directory.

58. Each process is contained in a single section of memory that is contiguous to the section containing the next process is called
प्रत्येक प्रक्रिया मेमोरी के एकल खंड में समाविष्ट होती है जो आने वाली प्रक्रिया के समाविष्ट खंड से संलग्न होती है, कहलाती है ।

(a) Contiguous memory protection.
संलग्न स्मृति रक्षण
(b) Contiguous memory allocation.
संलग्न स्मृति नियतन
(c) Contiguous memory relocation.
संलग्न स्मृति पुनरावस्थापन
(d) Contiguous memory utilization.
संलग्न स्मृति उपयोग

KVS PGT C.S. 2013

Ans. (b) : Contiguous memory allocation refers to a memory management technique in which whenever there occurs a request by a user process for the memory one of the sections of the contiguous memory block would be given to that process in accordance with its requirement.

Sarkari Resource

59. **systems have well defined fixed time constraints on processing of special events.**
विशेष इवेंट्स की प्रोसेसिंग के संबंध में _____ सिस्टमों में सुपरिभाषित फिक्स्ड टाइम कन्स्ट्रेन्ट्स होते है।
(a) Multiprocessor/मल्टीप्रोसेसर
(b) Time sharing/टाइम शेयरिंग
(c) Interactive/इंटरएक्टिव
(d) Real time/रीयल टाइम

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (d) : A Real time system is a time-bond system with well defined and fixed time constraints and process must be done within the defined constraints, otherwise, the system will fail.

60. **In a monolithic kernel, operating system runs in**
मोनोलिथिक कर्नेल में, ओपरेटिंग सिस्टम इसमें चलाता है
(a) User mode./यूजर मोड
(b) Supervisor mode./सुपरवाईजर मोड
(c) User/supervisor mode./यूजर/सुपरवाईजर मोड
(d) None of these/इनमें से कोई नहीं

KVS PGT C.S. 2013

Ans. (d) : None of these because of Monolithic kernel means that the whole operating system runs in kernel mode
(i.e. highly privileged by the hardware).
That is no part of the OS runs in user mode (lower privileged). Only applications an top of the OS run in user mole .

61. **One scheme for communication between user-thread library and the kernel is known as**
यूजर-थ्रेड लाइब्रेरी और कर्नेल के बीच कम्यूनिकेशन की एक योजना को कहते है
(a) Lightweight process./लाइटवेइट प्रोसेस
(b) Upcall handler./अपकॉल हेन्डलर
(c) Scheduler activation./शिड्यूलर ऐक्टीवेशन
(d) Cleanup handler./क्लीनअप हेन्डलर

KVS PGT C.S. 2013

Ans. (c) : Scheduler activation works as likes the kernel provides an application with a set of virtual processors, and the application can scheduler user threads onto an available virtual processor.

62. **_____is a sequence of memory read-write operations that are atomic.**
_____ मेमोरी रीड-राइट संक्रिया का एक क्रम है जो एटोमिक है ।
(a) Critical section object/क्रिटीकल सेक्शन ओब्जेक्ट
(b) Adaptive mutex/एडेप्डीव म्यूटेक्स
(c) Turnstile/टर्नस्टाइल
(d) Memory transaction/मेमोरी ट्रांजेक्शन

KVS PGT C.S. 2013

Ans. (d) : A memory transaction is a sequence of read write operation to memory that are performed automatically.
Void update (){
read | write memory
}

63. **Which of the following concurrency control protocols ensure(s) both conflict serializability and freedom from deadlock?**
निम्न में से कौन-सा/से समवर्ती नियंत्रण (Concurrency Control) प्रोटोकॉल, कॉन्फ्लिक्ट सिरियलाइजेबिलिटि और डेडलॉक से स्वतंत्रता, दोनों सुनिश्चित करता है/करते हैं?
I. 2-phase locking/2-फेज लॉकिंग
II. Timestamp ordering/टाइमस्टैम्प ऑर्डरिंग
(a) I only/ केवल I
(b) II only/केवल II
(c) Both I and II/I और II दोनों
(d) Neither I nor II/न तो I, न ही II

LT. Grade C.S. 2018

Ans. (b) : 2 phase locking (2PL) is a concurrency control method that guarantees serializability. The protocol utilizes locks applied by a transaction to data, which may block other transaction from accessing the same data during the transaction's life. 2PL may be lead to deadlocks that result from the mutual blocking of two or move transactions.

64. **What is a page in paging?**
पेजिंग में पेज क्या है?
(a) A fixed-size block of virtual memory
वर्चुअल मेमोरी का एक निश्चित आकार का ब्लॉक
(b) A fixed-size block of physical memory
भौतिक मेमोरी का एक निश्चित आकार का ब्लॉक
(c) A type of memory allocation
मेमोरी आवंटन का एक प्रकार
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) **:** Paging is a memory management technique used to divide virtual memory into small blocks of physical memory. In this virtual memory is divided into pages of equal size and physical memory is divided into frames.

65. **In a system with a page size of 4 KB and a physical address space of 64 GB, what is the number of bits required for the page offset?**
4 KB पृष्ठ आकार और 64 GB भौतिक पता स्थान वाले सिस्टम में, पृष्ठ ऑफसेट के लिए आवश्यक बिट्स की संख्या क्या है?
(a) 10 bits/10 बिट्स
(b) 12 bits/12 बिट्स
(c) 14 bits/14 बिट्स
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) **:** To determine the number of page offset we have to calculate the number of bytes based on the pages size.
Page size = 4 KB
1 KB = 1024 bytes

$4 \times 1024 = 4096$ bytes.
Number of page offset bits–
Number of page offset bits = log : (Page size)
Number of page offset bits = log2 (4096)
$4096 = 2^{12}$ So log2 (4096) = 12
Hence, number of bits required for page offset is 12 bits.

66. What is a segment base in segmentation?
विभाजन में खंड आधार क्या है?

(a) The starting address of a segment in physical memory
भौतिक मेमोरी में किसी खंड का प्रारंभिक पता
(b) The starting address of a segment in virtual memory
वर्चुअल मेमोरी में एक सेगमेंट का प्रारंभिक पता
(c) The ending address of a segment in physical memory
भौतिक मेमोरी में किसी खंड का अंतिम पता
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) **:** In segmentation, the segment base refers to the starting address of a segment in physical memory, which is used as a reference point to locate the contents of the segment in memory.

67. What causes thrashing to occur?
Thrashing किस कारण से होती है?

(a) Excessive paging activity
अत्यधिक पेजिंग गतिविधि
(b) Insufficient disk space/अपर्याप्त डिस्क स्थान
(c) Hardware failures/हार्डवेयर विफलता
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) **:** Thrashing is caused by excessive paging activity. Thrashing occurs when the operating system spends a lot of time paging and swapping pages too often, which reduce system performance.

68. Consider a system with a total physical memory of 8 GB and total virtual memory of 16 GB. If the system is using 6 GB of physical memory and 10 GB of virtual memory, how much memory is being swapped out?
एक सिस्टम पर विचार करें जिसमें कुल भौतिक मेमोरी 8 जीबी और कुल वर्चुअल मेमोरी 16 जीबी है। यदि सिस्टम 6 जीबी भौतिक मेमोरी और 10 जीबी वर्चुअल मेमोरी का उपयोग कर रहा है, तो कितनी मेमोरी स्वैप की जा रही है?

(a) 2 GB/2 जीबी
(b) 4 GB/4 जीबी
(c) 6 GB/6 जीबी
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) **:** The system is using 6GB out of the total physical memory of 8GB, so 2GB of physical memory is free now, since 10GB virtual memory is in used and out of this only 8GB can be loaded into physical memory. So, the remaining 2GB (10GB-8GB) will be kept in swap space (disk). So 2GB of memory is being swapped.

69. A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.
एक डेडलॉक परिहार एल्गोरिथम गतिशील रूप से यह सुनिश्चित करने के लिए _____ की जांच करता है कि एक परिपत्र प्रतीक्षा स्थिति कभी मौजूद नहीं हो सकती है।

(a) Resource allocation state
(b) Resources
(c) System storage state
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : Resource allocation state is used to maintain the availability of previously available and current resources. A deadlock avoidance algorithm dynamically checks the resource allocation state to ensure that a circular wait condition can never exist.

70. The operating system maintains a _____table that keeps track of how many frameshave been allocated, how many are there and how many are available
ऑपरेटिंग सिस्टम एक _____ तालिका रखता है जो ट्रैक करता है कि कितने फ्रेम आवंटित किए गए हैं, कितने हैं और कितने उपलब्ध हैं।

(a) Frame
(b) Mapping
(c) Page
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : The operating system uses a frame table to keep track of memory frames. The from table records how many frames have been allocated, how many are in use, and how many are available for allocation. Each entry in the frame table corresponds to one physical memory frame.

71. For real time operating systems, interrupt latency should be _____.
वास्तविक समय ऑपरेटिंग सिस्टम के लिए, इंटरप्ट लेटेंसी _____ होनी चाहिए।

(a) Zero
(b) Minimal
(c) Maximum
(d) More than one of the above
(e) None of the above

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Ans. (b) : In real-time systems, it is extremely important to keep interrupt latency to a minimum in order to respond to critical events in a timely manner.

Sarkari Resource

72. On systems where there are multiple operating system, the decision to load a particular one is done by _____.
उन प्रणालियों पर जहां कई ऑपरेटिंग सिस्टम होते हैं, किसी विशेष को लोड करने का निर्णय _____ द्वारा किया जाता है

(a) Process control block
(b) File control block
(c) Boot loader
(d) More than one of the above
(e) None of the above

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Ans. (c) : The boot loader is responsible for loading and starting the operating system when the computer. In system that have multiple operating systems, the boot loader gives users the option to choose which operating system to load.

73. Thread shares with other threads belonging to the same process its.
Thread उसी प्रक्रिया से संबंधित अन्य threads के साथ अपनी जानकारी साझा करता है।

(a) thread ID/थ्रेड आईडी
(b) program counter/प्रोग्राम काउंटर
(c) code section and data section/कोड अनुभाग और डेटा अनुभाग
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : Threads are part of the same process so they share code and data section. However, each thread has its own program counter and thread ID.

74. What does OS X has?
ओ. एस. एक्स में क्या हैं ?

(a) Monolithic kernel with modules
(b) Micro kernel
(c) Hybrid kernel
(d) More than one of the above
(e) None of the above

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Ans. (c) : OS X has a hybrid kernel. A hybrid kernel is a combination of two different kernels. OS X is developed by Apple and is originally known as MacOS X.

75. In a timeshare operating system, when the time slot assigned to a process is completed, the process switches from the current state to?
टाइमशेयर ऑपरेटिंग सिस्टम में, जब किसी प्रक्रिया को सौंपा गया समय स्लॉट पूरा हो जाता है, तो प्रक्रिया वर्तमान स्थिति से स्विच हो जाती है?

(a) Suspended state
(b) Blocked state
(c) Ready state
(d) More than one of the above
(e) None of the above

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Ans. (c) : In a time-sharing operating system, when the time slot given to a process is completed the process goes from running state to ready state. In a time sharing operating system, a unit of time is determined for sharing the CPU, this is called a time quantum of time slice. If a process takes less than the process itself releases the CPU.

76. Which one of the following is not true?
निम्नलिखित में से कौन-सा सत्य नहीं है?

(a) Kernel remains in the memory during the entire computer session
(b) Kernel is made of various modules which can not be loaded in running operating system
(c) Kernel is the first part of the operating system to load into memory during booting
(d) More than one of the above
(e) None of the above

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Ans. (b) : The kernel is the first program that is loaded into memory when the Operating System (OS) is loaded and also it remains in memory as long as the OS is running. The kernel is the main part of the OS that is responsible for managing resources allowing multiple processes to use resources and providing services to different processes. Kernel modules can be loaded at run-time (i.e. from the running OS).

77. Which of the following wait is undesirable for a process?
निम्न में से कौन-सा इन्तजार (wait) किसी प्रोसेस के लिए वांछित नहीं है?

(a) Waiting in ready queue
रेडी क्यू में इन्तजार (wait)
(b) Waiting for input/output
इनपुट/आउटपुट हेतु इन्तजार (wait)
(c) Waiting for semaphore
सीमाफोर हेतु इन्तजार (wait)
(d) Busy waiting/बिजी वेटिंग

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Ans. (d) : Busy waiting occurs when a process continuously checks for a condition to be met, effectively consuming CPU cycles without doing any productive work. This is inefficient because the processor is occupied with repeatedly polling rather than executing other useful tasks. In contrast, waiting in the ready queue, waiting for input/output, or waiting for a semaphore are more efficient forms of waiting where the process is either scheduled for execution or blocked until the condition is met, thereby freeing up the CPU for other process.

78. On a system using fixed partition with sizes 2^{16}, 2^{24}, 2^{32}, how many bits must the limit register have?
एक सिस्टम जो की फिक्स्ड पार्टीशन का इस्तेमाल करता है, जिनका आकार 2^{16}, 2^{24}, 2^{32} है, इसके लिए लिमिट रजिस्टर में कितनी बिट होनी चाहिए?

(a) 32 Bits/32 बिट
(b) 24 Bits/24 बिट
(c) 16 Bits/16 बिट
(d) All of these/इनमें से सभी

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Ans. (a) : The limit register stores the maximum size of a partition (in bytes), so it must be able to represent the largest partition size, which is 2^{32} bytes. To store the number 2^{32}, you need 32 bits (since 2^{32} requires 32 bits to represent). Therefore only 32 bits is sufficient to represent the largest partition size.

79. Which is false statement:
इनमें से कौन-सा गलत कथन है?

(a) The dispatcher module gives control to CPU, to process the selected short term scheduler (STS)/डिस्पैचर मोड्यूल, CPU, को नियंत्रण देता है, चयनित अल्पकालिक अनुसूचक (STS) प्रक्रिया को करने के लिए

(b) The dispatcher may invoke during process context switch/डिस्पैचर को आहवान कर सकते है, प्रक्रिया प्रसंग स्विच के दौरान

(c) Dispatch latency, stop one process and start another process/डिस्पैचर लेटेंसी, एक प्रक्रिया को बंद कर के दूसरी प्रक्रिया आरम्भ करता है

(d) The dispatcher is not used in CPU scheduling/डिस्पैचर, CPU शेड्यूलिंग में उपयोग नहीं आता

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Ans. (d) : The dispatcher is an essential part of CPU scheduling. It is responsible for-
⇒ Switching context between processes.
⇒ Switching to user mode.
⇒ Jumping to the proper location in the user program to restart it.

80. The process arrive in a system at the rate of 10 processes/minute and average service time for each job is 3 seconds. What will be the system load?
10 प्रोसेस प्रति मिनट की दर से किसी सिस्टम में प्रोसेस आते है एवं औसत सर्विस समय 3 सेकंड प्रति जॉब है, तो सिस्टम का लोड बताए:

(a) $\rho = 0.5$ (b) $\rho = 3.3$
(c) $\rho = 0.3$ (d) $\rho = 1$

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Ans. (a) : To calculate system load (ρ), we use the formula:

$$\rho = \lambda \times T_8$$

Where–
$\lambda \rightarrow$ Arrival rate (in jobs per second)
$T_8 \rightarrow$ Average service time per job (in seconds)
Given–
- Arrival rate = 10 processes per minute $= \frac{10}{60} = \frac{1}{6} \approx 0.1667$ processes/second.
- Service time = 3 seconds.

Now, put into the formula:
$\rho = 0.1667 \times 3 \approx 0.5$

81. On a system using non preemptive scheduling, process with expected runtime of 5, 18, 9, 12 are in ready queue. In when order they should be run to minimize wait time?
नॉन प्रिएमटीव शेड्यूलिंग आधारित सिस्टम की रेडी क्यू में प्रोसेस जिनका अनुमानित रन टाइम 5, 18, 9, 12 है, रखे गए है। इन्हें किस क्रम में रन किया जाए जिससे वेटिंग न्यूनतम आये?

(a) 5, 18, 9, 12 (b) 5, 9, 12, 18
(c) 18, 12, 9, 5 (d) 18, 5, 9, 12

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Ans. (b) : To minimize wait time on a non-preemptive scheduling system, the best strategy is to use Shortest Job First (SJF). This means processes with the shortest expected runtime should be executed first.
Given runtimes– 5, 18, 9, 12
Sorted in ascending order– 5, 9, 12, 18
So the correct order is– 5, 9, 12, 18

82. RAID stands for?
RAID से क्या तात्पर्य है?

(a) Redundant array of index pensive disks
रिडन्डेंट ऐरै ऑफ इनएक्सपेंसिव डिस्क

(b) Relative array of inexpensive disks
रिलेटिव ऐरै ऑफ इनएक्सपेंसिव डिस्क

(c) Redundant array of inter-related disks
रिडन्डेंट ऐरै ऑफ इन्टर-रिलेटेड डिस्क

(d) Relative array of inter-related disks
रिलेटिव ऐरै ऑफ इन्टर-रिलेटिव डिस्क

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Ans. (a) : RAID (Redundant Array of Inexpensive Disks) is a technique that makes use of a combination of multiple disks for storing the data instead of using a single disk for increased performance, data redundancy, or to protect data in the case of a drive failure. The term was defined by 'David Patter Son' : 'Garth A. Gibson', and 'Randy Katz' at the University of California, Berkeley in 1987.

83. Which of the statement is not true related to threads?
थ्रेड से सम्बंधित कौन-सा कथन सही नहीं है?

(a) Thread is a light weight process
थ्रेड एक लाइट वेट प्रोसेस है

(b) User level threads are managed by thread library
यूजर लेवल थ्रेड, थ्रेड लाइब्रेरी से मैनेज होती है

(c) Kernel level threads are faster to create and manage than user level threads
करनल लेवल थ्रेड, यूजर लेवल थ्रेड से तेजी से तैयार एवं मैनेज होती है

(d) Thread minimize context switching time
थ्रेड कन्टेक्स्ट स्विचिंग समय को कम करती है

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Ans. (c) : Kernel level threads require support from the operating system and involve more overhead, making them slower to create and manage compared to user space without kernel intervention.
User level threads are generally faster to create and manage, though they have other limitations.

84. Which of the following instruction need not to be executed in kernel mode?
निम्न में से किस अनुदेश को करनल मोड में क्रियान्वित करने की आवश्यकता नहीं है?

(a) Change memory management registers
स्मृति प्रबंधन रजिस्टर में बदलाव

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(b) Set the time of day clock
घड़ी में दिन का समय सेट करना
(c) Change processor priority
प्रोसेसर की प्राथमिकता में बदलाव
(d) Write the program counter
प्रोग्राम काउंटर को लिखना

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Ans. (d) :
• **Kernel Mode–** CPU has access to all system resources. Privileged instructions. (Those that can affect system-wide operations) must be executed in this mode.
• **User Mode–** Restricted access; used for running user applications.
Write the program counter, it does not need to be explicitly executed in kernel mode.
⇒ The Program Counter (PC) holds the address of the next instruction. It's automatically updated by the processor during instruction execution, including in user mode. So, while you don't directly "Write" to if, it changes in user mode as a program runs.

85. Which of the following process state transition is not possible?
निम्न में से किस प्रोसेस स्टेट में कौन-सा बदलाव संभव नहीं है?
(a) Run → Ready
रन → रेडी
(b) Run → Blocked
रन → ब्लॉक्ड
(c) Blocked → Run
ब्लॉक्ड → रन
(d) Run → Terminated
रन → टर्मिनेटेड

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Ans. (c) : A process can move from a "Running" state to a "Blocked" state if it needs to wait for something like an I/O operation or a resource to become available. However, a process in a "Blocked" state cannot directly transition to a "Running" state. It must first move to the "Ready" state, where it is queued to be scheduled for execution again. The CPU scheduler will then select a process from the ready queue to run.

86 Following are applications and scheduling algorithms:
निम्नलिखित अनुप्रयोग एवं शेड्यूलिंग अल्गोरिथम है:

A.	**Guaranteed Scheduling/गारंटीड शेड्यूलिंग**	**1.**	**Gang Scheduling Algorithm/गैंग शेड्यूलिंग अल्गोरिथम**
B.	**Real Time Scheduling/रियल टाइम शेड्यूलिंग**	**2.**	**Rate Monotonic Scheduling Algorithm रेट मोनोटोनिक शेड्यूलिंग अल्गोरिथम**
C.	**Thread Scheduling थ्रेड शेड्यूलिंग**	**3.**	**Fare Share Scheduling Algorithm/फेयर शेयर शेड्यूलिंग अल्गोरिथम**

Which of the below combination suits better.
निम्न में से कौन-सा मेल उपयुक्त है:
(a) A-1, B-2, C-3 (b) A-3, B-2, C-1
(c) A-2, B-3, C-1 (d) A-3, B-1, C-2

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Ans. (b) :
A. Guaranteed Scheduling– Fare share scheduling ensures each user or group gets a guaranteed share of C.P.U time aligns with Guaranteed Scheduling.
B. Real Time Scheduling– Rate monotonic scheduling is a Real Time Scheduling Algorithm used in systems where timing is critical.
C. Thread Scheduling– Gang Scheduling is used for scheduling threads/processes in parallel computing systems.

87. On a System with 2^{24} bytes of memory and fixed partitions, all of size 65,536 byte, how many bits must the limit register have?
लिमिट रजिस्टर में कितनी बिट होनी चाहिए यदि 2^{24} बाइट स्मृति एवं 65,536 बाइट के फिक्स्ड पार्टीशन हैं?
(a) 8 (b) 16
(c) 24 (d) 32

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Ans. (b) : Total Memory = 2^{24} bytes
Each partition size = 65,536 bytes = 2^{16} bytes.
The memory is divided into fixed size partitions.
Step 1– Calculate the number of partitions

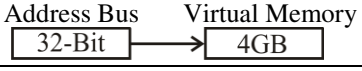
$$\frac{2^{24}}{2^{16}} = 28 = 256 \text{ partitions}$$

Each partition is 65,536 bytes and the limit register holds the size of a partition (or the maximum size a process can use in its partition)
Step 2– Bits are needed to represent the size 65,536 bytes–
⇒ $65,536 = 2^{16}$
So, to store the value 65,536 (or up to that value), the limit register must be 16 bits.

88. Size of virtual memory depends on the size of?
अप्रत्यक्ष स्मृति का आकार निम्न में से किसके आकार पर निर्भर है?
(a) Data bus/डाटा बस
(b) Address Bus/एड्रेस बस
(c) Main Memory/मेन मेमोरी
(d) None of these/इनमें से कोई नहीं

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Ans. (b) : Virtual memory size is primarily determined by the size of the address bus, because the address bus defines how many unique memory addresses the CPU can generate for example, a 32-bit address bus can address up to 4 GB of memory (2^{32} addresses), regardless of the size of the physical memory (main memory)
Example–



89. In round robin CPU scheduling, as the time quantum is increased, the average turnaround time–
राउंड रोबिन सी.पी.यू. शेड्युलिंग प्रक्रिया में, जब टाइम क्वांटम को बढ़ाया जाता है तो औसत टर्न अराउंड समय

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(a) Increases/बढेगा
(b) Decreases/घटेगा
(c) Remain constant/नियत रहेगा
(d) Varies irregularly/अनियमित बदलाव

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Ans. (d) : As the time quantum in round robin scheduling increases, the average turnaround time can either increase or decrease depending on the specific scheduling patterns and job characteristics.
⇒ The Round-Robin method mainly used the time sharing method. The time frame for which a method or task is permitted to run in a pre-emptive technique is known as time Quantum.
Characteristics of Round-Robin method–
⇒ It is very easy to implement in the system.
⇒ In this process, time is shared equally for each process.

90. In operating system which one of the following is not a scheduling algorithm?
ऑपरेटिंग सिस्टम में, इनमें से कौन शेड्युलिंग अल्गोरिथम नहीं है?
(a) FCFS
(b) SJF
(c) Round Robin/राउंड रोबिन
(d) Starvation/स्टारवैशन

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Ans. (d) :
- **FCFS (First-Come, First Served)–** A basic scheduling algorithm that executes processes in the order they arrive.
- **SJF (Shortest Job First)–** A scheduling algorithm that selects the process with the smallest execution time.
- **Round Robin–** A scheduling algorithm that assigns time slices to each process in equal portions and in circular order.
- Starvation is not a scheduling algorithm it is a problem that can occur in scheduling (e.g. when a process waits indefinitely because other processes are continuously given preference).

91. Fence register is used for?
फेंस रजिस्टर का प्रयोग किया जाता है?
(a) Memory Protection/स्मृति सुरक्षा
(b) File Protection/फाइल सुरक्षा
(c) CPU Protection/सी.पी.यू. सुरक्षा
(d) All of these/इनमें से सभी

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Ans. (a) : A fence register is used in memory protection mechanisms to restrict a program's access to memory. It sets a boundary (or "fence") so that the currently running user program cannot access memory locations below a certain address. This prevent it from interfering with the operating system code & data, which typically reside in the lower part of memory.
• File protection is handled by file system permissions not fence registers.
• CPU protection is handled by modes of operation (user Vs kernel mode), not by memory fences.

92. Amount of work completed in a unit of time is?
इकाई समयावधि में निष्पादित कार्य को कहा जाता है?
(a) Throughput/थ्रू-पुट
(b) Response/रिस्पांस
(c) Turn Around/टर्न अराउंड
(d) Latency/लेटेंसी

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Ans. (a) : Throughput refers to the number of tasks or processes completed in a unit of time by a system. It measures the efficiency or productivity of a system.
- **Response time–** Response time is taken to start responding after a request.
- **Turn around time–** Turn around time is taken from submission to completion of a process.
- **Latency–** It is delay before a transfer of data begins following an instruction.

93. Which one the of following is a technique to prevent starvation?
निम्न में से कौन-सी तकनीक स्टारवेशन को रोकने के लिए है?
(a) Context Switching/कॉन्टेक्स्ट स्वीचिंग
(b) Flooding/फ्लोडिंग
(c) Aging/एजिंग
(d) Dispatching/डिस्पेचिंग

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Ans. (c) : Aging is a technique used in operating systems to prevent starvation.
In priority-based scheduling, low-priority processes may starve if high-priority processes keep executing.
• Aging gradually increases the priority of a waiting process over time.
• Eventually the low priority process gets high enough priority to be executed, preventing starvation.

94. Which of the following is not a memory allocation method?
निम्न में से कौन सा स्मृति आवंटन का तरीका नहीं है?
(a) First fit/फर्स्ट फिट (b) Best fit/बेस्ट फिट
(c) Worst fit/वर्स्ट फिट (d) Not fit/नॉट फिट

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Ans. (d) : In an operating system (OS), memory management is a critical function that ensures efficient allocation and utilization of memory resources. When processes are initiated, they require execution time & storage space, which is allocated in memory blocks. The OS uses various memory allocation algorithms to manage the assignment of memory to these incoming processes.
(1) First Fit (2) Best Fit
(3) Worst Fit (4) Next Fit
These are contiguous memory allocation techniques.

95. Which of the following system objects that are neither files nor directories?
निम्न में से कौन से सिस्टम ऑब्जेक्ट ना तो फाइल है ना ही डायरेक्टरी?
(a) Semaphore/सीमाफोर
(b) Monitor/मॉनिटर

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(c) Both Semaphore and Monitor
सीमाफोर और मॉनिटर दोनों
(d) None of these/इनमें से कोई नहीं

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Ans. (c) : Semaphore and Monitor are synchronization constructs used in operating systems and multithreaded programming.
• They are not files or directories, but rather system-level objects used to manage concurrent processes or threads.
• These objects help control access to shared resources and avoid race conditions.

96. Solution to critical section problem?
क्रिटिकल सेक्शन समस्या का उपाय है-
(a) Mutual exclusion, Progress, Bounded waiting
म्यूच्यूअल एक्सक्लूशन, प्रोग्रेस, बाउन्डेड वेटिंग
(b) Mutual exclusion, Progress, Race Condition
म्यूच्यूअल एक्सक्लूशन, प्रोग्रेस, रेस शर्त
(c) Bounded Waiting, Progress, Unbounded Buffer/बोउन्डेड वेटिंग, प्रोसेस, अनबाउन्डेड बफर
(d) Race Condition, Unbounded buffer, Progress
रेस शर्त, अनबाउन्डेड बफर, प्रोग्रेस

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : A solution to the critical section problem must satisfy the following three conditions:
(1) Mutual Exclusion– Only one process can enter the critical section at a time.
(2) Progress– If no process is in the critical section, and some processes wish to enter, then the selection of the next process to enter cannot be postponed indefinitely.
(3) Bounded Waiting– There must be a limit on the number of times other processes can enter the critical section before a waiting process gets its turn.

97. Which of the following methods are used to pass any number of parameters to the operating system through system calls?
(a) Registers
(b) Block or table in main memory
(c) Stack
(d) Block in main memory and stack

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (d) : Following are the methods used to pass parameters to the OS.
1-Registers : Parameters are passed via registers but the limitation of this method is that the number of parameters cannot exceed the number of register.
2-Blocks : When there are more parameters than registers, these parameters are stored on a main memory block and the pointer to this block (block address) is passed as a parameter to a register.
3-Stack : Parameter can also be pushed on and popped off the stack by the OS. This method can be used when we don't certainly know the number of parameters be forehand. Therefore, methods used to pass any number of parameters to the operating system through system calls are via block and stack. Register cannot pass. any number of parameters.

98. Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves. In this case, which of the following is false ?
(a) The administrator enjoys more control on the grant option.
(b) It is difficult to differentiate among the update, delete and insert authorizations.
(c) Cannot store more than one relation in a file.
(d) Operations on the database are speeded up as the authorization procedure is carried out at the operating system level.

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Ans. (d) : Database systems that store each relation in a separate operating system file may use the operating system's authorization scheme, instead of defining a special scheme themselves.
(a) The administrator enjoys more control on the grant option.
(b) It is difficult to differentiate among the update, delete and insert authorizations.
(c) Cannot store more than one relation in a file.

99. Which of the following statements are true ?
(a) External Fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous.
(b) Memory Fragmentation can be internal as well as external.
(c) One solution to external Fragmentation is compaction.
Code :
(a) (a) and (b) only (b) (a) and (c) only
(c) (b) and (c) only (d) (a), (b) and (c)

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Ans. (c) : External fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous is fail.
External fragmentation exists when there is enough total memory space to satisfy a request but travel spell available space is continuous in so this is false the definition of external fragmentation is that whenever you have canting venom or any process want some data for some space from the main memory space is available in the main memory but it is not contiguous ok for example let us.

100. Page information in memory is also called as Page Table. The essential contents in each entry of a page table is/are _____.
(a) Page Access information
(b) Virtual Page number
(c) Page Frame number
(d) Both virtual page number and Page Frame Number

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Ans. (c) : Page information in memory is also called as Page Table. The essential contents in each entry of a Page Table is Page Frame number.

Sarkari Resource

101. In which of the following scheduling criteria, context switching will never take place ?

(a) ROUND ROBIN (b) Preemptive SJF
(c) Non-preemptive SJF (d) Preemptive priority

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Ans. (c) : Shortest job first or shortest job next, is a scheduling policy that selects the waiting process with the smallest execution time to execute next SJF is non-preemptive algorithm.
Except Non-Preemptive SJF, all are preemptive. CPU will have leave the shortest process only after its full execution, once process is started to execute.

102. Which of the following statements are CORRECT?

(A) A process always check state of currently executing process to enter critical schema.
(B) Spin locks uses busy waiting.
(C) Periodically testing a variable until some value appear is known as busy waiting.
(D) Critical region is a part of program, where shared memory is kept.
(E) Printer daemon, continuously checks to see if there are any file to be printed.

Choose the correct answer from the options given below:

(a) (A) and (B) Only (b) (B) and (C) Only
(c) (B) and (D) Only (d) (B) and (E) Only

UGC NET C.S. December-2023

Ans. (c) :
(A) A process does not always need to check the state of the currently executing process to enter the critical section.
(B) Spin locks uses busy waiting.
(C) Busy waiting, also known as spinning, is a technique where a process continuously checks for a condition to be true in a tight loop, without yielding the CPU to other processes.
(D) Critical region is a part of program, where shared memory is kept.

103. In a multiuser operating system, 20 request are made to use a particular resource per hour, on an average. The probability that no request is made in 45 minutes is

(a) e^{-15} (b) e^{-5}
(c) $1-e^{-5}$ (d) $1-e^{-10}$

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Ans. (a) : To calculate the probability that no request is made in 45 minutes, we need to convert the time frame to hours.

Since, 4 minutes is equivalent to $\frac{45}{60}=\frac{3}{4}$ hours.

Now, if 20 requests are made per hour on average.
then the average rate of requests per hour would be $\frac{3}{4}\times 20 = 15$ request per hour.
using the Poisson distribution formula-

$$P(X=k)=\frac{e^{-\lambda}\lambda^{k}}{k!}$$

Where, λ is the average rate of events per interval k is the number of occurrences.
So, $\lambda = 15$ $K = 0$

$$P(X=0)=\frac{e^{-15}15^{\circ}}{0!}=e \qquad \because 0!=1$$
$$= e^{-15}$$

104. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : A process involves a library function to create a thread.
Reason R : The threads make system calls to convey their resource and I/O requirement to the Kernel.
In the light of the above statements choose the correct answer from the options given below.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

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Ans. (b) : Many programming languages and environment provide library functions to create threads, making it easier for developers to implement multithreading in their application.
Threads can indeed make system calls to convey their resource and I/O requirements to the kernel. However, this is not the primary purpose of using a library function to create a thread. The primary purpose is to enable concurrent execute within the same process. The system calls for resource and I/O requirements typically involve interactions with the operating system, but they are not directly related to the creation of threaded via library functions. So, Assertion A and Reason R are true but R is not the correct explanation of A.

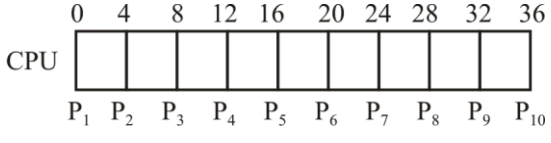
105. An OS follows Round-Robin Scheduling with time quantum of 4ms. Assuming that the CPU is free now and there are 20 processes waiting in the ready queue, the maximum amount of time that a process waits before getting into the CPU is _____.

(a) 80ms (b) 76ms
(c) 84ms (d) None of the above

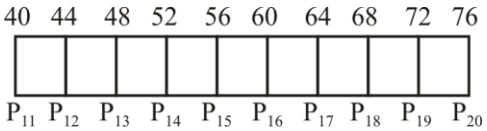
UGC NET C.S. December-2022

Ans. (b) : Given that
time quantum = 4 ms
Processes = 20
Ready queue $P_1\ P_2\ P_3\ P_4\ P_5\ P_6\ P_7\ P_8\ P_9\ P_{10}\ P_{11}$

$P_{12}\ P_{13}\ P_{14}\ P_{15}\ P_{16}\ P_{17}\ P_{18}\ P_{19}\ P_{20}$



Sarkari Resource



The maximum amount of time that a process waits before getting into the CPU is 76

106. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is first come first served (FCFS). If FCFS is replaced by shortest seek time first (SSTF) and the vendor claims 50% better benchmark results. What is the expected improvement in the I/O performance of user programs?

(a) 50% (b) 100%
(c) 25% (d) 0%

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Ans. (d) : "Single sequential user process". So all the requests to disk scheduler will be in sequence and each one will be blocking the execution and hence there is no use of any disk scheduling algorithm. Any disk scheduling algorithm gives the same input sequence and hence the improvement will 0% for SSTF over FCFS.

107. In distributed system, the capacity of a system to adapt the increased service load is called_____

(a) Tolerance (b) Scalability
(c) Capability (d) Loading

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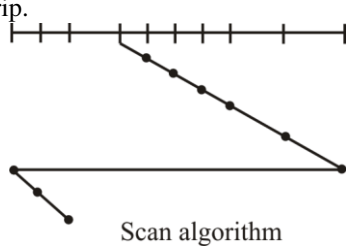
Ans : (b) Scalability is an attribute that describes the ability of a process, network, software or organization to grow and manage increased demand a system, business that is described as scalable has an advantage because it is more adaptable to the changing needs & even presence of its users.

108. In _____ disk scheduling algorithm, the disk head moves from one end to other end of the disk, serving the request along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without serving any requests on the return trip.

(a) LOOK (b) SCAN
(c) C – LOOK (d) c – SCAN

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Ans : (b) In scan disk scheduling algorithm, the disk head moves from one end to another end of disk, serving the requests along the way. When the head reach the other end, it immediately returns to the beginning of the disk, without serving any request on the return trip.



Scan algorithm

109. Suppose there are six files F1, F2, F3, F4, F5, F6 with corresponding sizes 150 KB, 225 KB, 75 KB, 60 KB, 275 KB and 65 KB respectively. The files are to be stored on a sequential device in such a way that optimizes access time. In what order should the files be stored?

(a) F5, F2, F1, F3, F6, F4
(b) F4, F6, F3, F1, F2, F5
(c) F1, F2, F3, F4, F5, F6
(d) F6, F5, F4, F3, F2, F1

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Ans : (b) Files with sizes:
F1 = 150 KB, F2 = 225 KB, F3 = 75 KB,
F4 = 60 KB, F5 = 275 KB, F6 = 65 KB
So, to optimize access time merge two smallest files and put back in same badge. Repeatedly i.e.
(F4, F6) = 125 KB
Then ((F4, F6), F3) = 200 KB
Then, (((F4, F6), F3), F1) = 350 KB
Then, (F2, F5) = 500 KB
Then finally ((F4, F6, F3, F1), (F2, F5)) = 700 KB
So, order will be F4, F6, F3, F1, F2, F5.

110. Which module gives control of the CPU to the process selected by the sort - term scheduler?

(a) Dispatcher (b) Interrupt
(c) Scheduler (d) Threading

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Ans : (a) Dispatcher is the module that gives control of the CPU to the process selected by the sort term scheduler.

111. Two atomic operations permissible on Semaphores are _____ and _____

(a) Wait, stop (b) Wait, hold
(c) Hold, signal (d) Wait, signal

UGC NET C.S. November-2017 (Paper-II)

Ans : (d) Wait & signal operations are the two atomic operations performed on a semaphore.

112. Match List - I with List - II.

List - I (Algorithms)		List - II (Characteristics)	
A.	**First Come First Served (FCFS)**	**I.**	**Each process is assigned a priority.**
B.	**Shortest Job First (SIF)**	**II.**	**Ensures fair allocation of CPU time by assigning time slice.**
C.	**Round Robin (RR)**	**III.**	**Processes are executed in the order they arrive.**
D.	**Priority Scheduling**	**IV.**	**Select the process for execution with smallest next Burst time.**

Choose the correct answer from the options given below :

(a) A-II, B-III, C-IV, D-I
(b) A-II, B-IV, C-III, D-I
(c) A-III, B-IV, C-II, D-I
(d) A-III, B-IV, C-I, D-II

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Ans. (c) : Correct match-

List -I (Algorithms)			List -II (Characteristics)
A.	First come first served (FCFS)	–	Process are executed in the order they arrive.
B.	Shortest Job first (SJF)	–	Select the process for execution with smallest next.
C.	Round Robin (R.R)	–	Ensures fair allocation of CPU time by assigning time slice.
D.	Priority Scheduling	–	Each process is assigned a priority.

113. A multi processor system with 16 processors is used to execute a parallelizable task. If the serial portion of the task takes 200 clock cycles and the parallel portion take 800 clock cycles. When all 16 processor are used how many total clock cycles are required to complete the task ?

(a) 250 (b) 300
(c) 400 (d) 450

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Ans. (a) : The serial portion of the task takes 200 clock cycles, and this portion cannot be parallelized, So it must be completed by a single processor. The parallel portion of the task takes 800 clock cycles, and this portion can be distributed among the 16 processors.
To calculated the time taken for the parallel portion–

$$\frac{800 \text{ clock cycles}}{16 \text{ Processors}} = 50 \text{ clock cycles}$$

Therefore, the total time to complete the task is the sum of the serial portion and the parallel portion–
200 (serial) + 50 (Parallel) = 250 clock cycles.

114. Which of the following statements are TRUE about mutual exclusion in concurrent programming?

A. Mutual exclusion ensures that only one process can be in a critical section at any given time.
B. Mutual exclusion are designed to prevent conflicts and ensure that only one process can access shared resources at a time
C. Mutual exclusion can use various algorithms to ensure that processes do not enter the critical section simultaneously.
D. Mutual exclusion allows multiple processes to access the critical section simultaneously to improve performance.

Choose the correct answer from the options given below :

(a) (A), (B), (C) Only (b) (B), (C), (D) Only
(c) (B), (D), (A) Only (d) (A), (C), (D) Only

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Ans. (a) : Mutual exclusion is a property of process synchronization that states that 'no two processes can exist in the critical section at any given point of time' Any process synchronization technique being used must satisfy the property of mutual exclusion, without which it would not be possible to get rid of a race condition. Mutual exclusion are designed to prevent conflicts and ensure that only one process can access shared resources at a time. Mutual exclusion can use various algorithms to ensure that processes do not enter the critical section simultaneously.

115. Which of the following statements are TRUE about Privileged Instructions?

A. It can only be executed by the Operating System kernel and not by user applications.
B. It is designed to perform operations that can directly affect the hardware or system state such as I/O operations or changing memory management setting.
C. User application can execute privileged instructions if they have to correct permissions, set by the Operating System.
D. It usually executed in user mode to ensure the safety and security of the system.

Choose the correct answer from the options given below :

(a) (A) and (B) Only (b) (A), (B) and (C) Only
(c) (B) and (C) Only (d) (B), (C) and (D) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (a) : Privileged instructions cannot be executed by user application because they could harm the system by accidentally or maliciously modifying hardware or critical setting. They are restricted to kernel mode, where only the OS can execute them. Privileged instructions are designed to perform operations that can directly affect the hardware or system state such as I/O operations or changing memory management settings.
The statement (C) is incorrect because user applications cannot directly execute privileged instructions. The statement (D) is incorrect because privileged instructions run in kernel mode, not user mode.

116. There are three processes P1, P2 and P3 sharing a semaphore for synchronizing a variable. Initial value of semaphore is one. Assume that negative value of semaphore tells us how many processes are waiting in queue. Processes access the semaphore in following order :

(A) P2 needs to access
(B) P1 needs to access
(C) P3 needs to access
(D) P2 exits critical section
(E) P1 exits critical section

The final value of semaphore will be :

(a) 0 (b) 1
(c) –1 (d) –2

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) Initial value of semaphores S = 1
(1) P2 needs to access decrease it to 0 (no one is waiting)
(2) P1 needs to access decreases it to –1 (one process in waiting)
(3) P3 needs to access decreases it to – 2 (2 process are waiting
(4) P2 exits critical section increases it to –1 (one process is waiting)
(5) P1 exits critical section increases it to 0 (no process is waiting)

Sarkari Resource

117. In a paging system, it takes 30 ns to search translation Look-a-side buffer (TLB) and 90 ns to access the main memory. If the TLB hit ratio is 70%, the effective memory access time is :

(a) 48ns (b) 147ns
(c) 120ns (d) 84ns

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) 70% TLB hit ratio 30 ns to search TLB
(0.7 * 30) in hit case : to find the page
0.3*(90 +30) in miss case : to find the page
90 ns to access the data from main memory
So total time : (0.7 * 30) + (0.3 * (90 + 30)) + 90 = 147

118. Match the following w.r.t. Input/Output management :

List-I		List-II	
A.	**Device controller**	**i.**	**Extracts information from the controller register and store it in data buffer**
B.	**Device driver**	**ii.**	**I/O scheduling**
C.	**Interrupt handler**	**iii.**	**Performs data transfer**
D.	**Kernel I/O subsystem**	**iv.**	**Processing of I/O request**

Codes :

	A	**B**	**C**	**D**
(a)	iii	iv	i	ii
(b)	ii	i	iv	iii
(c)	iv	i	ii	iii
(d)	i	iii	iv	ii

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Ans : (d) The correct match–

List-I		List-II	
A.	Device controller	**i.**	Extracts information from the controller register and store it in data buffer
B.	Device driver	**iii.**	Performs data transfer
C.	Interrupt handler	**iv.**	Processing of I/O request
D.	Kernel I/O subsystem	**ii.**	I/O scheduling

119. Which of the following scheduling algorithms may cause starvation?

A. First-come-first-served
B. Round Robin
C. Priority
D. Shortest process next
E. Shortest remaining time first

(a) A, C and E (b) C, D and E
(c) B, D, and E (d) B, C and D

UGC NET C.S. June-2017 (Paper-II)

Ans : (b)
(1) First come – first served – No starvation
(2) Round robin – No starvation
(3) Priority – starvation if higher priority process called again & again lower priority starves
(4) Shortest process next = starvation possible
(5) Shortest remaining time first = starvation possible

120. Distributed operating system consist of :

(a) Loosely coupled O.S. software on a loosely coupled hardware
(b) Loosely coupled O.S. Software on a tightly coupled hardware
(c) Tightly coupled O.S. Software on a loosely coupled hardware
(d) Tightly coupled O.S. software on a tightly coupled hardware

UGC NET C.S. June-2017 (Paper-II)

Ans : (c) A distributed operating system is usually defined as running on more loosely coupled hardware. There is no shared memory, but provides a single memory by tightly coupled software.

121. A memory management system has 64 pages with 512 bytes page size. Physical memory consists of 32 page frames. Number of bits required in logical and physical address are respectively :

(a) 14 and 15 (b) 14 and 29
(c) 15 and 14 (d) 16 and 32

UGC NET C.S. June-2017 (Paper-III)

Ans : (c) We know, number of page = virtual address space/page size
So virtual address space = number of page × page size
$= 2^6 \times 2^9 = 2^{15}$
So we need is bit of VA & physical address = page size × frame $= 2^9 \times 2^5 = 2^{15}$
So 14 bit required for PA

122. Some of the criteria for calculation of priority of a process are :

A. Processor utilization by an individual process
B. Weight assigned to a user or group of users
C. Processor utilization by a user or group of processes

In fair share scheduler, priority is calculated based on :

(a) Only (A) and (B) (b) Only (A) and (C)
(c) (A), (B), and (C) (d) Only (B) and (C)

UGC NET C.S. June-2017 (Paper-III)

Ans : (c) Fair-share scheduling is a scheduling algorithm for computer operating system is which the CPU usage is equality distributed among system users or groups, as opposed to equal distribution among processes.

123. One of the disadvantages of user level threads compared to Kernel level threads is

(a) If a user level thread of a process executes a system call, all threads in that process are blocked
(b) Scheduling is application dependent
(c) Thread switching doesn't require kernel mode privileges
(d) The library procedures invoked fro thread management is user level threads are local procedures

UGC NET C.S. June-2017 (Paper-III)

Sarkari Resource

Ans : (a) User level threads
Advantage
- Thread switching does not require kernel mode privileges
- User level thread can run on any OS.
- User level threads are fast to create & manage.

Disadvantage
- In a typical OS, most system calls are blocking
- Multithreaded application cannot take advantage of multiprocessing.

124. Which statements is not correct about "init" process in Unix?
(a) It is generally the parent of the login shell
(b) It has PID 1.
(c) It is the first process in the system.
(d) Init forks and execs a 'getty' process at every port connected to a terminal.

UGC NET C.S. June-2017 (Paper-III)

Ans : (c) "Init" is the second process first processed is schedule

125. The Bounded buffer problem is also known as_____
(a) Producer - consumer problem
(b) Reader - writer problem
(c) Dining philosophers problem
(d) Both (a) and (b)

UGC NET C.S. November-2017 (Paper-III)

Ans : (a) The producer consumer problem (or bounded buffer problem) describes two processes, the producer and the consumers which share a common, fixed size buffer used as a queue. Producer produces an item & put into buffer. If buffer is already full then producer will wait for an empty block in buffer consumer consumes an item from buffer.

126. Consider a file currently consisting of 50 blocks. Assume that the file control block and the index block is already in memory. If a block is added at the end (and the block information to be added is stored in memory), then how many disk I/O operations are required for indexed (single-level) allocation strategy?
(a) 1 (b) 101
(c) 27 (d) 0

UGC NET C.S. August-2016 (Paper-III)

Ans : (a) According to question, file control block and the index block is already in memory. So, 1, 3 and 1, I/O disk operations are required for contiguous, linked and indexed allocation respectively.

127. An operating system supports a paged virtual memory, using a central processor with a cycle time of one microsecond. It costs an additional one microsecond to access a page other than the current one. Pages have 1000 words, and the paging device is a drum that rotates at 3000 revolutions per minute and transfers one million words per second. Further, one percent of all instructions executed accessed a page other than the current page. The instruction that accessed another page, 80% accessed a page already in memory and when a new page was required, the replaced page was modified 50% of the time. What is the effective access time on this system, assuming that the system is running only one process and the processor is idle during drum transfers?
(a) 30 microseconds (b) 34 microseconds
(c) 60 microseconds (d) 68 microseconds

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Effective access time
= 0.99 (1 µsec + 0.008 × 2 µsec) + 0.002 × (10000 µsec + 1000 µsec) + 0.001 × (1000 µsec + 1000 µsec)
= (0.99 + 0.016 + 22.0 + 11.0) µsec
= 34.0 µsec

128. User level threads are threads that are visible to the programmer and are unknown to the kernel. The operating system kernel supports and manages kernel level threads. Three different types of models relate user and kernel level threads.
Which of the following statements is/are true?
(A) (i) The many to one model maps many user threads to one kernel thread
(ii) The one - to - one model maps one user threads to one kernel thread
(iii) The many model maps many user threads to smaller or equal kernel threads
(B) (i) Many - to- one model maps many kernel threads to one user thread
(ii) One - to - one model maps one kernel thread to one user thread
(iii) Many - to - many model maps many kernel threads to smaller or equal user threads
Code :
(a) (A) is true; (B) is false
(b) (A) is false; (B) is true
(c) Both (A) and (B) are true
(d) Both (A) and (B) are false

UGC NET C.S. November-2017 (Paper-III)

Ans : (a)
(a) (i) The way to one model maps many user thread to one kernel thread.
(ii) The one to one model maps one user thread to one kernel thread.
(iii) The many to many model maps many user thread to smaller or equal kernel threads.
(b) (i) Many to one model maps many kernel thread to one user thread.
(ii) One to one model maps one kernel thread to one user thread.
(iii) Many to many model maps many kernel thread to smaller or equal user thread.

129. In an operating system, indivisibility of operation means :
(a) Operation is interruptible
(b) Race – condition may occur
(c) Processor cannot be pre-empted
(d) All of the above

UGC NET C.S. December-2015 (Paper-III)

Sarkari Resource

Ans : (c) In an OS, an divisible or irreducible operation is an atomic operation that is performed entirely or not performed at all. Means processor can not be pre-empted. It will be performed entirely or not at all.

130. Function of memory management unit is :

(a) Address translation (b) Memory allocation
(c) Cache management (d) All of the above

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) A memory management unit (MMU), sometimes called paged memory management unit (PMMU), is a computer hardware unit having all memory references passed through itself, primarily performing the translation of virtual memory address to physical address.

131. Which of the following statements is not true for multi level feedback queue processor scheduling algorithm?

(a) Queues have different priorities
(b) Each queue may have different scheduling algorithm
(c) Processes are permanently assigned to a queue
(d) This algorithm can be configured to match a specific system under design

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Ans : (c) Processes are not permanently assigned to a queue they can vary between queues as their priorities change

132. For the implementation of a paging scheme, suppose the average process size be x bytes, the page size be y bytes, and each entry requires z bytes. The optimum page size that minimizes the total overhead due to the page table and the internal fragmentation loss is given by :

(a) $\frac{x}{2}$ (b) $\frac{xz}{2}$
(c) $\sqrt{2xz}$ (d) $\frac{\sqrt{xz}}{2}$

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Ans : (c) Average amount of internal fragmentation per segment is y/2
Average no. of pages per process segment is x/y each page requires 'z' bytes of page table.
So each process segment require page table size of xz/y bytes.
Total overload per segment, therefore, due to internal fragmentation to minimize the overhead, differentiate with respect to page size, P & equal to 0
$d(xz/y + y/2)/dy = -xz/y^2 + 1/2 = 0$
$\Rightarrow \quad y = \sqrt{2xz}$

133. In a demand paging memory system, page table is held in registers. The time taken to service a page fault is 8 m. sec. if an empty frame is available or if the replaced page is not modified, and it takes 20 m. sec. if the replaced page is modified. What is the average access time to service a page fault assuming that the page to be replaced is modified 70% of the time?

(a) 11.6 m. sec. (b) 16.4 m. sec.
(c) 28 m. sec. (d) 14 m. sec.

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Ans : (b) Service a page fault = 8 m.sec.
page is not modified = 20 m.sec.
page replace = 70% of time
then the average time of service a page fault
$= 0.7 \times 20 + 0.3 \times 8 = 14.4.4 = 16.4$ ms

134. A LRU page replacement is used with four page frames and eight pages. How many page faults will occur with the reference string 0172327103 if the four frames are initially empty?

(a) 6 (b) 7
(c) 5 (d) 8

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Ans. (b) :

1	2	3	4	5	hit	hit	hit	6	7
0	0	0	0	3	3	3	3	0	0
	1	1	1	1	1	1	1	1	1
		7	7	7	7	7	7	7	7
			2	2	2	2	2	2	3

Total Page fault = 7

135. Which of the following system call include 'wait for time' operation command in Operating System?

(a) Process Control (b) Device Management
(c) File Management (d) Communications

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : Process Control– In an operating system, Process Control system calls are used to create, terminate, schedule suspend and resume process. These calls also include operation where a process needs to "wait for a specific time" before resuming execution.

136. Which of the following statement (s) is/are correct for context switching in Operating System?

I. When a context switch occurs, the kernel saves the context of the old process in its PCB (Process Control Block).
II. Context-Switch time is pure overhead, because the system does no useful work while switching.

(a) Neither I nor II (b) Only I
(c) Only II (d) Both I and II

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Ans. (d) : Context switching in an operating system happens when the CPU shifts from one process to another. The kernel saves the state of the current process in its Process Control Block (PCB) and loads the state of the next process ensuring smooth continuation. This mechanism enable multitasking and four CPU sharing among processes.

137. Which of the following page replacement algorithms(s) does not suffer from Belady's anomaly in Operating System?

I. Least-Recently Used (LRU) page replacement.
II. First-In, First-Out (FIFO) page replacement.

(a) Only I (b) Neither I nor II
(c) Only II (d) Both I and II

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Ans. (a) : Belady's anomaly– In same page replacement algorithms like FIFO (First-In-First-Out) increasing the number of page frames can paradoxically increase the number of page faults. This strange behavior is known as Belady's anomaly.
FIFO– Suffers from Belady's aromaly.
LRU– Does not suffer from Belady's anomaly.

138. Suppose that a process is in 'BLOCKED' state waiting for some I/O service. When the service is completed, it goes to the _____ state.
मान लीजिए कि 'BLOCKED' स्थिति में प्रॉसेस कुछ I/O सर्विस के लिए प्रतीक्षारत है। जब सर्विस पूरी हो जाती है, तो यह _____ स्थिति में आ जाती है।
(a) Running/रनिंग (b) Ready/रेडी
(c) Suspended/सस्पेंडेड (d) Terminated/टर्मिनेटेड

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Therefore a process that was in the Blocked state waiting for I/O moves to the Ready state after I/O completion. From there, it will eventually move to the running state when the CPU scheduler selects it.

139. Consider a main memory with 3 page frames in the following sequence of page reference 3, 4, 2, 3, 9, 1, 6, 3, 4, 9. What is the number of page fault using FIFO page replacement algorithm?
तीन पेज वाली एक मुख्य मेमोरी में पेज रेफरेंस 3, 4, 2, 3, 9, 1, 6, 3, 4, 9 के निम्नलिखित क्रम को फ्रेम करता है। FIFO पेज रिप्लेसमेंट एल्गोरिथम का उपयोग करके पेज फॉल्ट की संख्या क्या है?
(a) 9 (b) 10
(c) 8 (d) 7

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Consider page reference string 3, 4, 2, 3, 9, 1, 6, 3, 4, 9 with 3 page frames using the FIFO page replacement algorithm.
Page 3, 4, 2, 3, 9, 1, 6, 3, 4, 9 reference

3	4	2	3	9	1	6	3	4	9
		2	2	2	2	6	6	6	9
	4	4	4	4	1	1	1	4	4
3	3	3	3	9	9	9	3	3	3
Fault	Fault	Fault	No fault	Fault	Fault	Fault	Fault	Fault	Fault

Total number of page fault = 9

140. What is Throughput in process scheduling?
प्रोसेस शेड्यूलिंग में थ्रूपुट क्या है?
(a) Time required to complete a process
एक प्रक्रिया को पूरा करने के लिए आवश्यक समय
(b) Number of processes completed per unit time
प्रति यूनिट समय में पूर्ण की गई प्रोसेसेस की संख्या
(c) Time required by a process to respond
प्रोसेस के लिए जवाब के लिए आवश्यक समय
(d) Time required to abort a process
एक प्रक्रिया को रद्द करने के लिए आवश्यक समय

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Ans. (b) : Throughput in process scheduling refers to the number processes that are completed and successfully executed within a specific period of time. It is a measure of the system is productivity and efficiency.
Higher throughput indicates that the system can handle more processes in given time frame, which is typically desirable in computing environment.

141. Information about each process in operating system is maintained by _____.
ऑपरेटिंग सिस्टम में प्रत्येक प्रक्रिया के बारे में जानकारी _____ द्वारा बनाए रखी जाती है।
(a) Scheduler/स्केडुलेर
(b) Process control block/प्रोसेस कण्ट्रोल ब्लॉक
(c) Central Processing Unit/सेंट्रल प्रोसेसिंग यूनिट
(d) Program Counter/प्रोग्राम काउंटर

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Ans. (b) : A Process Control Block (PCB) is a data structure used by computer operating system to store all the information about a process.
This information is essential for managing processes and ensuring that the operating system can switch between them efficiently.
A typical PCB includes–
(1) Program counter– Address of the next instruction to be executed.
(2) Process Identification (PID)– Unique identifier for the process.
(3) Process state– Current state of the process (e.g., running waiting, ready, terminated).

142. The instruction which are executed only in kernel mode is known as _____.
इंस्ट्रक्शन जो केवल कर्नेल मोड में निष्पादित होता है, _____ के रुप में जान जाता है।
(a) Memory instruction /मेमोरी इंस्ट्रक्शन
(b) Arithmetic instruction /अर्थमेटिक इंस्ट्रक्शन
(c) Privileged instruction /प्रीविल्लेज़ड इंस्ट्रक्शन
(d) Logical instruction /लॉजिकल इंस्ट्रक्शन

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Ans. (c) : In an operating system, same instructions can be dangerous if executed by a normal user process (for example I/O control interrupt handling, setting mode bits, managing memory). These are restricted and can only be executed in kernel mode. Such instruction are called Privileged Instructions.

143. Data structure maintained by the operating system to maintain information for each thread within a process is known as _____.
एक प्रक्रिया के अंतर्गत प्रत्येक थ्रेड के लिए जानकारी के प्रबंधन के लिए ऑपरेटिंग सिस्टम द्वारा संधृत डेटा संरचना को _____ के रुप मे जान जाता है ।
(a) Process control block /प्रोसेस कंट्रोल ब्लॉक
(b) Thread control block /थ्रेड कंट्रोल ब्लॉक
(c) Symbol table /सिम्बल टेबल
(d) Scheduling queue /शेड्यूलिंग क्यू

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Ans. (b) : Thread Control Block (TCB)⇒ The Thread Control Block is the OS data structure that maintains all information for each thread within a process. It is crucial for thread management, CPU scheduling and context switching other options like PCB symbol table or scheduling queue either manage processes, compiler-level information, or queues of threads, but they do not store per thread execution details.

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144. Consider a main memory with the capacity of 3 page frames which are initially empty. Page reference is 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. What is the number of page fault using optimal page replacement algorithm?
3पेज फ्रेम, जो आरंभ में खाली है, की क्षमता वाली एक मुख्य मेमोरी पर विचार करें। पेज रेफरेंस 4, 7, 6, 1, 7, 6, 1, 2, 7, 2 हैं। ऑप्टीमल पेज प्रतिस्थापन एल्गोरिथम का उपयोग करते हुए पेज फॉल्ट की संख्या क्या है?

(a) 7 (b) 6
(c) 5 (d) 4

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Ans. (c) : Main memory has 3 pages frames
Reference string = 4, 7, 6, 1, 7, 6, 1, 2, 7, 2
Use Optimal Page Replacement.
Step-by-step
1. 4 ⇒ fault (Frames : 4)
2. 7 ⇒ fault (Frames : 4, 7)
3. 6 ⇒ fault (Frames : 4, 7, 6)
4. 1 ⇒ fault (Replace 4, not needed again) ⇒ (1, 7, 6)
5. 7 ⇒ Hit
6. 6 ⇒ Hit
7. 1 ⇒ Hit
8. 2 ⇒ fault (Replace 1 or 6 both not needed again ⇒ Choose one say 1) ⇒ (2, 7, 6)
9. 7 ⇒ Hit
10. 2 ⇒ Hit

Total Page faults = 5.

145. Inner most layer of operating system which is always present in main memory is known as _____.
ऑपरेटिंग सिस्टम की सबसे भीतरी परत जो हमेशा मुख्य मेमोरी में उपस्थित रहती है, _____ के रुप मे जानी जाती है ।

(a) Hardware /हार्डवेयर
(b) User interface /यूजर इंटरफेस
(c) Application program /ऐप्लीकशन प्रोग्राम
(d) Kernel /कर्नेल

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Ans. (d) : The innermost layer of an operating system, which is always present in main memory, is the kernel.
Kernel– The care of OS, manages CPU, memory, devices it always stays in main memory to control the system.

146. When there is a memory wastage within the memory partition itself then it knows as _____.
जब मेमोरी पार्टीशन के भीतर ही मेमोरी वेस्टेज होती है तो इसे _____ के रूप में जाना जाता है।

(a) External fragmentation/एक्सटर्नल फ्रेगमेंटेशन
(b) Belady's Anomaly/बेलेडीज एनोमली
(c) Internal fragmentation/इंटरनल फ्रेगमेंटेशन
(d) Compaction/कम्पैक्शन

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Ans. (c) : Internal fragmentation is the space wasted inside of allocated memory blocks because of restrictions on the allowed sizes of allocated blocks. Allocated memory may be slightly larger than requested memory this size difference is memory internal to a partition, but not being used.

147. If a memory partition of size 80k is allocated to a process of size 60k then the 20k of the partition is wasted and cannot be allocated to any process. It is called _____ .
यदि आकार 80k का मेमोरी विभाजन 60k आकार की प्रक्रिया के लिए आवंटित किया जाता है तो विभाजन का 20k बर्बाद हो जाता है और किसी भी प्रक्रिया को आवंटित नहीं किया जा सकता है। इसको _____ कहा जाता है।

(a) External fragmentation/एक्सटर्नल फ्रेगमेंटेशन
(b) Best fit/बेस्ट फिट
(c) Worst fit/वर्स्ट फिट
(d) Internal fragmentation/इंटरनल फ्रेगमेंटेशन

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Ans. (d) : Internal fragmentation occurs when memory is allocated in fixed size blocks (partitions), and the allocated memory is larger than what the process actually needs. The unused space inside the partition is wasted.
Example– A process requests 60KB, but the system allocates a 80KB partition. The remaining 20KB is unused but cannot be assigned to another process.

148. If two non-continuous free partition of size 'a' and 'b' are available. A process of size 'c' cannot be allocated even when a + b > c, a < c and b < c. This problem is known as _____ .
यदि 'a' और 'b' परिमाण के दो गैर-निरंतर फ्री पारटिशन उपलब्ध हैं। a + b > c, a < c और b < c होने पर भी परिमाण 'c' की एक प्रक्रिया को आवंटित नहीं किया जा सकता है। इस समस्या को _____ के रूप में जाना जाता है।

(a) External fragmentation/एक्सटर्नल फ्रेगमेंटेशन
(b) Internal fragmentation/इंटरनल फ्रेगमेंटेशन
(c) Quick fit/क्विक फिट
(d) Worst fit/वर्स्ट फिट

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : This scenario describes external fragmentation, where the total free memory is sufficient to satisfy a request (a + b ≥ c), but the memory is not contiguous (non continuous partitions), so the request cannot be fulfilled.
The partitions are non-continuous (they're not adjacent)
Individually, each partition is two small (a < c and b < c)
Collectively, they have enough space (a + b ≥ c)
The process can't be allocated because the available memory is fragmented.

149. The rate at which page fault occurs is known as _____ .
जिस दर पर पेज फॉल्ट होता है उसे _____ के रूप में जाना जाता है।

(a) Swapping/स्वैपिंग (b) Paging/पेजिंग
(c) Thrashing/थ्रैसिंग (d) Compaction/कंपैक्शन

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Ans. (c) : Thrashing is a condition that occurs in virtual memory system when the computer spends more time paging than executing application instructions. This leads to a high page fault rate and a significant decrease in system performance.

150. _____ is a process of lifting the process from memory and placing it on disk.
_____ प्रोसेस को मेमोरी से उठाने और डिस्क पर रखने की एक प्रक्रिया है।
(a) Paging/पेजिंग
(b) Swapping out/स्वैपिंग आउट
(c) Swapping in/स्वैपिंग इन
(d) Thrashing/थ्रैसिंग

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : Swapping out refers to the process of moving a process from RAM (memory) to disk (swap space) to free up memory for other processes.
Swapping in is the reverse bringing a process back from disk into memory.
Paging is a memory management scheme where processes are divided into fixed size blocks called pages, but it doesn't necessarily involve moving entire processes to disk.

151. Consider a main memory with the capacity of 4 page frames which are initially empty. If the page reference string is 1, 3, 4, 4, 3, 2, 1, 7, 5, 6 then what is the number of page fault using FIFO page replacement algorithm?
4 पेज फ्रेम की क्षमता वाली एक मुख्य मेमोरी पर विचार करें जो प्रारंभ में खाली हैं। यदि पेज रिफरेन्स स्ट्रिंग 1, 3, 4, 4, 3, 2, 1, 7, 5, 6 है, तो एफ.आई.एफ.ओ पेज रिप्लेसमेंट एल्गोरिथम का उपयोग करके पेज फाल्ट की संख्या क्या है?
(a) 6 (b) 7
(c) 8 (d) 9

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : To calculate the number of page faults using the FIFO (First-in-First-Out) page replacement algorithm with 4 empty page frames, Let's simulate the process step by step with the reference string
Reference string– 1, 3, 4, 4, 3, 2, 1, 7, 5, 6
Frames -4 (initially empty)
Step by step (FIFO)
1 → Page fault → [1]
3 → Page fault → [1, 3]
4 → Page fault → [1, 3, 4]
4 → Already in memory→ no page fault
3 → Already in memory → no page fault
2 → Page fault → [1, 3, 4, 2]
1 → Already in memory→ no page fault
7 → Page fault → FIFO remove 1 → [3, 4, 2, 7]
5 → Page fault → FIFO remove 3 → [4, 2, 7, 5]
6 → Page fault → FIFO remove 4 → [2, 7, 5, 6]
Total Page Fault 1(1) + 1(3) + 1(4) + 0(4) + 0(3) + 1(2) + 0(1) + 1(7) + 1(5) + 1(6) = 7 page faults

152. Which page replacement algorithm has Belady's anomaly?
बेलेड़ी अनोमाली किस पेज रिप्लेसमेंट एल्गोरिथम में है?
(a) Optimal page replacement
ऑप्टीमल पेज रिप्लेसमेंट
(b) FIFO page replacement
एफ.आई.एफ.ओ. पेज रिप्लेसमेंट
(c) Least recent used page replacement
लीस्ट रीसेंट यूज्ड पेज रिप्लेसमेंट
(d) Not recently used page replacement
नॉट रिसेंटली यूज्ड पेज रिप्लेसमेंट

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : In computer storage, Belady's anomaly is the phenomenon in which increasing the number of page frames results in an increase in the number of page faults for certain memory access patterns. This phenomenon is commonly experienced when using the first in first out (FIFO) page replacement algorithm.

153. If P is the mean of Poisson distribution, then what is the standard deviation?
यदि P पॉइसन डिस्ट्रिब्यूशन का माध्य है, तो स्टैंडर्ड मानक क्या है?
(a) $\frac{P}{2}$ (b) P
(c) P^2 (d) $\sqrt{P}$

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : If mean = P
In the question, they say:
If P is the mean of a poison distribution…
that means:
$\lambda = P$
Variance = P
Standard Deviation = $\sqrt{P}$.

154. Which of the following is not a state of a process in operating system?
निम्नलिखित में से कौन सी ऑपरेटिंग सिस्टम में एक प्रक्रिया की एक अवस्था (स्टेट) नहीं होती है?
(a) Waiting/वेटिंग (b) Excite/एक्साइट
(c) Running/रनिंग (d) Ready/रेडी

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : The state 'Excite' is not a valid process state in an operating system. A process in an operating system can be in several states:
- **New :** The process is being created.
- **Ready :** The process is waiting to be assigned a CPU.
- **Running :** The process is currently executing on the CPU
- **Waiting :** The process is waiting for some event to occur.
- **Terminated :** The process has finished execution.

155. Which of the following does not interrupt a running process?
(a) Device (b) Timer
(c) Scheduler (d) Power Failure

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Ans. (c) : Device- Device interrupts are hardware interrupts that are triggered when a device needs attention.
Timer interrupts are used to trigger events based on time intervals, often for process scheduling or managing time slices.
Scheduler is not an interrupt, it is a part of the operating system responsible for deciding which process should run next. It does not interrupt a process but instead decides when to switch contexts.
Power failure is a type of hardware failure that results in an immediate system shutdown, which interrupts processes.
So, the scheduler does not interrupt a running process directly.

156. Which of the following is the correct sequence with regard to service routine?
A. Save contents of processor registers
B. Turn the interrupt facility on
C. Service the device whose flag is set
D. Check which flag is set
E. Restore contents of processor registers
Choose the correct answer from the options given below :
(a) A, B, C, D, E (b) D, B, C, A, E
(c) D, C, B, E, A (d) A, D, C, E, B
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Ans. (d) : The correct sequence for a service routine involves preserving the system state, identifying the interrupt source, handling the interrupt, restoring the state and re-enabling interrupts.
Save contents of processor-registers (A), This is critical to preserve the interrupted program's state.
Check which flag is set (D), Determine the Source of the interrupt to address it correctly.
Service the device whose flag is set (C), Resolve the interrupt by interacting with the specific device.
Restore contents of processor registers (E), Return the system to its pre interrupt state.
Turn the interrupt facility on (B), Renewable interrupts (if not handled automatically by the return instruction).

157. Suppose that the virtual address space has eight pages and physical memory with four page frames. If LRU page replacement algorithm is used, ... number of page faults occur with the reference string.
0 2 1 3 5 4 6 3 7 4 7 3 3 5 5 3 1 1 1 7 2 3 4 1
(a) 11 (b) 12
(c) 10 (d) 9
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Ans : (*)

0	2	1	3	5	4	6	3	7	4	7	3	3	5	5	3	1	1	1	7	2	3	4	1
0	0	0	0	5	5	5		7					7			7				7		7	1
	2	2	2	2	4	4		4					4			1				1		4	4
		1	1	1	1	6		6					5			5				2		2	2
			3	3	3	3		3					3			3				3		3	3

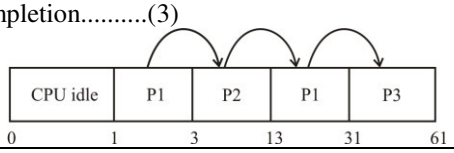
Total number of page fault = 13
So, no option is matching

158. Five jobs A, B, C, D and E are waiting in Ready Queue. Their expected runtimes are 9, 6, 3, 5 are x respectively. All jobs entered in Ready queue at time zero. They must run in ..... order to minimize average response time if 3 < x < 5.
(a) B, A, D, E, C (b) C, E, D, B, A
(c) E, D, C, B, A (d) C, B, A, E, D
UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) To minimize response time these must run in shortest job first order.
Therefore, C(3), E(3 < x < 5), D(5), B(6), A(9)

159. Consider three CPU intensive processes P1, P2, P3 which require 20, 10 and 30 units of time, arrive at times 1, 3 and 7 respectively. Suppose operating system is implementing shortest Remaining time first (preemptive scheduling) algorithm, then ..... context switches are required (suppose context switch at the beginning of ready queue and at the end of Ready queue are not counted).
(a) 3 (b) 2
(c) 4 (d) 5
UGC NET C.S. Augest-2016 (Paper-II)

Ans : (a) P1, P2, P3 are three processes
Arrival time of P1 is 1 so we made to run P1 at time 1 now P1 is run for 2 units, process arrived with less B.T, so switch happens to P2-(1) P2 run till completion since there are no shorter Jobs available, then switch happens to P1 .....(2)
Again P1 is run since it has burst time (is) < burst time of P3 (30) completion of P1, switch happens to P3 run till completion.....(3)



160. _____ is used to shift processes so they are contiguous, and all free memory is in one block.
_____ का उपयोग प्रक्रियाओं को स्थानान्तरित करने के लिए किया जाता है ताकि वे सन्निहित हों, और सभी मुक्त मेमोरी एक ब्लॉक में हों।
(a) Fragmentation
(b) Compaction
(c) External Fragmentation
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 1.0 2023 (11-12)

Ans. (b) : Compaction is used to relocate processes so that they are contiguous, and all free memory is in a single block. It reduce free space in memory by keeping part of memory management that helps ensure proper memory usage and performance.

161. The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by
(a) Page size
(b) Physical size of memory

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(c) The instruction set architecture
(d) Number of processes in memory

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : There are two important task in virtual memory management : a page replacement strategy and a frame-allocation strategy. Frame allocation strategy say gives the idea of minimum number of frames that a process must be allocated is dependent on system architecture, and corresponds to the number of pages that could be touched by a single (machine) instruction. So the instruction set architecture is right answer.

162. In _____ page replacement algorithm, page that will not be used for longest period of time is replaced whenever a page fault occurs.
_____ पेज रिप्लेसमेंट एल्गोरिथ्म में, जब भी पेज फॉल्ट होता है तो जो पेज लम्बे समय तक उपयोग में नहीं आयेगा उसे हटाया जाता है।

(a) LRU (b) Optimal
(c) FIFO (d) LFU

HTET PGT C.S. 2022

Ans. (b) : In the replacement algorithms, the goal is to decide which page in memory should be replaced when a page fault occurs.
The Optimal Page Replacement Algorithm (OPT) works by replacing the page that will not be used for the longest period of time in the future. Since it requires knowledge of future memory references, it cannot be implemented in real system but is used as a benchmark to compare the performance of other algorithms like FIFO, LRU, and LFU.

163. In a round robin scheduling, context switch time is 4 units and the average process running time before blocking is 6 units, then CPU efficiency is:

(a) 0.2 (b) 0.4
(c) 0.6 (d) 0.1

CGPSC C.S. 2020 Set-A

Ans. (c) :
Formula for CPU Efficiency in Round Robin–

$$\text{Efficiency} = \frac{\text{Useful CPU time}}{\text{Useful CPU time} + \text{Context switch overhead}}$$

Given–
Context switch time = 4 units
Average running time before blocking = 6 units
So, in one cycle:
Useful execution time = 6 units
Overhead (context switch) = 4 units

$$\text{Efficiency} = \frac{6}{6+4} - \frac{6}{10} = 0.6$$

164. Consider a set of n tasks with known run times r_1, r_2,.....4n to be run on a uni-processor scheduling algorithms will result in the maximum throughput?

(a) Round robin
(b) SJF
(c) Highest response ratio next
(d) First-come-first served

CGPSC C.S. 2020 Set-A

Ans.(b): Throughput– In operating system, throughput is the number of processes completed per unit time. A good scheduling algorithm should minimize waiting and turnaround times so that more processes finish in less time, thereby maximizing throughput.
SJF (Shortest Job First)– SJF always picks the process with the shortest runtime first. This allows many processes to finish quickly, reducing average waiting time and turnaround time.

165. Consider a logical address space of four pages of 2048 words each mapped into a physical memory of 32 frames. How many bits are there in the logical address?

(a) 12 (b) 14
(c) 13 (d) 11

CGPSC C.S. 2020 Set-A

Ans. (c) : Step-1: logical address structure–
A logical address = page number bits + page offset bits.
→ Page offset bits depend on the page size.
→ Page number bits depend on the number of pages in the logical address space.
Step-2 : Page size–
Each page = 2048 words
To address 2028 words, we need:
$\log_2 (2048) = 11$ bits
So page offset = 11 bits
Step-3 : No of pages–
Logical address space = 4 pages
To represent 4 pages, we need:
$\log_2(4) = 2$ bits
So, page number = 2 bits
Step-4 : Total bits in logical address–
Logical address bits = page number bits + page offset bits = 2 + 11 = 13 bits.

166. In which one of the following page replacement policies, Belady's anomaly may occur?

(a) FIFO (b) OPT
(c) LRU (d) MRU

CGPSC C.S. 2020 Set-A

Ans. (a) : Belay's anomaly refers to the counterintuitive situation where increasing the number of page foults instead of fever.
This anomaly is possible in the (First-in-first Out) page replacement policy, because FIFO evicts the oldest loaded page without considering how frequently or recently it has been used.

167. Amount of work that a system is able to do per unit time is called as:
कार्य की वह मात्रा जो एक प्रणाली प्रति इकाई समय में करने में सक्षम होती है, कहलाती है–

(a) Response Time/रेसपॉन्स टाइम
(b) Turnaround Time/टर्नअराउंड टाइम
(c) Throughput/थ्रूपुट
(d) Input Time/इनपुट टाइम

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Ans. (c) : In operating systems and computer performance evolution, Throughput refers to the amount of work or number of processes completed by the system in a unit of time.
Example– If a system finishes 10 jobs in 5 second, its throughput is 2 jobs per second.

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168. **_____is the program in execution.**
_____निष्पादन में प्रोग्राम है।
(a) Process/प्रोसेस
(b) Thread/थ्रेड
(c) Operating system/ऑपरेटिंग सिस्टम
(d) Application software/एप्लीकेशन सॉफ्टवेयर

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Ans. (a) : A program is just a passive set of instructions stored on disk. It becomes active only when it is loaded into memory and executed by the CPU. At that moment, the program in execution is called a process.
Program in execution = process.

169. **Which of the following need not be saved on a context switch between processes?**
(a) General purpose registers
(b) Translation look-aside buffer
(c) Program counter
(d) All of the above

CGPSC C.S. 2020 Set-A

Ans. (b) : During a context switch the operating system saves the state of the current process so it can resume later, and loads the state of the next process. Important information such as the program counter and the general purpose registers must be saved, otherwise the process will loss its progress. However, the Translation Look side Buffer (TLB) does not need to be saved. This is because the TLB is just a cache of page table entries. When a new process starts, the TLB is flushed and automatically refilled a needed.

170. **If only one job of 10 time units is present in a ready queue, and the time quantum is 1 time unit in round robin scheduling, then**
(a) 4 context switches will occur.
(b) Only 1 context switch will occur
(c) 9 context switches will occur
(d) No context switch will occur.
(e) Not attempted

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Ans. (c) : Round Robin Scheduling– Round Robin (RR) is a preemptive scheduling algorithm. Each process gets a fixed time slice (quantum). If the process does not finish in the quantum, the CPU scheduler takes it away and gives it to the next process in the ready queue.
Context switch = when the CPU switches from one process to another.
Execution–
- At time 0, the only process is loaded into the CPU.
- It runs for 1 unit. Time quantum expires.
- Now the scheduler checks → It there another process waiting.
- If yes → switch CPU to that process (context switch happens).
- If no → CPU keeps running the same process again (No context switch).

This repeats until the process completes all 10 units.

171. **Turnaround time in operating systems is**
(a) The sum of periods spent in waiting to get into memory, waiting in the ready queue, executing on the CPU, and doing I/O.
(b) The amount of time a job spends waiting in the ready queue.
(c) The time from the submission of a request unitl the first response is produced.
(d) The amount of time spent waiting in the device queue.
(e) Not attempted

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Ans. (a) : Turnaround Time– In operating system, turnaround time refers to the total time taken by a process from the moment it enters the system until it finishes execution. It tells us how long a process stays in the system, including all waiting and execution phases.
Formula– Turnaround Time (TAT) = Completion Time - Arrival Time This formula captures the total duration a process spends in the system.

172. **For each of the following transitions, between process states, which transition is not possible?**
(a) Running → Ready
(b) Blocked → Suspended
(c) Ready → Ready/Suspended
(d) Blocked → Running

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Ans.(d):
(a) Running → Ready– This happens when a running process is preempted by the scheduler (e.g. time quantum expires in Round Robin).
(b) Blocked → Suspended– A blocked process can be suspended if the OS wants to move it to disk to free up memory.
(c) Ready → Ready/Suspended– A ready process can be suspended (moved out of main memory) when memory is low.
(d) Blocked → Running– A blocked process is waiting for I/O/event, so it cannot directly move to running.

173. **Consider the following statements regarding the term 'aging' used in operating systems.**
(i) Aging is a technique used to preempt currently executing job if a new job with a shorter next CPU burst arrives.
(ii) It is a technique that gradually increases the lift-time of high-priority jobs.
(iii) Aging is used to solve the problem of starvation of low-priority jobs.
(iv) Aging is a technique that is used to gradually increase the priority of high-priority jobs.
Which of the statements given above are correct?
(a) (i) and (ii) only (b) (ii) and (ii) only
(c) (iii) only. (d) (i) and (iv) only
(e) Not attempted

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Ans. (c) : Aging in Operating System– Aging is a scheduling technique used to prevent starvation in priority scheduling. If a low priority job waits too long in the ready queue, its priority is gradually increased over time, so eventually it will get CPU.
That is the main purpose of aging.

174. Consider the following reference string and a memory with three frames:
7, 0, 2, 0, 0, 7, 1.
How many page faults will be produced by the LRU page replacement algorithm?
(a) 3 (b) 4
(c) 5 (d) 7
(e) Not attempted

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Ans. (b) : Quick LRU trace with 3 frames for reference string 7, 0, 2, 0, 0, 7, 1.
7 → Fault (Frames : [7])
0 → Fault (Frames : [7, 0])
2 → Fault (Frames : [7, 0, 2])
0 → Hit (Frames unchanged)
0 → Hit (Frames unchanged)
7 → Hit (Frames unchanged : LRU how is 2)
1 → Fault, evict LRU (2) → (Frames : [7, 0, 1])
Total page fault = 4.

175. In addition to the signal() operation, which other standard atomic operation can be used to access a semaphore?
(a) Block() (b) Fork()
(c) Wait() (d) Exec()
(e) Not attempted

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Ans. (c) : A semaphore is a synchronization tool in operating system used to control access to shared resources. It is an integer variable that can be incremented or decremented using atomic operations.
Wait()– This operation decreases the value of the semaphore by 1. If the resulting value is negative the process is blocked and added to the semaphore's waiting queue until the resources becomes available.

176. The technique of operating system to temporary remove inactive programs from memory of computer system:
कम्प्यूटर सिस्टम की मेमोरी से निष्क्रिय प्रोग्रामों को अस्थायी रूप से हटाने की ऑपरेटिंग सिस्टम की तकनीक कहलाती है–
(a) Scheduling (b) Slicing
(c) Swapping (d) Kernel

HTET PGT C.S. 30.07.2025

Ans. (c) : Swapping is the technique used to temporarily move inactive processes out of main memory and later bring them back when needed.

177. Critical region mechanism in operating system design is used for
(a) Memory management
(b) File management
(c) Process synchronization
(d) Disk management
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : A critical region is a section of code where only one process can access a shared resource at a time. This is used to prevent race conditions, which occur when multiple processes try to access the same resource simultaneously, leading to unpredictable result. The operating system uses synchronization mechanism like semaphores or mutexes to control access to critical regions.

178. Belady's anomaly may occur in
(a) Round-robin scheduling algorithm
(b) Page replacement algorithm
(c) SSTF disk scheduling algorithm
(d) SCAN disk scheduling algorithm
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : Belady's anomaly is a phenomenon in operating systems where increasing the number of page frames in memory can actually increase the number of page faults in certain page replacement algorithms, especially FIFO (First-In-First-Out).

179. Which of the following is a CPU scheduling algorithm?
(a) C-LOOK (b) SJF
(c) LRU (d) SCAN
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : CPU scheduling algorithms determine which process in the ready queue gets the CPU next. Example include FCFS, SJF, Priority, Round Robin etc.

180. In operating systems, a page fault means
(a) Some I/O error in page
(b) An error due to difference in size of a page
(c) Reference of address on a page, not present in the memory
(d) None of these
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : In operating systems a page fault occurs in virtual memory system, when a process tries to access a page that is not currently loaded in the main memory (RAM).

181. Consider the following statements in view of operating systems:
(i) The SJF scheduling algorithm can be either preemptive or non-preemptive.
(ii) The FCFS CPU scheduling algorithm is non-preemptive.
(iii) The priority scheduling algorithm can be either preemptive or non-preemptive.
How many of the statements given above are correct?
(a) Only one (b) Only two
(c) All three (d) None
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) :
(i) SJF (Shortest Job First)– Can be preemptive (SRTF) or non-preemptive.
(ii) FCFS (First-Come, First-Served)– Always non-preemptive.
(iii) Priority Scheduling– Can be preemptive or non-preemptive.
All three statement are correct.

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182. No. of programs processed by the system per unit time is called:
प्रति यूनिट समय में सिस्टम द्वारा संसाधित प्रोग्रामों की संख्या को कहा जाता है–

(a) Throughput (b) Scheduling
(c) Interrupt (d) Processing

HTET PGT C.S. 30.07.2025

Ans. (a) : Throughput refers to the number of processes completed by the system per unit of time. It is a key performance metric in operating systems, showing how efficiently the CPU, memory, and I/O devices are being used. For example if 20 jobs are completed in 10 seconds, the throughput is 2 jobs per second.

183. Suppose that a process is in "BLOCKED" state waiting for some I/O service, when the service is completed, it goes to
मान लीजिए कि एक प्रक्रिया ''ब्लॉक'' स्थिति में है जो कुछ I/O सेवा की प्रतीक्षा कर रहा है। जब सेवा पूर्ण हो जाती है, तो निम्न में चला जाता है:

(a) Ready state/रेडी स्टेट
(b) Running state /रनिंग स्टेट
(c) Terminated state/टर्मिनेटेड स्टेट
(d) Suspended state/सस्पेन्डेड स्टेट

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : Blocked state– When a process request an I/O operation (like reading from a disk, printer, or keyboard), it cannot continue execution until the I/O is finished. At this point, the process leaves the Ready/Running state and enters the Blocked (or waiting) state. In this state, the process is inactive because it is waiting for the I/O services to complete.

184. Mutual exclusion problem occurs
मुच्युअल एक्सक्लुजन समस्या उत्पन्न होती है

(a) Between two disjoint processes that do not interact/दो विघटित प्रक्रिया के बीच जो आपस में इन्टरैक्ट नहीं करती हैं।
(b) Among processes that share resources
प्रक्रियाओं के बीच जो संसाधनों को साझा करती हैं।
(c) Among processes that do not use the same resources./ऐसी प्रक्रियाओं के बीच, जो एक ही संसाधन का उपयोग न करें।
(d) None of the above /उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : Mutual Exclusion– Mutual exclusion means that when one process is using a shared resource (like a printer, file or memory location), no other process should access it at the same time. This ensure data consistency prevents race conditions.
The mutual exclusion problem occurs only when multiple processes share a common resource and try to access it simultaneously.

185. Suppose that a process is in "BLOCKED" state waiting for some I/O service, when the service is completed, it goes to
मान लीजिए कि एक प्रक्रिया ''ब्लॉक'' स्थिति में है जो कुछ I/O सेवा की प्रतीक्षा कर रहा है। जब सेवा पूर्ण हो जाती है, तो निम्न में चला जाता है:

(a) Ready state/रेडी स्टेट
(b) Running state /रनिंग स्टेट
(c) Terminated state/टर्मिनेटेड स्टेट
(d) Suspended state/सस्पेन्डेड स्टेट

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : Blocked state– When a process request an I/O operation (like reading from a disk, printer, or keyboard), it cannot continue execution until the I/O is finished. At this point, the process leaves the Ready/Running state and enters the Blocked (or waiting) state. In this state, the process is inactive because it is waiting for the I/O services to complete.

186. Mutual exclusion problem occurs
मुच्युअल एक्सक्लुजन समस्या उत्पन्न होती है

(a) Between two disjoint processes that do not interact/दो विघटित प्रक्रिया के बीच जो आपस में इन्टरैक्ट नहीं करती हैं।
(b) Among processes that share resources
प्रक्रियाओं के बीच जो संसाधनों को साझा करती हैं।
(c) Among processes that do not use the same resources./ऐसी प्रक्रियाओं के बीच, जो एक ही संसाधन का उपयोग न करें।
(d) None of the above /उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : Mutual Exclusion– Mutual exclusion means that when one process is using a shared resource (like a printer, file or memory location), no other process should access it at the same time. This ensure data consistency prevents race conditions.
The mutual exclusion problem occurs only when multiple processes share a common resource and try to access it simultaneously.

187. In process swapping in operating system, what is the residing location of swap space ?
ऑपरेटिंग सिस्टम में स्वैपिंग की प्रक्रिया के लिए स्वैप स्पेस कहाँ रहता है?

(a) RAM (b) Disk
(c) ROM (d) On-chip cache

HTET PGT C.S. 2020

Ans. (b) : The swap in an operating system is located on the disk. When the main memory (RAM) becomes full, the operating system uses the swap space on the disk to temporarily store inactive or less used processes. This process is called swapping. The purpose of swap space is to free up RAM for currently active processes so that the system can run smoothly.
Example–

- RAM = 4GB
- Three process (P1, P2, P3) are running.
- A fourth process (p4) needs memory, but RAM is full.
- OS select p2 (least used recently), moves it from RAM ⇒ swap space on Disk.
- Now, RAM has free space for P4.
- Later, when P2 is needed again, OS brings it back from Disk- RAM.

This cycle ensure efficient utilization of limited RAM.

Sarkari Resource

188. In a multiprogramming environment
एक मल्टीप्रोग्रामिंग वातावरण में

(a) The processor executes more than one process at a time./प्रोसेसर एक समय में एक से अधिक प्रक्रिया निष्पादित करता है।

(b) The programs are developed by more than one person./एक से अधिक व्यक्ति द्वारा प्रोग्राम विकसित किया जाता है।

(c) More than one process resides in the memory./एक से अधिक प्रक्रिया मेमोरी में रहती है।

(d) A single user can execute many programs at the same time./एक ही यूजर, एक ही समय में कई प्रोग्राम को निष्पादित कर सकता है।

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : Multiprogramming Environment– In multiprogramming, more than one process is kept in the main memory at the same time. The CPU executes one process, but when that process waits (e.g. for I/O), the CPU switches to another process.
This is the exact definition of multiprogramming multiple processes are loaded into memory so the CPU can switch between them.

189. Which of the following is not a CPU scheduling Algorithm?
निम्नलिखित में से कौन सा एक सी.पी.यू. शेड्यूलिंग एल्गोरिद्म नहीं है?

(a) FCFS (b) C-SCAN
(c) SJF (d) SJN

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : C-SCAN stands for circular SCAN. While it is a disk scheduling algorithm used to optimize disk access, it is not a CPU scheduling algorithm. CPU scheduling algorithms focus on managing the allocation of the CPU to different processes. FCFS (First Come First Served). SJF (Shortest Job First) and SJN (Shortest Job Next) are all recognized CPU scheduling algorithms.

190. Which of the following statement(s) is/are true about CPU scheduling algorithm?
सी.पी.यू. शेड्यूलिंग एल्गोरिथम के बारे में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. Shortest remaining time first scheduling may cause starvation./शॉर्टेस्ट रिमेनिंग टाइम फर्स्ट शेड्यूलिंग से स्टारवेशन हो सकती है

II. Round Robin is better than First ComeFirst Serve in terms of response time/प्रतिक्रिया समय के मामले में राउंड रॉबिन फर्स्ट कम फर्स्ट सर्व बेहतर है

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : Statement-I : Shortest Remaining time first (SRTF) is a preemptive version of Shortest Job First (SJF). In SRTF, a newly arriving process with a shorter remaining time than the current process will preempt the running process. This can cause starvation for longer processes if short processes keep coming.

Statement-II : Round Robin (RR) scheduling assigns a fixed time quantum & cycles through processes. It ensures that no process waits too long before getting CPU time, so response time (the time before the first response is given) is better than in First Come First Serve (FCFS), where a long job can delay all others. Therefore both statements are true.

191. Assume that the following jobs are to be executed on a single processor system. All jobs have arrived at 0 msec.

Job Id	CPU
a	4
b	1
c	7
d	2

If scheduling is round robin with time slice I msec, then what is the completion time for job 'a'?
मान लें कि निम्नलिखित कार्य एकल प्रोसेसर सिस्टम पर निष्पादित किए जाने हैं। सभी कार्य 0 msec. पर पहुंचे हैं।

जॉब आई.डी.	सी.पी.यू
a	4
b	1
c	7
d	2

यदि शेड्यूलिंग टाइम स्लाइस 1 msec. के साथ राउंड रॉबिन है, तो कार्य 'a' के लिए पूरा होने का समय कितना है?

(a) 4 (b) 10
(c) 7 (d) 11

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : Round Robin (RR) is a CPU scheduling algorithm where each process is assigned a fixed time unit called a time slice or quantum. In this problem the time slice is 1msec. The scheduler runs processes in the queue, moving to the next process once the current job's time slice expires.

192. Consider the following justifications for commonly using the two-level CPU scheduling:

I. It is used when memory is too small to hold all the ready processes

II. Because its performance is same as that of the FIFO.

III. Because it facilitates putting some set of processes into memory and a choice is made from that.

IV. Because it does not allow to adjust the set of in-core processes

Which of the following is true?

(a) I, III and IV (b) I and II
(c) III and IV (d) I and III

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Because, two level scheduling is a method to more efficiently perform process scheduling that involved swapped out process. Two level scheduling uses two different schedulers, one lower-level schedulers which can only select among those processes

Sarkari Resource

in memory to run. That scheduler could be a round Robin scheduler. The other scheduler is the higher level scheduler whose only concern is to swap in and swap out processes from memory.

193. Which of the following is the correct value returned to the operating system upon the successful completion of a program?

(a) 0
(b) 1
(c) – 1
(d) Program do not return a value.

UGC NET C.S. June-2014 (Paper-II)

Ans : (a) Because, 0 is the identifier for a program to inform the operating system (OS) about proper running of a program. Each of the non-zero value signifies unique error doing running of a program.

194. Consider a program that consists of 8 pages (from 0 to 7) and we have 4 page frames in the physical memory for the pages. The page reference string is :
1 2 3 2 5 6 3 4 6 3 7 3 1 5 3 6 3 4 2 4 3 4 5 1
The number of page faults in LRU and optimal page replacement algorithms are respectively (without including initial page faults to fill available page frames with pages) :

(a) 9 and 6 (b) 10 and 7
(c) 9 and 7 (d) 10 and 6

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Because, first we finds the LRU page replacement

1	2	3	2	5	6	3	4	6	3	7	3	1	5	3	6	3	4	2	4	3

then,

			5	5	5	5	5	7	7	7	7	7	6	6	6	6	6			
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	2	3	3	
	2	2	2	2	2	2	4	4	4	4	4	1	1	1	1	1	4	4	4	4
1	1	1	1	1	6	6	6	6	6	6	6	6	5	5	5	5	5	2	2	2
					F		F			F		F	F		F			F	F	

So page faults
$\Rightarrow$ 10, means total replacement at LRU is 10.
Now, we final the optimal page replacement of given above question.

			5	5	5	4	4	4	7	7	7	5	5	5	5	5	5			
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	
	2	2	2	2	6	6	6	6	6	6	6	6	6	6	6	4	4	4	4	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	
					F		F			F			F			F				

So total replacement at optimal page replacement is 7.
So the correct answer is (b).

195. Which of the following is an interrupt according to temporal relationship with system clock?

(a) Maskable interrupt
(b) Periodic interrupt
(c) Division by zero
(d) Synchronous interrupt

UGC NET C.S. June-2017 (Paper-III)

Ans : (d) Synchronous Interrupt : The source of interrupt is in phase to the system clock in called synchronous interrupt. In other words interrupts which are dependent on the system clock.
Example : Time service that uses the system clock.

196. Which of the following statements are true?

A. Shortest remaining time first scheduling may cause starvation
B. Preemptive scheduling may cause starvation
C. Round robin is better than FCFS in terms of response time

Choose the correct answer from the options given below :

(a) A only (b) B, C only
(c) A, B only (d) A, B, C

UGC NET C.S. December-2022

Ans. (d) : (A) Shortest Remaining Time First (SRTF) scheduling is pre-emptive version of shortest job scheduling. In SRTF, Job with the shortest CPU burst will be scheduled first. Because of this, starvation may-occur if shorter processes keep arriving and a long CPU burst process never gets the CPU.
(B) Pre-emptive just means that a process before completing execution stopped so that another process can start execution. The stopped process can later come back and continue from where it was stopped. In pre-emptive scheduling, suppose process P1 is executing in CPU and after some time process P2 with high priority then P1 is pre-empted and P2 will be brought into CPU for execution. In this way if a process which is arriving in ready queue of higher priority than P1, then P1 always be pr-empted and it may possible that it suffer from starvation.
(C) Round Robin will give better response time than FCFS. In FCFS, when a process is executing, it executed up to its complete burst time, but in Round Robin it will execute up to time quantum. So Round Robin Scheduling improves response time as all processes get CPU after a specified time.
So statements A, B and C is true.

197. Identify the circumstances under which pre-emptive CPU scheduling is used:

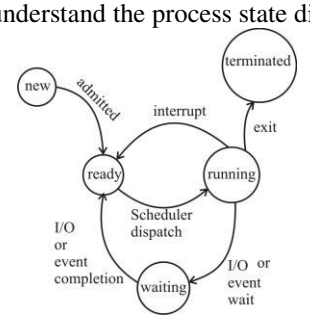
(A) A process switches from Running state to Ready state
(B) A process switches from Waiting state to Ready state
(C) A process completes its execution
(D) A process switches from Ready to Waiting state

Choose the correct option:

(a) (A) and (B) (b) (A) and (D)
(c) (C) and (D) (d) (A), (B), (C) only

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (a) : To understand pre-emptive scheduling we first need to understand the process state diagram.



Sarkari Resource

Scheduling is performed from when the process is taken from ready state on to the running state.
Hence CPU scheduling decisions take place under one of four conditions.
(1) Process switch from running state to waiting state.
(2) Process switch from running state to Ready State.
(3) Process switch from waiting state to ready state (statement b)
(4) Process termination.
Statement C- No need for preemptive scheduling as execution is complete.
Statement D- Process cannot switch from Ready state to waiting state.

198. The problem of indefinite blockage of low-priority jobs in general priority scheduling algorithm can be solved using :
'जनरल प्रयोरिटी शेड्यूलिंग अल्गोरिद्म' में ''लो – प्रायोरिटी जॉब'' की अनिश्चित ब्लाकेज की समस्या का समाधान _____की सहायता से किया जा सकता है।
(a) Swapping/स्वेपिंग
(b) Dirty bit/डर्टी बिट
(c) Aging/एजिंग
(d) Compaction/कम्पेक्शन

KVS PGT C.S. 2017

Ans. (c) : Aging is a scheduling technique used to avoid starvation. Fixed priority scheduling is a scheduling discipline, in which tasks queued for utilizing a system resource are assigned a priority each. A task with a high priority is allowed to access a specific system resource before a task with a lower priority is allowed to do the same. Aging is used to gradually increase the priority of a task, based on its waiting time in the ready queue.

199. A free list contains three memory areas of sizes 6 KB, 15 KB and 12 KB. The next three memory requests are for 10 KB, 2 KB and 14 KB. Which of the memory allocation strategy that would be able to accommodate all the three requests?
एक फ्री लिस्ट में 6 केबी, और 12 केबी आकार के तीन मेमोरी क्षेत्र हैं। अगले तीन मेमोरी अनुरोध 10 केबी, 2 केबी और 14 केबी के लिए हैं। कौन सी मेमोरी आवंटन रणनीति तीनों अनुरोधों को समायोजित करने में सक्षम होगी?
(a) Worst Fit/वर्स्ट फिट (b) Best Fit/बेस्ट फिट
(c) Next Fit/नेवस्ट फिट (d) First Fit/फर्स्ट फिट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : Best Fit Strategy– Allocates the smallest block of memory that is large enough for the requests, aiming to leave the smallest possible left over.
(i) 10 KB request– Fits into the 12 KB block (remaining 2 KB).
(ii) 2 KB request– Fits into the remaining 2 KB block (remaining 0 KB).
(iii) 14 KB request– Fits into the 15 KB block (remaining 1 KB).
Best Fit can accommodate all requests. The Best Fit strategy is the only one that can accommodate all three memory request.

200. If there are 64 segments, each of size 2KB, then how many bits should a logical address have?
यदि 64 सेगमेंट हैं, प्रत्येक का आकार 2KB होता है, लॉजिकल एड्रेस में कितने बिट्स होने चाहिए।
(a) 11 bits/ 11 बिट्स (b) 6 bits/ 6 बिट्स
(c) 5 bits/ 5 बिट्स (d) 17 bits/ 17 बिट्स

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : j We can solve this step by step.
We are asked: There are 64 segments each of 2kb. How many bits are needed for a logical address?
Step-1: Identify the fields of a logical address in segmentation
In segmented memory, a logical address is usually of the form:
Logical Address = Segment Number + Offset within segment
1. Segment Number : Selects which segment.
2. Offset : specifies the address inside that segment.
Step-2: Bits needed for segment number
• Number of segments = 64
• To represent 64 segments, we need:
Bits for segment number = $\log_2 64$ = 6bits
Step-3: Bits needed for offset
• Size of each segment = 2 KB = 2 × 1024 = 2048 bytes
• To address 2048 bytes inside a segment
• Size of each segment = 2 KB = 2 × 1024 = 2048 bytes
• To address 2048 bytes inside a segment:
Bits for offset = $\log_2$ 2048 = 11bits
Step 4: Total bits for logical address
Total bits = bits for segment number + bits for offset = 6 + 11 = 17 bits

201. When we apply Shortest Job First (SJF) scheduling Policy/algorithm, Process (P_2) faces partial starvation and waits for the longer time to execute. Which one of the following algorithms addresses this problem?
(a) Round Robin
(b) Priority
(c) Highest Response ratio Next
(d) Least Completed next

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Ans. (c) : The shortest job first (SJF) scheduling algorithm leads to partial starvation for process P_2. This occurs because P_2 has a shorter burst time (3 ms) than P_1 (SMS), but P_1 arrives earlier and is non-primitive, meaning it runs to completion before P_2 can start.
(a) Round Robin: Round Robin is a primitive algorithm that gives each process a small time slice (quantum). If a process does not finish with in its quantum, it's moved to the back of the queue. This prevents starvation as every process gets chance to run, but it does not specifically target the Issue of shorter jobs being delayed by longer ones like SJF.
(b) Priority: Priority scheduling assign a priority to each process. The highest priority processes run first. If P_2 were given a higher priority than P_1, it would solve the starvation problem in the specific instance.

(c) Highest Response Ratio Next (HRRN): HRRN is a scheduling algorithm that calculates a Response a ratio for each process.

• Response Ratio = (Waiting Time + Burst Time)/Burst Time

The process with the highest response ratio selected for execution.

(d) **Least Completed Next:-** This is not a standard Scheduling algorithm name. It might imply a policy where the process that is closest to completion is run next. However, this is essentially a variation of SJF and does not inherently solve the starvation problem.

Therefore, the correct answer is c, that is Highest Response Ratio Next.

202. What is the average turnaround time for these Processes with First Come First Serve (FCFS) scheduling Algorithms?

(a) 5.3 ms (b) 6.4 ms
(c) 7.0 ms (d) 8.2 ms

UGC NET C.S. December-2024

Ans. (c) : These processes with first come first serve (FCFS).

Understanding FCFS:- FCFS is an non-primitive scheduling algorithm This means that process are executed in the order they arrive, and once a process starts, it runs to completion before another process can be executed.

Given data:-

Process	Arrival	Burst time
P_1	0	5
P_2	2	3
P_3	2	2
P_4	5	3
P_5	6	1

Execution order (based on arrival time)

1. P_1 (arrival at 0)
2. P_2 (arrival at 2)
3. P_3 (arrival at 2)-Since P_2 and P_3
4. P_4 (arrival at 5)
5. P_5 (arrival at 6)

Calculating completion and turnaround Time:-

Process	Arrival time	Burst time	Completion time	Turnaround time (completion time arrival time)
P_1	0	5	5	5 – 0
P_2	2	3	8	8 – 2 = 6
P_3	2	2	10	10 – 2 = 8
P_4	5	3	13	13 – 5 = 8
P_5	6	1	14	14 – 6 = 8

Calculating Average Turnaround Time:-

Sum of turnaround Time/Number of processes
= (5 + 6 + 8 + 8 + 8)/5 = 35/5 = 7ms

203. Which one of the following is not a condition for process state transition from running state to wait state?
निम्नलिखित में से कौन-सी एक रनिंग स्टेट से वेटिंग स्टेट में प्रोसेस स्टेट ट्रांजिशन की शर्त नहीं है?

(a) If a process request an Input/Output operation/यदि कोई प्रोसेस इनपुट/आउटपुट ऑपरेशन का अनुरोध करती है
(b) If a higher priority process arrived
यदि उच्च प्राथमिकता वाला प्रॉसेस आ गया
(c) If a lower priority process arrived
यदि कम प्राथमिकता वाला प्रॉसेस आ गया
(d) If a process waits for some action performed by another process/यदि कोई प्रॉसेस किसी अन्य प्रॉसेस द्वारा किसी क्रिया को करने की प्रतीक्षा करता है

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The condition that is not for a process state transition from the running state to the wait state is it lower priority process arrived. A process typically transitions to a wait state if it is request an I/O operation, waits for some action performed by another process, or is preempted by a higher priority process. Arrival of a lower priority process does not cause the currently running process to enter a wait state.

204. Consider three processes P1, P2, P3, which require 24, 3, and 3 units of CPU time respectively. For time quantum of 4 units and Round - Robin scheduling policy, the sequence in which the processes be executed for each time quantum is _____.
तीन प्रोसेसों- P1, P2, P3 पर विचार कीजिए जिन्हें क्रमशः सीपीयू की 24, 3, और 3 यूनिटों की आवश्यकता है। 4 यूनिटों की समय मात्रा तथा राउंड रॉबिन शैड्यूलिंग नीति के लिए प्रत्येक समय मात्रा हेतु प्रोसेस किस क्रम में निष्पादित होंगे:

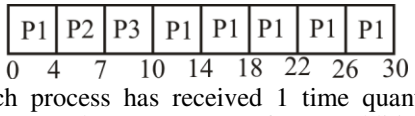
(a) P1, P2, P3, P1, P1, P1, P1, P1
(b) P1, P2, P3, P1, P1, P1, P1
(c) P1, P2, P3, P1, P1, P1
(d) P2, P3, P1, P1, P1

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (a) :

(Process)	Burst time
P1	24
P2	3
P3	3

If we use a time quatum of a 4 millisecond then process P1 gets the first 4 millisecond the resulting RR schedule is as follows.



once each process has received 1 time quantum, the CPU is returned to process P1 for an additional time quantum.

205. In which of the following state, the process is waiting for processor ?
निम्नलिखित में से किस दशा में 'प्रोसेस' 'प्रोसेसर' के लिए 'वेट' कर रहा होता है :

(a) Running/रनिंग (b) New/न्यू
(c) Ready/रेडी (d) Waiting/बेटिंग

KVS PGT C.S. 2017

Sarkari Resource

Ans. (c) : The state of a process is defined in part by the current activity of that process. process may be in one of the following states.
- New
- Running
- Waiting
- Ready
- Terminated.

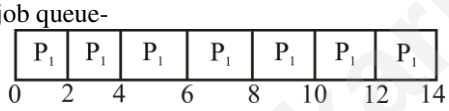
In Ready state, the process is waiting to be assigned to a processor.

206. If the time quantum size is 2 units of an there is only one job of 14 time unit in a ready queue, the round-robin scheduling algorithm will cause_____ context switches.
यदि समय क्वांटम आकार समय की 2 यूनिटें है और किसी रेडी क्यू में 14 समय यूनिटों का केवल एक 'जॉब' है, राउण्ड-रॉबिन शिड्यूलिंग एल्गोरिदम _____ कॉन्टेक्स्ट स्विचों को प्रेरित करेगा।

(a) 8 (b) 5
(c) 6 (d) 7

KVS PGT C.S. 2017

Ans. (c) : Round-Robin is a CPU scheduling algorithm where each process is cyclically assigned a fixed time slot. It is the preemptive version of First come first serve (FCFS) CPU scheduling algorithm. Round-Robin CPU Algorithm generally focuses on Time sharing technique.
Given time quantum = 2 units
There is only one job that requires 14 time units to execute completely.
So, the job queue-



context switches at 2, 4, 6, 8, 10, 12,
In this, there is a total of 6 context switches, as first and last will not be the context switch.

207. Let RR3 means Round Robin Scheduling algorithm with time quantum of 3 units and FCFS means First Come First Serve [FCFS] scheduling algorithm. Now if 3 processes P1, P2, P3 has arrival time in order is (0, 2, 4) and execution time is (8, 5, 4) respectively then the completion order of P1, P2, P3 under RR3 and FCFS scheduling is :
माना RR3 का तात्पर्य समय क्वांटम 3 वाली राउंड रॉबिन सेडयूलिंग एल्गोरिदम तथा FCFS का तात्पर्य पहले आओ पहले जाओ (First Come First Serve) सेडयूलिंग एल्गोरिदम है। अब यदि 3 प्रोसेसों P1, P2, P3 का अराइवल टाइम (आगमन समय) क्रमानुसार (0, 2, 4) तथा एक्सिक्यूशन टाइम (निष्पादन समय) क्रमशः (8, 5, 4) है, तब P1, P2, P3 का कम्प्लीशन आर्डर (पूर्णता कम) RR3 तथा FCFS सेडयूलिंग के अंतर्गत होगा:

(a) RR3 : P2, P1, P3 and FCFS : P3, P2, P1
(b) RR3 : P3, P2, P1 and FCFS : P1, P2, P3
(c) RR3 : P2, P1, P3 and FCFS : P1, P2, P3
(d) RR3 : P1, P2, P3 and FCFS : P3, P2, P1

HTET PGT C.S. 2018

Ans. (c) : We have 3 proscesses :
- Arrivial time : $P_1 = 0$, $P_2 = 2$, $P_3 = 4$
- Burst time : $P_1 = 8$, $P_2 = 5$, $P_3 = 4$
- RR quantum = 3

1. FCFS (first come first serve)
FCFS runs the process that arrives first.
- P_1 arrivess at 0 → runs for 8 → fineshes at 8.
- P_2 arrives at 2 → wait for P_1 → run for 5 → finishes 13
- P_3 arrives at 4 → wait for P_1 and P_2 → runs for 4 → finishes at 17

FCFS completion order : P_1, P_2, P_3
(2) RR3 (Round Robin with quantum = 3
Round Robin runs each process for 3 unite at a time, then moves it to the back of the queue if it's not finished.
Step by step :
1. 0 – 3 : P_1 runs 3 → remaining 5
2. 3 – 6 : P_2 has arrived → runs 3 → remaining 2
3. 6 – 9 : P_1 runs 3 → remaining 2
4. 9 – 12 : P_3 has arrived → runs 3 → remaining 1
5. 12 – 14 : P_2 runs remaining 2 → finishes.
6. 14 – 16 : P_1 runs remaining 2 → finishes
7. 16 – 17 : P_3 runs remaing 1 → finishes

RR3 completion order : P_2, P_1, P_3
RR3 : P_2, P_1, P_3 and FCFS : P_1, P_2, P_3

208. Priority Scheduling involves assigning a priority value to each process. Which scheduling algorithm is closely related to this concept?
प्रायोरिटी शेड्यूलिंग में प्रत्येक प्रोसेस को एक प्राथमिकता मान सौंपा जाता है। यह अवधारणा किस शेड्यूलिंग एल्गोरिदम से निकटता से संबंधित है?

(a) FCFS/एफसीएफएस (b) SJN/एसजेएन
(c) Multilevel Queue Scheduling
मल्टीलेवल क्यू शेड्यूलिंग
(d) First-Come, First-Served (FCFS)
फर्स्ट-कम, फर्स्ट-सर्व्ड (FCFS)

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : Priority scheduling implies that each process is assigned a priority, and the CPU is allocated to processes according to that priority.
In multilevel queue scheduling processes are divided into queues of different priorities and each queue has different priority. Therefore, this algorithm is called priority.

209. In processor P, non preemptive priority interrupts 4, 7, 1, 3, 0, 2, 6 arrive sequentially (low number high priority). If when one interrupt is being handled exactly two more interrupts arrive. What is order of these interrupts handled by P:
एक प्रोसेसर P में नॉन प्रिएम्पटिव प्रायोरिटी इंटरप्ट 4, 7, 1, 3, 0, 2, 6 क्रम से आते है (छोटा नंबर बड़ी प्रायोरिटी) यदि एक इंटरप्ट की हैंडलिंग के दौरान ठीक दो इंटरप्ट और आते है तो प्रोसेसर P द्वारा इन इंटरप्ट को हैंडल करने का क्रम होगा:

(a) 4, 7, 1, 3, 0, 2, 6 (b) 4, 1, 3, 0, 2, 6, 7
(c) 0, 1, 2, 3, 4, 6, 7 (d) 4, 1, 0, 2, 3, 6, 7

RPSC C.S. 2016 (Paper-I)

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Ans. (d) :
• Interrupts arrive in this sequence 4, 7, 1, 3, 0, 2, 6.
• Processor P uses non-preemptive priority interrupts
• Lower number = higher priority
• Once an interrupt starts, it runs to completion
• Exactly two new interrupts arrive while one is being handled.
Step-by-Step solution–
Step-1– First interrupt arrives – 4
• No other interrupts pending
• Processor start handling 4
while handling 4, two more arrive: 7, 1
→ Pending queue [7, 1]
Step-2– After 4 completes → choose from 7, 1
• priority 1 < 7 → choose 1
while handling 1, next two arrive : 3, 0
→ Pending [7, 30]
Step-3– After 1 completes → choose from 7, 3, 0
• Priority : 1 < 3 < 7 → choose 0
while handling 0, next two arrives : 2, 6
→ Pending [7, 3, 2, 6]
Step-4– After 0 completes → choose from 7, 3, 2, 6
• Priority : 2 < 3 < 6 < 7 → choose 2
No more interrupts arrive now (all 7 have come)
Pending : [7, 3, 6]
Step-5– After 2 → choose 3
Step-6– After 3 → choose 6
Step-7– After 6 → choose 7
Final order of execution–
4 → 1 → 0 → 2 → 3 → 6 → 7

210. A process residing in Main Memory and Ready and Waiting for execution, is kept on
(a) Execution Queue (b) Job Queue
(c) Wait Queue (d) Ready Queue

UGC NET C.S. December-2018 (Paper-II)

Ans. (d) : The processes that are residing in main memory and are ready and waiting to execute are kept on a list called the **ready queue**.

211. For the following set of processes scheduled using FCFS policy, determine the average waiting time. Assume that the process arrived in the order P1, P2, P3, P4.

Process	Burst Time (ms)
P1	8
P2	15
P3	10
P4	7

(a) 8 (b) 16
(c) 32 (d) 48

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Ans. (b) : Given that
the process arrived in the order P1, P2, P3, P4,

Process	Burst Time	C.T.	TAT	Waiting Time
			CT-AT	
P_1	8	8	8	0
P_2	15	23	23	8
P_3	10	33	33	23
P_4	7	40	40	33

Ready Queue $P_1\ P_2\ P_3\ P_4$

CPU | P_1 | P_2 | P_3 | P_4 |
0 8 23 33 40

CT = Completion Time
TAT = Turn Around Time
AT = Arrival Time

$$\text{Average waiting time} = \frac{0+8+23+33}{4}$$

$$= \frac{64}{4}$$

Average waiting time = 16

212. Consider the following statements :
निम्नलिखित कथनों पर विचार कीजिए :

S1 : a small page size causes large page tables
छोटा पृष्ठ आकार से बड़ी पृष्ठ तालिकाएँ बनती है।

S2 : internal fragmentation is increased with small pages.
छोटे पृष्ठों से आंतरिक विखण्डन में वृद्धि होती है।

S3 : I/O transfers are more efficient with large pages./बड़े पृष्ठों से I/O अंतरण और अधिक कुशल हो जाता है।

Which of the following is true?
निम्नलिखित में से कौन सा सही है?
(a) S1 is true and S3 is false
S1 सही है और S3 गलत है
(b) S1 and S2 are true/S1 और S2 सही हैं
(c) S2 and S3 are true/S2 और S3 सही हैं
(d) S1 is true and S2 is false
S1 सही है और S2 ग़लत है

KVS PGT C.S. 2017

Ans. (a) : Smaller page size means more pages required per process. It means large page tables are needed. Hence, the statement (S1) is correct.
Internal fragmentation means when process size is smaller than the available space. when pages are small, then available space becomes less and there will be less chances of internal fragmentation. Hence, the statement (S2) is incorrect.
An I/O system is required to take an application I/O requests and send it to the physical device transfering of I/O requests are more efficient with large pages. Hence the statement (S3) is correct.

213. A computer has 32 MB memory. How many bits are needed to access any single byte in the memory?
32 MB मेमोरी के एक कम्प्यूटर में एक सिंगल बाइट को मेमोरी में एक्सेस करने के लिए कुल कितने बिट की आवश्यकता होगी?
(a) 25 (b) 16
(c) 24 (d) 20

LT. Grade C.S. 2018

Ans. (a) : To determines the number of bits needed to access any single byte in a memory, calculate the total number of address and then find the number of bits required to represent those address-
1 MB = 2^{20} bytes
1 KB = 1024 bytes

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32 MB = 32×2^{20} bytes
= $2^5 \times 2^{20}$ bytes
= 2^{25} bytes
Since each byte can be accessed uniquely, we need 25 bits to address each byte in the 32 MB memory.

214. A computer has 128MB memory. Each word in this computer is of 8 bytes. How many bits are required to address any single word in the memory?
एक कम्प्यूटर में 128 MB की मेमोरी है और प्रत्येक वर्ड का साइज 8 बाइट है। किसी भी सिंगल वर्ड को मेमोरी में ऐड्रेस करने के लिए कितने बिट की आवश्यकता होगी?
(a) 27 (b) 24
(c) 23 (d) 25

LT. Grade C.S. 2018

Ans. (b) : To determine the number of bits needed to address a single word in memory. We need to calculate the total number of words in the memory.
1 MB = 2^{20} bytes
128 MB = 128×2^{20} bytes
= $2^7 \times 2^{20}$ bytes
= 2^{27} bytes
Each word is 8 bytes, the total number of words is
= 2^{27} bytes/8 bytes/ word
= 2^{27} bytes/2^3 bytes
= 2^{24} words
Since each word can be accessed uniquely
We need 24 bits to address each word in the 128 MB memory.

215. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track, 512 bytes of data are stored in a bit in serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively
16 सर्फ़ेस, प्रति सर्फेस 128 ट्रैक और प्रति ट्रैक 256 सेक्टर के एक बिट में 512 बाइट की डेटा क्रमानुसार संग्रहित है। डिस्क पैक की क्षमता और डिस्क में सेक्टर को निर्दिष्ट करने के लिए आवश्यक बिट की संख्या हैं, क्रमशः
(a) 256 MB, 19 bits/256 MB, 19 बिट
(b) 256 MB, 28 bits/256 MB, 28 बिट
(c) 512 MB, 20 bits/512 MB, 20 बिट
(d) 64 GB, 28 bits/64 GB, 28 बिट

LT. Grade C.S. 2018

Ans. (a) : Capacity of the disk = $16 \times 128 \times 256 \times 512$ bytes/sector
= 268,435,456 bytes
= 268.435 MB (approximately)
= 256 MB (rounded to nearest power of 2)
To calculate number of bits required to access a sector, we need to know total number of sectors. Total number of sectors = $16 \times 128 \times 256 = 2^{19}$.
So the number of bits required to access a sector is 19.

216. If the virtual memory size is 32MB and a physical memory size is 4MB with a page size of 2KB, Calculate the number of frames available in physical memory
(a) 1024 (b) 2048
(c) 3072 (d) 4096

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Ans. (b) : To calculate the number of frames available in physical memory, use the formula.

$$\text{Number of frames} = \frac{\text{Physical memory size}}{\text{page size}}$$

Given –
Physical memory size = 4MB
4×1024 = 4096 KB
Page size = 2kb

$$\text{Number of frames} = \frac{4096}{2} = 2048$$

217. How many address lines and input-output data are needed for the 64k × 8 memory unit?
64k × 8 मेमोरी यूनिट के लिए कितनी एड्रेस लाइन और इनपुट-आउटपुट डेटा लाइन की आवश्यकता होती है?
(a) 16 address line, 3 data line
16 एड्रेस लाइन, 3 डेटा लाइन
(b) 6 address line, 3 data line
6 एड्रेस लाइन, 3 डेटा लाइन
(c) 16 address line, 8 data line
16 एड्रेस लाइन, 8 डेटा लाइन
(d) 10 address line, 8 data line
10 एड्रेस लाइन, 8 डेटा लाइन

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : 64k × 8 memory unit
• 64k × 8 means 64k × 1024 = 65536 memory location.
• Each location holds 8 bits (which is on byte)
Address line–
• Each address line can represent two states (0 or1)
• To address 65,536 locations, you need enough lines such that.
$2^n = 65,536$
Taking $\log_2$ on both sides
$n = \log_2 (65,536) = 16$
is address line
Input-output (Data) lines
• Each location holds 8 bits.
• So, you need 8 data lines for input and output.

218. How many address lines and input–output data lines are needed in 2k× 16 memory unit?
2k× 16 मेमोरी यूनिट में कितनी एड्रेस लाइन और इनपुट-आउटपुट डेटा लाइन की आवश्यकता होती है?
(a) 10 address line, 16 data line
10 एड्रैस लाइन, 16 डेटा लाइन
(b) 10 address line, 4 data line
10 एड्रैस लाइन, 4 डेटा लाइन
(c) 11 address line, 4 data line
11 एड्रैस लाइन, 4 डेटा लाइन
(d) 11 address line, 16 data line
11 एड्रैस लाइन, 16 डेटा लाइन

DSSSB TGT C.S. 07.08.2021 (Shift-II)

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Ans. (d) : A 2k × 16 memory unit has 2048 addresses (2k) and each address can store 16bits of data.

1. **Address Lines:** To address 2048 Locations, you need Log_2 (2048) = 11 address Lines.

Data Lines: Since each memory Location contain 16 bits. You need 16 data lines.

219. Which of the following is a primary task of memory management system?
I. It keeps track of which parts of the memory are currently being used
II. When space is available, it allocates memory to the programs that are next to be loaded.
निम्नलिखित में से कौन सा मेमोरी प्रबंध प्रणाली का प्राथमिक कार्य है?
I. यह ट्रैक करता है कि वर्तमान में मेमोरी के किन हिस्सों का उपयोग किया जा रहा है
II. जब स्पेस उपलब्ध होता है, तो यह उन प्रोग्रामों को मेमोरी आवंटित करता है जो लोड होने वाले हैं

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Both of the statements describe primary and fundamental tasks of a memory management system.
Statement 1. A memory management system must keep track of the status of all memory location. By the operating system,
Statement 2. After tracking the available memory, the system must then allocate that space to new programs or processes that are about to be loaded into the main memory this involves deciding memory block to give to the new program, a process that can be handled using various strategies live first-fit, best-fit, or worst fit algorithms.

220. Memory buffer which stores the image definition is known as _____.
मेमोरी बफर जो इमेज डेफिनिशन को स्टोर करती है उसे _____ के रूप में जाना जाता है।

(a) Pixel Buffer/पिक्सेल बफर
(b) Frame Buffer/फ्रेम बफर
(c) Graphic Buffer/ग्राफिक बफर
(d) Picture Buffer/पिक्चर बफर

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : Memory buffer which stores the image definition is known as Frame Buffers.
The frame buffer holds the pixel data for an image and is used to store the information that is to be displayed on a screen.

221. A specific editor has 200 K of program text, 15 K of initial stack, 50 K of initialized data, and 70 K of bootstrap code. If five editors are started simultaneously, how much physical memory is needed if shared text is used?

(a) 1135 K (b) 335 K
(c) 1065 K (d) 320 K

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because, initial stack, initialized data and bootstrap code can be shared among the editors.
Because text is shared so no need to multiply 200 with 5 and other need each for running. Then the total memory is = 200 + 15 +50 + 70 = 335K

222. What is the most appropriate function of Memory management Unit (MMU)?

(a) It is an associative memory to store TLB
(b) It is a technique of supporting multiprogramming by creating dynamic partitions
(c) It is a chip to map virtual address to physical address
(d) It is an algorithm to allocate and deallocate main memory to a process

UGC NET C.S. June-2015 (Paper-III)

Ans : (c) Memory management unit (MMU), sometimes called paged memory management unit (PMMU), is a computer hardware unit having all memory references passing through itself, primarily performing the translation of virtual memory addresses to physical addresses. It is usually implemented as part of the central processing unit (CPU), but it also can be in the form of a separate integrated circuit.
So we can say it is a chip to map virtual address to physical address

223. A dynamic RAM is refresh cycle of 32 times per msec. Each refresh operation requires 100 nsec and a memory cycle requires 250 nsec. What percentage of memory's total operating time is required for refreshes?

(a) 0.64 (b) 0.96
(c) 2.00 (d) 0.32

UGC NET C.S. December-2015 (Paper-III)

Ans : (d) Given that in 1 msec (10^6 nsec) no. of refresh cycle = 32 cycle then we can find cycle in 250 nsec (which is 1 memory cycle no. of refresh cycle = (32 * 250) 10^6 cycle.
Now we can find total time taken by refresh operation in 1 memory cycle.
1 cycle of refresh operation required 100 nsec
(32 * 250)/10^6 cycle of refresh operation required = ((32 * 250)/(10^6)) * 100 = 0.8 nsec
Now find percentage of memory total operating time is required for refreshes (100/250)* 0.8 = 0.32%

224. Consider a virtual page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Suppose a demand paged virtual memory system running on a computer system such that the main memory has 3 page frames. Then..... page replacement algorithm has minimum number of page faults.

(a) FIFO (b) LIFO
(c) LRU (d) Optimal

UGC NET C.S. November-2017 (Paper-III)

Ans : (d) In optimal algorithm 9 page fault will occur

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
7	07	107	102	102	302	302	342	342	342	302	302	302	102	102	102	102	107	107	107
F	F	F	F		F		F			F			F			F			

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and if FIFO algorithm is fault will occur & in LIFO 11 page fault will occur & in LRU 12 page fault will occur. So the minimum page fault will occur in optimal algorithm.

225. Which of the following is incorrect for virtual memory?

(a) Large programs can be written
(b) More I/O is required
(c) More addressable memory available
(d) Faster and easy swapping of process

UGC NET C.S. June-2017 (Paper-III)

Ans : (b) Virtual Memory : Virtual memory is a memory management technique implemented using both software and hardware. It maps memory address used by program called virtual addresses into physical addresses in computer memory.
It provides more addressable memory, program of larger size than physical memory can be run on same computer but I/O cost is not increases.

226. Match List-I with List-II.

List-I (Operating system concepts)		List-II (Characteristics)	
A.	**Paging**	**I.**	**Evicts least recently used process**
B.	**LRU (Least Recently Used)**	**II.**	**Extends physical memory**
C.	**C-SCAN**	**III.**	**Logical to physical mapping**
D.	**Virtual Memory**	**IV.**	**Circular disk access**

Choose the correct answer from the options given below :

(a) A-III, B-IV, C-I, D-II
(b) A-III, B-I, C-IV, D-II
(c) A-I, B-III, C-IV, D-II
(d) A-I, B-IV, C-III, D-II

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : The correct matching is–

List-I (Operating system concepts)			List-II (Characteristics)
A.	Paging	–	Logical to physical mapping
B.	LRU (Least Recently Used)	–	Evicts least recently used process
C.	C-SCAN	–	Circular disk access
D.	Virtual Memory	–	Extends physical memory

227. Which of the statement is not true in regards to virtual memory?

(a) The main objective for using virtual memory is to increase the effective capacity of the memory system
(b) Size of the program can be as large as the size of the secondary memory
(c) Program and data are stored in the secondary memory
(d) None of these

UGC NET C.S. December-2022

Ans. (d) : Virtual Memory– Virtual memory is a storage allocation scheme in which secondary memory can be addressed as though it were part of the main memory. The addresses a program may use to reference memory are distinguished from the addresses the memory system uses to identify physical storage sites and the program-generated addresses are translated automatically to the corresponding machine addresses.
So all statement is correct.
So option(d) is correct.

228. The memory size for n address lines and m data lines is given by

(a) $2^m \times n$ (b) $m \times n^2$ (c) $2^n \times m$ (d) $n \times m^2$

UGC NET C.S. December-2022

Ans. (c) : We know that
ROM memory size = $2^n \times m$.
n = Number of address lines.
m = Number of data lines.

229. Dirty bit is used to show the

(a) Page with low frequency occurrence
(b) Wrong page
(c) Page that is modified after being loaded into cache memory
(d) Page with corrupted data

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : A dirty bit or modified bit is a bit that is associated with a block of computer memory and indicates whether the block of memory has been modified. The dirty bit is set when the processor writes to (modifies) this memory. Dirty bits are used by the CPU cache and in the page replacement algorithms of an operating system.

230. Let a computer has 512 KB of main memory having one byte and per word. If all the words of memory are operational then there are minimum____ bits in_____.
माना एक कंप्यूटर में मेन मेमोरी 512KB है तथा प्रत्येक वर्ड 1 बाइट का है। यदि मेमोरी के सभी वर्ड क्रियाशील है तो कंप्यूटर में न्यूनतम _____बिट की_____ है।

(a) 18, Data bus/18, डाटा बस
(b) 19, Data bus/19, डाटा बस
(c) 18, Address bus/18, एड्रेस बस
(d) 19, Address bus/19, एड्रेस बस

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : • The computer has 512 KB of memory.
• 1 KB = 1024 Bytes.
• So, 512 KB = 512 × 1024 = 524, 288 byte
since one byte per word is used, there are 524, 288 words.
Now you are asking about minimum byte required I-assume you mean the address lines needed to uniquely address each word Number of unique addresses = $2^{\text{address bits}}$ number of

$$2^{\text{address bits}} \geq 524, 288$$

Number of address lines = $\log_2$ (Number of words)

$$\log_2 (524, 288) = \log_2 (2^{19}) = 19$$

⇒ 19 address

Sarkari Resource

231. What is compaction?
संघनन क्या है?

(a) A technique for overcoming internal fragmentation
(b) A paging technique
(c) A technique for overcoming external fragmentation
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : Compaction is a memory management technique used to overcome external fragmentation.
Compaction work by shifting allocated memory blocks to one end of memory, creating a large contiguous free block.
It is not related to paging. paging deals with dividing memory into fixed-size page, which avoids external fragmentation,
compaction = solution for external fragmentation

232. In virtual memory paging, page size and frame size have which of the following relations?
वर्चुअल मेमोरी पेजिंग में पेज साइज व फ्रेम साइज में क्या सम्बन्ध होता है?

(a) Both are of equal size/दोनों समान माप की होती हैं
(b) Both are of unequal size
दोनों असमान माप की होती हैं
(c) The page size is exactly half of the frame size/पेज साइज, फ्रेम साइज की ठीक आधी होती है
(d) The page size is twice the frame size
पेज साइज, फ्रेम साइज की दोगुनी होती है

LT. Grade C.S. 2018

Ans. (a) : In virtual memory paging, the page size and frame size are equal- In virtual memory paging, each page is stored in a frame of the main memory. The size of a frame is the same as the size of a page for example- if a page size is 4096 bytes, then each frame will also hold 4096 bytes.

233. Sequential, direct, random and associative are access methods and key characteristics of computer ___ system.
क्रमिक, प्रत्यक्ष, यादृच्छिक और साहचर्य, कम्प्यूटर ___ सिस्टम की अभिगम विधियाँ और मुख्य लाक्षणिकताएँ है ।

(a) stack/स्टेक (b) counter/काउण्टर
(c) memory/मेमोरी (d) core/कोर

KVS PGT C.S. 2013

Ans. (c) : These are our types of memory access methods.
1. Sequential 2. Direct
3. Random 4. Associative
Sequential Access – memory accessed in a specific linear sequential manner.
Direct – Direct access is accomplished by direct access to reach a general vicinity plus sequential searching , counting or waiting to reach the find destination.
Random – Any location of the memory can be accessed read only like accessing in Array.
Associate Access – This access method is a special type of random access method.

234. _____ are used to access data on secondary, sequential-access stores, such as disks and tapes.
_____का उपयोग डिस्क और टेप जैसे द्वितीयक, अनुक्रमिक अभिगम भण्डार में डाटा अभिगम में होता है ।

(a) Sequences/अनुक्रम (b) Arrays/ऐरे
(c) Records/रिकार्ड्स (d) Registers/रजिस्टर्स

KVS PGT C.S. 2013

Ans. (a) : Sequential access memory is a class of data storage devices that read stored data in a sequence.
Sequential access devices are usually a form of magnetic storage or optical storage.

235. The mechanism that brings a page into memory only when it is needed is called _____.
वह पद्धति जो किसी पृष्ठ को स्मृति में केवल तभी लाती है जब उसकी आवश्यकता होती है, जिसे ____कहा जाता है।

(a) Page replacement/पेज रिप्लेसमेंट
(b) Segmentation/सेगमेन्टशन
(c) Fragmentation/फ्रेगमेन्टेशन
(d) Demand paging/डिमांड पेजिंग

KVS PGT C.S. 2017

Ans. (d) : The mechanism that brings a page into memory only when it is needed is called Demand paging. it allows the execution of programs larger than the size of physical memory.

236. Which of the following techniques allows execution of programs larger than the size of physical memory?
निम्नलिखित में से किस तकनीक में 'प्रोग्राम का निष्पादन ''भौतिक स्मृति'' के आकार से बृहतर हो सकता है?

(a) Thrashing/थ्रेशिंग
(b) DMA/डी एम ए
(c) Buffering/बफरिंग
(d) Demand paging/डिमांड पेजिंग

KVS PGT C.S. 2017

Ans. (d) : Demand paging is identical to the paging system with swapping. Demand paging allows execution of programs larger than the size of physical memory. It is combines the feature of simple paging and implement virtual memory as it has a large virtual memory. Lazy swapper concept is implemented in demand paging in which a page is not swapped into the memory unless it is required.

237. Copying a process from memory to disk to allow space for other processes is called _____.
अन्य प्रोसेसों को स्पेस प्रदान करने के लिए किसी प्रोसेस को मेमोरी से डिस्क में कॉपी करने को कहा जाता है_____।

(a) Demand paging/डिमांड पेजिंग
(b) Deadlock/डेडलॉक
(c) Pagefault/पेजफाल्ट
(d) Swapping/स्वैपिंग

KVS PGT C.S. 2017

Sarkari Resource

Ans. (d) : Swapping is mechanism in which a process can be swapped temporarily out of main memory (or move) to secondary storage (disk) and make that memory available to other processes. At some later time, the system swaps back the process from the secondary storage to main memory.

238. Magnetic disks are the most popular medium for
चुम्बकीय डिस्क सबसे लोकप्रिय माध्यम है
(a) Direct access
(b) Sequential access
(c) Both of these/इनमें से दोनों
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2020

Ans. (c) : Sequential Access→ Sequential access means that data is accessed in a fixed order, one after another
Direct Access → Direct Access means accessing data directly form a storage location. It is clear that magnetic disks are the most popular medium in both sequential accesses and direct accesses.

239. A storage area used to store data to compensate for the difference in speed at which the different units can handle data is
एक Storage area जिसे डेटा को उस गति के अंतर की भरपाई करने के लिए उपयोग किया जाता है जिस पर विभिन्न इकाइयाँ डेटा को संभाल सकती हैं
(a) Memory (b) Buffer
(c) Accumulator (d) Address

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : In computer science, a buffer is a temporary storage area used to hold data while It is being transferred from one place to another. It help smooth out differences in speed between devices or processes that are sending and receiving data.

240. The purpose of a page table in a paging system is to:/पेजिंग सिस्टम में पेज टेबल का उद्देश्य है:
(a) Store pages of memory
मेमोरी के पेजेस को स्टोर करना
(b) Manage input/output operations
इनपुट/आउटपुट ऑपरेशन्स का प्रबंधन करना
(c) Translate virtual addresses to physical addresses/ वर्चुअल एड्रेस को फिजिकल एड्रेस में ट्रांसलेट करना
(d) Store program instructions
प्रोग्राम इंस्ट्रक्शन्स को स्टोर करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The main purpose of a page table is to convert the virtual address generated by the CPU into the corresponding physical address in main memory (RAM).
• It works with the memory management unit.

241. In multiprogramming with fixed partitions, if a process requires more memory than is a available in a partition, it may lead to:
फिक्स्ड पार्टिशन के साथ मल्टीप्रोग्रामिंग में, यदि कोई प्रोसेस किसी पार्टिशन से अधिक मेमोरी मांगता है, तो यह किस समस्या को जन्म दे सकता है?
(a) Fragmentation/फ्रैगमेंटेशन
(b) Deadlock/डेडलॉक
(c) Priority inversion/प्रायोरिटी इनवर्जन
(d) Starvation/स्टारवेशन

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : Fragmentation in computer science means the inefficient use of memory that happens when free memory is divided into small scattered blocks, making it difficult to allocate large contiguous memory spaces to processes.

242. If the size of memory of 4K words and each word is of 8 bits, what is the size of address bus?
यदि मेमोरी की साइज 4K वर्ड्स है तथा प्रत्येक वर्ड 8 बिट का है तो एड्रेस बस की साइज क्या है?
(a) 256 (b) 12
(c) 3 (d) 32

UKPSC PGT C.S. Asstt. Prof. 2017

Ans(b) : 4K words → $4 \times 1024 = 4096$ words. These address bus requires enough lines to uniquely address each word
So $\log_2 (4096) = 12$

243. In memory a word is retrieved based on a portion of its content rather than its address. This memory is known as
मेमोरी में पता के बजाय उसके सामग्री के हिस्से के आधार पर शब्द पुनर्प्राप्त किया जाता है, यह मेमोरी _____ कहलाती है।
(a) Associative memory /एसोसिएटीव मेमोरी
(b) Retrieval memory /रिट्रीवल मेमोरी
(c) Read only memory /रीड ओनली मेमोरी
(d) None of the above /उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : Associative memory is known as CAM (Content-Addressable Memory), allows data retrieval by comparing content rather than using addresses.

244. Consider the following statements:
(i) The time required to position read/write head on appropriate track or cylinder of a hard disk is known as rotational latency.
(ii) The time required to position read/write head on appropriate track or cylinder on a hard disk is known as seek time.
(iii) A hard disk is a magnetic disk consisting of several platters.
(iv) A CD-ROM magnetizes spots on its surface in order to record information.
Which of the statements given above are correct?
(a) (i) and (ii) only (b) (ii) and (iii) only
(c) (iii) only (d) (i) and (iv) only.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : (i) The time required to position read/write head-appropriate track or cylinder of a hard disk is known as rotational latency. This statement is incorrect. rotational latency is the time it takes for the desired sector of the hard disk platter to rotate into position under the read/write head.

(ii) Time required to position read/write head on the appropriate track or cylinder on a hard disk is known as seek time this statement in correct.
(iii) A hard disk is a magnetic disk. Consisting of several platters. This statement is correct. Hard drives used rigid, spinning platters coated with a magnetic material to store data.
(iv) A CD-ROM magnetizes spots on its surface in order to record information. This statement is incorrect. A CD-ROM uses a laser to bum microscopic pits into a polycarbonate surface

245. How many 32 × 1 RAM chips are needed to provide a memory capacity of 256 kbytes?
(a) 8 (b) 32
(c) 64 (d) 128

CGPSC C.S. 2020 Set-A

Ans. (c) : You would need 64 32×1 RAM chips to provide a memory capacity of 256 kilobytes.
256 k bytes 256×1024 bytes {1 byte →8 bits}
Then, 256×1024×8 bits.
Each chip has capacity of 32 k × 1 bit
32 k bites = 32 × 1024.

Then $\frac{256\times1024\times8}{32\times1024}=64$

246. The average time required to search a storage location in memory and obtain its contents is called.
मेमोरी में किसी संग्रहण स्थान को खोजने और उसकी सामग्री प्राप्त करने में लगने वाले औसत समय को कहते हैं।
(a) Access time/पहुँच समय
(b) Latency time/विलंब समय
(c) Response time/प्रतिक्रिया समय
(d) Reading time/पठन समय

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Ans. (a) : The average time required to search a storage location in memory and obtain its contents is called Access time. Access time is the interval between the beginning of one storage device access and the beginning of the subsequent access.
Access time = Latency time + Transfer time.

247. Which term refers to the strategy of optimizing an algorithm's use of resources, often by making sacrifices in one aspect for improvements in another?
कौन-सा शब्द उस रणनीति को संदर्भित करता है जिसमें एल्गोरिदम के संसाधनों के उपयोग का अनुकूलन किया जाता है, अक्सर एक पहलू में बलिदान देकर दूसरे में सुधार किया जाता है?
(a) Greedy algorithm/ग्रीडी एल्गोरिदम
(b) Divide and conquer/डिवाइड एंड कॉन्कर
(c) Dynamic programming/डायनामिक प्रोग्रामिंग
(d) Tradeoff/ट्रेडऑफ

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : A trade of in algorithms refers to the situation where improving one aspect (like speed) requires sacrificing another (like memory) or vice versa. For example
- Using more memory to reduce computation time
- Compressing data to save space but increasing processing time.

248. How many page faults occur in LRU page replacement algorithm for the given reference string, with four page frames.
दी गई रेफरेन्स स्ट्रिंग के लिए LRU पेज रिप्लेसमेंट एल्गोरिथ्म में, चार पेज फ्रेम्स के साथ कितने पेज फॉल्ट होंगे?
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1
(a) 6 (b) 8
(c) 7 (d) 9

KVS PGT C.S. 2018

Ans. (c) : In least Recently used (LRU) algorithm is a Greedy algorithm where the page to be replaced in least recently used. Given reference string.
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1
with four page frames,
Initially all slots are empty, so when 7, 0, 1, 2 are allocated to the empty slots → 4 page faults
0 is already their so → 0 page fault.
when 3 came it will take the place of 7 because it is least recently used → 1 page fault
0 is already in memory so → 0 page fault
4 will takes place of 1 → 1 page fault.
Now, 2, 3, 0, 2, are already in memory
So → 0 page fault.
1 will takes place → 1 page fault
Now for the further page reference string → 0 page fault because they are already available in the memory.
So, total 7 page faults occurred.

249. How many page faults occur in LRU page replacement algorithm for the given reference string, with four page frames.
दी गई रेफरेन्स स्ट्रिंग के लिए LRU पेज रिप्लेसमेंट एल्गोरिथ्म में, चार पेज फ्रेम्स के साथ कितने पेज फॉल्ट होंगे?
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1
(a) 6 (b) 8
(c) 7 (d) 9

KVS PGT C.S. 2018

Ans. (c) : In least Recently used (LRU) algorithm is a Greedy algorithm where the page to be replaced in least recently used. Given reference string.
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1
with four page frames,
Initially all slots are empty, so when 7, 0, 1, 2 are allocated to the empty slots → 4 page faults
0 is already their so → 0 page fault.
when 3 came it will take the place of 7 because it is least recently used → 1 page fault
0 is already in memory so → 0 page fault
4 will takes place of 1 → 1 page fault.
Now, 2, 3, 0, 2, are already in memory
So → 0 page fault.

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1 will takes place → 1 page fault
Now for the further page reference string → 0 page fault because they are already available in the memory.
So, total 7 page faults occurred.

250. Consider the memory reference string given below, for a memory with 3 frames per process.
7, 0, 1, 2, 0, 3, 0, 4, 2, 3
The number of page faults for FIFO page replacement policy are _____.
3 फ्रेम प्रति सेकेंड वाले प्रोसेस वाली मेमोरी के लिए नीचे दी गई मेमोरी रेफरेंस स्ट्रिंग पर विचार कीजिए:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3
एफआईएफाओ पैज रिप्लेसमेंट पॉलिसी के लिए पेज फॉल्ट की संख्या कितनी है?

(a) Ten/दस (b) Seven/सात
(c) Nine/नौ (d) Eight/आठ

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : FIFO (First in first out)
⇒ it is the simplest replacement policy to implement.
⇒ Replace the page that has been brought into memory first or the page that has been in memory the longest is replaced.
3- frames per process

f3			1	1	1	1	0	0	0	3
f2		0	0	0	0	3	3	3	2	2
f1	7	7	7	2	2	2	2	4	4	4
	F	F	F	F		F	F	F	F	F

⇒ 7, 0, 1, 2, 0, 3, 0, 4, 2, 3
⇒ Page faults = 9

251. Which of the following statements is not correct?
निम्नलिखित में से कौन कथन सही नहीं है?

(a) Linked allocation of disk space allows only sequential access of data./डिस्क स्पेस के लिंक्ड एलोकेशन में डाटा के केवल सीक्वेंशियल एक्सेस की अनुमति है।
(b) Indexed allocation of disk space allows direct access to disk blocks./डिस्क स्पेस के इंडेक्स्ड एलोकेशन में डिस्क ब्लॉक के प्रत्यय एक्सेस की अनुमति होती है।
(c) There is no internal fragmentation in contiguous allocation of disk space./डिस्क स्पेस के कंटीगुअस एलोकेशन में कोई इंटरनल फ्रेग्मेंटेशन नहीं होता।
(d) There is no external fragmentation in linked allocation of disk space./डिस्क स्पेस के लिंक्ड एलोकेशन में कोई एक्सटर्नल फ्रेग्मेंटेशन नहीं होता।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : This method suffer from both internal External fragmentation. This makes it inefficient in terms of memory utilization.

252. Which of the following page replacement algorithms requires futuristic information about page references?
निम्न में से कौन-सा पेज रिप्लेसमेंट ऐल्गॉरिद्म, पेज रेफरेन्स की आगे की (भविष्य की) जानकारी पर काम करता है?

(a) FIFO (b) LRU
(c) Second chance/सेकेन्ड चान्स
(d) Optimal/ऑप्टिमल

LT. Grade C.S. 2018

Ans. (d) : The optimal page replacement algorithm requires knowledge of future page references to make the best decision about which page to replace. This algorithm chooses the page that will not be needed for the longest time in the future. The optimal algorithm is also known as the clairvoyant algorithm because it requires knowledge of future events to make decisions.

253. A process refers to 5 pages A, B, C, D, E in the order
A; B; C; D; A; B; E; A; B; C; D; E
If the page replacement algorithm is FIFO, the number of pages which transfer with an 'empty internal store of 3 frames' is
एक प्रक्रम 5 पृष्ठों A, B, C, D, E को
A; B; C; D; A; B; E; A; B; C; D; E
के क्रम में संदर्भित करता है। यदि पेज रिप्लेसमेंट ऐल्गॉरिद्म FIFO है, तो ऐसे पृष्ठों, जो 'empty internal store of 3 frames' के साथ स्थानांतरण करेंगी, की संख्या होगी।

(a) 8 (b) 10
(c) 9 (d) 7

LT. Grade C.S. 2018

Ans. (c) : From the table below, we can see that pages have been transferred into frames 9 times.

A	B	C	D	A	B	E	A	B	C	D	E
A	A	A	D	D	D	E			E	E	
	B	B	B	A	A	A			C	C	
		C	C	C	B	B			B	D	

According to the FIFO algorithm, the first page to enter is the first page that should be replaced.
Every time a page is required by a process, and existing page needs to be replaced from the memory with the page that by needed.

254. Consider a disk queue with requests for I/O to blocks on cylinders. 98 183 37 122 14 124 65 67 considering FCFS (First Come First Served) scheduling, the total number of head movements is, if the disk head is initially at 53 is?/सिलेंडरों पर ब्लॉकों के लिए I/O के अनुरोधों के साथ एक डिस्क कतार पर विचार करें। 98 183 37 122 14 124 65 67 FCFS (पहले आओ पहले पाओ) सेड्यूलिंग पर विचार करते हुए, हेड मूवमेंट की कुल संख्या क्या है, यदि डिस्क हेड शुरू में 53 पर है?

(a) 640 (b) 620
(c) 630
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Sarkari Resource

Ans. (a) : Start point of the disk heat : 53
Requests in order: 98, 183, 37, 122, 14, 124, 65, 67
In First Come First Serued (FCFS), we move the head to each request one by one, in the same order.
Step be step distance:
53 → 98 : move |98 – 53| = 45
98 → 183 : move |183 – 98| = 85
183 → 37 : move |37 – 183| = 146
37 → 122 : move |122 – 37| = 85
122 → 14 : move |14 – 122| = 108
14 → 124 : move |124 – 14| = 110
124 → 65 : move |65 – 124| = 59
65 → 67 : move |67 – 65| = 2
Add all noves:
45 + 85 + 146 + 85 + 108 + 110 + 59 + 2 = 640
So, the head travels a total of 640 cylinders.

255. What is the reason for using the MFU page replacement algorithm?
एम. एफ. यू. पेज रिप्लेसमेंट एल्गोरिथम का उपयोग करने का कारण क्या है?
(a) an actively used page should have a large reference count
(b) a less used page has more chances to be used again
(c) it is extremely efficient and optimal
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b): MFU page replacement algorithm is based on this principal based on the fact that pages that have been used more frequently in the post are likely ot have outlived their usefulness, while pages that have been used less frequently may still be needed. Therefore, MFU replaces pages with the highest reference counts, assuming they have been used frequently enough and may soon be less needed.

256. Valid sequence of moves to solve tower of Hanoi problem with 3 disks if disks are required from stand A to stand C using stand B?
तीन डिस्क की टावर ऑफ हनोई प्रॉब्लम के समाधान के लिए डिस्क विस्थापन का सही क्रम क्या होगा यदि डिस्कों को स्टैंड A से स्टैंड C पर स्टैंड B की सहायता से विस्थापित करना है
(a) A→C, A→B, C→B, A→C, B→C, B→C
(b) A→B, A→B, A→C, B→C, B→C
(c) A→B, A→C, B→A, C→B, A→C, B→C, A→C
(d) A→C, A→B, C→B, A→C, B→A, B→C A→C

RPSC C.S. 2016 (Paper-I)

Ans. (d) : To solve the 'Tower of Hanoi' problem with 3 disks, moving from stand A to C using B, you need to follow a specific sequence that obeys the rules:
(I) Only one disk can be moved at a time
(II) A larger disk can never be placed on top of a smaller disk.
(III) Only the top disk of a stack can be moved.
The correct minimal sequence for 3 disks:
• Disk 1 = Smallest
• Disk 2 = Medium
• Disk 3 = Largest
Standard 7-step solution for 3 disk (A→C via B):
(1) A→C (Move Disk 1)
(2) A→B (Move Disk 2)
(3) C→B (Move Disk 1)
(4) A→C (Move Disk 3)
(5) B→A (Move Disk 1)
(6) B→C (Move Disk 2)
(7) A→C (Move Disk 1)
The correct standard sequence of 7 valid moves:
A→C, A→B, C→B, A→C, B→A, B→C A→C

257. Process in memory also contain the heap section. What is the use of heap section?
(a) Contains temporary data
(b) Dynamically allocate the memory during process run time
(c) Contains Process priority and pointers to scheduling queues
(d) Contains amount of CPU and real time use.

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Ans. (b) : The heap section in memory is crucial for managing dynamically allocate data during a process's runtime. It allow programs to allocate memory as needed, making it possible to handle data with varying sizes and life spans efficiently.
However proper memory management is essential to prevent memory leaks and excessive memory consumption.
So option (b) is correct.

258. How many page faults occur for optimal algorithm for the following string with four frames?
1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8
(a) 8 (b) 7 (c) 6 (d) 5

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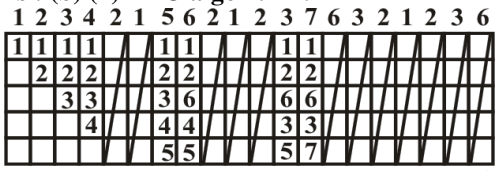
Ans. (a) : The given reference string :
1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8
Number of frames : 4
Let's simulate the algorithm step by step
1. Frame status [–, –, –,–] page fault! Page 1 is loaded into frame 1.
2. Frame status : [1, –, –, –] Page fault!
Page 2 is loaded into frame 2.
3. Frame status : [1, 2, –, –] page fault!
Page 3 is loaded into frame 3.
4. Frame status : [1, 2, 3, –] page fault!
Page 4 is loaded into frame 4.
5. Frame status : [1, 2, 3, 4] page fault!
Page 5 is loaded into frame 1.
6. Frame status : [5, 2, 3, 4] page fault!
Page 3 is already in a frame so no page fault occurs.
7. Frame status : [5, 2, 3, 4] page fault!
Page 4 is already in a frame, so no page fault occurs.
8. Frame status : [5, 2, 3, 4] page fault!
Page 1 is already in a frame, so no page fault occurs.

9. Frame status : [5, 2, 3, 4] page fault!
Page 6 is loaded into frame 2.
10. Frame status : [5, 6, 3, 4] page fault!
Page 7 is loaded into frame 3.
11. Frame status : [5, 6, 7, 4] page fault!
Page 8 is loaded into frame 4.
12. Frame status : [5, 6, 7, 8] page fault!
Now, when Page 7 is requested again, it will be replaced because it's the last occurrence in the reference string that requires a page replacement.
Total page faults : 8
So, for the given reference string and four frames the optimal algorithm result in 8 page faults.

259. Consider the following page reference string:
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6
Which of the following options, gives the correct number of page faults related of LRU, FIFO, and optimal page replacement algorithms respectively, assuming 05 page frames and all frames are initially empty?
(a) 10, 14, 8 (b) 8, 10, 7
(c) 7, 10, 8 (d) 7, 10, 7

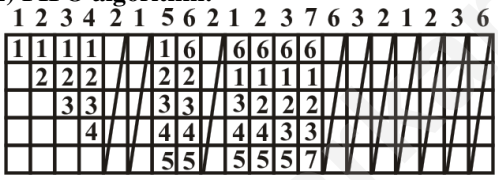
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Ans : (b) (1) LRU algorithm:



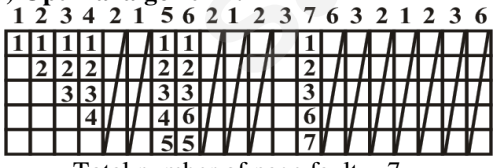
Total number of page fault = 8

(2) FIFO algorithm:



Total number of page fault = 10

(3) Optimal algorithm:



Total number of page fault = 7

260. Consider the following page reference string:
1, 2, 3, 2, 4, 1, 3, 2, 4, 1
Assume that there are three page frames which are initially empty. Then, which of the following expressions is true?
निम्नलिखित पेज रेफरेंस स्ट्रिंग पर विचार करें:
1, 2, 3, 2, 4, 1, 3, 2, 4, 1
मान लें कि तीन तीन पेज फ्रेम्स हैं जो प्रारंभ में खाली हैं। LRU, FIFO और OPTIMAL इसी पेज रिप्लेसमेंट रणनीति के तहत पेज फॉल्ट्स की संख्या को इंगित करने दें। फिर, निम्नलिखित में से कौन सा व्यंजक सत्य है?
(a) LRU = FIFO = OPTIMAL
(b) FIFO < OPTIMAL = LRU
(c) FIFO = OPTIMAL
(d) OPTIMAL < FIFO < LRU

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Ans. (d) : Reference string :–
FIFO →

string	1,2,3	4	1	2
	1	2	3	4
	2	3	4	1
	3	4	1	2
Number of page fault	3	1	1	1

Total page fault in FIFO = 3 +1 + 1 +1= 6
Reference string :– 1, 2, 3, 2, 4, 1, 3, 2, 4, 1
LRU →

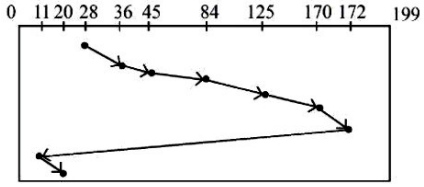
String	1, 2, 3	4	1	3	2	4	1
	1	2	2	4	1	3	2
	2	3	4	1	3	2	4
	3	4	1	3	2	4	1
Number of page fault	3	1	1	1	1	1	1

Total page fault in LRU = 3 + 1 + 1+ 1 + 1 + 1 + 1= 9
Refernce string :– 1, 2, 3, 2, 4, 1, 3, 2, 4, 1
Optimal →

String	1,2, 3	4	2
	1	1	1
	2	3	4
	3	4	2
Number of page fault	3	1	1

Total page fault with optimal = 3 + 1 + 1 = 5
Therefore,
OPTIMAL < FIFO < LRU

261. Consider the following Graph of disk scheduling and identify the disk scheduling algorithm representing by this graph.



(a) SSTF (b) SCAN
(c) C-SCAN (d) C-LOOK

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Ans. (d) : C- Look (Circular Look) is an optimized version of Look.
unlike Look, which moves the head back and forth servicing requests, C - Look services requests in one direction until the last request is reached, then jumps back to the lowest request without servicing cylinders in between.
Given problem:
Cylinders: 0 to 199
Initial head position : 50

Sarkari Resource

Pending request (arrival order) : 82, 170, 43, 140, 24, 16, 190
Initial head movement: Towards higher cylinder numbers.
Step-by-step execution C - Look:-
A sending order : 10, 24, 43, 82, 140, 170, 190
Greater than equal to : 50 : 43, 24, 16
Less than equal to : 50 : 43, 24, 16
Move from 50 → 82 → 140 → 170 →190
Jum back to the smallest request (C- look characteristic. no servicing while jumping Jump from 190 to 16
Move from 16 → 24 → 43
Total head movement calculation:-
50 → 82 = 32
82 → 140 = 58
140 → 170 = 30
170 → 190 = 20
Jump : 190 → 16 (not counted) = 0
16 → 24 = 8
24 → 43 = 19
Calculate:- 32 + 58 + 30 + 20+ 8+ 19 = 167 cylinders.
Final order of execution:-
82→140→170 → 190 → (Jum to 16) → 16 →24 → 43
This follows the C- Look pattern:- Moving in one direction, jumping back to the lowest pending request and continuing in that direction.

262. Consider the reference string:

0 1 2 3 0 1 4 0 1 2 3 4

If FIFO page replacement algorithm is used, then the number of page faults with three page frames and four page frames are _____ and _____ respectively.

(a) 10, 9 (b) 9, 9
(c) 10, 10 (d) 9, 10

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Ans : (d) Given reference string is 0, 1, 2, 3, 0, 1, 4, 0, 1, 2, 3, 4.
(i) FIFO (first in first out) algorithm with 3 page frames.

0	1	2	3	0	1	4	0	1	2	3	4
0	0	0	3	3	3	4			4	4	
	1	1	1	0	0	0			2	2	
		2	2	2	1	1			1	3	
M	*M*	*M*	*M*	*M*	*M*	*M*			*M*	*M*	*M*

Total number of page faults = 9
(ii) FIFO with 4 page frames.

0	1	2	3	0	1	4	0	1	2	3	4
0	0	0	0			4	4	4	4	3	3
	1	1	1			1	0	0	0	0	4
		2	2			2	2	1	1	1	1
			3			3	3	3	2	2	2
M	*M*	*M*	*M*			*M*	*M*	*M*	*M*	*M*	*M*

Total number of page faults = 10.

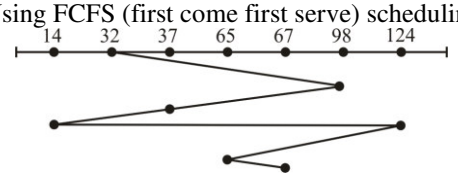
263. If the Disk head is located initially at track 32, find the number of disk moves required with FCFS scheduling criteria if the disk queue of I/O blocks requests are:

98, 37, 14, 124, 65, 67

(a) 320 (b) 322
(c) 321 (d) 319

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Ans : (c) Given I/O requests are:
98, 37, 14, 124, 65, 67
Using FCFS (first come first serve) scheduling:



Then total disk moves:
= (98 – 32)+(98 – 37)+(37 – 14)+(124 – 14)+(124 - 65)+(67 – 65) = 321

264. A scheduling Algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (lowest priority). The scheduler reevaluates the process priority for every '*T*' time units and decides next process to be scheduled. If the process have no I/O operations and all arrive at time zero, then the scheduler implements _____ criteria.

(a) Priority scheduling
(b) Round Robin Scheduling
(c) Shortest Job First
(d) FCFS

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Ans : (b) The scheduling algorithm works as **round robin algorithm with quantum time equals to *T*.** After a process's turn comes and it has executed for *T* units, its waiting time becomes least and its turn comes again after every other process has got the chance for *T* units.

(iv) Concurrent, Concurrent Processes

265. Time taken to switch between user and Kernel modes is _____ the time taken to switch between two processes.
यूजर तथा करनेल मोड के बीच अंतरित होने में लगने वाला समय दो प्रोसेसरों के मध्य अंतरित होने में लगने वाले समय से _____ है।

(a) More than/अधिक
(b) Independent of/स्वतंत्र
(c) Less than/कम
(d) Equal to/समान

KVS PGT C.S. 2018

Ans. (c) : Time taken to switch between user and kernel modes is less than the time taken to switch between two processes.

266. A semaphore is an integer variable that, apart from initialization, is accessed only through which of the following standard atomic operation?

(a) release( ) (b) lock( )
(c) acquire( ) (d) signal( )

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (d) : A semaphore is a synchronization tool used in operating systems to control access to shared resources by multiple processes. It is an integer variable that can only be modified using atomic operations to prevent race conditions.
Signal()– Increases the semaphore value. If the result is non-positive a waiting process is unblocked.

267. In which of the following condition of Deadlock Prevention, at least one resource must be non-sharable?
(a) Circular wait (b) Hold and wait
(c) Mutual Exclusion (d) No preemption

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : Mutual Exclusion– At least one resource must be non sharable (only one process can use it at a time). Example- a printer.
The Mutual Exclusion condition directly requires non sharable resources. If all resources were sharable deadlocks could not occur.

268. In _____, one thread immediately terminate the target thread.
____में, एक थ्रेड तुरंत लक्ष्य थ्रेड को समाप्त करता है।
(a) Asynchronous cancelation
अतुल्यकालिक रद्दीकरण
(b) Synchronous cancellation
तुल्यकालिक रद्दीकरण
(c) Deferred cancellation/स्थगित रद्दीकरण
(d) Both deferred and asynchronous cancellation
स्थगित (Deferred) और अतुल्यकालिक (Asynchronous) रद्दीकरण दोनों।

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Thread cancellation is the process of terminating a thread before it finishes execution.
Asynchronous Cancellation– One thread immediately terminates the target thread without waiting for it to reach a safe point.
It is immediate and can be dangerous if the target thread holds resources or locks.

269. Consider a system with ten units of resource R and three processes: X, Y, and Z. The maximum resource requirement and current allocation to these processes are as follows:
मान लीजिए रिसोर्सेस संसाधन (R) और तीन प्रॉसेस X, Y, और Z की दस यूनिट्स वाला एक सिस्टम है। इन प्रॉसेसके लिए अधिकतम रिसोर्स एलोकेशन रिक्वायरमेंट और करंट एलोकेशन निम्नानुसार हैं:

	X	Y	Z
Maximum Resource Requirement/अधिकतम रिसोर्स रिक्वायरमेंट	**8**	**7**	**5**
Current Allocation करंट एलोकेशन	**4**	**1**	**3**

Now, consider the following resource requests:
अब, निम्नलिखित रिसोर्स रिक्वेस्ट्स पर विचार करें:
(i) X makes a request for 2 resource units
X, 2 रिसोर्स यूनिट्स के लिए रिक्वेस्ट करता है
(ii) Y makes a request for 2 resource units
Y, 2 रिसोर्स यूनिट्स के लिए रिक्वेस्ट करता है
(iii) Z makes a request for 2 resource units
Z, 2 रिसोर्स यूनिट्स के लिए रिक्वेस्ट करता ह
For the system to be in safe state, which of the following must be granted?
सिस्टम को सेफ स्थिति में रखने के लिए, निम्नलिखित में से कौ सा अवश्य रूप से प्रदान किया जाना चाहिए?
(a) Only request (i) must be granted
केवल रिक्वेस्ट (i) प्रदान किया जाना चाहिए
(b) Only requests (i) and (ii) must be granted
केवल रिक्वेस्ट (i) और (ii) प्रदान किया जाना चाहिए
(c) Only request (iii) must be granted
केवल रिक्वेस्ट (iii) प्रदान किया जाना चाहिए
(d) Only request (i) and (iii) must be granted
केवल रिक्वेस्ट (i) और (iii) प्रदान किया जाना चाहिए

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Ans. (c) : Initial Data

Process	Maximum	Current Allocation	Need (=max-Allocation)
X	8	4	4
Y	7	1	6
Z	5	3	2

- Total resources (R) = 10
- Total Allocated = 4 +1 + 3 = 8

Available = 10 – 8 = 2
Check Each Request
1. X requests 2
- New allocation = X = 6 Available = 2 – 2 = 0
- Needs now: X = 2, Y = 6, Z = 2
- With work = 0, no process can finish (all needs 70)

→ Unsafe
2. Y requests 2
- New allocation: Y = 3, Available = 0
- Needs X = 4, Y = 4, Z = 2
- Work = 0, no process can finish

→ unsafe
3. Z requests 2
New allocation : Z = 5 Available = 0
Needs: X = 4, Y = 6, Z = 0
Z can immediately finish (need = 0) and release 5 units
Work = 0 + 5 = 5
- Now X can finish (need = 4)

release 6 → Y can finish (need = 6) →Safe sequence exists → safe
Conclusion:-
To keep the system in a safe state only Z'S request (iii) should be granted.

270. Let P_i and P_j be two processes, R be the set of variables read from memory, and W be the set of variables written to memory. For the concurrent execution of two processes P_i and P_j which of the following conditions is not true?
(a) $R(P_i) \cap W(P_j) = \Phi$
(b) $W(P_i) \cap R(P_j) = \Phi$
(c) $R(P_i) \cap R(P_j) = \Phi$
(d) $W(P_i) \cap W(P_j) = \Phi$

UGC NET C.S. June-2015 (Paper-II)

Sarkari Resource

Ans.(c) : Reading of two process won't create any problem,
So answer is (c)

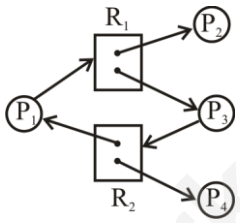
271. Which of the following conditions does not hold good for a solution to a critical section problem?

(a) No assumptions may be made about speeds or the number of CPUs.
(b) No two processes may be simultaneously inside their critical sections.
(c) Processes running outside its critical section may block other processes
(d) Processes do not wait forever to enter its critical section.

UGC NET C.S. December-2014 (Paper-II)

Ans : (c) Because, if a process is running outside the critical section we don't care whatever it is doing out there either it is blocking some processes or not it is not going to affect the processes which are running inside the critical section.

272. Given a set of four processes, two resources with two units, each. the following resource allocation graph exists at point in time:
चार प्रोसेस का एक समुच्चय दिया गया है। प्रत्येक में दो रिसोर्स जिनके पास दो यूनिट्स हैं। एक समय पर निम्न रिसोर्स आवंटन ग्राफ हैं:



The graph indicates:
यह ग्राफ संकेतिक करता है:

(a) cycle and deadlock/साइकिल तथा डेडलॉक
(b) no cycle but deadlock/कोई साइकिल नहीं किन्तु डेडलॉक
(c) neither cycle nor deadlock/न साइकिल न डेडलॉक
(d) cycle but no deadlock/साइकिल किन्तु कोई डेडलॉक नहीं

KVS PGT C.S. 2018

Ans. (d) : Deadlocks can be described more precisely in terms of a directed graph called a system resource allocation graph. This graph consists of a set of vertices V and a set of edges E. The set of vertices V is partitioned into two different types of nodes:
$P = \{P_1, P_2, \text{-----}P_n\}$ the set consisting of all the active processes in the system and
$R = \{R_1, R_2\text{----}R_m\}$ the set consisting of all resource types in the system.
In given resource-allocation graph we have
a cycle $P_1 \rightarrow R_1 \rightarrow P_3 \rightarrow R_2 \rightarrow P_1$
Hence the given graph indicates cycle but no deadlock.

273. Consider the following statements about deadlock :
डेडलॉक के बारे में निम्नलिखित कथनों पर विचार कीजिए:

A. Existence of a cycle in resource-allocation graph indicates deadlock.
वेट-फॉर ग्राफ में साइकिल का होना डेडलॉक का संकेत है।

B. Existence of a cycle in wait-for graph indicates deadlock.
वेट-फॉर ग्राफ में साइकिल का होना डेडलॉक का संकेत है।

C. Mutual-exclusion, hold and wait, no preemption and circular wait conditions, if occur simultaneously, may lead to deadlock.
म्यूचुअल-एक्सक्लूज़न, होल्ड तथा वेट, नो प्रीएम्पशन तथा सर्कुलर वेट की स्थितियाँ, यदि साथ-साथ हों, तो डेडलॉक की ओर ले जा सकती हैं।

Mark the correct option :
सही विकल्प पर निशान लगाइए:

(a) Only (B) is correct/केवल (B) सही है।
(b) Only (C) is correct/केवल (C) सही है।
(c) Only (A) and (C) are correct
केवल (A) तथा (C) सही है।
(d) Only (B) and (C) are correct
केवल (B) तथा (C) सही है।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (d) : Statement (B) : Existence of a cycle in wait-for graph indicates deadlock.
Statement (C) : Mutual-exclusion, hold and wait, no preemption and circular wait conditions, if occur simultaneously, may lead to deadlock.

274. Degree of concurrency in thread can be arranged in the manner.

(a) One-to-one > many-to-one > many-to-many
(b) One-to-one > many-to-many > many-to-one
(c) many-to-many > many-to-one>one-to-one
(d) None of the above

UGC NET C.S. December-2022

Ans. (b) : The correct arrangement of degree of concurrency from highest to lowest is One-to-one > many-to-many > many-to-one

275. Which of the following will be used for a system having multiple instances of a resource type?
निम्नलिखित में से किसका उपयोग रिसोर्स टाइप (संसाधन प्रकार) के बहु-दृष्टांत वाले सिस्टम के लिए किया जाएगा?

(a) Resource-allocation graph
रिसोर्स-अलोकेशन ग्राफ
(b) Banker's algorithm/बैंकर्स एल्गोरिदम
(c) Wait-for graph/वेट-फॉर ग्राफ
(d) Semaphores/सेमाफोर्स

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Sarkari Resource

Ans. (b) : Banker's algorithm is a dead lock avoidance algorithm. It is used when there are multiple instances of a resource in a system.

276. In context of database transactions, a 'wait-for-graph' is used for:-

(a) Concurrency control
(b) Dead lock prevention
(c) Recovery
(d) Dead lock detection

RPSC C.S. 2016 (Paper-II)

Ans. (d) : A wait for graph is one of the methods for detecting the deadlock situation. This method is suitable for smaller databases. In this method, a graph is drawn based on the transaction and its lock on the resource. If the graph created has a closed loop or a cycle, then there is a deadlock.

277. The resource matrix is a data structure used in deadlock detection algorithms to:
रिसोर्स मैट्रिक्स डेडलॉक डिटेक्शन एल्गोरिदम में किसके प्रतिनिधित्व के लिए उपयोग की जाती है?

(a) Represent resource allocation
संसाधन आवंटन को दर्शाने के लिए
(b) Represent the execution order of processes
प्रोसेस के निष्पादन क्रम को दर्शाने के लिए
(c) Represent the execution history of processes
प्रोसेस के निष्पादन इतिहास को दर्शाने के लिए
(d) Represent the priority of processes
प्रोसेस की प्राथमिकता को दर्शाने के लिए

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : A resource matrix is a data structure used in dead prevention algorithms to represent which resources are allocated to which processes. This allows us to determine whether a process is waiting for a resource, thus identifying a dead lock situation.

278. A computer has six tape drives with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

(a) 5 (b) 4
(c) 3 (d) 6

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (a) : Given tape drive = 6 and each process may need 2 drive.
When we give 1 drive to 1 process then total process will be 6 but in this case there will definitely deadlock occur because every process contain 1 drive and waiting for another drive which is hold by other process therefore when we reduce 1 process then system to be deadlock free.
Hence maximum value of n = 6 – 1 = 5

279. The Banker's algorithm is an example of a deadlock:/बैंकर्स एल्गोरिदम डेडलॉक की किस श्रेणी का उदाहरण है?

(a) Avoidance algorithm/ बचाव एल्गोरिदम
(b) Detection algorithm/पहचान एल्गोरिदम
(c) Prevention algorithm/रोकथाम एल्गोरिदम
(d) Termination algorithm/समापन एल्गोरिदम

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : The Bankers Algorithm is a well-known method in operating system used to deal with the problem of deadlock. Instead of allowing a deadlock to occurs and then handling it, this algorithm works in advance by checking whether the system is in safe state before allocating resources. If the allocation keeps the system safe, it is allowed etherwise, it is denied. In this way, the Banker's Algorithm avoids the possibilty dead lock altogether.
Therefore, it is an example of a deadlock avoidance algorithan.

280. A thread is also called :
थ्रेड को _____ भी कहा जाता है।

(a) A scheduler/शेड्यूलर
(b) A virtual process/वर्चुअल प्रोसेस
(c) A heavyweight process/हेवीवेट प्रोसेस
(d) A lightweight process/लाइटवेट प्रोसेस

KVS PGT C.S. 2017

Ans. (d) : A thread is also known as lightweight process. The idea is to achieve parallelism by dividing a process in to multiple threads. A thread is a path of execution within a process. A process can contain multiple threads.

281. Which one of the following statements are CORRECT?

(A) Granularity is the size of data item in a database.
(B) Two operations in a schedule are said to be conflict if they belong to same transaction.
(C) Two schedulers are said to be conflict equivalent if the order of any two conflicting operations is the same in both schedules.
(D) Write operations which are performed without performing the write operation are known as Blind Writes.

Choose the correct answer from the options given below.

(a) (A) and (B) Only
(b) (A), (B) and (C) Only
(c) (A), (B) and (D) Only
(d) (B) and (C) Only

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Ans. (b) : Performing the writing operation (updation), without reading operation a such write operations is known as a blind writes. Hence, the statement (D) is incorrect and statement (A), (B) and (C) are correct.

282. Which one of these is not a condition for Resource Deadlock?
इनमें से कौन सा रिसोर्स डेडलॉक के लिए एक शर्त नहीं है?

(a) Non-sharable resources/नॉन-शेयरबल रिसोर्सेज
(b) Hold-and-wait/होल्ड-एंड-वेट
(c) Preemption/प्रीएंपशन
(d) Circular waits/सर्कुलर वैट्स

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Condition for Resource Deadlock–
(1) Mutual Exclusion– At least one resource must be help in a non-shareable mode only one process can use the resource any given time.
(2) Hold and wait– A process holding at lest one resource is waiting to acquire additional resources currently held by other process.
(3) No Preemption– Resources cannot be forcibly removed from the processes holding them until the resources are used to completion.
(4) Circular wait– A set of processes must exist such that each process is waiting for a resource that is held by the next process in the set, forming a cycle. Preemption is indeed not a condition for a deadlock to occur.

283. Which of the following statement(s) is/are true regarding threads in operating system?
ऑपरेटिंग सिस्टम में थ्रेड्स के संदर्भ में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. The threads in a process are visible to other processes./एक प्रक्रिया में थ्रेड्स अन्य प्रक्रियाओ को दृश्यमान होते हैं।

II. A thread is strictly private resource of a process./एक थ्रेड़ एक प्रक्रिया का दृढ़तः निजी संसाधन होता है।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Statement-I : False
Threads are generally not directly visible or accessible to other independent process. Each process has its own set of threads. The operating system manages these threads within the context of their parent process. While there might be mechanisms for inter-process communication (IPC) that allow processes to coordinate or share information, the threads themselves remain encapsulated within the process that created them. One process cannot directly manipulate or control the threads of another unrelated process.
Statement-II : True
Threads are considered a resource that belongs exclusively to the process that created them. They operate within the memory space of that process and share its resources (like memory, open files, etc) other processes cannot directly access or manage these threads. The operating system ensures this isolation to maintain the integrity and stability of the system.

284. Which of the following statement(s) is/are true regarding processes and threads in operating system?
ऑपरेटिंग सिस्टम में प्रक्रिया और थ्रेड्स के संदर्भ में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. Threads encapsulates concurrency within a process./थ्रेडस एक प्रक्रिया के अंतर्मत कोनक्यूरेन्सी को प्रावरण करते हैं।

II. Process encapsulates protection within the system./प्रक्रिया सिस्टम के अंतर्मत सुरक्षा को प्रावरण करती है।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

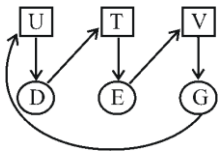
Ans. (c) : Statement-I : True
A process can have multiple threads running within its address space. These threads can execute concurrently meaning they appear to run simultaneously. This allows a single program to perform multiple tasks at the same time, improving responsiveness & efficiency. The operating system manages the scheduling and execution of these threads. So, the concept of concurrency, or the ability to perform multiple tasks seemingly at the same time, is confined and managed within the boundaries of a single process through the use of threads.
Statement-II : True
The operating system provides each process with its own isolated address space. This means that the memory allocated to one process is generally protected from access by other processes. This isolation is crucial for system stability & security. If one process crashes or behaves erratically, it is less likely to affect other independent processes because they operate in their own protected memory regions. The operating system enforces these protection boundaries.

285. Consider a system with seven processes A through G and six resources R through W.
Resource ownership is as follows :
process A holds R and wants T
process B holds nothing but wants T
process C holds nothing but wants S
process D holds U and wants S & T
process E holds T and wants S
process F holds W and wants S
process G holds V and wants U
Is the system deadlocked? If yes, ... processes are deadlocked.

(a) No (b) Yes, A, B, C
(c) Yes, D, E, G (d) Yes, A, B, F

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (c)



System is deadlock due to process D, E, G
So answer is option (c)

286. Suppose there are four processes in execution with 12 instances of a Resource R in a system. The maximum need of each process and current allocation are given below:

Process	Max. Need	Current Allocation
P_1	8	3
P_2	9	4
P_3	5	2
P_4	3	1

With reference to current allocation, is system safe? If so, what is the safe sequence?

(a) No (b) Yes, $P_1\ P_2\ P_3\ P_4$
(c) Yes, $P_4\ P_3\ P_1\ P_2$ (d) Yes, $P_2\ P_1\ P_3\ P_4$

UGC NET C.S. June-2016 (Paper-II)

Sarkari Resource

Ans : (c) Remaining need = Max need – Current allocation.

Process	Max need	Current allocation	Remaining need
P_1	8	3	5
P_2	9	4	5
P_3	5	2	3
P_4	3	1	2
		10	

Available resources = 12 – 10 = 2
Then, we can satisfy needs of P_4, then release total 4, then satisfy P_3, then release total 7, then satisfy P_1, and P_2 similarly.
Hence, it the safe sequence: P_4, P_3, P_1, P_2.

287. A computer system has 7 tape drives. There 'n' processes competing for them. Each process may need 2 tape drives. What is the maximum value of 'n' for which the system is guaranteed to be deadlock tree?
एक कंप्यूटर सिस्टम में 7 टेप ड्राइवर हैं। 'n' प्रक्रियाएं उनके लिए प्रतिस्पर्धा कर रही है। प्रत्येक प्रक्रिया को 2 टेप ड्राइवर की आवश्यकता हो सकती है। 'n' का अधिकतम मान क्या है जिस पर सिस्टम के डेडलॉक मुक्त होने की गारंटी है?

(a) 2 (b) 3
(c) 4 (d) 1

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Here the interpretation is slightly striker.
Each process maximum claim = 2 drives
To be deadlock-free, the system must always be able to satisfy at least one process maximum claim completely.
That means $n \times 2 \leq 7$
$n \leq 3$
So, n = 3. If there are 4 processes, each might demand 2 drives (total = 8), which is greater than 7, so the system cannot guarantee deadlock freedom.

288. In context of semaphores choose the option with valid codes for wait() and signal() functions :
सेमाफोर के संदर्भ में wait() तथा signal() फलनों के लिए उचित कोड वाले विकल्प का चुनाव कीजिए:

(a) Wait (s) { while (s) < = 0; s + + } and signal (s) { s – –; }
(b) Wait (s) { while (s) < = 0; s – – } and signal (s) { s + +; }
(c) Wait (s) { s – –; } and signal (s) { while (s) < = 0; s + + }
(d) Wait (s) { s + +; } and signal (s) { while (s) < = 0; s – –; }

HTET PGT C.S. 2022

Ans. (b) : A semaphore is a synchronization primitive used to control access to shared resources. It uses two atomic operations
1. Wait()–
• This decreases the value of the semaphore by 1.
• If the value becomes negative, the process is blocked until the semaphore is available.

```
c
Wait (s) {
  while (s < = 0);
  s – –;
}
```

2. Signal()–
• This increases the semaphore value.
• If there are processes waiting, one of them is allowed to proceed.

```
c
Signal (s) {
  s++;
}
```

289. Which of the following is not a valid deadlock prevention scheme ?
निम्नलिखित में से कौन एक वैध गतिरोध निवारण योजना नहीं है?

(a) Release all resources before requesting a new resource./नए संसाधन का अनुरोध करने से पहले सभी संसाधनों को छोड़ दें।
(b) Number the resources uniquely and never request a lower numbered resource than the last one requested./संसाधनों को विशिष्ट रूप से संख्या दें और कभी भी पिछले अनुरोध के मुकाबले कम संख्या वाले संसाधन का अनुरोध न करें।
(c) Never request a resource after releasing any resource./किसी भी संसाधन को जारी करने के बाद संसाधन का अनुरोध न करें।
(d) Request and all required resources be allocated before execution./निष्पादन से पहले अनुरोध और सभी आवश्यक संसाधन आवंटित किए जाएँ।

HTET PGT C.S. 2020

Ans. (c) : Deadlock prevention schemes work by ensuring that at least one of therefore necessary conditions of deadlock (Mutual Exclusion, Hold and Wait, No preemption, Circular Wait) is violated.
(a) Release all resources before requesting a new resource– This prevents Hold and Wait condition, so it is a valid scheme.
(b) Number the resources uniquely and never request a lower numbered resource than the last one requested– This prevents Circular Wait so it is a valid scheme.
(d) Request and allocate all required resources before execution– This also prevents Hold and Wait, so it is valid.
(c) Never request a resource after releasing any resources– This is not a recognized deadlock prevention scheme. It is impractical. It is impractical and not part of the standard methods.

290. Which of the following is not related to synchronization in Operating System ?
निम्न में से कौन-सा ऑपरेटिंग सिस्टम में सिंक्रोनाइजेशन से संबंधित नहीं है?

(a) Bounded Buffer problem
बाउन्डेड बफर प्रॉब्लम
(b) Readers-Writers problem
रीडर्स-राइटर्स प्रॉब्लम

Sarkari Resource

(c) Dining Philosophers problem
डाइनिंग फिलोसोफर्स प्रॉब्लम
(d) Belady's Anomaly
बिलेडी का अनॉमली

HTET PGT C.S. 2019

Ans. (d) : In operating systems, synchronization problems arise when multiple processes or threads need to coordinate their execution while sharing resources. Classic synchronization problem include.

- **Bounded Buffer Problem–** Synchronization needed between producer and consumer processes.
- **Readers-Writers Problem–** Synchronization needed between multiple readers and writers accessing a shared database.
- **Dining Philosophers Problem–** Synchronization needed to avoid deadlock when philosophers share chopsticks (resources).
- **Belady's Anomaly–** On the other hand, Belady's Anomaly is not related to synchronization.

291. Which of the following is not required condition for a deadlock ?
निम्न में से कौन-सी डेडलॉक के लिए आवश्यक शर्त नहीं है?
(a) Hold and Wait/होल्ड एवं वेट
(b) Preemptive/प्रीइम्पटिव
(c) Mutual Exclusion/म्यूचुअल एक्सक्लूज़न
(d) Cycle/साईकल

HTET PGT C.S. 2018

Ans. (b) :
1. Mutual Exclusion– At least one resource must be held in a non shareable mode (only one process can use it at a time).
2. Hold and Wait– A process is holding at least one resource and waiting to acquire additional resource that are being held by other processes.
3. No Preemption– Resources cannot be forcibly taken away from a process they must be released voluntarily.
4. Circular Wait– A set of processes exists such that each process is waiting for a resource held by the next process in the chain, forming a cycle.

292. For each thread of a multi-threaded process, there is a separate:
(a) Process control block
(b) User address space
(c) User and kernel stack
(d) Kernel space only

CGPSC C.S. 2020 Set-A

Ans. (c) : Multi-Threaded Process– All threads of a process share some resources.
1. Code section
2. Data section
3. Open files
4. User address space
But each thread must also maintain its own state for independent execution.

293. Semaphore are used to solve the problem of:
(a) Race condition (b) Multi-tasking
(c) Mutual exclusion (d) Both (A) and (C)

CGPSC C.S. 2020 Set-A

Ans. (c): Semaphores– Semaphores are synchronization tools used in operating system to control access to shared resources in concurrent programming.
(c) Mutual exclusion– Ensure that only one process thread at a time can access the critical section (shared resource)
(d) Multi tasking– No directly salved by semaphores multitasking is a broader OS scheduling concept.

294. The advantage of having multiple threads over multiple processes is:
(i) Less time for creation
(ii) Less time for termination
(iii) Less time for switching
(iv) Kernel not involved in communication among threads
(a) (i), (ii), (iii) (b) (i), (ii), (iv)
(c) (ii), (iii), (iv) (d) (i), (ii), (iii), (iv)

CGPSC C.S. 2020 Set-A

Ans. (d) : Multiple Threads– A thread is lighter : threads of the same process share memory and resources, but have separate registers, stack and program counter.
Multiple Processes– A process is heavier : it has its own memory space, PCB resources etc.
All four statements (i), (ii), (iii) and (iv) are correct advantage of threads over processes.

295. A processor while executing the instruction sequence of user mode process, received n interrupts. If no other activity is reported to processor during the execution of instruction sequence, what is the number of mode switches and process switches experienced?
(a) 2n, 2n (b) n, n
(c) 2n, 0 (d) n, 0

CGPSC C.S. 2020 Set-A

Ans. (d) :
1. CPU goes from user mode– Kernel mode (to handle interrupt).
2. After servicing CPU goes from kernel mode– user mode (to resume process) that make a node switches per interrupt.
• So for n interrupts → 2n mode switches.
Since the same process continues (no scheduling decision to replace it).
• 0 process switches.

296. In the case of parallelization, Amdahl's law states that if P is the proportion of a program that can be made parallel and (1–P) is the proportion that cannot be parallelized, then the maximum speed-up that can be achieved by using N processors is :
(a) $\frac{1}{(1-P)+N.P}$ (b) $\frac{1}{(N-1)P+P}$
(c) $\frac{1}{(1-P)+\frac{P}{N}}$ (d) $\frac{1}{P+\frac{(1-P)}{N}}$

UGC NET C.S. June-2015 (Paper-II)

Sarkari Resource

Ans.(c) : Amdahl's law states that if P is the propagation of a program that can be made parallel & (1 – P) is the propagation that cannot be parallelized, then the maximum speed up that can be achieved by using N processors is.

$$\frac{1}{(1-P)+\frac{P}{N}}$$

297. Peterson's solution is a classic software-based solution to the critical-section problem. What is the maximum number of processes to which it is restricted?

(a) 2 (b) 3
(c) 4 (d) 5
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Peterson's Solution– Peterson's solution is a software based solution on the critical section problem ensuring mutual exclusion, progress, and bounded waiting without using hardware instructions like test-and-set.
It uses two shared variables.
1. Flag[i]– indicates if process [i] wants to enter the critical section.
2. Turn– indicates whose turn it is.

298. Which of the following scheduling algorithms gives minimum average waiting time?
निम्नलिखित शेड्यूलिंग एल्गोरिथ्म में से कौन न्यूनतम औसत प्रतीक्षा समय देता है?

(a) FCFS (b) SJF
(c) Round Robin (d) Priority

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : SJF Scheduling– Shortest Job First (SJF) is a CPU scheduling algorithm where the process with the smallest CPU burst time (execution time) is selected next for execution. This means the CPU always executed the Job that will take the least time to finish among the available process.

(v) File and Secondary Storage Management

299. File manipulation operations in an operating system include:
ऑपरेटिंग सिस्टम में फाइल मैनिपुलेशन ऑपरेशन्स में क्या शामिल है?

(a) Allocating memory to files
फाइल्स को मेमोरी अलॉट करना
(b) Assigning file names to processes
फाइल नामों को प्रोसेसेस को असाइन करना
(c) Loading files into CPU registers
फाइल्स को CPU रजिस्टर में लोड करना
(d) Reading writing, and modifying file content
फाइल कंटेंट को पढ़ना, लिखना और संशोधित करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : File manipulation operations include opening, closing reading, writing, updating their contents, and deleting files. These operations allow a user or program to interact with files.

300. A large computer information system maintains many different computer files. Which amongst them is called a perpetual file?
एक बड़ा कंप्यूटर सूचना प्रणाली कई विभिन्न कंप्यूटर फ़ाइलों को बनाए रखता है। इनमें से कौन-सी 'परपेचुअल फाइल' कहलाती है?

(a) Specialized File/विशिष्ट फ़ाइल
(b) Master File/मास्टर फ़ाइल
(c) Log File/लॉग फ़ाइल
(d) Update File/अपडेट फ़ाइल

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : When there are many types of files in a large computer information system, them the file among them which is continually updated and is preserved for a long time is called Perpetual File or Master File.

301. What is coalescing ?
संलयन क्या है ?

(a) It is second strategy for allocating kernel memory./यह कर्नेल मेमोरी के आबंटन करने की एक दूसरी रणनीति है।
(b) The buddy system allocates memory from a fixed size segment consisting of physically contiguous pages.
एक बडी सिस्टम जो निश्चित आकार के भौतिक रूप से समीपस्थ पृष्ठों से मेमोरी का आबंटन करती है ।
(c) Kernel memory is often allocated from a free-memory pool different from the list used to satisfy ordinary user made processes./कर्नल मेमोरी बहुदा लिस्ट से भिन्न फ्री मेमोरी पुल का उपयोग, साधारण यूजर मोड के प्रोसेस को संतुष्ट करने के उपयोग में आबंटित होती है ।
(d) An advantage of the buddy system is how quickly adjacent buddies can be combined to form larger segments using this technique./बडी प्रणाली का लाभ यह है की कितनी शीघ्रता से बड़ी को इस तकनीक का उपयोग करने बृहदतर प्रखण्ड बनाने में संयोजित किया जा सकता है।

KVS PGT C.S. 2013

Ans. (d) : The buddy system allocates memory from a fixed size segment consisting of physically contiguous pages memory is allocated from the segment using a power of two allocator which satisfies requests in units sized as a power of 2 (4kb, 8kb, and so forth)

302. Physical memory is broken into fixed size blocks called?
भौतिक स्मृति के स्थिर आकार के विभाजित ब्लाक कहलाते है?

(a) Frames/फ्रेम्स (b) Pages/पेज
(c) Chunks/चंक (d) Buffers/बफर

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : In operating systems, physical memory is divided into fixed-size blocks called frames. Correspondingly, logical memory is divided into blocks of the same size called pages. This is part of the paging memory management scheme.

Sarkari Resource

303. What is the full form of ISAM in file organisation?
फाइल ऑर्गेनाइजेशन में आई.एस.ए.एम. का अंसक्षिप्त रुप क्या है ?

(a) Indirect Sequential Access Method
इनडायरेक्ट सिक्वेंशियल ऐक्सेस मेथड
(b) Indexed Sequential Access Method
इंडेक्स्ड सिक्वेंशियल ऐक्सेस मेथड
(c) Indirect Serial Access Method
इनडायरेक्ट सिरीयल ऐक्सेस मेथड
(d) Index Serial Access Method
इनडेक्स सिरीयल ऐक्सेस मेथड

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : ISAM stands for Indexed Sequential Access method. It's a method of file organization that combines the benefits of both sequential and direct file access.

Sequential Access:- Records are stored in a logical order, typically by a key field, which allows for efficient processing of records in a batch.

Indexed Access- An Index is created to provide a quick look up for specific records. This index acts like a table of contents, allowing the system to go directly to the location of a particular record without having to read through all the preceding records.

304. A disk consists of _____, _____, _____
एक डिस्क में _____, _____, _____ होते हैं

(a) track, segment and circle/ट्रैक, सेगमेंट और सर्किल
(b) track, circle and cone/ट्रैक, सर्किल और कोन
(c) drum, track and segment/ड्रम, ट्रैक और सेगमेंट
(d) cylinder, tracks and sectors/सिलेंडर, ट्रैक और सेक्टर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : A hard disk drive (HDD) stores data on spinning platters. To organize this data, the platters are divided into specific components.

Tracks: These are concentric circles on the surface of each platter. Data is recorded on these tracks.

Sectors: Each track is further divided into smaller arcs called sectors. A sector is the smallest unit of storage on a disk, typically 512 bytes or 4096 bytes.

Cylinders: A cylinder is a vertical stack of tracks that are at the same radial position on each platter. All the read/ write heads move together, so they are always positioned over a single cylinder. This organization is efficient for data access because it allows data to be read from multiple platters at the same time without moving the heads horizontally.

305. In a disk system with 100 cylinders, the request to access the cylinder occur in following sequence: 34, 4, 7, 19, 10, 83, 2, 6, 20, 15 Assuming that the head is currently at cylinder 50, then what is the time taken to satisfy all requests if it takes 1 second to move from one cylinder to another? (Shortest seek time first policy is used)
100 सिलेंडर वाले डिस्क सिस्टम में, सिलेंडर तक पहुँचने का अनुरोध निम्नलिखित क्रम में होता है: 34, 4, 7, 19, 10, 83, 2, 6, 20, 15 यह मानते हुए कि हेड वर्तमान में सिलेंडर 50 पर है, तो सभी अनुरोधों को पूरा करने में कितना समय लगता है यदि एक सिलेंडर से दूसरे सिलेंडर में जाने में 1 सेकंड लगता है? (शॉर्टेस्ट सीक टाइम फर्स्ट नीति का उपयोग किया जाता है)

(a) 119 seconds/ 119 सेकण्ड
(b) 129 seconds/ 129 सेकण्ड
(c) 109 seconds/ 109 सेकण्ड
(d) 139 seconds/ 139 सेकण्ड

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Ans. (b) : To determine the total time taken to satisfy all requests using the Shortest Seek Time First (SSTF) policy starting at cylinder 50, the SSTF policy will always move the head to the closest requested cylinder.

306. A disk drive has 100 cylinders, numbered 0 to 99. Disk requests come to the disk driver for cylinders 12, 26, 24, 4, 42, 8 and 50 in that order. The driver is currently serving a request at cylinder 24. A seek takes 6 msec per cylinder moved. How much seek time is needed for shortest seek time first (SSTF) algorithm?

(a) 0.984 sec (b) 0.396 sec
(c) 0.738 sec (d) 0.42 sec

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : Given requests is: 12, 26, 24, 4, 42, 8, 50
Using SSTF (shortest seek time first) algorithm:
Total distance covered
$= 2 + 22 + 46 = 70$
Total seek time is:
$= 70 \times 6$ msec
$= 70 \times 6 \times 10^{-3}$ sec $= 0.42$ sec

307. In direct file organization:-

(a) A collision always results into an overflow
(b) A collision never results into an overflow
(c) A collision may or may not lead to an overflow
(d) A collision never occurs

RPSC C.S. 2016 (Paper-II)

Ans. (c) : In Direct File Organization (DFO) (also known as hash file organization), a hash function is used to compute the address of the data record. If two records are assigned the same address by the hash function (i.e., a collision occurs), the system must find a way to store the second record. This extra storage is called overflow.
So, every collision requires storing the new record in an overflow area,
Hence– A collision always results in an overflow.

308. Sector interleaving in disks is done by?
डिस्क में सेक्टर इंटरलिविंग किसके द्वारा किया जाता है?

(a) Disk Manufacturer/डिस्क निर्माता
(b) Disk Controller/डिस्क नियंत्रक
(c) Operating System/ऑपरेटिंग सिस्टम
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Sarkari Resource

Ans. (c) : Sector interleaving is a technique used to optimize disk access by distributing sectors across the disk in a non-linear fashion, which improves read/write speeds. This is a task managed by the operating system to ensure efficient data access. The operating system determines the placement of sectors on the disk to minimize head movement and optimize access time.

309. In _____ allocation method for disk block allocation in a file system, insertion and deletion of blocks in a file is easy.

(a) Index (b) Linked
(c) Contiguous (d) Bit Map

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Just like linked list where insertion, deletion of blocks is easy. In linked allocation only pointer needs to be changed.
In linked allocation each file is a linked list of disk blocks. The directory contains a pointer to the first block of the file, for example, a file of 5 blocks which starts at block 4, might continuous at block 7, then block 16, block 10, 8 finally block 27. Each block constrains a pointer to the next block & the last block contains a NIL-pointer. The value-1 may be used for NIL to differential it from block 0.

310. _____rules used to limit the volume of log information that has to be handled and processed in the event of system failure involving the loss of volatile information:

(a) Write-ahead log (b) Check-pointing
(c) Log buffer (d) Thomas

UGC NET C.S. December-2014 (Paper-III)

Ans : (b) Check pointing is a technique to add fault tolerance into computer system. It basically consist of saving a snapshot of the application state, so that it can restart from that point in case of failures

311. K-mean clustering algorithm has clustered the given 8 observations into 3 clusters after Ist iteration as follows:
C1 : {(3, 3), (5, 5), (7, 7)}
C2 : {(0, 6), (6, 0), (3, 0)}
C3 : {(8, 8), (4, 4)}
What will be the Manhattan distance for observation (4, 4) from cluster centroid C1 in second iteration?

(a) 2 (b) $\sqrt{2}$
(c) 0 (d) 18

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (a) : Manhattan distance is the distance between two points measured at right angle along both the X-axis and Y-axis.
formula- It is calculated as : if there is a point $P(X_1, Y_1)$ and $Q(X_2, Y_2)$. Then Manhattan distance is
$|X_2 - X_1| + |Y_2 - Y_1|$
calculation :
C1 : {(3, 3), (5, 5), (7, 7)}
C2 : {(0, 6), (6, 0), (3, 0)}
C3 : {(8, 8), (4, 4)}
first point is (4, 4)
we have to find the Manhattan distance for cluster C1 in 2nd iteration :
for 2nd iteration : points will be
| (\left({\frac{1}{3}\left({3 + 5 + 7}\ right)}\right), \left ({\frac{1}{3}\left({3 + 5 + 7}\right)}\right) = \left({5,\;5}\right)\)
So Manhattan distance = $|5-4| + |5-4| = 1+1 = 2$

312. Which of the following statements is not true about disk-arm scheduling algorithms?

(a) SSTF (shortest seek time first) algorithm increases performance of FCFS.
(b) The number of requests for disk service are not influenced by file allocation method.
(c) Caching the directories and index blocks in main memory can also help in reducing disk arm movements.
(d) SCAN and C-SCAN algorithms are less likely to have a starvation problem.

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : Option (b) is NOT CORRECT as the number of request for disk drive can be greatly influenced by file allocation method.

313. Which of the following File operation need not involve any actual I/O (Input/Output) and is also known as 'File Seek'?

(a) Deleting a file
(b) Repositioning within a file
(c) Reading a file
(d) Truncating a file

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : In operating systems, file operations generally involve actual I/O with disk storage such a reading writing deleting or truncating. These require data to be transferred between memory and disk.

314. Which of the following statement is correct for a' Volume Control block' in File-System Implementation?

(a) It is used to organize the file
(b) It has a unique identifier number to allow association with a directory entry
(c) It contain the number of block in the partition
(d) It can contain information needed by the system to boot an operating system from that volume

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : In file system implementation, the Volume Control Block (VCB) stores essential information about a disk partition (Volume). This metadata helps the operating system manage and access files properly.

315. Which of the following is NOT a file manipulation system call in the Windows operating system?
विंडो ऑपरेटिंग सिस्टम में निम्नलिखित में से कौन सा फाइल मैनिपुलेशन सिस्टम कॉल नहीं है?

(a) CreateFile() (b) ReadFile()
(c) WriteFile() (d) CloseFile()

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

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Ans. (d) : File manipulation system calls are functions provided by the operating system that allow a program to create, read and deleting files. Close file() is not a file manipulation system call in windows operating system.

316. _____stores entries about files and their characteristics.
_____फाइलों और उनकी विशेषताओं के बारे में एंट्रीज़ संग्रहीत करता है।
(a) File application table/फाइल ऐप्लीकेशन टेबल
(b) File access table/फाइल ऐक्सेस टेबल
(c) Format allocation table/फॉर्मेट एलोकेशन टेबल
(d) File allocation table/फाइल एलोकेशन टेबल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : In an operating system, the File Allocation Table (FAT) is a data structure used to store entries a bout file and their characteristics. These characteristics include file's name, size, permissions, creation data, and the locations of the file's data blocks on the disk. FAT enables the operating system to efficiently locate, access, and manage files on the storage device.

317. What is the file extension for Batch files ?
बैच फाइल्स के लिए फाइल एक्सटेंशन क्या होता है?
(a) BAT (b) BCH
(c) CTH (d) BATCH

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : In Windows operating systems, batch files are scripts that contain a series of commands to be executed by the command line interpreter (CMD). The standard file extension for batch files is : BAT
Example : example bat.

318. _____ is a system to make a file or directory visible in multiple parts of the system's file tree in Linux operator system.
ऑपरेटर सिस्टम में सिस्टम के फाइल ट्री के कई भागों में फाइल या डायरेक्टरी को दृश्यमान बनाने के लिए _____एक सिस्टम है।
(a) Directories/डाइरेक्टरिज
(b) Special files/स्पेशल फाईल्स
(c) Links/लिंक्स
(d) Sockets/सोकेट्स

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : In Linux/Unix, if you want a file or directory to appear in multiple places in the file system tree, you use Links.
Hard Link– Another name for the same file (points to the same index node).
Soft Link– Work like a shorted (points to the pathname of the file).
This makes the same file/directory visible indifferent locations.

319. The process in an operating system in which the output of any program is not printed directly on printer but dumped onto a reserved area of a disk is known as _____.
एक ऑपरेटिंग सिस्टम में एक प्रक्रिया, जिसमें किसी प्रोग्राम का आउटपुट प्रत्यक्ष रुप में प्रिंटर पर प्रिंट नही होता है लेकिन एक डिस्क के आरक्षित क्षेत्र पर डंप किया जाता है, _____ के रुप में जानी जाती है ।
(a) System call /सिस्टम कॉल
(b) Teardown operation /टियरडॉउन ऑपरेशन
(c) Spooling /स्पूलिंग
(d) Swapping /स्वैपिंग

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Spooling– Spool (Simultaneous Peripheral Operations On-Line).
Example– When multiple users sent print jobs, all are placed in a spool (queue) on the disk. The printer reads one job at a time.
Advantage– Programs do not to wait for the slow printer; they can continue executing.

320. Rearrange steps involved in top-down design programming?
टॉप-डाउन डिज़ाइन प्रोग्रामिंग में शामिल चरणों को पुनर्व्यवस्थित करें?
I. Analyze the problem
समस्या का विश्लेषण करें
II. Revise and resequence
रिवाइज और रिसिकुएंस
III. Write the main module
मुख्य मॉड्यूल लिखें
IV. Connect the modules in proper order
मॉड्यूलों को उचित क्रम में कनेक्ट करें
V. Write the remaining modules
शेष मॉड्यूल लिखें
(a) II → IV → V → III → I
(b) III → I → V → IV → II
(c) V → III → I → IV → II
(d) I → III → V → IV → II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : The correct sequence of steps involved in top-down design programming is generally-
I- Analyze the problem– understand the overall problem and its requirements.
II- Write the main module– Design the highest level of the program, defining the main control flow and how it interacts with major sub-tasks.
III- Write the remaining modules– Break down the sub tasks identified in the main module into smaller, manageable modules and write the code for them. This step is recursive, breaking down further until modules are simple enough.
IV- Connect the modules in proper order– integrate the modules, ensuring they call each other correctly and pass data as needed.
V. Revise and resequence– Test debug and refine design and code based on the results. This step is often iterative.

321. Terms NTFS and FAT 32 are closely related to which action on computer ?
NTFS तथा FAT 32 टर्म्स का कम्प्यूटर की किस प्रक्रिया से निकट सम्बन्ध है?
(a) Defragmentation/डीफ्रेगमेंटेशन
(b) Scanning/स्कैनिंग
(c) Booting/बूटिंग
(d) Formatting/फॉर्मेटिंग

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Ans. (d) : NTFS and FAT 32
NTFS (New Technology File System) and FAT 32 (File Allocation Table 32) are file systems used by operating system to manage how data is stored and retrieved on a disk.
A file system decides how files are named, stored, organized and accessed.
Uses of NTFS or FAT 32– When you format a stored device (Like a hard disk, pen drive, or memory card) you must choose a file system.

322. Indexed/grouped allocation is useful as:
(A) It supports both sequential and direct access.
(B) Entire block is available for data.
(C) It does not require lots of space for keeping pointers.
(D) No external fragmentation.
Choose the correct answer from the options given below.
(a) (A) Only
(b) (B) and (C) Only
(c) (B) Only
(d) (A), (B) and (D) Only

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Ans. (d) : Indexed/grouped allocation typically supports both sequential and direct access. Entire block is available for data. It helps prevent external fragmentation as all files have their own index block that keeps track of the blocks associated with it. Indexed allocation actually needs space to store pointers in the index block for each file.

323. In context of Operating System File Management Function what is the full form of FAT ?
ऑपरेटिंग सिस्टम फाइल प्रबंधन फंक्शन के संदर्भ में FAT का अर्थ क्या है?
(a) File Attribute Table/फाइल एट्रिब्यूट टेबल
(b) File Allocation Table/फाइल एलोकेशन टेबल
(c) Format All Tabs/फॉर्मेट ऑल टैब्स
(d) File Attribute Type/फाइल एट्रिब्यूट टाइप

HTET PGT C.S. 2020

Ans. (b) : File Management in Operating System. FAT (File Allocation Table) is a data structure used by the file system to keep track of how disk space is allocated to files. It works like a map of the disk, showing which clusters are used by which files and which clusters are free.

(vi) UNIX and Shell Programming, Windows Programming

324. The command cat > x in unix
कमाण्ड cat > x यूनिक्स में
(a) Is invalid/अमान्य है।
(b) Creates a file x and displays an error message. एक फाइल x बनाता है और एक त्रुटि मेसेज प्रदर्शित करता है।
(c) Creates a file x and waits for the user to give input from the keyboard. /एक फाइल x बनाता है और यूजर द्वारा की-बोर्ड से इनपुट देने की प्रतीक्षा करता है।
(d) None of the above /उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : In Unix, the "cat" command is normally used to display the contents of a file. But when we write "cat>x", we are not giving any input file to "cat". Instead, the symbol ">" tells the shell to redirect the output of the command into a file named x.

325. Choose the incorrect statement.
गलत कथन का चयन करें:
(a) Shell scripts can accept arguments
शेल स्क्रिप्ट आर्गुमेंट्स स्वीकार कर सकते हैं।
(b) Shell scripts are interpreted
शेल स्क्रिप्ट अनुवादित होते हैं।
(c) Shell is a programming language
शेल एक प्रोग्रामिंग भाषा है।
(d) Shell scripts are compiled
शेल स्क्रिप्ट कम्पाइल्ड होते हैं।

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : Shell script are not compiled– C, C++ or java which require compilation before execution, shell scripts are not compiled. They remain plan text files and are executed by the shell interpreter at runtime. Therefore saying that shell script are compiled is incorrect.

326. Which of the following is not a WINDOWS version ?
निम्न में से कौन-सा WINDOWS का संस्करण नहीं है?
(a) WINDOWS-96 (b) WINDOWS-2000
(c) WINDOWS- XP (d) WINDOWS-98

HTET PGT C.S. 2018

Ans. (a) : Microsoft has released several versions of the windows operating system over the years. Some well known ones include Windows-95, Window-98, Windows-200, Windows-XP, Windows-7, Windows-10 and Windows-11.
- Windows-98, Windows-200 and Windows-XP are all genuine versions of Windows.
- But Window-96 was never released by Microsoft it does not exist as an official Windows version.

327. In windows which of the following, helps protect users from malware and other potentially unwanted software ?
विंडोज में निम्न में से कौन-सा, मैलवेयर तथा अन्य संभावित अनचाहे सॉफ्टवेयर से उपभोक्ताओं को सुरक्षित रखने में सहायक है?
(a) Windows Mobility Center
विंडोज मोबिलिटी सेन्टर
(b) Windows Defender/विंडोज डिफेंडर
(c) Indexing Options/इंडेक्सिंग ऑप्शन्स
(d) Windows Updates/विंडोज अपडेट्स

HTET PGT C.S. 2019

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Ans. (b) : Windows Defender is Microsoft's built in antivirus and anti malware tool. It continuously scans the system for malicious software, spyware, ransom ware, and other potentially unwanted programs, It provides real time protection, automatic updates and quick or full scans to ensure the system stays safe.

328. In windows, we go to _____ option in Control panel to uninstall a program from our computer.
विंडोज में, हम हमारे कम्प्यूटर से किसी प्रोग्राम को अन-इनस्टाल करने के लिए कंट्रोल पैनल में _____ विकल्प पर जाते है।

(a) Default Programs/डिफॉल्ट प्रोग्राम्स
(b) Action Center /एक्शन सेन्टर
(c) Device Manager /डिवाइस मैनेजर
(d) Programs and Features /प्रोग्राम्स एवं फीचर्स

HTET PGT C.S. 2019

Ans. (d) : In Windows, if you want to uninstall a program you go to the control panel and open the option "Programs and Features". This section lists all installed application on the system and provides options to uninstall change or repair them.

329. Which of the following command interface is normally used for windows operating system ?
निम्न में से कौन-सा कमाण्ड इंटरफेस आमतौर पर विंडोज ऑपरेटिंग सिस्टम के लिए उपयोग किया जाता है?

(a) Menu driven interface/मेनू संचालित इंटरफेस
(b) Command line interface/कमांड लाइन इंटरफेस
(c) Graphical user interface/ग्राफिकल यूजर इंटरफेस
(d) Icon interface/आइकन इंटरफेस

HTET PGT C.S. 2020

Ans. (c) : The Windows Operating System is primarily designed to be user friendly and easy to interact with. For this reason it mainly uses a Graphical User Interface (GUI), where users interact with the system using Windows, Icons, Menus and Pointer (WIMP). This allows even non-technical users to easily run applications, manage files and configure system settings.

330. Office Automation System deals with which of the following?
I. Word processing
II. Annual budgeting
ऑफिस ऑटोमेशन सिस्टम निम्नलिखित में से किससे संबंधित है?
I. वर्ड प्रोसेसिंग
II. एनुअल बजटिंग

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/न ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : Office automation system – office automation systems are designed to automate office tasks. Streamlining functions.

Word processing and Annual Budgeting – word processing is a key component of office automation systems, involving task such as creating, editing and printing documents. Annual budgeting, on the other hand, is a financial function that may be related to office automation systems but is primarily associated with financial management + software

331. What is the function of the following UNIX command?
WC – /<a > b&

(a) It runs the word count program to count the number of lines in its input, a, writing the result to b, as a foreground process.
(b) It runs the word count program to count the number of lines in its input, a, writing the result to b, but does it in the background.
(c) It counts the errors during the execution of a process, a, and puts the result in process b.
(d) It copies the '*l*' numbers of lines of program from file, a, and stores in file b.

UGC NET C.S. June-2016 (Paper-III)

Ans : (b) WC – 1<a > b&
wc for word cound-1 for line < a input a> b output b & for background.
It runs the word count program to count the number of lines in its input, a write result of b, but does it in the background. The & symbol at the end of the command signifies that the process runs in back ground.

332. Which of the following statement is not correct with reference to cron daemon in UNIX O.S.?

(a) The cron daemon is the standard tool for running commands on a pre-determined schedule.
(b) It starts when the system boots and runs as long as the system is up.
(c) Cron reads configuration files that contain list of command lines and the times at which they invoked.
(d) Cronatab for individual users are not stored.

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) Cron daemon is the standard tool for running command an a receteminced schedule. It starts when the system boots and run as long as the system is up. Crontabs for individual u>a are stored under /var/spool/cron. There is one crontab file per user.

333. In Unix, files can be protected by assigning each one a 9-bit mode called right bits. Now, consider the following two statements :

I. A mode of 641 (octal) means that the owner can read and write the file, other members of the owner's group can read it, and users can execute only.

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II. A mode of 100 (octal) allows the owner to execute the file, but prohibits all other access.

Which of the following options is correct with reference to above statements?

(a) Only I is correct
(b) Only II is correct
(c) Both I and II are correct
(d) Both I and II are incorrect

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) In UNIX the access control of file is done through chmod command. The common is octal, the there right most are for
First is for owner, second is for group, third is for other here rwx is read, write execute. This can take the value from 0 through 7 where

#	permission	rwx
7	read, write & execute	rwx
6	read, write	rw–
5	read & execute	r–x
4	read only	r– –
3	write & execute	–wx
2	write only	–w–
1	execute only	– –x
0	none	– –

641 represent → The owner can read and write the file, other member of the owner group can read it & user can execute only
100 → The owner to execute the file, but prohibits all other access

334. In UNIX, processes that have finished execution but have not yet had their status collected are known as .....

(a) Sleeping processes (b) Stopped processes
(c) Zombie processes (d) Orphan processes

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Ans : (c) **Zombie Process :** The UNIX system assumes that if child process terminate, the parent collects some data about the child, As soon as this collection is made, the system releases information bits about the child process and removes its PID from the process table. The UNIX is forced to keep the child's Process ID (PID) and termination data in the process table indefinitely. Such a terminated process whose status is not collected called a zombie process.

335. In Unix operating system, when a process creates a new process using the fork () system call, which of the following state is shared between the parent process and child process?

(a) Heap
(b) Stack
(c) Shared memory segments
(d) Both Heap and Stack

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Ans : (c) Parent and child share only the share memory segments everything else (like stack, heap etc) is duplicated.

336. Which of the following information about the UNIX file system is not correct?

(a) Super block contains the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks.
(b) An i-node contains accounting information as well as enough information to locate all the disk block that holds the file's data.
(c) Each i-node is 256-bytes long.
(d) All the files and directories are stored in data blocks.

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Ans : (c) **UNIX file system:**
Super block contains the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks.**Correct**
An i-node contains accounting information as well as enough information to locate all the disk blocks that holds the file's data.**Correct**
All the files and directories are stored in data blocks.**Correct**
Each i-node is 256-bytes long.**Incorrect**.It is 128 byte long.

337. Which of the following option with reference to UNIX operating system is not correct?

(a) INT signal is sent by the terminal driver when one types <Control> and it is a request to terminate the current operation.
(b) TERM is a request to terminate execution completely. The receiving process will clean up its state and exit.
(c) QUIT is similar to TERM, except that it defaults to producing a core dump if not caught.
(d) KILL is a block able signal.

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Ans : (d) UNIX operating system does not kills a block able signal.

338. _____ is a symbol used to represent special files in Linux OS.

_____ Linux OS में विशेष फाइलों को दर्शाने के लिए उपयोग किए जाने वाला एक प्रतीक है।

(a) d (b) e
(c) c (d) s

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : In Linux OS, special files (device files) are represented in the **Is - 1** listing by the symbol.
• c ⇒ Character special file
• b ⇒ Block special file.
Character devices (c) handle data one character at a time (e.g. keyboard, serial ports).

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339. What is the description of append keyword of visual Basic NET keyword?
विजुअल बेसिक NET कीवर्ड के अपैंड कीवर्ड का विवरण क्या है?

(a) Boolean operation/बूलियन ऑपरेशन
(b) Used in the declare statement
डिक्लेयर स्टेटमेंट में उपयोग किया जाता है
(c) Used as a symbolic constant in the file open function/फ़ाइल ओपन फलन में सिंबॉलिक कांस्टेंट के रूप में उपोयग कयिा जाता है
(d) Assembly – attribute specifier
असेंबली - एट्रिब्यूट स्पेसिफाइअर

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : In VB NET the keyword 'Append' is used when opening a file using the 'FileOpen' function. It tells the program to open the file in append mode, meaning that new data will be added to the end of the file without overwriting the existing content.
Example– File Open (1, "example.txt", OpenMode. Append)
- This opens example.txt in append mode using file number 1.
- Any new data written to this file will be added to the end.

340. The process of compiling Intermeidate Language programs to the actual machine language on the target platform prior to execution is knwon as _____.
इंटरमीडिएट लैंग्वेज प्रोग्राम को निष्पादन से पहले लक्ष्य प्लेटफॉर्म पर वास्तविक मशीन भाषा में संकलित करने की प्रक्रिया को_____ के रूप में जाना जाता है।

I. Just –In-Time compiling
जस्ट-इन-टाईम कम्पाइलिंग
II. First –In- Time compiling
फर्स्ट-इन-टाईम कम्पाइलिंग

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : Some programming language (like Java or C#) first compile source code into an Intermediate Language (IL) or byte code, which is platform-independent.
before execution on the target machine, this IL needs to be translated into actual machine code.
1. Just-In-Time (JIT) compiling– The IL is compiled into machine code at runtime, just before execution. This allows platform independence while still achieving good performance.

341. What does '/etc' directory contain in Linux operating system?
Linux ऑपरेटिंग सिस्टम में '/etc' डायरेक्टरी में क्या होती है?

(a) Home directories of the common user
आम उपयोगकर्ता की होम डायरेक्ट्रीज
(b) System configuration file
सिस्टम कॉन्फ़िगरेशन फ़ाइल
(c) Miscellaneous data/मिसलिनियस डेटा
(d) Common program shared by the system and the system administrator
सिस्टम द्वारा साझा किया गया सामान्य प्रोग्राम तथा सिस्टम एडमिनिस्ट्रेटर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : In Linux, the/etc directory is used to store system wide configuration files.
These files control how the system and installed application behave.
Example– /etc/passwd (user accounts), /etc/fstab (disk mounting), /etc/hosts (hostnames),/etc/ssh/(SSH configuration), etc.

342. The Unix Kernel maintains two key data structures related to processes, the process table and the user structure. Which of the following information is not the part of user structure?

(a) File descriptor table
(b) System call state
(c) Scheduling parameters
(d) Kernel stack

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Ans.(c) : The Unix Kernel maintains two key data structures related to processes, the process table and the user structure. In user structure there is no place for scheduling parameter process table store the information of all running process. The process table contains process ID's, memory usage of the process, what are file descriptor used in the process.

343. _____ maintains the list of free disk blocks in the Unix file system

(a) I-node (b) Boot block
(c) Super block (d) File allocation table

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Ans. (c) : Super block describes the state of file system, the total size of partition, the block size, pointers to a list of free blocks, the node number of root directory, magic number etc.

344. A part Windows 2000 operating system that is not portable is

(a) Device Management
(b) Virtual Memory Management
(c) Processor Management
(d) User Interface

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Ans. (b) : The design of virtual memory assumes that the underlying hardware supports virtual to physical mapping.

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345. Match the following with reference to Unix shell scripts :

List–I		List–II
A.	$?	i. File name of the current script
B.	$#	ii. List of arguments
C.	$0	iii. The number of arguments
D.	$*	iv. Exit status of last command

Codes :

	A	B	C	D
(a)	iii	ii	i	iv
(b)	ii	iii	i	iv
(c)	iv	iii	i	ii
(d)	i	iii	i	iv

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Ans. (c) :
$? : This command shows exit status of the last command executed.
$# : This command shows number of argument supplied to a script.
$O : This command shows filename of current script.
$* : This command shows all the arguments are double quoted.

346. The output generated by the LINUX command:
$ seq 1 2 10
will be
(a) 1 2 10
(b) 1 2 3 4 5 6 7 8 9 10
(c) 1 3 5 7 9
(d) 1 5 10

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Ans. (c) : In linux seq command is used to print sequence of number seq command print numbers from first to last, in steps of increment.
If first or incremented is omitted, it defaults to 1. That is, an emitted INCREMENT defaults to 1 even when last is smaller than FIRST. FIRST, INCREMENT, and LAST are interpreted as floating point values INCREMENTED is usually positive if FIRST is smaller than LAST, and INCREMENTED is usually negative if first is greater than LAST.
then,

$ seq	First	increment	last
	↑	↑	↑
	1	2	10

then the output sequence generated is = 13579

347. All the classes necessary for windows programming are available in the module :
(a) win.txt (b) win.main
(c) win.std (d) MFC

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Ans. (a) : win·txt is a group of classes that is use for windows programming.

348. Which of the following versions of window O.S. contain built-in partition manager which allows us to shrink and expand pre-defined drives?
(a) Windows vista (b) Window 2000
(c) Window NT (d) Window 98

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Ans : (a) Windows vista have logical disk manager that supports shrinking & expanding volumes vista also have a program that allow a user to upgrade their running vista to higher edition.

349. WOW32 is a :
(a) Win 32 API library for creating processes and threads
(b) Special kind of file system to the NT name space
(c) Kernel - mode objects accessible through Win 32 API
(d) Special execution environment used to run 16 bit Window applications on 32-bit machines

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Ans : (d) Wow 32 is a running environment used in windows to run 16 bit windows application on a 32 bit machine

350. The Unix command :
$ vi file1 file2
(a) Edits file1 and stores the contents of file1 in file2
(b) Both files i.e. file1 and file2 can be edited using 'ex' command to travel between the files
(c) Both files can be edited using 'mv' command to move between the files
(d) Edits file1 first, saves it and then edits file2

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Ans : (b) You can edit more than one file at a time with vi.
vi file1 file2 vi two (or more) files at the same time.

351. A Unix file may be of the type :
(a) Regular file
(b) Directory file
(c) Device file
(d) Any one of the above

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Ans : (d) A Unix file may be of any of the type like regular file, directory file, or device file.

352. In Unix, the login prompt can be changed by changing the contents of the file _____
(a) Contrab (b) Init
(c) Gettydefs (d) Inittab

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Ans : (c) The gettydefs file supplies the gety command with information about various terminal attributes, such as the default initialization & the login message. The

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gettydefs file contains single line entries that are separated by blank lines.
(i) There is nothing like contrab in unix.
(ii) The inittab file controls what happens whenever a Unix system is rebooted or forced to change run levels.
(iii) init (short for initialization) is the first process started during booting of the computer system.

353. The character set used in windows 2000 operating system is _____

(a) 8 bit ASCII (b) Extended ASCII
(c) 16 bit UNICODE (d) 12 bit UNICODE

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Ans : (c) 16 bit UNICODE character set used in Windows 2000 operating system. Extended ASCII character encoding are 8 bit or larger encoding that include 7 bit ASCII character. ASCII coding is 8 bit character encoding.

354. In Unix, the command to enable execution permission for file "mylife" by all is _____

(a) Chmod ugo + X myfile
(b) Chmod a + X myfile
(c) Chmod + X myfile
(d) All of the above

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Ans : (d) All three commands are used to enable execution permission CH mode X for execution ugo (user group other) and + for add
a (all) and + for add
only + for all.

355. What will be the output of the following Unix command?
$rm chap0\[1 – 3\]

(a) Remove file chap0[1 – 3]
(b) Remove file chap01, chap02, chap03
(c) Remove file chap\[1 – 3\]
(d) None of the above

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Ans : (a) If you use rm chap*, you will remove all files beginning with chap. The way to remove the file chap* is to use the backslash (\) as follows:
$ chap \ *
The use of the \ in removing any special character is called escaping or despecializing. For Ex-
to remove a file chap 0[1 – 3], use, use: $rm chap*\[1 – 3\].

356. In Unix operating system, special files are used to :

(a) Buffer data received in its input from where a process reads
(b) Provide a mechanism to map physical device to file names
(c) Store list of file names plus pointers associated with i-nodes
(d) Store information entered by a user application program or utility program

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Ans : (b) In the Unix file system, six types of file are distinguished.
– special, Named pipes, Links, Symbolic links, Regular or ordinary, Directory.
* Special- Contains no data, but provides a mechanism to map physical devices to file names. The file names are used to access peripheral devices, such as terminals and printers.

357. Consider the following statements :

(A) UNIX provides three types of permissions
*** Read**
*** Write**
*** Execute**

(B) UNIX provides three sets of permissions
*** permission for owner**
*** permission for group**
*** permission for others**

Which of the above statement/s is/are true?

(a) Only (A)
(b) Only (B)
(c) Both (A) and (B)
(d) Neither (A) nor (B)

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Ans : (c) UNIX provides read, write and execute permission on files UNIX provides there set of permission
- Permission for owner
- Permission for group
- Permission for others.

358. Match the following WINDOWS system calls and UNIX system calls with reference to process control and file manipulation.

Windows		UNIX	
A.	**Create - process ( )**	**i.**	**Open ( )**
B.	**Wait for single object ( )**	**ii.**	**Close ( )**
C.	**Create file ( )**	**iii.**	**Fork ( )**
D.	**Close Handle ( )**	**iv.**	**Wait ( )**

Code :

	A	B	C	D
(a)	iii	iv	i	ii
(b)	iv	iii	i	ii
(c)	iv	iii	ii	i
(d)	iii	iv	ii	i

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Ans : (b) 1. In UNIX wait ( ) is used to create process ()
2. In UNIX fork ( ) is used for wait for single object ( )
3. In UNIX create ( ) is used for create file ( )
4. In UNIX close ( ) is used for close handle ( )

359. Consider the following operations to be performed in Unix:
"The pipe sorts all files in the current directory modified in the month of "June" by order of size and prints them to the terminal screen.

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The sort option skips ten fields then sorts the lines in numeric order."
Which of the following Unix command will perform above set of operation?
(a) Is - *l* |grep "June" | sort + 10n
(b) Is - *l* |grep "June" | sort + 10r
(c) Is - *l* |grep -v "June" | sort + 10n
(d) Is - *l* |grep -n "June" | sort + 10x

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Ans : (a) Unix is a family of multitasking, multiuser operating system developed at AT & T's Bell Labs in the 1970s, known for its stability, flexibility, and powerful command line interface.
Correct Unix command is - Is - *l* |grep "June" | sort + 10n.

360. Which of the following statement(s) is/are True regarding 'nice' command of UNIX?
I. It is used to set or change the priority of a process.
II. A process's nice value can be set at the time of creation.
III. 'nice' takes a command line as an argument.
(a) I, II only (b) II, III only
(c) I, II, III (d) I, III only

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Ans : (c) A process's nice value can be set at the time of creation with nice command and adjusted later with renice command.
(i) To set or change the priority of a process.
(ii) It takes a command line as an argument.
(iii) It's value can be set at the time of creation.

361. Arrange the following levels of interrupt protection within the Linux Kernel, in the order of increasing priority.
(A) User mode programs
(B) Bottom half interrupt handlers
(C) Kernel system service routines
(D) Top half interrupt handlers
Choose the correct answer from the options given below:
(a) (A), (B), (D), (C)
(b) (A), (C), (B), (D)
(c) (A), (C), (D), (B)
(d) (D), (A), (C), (B)

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Ans. (b) : The correct order of increasing priority for the given levels of interrupt protection within the Linux kernel is-
(A) User mode programs
(C) Kernel system service routines
(B) Bottom half interrupt handlers
(D) Top half interrupt handlers

362. In Linux, where is he user password stored?
(a) / etc / password
(b) / root / password
(c) / etc / passwd
(d) / root / passwd

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Ans. (c) : In Linux systems, user passwords are typically stored in the '/etc/ passwd' file. This file is readable by all users on the system. It stores encrypted passwords and other security - related information for user accounts.

363. In Linux Operating System, when _____ is invoked, it is passed a set of flags that determine how much sharing is to take place between the parent and child tasks.
(a) fort() (b) clone()
(c) pthread() (d) thread()

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Ans. (b) : The main use of clone () is to implement threads : multiple threads of control in a program that run concurrently in a shared memory space.
The clone () system call is an upgraded version of the fork call. It's powerful science it creates a child process and provides more precise control over the data shared between the parent and child processes.

364. Which UNIX/Linux command is used to make all files and sub-directories in the directory "progs" executable by all users ?
(a) chmod − R a + x progs
(b) chmod − R 222 progs
(c) chmod − X a + x progs
(d) chmod − X 222 progs

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Ans. (a) : "chmod − R a + x progs" command is used to make all files and sub-directories in the directory "progs" executable by all users in the Linux.

365. In Linux operating system environment _____ command is used to print a file.
(a) Print (b) Lpr
(c) Pr (d) Ptr

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Ans. (b) : In Linux operating system environment Lpr command is used to print a file.
Lpr- Off line print

366. Which of the following statement/s is/are true?
(i) Windows XP supports both peer-peer and client-server networks.
(ii) Windows XP implements Transport protocols as drivers that can be loaded and unloaded from the system dynamically.
Choose the correct answer from the code given below :
Code :
(a) (ii) only (b) Neither (i) nor (ii)
(c) Both (i) and (ii) (d) (i) only

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Ans. (c) : Windows XP supports both peer-peer and client-server networks. In a Client-Server network, high-end servers, installed with the Network Operating System (Server Operating System) and the related software, serve the clients continuously on a network, by providing them with specific services upon request. Windows XP implements Transport protocols as drivers that can be loaded and unloaded from the system dynamically.

367. Suppose you have a Linux file system where the block size is 2K bytes, a disk address is 32 bits and in i-node contains the disk addresses of first 12 direct blocks of file, a single indirect block, and a double indirect block. Approximately, what is the largest file that can be represented by an i-node?

(a) 513 Kbytes (b) 513 MBytes
(c) 537 Mbytes (d) 537 KBytes

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Ans. (b) : Size of the disk block = 2048 bytes
disk address = 32 bits = 4 bytes

Number of addresses per block = $\frac{2048}{4} = 512 = 2^9$

Maximum file size = 12 direct blocks + 1 indirect Block + 1 double indirect block

$= 12 + 2^9 + 2^9 * 2^9$

$= 12 + 2^9 + 2^{18}$

the size of each block = 2^{11} bytes

Maximum size of the file = $(12 + 2^9 + 2^{18}) \times 2^{11}$ bytes

$\therefore$ Maximum size of the file = 5130.236 $\cong$ 513 MBytes

368. Which of the following UNIX/Linux pipes will count the number of lines in all the file having c and h as their extension in the current working directory?

(a) cat*ch| wc–1 (b) cat*[c–h]|wc–1
(c) cat*[ch]|1s–1 (d) cat*[ch]|wc–1

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Ans. (d) : Pipes in Linux/Unix : Pipe is a command which produces an output which serves as input for next command. It works as a pipeline structure in computer.

Explanation–

We have to find the number of lines in all the files having .c and .h as their extension in the current working directory.

WC- in Unix/Linux, "WC" is used for word count.

Cat- it is used to display the content of the files.

Option 4- Cat * .[ch] //it represent the content of files having .c and .h extension

Option 2 →

Cat * [c -h]/ WC-L //it also includes the files with c and .h extension but it also considers the other extensions that come between because it is a range which is not required in this question.

So cat * [ch]/WC-L

It will count the number of Lines in all the files having .c and .h as their extension in current working directory.

369. A successful "fork" system call in unix return_____ to child process.
यूनिक्स की एक सफल ''फॉर्क'' सिस्टम कॉल चाइल्ड प्रोसेस को _____लौटती है.

(a) PID (Parent)/पी.आई.डी. (पैरेंट)
(b) 0
(c) 1
(d) NULL/नल

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Ans. (b) : In Unix-like systems, the 'fork()' system call is used to create a new process. After a successful call to 'fork()':

- The child process receives a return value of 0.
- The parent process receives the PID of the child (a positive integer).

So, the child gets 0, and the parent gets the child's PID.

370. Number of arguments specified in a Unix command line argument is identified by?
यूनिक्स कमांड लाइन आर्ग्युमेंट में निर्दिष्ट आर्ग्युमेंट की संख्या की पहचान होती है?

(a) $$ (b) $?
(c) S* (d) $#

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : '$#'– It gives the number of command line arguments passed to a script.

'$'– It gives all the arguments as a single word.

'$?'– It gives the exit status of the last executed command.

'$$'– It gives the process ID of the current shell.

371. Which of the following is not a valid mode of file opening for "open" system call in Unix?
यूनिक्स की ''ओपन'' सिस्टम कॉल के सन्दर्भ में निम्न में से कौन-सा फाइल खोलने का मोड वैध नहीं है?

(a) O_RDONLY
(b) O_WRONLY
(c) O_WRRK
(d) O_RDWR

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : In Unix systems, the 'open' system call uses specific flags to determine how a file should be accessed. Common valid modes include:

- **O_RDONLY–** Open the file for read only.
- **O_WRONLY–** Open the file for write only.
- **O_RDWR–** Open the file for both reading & writing.
- **O_WRRK–** Is not a valid flag and does not exist in the standard Unix/Linux file open flags.

372. What is the return type of "write" system call in Unix?
यूनिक्स की "write" सिस्टम कॉल का रिटर्न टाइप कौन सा है?

(a) size_t (b) ssize_t
(c) int (d) char

RPSC C.S. 2016 (PAPER-I)

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Ans. (b) : In Unix-like systems, the write sysem call is declare as:
Syntax– ssize_t write (int fd, const void * buf, size_t count);
⇒ 'ssize_t' is the return type, which is a signed version of size_t.
⇒ It returns the number of bytes actually written, or -1 on error.

373. "Head" Unix command, when used without an option, displays how many lines of the file?
बिना कोई विकल्प दिए यूनिक्स का "Head" कमांड किसी फाइल की कितनी लाइने प्रदर्शित करता है?
(a) 3 (b) 5
(c) 10 (d) 100

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : When the 'head' command is used without any option, it displays the first '10' line of the file by default.
The 'head' command in Unix/Linux is used to view the beginning of a file or the output of a command.
Example– head filename.txt
This command will display the first 10 lines of filename.txt because the default number of lines is 10 if you don't specify any options.

374. What will be the output of the following Unix command - tail - c + 512 filename
निम्न यूनिक्स कमांड का आउटपुट क्या होगा - tall - c + 512 filename
(a) Copies last 512 bytes from file
फाइल से आखिरी 512 बाइट कॉपी करेगा
(b) Copies every byte after skipping 511 bytes
511 बाइट को छोड़कर बाकी सभी कॉपी करेगा
(c) Copies last 512 lines from file
फाइल से अंतिम 512 लाइने कॉपी करेगा
(d) Copies every line after skipping 511 lines
511 लाइने छोड़कर बाकी सभी लाइने कॉपी करेगा

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : The Unix command tail –c +512 filename means–
• **–c +512–** start from the 512th byte of the file and output everything from there to the end.
• The + sign means "starting from" instead of counting from the end.
So it skips the first 511 bytes and prints the rest.

375. Source code is available to view, modify and redistribute in-
स्रोत कोड देखने, संशोधित करने और पुनर्वितरित करने के लिए उपलब्ध है–
(a) Open source
(b) Licensed
(c) Proprietary
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) : There are many solutions available to view, modify and redistributed source code, the most prominent being open source in which the source code is publicly available. Anyone can view and modify it, for example- Linux, Apache, LibreOffice etc.

376. The data blocks of a very large file in UNIX file system are allocated using
UNIX फाइल सिस्टम में एक बहुत बड़ी फाइल के डेटा ब्लॉक आवंटित किए जाते हैं।
(a) Contiguous allocation
(b) Linked allocation
(c) Indexed allocation
(d) An extension of indexed allocation

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Ans. (d) : The Unix file system used an extension of indexed allocation. It uses direct blocks, single indirect blocks, double indirect blocks and triple indirect blocks. Indexed allocation is a file allocation method where a separate index block is used to store pointers to the data blocks of a file.

377. To change the command prompt in UNIX, one can use
UNIX में कमान्ड प्रॉम्प्ट बदलने के लिए प्रयोग किया जाता है
(a) Grep (b) LS
(c) PSL (d) PWD

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Ans. (c) : In UNIX the command prompt can be changed by modifying the value of the PS1 variable. PS1 stands for primary prompt string and is used to specify the format of the command prompt. To change the command prompt you can use the
PS1 = "new - prompt - string".

378. Which is the Linux?
इनमें से लिनक्स (Linux) कौन सी है?
(a) Open-source Operating System
ओपन-सोर्स ऑपरेटिंग सिस्टम
(b) Open-source Application Software
ओपन-सोर्स एप्लीकेशन सॉफ्टवेयर
(c) Windows based Application Software
विंडो आधारित एप्लीकेशन सॉफ्टवेयर
(d) Proprietary Operating System
प्रोप्रिएटरी ऑपरेटिंग सिस्टम

EMRS PGT C.S. 16.12.2024

Ans. (a) : The Linux operating system is a type of operating system that is similar to Unix, and it is built upon the Linux Kernel. The Linux Kernel is like the brain of the operating system because it manage how the computer interacts with its hardware and resources.

Sarkari Resource

07. Database Management System

(i) Database Systems, View of Data Models, Database Languages

1. 3NF is based on the concept of?
3NF किस संकल्पना पर आधारित है?
(a) Functional dependency/फंक्शनल डिपेन्डेन्सी
(b) Closure dependency/क्लोज़र डिपेन्डेन्सी
(c) Transitive dependency/ट्रान्जीटिव डिपेन्डेन्सी
(d) Normal dependency/नॉर्मल डिपेन्डेन्सी

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans, (c) : A relation is in third normal form if there is no transitive dependency for non-prime attributes as well as it is in second normal form.

2. Which forms have a relation that contains information about a single entity?
किन रूपों में एक संबंध है जिसमें एक इकाई के बारे में जानकारी है?
(a) 4NF
(b) 3NF
(c) 5NF
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) If a relation has no non-trivial multi- value dependencies, it is in 4 NF. This ensures that each record contains information about only one entity preventing redundancy or inconsistency in the data.

3. Which of the following is not the utility of DBMS?
निम्नलिखित में से कौन-सी DBMS की उपयोगिताएँ नहीं हैं?
I. Backup/बैकअप
II. Loading/लोडिंग
III. Process organisation/प्रक्रिया संगठन
IV. File organization/फाइल संगठन
(a) I, II and IV only/केवल I, II और IV
(b) I, II and III only/केवल I, II और III
(c) I, III and IV only/केवल I, III और IV
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (e): • Backup → DBMS can do backup .
• Loading → DBMS can load/ import data.
• Process organization → This is not done by DBMS, it is done by operating system.
• File organization → DBMS can organize files.
So, the wrong one is process organization.

4. Which data type can store unstructured data in a column?
कौन-सा डेटा टाइप असंरचित डेटा को एक कॉलम में संग्रहित कर सकता है?
(a) RAW
(b) Numeric
(c) VARCHAR
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a): We use RAW data type, although usually text (like an open text filed in a from) unstructured data can come in many forms to be stored as objects, images audio, video , text files and other file formats.

5. In relational databases, if relation R is in BCNF, then which of the following is true about relation R?
(a) R is in 4NF
(b) R is not in 1 NF
(c) R is in 2NF and not in 3NF
(d) R is in 2NF and 3NF

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : Relation is in 'N' Normal form it means it must be in 1NF, 2NF, ... N-1 NF.
standard Normal form hierarchy :→
1NF → 2NF → 3NF → BCNF
So,
Relation is in BCNF then it must be in 1NF, 2NF, & 3NF Hence option d is correct.

6. Match List I with List II

List I		List II	
(A)	**BCNF**	**(I)**	**It remove multivaled dependency**
(B)	**3 NF**	**(II)**	**It is not always dependency preserving**
(C)	**2 NF**	**(III)**	**It removes transitive dependency**
(D)	**4 NF**	**(IV)**	**It removes partial functional dependency**

	B	C	D	
(a)	(III)	(II)	(IV)	(I)
(b)	(II)	(IV)	(I)	(III)
(c)	(II)	(III)	(IV)	(I)
(d)	(II)	(I)	(IV)	(III)

UGC NET C.S. December 2021

Ans. (c) : Nomalization – Database normalization is a technique of organizing the data in the database.
Normalization is used for mainly two purposes.
- Eliminating redundant (useless) data
- Ensuring data dependencies make sense i.e. data is logically stored.

Sarkari Resource

First normal form (1NF) – The first normal form help to eliminate duplicate data and simplify queries.
Second normal form (2NF) – is based on the concept of fully functional dependency 2NF removes partial functional dependency.
Third normal form (3NF) – A relation is in the third normal form (3NF) if it is second normal form and no transitive dependencies exist.
Fourth normal form (4NF) – A group of tables that satisfies the first, scored and third normal forms are sufficiently well-designed 4NF removes multivalued dependency.
Boyce-Codd Normal Form BCNF is an extension to the third normal form also known as 3.5 normal form.
- BCNF is not always dependency preserving.

The correct answer is option (c).

7. Match List - I with List - II.

List - I		List - II	
A.	**BCNF iff**	**I**	**Every jd is implied by the candidate keys**
B.	**5 NF iff**	**II**	**All underlying domains contain scalar values only**
C.	**1 NF iff**	**III**	**Every MVD is implied by the candidate keys**
D.	**4 NF iff**	**IV**	**Every FD is implied by the candidate keys**

Choose the correct answer from the options given below:
(a) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(b) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(c) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
(d) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

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Ans (b) :

List - I		List - II
A.	BCNF iff	(IV). Every FD is implied by the candidate keys
B.	5 NF iff	(I). Every JD is implied by the candidate keys
C.	1 NF iff	(II). All underlying domains contain scalar values only
D.	4 NF iff	(III). Every MVD is implied by the candidate keys

8. What is DBMS?
DBMS क्या है?
(a) DBMS is a collection of queries
DBMS क्वेरीज का संग्रह है
(b) DBMS is a high-level language
DBMS एक हाई-लेवल भाषा है
(c) DBMS is a programming language
DBMS एक प्रोग्रामिंग भाषा है
(d) DBMS stores, modifies and retrieves data
DBMS डेटा को संग्रहित, संशोधित और पुनर्प्राप्त करता है

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : A DBMS (Database Management System) is software that stores, manages (e.g. modifies, retrieves data) and accesses databases, Example– MySQL, Grade.

9. What is CRUD an acronym for in database operations?
डेटाबेस संचालन में CRUD का संक्षिप्त नाम क्या है?
(a) Create, Read, Update, Delete
(b) Cache, Read, Update, Delete
(c) Copy, Replace, Update, Delete
(d) Collect, Read, Use, Discard

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : C-Create– Adding new data to a database is done using the INSERT statement in SQL.
R–Read– Retrieving data from a database is done using the SELECT statement in SQL.
U–Update– Modifying existing data in a database is done using the UPDATE statement in SQL.
D–Delete– Deleting existing data from a database is done using the DELETE statement in SQL.

10. Which SQL command is used to remove a table from a database?
डेटाबेस में तालिका हटाने के लिए किस SQL कमांड का उपयोग किया जाता है?
(a) DELETE TABLE (b) DROP TABLE
(c) REMOVE TABLE (d) ERASE TABLE

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : The DROP TABLE command completely removes the specified table from a database, including its data, structure, and associated indexed. The deleted table cannot be recovered unless the database has a backup or transaction log.

11. Knowledge discover and data mining are associated with?
नॉलेज डिस्कवरी एवं डाटा माइनींग किससे संबंधित है?
(a) Database (b) Oracle
(c) Internet (d) Data warehouse

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : Knowledge Discovery and Data Mining are processes used to extract important information, patterns and relationships from large data sets. This process typically begins in a Data Ware house, where large amounts of data are stored. Data Mining techniques are then applied to that data to extract hidden useful information.

12. Which of the following is a database?
निम्नलिखित में से कौन एक डेटाबेस है?
(a) MySQL
(b) Informix
(c) SQL Server
(d) All of the above/उपरोक्त सभी

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : All the given options are DBMS (Data Base Management System), which are used to store manage and access data.
MySQL is an open source database (developed by Informix), SQL Server (Developed by Microsoft) are popular databases.

13. What do you mean by one to many relationships?
'वन टू मेनी रिलेशनशिप' का क्या अर्थ होता है?
(a) Many teachers may have many classes
कई शिक्षक कई कक्षाओं को पढ़ा सकते हैं
(b) One class may have many teachers
एक कक्षा के कई शिक्षक हो सकते हैं

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(c) One teacher can have many classes
एक शिक्षक की कई कक्षाएँ हो सकती हैं
(d) Many classes may have many teachers
कई कक्षाओं के कई शिक्षक हो सकते हैं

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : A One-to-many Relationship means that one entity can be associated with many other entities. Example- A teacher can teach many classes, but each class can have only one main teacher.

14. What do data warehouses support?
डेटा वेयरहाउस किसे सपोर्ट करता है?

(a) Operational databases/ऑपरेशनल डेटाबेस
(b) OLAP
(c) OLTP
(d) OLAP and OLTP/OLAP और OLTP

KVS PGT C.S. 2017

Ans. (b) : Data warehousing is a process used to collect and manage data from multiple sources into centralized repository to drive actionable business insights. With all your data in one place , it becomes simpler to perform analysis and reporting at different aggregate levels.
A data warehouse includes OLAP (online analytical processing), ETL (extraction transformation and loading) other tools that manages the collection of data and deliver this data to business users. OLAP allow analysts to efficiently answer questions that require data from multiple source systems.

15. _____ is an intermediate storage area used for data processing during the extract, transform and load process of data warehousing.
_____एक मध्यवर्ती भण्डार क्षेत्र है जिसका उपयोग डेटा वेयरहाउसिंग के एक्स्ट्रेक्ट, ट्रांसफार्म और लोड प्रक्रिया के दौरान डेटा प्रोसेसिंग के लिए किया जाता है।

(a) Inter-storage area (b) Buffer
(c) Staging area (d) Virtual memory

KVS PGT C.S. 2017

Ans. (c) : Staging area is an intermediate storage area used for data processing during the extract, transform and load process of data warehousing. staging area is present between data source and destination which is mostly data marts.

16. The number of attributes in a relation is called the _____ of the relation.
एक रिलेशन (Relation) में अट्रिब्यूट्स (Attributes) की संख्या, रिलेशन की _____ कहलाती है।

(a) Degree/डिग्री (b) Domain/डोमेन
(c) Cardinality/कार्डिनेलिटी (d) Tuyples/टपल्स

EMRS PGT C.S. 16.12.2024

Ans. (a) : In a relational database, the number of attributes in a relation is called the degree of the relation.
The degree of the relation is the number of attributes it contains. For example if you have a relation student with attributes 'StudentId', 'Name' and 'Age' the degree of the student relation would be 3.

17. A data model is a collection of conceptual tools for describing
एक डेटा मॉडल, जो कि कॉन्सेपच्यूअल टूलों का समूह होता है, निम्न में से क्या वर्णित करता है?

(a) Data and data relationship/डेटा एवं डेटा सम्बन्ध
(b) Data semantics and consistency constraints
डेटा सिमैन्टिक्स एवं संगति व्यवरोध
(c) Data, data relationship, data semantics and consistency constraints/डेटा, डेटा सम्बन्ध, डेटा सिमैन्टिक्स एवं संगति व्यवरोध
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (c) : In conclusion, a data model is a collection of conceptual tools for describing data, data relationships, data semantics and consistency constraints. It is important in designing a database system that is efficient flexible and easy to maintain.

18. Which of the following is not a record-based data model?
निम्न में से कौन-सा रिकॉर्ड-बेस्ड डेटा मॉडल नहीं है?

(a) Relational model/रिलेशनल मॉडल
(b) Hierarchical model/हाइअरर्चिकल मॉडल
(c) E-R model/E-R मॉडल
(d) Network model/नेटवर्क मॉडल

LT. Grade C.S. 2018

Ans. (c) : An Entity relationship model uses a diagram called an ER diagram to represent the structure of a database. An ER model is a database architecture or blueprint that can be implemented as a database later.

19. Which of the following is a disadvantage of using a distributed database system?
निम्नलिखित में से कौन-सा वितरित डेटाबेस सिस्टम का उपयोग करने का नुकसान है?

(a) Improved data accessibility/बेहतर डेटा पहुँच
(b) Increased data security/बढ़ी हुई डेटा सुरक्षा
(c) Data inconsistency/डेटा असंगतता
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : The major disadvantage of using distributed database systems contain data indifferent locations, which may lead to data inconsistency problems if data synchronization and consistency are not managed properly.

20. Which of the following is not a function of the database?
निम्नलिखित में से कौन डेटाबेस का कार्य नहीं है?

(a) Managing and manipulating stored data
(b) Analysing Code
(c) Security for stored data
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) : Databases are primarily designed to manage and manipulate stored data, and they provide security features to protect that data. While analyzing code is not typically a function of a database.

Sarkari Resource

21. Which of the following is not the utility of DBMS?
निम्नलिखित में से कौन DBMS की उपयोगिता नहीं है?

(a) Backup
(b) Data Loading
(c) Process Organization
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)
BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c) : Database Management Systems (DBMS) usually handle task such as data backup and data loading, but process organization is not the main use of DBMS.
Process organization is not the main use of DBMS. Process organization is usually associated with operating systems and other applications.

22. Which type of data can be stored in the database?
डेटाबेस में किस प्रकार का डेटा संग्रहीत किया जा सकता है?

(a) Image oriented data
(b) Text, files containing data
(c) Data in the form of audio or video
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (d) : The reason for creating a database management system was to store large data and this data can be in any form like image, text, audio or video files etc. DBMS allows users to store and access data in any format.

23. Which of the following is not a feature of DBMS?
निम्नलिखित में से कौन DBMS की विशेषता नहीं है?

(a) Support ACID Property
(b) High Level of Security
(c) Single-user Access only
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : 'Single-user Access only' is not a feature of DBMS. DBMS allows multiple users to access and manipulate the database simultaneously. It provides mechanisms to handle simultaneous access and ensure consistency and integrity of data across multiple users.

24. The user IDs can be added or removed using which of the following fixed roles?
निम्नलिखित में से किस निश्चित भूमिका का उपयोग करके उपयोगकर्ता आईडी को जोड़ा या हटाया जा सकता है?

(a) db_sysadmin
(b) db_accessadmin
(c) db_securityadmin
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : Along with security, as the name suggests, the db_accessadmin role also handles access. db_sysadmin refers to the system administrator. As name suggest, db_securityadmin involves allowing or denying access to data ensuring security.

25. Which of the following is not a feature of DBMS?
निम्नलिखित में से कौन-सी DBMS की विशेषता नहीं है?

(a) Minimum duplication and redundancy of data
(b) Single user access only
(c) Support ACID property
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): Single user access only is not a feature of a DBMS. A DBMS allows multiple users to access and manipulate a database simultaneously. It provides mechanisms to handle concurrent access and ensure data consistency and integrity across multiple users.

26. Which of the following is not a function of the database?
निम्नलिखित में से कौन-सा डेटाबेस का कार्य नहीं है?

(a) Manipulating data
(b) Security for stored data
(c) Analysing code
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(c): It allows authorized users to update, store, manipulate or access data. Since the data is stored in tabular format, it is easy to access the data and perform the required tasks.

27. What is information about data called?
डेटा के बारे में जानाकारी क्या कहलाती है?

(a) Teradata (b) Metadata
(c) Relations
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(b): Meta data means "data about data". Metadata is defined as data that provides information about one or more aspects of data. It is used to summarize basic information about data.

1. Descriptive Metadata	2. Structural Metadata
3. Administrative Metadata	4. Reference Metadata
5. Statistical Metadata	6. Legal Metadata

28. The DBMS acts as an interface between_____ and _____of an enterprise class system.
DBMS एक एंटरप्राइज-क्लास सिस्टम के _____और _____के बीच एक इंटरफेस के रूप में कार्य करता है।

(a) Data, the DBMS
(b) Database application, the database
(c) The user, the software
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

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Ans.(b): The DBMS act as on interface between the database applications and the database of an enterprise class system.

29. What happens if a piece of data is stored in two places in the db?
यदि डेटा का एक टुकड़ा डीबी में दो स्थानों पर संग्रहीत किया जाता है, तो क्या होता है?

(a) Storage space is wasted and changing the data in one spot will cause data inconsistency
(b) Changing the data in one spot will cause data inconsistency
(c) Storage space is wasted
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) : If a piece of data is scheduled in two locations in the storage, the storage space is broken and reserving the data in one location will result in the data being reserved.

30. What is a database?
डेटाबेस क्या है?

(a) Organized collection of information that cannot be accessed, updated and managed
(b) Organized collection of data or information that can be accessed, updated and managed
(c) Organized collection of data that cannot be updated
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : A database is an organized collection of structured information or data, typically stored electronically in a computer system. A database is typically managed by a database management system.

31. Which type of data can be stored in the database?
डेटाबेस में किस प्रकार का डेटा संग्रहीत किया जा सकता है?

(a) Image-oriented data
(b) Text, files containing data
(c) Data in the form of audio or video
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d) : Computer databases are usually organized into data records or files. Databases store aggregated information such as sales transactions, customer data, financial and product information. Databases are used to store, maintain, and access any type of data, including text, data files, audio and video data, and image files.

32. Which of the following is not an example of DBMS?
निम्नलिखित में से कौन-सा DBMS का उदाहरण नहीं है?

(a) Microsoft access
(b) IBM Db2
(c) Google
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(c): Example of DBMS are as follows–
1. MySQL
2. SQL Server
3. Microsoft Access
4. SQL Lite
5. IBM Bb2

33. Which of the component module of DBMS does rearrangement and possible ordering of operations eliminate redundancy in query and use efficient algorithms and indexes during the execution of a query ?

(a) query compiler
(b) query optimizer
(c) Stored data manager
(d) Database processor

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (b) : There are many plans that a database management system can follow to process a query, but it is the amount of time that they need to run is given importance. For this, query optimization is required.
Query optimization examines all algebraic expressions that are equivalent to the one that is estimated to be the cheapest.
* Query optimization procedure either attempt to maximize of resources or to minimize the resource usage for a given output.
* It tries the minimize the response time for a given query language.
* It does rearrangement and possible ordering of operations, eliminate redundancy in query
* In order to allow fair comparison of efficiency, the functional capabilities of the query evaluation system to be compared must be similar.

34. In a Hierachical database, a hashing function is used to locate the _____.

(a) Collision
(b) Root
(c) Foreign Key
(d) Records

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : In a Hierachical database, a hashing function is used to locate the Root.

35. What are the drawbacks of using file systems to store data?
(A) Data inconsistency
(B) Difficulty in accessing data
(C) Data isolation
(D) Atomicity of updates
Choose the correct answer from the options given below :

(a) A, B only
(b) B, C, D only
(c) A, B, C only
(d) A, B, C, D

UGC NET C.S. December-2022

Ans. (d) : Using the file system to store data has several drawbacks.
- Distinguished and Isolated Data.
- Data Duplication/Data redundancy
- Dependence on data
- Data representation is challenging from the user's perspective
- Different file types

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- Data protection
- Issues with Transactions
- Concurrent issues.

So option (d) is correct.

36. Which of the following database model, we have a parent-child relationship?

(A) Hierarchical databases
(B) Network databases
(C) Relational databases

Choose the correct answer from the options below :

(a) A, B (b) A, C
(c) B, C (d) A, B, C

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Ans. (a) : Hierarchical databases have a parent - child relationship. In a Hierarchical databases model, data is organized in a tree-like structure where each record or data element has a parent and can have multiple children.
Network database is similar to hierarchical databases but allow more flexible relationships between records.
So option (a) is correct.

37. _____ refers loosely to the process of set automatically analyzing large databases find useful patterns

(a) Datamining (b) Data warehousing
(c) DBMS (d) Data mirroring

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) The practice of examining large pre-existing databases in order to generate new information is data mining. It refers loosely to the process of semi-automatically analyzing large database to find useful patterns.

38. Which of the following statement regarding the features of the object-oriented approach to databases are true?

(1) The ability to develop more realistic models of the real world
(2) The ability to represent the world in a non-geometric way
(3) The ability to develop databases using natural language approaches
(4) The need to split objects into their component parts
(5) The ability to develop database models based on location rather than state and behavior

Codes :

(a) (1), (2) and (3) (b) (2), (3) and (4)
(c) (1), (4) and (5) (d) (3), (4) and (5)

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) The following statements are true. The ability to develop more realistic models of the real world (a)

- The ability to represent the world in a non geometric way (b)
- The ability to develop databases using natural language approach (c)

Hence answer is (a)

37. _____ allows selection of the relevant information necessary for the data warehouse.

(a) The Top – Down view
(b) Data warehouse view
(c) Data source view
(d) Business query view

UGC NET C.S. December-2015 (Paper-II)

Ans : (a) Four views regarding the design of a data warehouse:

- **Top-down view:** Allows selection of the relevant information necessary for the data warehouse.
- **Data source view:** Exposes the information being captured, stored and managed by operational systems.
- **Data warehouse view:** Consists of fact tables and dimension tables.
- **Business query view :** Sees the perspectives of data in the warehouse from the view of end-user.

Top-down view allows selection of the relevant information necessary for the data warehouse.

38. Data which improves the performance and accessibility of the database are called :

(a) Indexes (b) User data
(c) Application metadata (d) Data dictionary

UGC NET C.S. December-2015 (Paper-II)

Ans : (a) Indexes are special lookup tables that the database search engine can use to speed up data retrieval. Simply put, an index is a pointer to data in a table. An indexed in a database is very similar to an index in the back of the book.

39. Database applications were built directly on top of file system to overcome the following drawbacks of using file-systems :

(1) Data redundancy and inconsistency
(2) Difficulty in accessing Data
(3) Integrity problems

(a) (1) (b) (1) and (4)
(c) (1), (2) and (3) (d) (1), (2), (3) and (4)

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : Traditional file system has all mentioned drawbacks. Database applications built by using tables are used to overcome them.
eg : Normalization and integrity constraints reduces redundancy & inconsistency.
Select * from table provide easy access to data references, joins etc removes data is action primary & foreign key along with check predicates solves integrity problem

40. A_____ index is an ordered file whose records are of fixed length with two fields primary key and a pointer to a disk block.

(a) Secondary (b) Multilevel
(c) Primary (d) Clustering

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : In database, a primary index is built on the primary key ensure each record is uniquely identified.
Example– In a table of employees, the employee_id column (acting as the primary key) could be the basis of a primary index.

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41. Which of the following data mining algorithms is used for association rule mining?
निम्नलिखित में से किस डेटा खनन एल्गोरिद् का उपयोग डेटा एसोसिएशन रूल माइनिंग के लिए किया जाता है?

(a) Decision tree/डिसीजन ट्री
(b) Naive Bayes/नाईव वेज
(c) Apriori/अप्रियोरी
(d) K-means/K-मिंस

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Apriori is the data mining algorithm specifically designed for association rule mining. Decision tree algorithms are primarily used for classification and regression tasks, not for association rule mining. Naïve Bayes is a classification algorithm commonly used for predictive modeling and text classification k-means is a clustering algorithm used to group data into clusters based on similarity.

42. With respect to data warehouse, OLAP stands for:
डेटा वेयरहाउस के संबंध में, OLAP का अर्थ क्या है?

(a) Overlapping Analytical Processing
ओवरलैपिंग एनालिटिक्ल प्रोसेसिंग
(b) Online Analytical Processing
ऑनलाइन एनालिटिक्ल प्रोसेसिंग
(c) Online Augmented Processing
ऑनलाइन ऑगमेंटेड प्रोसेसिंग
(d) Offline Analytical Processing
ऑफलाइन एनालिटिक्ल प्रोसेसिंग

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : In data warehousing, OLAP stands for Online Analytical Processing. This technology allows users to analyze large volumes of multidimensional data in real time, providing insights for business intelligence and decision making.

43. With respect to data warehouse, which of the following is not a valid type of OLAP technology?
डेटा वेयरहाउस के संबंध में, निम्नलिखित में से कौन सा OLAP तकनीक का मान्य प्रकार नहीं है?

(a) POLAP (b) MOLAP
(c) ROLAP (d) HOLAP

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In data warehouse, The type 'POLAP' is not a valid OLAP technology, the recognized types which are valid are MOLAP (Multidimensional OLAP), ROLAP (Relational OLAP), and HOLAP (Hybrid OLAP).

44. Which of the following statement(s) is/are correct regarding Data Dictionary in Database Management System?
डाटाबेस मैनेजमेंट सिस्टम में डाटा डिक्शनरी के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. Data Dictionary is a collection of definitions, rules and advisories of data designed to be used as a reference with the data warehouse.
डेटा डिक्शनरी डेटा की परिभाषाओं, नियमों और एडवाईसोरिस का एक एंग्रह है जिसे डेटा वेयरहाउस के संदर्भ के रूप में उपयोग करने के लिए डिजाइन किया गया है।

II. Data Dictionary is the area of memory allocated for data storage.
डेटा डिक्शनरी डेटा स्टोरेज के लिए आवंटित मेमोरी का क्षेत्र है।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Statement I- Correct.
A Data Dictionary is a centralized repository of information about data such as meaning, relationship to other data, origin, usage and format. It is used as a reference to manage and maintain the metadata of a database or data warehouse.
Statement II- Incorrect.
The Data Dictionary is not an area of memory allocated for data storage. That refers more to buffer memory or physical storage. The data dictionary stores metadata, not the actual data.

45. What is Customer Relationship Management?
ग्राहक संबंध प्रबंधन क्या है?

(a) Customer relationship management includes the methodologies, technology and capabilities that help an enterprise to manage customer relationships.
ग्राहक संबंध प्रबंधन में कार्यप्रणाली,प्रौद्योगिकी और क्षमताएं शामिल हैं जो एक उद्यम को ग्राहक संबंधी को प्रबंधित करने में मदद करती है।
(b) Customer relationship management is a corporate level strategy which focuses on creating new customers.
ग्राहक संबंध प्रबंधन एक कॉरपोरेट स्तर की रणनीति है जो नए ग्राहक बनाने पर केंद्रित है।
(c) Customer relationship management is the process of planning, implementation and controlling the operations of the supply chain.
ग्राहक संबंध प्रबंधन आपूर्ति श्रृंखला के संचालन की योजना, कार्यान्वयन और नियंत्रण की प्रक्रिया है।
(d) Customer relationship management is the process of daily production and distribution.
ग्राहक संबंध प्रबंधन दैनिक उत्पादन और वितरण की प्रक्रिया है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Customer Relationship Management (CRM) is a strategy and system that companies use to mange interactions with current and potential customers. It involves using data and technology to improve business relationships, streamline processes and enhance customer satisfaction and retention.

46. Which of the following is NOT an element of Database?
निम्नलिखित में से कौन डेटाबेस का एक तत्व नहीं है?

(a) Schema/स्कीमा
(b) Relationships/रिलेशनशिप्स
(c) Constraints/कंस्ट्रेंट्स
(d) Frontend/फ्रंटएंड

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Sarkari Resource

Ans. (d) : "Frontend" refers the user interface or client-side application not a core element a database. The other options- Schema, Relationships and Constraints are fundamental components of a database structure.

46. Primary index in sequential order file organization is also known as_____.
सिक्वेंशियल ऑर्डर फाइल में प्राइमरी इंडेक्स को_____के रुप में भी जान जाता है ।

(a) Dense index /डेंस इंडेक्स
(b) Clustering index /कलस्टरिंग इंडेक्स
(c) Non-clustering index /नॉन-कलस्टरिंग इंडेक्स
(d) Sparse index /स्पार्स इंडेक्स

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : A primary index in a sequentially ordered file is also known as a clustering index. This is because the data records in the file are physically sorted on the same field that the primary index is based on, meaning the ordering of the data is "Clustered" around the index's search keys.

47. Deductive model of DBMS is also known as _____.
डी.बी.एम.एस. के डिडक्टिव मॉडल को _____ के रुप में भी जान जाता है ।

(a) Object-oriented model
ऑबजेक्ट-ओरियेंटिड मॉडल
(b) Inference model /इंफ्रेंस मॉडल
(c) Interactive model /इंट्रेक्टिव मॉडल
(d) Differential model /डिफ्रेंशियल मॉडल

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : The deductive model of a Database Management System (DBMS) is also known as the inference model or a deductive database system.
A deductive (relational) database consists of an extensional database (the facts) and an intentional database (the rules) notes.

48. _____ is a system database that contains description of the data in the database.
_____एक सिस्टम डेटाबेस है जिसमें डेटाबेस में डेटा का विवरण होता है ।

(a) Data Catalog /डेटा कैटॉलौग
(b) Information Catalog /इंफौरमेशन कैटॉलौग
(c) Record Catalog /रिकॉर्ड कैटॉलौग
(d) File Catalog /फाइल कैटॉलौग

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : Data catalog is a system database that contains description of the data in the database. It serves as an organized inventory of data assets using metadata.

49. Which of the following statement(s) is/are correct regarding a relation in second normal form?
द्वितीय नॉर्मल फॉर्म में एक रिलेशन के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं ?

I. It is in first normal form
यह प्रथम नॉर्मल फॉर्म में होता है

II. Every non-key attribute is fully and functionally independent on the primary key/प्रत्येक गैर-की एट्रिब्यूट प्राथमिक की पर पूरी तरह से और कार्यात्मक रुप से स्वतंत्रा होता है

(a) Only I /केवल I
(b) Only II /केवल II
(c) Both I and II / I तथा II दोनो
(d) Neither I nor II /ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : It is a way to organize a database table so that it reduces redundancy and ensures data consistency. In 2NF form the table must be in first normal form (1 NF) and it should be fully functionally dependency. It means a non-key attribute depends on the entire primary key, not just part of it.

50. Which of the following is oldest database model?
निम्नलिखित में से कौन सा सबसे पुराना डेटाबेस मोडल है?

(a) Deductive/डिडक्टिव
(b) Relational/रिलेशनल
(c) Object–Oriented/ऑबजेक्ट ओरिएंटेड
(d) Hierarchical/हरैरिकल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : The Hierarchical model was developed in the 1960s by IBM and organizes data in a tree like structure with parent child relationships. It was widely used in early mainframe system such as IBM's IMS. Therefore the oldest database model is hierarchical model.

51. What is the full form of DDLC in database?
डेटाबसे में डी.डी.एल.सी. का असंक्षिप्त रूप क्या है?

(a) Distributed Database Log Clearing
डिस्ट्रीब्यूशन डाटाबेस लॉग क्लियरिंग
(b) Distributed Database Life Cycle
डिस्ट्रीब्यूटिड डाटाबेस लाइफ सांइकिल
(c) Database Development Life Cycle
डाटाबेस डेवलपमेंट लाइफ सांइकिल
(d) Deductive Database Life Cycle
डिडकटिव डाटाबेस लाईफ साइकिल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : The full form of DDLC in the context of database is "Database Development Life Cycle" The DDLC refers to the structured process of designing implementing a database in a systematic sequence, similar to what occurs in the Software Development Life Cycle (SDLC).

52. _____is used for management, maintenance and retrieval of the data stored in the database.
_____का उपयोग डेटाबेस में संग्रहीत डेटा के प्रबंधन रखरखाव और पुन:प्राप्ति के लिए किया जाता है।

(a) Operating system/ऑपरेटिंग सिस्टम
(b) Database Management System
डेटाबेस मैनेजमेंट सिस्टम
(c) System Software/सिस्टम सॉफ्टवेयर
(d) Processing Software/प्रोसेसिंग सॉफ्टवेयर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : A Database Management System is a software application designed specifically to manage databases. This includes functions like storing, retrieving, updating, and deleting data. It also ensures data integrity, security and consistency. A DBMS allows users to organize data into tables and relationships, making it easier to access and manipulates.

Sarkari Resource

53. Which of the following statement(s) is/are not true about magnetic tapes?
निम्नलिखित में से कौन सा/से कथन मैग्नेटिव टेप्स के बारे में सत्य नही है?

I. Direct access device/डायरेक्ट एक्सेस डिवाइस
II. Store large volume of data
बड़ी मात्रा में डेटा स्टोर करना
III. Use for backup purpose
बैकअप उद्देश्य के लिए उपयोग करना

(a) II and III (b) I and III
(c) Only I (d) Only II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : Magnetic tapes– In magnetic tape only one side of the ribbon is used for storing data it is sequential memory which contains thin plastic ribbon to store data and coated by magnetic oxide data read/write speed is slower because of sequential access. It is highly reliable which requires magnetic tape drive writing and reading data.
The width of the ribbon varies from 4mm to 1 inch and it has storage capacity 100MB to 200GB

Advantage	Disadvantage
(1) These are in expensive low cost memories (2) It provides backup or archival storage (3) It can be used for large files (4) It can be used for copying disc files. (5) It is a reusable memory (6) It is compact and easy to store on racks.	(1) Sequential access is the disadvantage, means it does not allow access randomly or directly (2) It requires caring to store vulnerable humidity dust free, and suitable environment (3) It stored data cannot be easily updated or modified difficult to make updates on data.

54. Which of the following is not key attribute of information?
निम्नलिखित में से कौनसा सूचना की प्रमुख विशेषता नहीं है?

(a) Accuracy/सटीकता
(b) Timeliness/समयबद्धता
(c) Relevancy/प्रासंगिकता
(d) Isolation/अलगाव

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Accuracy, timeliness, and relevancy are all considered key attributes of valuable information. They ensure information is correct, up-to date, and relevant to the user's needs. "Isolation", on the other hand, is not a recognized attribute of information quality. It might refer to keeping data separate for security reasons, but this is not a characteristic that makes information useful for decision-making.

55. What is the full form of DASD in database management system?
डेटाबेस प्रबंधन प्रणाली में DASD का असंक्षिप्त रूप क्या है ?

(a) Direct Access Storage Device
डायरेक्ट एक्सेस स्टोरेज डिवाइस
(b) Dynamic Access Storage Device
डायनामिक एक्सेस स्टोरेज डिवाइस
(c) Digital Access Storage Device
डिजिटल एक्सेस स्टोरेज डिवाइस
(d) Disk Access Storage Device
डिस्क एक्सेस स्टोरेज डिवाइस

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : DASD stands for Direct Access Storage Device, which refers to a storage system that allows direct access to data, rather than sequential access (like tape drives). Example Include Hard Disk Drives (HDDs) and Solid State Drives (SSDs).

56. The degree of a relation in database is

(a) The number of tuples in the relation
(b) The number of attribute in the relation
(c) The number of values in domain
(d) The number of keys in a relation

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Ans. (b) : The degree of a relation in a database refers to the number of attributes (columns) present in the relation.

57. _____ is known as the acquisition, storage, organization, retrieval, display and dissemination of information.
_____ को इंफॉर्मेशन के अधिग्रहण, भंडारण, संगठन, पुनर्प्राप्ति, प्रदर्शन तथा प्रसार के रूप में जाना जाता है।

(a) Data management/डेटा मैनेजमेंट
(b) Information processing/इंफॉर्मेशन प्रोसेसिंग
(c) Data structure/डेटा स्ट्रक्चर
(d) Programming management/प्रोग्रामिंग मैनेजमेंट

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Information Processing– Information Processing the acquisition, recording, organization of retrieval, display, and dissemination of information in recent years, the term has often been applied to computer-based operations specifically.
(1) Acquisition– refers to obtaining raw data from various sources.
(2) Storage– is the process of securely storing data for future use.
(3) Organization– involves structuring data to make it easily accessible and searchable.
(4) Retrieval– is the process of accessing and finding stored data.
(5) Display– is the presentation of data in a user friendly format, often through visualization.
(6) Dissemination– involves sharing data with relevant parties or stakeholders.

58. In Data mining, _____ is a method of incremental conceptual clustering.

(a) STRING (b) COBWEB
(c) CORBA (d) OLAD

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) COBWEB is an in rental system for hierarchal conceptual clustering. COBWEB incrementally organizes observation into a classification tree. Each node in classification tree represents a class (concept) and is labeled by a probalistic concept that summarizes the attribute value distribution of objects classified undo the node.

Sarkari Resource

59. _____ is subject oriented, integrated, time variant, nonvolatile collection of data in support of management decisions
(a) Data mining
(b) Web mining
(c) Data warehouse
(d) Database management system

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (c) Data warehouse is a subject oriented integrated, time variant and non volatile collection of data in support of management decision making process. A data warehouse is a copy of transaction data specifically structured for query & analysis.

60. Given the following statements:
S1: Strategic value of data mining is time stamping.
S2: Information collection is an expensive process in building an expert system.
Which of the following options is correct?
(a) Both S1 and S2 are false.
(b) Both S1 and S2 are true.
(c) S1 is true, S2 is false.
(d) S1 is false, S2 is true.

UGC NET C.S. June-2016 (Paper-II)

Ans : (b)
- Strategic value of data mining is time stamping → correct
- Information collection is an expensive process in building an expert system → correct
- An expert system is a computer system that emulates the decision making ability of a human expert.

61. In relational database, the data type of values in each column of a table is called
रिलेशनल डेटाबेस में, किसी टेबल के हरेक कॉलम के वैल्यू के डेटा टाइप को क्या कहा जाता है?
(a) Attribute/ऐट्रिब्यूट
(b) Domain/डोमेन
(c) Relation/रिलेशन
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : The data type of the values of each column of a table in a database is called the domain. The domain is the set of possible values of data, which is constant for a specific data type. It is called the set of all those values.

62. In relational database, the number of attributes of a relation is called the _____ of relation.
रिलेशनल डेटाबेस में, किसी रिलेशन के ऐट्रिब्यूटों की संख्या को रिलेशन का _____ कहा जाता है।
(a) Domain/डोमेन
(b) Degree/डिग्री
(c) Cardinality/कार्डिनलिटी
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : In a relational database, the number of attributes of a relation is called the degree of the relation. It is also called the degree of the relation which tells how many attributes are there in the relation.

63. Which of the following is not a feature of DBMS?
निम्नलिखित में से कौन-सा DBMS की विशेषता नहीं है?
(a) Minimum duplication and redundancy of data
डाटा का न्यूनतम दोहराव और अतिरेक
(b) High-level of security/सुरक्षा का उच्च स्तर
(c) Single-user access only
केवल एकल-उपयोगकर्ता पहुँच
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : The full form of DBMS is Database Management System. It is a software, with the help of which database is created, deleted, updated and managed. The following are the features of database–
1. Data is stored in a systematic manner in the database.
2. The security and privacy of data in the database is managed at a high level.
3. There is no duplicate of any data in the database, that is the database reduces redundancy.
4. Multiple users can access the data simultaneously
5. More data can be stored in less space in the database.

64. Which of the following is/are the open-some source software(s)?
निम्नलिखित में से कौन सा/से सॉटवेयर है/हैं?
I. PHP/पीएचपी
II. Apache/अपाचे
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Apache is a free-open source and cross platform web server software that delivers web pages and content to users over the Internet. It is developed & maintained by the Apache software foundation.

65. _____ describes the process of deciding about putting data to different sites in distributed database system.
_____ वितरित डेटाबेस सिस्टम में विभिन्न साइटों पर डेटा को डालने के बारे में निर्णय लेने की प्रक्रिया का वर्णन करता है।
(a) Data sharing/डेटा शेयरिंग
(b) Data allocation/डेटा एलोकेशन
(c) Data storing/डेटा स्टोरिंग
(d) Data mining/डेटा माइनिंग

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Data allocation describes the process of deciding about putting data to different sites in a distributed database system. It involves deciding how to distributing and place data fragments or replicas across multiple sites to optimize database performance availability and reliability. This crucial step impacts how fast and efficiently queries are processed and how well the database adapts to workload changes.

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66. **_____ is a technique of breaking up database into logical units, which may be assigned for storage at the various sites in distributed database system.**
_____ डेटाबेस को लॉजिकल इकाइयों जिसे वितरित डेटाबेस सिस्टम में निभिन्न साइटों पर स्टोरेज के लिए नियुक्त किया जा सकता है, में तोड़ने की एक तकनीक है।
(a) Data normalization/डेटा नॉर्मलाइजेशन
(b) Data mining/डेटा माइनिंग
(c) Data binding/डेटा बाइंडिंग
(d) Data fragmentation/डेटा फ्रैगमेंटेशन

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : Data fragmentation is a technique of breaking up database into logical units, which may be assigned for storage at the various sites in distributed database system.

67. **Which of the following is a component of Management Information System (MIS)?**
निम्नलिखित में सो कौन सा मैनेजमैंट इनपोर्मेशन सिस्टम (एमआईएश) का एक घटक है?
(a) External database/बाह्य डेटाबेस
(b) Knowledge engine/नौलेज़ इंजान
(c) Data entry/डेटा ऐंट्री
(d) Inquiry processing/इंकाईरी प्रोसेसिंग

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : External database is a core part of an MIS as it serves as a source of external data that can be integrated into the MIS structure for managerial analysis and decision making.

68. **In distributed database, location transparency allows for database users, programmers and administrators to treat the data as if it is at one location. A SQL query with location transparency needs to specify :**
(a) Inheritances (b) Fragments
(c) Locations (d) Local formats

UGC NET C.S. June-2016 (Paper-III)

Ans : (b) Fragmentation transparency is the highest level of transparency. The end user or programmer does not need to know that a database is partitioned. Therefore, neither prior to data access.

69. **A Database Management System (DBMS) is :**
एक डेटाबेस प्रबंधन प्रणाली (DBMS) है:
(a) Collection of interrelated data
परस्पर संबंधित डेटा का संग्रह
(b) Collection of programs to access data
डेटा तक पहुँचने के लिए कार्यक्रमों का संग्रह
(c) Collection of data describing one particular enterprise/एक विशेष उद्यम का वर्णन करने वाला डेटा का संग्रह
(d) All of the three/तीनों में सभी

HTET PGT C.S. 2020

Ans. (d) : A Database Management System (DBMS) is software that functions as an interface between users/applications and a database, enabling users to create store, retrieve, and manage data efficiently and in a structured way. It provides tools for data definitions, manipulation, and control ensuring data integrity, security, and concurrent access for multiple users. Key benefits include reduced data redundancy and inconsistency, improved data security, and enhanced performance through features like performance tuning and data recovery.

70. **A _____ operation combines the rows of the two relations and outputs a new relation that has both input relations rows in it.**
(a) Difference (b) Union
(c) Cartesian product (d) Project

UGC NET C.S. December-2022

Ans. (b) : The operation that combines the rows of two relations and outputs a new relation that has both input relation's rows in its is the "Union operation.
So the option (b) is correct.

71. **Intersection operation on two database relation is :**
दो डाटाबेस रिलेशन पर इंटर-सेक्शन ऑपरेशन है:
(a) Associative but not Commutative
सहचयं परन्तु विनिमयशील नहीं
(b) Commutative but not Associative
विनिमयशील परन्तु सहचर्य नहीं
(c) Associative and Commutative
सहचर्य और चिनिमयशील
(d) Neither Associative nor Commutative
न ही सहचर्य एवं न ही विनिमयशील

HTET PGT C.S. 2018

Ans. (c) : The intersection (n) operation in database relations returns only those tuples that are present in both relations, it has two important properties. First, it is commutative, which means the order of relations does not matter : RNS = SNR, second, it is associative, which means the grouping of operations does not affect the result: (RNS) NT = RN (SNT), for example, if Relations A = {1, 2, 3} and Relation B = {2, 3, 4} then ANB = {2, 3}. Hence intersection is both associative and commutative.

72. **In mass storage devices _____is process of rearranging/relocating many small non-contiguous blocks of data, into fewer contiguous big-sized blocks.**
वृहद् सटोरेज युक्तियों में _____ कई छोटे असतत डाटा ब्लॉक्स को, कुछ सतत बड़े आकार के डाटा ब्लॉक्स में पुनर्व्यवस्थित/पुनर्स्थापित करने की प्रक्रिया है।
(a) Defragmentation/डीफ्रेगमेंटेशन
(b) Paging/पेजिंग
(c) Booting/बूटिंग
(d) Restoring/रेस्टोरिंग

HTET PGT C.S. 2019

Ans. (a) : Defragmentation– It is a process that reduces the degree of fragmentation. it does this by physically organizing the contents of the mass storage device used to store files in to the smallest number of contiguous regions.
Paging– It is a memory management scheme by which a computer stores and retrieves data from secondary storage for use in the main memory.

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Booting– It is a startup sequence that starts the operating system of a computer when it is turned on. A boot sequence is the initial set of operations that the computer performs when it is switched on every computer has a boot sequence.
Data store– It is the process of copying backup data from secondary storage and restoring it to its original location or a new location. A restore is performed to return data that has been lost, stolen or damaged to its original condition or to move data to a new location.

73. To get information from users you use_____.
उपयोगकर्त्ताओं से जानकारी प्राप्त करने के लिए आप उपयोग में लेते हैं–
(a) Tables/टेबल्स (b) Queries/क्वेरीज़
(c) Forms/फॉर्म्स (d) Data/डेटा

HTET PGT C.S. 2023

Ans. (c) : Forms are the primary way to get information from users in applications, especially in web and database contexts, they provide input field, buttons and other controls for users to submit data.

74. A collection of concepts that can be used to describe the structure of database and provides the necessary means to achieve this abstraction:
कंसेप्ट्स का वह संग्रह जिसका उपयोग डेटाबेस की संरचना के वर्णन के लिए एवं एब्सट्रेक्शन प्राप्त करने के लिए किया जाता है–
(a) Data Model/डेटा मॉडल
(b) Relationship Model/रिलेशनशिप मॉडल
(c) Query Model/क्वेरी मॉडल
(d) Form Model/फॉर्म मॉडल

HTET PGT C.S. 2023

Ans. (a) : A data model is a collection of concepts used to describe the structure of a database and provides the necessary abstraction for hiding the specifics of data storage presenting users and designers with an organized and simplified logical view of the data.

75. The structure of data files is stored in:
(a) Metadata
(b) Database catalog
(c) Database schema
(d) Data model

CGPSC C.S. 2020 Set-A

Ans. (c) : Metadata → Describes data about data (like data type, size, creation time).
Database Catalog → Contain meta data about the database objects.
Database Schema → Defines the structure of the data files, i.e. how tables, fields relationships, constraints, etc. are organized.
Data model → Provides the abstract design/blue print (like relational, hierarchical network).
So the actual structure of data in files is defines by the database schema.

76. In which of the following records are maintained in the logical sequence of their primary key values?
निम्न में से किसमें रिकार्ड को उनकी प्राइमरी कुंजी मानों के लॉजिकल क्रम में व्यवस्थित किया जाता है?
(a) Index sequential file/इंडेक्स सिक्वेंशियल फाइल
(b) Sequential file/सिक्वेंशियल फाइल
(c) Relative file/रिलेटिव फाइल
(d) Direct file/डायरेक्ट फाइल

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (b) : Records are maintained in the logical sequence of their primary key values in a sequencial file. In sequential file organization, records are stored one after another, sorted based on the value of a primary key, ensuring that their order corresponding to the logical progression of those keys.

(ii) DBMS Architecture, Database Users and Data Independence

77. If the following functional dependencies holds on a relation R : A → B, A → C, BC → D, C → F, B → E, then which of the following dependency does not hold on R ?
यदि एक रिलेशन R पर निम्न कार्यात्मक निर्भरताएँ मौजूद है : A → B, A → C, BC → D, C → F, B → E, तब निम्न में से कौन-सी कार्यात्मक निर्भरता R पर मौजूद नहीं है?
(a) A → DE (b) BC → DEF
(c) A → EF (d) BD → EF

HTET PGT C.S. 2022

Ans. (d) :
(a) Check A → DE
Since A → D (from A → BC and BC → D)
And A → E (from A → B and B → E), A → DE holds
(b) BC → DEF; BC → D is given, and from BC, we can infer C, thus BC → F, and B → E, so BC → E holds.
Thus, BC → DEF holds.
(c) A → EF, we've established A → E (from A → B & B → E) and A → F (from A → C and C → F), so A → EF holds.
(d) BD → EF, for BD → EF to held, BD must determine both E and F. B → E is given, so BD → E holds. For F, since C → F, BD would need to determine C for BD → F to hold, which is not given directly.

78. Consider a relation R(A, B, C). Which of the following statement is not true according to inference rules for functional dependencies ?
एक रिलेशन R(A, B, C) पर विचार करें। फंक्शनल डिपेन्डेंसीज (कार्यात्मक निर्भरता) के इन्फेरेंस रूल्स (निष्कर्ष नियमों) के अनुसार निम्न में से कौन-सा कथन सत्य नहीं है?
(a) If A → B and B → C, then A → C
(b) If AB → C, then A → B and B → C
(c) If A → B and A → C, then A → BC
(d) If A → B, then AC → BC

HTET PGT C.S. 2018

Ans. (b) : (b) If AB → C we cannot inter A → B or B → C havings AB determine C gives no guarantee that A alone determines B or that B alone determines C.
Hence, not true.

Sarkari Resource

(a) If A → B and B → C then A → C this is Transitive. Hence true.
(C) If A → B and A → C then A → BC
This is union (Addetive) there true.
(d) If A → B then AC → BC
By and mentation, adding the same attribute C to both sides given AC → BC
Hence true.

79. Given an instance of the relation R(ABCD).
नीचे रिलेशन R(ABCD) दिया गया है।

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_2	c_2	d_2
a_2	b_2	c_2	d_3
a_3	b_3	c_4	d_3

Which of the following functional dependencies hold?
निम्नलिखित में से कौन-सी फंक्शनल डिपेन्डेन्सी बनी रहेंगी?

(a) {A B} → D and D → A
{A B} → D तथा D → A
(b) {A B → C} and B → D
{A B → C} तथा B → D
(c) {A B} → C and B → C
{A B} → C तथा B → C
(d) {A B} → D and A → D
{A B} → D तथा A → D

KVS PGT C.S. 2018

Ans, (c) : A functional dependency (FD) is a relationship between two attributes, typically between the primary key and non-key attributes within a table. Rules of defining functional dependencies (FD), If LHS of FD are all different, then we can directly say that FD holds true, If LHS of FD are same, then check the corresponding value of RHS, that should be same for all the matching value of LHS, otherwise FD will not hold true.
In option (c) {A B} → C and B → C
AB → C hold true. (As all the value of AB are different)
and B → C, in B all the values are not same.
So, we have to check value of C corresponds to b_2. Both the values in C are same for b_2.
Hence, B → C holds true.

80. Given a relational schema R(ABCDEFGH) in first normal form. For the set of dependencies F = {A→B, A→C, CG→H, B→H, G→F}, which dependency is logically implied?
फर्स्ट नॉर्मल फार्म में एक रिलेशन स्कीमा R(ABCDEFGH) दिया गया है। डिपेन्डेन्सी F = {A→B, A→C, CG→H, B→H, G→F}, के समुच्चय के लिए कौन-सी डिपेन्डेन्सी तर्कसंगत रूप से अन्तर्निहित है?

(a) AC→H (b) C→H
(c) G→H (d) A→H

KVS PGT C.S. 2018

Ans, (d) : Given, Relational schema R(ABC DEFGH) in first Normal form (1NF).
Given set of dependencies-
F = {A →B, A→C, CG→H, B→H, G →F},
if A → B
B → H
Then we can say that A → H which is logically implied.
So option (d) is the correct answer.

81. Let R (A,B, C,D,F) be a relational schema with following functional dependencies:
C→ F, E→ A, EC→ D, A→B
Which of the following is a key for R.
(a) CD (b) EC
(c) AE (d) AC

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Ans. (b) : To find a key for relation scheme R, we need to identify a minimal set of attributes that uniquely determine all other attributes in the reason.
Given the dependencies-
C → F
E → A
EC → D
A → B
We can start with attribute sets, and check if they satisfy the closure property to include all attributes.
Now, Let's try {E, C}:
$\{E, C\}^+ = \{C, E, A, F\}$
(from C → F and E → A)
It includes all attributes, So {E, C} is a candidate key.
We can verify that no subset of {E, C} uniquely determines all attributes, Hence, {E, C} is the key for relation scheme R.

82. For a database relation R(a, b, c, d) where the domains of a, b, c and d include only atomic values, and only the following functional dependencies and those that can be inferred from them hold :
a → c
b → d
The relation is in _____.
(a) First normal form but not in second normal form
(b) Second normal form but not in third normal form
(c) Third normal form
(d) BCNF

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : We can not inferred extra dependencies and transitive dependencies so key for a → c & b → d relation will be- ab
$ab^+ = \{a, b, c, d\}$, So, ab will be the key for this relation.
So, BCNF is says left hand side should be super key.
So, option (d) is wrong.
Second NF says that their should be not any partial dependencies i.e. PA → Non-key.
Drive attributes → Non-key.
So, option (b) is also wrong.
So, option (a) is correct.

Sarkari Resource

83. **In a schema R(A, B, C, D, E, F, G, H), each field of R contains only atomic values. F = {CH → G, A → BC, B → CFH, E → A,F → EG} is a set of functional dependencies so that F + is exactly the set of FDs that holds R. The relation R is :**

(a) In 1NF, but not in 2NF
(b) In 2NF, but not in 3NF
(c) In 3NF, but not in BCNF
(d) In BCNF

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Ans. (a) : Candidate keys are AD, BD, ED and FD. Partial dependency exists A → BC, B → CFH and F → EG etc. In the following FDs.
For example partial dependency A → C exists in A → BC and B → C and B → H in B → CFH Etc. So, given relation is in 1NF, but not in 2NF.

84. **Which of the following is a top-down approach in which the entity's higher level can be divided into two lower sub-entities?**
वह कौन-सा टॉप-डाउन दृष्टिकोण है जिसमें उच्च स्तर की एंटिटी को दो निम्न स्तरीय उप-एंटिटीज़ में विभाजित किया जा सकता है?

(a) Aggregation/एग्रीगेशन
(b) Generalization/जनरलाइजेशन
(c) Specialization/स्पेशलाइजेशन
(d) All of the above/उपरोक्त सभी

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Specialization – It is a top-down approach where a higher-level entity is divided into lower level Sub-entities.

85. **The top level of the hierarchy consists of_____each of which can contain_____**
पदानुक्रम के शीर्ष स्तर में_____होते हैं, जिनमें से प्रत्येक में_____हो सकता है।

(a) Schemas, catalogs
(b) Schemas, environment
(c) Catelogs, schemas
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): The top level of the hierarchy contains catalogs, each of which can contain a schema.
- **Catalog:**
- A catalog is the highest -level container
- It can hold multiple schemas.
- **Schema:**
- A schema is a nlogical collection of database objects such as tables views, index procedure etc.

Top level → Catalogs → Schemas → Tables/ Objects

86. **Which of the following is not a tier in three-tier architecture of data warehouse?**

(a) Data warehouse server
(b) OLE-DB
(c) OLAP server
(d) Front end client tool

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : A three-tier architecture in a data warehouse typically consists of-
1. **Bottom tier**– Data warehouse server, where data is loaded, cleansed, transformed, and stored.
2. **Middle tier**– OLAP server, provides multidimensional analysis of data.
3. **Top tier**– Front end client tools, used for querying, reporting and data analysis.

OLE-DB (Object Linking and Embedding, Database), is not tier, rather, it is a middleware/API used for accessing data from various sources, often used within the tiers for communication or data retrieval.

87. **Conceptual level, Internal level and External level are three components of the three- level RDBMS architecture. Which of the following is not part of the conceptual level?**
संकल्पनात्मक स्तर, आंतरिक स्तर और बह्य स्तर त्रि-स्तरीय आर डी बी एम एस संरचना के तीन घटक हैं। निम्नलिखित में से कौन सा संकल्पनात्मक स्तर का भाग नहीं है?

(a) Storage dependent details
भण्डार निर्भरता विवरण
(b) Entities, attributes, relationships
सत्ताएं (एन्टिटीस), गुण (एट्रिब्यूट्स), संबंध
(c) Constraints/प्रतिबंध
(d) Semantic information/सीमेन्टिक सूचना

KVS PGT C.S. 2017

Ans. (a) : The conceptual schema describes the design of a database at the conceptual level. conceptual level is also known as logical level. The conceptual level describes what data are to be stored in the database and also describes what relationship exists among those data. such as Entities, attributes, constraints, semantic information etc. while storage dependent details is not part of the conceptual level.

88. **If a function does not modify the database, the purity level is**
यदि एक फलन, डेटाबेस का परिवर्तन नहीं करता है, तो इसका शुद्धता-स्तर है

(a) RNDS (b) WNPS
(c) RNPS (d) WNDS

LT. Grade C.S. 2018

Ans. (d) : If a function does not modify the database, the purity level is WNDS. It covered code does not insert, update, delete or otherwise change data in the database.

89. **Which of the following is NOT a type of DBMS?**
निम्नलिखित में से कौन-सा DBMS का प्रकार नहीं है?

(a) Relational DBMS
(b) Object-Oriented DBMS
(c) Sequential DBMS
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Sarkari Resource

Ans. (c) **:** Sequential DBMS is not a recognized DBMS type. Common types of DBMS are relational, object oriented and others such as hybrid DBMS.

90. Which of the following is not a type of database?
निम्नलिखित में से कौन-सा डेटाबेस का एक प्रकार नहीं है?
(a) Network
(b) Distributed
(c) Decentralized
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c) : Different types of database–

1. Centralized	2. Distributed
3. Relational	4. NOSQL
5. Cloud	6. Object Oriented
7. Hierarchical	8. Network

91. Arrange the following phases of database design in the correct order :
A. Physical Design.
B. Conceptual Design.
C. Logical Design.
D. Requirement Analysis.
Choose the correct answer from the options given below :
(a) (B), (D), (A), (C) (b) (C), (A), (B), (D)
(c) (D), (B), (C), (A) (d) (A), (D), (C), (B)

UGC NET C.S. Re-Exam- June 2024

Ans. (c) : The correct order of the phases in database design is as follow:
(i) Requirement Analysis: Understand the needs and requirement of the system and users.
(ii) Conceptual Design: Create a high-level design, such as an ER diagram, to represent the data and its relationship.
(iii) Logical Design: Convert the conceptual model into a logical model that aligns with the database type.
(iv) Physical Design: Implement the logical model physically on the database system, optimizing for performance and storage.
Therefore, the correct answer is (D), (B), (C), (A).

92. The Global conceptual Schema in a distributed database contains information about global relations. The condition that all the data of the global relation must be mapped into the fragments, that is, it must not happen that a data item which belongs to a global relation does not belong to any fragment, is called:
(a) Disjointness condition
(b) Completeness condition
(c) Reconstruction condition
(d) Aggregation condition

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Completeness Condition: All the data of the global relation must be mapped into the fragments; it must not happen that a data item which belongs to a global relation does not belong to any fragment.

93. Manager's salary details are hidden from the employee. This is called as
(a) Conceptual level data hiding
(b) Physical level data hiding
(c) External level data hiding
(d) Local level data hiding

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, External level (highest level). It is the user's view of the database and it describes part of the database for Particular group of user and provides a powerful and flexible security mechanism by hiding part of the database from certain users. The user is not aware of the existence of any attribute that are missing from the view. It permits user to access data in a way that is customized to their needs. So that the same data can be seen by different user in different ways, at the same time.

94. Which of the following is not a valid category of database management system?
निम्नलिखित में से कौन सा डेटाबेस मैनेजमेंट सिस्टम की मान्य श्रेणी नहीं है?
(a) Hierarchical/हाइरार्कल (श्रेणीबद्ध)
(b) Sequential/सिक्वेंशियल (क्रमबद्ध)
(c) Object-Oriented/ऑब्जेक्ट-ओरिएंटेड
(d) Relational/रिलेशनल

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : In Database Management System (DBMS) the sequential model category is not the valid category of structure model. Hierarchical model, object oriented & Relational models are common category of DBMS.

95. Consider the following relation schema of students.
छात्रों के निम्नलिखित रिलेशन स्कीमा पर विचार करें।
STUDENT (Rollno, Name, DOB, Marks, Gender)
query is equivalent to this query in English? "Find the tuples of student having marks above 80".
दी गई क्वेरी में से कौन सी इस क्वेरी के तुल्यमान है? ''80 से अधिक अंक प्राप्त करने वाले विद्यार्थी के टुपल्स ज्ञात कीजिए''।
(a) sMarks>80 (STUDENT)
(b) ΠMarks>80 (STUDENT)
(c) sMarks<80 (STUDENT)
(d) ΠMarks<80 (STUDENT)

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : sMarks>80 (STUDENT) selects the tuples from the 'STUDENT' relation where the 'Marks' attributes is greater than. The selection operation 's' is used for filtering rows based on a condition.

96. Match the following formal relational term with informal equivalents.
निम्नलिखित फॉर्मल रिलेशनल पद का इनफॉर्मल एक्विवैलेंट्स से मिलान कीजिए।

i	**Relation/रिलेशन**	**1**	**Row/पंक्ति**
ii	**Tuple/टुपल**	**2**	**Value/मान**
iii	**Attribute/एट्रिब्यूट**	**3**	**Table/तालिका**
iv	**Domain/डोमेन**	**4**	**Column/स्तम्भ**

(a) i-3, ii-4, iii-1, iv-2 (b) i-3, ii-1, iii-4, iv-2
(c) i-1, ii-3, iii-2, iv-4 (d) i-4, ii-3, iii-2, iv-1

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Sarkari Resource

Ans. (b) : The matching of format relational term with their informal equivalents–

i	Relation	**3**	Table
ii	Tuple	**1**	Row
iii	Attribute	**4**	Column
iv	Domain	**2**	Value

97. Which of the following is Not a database model?
निम्नलिखित में से कौन एक डेटाबे मॉडल नहीं है?
(a) Network/नेटवर्क
(b) Relational/रिलेशनल
(c) Object-oriented/ओब्जेक्ट-ओरिएंटेड
(d) Inheritence model/इनहैरिटेंस मॉडल

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : Network, Relational and object-oriented are all well established types of database models.
Inheritance model is not a standalone database model; rather it is a concept used within object-oriented models and programming for defining relationships between classes.

98. When data is frequently added to or deleted from a file, the volatility of the file is_____.
ज़ब किसी फाइल में डेटा को बार-बार जोड़ा जाता है, तो फाइल की वोलैटिलिटी_____होती है।
(a) Low/निम्न (b) Moderate/मध्यम
(c) High/उच्च (d) Zero/शून्य

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : File volatility refers to how often a file is changed-specifically how often data is added, modified, or deleted. When data is frequently added to or deleted from a file, this means the file is subject to frequent changes making its volatility high.

99. The values of the attributes in a database describe a particular_____.
एक डेटाबेस में ऐट्रिब्यूट्स के मान एक विषेष_____ का वर्णन करते हैं।
(a) Entity/इकाई (b) Tuple/टपल
(c) Field/फ़ील्ड (d) View/व्यू

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : The values of attributes in a database describe a specific instance of an entity. Think of it like a table where each row represents an individual item (entity), and each column represents a characteristic (attribute) of that item. For example in a table of students, each row might represent a different student, and attribute like-Name, Age and Grade would contain the specific values describing that student.

100. Which of the following statement is/are correct regarding sequential file organization?
अनुक्रमिक फ़ाइल संगठन के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. Entire file must be read for new record insertion and record deletion.
नई रिकॉर्ड प्रविष्टि और रिकॉर्ड हटाने के लिए पूरी फाइल को पढ़ना चाहिए।
II. On an average, half of the records in the file must be processed to find the location of a record./एक रिकॉर्ड के स्थान का पता लगाने के लिए औसतन, फ़ाइल के आधे रिकॉर्डस क़ो संसाधित किया जाना चाहिए।
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Sequential file organization– Records are stored and accessed in a particular order sorted using a key field. Retrieval requires searching sequentially through the entire file record by record to the end.
Because the record in a file is sorted in a particular order, better file searching methods like the binary search technique can be used to reduce the time used for searching a file. Since the records are sorted, it is possible to known in which half of the file a particular record being searched is located, hence this method repeatedly divides the set of records in the file into two halves and searches only the half on which the records are found.

101. Which of the following is NOT an approach of database design in database management system?
निम्नलिखित में कौन सी डेटाबेस मैनेजमेंट सिस्टम में डेटाबेस डिज़ाइन की अप्रोच नहीं है?
(a) Top-down approach/टॉप-डाउन अप्रोच
(b) Bottom-up approach/बॉटम-अप अप्रोच
(c) Inside-out approach/इनसाइड-आउट अप्रोच
(d) UP-down approach/अप-डाउन अप्रोच

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) : In Database Management Systems (DBMS) a "NOT" an approach to **Up down approach** of database design would be something unstructured, ad-hoc, or careless, meaning it does not follow any logical method. The recognized approaches to database design.
- Top - Down approach
- Bottom - Up Approach
- Inside - Out Approach
- Mixed (hybrid) approach.

102. What will be the right sequence of the phases of database design-
A. Physical Design
B. Conceptual Design
C. Logical Design
D. Requirement collection and analysis
Choose the correct answer from the options given below :
(a) D, A, B, C (b) D, B, C, A
(c) D, B, A, C (d) D, A, C, B

UGC NET C.S. December-2024

Ans. (b) : The correct sequence of database design phases is- Requirement collection and Analysis - Gather user needs and business rules.
Conceptual design - Create high-level ER diagrams or VML models.
Logical design - Convert the conceptual model into a relation schema.
Physical Design - Implement the logical schema into a physical database.

103. In data mining, classification rules are extracted from _____
(a) Data (b) Information
(c) Decision tree (d) Database

UGC NET C.S. Augest-2016 (Paper-II)

Sarkari Resource

Ans : (c) A decision tree can be used to visually and explicitly represent decision & division making. In data mining, a decision tree describes data but not decisions; rather the resulting classification tree can be an input for decision making.

104. Discovery of cross sales opportunities is called as _____

(a) Association
(b) Visualization
(c) Correlation
(d) Segmentation

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (a) Association rule learning is a method for discovering intersecting relations between variables in large data base. It is intended to identify strong rules disconnected in database using some measures of interestingness.

105. In RDBMS, the constraint that no key attribute (column) may be NULL is referred to as:

(a) Referential integrity
(b) Multi-valued dependency
(c) Entity Integrity
(d) Functional dependency

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) The **entity integrity constraint** states that primary keys cannot be null. There must be a proper value in the primary key field.
The basis of referential integrity is foreign keys. A foreign key in one table references a primary key in another table.

106. Degree and cardinality of a relation in relational database are 4 and 3 respectively. If 3 attributes and 4 tuples are added to the relation, what will be its new cardinality and degree?
रिलेशनल डेटा बेस में एक रिलेशन का डिग्री और कार्डिनलिटी क्रमशः 4 और 3 हैं। अगर 3 ऐट्रिब्यूट और 4 ट्यूपल उस रिलेशन में जोड़ दिया जाए, तो उस रिलेशन का नया कार्डिनलिटी और डिग्री क्या होगा?

(a) 7, 6
(b) 7, 7
(c) 8, 6
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): The degree and cardinality of the relation can be determined after adding new attributes and tuples.
Degree– The degree of the relation will increase from 3 to 7 after adding new attributes.
Cardinality– After adding new tuples, if there is a primary key, then the cardinality will now increase from 4 to 7. If any non primary attributes is completely dependent only on the primary key, then the cardinality will remain the same.

107. Which one of the following would breach the integrity of a system?
निम्नलिखित में से कौन-सा एक सिस्टम की अखण्डता का उल्लंघन करेगा?

(a) Locking the room to prevent theft
(b) Full access rights for all users
(c) Fitting the system with an anti-theft device
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : A breach of integrity means violating the rules of honesty, ethics, and loyalty to a person, organization or system. This can be on a local or personal level such as betraying a user's full rights or publicly betraying an organization or government system.

108. The capacity to change the conceptual schema without having to change the external schemes or additional programs is called as_____.
एक्सटर्नल स्कीमा या अतिरिक्त प्रोग्राम्स को बदले बिना कन्सेप्चुअल स्कीमा को बदलना_____ कहलाता है।

(a) Physical Data Independence
फिजिटल डेटा इन्डिपेन्डेंस
(b) Lateral Data Independence
लेट्रल डेटा इन्डिपेन्डेंस
(c) Posterior Data Independence
पोस्टीरीयर डेटा इंडिपेन्डेंस
(d) Logical Data Independence
लॉजिकल डेटा इन्डिपेन्डेंस

HTET PGT C.S. 2023

Ans. (d) : Logical Data Independence refers to the capacity to change the conceptual schema like adding/removing new fields, relationship, constraints etc. Without heading to change the external schema. Physical Data Independence refers to changing the internal schema without affecting the conceptual schema. Since the question mentions changing the conceptual schema without affecting the external schema, the answer is Logical Data Independence.

109. A replica manager in gossip architecture contains the main state components is
गॉसिप तंत्र में प्रतिकृति प्रबंधक मुख्य स्टेट घटक धारित करता है

(a) Value time stamp/वैल्यू टाइम स्टैम्प
(b) Update log/अपडेट लॉग
(c) Replica time stamp/रिप्लीका टाइम स्टैम्प
(d) All of the above/उपरोक्त सभी

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (d) : In Gossip architecture a replica manager contains several main state components, including the value timestamp, update log, and replica timestamp, among others.

Sarkari Resource

(iii) ER Modelling Relational Model, Introduction to SQL Relational Database Design

110. After groups have been established, SQL applies predicates in the___clause, allowing aggregate functions to be used.
समूह स्थापित होने के बाद, SQL_____खंड में विधेय लागू करता है, जिससे समग्र कार्य करने की अनुमति मिलती है।
(a) HAVING
(b) GROUP BY
(c) WITH
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans,(a): In SQL, GROUP BY is used to create groups based on column values.
After grouping, if we want to filter those groups using conditions on aggregate functions (like SUM( ), COUNT( ) etc., we use the HAVING clause.
The WHERE clause is used before grouping (on individual rows), while HAVING is used after grouping)on grouped results.

111. The traditional storage of data organized by the customer, stored in separate folders in filing cabinets is an example of:
ग्राहक द्वारा व्यवस्थित डेटा का पारंपरिक भंडारण, फाइलिंग कैबिनेट में अलग-अलग फोल्डरों में संग्रहीत होता है, _____का एक उदाहरण है।
(a) Relational database management system
(b) Network database management system
(c) Hierarchical database management system
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): A filling cabinet has folders inside it, and each folder has papers/ documents . This is like a tree structure → one main item, with smaller items under it.
•This is the same as a Hierarchical Database management system (HDBMS).

112. Let R = (A, B, C) & r1 and r2 both be relation on schema R. The SQL expression Select * from r1 where (A, B, C) in (select * from r2) is equivalent to
(a) r1-r2 (b) r1∩r2
(c) r1∪r2 (d) r2–r1

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : The SQL expression SELECT * from r1 where (A, B, C) IN (SELECT * FROM R2) is equivalent to the intersection of r1 and r2. This is because it selects all tuples from r1 that also exist in r2.
Therefore, the correct option is (b) r1 ∩ r2.

113. Consider the following relational schema:
Employee (empId, empName, empDept)
Customer (custId, custName, salesRepId, rating)
SalesRepId is a foreign key referring to empId of the employee relation.
Assume that eachemployee makes a sale to at least one customer.
What does the following query return?
SELECT empName
FROM employee E
WHERE NOT EXIST (SELECT custId
FROM employee C
WHERE C. salesRepId = E. empId
AND C. rating <> 'GOOD')
(a) Names of all the employees with at least one of their customers having a 'GOOD' rating.
(b) Names of all the employees with at most one of their customers having a 'GOOD' rating
(c) Names of all the employees with none of their customers having a 'GOOD' rating.
(d) Names of all the employees with all their customers having a 'GOOD' rating.

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Ans. (d) : The query returns the names of all employees whose customers all hove a GOOD rating. The NOT EXISTS clause checks if there is any customer with a rating other than GOOD for each employee. If no such customer exists (i.e., all ratings are GOOD), the NOT EXISTS condition is true, and the employee's name is included in the result.

114. SELECT ecode FROM EMPLOYEE WHERE_____; is an SQL query to retrieve ecode of employees whose address contains 'RPSC'. Following completes the query:-
(a) address LIKE'_RPSC_'
(b) address = '%RPSC%'
(c) address AS '%RPSC_'
(d) address LIKE '%RPSC%'

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Ans. (d) : The SQL query to retrieve employee codes where addresses contain "RPSC" is SELECT ecode FROM EMPLOYEE WHERE address LIKE '%RPSC'.
SELECT ecode– This specifies that you want to retrieve the ecode (employee code) column.
FROM EMPLOYEE– This indicates that you are querying the EMPLOYEE table.
WHERE address LIKE '% RPSC %'– This is the filtering condition-
WHERE– This keyword is used to filter rows based on a specified condition.
Address– This refers to the address column in the EMPLOYEE table.
LIKE– This operator is used for pattern matching in strings.
'% RPSC%'– This is the pattern you are searching for.
%– This is a wildcard character that represents any sequence of characters (including zero characters).
RPSC– This is the literal string you are looking for.
The pattern '% RPSC' means any string that contains the substring 'RPSC'.

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115. What will be retrieved by following SQL query: SELECT ecode FROM EMPLOYEE WHERE Salary>= ALL (SELECT salary FROM EMPLOYEE);

(a) Ecode of employees getting maximum salary
(b) Ecode of all employees
(c) Ecode and salary of all employees
(d) Ecode and salary of employees getting maximum salary

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Ans. (a) : The given SQL query is-
SELECT ecode FROM EMPLOYEE WHERE Salary >=ALL (SELECT salary FROM EMPLOYEE) step by step explain-
SELECT salary FROM EMPLOYEE– This inner query selects all the salaries from the EMPLOYEE table.
Salary >= ALL (…..)– This condition checks for employees whose salary is greater than or equal to every salary in the table-effectively this means employees with the maximum salary.
The query retrieves– only the ecode (employee code) of these employee who have the maximum salary.

116. SQL command used to change attribute values of one or more selected tuples is:-

(a) MODIFY (b) UPDATE
(c) INSERT (d) ALTER

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Ans. (b) : SQL Command– It is used to change attribute values of one or more selected tuples (rows) in a table is-
The UPDATE command is specifically designed for modifying existing data in a table. It allows you to set new values for specific columns in one or more rows that match a specified condition.

117. To avoid duplicate tuples in result of an SQL query, we use:-

(a) DISTINCT Keyword in SELECT clause
(b) UNIQUE Keyword in SELECT clause
(c) DISTINCT clause after WHERE clause
(d) UNIQUE clause after WHERE clause

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Ans. (b) : One of the easiest ways to remove duplicate data in SQL is by using the DISTINCT keyword. You can use the Distinct keyword in a select statement to retrieve only unique values from a particular column.

118. Valid SQL query to know total number of tuples available is a relation 'Book' is:-

(a) SELECT SUM(*)FROM Book;
(b) SELECT NUMBER(*) FROM Book;
(c) SELECT COUNT(*) FROM Book;
(d) SELECT * FROM Book SUM(*);

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Ans. (c) : The correct SQL query to know the total number of tuples (i.e., rows) available in a relation (table) Book is– SELECT count (*) FROM Book.
Explain– COUNT (*) is the SQL aggregate function that count the total number of rows (tuples) in the specified table.
SUM(*) is invalid because SUM() is used with numeric columns not with *.
Number(*) is not a valid SQL function.
SELECT * FROM Book SUM(*) is a Syntactically incorrect query.
Hence, the correct option is- (c)

119. Which of the following operation requires two tables as input and the two table must have one common column?

(a) Division (b) Join
(c) Projection (d) Cartesian product

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Ans. (b) : A join operation combines data from two tables based on a shared column value, allowing you to retrieve related information from multiple tables.
Division– Requires two tables but the condition is more about the divisibility based an certain criteria, not necessarily having a common column.
Projection– Operates on a single table selecting specific columns.
Cartesian Product– Combines two tables by creating a new table that contains all possible combination of rows from the two input tables. It does not require a common column.

120. In E-R model when subgroups of a given entity type are defined, this is known as:-

(a) Aggregation (b) Inheritance
(c) Specialization (d) Generalization

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Ans. (c) : E-R Model Concepts– Analyze the given options in the context of the Entity-Relationship model.
1. Aggregation– Represents a relationship between a whole and its parts.
2. Inheritance– Not a direct concept in E-R modeling but is related to the idea of specialization and generalization in object oriented design.
3. Specialization– Involves defining subgroups within an entity set that have distinct attributes or relationships not shared by all entities in the set.
4. Generalization– The reverse process of specialization, where common attributes of entity sets are generalized into a higher-level entity set.
So, the correct option is (c).

121. In E-R diagram_____is represented by dashed ellipse:-

(a) Multivalued attribute
(b) Composite attribute
(c) Derived attribute
(d) Complex attribute

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Ans. (c) : In E-R diagram, derived attributes are represented by dashed ellipses.
Derived attribute that can be derived from other attribute is derived attributes. And it is represented by dotted oval shape.
There are different types of attributes–
Simple attribute
Composite attribute
Single-valued attribute
Multi-valued attribute
Derived attribute
Complex attribute
Stored attribute
Key attribute
NULL Attribute and Descriptive

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122. Which of the following is a primary key in a student table?
निम्न में से कौन छात्र तालिका में एक प्राथमिक कुँजी है?
(a) UID number/UID संख्या
(b) Admission number/Admission संख्या
(c) Account number/Account संख्या
(d) PAN

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Ans. (b) : A primary key is a field that uniquely identifies each record in a tale. In the STUDENT table this should be a column that:
is unique for each student
is not null
is always available.

123. Which of the following is a valid SQL command
निम्नलिखित में से कौन-सा एक वैध SQL command है?
(a) Cin (b) Cout
(c) Tuple (d) Select

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Ans. (d) : SQL (Structured Query Language) is a database language used to query, insert, update, delete, etc. data.

124. Which statement is used to retrieve data from a database in SQL?
SQL में डेटाबेस से डेटा प्राप्त करने के लिए किस कथन का उपयोग किया जाता है?
(a) SELECT (b) UPDATE
(c) DELETE (d) INSERT

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : The Select statement selects data from a table by retrieving specific data from the database and returns to the user and is used to extract data from a table based on columns, rows, or specific conditions.

125. Which one is a DCL command in SQL?
निम्न में कौन सा SQL में DCL Command है?
(a) UPDATE (b) SELECT
(c) DELETE (d) GRANT

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : DCL (Data Control Language) is used to control user access and permissions in a database. The main DCL commands in SQL are:
1. GRANT– Grants permissions to a user on an object.
2. REVOKE– Revokes permissions granted to a user.

126. Which of the following SQL commands is used to view data?
निम्नलिखित में से कौन सी SQL commands डेटा देखने के लिए उपयोग की जाती है?
(a) CREATE (b) INSERT
(c) SELECT (d) UPDATE

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Ans. (c) : The SELECT command is used to view or retrieve data in SQL. This command is most commonly used to query data from a database.
For example– SELECT*FROM student.
This command will display all data from the Student table.

127. Which of the following clause is used with SELECT command in MySQL, to avoid duplication of output rows.
MySQL में आउटपुट पंक्तियों (Rows) की आवृत्ति से बचने के लिए SELECT कमांड के साथ निम्नलिखित में से किस क्लॉज का प्रयोग किया जाता है?
(a) NO DUPLICATE (b) UNIQUE
(c) DISTINCT (d) NO REPEAT

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Ans. (c) : The DISTINCT keyword is used with the 'select' statement in MySQL to retrieve only unique rows from the result set. It eliminates duplicate rows from the output.

128. Which operator has highest precedence in */%?
निम्न में से */% की किस Operator मे सबसे अधिक पूर्वता है?
(a) *
(b) /
(c) %
(d) All have same precedence/सभी की समान पूर्वता

Bihar STET C.S. 2020

Ans. (d) : In most programming language like C, C++, Java, Python the following three operators *(Multiplication),/(Division), % (Modulus), all have the same precedence.
Meaning of Precedence-precedence determines which operator will be executed first when there is more than one operator in the same precedence.
When all have the same precedence.
Then associatively determines the direction in which the evaluation will take place. These three operators (*, /, %) are evaluated left-to-right associatively, manning they are solved from left to right.

129. Primary key is also called
प्राथमिक कुंजी को भी कहा जाता है।
(a) Candidate key (b) Super key
(c) Alternate key (d) Non key

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Ans. (a) : A Candidate key is a key that uniquely identifies all rows in a table. A table can have multiple candidate keys. When one of these is chosen as the primary key, it is still essentially a candidate key.

130. _____ offers the ability to query the data and insert, alter, and delete tuples.
_____ डेटा को क्वेरी करने और टपल्स को जोड़ने, बदलने तथा हटाने की क्षमता प्रदान करता है।
(a) Transaction Control Language (TCL)
ट्रांज़ैक्शन कंट्रोल लैंग्वेज (TCL)
(b) Data Control Language (DCL)
डेटा कंट्रोल लैंग्वेज (DCL)
(c) Data Definition Language (DDL)
डेटा डेफिनिशन लैंग्वेज (DDL)
(d) Data Manipulation Language (DML)
डेटा मैनिपुलेशन लैंग्वेज (DML)

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Ans. (d) : DML (Data Manipulation Language) is used to query, insert, modify (alter/update), and delete data in a database. Its main commands are:

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- SELECT (To view data)
- INSERT (To add new data)
- UPDATE (To change existing data)
- DELETE (To delete data)

131. Which of the following is DML command?
निम्नलिखित में से कौन सी एक DML कमांड है ?

(a) INSERT INTO (b) DROP TABLE
(c) CREATE TABLE (d) ALTER TABLE

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Ans. (a) : The DDL (Data Definition Language command among the options you've provided is DROP TABLE). DDL commands are used to define and manage the structure of database objects such as tables, command like up DATE SET, DELETE, and INSERT INTO are DML (Data manipulation Language) commands used to manipulate data with in tables.

132. An attribute or a set of attributes which are used to identify a tuple uniquely is known as
किसी टपल को अद्वितीय रूप से पहचानने के लिए प्रयुक्त गुण (अट्रीब्यूट) अथवा गुणों के समूह को क्या कहते हैं?

(a) Primary Key (b) Foreign Key
(c) Alternate Key (d) Candidate Key

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Ans. (a) : A primary key is a set of attributes that can uniquely identify the table. It is one of the candidate keys which is most suitable to identify a table. The value of the primary key in any row can't be null. There can be only one primary key in a table.

133. Which of the following is not a valid relational operator used on SQL?
निम्नलिखित में से कौन सा SQL में प्रयुक्त एक मान्य सम्बन्धपरक ऑपरेटर नहीं है?

(a) < = (b) < >
(c) = > (d) =

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Ans. (c) : All of the operators you've listed are listed are valid relational operators in SDL.
1. < = (Less than or equal to)
2. < > (Not equal to)
3. = (equal to)
4. > = (Greater than or equal to), in SQL, the greater than or equal to operator is represented as > = , not = >

134. Which of the following is using wrong syntax for a SELECT query in SQL ?
SQL में किसी SELECT के लिए गलत वाक्य - रचना (syntax) का प्रयोग निम्नलिखित में से किसमें किया गया है ?

(a) SELECT * WHERE RNO > 100 FROM STUDENT ;
(b) SELECT * FROM STUDENT WHERE RNO > 100;
(c) SELECT * FROM STUDENT WHERE RNO BETWEEN 100 AND 200;
(d) SELECT * FROM STUDENT WHERE RNO IN (100,101,105,104);

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Ans. (a) : The correct answer is (a) select * where RNO > 100 from student;
The correct syntax for a select query with a where clause in SQL is-
Select column 1. Column z------ from table-name where conditions;

135. Which of the following will be the correct SQL command to add a new column FEES in a table TEACHER ?
किसी TEACHER टेबल में एक नया FEES कॉलम जोड़ने के लिए सहीं SQL कमांड निम्नलिखित में से कौन सी है?

(a) ALTER TABLE TEACHER
ADD COLUMN FEES FLOAT;
(b) ADD COLUMN FEES FLOAT
INTO TEACHER;
(c) UPDATE TEACHER
ADD COLUMN FEES FLOAT;
(d) INSERT INTO TEACHER
ADD COLUMN FEES FLOAT;

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Ans. (a) : The correct SQL command to add a new column Fees in a table Teacher is.
Alter Table Teacher ADD Column Fees FLQAT, Here's a breakdown of the command-
• Alter table Teacher:- This specifies that you want to modify the structure of the teacher table.
• Add column Fees Float:- This adds a new column named fees to the Teacher table with a data type of float. The float data type is used for floating paint numbers, which are numbers that have a decimal paint.

136. Which of the following will display a list of tables in a database ?
किसी डाटाबेस (DATABASE) में टेबल (TABLES) सूची को किस प्रकार प्रदर्शित किया जाएगा ?

(a) DISPLAY TABLES; (b) SHOW TALES;
(c) SELECT TABLES; (d) VIEW TABLES ;

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Ans. (b) : The correct option to display a list of tables in a database depends on the database management system (DBMS) being used. For most poplar DBMS like MySQL, PostgreSQL, and SQLite, the correct command is show Tables;

137. Which of the following will display information about all the employees from employee table, whose names contains second letter as "A"?
किसी EMPLOYEE Table से सभी कर्मचारियों, जिनके नाम का दूसरा अक्षर 'A' है, के बारे में जानकारी दिखाने के लिए किया जाएगा ?

(a) SELECT * FROM EMPLOYEE WHERE NAME LIKE "_A% " ;
(b) SELECT * FROM EMPLOYEE WHERE NAME LIKE "%A_" ;
(c) SELECT * FROM EMPLOYEE WHERE NAME LIKE "_ _A%" ;
(d) SELECT * FROM EMPLOYEE WHERE NAME LIKE "–A%" ;

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Ans. (a) : To display information about all employees from the employee table whose names contain the second letter as "A", you can use the following SQL query.
Select * from employee where name LIKE '– A', This query uses the LIKF officiator with a pattern that matches any name where the second letter is 'A', The underscore (–) represents any single character in SQL patterns.

138. Which of the following SQL command will help in incrementing values of FEES column in STUDENT table by 10% ?
निम्नलिखित में से कौन सी SQL कमांड STUDENT टेबल में FEES कॉलम के बढ़ते हुए मान को 10% बढ़ाने में सहायता करेंगी?

(a) UPDATE STUDENT
ASSIGN FEES =FEES * 1.1;
(b) UPDATE STUDENT
SET FEES =FEES * 1.1;
(c) UPDATE STUDENT
SET FEES = FEES * 10%
(d) UPDATE STUDENT
SET FEES 10%

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Ans. (b) : The correct SQL command to increment the values of the 'fees' Column in the student table by 10% is

update student set fees = fees * 1.1;

- Update student- this specifies the table name (student) that you wont to update.
- Set fees = fees * 1.1 This is the key part of the command. It uses the set keyword to specify the column (fees) that you want to update and sets its value to itself (fees) multiplied by 1.1, which effectively increments it by 10%.

139. Which of the following is correct sequence in a SELECT query ?
SELECT प्रश्न में कौन सा क्रम सही है ?

(a) SELECT, FROM, WHERE, GROUP BY HAVING, ORDER BY
(b) SELECT,WHERE, FROM, GROUP BY, HAVING, ORDER BY
(c) SELECT, FROM, WHERE, HAVING, GROUP BY, ORDER BY
(d) SELECT, FROM, WHERE, GROUP BY, ORDER BY, HAVING,

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Ans. (a) : The correct sequence in a select query is (a) select, from, Group by, Having, order by
Here's a detailed description of each part,
1. **Select-** This is the first keyword used in a select query. It is followed by a comma-separated list of columns or expressions that you want to retrieve from the database.
2. **From**- After the select keyword, the from keyword is used to specify the table or tables from which you want to retrieve data. If you are retrieving data from multiple tables, you can use join clauses to specify how the tables are related.
3. **Where-** The where keyword is used to specify conditions that the retrieved data must meet Rows that do not meet the specified conditions are not included in the result set.
4. **Group by**- The Group by keyword is used to group the result set by one or more columns.
5. **Having-** The Having keyword is used to filter groups of rows based on specified conditions.
6. **Order by-** The order by keyword is used to sort the result set based on one or more columns.

Putting it all together,
Select column:- column 2, ------- from table where condition Group by column Having condition order by column ASC/DESE;

140. Consider the following tables and their respective degrees and cardinalities in a database called SCHOOL :
किसी (SCHOOL DATA BASE) में निम्नलिखित सारणी (table) और उसकी संबंधित डिग्री (degree) तथा कार्डिनेलिटी (cardinality) पर विचार कीजिए:

Database : SCHOOL		
Table	Degree	Cardinality
STUDENT	20	300
TEACHER	10	25

Select the degree and cardinality of the Cartesian product of the tables STUDENT × TEACHER from the following options :
निम्नलिखित विकल्पों में से STUDENT × TEACHER सारणी के कार्तीय उत्पाद (Cartesian Product) की डिग्री (Degree) और कार्डिनेलिटी (Cardinality) का चयन कीजिए:

(a) 30 7500
(b) 200 325
(c) 30 325
(d) 200 7500

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Ans. (a) : To find the degree and cardinality of the Cartesian product of two tables, you multiply the degrees and cardinalities of the individual tables.
For the Cartesian product of the tables Student and teacher-
Degree = Degree (student) + Degree (teacher)
= 20 + 10
= 30
Cardinality = Cardinality (student) * cordiality (teacher) = 300 * 25
= 7500
So, the correct option is (a).

141. Which of the following set of functions is a valid set of the aggregate functions in MySQL?
निम्नलिखित में से कौन सा कार्य (functions) समूह MySQL में एग्रीगेट फंक्शंस (Aggregate functions) का मान्य समूह है ?

(a) AVG ( ), ROUND ( ), COUNT ( )
(b) MIN ( ), UPPER ( ), AVG ( )
(c) COUNT ( ), MAX ( ), SUM ( )
(c) DATE ( ), COUNT ( ), LTRIM ( )

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Ans. (c) : My SQL aggregate function is used to perform calculations on multiple values and return the result in a single value like the average of all values.

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Syntax-
function-name (Distinct/All expression)
Aggregate function NAME-COUNT(), SUM(), AVG(), MIN(), MAX(), FIRST(), and LAST().

142. Find the output of the MySQL query based on the given Table-COACH (ignore the output header)
नीचे दी गई सारणी (Table-COACH) (आउटपुट शीर्ष को नजरअंदाज करें) पर आधारित MySQL प्रश्न के आउटपुट को ज्ञात कीजिए :

Table : COACH

CID	CNAME	GAME	SALARY
G03	VSHALI	NULL	60000
G05	YUVRAJA	CRICKET	NULL
G08	MALKHAN	ATHLATICS	80000
G09	SACINDRA	CRICKET	NULL

Query :
SELECT COUNT (GAME), AVG (SALARY)
FROM COACH :

(a) 3 70000 (b) 4 35000
(c) 4 70000 (d) 3 35000

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Ans. (a) : The query selects the count of non-null values in the GAME Column and the average of non-null values in the SALARY column from the COACH table. There are 3 non-null values in the GAME Column. The non-null values in the GAME column. The non-null salaries are 60000 and 80000, so the average salary 15 (60000 + 80000)/2 = 70000
So, the correct output is- (a) 3 70000

143. Find the output of the MySQL query based on the given Table-STUDENT (Ignore the output header)
दी गई सारणी (Table-STUDENT) पर आधारित MySQL प्रश्न के आउटपुट को ज्ञात कीजिए (आउटपुट शीर्ष को नजरअंदाज कों)

Table : STUDENT

S1D	SNAME	SEC	MARKS
101	Ameena	A	83
107	Arjun	B	86
112	Albert	B	80
120	Akram	A	85

Query :
SELECT SEC, AVG (MARKS) FROM STUDENT GROUP BY SEC HAVING MIN (MARKS) > 80 ;

(a) B 83
(b) A 84
(c) A 84
B 83
(d) A 83
B 80

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Ans. (b) : The query selects the section (SEC) and the average marks of students in that section, where the minimum marks are greater than 80. Based on the given table, the minimum marks for sections A and B are 83 and 80 respectively. Since the minimum marks for section B are not greater than 80, the query will only …….. the result for section A
So, the output option (b) A 84

144. Consider the following tables in a database called SPORTS.
निम्नलिखित सारणियों (Tables) में SPORTS नामक डाटाबोस पर विचार कीजिए :

Database : SPORTS
Table : GAMES

G1D	GNAME	FEES
G01	Football	300
G02	Cricket	1000
G03	Basketball	500

Table PLAYERS

P1 D	PNAME	SEX	G1D
P101	Amrita	F	G03
P103	Ameena	F	G02
P107	Angel	F	G02
P109	Aman	M	G03

Which is the best command from the following options to display the name of the Player (PNAME) and their respective games (GNAME)
खिलाड़ी (PNAME) और उनके संबंधित खेल (GNAME) के नाम को प्रदर्शित करने के लिए निम्नलिखित विकल्पों से बेहतर कमांड कौन सी है।

(a) SELECT PNAME, GID FROM PLAYERS:
(b) SELECT PNAME, GNAME FROM GAMES, PLAYERS WHERE GAMES G1D = PLAYERS. GID ;
(c) SELECT PNAME, GNAME FROM GAMES, PLAYERS;
(d) SELECT PNAME, GNAME FROM GAMES, PLAYERS WHERE P.GID = G. GID;

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Ans. (b) : The best command from the options provided to display the name of the player (PNAME) and their respective games (GNAME) is option (b)
Select PNAME, GNAME from GAME, PLAYERS where GAME. GID = PLAYERS. GID,
1. **Select-** The select statement is based to retrieve data from the database.
2. **From Clause-** The from clause specifies the tables from which the data will be retrieved. In this command, it mentions the tables GAMES and PLAYERS.
3. **Join condition-** The where clause contain the condition that links the rows from the GAMES table to the rows form the PLAYEPS table.
4. **Column selection-** The select statement specifies which columns from the tables should be included in the result set.

145. Following are the two tables in a MySQL database
MySQL डाटाबेस में दो सारणियाँ ( ) दी गई हैं :
डाटाबेस /Database : COMPARTMENT

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Table : SUBJECTS

SUBCODE	SUBNAME	MAXT	MAXP
065	Informatics Practices	70	30
083	Computer Science	70	30
301	English Core	80	20

Table : RESULT

ROLL	SUBCODE	TMARKS	PMARKS
10045	083	70	30
10135	065	56	29
10231	065	63	30
10345	083	56	29

Select the command from the following options to display subjects (SUBNAME) and number of students registered in that subject in table RESULT.
Subjects (SUBNAME) और सारणी-RESULT में उस विषय में दर्ज छात्रों की संख्या को दर्शाने के लिए निम्नलिखित विकल्पों में से कमांड का चयन कीजिए :

(a) SELECT SUBNAME, COUNT (*)
FROM SUBJECTS, RESULT GROUP BY SUBNAME;
(b) SELECT SUBNAME, COUNT (*)
FROM SUBJECTS, RESULT ORDER BY SUBNAME;
(c) SELECT SUBNAME, COUNT (*)
FROM SUBJECTS, NATURAL JOIN RESULT ORDER BY SUBNAME;
(d) SELECT SUBNAME, COUNT (*)
FROM SUBJECTS, NATURAL JOIN RESULT GROUP BY SUBNAME;

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Ans. (d) : The correct command to display the subjects (subname) and the number of students registered in that subject from the 'result' table is
Select surname, count (*) from subjects natural join Result group by subname,
This command used the 'Natural join' to join the 'subjects' and 'Result' tables on the common column 'subject', and then groups the result by 'Subname' to count the numbers of students registered in each subject.

146. Consider the database table "Persons" having persons_ID as the primary key:
डेटाबेस तालिका पर विचार कीजिए व्यक्ति आईडी रखने वाले 'व्यक्ति', 'प्राथमिक कुंजी' (प्राइमरी की) के रूप में है:

person _ ID	Name	Age
1000	Rajan	40
–	Jatin	–45
1001	Rajesh	35

What are the constraints violated by the above table?/उपर्युक्त तालिका द्वारा उल्लंघन किए जाने वाले प्रतिबंध क्या है?

(a) Relationship integrity/संबंध इंटेग्रिटी
(b) Referential integrity only/केवल रेफरेंशल इंटेग्रिटी
(c) Entity and domain integrities
एन्टिटी और डोमेन इंटेग्रिटी
(d) Referential and domain integrities
रेफरंशल और डोमेन इंटेग्रिटी

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Ans. (c) : According to entity integrity rule. The primary key or part of it in any relation can not be NULL. Primary key is Person_Id which does not follow entity integrity rule, as value in the second row of primary key is Null. Domain constraints means all data items in a column must follow within a defined set of valid values. Age has negative value which does not follow Domain integrity rule.

147. What result set is returned from the following SQL query ?
निम्नलिखित एस क्यू एल क्वेरी से क्या परिणाम सेट लौटाया गया है?

```
Select customer_name, telephone
From customers
Where city in ('Jaipur','Delhi','Agra');
```

(a) The customer_name of all customers who are not living in Jaipur, Delhi or Agra./सभी उपभोक्ताओं _नाम जो जयपुर, दिल्ली या आगरा में नहीं रहते हैं।
(b) The customer_name and telephone of all customers./उपभोक्ता_नाम और सभी उपभोक्ताओं के टेलीफोन ।
(c) The customer_name and telephone of all customers living in either Jaipur, Delhi or Agra./उपभोक्ता___नाम और जयपुर, दिल्ली में या आगरा में रहने वाले सभी उपभोक्ता के टेलीफोन ।
(d) The customer_name and telephone of all customers living in Jaipur, Delhi and Agra./उपभोक्ता__नाम और जयपुर, दिल्ली और आगरा में रहने वाले सभी उपभोक्ता के टेलीफोन।

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Ans. (c) : SQL query –

```
SELECT customer_name, telephone
FROM customers
WHERE city IN ('Jaipur', 'Delhi', 'Agra');
```

when we are using keyword IN there must be OR keyword. So, result is –
The customer_name and telephone of all customers living in either Jaipur, delhi or Agra.

148. Consider two database relations R and S having 3 tuples in R and 2 tuples in S. What is the maximum number of tuples that could appear in the natural join of R and S?
दो डेटाबेस संबंध R और S पर विचार कीजिए, R में 3 टपल्स है और S में 2 टपल्स है। R और S के नैसर्गिक जोड (नेचुरल ज्वाइन) में दिखाई दे सकने वाले अधिकतम टपल्स की संख्या क्या होगी?

(a) 6 (b) 2 (c) 3 (d) 5

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Ans. (a) : In Natural join, if two tables R and S with m and n number of tuples respectively. Then maximum number of tuples in their natural join will be m×n. Hence , given two relations R and S having tuples 3 and 2 respectively. So, maximum number of tuples in natural join of R and S is 3×2 = 6.

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149. Which of the following is DDL command ?
निम्नलिखित में से कौन सी एक DDL कमांड है?

(a) UPDATE SET (b) DELETE
(c) DROP TABLE (d) INSERT INTO

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Ans. (c) : The DDL (Data Definition Language) command among the options you've provided is DROP TABLE. DDL commands are used to define and mange the structure of database objects such as tables, command like UPDATE SET, DELETE, and INSERT INTO are DML (data manipulation) commands used to manipulate data with in tables.

150. In a relational database model, cardinality of a relation means :
संबंधात्मक डेटाबेस मॉडल में, संबंध की गणनात्मकता का अर्थ है :

(a) The number of constraints
प्रतिबंधों (कन्सट्रेन्ट्स) की संख्या
(b) The number of tuples/टपल्स की संख्या
(c) The number of attributes
विशेषताओं (एट्रिब्यूट्स) की संख्या
(d) The number of tables
तालिकाओं (टेबल्स) की संख्या

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Ans. (b) : In a relational database model, cardinality represents the number of times an entity of an entity set participates in a relationship set. Or we can say that the cardinality of a relationship is the number of tuples (rows) in a relationship.

151. An entity set that does not have sufficient attributes to form a key is termed as :
कोई एन्टिटी सेट जिसमें 'कुंजी' बनाने के लिए पर्याप्त गुण नहीं होते हैं को कहा जाता है :

(a) Primary entity set (b) Strong entity set
(c) Weak entity set (d) Simple entity set

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Ans. (c) : The entity sets which do not have sufficient attribute to form a primary key are known as weak entity sets and the entity sets which have a primary key are known as strong entity sets.

152. In relational databases, the natural join of two tables is :
रिलेशनल डेटाबेसों में, दो तालिकाओं की नैसर्गिक मिलन (नेचुरल ज्वाइन) है:

(a) Cartesian Product always.
सदैव कार्टेसियन उत्पाद।
(b) Combination of Union and filtered Cartesian Product.
यूनियन और छनित कार्टेसियन उत्पाद का संयोजन।
(c) Combination of Selection and filtered Cartesian Product.
चयन और छनित कार्टेसियन उत्पाद का संयोजन।
(d) Combination of Projection and filtered Cartesian Product.
प्रक्षेपण और छनित कार्टेसियन उत्पाद का संयोजन।

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Ans. (d) : In relational databases, a natural Join is a Join operation that creates an implicit join clause for you based on the common columns in the two tables being joined. Common columns are columns that have the same name in both tables. A Natural Join can be an Inner join, a Left outer Join, or a Right Outer Join. It is denoted by a symbol (⋈). It is the Combination of projection and filtered Cartesian Product.

153. With SQL, how can you return the number of records in the "Persons" table?
एस क्यू एल से, आप "Persons" तालिका में रिकार्डो की संख्या को कैसे वापस भेज सकते है?

(a) SELECT COUNT (*) FROM Persons
(b) SELECT COUNT () FROM Persons
(c) SELECT COUNT () FROM Persons
(d) SELECT COLUMN (*) FROM Persons

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Ans. (a) : SQL Count function is used to count the number of records in the selected table. It is used with the SELECT statement.
Syntax–
SELECT COUNT (*) FROM table_name;
Here table_name is given as 'Persons'
So, the number of records in the "Persons" table can find out as–
SELECT COUNT (*) FROM Persons ;

154. Which of the following statements is false with respect to Relational DBMS?
संबंधात्मक डी बी एम एस के संबंध में निम्नलिखित में से कौन सा कथन गलत है?

(a) A disadvantage of highly normalized tables is that queries may require too many time consuming joins./अत्यधिक प्रासामान्यीकृत तालिकाओं की एक कमी यह है कि प्रश्नों (क्वेरी) को बहुत से समय साध्य ज्वाइनों की आवश्यकता होगी।
(b) Foreign key constraints are referential integrity constraints./'फारेन की ' प्रतिबंध रेफरंशल इन्टेग्रिटी प्रतिबंध है।
(c) A primary key uniquely indentifies a row in a table./प्राथमिक कुंजी अद्वितीय रूप से तालिका में पंक्ति की पहचान करती है।
(d) Nulls reduce space requirements in tables./अमान्य (नल्स) तालिका में स्थान की आवश्यकता को कम करता है।

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Ans. (d) : A relational database managements System is a collection of programs and capabilities that enable IT teams and others to create, update, administer and otherwise interact with a relational database. There are some properties of a Relation DBMS is given below-
⇒Foreign key constraints are referential integrity constraints.
⇒A disadvantage of highly normalized tables is that queries may require too many time consuming joins.
⇒A primary key uniquely identifies a row in a table.
⇒Null does not reduce space requirements in tables.
⇒All attributes in a relation are atomic.

155. Which command in MySQL is used to list all (non-temporary) tables in a given database?
एक दिए गए डाटाबेस में सभी (गैर-अस्थायी) टेबलों को सूचीबद्ध करने के लिए MySQL में कौन सी कमांड प्रयोग की जाती है?

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(a) DESCRIBE TABLES
(b) DISPLAY TABALES
(c) VIEW TABLES
(d) SHOW TABLES

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Ans. (d) : To list all non-temporary tables in a given database in MySQL you can use show tables, command. This command will display a list of all tables in the current database.

156. Which of the following MySQL query is syntactically correct and most preferred one?
निम्नलिखित में से कौन सा MySQL query syntactically रूप से सही है और जिसे सर्वाधिक प्राथमिकता दी जाती है?

(a) SELECT SECTION, COUNT(*) FROM STUDENT
ORDER BY SECTION GROUP BY SECTION WHERE MARKS<33 AND COUNT(*)>0;
(b) SELECT SECTION, COUNT(*) FROM STUDENT
WHERE MARKS<33 HAVING COUNT(*)>0 GROUP BY SECTION ORDER BY SECTION;
(c) SELECT SECTION, COUNT(*) FROM STUDENT
WHERE MARKS<33 GROUP BY SECTION HAVING COUNT(*)>0 ORDER BY SECTION;
(d) SELECT SECTION COUNT(*) FROM STUDENT
GROUP BY SECTION WHERE MARKS<33 AND COUNT(*)>0 ORDER BY SECTION;

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Ans. (c) : Among the provided option the third query is the most syntactically correct and preferred one for mySQL
Select section, count (*) from student where marks <33 group by section having count (*)> 0 order by section.
Select section. count (*)- selects the section column and count the number of rows for each section.
From student - specifies the table from which to retrieve data.
Where books < 33 - filters the rows where the marks column is less than 33.
Group by section - Groups the result by the section column.
Having count (*)>0- specifies that only groups with a count greater than 0 should be included.
Order by Section- Orders the result by the section column.

157. Which of the following MySQL query will display all the NAME from the table STUDENT, which are having maximum five characters and starting with the alphabet 'A'?
निम्नलिखित में से किस MySQL query में STUDENT से उन सभी NAME को दर्शाया गया है। जिनमें अधिकतम पाँच करेक्टर हैं और _____ 'A' से शुरू होते हैं?

(a) SELECT NAME FROM STUDENT WHERE NAME LIKE 'A_____';
(b) SELECT NAME FROM STUDENT WHERE NAME = 'A_____';
(c) SELECT NAME FROM STUDENT WHERE NAME = 'A%%%%;
(d) SELECT NAME FROM STUDENT WHERE NAME LIKE 'A%%%%'

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Ans. (a) : The correct MySQL query to achieve the desired result would be.
select name from student where name like 'A——';
This query selects all names from the table student where the name starts will the letter 'A' and has exactly five characters in table. The underscore character(_) in the LIKE clause represents a single wild and character, So 'A——' ,matches any fine character string starting with 'A'.

158. Identify the output value (ignoring the output header) of the MySQL query.
MySQL query की आउटपुट वेल्यू (आउटपुट को नजरंदाज करते हुए) की पहचान कीजिए।
SELECT 6 MOD 20;

(a) 3 (b) 6
(c) 20 (d) 2

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Ans. (b) : We need to calculate the modules of 6 divided by 20.
The modules operator (%) returns the remainder of the division operation. So 6 MOD 20 is equal to 6 since 6 is less than 20 and doesn't divide evenly into 20. Therefore the current is a (6)

159. Using join, which of the following MySQL query is almost equivalent to:
ज्वांइन का प्रयोग करते हुए, निम्नलिखित में से कौन सा MySQL query निम्नलिखित के लगभग समतुल्य है?
SELECT NAME, MARKS FROM STUDENT, RESULT WHERE STUDENT.ROLL= RESULT.ROLL;

(a) SELECT NAME, MARKS FROM STUDENT JOIN RESULT;
(b) SELECT NAME, MARKS FROM STUDENT EQUI JOIN RESULT;
(c) SELECT NAME, MARKS FROM STUDENT NATURAL JOIN RESULT;
(d) SELECT NAME, MARKS FROM STUDENT, RESULT;

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Ans. (c) : The original query
Select name, marks from student, result where student Roll = result.Roll;
This query retrieves the 'Name' and 'marks' columns from the 'student' an 'result ' table, respectively, where the 'Roll' Column in 'Student' is equal to the 'Roll' column in 'Result '.
Now let's analyze the given all quires. The closest query to the original is option (c) select name, marks from student Natural join result. If there are no another columns with the same name between 'Student' and 'Result' Otherwise. you should use the explicit 'Join'.
Syntax-
Select name marks from student join result on student Roll = Result. Roll

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160. Which of the following best describes an attribute/column in a table ?
निम्नलिखित में से कौन सा विकल्प किसी टेबल में एट्रीब्यूट/कॉलम को सर्वश्रेष्ठ रूप से परिभाषित करता है?

(a) It is a set of numeric values in a table
यह किसी टेबल में संख्यात्मक मान का एक सेट है।

(b) It is a set of string values in a table
यह किसी टेबल में स्ट्रिंग मान का एक सेट है।

(c) It is a vertical subset of a table
यह किसी टेबल का ऊर्ध्वाधर सबसेट है।

(d) It is a horizontal subset of a table
यह किसी टेबल का क्षैतिज सबसेट है।

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Ans. (c) : A column is a collection of cells aligned vertically in a table.
So option (c) is correct.

161. TRUNCATE command of SQL is an example of
SQL में TRUNCATE कमांड किसका उदाहरण है?

(a) DML (b) DCL
(c) DDL (d) TCL

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Ans. (c) : SQL TRUNCATE is a data definition language (DDL) Command. It removes all rows in a table.

162. This question is based on following table :
यह प्रश्न निम्नलिखित तालिका पर आधारित है।

TABLE : DELIVERY

SNO	DBOY	ITEM	STAR	AMOUNT
1001	AMAN	FOOD	5	3000
1005	JONES	MEDICINE	3	5000
1008	AMAN	MEDICINE	2	4000
1009	RAZID	FOOD	4	4500
1010	JONES	GIFT	5	5000
1018	AMAN	FOOD	5	3000
1022	RAZID	MEDICINE	4	5000

Which of the following commands will produce the output from the table DELIVERY, as shown below ?
निम्नलिखित में से किस कमांड से तालिका DELIVERY में से नीचे दिया आउटपुट आएगा?

DBOY	SUM (AMOUNT)
AMAN	10000
JONES	10000
RAZID	9500

(a) SELECT DBOY, AMOUNT FROM DELIVERY;
(b) SELECT DBOY, SUM (AMOUNT) FROM DELIVERY;
(c) SELECT DBOY, SUM (AMOUNT) FROM DELIVERY GROUP BY AMOUNT;
(d) SELECT DBOY, SUM (AMOUNT) FROM DELIVERY GROUP BY DBOY;

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Ans. (d) : The GROUP by statement groups rows that have the same values into summary rows, like "find the number of DELIVERY in each SUM (AMOUNT)"
Syntax-
SELECT column_name(s) FROM table_name WHERE condition GROUP BY column_name(s) ORDER BY column_name(s).
Follow the given above table and fire the query,
SELECT DBOY, SUM(AMOUNT) FROM DELIVERY GROUP BY DBOY.
So produce the output

DBOY	SUM(AMOUNT)
Aman	1000
Jones	1000
Razid	9500

163. This question is based on following table :
यह प्रश्न निम्नलिखित तालिका पर आधारित है।

TABLE : DELIVERY

SNO	DBOY	ITEM	STAR	AMOUNT
1001	AMAN	FOOD	5	3000
1005	JONES	MEDICINE	3	5000
1008	AMAN	MEDICINE	2	4000
1009	RAZID	FOOD	4	4500
1010	JONES	GIFT	5	5000
1018	AMAN	FOOD	5	3000
1022	RAZID	MEDICINE	4	5000

Which of the following SQL commands will display the names of delivery boys (DBOY) without any repeatation of names ?
निम्नलिखित में से किस SQL कमांड से बिना दोहराव के डिलीवरी बॉयज़ (DBOY) के नाम प्रदर्शित होंगे?

(a) SELECT DISTINCT DBOY FROM DELIVERY;
(b) SELECT UNIQUE DBOY FROM DELIVERY;
(c) SELECT NOREPEAT DBOY FROM DELIVERY;
(d) SELECT DBOY DISTINCT FROM DELIVERY;

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Ans. (a) : The SQL DISTINCT keyword is used in conjunction with the SELECT statement to eliminate all the duplicate records and fetching only unique records.
Syntax.
SELECT DISTINCT column1, column2, ...
FROM table_name,
So fire the Query, and produce the output
SELECT DISTINCT DBOY FROM DELIVERY

SNO	DBOY	ITEM	STAR	AMOUNT
1001	Aman	Food	5	3000
1005	Jones	medicine	3	5000
1009	Razid	Food	4	4500

164. This question is based on following table :
यह प्रश्न निम्नलिखित तालिका पर आधारित है।

TABLE : DELIVERY

SNO	DBOY	ITEM	STAR	AMOUNT
1001	AMAN	FOOD	5	3000
1005	JONES	MEDICINE	3	5000
1008	AMAN	MEDICINE	2	4000
1009	RAZID	FOOD	4	4500
1010	JONES	GIFT	5	5000
1018	AMAN	FOOD	5	3000
1022	RAZID	MEDICINE	4	5000

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Which of the following commands will display the entire content of the table DELIVERY in descending order of STAR rating ?
निम्नलिखित में से किस कमांड से तालिका DELIVERY के सारे घटक स्टार रेटिंग के अवरोही क्रम में प्रदर्शित होंगे?

(a) SELECT * FROM DELIVERY ORDER BY STAR DESCENDING;
(b) SELECT * FROM DELIVERY ORDER BY STAR;
(c) SELECT * FROM DELIVERY ORDER BY STAR DESC;
(d) SELECT * FROM DELIVERY ORDER BY STAR REVERSE;

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Ans. (c) : The ORDER BY keyword is used to sort the result set in ascending or descending order. Ascending order BY default.
Syntax.
SELECT Column1, Column2, ... FROM table_name ORDER BY column1, column2 ..... ASC/DESC,
So, when fire this query
SELECT * FROM DELIVERY ORDER BY STAR DESC.

SNO	DBOY	ITEM	STAR	AMOUNT
1001	Aman	Food	5	3000
1010	Jones	Gift	5	5000
1018	Aman	Food	5	3000
1009	Razid	Food	4	4500
1022	Razid	medicine	4	5000
1005	Jones	medicine	3	5000
1008	Amens	medicine	2	4000

165. This question is based on following table :
यह प्रश्न निम्नलिखित तालिका पर आधारित है।

TABLE : DELIVERY

SNO	DBOY	ITEM	STAR	AMOUNT
1001	AMAN	FOOD	5	3000
1005	JONES	MEDICINE	3	5000
1008	AMAN	MEDICINE	2	4000
1009	RAZID	FOOD	4	4500
1010	JONES	GIFT	5	5000
1018	AMAN	FOOD	5	3000
1022	RAZID	MEDICINE	4	5000

Which of the following SQL commands will remove the rows, which have column value of STAR less than 5 ?
निम्नलिखित में से किस SQL कमांड से वह पंक्ति (Row) हट जाएगी जिसके STAR कॉलम का मान 5 से कम है?

(a) REMOVE FROM DELIVERY WHERE STAR < 5;
(b) DROP TABLE DELIVERY WHERE STAR < 5;
(c) DELETE FROM DELIVERY WHERE STAR < 5;
(d) DELETE * FROM DELIVERY WHERE STAR < 5;

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Ans. (c) : The DELETE statement is used to delete existing records in a table.
Syntax-
DELETE FROM table_name WHERE Condition;
So when fire the query,
DELETE FROM DELIVERY WHERE STAR < 5;

SNO	DBOY	ITEM	STAR	AMOUNT
1001	Aman	Food	5	3000
1010	Jones	Gift	5	5000
1018	Aman	Food	5	3000

Condition where star < 5, deleted all record who less than 5.

166. Consider the following tables in a database SPORTS :
SPORTS डाटाबेस में निम्नलिखित टेबलों पर विचार कीजिए।

Table 1 : STUDENTS

ADMNO	SNAME
101	AMAR
205	MARIA
410	MALALA

Table 2 : GAMES

GAMEID	GNAME
G03	CRICKET
G05	FOOTBALL

Which of the following SQL Command will produce an output as :
किस SQL कमांड से नीचे दिया गया आउटपुट आएगा:

SNAME	GNAME
AMAR	CRICKET
MARIA	CRICKET
MALALA	CRICKET
AMAR	FOOTBALL
MARIA	FOOTBALL
MALALA	FOOTBALL

(a) SELECT SNAME, GNAME FROM STUDENTS, GAMES;
(b) SELECT SNAME, GNAME FROM STUDENTS X GAMES;
(c) SELECT SNAME FROM STUDENTS, GNAME FROM GAMES;
(d) SELECT SNAME FROM STUDENTS, SELECT GNAME FROM GAMES;

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Ans. (a) : A FOREIGN KEY is a field (or collection of field) in one table, that refers to the PRIMARY KEY in another table. The table with the foreign key is called the child table and the table with the primary key is called the referenced or parent table.

167. Which of the following is not an example of Database Application ?
निम्नलिखित में से कौन सा डाटाबेस एप्लीकेशन का उदाहरण नहीं है?

(a) Payroll Management/पेरोल मैनेजमेंट
(b) Digital Drawing/डिजिटल ड्राइंग
(c) HR Management/एच आर मैनेजमेंट
(d) Invoicing System/इनवॉयसिंग सिस्टम

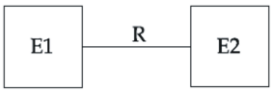
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Ans. (b) : Payroll Management, HR Management and Invoicing System are related to Database application while Digital Drawing refers to any artist work of practice that uses digital technology as part of the

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creative or presentation process, or more specifically compatational art that uses and engages with digital media.

168. Which of the following options appropriately represent the relationship shown in the diagram ?
नीचे दिए गए चित्र में प्रदर्शित संबंध को कौन सा विकल्प उपयुक्त रूप से निश्चित करता है?



(a) ONE - TO - ONE
(b) ONE - TO - MANY
(c) MANY - TO - ONE
(d) MANY - TO - MANY

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Ans. (a) : One to one relationship in DBMS is a relationship between an instance of an entity with another.

169. In order to add a new column to an existing table in SQL, we can use the command
SQL में वर्तमान टेबल में नया कॉलम को जोड़ने के लिए. किस कमान्ड (आदेश) का प्रयोग कर सकते है?

(a) MODIFY TABLE
(b) EDIT TABLE
(c) ALTER TABLE
(d) ALTER COLUMNS

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Ans. (c) : The ALTER TABLE Command is used to add, delete. Or modify columns in an existing table.

170. In an RDBMS relationships between tables are created by using
RDBMS में टेबलों के बीच संबंध का निर्माण किसके उपयोग से किया जाता है ?

(a) Alternate keys./अल्टरनेट की
(b) Foreign keys./फोरेन की
(c) Candidate keys./केन्डीडेट की
(d) Composite keys./कम्पोज़ीट की

KVS PGT C.S. 2013

Ans. (b) : Foreign key is a field in one table, that refers to the primary key in another table. The table with the foreign key is called the child table, and the table with the primary key is called the referenced or parent table.

171. The command used to see the fields of the table along with their data types in SQL is
SQL में कौन से कमांड के प्रयोग से सारिणी के फील्ड्स को उसके डेटाटाईप सहित देखा जा सकता है?

(a) Select fields from dual where table = "MANAGER";
(b) Select field_names, datatype from dual where table_ name = "MANAGER";
(c) Desc MANAGER;
(d) Select description from dual where table_ name = "MANAGER";

KVS PGT C.S. 2013

Ans. (c) : Describe is used to describe something. since in database we have tables that's why we use describe or Desc command to describe the structure of a table syntax
Describe one,
or
Desc one,
So Desc command used to see the fields with data type.

172. Which of the following statements is NOT true for Rollback statements in SQL ?
SQL में रोलबैक कथन के लिए निम्नलिखित में से कौन सा कथन सत्य नहीं है ?

(a) All save points marked after the save point to which you roll backed, are erased./सेव पोइन्ट के पश्चात् चिन्हित सभी सेव पोइन्टों को जिन्हें आपने रोल बैक किया है, मिटाया जाता है ।
(b) It does not free any resources held by the transaction./यह कार्य संपादन में रखे हुए कोई भी संसाधन को मुक्त नहीं करता है ।
(c) The save point to which you rollback is not erased./वह सेव पोईंट जिस पर आपने रोल बैक किया है नहीं मिटाया जाता ।
(d) The Rollback statement will erase all data modifications made from the start of the transaction to the save point./रोल बैक कथन, कार्य संपादन (ट्राजेक्शन) के आरम्भ से सेव पोईंट तक के सभी डेटा संशोधनों को मिटा देगा।

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Ans. (d) : Rollback in SQL is transactional Control language that is used to undo the transaction that have not been saved in the database.
The command is only been used to undo changes since the last commit.
So given option (d) is not trail about Rollback.

173. Which of The following statements is not true for views in SQL ?
SQL में दृश्यों के लिए निम्न में कौन सा कथन सत्य नहीं हैं ?

(a) Select statement used the view definition can not include ORDER BY clause./दृश्य परिभाषित करने में Select कथन का उपयोग ORDER BY वाक्यांस समाहित नहीं कर सकता।
(b) A view derives its data from the base tables (s)./एक दृश्य अपना डेटा आधार सारिणी से व्यत्पन्न करता है ।
(c) A view is updatable if it has been defined from a single relation./यदि एकल संबंध से परिभाषित किया गया हो तो कोई दृश्य अद्यतन हो सकता है ।
(d) A view contains a copy of the data.
एक दृश्य में डेटा की प्रतिकृति रहती है ।

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Ans. (c) : A view in SQL is simply a virtual table created based an a result set of another SQL statement.
A view to be updatable , there must be one to one relationship between the rows in the underlying table.

174. Consider the following EMP table and answer the question below:
निम्न EMP सारिणी पर मनन करें तथा नीचे लिखे प्रश्न का उत्तर दीजिए :

EMP_NO	SALARY	GRADE
E001	1000	E1
E002	2000	E4
E003	1500	E2
E004	4000	E4

Which of the following select statement should be executed if we need to display the average salary of employees who belong to grade "E4"?
निम्न में से किस चयन कथन को कार्यान्वित किया जाना चाहिये, यदि हमें "E4" ग्रेड के कर्मचारी का औसत वेतन दिखाना हों ?

(a) Select avg(salary) from EMP whose grade = "E4";
(b) Select avg(salary) from EMP having grade = "E4";
(c) Select avg(salary) from EMP group by grade where grade = "E4";
(d) Select avg(salary) from EMP group by grade having grade ="E4";

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Ans. (d) : Condition specified in where clause is used while fetching data from table and data which doesn't pass the condition will not be fetched into result set.
Having clause is used to filter summarized data or grouped data . we require average Salary of employees whose grade is E4. so we require grouped data.

175. Consider the relation Emp-Dept with SSn as key.
कुंजी के रूप में SSn के साथ Emp-Dept के संबंध पर विचार करे।

Ename	SSn	EAddr	Dept. No.	Dept. Name	Dept. Manager	SSn

Which of the following is (are) invalid operations (s)?
निम्नलिखित में से कौन सा अवैध संचालन है?

(i) Inserting an employee without name and address./नाम और पते के बिना किसी कर्मचारी को सम्मिलित करना।
(ii) Inserting an employee with only SSn.
केवल SSn वाले कर्मचारी को सम्मिलित करना।
(iii) Inserting a department with no employee./बिना कर्मचारी वाले विभाग को सम्मिलित करना।
(iv) Inserting an employee without SSn
SSn के बिना कर्मचारी को सम्मिलित करना।

(a) (i) (b) (iv)
(c) (iii) and (iv) (d) (ii) and (iii)

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Ans. (b) : In given relation employee and department both table are merged into one table so right most SSn id is foreign key and second entry from left side is key of employee table. Hence, Inserting an employee without SSn is invalid operation because SSn employee id cannot be null.

176. Which of the following statements is incorrect?
निम्नलिखित में से कौन सा कथन गलत है?

(a) Data definition languages is used by DBA and database designers to define schemas.
स्कीमा को परिभाषित करने के लिए DBA और डेटाबेस डिजाइनरों द्वारा डेटा परिभाषा भाषाओं का उपयोग किया जाता है।
(b) Storage definition language is used to specify the internal schema/आंतरिक स्कीमा को निर्दिष्ट करने के लिए स्टोरेज परिभाषा का उपयोग किया जाता है।
(c) Storage definition language is used to insert delete and update data /स्टोरेजे डेफिनिशन लैंग्वेज का इस्तेमाल डिलीट और अपडेट डेटा डालने के लिए किया जाता है।
(d) Data definition languages is used to retrieve data from the database./डेटा डिफिनिशिन लैंग्वेज का उपयोग डेटाबेस से डेटा को पुनः प्राप्त करने के लिए किया जाता है।

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Ans. (c) : Data Definition Language (DDL) actually consists of the SQL commands that can be used to define the database schema. It simply deals with descriptions of the database schema and is used to retrieve data from the database.
Storage Definition Language (SDL) is different in every DBMS which specifies anything to do with how or where data in relevant table is stored. It is used to define internal schema. It defines physical structure of database.
So, option (c) is incorrect statement.

177. Consider the following SQL query:
निम्न SQL क्वेरी पर विचार करें:

PROJ (Pnum, Dnum)
DEPT (Dnumber, Mgr_SSn)
EMP (fname, lname, SSn)
Works_ON (Pno, E SSn)
(SELECT DISTINCT Pnum FROM PROJ, DEPT,
EMP WHERE Dnum = Dnumber
AND Mgr_SSn = SSn AND Lname = 'XXXX')
UNION
(SELECT DISTINCT Pnum FROM PROJ, WORKS_ON,
EMP WHERE Pnum = Pno AND ESSn = SSn
AND Lname = 'XXXX')

(a) All project numbers for projects that involve an employee with last name 'XXXX', either as worker or as manager
(b) All project numbers for projects that involve an employee with last name 'XXXX' as manager.
(c) Records of all employees with last name as 'XXXX' working in a project as manager.
(d) All project number for projects in department Dnumber and where 'XXXX' is involved either as worker or as manager.

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Ans. (a) : The SQL UNION operator is used to combine the results of two or more SELECT statements without returning any duplicate rows. So, the union of both subquery displayed as a result, 'all project number for projects that involve an employee with last name 'XXXX', either as worker or a manager.'

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178. Given the following two statements about SQL.
नीचे SQL के बारे में दो कथन दिए गए हैं:

(A) An SQL query can contain HAVING clause only if it has GROUP BY clause.
किसी SQL क्वेरी में HAVING क्लॉज तभी हो सकता है यदि इसमें GROUP BY क्लॉज हो।

(B) In an SQL query "SELECT_FROM_WHERE_GROUP BY_HAVING". HAVING is executed before WHERE.
किसी SQL क्वेरी "SELECT_FROM_WHERE_GROUP BY_HAVING" में WHERE से पहले HAVING निष्पादित किया जाता है।

Which of the following is correct?
निम्नलिखित में से कौन-सा सत्य है?

(a) (A) and (B) both are true
(A) तथा (B) दोनों सत्य है?
(b) (A) is true, (B) is false
(A) सत्य है, (B) असत्य है।
(c) (A) is false, (B) is true
(A) असत्य है, (B) सत्य है।
(d) (A) and (B) both are false
(A) तथा (B) दोनों असत्य हैं।

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Ans. (b) : Statement (A) 'An SQL query can contain HAVING clause only if it has GROUP By clause', is correct because Having clause is applied after the aggregation phase and must be used if you want to filter aggregate results.
Statement (B) is saying that HAVING is executed before WHERE but according to order of execution of clauses, HAVING is executed after WHERE.
So, statement (B) is incorrect.

179. _____ operation preserves those tuples that would be lost in _____.
_____ ऑपरेशन उन टपल्स को संरक्षित करते है जो कि _____ में गुम जो जाती।

(a) Natural join, outer join
नैचुरल जॉइन, आउटर जॉइन
(b) Outer join, natural join
आउटर जॉइन, नैचुरल जॉइन
(c) Left outer join, right outer join
लैफ्ट आउटर जॉइन, राइट आउटर जॉइन
(d) Left outer join, natural join
लैफ्ट आउटर जॉइन, नैचुरल जॉइन

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Ans. (b) : A outer join operation preserves those tuples that would be lost in natural join. That is, a full outer join returns all matching and non-matching rows from the left and right table.

180. _____ key must satisfy referential integrity in a relation, white _____ key must satisfy entity integrity.
_____ 'की' को किसी रिलेशन में रेफरेन्सियल इंटीग्रिटी को संतुष्ट करना चाहिए जबकि _____ 'की' को एन्टिटी इंटीग्रिटी को संतुष्ट करना चाहिए।

(a) Candidate, primary/कैंडिडेट, प्राइमरी
(b) Foreign, primary/फॉरेन, प्राइमरी
(c) Primary, foreign/प्राइमरी, फॉरेन
(d) Foreign super key/फॉरेन, सुपर- 'की'

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Ans. (b) : Foreign key must satisfy referential integrity in a relation, while primary key must satisfy entity integrity. A primary key is a column or a set of columns in a table whose values uniquely identify a row in the table. A foreign key is a column or set of columns in a table whose value correspond to the value of the primary key in another table.

181. In RDBMS, different classes of relations are created using _____technique to prevent modification anomalies.
RDBMS में संशोधन विसंगतियों को रोकने के लिए _____ तकनीक का उपयोग करके सम्बन्धों के विभिन्न वर्ग बनाए जाते हैं।

(a) Functional dependencies/फंकशनल डिपेन्डेन्सिज
(b) Data integrity/डेटा इन्टेग्रिटि
(c) Referential integrity/रेफरेन्शियल इन्टेग्रिटि
(d) Normal forms/नॉर्मल फॉर्म्स

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Ans. (d) : In relational database management systems different classes of relations are created using normal forms to prevent modification anomalies. Normalization is the process of organizing data to reduce redundancy and improve data integrity.

182. Which SQL command is used to add new rows to a database table?
डेटाबेस तालिका में नई पंक्तियाँ जोड़ने के लिए किस SQL कमांड का उपयोग किया जाता है

(a) ADD/जोड़ें
(b) CREATE/बनाएँ
(c) INSERT/सम्मिलित करें
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : The SQL command INSERT is used to add new rows to a database table.

183. In a relational database, what is a primary key?
रिलेशनल डेटाबेस में प्राथमिक कुंजी क्या है?

(a) A key used to uniquely identify each record in a table/विशिष्ट पहचान के लिए उपयोग की जाने वाली कुंजी तालिका में प्रत्येक रिकॉर्ड
(b) A key used to establish relationships between tables/स्थापित करने के लिए उपयोग की जाने वाली कुंजी तालिकाओं के बीच संबंध
(c) A key used to sort records in a table/एक कुंजी जिसका उपयोग रिकॉर्ड्स को सॉर्ट करने के लिए किया जाता है
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : In a relational database, a primary key is the 'key' that is used to uniquely identify each record in a table.

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184. Which of the following is a common file operation?
निम्नलिखित में से कौन-सा एक सामान्य फाइल ऑपरेशन है?
(a) Create/बनाएँ
(b) Delete/हटाएँ
(c) Rename/नाम बदलना
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) **:** Common file operations are as follows.
1. Create- Creating a new file.
2. Read- Reading the contents of a file.
3. Write- adding or change data in the file.
4. Rename- Rename the file.
5. Remove- Remove the file comletely.

185. What is an Instance of a Database?
डेटाबेस का एक उदाहरण क्या है?
(a) The state of the database system at any given point of time
(b) The entire set of attributes of the Database put together in a single relation
(c) The initial values inserted into the Database immediately after its creation
(d) More than one of the above
(e) None of the above
BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) Instance of a database is the values of these variables at any given point of time. Instance is also called the current state or state of the database. Database schema is the design that defines the variables in the table that belong to a particular database there can be multiple instances for a certain database schema.

186. Which command is used to remove a relation from an SQL?
एस. क्यू. एल. से संबंध हटाने के लिए किस कमांड का उपयोग किया जाता है?
(a) Drop table
(b) Delete
(c) Purge
(d) More than one of the above
(e) None of the above
BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) In SQL, Drop table deletes the entire structure of the relation. While purge deletes the table which cannot be recovered.

187. _____is a set of one or more attributes taken collectively to uniquely identify a record.
_____किसी रिकॉर्ड को विशिष्ट रूप से पहचानने के लिए सामूहिक रूप से ली गई एक या अधिक विशेषताओं का एक सेट है।
(a) Primary key
(b) Foreign key
(c) Superkey
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(c): A 'Super key' is a set of one or more attributes, taken collectively. to uniquely identify a record.

188. Which of the following commands is correct to delete the values in the relation teaches?
संबंध सिखाने वाले मानों को हटाने के लिए निम्नलिखित में से कौन-सा कमांड सही है?
(a) Delete from teaches.
(b) Delete from teaches where Id = 'Null';
(c) Remove table teaches:
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं
BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) : In the database, the values of a table can be deleted using Delete command.
Delete from table name
Delete from teaches;
In this, the table named teaches has been deleted.

189. In relational database management, which of the following is/are property/properties of candidate key?
P : Uniqueness **Q : Irreducibility**
(a) P only (b) Q only
(c) Both P and Q (d) Neither P nor Q
UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : Candidate key : It uniquely identifies the tuples in a relation. A candidate key is a super key with no redundant attribute. In a relation, there can be more than one candidate key.
As, candidate uniquely identifies the tuples in a relation. So, first property of the candidate key is its uniqueness.
second, candidate key does not contain redundant attribute. We cannot further reduce a candidate key. If a candidate key is reduced, then it can cause partial dependency. second property is its irreducibility.
Example :
A relation with four attributes : R(A,B,C,D).
functional dependencies are :
$A \rightarrow C$
$B \rightarrow D$
In this, AB is the candidate key.
AB uniquely tuples in relation R also AB cannot be reduced further.

190. Which of the following statements are DML statements?
(A) Update [tablename]
Set [columnname] = VALUE
(B) Delete [tablename]
(C) Select * from [tablename]
(a) (A) and (B) (b) (A) and (D)
(c) (A), (B), and (C) (d) (B) and (C)
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Ans. (c) : A data manipulation language (DML) is a language that enables users to access or manipulate data as organized by the appropriate model.
Example of DML commands : select, insert, delete, update etc.

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therefore a, b and c are DML statements.
A database system provides a **data-definition language (DDL)** to specify the database schema example of DDL commands : Create, drop, alter, rename, truncate etc.
A **data control language (DCL)** is used to control access (Authorization) to data stored in a database example of DCL commands : grant and revoke.

191. Following table has two attributes Employee id and Manager id, where Employee is a primary key and manager id is a foreign key referencing Employee id with on delete cascade:

Employee_id	Manager_id
20	40
25	40
30	35
35	20
40	45
45	25

On deleting the table (20, 40), the set of other tuples that must be deleted to maintain the referential integrity of table is

(a) (30, 35) only
(b) (30, 35) and (35, 20) only
(c) (35, 20) only
(d) (40, 45) and (25, 40) only

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Ans. (b) : Manager-id is foriegn key referring to employee-id (primary key) of the same table

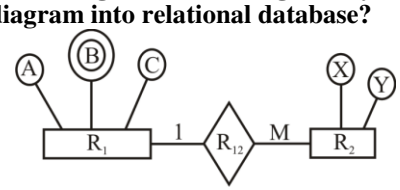
Employee_id	Manager_id
20 (Deleted 1)	40 (Deleted 1)
25	40
30 (Deleted 3)	35 (Deleted 3)
35 (Deleted 3)	20 (Deleted 2)
40	45
45	25

(20, 40) is deleted, that is employee-id = 20 is deleted therefore manager-id = 20 should be deleted to maintain the referential integrity of table.
If Manager-id = 20 row is deleted, then employee-id = 35 is also deleted
therefore manager-id = 35 should also be deleted to maintain the referential integrity of table.
If manager-id = 35 row is deleted then employee-id = 30 is also deleted since If manager-id ≠ 35 we stop here.
Table after deletion

Employee_id	Manager_id
25	40
40	45
45	25

A part from (20, 40) (30, 20) also gets deleted to maintain the referential integrity of table.

192. Find minimum number of tables required for converting the following entity relationship diagram into relational database?



(a) 2 (b) 4
(c) 3 (d) 5

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Ans. (c) : Rules for finding minimum number of tables required for an ER diagram :

1. A strong entity with single or composite attributes require one table.
2. A strong entity with multivalued attribute requires two tables.
3. In case of many to many relations between two entities, 3 tables are required.

There is one to many relationships between R1 and R2. So, two tables are required for two entities. But, entity R2 contains multivalued attribute B, due to which one table for this is also needed. So, minimum number of tables required for the given ER diagram are 3.

193. Which of the following key constraints is required for functioning of foreign key in the context of relational data bases?

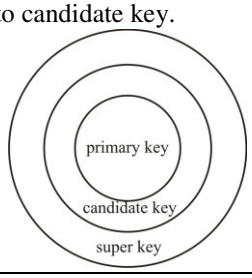
(a) Unique key (b) Primary key
(c) Candidate key (d) Check key

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) :

* **Candidate Key →**
- It can be null,
- It is null in that such case
- Foreign key point to primary key.

* **Alternate Key →** unique and null
* If candidate key is null in such case foreign key points to primary key.
though foreign keys can always point to primary key but not to candidate key.



194. Given two tables EMPLOYEE (EID, ENAME, DEPTNO) DEPARTMENT (DEPTNO, DEPTNAME)
Find the most appropriate statement of the given query :

```
Select count (*) 'total'
from EMPLOYEE
where DEPTNO IN (D1, D2)
group by EPTNO
having count (*) > 5
```

(a) Total number of employees in each department D1 and D2
(b) Total number of employees of department D1 and D2 if their total is > 5
(c) Display total number of employees in both departments D1 and D2
(d) The output of the query must have atleast two rows

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Ans. (b) : Given query is :

```
select count (*) total
from EMPLOYEE where
DEPTNO IN (D1, D2)
group by DEPTNO
having count (*) > 5
```

STEP1 :- It will first find the all the entries with department no. as D1 and D2 with the help of natural join from both the relations.

STEP2 :- Then in the resulting table, it groups the tuples with the department no. (DEPTNO)

STEP3 :- Then select all those tuples having count > 5

STEP4 :- Finally, it results in the total number of employees of department D1 and D2 if their total is > 5

195. Given the following STUDENT - COURSE scheme: STUDENT (Rollno, Name, Courseno) COURSE (Courseno, Coursename, Capacity), where Rollno is the primary key of relation STUDENT and Courseno is the primary key of relation COURSE. Attribute Coursename of COURSE takes unique values only. The number of records in COURSE and STUDENT tables are 3 and 5 respectively. Following relational algebra query is executed: R=STUDENT X COURSE

Match List-I with List-II in context to the above problem statement.

	List - I		List - II
A.	**Degree of table R**	**i.**	**15**
B.	Cardinality of table R	**ii.**	**NIL**
C.	**Foreign key of relation STUDENT**	**iii.**	**6**
D.	**Foreign key of relation COURSE**	**iv.**	**Courseno**

Choose the correct answer from the options given below:

(a) A - III, B - I, C - IV, D - II
(b) A - I, B - III, C - IV, D - II
(c) A - I, B - III, C - IV, D - II
(d) A - I, B - III, C - II , D - IV

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Ans. (a) :

	List -I		List-II
A.	Degree of table R	iii.	6
B.	Cardinality of table R	i.	15
C.	Foreign key of relation STUDENT	iv.	Courseno
D.	Foreign key of relation COURSE	ii.	NIL

196. Identify the incorrect statement(s).

(a) A candidate key is minimal set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set.
(b) A candidate key for which no proper subset is also a candidate key is called a super key.
(c) A super key is a set of one or more attributes that, taken collectively, allows us to uniquely identify any entity in the entity set.
(d) A super key for which no proper subset is also a super key is called a candidate key.

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Ans. (b) : Candidate key : The minimal set of attributes that can uniquely identify a tuple is known as a candidate key.

- It is a minimal super key.
- It most contain unique values.
- The minimal set of attributes that can uniquely identify a record.

Super key : The set of attributes that can uniquely identify a tuple in know a super key.

- A candidate key is a super key but vice versa is not true.

197. Which of the following statement is correct, according to the following instance of the relational schema R(X, Y, Z)?

X	Y	Z
1	**1**	**a**
1	**1**	**a**
2	**1**	**b**
3	**2**	**b**

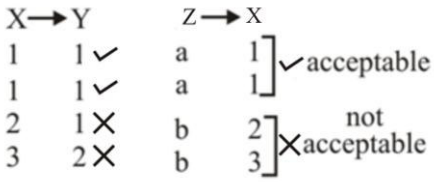
(a) X→Y, Z→X (b) Y→Z, Z→X
(c) X→Y, X→Z (d) Y→X, X→Z

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Ans. (c) : Given the relational scheme R (X, Y, Z)

X	Y	Z
1	1	a
1	1	a
2	1	b
3	2	b

Option (a) is in correct



Option (b) is incorrect because Y→ Z is not acceptable..
Option (c) is correct
X→ Y and X→ Z is acceptable.
Option (d) is incorrect because Y→ X is not acceptable.

198. In RDBMS, which type of Join returns all rows that satisfy the join condition ?

(a) Inner Join (b) Outer Join
(c) Semi Join (d) Anti Join

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Ans. (a) : An Inner join (Sometimes called a simple join) is a join of two or more tables that return only those rows that satisfy the join condition.

Outer Joins– An outer join extends the result of a simple join. An outer join returns all rows that satisfy the join condition and also return some or all of those rows from one table for which no rows from the other satisfy the join condition.

199. Consider a relation book (title, price) which contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list ?

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Select title
from book as B
where (select count (*)
from book as T
where T.price > B.price) < 7

(a) Titles of the six most expensive books.
(b) Title of the sixth most expensive books.
(c) Titles of the seven most expensive books.
(d) Title of the seventh most expensive books.

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Ans. (c) : Select title
from book as B
where (select count (*)
from book as T
where T.price > B.price) < 7
Titles of the seven most expensive books.

200. _____ command is used to remove a relation from an SQL database.

(a) Remove table (b) Delete table
(c) Update table (d) Drop table

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Ans. (d) : Drop table deletes the whole structure of the relation. purge removes the table which cannot be obtained again. The SQL UPDATE Query is used to modify the existing records in a table.

201. Identify the correct operation which produces that below given output based on two relations R1 and R2.

R1

Name	Class
a1	9
b1	5
c1	6
d1	11

R2

Name	Class
b1	5
f1	8

Output

Name	Class
b1	5

Choose the correct answer from the options below:

(a) R1 ∩ R2 (b) R1 – R2
(c) R1 – (R1 – R2) (d) R1 ∪ R2

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Ans. (a) : Given two relation

R_1

Name	Class
a1	9
b1	5
c1	6
d1	11

R_2

Name	Class
b1	5
f1	8

Output

Name	Class
b1	5

Intersection of set→The intersection of two given sets is the set that contains all the element that are common to both sets given relation.

$R_1 \{9, 5, 6, 11\}$

$R_2 \{5, 8\}$

$R_1 \cap R_2 = \{5\}$

So option (a) is correct. but commission answer were both(a & c).

202. Which of the following queries would retrieve the roll number and names of students who have enrolled for all the prerequisite course of the course number 324?

(a) Select s.rollno, s.name From student s, enrollment e Where e.rollno = s.rollno and e.courseid = ANY(select precouseied from prerequisite p where p.courseid = "324")
(b) Select s.rollno, s.name From student s, enrollment e Where e.rollno = s.rollno and e.courseid = ALL(select precouseied from prerequisite p where p.courseid = "324")
(c) Select s.rollno, s.name From student s Where NOT EXISTS (select * prerequisite p where p.courseid = "324") and NOT EXISTS (select * from enrollment e where e.courseid = p.precoursied and e.rollno = s.rollno.)
(d) Select s.rollno, s.name From student s Where EXISTS (select * prerequisite p where p.courseid = "324") and EXISTS (select * from enrollment e where e.courseid = p.precoursied and e.rollno = s.rollno.)

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Ans. (c) : The correct query to retrieve the roll number and name of students who have enrolled for all the prerequisite courses of the course number 324.
C. Select s.rollno, s.name from student s Where NOT EXISTS (select * prerequisite p where p.courseid = "324") and NOT EXISTS (select * from enrollment e where e.courseid = p.precourseid and e.rollno = s.rollno.)

203. Given query :
select s.rollno, s.name from student s where EXISTS (select c.courseid from course c Where c.deptno = 5 and EXISTS (select e.* from enrollment e where e.courseid = c.courseid and e.rollno = s.rollno)));
Which one of the following statement is correct with respect to the given query?

(a) It retrieves the roll number and names of students who have not enrolled for all course offered by department number 5.
(b) It retrieves the roll number and names of students who have enrolled for all course offered by department number 5.
(c) It retrieves the roll number and names of students who have enrolled for at least one course offered by department number 5.
(d) It retrieves the roll number and names of students who are such that there exists at least one course offered by department number 5 which is not enrolled by the student.

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Ans. (c) : Given query :
select s.rollno, s.name From student
Where EXISTS (select c.courseid from course c Where c.deptno = 5 and EXISTS (select e.* from enrollment e where e.courseid = c.courseid and e.rollno = s.rollno));
The following statement is correct with respect to the given query.

- It retrievs the roll number and name of student who have enrolled for at least one course offered by department number 5.

204. Given the basic ER diagram and relational model, which of the following is incorrect?

(a) An attribute of an entity can have more than one value
(b) An attribute of an entity can be composite
(c) In a row of relational table, an attribute can have more than one value
(d) In a row of a relational table, an attribute can have exactly one value or a NULL value

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Ans. (c) : The term 'entity' belongs to ER model and the term 'relational table' belongs to relational model. Statement of the option (a) and (b) both are correct. ER model supports both multivalued and composite attributes. Statement of the option (c) is incorrect and (d) is correct. In relation model, an entry in relational table can have exactly one value or a NULL value.

205. "CREATE TABLE T" in SQL is an example of :

(a) Normalization (b) DML
(c) DDL (d) Primary key

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Ans. (c) : SQL is a standard database language used to access and manipulate data in databases. SQL was developed by IBM computer scientists in the 1970s. There are several types of SQL statemants-
(i) Data Definition Language (DDL)
(ii) Data Manipulation Language (DML)
(iii) Data Query Language (DQL)
(iv) Data control Language (DCL)
(v) Transaction control Language (TCL)
In DDL, used to define, modify and delete database objects such as tables, indexes, and views. Examples include CREATE, ALTER, DROP
So, 'CREATE TABLE T' in SQL is an example of DDL.

206. What is SQL primarily used for in the context of relational databases ?

(a) To design user interfaces
(b) To create and manipulate databases
(c) To display data on web pages
(d) To format printed reports

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Ans. (b) : SQL (Structured Query Language) is a standard language used to store retrieves and manipulate data in relational databases. It allows end users to communicate with databases and perform tasks like creating, updating, and deleting databases.

207. In RDBMS, different classes or relations are created using _____ technique to prevent modification anomalies.

(a) Functional dependencies
(b) Data integrity
(c) Referential integrity
(d) Normal forms

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Ans : (d) Normal forms are defined as functional dependency, data integrity, referential integrity, normalization.

208. _____ SQL command changes one or more fields in a record.

(a) LOOK-UP (b) INSERT
(c) MODIFY (d) CHANGE

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) In DBMS, modify, a SQL command that changes one or more fields in a record.

209. How does a relational database ensure data integrity?

(a) By encrypting all data stored
(b) By enforcing rules defined in the schema
(c) By compressing data for efficient storage
(d) By allowing unrestricted access to all users

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : Relational database ensure data integrity by enforcing rules defined in the schema, such as constraints (e.g. primary keys, foreign keys, unique constraints, and check constraints). These rules help maintain accuracy and consistency of data within the database.

210. For a database relation R(A, B, C, D) where the domains of A, B and D include only atomic values, only the following functional dependencies and those that can be inferred from them are :
$A \rightarrow C$
$B \rightarrow D$
The relation R is in _____

(a) First normal form but not in second normal form
(b) Both in first normal form as well in second normal form
(c) Second normal form but not in third normal form
(d) Both in second normal form as well in third normal form

UGC NET C.S. June-2017 (Paper-III)

Ans : (a) Here $A \rightarrow C$ & $B \rightarrow D$ no. Singleton attribute derives all other attribute. But if we take $AC \rightarrow BD$ so closure of $AC = \{AC\}^+ = \{A, B, C, D\}$
So AB is the only candidate key of the relation.
Now as 2NF says, attributes should be fully dependent on the key. But here C & D are partially dependent.
So it is in 1NF

211. Consider the following relation :
Works (emp_name, company_name, salary)
Here, emp_name is primary key.
Consider the following SQL query

```
Select emp_name
From works T
where salary > (select avg (salary)
                from works S
                where T.company_name =
                      S.company_name)
```

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The above query if for following :

(a) Find the highest paid employee who earns more than the average salary of all employees of his company

(b) Find the highest paid employee who earns more than the average salary of all the employees of all companies

(c) Find all employees who earn more than the average salary of all employees of all the companies

(d) Find all employees who earn more than the average salary of all employees of their company

UGC NET C.S. June-2017 (Paper-III)

Ans : (d) It doesn't select the highest paid employees, as max is not used, It just select employees name who earns more than the average salary of all employee of the all companies will not be considered.

212. Which of the following is/are true with reference to 'view' in DBMS?

(A) A 'view' is a special stored procedure executed when certain event occurs

(B) A 'view' is a virtual table, which occurs after executing a pre-complied query.

Code :

(a) Only (A) is true

(b) Only (B) is true

(c) Both (A) and (B) are true

(d) Neither (A) nor (B) are true

UGC NET C.S. November-2017 (Paper-II)

Ans : (b) 'view' is a virtual table, through which a selective portion of the data from one or more table can be seen. A 'view' does not contain data of its own.

213. In SQL _____is an Aggregate function.

(a) SELECT (b) CREATE

(c) AVG (d) MODIFY

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) AVG function returns the average of the values in a group. It ignores null values. It is an Aggregate function.
Syntax : ([All Distinct] expression)

214. Match the following with respect to RDBMS :

A.	**Entity integrity**	**i.**	**Enforces some specific business rule that to not fall into entity or domain**
B.	**Domain integrity**	**ii.**	**Rows can't be deleted which are used by other records**
C.	**Referential integrity**	**iii.**	**Enforces valid entries for a column**
D.	**User defined integrity**	**iv.**	**No duplicate rows in a table**

Code :

	A	**B**	**C**	**D**
(a)	iii	iv	i	ii
(b)	iv	iii	ii	i
(c)	iv	ii	iii	i
(d)	ii	iii	iv	i

UGC NET C.S. November-2017 (Paper-II)

Ans : (b) : The correct match–

A.	Entity integrity	iv.	No duplicate rows in a table
B.	Domain integrity	iii.	Enforces valid entries for a column
C.	Referential integrity	ii.	Rows can't be deleted which are used by other records
D.	User defined integrity	i.	Enforces some specific business rule that to not fall into entity or domain

215. Which one is correct w.r.t. RDMBS?

(a) Primary key ⊆ super key ⊆ candidate key

(b) Primary key ⊆ candidate key ⊆ super key

(c) Super key ⊆ candidate key ⊆ primary key

(d) Super key ⊆ primary key ⊆ candidate key

UGC NET C.S. June-2017 (Paper-III)

Ans : (b) Super Key : An attribute that uniquely defines a tuple within a relation.
Candidate key : Super key such that no proper subset is a super key within in a relation so basically has two **Properties :** Each candidate key uniquely identifies tuple in the relation & no proper subset of the composite key has the uniqueness property.
Primary Key : Candidate key chosen to identify tuples uniquely within the relation.

216. Consider the following ORACLE relations:

One (*x*, *y*) = {<2, 5>, <1, 6>, <1, 6>, <1, 6>, <4, 8>, <4, 8>}

Two (*x*, *y*) = {<2, 55>, <1, 1>, <4, 4>, <1, 6>, <4, 8>, <4, 8>}

Consider the following two SQL queries SQ1 and SQ2:

SQ1: SELECT * FROM One)
EXCEPT
(SELECT * FROM Two);

SQ2: SELECT * FROM One)
EXCEPT ALL
(SELECT * FROM Two);

For each of the SQL queries, what is the cardinality (number of rows) of the result obtained when applied to the instances above?

(a) 2 and 1 respectively

(b) 1 and 2 respectively

(c) 2 and 2 respectively

(d) 1 and 1 respectively

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Except clause returns all records present in the first relation and absent in the second relation. But this clause eliminates duplicates tuples.
SQ1: SELECT * FROM One)
EXCEPT
(SELECT * FROM Two);
It will result into single tuple (2, 5) because all duplicate tuples will be removed.
Except all returns all records from the first table which are not present in the second table, leaving the duplicates tuples as it is.

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SQ2: SELECT * FROM One)
EXCEPT ALL
(SELECT * FROM Two);
It will result into 2 tuples < (2, 5), (1, 6) > because it will not remove duplicate.

217. Which of the following statements is/are true w.r.t. Enterprise Resource planning (ERP)?
(A) ERP automates and integrates majority of business processes
(B) ERP provides access to information in a Real Time Environment.
(C) ERP is inexpensive to implement.
(a) (A), (B) and (C) are false
(b) (A) and (B) false; (C) true
(c) (A) and (B) true; (C) false
(d) (A) true; (B) and (C) are false

UGC NET C.S. June-2017 (Paper-II)

Ans : (c)
(1) ERP automates and integrates majority of business – true
(2) ERP provides access to information in a real time environment – true
(3) ERP is inexpensive to implement – false

218. An attribute A of data type varchar (20) has value 'Ram' and the attribute B of data type char (20) has value 'Sita' in oracle. The attribute A has _____ memory spaces and B has _____ memory spaces.
(a) 20, 20 (b) 3, 20
(c) 3, 4 (d) 20, 4

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) The CHAR & VARCHAR type are similar, but differ in the way they are stored and retrieved. They also differ in maximum length & in whether railing space are retained.
The CHAR and VARCHAR types are declared with a length that indicates the maximum number of characters you want to store.
The length of a CHAR column is fixed to the length that you declare when you create the table. VARCHAR (20) can also hold up to 20 characters but if character is less it will take less space so a needs 3 space (for ram)
b need 20 (for sita).

219. Integrity constraints ensure that change made to the database by authorized users do not result into loss of data consistency. Which of the following statement (s) is (are) true w.r.t. the examples of integrity constraints?
(A) An instructor Id. No. cannot be null, provided Instructor Id No. being primary key.
(B) No two citizens have same Adhar-Id
(a) (A), (B) and (C) are true
(b) (A) false, (B) and (C) are true
(c) (A) and (B) are true; (C) false
(d) (A), (B) and (C) are false

UGC NET C.S. June-2017 (Paper-II)

Ans : (c) Integrity means something like "be right" & consistent. The data in a database must be right and in good condition.
Domain integrity means the definition of a valid set of values for and attribute you define data type, length etc.
An instructor Id. no. cannot be null, provided instructor Id. no. being the primary key is valid.
No. two citizens how same adhar-id. is valid but, the budget of company must be zero is not a valid constraint.

220. Let M and N be two entities in an E-R diagram with simple single value attributes. R_1 and R_2 are two relationship between M and N, where as :
R_1 is one-to-many and R_2 is many-to-many.
The minimum number of tables required to represent M, N, R_1 and R_2 in the relational model are _____
(a) 4 (b) 6
(c) 7 (d) 3

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Ans : (d) Two tables are created or entities M & N and third table is created for relationships R2 (many to many) for R1 no need of separate table (one to many) 1 – m relationships.
The primary key from the one side is placed as a foreign key in the 'many side'
- m – n relationships
  A new relation is created with primary key
- 1 – 1 relationship

221. Given two relations R_1(A, B) and R_2(C, D), the result of following query
Select distinct A, B
From R_1, R_2
is guaranteed to be same as R_1 provided one of the following condition is satisfied.
(a) R_1 has no duplicates and R_2 is empty
(b) R_1 has no duplicates and R2 is non - empty
(c) Both R_1 and R_2 have no duplicates
(d) R_2 has no duplicates and R_1 is non - empty

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Ans : (b) In this query first we will take Cartesian product R_1, R_2 for this 0–R_2 then select distinct A, B from Cartesian product of R1, R_2 and select A, B for R2 for this query A, B → No duplicate A, B by combination condition for both we will get the right condition.

222. Consider a schema R(A, B, C, D) and following functional dependencies.
A → B
B → C
C → D
D → B
Then decomposition of R into R1(A, B), R2(B, C) and R3 (B, D) is _____
(a) Dependency preserving and lossless join
(b) Lossless join but not dependency preserving
(c) Dependency preserving but not lossless join
(d) Not dependency preserving and not lossless join

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Ans : (a) Schema R(A, B, C, D) is decomposed into tree relation → R_1(A, B), R2 (B, C) & R3 (B, D)
Now dependencies derived from R_1 (A, B) are A →B, B→C but C is not attribute here is this relation, similarly do for R_2 & R_3 & in the last all the dependencies are preserved & it is a lossless decomposition.

223. Aliasing in the context of programming languages refers to:
(a) Multiple variables having the same location
(b) Multiple variables having the same identifier
(c) Multiple variables having the same value
(d) Multiple use of same variable

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Ans : (a) Aliasing: Describes a situation in which a data location in memory can be accessed through different symbolic names in the program. Thus, modifying the data through one name implicitly modifies the value associated with all aliased names, which may not be expected by the programmer.

224. Consider a "CUSTOMERS" database table having a column "CITY" filled with all the names of Indian cities (in capital letters). The SQL statement that finds all cities that have "GAR" somewhere in its name, is :
(a) Select * from customers where city = '%GAR%';
(b) Select * from customers where city = 'GAR'
(c) Select * from customers where city like '%GAR%';
(d) Select * from customers where city as '%GAR'

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Ans : (c) Select * from customers where city like '%GAR%';

225. Consider the following database table :

```
Create table test (
        one integer,
        two integer,
        primary key (one),
        unique (two),
        check(one > = 1 and < = 10),
        check(two > = 1 and < = 5)
```

How many data records/tuples atmost can this table contain?
(a) 5 (b) 10
(c) 15 (d) 50

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Ans : (a) Attribute 'one' can have 10 values from 1 to 10 and attribute 'two' can have 5 values from 1 to 5. But attribute 'one' is primary key it cannot have duplicate values, so attribute 'one' is allowed only 5 values from 1 to 10. Any 5 value from 1 to 5 of attribute 'two so, relation can have atmost 5 values.

226. Suppose ORACLE relation R(A, B) currently has tuples {(1, 2), (1, 3), (3, 4)} and relation S(B, C) currently has {(2, 5), (4, 6), (7, 8)}. Consider the following two SQL queries SQ1 and SQ2 :

```
SQ1 : Select *
      From R Full Join S
      On R.B = S.B;
SQ2 : Select *
      From R Inner Join S
      On R.B = S.B;
```

The numbers of tuples in the result of the SQL query SQ1 query SQ1 and the SQL query SQ2 are given by :
(a) 2 and 6 respectively
(b) 6 and 2 respectively
(c) 2 and 4 respectively
(d) 4 and 2 respectively

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Ans : (d) Here R.B = S.B holds for 2 records when B = 2 & B = 4 hence its result w hence 2
for SQ2 full join or full outer join gives the result of left outer join (all rows of left table whether there is match or not with the left) now both table has 3 records but there are 2 match and 2 mismatch.
Hence SQ1 will give 4 records.

227. Consider the following three SQL queries (Assume the data in the people table) :
(1) Select Name from people where Age > 21;
(2) Select Name from people where Height > 180;
(3) Select Name from people where (age > 21) or (Height > 180);
If the SQL queries (1) and (2) above, return 10 row and 7 rows in the result set respectively, then what is one possible number of rows returned by the SQL query (3)?
(a) 3 (b) 7
(c) 10 (d) 21

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Ans : (c) When set a and set b all rows are distains answer would be = 10 + 7 = 17
When set b is subset of set a than 10 + 7 – 7 = 10

228. Let E_1 and E_2 be two entities in E-R diagram with simple single valued attributes. R_1 and R_2 are two relationships between E_1 and E_2 where R_1 is one - many and R_2 is many - many. R_1 and R_2 do not have any attribute of their own. How minimum number of tables are required to represent this situation in the Relational model?
(a) 4 (b) 3
(c) 2 (d) 1

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Ans : (b) One table for each entity : 2
R_1 is one to many (No separate table): 0
R_2 is many to many and requires one separate table: 1
Total number of tables: 3

229. The STUDENT information in a university stored in the relation STUDENT (Name, SEX, Marks, DEPT _Name)
Consider the following SQL query SELECT DEPT_Name for STUDENT where SEX = 'M' group by DEPT_Name having avg (Marks) > SELECT avg (Marks) STUDENT. It Returns the Name of the Department for which :

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(a) The average marks of male students is more than the average marks of students in the same Department
(b) The average marks of male students is more than the average marks of students in the University
(c) The average marks of male students is more than the average marks of male students in the University
(d) The average marks of students is more than the average marks of male students in the University

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) The average marks of male students is more than the average marks of student in the university.

230. A clustering index is created when _____.
(a) Primary key is declared and ordered
(b) No key ordered
(c) Foreign key ordered
(d) There is no key and no order

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Ans : (a) Clustered index determines the order in which the rows of a table are stored on disk. If a table has a clustered index. Then the rows of that table will be stored on disk in the same exact order as the clustered index. In some cases, the index is created on non-primary key columns which may not be unique for each record. Clustering index is defined as an ordered data files.

231. For a weak entity set to be meaningful, it must be associated with another entity set in combination with some of their attribute values, is called as :
(a) Neighbor Set (b) Strong Entity Set
(c) Owner Entity Set (d) Weak Set

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Every weak entity must be associated with an identifying entity i.e. the weak entity set is said to be existence dependent on the identifying set. The identifying entity set is said to own the weak entity set that it identifies. It is also called owner entity set.

232. Which of the following is true?
I. Implementation of self-join is possible is SQL with table alias
II. Outer-join operation is basic operation in relational algebra
III. Natural join and outer join operations are equivalent.
(a) I and II are correct
(b) II and III are correct
(c) Only III is correct
(d) Only I is correct

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Only I is correct. Self join is implemented in a single table by alias names rest 2 are incorrect.

233. What kind of mechanism is to be taken into account for converting a weak entity set into strong entity set in entity relationship diagram?
(a) Generalization
(b) Aggregation
(c) Specialization
(d) Adding suitable attributes

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Weak entity is an entity which does not have sufficient attributes to form a primary key. Weak entity is represented by Double rectangle in an ER Diagram. It must be in a 1:M relation. A member of a weak entity is called a subordinate entity. For converting a weak entity set into strong entity set in entity-relationship diagram by simply adding appropriate attributes.

234. Drop Table cannot be used to drop a Table referenced by _____constraint.
(1) Primary key **(2) Sub key**
(3) Super key **(4) Foreign key**
(a) (1) (b) (1), (2) and (3)
(c) (4) (d) (1) and (4)

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : If you drop child table first, the foreign key will be dropped as well. If you attempt to drop percent table first you will get an "could not drop object "a" because it is referenced by a foreign key constraint error, Drop table can not be used to drop a table reference by a foreign key constraint. The referencing foreign key constraint or the referencing table must be dropped.

235. Which of the following statements is FALSE about weak entity set?
(a) Weak entities can be deleted automatically when their strong entity is deleted
(b) Weak entity set avoids the data duplication and consequent possible inconsistencies caused by duplicating the key of the strong entity
(c) A weak entity set has no primary keys unless attributes of the strong entity set on which it depends are included
(d) Tuples in a weak entity set are not partitioned according to their relationship with tuples in a strong entity set

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Ans.(d) : We have weak entities for several reasons :
- We want to avoid the data duplication and consequent entity. Possible inconsistencies caused by duplicating the key of strong.
- Weak entities be stored physically with their strong entities
- Weak entities can be detected automatically when their strong entity is detected.

So (d) is incorrect statement, Hence (d) is the answer.

236. An assertion is a predicate expressing a condition we wish database to always satisfy. The correct syntax for Assertion is :
(a) CREATE ASSERTION 'ASSERTION Name' CHECK 'Predicate'
(b) CREATE ASSERTION 'ASSERTION Name'
(c) CREATE ASSERTION, CHECK Predicate
(d) SELECT ASSERTION

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Ans.(a) : CREATE ASSERTION 'ASSERTION Name' CHECK 'Predicate' is the proper syntax for applying constraints.
For example : This SQL statement creates an assertion to demand that there's no more than a single president among the employees.

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Create assertion AT-MOST-ONE-PRESIDENT as CHECK
(Select count (*) from EMPe where e job = PRESIDENT +4) < I)

237. Which of the following operator is used as concatenation operator in SQL?

(a) + + (b) " "
(c) # # (d) | |

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Ans. (d) : The SQL concatenation operator (| |) is a powerful feature that allows us to merge two or more strings into a single output.

238. In Database Management System (DBMS), which of the following term defines a collection of entities that have the same attributes in the database?

(a) Entity type (b) Value set
(c) Entity set (d) Value type

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Ans. (a) : An entity set is a collection of entities that share the same set of properties or attributes, like a table of all students in a database, where each student entity has the same attributes (eg. student ID, name).

239. Which of the following is NOT an aggregate function in SQL?

(a) CROSS PRODUCT (b) MAXIMUM
(c) COUNT (d) SUM

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Ans. (a) : Aggregate functions in SQL perform a calculation on a set of values and return a single summary value.
Examples COUNT(), SUM(), AVG(), MAX() AND MIN().
Cross product (or Cartesian product) is a type of join in SQL.

240. Which of the following symbol is used with SELECT command of SQL, to retrieve all the attribute values of the selected tuples?

(a) # (b) %
(c) * (d) :

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Ans. (c) : '*' symbol is used with SELECT command of SQL, to retrieve all the attribute values of the selected tuples.

241. Which of the following maths function in SQL returns the smallest integer greater than, or equal to, the specified numeric expression?

(a) ROUND (b) CEILING
(c) ABS (d) FLOOR

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Ans. (b) : The CEILING function in SQL returns the smallest integer value that is greater than or equal to the given numeric expression. It essentially rounds a number up to the next whole number.

242. Which operation is provided in Pandas, when multiple tables can be concatenated both column wise and row wise as database?

(a) SUM (b) MERGE
(c) GROUP BY (d) SELECT

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Ans. (b) : The correct operation in pandas for concatenating multiple tables both column wise and row wise (similar to database operations) is not SUM, GROUP By, or SELECT, but the ability provided by the 'MERGE' and 'CONCAT' function.

243. Which of the following normal form of normalization in relational model, disallows relations within relations or relations as attribute values within tuples?

(a) Fourth (b) First
(c) Second (d) Third

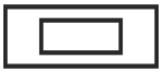
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Ans. (b) : The first normal form (INF) , which allows relations within relations or relations as attribute values within tuples.
First Normal Form (INF) requires that all attributes in a relation must have atomic (indivisible) values, meaning composite or multi-valued attributes (such as relations within relations) are not allowed.

244. Which of the following notation of ERD (Entity– Relationship Diagram) shows the 'Weak Entity'?

(a) 

(b) 

(c) 

(d) 

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Ans. (b) : Double rectangle '  ' represents the weak Entity; while double diamond '  ' represents the weak relationship between the entities.

245. Which of the following statement is correct for DATEDIFF function in SQL?

I. This function reports difference between the arguments 'startdate' and 'enddate'.
II. In this, 'datepart' argument/value can be specified in a variable.

(a) Only I (b) Neither I nor II
(c) Both I and II (d) Only II

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Ans. (a) : Statement (I) is correct. The 'Datediff' function in SQL is designed to calculate the difference between a start date and an end date based on a specified date part (e.g. year month day).
Syntax– DATEDIFF (datpart. start date, end date)

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246. Which of the following data function in SQL returns the current database system timestamp as a 'datetime' value without the database time zone offset?
(a) GET TIME (b) GET DAT E
(c) CAST DAT E (d) CAST T IME
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Ans. (b) : The GETDATE() function returns the current database system date and time in a "YYY-MMM-DD hh:mm:ss mmm" format.
This function is commonly used in SQL server and returns the current date and time of the computer.
Example– select GETSATE().

47. Which of the following operator of SQL is used in WHERE clause for match a character pattern in database?
(a) LIKE (b) IN
(c) IS NULL (d) OR
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Ans. (a) : The 'Like' operator is used in a WHERE clause to search for a specified pattern in a column. There are two wildcards often used in conjunction with the LIKE operator. The operator sing(%) represents zero, one or multiple characters. The underscore sign(-) represents one single character.

248. In SQL, which of the following is correct command for schema evolution, if the option is chosen than the schema is dropped only if schema has no element in it?
(a) DELETE (b) UPDATE
(c) ALTER (d) DROP
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Ans. (d) : The correct command for schema evolution in SQL, if the option is chosen that the schema is dropped only if it has no element in it is 'DROP'.

249. Which of the following term of ER model (Entity – Relationship) is corresponding to the foreign key (or relationship relation) in the relational model of database?
(a) Composite attribute
(b) Entity type
(c) 1:1 (one to one) or 1: N (one to many) relationship type
(d) M : N (many to many) relationship type
DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : 1:1 (one to one) or 1:N (one to many) relationship type, which in the relational model corresponds to the use of foreign keys to link related tables.

250. In relational databases, which of the following integrity constraint is specified between two relations and is used to maintain the consistency among tuples in the two relations?
(a) Entity integrity constraint
(b) Semantic integrity constraint
(c) Functional integrity constraint
(d) Referential integrity constraint
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Ans. (d) : A referential integrity constraint ensures that a foreign key in one table references a valid primary key in another table. This maintains consistency between the two tables by preventing references to non-existent rows.

251. State whether True or False?
यह बताएँ कि क्या सही या गलत है?
(i) In a database, low cardinality means more unique value are stored in a column and vice-versa.
डेटाबेस में निम्न गणनीयता का अर्थ है कि अधिक अद्वितीय मान कॉलम में संग्रहित होते हैं और उनके विपरीत।
(ii) If a standalone column of a table is connected to the primary key of the same table, then it is called a recursive relationship.
यदि किसी टेबल का एक स्टैंडअलोन कॉलम उसी टेबल की प्राइमरी से जुड़ा हुआ है, तो इसे रिकर्सिव रिलेशनशिप कहा जाता है।
(a) (i) True, (ii) True/ (i) सत्य, (ii) सत्य
(b) (i) True, (ii) False/ (i) सत्य, (ii) असत्य
(c) (i) False, (ii) True/ (i) असत्य, (ii) सत्य
(d) (i) False, (ii) False/ (i) असत्य, (ii) असत्य
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Ans. (c) : The first statement is false and the second statement is True.
Statement (i) False– In a database, low cardinality means fewer unique values are stored in a column; high cardinality means more unique values are stored in a column.
Statement (ii) True– If a standalone column of a table references the primary key of the same table, it is called a recursive (unary).

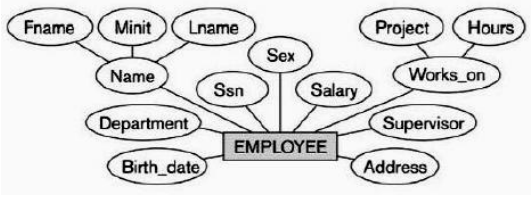
252. Match the following with respect to ER diagram:
ER आरेख के संबंध में निम्नलिखित का मिलान करें:

	List-1/सूची-1		List-2/सूची-2
I	Boxes बॉक्सेस	P	Attributes ऐट्रब्यूट्स (विशेषताएं)
II	Ovals ओवल्स	Q	Relationships रिलेशनशिप (संबंध)
III	Diamonds डायमंड्स (हीरे)	R	Entities एंटिटीज (निकाय)

(a) I-Q, II-P, III-R (b) I-P, II-R, III-Q
(c) I-R, II-Q, III-P (d) I-R, II-P, III-Q
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Ans. (d) : In ER diagram Boxes are used to represent Entity. Ovals are used to represent Attributes. while Diamonds are used to show the Relationships between the Entities.

253. How many attributes in the following Er diagram are both multi-valued as well as composite?
निम्नलिखित ईआर आरेख (ER diagram) में कितने एट्रिब्यूट बहु-मूल्य और समग्र, दोनों हैं?



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(a) 1 (b) 2
(c) 3 (d) 4

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Ans. (a) : Multi valued can hold multiple values for a single entity while composite attributes can be subdivided into smaller sub parts. Therefore in the given ER diagram 'Address' attributes satisfied both condition for multi valued as well as composite.

254. Which of the following keyword is used in Oracle SQL to sort the result-set in ascending order?
रिजल्ट-सेट को आरोही या अवरोही क्रम में क्रमबद्ध करने के लिए Oracle SQL में निम्नलिखित में से किस कीवर्ड उपयोग किया जाता है?

(a) MANGE (b) SORT
(c) ARRANGE (d) ORDER BY

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Ans. (d) : The 'ORDER BY' command is used to sort the result set in ascending or descending order.
The 'order by' command sorts the result set in ascending order by default.

255. Identify the SQL wildcard character in Oracle which represents zero, one, or multiple characters?
Oracle में SQL वाइल्डकार्ड वर्ण की पहचान करें जो शून्य, एक, या एकाधिक कैरेक्टर्स (वर्णों) को दर्शाता है?

(a) Percent (%)/प्रतिशत
(b) Underscore (_)/अंडरस्कोर
(c) Question mark (?)/क्वेश्चन मार्क
(d) Asterisk (*)/ऐस्टरिस्क

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Ans. (a) : Percent (%) is the SQL wildcard character in oracle which represents zero, one or multiple characters.

256. Which of the following Oracle SQL statements can be used to select the name of customers who have no address?
निम्नलिखित में से किस Oracle SQL स्टेटमेंट का उपयोग उन ग्राहकों के नाम का चयन करने के लिए किया जा सकता है जिनके पास कोई एड्रेस (पता) नहीं है?

(a) SELECT Name
FROM Customers
WHERE Address IS NULL;
(b) SELECT Name
FROM Customers
WHERE Address = NULL;
(c) SELECT Name
FROM Customers
WHERE No Address;
(d) SELECT Name
FROM Customers
WHERE Address IS EMPTY;

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : To select the name of customers who have no address in the customers table. The Query would be:

```
SELECT Name
FROM Customers
WHERE Address IS NULL;
```

257. Consider the following Oracle SQL query:
SELECT * FROM Products
WHERE Price BETWEEN 10 AND 20;
निम्नलिखित Oracle SQL क्वेरी पर विचार करें :
SELECT * FROM प्रॉडक्ट
जहां मूल्य 10 और 20 के बीच है;
Which of the following is the correct interpretation of this query?
निम्नलिखित में से कौन सा इस प्रश्न की सही व्याख्या है?

(a) Select all products whose price is greater than 10 but less then 20/उन सभी प्रॉडक्ट्स का चयन करें जिनकी कीमत 10 से अधिक है लेकिन 20 से कम है
(b) Select all products whose price is greater than or equal to 10 but less then 20/उन सभी प्रॉडक्ट्स का चयन करें जिनकी कीमत 10 से अधिक या बराबर है लेकिन 20 से कम है
(c) Select all products whose price is greater than 10 but less then or equal to 20/उन सभी प्रॉडक्ट्स का चयन करें जिनकी कीमत 10 से अधिक है लेकिन 20 से कम या उसके बराबर है
(d) Select all products whose price is greater than equal to 10 but less then or equal to 20/उन सभी प्रॉडक्ट्स का चयन करें जिनकी कीमत 10 से अधिक या बराबर है लेकिन 20 से कम या उसके बराबर है

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Ans. (d) : The SQL 'BETWEEN' operator is used in a WHERE clause to select values within a specified range. This range can apply to number, text, or dates.
• The Between operator is inclusive, meaning that value 1 (the lower bound) and value 2 (the upper bound) are included in the result set if they match the column_name.

258. Which of the following Oracle SQL statements is not correct?
निम्नलिखित में से कौन सा Oracle SQL कथन सही नहीं है?

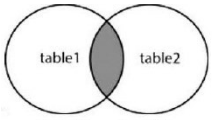
(a) SELECT * FROM Customers
ORDER BY Country;
(b) SELECT * FROM Customers
ORDER BY Country DESC;
(c) SELECT * FROM Customers
ORDER BY Country ASC, DESC Customer Name;
(d) SELECT * FROM Customers
ORDER BY Country Customer Name DESC;

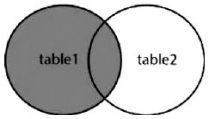
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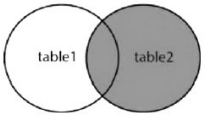
Ans. (c) : In the given SQL Query option (c) statement is wrong because the syntax for mate of Query is written in a wrong way.

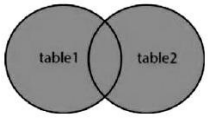
259. Which of the following represents the "inner" join operation?
निम्नलिखित में से कौन ''इनर'' जॉइन ऑपरेशन को दर्शाता है?

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(a) 

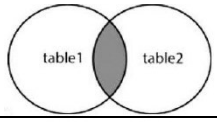
(b) 

(c) 

(d) 

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Ans. (a) : An Inner Join in a Database management system (DBMS) is a type of join operation used to combine rows from two or more tables based on a related column between them. It returns only the rows where there is a match in the specified join condition in both tables.
Example–



260. Which of the following category of join returns all records when there is a match in either left of right table?
बाएं या दाएं टेबल में कोई मिलान होने पर निम्नलिखित में से कौन सी श्रेणी में सभी रिकॉर्ड लौटाए जाते हैं?

(a) Inner join/इनर जॉइन
(b) Left outer join/लेफ्ट आउटर जॉइन
(c) Right outer join/राइट आउटर जॉइन
(d) Full outer join/फुल आउटर जॉइन

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Ans. (d) : A full outer join is a method of combining tables so that the result includes unmatched rows of both tables. If you are joining two tables and want the result set to include unmatched rows from both tables, use a full outer joint clause. The matching is based on join condition. If any column of the result table does not have a value, that column has the null value in the result table.

261. The SQL ALTER TABLE statement is used to:
SQL ALTER TABLE स्टेटमेंट का उपयोग इस प्रकार किया जाता है:

(a) Add, delete, or modify columns in an existing table./किसी मौजूदा टेबल में कॉलम जोड़ें, हटाएं या संशोधित करें
(b) Drop an existing table in a database.
डेटाबेस में मौजूद टेबल ड्रॉप करें
(c) Insert or modify records in a table
किसी टेबल में रिकॉर्ड्स डालें या संशोधित करें
(d) Delete existing records in a table.
टेबल में मौजूदा रिकॉर्ड हटाएं

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Ans. (a) : The SQL ALTER Table statement is used to Add, delete or modify columns in an existing table. The ALTER command is a DDL (Data Definition Language) statement used to modify the structure of an existing database object.

262. Which of the following is used single-line comment in SQL?
निम्नलिखित में से किसका उपयोग SQL में सिंगल-लाइन कमेंट के लिए किया जाता है?

(a) && (b) /*...*/
(c) -- (d) ##

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Ans. (c) : The double hyphen (--) is used for single line comments in SQL because it is defined by the ANSI SQL standard. Everything following '--' on that line is ignored by the SQL Engine, making it a clear and simple way to add notes or explanations directly within SQL code without affecting execution.

263. Which of the following command is used with SQL SELECT statement in Oracle to return only distinct (different) Values?
निम्नलिखित में से कौन सी कमांड का उपयोग Oracle में SQL SELECT स्टेटमेंट के साथ केवल विशिष्ट (भिन्न) मान वापस करने के लिए किया जाता है।

(a) UNIQUE (b) DIFFERENT
(c) DISTINCT (d) SINGLE

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Ans. (c) : The correct command used with the SQL SELECT statement in oracle to return only distinct (different) values is DISTINCT. The syntax is SELECT DISTINCT column_name from table_name; which retrieves only unique values, filtering out any duplicates.

264. Which of the following is TRUE?
निम्नलिखित में से कौन सत्य है?

(a) A relation (table) may have more than one foreign keys/रिलेशन (टेबल) में एक से अधिक फॉरेन की हो सकती है
(b) Primary key is not a candidate key
प्राइमरी की कैंडिडेट की नहीं है
(c) Every candidate key is a primary key
प्रत्येक कैंडिडेट की एक प्राइमरी की है
(d) Candidate key is maximal set of attributes whose values are unique/कैंडिडेट की उन विशेषताओं का अधिकतम सेट है जिनके मान विशिष्ट हैं

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Ans. (a) : A relation can have multiple foreign keys, each one establishes a link to a different parent table or references different sets of columns.

265. Which of the following is FALSE?
निम्नलिखित में से कौन सा गलत है?

(a) A relation (table) may have multiple secondary keys/एक रिलेशन (टेबल) में मल्टीपल सेकंडरी की हो सकती हैं
(b) The value of primary key can't be null
प्राइमरी का मान शून्य नहीं हो सकता है

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(c) Secondary key is also a candidate key
सेकंडरी की भी कैंडिडेट की है

(d) Super key satisfies the "minimality" condition
सुपर की ''न्यूनतमता''शर्त को पूरा करती है

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Ans. (d) : A super key is any set of attributes that can uniquely identify a row in a table, but it does not necessarily satisfy 'minimality' only a candidate key (including primary key) is a minimal super key, meaning it contains no unnecessary attributes. Therefore super keys can be non-minimal and may have redundant attributes.

266. If X_1, X_2, X_3,..., X_N are the domains in a relational database model, then the relation (table) is a subset of which of the following?
अगर X_1, X_2, X_3,..., X_N एक रिलेशनल डेटाबेस मॉडल में डोमेन हैं, तो रिलेशन (टेबल) निम्नलिखित में से किसका सबसेट (उप-समूह) है?

(a) $X_1 \times X_2 \times X_3 \times ... \times X_N$
(b) $X_1 + X_2 + X_3 + ... + X_N$
(c) $X_1 \cup X_2 \cup X_3 \cup ... \cup X_N$
(d) $X_1 \cap X_2 \cap X_3 \cap ... \cap X_N$

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Ans. (a) : The relation (table) in a relational database model is a subset of the Cartesian product of its domains.
$X_1 * X_2 * X_3 * ... * X_N$.
This represents the Cartesian product of all domains. A relation is mathematically defined as a subset of this product, meaning every tuple in the relation contains one value from each domain, and the entire table consists of only certain combinations chosen from all possible one.

267. If AB and C are the candidate keys in a relation schema R(A, B, C, D)
यदि AB और C किसी रिलेशन स्कीमा में कैंडिडेट की R(A, B, C, D), तो प्रमुख विशेषताएं क्या हैं?

(a) A, B, and C only/केवल A, B,और C
(b) A and B only/केवल A और B
(c) A and C only/केवल A और C
(d) B and C only/केवल B और C

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Ans. (a) : A candidate key is a set of attribute that can specifically identity the relationship. If there or AB and C candidate, it means that all the features (A, B, C and D) can be obtained from any combination of AB, and C can also achieve all features alone.

268. Which of the following statement(s) is/are correct regarding Structured Query Language (SQL)?
स्ट्रक्चर्ड क्वेरी लैंग्वेज (एसक्यूएल) के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. The datatype FLOAT can hold numbers with decimal points.
डेटाटाइप FLOAT दशमलव बिंदुओं युक्त संख्याएँ धारण कर सकता है।

II. The datatype BIGINT occupies I6 bytes.
डेटाटाइप BIGINT 16 बाइट्स ग्रहण करता है।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

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Ans. (a) :
(i) The 'FLOAT' data type in SQL is used to store approximate numeric values with floating point precision, which means it can indeed hold numbers with decimal points.
(ii) The 'BIGINT' data type in SQL is typically used to store large integer values, but it occupies 8 bytes (64 bits) not 16 bytes.
Hence statement II is false and statement I is true.

269. Which of the following statement is a part of Data Manipulative Language in Structured Query Language?
निम्नलिखित में से कौन सा कथन स्ट्रक्चर्ड क्वेरी लैंग्वेज में डेटा मैनिपुलेटिव लैंग्वेज का एक भाग है?

(a) Delete/डिलीट (b) Alter table/आल्टर टेबल
(c) Copy/कॉपी (d) Paste/पेस्ट

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Ans. (a) : 'Delete' is a part of Data Manipulation Language (DML) in SQL, which is used to modify the data within the database.
DML commands–
SELECT, INSERT, UPDATE and Delete.

270. Which of the following statement(s) is/are correct regarding COUNT functions in Structured Query Language of Relational Database Management System?
रिलेशनल डेटाबेस मैनेजमेंट सिस्टम की स्ट्रक्चर्ड क्वेरी लैंग्वेज में COUNT फंक्शन के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. COUNT(*) is used to count the number of values in a column./COUNT(*) का उपयोग किसी स्तम्भ में मानों की संख्या गिनने के लिए किया जाता है।

II. COUNT() is used to count the number of rows of the query result./COUNT() का उपयोग क्वेरी परिणाम (रिजल्ट) की पंक्तियों की संख्या गिनने के लिए किया जाता है।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

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Ans. (d) : The COUNT (*) Function returns the total number of rows in a table, including duplicate rows and null values.
The COUNT() function returns the numbers of rows that matches a specific criterion.

271. Which of the following operations CANNOT be done on a database?
निम्नलिखित में से कौन सा ऑपरेशन डेटाबेस पर नहीं किया जा सकता है?

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(a) Sorting/सॉर्टिंग (b) Ordering/ओर्डरिंग
(c) Grouping/ग्रुपिंग (d) Iteration/इटरेशन

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Ans. (d) : 'Iteration' operations cannot be done on a database, sorting, ordering and grouping all operations can be perform on the database.

272. Which of the following is NOT a part of Data Definitions Language in Relational Database Management System?
निम्नलिखित में से कौन सा रिलेशनल डेटाबेस मैनेजमेंट सिस्टम में डेटा डेफिनिशन लैंग्वेज का एक भाग नहीं है?

(a) Create table/क्रिएट टेबल
(b) Insert table/इन्सर्ट टेबल
(c) Alter table/आल्टर टेबल
(d) Drop table/ड्राप टेबल

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Ans. (b) : "Insert table" (Which should actually be "INSERT INTO") is not part of Data Definition language (DDL). It is a part of Data Manipulation Language (DML), which is used to inset, update or delete data within tables. The other options- create table, Alter table and Drop table are all DDL commands used to define or modify the structure of a database.

273. What is self join in Database Management System?
डेटाबेस मैनेजमेंट सिस्टम में सेल्फ जॉइन क्या है?

(a) In a self join, a table is joined to a number
सेल्फ जॉइन में, एक टेबल एक नंबर से जोड़ी जाती है
(b) In a self-join, a table is joined to itself/सेल्फ जॉइन में, एक टेबल अपने आप से जोड़ी जाती है
(c) In a self join, a table is joined with a query
सेल्फ जॉइन में, एक टेबल क्वेरी से जोड़ी जाती है
(d) In a self-join, two databases are joined
सेल्फ जॉइन में, दो डाटाबेस जोड़े जाते हैं

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Ans. (b) : A self join is a regular joint where a table is joined with itself. This is useful when you want to compare rows within the same table.

274. How many entities are associated in the quaternary relationship in E – R modeling?
E - R मॉडलिंग में क्वाटरनरी रिलेशनशिप में कितनी एनटीटीस जुड़ी हुई है?

(a) 2 (b) 3
(c) 4 (d) 5

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Ans. (d) : E - R (Entity - Relationship) modeling is a technique used in database design to visually represent the data and its relationships. It helps in structuring and organizing data efficiently before creating a relational database. In E - R modeling, a quaternary relationship involves four entities. The prefix "quaternary" comes from "quaternary = 4", meaning this relationship connect four different entity sets.
For example, a relationship like supplies involving entities Supplier, Product, Warehouse and Data would be quaternary relationship.

275. In Structured Query Language, what is the GROUP BY clause?
स्ट्रक्चर्ड क्वेरी लैंग्वेज में, GROUP BY क्लॉज क्या है?

(a) It groups the table with same name in the database./यह डेटाबेस में समान नाम वाली टेबल को समूहित करता है।
(b) It groups the rows together that contain the same values in a specific column.
यह उन रो को एक साथ समूहित करता है जिनमें एक विशिष्ट कॉलम में समान मान होते हैं।
(c) It combines data from two tables on specific conditions./यह विशिष्ट स्थितियों पर दो टेबल्स के डेटा को जोड़ती है।
(d) It groups the columns together that contains the same values in a specific rows.
यह उन कॉलम को एक साथ समूहित करता है जिनमें विशिष्ट रो में समान मा होते हैं।

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Ans. (b) : The GROUP BY clause in SQL is used to arrange identical data into groups. This is often used with aggregate function like COUNT(), SUM(), AVG() etc. to perform operations on each group of data.

276. Which of the following statement(s) is/are correct regarding single row functions in structured query language?
स्ट्रक्चर्ड क्वेरी लैंग्वेस में सिंगल रौ फंक्शन के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. INSERT() is a numeric function.
INSERT() एक नुमेरिक फंक्शन है।
II. ROUND() is a numeric function.
ROUND() एक नुमेरिक फंक्शन है।
III. LENGTH() is a character function.
LENGTH() एक कैरेक्टर फंक्शन है।

(a) Both I and II/ I तथा II दोनों
(b) Both II and III/ II तथा III दोनों
(c) Only III/ केवल III
(d) Both I and III/ I तथा III दोनों

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Ans. (b) :
(I) INSERT()– INSERT() is actually a character function not a numeric one. It is used to insert a substring into a string.
(II) ROUND()– ROUND() is used to round a number to a specified number of decimal places, so it is indeed a numeric function.
(III) LENGTH()– LENGTH() returns the number of characters in a string, making is a character function.
The correct answer is (b)- Both II and III.

277. All candidate keys other than primary keys are known as_____
प्राइमरी की के अलावा अन्य सभी कैंडिडेट की को _____ कहा जाता है।

(a) Foreign key/फॉरेन की
(b) Secondary key/सेकंडरी की
(c) Alternate key/अल्टर्नेट की
(d) Remaining key/रिमैनिंग की

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Ans. (c) : All candidate keys other than the primary key are known as alternate keys or secondary keys. These are attributes that could uniquely identify a row in a table but were not chosen as the primary key for the table but were not chosen as the primary key for the table.

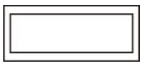
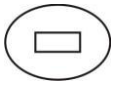
278. _____are some problems that may arise while creating the Entity relationship model.
_____ कुछ समस्याएं हैं जो एंटिटि रिलेशनशिप मॉडल बनाते समय उत्पन्न हो सकती हैं।

(a) Connection traps/कनेक्शन ट्रैप्स
(b) Transition traps/ट्रांसिशन ट्रैप्स
(c) Groups traps/ग्रुप्स ट्रैप्स
(d) Data traps/डाटा ट्रैप्स

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Ans. (a) : The most recognized problems that may arise while creating an Entity Relationship (ER) model are termed connection traps. Transition traps, groups, traps and data traps are not standard recognized problems in ER modeling but are sometimes mentioned in non-standard texts.

279. Which of the following is a symbol of multivalued attribute in Entity relationship diagram?
निम्नलिखित में से कौन सा एंटिटी रिलेशनशिप आरेख में मल्टीवेल्यूड एट्रिब्यूट का प्रतीक है?

(a) 
(b) 
(c) 
(d) 

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Ans. (a) : Double Ellipse '  ' symbol is used to present the multivalued attribute in ER Diagram. Double Rectangle '  ' symbol is used to represent weak entity '  ' symbol represent weak relationship.

280. Which of the following is not a structured Query Language Data type?
निम्नलिखित में से कौन एक स्ट्रक्चरड क्वेरी लैंग्वेज का डेटा प्रकार नहीं है?

(a) REAL (b) INTERVAL
(c) TIME STAMP (d) BASE

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Ans. (d) : REAL– This is a valid data type in SQL representing a floating point number.
INTERVAL– This is also a valid SQL data type, used to store duration information (like days, hours, minutes or seconds) between two dates or times.
TIME STAMP– This is a valid data type for storing a specific point in time.
BASE– base is not a recognized data type in most SQL implementations, it might refer to a database or table name in some specific contexts but not as a data type itself.

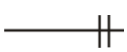
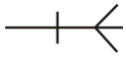
281. Which of the following is an Aggregate operator in structured Query Language?
निम्नलिखित में से कौन सा स्ट्रक्चर्ड क्वेरी लैंग्वेज़ में एग्रीगेट ऑपरेटर है?

(a) Minus (b) Intersect
(c) Union (d) Count

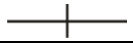
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Ans. (d) : SQL Aggregate Functions– SQL Aggregate Functions are used to perform calculation on a set of rows and return a single value. These functions are particularly useful when we need to summarize, analyze or group large database in SQL database whether you are working with sales data employee records, or product inventories, aggregate functions help us derive meaningful insights. They are often with the group by clause in SQL to summarize data for each group. Commonly used aggregate function include COUNT(), SUM(), AUG(), MIN() AND MAX(). In this article we will explain the most commonly used SQL Aggregate Function, their syntax, practical examples, and how to make the best use of them in your queries.

282. Which of the following symbol represent one to one cardinality relationship in Entity relationship diagram?
निम्नलिखित में से कौन सा प्रतीक एनटीटी रिलेशनशिप डाइग्राम में वन-टु-वन कार्डिनैलिटी रिलेशनशिप को दर्शाता है?

(a)  (b) 
(c)  (d) 

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Ans. (b) : Denoting a one to one Relationship in an ER– A one-to-one relationship in an ER diagram is denoted, like all relationship, with a line connecting the two entities, The "one" cardinality is denoted with a single straight line (The "Many" cardinality is denoted with a crows foot symbol)
• A single line between the entities.
• Cardinality Indicators such as "1" or a vertical bar (|) on both ends or the line, showing that one instance of an entity is associated with exactly one instance of the other entity.
Ex- Presentation:–
entity  A entity B.

283. INSERT command is used to
A. Add a single tuple to a relation
B. Add multiple tuples to a relation
C. Add values to specific attributes
D. Insert new table
Choose the correct answer from the options given below :

(a) A only (b) A, B only
(c) A, B, C only (d) B, C, D only

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Ans. (c) : Explanation-
(A) add a single tuple a relation.
The Insert statement is used to add a single row (tuple) to a table.
(B) add multiple tuples to a relation.
Using the Insert into--- values statement with multiple rows or Insert into----select, multiple tuples can be added.
(C) add values to specific attributes. The Insert statement allows inserting values into specific columns by specifying column names.
However, D is incorrect because Insert is for adding data, while creating a new table is done using the create table statement.

284. An alternate key in database table is also called
(a) Primary key (b) Candidate key
(c) Super key (d) Foreign key

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Ans. (b) : An alternate key is any candidate key that is not chosen as the primary key, it can uniquely identify a row in the table but is not the primary identifier selected by the database administrator.

285. Which of the following command can be used to modify data in an SQL Table?
A. INSERT **B. DELETE**
C. UPDATE **D. SELECT**
Choose the correct answer from the options given below :
(a) A, B, D only (b) A, C, D, only
(c) B, C, D only (d) A, B, C only

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Ans. (d) : The modify date in SQL table commands:
A. INSERT:- The Insert is used to add new rows to a table
B. DELETE :- The Delete is used to remove rows from a table.
C. UPDATE:- Update is the SQL command specifically used to modify existing date in a table. You used it to change the values in one or more columns of a row (or rows) that meet a certain condition.
Therefore, is correct option d: A, B, C only.

286. Consider a database table R with attributes A and B. Which of the following SQL queries is illegal?
(a) SELECT A FROM R;
(b) SELECT A, COUNT (*) FROM R;
(c) SELECT A, COUNT(*) FROM GROUP BY A;
(d) SELECT A, B, COUNT (*) FROM R GROUP BY A, B;

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Ans : (b) SELECT A, COUNT (*) FROM R GROUP By A, B, legal query.
When aggregation function (sum, avg, count, max and min) are used in select clause then we need to use group by clause for grouping the attributes.

287. DBMS provides the facility of accessing data from a database through
(a) DDL (b) DML
(c) DBA (d) Schema

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Ans : (b) Accessing the data from the database in DBMS done through DML (Data manipulation language). The DML command can access the data from a database.

288. Relational database schema normalization is NOT for :
(a) Reducing the number of joins required to satisfy a query
(b) Eliminating uncontrolled redundancy of data stored in the database
(c) Eliminating number of anomalies that could otherwise occur with inserts and deletes
(d) Ensuring that functional dependencies are enforced

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Ans : (a) Relational database schema normalization is not for reducing the number of joins required to satisfy a query.
Relational database schema normalization goal is
(1) eliminating reduclary
(2) function dependencies.

289. In a relational database model, NULL values can be used for all but which one of the following?
(a) To allow duplicate tuples in the table by filling the primary key column(s) with NULL.
(b) To avoid confusion with actual legitimate data values like 0 (zero) for integer columns and" (the empty string) for string columns.
(c) To leave columns in a tuple marked as "unknown" when the actual value in unknown.
(d) To fill a column in a tuple when that column does not really "exist" for that particular tuple.

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Ans : (a) The **SQL** null is the term used to represent a missing value. A null value in a table is a value in a field that appears to be blank. A field with a null value is a field with no value.

290. Consider the following two commands C1 and C2 on the relation R from an SQL database:
C1: drop table R;
C2: delete from R;
Which of the following statements is TRUE?
I. Both C1 and C2 delete the schema for R.
II. C2 retains relation R, but deletes all tuples in R.
III. C1 deletes not only all tuples of R, but also the schema for R.
(a) I only (b) I and II only
(c) II and III only (d) I, II and III

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Ans : (c) Drop command remove all tuples of the relation as well as schema also.
Delete command in SQL removes tuples from the relation but retain the schema of table in database.

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291. Consider the following database table having A, B, C and D as its four attributes and four possible candidate keys (I, II, III, IV) for this table:

A	*B*	*C*	*D*
a_1	b_1	c_1	d_1
a_2	b_2	c_3	d_2
a_3	b_3	c_1	d_3

I: {*B*} **II: {*B, C*}**
III: {*A, D*} **IV: {*C, D*}**

If different symbols stand for different values in the table (e.g., d_1 is definitely not equal to d_2), then which of the above could not be the candidate key for the database table?

(a) I and III only (b) III and IV only
(c) II only (d) I only

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Ans : (c) Attribute B has distinct tuples. So, B is candidate key and {*BC*} is superkey because a candidate key is minimal superkey.
Key {*A, D*} and {*C, D*} has distinct tuples, so, both are candidate key.

292. Which of the following statement(s) is/are FALSE in the context of Relational DBMS?

I. Views in a database system are important because they help with access control by allowing users to see only a particular subset of the data in the database.
II. E-R diagrams are useful to logically model concepts.
III. An update anomaly is when it is not possible to store information unless some other, unrelated information is stored as well.
IV. SQL is a procedural language.

(a) I and IV only (b) III and IV only
(c) I, II and III only (d) II, III and IV only

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Ans : (b)
(i) Views in a database system are important because:
- They help provide data independence.
- They allow the schema to change without forcing existing applications to be recompiled.
- They help with access control by allowing users to see only a particular subset of the data in the database.

(ii) A conceptual ER model may be used as the foundation for one or more logical data models.
(iii) That's not update anomaly but it is insertion anomalies.
(iv) SQL is not procedural language.
Only statements (iii) and (iv) are false.

293. Which SQL command is used to remove a table from a database?
किसी डेटाबेस से टेबल को हटाने के लिए किस SQL कमांड का प्रयोग किया जाता है?

(a) DELETE
(b) DROP
(c) ERASE
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b): 'DROP' command is used to remove a table from a database. By using 'drop' command user can remove table permanently, this cannot retain again after removing.
Example– DROP Table Table_name;
Here Table_name is that table name what you want to remove.

294. Which of the following SQL commands is used to change the attribute values of one or more rows in a table?
निम्नांकित में से किस SQL कमांड का किसी टेबल के एक या एक से अधिक रो (Rows) के ऐट्रिब्यूट वैल्यू को बदलने के लिए प्रयोग किया जाता है?

(a) Update/अपडेट
(b) Insert/इनसर्ट
(c) Alter/ऑल्टर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

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Ans. (a): The UPDATE command in SQL is used to change the attribute values of one or more rows of a table.

295. To retrieve information from a database, which of the following SQL commands is used?
किसी डेटाबेस से सूचना को निकालने के लिए निम्नांकित में से किस कमांड का प्रयोग किया जाता है?

(a) Display/डिस्प्ले (b) Show/शो
(c) Select/सिलेक्ट
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

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Ans. (c) : To retrieve information from a database. 'SELECT' command is used. This command is used to read and return the values to display the user.
For Example– Select * From employees;
In this command, all the rows and columns data will display from employees table.

296. Which of the following SQL commands is used to modify the structure of a table?
निम्नांकित में से किस SQL कमांड (Command) का प्रयोग किसी टेबल के स्ट्रक्चर (Structure) को बदलने के लिए प्रयोग किया जाता है?

(a) Alter
(b) Modify
(c) Update
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

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Ans. (a) : To modify the structure of table. 'Alter' command is used. With the help of this command, user can add the new columns, modify the existing column, or change the structure of table.

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297. Which of the following SQL keywords is used to display data from a table based on certain pattern?
निम्नांकित में से किस SQL कीवर्ड (Keywords) का प्रयोग किसी पैटर्न पर आधारित टेबल से डेटा को दर्शाने/दिखाने के लिए होता है?
(a) IN
(b) LIKE
(c) BETWEEN
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

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Ans. (b) : 'LIKE' SQL keyword is used to display data from a table based on certain pattern. This keyword is used with 'WHERE' clause so that user can filter and select the specific matched words.
Example- Select * from employees Where name LIKE 'A%'
In this Query those data would be selected which name's starts with 'A' letter. In Employees table.

298. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____ integrity constraint.
संदर्भ सम्बन्ध में किसी भी टपल की दी गई विशेषताओं में दिखाई देने वाले मान भी संदर्भित सम्बन्ध में कम-से कम एक टपल की निर्दिष्ट विशेषताओं में होने चाहिए, _____ अखण्डता बाधा के अनुसार।
(a) Referential/संदर्भात्मक
(b) Primary/प्राथमिक
(c) Referencing/संदर्भित
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : Referential Integrity is a property of data that states that all its references are valid. In relational databases, it requires that if the value of one attribute of a table refers to the value of another attribute, then the referenced value most exist.

299. An attribute of one table matching the primary key of another table is called as
एक टेबल का ऐट्रिब्यूट, जो दूसरे टेबल की प्राइमरी की से मैच (मिलता हो) करता हो, कहलाता है
(a) Foreign key/फॉरेन की
(b) Secondary key/सेकेन्डरि की
(c) Candidate key/कैन्डिडेट की
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

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Ans. (a) : When an attribute of one table matches the primary key of another table, it is called a 'foreign key' because it refers to the key of another table, called the primary key. The foreign key creates that relation that joins a table with another table. It is used to link the data in relational database,

300. Which of the following is the basic approach for joining tables?
Joining tables के लिए निम्नलिखित में से कौन-सा बुनियादी तरीका है?
(a) Union JOIN (b) Natural JOIN
(c) Subqueries
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (d) : SQL Join is an important part of a database query that is used to join data in two or more tables. Join is used to match the information of one table with another table in a particular condition. Some major JOIN are as follows– INNER JOIN, LEFT JOIN, RIGHT JOIN, UNION JOIN, Natural JOIN and FULL JOIN.

301. How many primary keys can be there in a table?
एक table में कितनी primary key/keys हो सकती है/हैं?
(a) Only 1/केवल 1
(b) Only 2/केवल 2
(c) Depends on number of columns
कॉलमों की संख्या पर निर्भर करती है
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : A Primary key is a field or combination of fields in a database table that uniquely identifies each record in the table. This ensures that each row in the table has a distinct and non-null value for the primary key; which helps maintain data integrity. A table can have only one primary key.

302. Which of the following is not constraint in SQL?
निम्नलिखित में से कौन-सा SQL में Constraint नहीं है?
(a) Primary key
(b) Not null
(c) Union
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c) : Constraint in SQL is a type of special rule that helps to ensure validity & security of data in the database. It defines the rules for inserting and updating data in a database table, such as the value of a column must be unique, the value of a column must not be null, or the value of a column must be as per the given conditions. Hence, Union is not a constraint in SQL.

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303. Which of the following is not a valid aggregate function?
निम्नलिखित में से कौन-सा वैध aggregate function नहीं है?
(a) COUNT
(b) COMPUTE
(c) SUM
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b) : Aggregate function is a type of function used to store and process data in database. It can be used for various types of data processing, such as combining information with numbers, finding the highest and lowest value, calculating data, etc. There are many types of aggregate functions in SQL such as SUM, AVG, COUNT, MAX, MIN, etc. Hence COMPUTE is not a valid aggregate function.

304. If we have not specified ASC or DESC after an SQL ORDER BY clause, which of the following is used by default?
यदि हमने SQL ORDER BY क्लॉज के बाद ASC या DESC को निर्दिष्ट नहीं किया है, तो निम्नलिखित में से कौन-सा डिफ़ॉल्ट रूप से उपयोग किया जता है?
(a) DESC
(b) ASC
(c) There is no default value
कोई default value नहीं
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b) : If we have not specified ASC or DESC after an SQL order by clause, then 'ASC' is used by default. 'ASC' is the short form of Ascending order.

305. Which of the following is true about the HAVING clause?
HAVING खण्ड के बारे में, निम्नलिखित में से कौन-सा सत्य है?
(a) Similar to the WHERE clause but is used for columns rather than groups
(b) Similar to WHERE clause but is used for rows rather than columns
(c) Similar to WHERE clause but is used for groups rather than rows
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : 'HAVING' clause is used to filter data from database like WHERE clause, but WHERE clause is used with grouping and aggregation functions in group by statement.

306. _____ clause creates temporary relation for the query on which it is defined.
_____ खण्ड उस क्वेरी के लिए अस्थायी सम्बन्ध बनाता है, जिस पर इसे परिभाषित किया गया है।
(a) WITH
(b) FROM
(c) WHERE
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : The WITH clause in a SQL query creates a temporary relation for the query on which it is defined. Its main purpose is to make the code cleaner and easier to read, and can be used to query special cases.

307. The SQL statement
SELECT ROUND (65.726, –1) FROM DUAL prints
SQL कथन SELECT ROUND (65.726, –1) FROM DUAL; प्रिन्ट
(a) 70
(b) Garbage
(c) 726
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (a) : In SQL the ROUND function is used to round a number to a specified number of decimal places. Using the ROUND function in the given SQL Query will round the number 65.726 to the nearest multiple of 10. So the result will be 70.

308. Which of the following is a comparison operator is structured Query language?
निम्नलिखित में से कौन सा स्ट्रक्चर्ड क्वेरी लैंग्वेज में एक कम्पेरीज़न ऑपरेटर है?
(a) +
(b) =
(c) /
(d) >=

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Ans. (d) : In Structured Query Language (SQL), the comparison operator are >, <, >=, <=, <>. These operators are used to compare the two values.

309. Match the building blocks in Entity relationship diagram.
एंटिटी रिलेशनशिप आरेख में बिल्डिंग ब्लॉक्स को सुमेलित कीजिए।

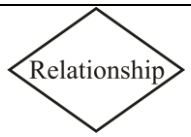
i		a	Relationship रिलेशनशिप
ii		b	Attributes एट्रिब्यूट
iii		c	Optional one ऑप्शनल वन
iv		d	Mandatory one मैनडेटरी वन

(a) i-b, ii-c, iii-a, iv-d
(b) i-b, ii-c, iii-d, iv-a
(c) i-c, ii-b, iii-d, iv-a
(d) i-b, ii-a, iii-d, iv-c

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Ans. (b) : Entity Relationship Diagram– It is a diagram that displays the relationship of entity sets stored in a database. ER diagrams are created based on three basic concepts: entities, attributes and relationship.
• **Relationship–** It is nothing but an association among two or more entities. It is represented by a Diamond

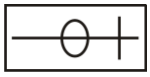
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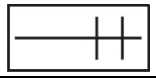
• **Attributes**– It is single valued property of either an entity-type or a relationship type.
It is represented by an oval



Optional One– It represents a relationship between two entities that is not mandatory to exist. It is represented by



Mandatory One– It represent a relationship between two entities that is mandatory to exist. It is represented by a line intersected by two small vertical lines at one end



310. Match the following operators with their definition in data definition language.
निम्नलिखित ऑपरेटरों के डेटा डेफिनेशन लैंग्वेज में उनकी डेफिनेशन से सुमेलित करें।

1	**{ }:**	**i**	**Optional data/ऑप्शनल डाटा**
2	**( ):**	**ii**	**Selection/सिलेक्शन**
3	**[, ,]:**	**iii**	**Iterative/ईटरेटिव**

(a) 1-iii, 2-i, 3-iii
(b) 1-i, 2-ii, 3-iii
(c) 1-ii, 2-iii, 3-i
(d) 1-iii, 2-ii, 3-i

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : { } : This operator denotes an iterative process, often used to define a set or an iterable object in programming and data languages, where each value can be processed in loops.
( ) **:** This operator represents optional data frequently used for function parameters or data fields that may or may not be supplied.
[, ,] : This operator indicates selection, typically utilized to define arrays or lists where elements are chosen or selected.

311. Consider the following ORACLE relations :
R (A, B, C) = {<1, 2, 3>, <1, 2, 0>, <1, 3, 1>, <6, 2, 3>, <1, 4, 2>, <3, 1, 4>}
S (B, C, D) + {<2, 3, 7>, <1, 4, 5>, <1, 2, 3>,<2, 3,4>, <3, 1, 4>}
Consider the following two SQL queries SQ_1 and SQ_2 :

SQ_1 : SELECT R. B, AVG (S. B)
FROM R, S
WHERE R.A = S.C AND S.D < 7
GROUP BY R.B;

SQ_2 : SELECT DISTINCT S.B, MIN (S.C)
FROM S
GROUP BY S.B
HAVING COUNT (DISTINCT S.D)>1;

If M is the number of tuples returned by SQ_1 and N is the number of tuples returned by SQ_2 then

(a) M = 4, N = 2 (b) M = 5, N = 3
(c) M = 2, N = 2 (d) M = 3, N = 3

UGC NET C.S. June-2016 (Paper-III)

Ans : (a)

Distinct S.B	Count(distinct S.d)
1	2
2	2
3	1

but the third tuple is not taken because count (distinct SD) >1

312. Semi-join strategies are techniques for query processing in distributed database system. Which of the following is a semi-join technique?

(a) Only the joining attributes are sent from one site to another and then all of the rows are returned
(b) All of the attributes are sent from one site to another and then only the required rows are returned
(c) Only the joining attributes are sent from site to another and then only the required rows are returned
(d) All of the attributes are sent from one site to another and then only the required rows are returned.

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) Semi-join strategies are techniques for query processing in distributed database system. In which only the joining attributes are sent from one site to another and then only the required rows are returned.

313. SQL command used to add a new column in an existing table is :
एक मौजूदा टेबल में एक नया कॉलम जोड़ने के लिए प्रयुक्त SQL कमांड है:

(a) ALTER (b) UPDATE
(c) CHANGE (d) ADD NEW

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Ans. (a) : The ALTER is used to add, delete, or modify columns in an existing table, and also used to add and drop various constraints on an existing table.

314. What is join operation in Structure Query Language (SQL)?
स्ट्रक्चर क्वेरी लैंग्वेज (एसक्यूएल) में जॉइन ऑपरेशन क्या है?

(a) Join is an operation used to select group rows of a table that contains the same values in a specific column.
जॉइन एक ऐसी तालिका, जिसमें एक विशिष्ट स्तम्भ में समान मान होते हैं, की पंक्तियों को सेलेक्ट करने के लिए उपयोग किये जाने वाला एक ऑपरेशन है।
(b) Join is an operation which is used to combine two different data keys.
जॉइन एक ऑपरेशन है जिसका उपयोग दो अलग-अलग डेटा को संयुक्त करने के लिए किया जाता है।

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(c) Join is an operation used to combine rows from two or more tables based on one or more common fields between them.
जॉइन एक ऑपरेशन है जिसका उपयोग दो या दो से अधिक तालिकाओं से पंक्तियों को उनके बीच एक या, अधिक सामान्य फील्ड के आधार पर संयोजित करने के लिए किया जाता है।

(d) Join is an operation which work on a single row of a table and return a single value.
जॉइन एक ऐसा ऑपरेशन है जो तालिका की एकल पंक्ति पर काम करता है तथा एक एकल मान लौटता है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : A join clause is used to compile rows from two or more tables based on one or more common fields between them.

315. The _____ of weak entity set distinguishes among all those entities in the weak entity set that depends upon one particular strong entity.
वीक एंटिटी सेट का _____ उस वीक एंटिटी सेट की उन सभी एंटिटीज में भेद करता है जो किसी एक विशिष्ट स्ट्रांग एंटिटी पर निर्भर है।

(a) Primary key
(b) Discriminator
(c) Identifying Relationship
(d) Candidate key

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Ans. (b) : A strong entity is not dependent on any other entity in the schema. A strong entity will always have a primary key strong entities are represented by a single rectangle. The relationship of two strong entities is represented by a single diamond various strong entities, when combined together, create a strong entity set.

316. SQL query and modification commands are represented in which of the following ?
SQL क्वेरी और संशोधन आदेशों को निम्न में से किसमें दर्शाया गया है?

(a) DDL (b) DML
(c) HTML (d) XML

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Ans. (b) : SQL structured Query Languages commonly abbreviated to SQL is a language used in Relational Database. This mainly focuses on creating a database and executing queries against a database.
SQL comprises both data definition and data manipulation language (DML) the commands used to manipulate data within a database, such as SELECT (for queering), INSERT (for adding new data),UPDATE (for modifying existing data), and DELETE (for removing data).

317. Choose the SQL commands to perform following operations on a relation R :
एक रिलेशन R पर निम्न क्रियाएँ करने के लिए SQL कमांड्स का चुनाव कीजिए:

I. Retain Relation R in database, but delete all tuples in R./रिलेशन R को डाटाबेस में रखें परन्तु उसके सारे टपल्स डिलीट करें।

II. Delete all information about R in Database. रिलेशन R के बारे में समस्त सूचनाएँ

(a) I : Drop table R; II : Delete from R
(b) I : Delete from R; II : Drop table R
(c) I : Delete table R; II : Drop from R
(d) I : Alter table R; II : Alter R Drop

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Ans. (b) :
(I) Retain Relation R in database, but delete all tuples in R–
This operation is achieved using the DELETE FROM command.
DELETE FROM R;
This command removes all rows (tuples) from the table R but keeps the table structure (Schema) and its definition within the database.
(II) Delete all information about R in Database– This operation is achieved using DROP TABLE command.
Now, substituting back
T total = 200 + 50 = 250
So, the total clock cycles required to complete the task is 250.
DROP TABLE R;
This command permanently removes the entire table R from the database, including its structure, data, and any associated indexes or constraints.

318. Which of the following is not a DDL command?
निम्नलिखित में से कौन डी डी एल कमांड नहीं है?

(a) TRUNCATE/ट्रंकेट (b) ALTER/आल्टर
(c) CREATE/क्रिएट (d) UPDATE/अपडेट

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Ans. (d) : Data Definition Language (DDL) statements are used to define, modify and delete database objects such as tables, views, indexes, etc. The DDL commands are CREATE, ALTER, TRUNCATE, DROP etc. But the UPDATE command is Data Manipulation Language (DDL) command, used to modify existing data in a table.

319. Which of the following is true about the HAVING clause ?
हैविंग क्लॉज के बारे में निम्नलिखित में से कौन-सा कथन सत्य है?

(a) Similar to the WHERE clause but is used for columns rather than groups.
वेयर क्लॉज के समान लेकिन यह समूहों के बजाय कॉलम के लिए उपयोग किया जाता हैं

(b) Similar to WHERE clause but is used for rows rather than columns.
वेयर क्लॉज के समान लेकिन कॉलम के बजाय रो के लिए उपयोग किया जाता है।

(c) Similar to WHERE clause but is used for groups rather than rows.
वेयर क्लॉज के समान लेकिन यह रो के बजाय समूहों के लिए उपयोग किया जाता है।

(d) Acts exactly like a WHERE clause.
बिल्कुल वेयर क्लॉज की तरह काम करता है।

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Ans. (c) : In SQL, the HAVING clause is used to filter grouped data after the GROUP BY operation. It is similar to the WHERE clause but applies to aggregated results rather than individual rows.

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Important Key Points–
1. Difference Between WHERE and HAVING : WHERE is used to filter rows before aggregation (works on individual records)
HAVING is used to filter group after aggregation (works on group records).

320. Which of the following is generally used for performing tasks like creating the structure of the relations, deleting relation ?
आमतौर पर रिलेशन की संरचना बनाने, रिलेशन को हटाने जैसे कार्यों को करने के लिए निम्नलिखित में से किसका उपयोग किया जाता है?
(a) DML (Data Manipulation Language) डाटा मैनीपुलेशन लैंग्वेज
(b) Query/क्वेरी
(c) Relational Schema/रिलेशनल स्कीमा
(d) DDL (Data Definition Language) डाटा डेफिनिशन लैंग्वेज

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Ans. (d) : In SQL, Data Definition Language (DDL) statements are used to define the structure of the database, including creating relating altering and dropping database objects such as tables, indexes, and views.

321. Which of the following provides the ability to query information from the database and insert tuples into, delete tuples from, and modify tuples in the database ?
निम्नलिखित में से कौन डेटाबेस से जानकारी को क्वेरी करने और डेटाबेस में टपल्स को सम्मिलित करने, टपल्स को हटाने और डेटाबेस में टपल्स को संशोधित करने की क्षमता प्रदान करता है?
(a) DML (Data Manipulation Language) डाटा मैनीपुलेशन लैंग्वेज
(b) DDL (Data Definition Language) डाटा डेफिनिशन लैंग्वेज
(c) Query/क्वेरी
(d) Relational Schema/रिलेशनल स्कीमा

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Ans. (a) :
- A DML (Data Manipulation Language) refers to a computer programming language that allows you to add (insert), delete, and after update data in a database.
- A DML is typically a sublanguage of a larger database language like SQL, with the DML containing some of the languages operators.
- The majority of SQL statements are categorized as DML (Data Manipulation Language) which includes SQL commands that deal with modifying data in database.

322. SQL Views are also known as :
SQL व्यूज को निम्न रूप में भी जाना जाता है:
(a) Simple Tables/सिंपल टेबल्स
(b) Virtual Tables/वर्चुअल टेबल्स
(c) Complex Tables/कॉम्प्लेक्स टेबल्स
(d) Actual Tables/एक्चुअल टेबल्स

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Ans. (b) :
- A view is a table that is derived from other tables. These other tables can be base tables or previously defined view.
- A view does not necessarily exist in physical form it is considered as virtual table in contrast to base tables whose tuples are actually stored in the database.
- In SQL, view are the virtual tables that can be created from one or more tables. A view can contain all rows or some selected rows of a table.

323. The outcome of a SQL SELECT statement is a:
SQL SELECT स्टेटमेंट का परिणाम है:
(a) File/फाइल (b) Table/तालिका
(c) Report/रिपोर्ट (d) Form/प्रपत्र

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Ans. (b) : The SQL SELECT statement is used to select a specific set of data from the database. The data returned by the SELECT statement is stored in the result table called the result set. A table is a collection of related data help in a table format within a database in relational data base and flat file database, a table is a set of data elements (values) using a model of vertical columns (identifiable by name) and horizontal rows, the cell being the unit where a row and column intersect.

324. Consider the following SQL query to retrieve ecode of employees from EMPLOYEE table, whose address contains 'INDIA' :
EMPLOYEE टेबल में से ऐसे एम्प्लॉइज के ecode प्राप्त करने के लिए, जिनके address में 'INDIA' आता हो, निम्न SQL क्वेरी पर विचार करे:

```
SELECT ecode
FROM EMPLOYEE
WHERE ..... ;
```

Which of the following is most suitable to complete the query ?
निम्न में से कौन-सा क्वेरी को पूर्ण करने के लिए सर्वाधिक उपयुक्त है?
(a) Address LIKE '_INDIA_'
(b) Address = %INDIA%
(c) Address AS '%INDIA_'
(d) Address LIKE '%INDIA%

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Ans. (d) :
- Like operator is used in a WHERE clause to search for a specified pattern in a column.
- There are two wildcards often used in conjunction with the LIKE operator.

Character	Description
% (percentage)	The percent sign represents zero, one, or multiple characters
– (underscore)	The underscore represents a single character

Syntax:–
SELECT column_name1, column_name2... FROM table_name
WHERE column_name LIKE '% match'
OR
SELECT column_name1, column_name2... FROM

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table_name
WHERE column_name LIKE '% match%'
SQL Command–
SELECT ecode
FROM EMPLOYEE
WHERE Address LIKE '% INDIA %'
Return code from EMPLOYEE table whose address part contains "INDIA" in any position
NOTE– Address is a column in EMPLOYEE table.

325. Which of the following set of operations is a valid set of aggregate operations in SQL ?
निम्न में से ऑपरेशनों का कौन-सा समूह SQL में एग्रीगेट (सकल/सामूहिक) ऑपरेशनों का उचित समूह हैं?
(a) COUNT, MAX, AVG, SUM
(b) MAX, AVG, SUM, SELECT
(c) UNION, COUNT, MIN, DESC
(d) AVG, MIN, MAX, ASC

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Ans. (a) : In SQL, AVG, MIN, MAX, SUM, COUNT is called as aggregate function. The aggregate functions return a single value after calculating from a group of values
- **AVG()–** It returns the average values
- **COUNT()–** It finds the number of rows that matches the specified condition.
- **MIN()–** It returns the minimum value.
- **MAX()–** It returns the maximum value.
- **SUM()–** It returns the sum of values.

326. SQL command used to modify attribute values of one or more selected tuples is :
एक या अधिक चुने हुए टपल/टपलों के एट्रिब्यूट मान बदलने में प्रयुक्त SQL कमांड है:
(a) MODIFY (b) UPDATE
(c) INSERT (d) ALTER

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Ans. (b) : SQL command used to modify attribute values of one or more selected types is UPDATE.
The ALERT TABLE statement is used to add, delete, or modify columns in an existing table. The ALTER TABLE statement is also used to add and drop various constraints on an existing table.
Syntax– The UPDATE statement is used to modify the existing records in a table
UPDATE table_name
SET column1 = value 1, column 2 = value 2,….
WHERE conditions;

327. Which one of the following is used to sort rows in SQL ?
निम्न में से किसका उपयोग SQL में पंक्तियों को सॉर्ट करने के लिए किया जाता है?
(a) SORTBY (b) ALIGNBY
(c) ORDERBY (d) GROUPBY

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Ans. (c) : The ORDER BY statement in SQL is used to sort the fetched data in either ascending or descending according to one or more columns.
- By default ORDER BY sorts the data in ascending order.
- We can use the keyword DESC to sort the data in descending order and the keyword ASC to sort in ascending order.

syntax of all ways of using ORDER BY is shown below.
Syntax– SELECT*FROm table_name ORDER BY column_name ASC/DESC.

328. Consider the following SQL query :
निम्न SQL क्वेरी पर विचार करें:
SELECT title, name
FROM Employee
ORDER BY title, name DESC;
In output of above query the different titles are listed in _____ order and names of employees having same title are listed in _____ order.
इस क्वेरी के आउटपुट में विभिन्न टाइटल _____ क्रम में सूचीबद्ध होंगे तथा समान टाइटल वाले कर्मचारियों के नाम _____ क्रम में सूचीबद्ध होंगे।
(a) Ascending, Ascending/आरोही, आरोही
(b) Ascending, Descending/आरोही, अवरोही
(c) Descending, Ascending/अवरोही, आरोही
(d) Descending, Descending/अवरोही, अवरोही

HTET PGT C.S. 2018

Ans. (b) : The ORDER BY command is used to sort the result set in ascending or descending order.
The ORDER BY command sorts the result set in ascending order by default. To sort the DESC keyword.
Syntax– SELECT*FROM table_name ORDER BY column_name ASC/DESC
- Table_name : name of the table
- Column_name : name of the column according to which the data is needed to be arranged.
- ASC : to sort the data in descending order.
- DESC : to sort the data in descending order.

329. Which SQL Query is used to remove a table and all its data from the database ?
डेटाबेस से एक तालिका और उसके सभी डेटा को निकालने के लिए कौन-सी SQL क्वेरी का उपयोग किया जाता है?
(a) Create Table
(b) Alter Table
(c) Drop Table
(d) None of three/तीनों में कोई नहीं

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Ans. (c) : Key points–
- **DROP–** It is a Data Definition Language (DDL) command which is used to delete, tables, views, triggers, etc from a database.
- A DROP statement in SQL ramous a component from a Relational Database Management System (RDBMS)

Additional Information–
- **Data Definition Language (DDL)–** Data Definition Language Consists of SQL command that can be used to define the database.

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■ List of DDL commands.

Command	Description
DROP	This command is used to delete objects from the database.
ALTER	This is used to alter the structure of the database.
TRUNCATE	This is used to remove all records from a table including all spaces allocated for the records are removed.
COMMENT	This is used to add comments to the data dictionary.
RENAME	This is used to rename on object existing in the database.

330. In context of a relation in database, choose a false statement :
डाटाबेस में एक रिलेशन के सन्दर्भ में असत्य कथन का चुनाव करें:

(a) There can be more than one super keys.
एक से अधिक सुपर कुंजी होना संभव है।

(b) A candidate key is a minimal super key.
कैंडिडेट कुंजी एक न्यूनतम सुपर कुंजी होती है।

(c) One of the candidate keys is designated as primary key./किसी एक कैंडिडेट कुंजी को प्राथमिक कुंजी बना लिया जाता है।

(d) Primary key is obtained by removing one or more attributes from a candidate key./कैंडिडेट कुंजी में से एक या अधिक एट्रिब्यूट हटाकर प्राथमिक कुंजी प्राप्त की जाती है।

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Ans. (d) : Super key–
- It is an attribute (or set of attributes) that is used to uniquely Identifies all attributes in a relation.
- All Super keys cannot be candidate keys but its reserve is true.
- There can be more than one super key.
- In relation, the number of super key is always greater than or equal to the number of candidate keys.
- There always exists at least one super key in table.

Candidate key–
- It is a minimal set of attributes necessary to identify a tuple; This is also called a minimal super key.
- candidate key can be more then one.
- One of the candidate keys is designated as the primary key.

Primary key–
- Candidate key from the table selected by the database administrator to uniquely identify tuples in a table known as the primary key.
- Since the candidate is a minimal set of attributes necessary to identify a tuple therefore the primary key is also a minimal set of attributes necessary to identify a tuple and hence primary key cannot be obtained by removing one or more attributes from a candidate key. Therefore option 4 is false.

331. Consider a weak entity set in DBMS. Cardinality ratio 'strong entity set : weak entity set' in their binary identifying relationship is :
DBMS में एक वीक एन्टिटी सेट पर विचार करें। अपने बाइनरी आइडेन्टीफाइंग रिलेशनशिप में गणन संख्यात्मक अनुपात (कार्डिनेलिटी रेशियो) 'स्ट्रांग एंटिटी सेट : वीक एंटिटी सेट' होता है:

(a) One to one/एक से एक
(b) One to many/एक से अधिक
(c) Many to one/अधिक से एक
(d) Many to many/अधिक से अधिक

HTET PGT C.S. 2019

Ans. (b) : Rules for Weak Entity Set Relationship–
- A weak entity set has one or more many one relationships to other entity sets.
- Not every man one relationship from a weak entity set needs to be supporting but supporting relationships must have a rounded arrow.
- The key for a weak entity set is its own underlined attributes and the keys for the supporting entity sets.
- Weak entity sets depend on at least one strong entity set.
- Relation schema for weak entity set already include a primary key for strong entity set.
- If strong entity se exists → weak entity set exists.
- If (strong entity set) not exists → weak entity set not exists.

So, the one-to-many relationship will be there.

332. Let for any relation R, IR represents operation insert into R and DR represents operation delete from R. Assume that a relation R2 has a foreign key that refers to primary key of another relation R1. Two operations that may cause violation of referential integrity constraints are :
माना किसी रिलेशन R के लिए R में सन्निविष्ट क्रिया (Insert Operation) को I_R तथा R से विलोपन क्रिया (Delete Operation) को D_R से निरूपित किया जाता है। मान लें कि एक रिलेशन R2 में एक बाह्य कुंजी है जो रिलेशन R1 की प्राथमिक कुंजी से सम्बद्ध है। वे दो ऑपरेशन जो सन्दर्भ समग्रता बंधकारकों (Referential Integrity Constraints) के उल्लंघन का कारण हो सकते हैं

(a) I_{R1} and I_{R2} (b) D_{R1} and D_{R2}
(c) I_{R1} and D_{R2} (d) I_{R2} and D_{R1}

HTET PGT C.S. 2018

Ans. (d) :
- Insert into R1 cannot cause any violation.
- Insert into R2 can cause violation if any value is inserteed into D of R2 which value is not in A in R1
- Delete from R1 can cause violation if any tuple is deleted which is refrenced to by D in R2.
- Delete from R2 cannot cause any violation.

Violation of Referential integrity constraints.

333. This command is used to modify attribute values of one or more selected tuples:
इस कमांड का उपयोग एक या अधिक चयनित टपल्स की एट्रीब्यूट वैल्यूज़ को संशोधित करने के लिए किया जाता है-

(a) Drop (b) Alter
(c) Update (d) Modify

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Ans. (c) : DROP ⇒ Used to remove a table, column, or database.
ALTER⇒ Used to change the structure of a table (e.g. add/modify columns).
UPDATE⇒ Used to modify attribute (column) values of one or more selected tuples (rows).
MODIFY⇒ Not a standard SQL command (some databases use it as part of ALTER TABLE).
So in SQL, to change values inside rows we use UPDATE.

334. A double ellipse represents _____ attribute in ER-diagram.
ई. आर. चित्रण में एक दोहरा दीर्घवृत्त _____ एट्रिब्यूट को निरूपित करता है।
(a) Multi-valued/बहुमान (b) Composite/समग्र
(c) Derived/व्युत्पन्न (d) Complex/मिश्रित
HTET PGT C.S. 2018

Ans. (a) : In an ER diagram–
Single ellipse → simple attribute
Double ellipse → multi valued attribute
Dashed ellipse → derived attribute
Ellipse connected with others → composite attribute.
So, a double ellipse represents a multi-valued attribute.

335. The select operation is used to choose a subset of tuples from a relation that satisfies a_____.
सेलेक्ट ऑपरेशन का उपयोग टपल्स के सबसेट को चयनित करने में किया जता है, जो संतुष्ट करता है–
(a) Check Condition/चेक कंडीशन
(b) Selection Condition/सेलेक्शन कंडीशन
(c) Insertion Condition/इंसर्शन कंडीशन
(d) Update Condition/अपडेट कंडीशन
HTET PGT C.S. 2023

Ans. (b) : In relational algebra, the select (σ) operate is used to choose a subset of tuples from a relation that satisfy a selection condition (a predicate or logical expression).

336. A field that gives each record a unique identify is called as:
एक फील्ड जो प्रत्येक रिकॉर्ड को एक विशिष्ट पहचान देता है, कहलाता है–
(a) Unique key (b) First key
(c) Primary key (d) Candidate key
HTET PGT C.S. 2023

Ans. (c) : Primary key is a unique key which is used uniquely identify the record from the table in database.

337. The query that is used to summarize the data from various tables is called as:
वह क्वेरी जिसका उपयोग विभिन्न टेबल से डेटा को सारांशित करने के लिए किया जाता है, कहलाता है–
(a) SQL query (b) Join query
(c) Selection query (d) Crosstab query
HTET PGT C.S. 2023

Ans. (d) : A query used to summarize data from various tables is commonly referred to as a aggregate query, a crosstab query (which adds a pivot-table-like structure to the summary), or a SQL GROUP BY query when using structured Query Language.

338. In database _____ type of field is used to insert various objects mode in other Windows applications into table.
डेटाबेस में _____ फील्ड का प्रयोग विंडोज एप्लीकेशन के ऑब्जेक्ट्स को टेबल में इंसर्ट करने के लिए किया जाता है।
(a) SQL query (b) Join object
(c) OLE object (d) Hyperlink
HTET PGT C.S. 2023

Ans. (c) : The database field type used to insert objects like pictures, graphs or other ActiveX objects from other windows application is called an OLE Object or ActiveX object field. This data type allows you to either embed or link these external objects directly into a Microsoft Access table enabling rich content to be stored alongside your structured data.

339. Degree of relationship type is the number of participating_____.
रिलेशनशिप टाइप की डिग्री है, पार्टिसिपेट करने वाली_____की संख्या।
(a) Keys types (b) Entity types
(c) Condition types (d) Attributes types
HTET PGT C.S. 2023

Ans. (b) : The degree of a relationship indicates the number of participating entity types in that relationship for example, a relationship with two participation entity types like "Customer" and "Account", is a binary relationship with a degree of two.

340. In SQL terminology a _____ is a single table that is derived from other tables.
SQL शब्दावली में _____ एक ऐसी टेबल है जो अन्य टेबल्स से बनाया जाता है।
(a) Table (b) View
(c) Base Table (d) Form
HTET PGT C.S. 2023

Ans. (b) : In SQL terminology a view is a single table that is derived from other tables. A view is a temporary table in database which is the subset of main table or we can it is a part of main table.

341. What type of key is used to link tables in a relational database?
(a) Candidate key (b) Foreign key
(c) Composite key (d) Super key
(e) Not attempted
Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : A foreign key is a column or set of columns in one table that refers to the primary key of another table establishing a link or relationship between them and ensuring referencial integrity.

342. Which SQL keyword is used to prevent duplicate records in a query result?
(a) DISTINCT (b) UNIQUE
(c) NO_DUPLICATE (d) SOLE
(e) Not attempted
Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : The Distinct Keyword is used in a SELECT statement to eliminate duplicate rows from the result set. When DISTINCT is specified, only unique combination of selected columns are returned.

Sarkari Resource

343. Which clause in SQL is used to restrict the rows returned by a query?
(a) WHERE (b) ORDER BY
(c) SEQ BY (d) GROUP BY
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : The SQL clause used to restrict the rows returned by a query is the 'WHERE' clause "WHERE' clause filters roués based on specified conditions. 'ORDER BY' clause sorts the result set in ascending or descending order based on one or more columns.

344. In the context of database management systems, a candidate key
(a) Is same as a primary key
(b) Can be a primary key
(c) Must be a composite key
(d) Must be a foreign key
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : A candidate key is a column or set of columns that uniquely identifies a record in a table and has the potential to be chosen as the primary key, a table can have multiple candidate keys, but only one is selected as the primary key.

345. Which of the following is wrong?
(a) An SQL query automatically eliminates duplicates
(b) An SQL query will not work if there are no indexes on relations
(c) SQL permits attribute names to be repeated on the same relation
(d) All of the above are wrong

CGPSC C.S. 2020 Set-A

Ans. (d) :
(a) By default, SQL returns duplicates (multi set semantics). To eliminate duplicates, we must use SELECT DISTINCT.
(b) An SQL query will not work if there are no indexes on relations wrong.
Indexes improve performance but are not required. Queries still work without indexes.
(c) SQL permits attribute names to be repeated on the same relation wrong.
In a relation (table), attribute (column) names must be unique. You cannot define a table with duplicate column names.

346. The weak entity set is represented by:
(a) Box
(b) Ellipse
(c) Diamond
(d) Double outlined box

CGPSC C.S. 2020 Set-A

Ans. (d) : In ER (Entity Relationship) diagrams:
Entity set → represented by a rectangle (box).
Weak entity set → represented by a double outlined rectangle (double box).
Attributes → ellipse
Relationship → diamond.
So the correct answer is (d) double outlined box.

347. In ER diagram, double lines indicate:
(a) Cardinality (b) Relationship
(c) Partial participation (d) Total participation

CGPSC C.S. 2020 Set-A

Ans. (d) : In an ER (Entity-Relationship) diagram, double lines are used to indicate total participation of an entity in a relationship.
So, the correct answer is (d) total participation.

348. Which one is correct?
(a) Primary key ⊂ Super key ⊂ Candidate key
(b) Candidate key Super key ⊂ Primary key
(c) Primary key ⊂ Candidate key ⊂ Super key
(d) Super key ⊂ Primary key ⊂ Candidate key

CGPSC C.S. 2020 Set-A

Ans. (c) : Super Key– Any of attributes that uniquely identifies a row.
Candidate Key– A minimal super key (no extra attributes).
Primary key– One chosen candidate key as the main identifier. So, the correct relationship is primary key ⊆ candidate key ⊆ superkey.

349. The correct order of SQL expression is:
(a) Select, group by, where, having
(b) Select, where, group by, having
(c) Select, group by, having, where
(d) Select, having, where, group by

CGPSC C.S. 2020 Set-A

Ans. (b) :
(1) From/Join - pick the tables.
(2) WHERE-filter rows.
(3) GROUP BY - group rows.
(4) HAVIND - Filter groups.
(5) SELECT - choose the columns/expressions.
(6) ORDER BY - sort there suits.
So the among given options, the correct order is (b) Select, where, group by, having

350. Which of the following are set operations?
(i) Union **(ii) Intersection**
(iii) Set difference **(iv) Cartesian product**
(a) (i), (ii), (iii) (b) (i), (iii), (iv)
(c) (i), (ii), (iii), (iv) (d) (i), (ii), (iv)

CGPSC C.S. 2020 Set-A

Ans. (c) : As all four options union, intersection set differences and Cartesian product are fundamental sent operations.

351. The ? wildcard character represents:
? वाइल्डकार्ड कैरेक्टर प्रतिनिधित्व करता है–
(a) Group of characters
(b) Any single character
(c) Date
(d) Multiple digit

HTET PGT C.S. 30.07.2025

Ans. (b) : The '?' wildcard character represents any single character. It functions as a placeholder, allowing a search to match a single, unknown character at that position. For example. "b ? g" would find words like "bag", "big" or "bug" etc.

352. Which operation is used to change the values of one or more attributes in a tuple (or tuples) of some relation R?

Sarkari Resource

किसी रिलेशन R के टपल (टपल्स) के एक या अधिक एट्रीब्यूट्स के मान को बदलने के लिए किस ऑपरेशन का उपयोग किया जाता है?

(a) Update (b) Delete
(c) Select (d) Where

HTET PGT C.S. 30.07.2025

Ans. (a) : The operation used to change the values of one or more attributes in a tuple (or tuples) of some relation 'R' is the 'UPDATE' Operation In SQL, this is typically performed using the 'UPDATE' statement.

353. **_____ are complete the select-from-where blocks with in another SQL query.**
_____ किसी अन्य SQL क्वेरी के भीतर कम्प्लीट सेलेक्ट-फ्रॉम-वेयर ब्लॉक है।

(a) Nested queries (b) SQL queries
(c) UPdate queries (d) Select queries

HTET PGT C.S. 30.07.2025

Ans. (a) : Nested queries (also called sub queries) are SQL queries that are embedded within another SQL query, typically inside the SELECT: FROM-WHERE block or even in other statements like INSERT, UPDATE or DELETE. The result of the inner (nested) query is used by the surrounding (outer) query enabling more complex data retrieval & manipulation.

354. **In SQL_____ is treated as a virtual table that is derived from other tables.**
SQL में _____ को वर्चुअल टेबल माना जाता है, जो अन्य टेबल्स से डेराइवड होता है।

(a) Table (b) View
(c) Base Table (d) Form

HTET PGT C.S. 30.07.2025

Ans. (b) : In SQL a view is treated as a virtual table that is derived from other tables. A view is a result set of a stored query, which can be treated as a table.

355. **The _____ states that no primary key value can be NULL.**
_____ बताता है कि किसी भी प्राइमरी 'की' का मान NULL नहीं हो सकता है।

(a) Entity Integrity Constraint
(b) Referential Integrity
(c) Primary Key Constraint
(d) Foreign Key Constraint

HTET PGT C.S. 30.07.2025

Ans. (a) : The constraint that states no primary key value can be NULL is called Entity Integrity Constraint. A primary key is designed to uniquely identify each row in a table, and a NULL value world negate this purpose. Therefore the constraint ensuring a primary key is always not null is termed "Entity Integrity constraint".

356. **Each row or record of a relational table is known as**
रिलेशनल टेबल की प्रत्येक पंक्ति या रिकार्ड कहलाती है

(a) Domain/डोमेन (b) Attribute/एट्रीब्यूट
(c) Tuple/ट्पल (d) Key/की

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (c) : Each row or record of a relational table is known as a tuple.
In a Relational database, a tuple refers to a single row in a table, representing one record.

357. **Which of the following operation of relational algebra, selects certain columns from the table and discards the other columns?**
निम्न में से कौन सा रिलेशनल एल्जेब्रा का ऑपरेशन किसी टेबल में से कुछ निश्चित कॉलम चुनता है तथा अन्य को अस्वीकृत कर देता है?

(a) SELECT (b) ROJECT
(c) DIVISION (d) None of the above

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (b) : The relational algebra operation that selects certain columns from the table and discards the other columns is the 'PROJECT' operation.
The PROJECT operation in relational algebra (denoted by the symbol (π) selects specific attributes. (columns) from a relation effectively.

358. **A table space is further broken down into____.**
एक टेबल स्पेस को आगे _____ में तोड़ा जाता है।

(a) Blocks/ब्लॉक (b) Extents/एक्सटेंट्स
(c) Files/फाइल्स (d) Segments/सेगमेन्ट्स

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (d) : A table space is further broken down into segments, which are then broken down into extends, and extents are made up of blocks.

(iv) Database Security, Transaction Management, introduction to Query

359. **Which of the following statements are NOT correct?**
I. Parallel processing is also known as loosely coupled architecture.
II. Distributed system is also known as tightly coupled architecture.
निम्नलिखित में से कौन सा कथन सही नहीं हैं?
I. पैरेलल प्रोसेसिंग को लुसली कपल्ड आर्किटेक्चर के रूप में भी जाना जाता है।
II. डिस्ट्रिब्यूटेड सिस्टम को टाइटली कपल्ड आर्किटेक्चर के रूप में भी जाना जाता है।

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Parallel processing is generally associated with a tightly coupled architecture. A distributed system, on the other hard, is characterized by a loosely coupled architecture.

360. **All changes made by a transaction are made permanent in the database only after transaction:-**

(a) FINAL (b) ROLL BACK
(c) ABORT (d) COMMIT

RPSC C.S. 2016 (PAPER-II)

Ans. (d) : The COMMIT statement ends the current transaction and makes all changes performed in the transaction permanent.

Sarkari Resource

Roll Back– Rolling back a transaction cancels all modifications made by the current transaction.
Abort– Aborting a transaction is the same as rolling it back; it discards all changes.

361. A schedule in which transactions can neither read nor write an item X until the last transaction that wrote x has committed or aborted. This type of schedule is called.
(a) Recoverable (b) Cascadless
(c) Strict (d) Non-recoverable

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : A strict schedule ensures that a transaction T can neither read nor write an item X until the last transaction that wrote X has committed or aborted. This strictness helps in maintaining the consistency of the database and avoids cascading rollbacks.

362. A multidimensional database schema with one central fact table and one table for each dimension is called:-
(a) Snow-flake schema
(b) Star schema
(c) Fact-constellation schema
(d) Hybrid schema

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : A star schema is used to denormalize business data into dimensions (like time and product) and facts (like transactions in amounts and quantities). A star schema has a single fact table in the center, containing business "facts" (like transaction amounts and quantities).

363. Which of the following is correct according to the technology deployed by DBMS?
DBMS द्वारा प्रयुक्त तकनीक के अनुसार निम्नलिखित में से कौन-सा सही है?
(a) Pointers are used to maintain transactional integrity and consistency
प्वाईटर्स का उपयोग लेनदेन की अखंडता और स्थिरता बनाए रखने के लिए किया जाता है
(b) Cursors are used to maintain transactional integrity and consistency
कर्सर का उपयोग ट्रांजेक्शनल अखंडता और स्थिरता बनाए रखने के लिए किया जाता है
(c) Locks are used to maintain transactional integrity and consistency
लॉक्स का उपयोग ट्रांजेक्शनल अखंडता और स्थिरता बनाए रखने के लिए किया जाता है
(d) Triggers are used to maintain transactional integrity and consistency
ट्रिगर्स का उपयोग ट्रांजेक्शनल अखंडता और स्थिरता बनाए रखने के लिए किया जाता है

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : In database management, locks are used to prevent problems (such as data inconsistency) that occur when multiple users access data at the same time. They are used to maintain Transactional integrity and consistency.

364. Which of the following is not a transaction management SQL command ?
निम्नलिखित में से क्या ट्रांजक्शन मैनेजमेंट एस क्यू एल कमांड नहीं है?
(a) Rollback/रोलबैक (b) Commit/कमिट
(c) Select/सलेक्ट (d) Savepoint/सेवप्वाइंट

KVS PGT C.S. 2017

Ans. (c) : The commands are used to control the transactions in SQL these commands are given below –
- COMMIT
- ROLLBACK
- SAVE POINT
- SET TRANSPORT.

Transactional commands are used with the help of DML commands. It cannot be applied when we are creating or dropping the tables.

365. If transaction A holds a shared lock R and if transaction B also requests for a shared lock on R, it will
यदि ट्रान्जैक्शन A एक शेयर्ड लॉक R रखता है और R पर ट्रान्जैक्शन B भी एक शेयर्ड लॉक की प्रार्थना करता है, तो यह
(a) Result in a deadlock situation
एक गतिरोध (Deadlock) की स्थिति उत्पन्न करता है
(b) Immediately be granted/तुरन्त स्वीकार की जाती है
(c) Immediately be rejected
तुरन्त अस्वीकार की जाती है
(d) Be granted as soon as it is released by A
A द्वारा मुक्त होने के उपरान्त तुरन्त स्वीकार की जाती है

LT. Grade C.S. 2018

Ans. (b) : Simple locking scheme
1. Shared lock (S) 2. An exclusive lock (X)
Shared lock (S) - If transaction locked data item in shared mode, allowed to read - only.
Exclusive lock (X) - If transaction locked data item in shared mode, allowed to read and write both.

REQUEST	Grant lock	
Lock	S	X
S	Yes	No
X	No	No

From the above table, we can see that if one transaction has a shared lock on a data item and if one other transaction request for a shared lock on the same data item it will be immediately granted.

366. Which of the following is correct according to the technology deployed by DBMS?
DBMS द्वारा तैनात तकनीक के अनुसार निम्नलिखित में से कौन-सा सही है?
(a) Pointers are used to maintain transactional integrity and consistency
(b) Cursors are used to maintain transactional integrity and consistency
(c) Locks are used to maintain transactional integrity and consistency
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Sarkari Resource

Ans. (c) Locks are the mechanisms used by the DBMS to control access to data. By using locks, the DBMS ensures that multiple transactions cannot modify the same data at the sometime, therapy maintaining the integrity and the consistency of the data.

367. Which one of the following is NOT a part of ACID properties of a database transaction?
(a) Atomicity (b) Consistency
(c) Isolation (d) Deadlock-freedom

UGC NET C.S. June-2023

Ans. (d) : A transaction is a single logical unit of work that accesses and possibly modifies the contents of a database. Transactions access data using read and write operations. In order to maintain consistency in a database, before and after the transaction, certain properties are followed. These are called ACID properties.
A = Atomicity
C = Consistency
I = Isolation
D = Durability

368. Consider the following set of functional dependencies defined over the relation schema R(A, B, C, D, E, F, G, H)
रिलेशन स्कीमा R(A, B, C, D, E, F, G, H) पर परिभाषित कार्यात्मक निर्भरताओं (फंक्शनल डिपेंडेंसी) के निम्नलिखित सेट पर विचार करें।
$F = \{AB \rightarrow C, C \rightarrow A, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$
How many candidate keys does the relation R have ?
रिलेशन R के पास कितनी कैंडिडेट की हैं?
(a) 3 (b) 4
(c) 5 (d) 6

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : The relation R (A, B, C, D, E, F, G, H) with the given set of functional dependencies has four candidate keys.
Candidate keys list
- AD
- BD
- ED
- FD

Each of these attributes combinations can uniquely identify all attributes in relation R, as confirmed by their attribute closure. L, Attribute D does not appear on the left side of any functional dependency so it must be included in all candidate keys checking the closure of AD, BD, ED and FD using the given functional dependencies confirms that each combination can functionally determine all attributes in R.

369. Which of the following is also known as read-write conflict?
निम्नलिखित में से कौन सा रीड-राइड कनफ्लिक्ट के रूप में भी जाना जाता है?
(a) Dirty read/डर्टी रीड
(b) Lost update problem/लॉस्ट अपडेट प्रॉब्लम
(c) Unrepeatable read/अनरिपीटेबल रीड
(d) Uncommitted dependency/अनकमिटेड डिपेंडेंसी

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : The read write conflict occurs when a transaction reads a value that another concurrent transaction modifies before the first transaction commits, leading to inconsistencies and the phenomenon called unrepeatable reads.

370. What is Denormalization in Relational Database Management System?
रिलेशनल डेटाबेस मैनेजमेंट सिस्टम में डीनोर्मलाइजेशन क्या होता है?
(a) Denormalization is the process of increasing redundancy to improve performance.
प्रदर्शन को बेहतर बनाने के लिए अतिरिक्तता बढ़ाने की प्रक्रिया को डीनोर्मलाइजेशन कहा जाता है।
(b) Denormalization is used to decrease rredundancy in database.
डेटाबेस में अतिरिक्तता को कम करने के लिए डीनोर्मलाइजेशन का उपयोग किया जाता है।
(c) Denormalization is used to decrease the dependence on normal forms.
सामान्य फॉर्म्स पर निर्भरता को कम करने के लिए डीनोर्मलाइजेशन का उपयोग किया जाता है।
(d) Denormalization is used to develop a new schema in database.
डेटाबेस में एक नया स्कीमा विकसित करने के लिए डीनोर्मलाइजेशन का उपयोग किया जाता है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Denormalization is the process of increasing redundancy to improve performance.

371. Which one of the following is NOT a part of the ACID properties of database transaction?
निम्नलिखित में से कौन सा डेटाबेस ट्रांसएक्शन के ACID गुणों का हिस्सा नहीं है?
(a) Atomicity (b) Consistency
(c) Instance (d) Durability

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : The ACID properties of database transactions are a set of principles that ensure reliable processing of database transactions.
ACID stand for
Atomicity, Consistency, Isolation, Durability.

372. _____is the point of synchronisation between the database and the transaction log file in database management system.
डेटाबेस प्रबंधन प्रणाली में डेटाबेस और ट्रांजैक्शन लॉग फ़ाइल के बीच सिंक्रनाइजेंशन का बिंदु है।
(a) Checkpoint/चेकपॉइंट
(b) Shadow point/शेडो पॉइंट
(c) Mapping point/मैपिंग पॉइंट
(d) Key point/की पॉइंट

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : The checkpoint is the point of synchronization between the database and the transaction log file in a database management system (DBMS). A checkpoint ensures that all modifies data pages from memory (dirty pages) are written to physical disk.

Sarkari Resource

373. **_____ command makes the updates performed by the transaction permanent in the database.**
_____ कमान्ड Transaction द्वारा किए गए अपडेट का डेटाबेस में स्थायी बनाता है।फे

(a) ROLLBACK
(b) COMMIT
(c) TRUNCATE
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : 'COMMIT' command makes the updates performed by the transaction permanent in the database. After a transaction commits all database operations, their results are permanently stored in the database successfully. It is used to make database operations secure and dependable, ensuring database compatibility & data security.

374. **Which of the following statement(s) is/are correct regarding On-Line Transaction Procession (OLTP)?**
ऑन-लाईन ट्रांसेसिंग (OLTP) के संदर्भ निम्नलिखित में सा/से कथन सही है/हैं?

I. **Responses to the user inquiry are immediate./उपयोगकर्ता इंक्वाइरी को जवाब तत्काल होते हैं।**
II. **The associated cost is economical with efficient utilization of resources.**
संसाधनों के कुशल उपयोग के साथ संबद्ध लागत किफाती होती ही।
III. **The database is always up-to date.**
डेटाबेस हमेशा अप-टू-डेट रहता है।

(a) Only II (b) I and II
(c) Only II (d) I and II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : In data warehousing, an OLTP (Online Transaction Processing) System is a source system designed to handle day to day operational, real time transactions like order entry and banking, to using on fast data insertions updates and deletions at a granular, record level.
In OLTP-
- Responses to the user inquiry are immediate
- The associated cost is economical with efficient utilization of resources.

375. **In context of 2-phase locking protocol in DBMS, a transaction issues lock requests in _____ phase and unlock requests in _____ phase.**
डाटाबेस में 2-फेज लॉकिंग के संदर्भ में, एक ट्रांजेक्शन लॉक निवेदन _____ फेज में तथा अनलॉक निवेदन _____फेज में जारी करता है।

(a) Growing, Shrinking/ग्रोइंग, श्रिंकिंग
(b) Shared, Exclusive/शेअर्ड, एक्सक्लूसिव
(c) Wait, Wound/वेट, वुंड
(d) Explicit, Implicit/एक्सप्लिसिट, इम्प्लिसिट

HTET PGT C.S. 2022

Ans. (a) : In a two-phase locking (2PL) protocol, a transaction issues lock requests in the growing phase and unlock requests in the shrinking phase. The growing phase is when a transaction may acquire new locks but no locks can be released conversely the shrinking phase allows a transaction to release locks but not acquire any new ones.

376. **If one attribute is determinant of second, which in turn is determinant of third, then the relation cannot be :**
यदि एक एट्रिब्यूट दूसरे का निर्धारक है, जो बदले में तीसरे का निर्धारक है, तो यह संबंध क्या नहीं हो सकता?

(a) Well-structured/अच्छी तरह से संरचित
(b) 1NF
(c) 2NF
(d) 3NF

HTET PGT C.S. 2021

Ans. (d) : The relation cannot be in 3NF (Third Normal Form) in 3NF, every non-prime attribute is non transitively dependent on every superkey. Since the second attribute is determinant of the third, and the first attribute is determinant of the second, this violate the non-transitive dependency ruse of 3NF.

377. **A function that has no partial functional dependencies is in _____ form.**
एक फंक्शन जिसमें कोई आंशिक कार्यात्मक निर्भरता नहीं है _____के रूप में है।

(a) 4NF
(b) 2NF
(c) 3NF
(d) None of the above/उपरोक्त मे से कोई नहीं

HTET PGT C.S. 2021

Ans. (b) : A table with no partial functional dependencies is in 2NF (Second Normal Form). The definition of 2NF states that a relation must be in 1NF and have no partial dependencies meaning every non-prime attribute must be fully functionally dependent on the entire candidate key.

378. **A functional dependency is a relationship between or among :**
कार्यात्मक निर्भरता (फंक्शनल डिपेंडेंसी) निम्नलिखित में से किसके बीच में एक संबंध है?

(a) Table/टेबल (b) Row/पंक्ति
(c) Relation/संबंध (d) Attribute/गुण

HTET PGT C.S. 2021

Ans. (d) : A functional dependency is a relationship between attributes in the relational model, attributes are the columns in a table and functional dependencies describe how the value of certain attributes determine the values of other attributes.

379. **Which of the following refers to the correctness and completeness of the data in a database ?**
निम्नलिखित में से कौन एक डेटाबेस में डेटा की शुद्धता और पूर्णता को संदर्भित करता है?

(a) Data security/डेटा सिक्योरिटी
(b) Data integrity/डेटा इंटीग्रिटी
(c) Data constraint/डेटा कन्स्ट्रैन्ट
(d) Data independence/डेटा इंडिपेंडेंस

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Ans. (b) : Key Points–
- Data integrity refers to the correctness and completeness of data in a database.
- It ensures that the data is accurate, consistent and reliable over it's entire life cycle.
- Maintaining data integrity is crucial for making informed business decisions and enduring compliance with regulations.

380. What is ACID properties of DBMS Transactions?
DBMS ट्रांसैक्शन का ACID गुण क्या है?
(a) Atomicity, Consistency, Isolation, Database
(b) Atomicity, Consistency, Isolation, Durability
(c) Atomicity, Consistency, Inconsistent, Durability
(d) Automatically, Concurrency, Isolation, Durability

HTET PGT C.S. 2020

Ans. (b) : ACID Properties– A transaction in a database system must maintain automicity consistency, isolation and Durability commonly known as acid properties in order to ensure accuracy, completeness, and data integrity.
Automicity– This property states that a transaction must be treated as an atomic unit, that is , either all its operation are executed or none.
Consistency– The database must remain in a consistent state after any transaction no transaction should have any adverse effect on the data residing in the database.
Isolation– In a database system where more than one transaction is being executed simultaneously and in parallel, the property of isolation in that all the transactions will be carried out and executed as if it is the only transaction in the system.
Durability– The database should be durable enough to hold all its latest updates even if the system fails or restarts.
Therefore, option 2 is correct.

381. Let transaction T1 has obtained a shared mode lock S on data item Q and transaction T2 has obtained an exclusive-mode lock X on data item R.
माना ट्रांजेक्शन T1 ने डेटा आइटम Q पर साझा लॉक S प्राप्त कर लिया है तथा ट्रांजेक्शन T2 ने डेटा आइटम R पर विशिष्ट-मोड लॉक X प्राप्त कर लिया है। निम्न कथनों पर विचार करें:

Consider the following statements :
I : T1 can read Q but cannot write Q.
T1, को Q पढ़ सकता है परन्तु को लिख नही सकता।
II : T2 can read R but cannot write R.
T2, R को पढ़ सकता है परन्तु R को लिख नहीं सकता।

Which of the above statement(s) is/are valid ?
उपरोक्त में से उचित कथन है/हैं:
(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/I तथा II दोनों
(d) Neither I nor II/न तो I और न ही II

HTET PGT C.S. 2019

Ans. (a) : A lock is a variable associated with data item that describes the status of the data item with respect possible operations that can be applied to it. They synchronize the access by concurrent transactions to the database items. It is required in this protocol that all the data items must be accessed in a mutually exclusive manner
Shared Lock– If a transaction T1 has obtained a shared mode lock (denoted by s) on time Q, then T1 can read, but cannot write Q.
Exclusive Lock– If a transaction T1 has obtained an exclusive mode lock, then it can both read and write on the item on which it in operating.

382. In context of database, let T1 and T2 are two concurrent transactions. Consider the following sequence of operations on data
DBMS के सन्दर्भ में, माना T1 तथा T2 दो संगामी ट्रांजेक्शन हैं। डेटा X पर ऑपरेशन्स के निम्न अनुक्रम पर विचार करें:
X :
T1 : READ(X) T2 : READ(X)
T1 : WRITE(X) T2 : WRITE(X)
This is a _____ problem.
यह एक _____ समस्या है।
(a) Dirty Read/डर्टी रीड
(b) Lost Update/लॉस्ट अपडेट
(c) Incorrect Summary/इनकरेक्ट समरी
(d) Unrepeatable Read/अनरिपीटेबल रीड

HTET PGT C.S. 2019

Ans. (b) : Lost update– In the lost update problem, an update done to a data item by a transaction is lost as it is overwritten by the update done by another transaction.
Example–

T1	T2
read_item (X) X = X + N	 X = X + 10 write_item (X)

In the above example, transaction 2 changes the value of X but it will get overwritten by the write commit by transaction 1 on X (not shown in the image above). Therefore the update done by transaction 2 will be lost. Basically, the write commit done by the lost transaction will overwrite all previous write commits.

383. What does the ACID property stand for in the context of database transactions?
(a) Atomicity, consistency, isolation, durability
(b) Accuracy, concurrency, integrity, durability
(c) Accuracy, consistency, isolation, durability
(d) Atomicity, consistency, independence, durability
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : ACID Property Stands for Atomicity, Consistency, Isolation & durability.

384. The basic properties of transaction is called
लेन-देन का मूलभूत गुण कहलाता है
(a) ACID Property/ए.सी.आई.डी. गुण
(b) RAW Property/आर.ए.डब्ल्यू. गुण

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(c) BEEM Property/बी.ई.ई.एम. गुण
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (a) : The basic properties of a transaction are known as the ACID properties: Atomicity, Consistency, Isolation, and Durability. These properties ensures the correctness and reliability of database transactions.

v. Processing and Query Optimization, Concurrency Control, and Recovery Techniques

385. Two concurrent transaction T_1 & T_2 reads same data value x, both updates and write back x to database. This may cause _____problem.
(a) Lost update problem
(b) Direct read problem
(c) Incorrect summary problem
(d) Unrepeatable read problem

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : The main problem that can occur when two concurrent transaction (T_1 and T_2) read the same data value (x), both update if, and write it back to the database is the Lost Update Problem.
Lost Update Problem– Two transactions are accessing and modifying the same data at the same time. If one transaction reads the data, updates it, and then writes it back to the database, but before it completes, another transaction reads the original value of the data, updates it, and writes it back, the first transactions updates will be overwritten by the second transaction, causing the first transactions changes to be lost. This essentially means that the database will reflect the second transaction's changes and the first transaction's change and the first transaction's changes will be lost.

386. Commit ( ) is required to be used after the execution of certain queries in Python-MySQL connectivity applications. Identify one such MySQL command out of the following options:
Python-MySQL कनेक्टिविटी (connectivity) एप्लीकेशन में कुछ प्रश्नों के निष्पादन के बाद commit ( ) का प्रयोग किया जाना आपेक्षित है। निम्नलिखित विकल्पों में से ऐसी एक MySQL कमांड की पहचान कीजिए:
(a) CREATE (b) INSERT
(c) SELECT (d) DROP

KVS PGT 23.03.2023 (Shift-II)

Ans. (b) : In python MySQL connectivity applications the commit () method is required to be used after the exaction of queries that modify the database, such as INSERT, UPDATE, DELETE and DROP. This is because these queries change the state of the database, and the changes are not reflected immediately
INSERT is the MySQL Command that requires a commit () after its execution The INSERT command is used to add new rows of data into a table. After executing an INSERT query, calling commit () ensures that the inserted data is saved permanently in the database.

387. Concurrency control in RDBMS is important for which of the following reasons ?
आर डी बी एम एस में समापात (कनकरेंसी) नियंत्रण निम्नलिखित किस कारण से महत्वपूर्ण है?
(a) To ensure data integrity when reads occur to the database in a multi-user environment.
मल्टीयूजर वातावरण में डेटाबेस में 'रीड्स' होने पर डेटा इंटेग्रिटी सुनिश्चित करना।
(b) To ensure data integrity when updates occur to the database in a single-user environment.
एकल प्रयोक्ता वातावरण में डेटाबेस में अपडेट होने पर डेटा इंटेग्रिटी सुनिश्चित करना।
(c) To ensure data integrity when updates occur to the database in a multi-user environment.
मल्टीयूजर वातावरण में डेटाबेस में अपडेट होने पर डेटा इंटेग्रिटी सुनिश्चित करना ।
(d) To ensure data integrity when reads occur to the database in a single-user environment.
एकल प्रयोक्ता वातावरण में डेटाबेस में 'रीड्स' होने पर डेटा इंटेग्रिटी सुनिश्चित करना।

KVS PGT C.S. 2017

Ans. (c) : Concurrency control concept comes under the Transaction in database management System. it is a procedure in DBMS which helps us for the management of two simultaneous processes to execute without conflicts between each other, these conflicts occur in multiuser systems. Concurrency control in RDBMS is important reasons to ensure data integrity when updates occur to the data base in a multi-user environment.

388. Which of the following statements is NOT true with respect to database normalization?
डेटाबेस सामान्यीकरण के संबंध में निम्न में से कौन सा कथन सत्य नहीं है ?
(a) A 4NF table is more immune against logical inconsistencies than a 3NF table.
एक 4NF सारिणी तार्किक विसंगतियों के प्रति 3NF सारिणी से अधिक प्रतिरोधी होती है ।
(b) A 3NF table will have fewer anomalies than a 2NF. table./एक 3NF सारिणी में, एक 2NF सारिणी की तुलना में कमतर असंगतियाँ होती हैं ।
(c) A 3NF table is more vulnerable than a 2NF table./एक 3NF सारिणी (table) 2NF सारिणी से अधिक सुभेध होती है ।
(d) A database is said to be in 3NF if all its tables are in 3NF /एक डेटाबेस 3NF में है, ऐसा कहा जाता है यदि इसकी सभी सारिणी 3NF में हो ।

KVS PGT C.S. 2013

Ans. (a) : A 4NF table is more immune against logical inconsistencies than a 3NF table Above statement is Not true about respect to database normalization.

389. Which normal form ensures that every non-prime attribute in a table is fully functionally dependent on the primary key?
कौन-सा सामान्य रूप यह सुनिश्चित करता है कि तालिका में प्रत्येक गैर-प्रधान विशेषता प्राथमिक कुंजी पर पूरी तरह कार्यात्मक रूप से निर्भर है?
(a) First Normal Form (1NF)
पहला सामान्य रूप (1 NF)

Sarkari Resource

(b) Second Normal Form (2NF)
दूसरा सामान्य रूप (2 NF)
(c) Third Normal Form (3NF)
तीसरा सामान्य रूप (3 NF)
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : The rule of second normal form (2NF) ensure that every non-prime attribute in a table is fully functionally dependent on the primary key.
In 2NF, the table must be in first normal form 1(NF) and all partial dependencies must be eliminated.

390. For designing a normal RDBMS, which of the following normal forms is considered adequate?
सामान्य RDBMS को डिजाइन करने के लिए निम्नलिखित में से कौन-सा सामान्य रूप पर्याप्त माना जाता है?
(a) 4NF
(b) 3NF
(c) 2NF
(d) More than one of the above
/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : 3NF is sufficient for designing a general relational database, because most third normal form tables are free from deletion, update, and insertion anomalies.

391. Which of the DBMS component ensures that concurrent execution of multiple operations on the database results into a consistent database state?
(a) Logs
(b) Buffer manager
(c) File manager
(d) Transaction processing system

UGC NET C.S. December 2020

Ans. (d) :Transaction Processing System :- A transaction processing system (TPS) is a type of information system that collect, stores, modifies and retrieves the data transactions of an enterprise.
In order to qualify as a TPS, transactions made by the system must pass the ACID test. The ACID tests refers to the following four requisites : -
1) Atomicity
2) Consistency
3) Isolation
4) Durability
TPS provides, Rapid response, continuous availability data integrity, Ease of use.
ex - Online transaction processing (OLTP), Real time transaction.

392. What of the following scenario may lead to an irrecoverable error in a database system?
(a) A transaction writes a data item after it is read by an uncommitted transaction
(b) A transaction reads a data item after it is read by an uncommitted transaction
(c) A transaction reads a data item after it is written by a committed transactions
(d) A transaction reads a data item after it is written by an uncommitted transaction

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Ans. (d) : In database system, reading a data item after it has been written by an uncommitted transaction is an example of a phenomenon known as 'dirty read' or 'uncommitted dependency'. This can lead to inconsistent or incorrect result if the uncommitted transaction is later rolled back.

393. Consider the following statements:
(A) A database design is in BCNF if each member of the set relation schemas that constitutes the design is in BCNF
(B) A BCNF schema can have transitive dependency
(C) It is always possible to obtain a 3NF design without sacrificing a lossless join
(D) There are multivalve dependencies in 4NF
(a) A, B and C only (b) B, C and D only
(c) A, B and D only (d) A, C and D only

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Ans. (d) : A BCNF (Boyce-Codd Normal form) schema, cannot have any non-trivial dependencies other than superkeys. Transitive dependencies are a form of non-trivial dependencies. So, a BCNF schema cannot have transitive dependency. Hence, statement B is false while statement A, C and D are true.

394. Fifth normal form is concerned with :
(a) Join Dependency
(b) Domain-Key
(c) Multivalued dependency
(d) Functional dependency

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Ans. (a) : Fifth Normal Form (5NF) is primarily concerned with "Join Dependency". It deals with situations where there are multiple overlapping schema. In 5NF, the goal is to eliminate certain types of redundancies that arise from complex relationship and dependencies involving multiple tables in a database.

395. Consider a schema R(MNPQ) and functional dependencies M $\rightarrow$ N, P $\rightarrow$ Q. Then the decomposition of R into R_1 (MN) and R_2(PQ) is _____
(a) Dependency preserving but not lossless join
(b) Dependency preserving and lossless join
(c) Lossless join but not dependency preserving
(d) Neither dependency preserving nor lossless join

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) R(MNPQ) and functional dependencies M $\rightarrow$N, P $\rightarrow$ Q.
Then the decomposition of R into R1 (MN) & R2 (PQ) here clearly the dependencies are preserved but they are not lossless as both R1 & R2 cannot be formed back again as they do not have a common attribute.

396. If every non-key attribute is functionally dependent on the primary key, then the relation is in _____
(a) First normal form
(b) Second normal form

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(c) Third normal form
(d) Fourth normal form

UGC NET C.S. November-2017 (Paper-III)

Ans : (b) 2NF is a relation R is in second normal from (2NF) if and only if it is in 1NF and every non key attribute is fully dependent on primary key ex : relation R(xyz) with functional dependencies (x →y, y →z, x →z). The relation is in 2NF but not in 3NF because of every non key attribute is transitively dependent on the primary key. Here {X} will be candidate key.

397. Consider a relation R (A, B, C, D, E, F, G, H), where each attribute is atomic, and following functional dependencies exist

CH → G
A → BC
B → CFH
E → A
F → EG

The relation R is _____

(a) In 1NF but not in 2NF
(b) In 2NF but not in 3NF
(c) In 3NF but not in BCNF
(d) In BCNF

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Ans : (a) A+ → All attribute except (d)
similarly for other keys we can find closure, but (d) can't be derived from any key & it must be added to all keys to be derived from. That's why relation is in 1NF, Since there is no partial dependency, So this relation is not in 2NF.

398. Select the 'False' statement from the following statements about Normal forms :

(a) Lossless preserving decomposition into 3NF is always possible
(b) Lossless preserving decomposition into BCNF is always possible
(c) Any Relation with two attributes is in BCNF
(d) BCNF is stronger than 3NF

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Ans : (c) A BCNF decomposition which is both lossless as well as dependency preserving at the same time is not always possible.
Example: Relation R(a, b, c) with FDs$(ab \to c,\ c \to b)$ cannot be preserved when decomposition is done.

399. The Relation
Vendor Order (V_no, V_ord_no, V_name, Qty_sup, unit_price)
is in 2NF because :

(a) Non_key attribute V_name is dependent on V_no which is part of composite key
(b) Non_key attribute V_name is dependent on Qty_sup
(c) Key attribute Qty_sup is dependent on primary_key unit price
(d) Key attribute V_ord_no is dependent on primary_key unit price

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Ans : (a) Option (a) is which it is clearly mentioned that partial dependency exists is against 2NF principle.
Question seems incomplete, because there is nothing said about functional dependency.

400. Consider the schema
R = {S, T, U, V]
and the dependencies
S → T, T → U, U → V and V → S
If R = (R_1 and R_2) be a decomposition such that $R_1 \cap R_2 = \phi$ then the decomposition is

(a) Not in 2 NF
(b) In 2 NF but not in 3 NF
(c) In 3 NF but not in 2 NF
(d) In both 2 NF and 3 NF

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Ans. (d) : Because, $R_1 \cap R_2 \neq \phi$. This makes the decomposition lossless join, as all the attributes are keys, $R_1 \cap R_2$ will be a key of the decomposed relation Now, even the original relation R is in 3NF (Even BCNF) as all the attributes are prime attributes (in fact attribute is a candidate key) Hence, any decomposition will also be in 3NF and 2NF, So the answer is option (d).

401. Which of the following is false ?

(a) Every binary relation is never be in BCNF
(b) Every BCNF relation is in 3NF
(c) 1 NF, 2 NF, 3 NF and BCNF are based on functional dependencies
(d) Multivalued dependency (MVD) is a special case of join dependency (JD)

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Ans : (a) Every binary relation (a relation with only 2 attributes) is always in BCNF. Because if both attributes form the primary key, the relation is in BCNF. If one of the attribute is a primary key, the other must be determined by it and thus in BCNF.

402. Which of the following statement(s) is/are incorrect?

(1) Mapping the co-ordinates of the points and lines that form the picture into the appropriate co-ordinates on the device or workstation is known as viewing transformation
(2) The right-handed Cartesian co-ordinates system in whose co-ordinates we describe the picture is known as world co-ordinate system
(3) The co-ordinate that corresponds to the device or workstation where the image is to be displayed is known as physical device co-ordinate system
(4) Left - handed co-ordinate system in which the display area of the virtual display device corresponds to the unit $(|x|)$ square whose lower left-hand corner is at the origin of the co-ordinate system, is known as normalized device co-ordinate system

(a) (1) only (b) (1) and (2)
(c) (3) only (d) (4) only

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Ans : (d) Normalized device co-ordinate system make up a coordinate system that describes positions on a virtual plotting device. The left corresponds to (0,0) & the upper right corner corresponds to (1,1)
NDCs can be used when you want to position text, lines, markers or polygons anywhere on the plotting device.

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403. **Consider the following relational schemas for a library database :**
Book (Title, Author, Catalog–No, Publisher, Year, Price)
Collection (Title, Author, Catalog–no) with the following functional dependencies :
I. Title, Author → Catalog–no
II. Catalog–no → Title, Author, Publisher, Year
III. Publisher, Title, Year → Price
Assume (Author, Title) is the key for both schemas. Which one of the following is true?
(a) Both Book and Collection are in BCNF.
(b) Both Book and Collection are in 3 NF.
(c) Book is in 2NF and Collection in 3 NF.
(d) Both Book and Collection are in 2 NF.

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Ans. (c) : Because, 2NF says that all the non-key attributes should depend the keys, partial dependency (an attribute is defined by a part of a composite key) is not allowed 3NF says that any dependency between non-key attribute should be cleared.
BCNF says that all determinants in a table is key for every non trivial (no overlapping attributes) function. In the table book, (Title Author) identifies catalog-no. Which if turn identifies publisher and year further (publisher, Title, Year) identifies price. Now here the dependents are partially dependent on the key attributes on left hand side of (R). So the table book is in 2NF. In the table collection, (Title, Author) determines catalog-no., the only non-key attribute. So it is in 3NF as well as BCNF.

404. **Which of the following statements is false?**
(a) Any relation with two attributes is in BCNF.
(b) A relation in which every key has only one attribute is in 2NF.
(c) A prime attribute can be transitively dependent on a key in 3NF relation.
(d) A prime attribute can be transitively dependent on a key in BCNF relation.

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Ans : (d) Because, A relation is in BCNF, if and only if, every determinant is a candidate key i.e. the left hand side of every non trivial dependency is a super key or there is a trivial dependency or both. After 3NF, there may arise a situation that two candidate keys are having one attribute in common (overlapping) and this overlapping must be cleared to satisfy BCNF. A prime attribute (an attribute that is a part of a candidate key) can on depend on a super key. So a prime attribute cannot be transitively dependent on a key in BCNF relation. So the false statement is option (d).

405. **The best normal form of relation scheme R(A, B, C, D) along with the set of functional dependencies F = { AB → C, AB → D, C → A, D →B} is :**
(a) Boyce-Codd Normal form
(b) Third Normal form
(c) Second Normal form
(d) First Normal form

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because, keys are AB, CD, BC, AD
Here A, B, C, D all are prime attributes
So, no partial or transitive dependency are there
So it in 3NF
Hence the answer is (b)

406. **Identify the minimal key for relational scheme R(A, B, C, D, E) with functional dependencies F = {A → B, B → C, AC → D}**
(a) A (b) AE
(c) BE (d) CE

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because, {A} + = {A, B, C, D}
Here A can not be a key because we can't get attribute E from here.
{AE}= {A, B, C, D, E}
Here AE is the minimal key we can't remove anything from it
Hence the correct choice is (b)

407. **Which of the following concurrency protocol ensures both conflict serializability and freedom from deadlock?**
(1) z - phase Locking
(2) Time stamp - ordering
(a) Both (1) and (2) (b) (1) only
(c) (2) only (d) Neither (1) nor (2)

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : In basic two phase locking there is a chance for deadlock.
Conservative 2 phase is deadlock free
So answer is (2) only

408. **If a relation schema is in 2NF, then it is definitely in:**
यदि रिलेशन स्कीमा 2NF में है, तो यह निश्चित रूप से _____ में भी है।
(a) 1NF (b) 4NF
(c) 3NF (d) BCNF

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : If a relation schema is in 2NF (Second Normal Form), it is definitely in 1NF (First Normal Form), because 2NF is defined as being in 1NF with the added condition that all non-key attributes are fully functionally dependent on the primary key.

409. **The left-hand side and the right-hand sides of a functional dependency are sometimes called____ and ____ respectively.**
एक फंक्शनल डिपेंडेंसी के बाएँ तरफ और दाएँ तरफ को कभी-कभी क्रमशः ____ और ____कहा जाता है।
(a) Domain; attributes/डोमेन, एट्रीब्यूट्स
(b) Determinant; dependent/डेटरमिनेंट, डिपेनडेंट
(c) Range; table/रेंज, टेबल
(d) Tuple; relation/ट्यूपल; रिलेशन

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : In a functional dependency (X→Y), the left-hand side (X) is called the Determinant, and the right-hand side (Y) is called the Dependent. This terminology signifies that the determinant's attributes determine or uniquely establish the values of the dependent attributes which, in turn, are dependent on the determinant.

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410. In concurrency control, phantom problem may occur, when–

(a) Records are inserted (b) Records are deleted
(c) Records are modified (d) Records are indexed

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Ans. (a) : A phantom problem happens when a transaction reads a set of records based on a certain condition, but when it tries to read the same set again, new records that weren't there before appear, making it seem like "phantom" records have been added, which usually occurs due to another transaction inserting new data that meets the query criteria.

411. Which of the following statements are true about the sets.

A. $0 \in \emptyset$
B. $\emptyset \in \{0\}$
C. $\emptyset \in \{\emptyset\}$
D. $\{\emptyset\} \in \{\emptyset\}$
E. $\{\emptyset\} \subset \{\emptyset,\{\emptyset\}\}$

Choose the correct answer from the options given below :

(a) A, B, C, D, and E (b) A, B , C and E only
(c) A, C only (d) C and E only

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Ans. (d) : Let's analyze each statement-

(A) $0 \in \emptyset$, false. The empty set $\emptyset$ contains no elements, so 0 cannot be an element of it.

(B) $\emptyset \in \{0\}$ False. $\{0\}$ which contains only the elements 0. The empty set $\emptyset$ is not an element of $\{0\}$.

(C) $\emptyset \in \{\emptyset\}$ True, The Set $\{\emptyset\}$ contains exactly one element, which is the empty set $\emptyset$. Since $\emptyset$ is an element of this set, the statement is true.

(D) $\{\emptyset\} \in \{\emptyset\}$, false. Because $\{\emptyset\}$is a set containing $\emptyset$, but it does not contain itself as an element.

(E) $\{\emptyset\} \subset \{\emptyset,\{\emptyset\}\}$, True because $\{\emptyset\}$ is a subset of $\{\emptyset,\{\emptyset\}\}$ Since all elements of $\{\emptyset\}$ are also in $\{\emptyset,\{\emptyset\}\}$.

Since C and E are true, the correct option is (d) C and E only.

412. Which of the followings is/are FALSE statement?

A. An all key relation is always in BCNF since it has no FDs.
B. A relation that is not in 4NF due to nontrivial MVD must be decomposed to convert it into a set of relations in 4NF.
C. The decomposition removes the redundancy by the MVD.
D. 3NF is stronger than BCNF.

Choose the stronger than BCNF.

(a) A only (b) A, B only
(c) D only (d) C only

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Ans. (c) :

- An all-key relation, where all attributes from the primary key, is always in BCNF because it has no other determinants besides the entire key itself.
- A relation falling 4NF due to a non-trivial's MVD must be decomposed to eliminate the redundancy and achieve 4NF.
- Decomposing a relation to satisfy 4NF eliminates redundancy from multi-valued dependencies.
- BCNF is stronger than 3NF, addressing more redundancy issues.

413. Consider the following statement regarding relational database model :

(1) NULL values can be used to opt a tuple out enforcement of a foreign key
(2) Suppose that table T has only one candidate key. If Q is in 3NF, then it is also in BCNF
(3) The difference between the project operator (II) in relational algebra and the SELECT keyword in SQL is that if the resulting table/set has more than one occurrences of the same tuple, then II will return only one of them, while SQL SELECT will return all

One can determine that :

(a) (1) and (2) are true
(b) (1) and (3) are true
(c) (2) and (3) are true
(d) (1), (2) and (3) are true

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Ans : (d) **(i)** In Relational Database Model:
(a) NULL values can be used to opt a tuple out of enforcement of a foreign key correct.
(b) Suppose that table T has only one candidate key. If Q is in 3NF, then it is also in BCNF correct.
(c) The difference between the project operator (II) in relational algebra and the SELECT keyword in SQL is that if the resulting table/set has more than one occurrences of the same tuple, then (II) will return only one of them, while SQL SELECT will return al correct.

414. If every non-key attribute is fully dependent on the primary key, then the relation will be in
अगर प्रत्येक नॉन-की ऐट्रिब्यूट पूरी तरह से प्राइमरी की पर निर्भर रहता है, तो रिलेशन होगा

(a) First Normal Form (1NF)
प्रथम नॉर्मल फॉर्म (1NF) में
(b) Second Normal Form (2NF)
द्वितीय नॉर्मल फॉर्म (2NF) में
(c) Third Normal Form (3NF)
तृतीय नॉर्मल फॉर्म (2NF) में
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b): If a relation is in 1NF then it will be in 2NF and all non-key attributes are completely functionally dependent on the primary key.

415. Which of the following is NOT a variant of time stamp-based based method of concurrency control in database management system?
निम्नलिखित मे से कौन सा डेटाबेस मैनेजमैंट सिस्टम में कंकर्रेंसी कण्ट्रोल की टाइम स्टैम्प-आधारित पद्धति का एक प्रकार नहीं है/हैं?

(a) Total time stamp ordering
टोटल टाइम स्टैम्प ऑर्डरिंग

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(b) Partial time stamp ordering
पार्शियल टाइम स्टैम्प ऑर्डरिंग
(c) Multi version time stamp ordering
मल्टी वर्जन टाइम स्टैम्प ऑर्डरिंग
(d) Multilevel time stamp ordering
मल्टी लेवल टाइम स्टैम्प ऑर्डरिंग

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : Multilevel time stamp ordering is not variant of timestamp-based concurrency control in database management system.

Timestamp-Based Variants are–

- Total time ordering ensures all transactions are ordered strictly by their timestamps. and is a standard protocol.
- Partial timestamp ordering allows limited application or timestamp ordering and while rare, is referenced in DBMS literature.
- Multi version time stamp ordering uses multiple versions of data to allow greater concurrency and is widely studied.

416. A server crashes on the average once in 30 days, that is the Mean Time Between Failures (MTBF) is 30 days. When this happens, it takes 12 hours to reboot it, that is, the Mean Time to Repair (MTTR) is 12 hours. The availability of server with these reliability data values is approximately :

(a) 96.3% (b) 97.3%
(c) 98.3% (d) 99.3%

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Ans : (c) If we let A represent availability, then the simplest formula for availability
A = uptime/(uptime + downtime)
MTBF is mean time between failures
A = MTTR is mean time to repair
A = MTBF/(MTBF + MTTR)
So MTBF = 30 days = 30 × 24 = 720 hr
A = 720/(720 + 12)
= 98.3%

417. Consider the relations R(A, B) and S(B, C) and the following four relational algebra queries over R and S :

I. $\Pi_{A.B}$ **(R ⋈ S)**

II. $R \bowtie \Pi_{B}(S)$

III. $R \cap (A(R) \times \Pi_{B}(S))$

IV. $\Pi_{A.R.B}$ **(R × S) where R.B refers to the column B in table R.**

One can determine that :

(a) I, III and IV are the same query
(b) II, III and IV are the same query
(c) I, II and IV are the same query
(d) I, II and III are the same query

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Ans:(d) Let us take an example as R = {(10, 2) (20, 2)}
S = {(2, 30); (3, 40)}

(I) R ⋈ S = {(20, 2, 30)} so πA, B R ⋈ S = {(20, 2)}

(II) πB(S) = {(2), (3)}, so R ⋈ π B (S) = {(20)2)}

(III) πA (R) × πB(S) = {10, 20} × {2, 3} ={(10, 2), (10, 3), (20, 2), (20, 3)} R∩πA(R) × πB(S) = {(20, 2)}

(IV) R × S = {(10, 1, 2, 30), (10, 1, 3, 40), (20, 2, 2, 30), (20, 2, 3, 40)} πA, R, B R × S = {(10, 2), (10, 3), (20, 2), (20, 3)}

418. Which of the following statements is TRUE?

D_1 : The decomposition of the schema R(A, B, C) into R_1(A, B) and R_2 (A, C) is always lossless.

D_2 : The decomposition of the schema R(A, B, C, D, E) having AD →B, C →DE, B → AE and AE →C, into R_1 (A, B, D) and R_2 (A, C, D, E) is lossless.

(a) Both D_1 and D_2 (b) Neither D_1 nor D_2
(c) Only D_1 (d) Only D_2

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Ans : (d) Only D_2 Since AD is key and it present in both the tables not D_1
Since fd's not given
If we take B →A & C → A then it is lossy since no common attribute contain key from one of the table.

419. In context of deadlock prevention in DBMS, wait-die and wound-wait schemes are ..... techniques.

DBMS में डेडलॉक प्रिवेंशन के संदर्भ में, wait-die तथा wound-wait स्कीम ..... तकनीक है।

(a) Both preemptive/दोनों प्रिम्पटिव
(b) Both non-preemptive/दोनों नॉन-प्रिम्पटिव
(c) Preemptive and non-preemptive respectively
क्रमशः प्रिम्पटिव तथा नॉन-प्रिम्पटिव
(d) Non-preemptive and preemptive respectively
क्रमशः नॉन-प्रिम्पटिव तथा प्रिम्पटिव

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Ans. (d) : Wait-die Scheme– This is a non preemptive technique where a younger transaction must wait for an older transaction to finish if it requests a resource currently held by the older transaction. It the older transaction requests a resources held by the younger transaction, the younger transaction waits.

■ **Wound-wait Scheme–** This is a preemptive technique where a younger transaction can preemptive an older transaction if it needs the resource currently held by the older.

Transaction if the older transaction requests a resource held by the younger transaction, the older transaction is "wounded" (aborted) and must start over.

420. Which RAID level is appropriate for storing the log file of a ticket booking system where a huge database update is required?

(a) Level 0 (b) Level 1
(c) Level 2 (d) Level 3

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Ans. (b) : RAID (Redundant Array of Independent Disks) Level 1 provided mirroring, meaning that data is duplicated across two or more drives. In this configuration, all data written to one drive is also written to another drive in real-time. This redundancy ensures data integrity and availability making it a

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suitable choice for storing critical log file that must be preserved and protected during a huge database update. So option (b) is correct.

421. Given below are two statements:
Which of the following concurrency control protocol ensures both conflict serializability and freedom from deadlock?
Statement I: Two phase locking
Statement II: Timestamp ordering
In the light of the above statements, choose the most appropriate answer from the option given below:
(a) Both Statement I and Statement II are correct
(b) Both Statement I and Statement II are incorrect
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

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Ans. (d) : Two phase locking (2PL) is a concurrency control method that guarantees serializability. the protocol utilizes locks, applied by a transaction to data, which may block (interpreted as signals to stop) other transactions from accessing the same data during the transaction. 2PL may be lead to deadlocks that result from the mutual blocking of two or move transactions. timestamp. Based concurrency control algorithm is a non-lock concurrency control method. In Timestamp based method, deadlock cannot occur as no transaction ever waits. so, the statement I is incorrect but statement II is correct.

422. Which database normal form is considered satisfactory for relational database design ?
रिलेशनल डेटाबेस डिजाइन के लिए कौन सा सामान्य रूप पर्याप्त माना जाता है?
(a) 1 NF (b) 2 NF
(c) 3 NF (d) 4 NF

HTET PGT C.S. 2020

Ans. (c) : The Third Normal Form (3NF) is generally considered satisfactory for relational database design because it eliminates most common anomalies related to data insertion update, and deletion, providing a good balance of data integrity and normal design principles without the complexity of higher normal forms while higher forms like 4NF exists, they handle more specialized causes and are not typically necessary for most standard relational database.

423. In DBMS when a transaction completes its execution it is said to be :
DBMS में जब कोई लेन-देन अपने निष्पादन को पूरा करता है, तो यह कहा जाता है:
(a) Saved (b) Loaded
(c) Rolled (d) Committed

HTET PGT C.S. 2020

Ans. (d) : This state of transaction is achieved when all the transaction related operations have been executed successfully along with the commit operation, data is saved in to the database after the required manipulation in this state. This marks the successful completion of a transaction.

424. In a relational schema R whenever a non-trivial functional dependency X–>A holds in relation R, X is a superkey of relation R. Relation R is in _____.
एक रिलेशनल स्कीमा R में जब भी एक गैर - नगण्य (नॉन-ट्रिविअल) कार्यात्मक निर्भरता (फंक्शनल डिपेन्डेंसी) X->A रिलेशन R में आधारित होती है, तब X रिलेशन R में आधारित होती है, तब X रिलेशन R की एक सुपर-कुंजी है। R _____ में है।
The most suitable choice to fill in the blank is :
रिक्त स्थान भरने के लिए सर्वाधिक उपयुक्त विकल्प है:
(a) BCNF (b) 2NF
(c) 3NF (d) 4NF

HTET PGT C.S. 2019

Ans. (a) :
- It stands for Boyce Codd Normal Form.
- A relation R is in BCNF if whenever a non-trivial functional dependency X→A holds in R, then X is a superkey of R.
- Any relation with two attributes is always in BCNF. Because when a relation contains only two attributes then one attributes determines another and left side of the functional dependency will always be a candidate key in that case.
- BCNF is always lossless but not always dependency preserving.

425. A relation in which every non-key attribute is fully functionally dependent on the primary key and which has no transitive dependencies is in
(a) 1NF (b) 2NF
(c) 3NF (d) BCNF
(e) Not attempted

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HTET PGT C.S. 2018

Ans. (c) :
- The third normal form is based on the concept of transitive dependency A Functional dependency x→y in a relation schema R is a transitive dependency if there exits a set of attributed z in R that is neither a candidate key nor a subset of any key of R
- A relation is in 3NF if it satisfies 2NF and no prime attribute of R is transitively dependent on the primary key.
- If X → A is a functional dependency, then a should be a prime attribute or X should be a candidate key.
- Lossless join and dependency preservation is always possible in 3NF. 3NF decomposition is always lossless join and dependency preserving.

426. Which of the following normal form is based on the concept of full functional dependency on each candidate key of the relation?
निम्न में से कौन सा नार्मल फार्म रिलेशन की प्रत्येक कैन्डीडेट की पूर्ण फंक्शनल निर्भरता की अवधारणा पर आधारित है?
(a) 1 NF (b) 2 NF
(c) 3 NF (d) None of the above

UKPSC PGT C.S. Asstt. Prof. 2017

Sarkari Resource

Ans. (b) : The normal form based on the concept of full functional dependency on each. candidate key of the relation is second Normal form (2NF). 2NF ensures that every non-key attribute in a table depends on the entire primary key, not just part of it.

427. The _____ information is used to limit the amount of recovery operations to be done following a system crash resulting in the lose of volatile storage.
_____ सूचना रिकवरी ऑपरेशन्स को पाबंदी लगाने में प्रयोग होती है जिससे कि सिस्टम क्रैश हो जाता है
(a) Log/लॉग
(b) Data file/डाटा फाइल
(c) Check point/चेक प्वाइंट
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans. (c) : A checkpoint is used in database and system recovery to limit the amount of recovery operations needed after a crash that results in loss of volatile storage By marking a specific point in time, the system only needs to consider transactions and operations occurring after the last checkpoint during recovery.

428. Consider the given functional dependencies:
$A \rightarrow B$
$BC \rightarrow DE$
$AEF \rightarrow G$
Which of the following is true?
(a) Functional dependency ACF→DG implied by the set
(b) Functional dependency ACF→DG cannot be implied by the set
(c) Functional dependency AB→G implied by the set
(d) Both (B) and (C)

CGPSC C.S. 2020 Set-A

Ans. (a) :
(1) From A→B, ACF gives B so we have A,C, F, B.
(2) From BC→DE, having B and c gives D and E (how we have A, B, C, D, E, F).
(3) From AEF → G, having A, E, F gives G.
So ACF implies both D and G i.e. ACF → DF is implied.
AB → G is not implied because AB alone does not gives E or F so (c) is false.
Hence the correct choice is (a).

429. A relation will be in 2NF, if we:
(a) Remove repeating groups
(b) Remove partial dependency
(c) Remove transitive dependency
(d) Have overlapping candidate key

CGPSC C.S. 2020 Set-A

Ans. (b) : 1NF (First Normal Form)– Achieved when repeating groups are removed (so option a relates to 1NF).
2NF (Second Normal Form)– Achieved when the relation is in 1NF and there are 10 partial dependencies.
3NF (Third Normal Form)– Achieved when the relation is in 2NF and there no transitive dependencies (So, option creates to 3NF).
Overlapping candidate keys (d) : Not directly related to 2NF.
So, 2NF = 1NF + No Partial Dependency.

430. BCNF can be achieved from 3 NF by removing:
(a) Repeating groups
(b) Partial dependencies
(c) Transitive dependencies
(d) Overlapping dependencies

CGPSC C.S. 2020 Set-A

Ans. (d) :
1NF → 2NF : Remove partial dependencies.
2NF → 3NF : Remove transitive dependencies.
3NF → BCNF : Remove overlapping dependencies.
So the correct answer is (d) overlapping dependencies.

431. Which of the following is an altermotive to the log-based recovery schemes in database recovery system?
निम्नलिखित में से कौन सा डेटाबेस रिकवरी सिस्टम में लॉग-आधारित रिकवरी स्कीम्स का एक विकल्प है?
(a) Shadow Paging/शैडो पेजिंग
(b) Defragmentation/डिफ्रैगमेंटेशन
(c) Prepaging/प्रीपेजिंग
(d) Demand Paging/डिमांड पेजिं

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : The correct alternative to log based recovery schemes in a database recovery system is shadow raging shadow paging avoids the use of logs by maintaining a duplicate page table during transactions and switches pointers on commit, providing a non-log-based recovery mechanism.

432. Which of the following statements is/are True regarding some advantages that an object-oriented DBMS (OODBMS) offers over a relational database?
I. An OODBMS avoids the "impedance mismatch" problem.
II. An OODBMS avoids the "phantom" problem.
III. An OODBMS provides higher performance concurrency control than most relational database.
IV. An OODBMS provides faster access to individual data objects once they have been read from disk.
(a) II and III only (b) I and IV only
(c) I, II, and III only (d) I, III and IV only

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) I. An OODBMS avoids the "impedance mismatch" problem. This is true.
II. An OODBMS doesn't avoids the "Phantom" problem.
III. An OODBMS doesn't provides higher performance concurrency control than most relational database since it is distributed
IV. An OODBMS provides faster access to individual data objects once they have been read from disk. This is true.

Sarkari Resource

08. Computer Networks

(i) Network Definition, Topologies, Classifications and Protocol.

1. Which of the following are example of CSMA channel sensing methods?
(A) 1-persistent
(B) 2-persistent
(C) p-persistent
(D) o-persistent
Choose the correct answer from the options given below:
(a) (A), (B) and (D) Only
(b) (A), (C) and (D) Only
(c) (B), (C) and (D) Only
(d) (A), (B) and (C) Only

UGC NET C.S. December-2023

Ans. (b) : CSMA (Carrier Sense Multiple Access) has different access methods- including.
1. 1- persistent CSMA
2. Non- persistent CSMA
3. P - Persistent CSMA
4. Q- persistent CSMA

2. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Find the number of addresses in each subnet
(a) 32 (b) 64
(c) 128 (d) 16

UGC NET C.S. December-2022

Ans. (b) : Number of subnets = 2^n = 1024
Therefore, n = 10. The address given here is a class B, therefore the default mask is/16.10 bits are required for subnets and hence the required subnet mask is / (16 + 10) = / 26 In dotted decimal format, it shall be 192 and therefore, the subnet mask will be 255.255.255.192. The remaining bits must be used for the address in each subnet i.e. there will be 32 – 26 = 6 host bits available for the address component. There will be a total of 2^6 = 64 bits available.
So option (b) is correct answer.

3. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Find the subnet prefix.
(a) 130.56 (b) 136.0
(c) 136.255 (d) 136.56.255

UGC NET C.S. December-2022

Ans. (a) : 1024 = 2^n where n = 10
This is class B IP address and the default mask is 16 therefore we need 10 bits the subnets and now the subnet mask is /(16 + 10) = / 26 as slash notation with IP address prefix.
represents a group of IPV4 addresses Number the represents the group
If the prefix length is / P so P = 26
1. Copy the first P bits P = 5 bits
2. Change the rest of the bits to 0.
So, 130.56.0.0
So The subnet of prefix = 130.56

4. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Find first and last addresses of First subnet.
(a) 130.56.0.0 130.56.254.254
(b) 130.56.0.0 130.56.0.63
(c) 130.0.0.0 130.255.255.255
(d) 130.56.0.0 130.56.255.63

UGC NET C.S. December-2022

Ans. (b) : The first address is subnet 1 = 130.56.0.1
The last address is subnet 1 = 130.560.62 the first address of is the address 130.56.00 and subnet mask/26 connected by AND, and next can be estimated as 10000010 00000010.0.0
This address cannot be assigned, so consider the following 10000010 00111000 00000000 00000001(130.56.0.1
Similarly to the last address that can be assigned before the broadcast address is 130.56.0.62.

5. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Find first and last addresses of last subnet.
(a) 130.56.0.0 130.56.254.255
(b) 130.56.255.0 130.56.255.255
(c) 130.56.0.192 130.255.255.255
(d) 130.56.255.192 130.56.255.255

UGC NET C.S. December-2022

Ans. (d) : /16 'n need 1024= 2^{10} subnets, so borrow 10 bit 'n new prefix /26.
Each subnet size = $2^{(32-26)}$ = 2^6= 64 addresses.
The last (1024th) subnet starts at offset 1023× 64 = 65472 from 130.56.0.0 'n 130.255.192, and its last address (broadcast) is 130.56.255.255. (Usable hosts per subnets = 62)
(where; 'n = number of bits you need to borrow)

6. The IEEE-754 double-precision format to represent floating point number, has a length of _____ bits.
(a) 16 (b) 32
(c) 48 (d) 64

UGC NET C.S. June-2016 (Paper-II)

Ans : (d) Double precision floating point format is a computer number format that occupies 8 bytes (64 bits) in computer memory and represents a wide, dynamic range of values by using a floating point.
- Sign bit (1 bit)
- Exponent (11 bit)
- Significant precision (53 bits)/52 explicitly stored.

Sarkari Resource

7. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Find subnet mask.

(a) 130.255.255.255 (b) 130.56.255.255
(c) 130.56.0.255 (d) 130.56.155.192

UGC NET C.S. December-2022

Ans. (*) : $2^n = 1024$
So n = 10
Address is class B, so the default mask for it / 16. Since subnet and require 10 bits the correct subnet mask is / (16 + 10) = /26. The subnet mask is 255.255.255.192 because it is 192 in decimal format with dots.

8. SSID stands for
SSID किसका संक्षिप्त रूप है?

(a) Secure Service Identifier
(b) Secure Set Independent Device
(c) Service Set Identifier
(d) Service Set Independent Device

LT. Grade C.S. 2018

Ans, (c) : SSID stands for Service Set Identifier. It is the name assigned to a Wi-Fi network, allowing users to identify and connect to it essentially; it's the network's identifier that appears when you search for available Wi-Fi connections on your device.

9. In computer network nodes are
कंप्यूटर नेटवर्क में Nodes है

(a) The computer that originates the data
(b) The computer that routes the data
(c) The computer that terminates the data
(d) All of these/इनमें से सभी

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : In a computer network a node is any device connected to the network that that has an address. A node can be:
- A computer that originates the data (Source)
- A device that routes the data (like a router or switch)
- A computer that terminates the data (Destination)

So, a node can be anydevice that sends, receives or forwards data in the network.

10. A host is connected to a department network which is part of a university network. The university network, in turn, is part of the Internet. The largest network, in which the Ethernet address of the host is unique, is

(a) The department network
(b) The Ethernet
(c) The subnet to which the host belongs
(d) The university network

UGC NET C.S. December-2018 (Paper-II)

Ans. (b) : Ethernet address is basically the MAC address, which is supposed to be unique to a NIC. Thus it is unique over the Internet.

11. Which of the following networks is considered as the first network that initiated the internet ?
निम्नलिखित में से कौन सा नेटवर्क इंटरनेट शुरू करने वाला पहला नेटवर्क माना जाता है?

(a) ARPANET (b) UUCPNET
(c) AOL (d) PTRINET

HTET PGT C.S. 2020

Ans. (a) : The internet has become a basic necessity in economic activities and it is depending its root in social activities too. The past- the globalized era of the annals of human civilization with ICI getting a huge impetus as a consequence of the internet taking rapid, brisk and phenomenal strides, changing in the process the communication scenario the world over.

Arpanet:–
- The internet, then known as Arpanet was brought online in 1969 by the renamed Advanced Research Projects Agency (ARPA)
- Arpanet used a technology, called packet switching, which splits message in to tiny pieces to be further transmitted through different telephone routes to the desired destination.

12. Which of the following term is used for a computer network that typically connects personal computers within a limited geographical area, usually within a single building?
कंप्यूटर नेटवर्क के लिए उपयोग किया जाने वाला शब्द जो आमतौर पर कम्प्यूटर्स को एक सीमित भौगोलिक क्षेत्र या एक इमारत के भीतर जोड़ता है:

(a) LAN (b) WAN
(c) MAN (d) BAN

HTET PGT C.S. 2020

Ans. (a) : A computer network is a group of computers linked together for the purpose of communication and sharing resources the following are the various types of communication networks.
LAN:– Its stands for local area network. It is a computer network that spans a relatively small area usually confined to a single room or within a building or group of buildings.
One LAN can be connected to other LANs over any distance via telephone lines and radio waves.
Hence, it can be concluded from the above table that a computer network that is used within a building is called a LAN.

13. A token ring LAN uses differential Manchester encoding. If the LAN speed is 10 Mbps, what is the baud rate?

(a) 10 Mbaud (b) 20 Mbaud
(c) 5 Mbaud (d) 100 Mbaud

CGPSC C.S. 2020 Set-A

Ans.(b): Step-1 : Recall Differential Manchester Encoding.
- In differential Manchester, each bit period always has a transition in the middle (for clocking)
- Depending on the bit value (0 or 1), there may or may not be an extra transition at the start.
- Thus, each data bit requires 2 single changes per bit time.

that means–
Baud rate = 2 × bit rate
- Step 2 : given data
- LAN speed (bit rate) = 10 Mbps
- Encoding = Differential Manchester → requires 2 single elements per bit.

So, Baud rate = 2 × 10 mbps = 20 Mbaud.

Sarkari Resource

14. Consider the following two statements:
Statement-I: A mesh network with 10 nodes requires 45 links.
Statement-II: A mesh network with 4 nodes requires 6 links.
In light of the above statements, choose the correct answer from the options given below:
(a) Both statement I and statement II are true.
(b) Both statement I and statement II are false.
(c) Statement I is correct, but statement II is false.
(d) Statement I is incorrect, but statement II is true.
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : In a mesh topology, every node is directly connected to all other nodes in the network this ensures maximum reliability and fault tolerance, since even if one link fails, data can take an attractive route. To calculate the number of links in a fully connected mesh with n nodes, the formula used in:

$$\text{Links} = \frac{n(n-1)}{2}$$

This is because each node connects with n-1 others, but every connection is counted twice, so division by 2 is necessary.
For statement I, with n = 10

$$\frac{10(9)}{2} = \frac{90}{2} = 45$$

The requirement 45 links is correct.
For statement II, with n = 4

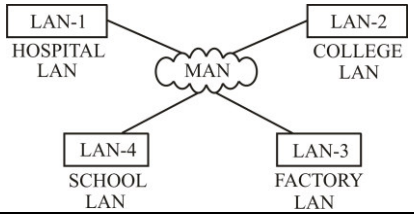
$$\frac{4(3)}{2} = \frac{12}{2} = 6$$

Thus, the requirement of 6 links is also correct.

15. MAN stands for:
MAN का मतलब है–
(a) Mobile Area Network
(b) MAN Area Network
(c) Metro Area Network
(d) Metropolitan Area Network

HTET PGT C.S. 30.07.2025

Ans.(d): A Metropolitan Area Network (MAN) is a type of computer network that spans over a metropolitan area, typically a city it provides high-speed data communication services such as video, audio, and data transfer between multiple LANs (Local Area Networks). The main purpose of a MAN is to connect different LANs in a city to share resources and exchange data as well as to provide internet access to users. A MAN typically covers a geographic area of several kilometers and is larger than a LAN but smaller than a WAN.



16. Which one of the following is not a network topology?
निम्नलिखित में से कौन एक नेटवर्क टोपोलॉजी नहीं है?
(a) Ring (b) Bus
(c) Stack (d) Star

HTET PGT C.S. 30.07.2025

Ans.(c): A network topology defines the physical or logical arrangement of devices (nodes) and communication paths in a computer network the common types of network topologies include Bus, Ring, Star, Mesh, Tree, and Hybrid.
- Bus topology uses a single backbone cable to which all devices are connected.
- Ring topology connects devices in a circular manner, where each devices has exactly two neighbors.
- Star topology connects all devices to a central hub or switch making it easy to manage and troubleshoot.

On the other hand, stack is not a network topology it is a data structure in computer science that follows the LiFO (Last in First Out). Principal since it does not describe the arrangement of network devices, it cannot be considered topology.

17. Which of the following statements are true?
A. X.25 is connection-oriented network
B. X.25 doesn't support switched virtual circuits.
C. Frame relay service provides acknowledgements.
D. Frame relay service provides detection of transmission errors.
Choose the correct answer from the options given below:
(a) A and D only (b) B and D only
(c) C and D only (d) B and C only

UGC NET C.S. December 2020

Ans. (a) :
(a) X.25 is connection-oriented network– This is true. X.25 is a protocol designed for establishing virtual circuits before data transmission making it a connection oriented network.
(d) This is true. Frame relay uses a Cyclic Redundancy Check (CRC) to detect transmission errors. However it does not guarantee error correction or retransmission.

18. _____topology requires highest number of ports and largest amount of cabling in computer networks.
कम्प्यूटर नेटवर्क्स में _____ टोपोलॉजी में सर्वाधिक संख्या में इनपुट आउटपुट पोर्ट तथा अधिकतम मात्रा में केबलिंग की आवश्यकता होती है।
(a) Star/स्टार (b) Ring/रिंग
(c) Mesh/मेश (d) Bus/बस

HTET PGT C.S. 2022

Ans. (c) : Each node in the mesh topology linked to at least three addition nodes. The Ring or Bus topology, in which each node is connected to only two other nodes, Star topology, in which each node is connected to a central hub or switch, necessitates additional cabling.
- The topology of a mesh network is the most fault-tolerant and redundant.

Sarkari Resource

- In a mesh network, Every device is directly connected to every other device on the network there is always another link available if one fails.

19. Which one of the following is not a network topology?
निम्नलिखित में से कौन-सा एक नेटवर्क टोपोलॉजी नहीं है?

(a) Peer-to-peer (b) Ring
(c) Bus
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : Network topology refers to the physical and logical structure of a network, including the devices in the network and their interconnections. This topology defines the flow of data and the pattern of communication in the network and helps to determine how data will be transmitted and received. Some of the major topologies are Star, Bus, Ring and Mesh topology etc. While peer-to-peer is a protocol.

20. Which network topology requires a central controller or hub?
किस नेटवर्क टोपोलॉजी के लिए, केन्द्रीय नियन्त्रक या हब की आवश्यकता होती है?

(a) Star/स्टार (b) Mesh/मेश
(c) Ring/रिंग
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : Star topology is a type of network topology in which many devices or computers are connected. In this topology all the devices are directly connected to the hub or switch and the hub or switch is used to transmit data packets between the devices. Whereas in Mesh and Ring topology there is no need for a hub.

21. The topology with highest reliability is
उच्चतम विश्वसनीयता वाली टोपीलॉजी है

(a) Star/स्टार
(b) Ring/रिंग
(c) Mesh/जाल
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : Mesh is a topology with highest reliability. It a network topology in which each node is directly connected to all other nodes. In this topology, each node has an independent network path, allowing the nodes to communication directly with each other. This type of topology is specially used in large network, where there are a large number of nodes and maximum availability and network performance is required.

22. A _____ topology uses a single connection to connect all devices together.
_____ टोपोलॉजी, एक ही जोड़ (कनेक्शन) का प्रयोग सभी युक्तियों (Devices) को एक साथ जोड़ने के लिए करता है।

(a) Bus/बस
(b) Star/स्टार
(c) Ring/रिंग
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a) : A bus topology uses a single connection to connect all devices together.

23. A _____ is used to connect networking devices that are in a very close geographic area such as a floor of a building.
_____ का उपयोग उन नेटवर्किंग उपकरणों को जोड़ने के लिए किया जाता है, जो किसी भवन के फर्श जैसे बहुत करीबी भौगोलिक क्षेत्र में होता है।

(a) WAN/वैन
(b) LAN/लैन
(c) MAN/मैन
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b) : LAN stands for a Local Area Network. It is used to connect networking devices that are in a very close geographic area such as a floor of a building.

24. In a fully-connected mesh network with 10 computers, total _____ number of cables are required and _____ number of ports are required for each device.

(a) 40, 9 (b) 45, 10
(c) 45, 9 (d) 50, 10

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Since, each in mesh network total number of cables are required

$${}^{n}C_2 = {}^{10}C_2 = 45$$

Each device needs to be connected to 9 other devices. So, each device needs to have 9 ports.

25. Which statements are true about local area network?
स्थानीय क्षेत्र नेटवर्क के बारे में कौन से कथन सत्य है/हैं?

I. Communication takes place by way of satellites, microwave.
उपग्रहों माइक्रोवेव के माध्यम से संचार होता है

II. Faster than wide area network
वृहत क्षेत्र नेटवर्क से तेज है

III. Covers limited area
सीमित क्षेत्र को कवर करता है

(a) I and II (b) II and III
(c) I and III (d) I, II and III

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Statement-I : False
LAN use wires (Ethernet cables) or Wi-Fi (Wireless signals) for Communication within a small area like a home, school or office.

Sarkari Resource

Statement-II : True
LANs are faster because they operate over a small distance & can use high-speed connections like:
- Ethernet (up to 1 Gbps or even 10 Gbps)
- High-speed Wi-Fi (Wi-Fi 6, Wi-Fi 7, etc)

WANs are slower because they cover large distance & often depend on multiple devices, routers & external providers (like telecom companies)
Statement-III : True
LAN's are designed for limited geographic areas such as A room, building, and college campus.

26. Which statements are true wide area network?
वृहत् क्षेत्र नेटवर्क के बारे में कौन से कथन सत्य हैं?

I. Covers wide geographical area
वृहत भौगोलिक क्षेत्र को कवर करता है

II. Communication take place way of satellites, microwaves
उपग्रहों माइक्रोवेव के माध्यम से संचार होता है

III. Relatively faster local area network
स्थानीय क्षेत्र नेटवर्क की तुलना में अपेक्षाकृत तेज़ है

(a) I and II (b) II and III
(c) I and III (d) I, II and III

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : Statement-I : True
Wide Area Networks (WANs) are designed to cover large geographical areas, such as cities, countries or even continents.
Statement-II : True
WANs often use satellite links, microwaves, fiber optics and other communication technologies.
Statement-III False
WANs are generally slower compared to LAN's because of the long distances & more complex communication equipment involved.

27. Which of the following statement(s) is/are not true about star topology?
स्टार टोपोलॉजी के बारे में निम्नलिखित में से कौन सा/से कथन सत्य नही है/हैं?

I. Failure of the central Computer shut down the entire network/सेंट्रल कंप्यूटर की विफलता संपूर्ण नेटवर्क को बंद कर देती है

II. Connection of additional computers increase the communication time/अतिरिक्त कंप्यूटरों के कनेक्शन से संचार समय बढ़ जाता है

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Statement I : True
In star topology, failure of the central computer/device shut down the entire network. Because in this topology all the nodes are connected with central device.
Statement II : False
In Star Topology, each computer has a direct connection to the central device. Adding more computers usually does not significantly increase communication time unless the central device becomes overloaded, which is rare with modern switches.

28. Which of the following statement(s) is/are correct?
निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. Metropolitan area network is a collection of Local area network/मेट्रोपॉलिटन एरिया नेटवर्क लोकल एरिया नेटवर्क का एक संग्रह है।

II. Metropolitan area network support both date and voices transmission
मेट्रोपॉलिटन एरिया नेटवर्क डेटा और वॉयस ट्रांसमिशन दोनों को सपोर्ट करता है

(a) Only I/ केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Statements-I, Metropolitan Area Network is a collection of Local Area Network– This is generally true. A MAN often connects multiple LAN with a larger geographical area, like a city or metropolitan region. Think of it as a bigger network that ties smaller network together.
Statement-II, Metropolitan area network supports both data and voice transmission- It's also true. Modern MAN are designed to handle various types of communication, including data transfer (like files and internet traffic) as well as voice.
Communication (Like VoIP)
So, the option (c) is correct.

29. Which of the following is a type of network topology?
निम्नलिखित में से कौन सा नेटवर्क टोपोलॉजी का एक प्रकार है?

I. Star topology/स्टार टोपोलॉजी

II. Bus topology/बस टोपोलॉली

III. Car topology/क़ार टोपोलॉजी

(a) I and II (b) I and III
(c) II and III (d) I, II and III

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Network topology refers to the arrangement of different elements in a computer network. It describes now devices are connected and now data flow between them.
Star topology– All devices are connected to a central hub or switch.
Bus Topology– All devices are connected to a single communication line (backbone).

30. Which protocol is used for basic transmission of multimedia content?
मल्टीमीडिया सामग्री के बुनियादी प्रसारण के लिए किस प्रोटोकॉल का उपयोग किया जाता है?

(a) Real –time transport protocol
रियल-टाइम ट्रांसपोर्ट प्रोटोकॉल

(b) Real –system transport protocol
रियल-सिस्टम ट्रांसपोर्ट प्रोटोकॉल

(c) Real –system transfer protocol
रियल-सिस्टम ट्रांसफर प्रोटोकॉल

(d) Real–software transport protocol
रियल-सॉफ्टवेयर ट्रांसपोर्ट प्रोटोकॉल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : RTP stands for Real Time Transport Protocol. It is specifically designed for transmitting multimedia data like audio and video in real-time, making it the most accurate choice for this question.

Sarkari Resource

31. Which of the following is not a example of Distributive computing?
निम्नलिखित में से कौन सा डिस्ट्रिब्यूटेड कम्प्यूटिंग का उदाहरण नहीं है?
(a) LAN network/लेन नेटवर्क
(b) Mobile network/मोबाईल नेटवर्क
(c) Parallel computing/पैरेलल कम्पयूटिंग
(d) Internet/इंटरनेट

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Distributed computing involves multiple computers working together on a task, typically communicating over a network.
Parallel computing, on the other hand usually refers to multiple process within a single machine or closely connected machines working simultaneously on a task. So, while LAN network, mobile network and the Internet can all be used to support distributed computing environments parallel computing is a distinct concept.

32. _____ is loosely coupled system and _____ is tightly coupled system.
_____ एक लूसेली कपल्ड सिस्टम है तथा _____ एक टाइटली कपल्ड सिस्टम है।
(a) Distributed processing; Uniprocessing
डिस्ट्रिब्यूटेड प्रोसेसिंग: युनीप्रोसेसिंग
(b) Batch processing; Parallel processing
बैच प्रोसेसिंग: पैरेलल प्रोसेसिंग
(c) Distributed processing; Parallel processing
डिस्ट्रिब्यूटेड प्रोसेसिंग; पैरेलल प्रोसेसिंग
(d) Uniprocessing; Real Processing
युनीप्रोसेसिंग रियल प्रोसेगिं

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Distributed processing is a loosely coupled system because it involves multiple independent computers (node) that work together to achieve a common goal, communicating over a network with minimal direct dependency on each other.
Parallel processing is a tightly coupled system because it involves multiple processors or cores within a single system that work closely together, sharing resources like memory and requiring synchronized operations.

33. What is the full form of MISD?
एम.आई.एस.डी. का असंक्षिप्त रूप क्या है?
(a) What is the full form of MISD?
मल्टीप्ल इंटरप्ट स्ट्रीम तथा सिंगल डाटा स्ट्रीम
(b) Multiple Instruction stream and single Data stream
मल्टीप्ल इंस्ट्रक्शन स्ट्रीम तथा सिंगल डाटा स्ट्रीम
(c) Many Interrupt stream and single digital stream
मेनी इंटरप्ट स्ट्रीम तथा सिंगल डिजिटल स्ट्रीम
(d) Many Instruction stream and Single Digital stream
मेनी इंस्ट्रक्शन स्ट्रीम तथा सिंगल डिजिटल स्ट्रीम

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : MISD stands for Multiple Instruct Stream and single Data stream, a type of parallel computing architecture where multiple instructions are executed on a single data stream this is often used in systems requiring thigh reliability, such as fault tolerant computing.

34. Which of the following is not a type of Network topology?
निम्न में से कौन सा नेटवर्क टोपोलॉजी का एक प्रकार नहीं है?
(a) Star Network/स्टार नेटवर्क
(b) Ring Network/रिंग नेटवर्क
(c) Hybrid Network/हाइब्रिड नेटवर्क
(d) Singleton Network/सिंगलटन नेटवर्क

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : Singleton Network is not a type of Network topology where as Ring Star and Hybrid are network topology.

35. How many maximum links can be created in a mesh topology network of n computer?
n कंप्यूटर के मेश टोपोलॉजी नेटवर्क में कितने अधिकतम लिंक बनाए जा सकते हैं?
(a) n (b) n-1
(c) n(n-1) (d) $\frac{n(n-1)}{2}$

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : In a mesh topology every computer is directly connected to every other computer.
for n computers-
Each computer connects to n-1 others.
So, total links = $\frac{n(n-1)}{2}$.

36. Which of the following is not a LAN technology?
इनमें से कौन सा लैन (LAN) तकनीक नहीं है?
(a) Fiber Distributed Data Interface
फाइबर डिस्ट्रिब्यूटेड डेटा इंटरफेस
(b) Token Ring/टोकन रिंग
(c) Ethernet/ईथरनेट
(d) Frame Relay/फ्रेम रिले

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : Analyze each option–
Fiber Distributed Data Interface (FDDI)– This is a standard for data transmission in a local area network (LAN) using optical fiber.
Token Ring– This is a LAN technology where all devices are connected in a ring or star topology and use a token passing mechanism to control access to the network.
Ethernet– This is the most widely used LAN Technology, defining wiring and signaling standards for the physical layer and data link layer.
Frame Relay– It's a wide area network technology that operates at the data link layer.
Therefore, the option that is not LAN technology is Frame Relay.

37. Suppose a CSMA/CD local Area Network of 1 Gbps is there over a 1 km cable. If the signal speed in the cable is 2×10^5 km/sec. will be the minimum frame size?
मान लीजिये कि 1 Gbps का CSMA/CD लोकल नेटवर्क 1 km केबल पर है। यदि केबल में सिग्नल की गति 2×10^5 km/sec है, तो मिनिमम (न्यूनतम) फ्रेम साइज (आकार) क्या होगा।

Sarkari Resource

(a) 4000 (b) 100000
(c) 10000 (d) 5000

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : The minimum frame size in a CSMA/CD network is determined by the round trip time of the signal over the cable.

Step by step calculation–

Network Bandwidth 1 Gbps = 1×10^9 bps

Cable Length 1 km = 1000 m

Signal speed 2×10^5 km/sec = 2×10^8 m/sec

Propagation Delay (t_p)– This is the time it makes for a signal to frame the length of the cable–

$$t_p = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{1000\,\text{m}}{2\times10^8\,\text{m/s}} = 5\times10^{-6}\,\text{sec}$$

Round-Trip Time– This is the time it takes for a signal to travel to the other end of the cable and for a collision signal to travel back.

RTT = $2 \times t_p = 2 \times (5 \times 10^{-6}) = 1 \times 10^{-5}$ seconds.

Minimum frame size– The Transmission time of the frame must be at least equal to the round-trip time.

MFS = RTT × B = $(1 \times 10^{-5}) \times (1 \times 10^9$ bps)

= 10,000 bits.

38. Which of the following is a communication pathway connecting two or more devices of a computer ?
दो या दो से अधिक कंप्यूटर डिवाइस को कनेक्ट करने वाला संचार पथ निम्नलिखित में से कौन सा है।

(a) Bus/बस
(b) Memory/मेमोरी
(c) UPS/यू.पी.एस.
(d) Control Unit (CU)/कंट्रोल यूनिट

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : A bus is a shared communication pathway that allows multiple devices in a computer system to connect and exchange data with each other. It essentially acts like a highway where different components can send and receive information.

39. Match the following :

	List–I		List–II
A.	**Call control protocol**	**i.**	**Interface between Base Transceiver Station (BTS) and Base Station Controller (BSC)**
B.	**A-bis**	**ii.**	**Spread spectrum**
C.	**BSMAP**	**iii.**	**Connection management**
D.	**CDMA**	**iv.**	**Works between Mobile Switching Centre (MSC) and Base Station Subsystem (BSS)**

Codes :

	A	B	C	D
(a)	iii	iv	i	ii
(b)	iii	i	iv	ii
(c)	i	ii	iii	iv
(d)	iv	iii	ii	i

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) The correct match–

	List–I		List–II
A.	Call control protocol	**iii.**	Connection management
B.	A-bis	**i.**	Interface between Base Transceiver Station (BTS) and Base Station Controller (BSC)
C.	BSMAP	**iv.**	Works between Mobile Switching Centre (MSC) and Base Station Subsystem (BSS)
D.	CDMA	**ii.**	Spread spectrum

40. Which of the following concurrency protocol ensures both conflict serializability and freedom from deadlock :
I. 2-phase locking
II. Time phase ordering

(a) Both I & II (b) II only
(c) I only (d) Neither I nor II

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : 2 phase locking is a concurrency control method that guarantees serializability. The protocol utilizes locks, applied by a transaction to a data, which may block other transaction from accessing the same data during the transactions life. 2PL may be lead to deadlocks that results from the mutual blocking of two or more transaction.

Timestamp-based concurrency control is a non lock concurrency method. In timestamp based method, deadlock occur as no transaction ever waits.

41. Which of the following services is not provided by wireless access point in 802.11 WLAN?

(a) Association (b) Disassociation
(c) Error correction (d) Integration

UGC NET C.S. December-2015 (Paper-II)

Ans : (c) Service provided by wireless access point in 802.11 WLAN"
(i) Integration
(ii) Disassociation
(iii) Association
(iv) Authentication
(v) Privacy
(vi) distribution
(vii) MSDU delivery
For error correction we have different policies, and correction does not provided by wireless access point in 802.11 WLAN.

42. _____ do not take their decisions on measurements or estimates of the current traffic and topology.

(a) Static algorithm
(b) Adaptive algorithm
(c) Non-adaptive algorithm
(d) Recursive algorithm

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Non adaptive algorithm Implements static routing non adaptive algorithm do not base their routing decision on measurements and estimates of the current traffic & topology. Instead the route to be taken in going from one node to the other in computed in advance.

Sarkari Resource

43. IP datagram has arrived with following partial information in the header (in hexadecimal) 45000054000300002006....
How many more routers can the packet travel to?
(a) 22 (b) 26
(c) 30 (d) 32
UGC NET C.S. June-2023

Ans. (d) : The Time-to-Live (TTL) field represents the number of routers, the packet can traverse before being discarded. In hexadecimal, the TTL field is typically represented by the 9th byte in the header. In the given partial information, the 9th byte is '20'. Which in hexadecimal equal 32 in decimal. So, the packet can travel through 32 more routers before being discarded.

44. IP datagram has arrived with following partial information in the header (in hexadecimal) 45000054000300002006....
What is the size datagram?
(a) 64 bytes (b) 74 bytes
(c) 84 bytes (d) 104 bytes
UGC NET C.S. June-2023

Ans. (c) : In hexadecimal, this field typically occupies bytes 3 and 4 of the header. From the partial information provided, the total length field is represented by '0054' in hexadecimal. Which in hexadecimal equals 84 in decimal. Therefore, the size of the datagram is 84 bytes.

45. IP datagram has arrived with following partial information in the header (in hexadecimal) 45000054000300002006....
What is the header size?
(a) 10 bytes (b) 20 bytes
(c) 30 bytes (d) 40 bytes
UGC NET C.S. June-2023

Ans. (b) : In the given partial information the internet header length (IHL) field is represented by the first nibble (4 bits) of the second byts. The second byte is '05' in hexadecimal, since the IHL field is measured in 32 bit words.
header length = 5 × 4 = 20 bytes
So, the header size is 20 bytes.

46. IP datagram has arrived with following partial information in the header (in hexadecimal) 45000054000300002006....
What is the efficiency of this datagram?
(a) 76.19% (b) 80.50%
(c) 82.24% (d) 85.45%
UGC NET C.S. June-2023

Ans. (a) : Given that-
Total length of the datagram = 84 bytes
The header size = 20 bytes
Payload size = Total length – Header size
= 84 – 20 = 64 byte
Now, calculate the efficiency-

$$\text{Efficiency} = \frac{\text{Payload size}}{\text{Total length}} \times 100$$

$$= \frac{64}{84} \times 100 = 76.19\%$$

So, the efficiency of the datagram is 76.19%.

47. Network uses star topology if:
नेटवर्क स्टार टोपोलॉजी का उपयोग करता है यदि
(a) All computer are arranged in local loop
सभी कंप्यूटर लोकल लूप में व्यवस्थित हो
(b) All computers are attached to a central point
सभी कंप्यूटर एक सेंट्रल पॉइंट से जुड़े हो
(c) All computers are attached to a single long cable/सभी कंप्यूटर एक लम्बी केबल से जुड़े हो
(d) All computers are attached to multiple hierarchical cable
सभी कंप्यूटर मल्टीप्ल वर्गीकृत केबल से जुड़े हो
RPSC C.S. 2016 (PAPER-I)

Ans. (b) : In a star topology, all computers (nodes) are connected to a central device, such as a hub, switch, or router. This central point manages the data traffic between nodes, and if one cable fails, it doesn't affect the rest of the network (unlike in a bus topology).

48. Which type of network shares the communication channel among all the machines?
किस प्रकार का नेटवर्क सभी मशीनों के बीच संचार चैनल साझा करता है?
(a) Broadcast network
(b) Multicast network
(c) Unicast network
(d) More than one of the above
(e) None of the above
BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(a) A broadcast network is a type of network where a machine can send data to all other machine on the network simultaneously. This means that when one device transmits a message, every other device connected to the same network receives it. This makes it ideal for situations like announcing important information or sending updates to all users.

49. Which topology requires a multipoint connection?
किस टोपोलॉजी के लिए मल्टीपॉइंट कनेक्शन की आवश्यकता होती है?
(a) Mesh Topology
(b) Bus Topology
(c) Star Topology
(d) More than one of the above
(e) None of the above
BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(b) In bus topology, all devices are connected a single central cable, which acts as a shared communication medium. This is called a multipoint connection.

50. Which of the following networks extends a private network across public networks?
निम्नलिखित में से कौन-सा नेटवर्क सार्वजनिक नेटवर्क में एक निजी नेटवर्क का विस्तार करता है?
(a) Virtual private network
(b) Local area network
(c) Enterprise private network
(d) More than one of the above
(e) None of the above
BPSC TRE 3.0 22.07.2024 (6-10)

Sarkari Resource

Ans.(a) VPN stands for Virtual Private Network. It is network extends a private network across a public network, and enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network. VPN provides enhanced security and online anonymity to users on the internet. It is also used to block websites which are unavailable in certain regions.

51. What type of protocol is HTTP?
HTTP किस प्रकार का प्रोटोकॉल है?
(a) Stateful
(b) Stateless
(c) Transfer protocol
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(b) HTTP is a stateless protocol, which means that each request sent from the client to the server is treated as an independent transaction, with no information or memory of previous requests.

52. Real time streaming protocol is used____
वास्तविक समय स्ट्रीमिंग प्रोटोकॉल का उपयोग किया जाता है_____
(a) To control streaming media servers
(b) For establishing and controlling media sessions between endpoints
(c) To provide real time control of playback of media files from the server
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(d) The Real Time Streaming Protocol is used for-
Controlling streaming media servers (like play, pause, stop, record).
Establishing and controlling media sessions between end points.
Providing real-rime control of playback of media from the server.
So, the correct answer is (d).

53. Which device is used to connect multiple devices within the same local area network (LAN)?
एक ही लोकल एरिया नेटवर्क (LAN) के भीतर कई डिवाइसों को जोड़ने के लिए किस डिवाइस का उपयोग किया जाता है?
(a) Router/राउटर
(b) Switch/स्विच
(c) Bridge/पुल
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : Switches are used to connect multiple devices within a single Local Area Network (LAN). Switches interconnect different devices in the network and help in routing data packets to their destination correctly.

54. X.21 is
X.21 है
(a) A method of determining which device has access to the transmission medium at any time/एक मेथड, जो बताता है कि कौन-सी डिवाइस ट्रान्समिशन मीडियम को किसी समय पर एक्सेस करेगी
(b) A method access control technique or multiple access transmission media
एक मेथड एक्सेस कन्ट्रोल टेकनिक या मल्टिपल एक्सेस ट्रान्समिशन मीडिया
(c) Very common but oriented at linked protocol issued to ISO/अति सामान्य लेकिन लिंक प्रोटोकॉल की ओर ओरिएन्टेड है जो कि ISO को इस्यू है
(d) A network access standard for connecting stations to a circuit switch network/एक नेटवर्क एक्सेस स्टैन्डर्ड जिससे स्टेशनों को सार्किट स्विच नेटवर्क से जोड़ा जाता है

LT. Grade C.S. 2018

Ans. (d) : X.21 is an ITU-T standard that defines a network access interface for connecting data terminal equipment (DTE) and data circuit terminating equipment (DCE) to a circuit switched network. It is primarily used in synchronous data transmission environments.

55. Which of the following is an advantage of computer networks?
निम्न में से कौन-सा कम्प्यूटर नेटवर्क का एक लाभ है?
(a) Resource sharing/संसाधन की हिस्सेदारी होना
(b) Improving reliability and availability
रिलायबिलिटी तथा एवलेबिलिटी को बढ़ाना
(c) Enhancing performance/परफॉर्मेन्स को बढ़ाना
(d) All of the above/उपर्युक्त सभी

LT. Grade C.S. 2018

Ans. (d) : Computer network offer several advantage including.
Resource sharing– computer networks allow users to share files and documents, printers and scanners, internet connections and software applications.
Improving reliability and availability– Redundancy and failure mechanisms in networks can improve system reliability and ensure services are available even in the event of a failure.
Enhancing performance– Networks can distribute tasks across multiple machines or locations, improving performance through load balancing and parallel processing.
By connecting devices and resources, computer networks can improve productivity, efficiency and overall performance.

56. A network spreading in different geographical locations is known as
एक नेटवर्क, जो विभिन्न भौगोलिक जगहों पर फैला हो, को कहते हैं।
(a) LAN
(b) WAN
(c) Both (a) and (b)/(a) और (b) दोनों
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Sarkari Resource

Ans. (b) : A WAN (Wide Area Network) is a communications network that spans a large geographic area such as across cities, states or countries.

57. Which of the following is a token-based protocol?
निम्न में से कौन-सा एक टोकेन-बेस्ड प्रोटोकॉल है?
(a) CSMA/CD
(b) Slotted ring/स्लॉटेड रिंग
(c) Both (a) and (b)/ (a) और (b) दोनों
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (b) : Slotted ring is a token based protocol where a special token circulates in the network, and the node that holds the token can transmit data. This is a token based access control method.

58. Select the correct full form of the term ARPANET from the given options.
नीचे दिए गए विकल्पों में से ARPANET पद का सही पूरा नाम बताएं।
(a) American Research Projects Advance Network
(b) American Research Programs Advance Netwrok
(c) Advance Research Programs Academic Netwrok
(d) Advanced Research Projects Agency Network

EMRS PGT C.S. 16.12.2024

Ans. (d) : The full form of ARPANET is Advanced Research Projects Agency Network. It was the first wide area packet switched network with distributed control and one of the first computer networks to implement the TCP/IP protocol suite. The ARPANET was established by the Advanced Research Project Agency of the United States Department of Defense.

59. Choose the best option to fill the blank correctly.
A _____ is a human-friendly text form of the IP addresses.
रिक्त स्थान को सही प्रकार से भरने के लिए सर्वोत्तम विकल्प का चयन कीजिए।
एक _____ आईपी एड्रेस का Human फ्रैंडली टेक्स्ट प्रकार है।
(a) Domain name/डोमेन नेम
(b) MAC address/मैक एड्रेस
(c) Memory address/मेमोरी एड्रेस
(d) URL/यू आर एल

EMRS PGT C.S. 16.12.2024

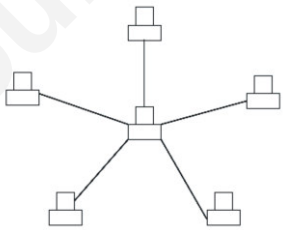
Ans. (a) : A domain name is a human readable text address that corresponds to the numeric IP address of a specific server on the internet. It's used to identify and locate resources, like websites on the World Wide Web. Domain names provide an easier way to people to remember and access websites compared to using IP addresses directly.

60. Identify the topology from the given options, which is a hierarchical topology, in which there are multiple branches and each branch can have one or more basic topologies and are usually connect multiple LAN.
नीचे दिए गए विकल्पों में से उस टोपोलॉजी की पहचान कीजिए जो एक हिरारचिकल (hierarchical) टोपोलॉजी है जिसमें मल्टीपल ब्रांच हैं और प्रत्येक ब्रांच की एक या अधिक बेसिक टोपोलॉजी हो सकती हैं और वह आमतौर पर मल्टीपल LAN से कनेक्ट होती है।
(a) Bus topology/बस टोपोलॉजी
(b) Tree topology/ट्री टोपोलॉजी
(c) Star topology/स्टार टोपोलॉजी
(d) Mesh topology/मेश टोपोलॉजी

EMRS PGT C.S. 16.12.2024

Ans. (b) : Tree or Hybrid Topology is a hierarchical topology in which there are multiple branches and each branch can have one or more basic topologies like Star, Ring and Bus. Such topologies are usually realised in WANs where multiple LANs are connected.

61. Which topology of network is represented in the following diagram ?
नीचे दिए गए आरेख में कौन सी नेटवर्क टोपोलॉजी निरूपित की गई है?



(a) STAR (b) BUS
(c) TREE (d) RING

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (a) : Star topologies are used in many networks large and small. In a star topology all nodes indirectly connect to each other through one or more switches. The acts as a central point through which all communication are passed. Large network using a star topology are usually controlled by one or more servers.

62. Frames from one LAN can be transmitted to another LAN via a device called
एक डिवाइस से एक LAN की फ्रेम का दूसरे LAN में प्रेषण किया जा सकता है, जिसे कहते है
(a) Router./रूटर (b) Bridge./ब्रिज
(c) Repeater./रिपीटर (d) Modem./मोडेम

KVS PGT C.S. 2013

Ans. (b) : The bridge is used at the data link layer in the OSI model to connect a network to another one. Bridge can Connect multiple LAN networks and separate network segments.

63. A____ is a communication pathway connecting two or more devices. Another of its' key characteristic is that it is a shared transmission medium. A signal transmitted by any one device is available for reception by all other devices attached to it.
____एक संचार पाथ-वे जो दो या अधिक युक्तियों को सम्पर्कित करता है। इसकी अन्य मुख्य लाक्षणिकता यह है कि यह एक साझा प्रसारण माध्यम है । किसी एक युक्ति द्वारा संप्रेषित सिग्नल इससे जुड़ी अन्य सभी युक्तियों द्वारा ग्रहण करने हेतु उपलब्ध होता है ।

Sarkari Resource

(a) Train/ट्रेन (b) Bus/बस
(c) Tram/ट्राम (d) Aeroplane/एयरोप्लेन

KVS PGT C.S. 2013

Ans. (b) : A bus is a communication pathway connecting two or more device. Another of it's key characteristic is that it is a shared by transmission medium. A signal transmitted by any one device is available for reception by all other device attached to it.

64. In which of the topology, each device has a dedicated point-to-point link to a central controller?
किस टोपोलॉजी में, प्रत्येक डिवाइस में एक केन्द्रीय नियंत्रक के लिए एक समर्पित पॉइंट-टू-पॉइंट लिंक होता है?

(a) Mesh/मेश (b) Bus/बस
(c) Ring/रिंग (d) Star/स्टार

KVS PGT C.S. 2018

Ans. (d) : Star topology is a network topology, in which each network component is physically connected to a central node such as a router, hub or switch. Star networks require a point-to-point connection between the central node and connecting devices. In a star topology, the central hub acts like a server and the connecting nodes act like clients. When the central node receives a packet from a connecting node, it can pass the packed on to other nodes in the network.

65. The communication between the components in a microcomputer takes place via the address and _____.
किसी माइक्रोकम्प्यूटर में घटकों के मध्य कम्यूनिकेशन एड्रेस तथा _____ के माध्यम से होता है।

(a) I/O bus/I/O बस
(b) Data bus/डाटा बस
(c) Address bus/एड्रेस बस
(d) Control lines/कन्ट्रोल लाइनें

KVS PGT C.S. 2018

Ans. (b) : The communication between the components in a microcomputer takes place via the address and Data bus. A data bus is a system within a computer or device, consisting of a connector or set of wires, that provides transpiration for data.

66. Which of the following topologies gives highest reliability ?
निम्नलिखित में से कौन सी टॉपोलॉजी अधिकतम विश्वासनीयता प्रदान करती है?

(a) Mesh topology/मेश टॉपोलॉजी
(b) Bus topology/बस टॉपोलॉजी
(c) Star topology/स्टार टॉपोलॉजी
(d) Ring topology/रिंग टॉपोलॉजी

KVS PGT C.S. 2017

Ans. (a) : A mesh topology is a network setup where each computer and network device is interconnected with one another. This topology setup allows for most transmissions to be distributed even if one of the connections goes down. The topology with highest reliability is 'mesh topology'.

67. Mechanism that is used to convert domain name into IP address is known as _____.
डोमेन नाम का आई पी एड्रेस में परिवर्तित करने के लिए उपयोग लाई जाने वाली प्रौद्योगिकी को _____ कहा जाता है।

(a) HTTP (b) URL
(c) FTP (d) DNS

KVS PGT C.S. 2017

Ans. (d) : The Domain Name System (DNS) is a naming database in which internet domain names are located and translated into Internet Protocol (IP) addresses. The domain name System maps the name people use to locate a website to the IP address that a computer uses to locate that website.

68. Garuda Skill Education has set up their new skill training campus in a place near Pune, The campus has three close by buildings connected with each other using CAT6 cable. Which type of network is established in the campus?
गरूड़ स्किल एजुकेशन ने पुणे के नजदीक एक स्थान पर अपना नया स्किल ट्रेनिंग कैम्पस स्थापित किया है। कैम्पस में तीन नजदीकी बिल्डिंग हैं जो CAT'6 केबल से आपस में जुड़ी हुई हैं। कैम्पस में किस प्रकार का नेटवर्क स्थापित किया गया है ?

(a) PAN (b) MAN
(c) LAN (d) WAN

KVS PGT 23.03.2023 (Shift-II)

Ans. (c) : A computer network that links device within a building or group of adjacent building, especially one with a radius of less than 1 km. This network is known as LAN (Local Area Network).

69. What is full form of ARPANET ?
ए आर पी ए एन ई टी (ARPANET) का पूरा नाम क्या है ?

(a) Advancement of Research Project Agency Network
(b) Advanced Research Project Agency Network
(c) Add on Research Project Agency Network
(d) American Research Project Agency Network

KVS PGT 23.03.2023 (Shift-II)

Ans. (b) : The full form of ARPANET is Advanced Research Project Agency Network. It was the first wide area packet switched network with distributed control and one of the first computer networks to implement the TCP/IP protocol suite.

70. Two branches of a financial organization are connected with each other on a network. If these two branches are established in two different countries, which of the following type of network is set-up ?
एक वित्तिय संगठन की दो शाखाएँ एक नेटवर्क पर एक दूसरे जुड़ी हुई है। यदि ये दो शाखाएँ दो अलग-अलग देशों में है तो किस प्रकार का नेटवर्क होगा?

(a) WAN/डब्ल्यू ए एन
(b) MAN/एम ए एन
(c) LAN/एल ए एन
(d) PAN/पी ए एन

NVS PGT- 16.12.2022 (Shift-II)

Sarkari Resource

Ans. (a) : WAN stands for Wide Area Network. It is a network that spans a large geographical area, such as a country, continent or ever globally. WANs are typically used to connect multiple smaller networks, such as Local Area or other regional networks. It's share resources, communicate, and collaborate across different locations.

71. What is a Firewall in Computer Network?
कंप्यूटर नेटवर्क में फायरवॉल क्या है?

(a) The physical boundary of Network
(b) An operating system of computer Network
(c) A system designed to prevent unauthorized access
(d) A web browsing Software

Bihar STET C.S. 2020

Ans. (c) : Firewall is a security system designed to prevent unauthorized access to a computer network. Firewall is a software or hardware that protect your computer or network from dangerous data, viruses, hackers etc. Coming from internet or other network. It filters the data coming and going in the network and decides which information can come in and which cannot.

72. Protocols are
प्रोटोकॉल हैं

(a) Agreements on how communication components and DTEs are to communicate
(b) Logical communication channels for transferring data
(c) Physical communication channels used for transferring data
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2020

Ans. (a) : Protocols are essentially a set of rules that govern how data is transmitted between different devices in a network. Think of them as the agreed upon language that computers, routers, and other communication devices use to understand each other. This is crucial for ensuring that information sent from one device is correctly received and interpreted by another.
The correct option (a)– A agreements on how communication components and DTE's are communicate.

73. A physical connection between the microprocessor memory and other parts of the microcomputer is known as
माइक्रोप्रोसेसर मेमोरी और माइक्रोकंप्यूटर के अन्य भागों के बीच एक Physical Connection को जाना जाता है

(a) Path (b) Address bus
(c) Route (d) All of these/इनमें से सभी

Bihar STET C.S. 2020

Ans. (b) : An element in a computer CPU that transmits the location of stored information the address bus allows the CPU to send the address to RAM and the data bus allow the actual data transfer to the CPU.

74. Which one is a communication channel security protocol?
निम्नलिखित में से कौन-सा एक संचार चैनल सुरक्षा प्रोटोकॉल है?

(a) SSL (b) SET
(c) S-HTTP (d) All of the above

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : SSL (Secure Sockets Layer)– It is a security protocol that creates a secure and encrypted connection between a web server and browser. It is mainly used in HTTPs.
SET (Secure Electronic Transaction)– It was developed by visa and Master card to secure online transacting using credit card-2.
S.HTTP (Secure Hyper Text Transfer)- It is a protocol that encrypts every HTTP message sent over the internet. It was an alternative to HTTPs.

75. Protocol used for the working with the internet?
इन्टरनेट के साथ उपयोग होने वाला प्रोटोकॉल है?

(a) CPU
(b) Magnetic tapes and disks
(c) Video terminal
(d) None of these

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : The internet uses a variety of protocols such as TCP/IP, UDP and DNS. A protocol is a set of rules that govern how data is exchanged between computers. A CPU is the brain of a computer and doesn't define the internet protocol. Magnetic tapes and disks are storage devices, not protocols. A video terminal is a device used to display information, not a protocol.

76. What is internet ?
इंटरनेट क्या है?

(a) A single network
(b) A vast collection of different networks
(c) Interconnection of local area networks
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : The internet is a global system of interconnected computer networks that use standardized communication protocol, such as TCP/IP, to link devices worldwide.
The internet is essentially a network of networks, encompassing public, private, academic business and government networks of local to global scope.

77. To join the internet, the computer has to be connected to a
इंटरनेट से जुड़ने के लिए कंप्यूटर को एक_____ से कनेक्ट करना होगा।

(a) Internet architecture board
(b) Internet society
(c) Internet service provider
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : To connect to the internet, the computer has to be connected to the internet service provider ISP is the company that provides internet service such as Jio, Airtel, BSNL etc.

78. Internet access by transmitting digital data over the wires of a local telephone network is provided by
एक स्थानीय टेलीफोन नेटवर्क के तारों पर डिजिटल डेटा संचारित करके इंटरनेट का उपयोग द्वारा प्रदान किया जाता है

Sarkari Resource

(a) Leased line
(b) Digital subscriber line
(c) Digital signal line
(d) None of the mentioned

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : Internet access by transmitting digital data over telephone network wires is provided by Digital Subscriber Line (DSL) technology. DSL uses existing capper telephone lines to deliver high- bandwidth internet connectivity to homes and small businesses allowing for simultaneous voice and data transmission over the sane lines.

79. In WLAN which of the following services is not provided by Distribution system?
(a) Association (b) Distribution
(c) Authentication (d) Integration

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : In WLAN (Wireless Local Area Network), the service that is not provided by the Distribution System is Authentication.
The Distribution System in WLAN is responsible for managing communication between access points and other parts of the network (like the wired LAN). It provides the following services.
1. Distribution
2. Integration
3. Association.

80. WLAN security standard make use of IEEE802.1x, port based network access protocol, for the purpose of :-
(a) Authentication protocol
(b) Generating Master Session key (MKY)
(c) Both Authentication protocol and Generating Master Session key (MSK)
(d) Neither Authentication protocol nor Generating Master Session key (MSK)

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : IEEE802.1x in WLAN security– IEEE802.1x, is a part based network access control protocol that is widely used in WLAN security for.
Purpose of IEEE802.1x–
1. Authentication protocol– IEEE802.1x provides a frame work for authenticating uses and devices before allowing them to access the network.
2. Generating Master Session Key (MSK)– As part of the authentication process, IEEE802.1x can facilitate the generation of keys, including the Master Session Key (MSK), which is used for encrypting and securing the communication.

81. According to the IEEE project 802.11, there are two types of wireless LAN. In an infrastructure-based network, what is a BSA (Basic Service Area):-
(a) A BSA is wireless station
(b) A BSA is a gateway connects a wireless station to a network
(c) A BSA is simply a cell
(d) A BSA is another word for server

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : In IEEE 802.11 (Wi-Fi) network.
BSA (Basic Service Area) refers to the coverage area provided by one Access Point in an infrastructure based network.
It is essentially a cell in which wireless devices can communicate with the network via the AP multiple BSAs can form an Extended service set.

82. Which of the following is NOT a network topology?
निम्नलिखित में से कौन-सी नेटवर्क टोपोलॉजी नहीं है?
(a) Star/स्टार (b) Ring/रिंग
(c) Disk/डिस्क (d) Mesh/मेश

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Network topology is the configuration of a network made up of nodes and connecting lines via sender and receiver.
The most common types are Star, Ring and Mesh.
Disk is a storage medium and not a network topology. It refers to a circular, flat plate used for storing data, such as a hard disk drive (HDD) or a compact disc (CD).

(ii) Network Architecture and OSI Model.

83. Which layer of the ISO OSI reference model delas with determining a route to be followed by a packet?
(a) Physical layer (b) Session layer
(c) Data link layer (d) Network layer
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : The network layer works for the transmission of data from one host to other located in different networks it also takes care of packet routing selection of the shortest path to transmit the packet, from the number of routes available the sender and receiver's IP address are placed in the header by the network layer segment in the network layer is referred to as packet.

84. Choose the correct statements from the following
A. In connection oriented service, the destination address is to be specified only during the setup.
B. Packet sequencing is not guaranteed in connection oriented service.
C. Flooding is method in which every incoming packets is sent out on every outgoing line except the one by which it arrived.
D. Presentation layer performs detection and recovery from errors in the transmitted data.
Choose the correct answer from the options given below :
(a) A, B, & C only (b) B, C & D only
(c) A & C only (d) A, C & D only

UGC NET C.S. December-2024

Ans. (c) : The correct option is (c).
A. In a connection- oriented service the destination address is only needed during the initial connection setup phase, as the established connection ensures data is sent along the same path.

Sarkari Resource

C. Flooding is a routing technique where every incoming packet is forwarded on every outgoing link except the one it arrived on.

B and D is incorrect, Packet sequencing is guaranteed in a connection- oriented service, as the data is delivered in order.

The responsibility of error detection and recovery typically lies with the lower layer like the Data Link Layer, not the presentation layer.

85. Which of the following is not Network device?
निम्नलिखित में से क्या एक नेटवर्क डिवाइस नहीं है ?

(a) Switch/स्विच (b) Router/राउटर
(c) Hub/ हब (d) RAM/आर ए एम

KVS PGT 23.03.2023 (Shift-II)

Ans, (d) : Network devices, also known as network hardware or physical devices that allow computer network to communicate and interact with one another. For example-repeater, hub, bridge, switch, routers, gateway, brovhers and NICs. RAM is a volatile memory where data is stored temporarily.

86. Which layer in OSI model is responsible for the dialogue control and token management?
संवाद नियंत्रण और टोकन प्रबंधन के लिए OSI मॉडल में कौन सी परत जिम्मेदार है?

(a) Transport layer/ट्रांसपोर्ट लेयर
(b) Session layer/सेशन लेयर
(c) Physical layer/फिजिकल लेयर
(d) Network layer/नेटवर्क लेयर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : The session layers primary function is to establish, manage and terminate connections between two application, ensuring that the dialogue between them is properly controlled. This involves controlling the flow of communication, synchronizing messages, and managing access to resources, including token management.

87. Which statement is/are true about the OSI model?
ओ.एस.आई. मॉडल के बारे में कौन-सा/से कथन सत्य है/हैं?

I. Layer 1 through 3 are end-to end
परत 1 से 3 तक एंड टू एंड हैं

II. Layer 4 through 7 are chained
परत 4 से 7 तक चैन्ड हैं

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : The OSI model layers are structured in a hierarchical manner, but they are not strictly "chained" in the sense that each layer directly interacts only with the layer above and below it. The terms "end to end" and "chained" refer to different concepts in the context of network communication.

88. Layer five of OSI model is _____ .
औ.एस.आई. मॉडल की परत पांच _____ है।

(a) Network layer/नेटवर्क लेयर
(b) Transport layer/ट्रांसपोर्ट लेयर
(c) Session layer/सेशन लेयर
(d) Presentation layer/प्रेजेंटेशन लेयर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Session Layer in the OSI model is responsible for the establishment of connections, management of connections, terminations of sessions between two devices. It also provides authentication and security. Protocols used in the Session Layer are NetBIOS, PPTP.

89. Which of the following statement(s)is/are correct regarding Peer-to-Peer Network Architecture?
पीयर-टू-पीयर नेटवर्क आर्किटेक्चर के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. Any peer can access any other peer on the network for sharing and retrieving data/कोई भी पीयर डेटा साझा करने और पुनर्प्राप्त करने के लिए नेटवर्क पर किसी अन्य पीयर तक पहुंच सकता है

II. Needs a dedicated server to provide various services to client
क्लाइंट को विभिन्न सेवाएं प्रदान करने के लिए एक समर्पित सर्वर की आवश्यकता होती है

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : Check both statements about peer-to-peer Network Architecture.
Statement-1– Any peer can access any other peer on the network for sharing and retrieving data. It's correct. In a P2P network, every node cannot as both a client and a server, meaning they can share and access resources directly without hierarchy.
Statement-II– Needs a dedicated server to provide various services to client incorrect. This describes a client server Architecture, not a peer-to-peer network. P2P does not require a dedicated server.

90. Which of the following statement(s) is/are true regarding OSI layers in computer networks?
कंप्यूटर नेटवर्क में ओ.एस.आई. परतों के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. The session layer provides means of establishing, maintaining and terminating a session between two end users.
सेशन परत दो उपयोगकर्ताओं के बीच सत्र को स्थापित करने, बनाए रखने और समाप्त करने का साधन प्रदान करती है।

II. Data Link layer sets up a logical path between the source and destination computer of the network./डेटा लिंक परत नेटवर्क के स्रोत और गंतव्य कम्प्यूटर के बीच एक तार्किक पथ स्थापित करती है।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

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Ans. (a) :
(i) Statement (I) is true.
The session layer (layer 5) is responsible for establishing, managing and terminating session between two communicating end points in a network.
(ii) Statement (II) is false.
The Data Link layer (Layer 2) is responsible for the node-to-node delivery of data and handles error corrections from the physical layer. It does not set up a logical path between the source and destination, that function is handled by the network layer (Layer 3).

91. Which of the following is not associated with the session layer?
(a) Dialog control
(b) Token management
(c) Semantics of the information transmitted
(d) Synchronization

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Semantics of the information transmitted is not associated with the session layer but associated with presentation layer
So correct option is (c).

92. Which layers of the OSI reference model are host-to-host layers ?
(a) Transport, Session, Presentation, Application
(b) Network, Transport, Session, Presentation
(c) Data-link, Network, Transport, Session
(d) Physical, Data-link, Network, Transport

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : The host-to-host layers main purpose is to shield the upper layer application from the complexity of network.

	Layer	Protocol data unit (PDU)
40St layers	7. Application	Data
	6. Presentation	
	5. Session	
	4. Transport	Segment (TCP) Datagram (UDP)
Media layers	3. Network	Packet
	2. Data link	Frame
	1. Physical	Bit

So the answer is (a).

93. Which of the following layer of OSI Reference model is also called End-to-End layer?
(a) Network layer (b) Datalink layer
(c) Session layer (d) Transport layer

UGC NET C.S. November-2017 (Paper-II)

Ans : (d) Transport layer is the layer in the OSI model responsible for end to end communication over a network. It provides logical communication between application a layered architecture of protocols and other network component.

94. Rearrange the following sequence in the context of OSI Layers:
A. Transforming the raw bits in the form of frame for transmission
B. Transmission of raw bits over communication channel
C. Handling user interfaces
D. Control and monitoring of subnet
E. Transmission data through connection oriented or connection less using datagrams
Choose the correct answer from the options given below :
(a) A-B-C-D-E (b) B-C-D-E-A
(c) B-A-D-E-C (d) A-B-D-E-C

UGC NET C.S. December-2022

Ans. (c) : Sequence in the context of OSI layers–
(1) Physical layer
The function of the Physical Layer
- Bit synchronization
- Bit rate control
- Physical topologies
- Transmission Mode

(2) Data Link Layer
- Framing
- Physical Addressing
- Error Control
- Flow Control
- Access Control

(3) Network Layer
The function of the network layer
- Routing
- Logical Addressing

(4) Transport Layer
The function of Transport Layer
- Segmentation and Reassembly
- Service Point Addressing

(5) Session Layer
The function of Session Layer
- Synchronization
- Dialog Controller

(6) Presentation Layer
The function of Presentation Layer
- Translation
- Encryption / Decryption
- Compression

(7) Application Layer
The function of Application Layer
- Network Virtual Terminal
- FTAM
- Mail Services
- Directory Services

So option (c) is correct answer.

95. The full from of ICANN is
(a) Internet Corporation for Assigned Name and Numbers
(b) Internet Corporation for Assigned Number and Name
(c) Institute of Corporation for Assigned Name and Numbers
(d) Internet Connection for Assigned Name and Numbers

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (a) : ICANN → Internet corporation for assigned Names and Numbers.

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96. In OSI Model match the following :
OSI मॉडल में निम्न की जोड़ी बनाये:

A.	**Physical Layer फिजिकल लेयर**	**1.**	**Dialog control & synchronization डायलॉग कण्ट्रोल एवं सिंक्रोनाइजेशन**
B.	**Data Link Layer डाटा लिंक लेयर**	**2.**	**Synchronization of bits and transmission mode सिंक्रोनाइजेशन ऑफ बिट्स, ट्रांसमिशन मोड**
C.	**Session layer सेशन लेयर**	**3.**	**Translation and Compression ट्रांसलेशन एवं कम्प्रेशन**
D.	**Presentation layer प्रेजेंटेशन लेयर**	**4.**	**Error Control and Framing/एरर कण्ट्रोल एवं फ्रेमिंग**

(a) A-2, B-4, C-1, D-3
(b) A-1, B-3, C-2, D-4
(c) A-3, B-1, C-4, D-2
(d) A-4, B-2, C-3, D-1

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Physical Layer– Synchronization of bits and transmission mode
- The Physical Layer is responsible for bit level transmission over physical mediums. It defines hardware specifications, transmission modes (simplex, half-duplex, full-duplex) and synchronization of bits (ensuring sender & receiver are aligned in terms of clock timing)

Data Link Layer– Error control and Framing The Data Link Layer ensures reliable transmission over a physical link. It handles framing (organizing data into frames) and error control (Detecting and correcting errors using checksums, CRC etc).
Session Layer– Dialog control & Synchronization
The Session Layer manages sessions (Connections) between applications. It provides dialog control (Who sends and receives at what time) and synchronization (placing checkpoints) to resume data transfer after a crash.
Presentation Layer– Translation & Compression
The Presentation Layer deals with data format translation (e.g., converting between different character encoding schemes), encryption/decryption, and data compression to reduce transmission size.

97. In which layer of OSI model does the repeater work?
रिपीटर OSI मॉडल की किस परत पर काम करता है?
(a) Physical Layer/फिजिकल परत
(b) Data Link Layer/डाटा लिंक परत
(c) Network Layer/नेटवर्क परत
(d) Transport Layer/ट्रांसपोर्ट परत

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : A repeater is a network device used to regenerate & amplify signals in a network. Over long distances, digital or analog signals can weaken (a phenomenon called attenuation) A repeater boosts these signals so they can travel further without degradation. The physical layer (Layer 1) of the OSI model is responsible for the transmission & reception of raw bit streams over a physical medium (like cables, fiber optics etc).

98. In the OSI model, which layer is responsible for routing and forwarding data packets?
OSI model में, कौन-सी परत डेटा पैकेट को रूट करने और अग्रेषित करने के लिए जिम्मेदार है?
(a) Data Link Layer/डेटा लिंक परत
(b) Transport Layer/ट्रांसपोर्ट लेयर
(c) Physical Layer/भौतिक परत
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (e) : In the OSI model, the network layer is responsible for routing and forwarding data packages. This layer performs routing and forwarding to send data packages from one network to another

99. The media access control sublayer resides in which OSI layer?
मीडिया एक्सेस कन्ट्रोल सबलेयर किस OSI लेयर में होता है?
(a) Transport/ट्रान्सपोर्ट (b) Network/नेटवर्क
(c) Physical/फिजिकल (d) Data link/डेटा लिंक

LT. Grade C.S. 2018

Ans. (d) : The Media Access Control (MAC) sublayer resides in the data link layer of the open systems interconnection (OSI) model.
The Data Link Layer is layer 2 of the OSI model. It ensures that data is sent to the correct device on a local network segment. The MAC sublayer is responsible for managing access and permissions for transmitting data between devices.

100. The number of layers in ISO-OSI reference model is
ISO-OSI रेफरेन्स मॉडल में कितनी परतें होती हैं?
(a) 5 (b) 7
(c) 9 (d) 6

LT. Grade C.S. 2018

Ans. (b): OSI stands for 'Open System Interconnection'. It is a seven layers in ISO-OSI reference model is Application, Presentation, Session, Transport, Network, Data Link and Physical layer. Each layer uses a protocol to perform its designated function, for exa - the data link layer uses error detection protocols for error control functions.

101. In the following pairs of OSI reference layers/sublayers and their functionality, the incorrect pair is
OSI रेफरेन्स परत/उपपरत और इसकी कार्यक्षमता के निम्न जोड़े में से बेमेल जोड़ा हैं।
(a) Network layer and routing
नेटवर्क परत और रूटिंग
(b) Data link layer and bit synchronization
डेटा लिंक परत और बिट सिंक्रोनाइजेशन
(c) Transport layer and end-to-end process communication/ट्रांसपोर्ट परत और एन्ड-टू-एन्ड प्रक्रिया संचार
(d) Medium access control sublayer and channel sharing/मध्यम पहुँच नियंत्रण उपपरत और चैनल साझाकरण

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Ans. (b) : One of the main functionality of network layer is routing. So, option (a) is correct.
Bit synchronization is always handled by physical layer of OSI model but not data link layer so, option (b) is incorrect.
End to End process communication is handled by transport layer so, option (c) is correct. MAC sub layer have 3 types of protocols (Random, Controlled and channelized Access).

102. In OSI network architecture, routing is performed by the
OSI नेटवर्क आर्किटेक्चर मे, रूटिंग किसके द्वारा निष्पादित होता है
(a) Network layer./नेटवर्क लेयर
(b) Data link layer./डाटा लिंक लेयर
(c) Transport layer./ट्रांसपोर्ट लेयर
(d) Session layer./सेशन लेयर

KVS PGT C.S. 2013

Ans. (a) : OSI stands for Open System Interconnection is a reference model that describes how information form a software application in one computer move through a physical medium to the software application in another computer. There are the seven OSI layer.
1. Physical layer **2.** Data link layer
3. Network layers **4.** Transport layer
5. Session Layer **6.** Presentation layer
7. Application layer .
The Network layer works for the transmission of data from one host to the other located in different network . It also take care of packet routing i.e. selection of the shortest path to transmit the packet from the number of routes available.

103. Bridge works in which layer of the OSI model ?
ब्रीज OSI मॉडल के किस लेयर में कार्य करता है ?
(a) Application Layer/ऐप्लीकेशन लेयर
(b) Transport Layer/ट्रांसपोर्ट लेयर
(c) Network layer/नेटवर्क लेयर
(d) Datalink Layer/डाटालिंक लेयर

KVS PGT C.S. 2013

Ans. (d) : Bridge works in the data link layer of the OSI model. The data link layer is the protocol layer in a program that handles the moving of data into and out of a physical link in a network.

104. Which layers of the OSI model are host-to-host layers ?
OSI मॉडल की कौन सी परत मेजबान-मेजबान परत हैं ?
(a) Transport, Session, Presentation, Application./परिवहन, सत्र, प्रस्तुति, प्रयुक्ति
(b) Network, Transport, Session, presentation. नेटवर्क, परिवहन, सत्र, प्रस्तुति
(c) Datalink, Network, Transport, session. डेटालिंक, नेटवर्क, परिवहन, सत्र
(d) Physical, Datalink, Network, Transport. भौतिक, डेटालिक, नेटवर्क, परिवहन

KVS PGT C.S. 2013

Ans. (a) : In OSI model physical, Datalink and network layer are hop to hop layer when Transport, Session, Presentation and Application layer are end to end or host to host layer.

105. The device bridge is used at _____ layer of OSI reference model.
डिवाइस ब्रिज का उपयोग OSI रेफरेंस मॉडल की लेयर पर किया जाता है?
(a) Data link/डाटा लिंक
(b) Network/नेटवर्क
(c) Transport/ट्रांसपोर्ट
(d) Application/एप्लीकेशन

KVS PGT C.S. 2018

Ans. (a) : A bridge in a computer network is a device used to connect multiple LAN, together with a larger Local Area Network (LAN). The bridge is a physical or hardware device but operates at the OSI model's data link layer and is also known as a layer of two switches.

106. OSI stands for
OSI का पूरा नाम है
(a) Open Soft ID
(b) Open System ID
(c) Open System Interconnection
(d) Open Source Information

Bihar STET C.S. 2020

Ans. (c) : OSI stands for Open System Interconnection. It is a conceptual framework and reference model that defines how different communication systems can exchange data by breaking down the process into seven obstruct layers.

107. Which of the following is not a network edge device?
निम्न में से कौन एक नेटवर्क एज डिवाइस नहीं है?
(a) PC (b) Smartphones
(c) Servers (d) Switch

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : Network Edge Devices are devices that are located at the edge of the network and directly interact with users. These devices generate, receive or process data.
Example– PC, Smart Phones, Servers, etc.
• Switch is a network internal device that forwards data traffic within the network.

108. Data transmission using multiple pathways simultaneously is known as:
एक साथ कई मार्गों का उपयोग करके डेटा ट्रांसमिशन को क्या कहा जाता है?
(a) Parallel Transmission/ पैरेलल ट्रांसमिशन
(b) Serial Transmission/सीरियल ट्रांसमिशन
(c) Duplex Transmission/डुप्लेक्स ट्रांसमिशन
(d) Half-duplex Transmission/हाफ-डुप्लेक्स ट्रांसमिशन

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : Data transmission using multiple pathways simultaneously is known as Parallel Transmission. In parallel transmission, data bits are sent concurrently over separate wires or channels, allowing for faster data transfer over short distances.

Sarkari Resource

(iii) Data Communication Fundamental and Techniques

109. What is the maximum data rate of Bluetooth version 5.0?
ब्लूटूथ संस्करण 5.0 की अधिकतम डेटा दर क्या है?
(a) 1 Mbps
(b) 2 Mbps
(c) 5 Mbps
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : Bluetooth version 5.0 has a maximum data rate of 2 Mbps. This means Bluetooth 5.0 can transmit data at a maximum rate of 2 Mbps which is twice the speed compared to Bluetooth 4.2.

110. The maximum data rate that a medium of 3 KHz and signal to noise ratio of 30 dB is roughly
अधिकतम डेटा दर जो 3 किलो हर्टज का एक माध्यम और 30 डी बी के सिग्नल/न्वायज पर अर्जित हो सकता है
(a) 10 kbps (b) 100kbps
(c) 3 kbps (d) 30 kbps

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (d) : To find the maximum data rate for a 3KHz channel with a 30 dB Signal to Noise Ratio (SNR)m we apply the Shannon - Hartley theorem $C = B \log_2 (1 + SNR)$. Here, bandwidth (B) is 3 KHZ = 3000 HZ. The SNR of 30dB converts to linear scale ; $10^{30}/10 = 10^3 = 1000$ thus, 1 + SNR = 1001, and $\log_2(1001) \approx 10$ (Since 2^{10} - 1024 ≈ 1001), calculating : C = 3000.10 = 30000 bbs = 30kbps. For precision, $\log_2 (1001) \approx 9.966$, so, C ≈ 3000 9.966 ≈ 29,898 bps, still closest to 30 kbps.

111. An analog signal has a bit rate of 6000 bps and a baud rate of 2000 baud. How many data elements are carried by each signal element?
(a) 0.336 bits/baud
(b) 3 bits/baud
(c) 120,00,000 bits/baud
(d) None of the above

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because the analog signal 2000 bauds or element in one second.
- To send this 6000 bits are transmitted second
- Thus 6000/2000 = 3 bits/baud is transmitted

112. In which of the following, the density of the core remains constant from the center to the edges?
निम्नलिखित में से किस में कोर की सघनता केन्द्र..... किनारों तक एकसमान रहती है?
(a) Single mode fiber/एकल-मोड फाइबर
(b) Multimode step-index fiber
बहुमोड स्टेप-इन्डैक्स फाइबर
(c) Multimode graded-index fiber
बहुमोड ग्रेडेड-इन्डैक्स फाइबर
(d) Single mode step-index fiber
एकल-मोड स्टेप-इन्डैक्स फाइबर

KVS PGT C.S. 2018

Ans, (b) : Multimode step-index fiber is the most widely used in optical fiber communication. In this mode, the light ray is propagated using the principle of total internal reflection. Since the core index of refraction is to much higher than the cladding index of refraction, so the light enters at less than the critical angle is guided along the fiber. The density of the core remains constant from the center to the edges.

113. Consider a discrete memory less channel and assume that H(x) is the amount of information per symbol at the input of the channel; H(y) is the amount of information per symbol at the output of the channel; H(x | y) is the amount of uncertainty remaining on x knowing y; and I(x; y) is the information transmission.
Which of the following does not define the channel capacity of a discrete memoryless channel?
(a) $\max_{p(x)} I(x ; y)$
(b) $\max_{p(x)} [H(y) - H(y| x)]$
(c) $\max_{p(x)} [H(x) - H(x| y)]$
(d) $\max_{p(x)} H(x| y)$

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) The channel capacity of a discrete memory less channels is defined by :
C = max I (x; y) p (x)
= max [H(y) – H(y| x)] p(x)
in ax [H(x) – H(x)] p (x)

114. The bit rate of a signal is 3000 bps. If each signal unit carries 6 bits, the baud rate of the signal is _____.
(a) 500 baud/sec (b) 1000 baud/sec
(c) 3000 baud/sec (d) 18000 baud/sec

UGC NET C.S. June-2014 (Paper-II)

Ans : (a) Because, Bit rate = Numbers of bits per unit time
Baud rate = Number of signal element or signal units per unit time .
Here, baud rate = (1/6)th bit rate
Therefore, baud rate = (1/6) * 3000 = 500 baud /sec

115. Assume that we need to download text documents at the rate of 100 pages per minute. A page is an average of 24 lines with 80 characters in each line and each character requires 8 bits. Then the required bit rate of the channel is .....
(a) 192 mbps (b) 512 mbps
(c) 1.248 mbps (d) 1.536 mbps

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) Downloading rate = 100 page
⇒ 100 × 24 × 80 × 8 = 1536000 bit rate per second
⇒ 1.536 mbps

116. In Huffman's coding, if the symbol A has probability 0.3, B has probability 0.15, C has probability 0.1, D has probability 0.25 and E has probability 0.2, then the minimum number of bits required to represent all the symbols together is

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यदि प्रतीक A की प्रायिकता 0.3, प्रतीक B की प्रायिकता 0.15, C की 0.1, D की 0.25 एवं E की 0.2 है, तो हफमैन के कोड में सभी प्रतीकों को एकसाथ निरूपित करने के लिए न्यूनतम कितने बिट की आवश्यकता होगी?

(a) 14 (b) 11
(c) 12 (d) 15

LT. Grade C.S. 2018

Ans, (c) : To determine the minimum number of bits required to represent all the symbols using Huffman's coding, we need to build a Huffman tree based on the given probabilities.

Now, construct the Huffman tree:

(i) Combine the smallest probabilities (C and B):

$0.1 + 0.15 = 0.25$

(ii) Combine this new node with the next smallest probability (E):

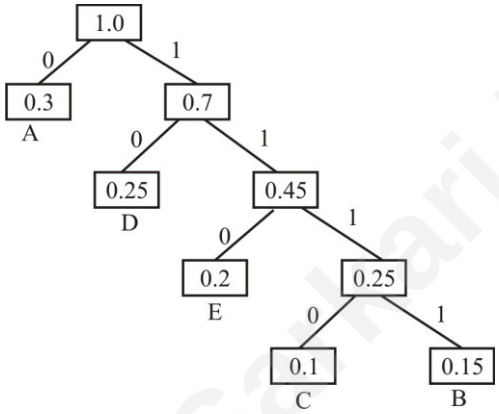
$0.25 + 0.2 = 0.45$

(iii) Combine this new node with the next smallest probability (D):

$0.45 + 0.25 = 0.7$

(iv) Finally, combine this new node with the remaining probability (A):

$0.7 + 0.3 = 1.0$



Now, Assign codes to each symbol:

Symbol	Code	Code Length
A	0	1
B	1111	4
C	1110	4
D	10	2
E	110	3

Calculate the total number of bits:

$$\text{Total Bits} = \sum_i \text{Probability of symbol} \times \text{Length of code}$$

$= 0.3 \times 1 + 0.15 \times 4 + 0.1 \times 4 + 0.25 \times 2 + 0.2 \times 3$

$= 0.3 + 0.6 + 0.4 + 0.5 + 0.6$

$= 2.4$

Since there are 5 symbols, multiply by the total number of symbols to get:

Minimum Total Bits $= 2.4 \times 5$

$= 12$

117. If link transmits 4000 frames per second and each slot has 8 bits, the transmission rate of circuit of this TDM is .....

(a) 64 Kbps (b) 32 MbpS
(c) 32 Kbps (d) 64 MbpS

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Ans : (c) Transmission rate = frame rate * no of bits in a slot. Frame rate = 4000 frames per second # of bits in a slot = 8 bit TDM = 4000 * 8 bps = 32 Kbps. So, option (C) is correct.

118. Given the following statements :

(1) Frequency division multiplexing is a technique that can be applied when the bandwidth of a link is greater than combined bandwidth of signals to be transmitted.
(2) Wavelength division multiplexing (WDM) is an analog multiplexing technique to combine optical signals.
(3) WDM is a digital multiplexing technique.
(4) TDM is digital multiplexing technique.

Which of the following is correct?

(a) (1), (2), (3) and (4) are true
(b) (1), (2), (3) and (4) are false
(c) (1), (2) and (4) are false (3) is true
(d) (1), (2) and (4) are true (3) is false

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Ans : (d) Multiplexing can be of three types
(1) Frequency division multiplexing (FDM) : analog signal
(2) Wave division multiplexing (WDM) : analog signal (used in optical fiber)
(3) Time division multiplexing (TDM) : digital signals : : In TDM, a single frame is made with interleaved digital signals from n devices. In TDM, the transmission rate of the multiplexed path is greater than the sum of transmission rates of all sources.

119. A pure ALOHA network transmits 200 bit frames using a shared channel with 200 Kbps bandwidth. If the system (all stations put together) produces 500 frames per second, then the throughput of the system is .....

(a) 0.384 (b) 0.184
(c) 0.286 (d) 0.586

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Ans : (b) Throughput for pure ALOHA is $s = G \times e^{-2}$, where G is the load factor and take e = 2.71
The throughput is maximum, when G = 1/2 which is S_{max} = 0.184 Transmission time of rate = 200 bits/200 kbps = 1 ms
Now, system as taken all together produce 500 frames per seci ; e
Now, as load factor G = 1/2, then throughput is maximum for pure aloha
So answer is (b)

120. _____ mode allows transmission in both directions simultaneously.
_____ मोड दोनों दिशाओं में एक साथ संचारण की अनुमति देता है।

(a) Duplex/डुप्लेक्स
(b) Half duplex/हाल्फ डुपलेक्स

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(c) Simplex/सिम्पलेक्स
(d) Half simplex/हाल्फ सिम्पलेक्स

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : Simplex– One-way communication only (e.g., keyword to computer).
Half Duplex– Communication in both directions, but only one at a time (e.g., walkie-talkie).
Duplex (Full Duplex)– Communication in both directions simultaneously (e.g., telephone).

121. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interference?
विद्युत् हस्तक्षेप के सम्पर्क में आने वो कम्प्यूटर नेटवर्क में डेटा ले जाने के लिए किस प्रकार का ट्रान्समिशन माध्यम सबसे उपयुक्त है?
(a) Unshielded twisted pair
(b) Optical fiber
(c) Coaxial cable
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : In computer network, transmission medium is the physical medium which is used to transfer data from one place to another. It provides a type of connectivity through which data is transferred. The major transmission medium are wireless transmission and cable transmission optical fiber transmission medium is most suitable for carrying data in computer networks that come under electrical interference.

122. A high-speed communication equipment typically would not be needed for
_____ के लिए आमतौर पर उच्च-गति संचार उपकरण की आवश्यकता नहीं होगी।
(a) E-mail/ई-मेल
(b) Transferring large volume of data
बड़ी मात्रा में डेटा स्थानांतरित करने
(c) Supporting communication between nodes in a LAN/एक लैन (LAN) के नोडों के बीच संचार का समर्थन करने
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a) : E-mail, while often consisting of text and attachments, generally doesn't require high speed communication. The relatively small size of emails and the intermittent nature of sending then means that standard internet speeds are sufficient. High speed equipment is typically reserved for large data transfers or real time applications.

123. Optical fiber uses reflection to guide light through a channel, in which angle of incidence is _____ the critical angle.
(a) Equal to (b) Less than
(c) Greater than (d) Less than or equal to

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) If the **angle of incidence** is equal to the critical angle, the light beam bends along the interface. When a **beam of light travels** through media of two different densities, if the angle of incidence is greater than the critical angle, reflection occurs.

124. In a fast Ethernet cabling, 100 base-TX uses ..... cable and maximum segment size is .....
(a) Twisted pair, 100 meters
(b) Twisted pair, 200 meters
(c) Fibre optics, 1000 meters
(d) Fibre optics, 2000 meters

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (a) **Fast Ethernet:-** Cabling, 100 Base- Tx uses twisted pair cable and maximum segment size is 100 metres.

125. A network with bandwidth of 10 Mbps can pass an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
(a) 1 Mbps (b) 2 Mbps
(c) 10 Mbps (d) 12 Mbps

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) 12000 * 10000/60 = 2 Mbps
So answer is option (b)

126. Consider a noiseless channel with a bandwidth of 5000Hz transmitting a signal with two signal levels. The maximum bit rate is:
(a) 2500 bps (b) 10000 bps
(c) 5000 bps (d) 20000 bps

UGC NET C.S. December-2024

Ans. (b) : The maximum bit rate for a noiseless channel can be calculated using NY Quist's formula.

$$\text{Bit rate} = 2 \times B \times \log_2 (L)$$

Where:-
- B = Bandwidth of channel (Hz)
- L = Number of signal levels
- The factor 2 accounts for the maximum signaling rate is a noiseless channel.

Given:-
- B = 5000 Hz
- L = 2 (Two signal levels)

Bit rate = $2 \times 5000 \times \log_2 (2)$
Since $\log_2(2) = 1$
Bit rate = $2 \times 5000 \times 1 = 10000$ bps

127. Which data transmission technique in a computer network required dedicated communication channel?
कंप्यूटर नेटवर्क में किस डेटा ट्रांसमिशन तकनीक के लिए समर्पित संचार चैनल की आवश्यकता होती है?
(a) Circuit Switching /सर्किट स्विचिंग
(b) Packet Switching/पैकेट स्विचिंग
(c) Message Switching/मैसेज़ स्विचिंग
(d) Frame Switching/फ्रेम स्विचिंग

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : Circuit switching requires a dedicated communication channel between the sender & receiver for the duration of the communication session. This technique is commonly used in traditional telephone networks.

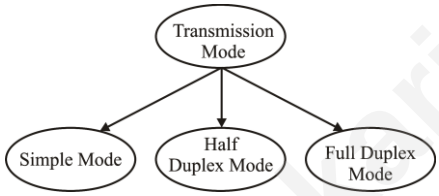
Sarkari Resource

Comparison Table				
Technique	**Dedicated Channel**	**Real-time Support**	**Efficiency**	**Example use**
Circuit Switching	Yes	Yes	Low (idle time)	Telephone networks
Packet Switching	No	Limited	High	Internet data transfer
Message Switching	No	No	Medium	Early network system
Frame Switching	No	Yes (to some extent)	High	Modern LANs (ATM etc).

128. ____ mode allows transmission in either direction but no simultaneously.
_____ मोड किसी भी दिशा में ट्रांसमिशन की अनुमति देता है लेकिन एक साथ नहीं।
(a) Duplex/डुप्लेक्स
(b) Half duplex/हाल्फ डुप्लेक्स
(c) Simplex/सिम्पलेक्स
(d) Half-simplex/हाल्फ सिम्पलेक्स

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : Transmission modes– Transmission modes means trans furring data between two devices it is also known as a communication mode and networks are designed to allow communication to occur between individual devices that are interconnected.



Half Duplex Mode– in half duplex mode, each station can both transmit and receive but not at the sometime when one device is sending the other can only receive and vice versa. Half-duplex mode is used in cases is no need for communication in both directions at the sometime the entire capacity of the channel can be utilized for each direction.
Example– walkie-talkie in which massage is sent one at a time and massage are sent in both directions.

129. _____is a device that combines a large number of low speed transmission lines into a high speed line.
_____ एक उपकरण है जो बड़ी संख्या में कम गति संचरण वाली लाइनों को उच्च गति की एक लाइन में जोड़ता है।
(a) Multiplexer/मल्टीप्लेक्सर
(b) Demultiplexer/डीमल्टीप्लेक्सर
(c) Repeater/रिपीटर
(d) Switch/स्विच

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : A multiplexer is a device that combines multiple low speed transmission lines or signals into a single high speed line. This allows more efficient use of the communication medium by aggregating bandwidth. Demultiplexer does the opposite it separates a high speed line into multiple low speed lines.
Repeater regenerates signals to extend transmission distance.
Switch directs data packets to specific devices within a network.

130. Which of the following is a type of transmission mode?/निम्नलिखित में से कौन सा ट्रांसमिशन मोड का एक प्रकार है?
I. Simplex mode/सिम्प्लेक्स मोड
II. Half simplex mode/हाफ सिम्प्लेक्स मोड
III. Duplex mode/डूप्लेक्स मोड
(a) I and II (b) II and III
(c) I and III (d) I, II and III

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Transmission modes refer to now data is transmitted between two devices. The types are-
Simplex Mode– This is a one-way communication. Think of a radio broadcast- information flows in only one direction, from the transmitter to the receiver.
Half duplex mode– In this made, communication can happen in both directions, but only one at a time. A walkie-talkie is a classic example. One person speaks. Then says "Over" to indicate they are done, and then the other person can respond.
Duplex mode– This allows for simultaneous two way communication. A telephone conversation is a good example both parties can talk and listen at the same time.

131. The loss of energy as signal propagates outward is known as _____.
सिग्नल के बाहर की ओर प्रसारित होते हुई ऊर्जा की हानि को _____ के रुप में जान जाता है।
(a) Noise/रव (नॉइज़) (b) Distortion/विरुपण
(c) Attenuation/क्षीणन (d) Scattering/प्रकीर्णन

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Attenuation refers to the gradual loss of signal strength (energy) as it propagates through a medium, such as a cable, fiber, or free space.

132. Match the following.
निम्नलिखित को सुमेलित करें ।

A.	**Repeater/रिपीटर**	**i.**	**Network layer नेटवर्क लेयर**
B.	**Bridge/ब्रिज**	**ii.**	**Transport layer ट्रांसपोर्ट लेयर**
C.	**Routers/राउट**	**iii.**	**Data link layer डेटा लिंक लेयर**
D.	**Gateway/गेटवे**	**iv.**	**Physical layer फिजिकल लेयर**

(a) A-ii, B-i, C-iii, D-iv
(b) A-iii, B-ii, C-i, D-iv
(c) A-iv, B-iii, C-i, D-ii
(d) A-i, B-iii, C-iv, D-ii

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : The correct match–

A.	Repeater	**iv.**	Physical layer
B.	Bridge	**iii.**	Data link layer
C.	Routers	**i.**	Network layer
D.	Gateway	**ii.**	Transport layer

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133. A connection that allows traffic either way, but only one way at a time is known as _____.
एक कनेक्शन जो किसी भी तरह से ट्रैफिक की अनुमति देता है, लेकिन एक साथ समय में केवल एक ही तरह से देता है, _____के रुप में जान जाता है ?
(a) Full duplex/फूल डुप्लैक्स
(b) Half duplex/हाल्फ डुप्लैक्स
(c) Duplex/डुप्लैक्स
(d) Simplex/सिम्पलैक्स

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : Half duplex allows communication in both direction, but not simultaneously. Only one device can transmit at a time, while the other receives, and then they can switch roles.
Full duplex allows simultaneously two way communication (Both devices can transmit and receive at the same time).
Duplex is a general term that can refer to either half or full duplex, but is not specific. Simplex allows communication in only one direction.

134. Which of the following is not an example of Guided transmission media?
निम्नलिखित में से कौन सा गइडेड ट्रांसमिशन मीडिया का एक उदाहरण नहीं है?
(a) Fiber optic/फाइबर ऑप्टिक
(b) Infrared wave/इन्फ्रारेड वेव
(c) Twisted pair/टिव्स्टिड पेयर
(d) Coaxial cable/कोएकसिअल केबल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : Guided transmission media means the signal travels through a physical medium (like wire, cable, or fiber)
Unguided media means signals propagate through the air (Wireless).
Fiber optic - guided
Twisted pair - guided
Coaxial cable - guided
Infrared ware -unguided.

135. Which of the following statement(s) is/are true regarding Data Communication?
डेटा संचार के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. Half duplex communication is multidirectional communication.
हाफ डुप्लेक्स संचार मल्टीडायरेक्शनल संचार है।
II. Full duplex communication is bidirectional communication.
फुल डुप्लेक्स संचार बाईडायरेक्शनल संचार है।
(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : Statement (I) is false.
A half-duplex system is a communication system that allows data transmission in both direction but not at the same time.
(II) Full duplex communication is bidirectional communication So, the correct answer is option (ii) only II.

136. Which of the following statement(s) is/are true regarding switching techniques in Data communication?
डेटा संचार में स्विचिंग तकनीक के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. In circuit switching a dedicated path is identified between the sender and the receiver.
सर्किट स्विचिंग में प्रेषक तथा रिसीवर के बीच एक समर्पित पथ की पहचान की जाती है।
II. In packet switching each information to be transmitted between sender and receiver is broken down into smaller packets.
पैकेट स्विचिंग में प्रेषक तथा रिसीवर के बीच संचारित की जाने वाली प्रत्येक सूचना को छोटे पैकेटों में विभाजित किया जाता है।
(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Both statement are true–
(i) In circuit switching a dedicated communication path is established between the sender and the receiver for the duration of the communication session this path remains reserved and is exclusively used by the two communicating parties until the session ends.
(ii) In packet switching, the data to be transmitted is broken down into smaller packet. Each packet is then sent indendently over the network .

137. Which of the following connector is used by Unshielded Twisted Pair (UTP)?
अनशील्डेड ट्विस्टेड पेयर (UTP) द्वारा निम्नलिखित में से किस कनेक्टर का उपयोग किया जाता है?
(a) RJ45 (b) RJ55
(c) RS45 (d) RS55

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : UTP stands for Unshielded Twisted Pair. RJ45 is the standard connector for UTP cables, commonly used in Ethernet Networking.

138. Suppose a 100 km long cable has a speed of 1.536 Mbps. The signal speed in cable is 2/3 speed of the light. What will be the number of bits in the cable?
मान लें कि 100 km लंबे केबल की गति 1.536 Mbps है। केबल में सिग्नल की गति प्रकाश की गति पर 2/3 है। केबल में बिट्स की संख्या क्या होगी?
(a) 500 (b) 768
(c) 307 (d) 400

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Cable length = 100 km = 100×10^3 m
Data rate = 0.536 mbps = 1.536×10^6 bits per sec
Signal speed = 2/3 times the speed of light
Speed of light (c) = 3×10^8 m/m.
The number of bits in the cable is the product of the data rate and the propagation delay. The propagation delay (tp) is the time it takes for a signal to travel the length of the cable.
Number of bits in the cable = R × tp

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First, compute the signal speed.

$$\mu = \frac{2}{3} \times c = \frac{2}{3} \times 3 \times 10^8 = 2 \times 10^8 \text{ m/s}$$

Now, compute the propagation delay (tp).

$$tp = \frac{2}{u} = \frac{100 \times 10^3}{2 \times 10^8} = \frac{10^5}{2 \times 10^8} = \frac{1}{2 \times 10^3} = 0.5 \times 10^{-3}$$

= 0.5 milliseconds.

Now, compute the number of bits.

Number of bits = R × tp = $(1.536 \times 10^6) \times (0.5 \times 10^{-3})$

Calculate–

$1.536 \times 10^6 \times 0.5 \times 10^{-3} = 1.536 \times 0.5 \times 106^{-3} = 0.768 \times 10^3 = 768$

So, the number of bits in the cable is 768.

139. State whether True of False?
बताएं कि क्या सच है या गलत?

I. Radio waves can penetrate the walls and used for multicast communication.
रेडियो तरंगें दीवारों में प्रवेश कर सकती हैं और मल्टीकास्ट संचार के लिए उपयोग की जा सकती हैं।

II. Infrared signals are used for short range communication.
इन्फ्रारेड सिग्नल का उपयोग शॉर्ट रेंज संचार के लिए किया जाता है।

(a) I-True, II-False/ I-सत्य, II-असत्य
(b) I-True, II-True/ I-सत्य, II-सत्य
(c) I-False, II-True/ I-असत्य, II-सत्य
(d) I-False, II-False/ I-असत्य, II-असत्य

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Evaluate each statement–
(I) Radio waves can penetrate walls and be used for multicast communications.
Radio waves are a type of electromagnetic radiation with long wavelengths, which allow them to penetrate solid objects like walls to some extent. They are indeed used for multicast communications so, this is True.
(II) Infrared signals are used for short range communication.
Infrared signals are electromagnetic waves with shorter wavelengths than radio waves. There are commonly used for short range communication, such as in TV remotes, infrared data association for device to device data transfer, and some wireless keyboards/mice. IR signals typically require line of sight and have a limited range. So, this is True.
So, both statements are True.

140. Which of the following is correct formula for calculating wavelength (λ) of a signal in the data communication?

(a) λ = propagation speed + frequency
(b) λ = propagation speed × period
(c) λ = propagation speed + period
(d) λ = propagation speed × frequency

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : The basic relation in data communication

$$\lambda = \frac{\text{Propagation speed}}{\text{Frequency}}$$

And since period (T) = 1/Frequency, we can also write.
λ = propagation speed × period.

141. In periodic analog signals, which of the following is NOT a parameter to describe a sine wave in data communication?

(a) Wavelength (b) Phase
(c) Frequency (d) Peak amplitude

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : In data communication, a sine wave is typically described by parameters such as peak amplitude, frequency, and phase. These parameters are fundamental to characterizing the signal.
Peak amplitude– The maximum value of the signal.
Frequency– The number of cycles per second, measured in Hertz (Hz).
Phase– The position of the waveform relative to a reference point in time, measured in degrees or radians.
So, the correct answer is wavelength.

142. Which of the following is/are restriction (s) in classless addressing?

(a) The number of addresses needs to be power of 2
(b) The mask needs to be include in the address to define the block
(c) The starting address must be divisible by the number of addresses in the block
(d) All of the above

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : In class addressing : (1) the number of addresses needs to be power of 2
(2) The mask needs to be included in the address to define the block
(3) The starting address must be divisible by the number of addresses in block
So answer is (d)

143. A terminal multiplexer has six 1200 bps terminals and 'n' 300 bps terminals connected to it. If the outgoing line is 9600 bps, what is the value of n?

(a) 4 (b) 8
(c) 16 (d) 28

UGC NET C.S. June-2014 (Paper-III)

Ans. (b) : The outgoing line = 9600 pbs terminal multiplexer has six 1200 pbs so total traffic (1200 * 6) = 7200 bps and then are n 300 bps terminals
So 7200 + 300 * n = 9600 ⇒ 300n = 2400 ⇒ n = 8
So the answer is (b).

144. In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Mbps?

(a) 10 bps (b) 100 bps
(c) 1000 bps (d) 10000 bps

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Receiver clock is 0.1% faster than sender clock. At 1 mbps bits send by sender = 1000000 receiver clock is 0.1% faster than sender clock so extra

$$\text{bit received by receiver} = 1000000 \times \frac{0.1}{100}$$

= 1000 bits
total bits received by receiver = 1001000 bits
Extra bit per second received by receiver = 1000 bps
Hence option (c) 1000 bps is the correct.

Sarkari Resource

145. Which of the following routing technique/ techniques is/are used in distributed systems?

(A) Fixed Routing
(B) Virtual Routing
(C) Dynamic Routing
Code :

(a) (A) only (b) (A) and (B) only
(c) (C) only (d) All (A), (B), (C)

UGC NET C.S. November-2017 (Paper-III)

Ans : (a) Distributed system uses fixed routing technique dynamic & virtual routing are not used is distributed system.

146. Quadrature Amplitude Modulation means changing both :

(a) Frequency and phase of the carrier
(b) Frequency and amplitude of the carrier
(c) Amplitude and phase of the carrier
(d) Amplitude and wavelength of the carrier

UGC NET C.S. November-2017 (Paper-III)

Ans : (c) Quadrature amplitude modulation means changing both amplitude and phase of the Carrier.

147. If a file consisting of 50,000 characters takes 40 seconds to send, then the data rate is .....

(a) 1 kbps (b) 1.25 kbps
(c) 2 kbps (d) 10 kbps

UGC NET C.S. November-2017 (Paper-III)

Ans : (d) 50,000 character takes 40 seconds i.e. 50,000*8 = 4 second (each character is of 8 bits)
In one second → (50,000*8)/40 bits/sec
10,000 bits/sec i.e. 10 kbps

148. Suppose transmission rate of a channel is 32 kbps. If there are '8' routes from source to destination and each packet p contains 8000 bits. Total end to end delay in sending packet *P* is _____ .

(a) 2 sec (b) 3 sec (c) 4 sec (d) 1 sec

UGC NET C.S. August-2016 (Paper-III)

Ans : (a) Each router takes

$$= \frac{8000}{32000} \text{ second}$$

All 8 routes takes

$$= 8 \times \left(\frac{8000}{32000} \right) = 2 \text{ second}$$

149. Consider a CSMA/CD network that transmits data at the rate of 100 Mbps over a 1 kilometer cable with no repeater. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(a) 8000 (b) 16000
(c) 12000 (d) 20000

UGC NET C.S. Re-Exam- June 2024

Ans. (d) : For collision to be detected , the frame size should be such that the transmission time of the frame should be greater than twice the propagation delay (So, before greater)
Data should be transmitted at the rate of Mbps.
Transmission Time>=2* Propagation Time
$1250*8/(100*10^6) <= 2*$ Length/signal_speed
signal_speed<= $(2*10^3*100*10^6)/(1250*8)$
$= 2*8*(10^3)$km/sec = 20000

150. Baud is:
बॉड है:

(a) Number of bits transmitted per unit time
नंबर ऑफ बिट्स ट्रांसमिटेड पर यूनिट टाइम
(b) Number of bytes transmitted per unit time
नंबर ऑफ बाइटस ट्रांसमिटेड पर यूनिट टाइम
(c) Number of signal changes per second
नंबर ऑफ सिग्नल चेंज्स पर सेकंड
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : Baud rate refers to the number of signal units (Symbols) transmitted per second in a communication channel.
→ Each signal change, or symbol, can represent one or more bits depending on the modulation technique.
Baud rate # bit rate, unless each signal carries exactly one bit.
For example– if each symbol represents
2bits, then,
Baud rate = 1000
Bit rate = 2000 bits per second So, baud is not about the number of bits or bytes, but about how many times he signal changes (modulates) per second.

151. In cycle redundancy check, CRC is:
साइकिल रिडनडेंसी चेकिंग में, सी.आर.सी है–

(a) Divisor/विभाजन (b) Quotient/भागफल
(c) Dividend/भाज्य (d) Remainder/शेषफल

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : Cyclic Redundancy Check (CRC) is an error-detecting code. In CRC, data is divided by a fixed polynomial (divisor) and the remainder of this division is the CRC.
⇒ Data (with added zeros) → Dividend
⇒ Generator Polynomial → Divisor
⇒ Result of division → Quotient (not used)
⇒ CRC → Remainder
This CRC remainder is added to the data before transmission. At the receiver end, the same division is done, and if the remainder is 0, the data is considered error-free.

152. Loss in signal power as light travel down the fiber is called
प्रकाश जब फाइबर से गुजरता है तो सिग्नल के पॉवर में हानि को कहा जाता है?

(a) Attenuation/एटीन्यूएशन
(b) Propagation/प्रोपोगेशन
(c) Scattering/स्कैटरिंग
(d) Interruption/इंटरप्शन

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Attenuation refers to the loss of signal power as light travels through an optical fiber.
→ It happens due to absorption, scattering and other imperfections in the fiber.
→ It is measured in decibels per kilometer (dB/km).
→ The longer the fiber or the power the quality, the more the signal weakens.

Sarkari Resource

153. Four 1 KbPS connections are multiplexed together, if unit is 1 bit, find the duration of a frame-
1 KbPS के चार कनेक्शन एक साथ मल्टीप्लेक्सड किये गए है, यदि यूनिट 1 बिट है तो फ्रेमिंग की अवधि क्या होगी?

(a) 250 μs (b) 1 ms
(c) 1 μs (d) 250 ms

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : In Time Division Multiplexing (TDM)– 1 frame includes 1 bit from each connection so, 1 frame = 4 (since there are 4 connections) Total data rate = 4 bits per frame each connection sends 1000 bits per second 1000 frames per second needed (since 1 bit per frame per connection)
Therefore, frame rate = 1000 frames/sec
Now, duration of one frame = 1/1000 seconds = 0.001 seconds = 1 millisecond (ms)

154. Communication between a computer and a keyboard involves_____ transmission.
कंप्यूटर और कीबोर्ड के बीच संचार में_____ ट्रांसमिशन शामिल होता है।

(a) Simplex (b) Half-dulpex
(c) Full-duplex
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) : Simplex transmission is a type of transmission made. In this data can be sent though only one direction. That is the communication is unidirectional.

155. Wireless transmission of signals can be done via
सिग्नलों का वायरलेस ट्रांसमिशन_____ के माध्यम से किया जा सकता है।

(a) Radio waves (b) Microwaves
(c) Infrared
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d) : Wireless transmission of signals can be done using radio waves, microwaves, and infrared. All three are forms of electromagnetic radiation that can propagate through the air without the need for a physical cable.

156. In TDM, slots are further divided into:
टीडीएम में, स्लॉट को_____में विभाजित किया गया है।

(a) Frames
(b) Packets
(c) Bits
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) : In TDM the slots can be divided into frames. TDM is the short form of time division multiplexing. It is a techniques of combining several low rate channels into one high rate channel. Several channels can use the maximum bandwidth for a given time slot.

157. What type of transmission is involved in communication between a computer and a keyboard?
कम्प्यूटर और कीबोर्ड के बीच संचार में किस प्रकार का ट्रांसमिशन शामिल है?

(a) Half-duplex (b) Full-duplex
(c) Simplex
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c) The communication between a computer and a keyboard involves simplex transmission, because data flows in only one direction, from the keyboard to the computer, without any data being sent back to the keyboard.

158. The expansion for MIDI is
MIDI के लिए विस्तार है

(a) Musical Instrument Digital Interface
(b) Musical Instrument Data Interface
(c) Musical Instructions Digital Interface
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

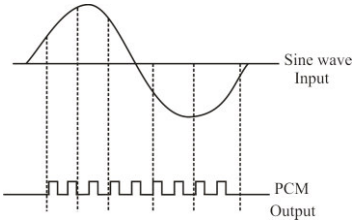
Ans.(a) MIDI stands for 'Musical Instrument Digital Interface'. It is standard protocol used to connect send control electronic musical instruments, computers and other related devices.

159. Which of the following techniques is used for transmitting digital signal?
डिजिटल सिग्नल को प्रसारित करने के लिए निम्न में से किस विधि का प्रयोग होता है?

(a) Amplitude modulation/ऐम्प्लिटयूड मॉड्यूलेशन
(b) Frequency modulation/फ्रीक्वेन्सी मॉड्यूलेशन
(c) Pulse code modulation/पल्स कोड मॉड्यूलेशन
(d) Amplitude shift keying/ऐम्प्लिटयूड शिफ्ट कीइंग

LT. Grade C.S. 2018

Ans. (c) : When a digital signal undergoes pulse code modulation, it converts the analog information into a binary sequence (1 to 0). Through the demodulation process, we can obtain the original analog signal. The figure below represents the output of the PCM signal with respect to the sine wave.



Pulse code modulation techniques are used to produce a series of number or digits in binary form.

160. Which one of the following transmission systems provides the highest data rate to an individual device ?

Sarkari Resource

निम्नलिखित में से कौन सा ट्रांसमिशन सिस्टम प्रत्येक प्रयुक्ति को उच्चतम डाटा रेट प्रदान करता है ?

(a) Computer bus/कम्प्यूटर बस
(b) Telephone bus/टेलीफोन बस
(c) Voice and mode/वोइस एंड मोड
(d) Lease lines/लिज़ लाइन

KVS PGT C.S. 2013

Ans. (a) : The Computer bus is a communication link used in a computer system to send data, addresses, control signals and power to various components in a computer System.

161. Which of the following transmission media works on the principle of total internal reflection ?
निम्नलिखित में से कौन सा संरचण माध्यम 'टोटल इंटरनल रिफलेक्शन' के सिद्धांत पर कार्य करता है?

(a) Optical fiber cable/आप्टिकल फाइबर केबल
(b) Shielded twisted pair cable
प्रतिरक्षित व्यावर्तित युग्म केबल
(c) Unshielded twisted pair cable
अप्रतिरक्षित व्यावर्तित युग्म केबल
(d) Coaxial cable/समाक्ष केबल

KVS PGT C.S. 2017

Ans. (a) : Optical fiber cable, refers to the technology that transmits information as light pulses along a glass or plastic fiber. Fiber optics transmit data in the form of light particles (photons) that pulse through a fiber optic cable. Optical fiber Cable works on the principle of total internal reflection.

162. The 10Base5 cabling is also known as _____.
10Base5 केबलिंग को _____ भी कहा जाता है।

(a) Fast Ethernet/फास्ट इथरनेट
(b) Thick Ethernet/थिक इथरनेट
(c) Thin Ethernet/थिन इथरनेट
(d) Gigabit Ethernet/गीगाबिट इथरनेट

KVS PGT C.S. 2017

Ans. (b) : The 10Base5 cabling is also known as Thick Ethernet . It was the first commercially available variant of Ethernet. The technology was standardized in 1982 as IEEE 802.3. 10Base5 uses a thick and stiff coaxial cable up to 500 meters in length.

163. Which data communication method is used to transmit the data over a serial communication link?
Serial Communication Link पर डेटा संचारित करने के लिए किस डेटा संचार पद्धति का उपयोग होता है?

(a) Simplex (b) Half-duplex
(c) Full-duplex (d) All of these

Bihar STET C.S. 2020

Ans. (d) : 'Serial Communication Link' is a communication medium in which data is sent one bit at a time one after the other. It is very useful for controlling the spaced of the communication and for long distance communication. Serial communication supports three types of data methods.
(i) Simplex
(ii) Half-Duplex
(iii) Full-Duplex

164. Which of the following is not the possible ways of data exchange?
निम्नलिखित में से कौन डेटा विनिमय के संभावित तरीके नहीं है?

(a) Simplex (b) Multiplex
(c) Half-duplex (d) Full-duplex

Bihar STET C.S. 2020

Ans. (b) : In the context of data transmission, modes include.
Simplex–Communication only happens in one direction the receiver cannot send message back to the sender.
Half-duplex– Communication can happen in both directions, but only one way at a time.
Full duplex– Communication can take place in both directions simultaneously.

165. The management of data flow between computers or devices or between nodes in a network is called
कंप्यूटर या उपकरणों के बीच या नेटवर्क में नोड्स के बीच डेटा प्रवाह का प्रबंधन कहा जाता है

(a) Flow Control (b) Data Control
(c) Data Management (d) Flow Management

Bihar STET C.S. 2020

Ans. (a) : Flow Control is a technique in data communication used to control the flow of data between computers or network nodes, so that the receiver is not over/loaded and data is transferred efficiently. It is particularly used in networking protocols such as TCP.

166. A technique used by codes to convert an analog signal into a digital bit stream is known as
कोड द्वारा उपयोग की गई एक तकनीक जिसे एनालॉग सिग्नल को डिजिटल बिट स्ट्रीम में बदलने के लिए उपयोग किया जाता है, वह है

(a) Pulse code modulation (b) Pulse stretcher
(c) Query processing (d) Queue management

Bihar STET C.S. 2020

Ans. (a) : The technique used to convert an analog signal into a digital bit stream is Pulse Code Modulation PCM involves sampling the analog signal at regular intervals, quantizing each sample into a discrete value, and then encoding those values as binary digits.

167. Upward multiplexing involves:
अपवर्ड मल्टीप्लेक्सिंग में क्या होता है?

(a) Combining multiple data streams into one
कई डेटा स्ट्रीम्स को एक में जोड़ा जाता है
(b) Distributing data to different physical channels/डेटा को विभिन्न फिजिकल चैनलों में वितरित किया जाता है
(c) Separating data into smaller units
डेटा को छोटे भागों में विभाजित किया जाता है
(d) Reordering data packets
डेटा पैकेट्स को पुनः क्रमबद्ध किया जाता है

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : Up ward Multiplexing means combining several data streams and sending them through a single connection (or channel).

Sarkari Resource

168. Communication between a computer and a keyboard involves _____ transmission.
कंप्यूटर और कीबोर्ड के बीच संचार_____ ट्रांसमिशन में शामिल है।

(a) Automatic (b) Half-duplex
(c) Full-duplex (d) Simplex

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) : In a simplex transmission, data flows in only one direction. The keyboard sends data to the computer, but computer doesn't sent data back to the keyboard in a way that controls the keyboards function. The keyboard is the sender and the computer is the receiver.

(iv) Networks Switching and Data Link Layer Protocol.

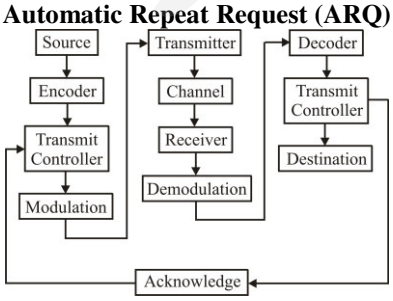
169. ARQ (Automatic Repeat Request) is:
ए. आर. क्यु. (ऑटोमैटिक रिपीट रिकेस्ट) है:

(a) Asking question and getting the reply
सवाल पूछना और जवाब मिलना
(b) Protocol in which the sends for positive acknowledgement before advancing to the next data item/प्रोटोकॉल जिसमें अगले डेटा आइटम पर आगे बढ़ने से पहले सकारात्मक पावती भेजता है
(c) Protocol allows a newly booted workstation to broadcast its Ethernet address/प्रोटोकॉल एक नए बूट वर्कस्टेशन को अपने ईथरनेट पते को प्रसारित की अनुमति देता है
(d) Protocol allows both manual IP address assign infant and automatic assignment
प्रोटोकॉल शिशु और स्वचालित असाइनमेंट दोनों को मैनुअल आई.पी.एड्रेस असाइन करने की अनुमति देता

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : ARO is the flow control approach used in a two-way communication system that encounters positive and negative acknowledgement and time out for error control if no error occurs at the receiver end then it sends a positive acknowledgement to the sender for addressing the next data.

Automatic Repeat Request (ARQ)



- If an error detects it sends a negative acknowledgement to the sender else sends a positive acknowledgement for Introducing the next data process
- While timeout returns when an unreliable communication link is produced through data transmission.

170. In Ethernet, when Manchester encoding is used, then the pulse width are _____ of the straight binary encoding.
ईथरनेट में, जब मैनचेस्टर एन्कोडिंग का उपयोग किया जाता है, तो पल्स चौड़ाई स्ट्रेट बाइनरी एन्कोडिंग के/की _____ होती है।

(a) Same/समान (b) Half/आधी
(c) Twice/दोगुनी (d) One third/तिहाई

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : In Manchester encoding (used in Ethernet), each bit period is divided into two half's:
- A logic '1' is represented by a high-to-low transition.
- A logic '0' is represented by a low-to-high transition.

This means that there is always a transition in the middle of each bit period, effectively making the pulse width half compared to straight binary encoding (NRZ-Non-Return-to-zero), where the signal stays constant for the entire bit duration.

171. The three aspects of quantization, programmers generally concerned with are :

(a) Coding error, sampling rate and Amplification
(b) Sampling rate, Coding error and Conditioning
(c) Sampling rate, Aperture time and Coding error
(d) Aperture time, Coding error and Strobing

UGC NET C.S. November-2017 (Paper-III)

Ans : (c) The three aspects of quantization, programmers generally concerned with are-
(1) Sampling rate = The number of samples per second.
(2) Aperture time = Aperture time is the sampling rate defined as the number of the second per sample.
(3) Coding error = The difference b/w the conditional signal and the quantized signal.
So option (c) is correct.

172. The Hamming (7 , 4) code for 0000 using even parity is
0000 के लिए सुसमता का प्रयोग करके हेमिंग (7 ,4) कोड है

(a) 0000000 (b) 1111111
(c) 2222222 (d) 1212121

KVS PGT C.S. 2013

Ans, (a) : Hamming code is an error correction system that can detect and correct errors when data is stored or transmitted. It requires adding additional parity bits with the data. It is Commonly used in error correction code RAM

About the Question , The four data bits assembled as a vector P is pre multiplied by G and taken modulo 2 to yield the encoded value that is transmitted. The original 4 data bits are converted to seven bits with three parity bits added to ensue even parity using the about data bit corerages. So given data 0000 by using hamming (7, 4) we are sending 0000000.

173. Which of the following is error detection code used in digital logic?/निम्नलिखित में से कौन-सा डिजिटल लॉजिक में प्रयुक्त त्रुटि पहचान कोड है?

(a) Hamming code/हैमिंग कोड
(b) Cyclic redundancy check (CRC)
चक्रीय अतिरेक जाँच (सीआरसी)
(c) Checksum चेकसम

Sarkari Resource

(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : Hamming code: This is an error-correction code that can detect and correct single bit errors. It is often used in digital systems for error detection and correction.
Cyclic redundancy cheek (CRC): This is a widely used method for detecting errors in digital data. It is particularly useful for detecting error in data transmission and storage.
Checksum: This is a simple error detection method that involves calculating a value based on data being transmitted or stored and verifying it at the receiving end to detect any errors.
So, the correct option is (d) more than one of the above.

174. Consider a code with only four valid code words 0000000000, 0000011111, 1111100000, and 1111111111. This code has distance 5. If the code word arrived is 0000000111 then the original code word must be ____ .
(a) 0000011111 (b) 0000000000
(c) 1111100000 (d) 1111111111

UGC NET C.S. June-2020 (Paper-II)

Ans. (a) : Arrange one below the other the XOR operations.
for correcting the d-bit error hamming distance should be 2d + 1.
So, 2d + 1 = 5
d = 2. It means the hamming distance b/w the code word 0000000111 and given options should be 2-bits only options 1 match.

175. In CRC checksum method, assume that given frame for transmission is 11010111011 and the generator polynomial is
$$G(x) = x^4 + x + 1.$$
After implementing CRC encoder, the encoded word sent from sender side is ____ .
(a) 11010110111110 (b) 11101101011011
(c) 110101111100111 (d) 110101111001111

UGC NET C.S. August-2016 (Paper-III)

Ans : (a) Given, message is: 1101011011 and generator polynomial is $G(x) = x^4 + x + 1$, which is equivalent to 10011. Now we add $(n - 1)$ 0's to the LSB of message, if CRC generator is of n-bits.
Therefore, message will be 11010110110000. Now we divide it by 10011, we will get 1110 as remainder, therefore transmitted message will be:
11010110111110.

176. In "bit stuffing", each frame begins and ends with a bit pattern in hexadecimal?
(a) Ox8C (b) Ox6F
(c) OxFF (d) Ox7E

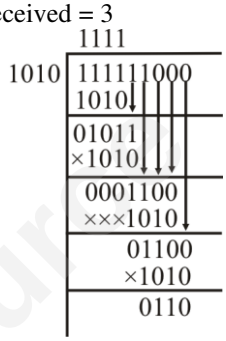
UGC NET C.S. December-2023

Ans. (d) : In bit stuffing, each frame typically begins and ends with a specific bit pattern, not necessarily in hexadecimal. The purpose of this pattern is to enable the receiver to identify the start and end of the frame. In some networking protocols, Such as HDLC (High-Level Data Link Control), each frame begins and ends with a special bit pattern, 01111110 or Ox7E in hexadecimal. This pattern is a flag byte.

177. In CRC coding if the data word is 111111, divisor 1010 and the remainder is 110. Which of the following code is true?
(a) 011111101 (b) 001010110
(c) 111111110 (d) 110111111

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Ans. (c) : Given that
data word = 111111
divisor = 1010
Data plus CRC received = n – 1
n = number of divisor
n = 4
Data plus CRC received = 3



So add the data word is 111111110
So option (c) is correct.

178. Given Ethernet address 0101101000010001 01010101 00011000 10101010 00001111 in binary, what is the address in hexadecimal notation?
(a) 5A:88:AA:18:55:F0
(b) 5A:81:BA:81:AA:08
(c) 5A:18:5A:18::55:0F
(d) 5A:11:55:18:AA:0F

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Ans. (d) : Given Ethernet address
01011010 00010001 010100101 00011000
10101010 00001111

Decimal	Binary	Hexa
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

So by table 5A : 11 : 55 : 18 : AA : 0F

Sarkari Resource

179. Consider channel capacity is 100 Mbps, mean frame length is 10000 bits and average frame arrival rate is 5000 frames/sec. Applying simple queuing theory, the mean frame delay to send a frame onto the channel is :
माना चैनल क्षमता 100 एम.बी.पी.एस औसत फ्रेम लम्बाई 10000 बिट्स तथा औसत फ्रेम आगमन दर 5000 फ्रेम्स/सेकण्ड है। साधारण क्युइंग सिद्धांत लागू करते हुए, चैनल पर एक फ्रेम भेजने में औसत फ्रेम विलम्ब होगा:

(a) 200 micro seconds/200 माइक्रो सेकण्ड्स
(b) 100 micro seconds/100 माइक्रो सेकण्ड्स
(c) 0.5 seconds /0.5 सेकण्ड्स
(d) 2 mili seconds /2 मिली सेकड्स

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Ans. (a) : Let's carefully solve step by step using queuing theory (M/M/1 model)
Given,
• Channel capacity
$C = 100 \text{ Mbps} = 100 \times 10^6$, bits/Sec
• Mean frame length L = 10000, bits
• Average frame arrival rate
$\lambda = 5000$, frames/sec
Step - 1 : Service rate

$$\mu = \frac{\text{channel capacity}}{\text{frame length}} = \frac{100 \times 10^6}{10000} = 10000 \text{ frames/sec}$$

Step - 2 : Utilization factor

$$\rho = \frac{\lambda}{\mu} = \frac{5000}{10000} = 0.5$$

Step - 3 : Mean delay (queuing + service time)
For M/M/1 system:

$$\text{Mean delay } (W) = \frac{1}{\mu - \lambda}$$

$$W = \frac{1}{10000 - 5000} = \frac{1}{5000} \text{ sec}$$

Stem - 4: Convert to milliseconds

$$\frac{1}{5000} = 0.0002 \text{ Sec} = 0.2 \text{ ms} = 200 \ \mu\text{s}$$

180. What is the relationship between the distance between parity bits and their error detection and correction capability in Hamming Code?
हैमिंग कोड में पैरी बिट्स के बीच की दूरी और उनके त्रुटि पहचान और सुधार क्षमता के बीच क्या संबंध है?

(a) Closer parity bits provide higher error correction capability/पास-पास पैरी बिट्स अधिक त्रुटि सुधार क्षमता प्रदान करते हैं
(b) Closer parity bits provide lower error correction capability/पास-पास पैरी बिट्स कम त्रुटि सुधार क्षमता प्रदान करते हैं
(c) Distance between parity bits has no impact on error correction capability/पैरी बिट्स के बीच की दूरी का त्रुटि सुधार क्षमता पर कोई प्रभाव नहीं पड़ता
(d) The position of parity bits is irrelevant in Hamming Code/ हैमिंग कोड में पैरी बिट्स की स्थिति अप्रासंगिक है

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : In Hamming code, parity bits are not placed randomly but at positions that are powers of 2 (112, 4, 8, …). This spacing ensures that each parity bit cheks a unique combination of data, bits. If the parity bits are closer together, they overlap more and provide less unique coverage, which reduces error correction capability. By maintaining proper distance between parity bits, Hamming code achieve reliable single-bit error correction and double-bit error detection.

181. IEEE 802.11 standard belongs to :
IEEE 802.11 मानक .....से संबंधित है।

(a) Wireless LAN/वायरलेस LAN
(b) Wired LAN/वायर्ड LAN
(c) Bluetooth Certification/ब्लूटूथ सर्टिफिकेशन
(d) Token Bus/टोकेन बस

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Ans. (a) : The IEEE 802.11 standard specifies the protocols for the physical and data link layers of wireless networks
• It provides the foundation for various wireless networks.
• It provides the foundation for various wireless network products that allow devices like laptops Smartphone, and printers to connect and communicate without wires.
• Wi-fi is the marketing name for the IEEE 802.11 standard, highlighting its role in wireless data transmission.

182. Datagram networks are also referred to as ..... networks.
डाटाग्राम नेटवर्क्स को..... नेटवर्क्स भी कहा जाता है।

(a) Connection oriented/कनेक्शन ओरियेन्टेड
(b) Bluetooth/ब्लूटूथ
(c) Connectionless/कनेक्शनलेस
(d) Wi-Fi/वाई-फाई

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Ans. (c) : Computer networks that provide connection oriented services are called with circuits while those providing connectionless services are called datagram network for prior knowledge, the internet that we use is based on a datagram network (connectionless) at the network level as all packets from a source to a destination do not follow the same path.

183. A network using CSMA/CD has a bandwidth 10 Mbps. If the maximum propagation time (including delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 micro seconds. What is the minimum size of the frame ?
CSMA/CD प्रयोग करने वाले एक नेटवर्क की बैडविड्थ 10 Mbps है। यदि अधिकतम प्रचार (प्रोपेगेशन) समय (युक्तियों में विलम्ब सहित तथा जैमिगं संकेत भेजने में लगने वाले समय को नजरअंदाज करते हुए) 25.6 माइक्रो सेकण्ड्स है। फ्रेम का न्यूनतम आकार क्या होगा?

(a) 512bytes/512बाइट्स (b) 1024bytes/1024बाइट्स
(c) 64 bytes/64 बाइट्स (d) 32 bytes/32 बाइट्स

HTET PGT C.S. 2019

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Ans. (c) : Data:–
L = size of frame
Bw → Bandwidth = 10 mbps = 10×106 b/s
TP → propagation time = 25.6us = 25.6 ×10 – 65
Tt → transmission time
Formula–
For CSMA/CD
$Tt \geq 2 \times TP$
$Tt = \frac{L}{BW}$
Calculation:–
$\frac{L}{BW} \geq 2 \times T$
$L \geq 2 \times Tp \times BW$
$L \geq 2 \times 2.56 \times 10^{-6} \times 10 \times 106$
$L \geq 512$ bits
$L \geq 64 \times 8$ bits // 8 bit = 1 byte
L min = 64 bytes
Therefore, the minimum size of a frame in the network is 64 bytes.

184. In context of error detection and correction in computer networks, CRC stands for :
कम्प्यूटर नेटवर्क्स में, त्रुटि खोज तथा सुधार के सन्दर्भ में, CRC का तात्पर्य है:
(a) Cyclic Reduction Code
साइकिलिक रिडक्शन कोड
(b) Cyclic Redundancy Check
साइकिलिक रिडनडेंसी चेक
(c) Cyber Request Check
साइबर रिक्वेस्ट चेक
(d) Cyber Repetition Code
साइबर रिपिटीशन कोड

HTET PGT C.S. 2019

Ans. (b) : A Cyclic Redundancy Check (CRC) is an error-detecting code commonly used in digital networks and storage devices to defect accidental changes to raw data. Blocks of data entering these systems get a short check value attached, based on the remainder of a polynomial division of their contents.
Example– Polynomial is– $x^5 + x^4 + x^3 + x^2 + 1$
A polynomial higher degree is 5, so append 5 O's to the message bits
Message– 101000110100000
Divisor polynomial in bits = 110101
⇒ CRC ⇒ 01110

185. _____is used to detect the errors in the headers to avoid miss delivery.
गलत डिलिवरी से बचने के लिए हेडर में त्रुटियों का पता लगाने के लिए ____ का उपयोग किया जाता है।
(a) Checksum (b) Fault
(c) Error (d) Padding

HTET PGT C.S. 2023

Ans.(a): In networking, particularly in the IP header, a checksum is used to detect errors it ensures that the header information is not corrupted during transmission. When the packet reaches the destination, the checksum is recalculated and compared with the transmitted value, if they do not match it means there's an error, and the packet is discarded to avoid miss delivery.

186. What does the routing algorithm perform?
(a) Decides if incoming packet should be further corrected for transmission errors
(b) Adds checksum bits to packets
(c) Encrypts the packets
(d) Decides the output line on which the packet should be transmitted

CGPSC C.S. 2020 Set-A

Ans.(d): A routing algorithm is a fundamental function of the network layer in the OSI model its primary job is the determined the most suitable path for data packets to travel from the source to the destination across interconnected networks when a packet arrives at a router, the routing algorithm checks its destination address and consults the routing table. Based on this, it decides the output line (next hop) through which the packet should be forwarded.
It is important to note what a routing algorithm does not do, it does not handle error detection or correction that is the responsibility of the data link and transport layers it does not add checksum bits; This function also belong to lower layers for error control similarly encryption is handled at the presentation or application layer, not by routing.

187. How many parity bits are required to construct a hamming code for 32 data bits?
(a) 4 (b) 5
(c) 6 (d) 7
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (c) : In hamming code, the number of parity bits (r) required for a given number of data bits (m) is determined by the inequality $2^r \geq m + r + 1$
This rate ensures that each bit positions in the code word can be uniquely identified including one state for "no error"
m = 32 substituting values step by step
• For r = 4 : $2^4 = 16$, while 32 + 4 + 1 = 37
Since 16 < 37, not possible
• For r = 5 : $2^5 = 32$, while 32 + 5 + 1 = 38
Since 32 < 38, still not possible
• For r = 6 : $2^6 = 64$, while 32 + 6 + 1 = 39 39, since 64 ≥ 39, condition satisfied.
Thus, the minimum panty bits required = 6.

188. Which of the following transmission media works on the principle of total internal reflection?
(a) Baseband coaxial cable
(b) Optical fiber
(c) Twisted pair wires
(d) UTP
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : An optical fiber is a thin, hair-like strand of glass or plastic that transmits data in the form of light pulses over long distances, a technology known as fiber optics it works by utilizing the principle of total internal reflection, where light reflects off the core-cladding boundary to stay within the fiber. Optical fibers are widely used in modern telecommunication and internet networks for their ability to carry high bandwidths with minimal signal loss and resistance to electromagnetic interference.

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Component:–
(1) Core– The central part of the fiber, usually made from silica glass, they carries the light signals.
(2) Cladding– A layer surrounding the core with a slightly lower refractive index, which reflects the light back into the core preventing it from escaping.
(3) Buffer coating– One or more protective layers of plastic that shield the delicate core and cladding from external damage and crosstalk.

189. Consider a subnet with 630 routers. Assuming a three-level hierarchy, how many entries will be required in the routing table of each router, if the subnet is partitioned into 7 clusters, each containing 9 regions of 10 routers?
(a) 24 (b) 25
(c) 26 (d) 27
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : In hierarchical routing, routers are classified in groups known as regions. In three level hierarchical routing, network is classified into a number of clusters. Each cluster is made up of number of regions and each region contain number of routers.
Given- number of routers = 720
Clusters = 8
Regions = 9 of 10 raters
Total number of entries in hierarchical table of each router = (number of clusters – 1) + (number or regions – 1) + routers in each region.
= (8 – 1) + (9 – 1) + 10 = 7 + 8 + 10 = 25.

190. Consider the following statements:
(i) In a hamming code, the parity bit positions are powers of 2.
(ii) The hamming code can detect and correct multiple errors.
(iii) In a hamming code, the data bits must be located at even numbered positions.
How many of the statements given above are correct?
(a) Only one (b) Only two
(c) All three (d) None
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : Statement (I):– In a hamming code, the parity bit positions are powers of 2.
- in hamming code, parity bits are placed at positions 1, 2, 4, 8….(2^0, 2^1, 2^2, 2^3).
- These positions are reversed for parity calculation, while the remaining positions are for data bits.
  Example– if we have a 7-bit hamming code (4 data bits + 3 parity bits)
- Parity bits : positions 1, 2, 4
- Data bits : Positions 3, 5, 6, 7

Statement (II) :– The hamming code can detect and correct multiple errors.
- Hamming code is designed as SEC (Single Error Correction) it can detect and correct only 1 bit error.
- It cannot correct multiple errors (2 or more errors), although it can sometimes detect double errors if extended (SECDED variant)
- This statement is incorrect.

Statement (III) :– in a hamming code, the data bits must be located at even numbered positions.
- Data bits are placed at all positions that are not powers of 2 (not reserved for parity).
- There's no rule that data bits must be at even positions. They can be odd or even depending on the size of the code.
- This statement is incorrect.

191. Fibre optics is an example of_____ .
फाइबर ऑप्टिक्स_____का उदाहरण है।
(a) Guided Media (b) Unguided Media
(c) Wireless Media
(d) All options are wrong/सभी विकल्प गलत है

HTET PGT C.S. 30.07.2025

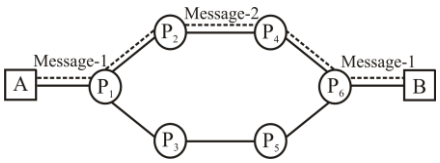
Ans. (a) : Guided or wired media allows signal energy enclosed and guided within a physical medium. This media is used either for point-to-point links or a shared link with various connections in guided media, interruption is generated by outputs in the adjacent cables proper covering of guided media is required to reduce the interruption problem
- **Twisted pair cable–** it is made up of insulated copper wires twisted together that's why it called is as twisted pair cable mostly used in telephone networks and Ethernet cables it is affordable.
- **Coaxial cable–** in coaxial cable one control conductor surrounded by insulation, a metallic shield, and an exterior cover. This type of cable used in television networks and for long distance communication lines. It provides better protection against interruption than twisted pair cables.
- **Optical Fiber–** This cable uses light signals to transfer data mode up of glass or plastic fibers, it provides very high bandwidth and low signal reduction making it ideal for long distance and high speed data transmission it immune to electromagnetic obstruction but more expensive than copper cables.

162. Which switching technique establishes a dedicated physical path between source and destination nodes through intermediate nodes of communication network?
(a) Packet Switching (b) Message Switching
(c) Physical Switching (d) Circuit Switching

HTET PGT C.S. 30.07.2025

Ans.(d): Circuit switching is a type of switching in which a connection is established between the source and destination before communication this connection receives the complete this bandwidth of the network until the data is transferred completely. However circuit switching can be inefficient and costly due to its requirements for dedicated resources, making it less suitable for high-traffic or large scale networks in this article, we will discuss every point about circuit switching.
Circuit Switching– In circuit switching network resources (bandwidth) are divided into pieces and the bit delay is constant during a connection the dedicated path/circuit established between the sender and receiver provides a guaranteed data rage. Data can be transmitted without any delays.
Once the circuit is established

Sarkari Resource



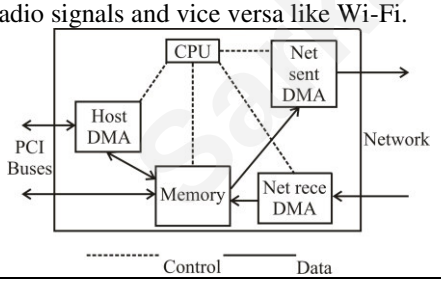
The telephone system network is one of the example of circuit switching. FDM (Frequency Division Multiplexing) and TDM (Time Division Multiplexing) are two methods of multiplexing multiple signals into a single carrier.

193. NIC stands for:
NIC का मतलब है–

(a) Network Interface Card
(b) Network Interface Controller
(c) New Interface Controller
(d) New Interface Card

HTET PGT C.S. 30.07.2025

Ans.(a): Network Interface Card– NIC stands for Network Interface Card (NIC) is additionally called Ethernet or physical or network card. NIC is one of the major and imperative components of associating a gadget with the network each gadget that must be associated with a network must have a network interface card. Even the switches comprise of NIC in arrange to associate to the system it is the fore most essential network component without which you cannot interface any gadget to a network. It could be a circuit board or a card that is introduced in the computer or PC so that it acts as an arbiter for organizing and computer or PC. It is shown on layer 2 of the OSI show which is an information interface layer that employs MAC address in a straight forward frame it can be clarified as the programmable card that changes over the address to electrical signals through the cables to the web and vice versa like Ethernet, and for remote, it changes over the bits to radio signals and vice versa like Wi-Fi.



194. WPA2 is used for security in_____.
_____ में डब्ल्यू पी ए-2 का प्रयोग सुरक्षा के लिए होता है।

(a) Ethernet/इथरनेट (b) Bluetooth/ब्लूटूथ
(c) Wi-Fi/वाई-फाई
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (c) : WPA2 replaced WPA. Which requires testing and certification by the Wi-Fi alliance, implemented the mandatory elements of IEEE 802.11 particularly it included mandatory support for CCMP (Counter mode CBC MAC Protocol), an AES (Advanced Encryption standard) based encryption mode certification began in September 2004. WAP2 certification is mandatory for all new devices to bear the wi-fi trademark from march 13, 2006.

New Features of WPA2 Protocol Offer:– WPA was replaced in 2004 with WPA2 employs the block chaining message authentication code protocol (CCMP), a counter mode cipher. The Advance Encryption Standard (AES) algorithm, which verifies the Authenticity and integrity of message, forms the foundation of the CCMP protocol compared to the original temporal key integrity protocol (Tkip) used by WPA CCMP is more rebuts and dependable.

195. TDD-TDMA is a kind of _____communication.
टी डी डी-टी डी एम ए _____ संचार का प्रकार है।

(a) Full duplex/फुल डुप्लेक्स
(b) Half duplex/हाफ डुप्लेक्स
(c) Semi duplex/सेमी डुप्लेक्स
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (b) : TDD-TDMA Communication– TDD-TDMA stands for Time Division Duplex - Time Division multiple Accesses. In this system, both uplink (transmission) and downlink (reception) use the same frequency band, but they operate in different time slots. This means the channel is divided in to time travels, where some slots are used for sending data and others for receiving data multiple users can also share the channel by getting their own time slots through TDMA. Because transmission and reception do not occur simultaneously TDD is not a full duplex system instead it allows only one direction of communication at a time which makes it a form of half duplex communication.

196. What is difference between a Router and a Brouter?
एक राउटर और ब्राउटर के बीच क्या अन्तर होता है?

(a) A brouter routinely broadcasts routing information.
एक ब्राउटर नियमित राउटिंग सूचना प्रसारित करता है।
(b) A brouter is a basic functionality of router
एक ब्राउजर राउटर की बुनियादी कार्यक्षमता है।
(c) A brouter can function as a bridge as well as a router./एक ब्राउटर, एक ब्रिज के साथ-साथ एक राउटर के रूप में भी कार्य करता है।
(d) A brouter is a router that routes signals on broadband network./एक ब्राउटर, एक राउटर है जो कि ब्राडबैण्ड नेटवर्क पर रुटों का संकेत देता है।

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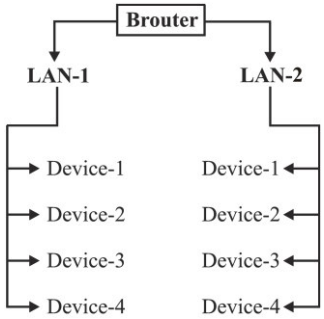
Ans (b) : Router– A router is a networking device that is designed to receive, analyze and forward data packets between computer networks. A router is used to connect LAN (Local Area Network) and WAN (Wide Area Network).

Advantages of router:–
- It improves security and reduces broadcast traffic.
- Routers connect different networks globally.
- Many routers come with built in security features like firewall's instruction detection system's and VPNs, which help to protect networks from unauthorized access.

Disadvantages:–
- Routers are hard to set up.
- They can be hacked.

Sarkari Resource

• Routers can break the whole network if single point of failure.
• It slow down internet, making video calls less smooth.
Brouter:– A brouter is a networking device that functions both as a bridge and a router it can forward data between networks (serving as a bridge), but can also route data to individual systems within a network (serving as a router).



Advantages:– Brouter can do both routing and bridging, making networks more flexible
• It makes the network simpler by doing two jobs at once.
• They choose the best way to send data, making the network faster.
• These are cast-effective.
Disadvantages:–
• Brouters are harder to set up then routers or bridges.
• This can break the entire network if they fail.
• These might be slower than routers.
• Availability is less.

197. In distance vector routing, a router sends out information of estimated delays.
दूरस्त वेक्टर राउटिंग में, एक राउटर आंकलित विलम्ब सूचना भेजता है
(a) Only when there is a change in its table.
तभी जब इसकी तालिका में कोई बदलाव होता है।
(b) Only when, a new host is added.
केवल जब नया होस्ट जोड़ा जाता है।
(c) At regularly scheduled intervals.
नियमित रूप से निर्धारित अंतराल पर
(d) Only when a new network is added.
केवल तब जब एक नया नेटवर्क जोड़ा जाता है।

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (c) : Distance Vector Routing (DVR) Protocol is a method used by routers to find the best path for data to travel across a network each router keeps a table that shows the shortest distance to every other router, based on the number of hops (or steps) needed to reach them. Routers share this information with their tables and find the most efficiency router this protocols helps ensure that data move quickly and smoothly through the network.

198. A jam signal sent through a network means
एक जाम संकेत, एक नेटवर्क के माध्यम से भेजने का मतलब होता है
(a) The network traffic is too congested
एक नेटवर्क यातायात बहुत कन्जेस्टेड है।
(b) There has been a data collision
यहाँ डाटा कोलीजन हुआ है।
(c) It is time to go/यह जाने का समय है।
(d) The printer's paper feeder is jammed
प्रिंन्टर का पेपर फीडर जाम है।

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : In computers networks that use CSMA/CD (Carrier Sense Multiple Access with Collision Detection), like traditional Ethernet, a jam signal is sent when two or more devices try to transmit data at the same time.
• This causes a collision (their signals overlap and get corrupted)
When a collision is detected:–
1. The network device immediately sends a jam signal.
2. This jam signal is broadcast to all devices on the network segment.
3. It tells them:
  • "A collision has occurred"
  • "Stop transmitting immediately"
4. After that each device waits for a random buck off time before trying again.

199. Bit stuffing is used to
बिट स्टफिंग का प्रयोग किया जाता है
(a) Pad insufficient information
पैड अपर्याप्त सूचना
(b) Distinguish beginning and ending flags from information
सूचना के फ्लैग की शुरूआत और अंत में भेद
(c) Convert 8 bit words into 16 bit words
8 बिट शब्दों को 16 बिट शब्दों में परिवर्तित करना
(d) Fill a data turkey/डेटा टर्की भरना

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : In data communication protocols like HDLC (High-level Data Link Control), a special bit pattern called a flag is used to mark the start and end of a frame.
The flag pattern is always : 01111110 (8 bits)
Uses of bit staffing–
(1) Frame Boundary Detection:–
• in communication, a special flag sequence (01111110) marks the start and end of a frame
• Bit scuffing ensures that this flag sequence only appears at the boundaries, not inside data.
(2) Error free communication:-
• If the same flag pattern uppers inside the data, the receiver could get confused
• Bit suffering percents this confusion by inserting extra bits, so the receiver can clearly separate data from control information.
(3) Data Transparency:–
• Any kind of bit pattern can appear inside the data, even the same as the flag.
• With bit staffing, the sender inserts a 0 after five consecutive 15. making the data "transparent".

200. Which sublayer of the data link layer performs data link functions that depend upon the type of medium?
डेटा लिंक परत का कौन-सा सबलेयर डेटा लिंक फ़ंक्शन करता है, जो माध्यम के प्रकार पर निर्भर करता है?

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(a) Logical Link Control Sublayer
(b) Media Access Control Sublayer
(c) Network Interface Control Sublayer
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : MAC stands for Media Access Control. It is deals with transmission of data packets to and from the network- interface card, and also to and from another remotely shared channel. The MAC sublayer also prevents collision using protocols like CSMA/CD.

201. The _____ measures the number of lost or garbled messages as a fraction of the total sent in the sampling period.
_____ नमूना अवधि (Sampling Period) में भेजे गए कुल संदेशों के एक अंश के रूप में खोए या विकृत संदेशों की संख्या को मापता है।

(a) Residual error rate/अवशिष्ट त्रुटि दर
(b) Transfer failure probability
स्थानांतरण विफलता की संभावना
(c) Connection release failure probability
कनेक्शन रिलीज विफलता की संभावना
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (a) : The residual error rate measures the number of lost or garbled message as a fraction of the total sent in the sampling period.

202. What is the size of MAC address in computer network?
कम्प्यूटर नेटवर्क में MAC address का साइज कितना होता है?

(a) 32 bits/32 बिट
(b) 16 bits/16 बिट
(c) 48 bits/48 बिट
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (c) : MAC stands for Media Access Control. In computer networks, the size of MAC address is 48 bits. This means that MAC address is usually written as 12 hexadecimal numbers (i.e., 6 layer), such as 00: 1A:2B:3C:4D:5E

203. Sequence the following stages of the sliding window protocol for flow control.
A. Sender moves the window to the next set of frames.
B. Sender sends a window of frames.
C. Receiver acknowledges the received frames.
D. If acknowledgment is not received, Sender retransmits the frame.
Choose the correct answer from the options given below :

(a) B, A, C, D (b) B, C, D, A
(c) B, D, C, A (d) B, C, A, D

UGC NET C.S. December-2024

Ans. (d) : The sliding window protocol is used for reliable and sequential delivery of data frames. This protocol controls the data flow between the sender and the receiver step by step.
B. Sender sends a window a frames.
C. Receiver acknowledge the receive frames
A. Sender move the window the next set of frames.
D. If acknowledgement is not received, sender retransmits the frame.
Therefore, is the correct B, C, A, D.

204. Which of the following switching technique was used in ARPANET?
ARPANET में निम्नलिखित में से कौन से स्विचिंग तकनीक का उपयोग किया जाता है?

(a) Message Switching/मेसेज स्विचिंग
(b) Circuit Switching/सर्किट स्विचिंग
(c) Packet Switching/पैकेट स्विचिंग
(d) Data switching/डेटा स्विचिंग

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : ARPANET stands for Advanced Research Projects Agency Network. It was a precursor to the modern internet, utilized packet switching. This technique involves breaking data into small packets that can be transmitted independently across a network allowing for efficient and flexible data transmission.

205. Which of the following protocol is used to convert the IP address to MAC address?
आईपी एड्रेस को मैक (MAC) एड्रेस में बदलने के लिए निम्नलिखित में से किस प्रोटोकॉल का उपयोग किया जाता है?

(a) Border Gateway protocol/बॉर्डर गेटवे प्रोटोकॉल
(b) Address Resolution Protocol
एड्रेस रेजोल्यूशन प्रोटोकॉल
(c) Reverse Address Resolution Protocol
रिवर्स एड्रेस रेजोल्यूशन प्रोटोकॉल
(d) Dynamic Host Routing Protocol
डायनामिक होस्ट रूटिंग प्रोटोकॉल

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : The Address Resolution Protocol is used to convert IP addresses to MAC addresses, which are the physical addresses found on a network.

206. Which of the following is used to alarm about the collision detection in a network using the CSMA/CD?
CSMA/CD का उपयोग कर नेटवर्क में कोलिजन (टकराव) का पता लगाने के बारे में अलार्म के लिए निम्नलिखित में से किसका उपयोग किया जाता है?

(a) Backoff value/बैकऑफ वैल्यू
(b) Jamming signal/जैमिंग सिग्नल
(c) Frame/फ्रेम
(d) Interframe space/इंटरफ्रेम स्पेस

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : When multiple stations attempt to transmit data simultaneously on the same network segment, a collision occurs. To inform other stations about the collision and prevent further attempts, the collising stations immediately step transmitting and send a jamming signal. This signal alerts all other stations that the channel is busy and they should wait before attempting to transmit.

Sarkari Resource

207. What if the value of Start Frame Delimiter in Ethernet frame?
ईथरनेट फ्रेम में स्टार्ट फ्रेम डिलीमीटर का मान क्या होगा?
(a) 10101011 (b) 10101010
(c) 01101011 (d) 10101111
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In Ethernet, the Start Frame Delimiter (SFD) is the special byte that comes immediately after the 7-byte preamble (10101010 repeated).
The value of SFD is 10101011.

208. In the data link layer, _____ is based on automatic repeat request, which is the retransmission of data.
(a) Framing (b) Physical addressing
(c) Error control (d) Flow control
DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : In the data link layer. Error control is based on automatic repeat request, which is the retransmission of data to ensure reliable data transfer by recovering from lost or damaged frames.

209. The minimum frame length for 10 Mbps Ethernet is _____ bytes and maximum is _____ bytes.
(a) 64 & 128 (b) 128 & 1518
(c) 1518 & 3036 (d) 64 & 1518
UGC NET C.S. June-2014 (Paper-II)

Ans : (d) The original ethernet IEEE 802.3 standard define the minimum ethernet frame size as 64 bytes and maximum as 1518 bytes.
Where payload size in ethernet defined the minimum 46 bytes and maximum 1500 bytes.

210. Four bits are used for packed sequence numbering in a sliding window protocol used in a computer network. What is the maximum window size?
(a) 4 (b) 8
(c) 15 (d) 16
UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : In case of Go Back N protocol sender window size is n – 1 and receiver window size is 1 the size is (2 ^ n) – 1
So the answer is 15.

211. _____ is a bit-oriented protocol for communication over point-to-point and multipoint links.
(a) Stop-and-wait (b) HDLC
(c) Sliding window (d) Go-back-N
UGC NET C.S. December-2014 (Paper-III)

Ans : (b) HDLC (High level data link control) is a classical bit oriented code transparent synchronous data link layer protocol by ISO protocol.
HDLC can be used for point to multipoint connections. But is now used almost exclusively to connect the one device to another.

212. Which transmission technique guarantees that data packets will be received by the receiver in the same order in which they were sent by the sender?
(a) Broadcasting (b) Unicasting
(c) Packet switching (d) Circuit switching
UGC NET C.S. June-2015 (Paper-III)

Ans : (d) Circuit switching transmission technique guarantees the data packets will be received by the receiver in the same order in which they were sent by the sender because path is fixed and all packet goes through that path in same order.

213. Consider the following statements:
I. High speed Ethernet works on optic fiber.
II. A point to point protocol over Ethernet is a network protocol for encapsulating PPP frames inside Ethernet frames.
III. High speed Ethernet does not work on optic fiber.
IV. A point to point protocol over Ethernet is a network protocol for encapsulating Ethernet frames inside PPP frames.
Which of the following is correct?
(a) I and II are true; III and IV are false.
(b) I and II are false; III and IV are true.
(c) I, II, III and IV are true.
(d) I, II, III and IV are false.
UGC NET C.S. August-2016 (Paper-III)

Ans : (a) Statement I– True, High speed Ethernet can utilize various media, including optical fiber. It's not limited to just copper cables.
Statement II– True, the definition of PPPoE is to encapsulate PPP frames within Ethernet frames.
Statement III– False Since high speed Ethernet can work on optic fiber, this statement in incorrect.
Statement IV– False, The correct encapsulation is PPP frames inside Ethernet frames.

214. Which of the following devices takes data sent from one network device and forwards it to the destination node based on MAC address?
(a) Hub (b) Modem
(c) Switch (d) Gateway
UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Switch takes data sent from one network device & forwards it to the destination node based on MAC address.
A switch is a multi port bridge with a buffer and a design that can boost its efficiency and performance. Switch is data link layer device. It can perform error checking before forwarding data.

215. The number of bits used for addressing in Gigabit Ethernet is _____
(a) 32 bits (b) 48 bits
(c) 64 bits (d) 128 bits
UGC NET C.S. November-2017 (Paper-II)

Ans : (b)

Ethernet	Frame
62 bits	Preamble used for bit synchronization
2 bits	Start or frame delimeter
48 bits	Destination Ethernet address
48 bits	Source ethernet address
16 bits	Length
46 - 1500 bytes	Data
32 bits	Frame check sequence

Sarkari Resource

The Ethernet frame is the packet that delivers data between stations. Each frame is made up of bits. Which are organized into different fields. The four main fields are the destination Ethernet address field, source. Ethernet address field, data field & frame check sequence filed. The address fields are us bit long.

216. Given below are two statements:
Statement (I) : In datagram networks - routers hold state information about connections
Statement (II) : In virtual circuit network - each virtual circuit requires router table space per connection
In the light of the above statements, choose the most appropriate answer from the options given below:
(a) Both Statement I and Statement II are correct
(b) Both Statement I and Statement II are incorrect
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

UGC NET C.S. December-2023

Ans. (d) : In datagram networks, routes do not hold state information about connection because each packet is treated independently. so statement (I) is incorrect.
However, in virtual circuit networks, each virtual circuit does require routers need to maintain state information for each virtual circuit. So statement (II) is correct.

217. The work done by UDP is/are:
(A) Congestion control
(B) Flow control
(C) Retransmission
(D) Segment transmission
Choose the correct answer from the options given below:
(a) (A) and (D) Only (b) (C) Only
(c) (D) Only (d) (B) and (C) Only

UGC NET C.S. December-2023

Ans. (c) : UDP (User Datagram Protocol) is a connectionless protocol that operates at the Transport Layer of the a network without the need for establishing a connection. UDP does not provide features like congestion control, Flow control or retransmission of lost packets. It simply encapsulates the data into datagram's (segments) and transmits them without any guarantee of delivery or ordering.

218. In the standard Ethernet with transmission rate of 10 Mbps, assume that the length of the medium is 2500m and size of a frame is 512 bytes. The propagation speed of a signal in a cable is normally 2 x 10^8 m/s. Calculate Transmission delay and propagation delay.
(a) 25.25μs and 51.2μs
(b) 51.2μs and 12.5μs
(c) 10.24μs and 50.12μs
(d) 12.5μs and 51.2μs

UGC NET C.S. December-2022

Ans. (b) : Given that
Bandwidth = 10 Mbps
Distance = 2500 m
Propagation speed = 2×10^8 m/s
packet size = 512 bytes
We know that
Transmission delay (T_t)

$$= \frac{\text{Packet size}}{\text{Bandwidth}}$$

$$T_t = \frac{512}{10 \times 10^6}$$

$\boxed{T_t = 51.2\mu s}$ $\boxed{1\mu = 10^{-6}}$

Propagation delay (T_p) = $\frac{\text{Distance}}{\text{Speed}}$

$$T_p = \frac{2500}{2.0 \times 10^8}$$

$\boxed{T_p = 12.5\ \mu s}$

219. Which one is the connectionless transport – layer protocol that belongs to the Internet protocol family?
(a) Transmission Control Protocol (TCP)
(b) User Datagram Protocol (UDP)
(c) Routing Protocol
(d) Datagram Control Protocol (DCP)

UGC NET C.S. December 2021

Ans. (b) : User Datagram Protocol (UDP) is also a transport – layer protocol and an alternative to TCP. It provides a connectionless data transmission service between application. UDP has less error checking than TCP. If UDP users want to be able to respond to errors, the communicating programs must establish their own protocol for error handling with high quality transmission network, UDP errors are of little concern.

220. Match List I with List II

	List - I		**List - II**
A.	**Serial Line IP (SLIP)**	**i.**	**Application layer**
B.	**Border Gateway Protocol (BGP)**	**ii.**	**Transport layer**
C.	**User Data Protocol (UDP)**	**iii.**	**Data Link layer**
D.	**Simple Network management protocol**	**iv.**	**Network layer**

Choose the correct answer form the options given below:
(a) A-I, B-II, C-III, D-IV
(b) A-III, B-IV, C-II, D-I
(c) A-II, B-III, C-IV, D-I
(d) A-III, B-I, C-IV, D-II

UGC NET C.S. June-2020 (Paper-II)

Ans. (b) : The correct match–

	List - I		**List - II**
A.	Serial Line IP (SLIP)	**iii.**	Data Link layer
B.	Border Gateway Protocol (BGP)	**iv.**	Network layer
C.	User Data Protocol (UDP)	**ii.**	Transport layer
D.	Simple Network management protocol	**i.**	Application layer

Sarkari Resource

221. In which switching each packet of a message follow the same path from sender to receiver?
किस स्विचिंग प्रणाली में, किसी मेसेज के हर पैकेट को सेंडर से रिसीवर तक जाने के लिए समान पथ का उपयोग किया जाता है?

(a) Circuit Switching/सर्किट स्विचिंग
(b) Message Switching/मेसेज स्विचिंग
(c) Virtual approach to packet switch
वर्चुअल एप्रोच टू पैकेट स्विचिंग
(d) Datagram approach to packet switching
डाटाग्राम एप्रोच टू पैकेट स्विचिंग

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Circuit Switching– A dedicated path is established before data transfer (like in telephone calls)
⇒ All data follows that path, but not in packets- it's a continuous stream.
- In message switching, the entire message is sent as a whole, stored at intermediate devices, and then forwarded (store-and-forward).
- In Packet Switching each packet is treated independently and may take different paths.

222. Sniffers can be prevented by using:
स्निफरो के निम्नलिखित का उपयोग करके रोका जा सकता है:

(a) Wired environment
(b) Ethernet LAN
(c) Switched network
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c) : Switched make sure that the packet is sent to the intended receiver and no one else, thus preventing sniffers to perform their function intelligent switches are hence used preferably for the network.

223. Which layer does the data link layer take packets from and encapsulate them into frames for transmission?
डेटा लिंक परत किस परत से पैकेट लेती है और उन्हें ट्रांसमिशन के लिए फ्रेम में समाहित करती हैं?

(a) Transport layer
(b) Application layer
(c) Network layer
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c) In computer networks, data from the application layer is sent to the transport layer and converted into segments. These segments are then transferred to the network layer and are called packets. These packets are then sent to the data processing layer where they are encapsulation into frames. These frames are then transferred to the physical layer where the sequences are converted to bits.

224. The number of crosspoints needed for 10 lines in a crosspoint switch, which is full duplex in nature and there is no self-connection, is
एक क्रॉसप्वॉइंट स्विच, जो पूर्णतया duplex है एवं जिसमें कोई स्वतः सम्बन्ध नहीं है, में 10 लाइनों के लिए क्रॉसप्वॉइंट की संख्या होगी

(a) 100
(b) 45
(c) 50
(d) 90

LT. Grade C.S. 2018

Ans. (b) : In a full duplex crosspoint switch with no self connections the number of crosspoints required for 10 lines can be calculated using the formula.

$$\text{Number of crosspoints} = \frac{n \times (n-1)}{2}$$

Where n is the number of lines
For n = 10

$$\text{Number of crosspoints} = \frac{10 \times (10-1)}{2} = 45$$

Thus, 45 crosspoints are needed for a full - duplex switch with 10 lines and no self - connections.

225. Identify the communication protocol which establishes a dedicated and direct connection between two communication devices. This protocol defines how two devices will authenticate each other and establish a direct link between them to exchange data.
उस कम्युनिकेशन प्रोटोकॉल की पहचान कीजिए जो दो कम्युनिकेटिंग डिवाइस के बीच एक डेडिकेटेड और डायरेक्ट कनेक्शन स्थापित करता है। यह प्रोटोकॉल यह बताता है कि ये दोनों डिवाइस किस प्रकार एक दूसरे को अथेन्टिकेट करेंगे और डाटा एक्सचेंज करने के लिए अपने बीच एक डायरेक्ट लिंक स्थापित करेंगे।

(a) PPP/पी पी पी
(b) SMTP/एस एम टी पी
(c) TELNET/टेलनेट
(d) FTP/एफ टी पी

EMRS PGT C.S. 16.12.2024

Ans. (a) : In computer networking, Point to Point Protocol is a data link layer communication protocol between two routes directly without any host or any other networking in between. It can provide loop detection authentication transmission encryption and data compression.

226. Identify the switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually. The pieces are given a unique number and addresses to identify their order at the receiving end.
उस स्विच तकनीक की पहचान कीजिए जिसमें एक बार में मैसेज भेजा जाता है, किंतु उसे छोटे-हिस्सों में विभाजित किया जाता है, और उन्हें अलग-अलग भेजा जाता है। उन हिस्सों को एक अनूठा नंबर और एड्रेस दिया जाता है ताकि प्राप्तकर्त्ता की ओर से उनके क्रम (आर्डर) की पहचान की जा सके।

(a) Message Switching/मैसेज स्विचिंग
(b) Packet Switching/पैकेट स्विचिंग
(c) Binary Switching/बाइनरी स्विचिंग
(d) Circuit Switching/सर्किट स्विचिंग

EMRS PGT C.S. 16.12.2024

Ans. (b) : The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces and they are sent individually. The messages splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.

Sarkari Resource

227. GPRS is basically the standard-bearer of:
बुनियादी रूप से GPRS किसका एक मानक वाहक (Standard-Bearer) है?
(a) 2.5G technology/2.5जी तकनीक
(b) 3G technology /3जी तकनीक
(c) 4G technology4जी तकनीक
(d) 2G technology /2जी तकनीक

EMRS PGT C.S. 16.12.2024

Ans. (a) : GPRS is basically the standard bearer of 2.5G technology. It is a packet switched type of network in which the billing depends on the total data transferred. It lets its users develop a direct connection over the internet.

228. The _____ houses the switches in token ring.
_____ स्विचों को टोकन रींग में रखता है
(a) Transceiver/ट्रांसिवर
(b) Nine-pin connector/नाइन-पीन कनेक्टर
(c) MAU
(d) NIC

KVS PGT C.S. 2013

Ans. (c) : MAU stands for Media Access Unit. A MAU is also known as an Ethernet transceiver or MSAU (Multistation Access Unit) is an adapter. Connector or stand also device that enables a network device to be connected to a token ring network.

229. Which of the following devices understands the format and contents of the data and translate messages from one format to another?
निम्नलिखित में से कौन सी युक्ति डेटा के फार्मेट और अन्तर्वस्तु को समझती है और संदेश को एक फार्मेट से दूसरे फार्मेट में परिवर्तित करती है?
(a) Gateway/गेटवे (b) Hub/हब
(c) Switch/स्विच (d) Router/राउटर

KVS PGT C.S. 2017

Ans. (a) : Gateway devices understands the format and contents of the data and translate messages from one format to another. It is a passage to connect two networks that may work upon different networking models. Gateways are also called protocol converters. and can operate at any network layer.

230. Ethernet is a standard for
Ethernet का स्टैंडर्ड है।
(a) HDD (b) CDs
(c) USB (d) NIC

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : Ethernet is a networking standard that defines how data is transmitted over a wired network. A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network using the Ethernet standard. It's the physical card that you ping an Ethernet cable into.

231. Which of the following is done by Secure Sockets Layers?
सिक्योर सॉकेट लेयर (SSL) क्या करता है?
(a) Creates a secure, private connection to a web server/एक सुरक्षित, निजी कनेक्शन बनाता है
(b) Encrypts information
सूचना को एन्क्रिप्ट करता है
(c) Sends information over the internet
इंटरनेट पर जानकारी भेजता है
(d) All of the above/उपरोक्त सभी

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : SSL stands for Secure Socket Layer. It is a technology that creates a secure and private connection to a web server, encrypts information and sends it securely over the internet.

232. Contention-based MAC protocols are commonly used in:
कंटेंशन-आधारित MAC प्रोटोकॉल सामान्यतः कहाँ उपयोग किए जाते हैं?
(a) Ethernet network/ईथरनेट नेटवर्क
(b) Token Ring networks/टोकन रिंग नेटवर्क
(c) ATM networks/एटीएम नेटवर्क
(d) Point-to-Point networks/पॉइंट-टू-पॉइंट नेटवर्क

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : A contention-based MAC protocol allows multiple devices to share a communication channel or medium. These protocols operate on the principle of listen before you talk. Before transmitting data a device listens to see if the channel is true. If it is the device can send its data.
Token Ring networks use token passing (not contention).
ATM networks use reservation-based methods (not contention).
Point to point networks don't require contention since only two devices are connected so, contention-based MAC protocols are commonly used in Ethernet.

233. What is the purpose of ARP (Address Resolution Protocol)?
ARP (एड्रेस रेजोल्यूशन प्रोटोकॉल) का उद्देश्य क्या है?
(a) To assign IP addresses to devices
उपकरणों को IP एड्रेस असाइन करना
(b) To resolve domain names to IP addresses
डोमेन नाम को IP एड्रेस में परिवर्तित करना
(c) To map MAC addresses to IP addresses
MAC एड्रेस को IP एड्रेस से मैप करना
(d) To manage network congestion
नेटवर्क भीड़ का प्रबंधन करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : ARP stands for Address Resolution Protocol. The primary purpose of ARP is to map a device's logical IP address to its physical MAC address on a local network, enabling devices to communicate with each other by finding the correct hardware address to send data packets to the right destination within the same network segment without ARP, devices on a local network wouldn't be able to discover the MAC addresses needed to deliver data packets making local communication impossible.

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(v) Multiple Access Protocol and Networks.

234. Which of the following is not an example of unguided transmission media?
निम्नलिखित में से कौन सा अनगाइडेड ट्रांसमिशन मीडिया का एक उदाहरण नहीं है?
(a) Radio Transmission/रेडियों ट्रांसमिशन
(b) Microwave transmission/माइक्रोवेव ट्रांसमिशन
(c) Coaxial cable/कोएकसिअल केबल
(d) Light transmission/लाइट ट्रांसमिशन

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : Unguided medium is transmission medium which travel through wireless communication channel and unbounded transmission media. It is generally suited for ratio broadcasting in all directions example of unguided medium is include microwave, radio or infrared.

235. When a mobile telephone physically moves from one cell to another cell, the base station transfers ownership to the cell, getting strongest signal. This process is known as ——.
जब एक मोबाइल टेलीफोन भौतिक रूप से एक सेल से दूसरे सेल में जाता है, तो बेस स्टेशन सबसे मजबूत सिग्नल प्राप्त करने वाले सेल को स्वामित्व हस्तांतरित करता है। इस प्रक्रिया को .....के रूप में जाना जाता है।
(a) Handoff (b) Mobile routing
(c) Mobile switching (d) Cell Switching

KVS PGT C.S. 2018

Ans. (a) : When a mobile telephone physically moves from the one cell to another cell, the base station transfers ownership to the cell getting strongest signal. This process is known as Handoff. A Handoff refers to the process of transferring an active call or data session from one cell in a cellular network to another or from one channel in a cell to another. A Well-implemented Handoff is important for delivering uninterrupted service to a caller or data session user.

236. Which of the following are key features of Mobile Adhoc Networks (MANETs)
A. Self-Organizing and decentralized.
B. High Mobility of nodes.
C. Requires fixed Infrastructure for operation.
D. Network Topology is dynamic a-nd constantly changing.
E. It requires a central controller to manage network traffic.
Choose the correct answer from the options given below :
(a) A, B & E only (b) B, C, D & E only
(c) A, B & D only (d) A, B & C only

UGC NET C.S. December-2024

Ans. (c) : Key features of Mobile Adhoc Network (MANETs) Include self organizing and decentralized, dynamic network topology, high mobility of nodes.
Self organization and decentralized:- MANETs automatically from networks without the need for a preconfigured infrastructure, with device dynamically connecting.
Dynamic Network topology:- As device move, the network structure constantly changes, requiring flexible routing protocols to adapt to new connections.
Mobile nodes:- Device with in a MANETs are able to move freely, impacting network connectively.

237. Which device is needed to connect two computer systems, if the systems use different protocols ?
यदि सिस्टम अलग-अलग प्रोटोकॉल का उपयोग करता है, तो दो कंप्यूटर सिस्टम को जोड़ने के लिए कौन-सी डिवाइस की आवश्यकता होती है।
(a) Repeater (b) Gateway
(c) Bridge (d) Hub

HTET PGT C.S. 2020

Ans. (b) : Gateway:–
- A gateway is a network device that connects two networks with different protocols.
- It acts as a translator, enabling communication between networks that use different communication protocol
- Gateways are crucial for communication between different network architecture between a local area network (LAN) and the Internet.
- They operate at any network layer, usually focusing on the application layer for protocol translation layer for protocol translation

238. The traffic which can adjust over wide ranges, to changes in delay and throughput across an internet and still meets the needs of an application, is called as:
ट्रैफिक जो इंटरनेट पर देरी और थ्रूपुट में बदलाव के लिए व्यापक रेंज में समायोजित हो सकता है और फिर भी एक एप्लीकेशन की जरूरतों को पूरा करता है–
(a) Non Elastic traffic/नॉन इलास्टिक ट्रैफिक
(b) Elastic traffic/इलास्टिक ट्रैफिक
(c) Semi Elastic traffic/सेमी इलास्टिक ट्रैफिक
(d) Hybrid traffic/हाईब्रिड ट्रैफिक

HTET PGT C.S. 2023

Ans.(b): Elastic traffic can adjust, over wide ranges, to changes in delay and throughput across the internet and still meet the needs of its applications, in elastic traffic does not easily adopt, if at all, to changes in delay and throughput across the internet. The prime example is real time traffic, such as voice and video.
Elastic traffic has the ability to make adjustments to wide ranging changes in delay and throughput across the internet and still meets the needs of the applications, it adjust its throughput between end hosts in response to network conditions network load or congestion may cause packet loss.

239. Real time transport control protocol informs sender of multimedia stream about _____.
वास्तविक समय ट्रांसपोर्ट कन्ट्रोल प्रोटोकाल _____ के बारे में मल्टीमीडिया स्ट्रीम के प्रेषक को सूचित करता है।
(a) Length of multimedia data
मल्टीमीडिया डाटा की लम्बाई
(b) Network performance/तंत्रिका प्रदर्शन

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(c) Voice and data quality/वॉइस एवं डाटा गुणवत्ता
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (b) : RTCP (Real Time-transport Control Protocol):–
- RTCP works in conjunction with RTP to provide control service for media streaming by offering feedback on the quality of services being provided.
- RTCP collects and sends control data to participants in a streaming media session with the primary function of providing feedback on the quality of the data distribution.
- This Protocol is used to monitor transmission statistics and quality of services and aids in controlling congestion.
- RTCP provides a standard way to carry statistical and control data, such as sender or receiver reports that describe the quality of services, bandwidth media characteristics, and network delay.

240. If there are *N* people in the world and are using secret key encryption/decryption for privacy purpose, then number of secret keys required will be:
(a) N (b) $(N-1)$
(c) $\frac{N(N-1)}{2}$ (d) $\frac{N(N+1)}{2}$

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Each pair of users must have a secrete key. If N people in world want to use this method, there needs to be ${}^{n}C_{2} = \frac{n(n-1)}{2}$ secrete keys.

241. The network using CSMA/CD has a bandwidth of 20 mbps. If the maximum propagation time is 25μ sec (microsecond), what is the minimum size of the frame?
(a) 500 bits (b) 1000 bits
(c) 1500 bits (d) 2000 bits

UGC NET C.S. December-2024

Ans. (b) : The minimum frame size in a CSMA/CD network, we use the following formula:
Minimum Frame size = 2 × propagation time × Bandwidth.
Given data:-
• Bandwidth (B) = 20m bps = 20×10^6 mbps
• Maximum propagation time (Tp) = 25.μs
$= 25 \times 10^{-6}$ μs
Step by step calculation:-
1. Calculate round trip time:-
$2 \times T_p = 2 \times 25 \times 10^{-6}$ s.
2. Compute minimum frame size:
Frame size = $50 \times 10^{-6} \times 20\ 10^6$
= 1000 bits

242. Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as _____.
सिस्टम, जिनमें कई उपयोगकर्ता एक सामान्य चैनल को इस तरह से साझा करते हैं की जिससे कॉनफ्लिक्ट हो सकते हैं, को व्यापक रुप से _____ के रुप में जाना जाता है ।
(a) Carrier/कैरियर
(b) Communication network/कम्यूनिकेशन नेटवर्क
(c) Contention/कंटैंशन
(d) Bus/बस

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : A contention system is a network where multiple users share a common communication channel without any predefined schedule. This means any predefined schedule. This means that users can transmit data whenever they want, which can need to collisions if multiple users try to transmit at the same time. This creates a conflict situation, as the data packet might interface with each other, causing transmission errors.

243. _____is a private network that uses a public network to connect remote sites or users together.
_____एक निजी नेटवर्क है जो रिमोट साइट्स या उपयोगकर्ताओं को एक साथ जोड़ने के लिए सार्वजनिक नेटवर्क का उपयोग करता है।
(a) Visual public network/विज़ुअल पब्लिक नेटवर्क
(b) Virtual private network/वर्चुअल प्राइवेट नेटवर्क
(c) Virtual public network/वर्चुअल पब्लिक नेटवर्क
(d) Visual private network/विज़ुअल प्राइवेट नेटवर्क

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : VPN stands for virtual private network. It is a technology that creates a secure, private connection over a public network like the internet, allowing remote users to access private networks or share data securely.

244. Which of the following type of guided transmission media are categorized by their Radio Government (RG) ratings in computer networking?
(a) Fiber-Optic Cable (b) Coaxial Cable
(c) Metallic Cable (d) Twisted-Pair Cable

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : In computer networking the type of guided transmission media are categorized by their Radio Government (RG) ratings is coaxial cable.
Coaxial cables are assigned RG ratings (e.g., RG-6, RG-11, RG-58, RG-59) which specify their characteristics such as impedance, thickness, and shielding.

245. Which of the following statement is correct regarding unguided media (wireless)?
I. Unguided media transport electromagnetic waves without using a physical conductor.
II. Unguided signals can travel from the source to destination in ground propagation way.
(a) Only II (b) Both I and II
(c) Only I (d) Neither I nor II

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (b) : Unguided media, also known as wireless communication, transmits data via electromagnetic waves through a medium like air or space.
So, both statement I and II are correct regarding unguided media.

246. Which of the following statement is incorrect for parallel virtual machine (PVM)?
(a) The PVM communication model asynchronous blocking send, asynchronous blocking receive, and non-blocking receive function.

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(b) Message buffers are allocated dynamically
(c) The PVM communication model assumes that any task can send a message to any other PVM task and that there is no limit to the size or number of such messages
(d) In PVM model, the message order is not preserved.

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : Fault tolerance through dynamic addition & deletion of hosts/processes.
A widely used parallel message passing library defines
• How takes communicative or synchronize with each other.

247. Which of the following algorithms is not a broadcast routing algorithm?
(a) Flooding
(b) Multi destination routing
(c) Reverse path forwarding
(d) All of the above

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Flooding is suited for ordinary point to point communication, for broadcasting it might be good solution. The problem with flooding is it generates too may packets & consumes too much bandwidth.
Multi destination routing have each packet contains either a list of destination and reverse path forwarding is an attempt to approximate the spanning tree broadcast even when the routers do not know anything at all about spanning tree.

248. A message "COMPUTER NETWORK" encrypted (ignore quotes) using columnar transposition cipher with a key "LAYER". The encrypted message is :
(a) CTTOEWMROPNRUEK
(b) MROUEKCTTPNROEW
(c) OEWPNRCTTUEKMRO
(d) UEKPNRMROOEWCTT

UGC NET C.S. June-2015 (Paper-III)

Ans : (c)

3	1	5	2	4
C	O	M	P	U
T	E	R	N	E
T	W	O	R	K

Now, read column by column; first 1st, 2nd, 3rd, 4th then 5th. Therefore, encrypted message is:-
OEW PNR CTT UEK MRO.

249. Which of the following statement/s is/are true?
(i) Firewalls can screen traffic going into or out of an organization.
(ii) Virtual private networks can simulate an old leased network to provide certain desirable properties.
Choose the correct answer from the code given below :
(a) (ii) only (b) Neither (i) nor (ii)
(c) (i) only (d) Both (i) and (ii)

UGC NET C.S. December-2018 (Paper-II)

Ans. (d) : Network traffic passes through the firewall, the firewall decides which traffic to forward and which traffic not to forward, based on rules that you have defined. All firewalls screen traffic that comes into your network, but a good firewall should also screen outgoing traffic.
A network is often considered private even though an organization leases rather than owns all the channels that connect its sites. This is because the technical effects of traffic transmission are the same whether physical channels are owned or leased, as these channels always have a known and fixed bandwidth. By contrast, when an organization uses a public data network to connect its sites, traffic goes through shared public channels and receives an unknown share of the channels' bandwidth.

250. Consider the following statements with respect to network security:
(A) Message confidentiality means that the sender and the receiver expect privacy.
(B) Message integrity means that the data must arrive at the receiver exactly as they were sent.
(C) Message authentication means the receiver is ensured that the message is coming from the intended sender.
Which of the statements is (are) correct?
(a) Only (A) and (B) (b) Only (A) and (C)
(c) Only (B) and (C) (d) (A), (B) and (C)

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (d) : (a) message confidentiality means that the sender and the receiver expect privacy.
Message confidentiality means that only sender and intended receiver should be able to access the contents of a message.
Confidentiality gets compromised if an unauthorized person is able to access a message. It means sender and receiver expect privacy.
(b) Message integrity means that the data must arrive at the receiver exactly as they were sent.
When the contents of a message are changed after the sender sends it, but before it reaches the intended recipient, we say that the integrity of the message is lost. It means that the data must arrive at the receiver exactly as they were sent.
(c) Message authentication means the receiver is ensured that the message is coming from the intended sender.
Authentication mechanisms help establish proof of identities. The authentication process ensures that the origin of an electronic message or document is correctly identified. It means that receiver is ensured that the message is coming from the intended sender.

251. The ability to inject packets into the Internet with a false source address is known as
(a) Man-in-the-middle attack
(b) IP phishing
(c) IP sniffing
(d) IP spoofing

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : IP spoofing :-
IP address spoofing or IP spoofing is the creation of Internet protocol packets with a false source IP address, for the purpose of impersonating another computing system.

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Main-in-the-Middle Attack : It is an attack where the attacker secretly relays and possibly alters the communications between two parties who believe they are directly communicating with each other three security issue that are particularly applicable to the IP protocol : **packet sniffing, packet modification, and IP spoofing**.
Packet sniffing : In packet sniffing, an intruder tries to intercept the IP packet and make a copy of it is a passive attack, in which the attacker does not change the contents of the packet.
IP spoofing : In IP spoofing, an attacker can masquerade (Pretense) somebody else and creates an IP packet that carries the source address of another computer. An attacker can send an IP packet to a bank pretending that is coming from one of the customers.

252. What does MIME stand for?
MIME का क्या अर्थ है?
(a) Multipurpose Internet Messaging Extension
(b) Multipurpose Internet Mail Extension
(c) Multipurpose Internet Multimedia Extension
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(b) MIME stands for 'Multipurpose Internet Mail Extension'. It is a standard used to comprise different types of data such as text, images, audio, video etc. on the Internet.

253. What is the primary function of a router in a computer network?
कम्प्यूटर नेटवर्क में राउटर का प्राथमिक कार्य क्या है?
(a) To connect devices within the same network
के भीतर उपकरणों को जोड़ने के लिए वही नेटवर्क
(b) To connect multiple networks together
एकाधिक नेटवर्क को जोड़ने के लिए एक साथ
(c) To filter incoming traffic based on IP addresses/आने वाले ट्रैफिक को फिल्टर करने के लिए आईपी पते
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : The primary function of a router in a computer network is to connect multiple networks together. A router connects different networks and guides data packets from one network to another.

254. Identify the term, which is an unauthorized real-time interception or monitoring of private communication between two entities over a network.
उस पद की पहचान कीजिए, जो एक नेटवर्क पर दो व्यक्तियों के बीच के निजी वार्तालाप को अनाधिकृत रूप से रीयल-टाइम इंटरसेप्ट या मॉनीटर करता है।
(a) Eavesdropping/इव्सड्रापिंग
(b) Phishing/फिशिंग
(c) Snooping/स्नूपिंग
(d) Cyberstalking/साइबरस्टाकिंग

EMRS PGT C.S. 16.12.2024

Ans. (a) : The term for unauthorized real time interception or monitoring of private communication between two entities over a network is eavesdropping.

255. Which type of cable has high bandwidth supports high data transfer rate, signals can travel longer distances and electromagnetic noise cannot affect the cable, however expensive. Identify the transmission media.
किसी प्रकार के केबल की बैंडविड्थ अधिक होती वह हाई डाटा ट्रांसफर रेट को सपोर्ट करता है, उसके सिग्नल काफी दूरी तक जाते हैं और उसकी इलेक्ट्रोमैग्नेटिक ध्वनि केबल को प्रभावित नहीं करती है, तथापि यह महंगी होती है। ट्रांसमिशन मीडिया की पहचान कीजिए।
(a) Coaxial Cable (b) Optical Fiber
(c) Twisted Pair Cable (d) Cat 6 Cable

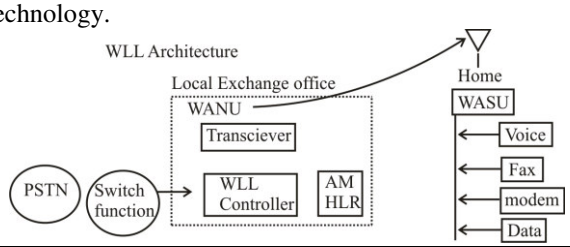
EMRS PGT C.S. 16.12.2024

Ans. (b) : Fiber optical cable has the ability to transmit signals over much longer distances than coaxial and twisted pair. It also has the capability to carry information at vastly greater speeds.

256. What is the full form of WLL ?
डब्ल्यू एलएल (WLL) का पूरा नाम क्या है?
(a) Wireless Local Line/वायरलेस लोकल लाइन
(b) Wireless Local Loop/वायरलेस लोकल लूप
(c) Wireless Local Live/वायरलेस लोकल लाइव
(d) Wireless Low Line/वायरलेस लो लाइन

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (b) : WLL stands for Wireless Local Loop. WLL is a generic word for an access system that connects users to the local telephone company's switch via wireless links rather than traditional copper cables. This system also known as fixed wireless access or fixed radio, provides telephone, and data services to business and residential subscribers using analog or digital radio technology.



257. The term ____ refers to the use of electronics and software within a product, as opposed to a general-purpose computer, such as a laptop or desktop system.
_____पद प्रोडक्ट में इलेक्ट्रोनिक्स और सॉफ्टवेयर के उपयोग से संबंधित है, जो सामान्य हेतु कम्प्यूटर्स जैसे लेपटोप या डेस्कटोप सिस्टम के विपरीत है ।
(a) Complex system/काम्प्लेक्स सिस्टम
(b) Simple system/सिम्पल सिस्टम
(c) Fuzzy logic/फ़ज़ी लॉजिक
(d) Embedded system/एम्बेडेड सिस्टम

KVS PGT C.S. 2013

Ans. (d) : An embedded device is a part of an integrated system that is formed as a combination of computer hardware and software for a specific function and which can operate without human interaction.

Sarkari Resource

258. The x.25 standard specifies a
x.25 मानक निर्दिष्ट करता है

(a) Technique for start-stop data.
स्टार्ट-स्टोप डाटा की प्रविधि
(b) Technique for dial access.
डायल एक्सेस की तकनीक
(c) DTE/DCE interface./DTE/DCE इंटरफेस
(d) Data bit rate./डाटा बिट रेट

KVS PGT C.S. 2013

Ans. (c) : An X.25 network is an interface b/w data terminal equipment (DTE) and data circuit – terminating equipment (DCE) that operates in the packet mode. An X .25 network connects to public data networks by dedicated circuits. X.25 networks use the connection mode network services.

259. What is the meaning of bandwidth in a network ?
नेटवर्क में बैंडविड्थ का क्या अर्थ है ?

(a) Transmission capacity of a communication channel./एक कम्युनिकेशन चैनल की ट्रांसमिशन क्षमता
(b) Connected computers in a network.
नेटवर्क में संयोजित कम्प्यूटर्स
(c) Class of IP used in Network.
नेटवर्क में प्रयुक्त IP का क्लास
(d) Interconnected by communication channels.
कम्युनिकेशन चेनलों द्वारा अंतःसंयोजित

KVS PGT C.S. 2013

Ans. (a) : Network bandwidth is a measurement indicating the maximum capacity of a wired or wireless communications link to transmit data over a network connection in a given amount of time. Bandwidth is represented in the number of bits, kilobits, megabits or gigabits that can be transmitted in 1 second .

260. The unit receiving the data item responds with another control signal to acknowledge receipt of the data. This type of agreement between two independent units is known as____.
डेटा मद को प्राप्त करने वाली इकाई डेटा की प्राप्ति की सूचना के लिए अन्य नियंत्रण सिंगनल के साथ अनुक्रिया करता है। दो स्वतंत्र इकाइयों के बीच इस प्रकार के संबंध को _____ कहा जाता है।

(a) Strobe control/स्ट्रोब कंट्रोल
(b) Multitasking/मल्टीटास्किंग
(c) Handshaking/हैंडशेकिंग
(d) Piggybacking/पिग्गीबेकिंग

KVS PGT C.S. 2017

Ans. (c) : The units receiving the data item responds with another control signal to acknowledge receipt of the data. This type of agreement between two independent units is known as handshaking. The handshaking process means communication between two units or the transfer of data between the two Systems.

261. Ten signals, each requiring 3000Hz, are multiplexed onto a single channel using FDM. How much minimum bandwidth is required for the multiplexed channel? Assume that the guard bands are 300 Hz wide.
दस सिग्नल जिनमें प्रत्येक की 3000 अपेक्षित है, एफ डी एम का उपयोग करते हुए एक एकल चैनल पर बहु संकेतिक (मल्टीप्लेक्स) किया गया है। बहु संकेतिक चैनल के लिए कम से कम कितने बैण्डविड्थ की आवश्यकता होगी? यह मान ले कि गार्ड बैण्ड 300Hz विस्तीर्य (वाइड) है।

(a) 33,700 (b) 30,000
(c) 32,700 (d) 33,000

KVS PGT C.S. 2017

Ans. (c) : 10 signals with having 3000 Hz frequency.
if there are n signals multiplexed on to a single channel. Then the number of guard bands between each signal required = n–1
Hence, guard bands required = 10–1 = 9
guard bands = 300 Hz wide.
minimum bandwidth required for the multiplexed channel = 3000×10+300×9
= 30000+2700
= 32700Hz

262. The IETF standards of documents are called
IETF मानकों के दस्तावेज कहलाते हैं

(a) RFC (b) RCF
(c) ID (d) None of these

Bihar STET C.S. 2020

Ans. (a) : RFC stands for Request for comments and they are documents that describe methods, behaviors, research, or innovations applicable to the working of the internet.

263. Two devices are in network is
दो डिवाइस नेटवर्क में हैं, अगर

(a) A process in one device is able to exchange information with a process in another device
(b) A process is running on both devices
(c) PIDs of the processes running of different devices are same
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2020

Ans. (a) : A computer network or data network is a digital telecommunications network which allow nodes to share resources. In computer networks, computing devices exchange data with each other using connections between nodes. The nodes have certain processes which enable them to share a specific type of data using a distinct protocol.

264. _____ is a combination of hardware and software that facilitates the sharing of information between computing devices.
_____ हार्डवेयर और सॉफ्टवेयर का एक संयोजन है जो कंप्यूटिंग डिवाइसों के बीच सूचना के आदान-प्रदान की सुविधा प्रदान करता है।

(a) Network (b) Peripheral
(c) Expansion board (d) Digital device

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : Network is a combination of hardware and software that facilitates the sharing of information between computing devices.

Sarkari Resource

265. ISP exchanges internet traffic between their networks by
ISP अपने नेटवर्कों के बीच इंटरनेट ट्रैफिक का आदान-प्रदान करता है
(a) Internet exchange point
(b) Subscriber end point
(c) ISP end point
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : When different internet service providers exchange internet traffic between their networks, it is done through an Internet Exchange Point (IEP). It is a physical location where ISPs share data traffic among themselves to improve internet speed and reduce cost.

266. Which one of the following protocols is not used in internet?
निम्नलिखित में से कौन सा प्रोटोकॉल इंटरनेट में उपयोगी नहीं किया जाता है?
(a) HTTP (b) DHCP
(c) DNS
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (d) :
- **HTTP (Hyper Text Transfer Protocol)–** To local web pages.
- **DHCP (Dynamic Host Configuration Protocol)–** To assign an IP address to the device.
- **DNS (Domain Name System)–** To convert domain to IP address.

267. Internet works on
Internet काम करता है
(a) Packet switching
(b) Circuit switching
(c) Both Packet switching and circuit switching
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : The internet works on packet switching technology in which data is divided into small packets and sent through different routes and reassembled at the destination.

268. Which protocol assigns IP address to the client connected in the internet?
कौन सा प्रोटोकॉल, इंटरनेट में जुड़े Client को IP Address प्रदान करता है?
(a) DHCP (b) IP
(c) RPC
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : DHCP (Dynamic Host Configuration Protocol) is a protocol that automatically provides IP Address, Subnet Mask, Gateway etc. to the client devices connected to the internet.

269. Which one of the following is not used in media access control?
Media Access Control में निम्नलिखित में से किसका उपयोग नहीं किया जाता है?
(a) Ethernet
(b) Digital subscriber line
(c) Fiber distributed data interface
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : Media Access Control (MAC) deals with controlling how devices in a network gain access to the medium (cable/wireless) and permission to transmit data.
(a) Ethernet– It uses MAC for collision detection and access control (CSMA/CD, CSMA/CA).
(b) Digital Subscriber Line (DSL)– It is a point to point communication technology for internet over telephone lines. It does not use media access control, since there is no contention (dedicated line between subscriber and provider).
(c) Fiber Distributed Data Interface (FDDI)– It uses token passing which is a media access control method.

270. Two devices are in network if
दो डिवाइस नेटवर्क में है, अगर
(a) A process in one device is able to exchange
एक प्रक्रिया में एक डिवाइस एक्सचेंज कर सकती है
(b) A process is running on both devices
दोनों डिवाइसों में एक प्रक्रिया रन कर रहे हैं
(c) PIDs of the processes running on different devices are same/विभिन्न डिवाइसों में चलने वाली प्रक्रियाओं के PID एक ही है
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : A computer network, or data network is a digital telecommunications network which allows nodes to share resources. In computer network, computing devices exchange data with each other using connections between nodes. The nodes have certain processes which enable them to share a specific type of data using a distinct protocol.

271. Communication channel is shared by all the machines on the network in
संचार चैनल नेटवर्क पर सभी मशीनों द्वारा साझा किया जाता है
(a) Broadcast network
(b) Unicast network
(c) Multicast network
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : In a broadcast network, information is sent to all stations in a network whereas in a multicast network the data or information is sent to a group of stations in the network. In unicast network, information is sent to only one specific station. The broadcast address of the network is the bust assigned address of the network.

272. Which transmission media has the highest transmission speed in a network?
किस Transmission मीडिया के एक नेटवर्क में सबसे अधिक transmission speed है?
(a) Coaxial cable (b) Twisted pair cable
(c) Optical fiber (d) Electrical cable

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : Optical Fiber cables transmit data as pulses of light through thin strands of glass or plastic. This method allows for significantly higher data transfer roots and longer transmission distances compared to other media.

Sarkari Resource

273. In teardown state of real time streaming protocol is _____
रियल टाइम स्ट्रीमिंग प्रोटोकॉल की 'Teardown' स्थिति में क्या होता है?
(a) The server resources for client
सर्वर क्लाइंट के लिए संसाधन मुक्त करता है
(b) Server delivers the stream to client
सर्वर क्लाइंट को स्ट्रीम भेजता है
(c) Server suspends delivery of stream
सर्वर स्ट्रीम डिलीवरी को निलंबित करता है
(d) Server breaks down the connection
सर्वर कनेक्शन तोड़ देता है

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : In the Teardown state of the real time streaming protocol (RTSP), the server terminates the streaming session by breaking down the connection and freeing up any resources that were allocated for that specific stream. This is a final phase of the session, where the connection is gracefully terminated, stopping all media streams and freeing all session related data on the server.

274. In Real Time Interactive Audio Video, Jitter is introduced in real-time data by delay between…
रियल टाइम इंटरएक्टिव ऑडियो वीडियो में जिटर किसके बीच की देरी से उत्पन्न होता है?
(a) Pixels/पिक्सल्स (b) Layers/लेयर्स
(c) Frames/फ्रेम्स (d) Packets/पैकेट्स

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : Jitter is the condition when network packets (small pieces of data) do not arrive in the same order and at the sometime. When there is a delay between packets, it causes jitter, which can cause interruptions or distortion in real time audio or video.

(vi) TCP/IP Protocol, Network Layer, Transport Layer and Application Layer Protocol.

275. What is the full form of WDMA protocol?
डब्लयु.डी.एम.ए. प्रोटोकॉल का असंक्षिप्त रूप क्या है?
(a) Wide Division Multiple Access protocol
वाईड डिवीजन मल्टीपल एक्सेस प्रोटोकॉल
(b) Wide Digital multiple Access Protocol
वाईड डिजिटल मल्टीपल एक्सेस प्रोटोकॉल
(c) Wavelength Digital Multiple Access Protocol
वेवलेंथ डिजिटल मल्टीपल एक्सेस प्रोटोकॉल
(d) Wavelength Division Multiple Access Protocol
वेवलेंथ डिवीजन मल्टीपल एक्सम्स प्रोटोकॉल

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : WDMA stands for Wavelength Division Multiple Access. It is multiple access technique derived from Wavelength Division Multiplexing (WDM)
In fiber optic communication, WDM allows multiple signals to be transmitted simultaneously over a single optical fiber using different wavelengths (colors) of light. WDMA extends this concept to provide multiple users with access to the network resources based on assigning different wavelength for their communication.

276. Socket address is :
सॉकेट एड्रेस है:
(a) IPv4 address translated to IPv6 address
IPv4 में अनुवादित IPv6 एड्रेस
(b) Combination of IP address and port address
IP एड्रेस तथा पोर्ट एड्रेस का संयोजन
(c) Combination of MAC address and domain address/MAC एड्रेस तथा डोमेन एड्रेस का संयोजन
(d) Application layer address
एप्लीकेशन लेयर एड्रेस

HTET PGT C.S. 2022

Ans. (b) : A socket address is formed by combining an IP address (which identifies a specific processor or service on that host), this combination uniquely identifies a connection endpoint in networking.
• Correct socket address = IP address + port address.

277. The protocol which provides e-mail facility is :
ई-मेल सुविधा प्रदान करने वाला प्रोटोकॉल है
(a) FTP
(b) SMTP
(c) TELNET/टेलनेट
(d) None of these/इनमें से कोई नहीं

HTET PGT C.S. 2021

Ans. (b) : SMTP (Simple Mail Transfer Protocol) is a TCP/IP Protocol used in sending and receiving e-mail. However, since it is limited in its ability to queue message at the receiving end, it is usually used with one of two other protocols, PoP_3 or IMAP that let the user save massage in a server mail box and download them periodically from the server. SMTP usually is implemented to operate over internet port 25 many mail servers now support Extended Simple Mail Transfer Protocol (ESMTP). Which allows multimedia files to be delivered as e-mail.

278. Arrange the following steps in the correct order for a DHCP Client to renew its IP lease with a DHCP server :
(A) DHCP client sends a DHCPREQUEST message
(B) DHCP server acknowledges the renewal with a DHCPACK message
(C) DHCP client checks the local lease timer and initiates renewal
(D) DHCP server updates its lease database
Choose the correct answer from the options below :
(a) (A), (B), (C), (D) (b) (C), (D), (B), (A)
(c) (C), (B), (A), (D) (d) (C), (A), (B), (D)

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Ans. (d) : The correct order for a DHCP client to renew its IP lease with a DHCP server is-
(C) The DHCP client checks the local lease timer and initiates renewal.
(A) The DHCP client lends a DHCPREQUEST message to the DHCP server requesting to renew its lease.
(B) The DHCP server acknowledge the renewal with a DHCPACK message.
(D) The DHCP server updates its lease database to reflect the renewed lease.

Sarkari Resource

279. Which of the following terms is used to represent "a set of rules" in computer networks?
कम्प्यूटर नेटवर्क में ''नियमों का एक सेट'' का प्रतिनिधित्व करने के लिए निम्नलिखित में से किस शब्द का उपयोग किया जाता है?
(a) Resource locator/संसाधन लोकेटर
(b) Protocol/प्रोटोकॉल
(c) Domain/डोमेन
(d) URL/यूआरएल

HTET PGT C.S. 2020

Ans. (b) : A protocol is set of guidelines that govern how devices communicate on a network, dictating how data is transmitted, received and interpreted.

280. Which of the following protocol used to transmit information on the World Wide Web?
निम्नलिखित में से कौन सा प्रोटोकॉल वर्ल्ड वाइड वेब पर जानकारी प्रसारित करने के लिए उपयोग किया जाता है?
(a) HTT (b) HPP
(c) HTTP (d) HPPT

HTET PGT C.S. 2020

Ans. (c) : HTTP is used to transmit information on the World Wide Web.
• The Hypertext Transfer Protocol (HTTP) is an application layer protocol for distribute, collaborative, hypermedia information systems.
• HTTP is the foundation of data communication for the World Wide Web, where hypertext documents include hyperlinks to another resources that the user can easily access, for example by a mouse click or by tapping the screen in a web browser.

281. IP address 10.231.241.1 belongs to which of the following class ?
आई. पी. एड्रेस 10.231.241.1 निम्न में से किस क्लास से संबंधित है?
(a) Class A/क्लास A (b) Class B/क्लास B
(c) Class C/क्लास C (d) Class D/क्लास D

HTET PGT C.S. 2018

Ans. (a) : Looking at the first octet of an IP address, you can Identify the class of that address.
- Class A : 0 – 127
- Class B : 128 – 191
- Class C : 192 – 223
- Class D : 224 – 239
- Class E : 240 – 254

In IP address 10, 231, 241, 1, first octet value of lies between 0 to 127
Therefore it belongs to class A.

282. _____ is an application layer protocol while _____ is internet layer protocol in TCP/IP reference model.
TCP/IP रेफरेन्स मॉडल में _____ एक एप्लीकेशन लेयर प्रोटोकॉल है जबकि _____ एक इन्टरनेट लेयर प्रोटोकॉल है।
(a) RTP, ICMP (b) DSL, IP
(c) SONET, DSL (d) DNS, TCP

HTET PGT C.S. 2018

Ans. (a) : Application Layer Protocol:– Application layer is the top most layer of TCP/IP model. This layer performs all the task that are performed by the session layer, presentation layer and application layer the protocols used at application layer conveys the user request to transport layer. These protocols help to transfer mail, sharing of file and terminal login. Below are the protocol used at application layer of TCP/IP model.
(a) RTP (Application Layer).
Internet layer :– Internet layer is also known as network layer. internet layer ensures that data is sent accurately and fastly by controlling the flow and routing of traffic if the internet traffic is more it take more time to transfer the data.
(a) ICMP (Internet Layer).

283. Consider the following statements:
दिये गये कथनों पर विचार कीजिए:-
(i) The distance vector routing algorithm may cause "Count to infinity" problem./डिस्टेंस वेक्टर रूटिंग एल्गोरिथ्म में ''काउंट टू इन्फिनिटी'' समस्या हो सकती है।
(ii) Flooding is a static routing algorithm./फ्लाडिंग एक स्थिर रूटिंग एल्गोरिथ्म है।
(iii) The link state routing algorithm is also known as bellman-ford routing algorithm./लिंक स्टेट रूटिंग एल्गोरिथ्म को बेलमैन- फोर्ड रूटिंग एल्गोरिथ्म भी कहा जाता है।
How many of the statements given above are correct?
ऊपर दिए गये कथनों में से कितने कथन सही है?
(a) Only one
(b) Only two
(c) All three
(d) None
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : Flooding is a static routing algorithm is correct. This statement where every incoming packet is sent out on every out going line except the one it arrived on.
Therefore, two statements are correct

284. Which of the following statement/s is/are NOT CORRECT?
(A) OSPF is based on distance-vector routing protocol.
(B) Both link-state and distance-vector routing are based on the least cost goal.
(C) BGP4 is based on the path-vector algorithm.
(D) The three-node instability can be avoided using split horizon combined with poison reverse.
(E) RIP is based on link state algorithm.
Choose the correct answer from the options given below:
(a) (A), (D) and (E) Only
(b) (A) and (B) Only
(c) (B) and (C) Only
(d) (B), (C) and (E) Only

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Sarkari Resource

Ans. (a) : OSPF is based on distance- vector routing protocol. This statement is incorrect. OSPE (Open shortest path first) is a link- state routing protocol, not distance- vector.
The three node instability can be avoided using split horizon combined with poison reverse. This statement is incorrect split horizon and poison reverse are techniques used to avoid the count to infinity problem in distance vector routing protocols.
RIP is based on link state algorithm. This statement is also incorrect. RIP (Routing Information Protocol) is a distance vector routing protocol, not a link state.

285. Transmission control protocol is:
ट्रांसमिशन कंट्रोल प्रोटोकॉल है–
(a) Collision sensitive protocol
(b) Collision less property
(c) Connection less protocol
(d) Connection oriented protocol

HTET PGT C.S. 2023

Ans.(d): Transmission control protocol (TCP) is a connection-oriented protocol for communication that helps in the exchange of messages between different device over a network it is one of the main protocols of TCP/IP suite in OSI model, it operators at the transport layer (Layer 4). It lies between the application and network layers which are used in providing reliable delivery services the internet protocol (IP), which establishes the techniques for sending data packets between computer works with TCP.

286. TTL field in IP header is used:
(a) To avoid infinite loops
(b) To fragment the packets in the subnet
(c) To calculate the shortest path from source to destination
(d) None of the above

CGPSC C.S. 2020 Set-A

Ans.(a): Time To Live (TTL)– This field in an IP packet header specifies the number of hops a packet can traverse before being.
• **Preventing in finite loops–** if a packet encounters a routing error and keeps looping back through the network, the TTL value decreases with each hop when the TTL reaches zero, the packet is dropped, preventing it from endlessly circulating.

287. The acknowledgement number field in a TCP segment header specifies
(a) Next byte expected
(b) Last byte correctly received
(c) Number of bytes expected next
(d) Number of bytes correctly analyzed
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (a) : TCP is a byte oriented reliable protocol.
• Each type of data is given a sequence number.
• The Acknowledgement (ACK) number field in the TCP header tells the sender what the receiver is expecting next.
So the ACK number is one more than the last successfully received byte number.
• In simple words, it does not repeat what is received, but instead tells send me the next byte starting from here.

288. Which of the following is an invalid IPv4 address?
(a) 240.132.129.64
(b) 153.254.287.83
(c) 177.250.154.171
(d) 202.127.45.109
(e) Not attempted

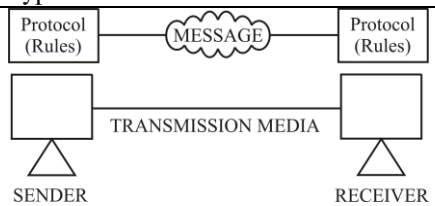
Haryana PGT C.S. Asstt. Prof. 2024

Ans. (b) : An IPv4 address has 4 octets each ranging from 0 to 255.
So, if any octet is less than 0 or greater than 255 the address is invalid.
(a) 240.132.129.64
• octet ; 240, 132, 129, 64 → all within 0–255
(b) 153.254.287.83
• octet ; 153, 254, 287, 83
• Here 287 > 255 → Invalid
(c) 177.250.154.171
• octets ; 177, 250, 154, 171 → all within 0–255.
(d) 202.127.45.109
• octets ; 202, 127, 45, 109 → all within 0–255.

289. A _____ is a set of formal operating rules, procedure or conventions that govern a given process.
_____ औपचारिक संचालन नियमों, प्रक्रिया या सम्मेलनों का एक सेट जो किसी दी गई प्रक्रिया को नियंत्रित करता है।
(a) Routing (b) Protocol
(c) Layer (d) Switching

HTET PGT C.S. 30.07.2025

Ans. (b) : A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly protocols help make sure that data moves smoothly and securely between devices on a network. To make communication successful between devices, some rules and procedures should be agreed upon at the sending and receiving ends of the system such rules and procedures are called Protocols. Different types of protocols are used for different types of communication.



In the above diagram, protocols are shown as a set of rules, communication between the sender and receiver is not possible without protocol.

290. Consider the following two statements:
Statement I: The size of the fixed part of the IP datagram header is 50 bytes.
Statement II: An IP datagram is transmitted in little endian order.
In light of the above statements, choose the correct answer from the options given below:
(a) Both statement I and statement II are true.
(b) Both statement I and statement II are false

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(c) Statement I is correct, but statement II is false
(d) Statement I is incorrect, but statement II is true.
(e) Not attempted

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Ans. (b) : Statement I : The size of the fixed part of the datagram header is 50 bytes.
- In IPv4, the fixed part of the IP header is 20 bytes.
- The header can be longer (up to 60 bytes) if options are present, but the fixed portion is always 20 bytes.
- so, 50 bytes is not the size of the fixed part.
- Therefore statement is false.

Statement II : An IP datagram is transmitted in little endian order.
- Internet protocols (Including IP, ICP, UDP) use network byte order, which is big endian.
- This means the most significant byte is transmitted first.
- Therefore, statement II is also false.

300. Flow control in TCP is handled using the
(a) SYN flag
(b) FIN flag
(c) Window size field
(d) Sequence number field
(e) Not attempted

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Ans. (c) : In TCP, flow control is a techniques that ensures the sender does not transmit more data than the receiver can handle, if the sender sends data too quickly the receiver buffer ray overflow, leading to packet loss. To present this, TCP uses the sliding window protocol, controlled by the window size field in the TCP header.
The receiver specifies in the window size field how many bytes it is currently able to accept. Based on this, value. The sender limits its transmission rate if the receiver is busy or its buffer is almost full, it advertises a smaller window size, which slows down the sender if the buffer is free again the receiver increases the window size, allowing more data to flow.
Other field like SYN (for connection established) FTN (for termination), and sequence numbers (for ordering) are important but flow control specifically depends on the window size field.

301. Consider the following two statements:
Statement I: The IP address 197.107.56.205 belongs to class C.
Statement II: The IPv4 address consists of 64 bits.
In light of the above statements, choose the correct answer from the options given below:
(a) Both statement I and statement II are true.
(b) Both statement I and statement II are false
(c) Statement I is correct, but statement II is false.
(d) Statement I is incorrect, but statement II is true.
(e) Not attempted

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Ans. (c) : The IP address given is 197.107.56.205. To Identify its class, we check the value of the first octet (197), in IPv4, the class ranges are:–
- Class A : 0 – 127
- Class B : 128 – 191
- Class C : 192 – 223
- Class D : 224 – 239 (Multicast)
- Class E : 240 – 255 (Experimental)

Since 197 false in the range 192 – 223 this address belongs to class C. Therefore this address belong to C, making statement 1 true.
Regarding statement 11, it is incorrect because an. IPv4 address is 32 bits long, divided into four octets of 8 bits each. This 32-bit value is usually expressed in dotted decimal format such as 197.107.56.205. By contrast, and IPv6 address is 128 bits 10mg, not 64 bits. Hence, the claim that an IPv4 address has 64 bits is false.

302. Which of the following is solution of mobile IP problem?
निम्न में से कौन मोबाइल आई.पी. समस्या का समाधान है?
(a) Care of address/केयर ऑफ एड्रेस
(b) Carefree address/केयरफ्री एड्रेस
(c) Don't care address/डोन्ट केयर एड्रेस
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asstt. Prof. 2017

Ans (a) : Mobile IP is standard protocol that enables a mobile device to maintain its permanent IP Address while roaming between different networks the key problem is now to route packets to the mobile device when it is away from its home network, since the permanent IP address is tied to home.

303. In tunnel mode, IP Sec protect:
टनल मोड में, आई.पी. सेक सुरक्षा करता है:
(a) Entire packet/पूरे पैकेट
(b) IP header/आई.पी. हेडर
(c) IP Payload/आई.पी.पेलोड
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : IP Sec Tunnel mode is used primarily for network-to-network communications, such as between two routers or gateways. In tunnel mode, the entire original IP packet (Header + Payload) is:
→ Encrypted
→ Encapsulated
→ Then a new IP header is added.
This means the entire packet (original IP header + IP payload) is protected by IP sec.
Example– If Host A wants to send data to Host B via two routers using IP sec in tunnel mode:
(1) The original IP packet is created (with IP header & data)
(2) IP sec encrypts and encapsulates the whole packets.
(3) A new IP header is added for delivery between the IP sec gateways (routers).
This ensures–
- The original source and destination IP addresses are hidden.
- The entire original packet is secure during transit.

304. Which of the following is true with regard to the ping command?
पिंग कमान्ड के सम्बन्ध में, निम्नलिखित में से कौन-सा सत्य है?

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(a) Ping stands for Packet Internet Generator.
(b) The ping command checks the port level connectivity between source destinations endpoints.
(c) Ping summarizes the packet loss and round-trip delay between two IP endpoints.
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : The ping command is used to check the presence and operation of a network device. It is useful for checking the states of network connectivity. It pings a remote device to see if it is available and how long it takes to respond. It summarizes packet loss and round trip delay between two IP and points.

305. When the mail server sends mail to other mail servers it becomes
जब मेल सर्वर, अन्य मेल सर्वरों को मेल भेजता है, तो यह ____ हो जाता है।

(a) SMTP client
(b) SMTP server
(c) Peer
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : When a mail server sends mail to other mail servers it becomes an SMTP Client.
SMTP Client– When a server initiates the communication to send an email to another server, it acts as the client in the SMTP (Simple mail Transfer protocol) process.
SMTP Server– The server receiving the email is considered the SMTP server.
Peer– While both servers and client are involved in the communication, they are not peers in the traditional sense. An SMTP Client initiates the connection and sends the data, making it the client in this interaction.

306. The length of an IPv6 address is
IPv6 address की लम्बाई होती है

(a) 32 bits
(b) 64 bits
(c) 256 bits
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (e) : IPv6 is an Internet Protocol used to identify devices on the internet. The length of an IPv6 address is 128-bits. Where as the length of an IPv4 address is 32-bits. Hexadecimal format is usually used as IPv6 address.

307. Which layer of the TCP/IP stack corresponds to the OSI model transport layer?
TCP/IP स्टैक की कौन-सी परत, OSI मॉडल ट्रान्सपोर्ट परत से मेल खाती है?

(a) Host-to-host
(b) Application
(c) Internet
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : The four layers of the TCP/IP stack (also called the DoD model) are Application/process. Host to Host, Internet, and Network Access. The Host to Host layer is equivalent to the Transport layer of the OSI model.

308. Which of the following is not a step in the Header translation procedure?
हेडर अनुवाद प्रक्रिया में, निम्नलिखित में से कौन-सा एक चरण नहीं है?

(a) The IPv6 mapped address is changed to an IPv4 address by extracting the rightmost 32 bits/ IPv6 मैप किए गए पते को सबसे दाएँ 32 bits निकालकर IPv4 पते में बदल दिया जाता है
(b) The value of the IPv6 priority field is discarded/ IPv6 प्राथमिकता फ़ील्ड का मान त्याग दिया जाता है
(c) The IPv6 flow label is considered
IPv6 प्रवाह लेबल पर विचार किया जाता है।
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : In the header translation procedure, first the IPv6 mapped address is changed to an IPv4 address by extracting the rightmost 32 bits, then the value of the IPv6 priority field is discarded, and finally the ToS field in the IPv4 header is set to zero. IPv6 flow label is ignored in the procedure.

309. End-to-end connectivity is provided from host-to-host in
होस्ट-से-होस्ट तक एंड-टू-एंड (End-to-End) कनेक्टिविटी किस लेयर में प्रदान की जाती है?

(a) Network layer/नेटवर्क लेयर
(b) Transport layer/ट्रान्सपोर्ट लेयर
(c) Session layer/सेशन (सत्र) लेयर
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

Ans. (b): End-to-end connectivity from host to host is provided at the highest layer of any network, i.e., the transport layer. The transport layer establishes the End-to-end connectivity of the network and is responsible for transmitting data from the source to the users.

310. In computer network, what is the size of IP address in IPv4?
कम्प्यूटर नेटवर्क के IPv4 में IP एड्रेस का साइज कितना होता है?

(a) 8 bits/8 बिट (b) 16 bits/16 बिट
(c) 32 bits/32 बिट
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above
उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 2023 (11-12)

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Ans. (c) : In IPv4, the size of an IP address is 32 bits. Which is usually written as a decimal number. It represents a binary number of 4 bytes for example 192.166.1.1 is an IPv4 address.

311. Match the following :

List - I		List – II	
A.	**Session layer**	**i.**	**Virtual terminal software**
B.	**Application layer**	**ii.**	**Semantics of the information transmitted**
C.	**Presentation layer**	**iii.**	**Flow control**
D.	**Transport layer**	**iv.**	**Manage dialogue control**

Codes :

	A	B	C	D
(a)	iv	i	ii	iii
(b)	i	iv	ii	iii
(c)	iv	i	iii	ii
(d)	iv	ii	i	iii

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Ans : (a) The correct match–

List - I		List – II	
A.	Session layer	**iv.**	Manage dialogue control
B.	Application layer	**i.**	Virtual terminal software
C.	Presentation layer	**ii.**	Semantics of the information transmitted
D.	Transport layer	**iii.**	Flow control

312. Which of the following protocols is used by email server to maintain a central repository that can be accessed from any machine?

(a) POP3 (b) IMAP
(c) SMTP (d) DMSP

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) IMAP : The basic idea behind IMAP is for all email server to maintain a central repository that can be accessed from any machine

313. Match the List-I with List-II

	List-I (Process)		List-II (Functions)
A.	**Internet message Access Protocol (IMAP)**	**I.**	**A protocol that allows email clients to retrieve emails from a mail server**
B.	**Address Resolution Protocol (ARP)**	**II.**	**Used for remove login to another computer over a network**
C.	**Post Office Protocol (POP)**	**III.**	**A protocol that maps IP addresses to MAC addresses on a local area network**
D.	**TELNET**	**IV.**	**A protocol that allows accessing email on the server without downloading them**

Choose the correct answer from the options given below :

(a) A-II, B-I, C-III, D-IV
(b) A-III, B-IV, C-I, D-II
(c) A-IV, B-III, C-I, D-II
(d) A-IV, B-III, C-II, D-I

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Ans. (c) : Correct match-

	List-I (Process)		List-II (Functions)
A.	Internet message Access Protocol (IMAP)	–	A protocol that allows accessing email on the server without downloading them
B.	Address Resolution Protocol (ARP)	–	A protocol that maps IP addresses to MAC addresses on a local area network
C.	Post Office Protocol (POP)	–	A protocol that allows email clients to retrieve emails from a mail server
D.	TELNET	–	Used for remove login to another computer over a network

314. Consider the following statements regarding multiple access techniques & identify the CORRECT ones.

A. ALOHA is used for multiple access on a shared medium
B. TDMA uses WALSH TABLES for time slot allocation
C. CSMA/CD was invented for wireless networks
D. CDMA codes are sequence of number generated by orthogonal codes.

Choose the correct answer from the options given below :

(a) A & B only (b) B % C only
(c) C & D only (d) A & D only

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Ans. (d) :

A. ALOHA is used for multiple access on a shared medium.

- This is true ALOHA is a multiple access technique that allows several users to transmit data over a shared communication medium.

D. CDMA Codes are sequences of number generated by orthogonal codes.

- This is also true CDMA (Code Division Multiple Access) uses orthogonal codes for separating multiple users signals in the same frequency spectrum.

B. TDMA uses WALSH TABLES for time slot allocation.

- This is incorrect- TDMA (Time Division multiple Access) typically uses a fixed time division approach for allocating time slots not walsh tables.

C. CSMA/CD was invented for wireless networks.

- This is incorrect CSMA/CD (Carrier sense multiple Access with collision Detection) was originally designed for wired networks, not wireless networks use variations like CSMA/CA (Carrier sense multiple Access with collision.

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315. Arrange the following steps of the DHCP (Dynamic Host Control Protocol) process in the correct sequence.
A. DHC PACK
B. DHCP DISCOVER
C. DHCP OFFER
D. DHCP REQUEST
E. Client receives IP configuration
Choose the correct answer from the options given below :
(a) A, B, C, D, E (b) B, C, D, A, E
(c) C, B, A, E, D (d) D, A, E, C, B

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Ans. (b) : The correct sequence of the DHCP process is-
DHCP DISCOVER- The client broadcasts a request to find a DHCP Server.
DHCP OFFER- The DHCP Server responds with an available IP address.
DHCP REQUEST- The Client requests to lease the offered IP address.
DHCP Ack- The DHCP server acknowledges the request and assigns the IP.
Client receives IP Configuration- The Client gets the IP and network configuration.
Thus the correct option is (b). B, C, D, A, E

316. CDMA is a example of which generation of cellular network?
सी.डी.एस.ए. सेलुलर नेटवर्क की किस जेनरेशन का उदाहरण है?
(a) 2G (b) 5G
(c) 3.5G (d) 4G

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : CDMA (Code Division Multiple Access) is a multiple access technology used in 2G mobile networks primarily in IS-95 (also known as CDMA one), developed by Qualcomm.
2G Network–
- The 2nd Generation (2G) of cellular networks introduced digital communication (as opposed to analog in 1G)
- Main technologies: GSM (using TDMA) and IS-95 (using CDMA)
- CDMA allowed multiple users to share the same frequency band by assigning them unique codes, instead of dividing the signal by time of frequency.

317. What does the acronym of ICMP protocol?
आई.सी.एम.पी. का असंक्षिप्त रूप क्या है?
(a) Internet Contention Message Protocol
इंटरनेट कंटेनशन मैसेज़ प्रोकोकॉल
(b) Internet Communication message Protocol
इंटरनेट कम्यूनिकेशन मैसेज़ प्रोटोकॉल
(c) Internet Control Message Protocol
इंटरनेट कंट्रोल मैसेज़ प्रोटोकॉल
(d) Internet Carrier Message Protocol
इंटरनेट कैरियर मैसेज़ प्रोटोकॉल

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : ICMP (Internet Control Message Protocol) is a network layer protocol used by network devices, like routers, to send error messages and operational information ICMP is mainly used for diagnostic or control purposes and is not used to exchange data between systems like TCP or UDP.
TCP or UDP.
→ It is defined in RFC 792.
→ It works closely with the IP protocol.
→ It helps in error reporting and testing network connectivity.
Key functions of ICMP.
- Reporting errors when IP packets fail to reach their destination.
- Testing the reach ability of a host (e.g. ping command)
- Informing about network congestion or unavailable services.

318. _____ protocol is used for uploading and downloading files between different computers.
विभिन्न कम्प्यूटरों के बीच फाइल अपलोडिंग तथा डाउनलोडिंग के लिए _____ प्रोटोकॉल का उपयोग किया जाता है।
(a) X.12 (b) XMODEM
(c) X.75 (d) XON

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : XMODEM File Transfer Protocol–
XMODEM is a file transfer protocol which enables users to transmit files between different computers.
Functioning– XMODEM breaks up the original data into a series of "Packets" which are sent to the receiver. It also contains additional information which allows the receiver to determine whether that packet was correctly received or not. The packet is resent after requested by the receiver if an error encounters. The transfer is aborted because of a string of bad packets.
Method of Transfer– Transfer are receiver driven the transmitter will not send any date until an initial Nak is sent by the receiver sends a NAK byte. The NAK byte is the signal that the receiver is ready to start this is the initial byte which displays that the file transfer is ready to being if the receiver takes too long or an error occurs then the transmitter will stop waiting or "Time Out".

319. What is the full form of DVMRP Protocol?
डी. वी. एम. आर. पी. प्रोटोकॉल का असंक्षिप्त रुप क्या है?
(a) Distance Vector Multiple Routing Protocol
डिस्टेंस वेक्टर मल्टीपल रुटिंग प्रोटोकॉल
(b) Distance Vector Multicast Routing Protocol
डिस्टेंस वेक्टर मल्टीकास्ट रुटिंग प्रोटोकॉल
(c) Distance Vector Multitask routing Protocol
डिस्टेंस वेक्टर मल्टीटास्क रुटिंग प्रोटोकॉल
(d) Distance Vector Message Routing Protocol
डिस्टेंस वेक्टर मैसेज़ रुटिंग प्रोटोकॉल

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : DVMRP stands for Distance Vector Multicast Routing Protocol. It is a routing protocol used for efficiently routing multicast data packets from a single source to multiple destinations simultaneously, using a distance vector approach for routing decisions and techniques like reverse path flooding and pruning to manage multicast treas.

Sarkari Resource

320. **_____ protocol is used for electronic data interchange. (Standard developed for the electronic exchange of machine – readable information between businesses.)**
इलेक्ट्रॉनिक डेटा इंटरचेंज के लिए _____ प्रोटोकॉल का उपयोग किया जाता है (व्यवसायों के बीच मशीन-पठनीय जानकारी के इलेक्ट्रॉनिक आदान-प्रदान के लिए विकसित किया गया मानक।)

(a) X.12 (b) X.75
(c) XON (d) XL2.5

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : Based on the analysis, the X.12 standard is the one specifically developed and used for Electronic Data Later change, providing the structure for the electronic exchange of machine readable business information between businesses.

321. **GPRS offers end to end packet switched data transfer service in_____**
जी.पी.आर.एस. _____में एंड टू एंड पैकेट स्विच्ड डेटा ट्रांसफर सेवा को प्रदान करता है।

I. Point to point/पॉइंट टू पॉइंट
II. Point to multipoint/प्वॉइंट टू मल्टीप्वॉइंट

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Point to Point (PTP) service is a data transfer mode where packets are sent from a single source to a single destination. In the context of GPRS, this is now most internet data sessions work, such as when a mobile phone accesses a websites or sends an email. The connection is established between a specific mobile device and a specific server on the network.
Point to Multipoint (PTM) service, on the other hand, allows data packets to be sent from a single source to multiple destinations simultaneously. This is often used for applications like multicast or broadcast services, where the same data needs to be delivered to a group of users at the same time.

322. **Match the following TCP/IP protocol with their use./निम्नलिखित TCP/IP प्रोटोकॉल को उनके उपयोग से सुमेलित करें।**

A.	**FTP**	**i.**	**Remote login** **रिमोट लोगिन**
B.	**TFTP**	**ii.**	**Send datagram** **सेन्ड डेटाग्राम**
C.	**Telnet**	**iii.**	**File transfer** **फाईल ट्रांसफर**
D.	**UDP**	**iv.**	**Trivgial file transfer** **ट्रिवियल फाइल ट्रांसफर**

(a) A-ii, B-iv, C-i, D-iii
(b) A-1, B-ii, C-iii, D-iv
(c) A-iii, B-iv, C-ii, D-i
(d) A-iii, B-iv, C-i, D-ii

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : The correct match–

A.	FTP	**iii.**	File transfer
B.	TFTP	**iv.**	Trivgial file transfer
C.	Telnet	**i.**	Remote login
D.	UDP	**ii.**	Send datagram

323. **DHCP (Dynamic Host Configuration Protocol) is a _____.**
डी.एच.सी.पी. (डाइनेमिक होस्ट कॉनफिग्रेशन प्रोटोकॉल) _____ है।

(a) Asking question and getting the reply
सवाल पूछना और जवाब पाना
(b) Protocol in which the sends for positive acknowledgement before advancing to the next data item/प्रोटोकॉल जिसमें अगले डेटा आइटम पर आगे बढ़ने से पहले सकारात्मक पावती भेजता है
(c) Protocol allows a newly booted workstation to broadcast its Ethernet address
प्रोटोकॉल एक नये बूट वर्कस्टेशन को अपने ईथरनेट पते को प्रसारित करने की अनुमति देता है
(d) Protocol allows both manual IP addres assign infant and automatic assignment
प्रोटोकॉल शिशु और स्वचालित असाइनमेंट दोनों को मैनुअल वाईपी एड्रेस असान करने की अनुमति देता

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : DHCP stands for Dynamic Host Configuration Protocol. It is used to automatically assign IP addresses and other network configuration parameters (like subnet mask, default gateway, DNS server) to hosts on a network.

324. **Sending incoming packet to every outgoing line except the one it arrived is known as _____.**
आने वाले पैकेट को हर आउटगोइंग लाइन को भेजना _____ के रूप में जाना जाता है।

(a) Congestion/कंजेस्शन
(b) Scratching/स्क्रेचिंग
(c) Flooding/फ्लडिंग
(d) Message calling/मैसेज कॉलिंग

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Flooding is a network routing technique where an incoming packet is sent out on all outgoing lines except that one it arrived on, ensuring that it reaches all the parts of the network.

325. **Routing algorithm which do not base their routing decisions on measurements or estimates of the current traffic and topology belongs to which category?**
रूटिंग एल्गोरिथम जो वर्तमान ट्रैफिक और टोपोलॉजी के माप या अनुमानों पर उनके रूटिंग निर्णयों को आधार नहीं बनाता है, किस श्रेणी से संबंधित है?

(a) Adaptive algorithm/अडाप्टिव अल्गोरिथम
(b) Greedy algorithm/ग्रीडी अल्गोरिथम
(c) Back propagation algorithm
बैक प्रोपगेशन अल्गोरिथम
(d) Non-adaptive algorithm
नॉन-अडाप्टिव अल्गोरिथम

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Ans. (d) : Adaptive Algorithms– Make routing decisions based on the current network traffic and topology.
Non-adaptive algorithms– Also called static routing algorithms, make decisions without considering current traffic or topology changes. The routes are fixed in Advance.
Greedy Algorithm– A problem solving approach that makes the locally optimal choice at each step, not a routing category.
Back Propagation Algorithm– Used in neural networks for training, not in routing.

326. Twisted-pair cables transmit signals in the form of:
ट्विस्टेड-पेयर केबल किस रूप में संकेत भेजते हैं?
(a) Light/प्रकाश (लाइट)
(b) Electric current/विद्युत प्रधारा
(c) Radio waves/रेडियो वेव (तरंगें)
(d) Infrared light/अवरक्त (इन्फारेड) किरणे

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : Twisted-pair cables transmit signals in the form of electric current, not light. The signals are carried by metallic conductors, and the twisting of the wires is a technique used to minimize electromagnetic interference and ensure that any noise is equally induced in both wires of the pair, allowing it to be canceled out at the receiving and.

327. The class-C IPv4 address must start with _____.
क्लास-सी आईपीवी 4 एड्रेस (class-C IPv4 address) _____ के साथ शुरू होना चाहिए।
(a) 10 (b) 101
(c) 110 (d) 011

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : IPv4, the class of an address is determined by the first few bits of the first octet.
Class A → Starts with 0 → range 0-127
Class B → Starts with 10 → range 128-191
Class C → Starts with 110 → range 191-223
Class D → Starts with 1110 → range 224-239
Class E → Start with 1111 → range 240-255
So, a class C IPv4 address must start with 110.

328. Which of the following statement is FALSE?
निम्नलिखित में से कौन सा कथन गलत है?
(a) A source address is always unicast/ एक सोर्स एड्रेस हमेशा यूनिकास्ट होता है
(b) A destination address can be unicast, multicast, or broadcast/डेस्टिनेशनल एड्रेस यूनिकास्ट, मल्टीकास्ट, या ब्रॉडकास्ट हो सकता है
(c) The multicast address is the special case of the broadcast address in which all the bits are 1/मल्टीकास्ट एड्रेस ब्रोडकास्ट केस का विशेष केस है जिसमें सभी बिट्स 1 होते हैं
(d) The least significant bit(s) of the first byte determines the class of the address
पहले बाइट के कम से कम महत्वपूर्ण बिट एड्रेस के क्लास को निर्धारित करता है

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Ans. (c) : The multicast address is the special case of the broadcast address in which all the bits are 1, is the false statement.
While a broadcast address has all bits set to 1, a multicast address is not simply a subset of broadcast address multicast addresses are used to target specific group of devices on a network, whereas a broadcast address targets all devices on the network.

329. State whether True or False?
बताएं कि क्या सच या गलत है?
I. The default subnet mask for Class-C IP address is 255.255.255.x.
क्लास-सी आईपी एड्रेस के लिए डिफॉल्ट सबनेट मास्क 255.255.255.x है।
II. Very first four bits of the first octet in Class-D IP addresses are set to 1100.
क्लास-डी आईपी एड्रेस में पहले ऑक्टेट के पहले चार बिट्स 1100 पर सेट किए गए हैं।
(a) I-True, II-False/I-सत्य, II-असत्य
(b) I-True, II-True/I-सत्य, II-सत्य
(c) I-False, II-True/I-असत्य, II-सत्य
(d) I-False, II-False/I-असत्य, II-असत्य

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Ans. (a) : Evaluate each statement–
(I) The default subnet mask for a class C IP address is 255.255.255.x.
For class C, the default subnet mask is 255.255.255.0. So, this is True.
(II) The first four bits of the first octet in a class-D IP address are set to 1100.
Class-D addresses range from 224.0.0.0 to 239.255.255.255.
Their first four bits are 1110, not 1100. So, this is false.

330. The IP address range _____ to _____ is reserved for loopback.
आईपी एड्रेस रेंज ___से___ लूपबैक के लिए आरक्षित है।
(a) 127.10.10.0, 127.255.255.255
(b) 127.0.0.0, 127.255.255.255
(c) 127.10.10.0, 127.255.255.255
(d) 127.0.0.0, 127.255.0.0

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : A loopback address is a special purpose IP address used for testing and diagnosing network connections on a computer. It's essentially a virtual network interface that allows a device to sent a network packet to itself without it ever leaving the devices physical hardware. The most well known loopback address 127.0.0.1, which is also referred to as local host.

331. In computer networking, a Repeater is a device that operates only in the _____ layer.
(a) Physical (b) Network
(c) Data link (d) Session

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : This layer is responsible for transmitting raw bit streams over a physical medium, and repeaters work by receiving, regenerating, and retransmitting these bit streams.
Repeaters extend the physical reach of a network by amplifying or regenerating weak or distorted signals before retransmitting.

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332. Which of the following protocols is an application layer protocol that establish, manages and terminates multimedia sessions?
(a) Session maintenance protocol
(b) Real – time streaming protocol
(c) Real – time transport control protocol
(d) Session Initiation protocol

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) : Session initiation protocol is an application layer control protocol that can establish, modify & terminate multimedia sessions such as internet telephony calls.

333. Match the following port numbers with their uses :

List - I		List - II	
(A)	23	(i)	World wide web
(B)	25	(ii)	Remote login
(C)	80	(iii)	USENET news
(D)	119	(iv)	E-mail

Codes :

	A	B	C	D
(a)	(iv)	(i)	(ii)	(iii)
(b)	(ii)	(i)	(iv)	(iii)
(c)	(ii)	(iv)	(iii)	(i)
(d)	(ii)	(iv)	(i)	(iii)

UGC NET C.S. June-2015 (Paper-II)

Ans.(d) :
23 port number is (TELNET) that uses by remote login
25 port number is (SMTP) that uses by E-mail
80 port number is (HTTP) that uses by world wide web
119 port number is used by USENET news

334. What is the size of the 'total length' in IPv4 datagram?
(a) 4 bits (b) 8 bits
(c) 16 bits (d) 32 bits

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Ans.(c) : Total length 16 bit field define the entire packet size in IPv4, including header and data, in bytes. The minimum length packet is 20 bytes (20 byte header. + 0 byte data) and maximum is 65, 535 bytes the maximum value of a 16 bit word.

335. A bridge has access to _____ address in the same network.
(a) Physical (b) Network
(c) Datalink (d) Application

UGC NET C.S. June-2014 (Paper-II)

Ans : (a) A function of a Bridge take place in the Data link layer that means it cannot read ID address so giving it access to the physical address of all stations connected to it.

336. Which of the following is used in the options field of IPv4?
(a) Strict source routing
(b) Loose source routing
(c) Time stamp
(d) All of the above

UGC NET C.S. June-2014 (Paper-III)

Ans. (d) : IPv4 is the fourth version in the development of the IP. and IPv4 use loose source for an IP option can be used for address translation and IPv4 also use strict source routing for the contrast with loose source routing, in every step of the route is decided in advance where the packet is send and IPv4 also time stamp.
So the correct answer is (d).

337. Which one of the following is not a Client-Server application?
(a) Internet chat (b) Web browser
(c) E-mail (d) Ping

UGC NET C.S. June-2014 (Paper-III)

Ans. (D) : Ping is not a client server application. Ping is a computer network administration unility used to test the reach ability of a host on an Internet protocol ping there is no server that provides a service.

338. Which of the following control fields in TCP header is used to specify whether the sender has no more data to transmit?
(a) FIN (b) RST
(c) SYN (d) PSH

UGC NET C.S. June-2015 (Paper-III)

Ans : (a) FIN control fields in TCP header is used to specify whether the sender has no more data to transmit i.e., close the connection from sender side.

339. Which are the two modes of IP security?
(a) Transport and certificate
(b) Transport and tunnel
(c) Certificate and tunnel
(d) Preshared and transport

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) **Transport mode:-** IP header, IPSec, headers IP payload.
Tunnel Mode:- New IP header, IPSec, headers, old IP header, IP payload.
IP Sec can be implemented in a host-to-host transport mode, as well as in network tunning mode.

340. The maximum payload of a TCP segment is :
(a) 65, 535 (b) 65, 515
(c) 65, 495 (d) 65, 475

UGC NET C.S. June-2015 (Paper-III)

Ans : (c) The maximum payload of TCP segment is 65, 495 bytes. Because maximum IP packet is 65, 535 bytes out of which 20 is used for header. TCP header is 20 bytes.
So, $2^{16} - 20 - 20 = 65535 - 40 = 65495$ bytes left for TCP.

341. Which of the following is not a congestion policy at network layer?
(a) Flow control policy
(b) Packet discard policy
(c) Packet lifetime management policy
(d) Routing algorithm

UGC NET C.S. December-2015 (Paper-II)

Ans : (a) Flow control is function of data link layer and transport layer but not by network layer. Flow control is the process of managing the rate of data transmission between two nodes to prevent a fast sender from over whelming a slow receiver. Routing Algorithm, Packet Lifetime Management Policy and Packet - Discard. Policy are congestion policies used in network layer. Network layer congestion policy are–

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(1) Virtual circuits vs datagram inside the sub net
(2) Packet sequencing & service policy
(3) Packet discard policy
(4) Routing algorithm
(5) Packet lifetime management.

342. Which of the following fields in IPv4 datagram is not related to fragmentation?
(a) Type of service (b) Fragment offset
(c) Flags (d) Identification
UGC NET C.S. December-2015 (Paper-II)

Ans : (a) Service types are not in the field of IPv4 datagram. While flags, identification and fragmentation offset are related to IPv4 datagram.

343. The combination of an IP address and a port number is known as .....
(a) Network number
(b) Socket address
(c) Subnet mask number
(d) MAC address
UGC NET C.S. December-2015 (Paper-III)

Ans : (b) A socket address is the combination of an IP address & a port number, much like one end of a telephone connection is the combination of a phone number and a particular extension.

344. Consider a subnet with 720 routers. If a three-level hierarchy is choosen with eight clusters, each containing 9 regions of 10 routers, then total number of entries in the routing table is.....
(a) 25 (b) 27
(c) 53 (d) 72
UGC NET C.S. December-2015 (Paper-III)

Ans : (a) Eight clusters, each containing 9 regions of 10 routers is a 3 level hierarchy. Each router needs 10 entries for local, 8 entries for routing the other regions and for distant cluster it need 7 entries i.e. total 10 + 8 + 7 = 25.

345. In a classful addressing, the IP addresses with 0 (zero) as network number:
(a) Refers to the current network
(b) Refers to broadcast on the local network
(c) Refers to broadcast on a distant network
(d) Refers to loopback testing
UGC NET C.S. December-2015 (Paper-III)

Ans : (a) If address belongs to any network with a host number of 0 (zero) is called the network address of the network. It does not denote any host.
Addresses with a network number of 0 are used refer to the current network, allowing a host to send to a neighbor without needing to know the network number of its local network (IP packets with destination network address of 0 are not routed).

346. In electronic mail, which of the following protocols allows the transfer of multimedia messages?
(a) IMAP (b) SMTP
(c) POP 3 (d) MIME
UGC NET C.S. December-2015 (Paper-III)

Ans : (d) Multipurpose Internet Mail Extensions (MIME) is an internet standard that extends the format of email to support. Text in character sets other than ASCII. No-text attachments: Audio, Video, Image, Application programs etc. message bodies with multiple parts.

347. Match the following :

	List-I		List-II
A	**Data link layer**	**i**	**Encryption**
B	**Network layer**	**ii**	**Connection control**
C	**Transport layer**	**iii**	**Routing**
D	**Presentation layer**	**iv**	**Framing**

Code :

	A	B	C	D
(a)	iv	iii	i	ii
(b)	iii	iv	ii	i
(c)	iv	ii	iii	i
(d)	iv	iii	ii	i

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Ans : (d) The correct match–

	List-I		List-II
A	Data link layer	**iv**	Framing
B	Network layer	**iii**	Routing
C	Transport layer	**ii**	Connection control
D	Presentation layer	**i**	Encryption

348. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet?
(a) 62 subnets and 1022 hosts
(b) 64 subnets and 1024 hosts
(c) 62 subnets and 254 hosts
(d) 64 subnets and 256 hosts
UGC NET C.S. November-2017 (Paper-III)

Ans : (a) Network Mask of class B = 255:255:0:0
Subnet bits carried from host portion:

$$255.255.\underbrace{111111}_{\text{Subnet}}\ \underbrace{00}_{\text{Host}}:00000000$$

Number of subnet = $2^6 - 2 = 64 - 2 = 62$
Number of host = $2^{10} - 2 = 1024 - 2 = 1022$

349. Which of the following statements are true?
(A) The fragmentation field in the base header section of IPv4 have moved to the fragmentation extension header in IPv6
(B) The authentication extension header is new in IPv6
(C) The record route option is not implemented in IPv6
Code :
(a) (A) and (B) only (b) (B) and (C) only
(c) (A) and (C) only (d) (A), (B) and (C)
UGC NET C.S. November-2017 (Paper-III)

Ans : (d) Fragmentation fields in the base header section of IPv4 have moved to the fragmentation extension header in IPv6

0	8	16	29	31
Next header (8 bits)	Reserved (0) (8 bits)	Fragmentation offset (13 bits)	Res (2 bits)	M
Identification (32 bits)				

So all the given statements are correct.

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350. Match the following layers and Protocols for a user browsing with SSL :

A.	**Application of layer**	**i.**	**TCP**
B.	**Transport layer**	**ii.**	**IP**
C.	**Network layer**	**iii.**	**PPP**
D.	**Datalink layer**	**iv.**	**HTTP**

Codes :

	A	**B**	**C**	**D**
(a)	iv	i	ii	iii
(b)	iii	ii	i	iv
(c)	ii	iii	iv	i
(d)	iii	i	iv	ii

UGC NET C.S. June-2017 (Paper-II)

Ans : (a) The correct match–

A.	Application of layer	**iv.**	HTTP
B.	Transport layer	**i.**	TCP
C.	Network layer	**ii.**	IP
D.	Datalink layer	**iii.**	PPP

351. A packet whose destination is outside the local TCP/IP network segment is sent to_____

(a) File server (b) DNS server
(c) DHCP server (d) Default gateway

UGC NET C.S. June-2017 (Paper-II)

Ans : (d) Default gateway serves an access point or IP router that a networked uses to send information to a computer in another network or the internet. Default simply means that this gateway is used by default, unless an application specifies another gateway.

352. Distance vector routing algorithm is a dynamic algorithm. The routing tables in distance vector routing algorithm are updated _____

(a) Automatically
(b) By server
(c) By exchanging information with neighbor nodes
(d) With back up database

UGC NET C.S. June-2017 (Paper-II)

Ans : (c) A distance vector routing protocol requires that a router inform its neighbors about its topology changes periodically.

353. The IP address _____ is used by hosts when they are being booted.

(a) 0.0.0.0 (b) 1.0.0.0
(c) 1.1.1.1 (d) 255.255.255.255

UGC NET C.S. November-2017 (Paper-II)

Ans : (a) 0.0.0.0 is also called DHCP address used when host was booted first time.
In IPv4 0.0.0.0 is used for the default route.

354. Match List-I with List-II.

	List-I (IP Address)		**List-II (Class)**
A.	**10.20.30.40**	**I.**	**Class E**
B.	**210.20.30.3**	**II.**	**Class B**
C.	**180.30.100.10**	**III.**	**Class A**
D.	**252.5.15.11**	**IV.**	**Class C**

Choose the correct answer from the options given below :

(a) A-II, B-III, C-I, D-IV
(b) A-I, B-IV, C-II, D-III
(c) A-I, B-II, C-IV, D-III
(d) A-III, B-IV, C-II, D-I

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Ans. (d) : IP Address Classes–
Class A - IP range is 0.0.0.0 to 127.255.255.255
Class B– IP range is 128.0.0.0 to 191.255.255.255
Class C – IP range is 192.0.0.0 to 223.255.255.255
Class D – IP range is 224.0.0.0 to 239.255.255.255
Class E – IP range is 240.0.0.0 to 255.255.255
So, the correct matching List-I to List-II is–

(A) 10.20.30.40	(iii) Class A
(B) 210.20.30.3	(iv) Class C
(C) 180.30.100.10	(ii) Class B
(D) 252.5.15.11	(i) Class E

355. Match List-I with List-II.

	List-I		**List-II**
A.	**Representation of bits**	**I.**	**Transport layer**
B.	**Physical Address**	**II.**	**Physical layer**
C.	**Logical Address**	**III.**	**Network layer**
D.	**Segmentation and reassembly**	**IV.**	**Data link layer**

Choose the correct answer from the options given below :

(a) A-I, B-II, C-III, D-IV
(b) A-II, B-IV, C-III, D-I
(c) A-III, B-II, C-IV, D-I
(d) A-I, B-IV, C-III, D-II

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Ans. (b) : The correct matching is–

	List-I		**List-II**
A.	Representation of bits	**ii**	Physical layer
B.	Physical Address	**iv**	Data link layer
C.	Logical Address	**iii**	Network layer
D.	Segmentation and reassembly	**i**	Transport layer

356. Host A (on TCP/IPv4 Networks) send an IP Datagram D to host B (also on TCP/IPv4 network). Assume that no error occurred during transmission of D, when D reaches B. Which of following header field may differ from that of original datagram D ?

A. TTL **B. Checksum**
C. Fragment offset **D. Source IP**
E. Destination IP

Choose the correct answer from the options given below :

(a) (A) and (B) Only
(b) (A), (B) and (C) Only
(c) (A), (B), (C) and (D) Only
(d) (A), (B), (C) and (E) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : Examine each of the header fields and which ones may differ when an IP datagram reaches host B:

- The TTL (Time to Live) Value decrements by one at each hop (i.e., router) the datagram passes through. Therefore, the TTL Value will differ from the original datagram.
- The IP header checksum will be recalculated at each hop because the TTL value charges.

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Hence, The checksum value may differ from the original datagram.
- If the datagram is fragmented along the route, the fragment offset field may change. However, if no fragmentation occurs, this field remains the same.
- The source IP address and destination IP address remains the same.

357. Two computers A and B are configured as follows : A has IP address 203.197.2.53 and subnet mask 255.255.128.0, B has IP address 203.197.75.201 and subnet mask 255.255.192.0. What one of the following statement is true?

(a) A and B both assume they are on the same network.
(b) B assumes, A is on same network, A assumes B is on different network.
(c) A assumes, B is on same network, but B assumes, A is on a different network.
(d) A and B both assume they are on different network.

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Ans. (c) : Network Id of A = bitwise '&' of IP of A and subnet mask of A
= (203.197.2.53) & (255.255.128.0)
= 203.197.0.0
A sees network ID of B as bitwise '&' of IP of B and subnet mask of A
= (203.197.75.201) & (255.255.128.0)
= 203.197.0.0
Which is same as Network ID of A.
Network ID of B = bitwise '&' of IP of B and subnet mask of B.
= (203.197.75.201) & (255.255.192.0)
= 203.197.64.0
B sees network ID of A as bitwise '&' of IP of A and Subnet mask of B.
= (203.197.2.53) & (255.255.192.0)
= 203.197.0.0
Which is different from network ID of B.
Therefore, A assumes, B is on same network, but B assumes, A is on a different network.

358. A class B network on the Internet has a subnet mask of 255. 255. 248.0, what is the maximum number of hosts per subnet?

(a) 1022 (b) 1023
(c) 2046 (d) 2047

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Ans. (c) : The subnet mask is 255.255.248.0, convert it to binary: 11111111.11111111.11111000.00000000
The subnet mask indicates the number of fixed bits (subnet bits) and the number of variable bits (host bits).
Here, there are:
- 21 subnet bits (count the 1s)
- 11 host bits (count the 0s)

The formula for the maximum number of hosts per subnet is
$2^{\text{number of host bits}} - 2$
(Subtract 2 to account for the network address and the broadcast address)
$2^{11} - 2 = 2048 - 2 = 2046$
Therefore, the maximum number of hosts per subnet is 2046.

359. A. In IPv4 addressing, a block of address can be define one of the address and /n define the mask.
B. The first address in the block can be found by setting the rightmost 32-n bits to 0s.
C. Address of class C is used for multicast communication.
D. There are five classes in IPv4 address.
E. Super netting combine several networks into one large network.
Choose the correct answer from the options given below :

(a) (A), (B) and (C) Only
(b) (A), (B), (D) and (E) Only
(c) (A), (C), (D) and (E) Only
(d) (C), (D) and (E) Only

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Ans. (b) : Analyze each statement–
(A) In IPv4 addressing, a block of address can be defined as x.y.z t/x in which x.y.z.t define one of the address and in define the mask. This notation is known as CIDR (Classless Inter-Domain Routing).
(B) The first address in the block can be found by setting the rightmost 32-n bits to 0s. This is new you calculate the network address in CIDR notation.
(C) Class D addresses (224.0.0.0 to 239. 255.255) are used for multicast communication.
(D) IPv4 addresses are divided into five classes: A, B, C, D and E.
(E) Super netting aggregates multiple smaller networks into a larger one, which is the opposite of subletting.
So, statements (A), (B), (D) and (E) are correct. Statement, (C) is incorrect because class C is not used for multicast communication.

360. Which of the following is not a field in TCP header?

(a) Sequence Number (b) Checksum
(c) Fragmentation offset (d) Window size

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Ans. (d) : Fragmentation offset is a field in the IP header, not in the TCP header. In the TCP header, the fields are sequence Number, checksum, and window size, among others, but not fragmentation offset. TCP operates at the transport layer, while fragmentation occurs at the network layer (IP layer), where packets are divided into smaller fragments to fit into the maximum transmission unit (MTU) of the network.

361. IP datagram has arrived with following partial information in the header (in hexadecimal) 45000054000300002006....
What is the protocol of the payload being carried by the packet?

(a) ICMP (b) SCTP
(c) TCP protocol (d) IGMP

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Ans. (c) : In hexadecimal, the protocol field is typically represented by the 10th byte in the header. In the given partial information, the 10th byte is '06', which in hexadecimal equals 6 in decimal. The value '6' corresponds to the TCP protocol. Therefore, the payload being carried by the packet is using the TCP protocol.

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362. **A TCP server application is programmed to listen on port P on host S. A TCP client is connected to the TCP server over the network. Consider that while TCP connection is active the server is crashed and rebooted. Assume that the client does not use TCP keepalive timer.**
Which of the following behavior is are possible?
Statement I : The TCP application server on S can listen on P after reboot.
Statement II : If client sends a packet after the server reboot, it will receive the RST segment.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

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Ans. (a) : The TCP application server on S can listen on P after reboot. when a server crashes and reboots, it loses its existing connections but can start listening on the same port again once its back online. If client sends a packet offer the server reboot. it will receive the RST segment. When the server reboots., its TCP stack might send a RST segment to any incoming packets from the client, indicating that connection is no longer valid. So, the both statement I and II are true.

363. **Which protocol is used to acquire the MAC address of a host whose IP address is known?**
(a) Dynamic Host control protocol
(b) Address Resolution Protocol
(c) Reverse Address Resolution Protocol
(d) User Datagram Protocol

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Ans. (b) : Address Resolution Protocol (ARP) is a procedure for mapping a dynamic IP address to permanent physical machine address in a local area network (LAN). The physical machine address is also know as a media access control (MAC) address.

364. **A TCP Server application is programmed to listen on port P on Host S. A TCP Client is connected to the TCP Server over the network. Considered that while TCP Connection is active the server is crashed and rebooted. Assume that the client does not use TCP keepalive timer. Which of the following behaviors is/are possible?**
Statement I : If client is waiting to receive a packet, it may wait indefinitely
Statement II : If the client sends a packet after the server reboot, it will receive the FIN segment. In the light of the above statements, choose the correct answer from the options given below
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

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Ans. (c) : A Keepalive timer is used to prevent a long idle connection between two TCPs
So, in absence of TCP Keepaline timer, Statement I is correct.
But statement(II) is false.
So option (c) is correct answer.

365. **Consider two hosts P and Q are connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes and between R and Q is 820 bytes. A TCP segment of size of 1400 bytes is transferred from P to Q through R with IP identification value of 0x1234. Assume that IP header size is 20 bytes. Further the packet is allowed to be fragmented that is Don't Fragment (DF) flag in the IP Header is not set by P. which of the following statement is/are true**
A. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes
B. If the second fragment is lost then R resend the fragment with IP Identification value of 0x 1234
C. If the second fragment is lost then P required to resend the entire TCP segment
D. TCP destination port can be determined by analyzing the second fragment only
Choose the correct answer from the options given below :
(a) A, B and C Only (b) A and C Only
(c) C and D Only (d) B and D Only

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Ans. (b) : Given that
TCP segment of size = 1400 bytes
IP header size = 20 bytes
At R packet will be Fragmented
MTU = 800 + 20 (Header)
Packet size = segment size + IP Header Size
= 1400 + 20
= 1420 bytes
Hence Packet's Data (1400) will be divided in two packets = (800 + 600)
First Fragment = (800 + 20) // Data + Header
Second Fragment = (600 + 20) bytes
A. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes.
It is correct.
B. It is second fragment is lost then R resend the fragment with IP Identification value of 0x1234.
It is incorrect As if any of the fragments is lost then the entire packet/TCP segment has to be retransmitted.
C. If the second fragment lost then P required to resend the entire TCP segment.
It is correct. If any of the fragments is lost then the entire packet TCP segment has to be retransmitted this one of the drawbacks of fragmentation.
D. TCP destination port can be determined by analyzing the second fragment only.
It is incorrect as Destination port number can be identified by looking at any packet or fragment not just only from second.
So correct option is (b).

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366. The Solution to Silly Window Syndrome problem is/are:
A. Nagle's Algorithm
B. Clark's Algorithm
C. Jacobson's Algorithm
D. Piggy backing Algorithm
Choose the correct answer form the options given below:
(a) A and B Only (b) A and C Only
(c) C and D Only (d) B and D Only
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Ans. (a) : Silly Window Syndrome is a problem that arises due to poor implementation of TCP.
Cause 1 : Sender window transmitting one byte of Data repeatedly–
This problem is solved by Nagla's algorithm.
Cause 2 : Receiver window accepting one byte of data repeatedly–
This problem is solved by problem is clark's solution or clark"s algorithm.
So option (1) is correct answer.

367. A classeless address is given as 167.199.170.82/27. The number of addresses in the network is
(a) 64 Addresses (b) 32 Addresses
(c) 28 Addresses (d) 30 Addresses
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Ans. (b) : To calculate the number of addresses in the given network you can use the formula : $2^{(32-\text{subnet_mask})}$.
In this case, the subnet mask is 27
(Which means 27 bits are used for the network portion).
So, the number of addresses = $2^{(32-27)}$
$= 2^5$
= 32 addresses
Therefore, the correct answer is option (b) 32 addresses.

368. The dotted-decimal notation of the following IPv4 address in binary notation is _____. 10000001 00001011 00001011 11101111
(a) 111.56.45.239 (b) 129.11.10.238
(c) 129.11.11.239 (d) 111.56.11.239
UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : $10000001 = 2^7 + 2^0 = 129$
$00001011 = 2^0 + 2^3 + 2^1 = 11$
$11101111 = 2^0 + 2^1 + 2^2 + 2^3 + 2^5 + 2^6 + 2^7$
$= 239$
Hence in dotted decimal notation, it would be 129.11.11.239.

369. Consider the following two statements :
S1 : TCP handles both congestion and flow control
S2 : UDP handles congestion but not flow control
Which of the following options is correct with respect to the above statements (S1) and (S2)?
Choose the correct answer from the code given below :
Code :
(a) S1 is correct but S2 is not correct
(b) Both, S1 and S2 are correct
(c) Neither S1 nor S2 is correct
(d) S1 is not correct but S2 is correct
UGC NET C.S. December-2018 (Paper-II)

Ans. (a) : S1: with the help of TCP window=>flow control and with the help of congestion window=> Congestion control
S2: No field are there in UDP header to control flow or congestion
So, option (a) is correct, statement S1 is correct but S2 is not correct.

370. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.
(a) HTTP GET request, DNS query, TCP SYN
(b) DNS query, TCP SYN, HTTP GET request
(c) DNS query, HTTP GET request, TCP SYN
(d) TCP SYN, DNS query, HTTP GET request
UGC NET C.S. December-2018 (Paper-II)

Ans. (b) : Step 1 : Whenever the client request for a webpage, the query makes the **DNS query** to identify the Domain Name Space. DNS query is the process to identify the IP address of the DNS such as www.org. The client's computer will make a DNS query to one of its internet service provider's DNS server.
Step 2 : As soon as DNS server is located a TCP connection is to be established for the further communication. The TCP protocol requests the server to establishing a connection by sending a **TCP SYN** message. Which is further responded by the server using SYN_ ACK from server to client and then ACK back to server from client (3- way hand shaking protocol).
Step 3 : Once the connection has been established the HTTP protocol comes into picture. It requests for the webpage using its GET method and thus, sending an **HTTP GET request**.
Hence, the correct sequence for the transmission of packets is DNS query, TCP SYN, HTTP GET request.

371. The four byte IP Address consists of
(a) Network Address
(b) Host Address
(c) Neither Network nor Host Address
(d) Both Network and Host Addresses
UGC NET C.S. December-2018 (Paper-II)

Ans. (d) : An IP address which is 32 bits long, that means it is of 4 bytes and is composed of a network and host portion and it depends on address class.

372. Consider ISO-OSI network architecture reference model. Session layer of this model offers dialog control, token management and _____ as services.
(a) Flow Control (b) Asynchronization
(c) Errors (d) Synchronization
UGC NET C.S. December-2018 (Paper-II)

Ans. (d) : Consider ISO-OSI network architecture reference model. Session layer of this model offers dialog control, token management and Synchronization as services.

373. In electronic mail, which of the following protocols allows the transfer of multimedia?
(a) IMAP (b) SMTP
(c) POP3 (d) MIME
UGC NET C.S. December 2020

Ans. (d) : IMAP :- is for the retrieval of emails and SMTP is for the sending of emails. That means IMAP talks to both the client and server to get emails, and SMTP talks only to servers to send emails.
POP3/IMAP :- Post office protocol (POP) is an application layer Internet standard protocol used by local e-mail clients to retrieve email from a remote server over a TCP/IP connection. Virtually all modern e-mail clients and servers support POP3, and it along with IMAP (Internet Message Access Protocols) are the two most prevalent Internet standard protocols for email retrieval, with IMAP (Internet Message Access Protocol) are the two most prevalent Internet standard protocols for emails retrieval, with many webmail service provides such as Google mail, Microsoft mail and Yahoo!
MIME :- MIME stands for multipurpose Internet mail Extensions. It is used to extend the capabilities of Internet e-mail protocols such as SMTP. The MIME protocols allows the users, to exchange various types of digital content such as pictures, audio, video, and various types of documents and files in the e-mails.

374. Match List-I with List-II

	List-I		List-II
A.	**Data Link Layer**	**i.**	**True end-to-end layer**
B.	**Network Layer**	**ii.**	**Token Management**
C.	**Transport Layer**	**iii.**	**Produce billing information**
D.	**Session Layer**	**iv.**	**Piggybacking**

Choose the correct answer from the options given below:
(a) A-IV, B-III, C-I, D-II
(b) A-IV, B-II, C-I, D-III
(c) A-II, B-III, C-I, D-IV
(d) A-IV, B-I, C-III, D-II

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Ans. (a) : Data Link Layer : pigybacking is sometimes referred to as "wi-fi squatting". The usual purpose of piggybacking is simply to gain free network access rather than any malicious intent, but it can slow down data transfer for legitimate users of network.
Transport Layer :- Responsible for true end-to-end layer from source to destination. It is termed as and end-to-end layer because it provides a point to point connection rather than hop to hop. It reliably deliver the services between the source-host and destination host.
Session Layer :- is responsible for establishing managing, synchronizing and termination session between end-user application processes. It works as a dialog, controller. It allows the system to communicate in either. Half duplex or full duplex mode of communication. It is responsible for token management.
Network Layer :- subnet usage accounting : has accounting functions to keep track of frames forwarded by subnet intermediate systems, to produce billing information.

375. Firewall is a device that filters access to the protected network from the outside network. Firewalls can filter the packets on the basis of
(A) Source IP Address
(B) Destination IP Address
(C) TCP Source port
(D) UDP Source port
(E) TCP Destination port
Choose the correct answer from the option given below:
(a) (A), (B) and (C) only
(b) (B) and (E) only
(c) (C) and (D) only
(d) (A), (B), (C), (D) and (E) only

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Ans. (d) : Network layer firewall can filter on source and destination IP address source and Destination port no for both TCP and UDP packet.
• Firewall :- A firewall is a form of cyber security device that filters networks traffic. Network nodes can be separated from external traffic source, Internal traffic sources or even individual applications using firewall.

376. Which of the following class of IP address has the last address as 223.255.255.255?
(a) Class A (b) Class B
(c) Class C (d) Class D

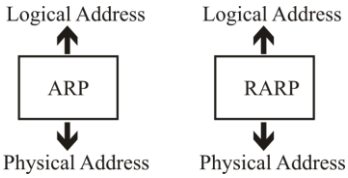
UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) : Class A 0.0.0.0 to 127.255.255.255
Class B 128.0.0.0 to 191.255.255.255
Class C 192.0.0.0 to 223.255.255.255
Class D 224.0.0.0 to 239.255.255.255
Class E 240.0.0.0 to 255.255.255.255
So Class of IP address 223.255.255.255 is belongs to class C.

377. What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?
(a) ARP (b) DNS
(c) RARP (d) ICMP

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Ans. (c) : What is the name of the protocol that allows a client to send a broadcast message with its MAC address and receive an IP address in reply?
Reverse Address Resolution Protocol - It is used to get IP address with the help of physical address (MAC Address)



ARP (Address resolution protocol) is a protocol used by the Internet protocol (IP), specifically IPv4, to map IP network addresses to the hardware addresses used by a data link protocol, that is MAC address.
DNS (Domain Name System) DNS is a host name to IP address translation service. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers.
RARP :- The Reverse Address Resolution Protocol (RARP) is a computer networking protocol used by a client computer to request its Internet protocol (IP) address from a computer network, when all it available

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is its link layer or hardware address, such as a MAC address. RARP protocol broadcast message with its MAC address and receive an IP address in reply.
The Internet Control Message Protocol (ICMP)- The internet control Message protocol (ICMP) is a supporting protocol in the Internet protocol suite. It is used by network devices, including routers, to send error messages and operational information indicating success or failure when communicating with another Host IP address.

378. You need 500 subnets, each with about 100 usable host addresses per subnet. What network mask will you assign using a class B network address?
(a) 255.255.255.252 (b) 255.255.255.128
(c) 255.255.255.0 (d) 255.255.254.0
UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) :
* In class B addressing : 16 - bit network ID and 16-bits for host ID.
* 500 subnets are required and $2^9 = 512$, hence 9 bits are needed from host id part of class B addressing in order to provide 500 subnets.
* Remaining 7 bits numbers of host possible per subnet is $128 - 2 = 126$

subnet - id (25 bits)/ Host - id (7-bits)
Subnet mask contains is in subnet id part and the os in host ID part. Hence subnet mask is :
11111111.11111111.11111111.10000000
In decimal form it can be represented as :
255.255.255.128

379. In the TCP/IP model, encryption and decryption are functions of _____ layer.
(a) Data link (b) Network
(c) Transport (d) Application
UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : TCP/IP model

Layers	Functions
Application Layer	Authentication, Authorization, Encryption, Decryption etc.
Transport Layer	End to end connectivity segmentation, Multiplexing, demultiplexing, congestion control etc
Internet Layer	Host to host connectivity fragmentation etc
Link layer (data link)	flow control, framing etc

380. CIDR stands for:
CIDR से तात्पर्य है
(a) Classified Internet Domain Routing
क्लासिफाइड इन्टरनेट डोमेन राऊटिंग
(b) Classless Inter Domain Routing
क्लासलेस इंटर डोमेन राऊटिंग
(c) Classless Internet Domain Routing
क्लासलेस इंटरनेट डोमेन राऊटिंग
(d) Classified Inter Domain Routing
क्लासिफाइड इंटर डोमेन राऊटिंग
RPSC C.S. 2016 (PAPER-I)

Ans. (b) : Classless Inter Domain Routing (CIDR) is a method for allocating IP addresses and routing Internet Protocol Packets, introduced to improve the scalability of IP address allocation and routing. CIDR is a method introduced in 1993 to replace the older system of IP address classification (Class A, B, C).

381. What is the address size of IPv6?
IPv6 में एड्रेस की साइज क्या होती है?
(a) 32 bit/32 बिट (b) 128 bit/128 बिट
(c) 256 bit/256 बिट (d) 512 bit/512 बिट
RPSC C.S. 2016 (PAPER-I)

Ans. (b) : The address size of IPv6 is 128 bit.
Because,
• **Larger address space–** IPv4 uses 32 bits, giving about 4.3 billion unique addresses. To cope with the explosive growth of devices, the IETF designed IPv6 with 128 bits, yielding 2^{128} possible addresses roughly 3.4×10^{38}. That's enough to given trillions of addresses to every person on Earth.
• **Representation–** A 128-bit IPv6 address is written as eight groups of four hexadecimal digits, separated by colons.
Example- 2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370: 7334 (we can shorten runs of zeros)
e.g.- 2001 : db8 : 85a3 : 8a2e : 370 : 7334
Benefits of bigger size–
⇒ Enables hierarchical addressing and efficient routing tables.
⇒ Provides room for specialized address types (Multicast, Any cost, Link-Local, Unique Local etc).

382. "Count to infinity", problem occurs in:
''काउंट टू इनफिनिटी'' समस्या आती है:
(a) Distance Vector Routing/डिस्टेंस वेक्टर राऊटिंग में
(b) Link State Routing/लिंक स्टेट राऊटिंग में
(c) Multicast Routing/मल्टीकास्ट राऊटिंग में
(d) Hierarchical Routing/हिरेरिकल राऊटिंग में
RPSC C.S. 2016 (PAPER-I)

Ans. (a) : In Distance Vector Routing. routers share their routing tables with their immediate neighbors. If a route becomes unreachable (e.g. due to a failure), it can take a long time for all routers to update their tables correctly. During this time incorrect routing information may propagate, causing routers to increment the distance metric for the unreachable destination-counting to infinity.

383. In distance vector routing algorithm, each router maintain a separate routing table with a following entries:
डिस्टेंस वेक्टर राऊटिंग अल्गोरिथम में प्रत्येक राऊटर एक अलग राऊटिंग टेबल बनाये रखता है जिनमें निम्न प्रविष्टियाँ होती है:
(a) Preferred input line, estimated time
अधिमानित इनपुट लाइन, अनुमानित टाइम
(b) Preferred input line, estimated distance
अधिमानित इनपुट लाइन, अनुमानित डिस्टेंस
(c) Preferred output line, estimated time
अधिमानित आउटपुट लाइन, अनुमानित टाइम
(d) Preferred output line, estimated distance
अधिमानित आउटपुट लाइन, अनुमानित डिस्टेंस
RPSC C.S. 2016 (PAPER-I)

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Ans. (c) : In the Distance Vector Routing Algorithm each router maintains a routing table (also known as a distance vector) that includes.
⇒ The preferred output line (next hop) to reach a particular destination.
⇒ The estimated distance (cost, hop, count or metric) to reach that destination.
The router updates its table based on information received from its neighbors, choosing the route with the shortest distance.

384. Which of following field in IPv4 datagram is not related with fragmentation?
निम्न में से कौन-सा IPv4 डाटाग्राम का फील्ड, फ्रेगमेंटेशन से सम्बंधित नहीं है?
(a) Flag/फ्लैग (b) Offset/ओफ्सेट
(c) TOS/टी ओ एस (d) Identifier/आइडेंटिफायर

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : In IPv4, the fields related to fragmentation are–
• **Flags–** Includes a 'More fragments' (MF) flag and a "Don't fragment" (DF) flag.
• **Fragment Offset–** Indicates the position of a fragment in the original data.
• **Identification (Identifier)–** Used to uniquely identify fragments of the same datagram TOS (Type of Service), on the other hand, is used to indicate the desired quality of service (like delay, throughput, reliability) and is not related to fragmentation.

385. DNS Client is also known as:
DNS क्लाइंट यह भी कहलाता है?
(a) DNS Updater/DNS अपडेटर
(b) DNS Resolver/DNS रिसोल्वर
(c) DNS Handler/DNS हैंडलर
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : A DNS client is also known as a DNS resolver. A DNS client initiates the process of DNS lookup, or DNS resolution, by sending a query to a DNS server to translate a domain name into an IP address.

386. The header length of IPv6 datagram is:
IPv6 डाटाग्राम के हैडर की लम्बाई होती है?
(a) 10 byte/10 बाइट (b) 20 byte/20 बाइट
(c) 40 byte/40 बाइट (d) 60 byte/60 बाइट

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : The header length of an IPv6 datagram is fixed at 40 bytes. Unlike IPv4, which has a variable header length, IPv6 simplifies the header by making it a fixed size to improve processing efficiency.

387. Transmission control protocol:
प्रसारण नियंत्रण प्रोटोकॉल–
(a) Is a connection-oriented protocol
(b) Uses a three-way handsake to establish a connection
(c) Receives data from application as a single stream
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(d): TCP is a connection oriented protocol that uses a three way handshake to establish a connection. It receives data from the application as a single stream, ensuring reliable and ordered delivery. Hence more than one option in the answer will be correct.

388. An endpoint of an inter-process communication flow across a computer network is called:
कंप्यूटर नेटवर्क में अंतर-प्रक्रिया संचार प्रवाह के समापन बिंदु को_____कहा जाता है।
(a) Socket
(b) Pipe
(c) Port
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(a): An endpoint of an inter process communication flow across a computer network is called socket.

389. Which transmission media provides the highest transmission speed in a network?
कौन-सा ट्रांसमिशन मीडिया किसी नेटवर्क में उच्चतर ट्रांसमिशन गति प्रदान करता है?
(a) Coaxial cable
(b) Twisted-pair cable
(c) Optical fiber
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(c): Optical fiber uses light signals to transmit data, which allows for significantly faster transmission speeds compared to electrical signals used in coaxial and twisted pair cables.

390. Which protocol is used to send email over the Internet?
इंटरनेट पर ईमेल भेजने के लिए किस प्रोटोकॉल का उपयोग किया जाता है?
(a) FTP
(b) SMTP
(c) HTTP
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : SMTP stands for Simple Mail Transfer Protocol. It is used to send emails over the internet. SMTP is a standard protocol for sending email messages from one mail server to another. POP3 and IMAP are used to receive emails.

391. What is the purpose of a subnet mask in TCP/IP networking?
TCP/IP नेटवर्किंग में सबनेट मास्क का उद्देश्य क्या है?
(a) To identify the network portion of an IP address/के नेटवर्क हिस्से की पहचान करने के लिए एक आईपी पता
(b) To identify the host portion of an IP address/किसी वस्तु के मेजबान हिस्से की पहचान करना आईपी पता

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(c) To convert IP addresses into domain names
आईपी पते को परिवर्तित करने के लिए कार्यक्षेत्र नाम
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : The main purpose of subnet mask in TCP/IP networking is to divide a network into smaller sub-networks. This helps in effectively managing and securing the networks.

392. Which type of support for TFTP is available from third parties, but is not included in Windows NT?
TFTP के लिए कौन-सा सपोर्ट, थर्ड पार्टी से उपस्थित है लेकिन Windows NT में, शामिल नहीं है?
(a) As client/क्लाइंट की भाँति
(b) As server/सर्वर की भाँति
(c) Windows NT includes full TFTP support
Windows NT पूरे TFTP सपोर्ट के साथ होता है
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (b) : Windows NT includes TFTP support as a client, but third party solutions are needed for TFTP server functionality.

393. In which layer of network architecture, the secured socket layer (SSL) is used?
नेटवर्क आर्किटेक्चर की किस परत में सुरक्षित सॉकेट परत (SSL) का उपयोग किया जाता है?
(a) Physical layer/फिजिकल परत
(b) Session layer/सेशन परत
(c) Application layer/ऐप्लीकेशन परत
(d) Presentation layer/प्रेजेन्टेशन परत

LT. Grade C.S. 2018

Ans. (d) : Secure Socket Layer is networking protocol used at transport layer to provide secure connection between client and server over internet. It places itself as and application layer protocol in the TCP/IP reference model and as presentation layer protocol in the OSI model.

394. The address resolution protocol (ARP) is used for
एड्रेस रिजोल्यूशन प्रोटोकॉल (ARP) का उपयोग किया जाता है
(a) Finding the IP address from DNS
DNS से IP पता ढूँढ़ने के लिए
(b) Finding the IP address of the default gateway
डिफॉल्ट गेटवे का IP पता ढूँढ़ने के लिए
(c) Finding the IP address that corresponds to a MAC address/एक MAC पते के अनुकूल IP पता ढूँढ़ने के लिए
(d) Finding the MAC address that corresponds to an IP address/एक IP पते के अनुकूल MAC पता ढूँढ़ने के लिए

LT. Grade C.S. 2018

Ans. (d) : The address resolution protocol (ARP) is a protocol used by the Internet Protocol (IP) specifically IPv4, to map IP network addresses to the hardware addresses used by a data link protocol. So the option (d) is correct.

395. With an IP address of 100, you currently have 80 subnets. What subnet mask should you use to maximize the number of available hosts ?
100 के IP एड्रेस के साथ आपके पास अभी 80 सबनेट है । आपको उपलब्ध मेजबान की संख्या को अधिकतम करने के लिए कौन सा सबनेट मास्क प्रयुक्त करना चाहिए ?
(a) 192 (b) 224
(c) 248 (d) 252

KVS PGT C.S. 2013

Ans. (d) : Given an IP address 100, it means the IP address is 100.0.0.0 ? Then in order to have 80 subnets we need 7 additional bits, and in that case the subnet mask would be 255. 254.0.0.

But if it is class less addressing then we can count the 7 bits required for 80 subnets starting from the LSB of the first active and in that case the required subnet mask would be

255. 252. 0.0.

396. Internet Control Message Protocol (ICMP)
इन्टरनेट कंट्रोल मैसेज प्रोटोकॉल (ICMP)
(a) Allows gateways to send error control messages to other gateways or hosts./गेटवेज को अन्य गेटवेज या मेजबान की त्रुटी नियंत्रण संदेश प्रेषित करने देता है ।
(b) Provides communication between the Internet Protocol Software on one machine and the Internet Protocol Software on another./एक मशीन में इंटरनेट प्रोटोकॉल सॉफ्टवेयर तथा अन्य के इन्टरनेट प्रोटोकॉल सॉफ्टवेयर के बीच संचार प्रदान करता है।
(c) Only reports error conditions to the original source, the source must relate errors to individual application programs and take action to correct the problem./मूल स्रोत को केवल त्रुटियों की परिस्थितियाँ सूचित करता है । स्रोत को त्रुटियों को पृथक-पृथक प्रयुक्ति प्रोग्रामों से जोड़ना होता है और समस्या को सुधारने के लिए कदम बढ़ाता है।
(d) All of these/उपरोक्त सभी

KVS PGT C.S. 2013

Ans. (d) : ICMP stands for Internet Control message Protocol. It is one of the main protocols of the internet protocol suite. It is used by network devices, like routers , to send error messages indicating. that a requested service is not available or that a host or routers could not be reached.

397. Which of the following protocol is used for transferring electronic mail messages from one machine to another?
एक मशीन से दूसरी मशीन तक इलेक्ट्रॉनिक मेल संदेश भेजने के लिए निम्नलिखित में से किस प्रोटोकॉल का उपयोग किया जाता है?
(a) FTP (b) SMTP
(c) SNMP (d) STTP

KVS PGT C.S. 2017, 2018

Ans. (b) : The Simple Mail Transfer Protocol (SMTP) is an Internet standard communication protocol for electronic mail transmission. Mail severs and other message transfer agents use SMTP to send and receive mail messages.

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398. What is the length of an IP address in bytes (Pre-IPv6) ?
बाइट (पूर्व IPv6) में IP पता की लंबाई क्या है?
(a) 8 (b) 1
(c) 2 (d) 4

KVS PGT C.S. 2017

Ans. (d) : Pre-IPv6 is also known as IPv4. An IPv4 address is 32 bits. An IP address is shown as 4 decimal numbers representing 4 bytes : d. d. d. d where d = decimal number (0-225). So, the length of an IPv4 address (pre-IPv6) in bytes is 4.

399. Which of the following protocol is used to upload and download files from remote server/computers?
रिमोट सर्वर/कम्प्यूटर से फाइलें अपलोड और डाउनलोड करने के लिए निम्नलिखित में से किस प्रोटोकॉल का प्रयोग किया जाता है?
(a) SMPT (b) VoIP
(c) HTTP (d) FTP

KVS PGT 23.03.2023 (Shift-II)

Ans. (d) : FTP is a way to download, upload and transfer files from one location (remote server/computers) to another on the internet and between computer systems. FTP stands for File Transfer Protocol. It enables the transfer of files back and forth between computers or through the cloud.

400. Which of the following wired medium will be able to have highest speed data communication possible?
निम्नलिखित में से किस तार माध्यम के द्वारा उच्चतम गति डेटा संप्रेषण संभव है ?
(a) Single pair telephone cable
एकल युग्म टेलीफोन केबल
(b) Fiber optic cable/फाइबर ऑप्टिकल केबल
(c) Co-axial cable/समाक्ष (को-एक्सियल) केबल
(d) Twisted pair cable/व्यावर्तित (ट्विस्टेड) केबल

KVS PGT 23.03.2023 (Shift-II)

Ans. (b) : A fiber optic cable contains anywhere from a few to hundreds of optical fiber within a plastic casing, also known as optic cables or optical fiber cables, they transfer data signal in the form of light and travel hundreds of miles significantly faster than those used in traditional electrical cable. This medium is able to have highest speed data communication possible than coaxial cable and twisted pair cable.

401. Which of the following is an invalid IP address?
निम्नलिखित में से कौन सा एक वैध आई पी (IP) एड्रेस नहीं है?
(a) 100.0.10.32 (b) 255.255.255.10
(c) 192.168.10.15 (d) 256.10.12.5

KVS PGT 23.03.2023 (Shift-II)

Ans. (d) : The range of the full IP address belongs from 0.0.0.0 to 255.255.255.255, so 100.0.10.32,255.255.255.10 and 192.168.10.15 are the valid IP addresses, But 256.10.12.5 is an invalid IP address as 256 does not fall into the range.

402. What is the minimum header size of an IP packet?
IP पैकेट का न्यूनतम हेडर आकार क्या है?
(a) 16 bytes (b) 10 bytes
(c) 20 bytes (d) 32 bytes

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Ans. (c) : The minimum header size of an IP packet is 20 bytes. The header of an IP (Internet Protocol) packet is the part that gives information to the network about how and where to send the data. The IP header has many fields such as-
- Version
- Header Length
- Source Address
- Destination Address
- Time to Live (TTL)
- Protocol
- Checksum.

403. Routing table of a router keeps track of
एक राउटर का राउटिंग टेबल नजर रखता है
(a) MAC Address Assignments
(b) Port Assignments to network devices
(c) Distributed IP addresses to network devices
(d) Routes to use for forwarding data to its destination

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Ans. (d) : Routing Table– A type of data table present in a router, which decides which route will be used to deliver a data packet to its destination. When any data leaves a device on the network, it may pass through many networks. The router chooses the most appropriate path to deliver that data to its destination and these paths are recorded in the routing table.
The grading table contains the following information–
- Destination IP Address
- Subnet mask
- Next Hop
- Interface
- Metric.

404. What is the IP Address range of APIPA?
APIPA की IP पता प्रसार क्या है?
(a) 169.254.0.1 to 169.254.0.254
(b) 169.254.0.1 to 169.254.0.255
(c) 169.254.0.1 to 169.254.255.254
(d) 169.254.0.1 to 169.254.255.255

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Ans. (c) : APIPA stands for Automatic Private IP Addressing. It is used when a computer cannot get an IP from a DHCP server. It automatically assigns itself an IP from the 169.254.0.1 to 169.254.255.254 range.

405. What does the port number in a TCP connection specify?
TCP कनेक्शन में, पोर्ट नंबर क्या निर्दिष्ट करता है?
(a) It specifies the communication process on the two-and systems
(b) It specifies the quality of the data & connection
(c) It specifies the size of data
(d) All of these./इनमें से सभी

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Ans. (a) : In a TCP connection, the port number is used to determine which service or application on a given system is sending or receiving data.
Example–
- Port 80 = HTTP (Web browsing)
- Port 443 = HTTPS (Secure wave)
- Port 25 = SMTP (Sending email)

406. TCP/IP is a
TCP/IP है एक
(a) Network Hardware
(b) Network Software
(c) Protocol
(d) None of these/इनमें से कोई नहीं

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Ans. (c) : TCP/IP (Transmission Control Protocol/Internet Protocol) is a network protocol used to send and receive data over computer network especially the internet. It is a group of two main protocols– TCP/IP which together make data transmission secure and organized.

407. What does subneting allow for-
सबनेटिंग किसके लिए अनुमति प्रदान करता है?
(a) Faster data transmission/तेज डेटा ट्रांसमिशन
(b) Improved error control/बेहतर त्रुटि नियंत्रण
(c) Efficient use of IP addresses
IP पतों का कुशल उपयोग
(d) Increased network security
नेटवर्क सुरक्षा में वृद्धि

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : Subnetting divides a large network into smaller subnetworks. This allows for more efficient allocation of IP addresses. Each subnet can have its range of IP Addresses, which means that a large organization doesn't need to use up all of its available IP addresses on a single network.

408. Process-to-process communication involves communication between:
प्रोसेस-टू-प्रोसेस कम्युनिकेशन का अर्थ है:
(a) Different network layers
विभिन्न नेटवर्क लेयर्स के बीच संचार
(b) Different devices on the same network
एक ही नेटवर्क पर विभिन्न डिवाइसों के बीच संचार
(c) Devices using different protocols
विभिन्न प्रोटोकॉल का उपयोग करने वाले डिवाइसों के बीच संचार
(d) Processes on different devices
विभिन्न डिवाइसों पर प्रक्रियाओं के बीच संचार

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : Process to Process Communication refers to the exchange of data between two processes that may be running on the same device or on different devices over a network.

409. The HTTP protocol is used for:/HTTP प्रोटोकॉल का उपयोग किसके लिए किया जाता है?
(a) Email communication/ईमेल संचार
(b) File transfers/फाइल ट्रांसफर
(c) Web browsing/वेब ब्राउज़िंग
(d) Remote terminal access/रिमोट टर्मिनल एक्सेस

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : HTTP is used for web browsing when you open a website, your browser and the web server exchange data with each other using the HTTP protocol.

410. The term HTTP stand for:
HTTP का पूर्ण रूप क्या है?
(a) Hyper Terminal Tracing Program
हाइपर टर्मिनल ट्रेसिंग प्रोग्राम
(b) Hypertext Tracing Program
हाइपरटेक्स्ट ट्रेसिंग प्रोग्राम
(c) Hypertext Transfer Protocol
हाइपरटेक्स्ट ट्रांसफर प्रोटोकॉल
(d) Hypertext Tracing Protocol
हाइपरटेक्स्ट ट्रेसिंग प्रोटोकॉल

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : HTTP stands for Hyper Text Transfer Protocol. It is the foundation of the World Wide Web and is used to load web pages using hypertext links.

411. The length of an IPv6 address is:
IPv6 पते की लंबाई होती है:
(a) 32 bits/32 बिट्स
(b) 64 bits/64 बिट्स
(c) 128 bits/128 बिट्स
(d) 256 bits/256 बिट्स

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : An Internet protocol version 6 (IPv6) address is a unique numerical identifier assigned to a device connected to a computer network that uses the Internet Protocol for Communication. It's the successor to the earlier version IPv4.
An IPv6 address is 128 bit long and IPv4 address is 32 bit long.

412. The maximum length (in bytes) of an IPv4 datagram is:
एक IPv4 डाटाग्राम की अधिकतम लंबाई (बाइट्स में) होती है:
(a) 32 (b) 256
(c) 1024 (d) 65535

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : The maximum length of an IPv4 datagram is 65535 bytes. This limit is for the entire datagram including the header, and is set by the 16 bit field "Total Length".

413. _____is used to transfer files from one computer to another computer.
_____का उपयोग File को एक कम्पयूटर से दूसरे कंप्यूटर में Transfer करने के लिए करते है।
(a) FTP (b) Telnet
(c) Chat (d) DNS

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : FTP stands for File Transfer Protocol. It is used to upload or download files from one computer to another. It allows file transfer over the internet or local Network.

414. TCP/IP is a_____
टी सी पी/आई पी एक _____है।
(a) Web Browser
(b) System Software

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(c) Application Software
(d) Protocol

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (d) : A protocol is a set of rules and guidelines for communication data. It dictates how data is formatted, transmitted and received ensuring different systems can understand each other.

415. FTP stands for?
FTP का अर्थ है?
(a) File Transfer Protocol
(b) Fixed Transfer Protocol
(c) Fixed transfer Port
(d) None of the above

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : The full form of FTP is File Transfer Protocol. It is a network protocol used to send and receive files from one computer to another over the internet.

416. Which one of the following is not an application layer protocol used in internet?
निम्नलिखित में से कौन एक इंटरनेट में उपयोग की जाने वाली Application layer प्रोटोकॉल नहीं है?
(a) Remote procedure call
(b) Internet relay chat
(c) Resource reservation protocol
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : Remote Procedure Call (RPC)– This protocol is used by the internet layer (also known as the network layer) and the transport layer. It's designed to reserve resources across a network for an integrated services model, ensuring a specific quality of service for data flows. It does not operate at the application layer.
Resource reservation protocol– It is actually a transport layer protocol, not an application layer protocol. It's used to reserve network resources for specific data streams, like video conferencing.

417. _____ is a device that forwards packets between networks by processing the routing information included in the packet.
_____ एक उपकरण है जो Packets में Routing जानकारी को Processing करके नेटवर्क के बीच Packets को Forward करता है।
(a) Bridge
(b) Firewall
(c) Router
(d) All of these/इनमें से सभी

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : The device that forwards packets between networks by processing routing information is a router. A router reads the network address in a packet's header and uses its routing table to determine the best both to sent the packet to its destination network.

418. Which protocol is commonly used for sending emails?
ईमेल भेजने के लिए आमतौर पर कौन-सा प्रोटोकॉल उपयोग किया जाता है?
(a) .HTTPS (b) FTP
(c) SMTP (d) TCP

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : SMTP stands for Simple Mail Transfer Protocol. It is a network protocols used to send emails. Whenever you send an email to someone, your email client like-Gmail, Outlook etc. uses this protocol to transmit your message to the email server.

419. What is the primary function of routing in the network layer?
नेटवर्क लेयर में राउटिंग का प्राथमिक कार्य क्या है?
(a) Data framing/डेटा फ्रेमिंग
(b) Error correction/त्रुटि सुधार (Error correction)
(c) Finding the best path for data
डेटा के लिए सर्वोत्तम मार्ग खोजना
(d) Flow control/फ्लो कंट्रोल

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The primary function of routing in the network layer is finding the best path for data.
Routing is the process of selecting a path for network traffic. It determines the most efficient route for data packets to travel from a source to a destination across different networks. This is crucial for ensuring that data reaches its intended recipient in a timely and efficient manner.

420. What is a socket in the context of process-to-process communication?
प्रोसेस-टू-प्रोसेस संचार के संदर्भ में सॉकेट क्या होता है?
(a) A hardware device for data transmission
डेटा ट्रांसमिशन के लिए एक हार्डवेयर डिवाइस
(b) A software interface for network communication
नेटवर्क संचार के लिए एक सॉफ़्टवेयर इंटरफ़ेस
(c) A type of cable used for high-speed data transmission
हाई-स्पीड डेटा ट्रांसमिशन के लिए एक प्रकार की केबल
(d) type of network topology
एक प्रकार की नेटवर्क टोपोलॉजी

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : Software is a software interface that enables data transmission over a network between two processes (client and server). It is not a hardware interface but a programming concept that uses IP addresses and port number.

421. Routing involves:
राउटिंग में क्या शामिल है?
(a) Dividing data into frames
डेटा को फ्रेम्स में विभाजित करना
(b) Managing flow control
फ्लो कंट्रोल का प्रबंधन
(c) Determining the best path for data packets
डेटा पैकेट्स के लिए सर्वोत्तम मार्ग निर्धारित करना
(d) Correcting errors in data transmission
डेटा ट्रांसमिशन में त्रुटियों को ठीक करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : Routing is a network layer function that involves selecting the optimal path through a network so that data packets can travel from the source to the destination efficiently.

Sarkari Resource

09. Software Engineering

(i) Definition, Software development, and life-cycle models, CMM

1. The characteristics of modern programming are:

A. Each module should do only one thing.

B. Communication between modules is allowed only by a calling module.

C. Communication can take place directly between modules that do not have calling–called relationship.

D. All modules are designed as single-entry, single exit systems using control structure.

Choose the correct answer from the options given below :

(a) A, B, & C only (b) A, B & D only

(c) A, C & D only (d) C & D only

UGC NET C.S. December-2024

Ans. (b) :

(A) Each module should do only one thing- This aligns with the core principle of modular programming where each module has a focused, single responsibility promoting code clarity and maintainability.

(B) Communication between modules is allowed only by a calling module. This emphasizes the concept of controlled interactions between modules, where one module explicit calls another to access its functionality.

(C) Communication can take place directly between modules that do not have calling called relationship. This is not a characteristic of modern programming as it can lead to complex dependencies and harder to manage code.

(D) All modules are designed as single entry single exit systems using control structure- This refers to the idea that a module should have a clear entry point and a single exit point, making code flow more predictable and easier to understand.

2. Arrange the following activities to form a general software engineering process model :

एक सामान्य सॉफ्टवेयर इंजीनियरिंग प्रक्रिया मॉडल बनाने के लिए निम्नलिखित गतिविधियों का सुव्यवस्थित मिलान करें:

(A) Modelling/मॉडलिंग

(B) Planning/प्लानिंग

(C) Deployment/डिप्लॉयमेंट

(D) Construction/कंस्ट्रक्शन

(E) Communication/कम्युनिकेशन

(a) A, B, C, D, E (b) B, C, E, A, D

(c) E, B, A, D, C (d) A, E, D, B, C

HTET PGT C.S. 2021

Ans. (c) : The activities in the correct sequence of a general Software Engineering Process Model.

1. **Communication (E)–** Discuss with stakeholders to gather requirements.
   Example– Meeting, requirement elicitation.
2. **Planning (B)–** Define project scope, resources, schedules, risks.
   Example– Project management planning.
3. **Modeling (A)–** Create system models (SRS, DFDs, UML Diagrams) for analysis and design.
4. **Construction (D)–** Actual coding + unit/integration testing.
5. **Deployment (C)–** Deliver the software to the customer, get feedback and maintain.

3. Which of the following is not covered by the vital philosophies of software engineering ?

सॉफ्टवेयर इंजीनियरिंग की मूल धारणा निम्नलिखित में से किसके लिए नहीं है?

(a) Software security/सॉफ्टवेयर सुरक्षा

(b) Software reuse/सॉफ्टवेयर का पुनः उपयोग

(c) Software processes/सॉफ्टवेयर प्रक्रियाएँ

(d) Software validation/सॉफ्टवेयर सत्यापन

HTET PGT C.S. 2020

Ans. (a) :

(b) Software Reuse– Encourages using existing software component to save time, cost, and effort.

(c) Software Processes– Defines structured steps, model and methodologies to develop quality software systematically.

(d) Software Validation– Ensure that the developed software meets user requirements and works as intended.

⇒ These three are core philosophies of Software Engineering.

(a) Software Security– Security is very important, it is considered a quality attribute or non-functional requirement not a core philosophy of software engineering itself.

4. Which software development process model assumes that the requirements of a system can be frozen before the design begins?

(a) Waterfall model

(b) Spiral model

(c) Iterative enhancement model

(d) Time boxing model

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : Waterfall Model– This is a linear sequential model. It assumes that all requirements are known and fixed before starting the design phase. Each phase (requirements ⇒ design ⇒ implementation ⇒ testing ⇒ deployment) is completed one after another without going back.

5. Which software development process model gives highest priority of satisfy customers through early and continuous delivery of working software by minimizing work products?

Sarkari Resource

(a) Waterfall model (b) Spiral model
(c) Agile process model (d) Time boxing model

Haryana PGT C.S. Asist. Prof. 2024

Ans. (c) : Agile Process Model– Agile emphasizes customer satisfaction by delivering small but working parts of the software frequently. It reduce unnecessary documentation and focuses on collaboration with customers, so they can sec progress early and continuously.

Waterfall Model– It is a traditional linear model where each phase (requirement, design, coding, testing, deployment) is completed step by step.

Spiral Model– This model combines iterative development with risk analysis. Each cycle (Spiral) include planning, risk evaluation, development, and testing.

Time Boxing Model– This model, tasks are given strict time limits. The focus is on delivering features within fixed deadlines, even if scope is reduced. It values speed and control.

6. _____is a collection of methodologies that emerged in response to the weakness of waterfall model development and its variations.
यह पद्धतियों का एक संग्रह है, जो वॉटरफॉल मॉडल की कमजोरियों और विविधताओं के जवाब में उभरा है–

(a) Break-even Point
(b) Rapid Application Development
(c) Acceptance Test Design
(d) Throwaway Prototyping

HTET PGT C.S. 30.07.2025

Ans.(b): The waterfall model is a linear and rigid approach to software development. Once a phase is completed, going back to make changes is very costly and time consuming.

Rapid Application Development (RAD)– RAD is a collection of methodologies that focus on prototyping iterative development and user involvement. It emphasizes quick delivery of functional component, frequent user feedback and shorter development cycles compared to the waterfall model.

7. Full form of SDLC is:
SDLC का पूर्ण रूप है–

(a) System Development Life Cycle
(b) Software Data Life Cycle
(c) Software Design Life Cycle
(d) System Design Life Cycle

HTET PGT C.S. 30.07.2025

Ans.(a): SDLC stands for System Development Life Cycle. It is a structured process used for planning, creating, testing deploying, and maintaining an information system or software.

Phases of SDLS–

- Requirement gathering & analysis
- System design
- Implementation (coding)
- Testing
- Deployment
- Maintenance.

8. Which of the following is/are aspect (s) of software evolution ?
निम्नलिखित में से कौन सा/से सॉफ्टवेयर विकास का/के पहलू है?

I. Software Architecture Aspect
सॉफ्टवेयर आर्किटेक्चर पहलू

II. Programming Language Aspect
प्रोग्रामिंग लैंग्वेज पहलू

(a) Only I /केवल I
(b) Only II /केवल II
(c) Both I and II / I तथा II दोना
(d) Neither I nor II / I नी ही II ना ही

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : Software evolution refers to the process of modifying and updating software after its initial development to correct fault, improve performance or adapt to changing requirements.

9. Which of the following is the most important feature of spiral model?
स्पॉइरल मॉडल का सबसे मुख्य गुण निम्न में से कौन सा है?

(a) Quality management/गुणवत्ता प्रबन्धन
(b) Performance management/परफॉरमेंस प्रबन्धन
(c) Efficiency management/दक्षता प्रबन्धन
(d) Risk management/जोखिम प्रबन्धन

UKPSC PGT C.S. Asist. Prof. 2017

Ans (d) : Spiral Model– The Spiral Model is a software development life cycle (SDLC) model introduced by Barry Boehm. It combines the iterative nature of prototyping with the systematic aspects of the waterfall model.

Risk Management– Software projects often face risks such as cost overruns, unclear requirements, or technology failures. The Spiral Model focuses on addressing these risks early, making it suitable for large, complex, and high risk projects.

10. Which of the following sets represent five stages defined by Capability Maturity Model (CMM) in increasing order of maturity?

(a) Initial, Defined, Repeatable, Managed, Optimized.
(b) Initial, Repeatable, Defined, Managed, Optimized.
(c) Initial, Defined, Managed, Repeatable, Optimized.
(d) Initial, Repeatable, Managed, Defined, Optimized.

UGC NET C.S. June-2016 (Paper-III)

Ans : (b) There are five levels defined along the continuum of the model and according to the SEI.

(1) Initial : The starting point for use of a new repeat process
(2) Repeatable : The process is at least documented sufficiently such that repeating the same steps may be attempted.
(3) Defined : The process is defined as a standard business process.
(4) Managed : This process is quantitatively managed in accordance with agreed upon metrics
(5) Optimizing : Process management includes deliberate process improvement.

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11. What is the other name of Linear Sequential model in software engineering?
सॉफ्टवेयर इंजीनियरिंग में लीनियर सिक्वेंशियल मॉडल का दूसरा नाम क्या है?
(a) Spiral model/स्पाइरल मॉडल
(b) Waterfall model/वाटर फॉल मॉडल
(c) Prototype model/प्रोटोटाइप मॉडल
(d) RAD model/आर.ए.डी. मॉडल

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : The Linear Sequential Model is also called the Waterfall Model.
In Software engineering the Waterfall Model is sequential software development process where progress flows steadily downwards through distinct phases–
Requirement analysis → System design → Implementation→Testing→Deployment→Maintenance
Each phase must be completed before moving to the next, and there is little to no overlap.

12. _____ model is a "high speed" adaption of the linear sequential model in software engineering.
_____ मॉडल सॉफ्टवेयर इंजीनियरिंग में रैखिक अनुक्रमिक मॉडल का ''हाई स्पीड'' अनुकूलन है।
(a) Prototype model/प्रोटोटाइप मॉडल
(b) Rapid Application development model रैपिड ऐप्लीकेशन डेवलपमेंट मॉडल
(c) Spiral model/स्पाइरल मॉडल
(d) Waterfall model/वाटर फॉल मॉडल

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : The Rapid Application Development (RAD) model is a fast paced adoption of the Linear Sequential (Waterfall) model, designed to speed up software development without sacrificing quality. It emphasized quick prototyping iterative delivery and active user involvement rather than strictly following long sequential phases.
RAD is best suited for projects with clearly defined objective modular components and users available for continuous feedback.

13. The _____ model is preferred for software development when the requirements are not clear.
(a) Rapid Application Development
(b) Rational Unified Process
(c) Evolutionary Model
(d) Waterfall Model

UGC NET C.S. June-2016 (Paper-II)

Ans : (c) Evolutionary Model is an iterative and incremental approach to software development. It is preferred for software development when requirements are not clear.

14. Which of the following is not included in waterfall model?
(a) Requirement analysis (b) Risk analysis
(c) Design (d) Coding

UGC NET C.S. June-2016 (Paper-II)

Ans : (b) The waterfall model is a sequential (non-iterative) process, it includes requirements, design, implementation, verification and maintenance.
Risk analysis used in spiral model.

15. The major shortcoming of waterfall model is :
(a) The difficulty in accommodating changes after requirement analysis
(b) The difficult in accommodating changes after feasibility analysis
(c) The system testing
(d) The maintenance of system

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (a) The main drawback of the waterfall model is the difficulty of accommodating change after the process is under way.
(1) Difficult to address change: Inflexible partitioning of the project into distinct stage makes it difficult to respond to changing customer requirements. Therefore, this model is only appropriate when the requirement are well understood and changes will be fairly limited during the design process

16. The quick design of a software that is visible to end users leads to _____
(a) Iterative model (b) Prototype model
(c) Spiral model (d) Waterfall model

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (b) Prototype typically simulates only a few aspects of, and may be completely different from, the final product.
Prototyping has several benefits. The software designer and implementer can get valuable feedback from the user early in the project.

17. Identify the correct statements from the following regarding spiral model
A. Spiral model is an incremental process model.
B. Spiral model is a risk - driven process mode.
C. Spiral model is an acyclic balancing approach.
D. Spiral model involves a set of anchor point milestone.
Choose the correct answer from the options given below :
(a) A & B Only (b) B & C only
(c) A & D Only (d) B & D only

UGC NET C.S. December-2024

Ans. (d) : (B) Spiral model is a risk- driven process mode:-
- The spiral model emphasizes Indentifying and managing risks early in the development process, with iterations that allow for ongoing risk management.

(D) Spiral model involves a set of anchor point milestones.
- The spiral model includes milestones (also called anchor points) at each iteration, which are used to assess the project's progress and determine if the next integration should continue.

The spiral model is not strictly Incremental (as in statement A) and does not focus on acyclic balancing (as in statement C).

18. As per the software Engineering Institute (SEI), the correct sequence is
A. Initial **B. Repeatable**
C. Managed **D. Optimizing**
E. Defined

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Choose the correct answer from the options given below :

(a) A, B, C, D, E (b) A, B, E, C, D
(c) A, E, C, D, B (d) A, E, B, C, D

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Ans. (b) : The correct sequence of the Software Engineering Institute Capability Maturity Model levels is as follows.
Initial– Ad hoc processes with unpredictable outcomes.
Repeatable– Basic project management processes established to track cost, schedule, and functionality.
Defined– Processes standardized, documented and integrated across the organization.
Managed– Processes quantitatively measured and controlled using data.
Optimizing– Continuous process improvement through incremental and innovative charges.

19. Spiral model of software development is also known a
सॉफ्टवेयर विकास के स्पाइरल मॉडल को _____ के रूप में भी जाना जाता है।

(a) Successive model/सक्सेसिव मॉडल
(b) Meta modal/मेटा मॉडल
(c) Prototype model/प्रोटोटाइप मॉडल
(d) Risk model/रिस्क मॉडल

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : The Spiral model of SDLC is also called the 'Meta Model'. This is because it's a framework that can incorporate and adapt to various SDLC models, including the waterfall and iterative models, making it a versatile approach to software development.
Key reasons why it's called the Meta Model–
→ Flexibility & Adaptability
→ Risk Management Focus
→ Iterative & Incremental
→ Combination of Models.

20. In _____ model, software is developed in a series of incremental releases.
_____ मॉडल में, सॉफ्टवेयर को इंक्रीमेंटल रीलीजेस की एक श्रृंखला में विकसित किया जाता है।

I. Spiral/स्पाइरल
II. Sequential/सिक्वैंशियल

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Spiral Model is an iterative and incremental approach to software development. It combines elements of both prototyping and sequential development (like the waterfall model). Development progresses in a series of incremental releases, where each spiral cycle refines and expands the software based on feedback and risk analysis.
Sequential Model– This is traditional waterfall model. Sequential model follows a linear and rigid approach, where each phase (requirements, design, implementation, testing, and maintenance) is completed before moving to the next. There are no incremental releases, instead the entries system is delivered at once at the end.

21. In _____ model, software is developed in a series of incremental releases.
मॉडल में, साफ्टवेयर को इंक्रिमेंटल रीलीजेस की एक श्रृंखला में विकसित किया जाता है।

(a) Spiral/स्पाइरल
(b) Sequential/सिक्वैंशियल
(c) Concurrent/कोंकुरैंट
(d) RAD/रैड (RAD)

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (a) : In the spiral model software is built through a series of incremental releases, with each cycle (or spiral) adding more features and improvements. Every loop of the spiral goes through four main phases– Planning, risk analysis, engineering and evaluation.
The first cycle may produce prototype, while later deliver more complete versions of the software.

22. Which process model is also called as classic life cycle model?

(a) Waterfall model (b) RAD model
(c) Prototyping model (d) Incremental model

UGC NET C.S. June-2015 (Paper-II)

Ans.(a) : The waterfall model was first process model to be introduce.
- It is also referred as linear sequential life cycle model
- It is very simple to understand and use. In this each phase must be complicated fully before the next phase can begin.
- In this model the testing stands only after the development is complete
- In this model phases do not overlap

23. Which e-business model allows consumers to name their own price for products and services?

(A) B2 B (B) B2 G
(C) C2 C (D) C2 B

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Because, C2B model called a reverse auction or demand collection model, enables buyers to name or demand their own price, which is often binding, for a specific good or service. The website collects the demand bids then offers the bids to participating sellers and C2C is a consumer to consumer business model and B2B and B2G has nothing to concern about consumer price setting.

24. _____ model is designed to bring prices down by increasing the number of customers who buy a particular product at once.

(A) Economic Order Quantity
(B) Inventory
(C) Data Mining
(D) Demand-Sensitive Pricing

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Because, Inventory and data mining has nothing to do with price. Economic order quality (ECQ) is the order quantity that minimizes the total holding costs and ordering cost it is one of the oldest classical production scheduling models. So these three options are get neglect so the correct is (d)

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25. _____are applied throughout the software process.

(a) Framework activities
(b) Umbrella activities
(c) Planning activities
(d) Construction activities

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) The **umbrella activities** in a software development life cycle process include the following: software project management, formal technical reviews, software quality assurance and software configuration management.
Umbrella activities occur throughout the software process and focus primarily on project management, tracking and control.

26. KPA in CMM stands for

(a) Key Process Area
(b) Key Product Area
(c) Key Principal Area
(d) Key Performance Area

UGC NET C.S. June-2014 (Paper-II)

Ans : (a) Because, Key Process Area (KPA) used in CMM identifies a cluster of related activities that, when performed collectively, achieve a set of goals considered important.

27. Which one of the following is not a key process area in CMM level 5?

(a) Defect prevention
(b) Process change management
(c) Software product engineering
(d) Technology change management

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : CMM (Capability maturity model)– It is framework that describes an evolutionary improvement path for software organizations from an immature process to a mature, discipline one.
The software process is at CMM level 5, if there is continuous process improvement, if there is quantitative feedback from the process and from piloting innovative ideas and technologies. Software processes are the CMM level 5, if there are process change management, detect prevention and technology change management, So software product engineering in not a key process area in CMM level 5

28. Which of the following is not a software process model?

(a) Prototyping (b) Iterative
(c) Timeboxing (d) Glassbosing

UGC NET C.S. December-2015 (Paper-II)

Ans : (d) Prototyping, iterative are well known software process model
In time boxing model, development is done iteratively as in iterative enhancement model each iteration is done in a timebox of fixed duration.
⇒ Glass boxing model is software testing pattern.

29. Statistical software quality assurance in software engineering involves_____

(a) Using sampling in place of exhaustive testing of software
(b) Surveying customers to find out their opinions about product quality
(c) Tracing each defect to its underlying cause, isolating the vital few causes, and moving to correct them
(d) Tracing each defect to its underlying causes, and using the pare to principle to correct each problem found

UGC NET C.S. November-2017 (Paper-III)

Ans : (c) Statistical software quality assurance in software engineering involves tracing each defect to its underlying cause, isolating the vital few causes, and moving to correct them.

30. Match each software lifecycle model in List-I to its description in List-II:

List-I		List-II	
A.	Code-and-Fix	**1.**	Assess risks at each step; do most critical action first.
B.	Evolutionary prototyping	**2.**	Build an initial small requirement specifications, code it, then "evolve" the specifications and code as needed.
C.	Spiral	**3.**	Build initial requirement specification for several releases, then design-and-code in sequence.
D.	Staged Delivery	**4.**	Standard phases (requirements, design, code, test) in order
E.	Waterfall	**5.**	Write some code, debug it, repeat (i.e. adhoc)

Codes:

	A	**B**	**C**	**D**	**E**
(a)	5	2	1	3	4
(b)	5	3	1	2	4
(c)	4	1	2	3	5
(d)	3	5	1	2	4

UGC NET C.S. August-2016 (Paper-III)

Ans : (a)
(i) Code and fix: write some code, debug it repeat.
(ii) Evaluation prototype: system develop incrementally.
(iii) Spiral model: access risk at each step.
(iv) Staged delivery: is an evolutionary software release.
(v) Water fall model: sequential (non-iterative design process.

31. The ISO quality assurance standard that applies to software Engineering is

(a) ISO 9000: 2004 (b) ISO 9001: 2000
(c) ISO 9002: 2001 (d) ISO 9003: 2004

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) The 9001 : 2000 is the quality assurance standard that applies to software engineering. The standard contains 20 requirement that must be present for an effective quality assurance system.

32. Software Engineering is an engineering discipline that is concerned with :

(a) How computer systems work
(b) Theories and methods that underline computers and software systems

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(c) All aspects of software production
(d) All aspects of computer-based system development, including hardware, software and process engineering

UGC NET C.S. June-2017 (Paper-II)

Ans : (c) Software engineering is an engineering discipline that is concerned with all aspects of software production.

33. The prototyping model of software development is :
(a) A reasonable approach when requirements are well-defined
(b) A useful approach when a customer cannot define requirements clearly
(c) The best approach to use for projects with large development teams
(d) A risky model that rarely produces a meaningful product

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) This model allows for the creation of a preliminary version (prototype) of the software, which can be refined iteratively based on user feedback until satisfactory version is achieved.
Clarifies Requirements– The prototyping process helps customers visualize and understand what the final product will look like and how it will function.
Reduce Risk– By identifying potential issues early on through prototyping, developers can avoid costly rework and delays later in the development cycle.

34. Software not wear-out in the traditional sense of the term, but software does tend to deteriorate as it evolves, because :
(a) Software suffers from exposure to hostile environments
(b) Defects are more likely to arise after software has been used often
(c) Multiple change requests introduce errors in component interactions
(d) Software spare parts become harder to order

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) As time passes software is likely to get multiple change request while using software by users & one does not identify all aspects of software when it is developed.

35. Software re-engineering is concerned with :
(a) Re-constructing the original source code from the existing machine (low - level) code program and modifying it to make it more user - friendly
(b) Scrapping the source code of a software and re-writing it entirely from scratch
(c) Re-organizing and modifying existing software systems to make then more maintainable
(d) Translating source code of an existing software to a new machine (low - level) language

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Software re-engineering is a process of software development which is done to improve the maintainability of a software system.

36. Which of the following is not a key issue stressed by an agile philosophy of software engineering?
(a) The importance of self-organizing teams as well as communication and collaboration between team members and customers
(b) Recognition that change represents opportunity
(c) Emphasis on rapid delivery of software that satisfies the customer
(d) Having a separate testing phase after a build phase

UGC NET C.S. November-2017 (Paper-II)

Ans : (d) Principles of agile software development:
1. Customer satisfaction by early and continuous delivery of valuable software.
2. Welcome changing requirements, even in late development.
3. Working software is delivered frequently.
4. Close, daily cooperation between business people and developers.
5. Face to face conversation is the best form of communication.
6. Projects are build around motivated individuals who should we trusted.
7. Working software is primary measure of progress.
8. Sustainable development, able to maintain a constant pace.
9. Continuous attention to technical excellence and good design.
10. Simplicity the art of maximizing the amount of work not done is essential.

37. Which of the following are Agile Process Models?
A. Extreme Programming (XP)
B. Waterfall
C. Scrum
D. Spiral
E. Incremental
Choose the correct answer from the options given below :
(a) (A) and (C) Only (b) (B) and (C) Only
(c) (A) and (D) Only (d) (B) and (E) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (a) : The Agile process model was primarily designed to help a project adopt quickly to charge requests. So, the main aim of the Agile model is to facilitate quick project completion. Given below are some Agile SDLC models:
(i) Extreme Programming (XP)
(ii) Scrum
(iii) Crystal Agile methodology
(iv) Dynamic systems development method (DSDM).
(v) Feature – driven development (FDD).
(vi) Lean Development
(vii) Unified Process

38. Which of the following are typical activities in the software process lifecycle?
A. Requirement Analysis **B. System Design**
C. Code refactoring **D. Deployment**
E. Substructure

Sarkari Resource

Choose the correct answer from the options given below :

(a) (A), (B), (C) Only (b) (B), (C), (D) Only
(c) (A), (B), (D) Only (d) (A), (D), (E) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (c) : Software development life cycle (SDLC) is a process followed for software building within a software organization. SDLC consists of a precise plan that describes how to develop, maintain, replace and enhance specific software. The life cycle defines a method for improving the quality of software and the all around development process.
The SDLC involves six phases or stages while developing any software.
(i) Planning & Requirement Analysis
(ii) Defining Requirements
(iii) Design
(iv) Development
(v) Testing
(vi) Deployment & Maintenance

39. The incremental model of software development is _____

(a) A reasonable approach when requirements are well defined.
(b) A good approach when a working core product is required quickly.
(c) The best approach to use for project with large developments teams.
(d) A revolutionary model that is not used for commercial products.

UGC NET C.S. December-2022

Ans. (b) : The incremental model is a software development process in which the product is designed, implemented, and tested in increments (small parts) until the entire system is completed.
Key Characteristics–
1. Early delivery- A core working product is delivered quickly.
2. Phased development- New features are added
3. Flexibility
4. Risk managements.

40. RAD software process model stands for

(a) Rapid Application Development
(b) Relative Application, Development
(c) Rapid Application Design
(d) Recent Application Development

UGC NET C.S. December 2021

Ans. (a) : The RAD (Rapid Application Development) is a software development methodology that uses minimal planning in favour of rapid prototyping. A prototype is working model that is functionally equivalent to a component of the product.
So RAD software process model stands for Rapid Application Development.

41. Relations produced from E - R Model will always be in _____.

(a) 1 NF (b) 2 NF
(c) 3 NF (d) 4 NF

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : Simple atomic values non multivalued attributes. Generally ER diagram is design level task and relational model is implementation. Whenever relational model is implemented from ER diagram. We get rid of multi valued attribute.
Multi valued attributes are removed as M : N binary relation by repeating becomes 1NF. We cannot guarantee any higher normalization.

42. Reasons to re-engineer a software include :

P : Allow legacy software to quickly adapt to the changing requirements
Q : Upgrade to newer technologies / platforms / paradigm (for example, object-oriented)
R : Improve software maintainability
S : Allow change in the functionality and architecture of the software

Code :

(a) P, R and S only (b) P and R only
(c) P, Q and S only (d) P, Q and R only

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : Allow legacy software to quickly adapt to the changing requirements. **True**
Upgrade to newer technologies / platforms / paradigm (for example, object-oriented). **True**
Improve software maintainability. **False**
Allow change in the functionality and architecture of the software. **True**

43. Which of the following statements is/are true?

P : Software Reengineering is preferable for software products having high failure rates, having poor design and/or having poor code structure.
Q : Software Reverse Engineering is the process of analyzing software with the objective of recovering its design and requirements specification.

Choose the correct answer from the code given below :

(a) Both P and Q (b) Neither P nor Q
(c) P only (d) Q only

UGC NET C.S. December-2018 (Paper-II)

Ans. (a) : Re-engineering is the examination, analysis and alteration of an existing software system to reconstitute it in a new form, and the subsequent implementation of the new form.
Software Reengineering is preferable for software products having high failure rates, having poor design and/or having poor code structure is correct.
Reverse engineering is the process of analyzing a subject system to identify the system's components and their interrelationships and create representations of the system in another form or at a higher level of abstraction.
Software Reverse Engineering is the process of analyzing software with the objective of recovering its design and requirements specification.

44. Which of the following is not one of the principles of agile software development method?

(a) Embrace Change
(b) Incremental Delivery
(c) Customer Involvement
(d) Following the Plan

UGC NET C.S. December-2018 (Paper-II)

Sarkari Resource

Ans. (d) : Embrace Change is the principle says, "Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage."
Incremental Model is a process of software development where requirements are broken down into multiple standalone modules of software development cycle.
customer involvement in the production of goods and services is fundamental to making significant improvements in the product output.

45. Match List-I with List-II

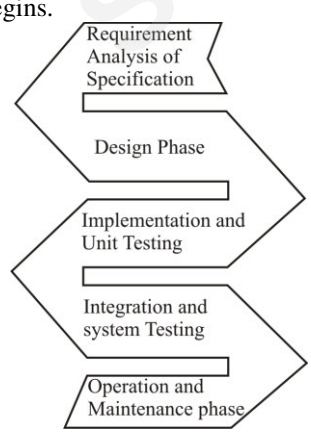
	List - I (Software Process Model)		List - II (Description)
A.	**Waterfall Model**	**i.**	**Software can be developed incrementally**
B.	**Evolutionary Model**	**ii.**	**Requirement compromises are inevitable**
C.	**Component - based Software Engineering**	**iii.**	**Explicit recognition of risk**
D.	**Spiral Development**	**iv.**	**Inflexible partitioning of the project into stages**

Choose the correct answer from the options given below:

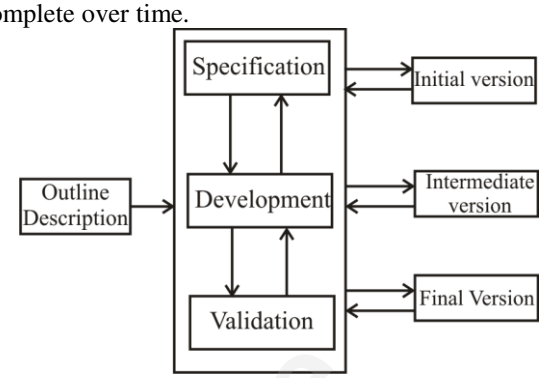
(a) A - IV, B - I, C - III, D - II
(b) A - I, B - IV, C - II, D - III
(c) A - II, B - III, C - I, D - IV
(d) A - IV, B - I, C - II, D - III

UGC NET C.S. December 2020

Ans. (d) : Waterfall Model :-This model has five phases: Requirements analysis and specification, design, implementation and unit testing, integration and system testing and operation and maintenance. The steps always follow in this order and do not overlap. The developer must complete every phase before the next phase begins.



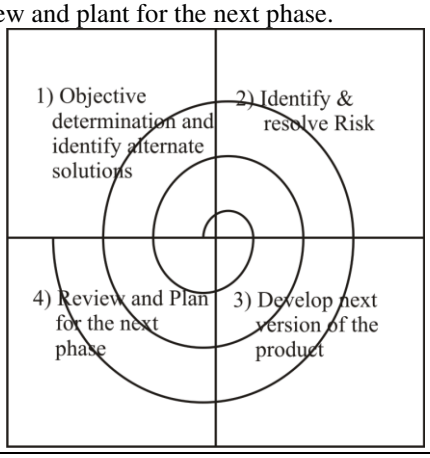
Evolutionary Model :- The evolutionary model is an iterative model as they are characterized in a manner that enable software engineering to develop a complete version of the software. Such models are applied because the requirement often change. So the end product will be unrealistic for a complete version is impossible due to tight market deadlines. It is better to introduce a limited version. Thus, software engineering can follow a process model that has been explicit designed to accommodate a product that gradually complete over time.



Component Based Development :- Component - based software engineering (CBSE) is an approach to software development that relies on the reuse of entities called 'software components'.

- It emerged from the failure of object - oriented development to support effective reuse single object classes are too detailed and specific.
- Components are more abstract than object classes and can be considered to be stand alone service providers. They can exist as stand-alone entities.

Spiral Development:- This software development Life Cycle models, which provides support for risk handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project.
Four quadrants are
1) Objective determination and identify alternative solutions
2) Identify and resolve Risks
3) Develop next version of the product.
4) Review and plant for the next phase.



46. Match List I with List II
With reference to CMM developed by Software Engineering Institute (SEI).

List I		List II
(A) INITIAL	**I.**	**Process measurement**
(B) REPEATRBLE	**II.**	**Process definition**
(C) DEFINED	**III.**	**Project management**
(D) MANAGED	**IV.**	**ADHOC**

Sarkari Resource

Choose the correct answer form the options given below:

(a) A-III, B-IV, C-II, D-I
(b) A-IV, B-III, C-I, D-II
(c) A-IV, B-III, C-II, D-I
(d) A-III, B-IV, C-I, D-II

UGC NET C.S. June-2020 (Paper-II)

Ans. (c) : Initial (Chaotic, adhoc, individual heroics):- the stage at which a new or undocumented repeat mechanism may be used.
Repeatable:- At the very least, the procedure is well recorded such that the same steps can be tried again.
Defined:- As a normal business procedure the process has been described and verified.
Managed (capable):- The method is handled quantitatively according to agreed-upon metrics.

47. Which of the following software process model gives explicit recognition of risk?
निम्न में से कौन-सा सॉफ्टवेयर प्रोसेस मॉडल रिस्क को स्पष्ट पहचान देता है?

(a) Waterfall Model/वॉटरफॉल मॉडल
(b) Incremental Development Model
इंक्रीमेंटल डेवलपमेंट मॉडल
(c) Spiral Model/स्पाइरल मॉडल
(d) Prototyping Model/प्रोटोटाइपिंग मॉडल

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : The Spiral Model gives explicit recognition of risk at each iteration. It incorporates risk analysis as a key part of its cycles, which distinguishes it from other models like waterfall or Incremental. Each phase in the spiral includes risks, making it highly suitable for large, complex and high-risk projects.

48. Which model in system modeling indicate "how software will respond to external event"?
सिस्टम मॉडलिंग में कौन-सा मॉडल यह बताता है कि सॉफ्टवेयर कैसे बाहरी घटना को प्रतिमिक्रया करेगा?

(a) Context Model/कॉन्टेक्स्ट मॉडल
(b) Data Model/डाटा मॉडल
(c) Behavior Model/बिहेवियर मॉडल
(d) Object Model/ऑब्जेक्ट मॉडल

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : The Behavior Model is system modeling is used to represent how software responds to external events. It focuses on the dynamic aspects of the system, such as the flow of control, state changes and interactions triggered by events.

49. Which model of software development is best suited if requirements are frequently changing?
सॉफ्टवेयर डेवलपमेंट का कौन-सा मॉडल अनुकूल रहेगा यदि आवश्यकता लगातार बदल रही हो?

(a) Incremental Model/इंक्रीमेंटल मॉडल
(b) Spiral Model/स्पाइरल मॉडल
(c) Prototype Model/प्रोटोटाइप मॉडल
(d) RAD Model/रेड मॉडल

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : The prototype model is best suited for software development when requirements are frequently changing. This model allows for the creation of a working model early in the process, which can be refined based on user feedback as the requirements evolve.

- Incremental model suited for projects where requirements are clear but need to be delivered in stages.
- Spiral model best for large scale project with complex and evolving requirements, particularly where risk management is critical.
- RAD model (Rapid Application Development) Focuses on speed & flexibility, suitable for projects with well defined requirements and short development cycles.

50. RAD Model is not suitable when-
RAD मॉडल अनुकूल नहीं है जब-

(a) Technical risks are high/टेक्निकल रिस्क ज्यादा हो
(b) Testing is not needed/टेस्टिंग की जरुरत ना हो
(c) Technical risks are high and testing is not needed/टेक्निकल रिस्क ज्यादा हो और टेस्टिंग की जरुरत ना हो
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : The RAD model is not suitable when technical risks are high & testing is not needed.
The RAD model or Rapid Application Development, is a software development life cycle model that emphasizes speed and iterative development. It's particularly well suited for situations where requirements are relatively stable, and rapid feedback to ensure the final product closely aligns with user expectations.

51. According to Brooks, if n is the number of programmers in a project team then number of communication path are:
ब्रुक्स के अनुसार, यदि प्रोजेक्ट टीम में n प्रोग्रामर है तो कम्युनिकेशन पाथ होंगे

(a) $\frac{n(n-1)}{2}$ (b) $\log_2 n$
(c) $\frac{n(n+1)}{2}$ (d) 2^n

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : According to Brooks Law (From the Mythical Man-Month by Fred Brooks), as the number of programmers increases, the number of communication paths increases rapidly. The Formula for the number of communication paths between people (programmer) is :

$$\frac{n(n-1)}{2}$$

52. What is prototype in Prototype Model?
प्रोटोटाइप मॉडल में प्रोटोटाइप क्या है?

(a) Mini-model of existing system
मौजूदा सिस्टम का मिनी मॉडल
(b) Mini-model of proposed system
प्रस्तावित सिस्टम का मिनी मॉडल
(c) Working model of existing system
मौजूदा सिस्टम का वर्किंग मॉडल
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Sarkari Resource

Ans. (b) : In software development the prototype model is a process where a prototype (An early sample or mock version) of the software is built before the actual system is developed. This prototype is not a full system. It's a mini-model of the proposed system designed to understand user requirements better.

• **Purpose of the prototype:**
(1) Understand Requirements
(2) Get Feedback Early
(3) Improve Communication
(4) Reduce Risk

53. According to Boehm's, what are the definitions of verification and validation?
बोहें के अनुसार, प्रमाणीकरण (वैरिफिकेशन) तथा मान्यकरण (वैलिडेशन) की परिभाषाएँ क्या है?

(a) Verification: Is the software free of errors ? Validation: Is the software commercially viable ?/प्रमाणीकरणः क्या सॉफ्टवेयर त्रुटियों से मुक्त है मान्यकरणः क्या सॉफ्टवेयर व्यावसायिक रुप से अर्थक्षम है

(b) Verification: Is the software commercially viable ? Validation: Is the software free of errors ?/प्रमाणीकरणः क्या सॉफ्टवेयर व्यावसायिक रुप से अर्थक्षम है मान्यकरणः क्या सॉफ्टवेयर त्रुटियों से मुक्त है

(c) Verification: Are we building the product right? Validation: Are we building the right product?/प्रमाणीकरणः क्या मह उत्पाद का सही से निर्माण कर रहे है मान्यकरणः क्या हम सही उत्पाद का निर्माण कर रहे हैं

(d) Verification: Are we building the right product? Validation: Are we building the product right ?/प्रमाणीकरणः क्या हम सही उत्पाद का निर्माण कर रहै हैं मान्यकरणः क्या हम उत्पाद का सही से निर्माण कर रहै हैं

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : According to Barry Boehm, verification is about building the product right, ensuring it adheres to specifications and requirement while validation is about building the right product ensuring it meets user needs and the intended purpose.

54. Who proposed the spiral model?
स्पाइरल मॉडल का प्रस्ताव किसने दिया?

(a) Barry Boehm (b) Pressman
(c) Royce (d) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : Barry Boehm proposed the spiral model in 1986. It is a software development process model that represents the elements of design and prototyping in stages.

55. 4GT Model is a set of _____
4GT मॉडल _____ का एक सेट है

(a) Programs (b) CASE Tools
(c) Software tools (d) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c) 4GT Includes a wide range of software tools that enable the software engineering to specify features at a high level, from which source code is automatically generated based on these specifications.

56. Which of the following is a software development life cycle model?
निम्नलिखित में से कौन-सा सॉफ्टवेयर विकास जीवन चक्र मॉडल है?

(a) Waterfall model/झरना मॉडल
(b) Agile model/एजाइल मॉडल
(c) Spiral model/स्पाइरल मॉडल
(d) All of the above/उपर्युक्त में से सभी

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : The Waterfall Model, Spiral Model, and Agile Model are all software development life cycle models.

57. What is the primary goal of software engineering?
सॉफ्टवेयर इंजीनियरिंग का प्राथमिक लक्ष्य क्या है?

(a) To write code efficiently
कुशलतापूर्वक कोड लिखना

(b) To develop software that meets user requirements within budget and schedule constraints/ऐसे सॉफ्टवेयर का विकास करना जो बजट के भीतर उपयोगकर्ता की आवश्यकताएँ और शेड्यूल संबंधी बाधाएँ

(c) To develop software with the latest technologies
सॉफ्टवेयर विकसित करना नवीनतम तकनीकें

(d) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) : The primary goal of software engineering is to develop software that meets the user's requirements within budget and schedule constraints. The objective is to ensure that the software is developed within these limits while fulfilling the user's expectations, where as other factor such as café quality and the use of the latest technology may be helpful but are not the primary goals.

58. Which of the following activities does not belong to implementation phase of SDLC?
निम्न में से कौन-सी गतिविधि SDLC के क्रियान्वयन फेज में नहीं होती है?

(a) Program testing/प्रोग्राम परीक्षण
(b) File conversion/फाइल परिवर्तन
(c) User training/उपयोगकर्ता प्रशिक्षण
(d) New changes to the system
प्रणाली में नए बदलाव

LT. Grade C.S. 2018

Ans. (d) : The implementation phase of the Software Development Life Cycle (SDLC) involves the following activities.
1. Program testing– Verifying that the software meets the requirements and works as expected.
2. File conversion– Converting existing from the old system to the new system.
3. User training– Educating users on how to use the new system on the other hand, New changes to the system is an activity that typically belongs to the maintenance phase, which occurs after the implementation phase. The maintenance phase involves making updates, fixes and changes to the system after it has been deployed.

Sarkari Resource

59. Project risk factor is considered in
प्रोजेक्ट रिस्क फैक्टर निम्न में से किस मॉडल में विचारणीय होता है?
(a) Waterfall model/वाटरफॉल मॉडल
(b) Prototyping model/प्रोटोटाइपिंग मॉडल
(c) Spiral model/स्पाइरल मॉडल
(d) Iterative enhancement model
इटरेटिव एन्हैन्समेन्ट मॉडल

LT. Grade C.S. 2018

Ans. (c) : Spiral model works as with the combination of interative development process model and linear development model. Which has a major level of risk analysis involved in the process.

60. Which of the following represents the life-cycle of software development ?
निम्नलिखित में से कौन सा सॉफ्टवेयर विकास का जीवन-चक्र दर्शाता है?
(a) Design → Analysis → Coding → Operation and maintenance → Testing
(b) Analysis → Design → Coding → Testing → Operation and Maintenance
(c) Design → Analysis → Coding → Testing → Operation and Maintenance
(d) Analysis → Design → Coding → Operation and Maintenance → Testing

KVS PGT C.S. 2017

Ans. (b) : Software Development Life Cycle (SDLC) is a framework that defines the steps involved in the development of software at each phase, it covers the detailed plan for building, deploying and maintaining the software. Various phases of the software development life cycle are–
1. Analysis 2. Design
3. Coding 4. Testing
5. Operation and maintenance.

61. Which of the following is not an example of cloud computing services ?
निम्नलिखित में से कौन सा क्लाउड कम्प्यूटिंग सर्विसेज का एक उदाहरण नहीं है ?
(a) SaaS (b) IaaS
(c) PaaS (d) SaaP

KVS PGT 23.03.2023 (Shift-II)

Ans. (d) : The main types of cloud computing services include software as a service (SaaS), Platform as a service (PaaS), Infrastructure as a service (IaaS) Server less computing, also known as function as a service (FaaS), is also as popular method of cloud computing for businesses. Hence SaaP is not an example of cloud computing services.

62. Which software life cycle model is based on iterative.
कौन सा सॉफ़्टवेयर जीवन चक्र मॉडल पुनरावृत्त (Iterative) पर आधारित है?
(a) Waterfall model/वॉटरफॉल मॉडल
(b) Spiral model/स्पाइरल मॉडल
(c) Agile model/एजाइल मॉडल
(d) Prototype model/प्रोटोटाइप मॉडल

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : The Spiral model is a software development model that combines both iteration (repetition) and prototyping.

63. The framework that provides guidelines for achieving consistent and high-quality software is:
वह फ्रेमवर्क जो लगातार उच्च गुणत्ता वाला सॉफ्टवेयर प्राप्त करने के लिए दिशा निर्देश प्रदान करता है
(a) COCOMO (b) ISO 9000
(c) Delphi Method (d) Agile Manifesto

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b): ISO 9000 is an international standard that provides guidelines for a Quality Management System (QMS). Its purpose is to ensure that a product r service consistently maintain high quality. Therefore, ISO 9000 help in maintaining the quality and reliability of software.

64. Which software life cycle model allows for iterative development and incorporates risk analysis?
कौन-सा सॉफ्टवेयर जीवन चक्र मॉडल पुनरावृत्त विकास की अनुमति देता है और जोखिम विश्लेषण को शामिल करता है?
(a) Waterfall model/वॉटरफॉल मॉडल
(b) Prototype model/प्रोटोटाइप मॉडल
(c) Spiral model/स्पाइरल मॉडल
(d) Agile model/एजाइल मॉडल

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : In the Spiral Model, software is developed in small phase and risk analysis is carried out in each phase to minimize risks.

(ii) Software Quality, role of metrics and measurement

65. In context of dependability in software engineering, the ability of the system to deliver services when requested, is called _____and the ability of the system to deliver services as requested, is called _____.
सॉफ्टवेयर इंजीनियरिंग में डिपेंडेबिलिटि (निर्भरता) के संदर्भ में, चाहे जाने पर सेवा प्रदान करने की सिस्टम की क्षमता को _____ कहा जाता है तथा जैसी चाही गई वैसी सेवा प्रदान करने की सिस्टम की क्षमता को _____ कहा जाता है।
(a) Availability, Reliability/उपलब्धता, विश्वसनीयता
(b) Reliability, Availability/विश्वसनीयता, उपलब्धता
(c) Security, Reliability/सुरक्षा, विश्वसनीयता
(d) Availability, Security/उपलब्धता, सुरक्षा

HTET PGT C.S. 2018

Ans. (a) : Availability– It refers to the system's ability to deliver services when requested. For example- if a user wants to access an online banking system, availability ensure that the system is up and running at that moment.
Reliability– It means the system's ability to deliver services as requested i.e. correctly and consistently without errors. For example, when you transfer money online, reliability ensure that the transaction is completed accurately without corruption.

Sarkari Resource

66. Size and complexity are a part of

(a) People Metrics (b) Project Metrics
(c) Process Metrics (d) Product Metrics

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Ans. (d) : Software metrics can be classified into three categories.
Product metrics–Describes the characteristics of product such as **size complexity**, design features, performance and quality level.
Process metrics–These characterstics can be used to improve the development and maintenance activities of the software.
Project metrics–This metrics describe the project characteristics and execution.
So correct answer is product metrics.

67. Which of the following statement(s) is/are correct regarding software metrics in software engineering?
सॉफ्टवेयर इंजीनियरिंग में सॉफ्टवेयर मेट्रिक्स के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/है?

I. Direct measures of the software product include functionality quality, efficiency etc. of the software./सॉफ्टवेयर उत्पाद के प्रत्यक्ष उपायों में सॉफ्टवेयर की कार्यक्षमता गुणवत्ता, दक्षता आदि शामिल हैं।

II. Indirect measures of the software product includes lines of code produces, execution speed, memory size etc, of the software product./सॉफ्टवेयर उत्पाद के अप्रत्यक्ष उपायों में सॉफ्टवेयर उत्पाद की उत्पादित कोड की लाइने, निष्पादन गति, मेमोरी साइज अदि शमिल हैं।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) : Statement-I
• **Incorrect :** Direct measures are quantitative and easily countable attributes of a product, such as:
→ Lines of code (LOC)
→ Execution speed
→ Memory usage
→ Number of errors
Statement-II
• **Incorrect :** These are actually direct measures because they can be measured precisely and directly.
So, both statements are incorrect.

68. What is the full form of SQA in software engineering?
सॉफ्टवेयर इंजीनियरिंग में एस. क्यू. ए. का असंक्षिप्ति रूप क्या है?

(a) Software Quality Activity
सॉफ्टवेयर क्वालिटी एक्टिविटी
(b) System Quality Assurance
सिस्टम क्वालिटी अशुरेंस
(c) Service Quality Activity
सर्विस क्वालिटी एक्टिविटी
(d) Software Quality Assurance
सॉफ्टवेयर क्वालिटी अशुरेंस

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : SQA stands for software Quality Assurance in software engineering. It refers to the process of ensuring that software products meet defined quality standards and requirements. SQA involves monitoring the software engineering process and methods used to ensure quality throughout the software development lifecycle.

69. Due to which property a computer can work for hours without creating any error and without grumbling ?
किस गुण के कारण कंप्यूटर बिना कोई त्रुटि और ग्रम्बलिंग को घंटों तक काम कर सकता है?

(a) No I.Q/नो आई. क्यु
(b) Versatility/बरसेटेलिटी
(c) Diligence/डिलिजेन्स
(d) Speed/स्पीड

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (c) : Diligence is the capacity of the computer of performing repetitive tasks without getting tired and computer is free from tiredness, look of concentration, fatique, etc. Therefore it can work for hours without creating any errors.
Characteristics of Computer System–
1. Speed 2. Versatility 3. Automation
4. Accuracy 5. Reliability 6. Memory
7. Diligence

70. Match the following :

List - I		List - II	
(A)	**Size-oriented metrics**	**(i)**	**Uses number of external interfaces as one of the measurement parameter**
(B)	**Function-oriented metrics**	**(ii)**	**Originally designed to be applied to business information system**
(C)	**Extended function point metrics**	**(iii)**	**Derived by normalizing quality and/or productivity measures by considering the size of the software**
(D)	**Function point**	**(iv)**	**uses algorithm characteristics as one of the measurement parameter**

Codes :

	A	**B**	**C**	**D**
(a)	(iii)	(iv)	(i)	(ii)
(b)	(ii)	(i)	(iv)	(iii)
(c)	(iv)	(ii)	(iii)	(i)
(d)	(iii)	(i)	(iv)	(ii)

UGC NET C.S. June-2015 (Paper-II)

Sarkari Resource

Ans.(d) :
(a) Size oriented metrics is directly implied with (iii)
(b) Function oriented metrics is number of user inputs, number of user outputs, number of files, number of external interfaces. So it satisfies (i)
(c) Extended function point metrics is a superset of the function point measure that can be applied to system. It uses algorithm characteristics as one of the measurement parameter.
(d) Function point is a "unit of measurement" to express the amount of business functionality an information provides to a user.

71. Which one of the following set of attributes should not be encompassed by effective software metrics?
(a) Simple and computable
(b) Consistent and objective
(c) Consistent in the use of units and dimensions
(d) Programming language dependent

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Because, effective software metrics should be simple, computable, consistent, objective and consistent in the use of units and dimensions. But it should not be programming language dependent as them it will not be consistent in measuring the software.

72. Which one of the following is used to compute cyclomatic complexity?
(a) The number of regions–1
(b) E – N + 1 , where E is the number of flow graph edges and N is the number of flow graph nodes
(c) P – 1, where P is the number of predicate nodes in the flow graph G.
(d) P + 1, where P is the number of predicate nodes in the flow graph G.

UGC NET C.S. December-2014 (Paper-II)

Ans : (d) Because, cyclomatic complexity is a metric to measure logical complexity of a program.
- Cyclomatic complexity defines the number of independent paths in a program
- It gives an upper bound for the number of tests that must be conducted to ensure that all statements have been executed at least once.

⇒ Cyclomatic complexity is computed in one of three ways.
(1) Number of regions of the flow graph
(2) Cyclometic complexity, V(G), for a flow graph G, is defined as V(G) = E – N + 2 K where E is the number of flow graph edges, N. is the number of flow graph and K is the number of connected components
(3) V(G) = P + 1 where P is the number of predicate nodes contain in the flow graph G.

73. A software metric that is a measure of the average length of words and sentences in documents is caused-
एक सॉफ्टवेयर मेट्रिक जो डॉक्यूमेंट में शब्दों तथा वाक्यों की औसत लम्बाई का मापक है, कहलाता है-
(a) Fan-in/फैन-इन
(b) Fan-out/फैन-आउट
(c) Cyclomatic complexity
सायक्लोमेट्रिक कॉम्प्लेक्सिटी
(d) Fog Index/फोग इंडेक्स

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : The Fog Index is a readability metric that estimates the years of formal education needed to understand a text on the first reading. It is based on the average sentence length and the number of complex words (words with three or more syllables).

74. Which design metric is used to measure the compactness of the program in terms of lines of codes ?
(a) Consistency (b) Conciseness
(c) Efficiency (d) Accuracy

UGC NET C.S. June-2015 (Paper-III)

Ans : (a) **Consistency:-** The use of uniform design and documentation techniques through the software development project.

75. Which of the following are external qualities of a software product?
(a) Maintainability, reusability, portability, efficiency, correctness.
(b) Correctness, reliability, robustness, efficiency, usability.
(c) Portability, interoperability, maintainability, reusability.
(d) Robustness, efficiency, reliability, maintainability, reusability.

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) External qualities of a software product:
- Correctness
- Reliability
- Robustness
- Efficiency
- Usability

Internal qualities of a software product
- Maintainability
- Reusability
- Portability
- Interoperability

76. Which of the following is not one of three software product aspects addressed by Mccall's software quality factors?
(a) Ability to undergo change
(b) Adaptability to new environments
(c) Operational characteristics
(d) Production costs and scheduling

UGC NET C.S. June-2017 (Paper-II)

Ans : (d) Mccalls's software quality factors.
(1) ability to undergo change
(2) Adaptability to new environments
(3) Operational characteristics
but production costs & scheduling is not the aspect of quality factor.

77. Which of the following is an indirect measure of product?
(a) Quality (b) Complexity
(c) Reliability (d) All of these

UGC NET C.S. December 2021

Ans. (d) : Software Measurement
Direct measures of software engineering process include cost and effort.

Sarkari Resource

- Direct measures of the product include lines of code (LOC), execution speed, memory size, defects per reporting time period.
- Indirect measures, examine the quality of the software product itself, e.g. functionality, complexity, efficiency reliability maintainability.

78. Coupling is a measure of the strength of the interconnections between software modules. Which of the following are correct statements with respect to module coupling ?

P : Common coupling occurs when one module controls the flow of another module by passing it information on what to do.

Q : In data coupling, the complete data structure is passed from one module to another through parameters.

R : Stamp coupling occurs when modules share a composite data structure and use only parts of it.

Code :

(a) P and Q only (b) P and R only
(c) Q and R only (d) All of P, Q and R

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : According to Myer's classification, which defines seven levels of coupling, from tightest to loosest.

1. **Content Coupling–** Modules rely on each other's internal data or internal organization.
2. **Common Coupling–** Modules shares the same global data.
3. **External Coupling–** Modules share an externally imposed data format, communication protocol or device interface.
4. **Control Coupling–** One module controls the flow of another, such as by passing it a flag or other information.
5. **Stamp Coupling–**Modules share a composite · data structure but use different parts of it.
6. **Data Coupling–** Modules share data through parameters such as in a subroutine cell.
7. **Message Coupling–** Modules communicate by passing messages.

79. Which one is not a part of software quality attribute FURPS developed by Hewlett-Packard?
निम्न में से कौन सा एक सॉफ्टवेयर क्वालिटी एट्रिब्यूट FURPS का भाग नहीं है जो की हेवलेट-पेकार्ड द्वारा बनाया गया है?

(a) Functionality/फंक्शनलिटी
(b) Portability/पोर्टेब्लिटी
(c) Reliability/रिलायब्लिटी
(d) Supportability/सपोर्टेब्लिटी

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : FURPS is a model developed by 'Hewlett-Packard' to classify software quality attributes. It stands for:

- **F-Functionality–** Features, capabilities, and security.
- **U-Usability–** User interface, ease of use documentation.
- **R-Reliability–** Fault tolerance, recoverability and availability.
- **P-Performance–** Speed, efficiency, resource usage.
- **S-Supportability–**Maintainability, testability adaptability.

Portability is not part of the 'FURPS' model. It refers to the ease with which software can be transferred from one environment to another.

80. Which is not a size metric?
निम्न में से कौन-सा साइज मेट्रिक नहीं है?

(a) Program Length/प्रोग्राम लेंग्थ
(b) Cyclomatic Complexity
सायक्लोमेट्रिक कोम्प्लेक्सिटी
(c) LOC (Line of Code)/LOC (लाइन ऑफ कोड)
(d) FP (Function Point)/FP (फंक्शन पॉइंट)

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : Cyclomatic Complexity is a complexity metric, not a size metric. It measures the number of linearly independent paths through a program's source code, indicating its control flow complexity.

Size metrics (used to measure the size of software):

(1) Program Length– size metric (number of tokens or statements)

(2) LOC (Line of Code)– size metric

(3) FP (Function Point)– size metric (based on functionality).

81. Which one of the following is not a software process quality?
निम्नलिखित में से कौन-सा एक सॉफ्टवेयर प्रक्रिया गुणवत्ता नहीं है?

(a) Visibility (b) Timeliness
(c) Portability (d) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans.(c) Portability is a quality attribute of a software product, not of the software process. Portability means that the software can run on different hardware platforms or software environments.

82. What is the main goal of software quality assurance?
सॉफ्टवेयर क्वालिटी एश्योरेंस का मुख्य उद्देश्य क्या है?

(a) To eliminate all defects from the software
सॉफ्टवेयर से सभी दोषों को समाप्त करना
(b) To ensure the software is defect - free before release
रिलीज़ से पहले सॉफ्टवेयर को दोषमुक्त बनाना
(c) To establish and enforce standards and processes to improve software quality
सॉफ्टवेयर गुणवत्ता में सुधार के लिए मानकों और प्रक्रियाओं को स्थापित करना और लागू करना
(d) To test software under different conditions
विभिन्न परिस्थितियों में सॉफ्टवेयर का परीक्षण करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The purpose of Software Quality Assurance (SQA) is to improve software quality by implementing processes and standards, not just to find or eliminate defects.

Sarkari Resource

83. Which metric is used for estimating the size of a software project?
किसी सॉफ़्टवेयर प्रोजेक्ट के आकार का अनुमान लगाने के लिए कौन-सा मीट्रिक उपयोग किया जाता है?
(a) Cyclomatic Complexity
साइक्लोमैटिक कॉम्प्लेक्सिटी
(b) Function Points (FP)/फंक्शन पॉइंट्स (FP)
(c) Defect Density/डिफेक्ट डेंसिटी
(d) Software Reliability Index
सॉफ़्टवेयर रिलायबिलिटी इंडेक्स

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : Function Points (FP) is a metric used to estimate the size of a software project. It is used to determine how much work will be required in a project and is based on the user's requirements.

(iii) Requirements Analysis and Specification

84. Which is the step in waterfall model that involves a meeting with the customer to understand the requirements ?
वाटरफॉल मॉडल में कौन सा एक चरण जिसमें आवश्यकताओं को समझने के लिए ग्राहक के साथ एक मीटिंग शामिल होती है?
(a) Requirement Gathering/रिक्वायरमेंट गैदरिंग
(b) SRS/एस आर एस
(c) Implementation/इम्प्लीमेंटेशन
(d) Customer review/कस्टमर रिव्यू

HTET PGT C.S. 2021

Ans. (a) : Requirement Gathering– This step directly involves meeting with the customer to collect and analyze requirements.
SRS– The software Requirement Specification is a document prepared after Requirement Gathering, not meeting itself.
Implementation– This is the coding step, not direct requirement discussion with the customer.
Customer review– Customer reviews usually happen after delivering a version are prototype not in the requirement collection stage.

85. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Validity checks real need of system users
Reason R : Completeness checks system defined requirements.
In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

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Ans. (b) : Validity checks are essential to ensure that the system meets the real needs and requirements of the system users. Completeness checks are indeed about verifying that the system user-defined requirements are fully captured, but this doesn't directly explain the concept of validity checks. So, Assertion A and reason R both are correct but Reason R is not the correct explanation of Assertion A.

86. Software Requirement Specifications Document (SRSD) does not include which of the following?
निम्नलिखित में से कौन-सा एक अच्छा सॉफ्टवेयर आवश्यकता विनिर्देश दस्तावेज (SRSD) में वांछित नहीं है?
(a) Algorithms for Software Implementation
सॉफ्टवेयर कार्यान्वयन के लिए एल्गोरिथ्म
(b) Non-Functional Requirements
गैर-कार्यात्मक आवश्यकताएँ
(c) Functional Requirements/कार्यकारी आवश्यकताएँ
(d) Goals of Implementation/कार्यान्वयन के लक्ष्य

HTET PGT C.S. 2020

Ans. (a) : A Software Requirement Specifications Document (SRSD) is a detailed description of what a software system should do. It defines the functional requirement (features and functions the software must provide) and non functional requirement (performance security, usability etc) along with the goals of implementation such as reliability, maintainability and portability.

87. Which of the following is not a phase or activity of requirement engineering in context of software engineering ?
सॉफ्टवेयर इंजीनियरिंग के संदर्भ में, निम्न में से कौन-सा रिक्वायरमेंट इंजीनियरिंग की एक अवस्था या गतिविधि नहीं है?
(a) Feasibility study/व्यवहार्यता विश्लेषण
(b) Requirements elicitation and analysis
रिक्वायरमेंट्स निकलना तथा विश्लेषण
(c) Requirement validation
रिक्वायरमेंट्स अभिपोषण
(d) Component specification
अवयव विनिर्देशन

HTET PGT C.S. 2019

Ans. (d) : Requirement Engineering is the process of identifying analyzing documenting and validating what a software system should do.
Component specification– Component specification is not part of requirement engineering. It is part of software design phase.
Requirement Engineering only focuses on what the system should do, not how it will be done.

88. In context of requirement analysis in software engineering, which of the following is not a type of 'nonfunctional requirements' ?
सॉफ्टवेयर इंजीनियरिंग में रिक्वायरमेंट एनालिसिस के संदर्भ में, निम्न में से कौन-सा 'नॉन फैन्कशॅनल रिक्वायरमेंट्स' का प्रकार नहीं है?
(a) Product Requirements
प्रोडक्ट रिक्वायरमेंट्स
(b) Organizational Requirements
ऑर्गेनाइजेशनल रिक्वायरमेंट्स
(c) External Requirements
एक्सटर्नल रिक्वायरमेंट्स

Sarkari Resource

(d) Umbrella Requirements
अॅमब्रेला रिक्वायरमेंट्स

HTET PGT C.S. 2018

Ans. (d) : In requirement analysis, software requirements are mainly divided into functional and non-functional requirements.
Umbrella Requirements– Umbrella Requirements are not a formal category of software requirements but are often used to describe general activities or supporting requirements that apply across all phases of the software development life cycle. They are called "Umbrella" because they cover and influence the entire project rather than a single phase.
Example–
1. Documentation
2. Configuration management
3. Quality assurance
4. Project management
5. Standards compliance
6. Risk management.

89. Which of the following is used to determine the specificity of requirements?

(a) $\frac{n_1}{n_2}$ (b) $\frac{n_2}{n_1}$
(c) $n_1 + n_2$ (d) $n_1 - n_2$

Where n_1 is the number of requirements of which all reviewers have identical interpretations, n_2 is number of requirements in a specification.

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (a) Choice 1 & 2 must be different assuming choice 2 as ($n_1 | n_2$). So the answer is option (a)

90. Project size is related with:
प्रोजेक्ट का आकार संबंधित है–
(a) Technical feasibility/टेक्निकल फिजिबिलिटी
(b) Economical feasibility/इकोनॉमिकल फिजिबिलिटी
(c) Organizational feasibility
ऑर्गेनाइजेशनल फिजिबिलिटी
(d) Testing feasibility/टेस्टिंग फिजिबिलिटी

HTET PGT C.S. 2023

Ans. (a) : The size of project (small, medium or large) is directly related to technical feasibility. Technical feasibility checks whether the current technology, tools, hardware, software and expertise are sufficient to handle the project.

91. Which phase answers the questions of who will use the system, what the system will do and where and when it will be used?
कौन-सा फेज इस प्रश्न का उत्तर देता है कि सिस्टम का उपयोग कौन करेगा, सिस्टम क्या करेगा और इसका उपयोग कहाँ और कब किया जाएगा?
(a) Analysis/एनालिसिस
(b) Design/डिज़ाइन
(c) Implementation/इम्प्लीमेंटेशन
(d) Planning/प्लानिंग

HTET PGT C.S. 2023

Ans. (a) : The Analysis phase of the Software Development Life Cycle (SDLC) is the stage where the main goal is to understand the problem clearly and define the requirements of the system. In this phase, questions like who will use the system, what the system will do, and where and when it will be used are answered.
It focuses on gathering and analyzing user needs so that the developers can design the right solution.

92. A group technique that may be used during requirement elicitation to understand the requirements is
रिक्वायरमेंट्स को समझने हेतु रिक्वायरमेंट इलिसिटेशन में प्रयोग की जा सकने वाली समूह विधि है
(a) System prototyping/सिस्टम प्रोटोटाइपिंग
(b) Software reengineering/सॉफ्टवेयर रीइंजीनियरिंग
(c) Brain storming/ब्रेन स्टार्मिंग
(d) System designing/सिस्टम डिजाइनिंग

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) : Requirement elicitation is the process of gathering requirements from stakeholders (customer, users, managers etc.) to understand what the system should do.
Brain storming– Brain storming is a group technique where participants come together and share ideas freely. The aim is to generate as many ideas as possible without criticism in the beginning, which later helps in refining and finalizing requirements.

93. SRS stands for
SRS का तात्पर्य है
(a) Software Requirements Specification
सॉफ्टवेयर रिक्वायरमेंट्स स्पेसिफिकेशन
(b) System Requirements Specification
सिस्टम रिक्वायरमेंट्स स्पेसिफिकेशन
(c) Systematic Requirements Specification
सिस्टमेटिक रिक्वायरमेंट्स स्पेसिफिकेशन
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asist. Prof. 2017

Ans (b) : SRS stands for System Requirements Specification. It is a well structured document that describe the complete behavior of a system.
Purpose of SRS– The main goal of SRS is to clearly capture functional and non-functional requirements of the software so that both developers and customers have a mutual understanding of what is to be built.
Content of SRS–
- Functional Requirements
- Non-Functional Requirements
- Constraints and Assumptions
- External Interfaces.

94. Match the software maintenance activities in List-I to its meaning in List-II

List-I		List-II	
i.	**Corrective**	**A.**	**Concerned with performing activities to reduce the software complexity thereby improving program understandability and increasing software maintainability**
ii.	**Adaptive**	**B.**	**Concerned with fixing errors that are observed when the software is in use.**

iii.	**Perfective**	**C.**	**Concerned with the change in the software that takes place to make the software adaptable to new environment (both hardware and software)**
iv.	**Preventive**	**D.**	**Concerned with the change in the software that takes place to make the software adaptable to changing user requirements.**

Code :

	(i)	(ii)	(iii)	(iv)
(a)	(B)	(D)	(C)	(A)
(b)	(B)	(C)	(D)	(A)
(c)	(C)	(B)	(D)	(A)
(d)	(A)	(D)	(B)	(C)

UGC NET C.S. June-2016 (Paper-III)

Ans : (b)

(I) Corrective maintenance is concerned with fixing errors that are observed when the software is in use (b)

(II) Adaptive maintains is concerned with the change in the S/w that take place to the S/w adaptable to new environment (c)

(III) Perfective maintenance is concerned with implementing new or changed user requirements (c)

(IV) Preventive maintenance involves performing activities to prevent the occurrence of error (a)

95. What is the full form of MTTF in software reliability?
सॉफ्टवेयर रिलाईबिलिटी में MTTF का असंक्षिप्त रूप क्या है?

(a) Minimum time to failure
मिनियम टाईम टू फेलियर

(b) Maximum time to failure
मैक्सिमम टाईम टू फेलियर

(c) Mean time to failure/मीन टाईम टू लियर

(d) Median time to failure/मीडियन टाईम टू फेलियर

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : MTTF = Mean Time To Failure
In software reliability engineering. MTTF is the average time a non-repairable system operates before it fails. It is calculated by summing up the total operational time of the system and dividing it by the number of observed failure.

$$\text{MTTF} = \frac{\text{Total operational time}}{\text{Number of failures}}$$

A higher MTTF indicates greater reliability for software it helps estimate how long the system can run without crashing or producing incorrect results.
MTTF is mainly used for predicting reliability, planning, maintenance and comparing system robustness.

96. Which one of the following is not the part of requirement management?

(a) Features Traceability Table
(b) Dependency Traceability Table
(c) Interface Traceability Table
(d) Page Traceability Table

UGC NET C.S. December-2024

Ans. (d) : Requirement management involves tracking and managing requirements throughout the projects lifecycle. The commonly used traceability table includes.

- Features Traceability Table-Links requirements to system features.
- Dependency Traceability Table – Tracks dependences between requirements.
- Interface Traceability Table-Ensure correct interaction between system components.

However, page Traceability Table is not a standard part of requirement management, as it does not directly contribute to tracking or managing requirements in software engineering.

97. In Software Engineering Jackson's Principle bases on
A. Designation
B. Definitions
C. Refutable Assertions
D. Formal Review
E. Requirement Elicitation
Choose the correct answer from the options given below :

(a) A, B & C only (b) B, C & D only
(c) C, D & E only (d) B, D & C only

UGC NET C.S. December-2024

Ans. (a) : The software Engineering Jackson's Principle rule is based on Designation, definition, refutable assertion.
Designation:- The designation refers to a specific job title within a company that indicates an engineer's level of experience and responsibility.
Definition:- The process of designing, developing, testing, and maintaining software applications.
Refutable Assertion:- The assertion Testing is used to verify a logic with Boolean expressions. It gives a true value if the logic being verified is correct, else it return false. In case true is obtained, in Indicates are error to code vice-versa.

98. Collaborative document inspection technique used for?

(a) Requirement Validation Process
(b) Requirement Verification Process
(c) Integration Testing
(d) Black box Testing

UGC NET C.S. December-2024

Ans. (a) : The requirement validation process collaborative document inspection, often referred to as a formal review or inspection, is a key technique used in the requirement validation process.

- **Requirement Verification**– requirement verification is about checking that the software being built meets the specified requirement. It's a development time activity inspections primarily focus on the requirements document itself not the software.
- **Integration Testing & Black box Testing:-** There are testing techniques performed on the developed software. Inspections happen much earlier, during the requirements phase.

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99. The SRS document produced at the end of requirement analysis and specification is known as _____ .
आवश्यकता विश्लेषण और स्पेसिफिकेशन के अंत में तैयार किए गए SRS दस्तावेज़ को _____ के रूप में जाना जाता है।

(a) White-box specification
व्हाईट-बॉक्स स्पेसिफिकेशन
(b) Black-box specification
ब्लैक-बॉक्स स्पेसिफिकेशन
(c) Design plan/डिज़ाइन प्लान
(d) Architectural design/आर्केटेक्चरल डिज़ाइन

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (b) : The Software Requirements Specification (SRS) document describes what the software system should do, not how it will do it.
→ It focuses on the external behavior of the system-inputs, outputs, functions, performance and constraints without describing the internal working or code.
→ In software engineering a black-box means you know the inputs and expected outputs but not the internal implementation.
Therefore, SRS is considered a black-box specification because it specifics requirements from the user's point of view, without delving into system internals.

100. Process of recovering the design and the requirement specification of a product an analysis of its code is called as _____ .
किसी उत्पाद के डिजाइन और आवश्यकता स्पेसिफिकेशन को उसके कोड के विश्लेषण से पुनर्प्राप्त करने की प्रक्रिया को _____ कहा जाता है।

(a) Software analysis/सॉफ्टवेयर ऐनालिसिस
(b) Software engineering/सॉफ्टवेयर इंजीनियरिंग
(c) Software reverse engineering
सॉफ्टवेयर रिवर्स इंजीनियरिंग
(d) Software return engineering
सॉफ्टवेयर रिटर्न इंजीनियरिंग

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : Software reverse engineering is the process of analyzing software code to understand its design and requirement specifications, especially when this information is not available or lost. This is commonly used in cases like:
- Legacy software where documentation is missing
- Analyzing malware or competitors products
- Updating or improving old systems.

In this process, the software is deconstructed to:
- Understand its components and how they work.
- Extract design elements, architecture, and requirements.

101. What is the other name of Data Flow Diagram (DFD)?
डेटा फ्लो डाइग्राम का दूसरा नाम क्या है?

(a) Flow chart /फ्लो चार्ट
(b) Pictorial graph /पिकटोरियल ग्राफ
(c) Bubble chart /बबल चार्ट
(d) Di graph /डाइ ग्राफ

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : A Data Flow Diagram (DFD) is also called a Bubble chart because in the diagram the process are drown as circles (bubbles) and arrows show how data moves between them.
- Input
- Process
- Output

The process look like bubbles, people also call it a bubble chart.

102. Match the symbol used in DFD with tehri function.
डी.एफ.डी. में प्रयुक्त प्रतीक को उनके कार्य के साथ सुमेलित कीजिए।

1	▭	i	Output
2	—	ii	Process
3	⌐	iii	Data store
4	◯	iv	External entity

(a) 1 – iv, 2–iii, 3–i, 4–ii
(b) 1 – ii, 2–iii, 3–i, 4–ii
(c) 1 – iv, 2–i, 3–iii, 4–ii
(d) 1 – i, 2–iii, 3–iv, 4–ii

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : Data Flow Diagram (DFD) is just a simple picture showing how information moves in a system where it starts what changes are made are mode to it, and where it ends up.
Symbols–
- Circle/Rounded box → Process (work done on data)
- Arrow → Data Flow (Movement of data)
- Open ended rectangle → Data store
- Square → External Entity.

103. What is another name of level 0-DFD?
लेवल0-डी.एफ.डी. का दूसरा नाम क्या है?

(a) E-R Diagram/ E-R डाइग्राम
(b) Context diagram/कंटैक्स्ट डाइग्राम
(c) Flow diagram/फ्लो डाइग्राम
(d) Structure chart/स्ट्रक्चर चार्ट

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : A Level 0-DFD, also called a Context Diagram, is the simplest form of a data flow diagram. It shows the whole system as one single process (often drown as a big circle or rounded rectangle) and illustrates how this system communicates with external entities like users, other system.
It focuses only on–
- Input (Data coming into the system)
- Output (Data leaving the system)
- External entities (Sources and destination of Data).

104. Which of the following statement(s) is/are correct regarding code walk-through in software engineering?
सॉफ्टवेयर इंजीनियरिंग में कोड वॉक-थ्रू के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. It is an informal technique for analysis of the code./यह कोड के विश्लेषण के लिए एक अनौपचारिक तकनीक है।

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II. It is undertaken just before the coding of the module./यह मॉड्यूल की कोडिंग से ठीक पहले किया जाता है

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ना ही I न ही II

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : A code walkthrough is an informal technique for analyzing code where developers (and sometimes testers or peers) go through the source code together to find errors, suggest improvements and ensure coding standards are followed.
It's informal because it doesn't require strict documentation or formal procedures like inspections.
The main goal error detection and knowledge sharing rather than proving code correctness.

105. Requirement Development, Organizational process focus, organizational training, risk management and integrated supplier management are process areas required to achieve maturity level

(a) Performed (b) Managed
(c) Defined (d) Optimized

UGC NET C.S. December-2014 (Paper-II)

Ans : (c) Because, maturity level 3 – defined
ATM : Acquisition technical management
AVAL : Acquisition validation
IPM : Integrated project management
OPF : organizational process focus
OT : organizational training
RSKM : Risk management

106. In function point analysis, the number of complexity adjustment factors is

(a) 10 (b) 12
(c) 14 (d) 20

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) There are 14 relative complexity adjustment factor are
(1) Requirement for reliable backup and recovery
(2) Require for data communication
(3) Extent of distributing processing
(4) Performance requirement
(5) Expected operational environment
(6) Extent of online data entries
(7) Extent of multi-operation online data input.
(8) Extent of online updating of master files.
(9) Extent of complex inputs, outputs.
(10) Extent of complex data processing.
(11) Extent of currently developed code can be designed for reuse.
(12) Extent of conversion and installation in the design.
(13) Extent of multiple installation in an organization.
(14) Extent of change and focus on ease of use.

107. Which one of the following statements, related to the requirements phase in software Engineering, is incorrect?

(a) "Requirement validation" is one of the activities in the requirements phase
(b) "Prototyping" is one of the methods for requirement analysis
(c) "Modeling-oriented approach" is one of the methods for specifying the functional specifications
(d) "Function points" is one of the most commonly used size metric for requirements

UGC NET C.S. December-2015 (Paper-III)

Ans : (c) Requirement validation– This is a crucial activity in the requirements phases where the gathered requirement are checked for completeness, consistency, and correctness against the project objectives.
Prototyping– Prototyping is a technique used to create a working model of the system to get feedback from users and refine the requirements.
Modeling-oriented approach– This is a method for specifying functional specifications by using formal models. These models help in understanding the system behavior and relationships between different components.
Function points– While function points are used to measure the size of a software system based on its functionality. They are not as prevalent as other metrics due to their limitations in capturing the complexity of the system.

108. _____ is intended to show that a system both conforms to its specifications and meets the expectation of the system customer.

(a) Software specification
(b) Software design
(c) Software evaluation
(d) Software validation

UGC NET C.S. June-2023

Ans. (d) : Software validation is the process of ensuring that a software system meets the requirement and specifications of its intended users or stakeholders. It involves evaluating the software at the end of the development process to determine whether it satisfies the specified requirements and functions correctly in its intended environment. validation ensures that software fulfills its intended purpose and delivers value to the users.

109. How do you create agile processes to manage unpredictability?

A. Requirements gathering must be conducted very carefully.
B. Risk analysis must be conducted before planning takes place
C. Software increments must be delivered in short time periods.
D. Software processes must not adapt to changes incrementally

Choose the correct answer from the options given below :

(a) A only (b) B only
(c) C only (d) D only

UGC NET C.S. December-2022

Ans. (c) : Manage unpredictability
- Process adaptability (to rapidly changing project and technical conditions.)
- Agile software process must adapt incrementally.
- To accomplish incremental adaptation, an agile team requires customer feedback.

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- An effective catalyst for customer feedback is an operational prototype or a portion of an operational system.
- Software increments (executable prototypes or portions of an operational system) must be delivered in shoot time periods so that adaptation keeps pace with change (unpredictability).

110. Which of the following is not a Non functional requirement?
(a) Portability. (b) Security.
(c) Scalability (d) User interaction

UGC NET C.S. December-2022

Ans. (d) : Non function requirement or NFRs are a set of specification that describe the system's operation capabilities and constraints and attempt to improve its functionality. These are basically the requirements that outline how well it will operate including thing like speed, security, reliability, data integrity, scalability etc. But user interaction is not NFRs.

111. A good software requirement specification does NOT have the characteristic
(a) Completeness (b) Consistency
(c) Clarity (d) Reliability

UGC NET C.S. December 2021

Ans. (d) : Characteristics of software requirement specification (SRS)
Quality characteristics of a good software requirement specification (SRS) document include.
1. Complete 2. Consistent
3. Unambiguous 4. Traceable
5. Verifiable 6. Modifiable
7. Prioritized 8. Testable
9. High-level and low level 10. Relevant
11. Human readable
12. Aligned with business goals
13. Agile methodologies
So reliability is not characteristics of good (SRS).

112. If every requirement can be checked by a cost - effective process, then SRS is called
(a) Verifiable (b) Tracable
(c) Modifiable (d) Complete

UGC NET C.S. December 2021

Ans. (a) : Quality characteristics of a good software requirement specification (SRS) document include.
1. Completeness 2. Clarity 3. Correctness
4. Consistency 5. Verifiability 6. Ranking
7. Modifiability 8. Tracebility 9. Feasibility
So if every requirement can be checked by a cost effective process, then SRS is called verifiable.

113. The process to gather the software requirements from client, analyze and document is known as–
(a) Software Engineering Process
(b) User Engineering Process
(c) Requirement Elicitation Process
(d) Requirement Engineering Process

UGC NET C.S. December 2021

Ans. (d) : The Requirement Engineering process to gather the software requirements from client, analyze and document them is known as requirement engineering.
Requirement Engineering process it is a four steps process.
- Feasibility.
- Requirement gathering.
- Software requirement specification.
- Software requirement validation.

114. The Software Requirement Specification (SRS) is said to be _____ if and only if no subset of individual requirements described in it conflict with each other.
(a) Verifiable (b) Correct
(c) Consistent (d) Unambiguous

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : The Software Requirement Specification (SRS) is said to be Consistent if and only if no subset of individual requirements described in it conflict with each other.

115. In software engineering, what kind of notation do formal methods predominantly use?
(a) Textual (b) Diagrammatic
(c) Mathematical (d) Computer code

UGC NET C.S. December 2020

Ans. (c) : Mathematical
Formal method:- System design techniques that use rigorously specified mathematical model to build software, hardware system.

116. If every requirement stated in the Software Requirement Specification (SRS) has only one interpretation, then SRS is said to be
(a) Correct (b) Consistent
(c) Unambiguous (d) Verifiable

UGC NET C.S. December 2020

Ans. (c) : An SRS is said to be unambiguous if all the requirements stated have only 1 interpretation.

117. Which of the following is a functional requirement?
निम्न में से कौन सी फंक्शनल आवश्यकता है?
(a) Maintainability/मेन्टेनेबिलीटी
(b) Portability/पोर्टेबिलिटी
(c) Robustness/रोबस्टनेस
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : Maintainability, Portability and Robustness are all non-functional requirements.
- Functional requirements describe what the system should do-specific behaviors or functions (e.g. "The system shall allow users to log in with a username and password")

118. Which of these is not an element of requirement model?
निम्न में से कौन सा आवश्यकता मॉडल का तत्त्व नही है?
(a) Behavior element/बिहेवियर तत्त्व
(b) Usage based element/यूसेज आधारित तत्त्व
(c) Flow oriented element/फ्लो ओरिएंटेड तत्त्व
(d) All of these/इनमें से सभी

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : A requirement model in software engineering is used to represent and analyze what the system should do. It includes different elements to capture different perspectives of the system requirements. The commonly known elements are–

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(1) Behavior Elements– These describe how the system behaves in response to certain inputs (e.g. state diagrams sequence diagrams).
(2) Flow-Oriented elements– These focus on how data moves through the system (e.g., data flow diagrams, control flow)
(3) Usage-based elements– This term is not a standard element in the requirement model. It may relate to use cases or user scenarios, but it's not formally categorized as a core element in requirement modeling.

119. The important concept of W^5HH principle is:
W^5HH नियम का महत्वपूर्ण सिद्धांत कौन सा है:
(a) Project Objective/प्रोजेक्ट ऑब्जेक्टिव
(b) Responsibilities/रेस्पोंसिब्लिटीज
(c) Schedule/शेड्यूल
(d) All of these/इनमें से सभी

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : The W^5HH principle, proposed by 'Barry Boehm', is a project managements approach that addresses key project questions.
- Why is the system being developed? (Project objective)
- What will be done? (Project Scope)
- When it be done? (Schedule)
- Who is responsible? (Responsibilities)
- Where are they organizationally located?
- How will the job be done?
- How much of each resource is needed?

So, project objective, responsibilities, and schedule are all important concepts in the W^5HH principle.

120. A data store in DFD represents?
DFD में डाटा स्टोर दर्शाता है?
(a) Sequential file/सिक्वेंशियल फाइल
(b) Disk store/डिस्क स्टोर
(c) Repository of data/रिपोजिटॉरी ऑफ डाटा
(d) Random access memory/रैंडम एक्सेस मेमोरी

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : In a Data Flow Diagram (DFD), a data store represents a repository of data that can be used by one or more processes within the system. It is a place where data is held for later use. It doesn't specify how the data is stored (like sequential file. RAM or disk); instead it focuses on the logical storage of data.
- Sequential file and disk store refer to physical storage types.
- Random Access Memory (RAM) is a temporary memory not typically the focus in a DFD.
- DFDs are logical models, so the abstract concept of a repository of data is most accurate.

121. Software feasibility study is based on:
सॉफ्टवेयर फिजेब्लिटी स्टडी आधारित है-
(a) Business and resource constraint only
केवल बिजनेस एवं रिसोर्स कंस्ट्रैन्ट
(b) Scope, business constraint only
केवल स्कोप, बिजनेस कंस्ट्रैन्ट
(c) Scope, technology, time constraint only
केवल स्कोप, टेक्नोलॉजी, टाइम कंस्ट्रैन्ट
(d) Technology, finance, time, resource constraint only
केवल टेक्नोलॉजी, फाइनेंस, टाइम, रिसोर्स, कंस्ट्रैन्ट

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : A software feasibility study evaluate whether a proposed software project is practically achievable within various constraints. It typically includes.
- **Technical feasibility–** Do we have the technology and technical skills?
- **Economic/Financial Feasibility–** Is it cost-effective and within budget?
- **Time Feasibility–** can it be completed within the required time?
- **Resource Feasibility–** Are human and material resource available?

122. SRS stands for:
SRS का तात्पर्य:
(a) Software Requirement Specification
सॉफ्टवेयर रिक्वायरमेंट स्पेसिफिकेशन
(b) Software Requirement Standard
सॉफ्टवेयर रिक्वायरमेंट स्टैण्डर्ड
(c) System Requirement Standard
सिस्टम रिक्वायरमेंट स्टैण्डर्ड
(d) Software Requirement Set
सॉफ्टवेयर रिक्वायरमेंट सेट

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : SRS (Software Requirement Specification) is a detailed document that describes:
- What the software will do.
- How it is expected to perform.
- The functional & non-functional requirements.
- Any constraints or limitation.

In acts as a contract between the client and the development team, ensuring both understand what is to be built. It is used as a reference throughout the development process.

123. Which of the following is NOT a software development requirement?/निम्नलिखित में से कौन-सी सॉफ्टवेयर विकास आवश्यकता नहीं है?
(a) Functional requirements/कार्यात्मक आवश्यकताएँ
(b) System requirements/सिस्टम आवश्यकताएँ
(c) Physical requirements/शारीरिक आवश्यकताएँ
(d) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : In software development physical requirement are generally not needed because they are mostly related to hardware or infrastructure.

(iv) Software Project Planning, Software Architecture

124. LOC oriented software estimation model E = 3.2 * $(KLOC)^{1.05}$ is called :
LOC उन्मुख सॉफ्टवेयर एस्टीमेशन मॉडल E = 3.2 * $(KLOC)^{1.05}$ कहलाता है:
(a) Walston – Felix model/वाल्स्टन – फेलिक्स मॉडल
(b) Bailey – Basili model/बेली – बेसिली मॉडल
(c) Boehm simple model/बोहम सिम्पल मॉडल
(d) Doty model for KLOC > 9
KLOC > 9 के लिए डॉटी मॉडल

HTET PGT C.S. 2022

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Ans. (c) : Walston - Felix model
Formula:
$$E = 3.2 \times (KLOC)^{1.05}$$
E = Effort (person-month) power of slightly > 1 (1.05) This is proposed by Walston and Felix (IBM, 1977). It is used for general purpose software projects.

125. Fill in the blank in context of software system architecture :
सॉफ्टवेयर सिस्टम आर्किटेक्चर के संदर्भ में रिक्त स्थान की पूर्ति करें:
In _____architecture, the processing of the data in a system is organized so that each processing component is discrete and carries out one type of data transformation. The data flows from one component to another for processing.
_____ आर्किटेक्चर में सिस्टम में डाटा प्रसंस्करण इस प्रकार व्यवस्थित किया जाता है कि प्रत्येक प्रसंस्करण घटक भिन्न तथा एक प्रकार का डाटा रूपान्तरण करता है। डाटा एक घटक से दूसरे घटक में प्रसंसकरण के लिए प्रवाहित होता है।
(a) Pipe and filter/पाइप तथा फिल्टर
(b) Repository/रिपोजिटरी
(c) Layered/लेयर्ड
(d) Client-Server/क्लाइंट सर्वर

HTET PGT C.S. 2019

Ans. (a) : Pipe and filter architecture– The system is organized as a series of processing components (Filters) connected by data streams (pipes).
Pipes– Carry the output of one filter as the input to the next filter. The data flows sequential from one component to another, without feedback loops.
Each Filter– Performs a single, well defined transformation on the data (e.g. parsing, sorting, formatting).

126. Gantt charts are used in_____phase.
गैंट चार्ट का उपयोग_____चरण में किया जाता है।
(a) Analysis (b) Planning
(c) Design (d) Implementation

HTET PGT C.S. 30.07.2025

Ans.(b): Gantt Chart– A Gantt Chart is a project management tool that represent a project schedule visually. It shows tasks, their duration, start and end dates, and dependencies in the form of a bar chart.
Planning phase– The planning phase of SDLC involves estimating time, cost, resources, and task scheduling. Gantt Charts help managers assign tasks to team members track progress and ensure deadlines are met.

127. Full form of MVC is:
MVC का फुल फॉर्म है–
(a) Memory Virtual Controller
मेमोरी वर्चुअल कंट्रोलर
(b) Model View Container/मॉडल व्यू कंटेनर
(c) Maximum View Container/मैक्सिमम व्यू कंटेनर
(d) Model View Controller/मॉडल व्यू कंट्रोलर

HTET PGT C.S. 2023

Ans.(d): It is a software architectural pattern commonly used for developing used interfaces dividing the application into three interconnected components. The model (Data) the view (user interface) and controller (logic handling user input)

128. The number of function points of a proposed system is calculated as 500. Suppose that the system is planned to be developed in Java and the LOC/FP ratio of Java is 50. Estimate the effort (E) required to complete the project using the effort formula of basic COCOMO given below :
$$E = a(KLOC)^b$$
Assume that the values of a and b are 2.5 and 1.0 respectively.
(a) 25 person months
(b) 75 person months
(c) 62.5 person months
(d) 72.5 person months

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) E = a (KLOC)b
a = 2.5, b = 1.0
LOC| Fp = 50 ⇒10 c = 25000 i.e. 25 KLOC
$E = 2.5 (25)^1 = 62.5$

129. Which of the following is not a project estimation technique in software engineering?
निम्नलिखित में से कौन सी सॉफ्टवेयर इंजीनियरिंग में प्रोजेक्ट एस्टिमेशन तगकनीक नहीं है?
(a) Empirical techniques/एम्पिरिकल तकनीक
(b) Heuristic techniques/हेयररिस्टिक तकनीक
(c) Analytical techniques/एनालिटिकल तकनीक
(d) Prototyping techniques/प्रोटोटाइपिंग तकनीक

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : In software engineering common project estimation techniques are:
1. Empirical Techniques– Based on past project data and experience (Example– Cocomo, Putnam Model).
2. Heuristic Techniques– Based on expert judgment, rules of thump or experience driven formula.
3. Analytical Techniques– Based on mathematical model and detailed analysis.
4. Prototyping Techniques– This techniques are not project estimation method- they are part of a software development approach used to create a working model of the system for requirement clarification and user feedback.

130. Which of the following statement(s) is/are true regarding PERT chart in software project management?
सॉफ्वेयर परियोजना प्रबंधन में PERT चार्ट के संदर्भ में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?
I. PERT charts consists of a network of boxes and arrows./ PERT चार्ट बॉक्स और एर्रोस के एक नेटवर्क से युक्त होता है।
II. In PERT chart, the boxes represent activities and the arrow represents task dependencies./ PERT चार्ट में, बॉक्स गतिविधियों को दर्शाते है और एरो कार्य निर्भरता का दर्शाते हैं।
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Sarkari Resource

Ans. (c) : A PERT (Program Evaluation and Review Technique) chart is a project management tool used to plan, schedule and control complex software projects. It represents project activities as a network diagram showing the sequence of tasks, their dependencies and the estimated time to complete each task.
In PERT each task's time is estimated using three values:
Optimistic time (O) – Best case scenario
Most likely time (M) – Realistic estimate
Pessimistic time (P) – Worst case scenario
The expected time is calculated as.

$$TE = \frac{O + 4M + P}{6}$$

131. Which of the following statements are not true about CASE tool?
CASE टूल के बारे में निम्नलिखित में से कौन से कथन सत्य नहीं है?

I. It degrades the quality
यह गुणवत्ता को कम करता है

II. Reduce the drudgery in software engineer's work/सॉफ्टवेयर इंजीनियर के काम में कठिनाइयां कम करता है

III. Increase maintenance cost
रखरखाव लागत को बढ़ाता है

(a) II and III (b) II and III
(c) I and II (d) I, II and II

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (a) : Statement-I : Not true
• CASE tools are actually designed to improve software quality by automating tasks, ensuring consistency and supporting design validation.
Statement-II : True
• CASE tools help automate repetitive tasks like documentation, code generation etc reducing manual effort.
Statement-III : Not true
• CASE tools can actually reduce maintenance cost by producing better documented, more consistent, and modular code.

131. Which of the following statement(s) is/are correct regarding software project management?
सॉफ्टवेयर परियोजना प्रबंधन के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. PERT charts can be derived automatically form Gantt charts.
पि.ई.आर.टी चार्ट स्वचालित रूप से गेंट चार्ट से प्राप्त किए जा सकते हैं।

II. Gantt charts cannot be automatically derived from PERT charts.
गेंट चार्ट स्वचालित रूपस से पि.ई.आर.टी. चार्ट से प्राप्त नहीं किए जा सकते।

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (d) : Statement-I – PERT charts can be derived automatically from Gantt charts.
Incorrect, PERT (Program Evaluation Review Technique) charts focus on task dependencies and the sequence of tasks (network based) while Gantt charts focus on the timeline. Gantt charts typically don't show dependencies in a way that allows direct and automatic derivation of a PERT chart.
Statement-II – Gantt charts can't be automatically derived from PERT charts.
Incorrect. In fact Gantt charts can be automatically derived from PERT charts because PERT charts contain task durations and dependencies, which are the necessary inputs for generating a Gantt chart.

133. What is the full form of CASE tool in software engineering?
सॉफ्टवेयर इंजीनियरिंग में (सीएएसई) टूल का अंसक्षिप्त रुप क्या है ?

(a) Computer Application Software Engineering
कंप्यूटर ऐप्लीकेशन सॉफ्टवेयर इंजीनियरिंग
(b) Control Application Software Engineering
कंप्यूटर ऐप्लीकेशन सॉफ्टवेयर इंजीनियरिंग
(c) Code Aided Software Engineering
कंट्रोल ऐप्लीकेशन सॉफ्टवेयर इंजीनियरिंग
(d) Computer Aided software Engineering
कोड ऐडिड सॉफ्टवेयर इंजीनियरिंग

DSSSB TGT C.S. 01.08.2021 (Shift-I)

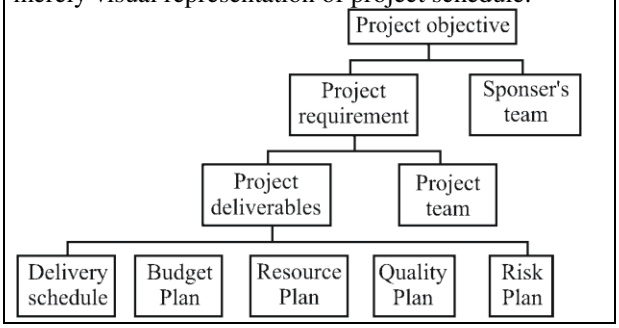
Ans. (c) : Case tool are software applications that help automates and support various phases of the software development life cycle (SDLC). They provide features for designing, coding, testing documentation, project management and maintenance.
CASE tools are generally divided into–
UPPER CASE tools– Used in early phases (planning, analysis, design)
Lower CASE tools– Used in later phases (Implementation, testing, maintenance)

134. Which one from the following is highly associated activity of project planning?

(a) Keep track of the project progress
(b) Compare actual and planned progress and costs
(c) Identify the activities, milestones and deliverables produced by a project
(d) Both (b) and (c)

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : Project planning is commonly perceived as creating 'Gantt chart', which is incorrect. Gantt chart is merely visual representation of project schedule.



Sarkari Resource

135. The software _____ of a program of a computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships among them.

(a) Design (b) Architecture
(c) Process (d) Requirement

UGC NET C.S. December-2014 (Paper-II)

Ans : (b) Because, software architecture encompasses the set of significant decisions about the organization of a software system including the selection of the structural elements and their interfaces by which the system is composed; behavior as specified an collaboration among those elements, composition of these structural and behavioral elements into larger subsystems : software architecture also involves functionality, usability, performance etc.

136. _____ of a system is the structure or structures of the system which comprise software elements, the externally visible properties of these elements and the relationship amongst them.

(a) Software construction
(b) Software evolution
(c) Software architecture
(d) Software reuse

UGC NET C.S. June-2014 (Paper-II)

Ans : (c) Because, Software architecture refers to the high level structure of software system, the discipline of creating such structures, and the documentation.

137. Requirements prioritization and negotiation belongs to :

(a) Requirements validation
(b) Requirements elicitation
(c) Feasibility study
(d) Requirements reviews

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Elicitation is the process of gathering, understanding and documenting requirement. Prioritization and negotiation are key activities within the elicitation process to ensure a feasible and well defined set of requirements for the project.

Requirements Validation– This is the process of checking if the gathered requirements are correct, complete and consistent.

Feasibility study– This is an assessment of whether a project is viable and achievable, considering factors like cost, resources and technology.

Requirements Reviews– These are formal assessments of the requirements documentation, typically conducted by stakeholders and experts, to identify potential issues and ensure quality.

138. Which one of the following non-functional quality attributes is not highly affected by the architecture of the software?

(a) Performance (b) Reliability
(c) Usability (d) Portability

UGC NET C.S. December-2015 (Paper-III)

Ans : (c) Usability should not be highly affected by the architecture of the software as it is basic things rest options are extra features offered by s/w & may be highly affected by the architecture of the s/w.

139. Which of the following statement(s) is/are true with respect to software architecture?

S1 : Coupling is a measure of how well the grouped together in a module belong together logically.
S2 : Cohesion is a measure of the degree of interaction between software modules
S3 : If coupling is low and cohesion is high then it is easier to change one module without affecting others.

(a) Only S1 and S2 (b) Only S3
(c) All of S1, S2 and S3 (d) Only S1

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) S3 is true because, If coupling is low and cohesion is high then it is easier to change one module without affecting others.

140. In Software configuration management, what is the primary purpose of version control?

(a) To control the changes made to software
(b) To document user requirements
(c) To estimate project cost
(d) To design the software architecture

UGC NET C.S. Re-Exam- June 2024

Ans. (a) : Version control is a core aspect of Software Configuration Management (SCM). It is primarily used to track and manage changes made to software code, documents, or other components throughout the development lifecycle. They allow multiple people to work on the same project simultaneously, track revisions, and maintain a history changes. This ensures that changes can be reviewed, reverted if necessary and integrated smoothly.

141. Software project manager is responsible for the following tasks:

A. Project Planning
B. Project status tracking
C. Resource management
D. Risk management
E. Project delivery within time & budget

Choose the correct answer from the options given below:

(a) All the statements are correct.
(b) Only B & C are correct
(c) Only A & D are correct
(d) All the statements are not correct.

UGC NET C.S. December-2022

Ans. (a) : Software project
→ A software project is the complete procedure of software development from requirement gathering to testing and maintenance, carried out according to the execution methodologies, in a specified period of time to achieve intended software product.

Need of software project management.

- Project Planning
- Scope Management
- Project Estimation
- Resourse management
- Project risk management
- Project Execution & Monitoring
- Project Communication Management
- Project Management Tools

Sarkari Resource

142. COCOMO stands for _____

(a) Consumed cost model
(b) Constructive cost model
(c) Common control model
(d) Composition cost model

UGC NET C.S. December-2022

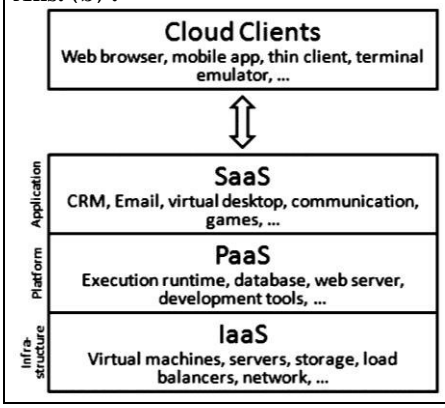
Ans. (b) : COCOMO (Constructions Cost Model) is a regression model based on LOC, i.e. Number Lines of Code. It is a procedural cost estimate model for software projects and is often use as a process of reliably predicting the various parameters associated with making a project such as size, effort cost, time and quality.

143. Suppose a cloud contains software stack such as Operating systems, Application softwares, etc. This model is referred as _____ model.

(a) MaaS (b) SaaS
(c) LaaS (d) PaaS

UGC NET C.S. December-2018 (Paper-II)

Ans. (b) :



144. In software project risk management, accessing the likelihood and consequences of identified risks is called -
सॉफ्टवेयर प्रोजेक्ट में जोखिम प्रबंधन के अंतर्गत चिन्हित जोखिमो की संभावनाओं तथा परिणामो को मूल्यांकन कहलाता है?

(a) Risk Identification/रिस्क आइडेंटिफिकेशन
(b) Risk analysis/रिस्क एनालिसिस
(c) Risk Planning/रिस्क प्लानिंग
(d) Risk monitoring/रिस्क मॉनिटरिंग

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : In software project risk management, once risks, identified, the next step is to asses (analyze) them. This involves evaluating.

- **Likelihood (Probability)**– How likely is it that the risk will occur.
- **Consequences (Impact)**– What will be the effect on the project if the risk occurs?

This process is known as Risk Analysis. It helps in prioritizing risks & deciding how to handle them.

145. COCOMO model stands for?
COCOMO मॉडल है?

(a) Common Cost Model
कॉमन कॉस्ट मॉडल
(b) Constructive Cost Model
कन्स्ट्रक्टिव कॉस्ट मॉडल
(c) Cost Constructive Comprehensive Model
कॉस्ट कन्स्ट्रक्टिव कॉम्प्रेहेन्सिव मॉडल
(d) Comprehensive Cost Model
कॉम्प्रेहेन्सिव कॉस्ट मॉडल

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : It is used to estimate the cost, effort, and time required to develop a software project based on:

- The size of the software (usually in KLOC thousands of lines of code).
- Project complexity.
- Developer experience, tools used, etc COCOMO model developed by 'Barry Boehm' in the 1980's.

146. Which of the following is not a decomposition method of software project estimation?
निम्न में से कौन-सा सॉफ्टवेयर प्रोजेक्ट एस्टीमेशन का डिकम्पोजीशन तरीका नहीं है?

(a) COCOMO model estimation
COCOMO मॉडल एस्टीमेशन
(b) LOC estimation/LOC एस्टीमेशन
(c) FP estimation/FP एस्टीमेशन
(d) All of these/इनमें से सभी

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Decomposition methods estimate effort by breaking down the project into smaller components–
Example:

- LOC (Lines of Code) estimation.
- FP (Function Point) estimation.

These are decomposition based because they involve analyzing the software's structure or functionality.
However, COCOMO (Constructive Cost Model) is an algorithmic estimation model, not a decomposition method. It uses mathematical formulas based on project size and other factors.

147. Effort is measured in terms of:
एफर्ट मापा जाता है-

(a) Person-months/पर्सन-मंथस
(b) Persons/पर्सन्स
(c) Rupees/रुपीज
(d) Months/मंथस

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Effort in software engineering is typically measured in person-months, which represents the amount of work one perso ह्मि म्दस्ज्ञा ग्ह दहा स्दहू.
ऊपे ल्हग् पत्जे ोर्ग्सूं ज्दराू ग्सत्हो म्दूं ह्मी दल्मि र्ज्हहग्हु.

148. According to PUTNAM, the project effort is inversely proportional to the fourth power of the development time. Doubling the development schedule for a 100 Person-Month (PM) project would reduce the project effort to
PUTNAM क्रे अनुसार परियोजना का प्रयास, विकास समय की चौथे घात (पावर) के विपरीत आनुपातिक है। 100 व्यक्ति-महीने (इश) परियोजना के विकास कार्यक्रम को दोगुना करने से इळऊँश के लिए परियोजना प्रयास कम हो जाता है।

Sarkari Resource

(a) 50 PM (b) 7.16 PM
(c) 25 PM (d) 6.25 PM

LT. Grade C.S. 2018

Ans. (d) : According to the PUTNAM model, the project effort (E) is inversely proportional to the fourth power of the development time (T) expressed as-

$$E \propto \frac{1}{T^4}$$

If the development schedule is doubled, then the effort can be calculated as -

$$E_2 = E_1 \times \left(\frac{T_1}{T_2}\right)^4$$

Given that $T_2 = 2T_1$, We substitute,

$$E_2 = E_1 \times \left(\frac{1}{2}\right)^4 = E_1 \times \frac{1}{16}$$

If the initial effort E_1 = 100 person - months the new effort becomes

$$E_2 = 100 \times \frac{1}{16} = 6.25PM$$

149. What is the purpose of staffing level estimation in software project management?
सॉफ़्टवेयर प्रोजेक्ट प्रबंधन में स्टाफिंग लेवल का अनुमान लगाने का उद्देश्य क्या है?

(a) To determine the number of hardware components required
आवश्यक हार्डवेयर घटकों की संख्या निर्धारित करना
(b) To identify potential risks in the project
प्रोजेक्ट में संभावित जोखिमों की पहचान करना
(c) To allocate appropriate resources to the project
प्रोजेक्ट के लिए उपयुक्त संसाधन आवंटित करना
(d) To estimate the maintenance costs of the software/सॉफ़्टवेयर के रखरखाव की लागत का अनुमान लगाना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The main objective of estimating the staffing level is to allocate appropriate resources (Such as developers, testers, etc) for the project so that work can be completed on time and efficiently.
Note– The commission has considered option (b) as the correct answer.

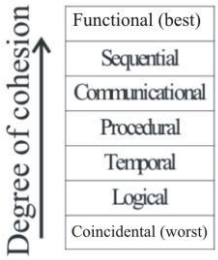
v. Software Design and implementation

150. Arrange the following types of Cohesion form the to worst type.
(A) Logical Cohesion
(B) Sequential Cohesion
(C) Communication Cohesion
(D) Temporal Cohesion
(E) Procedural Cohesion
Choose the correct answer from the given below:

(a) A → D → E → C → B
(b) A → E → D → C → B
(c) B → E → C → D → A
(d) B → C → E → D → A

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) : Cohesion order from worst to Best
(A) Logical Cohesion ⇒ (D) Temporal Cohesion ⇒ (C) Communication Cohesion ⇒ (B) Sequential Cohesion



151. The_____ phase of the system decides how the system will operate in terms of hardware, software and network infrastruction.
सिस्टम का_____फेज यह तय करता है कि सिस्टम हार्डवेयर, सॉफ्टवेयर और नेटवर्क इंफ्रास्ट्रक्चर के संदर्भ में कैसे काम करेगा–

(a) Analysis/एनालिसिस
(b) Design/डिजाइन
(c) Implementation/इम्प्लीमेंटेशन
(d) Planning/प्लानिंग

HTET PGT C.S. 2023

Ans.(b): The Design phase of the System Development Life Cycle (SDLC) focuses on how the system will work in practice. After the requirements are collected during the analysis phase, the design phase translates those requirements into a blueprint of the system.

152. In which coding approach, a programmer first writes the test scripts, and then writes the code to pass the tests?

(a) Pair programming
(b) Incremental coding process
(c) Test-driven development
(d) Source code control system and build

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Ans. (c) : Test- Driven Development (TDD) is a coding approach in which a programmer first writes test cases (test scripts) before writing the actual code. These test cases define what the code is supposed to do.
(a) Pair programming– Tow programmers work together at one computer one writes code, the other reviews it.
(b) Incremental coding process– Code is developed in small increments, adding functionality step by step.
(d) Source code control system– Tools like Git are used to manage versions of source code, not for writing tests first.

153. In an activity diagram used in software engineering, a _____ is used to split a single incoming flow into multiple concurrent flows, which is represented as a straight, slightly thicker horizontal line.

(a) Fork node (b) Join node
(c) Sync node (d) Master mode

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Ans. (a) : A fork node is represent as a straight, slightly thicker horizontal (or vertical) bar. It has one incoming arrow and multiple outgoing arrows, showing that the process branches out into parallel paths.
(b) Join Node– Opposite of fork, it combines multiple concurrent flows back into a single flow.
(c) Sync Node– Not a standard UML term, sometimes used informally for synchronization, but UML officially uses "Join Node".
(d) Master Node– Not related to UML activity diagrams.

154. _____ is a process of changing a software system in such a way that it does not alter the external behavior of the code or design, yet improves the internal structure.
(a) Refinement (b) Refactoring
(c) Regeneration (d) Reconstruction

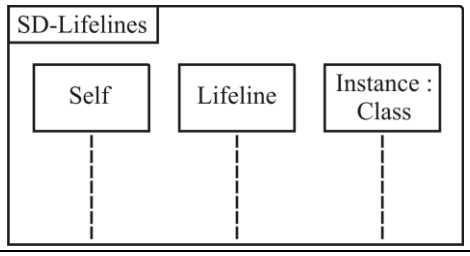
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Ans. (b) : The process described here is about changing the internal structure of the software (code of design) without changing its external behavior.
Refactoring– This exactly means restructuring existing code without changing its external behavior.
(a) Refinement– Usually means making a design or requirement more detailed, not specifically restructuring code.
(c) Regeneration– Usually refers to automatically generating code a gain from specification or models.
(d) Reconstruction– Rebuilding a system, often with major changes.

155. In a UML sequence diagram, an object's lifeline is shown using
(a) A U shaped arrow
(b) An open arrowhead with a dotted horizontal line
(c) A solid line
(d) A vertical dashed line

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Ans. (d) : Sequence Diagram– A sequence Diagram is a key component of Unified Modeling Language (UML) used to visualize the interaction between object in a sequential order.
A UML Sequence Diagram is used to show how interact with each other in a time sequence.
Lifelines– A lifeline represents and individual participant in a sequence diagram. A lifeline will usually have a rectangle containing its object name. If its name is "Self", that indicates that the lifeline represent the classifier which owns the sequence diagram.



156. The process through which an existing system undergoes an alteration, to be reconstituted in a new form, is called
वर्तमान सिस्टम को नवीन रूप में पुनर्संगठित करने के लिए किए जाने वाले बदलाव का प्रक्रम कहलाता है
(a) Engineering/इंजीनियरिंग
(b) Prototyping/प्रोटोटाइपिंग
(c) Re-Engineering/री-इंजीनियरिंग
(d) Debugging/डिबगिंग

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) : Software Re-Engineering is the process of analyzing and modifying an existing system to reconstitute it in a new and improved form. It is usually done when the old system is outdated hard to maintain, or needs to be adopted to new technologies.
(a) Engineering– This is the general discipline of creating systems, not specifically about modifying old ones.
(b) Prototyping– Used to build a model of the system to understand or validate requirements not to reconstitute an existing system.
(d) Debugging– This only involves finding and fixing errors (bugs) in software not redesigning it.

157. A software program that infers and manipulates existing knowledge in order to generate new knowledge is known as :
(a) Data dictionary
(b) Reference mechanism
(c) Inference engine
(d) Control strategy

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) A software program that infers and manipulates existing knowledge in order to generate new knowledge is known as **Inference engine**.
A data dictionary is a collection of descriptions of the data objects or items in a data model for the benefit of programmers and others who need to refer to them.
Control strategies are specific action plans for bringing a process back into control.
Reference is what relates words to the world of objects on whose condition the truth of sentences hinges and it is natural to wonder what sorts of relations underlie the reference relation-to wonder, that is, what constitutes the mechanism of reference.

158. Match each application/software design concept in List-I to its definition in List-II.

List-I		List-II	
i.	**Coupling**	**A.**	**Easy to visually inspect the design of the software and understand its purpose**
ii.	**Cohesion**	**B.**	**Easy to add functionality to a software without having to redesign it**
iii.	**Scalable**	**C.**	**Focus of a code upon a single goal.**
iv.	**Readable**	**D.**	**Reliance of a code module upon other code modules.**

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Code :

	(i)	(ii)	(iii)	(iv)
(a)	(B)	(A)	(D)	(C)
(b)	(C)	(D)	(A)	(B)
(c)	(D)	(C)	(B)	(A)
(d)	(D)	(A)	(C)	(B)

UGC NET C.S. June-2016 (Paper-III)

Ans : (c) Coupling, Cohesion, Scalable and Readability are key principle in software design and implementation that ensure a system is maintainable, efficient and adaptable

(i) Coupling– Measures how dependent modules are on each other with low coupling preferred for flexibility.

(ii) Cohesion– Reflect how well a modules elements work toward a single purpose with high cohesion improving clarity.

(iii) Scalability– Ensure the system can handle growth in workload without performance lass, through proper architecture and resource planning.

(iv) Readable– Readable focuses on writing clean, understandable code for easier maintenance.

159. What is COHESION is software design?
सॉफ्टवेयर डिजाइन में कोहेशन क्या होता है?

(a) Cohesion is a measure of complexity of module of a software/कोहेशन एक सॉफ्टवेयर के मॉड्यूल की जटिलता का एक माप होता है

(b) Cohesion is a measure of functional strength of a module of a software/कोहेशन एक सॉफ्टवेयर के मॉड्यूल की कार्यात्मक प्रबलता का एक माप होता है

(c) Cohesion is a measure of functional weakness of a module of a software/कोहेशन एक सॉफ्टवेयर के एक मॉड्यूल की कार्यात्मक कमजोरी का एक माप होता है

(d) Cohesion is a measure of efficiency of a module of a software/कोहेशन एक सॉफ्टवेयर के एक मॉड्यूल की दक्षता का एक माप होता है

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (b) : Cohesion in software design is the measure of how closely related and focused the responsibility of a single module or component are.

- High cohesion means a module performs one well defined task or set of related tasks, making it easier to understand, maintain, and reuse.
- Low cohesion means a module handles unrelated tasks, which makes the code more complex harder to maintain. In good software design we aim for high cohesion so that each part of the system has a clear singular propose which improve clarity and maintainability.

160. SQL views are also known as
SQL views को _____ नाम से भी जाना जाता है।

(a) Simple tables
(b) Virtual tables
(c) Complex tables
(d) More than one of the above
उपर्युक्त में से एक अधिक

BPSC TRE 1.0 2023 (11-12)

Ans. (b) : SQL Views are also known as virtual tables, which are presented as a representation of one or more base tables but do not actually store real data. They are used to present spared data in the database in different ways, making data retrieval and presentation simpler.

161. A module of a software is said to be logically cohesive if _____.
एक सॉफ्टवेयर के मॉड्यूल को तार्किक रूप से कोहीसिव कहा जाता है यदि _____।

(a) All the elements of the module perform similar operations like error handling, data input etc/मॉड्यूल के सभी तत्व समान संचालन करते हैं जैसे त्रुटि प्रबंधन डेटा इनपुट आदि

(b) Different elements of the module cooperate to achieve a single function for example managing and employee's pay roll/मॉड्यूल के विभिन्न तत्व एकल कार्य को प्राप्त करने के लिए सहयोग करते हैं उदाहरण के लिए किसी कर्मचारी के वेतन रोल का प्रबंधन

(c) If the set of functions of the module are all part of a procedure/यदि मॉड्यूल के कार्यों का सेट एक प्रक्रिया का हिस्सा है

(d) All the elements of the module performs different operations alternatively/मॉड्यूल के सभी तत्व वैकल्पिक रूप से अलग-अलग ऑपरेशन करते है

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : A logically cohesive module is one in which the elements of the module perform similar activities, even if they are not related functionally. Examples include a module that handles various types of input or a module for different error handling routines. They are grouped because they are of the same logical category, not because they work together toward a specific task.

162. Two modules are said to be common coupled in software design if _____.
सॉफ्टवेयर डिजाइन में दो मॉड्यूल को कॉमन कपल्ड कहा जाता है यदि _____।

(a) They share their code
वे अपना कोड साझा करते है

(b) They share same global data areas
वे समान ग्लोबल डेटा क्षेत्र साझा करते है

(c) They communicate via a composite data
वे एक कम्पोजिट डेटा माध्यम से संवाद करते है

(d) They communicate through an elementary data item/वे एक एलीमेंट्री डेटा आइटम के माधयम से संवाद करते है

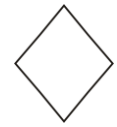
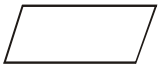
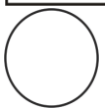
DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : Two modules are said to be "Common Coupled" in software design if they share the same global data areas.

Common coupling refers to a situation where multiple modules directly access and modify a shared global variable, leading to tight coupling and making it difficult to maintain and change one module without affecting the others.

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163. Which symbol in flowchart is used for processing?
फ्लोचार्ट में किस प्रतीक का उपयोग प्रक्रमण के लिए किया जाता है?

(a) 
(b) 
(c) 
(d) 

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : In a flowchart, the processing symbol is represented by a rectangle and is used to show any step where actual work or computation is performed. This can include tasks like calculating values, sorting data, updating variable or performing any kind of transformation on the input data.
Example– Calculate total, Sort List, Update record.

164. According to structured partitioning in software engineering a program_____
सॉफ्टवेयर इंजीनियरिंग में स्ट्रकचर्ड पार्टिशनिंग के अनुसार एक प्रोग्राम_____।

(a) Should be partitioned both horizontally and diagonally/हॉरिज़ॉन्टली और डायगोनली दोनों तरह से पार्टिशन किया जाना चाहिए
(b) Should be partitioned both vertically and diagonally/वर्टिकली और डायगोनली दोनों तरह से पार्टिशन किया जाना चाहिए
(c) Should be partitioned both horizontally and vertically/हॉरिज़ॉन्टली और वर्टिकली दोनों तरह से पार्टिशन किया जाना चाहिए
(d) Should be partitioned diagonally/डायग्रली पार्टिशन किया जाना चाहिए

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : According to structured partitioning in software engineering a program should be divided both horizontally and vertically.
• Horizontal partitioning means dividing the system into different branches or modules based on major function (top-down).
This reduces the complexity of each module and make maintenance easier.
• Vertical partitioning means separating control modules from worker modules, so higher level modules handle decision making and lower level module handle detailed processing.

165. What is COUPLING in software design?
सॉफ्टवेयर डिज़ाइन में COUPLING क्या है?

(a) Coupling is measure of degree of functional interdependence between the two modules
दो मॉड्यूल के बीच कार्यात्मक इंटरडेपेंडेन्स की डिग्री का मापन कपलिंग है
(b) Coupling is measure of degree of functional non-dependence between the two modules
दो मॉड्यूल के बीच कार्यात्मक नॉन-डेपेंडेन्स की डिग्री का मापन कपलिंग है
(c) Coupling is measure of degree of functional connected dependence between the two modules/दो मॉड्यूल के बीच कार्यात्मक कनेक्टेड डिपेंडेन्स की डिग्री का मापन कपलिंग है
(d) Coupling is measure of the functional strength of the module of the software
सॉफ्टवेयर के मॉड्यूल की कार्यात्मक प्रबलता का मापन कपलिंग है

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (a) : In software design, coupling is the degree of interdependence between software modules– that is how closely connected or reliant they are on each other's details. Lower coupling is preferred because it makes the system more flexible, maintainable and easier to modify.

166. _____ has been developed to incorporate all the character of every written language of the world. It provides a unique number for every character, irrespective of device [server, desktop, mob], OS [linux, win] or software application

(a) UNICODE (b) Binary Code
(c) ISCII (d) ASCII

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : Unicode is a universal character encoding standard that assigns a unique numeric value (code paints) to each character in every written language.
Binary Code– Binary code is the fundamental language of computer, using 0s and 1s to represent data, but it doesn't inherently handle character encoding in a standardized way for all languages.
ISCII– ISCII (Indian Script Code for Information Interchange) is a character encoding standard specifically for Indian languages.
ASCII– American Standard Code for Information Interchange (ASCII) was an early standard mainly used for English and some Western European languages.

167. Cohesion is an extension of :

(a) Abstraction concept
(b) Refinement concept
(c) Information hiding concept
(d) Modularity

UGC NET C.S. June-2015 (Paper-II)

Ans.(c) : A cohesion module performs a single task within a software procedure, requiring little interaction with little procedures in other parts of software.
Type of cohesion :
- A module that performs takes that are related logically is logically cohesion
- When a module contains takes that are related by the fact that all must be executed with the same span of time, the exhibits temporal cohesion
- At the low end of the spectrum, a module that performs a set of takes that relate to each other loosely called coincidentally cohesive.

168. Object Request Broker (ORB) is

I. A software program that runs on the client as well as on the application server.
II. A software program that runs on the client side only.
III. A software program that runs on the application server, where most of the components reside.

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(a) I, II & III (b) I & II
(c) II & III (d) I only

UGC NET C.S. June-2014 (Paper-II)

Ans : (d) Because, object request broker (ORB) is middle wave that uses the COBRA (Common Object Request Broker Architecture) specification. The ORB takes care of all of the details in routing a request from client to object and the response to its destination. ORB runs on clients and as well as server application.

169. A software agent is defined as

I. A software developed for accomplishing a given task.
II. A computer program which is capable of acting on behalf of the user in order to accomplish a given computational task.
III. An open source software for accomplishing a given task.

(a) I (b) II
(c) III (d) All of the above

UGC NET C.S. June-2014 (Paper-II)

Ans : (b) Because, A software agent is a piece of software that function as a agent for a user or another program, working autonomously and continuously in a particular environment. It is inhibited by other processes and agents, but it is also able to learn from its experience in functioning in an environment over a long period of time.

170. Match the following with respect to the jump statements :

List–I		List–II
A. Return	**i.**	**The conditional test and increment portions**
B. Goto	**ii.**	**A value associated with it**
C. Break	**iii.**	**Requires a label for operation**
D. Continue	**iv.**	**An exit from only the innermost loop**

Codes :

	A	B	C	D
(a)	ii	iii	iv	i
(b)	iii	iv	i	ii
(c)	iv	iii	ii	i
(d)	iv	iii	i	ii

UGC NET C.S. June-2014 (Paper-III)

Ans. (a) : Because,
(a) Return : Return the content of a variable after completion of the function body.
(b) go to : Control jumps from one statement to another by following the label.
(c) Break : Stops looping where the break statement appears.
Continue : It increments the loop controlling variable without executing the body of the loop.

171. A software design is highly modular if :

(a) Cohesion is functional and coupling is data type
(b) Cohesion is coincidental and coupling is data type
(c) Cohesion is sequential and coupling is content type
(d) Cohesion is functional and coupling is stamp type

UGC NET C.S. June-2015 (Paper-III)

Ans : (a) We know our aim is to maximize cohesion & minimum coupling
Strong form of cohesion is functional cohesion weakest form of coupling is data coupling.

172. A signal processor software is expected to operate for 91.25 days after repair, and the mean software repair time is expected to be 5 minutes. Then, the availability of the software is :

(a) 96.9862% (b) 97.9862%
(c) 98.9962% (d) 99.9962%

UGC NET C.S. November-2017 (Paper-III)

Ans : (d)

$$\text{Availability} = \frac{\text{MTBF}}{(\text{MTBF}+\text{MTRR})}$$

$$= \left(\frac{91.25\times 24\times 60}{91.25\times 24\times 60+5}\right)$$

Availability = 99.9962%

173. A software design pattern used to enhance the functionality of an object at run-time is :

(a) Adapter (b) Decorator
(c) Delegatio (d) Proxy

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) In object oriented programming, the decorate or pattern (also known as wrapper, an alternative naming shared with adapter pattern) is a design pattern that allows behavior to be added to an in dividable object, either statically or dynamically, without effecting the behavior of other object from the same class.

174. Which of the following is not one of the principles of good coding?

(a) Create unit tests before you begin coding.
(b) Create visual layout that aids understanding.
(c) Refractor the code after you completes the first coding pass.
(d) Write self-documenting code not program documentation.

UGC NET C.S. December-2022

Ans. (d) : Good principles of coding is
1. Create unit tests before you begin coding.
2. Create visual layout that aids understanding.
3. Refractor the code after you completes the first coding pass.
4. DRY (Don't repeat yourself)
5. KISS (Keep It short and simple)
6. Refactor
7. Creation over legacy
8. Open/closed

175. Given below are two statements: one is labeled as Assertion A and the other is labelled as Reason R.

Assertion A : Cohesion is a qualitative indication of the degree to which a module can be written more compactly and is able to complete its function in a timely manner.

Reason R : Cohesion is a qualitative indication of the degree to which a module is connected to other modules and the outside world.

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In the light of the above statements, choose the most appropriate answer from the options givens given below.

(a) Both A and R are correct and R is the correct explanation of A.
(b) Both A and R are correct but R is NOT the correct explanation of A.
(c) A is correct but R is not correct.
(d) A is not correct but R is correct.

UGC NET C.S. December-2022

Ans. (c) : Assertion A : Cohesion is a qualitative indication of the degree to which a module can be written more compactly and is able to complete its function in a timely manner.
Coupling between modules/components is their degree of mutual under dependence.

176. Match List I with List II

	List-I		List-II
A.	**Product complexity**	**i.**	**System Requirements Design**
B.	**Structured System analysis**	**ii.**	**Software design.**
C.	**Coupling & Cohesion**	**iii.**	**Validation techniques.**
D.	**Symbolic Execution**	**iv.**	**Software cost estimation.**

Choose the correct answer from the options given below :

(a) A-II, B-III, C-IV, D-I
(b) A-III, B-I, C-IV, D-II
(c) A-IV, B-I, C-II, D-III
(d) A-III, B-IV, C-I, D-II

UGC NET C.S. December-2022

Ans. (c) : Product complexity→
Product complexity is related with software complexity. Complexity of any software is measured in terms of software cost estimation.
(iv) Product complexity———→Software cost estimation
(i) Structured system analysis–
————→System Requirement Design.
Coupling and Cohesion–
Coupling and Cohesion are two key concepts in software engineering that are used to measure the quality of software system's design.
C. (ii) Coupling & Cohesion
————→Software design.
Symbolic Execution→Symbolic execution is a validation technique
or It is a software testing technique.
D. (iii) Symbolic Execution——→ Validation techniques.

177. In design protocol of critical section problem, each process must ask permission to enter critical section in _____ code; it then executes in the critical section; once it finishes executes in the critical section it enters the _____ code. The process then enters the _____ code.

(a) Entry section, remainder section, exit section
(b) Entry section, exit section, remainder section
(c) Remainder section, entry section, exit section
(d) Remainder section, exit section, entry section

UGC NET C.S. December-2022

Ans. (b) : Critical section problem in OS

```
Do {
        Entry section;
          Critical section
        Exit Section
          Remainder Section
      } while (true)
```

178. Mr. X designed open source software which must comply with some criteria. Choose the false statement in respect of above.

A. No restriction on redistribution of the software as part or whole.
B. The integrity of the author's source code must be maintained.
C. The software can be sold after distribution.

Choose the correct answer from the options given below :

(a) A only (b) B only
(c) C only (d) All statements are false

UGC NET C.S. December-2022

Ans. (c) : Open source doesn't just means access to the source code. The distribution terms of open source software must comply with the following criteria.

- Free Redistribution
- Derived work
- Integrity of the Author's Source Code
- No Discrimination Against persons or Groups
- No Discrimination Against Fields of Endeavor
- Distribution of license
- License must not be specific to a product
- License must be technology Neutral.

179. A software design pattern often used to restrict access to an object is :

(a) Adapter (b) Decorator
(c) Delegation (d) Proxy

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : A proxy is a wrapper or agent ·object· that is being called by the client to access the real serving object behind the scenes. Use of the proxy can simply be forwarding to the real object or can provide additional logic.
Adapter– Match interface of different classes.
Decorator– Add responsibilities objects dynamically. Effectively in the instantiation process, object-creation patterns use delegation effectively to get the job done.

180. Software reuse is

(a) The process of analysing software with the objective of recovering its design and specification
(b) The process of using existing software artifacts and knowledge to build new software
(c) Concerned with reimplementing legacy system to make them more maintainable
(d) The process of analysing software to create a representation of higher level of a abstraction and breaking software down into its parts to see how it works

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) : "Software reuse is the process of creating new software system from predefined software components."

Sarkari Resource

181. UML stands for?
UML का तात्पर्य है?
(a) Uniform Modeling Language
यूनिफार्म मॉडलिंग लैंग्वेज
(b) Unified Modeling Language
यूनिफाइड मॉडलिंग लैंग्वेज
(c) Universal Modeling Language
यूनिवर्सल मॉडलिंग लैंग्वेज
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : UML (Unified Modeling Language) is a standardized model Language used in software engineering to visualize, specify, construct, and document the structure and behavior of software systems.

182. The degree of interaction between two modules is known as:
दो मोड्यूल के मध्य डिग्री ऑफ इंटरेक्शन कहलाता है?
(a) Cohesion/कोहेसन
(b) Coupling/कपलिंग
(c) Strength/स्ट्रेंग्थ
(d) All of these/इनमें से सभी

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : Coupling refers to the degree of interdependence or interaction between two software modules.
Cohesion refers to how closely related and focused the responsibilities of a single module are.
Strength is not a standard term used in this context.

183. Which maintenance process involves documenting the software's structure and functionality?
सॉफ्टवेयर की संरचना और कार्यक्षमता को दस्तावेज करने वाली रखरखाव प्रक्रिया कौन-सी है?
(a) Reverse Engineering
(b) Preventive Maintenance
(c) Perfective Maintenance
(d) Adaptive Maintenance

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (a) : Reverse Engineering is a process in which an existing software is analyzed to understand its structure and working mechanism. The purpose is to redesign the system or prepare its documentation especially when the original design information is not available.

184. Reverse engineering is primarily used for:
रिवर्स इंजीनियरिंग का मुख्य उपयोग किसके लिए किया जाता है?
(a) Creating new software from scratch
शुरुआत से नया सॉफ्टवेयर बनाना
(b) Improving software performance
सॉफ्टवेयर प्रदर्शन में सुधार करना
(c) Understanding and documenting existing software
मौजूदा सॉफ्टवेयर को समझना और प्रलेखित करना
(d) Testing software components
सॉफ्टवेयर घटकों का परीक्षण करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : In reverse engineering, the structure and functionality of existing software are understood and its documentation is prepared.

(vi) Software Testing and Reliability

185. Term 'bath tub curve' is related to _____ .
टर्म 'बाथ टब कर्व' _____से संबंधित है।
(a) Software wear out rate/सॉफ्टवेयर क्षरण दर
(b) Software failure rate/सॉफ्टवेयर विफलता दर
(c) Requirement analysis/रिक्वायरमेंट विश्लेषण
(d) Hardware failure rate/हार्डवेयर विफलता दर

HTET PGT C.S. 2022

Ans. (d) : The bath tub curve is a well known concept that represents the failure rate of a system or component over time. It has three phase.
1. Infant mortality phase
2. Normal life phase (Useful life)
3. Wear-out phase.

186. Which is not the right approach of incremental testing approach ?
निम्नलिखित में से इंक्रीमेंटल टेस्टिंग का कौन-सा तरीका या अप्रोच सही नहीं है?
(a) Big bang approach/बिग बैंग दृष्टिकोण (अप्रोच)
(b) Top-down approach/टॉप डाउन दृष्टिकोण (अप्रोच)
(c) Functional incrimination/कार्यात्मक इंक्रीमिनेशन
(d) Bottom-up approach
बॉटम-अप दृष्टिकोण (अप्रोच)

HTET PGT C.S. 2021

Ans. (a) : Incremental testing means testing the software in small increments adding and testing modules step by step.
(b) Top-down approach– Modules are integrated from top (main module) to bottom (Sub modules) testing as we go.
(d) Bottom-up approach– Modules are integrated from the bottom (low level) to the top (mainmodule) testing step by step.
(c) Functional Incrimination– Adding functionality piece by piece and testing incrementally.
(a) Big Bang approach– This is not an incremental approach because is no step-by-step testing.

187. In which stage individual components are integrated and ensured that they are error free to meet customer requirements ?
वह कौन सा चरण है जिसमें सभी घटकों को एकीकृत किया जाता है और सुनिश्चित किया जाता है कि वे ग्राहकों की आवश्यकताओं को पूरा करने के लिए त्रुटि मुक्त है?

Sarkari Resource

(a) Coding/कोडिंग
(b) Testing/टेस्टिंग
(c) Design/डिजाइन
(d) Implementation/इम्प्लीमेंटेशन

HTET PGT C.S. 2021

Ans. (b) : Testing– After coding, individual module are integrated and tested together to ensure that they work correctly and meet customer requirements. This includes integration testing + system testing.

188. What is the alternate name for white box testing ?
व्हाइट बॉक्स परीक्षण का वैकल्पिक नाम क्या है?

(a) Graph Testing/ग्राफ टेस्टिंग
(b) Basic Path Testing/बेसिक पाथ टेस्टिंग
(c) Glass Box Testing/ग्लास बॉक्स टेस्टिंग
(d) Dataflow Testing/डेटाफ्लो टेस्टिंग

HTET PGT C.S. 2020

Ans. (c) : Glass box testing (also called white Box testing) is a software testing technique where the internal structure, logic and code of the program are tested. The tester knows the program's internal details and design test cases accordingly.

189. In context of software testing, which of the following testing is not a level or type of development testing ?
सॉफ्टवेयर टेस्टिंग के संदर्भ में, निम्न में से कौन-सी टेस्टिंग डेवलपमेंट टेस्टिंग का एक स्तर अथवा प्रकार नहीं है?

(a) System testing/सिस्टम टेसिटंग
(b) Unit testing /यूनिट टेस्टिंग
(c) Release testing/रिलीज टेस्टिंग
(d) Component testing/कम्पोनेंट टेस्टिंग

HTET PGT C.S. 2019

Ans. (c) : Release Testing is performed after development testing, usually a part of acceptance testing. Its goal is to check whether the system is ready for delivery to customers. It focuses on validating the product in a production like environment, not during development.

190. Choose invalid statement in context of software testing :
सॉफ्टवेयर टेस्टिंग के संदर्भ में अनुचित कथन चुनें:

(a) System testing is usually done by the development team./सिस्टम टेस्टिंग सामान्यतः डेवलपमेंट टीम द्वारा की जाती है।
(b) Alpha testing is a type of user testing. अल्फा टेटिटंग यूजर टेस्टिंग का एक प्रकार है।
(c) Release testing is usually a blackbox testing. रिलीज टेस्टिंग सामान्यतः एक ब्लैक-बॉक्स टेस्टिंग होती है।
(d) Release testing is performed after acceptance testing./रिलीज टेस्टिंग ऐक्सेटेंस टेस्टिंग के बाद निष्पादित की जाती है।

HTET PGT C.S. 2018

Ans. (d) : System Testing– System testing is generally done by the development/testing team to verify the whole integrated system.
Alpha Testing– It is a form of user testing but conducted in the development environment with limited users.
Release Testing– It is black-box in nature, focusing on system behavior without internal code knowledge.

191. The goal of _____ testing is test the internal logic of the modules.

(a) Beta (b) Acceptance
(c) Black-box (d) Unit

Haryana PGT C.S. Asist. Prof. 2024

Ans. (d) : Unit Testing focuses on testing individual modules or components of the software in isolation. Its main goal is to check the internal logic, correctness, and functioning of code segments such as functions, methods, or classes.

192. Fault base testing technique is

(a) Unit testing (b) Beta testing
(c) Stress testing (d) Mutation testing

UGC NET C.S. December 2021

Ans. (d) : Mutation testing purpose is to evaluate the quality of the case that should be able to fail the mutant code hence this method is also known as fault based testing as it used to produce an error in the program that why we can say that the mutation testing is performed to check the efficiency of the test cases.

193. Which testing is used to validate that a change has not affected some old functionality of the system?

(a) Unit testing (b) Regression testing
(c) Integration testing (d) White-box testing

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : Regression testing– performed after modification, bug fixes, or enhancements. Its purpose is to ensure that new changes have not broken or affected previously working functionality. Testers rerun previously passed test cases to confirm the system still behaves correctly.

194. In _____ the test plan is developed directly from the program specification and several test cases are developed from it.
_____ में टेस्ट प्लान सीधे प्रोग्राम विनिर्देश से विकसित किए जाते हैं और इससे कई टेस्ट केसेज़ भी विकसित किए जाते हैं।

(a) White Box Testing (b) Black Box Testing
(c) Grey Box Testing (d) Blue Box Testing

HTET PGT C.S. 30.07.2025

Ans.(b): Black Box Testing– Black Box Testing is done without looking at the internal code. The tester only focuses on the input based on the requirements. A test plan is created directly from the program specification, and test cases are written from it.
White Box Testing– depends on analyzing the internal code and structure.
Grey Box Testing– is a mix of both, where particle knowledge of the system is used.
Blue Box Testing– is not a standard method in software development.

Sarkari Resource

195. Let A represent the number of errors detected and B be the number of test cases used. what are the two fundamental goals of the testing activity?

(a) Maximize A and minimize B.
(b) Minimize A as well as B.
(c) Maximize A as well as B.
(d) Minimize A and maximize B.

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : First Goal of Testing- Maximizing Error Detection– The primary aim of software testing is to detect as many errors as possible before the system is delivered to the customer. It (A) (errors found) is high, it means the testing process is successful in uncovering defects that could otherwise cause failure in real-world usage. Hence, A should be maximized.

Second Goal of Testing-Minimizing the Effort– While finding errors is essential, the testing process should also be efficient. This means we should not needlessly use a very large number of test cases. If fever test cases (B) can detect more errors, then the testing process is considered efficient. Therefore B should be minimized.

A = Number of errors detected
B = Number of test cases used.

196. Fault tolerance achieved in distributed system by_____
_____ के द्वारा डिस्ट्रीब्यूटेड सिस्टम दोष सहिष्णुता प्राप्त की जाती है।

(a) Hardware redundancy/हार्डवेयर अतिरेक
(b) Software recovery/सॉफ्टवेयर रिकवरी
(c) Both (a) and (b)/दोनों (a) और (b)
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) : Fault Tolerance means the ability of a distributed system to continue functioning correctly even if some of its components fails.

Hardware Redundancy and Software Recovery– In practice both hardware redundancy and software recovery are used together to ensure strong fault tolerance.

For example– a cloud service may use redundant servers (hardware redundancy) and also keep regular backups with recovery protocols (software recovery).

197. System testing performed by the development team is called _____ where as the system testing performed by a friendly set of customers is called _____.
डेवलपमेंट टीम द्वारा की जाने वाली सिस्टम टेस्टिंग _____ कहलाती है जबकि ग्राहक के मित्रवत समूह द्वारा की जाने वाली सिस्टम टेस्टिंग _____ कहलाती है।

(a) β-testing, acceptance testing
β-टेस्टिंग, एक्सेप्टेंस टेस्टिंग
(b) α-testing, acceptance testing
α- टेस्टिंग, एक्सेप्टेंस टेस्टिंग
(c) α-testing, β-testing/α-टेस्टिंग, β-टेंस्टिंग
(d) β-testing, α-testing/β-टेस्टिंग, α-टेस्टिंग

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) :
1. Alpha Testing (α-testing)– Alpha testing is the process performed by the development team inside the company (in house testing). Its main goal is to ensure that the software works. Correctly and that major bugs do not reach customer.
2. Beta Testing (β-testing)– Beta testing is performed by a selected group of real customer or friendly users outside the development team.
3. Key- Difference–
Alpha testing– Done in-house by the development team.
Beta testing– done by external customers in real-world conditions.

198. Software safety is quality assurance activity that focuses on hazards that

(a) Affect the reliability of a software component
(b) May cause an entire system to fail
(c) May result from user input errors
(d) Prevent profitable marketing of the final product

UGC NET C.S. June-2016 (Paper-III)

Ans : (b) Software safety is a software quality assurance activity that focuses on the identification and assessment to potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software process, software design factures can be specified that will either eliminate or control potential hazards.

199. For a program of k variables, boundary value analysis yields _____ test cases.

(a) $4k - 1$ (b) $4k$
(c) $4k + 1$ (d) $2^k - 1$

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (c) A program of k variables boundary value analysis yields 4*k + 1 test Robustness testing yields 6*k + 1 worst case testing yields 5*k.

200. The extent to which a software performs its intended functions without failures, is termed as :

(a) Robustness (b) Correctness
(c) Reliability (d) Accuracy

UGC NET C.S. Augest-2016 (Paper-II)

Ans : (c) Reliability is the probability of failure free operation of a system over a specified time within a specified environment for a specified purpose.

201. Software Testability is simply how easily a software program can be tested. Which one characteristics does not lead to testable software?

(a) Observability (b) Controllability
(c) Repairability (d) Understandability

UGC NET C.S. December-2024

Ans. (c) : Repairability:- On the other hand is about how easily you can fix defects once they are found while important for software in the first place.

- **Observability:-** If you can easily observe the software's behaviour (through logs, outputs, or internal states), you can more easily verify if it's working as expected.
- **Controllability:** If you can easily control the inputs and environment of the software, you can perform more comprehensive test by simulating various scenario.

Sarkari Resource

• **Understandability:-** If the software is well documented and easy to understand, testers can more quickly grasp its functionality and design test case effectively.

202. What is the other name of white-box testing?
व्हाइट-बॉक्स टेस्टिंग का दूसरा नाम क्या है?

(a) Red-box testing/रेड-बॉक्स टेस्टिंग
(b) Glass-box testing/ग्लास-बॉक्स टेस्टिंग
(c) Open-box testing/ओपन-बॉक्स टेस्टिंग
(d) Close-box testing/क्लोज-बॉक्स टेस्टिंग

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : White-box Testing is a Software Testing Technique that involves testing the internal structure and working of a software Application. The tester has access to the source code and uses this knowledge to design test cases that can verify the correctness of the software at the code level.
Other names for white box testing include clear box testing, transparent box testing, structural testing and Glass Box testing.
Red-box texting– This is not a common term for white box testing. Read box testing might refer to error detection or other specific uses of the color red in testing contexts.
Open-box testing– While sometimes used interchangeably with white box testing, open box can also imply a broader approach where the tester has access to more information than just the source code, such as design documents or architecture.
Close-box testing– This is actually the term used to describe black box testing, where the tester only knows the functionality of the system from a user's perspective without access to the internal workings.

203. Equivalence partitioning is a _____.
एक्विवैलेन्स पार्टिशनिंग एक_____ है।

(a) White box testing method
व्हाईट बॉक्स टेस्टिंग मेथड
(b) Black box testing method
ब्लैक बॉक्स टेस्टिंग मेथड
(c) Comparison testing method
कंपैरिजन टेस्टिंग मेथड
(d) Loop testing Method
लूप टेस्टिंग मेथड

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : Equivalence Partitioning is a black box testing technique used to divide input data into different partitions (or classes) where test cases from each partition are expected to behave similarly. They idea is that if one test case in a partition works all others in that partition should also work.
(a) White Box Testing Method– Examines the internal structure, logic and code of the software.
(c) Comparison Testing Method– A method where two or more versions of the same software are sun with the same input and their output are compared.
(d) Loop Testing Method– Loop testing deals with control structured not input range partitioning.

204. What is the full form of ROCOF in software Reliability ?
सॉफ्टवेयर रिलाइबिलिटी में आर.ओ.सी.ओ.एफ का असंक्षिप्त रूप क्या है?

(a) Ratio of occurrence of fault
रेशियो ऑफ ऑकरन्स ऑफ फॉल्ट
(b) Rate of occurrence of fault
रेट ऑफ ऑकरन्स ऑफ फॉल्ट
(c) Rate of occurrence of failure
रेट ऑफ ऑकरन्स ऑफ फेलियर
(d) Ratio of occurrence of failure
रेशियो ऑफ ऑकरन्स ऑफ फेलियर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : ROCOF⇒ Rate of Occurrence of Failure.
In software reliability engineering, ROCOF measures how often failures occur in a given time period during system operation. It is usually expressed as the number of failure per unit time (Example– Failures / hour).
A high ROCOF means failures are frequent– Low reliability
A low ROCOF means failures are rare– High reliability.

205. In boundary value analysis testing method, if we want to design a test case that computes the square root of integer values in the range from 1 to 4000, then the test case must include what values?
बाउंड्री वैल्यू एनालिसिस टेस्टिंग मेथड में, यदि हम एक टेस्ट केस डिजाइन करना चाहते हैं जो 1 से 4000 तक के पूर्णांक मानों का वर्गमूल कंप्यूट करता है, तो टेस्ट केस में कौन से मान शामिल होने चाहिए?

(a) [0, 1, 3999, 4000] (b) [1, 2, 3999, 4000]
(c) [0, 1, 4000, 4001] (d) [1, 2, 4000, 4001]

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : Boundary Value Analysis (BVA) focuses on testing at the edges of the valid input range and just outside those edges.
Here–
Valid rant = 1 to 4000
Lower boundary :
Minimum valid → 1
Just above minimum → 2
Upper boundary
Maximum valid → 4000
Just above maximum → 4001
So the values are 1, 2, 4000, 4001

206. What is the other name of software updates?
सॉफ्टवेयर अपडेट का दूसरा नाम क्या है?

(a) Patching/पैचिंग (b) Dropping/ड्रॉपिंग
(c) Gaining/गेनिंग (d) Sandboxing/सैंडबॉक्सिंग

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : Patching is another name for software updates, where patches are applied to fix bugs, improve functionality or address security vulnerabilities.

207. In which testing strategy requirements established during requirements analysis are validated against developed software?

(a) Validation testing (b) Integration testing
(c) Regression testing (d) System testing

UGC NET C.S. June-2015 (Paper-II)

Ans. (a) : Validation is "Are we building the right product? According to the capability maturity model
Software validation : The process of evaluating software during or at the end of the development process to determine whether it satisfies specified requirements.

Sarkari Resource

208. Regression testing is primarily related to

(a) Functional testing
(b) Development testing
(c) Data flow testing
(d) Maintenance testing

UGC NET C.S. June-2014 (Paper-II)

Ans : (d) Because, regression testing is the process of testing changes to computer program to make sure that the older programming still works with the new changes. This happens only in case of maintenance. So it relates to the maintenance testing, hence the answer is option (d).

209. Which one of the following is not a definition of error?

(a) It refers to the discrepancy between a computed, observed or measured value and the true., specified or theoretically correct value.
(b) It refers to the actual output of a software and the correct output.
(c) It refers to a condition that causes a system to fail.
(d) It refers to human action that results in software containing a defect or fault.

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, Error mathematical is the difference between the observed value and approximate value and true value of a quantity.
Error is used to refer to human action that results in software containing a defector fault whereas failure refers to a condition that causes a system to fail.

210. Software testing is

(a) The process of establishing that errors are not present.
(b) The process of establishing confidence that a program does what it is supposed to do.
(c) The process of executing a program to show that it is working as per specifications.
(d) The process of executing a program with the intent of finding errors

UGC NET C.S. June-2014 (Paper-III)

Ans. (d) : Because, Software testing is a process of executing a program or application with the intent of finding errors (the software bugs) and the main objective of software testing is to find any bug present in the software program. Besides, through the process of testing. One can determine the strengths and weaknesses of the program in other criteria such as path testing, dependency testing etc.

211. Which of the following is correct?

I. Two phase locking is an optimistic protocol
II. Two phase locking is pessimistic protocol
III. Time stamping is an optimistic protocol
IV. Time stamping is pessimistic protocol

(a) I and III (b) II and IV
(c) I and IV (d) II and III

UGC NET C.S. December-2014 (Paper-III)

Ans : (d) Two phase locking is pessimistic protocol (assumes bad thing going to happen) since it applies locks (for safety)
Time stamping is an optimistic protocol (assumes good thing going to happen) since it uses no locking. If bad things happen, then they are handled later.

212. Adaptive maintenance is a maintenance which _____

(a) Correct errors that were not discovered till testing phase
(b) Is carried out to port the existing software to a new environment
(c) Improves the system performance
(d) Both (b) and (c)

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Maintenance is a modification of a software product performed after delivery to keep a software product usable in a changed or changing environment.

213. Which of the following statements is/are FALSE with respect to software testing?

S1 : White-box tests are based on specifications; better at telling whether program meets specification, better at finding errors of omission
S2 : Black-box tests are based on code; better for finding crashes, out of bounds errors, file not closed errors.
S3 : Alpha testing is conducted at the developer's site by a team of highly skilled testers for software that is developed as a product to be used by many customers

(a) Only S1 and S2 are FALSE
(b) Only S1 and S3 are FALSE
(c) Only S2 and S3 are FALSE
(d) All of S1, S2 and S3 are FALSE

UGC NET C.S. November-2017 (Paper-III)

Ans : (a) Alpha testing is conducted at the developer's site by a team of highly skilled testers for software that is developed as a product to be used by many customers statement about white box testing and black box testing are not correct. Only the last statement is correct.

214. Which of the following statements(s) is/are TRUE with regard to software testing?

I. Regression testing technique ensures that the software product runs correctly after the changes during maintenance.
II. Equivalence partitioning is a white-box testing technique that divides the input domain of a program into classes of data from which test cases can be derived.

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

UGC NET C.S. June-2017 (Paper-III)

Ans : (a) Regression testing is a type of s/w testing that verifies that s/w previously developed & tested still performs correctly even after it was changed with other software.
Equivalence portioning : is Black box testing method to divide input domain into categories for better test coverage.

215. Which of the following are facts about a top-down software testing approach?

I. Top-down testing typically requires the tester to build method stubs.
II. Top-down testing typically requires the tester to build test drivers.

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

UGC NET C.S. June-2017 (Paper-III)

Sarkari Resource

Ans : (a) Top Down Testing : In this approach testing is conducted from main module to sub module. If the sub module is not developed a temporary program called STUB, used fro simulate the sub module.
Bottom up testing : In this approach testing is conducted from sub module to main module, if the main module is not developed a temporary program called DRIVERS is used to simulate main module.

216. What is the normal order of activities in which traditional software testing is organized ?
(A) Integration Testing
(B) System testing
(C) Unit Testing
(D) Validation Testing
Code :
(a) (C), (A), (B), (D) (b) (C), (A), (D), (B)
(c) (D), (C), (B), (A) (d) (B), (D), (A), (C)

UGC NET C.S. November-2017 (Paper-II)

Ans : (a)
Bussiness need $\xleftarrow[\text{validation}]{}$ Solution assesment
Production vision $\xleftarrow[\text{validation}]{}$ Product owner acceptance
Bussines analysis document $\xleftarrow[\text{validation}]{}$ User testing
Technical specification $\xleftarrow[\text{validation}]{}$ System testing
Technical Design $\xleftarrow[\text{validation}]{}$ Integration testing
Unit specification $\xleftarrow[\text{validation}]{}$ Unit testing

217. Which of the following testing techniques ensures that the software product runs correctly after the changes during maintenance?
(a) Path Testing (b) Integration Testing
(c) Unit Testing (d) Regression Testing

UGC NET C.S. November-2017 (Paper-II)

Ans : (d) Regression testing is a type of software testing which verifies that software which was previously developed and tested still performs the same way it was changed with other software. Changes may include software enhancements, patches etc.

218. In what sequence the initialization, testing and execution of body is done while using do-while loop?
A. Commenting
B. Execution of the body
C. Initialisation
D. Testing the condition
Choose the correct answer form the following
(a) D, B, C (b) D, C, B
(c) C, A, B (d) C, B, D

UGC NET C.S. December-2022

Ans. (d) : Do while loop is a control flow statement that executes a block of code and then either repeats the block or exits the loop depending on a given Boolean condition.
So the sequence in a do while loop is option (d) Initialization, execution of body, testing the condition.

219. Modifying the software to match changes in the ever changing environment is called as
(a) Adaptive maintenance
(b) Corrective maintenance
(c) Perfective maintenance
(d) Preventive maintenance

UGC NET C.S. December-2022

Ans. (a) : Type of software maintenance corrective maintenance
Corrective maintenance aims to correct any remaining errors regardless of where they may cause specifications, design, coding testing and documentation etc.
- **Adaptive maintenance**–It contains modifying the software to match changes in the ever changing environment.
- **Preventive Maintenance**–It is the process by which we prevent our system from being obsolete. It involves the concept of reengineering & reverse engineering in which an old system with old technology is re-engineered using new technology. This maintenance prevents the system from dying out.
- **Perfective Maintenance**–It defined improving processing efficiency or performance or restricting the software to enhance changeability. This may contain enhancement of existing system functionality improvement in computational efficiency.

220. For a function of two variables, boundary value analysis yields,
(a) 4n+3 test cases (b) 4n+1 test cases
(c) n+4 test cases (d) 2n+4 test cases

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Ans. (b) : A program of n variables Boundary value analysis yields 4n + 1 test cases Robustness testing yields 6n + 1 test cases worst case testing yields 5n.

221. Which one of the following is not typically provided by Source Code Management Software ?
(a) Synchronisation
(b) Versioning and Revision history
(c) Syntax highlighting
(d) Project forking

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : Source code management software provides–
Synchronization, project forking, versioning, revision hystory·
But Syntax highlighting is not part of source code management software.

222. In software testing, beta testing is the testing performed by _____.
(a) Potential customers at the developer's location
(b) Potential customers at their own locations
(c) Product developers at the customer's location
(d) Product developers at their own locations

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Ans. (b) : Beta testing is a type of user acceptance testing among the most crucial testing, which performed before the release of the software. Beta testing is a type of field test. This testing performs at the end of the software testing life cycle. This type of testing can be considered as external user acceptance testing.
Beta testing always is done after the alpha testing, and before releasing it into the market. Beta testing is a black-box testing Beta testing performs in the absence of tester and the presence of real users. Beta testing generally is done for testing software product like utilities, operating systems, and applications, etc.

Sarkari Resource

223. Software validation mainly checks for inconsistencies between
(a) Use cases and user requirements
(b) Implementation and system design blueprints
(c) Detailed specifications and user requirements
(d) Functional specifications and use cases

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : Software validation means "are we building the right product".
Validation means that product meets the customers expectations. It goes beyond checking it meats its specification as system specifications don't always accurately reflect the real need of users.
Validation refers to the process of evaluating software at the end of its development to ensure that it is free from failure and implies with its requirement A failure is defined as incorrect product behaviors often the validation occurs through the utilization of various testing approaches.
There are two ways to perform software validation internal and external.
1– Internal validation assumes that goals of stakehoding were correctly understood.
2– External validation is performed by asking the stake holders if the software meets their needs.

224. Software products need adaptive maintenance for which of the following reasons?
(a) To rectify bugs observed while the system is in use
(b) When the customers need the product to run on new platforms
(c) To support the new features that users want it to support
(d) To overcome wear and tear caused by the repeated use of the software

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (b) : Maintenance :- Maintenance can be referred as a process of enhancement in the software product according to the changing requirements of users.
Four types of maintenance :
1– Adaptive : Modifying the system to cope with changes in the software environment (DBMS, OS),
2– Perfective : implementing new or changed user requirements which concern functional enhancements to the software.
3– Corrective : diagnosing and fixing errors, possibly ones found by users.
4– Preventive : Increasing software maintainability or reliability to prevent problems in the future.

225. In context of software testing, which of the following is not a type of user testing?
सॉफ्टवेयर टेस्टिंग के सन्दर्भ में, निम्न में से कौन-सा यूजर टेस्टिंग का एक प्रकार नहीं है?
(a) Performance Testing/परफॉरमेंस टेस्टिंग
(b) Alpha Testing/अल्फा टेस्टिंग
(c) Beta Testing/बिटा टेस्टिंग
(d) Acceptance Testing/एक्सेप्टेंस टेस्टिंग

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Performance Testing is a type of non functional testing that evaluates the speed, responsiveness, and stability of a system under a workload. It is not categorized as user testing.
• Alpha Testing, Beta Testing and Acceptance Testing all involve end-users or user perspectives in some form and are types of user testing.

226. Which of the following is not a category of maintenance?
निम्न में से कौन सी श्रेणी मेंटेनेंस की नहीं है?
(a) Corrective/करेक्टिव (b) Effective/इफेक्टिव
(c) Adaptive/एडपटिव (d) Perfective/परफेक्टिव

RPSC C.S. 2016 (PAPER-I)

Ans. (b) : The main categories of software maintenance are:
(i) Corrective maintenance– Fixing bugs or defect.
(ii) Adaptive Maintenance– Modifying the software to keep it usable in a changed or changing environment.
(iii) Perfective Maintenance– Enhancing performance or maintainability
Hence Effective is not a recognized category of software maintenance.

227. Which of the following is the method of black box testing?
निम्न में से कौन सी ब्लैक बॉक्स टेस्टिंग की विधि है?
(a) Boundary value analysis/बाउंड्री वैल्यू एनालिसिस
(b) Basic path testing/बेसिक पाथ टेस्टिंग
(c) Code path testing/कोड पाथ टेस्टिंग
(d) None of these/इनमें से कोई नहीं

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Boundary Value Analysis (BVA) is a black box testing technique where test cases are designed based on boundary values of Input domains.
• Basic Path Testing and Code Path Testing are white box texting methods because they require knowledge of the internal code structure.

228. Which of the following is fault base testing technique?
निम्न में से कौन सी फाल्ट बेस टेस्टिंग तकनीक है?
(a) Unit Testing/यूनिट टेस्टिंग
(b) Beta Testing/बीटा टेस्टिंग
(c) Stress Testing/स्ट्रेस टेस्टिंग
(d) Mutation Testing/म्युटेशन टेस्टिंग

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : Mutation Testing is a fault-based testing technique. It involves intentionally introducing small faults (Mutations) into the code to check if the existing test cases can detect them. The goal is to evaluate the effectiveness of the test cases in findings bugs.
Unit Testing is a white-box technique, not specifically fault-based.
Beta Testing is performed by end-users, focuses on usability and feedback, not fault based.
Stress Testing is performance testing under extreme conditions, not fault-based.

Sarkari Resource

229. Defect removal efficiency is
डिफेक्ट रिमूवल इफिसिएंसी (DRE) है-

(a) $\frac{E}{E+D}$ (b) E X D

(c) $\frac{E+D}{E}$ (d) E – D

RPSC C.S. 2016 (PAPER-I)

Ans. (a) : Defect Removal Efficiency (DRE) is calculated by dividing the number of defects removed before release (E) by the total number of defects found; both before & after release (E + D).

Therefore, DRE = $\frac{E}{E+D}$.

230. Which of the following activities is not involved in configuration management of a software system product?
निम्न में से कौन सी क्रिया सॉफ्टवेयर सिस्टम प्रोडक्ट के कान्फ्यूगुरेशन मैनेजमेंट में शामिल नहीं है?

(a) Change Management/चेंज मैनेजमेंट
(b) Version Management/वर्शन मैनेजमेंट
(c) Release Management/रिलीज मैनेजमेंट
(d) Quality Management/क्वालिटी मैनेजमेंट

RPSC C.S. 2016 (PAPER-I)

Ans. (d) : Configuration Management in software engineering primarily includes–
(1) Change Management– Handling changes in the software system.
(2) Version Management– Keeping track of different versions of software components.
(3) Release Management– Managing software releases and deployment.
Quality Management, however is a broader process focused on ensuring the product meets required standards and customer expectations.

231. Statement I : Corrective maintenance is universally used to refer to maintenance for fault repair.
कथन I : सुधारात्मक देखभाल, सार्वभौमिक रूप से गलती की मरम्मत करने के लिए प्रयुक्त किया जाता है।
Statement II : Perfective Maintenance sometimes means perfecting the software by Implementing new requirement. Which of the following is correct?
कथन II : परफेक्टीव देखभाल का मतलब कभी-कभी नयी आवश्यकता को लागू कर सॉफ्टवेयर को उत्तम करने से है। इनमें से कौन सा सही है?

(a) Statement I is true and Statement II is false
कथन I सही है और कथन II गलत है
(b) Statement I is false and Statement II is true
कथन I गलत है और कथन II सही है
(c) Statement I is true and Statement II is true
कथन I सही है और कथन II सही है
(d) Statement I is false and Statement II is true
कथन I गलत है और कथन II गलत है

RPSC C.S. 2016 (PAPER-I)

Ans. (c) : Corrective maintenance refers to fixing faults or defects in the software after it has been delivered. This is indeed the most common understanding of corrective maintenance perfective maintenance involves improving or enhancing the software by implementing new or changed user requirements, which can include performance improvements or adding new features.

232. Software patch is defined as _____
सॉफ्टवेयर पैच को _____ के रूप में परिभाषित किया गया है

(a) Daily or routine Fix
(b) Required or Critical Fix
(c) Emergency Fix
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) A software patch is an emergency fix that works on an outdated version when any vulnerability in encountered.

233. Semaphores are used to solve the problem of _____.
सेमाफोर्स का उपयोग _____की समस्या को हल करने के लिए किया जाता है।

(a) Mutual exclusion/म्युच्युअल एक्सक्लूसन
(b) Race condition/रेस (Race) दशा
(c) Process synchronization/प्रोसेस सिंक्रोनाइजेशन
(d) The belady problem/बेलाडी समस्या

KVS PGT C.S. 2017

Ans. (c) : Semaphores are integer variables that are used to solve the critical section problem by using two atomic operations, wait and signal that are used for process synchronization.

234. Which testing approach involves testing individual components or units of code?
कौन-सा परीक्षण दृष्टिकोण कोड के व्यक्तिगत घटकों या यूनिट्स के परीक्षण में शामिल होता है?

(a) Integration Testing/इंटीग्रेशन टेस्टिंग
(b) System Testing/सिस्टम टेस्टिंग
(c) Unit Testing/यूनिट टेस्टिंग
(d) Acceptance Testing/एक्सेप्टेंस टेस्टिंग

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : In unit testing, small parts of the software (such as functions or modules) are tested separately to ensure their functionality.

235. What is the main goal of System Testing?
सिस्टम टेस्टिंग का मुख्य उद्देश्य क्या है?

(a) Testing individual components in isolation
व्यक्तिगत घटकों का अलग-अलग परीक्षण
(b) Testing the integration between different software components/विभिन्न सॉफ़्टवेयर घटकों के बीच एकीकरण का परीक्षण
(c) Testing software functionally from the user's perspective/उपयोगकर्ता के दृष्टिकोण से सॉफ़्टवेयर को कार्यात्मक रूप से परीक्षण करना
(d) Identifying defects in the source code
स्रोत कोड में दोषों की पहचान करना

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : The main goal of system testing is to ensure that entire software system works according to the user's requirements. In this the system is tested as a whole unit rather than as separate parts.

Sarkari Resource

10. Internet Technology, Web Design and Web Technology

(i) Internet Technology and Protocol, Internet Connectivity, Internet Network, Services on Internet Electronic Mail, Current Trends on Internet

1. Which of the following is the best description of the word internet ?
निम्नलिखित में से कौन-सा शब्द इंटरनेट का सबसे अच्छा वर्णन है?

(a) A personal communication system
एक व्यक्तिगत संचार प्रणाली
(b) A corporate communication system
एक कॉर्पोरेट संचार प्रणाली
(c) Indian Government communication system
भारत सरकार की संचार प्रणाली
(d) A large network of networks
नेटवर्क्स का एक बड़ा नेटवर्क

HTET PGT C.S. 2020

Ans. (d) : Internet is a global Network which is the network of networks. In internet every devices are connected to each other in order to share information & resources.

2. A URL contains :
एक यू आर एल में शामिल है?

(a) Protocol identifier/प्रोटोकॉल पहचानकर्ता
(b) www
(c) Unique registered domain name
अद्वितीय पंजीकृत डोमेन नाम
(d) All of these/इनमें से सभी

HTET PGT C.S. 2021

Ans. (d) : A URL stands for 'Uniform Resource Locator'. A URL contain protocol, Host name, domain name with 'WWW'. It is used to uniquely identify the website on web server.

3. Specialized programs that assist users in locating information on the Web are called _____
वेब पर जानकारी का पता लगाने में उपयोगकर्ताओं की सहायता करने वाले विशिष्ट प्रोग्राम _____कहलाते है।

(a) Locator engines/लोकेटर इंजन
(b) Web browsers/वेब ब्राउजर
(c) Resource locators/रिसोर्स लोकेटर
(d) Search engines/सर्च इंजन

HTET PGT C.S. 2021

Ans. (d) : Specialized programs that assist users in locating information on the web are called Search Engines. A search engine is a kind of website through which user can search the content available on the Internet.

4. Which technique allows a group of users located around the globe to talk and interact with other?
कौन-सी तकनीक दुनिया भर में स्थित उपयोगकर्त्ताओं के एक समूह को एक-दूसरे से बातचीत करने की अनुमति देती है?

(a) Video Conferencing
(b) Web Hosting
(c) Customer Support Service
(d) Website Exploration

HTET PGT C.S. 30.07.2025

Ans. (a) : A technique called video conferencing allows group of users worldwide to talk and interact with each other in real-time, bridging geographical gaps by enabling face-to-face communication through visual and audio cues.

5. Which of the following is not a web browser ?
निम्न में से कौन सा एक वेब ब्राउजर नहीं है?

(a) MOSAIC (b) SAFARI
(c) MINIX (d) OPERA

HTET PGT C.S. 2022

Ans. (c) : A web browser is an application software which is used to run the websites. It basically interprets the HTML language some examples of Web Browser, Crome, Opera, Safari, Mosaic etc.

6. _____ are used to identify a user who returns to a website.
_____ का प्रयोग एक वेबसाइट पर पुनः आने वाले उपभोक्ता की पहचान के लिए किया जाता है।

(a) Blogs/ब्लॉग्स
(b) Identifiers/आइडेंटिफायर्स
(c) Scripts/स्क्रिप्ट्स
(d) Cookies/कूकीज

HTET PGT C.S. 2022

Ans. (d) : Cookies are used to identify a user who returns to a website. It is a small text storage area which stores the user information when he visits the website in browser.

7. _____ gives the location of a resource on the internet.
_____ इंटरनेट पर किसी रिसोर्स की लोकेशन देता है।

(a) URL
(b) TCP/IP

(c) Domain name/डोमेन नेम
(d) IP address/आई. पी. एड्रेस

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : URL stands for Uniform Resource Locator. It locates the resources or websites on the web server & Internet so that user can access by URL any particular website from server later on.

8. ____ is a set of rules that governs how two computer system interact.
_____ नियमों का एक सेट है जो नियंत्रित करता है कि दो कंप्यूटर सिस्टम कैसे इंटरैक्ट करते हैं।

(a) Syntax/सिंटैक्स (b) Rules/रूल्स
(c) Protocol/प्रोटोकॉल (d) Grammar/ग्रामर

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : Protocol is the set of rules that governs how two system interact to each other and communicate. These are many protocols such as HTTP, HTTPS, SMTP, TCP, IP, etc.

9. _____ lets users to show their location with websites they trust.
_____ उपयोगकर्ताओं को उन वेबसाइटों को अपना लोकेशन दिखाने देता है जिन पर वे भरोसा करते हैं।

(a) Geolocation/जियोलोकेशन
(b) Geotrust/जियोट्रस्ट
(c) Loctrust/लोकट्स्ट
(d) OLTrust/OL ट्रस्ट

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : Geolocation is the technology that lets users share their location with websites they trust, typically through browser prompts and based on GPS, wifi, or IP address data.

10. _____ URL is used to send electronic mail.
इलेक्ट्रानिक मेंल भेजने के लिए _____ URL का उपयोग किया जाता है।

(a) Mailto (b) Eto
(c) Email UR (d) EURL

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : A mailto URL is used to create links that open the user's default email client to send an email.
For Example– mail to : someone@example.com

11. Which of the following is not a search engine?
निम्नलिखित में से कौन-सा एक सर्च इंजन नहीं है?

(a) Yahoo/याहू
(b) Google/गूगल
(c) Internet explorer/इंटरनेट एक्सप्लोरर
(d) Lycos/लाइकॉस

HTET PGT C.S. 2023

Ans.(c): A search engine is a software system designed to find information on the internet or within a database, allowing users to enter keywords and receive a list of relevant web pages, documents, or other resources, some examples are, Google, Bing, Yahoo, Lycos etc.

12. _____is one of the gateway of internet.
_____इंटरनेट का एक गेटवे है।

(a) Intranet/इंट्रानेट
(b) Internet protocol/इंटरनेट प्रोटोकॉल
(c) Internet service provider/इंटरनेट सर्विस प्रोवाइडर
(d) Web/वेब

HTET PGT C.S. 2023

Ans.(c): Internet Service Provider, as an ISP provides the connection and services that allow users to access and use the Internet.

13. Initially internet was used for_____.
प्रारंभ में इंटरनेट का उपयोग_____किया जाता था।

(a) Medicine/मेडिसिन में
(b) Military intelligence/मिलिटरी इंटलीजेंस में
(c) Communication/कम्युनिकेशन में
(d) Education/एजुकेशन में

HTET PGT C.S. 2023

Ans.(b): The internet was initially used for military intelligence. It was first developed in the late 1960s as ARPANET by the US Military Department ARPA.

14. Which of the following statement(s) is/are true about wire framing of website?
वेबासाइट के वायर फ्रेमिंग के बारे में निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. Showing which content will appear on which page/किस पेज पर कौन सी अंतर्वस्तुयें प्रदर्शित होगी यह दर्शाना

II. Showing navigator between pages
पेजों के बीच में नैविगेटर दर्शाना

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (c) : A website wireframe is a visual guide that outlines the basic structure and layout of a webpage. It's like a blueprint for a website, showing the placement of elements, navigation and content.

15. _____produces a rough outline of what the website will look like when it is done
_____एक मोटे तौर पर रूपरेखा तैयार करता है कि जब यह किया जाएगा तो वेबसाइट कैसी दिखेगी।

(a) Wirefalls/वायरफॉल्स
(b) Wireframes/वायरफ्रेम्स
(c) Webframes/वेबफ्रेम्स
(d) Webflows/वेबफ्लोज

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (b) : Wireframe produces a rough outline of what the website will look like when it is done.

16. Mr. Janaki Raman has joined a financial organization as a chief of operations. On a regular basis, he has to conduct a number of Interactive training sessions for the employees spread across in 13 Countries. Which of the following protocols he must be using for the same ?

Sarkari Resource

श्री जानकी रमन ने 'चीफ ऑफ ऑपरेशन्स' के तौर पर एक वित्तिय संगठन में कार्यभार संभाला है। उन्हें नियमित तौर पर 13 देशों में फैले कर्मचारियों के लिए अनेक इंटरेक्टिव प्रशिक्षण सत्र आयोजित करने होते हैं, वे इसके लिए निम्नलिखित में से किस प्रोटोकॉल का अनिवार्यतः उपयोग कर रहे होंगे?

(a) FTP (b) VOIP
(c) SMTP (d) PPP

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (b) : VOIP stands for Voice Over Internet Protocol. VOIP is a technology that allowing you to make voice calls over a broadband Internet connection instead of an analog (regular) phone line. Some VOIP services allow you to call people using the same service, but others may allow you to call anyone.

17. Which of the following is a part(s) of a URL?
निम्नलिखित मे से कौन सा/से यू.आर.एल का/के भाग है/हैं?

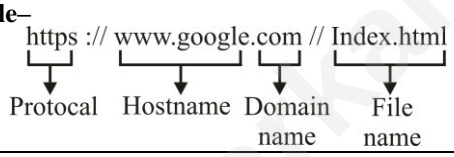
I. Protocol/प्रोटोकॉल
II. Hostname/होस्टनेम
III. Directory and Filename
डाइरेक्ट्री एंड फाइलनेम

(a) I and III/ I तथा III
(b) II and III/ II तथा III
(c) I and II/ I तथा III
(d) I, II and III/ I, II तथा III

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : URL is the unique address of any website which is located on the web server. URL has many parts like protocol, Host name, Domain name and Directory & file name.

Example–



18. What is the full form of SSL in web development?
वेब डेवलपमेंट में एस.एस.एल का संक्षिप्त रूप क्या है?

(a) Secure Sockets Layer/सिक्योर सॉकेट्स लेयर
(b) Secure Server Layout/सिक्योर सर्वर लेआउट
(c) Security Socket Layout/सिक्योर सॉकेट लेआउट
(d) Secure Server Layer/सिक्योर सर्वर लेयर

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : The correct Full form of SSL is web development is secure sockets layer. It is a standard security technology used to establish an encrypted link between a web server and a browser. This ensures that all data passed between them remains private and secure.

19. Which of the following protocols may be used for accessing World Wide Web?
वर्ल्ड वाइड वेब तक पहुँचने के लिए निम्न में से किस प्रोटोकॉल का प्रयोग होता है?

(a) HTTP
(b) SMTP
(c) SLIP
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (a) : HTTP stands for Hyper Text Transfer Protocol, it is used by the web. It is the set of rules for transferring files like text, graphic image, sound, video and other multimedia files etc. on the world wide web.

20. Web pages are created in a language called
जिस भाषा में वेब पेज बनता है, उस भाषा को क्या कहते हैं?

(a) SLIP (b) HTML
(c) HTTP (d) WWW

LT. Grade C.S. 2018

Ans. (b) : HTML stands for Hypertext Markup Language. It is the standard language used to create and design web pages. It defines the structure and layout of content on the web.

21. Which of the following is a valid domain name?
निम्न में से वैध डोमेन का नाम है

(a) .com
(b) .gov
(c) .net
(d) All of the above/उपर्युक्त सभी

LT. Grade C.S. 2018

Ans. (d) : All the options listed are valid top level domains. These are commonly used for different purposes.
.com– It's used for commercial purposes, but widely used for various types of websites.
.gov– Reserved for government related institutions (restricted to authorized entities).
.net– Originally intended for network related entities but now used broadly.

22. MIME stands for
MIME किसका संक्षिप्त रूप है?

(a) Multipurpose Internet Mail Exchange
(b) Multiple Information Main Exchange
(c) Multipurpose Internet Mail Extensions
(d) Multipurpose Information Main Extensions

LT. Grade C.S. 2018

Ans. (c) : MIME stands for Multipurpose Internet Mail Extensions. It's an internet standard that allows users to exchange a variety of data files via email and the web including image, audio, video, and application programs. MIME is an extension of the Simple Mail Transfer Protocol (SMTP) email protocol.

23. URL stands for
URL किसका संक्षिप्त रूप है?

(a) Unique Resource Locator
(b) Universal Resource Locator
(c) Uniform Resource Locator
(d) United Resource Locator

Bihar STET C.S. 18.09.2020 (Shift-I)
LT. Grade C.S. 2018
HTET PGT C.S. 2018

Sarkari Resource

Ans. (c) : A Uniform Resource Locator, Colloquially known as an address on the web, is a reference to a resource that specifies its location on a computer network and a mechanism for retrieving it.

24. Search engine is used
सर्च इंजन का प्रयोग होता है

(a) To search document on web
वेब पर दस्तावेज ढूँढ़ने के लिए
(b) To search the file on host computer
होस्ट कम्प्यूटर पर फाइल ढूँढ़ने के लिए
(c) To search data in a file
एक फाइल में डेटा ढूँढ़ने के लिए
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018

Ans. (a) : A search engine is a software system that is designed to carry out web searches, which means to search the world wide web in a systematic way for particular information specified in a textual web search query. Therefore search engines are used to search documents, video and images.

25. On what type of switching does Internet work?
इंटरनेट किस प्रकार की स्विचिंग पर काम करता है?

(a) Circuit/सर्किट (b) Bit/बिट
(c) Packet/पैकेट (d) Line/लाइन

LT. Grade C.S. 2018

Ans. (c) : Packet switching is used in the internet and most local area networks. The internet is implemented by the internet protocol suite using a variety of link layer technologies. For example Ethernet and frame relay are common. Newer mobile phone technologies also use packet switching.

26. Which of the following is not an Internet Browser ?
निम्नलिखित में से कौन सा एक इंटरनेट ब्राउजर नहीं है?

(a) Acrobat Reader/एक्रोबेट रीडर
(b) Edge/एज
(c) Chrome/क्रोम
(d) Firefox/फायरफॉक्स

NVS PGT- 16.12.2022 (Shift-II)

Ans. (a) : Acrobat Reader– This is not an Internet Browser. In Acrobat Reader we can open PDF documents and print it and also can fill up the PDF form.
Edge–It is a Microsoft Browser. Which is used to visit the website and see the photos videos and documents on the Internet chrome and fire fox are also Internet Browser.

27. Social network analysts use _____ to access Facebook data.
सोशल नेटवर्क विश्लेषण फेसबुक डाटा प्राप्त करने के लिए _____ का उपयोग करते है।

(a) Facebook System Calls/फेसबुक सिस्टम कॉल्स
(b) Facebook APIs/फेसबुक ए पी आई
(c) Facebook Scripts/फेसबुक स्क्रिप्ट्स
(d) Facebook System Libraries
फेसबुक सिस्टम लाइब्रेरीज

KVS PGT C.S. 2018

Ans. (b) : Social network analysis use facebook APIs to access facebook data. Facebook APIs are the primary way to connect with the massive Facebook platform programmatically. Third-party developers create cool facebook apps, games, steam live programmatically, run automated marketing campaigns, manage their business pages all with the help of the facebook APIs.

28. A small piece of text containing login information while working on a browser is known as :
एक ब्राउजर पर काम करते समय लॉगइन की सूचना वाला थोड़ा सा टेक्स्ट क्या कहलाता है?

(a) Antivirus/एंटीवायरस (b) Cookies/कुकीज
(c) Webpage/वेबपेज (d) Website/वेबसाइट

NVS PGT- 16.12.2022 (Shift-II)

Ans. (b) : Cookies is a small text file, which contains login information of user while he works on the browser by Internet. With the help of cookies browser remembers the information of user each time when he visits.

29. The computer jargon WWWW, stands for
कम्प्यूटर शब्दजाल WWWW का पूरा नाम है

(a) World Wide Web Worm
(b) World Wide Wildlife Web
(c) World Wide Women's Web
(d) World Wide Women's Week

Bihar STET C.S. 2020

Ans. (a) : WWWW (World Wide Web Worm) was a initial search engine which was developed in 1994.
Note– It is different from WWW (World Wide Web) which are the networks of web pages on Internet.

30. _____ is an online storage drive which can be browsed and from which items can be shared?
एक ऑनलाइन स्टोरेज ड्राइव है जिसे ब्राउज किया जा सकता है और किस आइटम के साथ साझा किया जा सकता है?

(a) Find my iPhone (b) iWeb Publish
(c) Mobile Me Gallery (d) iDesk

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : iDesk is an online storage drive offered by Apple Mac which can be browsed and from which items can be shared.

31. Network of networks is called?
नेटवर्क का नेटवर्क कहलाता है?

(a) INTERANET
(b) INTERNET
(c) NETWORKING
(d) COMMUNICATION

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (b) : Internet is basically called "Networks of Network". It is the global network where every devices are connected to each other for the purpose of share information and resources.

Sarkari Resource

32. The term "NTFS" refers to which one of the following?
'NTFS' का पूर्ण रूप क्या है?
(a) New Technology File System
न्यू टेक्नोलॉजी फाइल सिस्टम
(b) New Tree File System
न्यू ट्री फाइल सिस्टम
(c) New Table type File System
न्यू टेबल टाइप फाइल सिस्टम
(d) Both A and C
उपरोक्त में से दोनों A और C

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (a) : NTFS (New Technology File System) is a file system used in operating system which is known as support to large file and its security & reliability.

(ii) Web Publishing and Browsing, HTML, Programming Basics, Interactivity Tools Internet

33. Code of webpage is written by using _____.
वेबपेज का कोड___का प्रयोग करके लिखा जाता है।
(a) Hyper text markup language
हाइपर टेक्स्ट मार्कअप लैंग्वेज
(b) 5th generation language/5वीं पीढ़ी की भाषा
(c) Winzip/विनजिप
(d) Perl/पर्ल

HTET PGT C.S. 2021

Ans. (a) : Code of webpage is written by using HTML (Hyper Text Markup Language). HTML is a markup and tag based language. It was developed in 1990's.

34. On clicking on a _____, opens another document/webpage.
_____पर क्लिक करने पर एक और वेबपेज/दस्तावेज खुलता है।
(a) Anchor/एंकर (b) Hyperlink/हाइपरलिंक
(c) Reference/संदर्भ (d) E-mail/ई-मेल

HTET PGT C.S. 2021

Ans. (b) : A Hyperlink is a link which contains the link of another document. By clicking on it user can reach out to another web page or document in HTML language.

35. The root tag in HTML document is called :
HTML डॉक्यूमेंट में रूट टैग का नाम क्या है?
(a) HTML (b) BODY
(c) TITLE (d) HEAD

HTET PGT C.S. 2020

Ans. (a) : The root tag in HTML is '<HTML>'. It is placed at the top of the HTML document.

Example–
```
<HTML>
<HEAD>——</HEAD>
(BODY)
_______
_______
</BODY>
</HTML>.
```

36. Primary author of HTML (Hyper Text Markup Language) is ?
HTML (हाइपर टेक्स्ट मार्कअप भाषा) का प्राथमिक निर्माता कौन था?
(a) Brendan Eich/ब्रेंडन ईच
(b) Tim Berners-Lee/टिम बर्नेर्स ली
(c) Google/गूगल
(d) Peter Norton/पीटर नॉर्टन

HTET PGT C.S. 2020

Ans. (b) : HTML (Hyper Text Markup Language) was developed by sir "Tim Berner Lee" in 1990 as part of his work to create the world wide web. He invented HTML to provide a way to share and structure documents over the internet.

37. To insert images we use_____tag.
इमेज को सम्मिलित करने के लिए हम _____टैग का उपयोग करते हैं।
(a) Picture (b) Pic
(c) Img (d) Image

HTET PGT C.S. 2023

Ans.(c): To insert the image in HTML, <img> tag is used. Img tag have attributes like height, width, alt etc.

38. Which attribute of table sets the width, in pixel, between the edge of cell and its contents?
टेबल की कौन-सी विशेषता सेल के किनारे और उसके कंटेंट के बीच की चौड़ाई पिक्सल में निर्धारित करती है?
(a) Cell padding (b) Cell spacing
(c) Cols (d) Rows

HTET PGT C.S. 30.07.2025

Ans.(a): Cell padding is a space between cell content and its border. We can manage this space width by using cell padding attributes in table.
Syntax– <table cell padding = "5px">------</table>.

39. Choose the correct HTML tag for a large title.
(a) Head (b) H20
(c) H1 (d) H6
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (c) : Headings tag are used to write large text or to write the heading. There are six types of headings are used in HTML. It starts with <h1> tag and goes till <h6> tag. It means <h1>, <h2>, <h3>, <h4>, <h5> and <h6>. Here <h1> tag is the largest heading tag while <h6> tag is the smallest heading tag.

40. Which of the following HTML codes display the content in bold face?
निम्नलिखित में से कौन-सा HTML कोड कन्टेंट को बोल्ड केस में प्रदर्शित करता है?
(a) This text is big
(b) This text is big
(c) This text is big
(d) This text is big <\b>

HTET PGT C.S. 2023

Sarkari Resource

Ans.(b): Bold () tag is used to make the text in Bold format. It makes the larger text compare to the normal text.
Syntax– welcome in HTML .

41. _____lists can be preceded by one of the several bullet style – a closed circle, an open circle or a square.
_____लिस्ट को कई बुलेट स्टाइल जैसे–क्लोज्ड सर्कल, ओपन सर्कल या स्क्वायर में लिखा जा सकता है।

(a) Ordered list/ऑर्डर्ड लिस्ट
(b) Unordered list/अन-ऑर्डर्ड लिस्ट
(c) Vertical list/वर्टिकल लिस्ट
(d) Horizontal list/हॉरिजोन्टल लिस्ट

HTET PGT C.S. 2023

Ans.(b): Unordered list is used to make the list in bullet forms. HTML lists are two type ordered list & Unordered list There are different types of bullet point, we can use in unordered list like a closed circle (•), an open circle (○) or a square (■).

42. In HTML link for e-mail can be created using which of the following ?
एच.टी.एम.एल. में ई-मेल के लिए लिंक निम्न में से किसके प्रयोग से बनाया जा सकता है?

(a) <email = pqr@example.com> This is link to write me an email </email>
(b) <email> <SetID> pqr@example.com </SetID> This is link to write me an email </email>
(c) This is link to write me an email
(d) <a herf = "<SetID> pqr@example.com </SetID"> This is link to write me an email

HTET PGT C.S. 2018

Ans. (c) : HTML <a> tag provides you the option to specify an email address to send an email. While using <a> tag as an email tag we will use mail to : email address along with href attribute.
Example– This is link to write me an email

43. HTML tags must be written within _____.
HTML tags को_____ के अंदर लिखा जाता है।

(a) ( ) (b) [ ]
(c) { } (d) < >

HTET PGT C.S. 2023

Ans. (d) : HTML tags must be written within Angular bracket (< >). Tags are predefined and pre-stored commands in the Web Browser.

44. In HTML the _____ element is used to contain characters that should be super script :
एच.टी.एम.एल. में _____ एलिमेंट, उन वर्णों को समाविष्ट करने में प्रयुक्त होता है, जिनको की सुपर स्क्रिपट होना हो:

(a) <SUPER> (b) <SUP>
(c) <SUPER SCRIPT> (d) <SUP SCR>

HTET PGT C.S. 2018

Ans. (b) : '<sup>' Element is used in HTML to provide the superscript of the text. It places the text above from the normal text.
Example– x^2
x ².

45. In HTML Version 5 the _____ element is appropriate for marking up the titles of works of art, music, movies, books etc.
एचटीएमएल संस्करण-5 में _____ तत्व कला, संगीत, चलचित्र, पुस्तक आदि कार्यो के शीर्षकों को चिह्नित करने के लिए उपयुक्त हे।

(a) <cite > (b) <refer>
(c) <highlight> (d) <bold>

HTET PGT C.S. 2019

Ans. (a) : In HTML-5, the <cite> element is appropriate for marking up the titles of creative works such as books, movies, songs, and paintings. The content within the <cite> tags typically renders in italics by default, but this can be customized with CSS.

46. Consider the following HTML code :
निम्न एचटीएमएल कोड पर विचार करें:
<html> Task Completion Status :
<meter value = "0.8"> 80% Completed
</meter> </html>
Choose true statement(s) about the output displayed on web page :
वेब पेज पर प्रदर्शित आउटपुट के बारे में सत्य कथन/कथनों का चुनाव करें:

I. Text 'Task Completion Status:' is displayed./टेक्सट 'Task Completion Status:' प्रदर्शित होगा।
II. Text '80% Completed' is not displayed. टेक्स्ट '80% Completed' प्रदश्रित नहीं होगा।
III. A graphical gauge, 80% filled with color is displayed./80% रंग भरा हुआ, एक ग्राफिकल गेज प्रदर्शित होगा।

(a) I and II (b) II and III
(c) I and III (d) I, II and III

HTET PGT C.S. 2019

Ans. (d) : The '<meter>' tag defines a scalar measurement within a known range, or a fractional value. This is also known as gauge. The meter tag have some attributes like value, min, max, low, high, and optimum.

47. In HTML Version-5, the _____ declaration tag is used by the web browser to understand the version of the HTML :
एच.टी.एम.एल. संस्करण-5 में डिक्लेरेशन टैग _____ का प्रयोग वेब-ब्राउजर द्वारा एच.टी.एम.एल. के संरकरण को समझने के लिए किया जाता है।

(a) <!DOCTYPE html>
(b) <$DOCUMENT html>
(c) <#DOCTYPE html>
(d) <#DOC html>

HTET PGT C.S. 2018

Sarkari Resource

Ans. (a) : In HTML version-5, the <!DOCTYPE html> declaration tag is used by the web browser to understand the version of the HTML.

48. In HTML the _____ element is used to contain characters that should be sub-script.
एचटीएमएल में _____ एलिमेंट, उन वर्णो को समष्टि करने में प्रयुक्त होता है, जिनको की सबस्क्रिप्ट होना हो।

(a) < SCRIPT > (b) <SUB>
(c) <SUB SCRIPT> (d) < SUB SCR>

HTET PGT C.S. 2019

Ans. (b) : In HTML the <SUB> element is used to contain characters that should be sub-script. In sub-script the text is printed below to the normal text.
Example ⇒ CH_4
⇒ CH₄

49. In context of HTML what will be the result of the following ?
एच.टी.एम.एल. के सन्दर्भ में निम्न का क्या परिणाम होगा?
**<P> This is <del> WEB</del> <ins> HTML</ins> Page **

(a) This is WEB HTML Page
(b) This is HTML Page
(c) This is ~~WEB~~ HTML Page
(d) This is WEB (deleted) HTML (inserted) Page

HTET PGT C.S. 2018

Ans. (c) : The word 'WEB' with a line though it (strike-through, using <del>) and the word HTML with an underline (using <ins>) Therefore the output would be like
O/P ⇒ "This is ~~WEB~~ HTML page."

50. Which of the following is empty or void element in HTML ?
निम्न में से कौन-सा एचटीएमएल में रिक्त या शून्य तत्व है?

(a) (b)
(c) <abbr> (d) <sup>

HTET PGT C.S. 2019

Ans. (b) : An empty tag (Element) is that tag which does not have closing part of the tag for example
 <img> <hr> tag etc.
Syntax–

51. Which of the following is appropriate option to display on web page 'We Love BSF Very Much' with a tool tip (full form) for the word 'BSF' ?
वेब पेज पर BSF शब्द के लिए टूल टिप (फुल फार्म) के साथ 'We Love BSF Very Much' दर्शाने के लिए निम्न में से कौन-सा उपयुक्त विकल्प है?

(a) <html> We Love <FF = "Border Security Force"> BSF </FF> Very Much </html>
(b) <html> We Love <BSF = "Border Security Force"> BSF </BSF> Very Much </html>
(c) <html> We Love <abbr title = "Border Security Force"> BSF </abbr> Very Much </html>
(d) <html> We Love <Full "Border Security Force"> BSF </Full> Very Much </html>

HTET PGT C.S. 2019

Ans. (c) : In HTML the 'abbr' tag is used to represent an abbreviation or acronym. Its primary purpose is to provide semantic meaning to the marked text and to offer a full text explanation of the abbreviation when a user hover over it.
Example– <html> we love <abbr title = "Border Security Force"> BSF </abbr> very much </html>.

52. Which of the following elements is empty tag in HTML?
निम्नलिखित में से कौन-से HTML में खाली तत्व होते है/हैं?

(a)
(b) <hl>
(c) <head>
(d) All options are correct/सभी विकल्प सही हैं

HTET PGT C.S. 30.07.2025

Ans.(a): '
' tag is an Empty Element/tag. This is used to break the line in HTML document.
Syntax–
.

53. This element is used to change the current directionality of the text, in HTML :
HTML में यह एलिमेंट टेक्स्ट की मौजूदा दिशात्मकता को परिवर्तित करने में प्रयुक्त होता है:

(a) <dd> (b) <dfn>
(c) <pre> (d) <bdo>

HTET PGT C.S. 2022

Ans. (d) : '<bdo>' element is used to change the current directionality of the text, in HTML. The 'bdo' stands for Bidirectional override. It is used to explicitly override the default text directionality in a document.

54. In HTML version-5, the purpose of this element is to encapsulate group of links :
HTML संस्करण-5 में इस तत्त्व का उद्देश्य लिंक्स के समूह को संपुटित करना है:

(a) <web> (b) <link>
(c) <nav> (d) <a href>

HTET PGT C.S. 2022

Ans. (c) : In HTML, version 5, the purpose of <nav> element is to encapsulate group of links. It defines the navigation bar on the web page.
Syntax– <nav>————</nav>

55. In HTML5 the _____ element is used to include images, graphs and compound objects that contain text and images, in webpage.
HTML 5_____ एलिमेंट वेबपेज में इमेज, ग्राफ एवं कंपाउंड ऑब्जेक्ट्स जिसमें टेक्स्ट तथा इमेज होते हैं, को समविष्ट करने में प्रयुक्त होता है।

(a) <figure> (b) <fig>
(c) <photo> (d) <pic>

HTET PGT C.S. 2022

Ans. (a) : In HTML-5 the <figure> element is used to include the images graphs and compound objects that contain text and images in webpage.

Sarkari Resource

56. Which of the following statements is not correct?
(a) HTML is not screen precise formatting language
(b) HTML does not specify a logic
(c) DHTML is used for developing highly interactive web pages
(d) HTML is a programming language

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) HTML is not programming language Programming Language means where writing logic is possible which is not supported by HTML.

57. Golive is a_____.
गोलाइव है एक–
(a) Website/वेबसाइट
(b) Web Publishing Tool/वेब पब्लिशिंग टूल
(c) Web Database/वेब डेटाबेस
(d) Web Browser/वेब ब्राउजर

HTET PGT C.S. 2023

Ans.(b): Golive is a Web Publishing Tool. Specifically, Adobe Golive was WYSIWYG HTML editor and website management application from Adobe systems, used for creating, designing and publishing websites.

58. A collection of hyperlinked documents on the Internet forms the
इन्टरनेट पर हाइपरलिंक किए गए दस्तावेज़ों का एक संग्रह बनता है, जो है
(a) World Wide Web (WWW)
(b) E-mail system
(c) Mailing list
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : World Wide Web (WWW) is a huge collection of Hyperlinked Documents. User can reach on them by using Internet through web Browser. To access the content of WWW some specific web instructions or search engines are used.

59. Which language does a browser use to display information from the World Wide Web?
World Wide Web से जानकारी प्रदर्शित करने के लिए ब्राउज़र किस भाषा का उपयोग करता है?
(a) Machine code
(b) Assembly language
(c) C++
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (e) : A browser uses HTML (Hypertext Markup Language) to display information from the World Wide Web (WWW). HTML is a markup language to design web pages and web documents. Brower interpretates the HTML language and show the content of web sites.

60. Which attribute in HTML table tag defines the number of columns in a column group?
HTML टेबल टैग में कौन सा एट्रिब्युट कॉलम समूह के कॉलम की संख्या को परिभाषित करता है?
(a) Border/बॉर्डर (b) Colspan/कॉलस्पैन
(c) Rowspan/रॉस्पैन (d) Span/स्पैन

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (d) : In HTML, the <colgroup> and <col> tags are used to group & style columns in a table. The "span" attribute in the <col> tag specifies how many columns the <col> element should span.
- **Border**– sets the border around the table not related to columns.
- **Colspan**– Used in <td> or <th> to make a cell span multiple columns.
- **Rowspan**– used in <td> or <th> to make a cell span multiple rows.

61. How many different levels of heading does HTML support?
एच.टी.एम.एल. कितने विभिन्न स्तरों के शीर्षक का समर्थन करता है?
(a) 6 (b) 7
(c) 5 (d) 9

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : HTML supports 6 different levels of heading. These are represented by the tags <h1>, <h2>, <h3>, <h4>, <h5>, and <h6>, with <h1> being the highest (most important) level and <h6> being the lowest.

62. Which HTML tag is used to embed Java script scripts in a page?
किसी पेज में JavaScript को एम्बेड करने के किस एच.टी.एम.एल. टैग का उपयोग किया जाता है?
(a) <script>tag (b) <embed>tag
(c) <java>tag (d) <enable>tag

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : The script tag is specifically designed to include JavaScript code within an HTML page. This tag can either directly embed the JavaScript code within the tag itself or it can reference an external file containing the JavaScript code.

63. What is the CMY co-ordinates of a color at (0.2, 1, 0.5) in the RGB space?
आरजीबी (RGB) स्पेस में (0.2, 1, 0.5) पर एक रंग का सीएमवाई (CMY) निर्देशांक क्या होता है ?
(a) (0.8,1, 0.5) (b) (0.8, 0, 0.5)
(c) (1.2, 0, 1.5) (d) (0.2, 0.8, 0.3)

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : CMY is a subtractive color model, while RGB is an additive model CMY values are complements of RGB values.
Convert RGB to CMY by subtracting each RGB component from 1 :-
Step-1– Calculate the cyan (c) component
C= 1 – R
C = 1 – 0.2 = 0.8

Sarkari Resource

Step-2– Calculate the magenta (M) component
$M = 1 - G$
$M = 1 - 1 = 0$
Step-3– Calculate the yellow (Y) component
$Y = 1 - 13$
$Y = 1 - 0.5 \Rightarrow 0.5$
Hence CMY coordinates are (0.8, 0, 0.5)

64. Which property of CSS specifies the degree to which text should be emboldened ?
CSS की कौन सी प्रोपर्टी उस डिग्री को निर्दिष्ट करती है जिस तक टैक्स्ट को एमबोल्डेन्ड किया जाना चाहिए?
(a) Font – style/फॉन्ट–स्टाइल
(b) Font – family/फॉन्ट–फैमिली
(c) Font – weight/फॉन्ट– वेट
(d) Font – size/फॉन्ट–साइज़

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : The font-weight property sets how thick the characters in text should be displayed.
Syntax– Font-Weight- normal / bold / bolder / lighter / number / initial / inherit;

65. Which property of CSS enables to set multiple list properties at once ?
CSS की कौन सी प्रोपेर्टी एक साथ कई सूची प्रोपर्टियों को सेट करने में सक्षम बनाती है?
(a) List – style – type/लिस्ट-स्टाइल-टाईप
(b) List – style – image/लिस्ट-स्टाइल-इमेज़
(c) List – style – position/लिस्ट-स्टाइल-पोज़िशन
(d) List – style/लिस्ट-स्टाइल

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (d) : The CSS property that enables setting multiple list properties at once is list-style. This is a shorthand property that combines the functionalities of three individual list-related properties:
- List-style-type
- List-style-position
- List-style-image

66. Which of the following is false with reference to static website?
हर बार पेज लोड होने पर सामान कंटेंट लोड होता है?
(a) Same content is loaded every time the page is loaded/हर बार पेज लोड होता है
(b) It is loaded at a faster speed
यह तेज गति से लोड होता है।
(c) Content change at runtime
कंटेंट रनटाइम पर बदलता है
(d) No interaction/कोई इंटरेक्शन नहीं है

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Content change at runtime is false with reference to static website because static websites has fixed content.

67. Which of the following statement(s) is/are correct regarding <title> tag in HTML?
HTM के <title> टैग के सन्दर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?
I. This tag always goes outside to the <head> tag.
यह टैग हमेशा <head> टैग के बाहर आता है।
II. This tag is used to indicate the title of the webpage./इस टैग का उपयोग वेबपेज के शीर्षक को इंगित करने के लिए किया जाता है।
III. It is also used by search engines.
इसका उपयोग सर्च इंजन द्वारा भी किया जाता है।
(a) I and II (b) II and III
(c) Only II (d) I, II and III

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) :
(I) Title tags are only valid when found within the <head> section of a page. A title tag outside the <head> tag is invalid and will be ignored by search engines. Therefore this statement is incorrect.
(II) The <title> tag defines the title of the document. Hence statement (II) is correct.
(III) Title tags are used by search engines to understand a WebPages content and display it in search result so this statement is also correct.

68. Which color specification in CSS use three numbers representing the hue (form 0 to 360 degree), saturation and lightness percentage?
सी.एस.एस में कौनसा रंग स्पेसिफिकेशन तीन संख्याओं का उपयोग करता है जो हयु (0 से 360) सैचुरेशन प्रतिशत और लाईट और लाइट नैस प्रतिशत को दर्शाती है?
(a) Hexadecimal/हेक्साडेसिमल
(b) RGB/आर.जी.बी.
(c) HSL/एच.एस.एल.
(d) Alpha/अल्फा

DSSSB TGT C.S. 08.08.2021 (Shift-I)

Ans. (c) : HSL stands for Hue, Saturation, Lightness.
- Hue: Angle from 0 to 360 digress.

(0 = red, 120 = green, 240 = blue, etc)
- Saturation: 0% (Gray) to 100% (Full color)
- Lightness: 0% (Black) to 100% (white)

Example :
Color: HSL (120, 100%, 50%);

69. What is tag in HTML?
HTML में टैग क्या होता है?
(a) It contains the address of the webpage.
इसमें वेबपेज का पता होता है।
(b) It contains the number of nested tags in the page./इसमें पेज में नेस्टेड टैग्स की संख्या होती है।
(c) It contains information about the webpage.
इसमें वेबपेज के बारे में जानकारी होती है।
(d) It contains the number of tags in the web page./इसमें वेब पेज में टैग्स की संख्या होती है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In HTML a tag is a keyword used to define elements within a webpage. taps are used to create the structure and content of the page.

Sarkari Resource

70. How many forms can we have on a webpage at the maximum?
वेबपेज पर हमारे पास अधिकतम कितने फॉर्म हो सकते हैं?

(a) 1 (b) 5
(c) 10 (d) No limit

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : There is no maximum limit on the number of form on a webpage.

72. What is image map in HTML?
HTML में इमेज मैप क्या है?

(a) It is the backup copy of the image.
यह इमेज की बैकअप कॉपी है।
(b) It is a special image in which different area point to different locators on the web.
यह एक विशेष इमेज है जिसमें विभिन्न क्षेत्र वेब पर विभिन्न स्थानों की ओर पॉइंट करते हैं।
(c) It contains the properties of the image.
इसमें छवि की प्रॉपर्टीज होती हैं।
(d) It contains the number of pixels in the image.
इसमें छवि में पिक्सेज की संख्या होती है।

DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : An image map in HTML is a feature that allows you to checkable areas within an image. Where each area can link to different URLs or perform different actions.

73. In DHTML, the letter 'D' stands for:
DHTML में, अक्षर 'D' का अर्थ क्या है?

(a) Distributed/डिस्ट्रिब्यूटेड
(b) Dynamic/डायनमिक
(c) Daily/डेली
(d) Different/डिफरेंट

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : In DHTML, D stands for Dynamic DHTML is not a distinct language but rather a collection of web technologies used to create dynamic & interactive web pages. It emerged in the café 1990's as a way to enhance the capabilies of static HTML.

74. Which of the following HTML element is used to define table data/cell?
टेबल डेटा/सेल को निर्धारित करने के लिए निम्नलिखित में से कौन सा HTML एलीमेंट उपयोग किया जाता है?

(a) <table> (b) <td>
(c) <tr> (d) <th>

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : The <table> tag defines an HTML table. An HTML table consists of one <table> element and one or more <tr>, <th> and <td> elements. The <tr> element defines a table row, the <th> element defines a table header and the <td> element defines a table cell or table data.

75. Which of the following can be used to display black color in HTML?
HTML में रंग को प्रदर्शित करने के लिए निम्नलिखित में किसका उपयोग किया जा सकता है?

(a) RGB (0, 0, 0)
(b) RGB (255, 255, 255)
(c) RGB (255, 0, 0)
(d) RGB (0, 255, 0)

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In RGB color, RGB stands for Red, Green and Blue. In HTML we can use RGB color method to provide the element color. In RGB color method each color lies between the value (0-255). 0 is the lowest value while 255 is the highest value. Therefore to find the black color we write each color value to 0 i.e. RGB (0, 0. 0)

76. Which of the following category of CSS is defined in the <head> section of an HTML page, within a <style> element?
निम्नलिखित में से CSS की किस श्रेणी को <style> एलीमेंट के अंतर्गत HTML पेज के <head> सेक्सन में निर्धारित किया गया है?

(a) Inline CSS
(b) Internal CSS
(c) External CSS
(d) Both Internal and External CSS
Internal और External CSS दोनों

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : There are three method to write CSS code in HTML page.
(1) Inline CSS
(2) Internal CSS
(3) External CSS
In internal CSS, we write the CSS code in <head> tag of HTML page within <Style> Tag/Element.

77. Which of the following statement about HTML attributes is false?
HTML विशेषताओं के बारे में निम्नलिखित में से कौन सा कथन गलत है?

(a) HTML element can have attributes
HTML एलीमेंट का एट्रिब्यूट हो सकता है
(b) Attributes provide additional information about an element/एट्रिब्यूट एलीमेंट के बारे में अतिरिक्त जानकारी प्रदान करती है
(c) Attributes are always specified in the end tag
एट्रिब्यूट हमेशा अंत टैग में निर्दिष्ट होते हैं
(d) Attributes usually come in name/value pairs like: name = "value"/एट्रिब्यूट आमतौर पर name/value (नाम/मूल्य) युग्म में होते हैं जैसेः name = "value"

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : In HTML, Attributes of Element always define into the opening tag.
For example– <h1 style = "color : Red":>
This is heading </h1>

Sarkari Resource

78. What will be the output of the following segment of the HTML codes?
HTML कोड के निम्नलिखित खंड का आउटपुट क्या होगा?

```
<ol>
<li>Coffee<li>
<li>Tea<li>
<li>Milk<li>
<ol>
```

(a) 1. Coffee/कॉफी
2. Tea/चाय
3. Milk/दूध

(b) 1. Milk/दूध
2. Tea/चाय
3. Coffee/कॉफी

(c) 1. Coffee/कॉफी
2. Milk/दूध
3. Tea/चाय

(d) 1. Coffee/कॉफी
2. Tea/चाय
3. Milk/दूध

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Given HTML code is related to HTML listing which is in the form of Ordered list <ol> Element represents the ordered list and list items are written within <li> tag. Therefore Output would be:
1. Coffee
2. Tea
3. Milk

79. DHTML is a programming language
गलत कथन की पहचान करें।

(a) DHTML is a programming language
DHTML एक प्रोग्रामिंग भाषा है

(b) JavaScript is the scripting standard for HTML
JavaScript HTML के लिए स्क्रिप्टिंग मानक है

(c) DHTML is the art of combining HTML, JavaScript, DOM, and CSS
DHTML HTML, JavaScript, DOM, और CSS के संयोजन की कला है

(d) DHTML is a term used to describe the technologies used to make web pages dynamic and interactive/DHTML एक शब्दावली है जो वेब पेजों को गतिशील और इंटरैक्टिव बनाने के लिए उपयोग की जाने वाली तकनीकों का वर्णन करने के लिए प्रयोग किया जाता है

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : DHTML is not a programming language but rather a collection of web technologies used to create dynamic & interactive web pages.

80. Which of the following category of style sheet can be used to change the look of an entire website, by changing one file (CSS)?
फाइल (CSS) को बदलकर, संपूर्ण वेबसाइट के रूप को बदलने के लिए निम्नलिखित में से किन स्टाइल शीट श्रेणी का उपयोग किया जा सकता है?

(a) Inline CSS
(b) Internal CSS
(c) External CSS
(d) Both Inline and Internal CSS
Inline और Internal CSS दोनों

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Ans. (c) : In External CSS, we create a separate styling sheet (CSS file) with a .CSS extension. In this file we put all the formatting and designing property of HTML element. In order to apply all CSS property we link that external CSS file with HTML document page. If we want to change the Entire look of website then we only change only CSS file not all the HTML document pages.

81. In HTML, a color can be specified in the form of ha1(hue, saturation, lightness), where hue is a degree on the color wheel from 0 to 360. What are the hue values representing red, green, and blue?
HTML में, (hsl) (ह्यू (रंग), सेचुरेशन (संतृप्ति), लाइटनेस (हल्कापन) के रूप में एक रंग निर्दिष्ट किया जा सकता है, जहां रंग 0 से 360 तक कलर ह्वील पर आधारित डिग्री होती है। लाल, हरे और नीले रंग को दर्शाने वाले रंग का मान क्या है?

(a) 0: Red, 120: Green, 240: Blue
0: लाल, 120: हरा, 240: नीला

(b) 0: Red, 130: Green, 260: Blue
0: लाल, 130: हरा, 260: नीला

(c) 100: Red, 200: Green, 300: Blue
100: लाल, 200: हरा, 300: नीला

(d) 120: Red, 240: Green, 360: Blue
120: लाल, 240: हरा, 360: नीला

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Ans. (a) : In CSS to provide color we can use HSL method where HSL stands for Hue, saturation and lightness. In this method Hue lies between 0 to 360 on the color wheel where 0 is red 120 is green and 240 is Blue.

82. Any time you include JavaScript verbiage in HTML document, you must enclose those lines inside a _____ tag pair.
किसी भी समय जब आप किसी HTML दस्तावेज में JavaScript वर्बेज शामिल करते हैं, तो आपको उन पंक्तियों को _____ टैग के अंदर संलग्न करना होगा।

(a) <HTML> and </HTML>
<HTML> और </HTML>

(b) <Body> and <\Body>
<Body> और <\Body>

(c) <Div> and <\Div>
<Div> और <\Div>

(d) <Script> and </script>

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Ans. (d) : To write JavaScript code in html page/document we use <Script> tag.
Syntax :– <Script> and </script>

83. Which of the following HTML tag is used to puts a line break in the text?
(a) <B> (b) <BL>
(c) <L> (d)
DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (d) :
 tag is used to break the line in the HTML document. It can be used to provide line space in HTML page.

84. Which of the following rule is needed to be followed while declaring the XML (Extensible Markup Language) elements name?
(a) Name must start with a number
(b) Name can contain spaces
(c) Name must not start with the letter xml
(d) Name cannot contain letter, numbers and other characters
DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : In XML, element naming rules are–
(i) Name must start with a letter or underscore.
(ii) Names cannot start with "Xml" (or any case variation like "XML", "Xml").
(iii) Names can contain letters, digit, hyphens, underscores and periods.
(iv) Names can not contain spaces.

85. Which of the following form is correct to show a 'comments' in the HTML language?
(a) < – – T his is a comment – – >
(b) < – – T his is a comment – –! >
(c) < !– – T his is a comment – –! >
(d) < !– – T his is a comment – – >
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Ans. (d) : In HTML, to mark the comment we use '<! - - contents - - >' syntax.
For example– <! - - <h2> This is Heading </h2> - - >

86. Which of the following tag of HTML language is a container for the tags that contain information about the page?
(a) <title> (b) <html>
(c) <head> (d) <body>
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Ans. (c) : <Head> tag of HTML language is a container for the tags that contains information about the page.
Syntax– <Head>----------</Head>

87. Which of the following is NOT an attribute that included by the HR tag which puts a horizontal line on the page?
(a) SHADE (b) SIZE
(c) ALIGN (d) WIDT H
DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : The <hr> tag is used to create a horizontal line on a webpage. The attributes associated with this tag are; Size, Align, width, No shade. Hence 'Shade' is not an attribute for the <hr> tag.

88. In HTML body tag, which of the following attribute indicates the colour of the hyperlinks after the mouse is clicked on it?
(a) VLINK (b) ALINK
(c) LINK (d) HLINK
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Ans. (a) : The attributes in the HTML <body> tag that indicate the color of hyperlinks after the muse is clicked on them (i.e. after the link has been visited) is VLINK.
- **VLINK : (Visited Link)–** It specifies the color of a link that has already been clicked and visited by the user.
- **ALINK : (Active Link)–** It specifies the color of Link while it is being clicked or activated by the user.
- **LINK :** It specifies the default color of an unvisited Link.

89. Linking to another place in the same or another webpage require two A (Anchor) tags, the first with the _____ attribute and the second with the _____ attribute.
(a) NAME & LINK
(b) LINK & HREF
(c) HREF & NAME
(d) TARGET & VALUE
UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, Name attribute specifies the name of an anchor and it is used to create a bookmark inside a document.
Syntax : <a name = "value">
⇒ href attribute specifies the URL of the page the link goes to, and if the href attribute is not present, the <a> tag is not a hyperlink.
Syntax : <a href = "URL">

90. ..... tag is an extension to HTML that can enclose any number of JavaScript statements.
(a) <SCRIPT> (b) <BODY>
(c) <HEAD> (d) <TITLE>
UGC NET C.S. December-2015 (Paper-III)

Ans : (a) <Script> tag is an extension to HTML that can enclose any number of JavaScript statements. Syntax- <script>……</script>

91. JavaScript and Java similar name because ..... is/are true.
(1) JavaScript's syntax is loosely based on Java's syntax
(2) JavaScript is stripped down version of Java
(3) Java and JavaScript are originated from Island of Java

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Codes :

(a) (1) only (b) (1), (2) and (3)

(c) (1) and (2) (d) (1) and (3)

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) Statement (2) and Statement (3) are not true because JavaScript is not stripped down version of Java and they are not originated from Island of Java.
Although they are different languages, JavaScript's syntax is loosely based on Java.

92. Which of the following tag in HTML is used to surround information, such as signature of the person who created the page?

(a) <body> </body>

(b) <address> </address>

(c)

(d)

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) The **<address> tag** is used to surround information, such as the signature of the person who created the page or the address of the organization the page is about.
The **<body> tag** defines the document is body.
The ** tag** defines emphasized text.
The ** tag** defines important text.

93. Which of the following HTML code will affect the vertical alignment of the table content?

(a) <td style = "vertical-align : middle" > Text Here </td>

(b) <td valign = "centre"> Text Here </td>

(c) <td style = "text-align : center"> Text Here </td>

(d) <td align = "middle" > Text Here </td>

UGC NET C.S. June-2017 (Paper-III)

Ans : (b) Vertical alignment of the table contents:
<td valign = 'top |middle | bottom | baseline'>
Text here </td>
So, no option matching if middle = Centre, then option (b) is matching.

94. Match List-I with List-II.

List-I	List-II
(A) HTML	**(I) Allow for dynamic and interactive web pages.**
(B) DHTML	**(II) Defines the structure of web pages.**
(C) XML	**(III) Object-oriented programming language for web applications.**
(D) JAVA	**(IV) Used for data storage and transport.**

Choose the correct answer from the options given below :

(a) A-I, B-III, C-II, D-IV

(b) A-II, B-III, C-IV, D-I

(c) A-II, B-I, C-IV, D-III

(d) A-III, B-I, C-IV, D-II

UGC NET C.S. Re-Exam- June 2024

Ans. (c) : Correct match–

List-I			List-II
A.	HTML	–	Defines the structure of web page
B.	DHTML	–	Allow for dynamic and interactive web page
C.	XML	–	Used for data storage and transport.
D.	JAVA	–	Object oriented programming language for web applications

95. Which of the following lists defined in HTML share similar elements?

(A) Unordered lists (B) Ordered lists

(C) Definition lists (D) Correlated lists

(E) Enumerated lists

Choose the most appropriate answer from the options given below :

(a) A, B, D only (b) A, C, E only

(c) A, B, C only (d) A, C, D only

UGC NET C.S. December-2022

Ans. (*) : HTML lists→ are used to present list of information in well formed and semantic way. There are three different types of list in HTML and each one has a specific purpose and meaning.

- **Unordered list–** Used to create a list of related items, in no particular order.
- **Ordered list–** Used to create a list of related items in a specific order.
- **Description list–** Used to create a list of terms and their descriptions.

96. Which of the following HTML5 codes will affect the horizontal as well as vertical alignment of the table content ?

(a) <td style ="horizontal-align : center; vertical-align : middle;"> BASH </td>

(b) <td halign ="middle" valign="center" > BASH </td>

(c) <td style="text-align : center; vertical-align : middle;" > BASH </td>

(d) <td align="middle" valign="center" > BASH </td>

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : Align = left|center|right|justify|char [CI]
This attribute specifies the alignment of data and the justification of text in a cell. Possible values:

- left: Left-flush data/Left-justify text. This is the default value for table data.
- center: Center data/Center-justify text. This is the default value for table headers.
- right: Right-flush data/Right-justify text.
- justify: Double-justify text.
- char: Align text around a specific character. If a user agent doesn't support character alignment, behavior in the presence of this value is unspecified.

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97. **In HTML <map> tag is used for**

(a) Defining a path between two nodes in an image
(b) Defining clickable region in an image
(c) Highlighting an area in an image
(d) Defining the site amp of a web-site

UGC NET C.S. June-2020 (Paper-II)

Ans. (b) :
- The <map> tag is used to define an image map. An Image map is an image with clickable areas.
- The required name attribute of the <map> element is associated with the <img>'s use map attribute and creates a relationship b/w the image and the map.
- An image map is made up of an image with clickable areas, When you click on the image it will open to a new or previously selected destination.

98. **Match List-I with List-II**

List-I		List-II	
A.	**Frame attribute**	**1.**	**to create link in HTML**
B.	**<tr>tag**	**2.**	**for vertical alignment of content in cell**
C.	**Valign attribute**	**3.**	**to enclose each row in table**
D.	**<a>tag**	**4.**	**to specify the side of the table frame that display border**

Choose the correct option from those given below:

(a) A-1, B-2, C-3, D-4
(b) A-2, B-1, C-3, D-4
(c) A-4, B-3, C-2, D-1
(d) A-3, B-4, C-2, D-1

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (c) :

	List-I		List-II
A.	Frame attribute	4.	to specify the side of the table frame that display border
B.	<tr>tag	3.	to enclose each row in table
C.	Valign attribute	2.	for vertical alignment of content in cell
D.	<a>tag	1.	to create link in HTML

99. **What is the purpose of HTML in a web application?**
वेब एप्लिकेशन में HTML का उद्देश्य क्या है?

(a) To define the structure of web pages
Web pages की संरचना को परिभाषित करना
(b) To style web pages
Web pages की स्टाइल करने के लिए
(c) To add interactivity to web pages
Web pages में अन्तरक्रियाशीलता जोड़ने के लिए
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : The purpose of HTML in a web application is to define the structure of web pages. It create only static web pages.

100. **To jump from one page to another page, which of the following is used?**
एक पेज से दूसरे पेज पर सीधे पहुँचने के लिए निम्न में से किसका प्रयोग करेंगे?

(a) <A HREF> .....</A>
(b) <link>
(c) <jump>
(d) None of the above/उपर्युक्त में से कोई नहीं

LT. Grade C.S. 2018
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : To jump from one page to another page the HTML anchor tag <a href = " " > is used.
The <a href = " "> tag defines a hyperlink that links one page to another. The most important attribute of the <a href = " " > tag is the href attribute, which indicates the link's destination.

101. **Name the tag and the attribute to be used to change the sequence of the list items in an ordered list from numbers to upper case letters**
किसी व्यवस्थित सूची में लिस्ट आइटम की सूची के क्रम को संख्या से अप्पर केस में परिवर्तित करने के लिए किस टैग तथा एट्रीब्यूट का उपयोग किया जाएगा?

(a) Tag-<OL>, Attribute - START
(b) Tag-<OL>, Attribute - VALUE
(c) Tag-<OL>, Attribute - TYPE
(d) Tag-<OL>, Attribute - CASE

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : The <OL> element represents an ordered list of items.
Type:- the type attribute specifies the kind of marker to use in the list (letters or numbers)

102. **Study the 3 statements given below and identify the tag which satisfies these 3 statement conditions :**
नीचे दिए गए 3 कथनों का अध्ययन कीजिए और इन कथनों की शर्तों का पूरा करने वाले टैग की पहचान कीजिए?

(A) This tag is used to create a section break
इस टैग का उपयोग सेक्शन ब्रेक देने के लिए किया जाता है।

(B) This tag has attributes ALIGN, WIDTH, COLOR and SIZE
इस टैग के एट्रीब्यूट हैं ALIGN, WIDTH, COLOR तथा SIZE

(C) This tag does not have a closing tag
इस टैग का क्लोजिंग टैग नहीं होता।

(a) <P> (b)
(c) <HR> (d) <META>

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Ans. (c) : The <hr> tag defines a thematic break in an HTML page. The <hr> element is most often displayed as a horizontal rule that is used to separate content in an HTML page.

103. Which of the following HTML tag does not have a name attribute associated with it ?
निम्नलिखित में से किस HTML टैग के साथ नेम एट्रीब्यूट जुड़ा नहीं होता?

(a) <FORM> (b) <SELECT>
(c) <A> (d) <EMBED>

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Ans. (d) : The <embed> HTML element embeds external content at the specified point in the document. This content is provided by an external application.
<embed> tag attributes
height, src, type, width.
So <embed> tag does not have a name attribute.

104. Observe the code snippet given below :
नीचे दिए गए कोड स्निपेट को देखिए:
<A HREF = "# TOP "> Back to Top </A>
The # sign is used to :
चिह्न का प्रयोग किसके लिए किया गया है?

(a) Create a link to another webpage
किसी अन्य वेबपेज का लिंक बनाने के लिए।
(b) Create a link to an Email address
किसी ई-मेल एड्रेस का लिंक बनाने के लिए
(c) Create an internal link and instruct browser to look for a named anchor in current document
आंतरिक लिंक बनाने तथा ब्राउज़र को वर्तमान डॉक्यूमेंट में 'नेम्ड एंकर' को खोजने का निर्देश देने के लिए।
(d) Create an external link and instruct the browser to look for a named anchor in another HTML document/बाह्य लिंक बनाने तथा ब्राउज़र को अन्य HTML डॉक्यूमेंट में 'नेम्ड एंकर' का खोजने का निर्देश देने के लिए।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : The # sign is used to create an internal link and instruct browser to look for a named anchor in current document.

105. Which attribute of the <IMG> tag tells the browser what text to display if it cannot show the image ?
<IMG> टैग का कौन सा एट्रीब्यूट ब्राउज़र को यह बताता है कि यदि वह चित्र प्रदर्शित नहीं कर सकता तो उसे क्या टेक्स्ट प्रदर्शित करना है?

(a) TEXT (b) ALT
(c) NOSHOW (d) NOIMAGE

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (b) : The <img> HTML element embeds an image into the document
<img> tag attributes
alt, crossorigin, height, ismap, loading, longdese, referrerpolicy, sizes, src, srcset, usemap, width.
Alt:- Attribute specifies an alternate text for an image. If the image cannot be displayed.

106. Which of the following attribute of the <TABLE> tag specifies the amount of space between the border and the contents of the cell?
<TABLE> टैग का निम्नलिखित में से कौन सा एट्रीब्यूट सेल के बार्डर और कन्टेन्ट के मध्य स्पेस की मात्रा को विनिर्दिष्ट करता है?

(a) PADDING (b) SPACING
(c) CELL PADDING (d) CELL SPACING

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : <table>, tag in HTML consists of table cells inside rows and columns.
Cell padding is the space between the cell edges and the cell content. To add padding on table cells, use the css padding property.

107. What is the correct usage of the FRAMEBORDER attribute of the <FRAME> tag in HTML ?
HTML में <FRAME> टैग के FRAMEBORDER एट्रीब्यूट का सही उपयोग क्या है?

(a) Specifies the color of the border of the frame
फ्रेम के बॉर्डर का रंग विनिर्दिष्ट करता है।
(b) Specifies the status of scroll bars of a frame
फ्रेम के स्क्रॉल बार की स्थिति विनिर्दिष्ट करता है।
(c) Specifies the distance between top and bottom edge of the frame and frame contents
फ्रेम के ऊपरी तथा निचले किनारों तथा फ्रेम की सामग्री के बीच की दूरी विनिर्दिष्ट करता है।
(d) Specifies whether the border should be visible or not/यह विनिर्दिष्ट करता है कि बॉर्डर दृष्टिगोचर होता या नहीं।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (d) : The HTML <frame> frameborder attribute is used to specify whether or not a border should be displayed between the frames.
Syntax-
<frame frameborder = "1/0">
0: It has a default value It sets the border an one state.
1: It sets the border off state.
<frame> attributes.
bordercolor, frameborder, longdesc, marginheight, marginwidth, name noresize, scrolling, src.
The <frame> element also supports the Global attributes.

108. The style definitions are normally saved in external files that have the extension as :
स्टाइल डेफिनिशन्स आमतौर पर एक बाहरी फाइस में सेव होती हैं जिसका एक्सटेंशन है:

(a) .html (b) .css
(c) .js (d) .stl

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (b) : CSS external files saved .css extension name while HTML used to .html file name, and .js used to JavaScript file.

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109. Internal linking with respect to HTML means
HTML के संदर्भ में आंतरिक सम्पर्कन (Linking) का अर्थ है

(a) Linking one part of the document on one computer , to another part of the document on some other computer with the same file name./एक ही फाईल नाम से एक कम्प्यूटर में दस्तवेज के एक भाग को किसी दूसरे कम्प्यूटर में दस्तावेज के अन्य भाग से जोड़ना ।

(b) Linking one part of the document on one computer, to another part of the document on some other computer in the same LAN./एक ही LAN में कम्प्यूट दस्तावेज के एक भाग को किसी दूसरे कम्प्यूटर में दस्तावेज के अन्य भाग से जोड़ना ।

(c) Linking one part of the document to another part of the same document./दस्तावेज के एक भाग को उसी दस्तावेज के अन्य भाग से जोड़ना

(d) Linking one part of the document to another document on the same computer.
एक ही कम्प्यूटर में किसी दस्तावेज के एक भाग को अन्य दस्तावेज से जोड़ना ।

KVS PGT C.S. 2013

Ans. (c) : Internal linking occurs when a site links to other URLS on the same site. While external linking occurs when a site links to URLs on a different site.

110. If a user is not able to see the image on the web page, the text that is displayed in place of the image, is specified by which of the following attributes in HTML ?
यदि कोई उपयोगकर्त्ता वेब पेज पर इमेज (बिम्ब) देखने में सक्षम नहीं होता है तो ईमेज के स्थान पर अवतरण (टेक्स्ट) प्रदर्शित हो तो HTML में निम्न लक्षणों में से किसके द्वारा विनिर्दिष्ट की जाएगी ?

(a) ALTERNATE (b) ALTLINK
(c) ALT_TEXT (d) ALT

KVS PGT C.S. 2013

Ans. (d) : The Alt attribute of <img> tag is used to display the text in place of image for the users when browser don't support image or image link not found.

111. The _____ attribute of <TABLE> tag is used to control the distance between the data in a cell and the boundaries of the cell.
<TABLE> टैग के _____ एट्रीब्यूटर का उपयोग सैल में डाटा तथा सैल की परिसीमाओं के मध्य दूरी को नियंत्रित किया जाता है।

(a) Colspan (b) Rowspan
(c) Cellpadding (d) Cellspacing

KVS PGT C.S. 2018

Ans. (c) : The cellpadding attribute of <TABLE> tag is used to control the distance between the data in a cell and the boundaries of the cell. By default the padding is set to 0. The cellpadding attribute is set in terms of pixels.
Syntax–
<table cellpadding = "pixels">

112. While making bulleted lists, which of the following options are available?
बुलेटिड लिस्ट बनाने के लिए निम्नलिखित में से विकल्प उपलब्ध है?

(a) Square, disc, triangle/स्क्वायर, डिस्क, ट्राएंगल
(b) Triangle, disc, circle/ट्राएंगल, डिस्क, सर्कल
(c) Triangle, square, circle/ट्राएंगल, स्क्वायर, सर्कल
(d) Disc, square, circle/डिस्क, स्क्वायर, सर्कल

KVS PGT C.S. 2018

Ans. (d) : HTML Unordered list or Bulleted list displays elements in bulleted format. We can use unordered list where we do not need to display items in any particular order. The HTML ul tag is used for the unordered list. There can be 4 types of bulleted list:
1. disc
2. circle
3. Square
4. None

113. Given below are several usages of the anchor tag in HTML.
नीचे HTML में anchor टैग के अनेक उपयोग हैं:

(A) <A href = "http://w3c.org/test/html">Test </A>
(B) <A href = "/HTML/test.html">Test</A>
(C) <A href = "test.html">Test</A>
(D) <A href = "test.html#top">Test</A>
Which of the above are valid?
उपरोक्त में से कौन-सा वैध है?

(a) (A) and (B) only/केवल (A) तथा (B)
(b) (A), (B) and (C) only/केवल (A), (B) तथा (C)
(c) (B), (C) and (D) only/केवल (B), (C) तथा (D)
(d) (A), (B), (C) and (D)/(A), (B), (C) तथा (D)

KVS PGT C.S. 2018

Ans. (d) : The < a > tag (anchor tag) in HTML is used to create a hyperlink on the webpage. This hyperlink is used to link the webpage to other webpage or some section of the same webpage. It is either used to provide an absolute reference or a relative reference as its "href" value.
Syntax:
< a href = "link" > Link Name < /a >
So, the given, all anchor tag usages in HTML are valid.

114. What is the correct HTML tag for the largest heading?
सबसे बड़े 'हेडिंग' के लिए सही एच टी एम एल टैग क्या है?

(a) <h1> (b) <head> = large
(c) <heading> = large (d) <h6>

KVS PGT C.S. 2017

Ans. (a) : An HTML heading tag is used to define heading of a page. Headings are defined with the<h1>to <h6> tags. The size of the text depends upon the type of heading tag. h1 is the largest heading tag and h6 is the smallest one. (h1>h2>>h3>h4>h5>h6).

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115. HTML is used to create
HTML____को बनाने के लिए उपयोग किया जाता है।
(a) Machine language program
(b) High level program
(c) Web page
(d) Web server

Bihar STET C.S. 2020

Ans. (c) : HTML language is used to create the web pages. It is a markup language which creates the structure of web page.
Example– Heading, Paragraph, Link, Image, etc.

116. Full form of HTML is
HTML का पूरा रूप है
(a) Hyper Text Markup Language
(b) Hyper Text Manipulation Language
(c) Hyper Text Managing Links
(d) Hyper Text Manipulating Links

Bihar STET C.S. 2020

Ans. (a): HTML Full form of HTML is "Hyper Text Markup Language". It was developed by 'Tim Berner Lee' in 90's.

117. Unclassified email is called?
अवर्णीकृत (Unclassified) ईमेल को क्या कहा जाता है?
(a) Extra mail (b) Spam
(c) Junk mail (d) None of the above

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : Those emails which are unwanted & unclassified, called Junk Email. These emails generally come from Advertisement Promotion & unknown resources.

118. _____is used to send and receive mail.
_____का उपयोग Mail Send & Receive करने के लिए उपयोग करते हैं।
(a) E-mail (b) Internet Explorer
(c) Gopher (d) Telnet

Bihar STET C.S. 18.09.2020 (Shift-I)

Ans. (a) : E-mail stands for Electronic Mail. E-mail used to send and receive message, document, photo and files by using Internet.

119. What does CSS stand for?
CSS का पूर्ण रूप क्या है?
(a) Cascading Style System
(b) Creative Styling Script
(c) Computer Style Sheets
(d) Cascading Style Sheets

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (d) : CSS stands for 'Cascading Style Sheets'. It is a styling sheet which is used to design the website and it makes attractive WebPages.

120. How does the DOM tree structure reflect the hierarchy of HTML elements?
DOM ट्री स्ट्रक्चर HTML एलिमेंट्स की श्रेणीबद्धता को कैसे दर्शाता है?
(a) In alphabetical order/वर्णानुक्रम में
(b) Based on the element's size
एलिमेंट के आकार के आधार पर
(c) In a parent-child relationship
पैरेंट-चाइल्ड संबंध में
(d) According to the element's color
एलिमेंट के रंग के अनुसार

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (c) : DOM (Document Object Model) is like tree structure model that is why it is also called tree structure. In this model all the HTML Elements have relationship as parent-child.

(iii) Web Technology: protocols, development strategies, applications, Web project and team

121. From which of the following is HTML derived?
निम्न में से HTML किससे ली गई है?
(a) URL (b) TCP/IP
(c) SGML (d) XML

HTET PGT C.S. 2020

Ans. (c) : HTML is derived from SGML (Standard Generalized Markup Language). SGML is a meta language used for defining markup languages, and HTML is an application of SGML, meaning it adheres to the rules and structure defined by SGML.

122. A set of style rules that describes a document's display characteristics is:
स्टाइल रूल्स का एक सेट जो डॉक्यूमेंट की प्रदर्शन विशेषताओं का वर्णन करता है, है–
(a) Web Page/वेब पेज (b) Style Sheet/स्टाइल शीट
(c) Table/टेबल (d) List/लिस्ट

HTET PGT C.S. 2023

Ans.(b): Style sheet is a set of style rules that design and provide a better layout of web page or document.

123. What is AJAX in ASP.NET?
ASPNET में AJAX क्या हैं?
(a) Asynchronous JavaScript and XML
असिंक्रोनोस JavaScript एंड XML
(b) Associated JavaScript and XML
एसोसिएटेड JavaScript एंड XML
(c) Asynchronous java script and HTML
असिंक्रोनोस JavaScript एंड HTML
(d) Associated JavaScript and HTML
एसोसिएटेड JavaScript एंड HTML

Bihar STET C.S. 12.09.2023 (Shift-I)
DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : AJAX stands for Asynchronous JavaScript and XML. It's a technique that allows web applications to communicate with a server in the background without requiring a full page refresh. This is done using

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JavaScript the XMLHttp Request object, and often XML (though other data formats like JSON are also common for data exchange).

124. Which of the statements given below is/are correct?
It is always important and useful to include an 'alt' attribute on 'img' tag in HTML because
A. Users who cannot see the image due to vision impairment can have a textual description of the image (which can be spoken aloud by a screen reader).
B. If the image fails to load (slow connection, broken path, etc.) then alt text is displayed instead.
C. SEO (Search Engine Optimization) benefits.
Choose the correct answer from the options given below:
(a) A only (b) B and C only
(c) C only (d) A, B, and C

UGC NET C.S. December 2020

Ans. (d) : When an image contains words that are important to understanding the context, the all text should include those words. This will allow the alt text to play the same junction on the page as the image. Note that it does not necessarily describe the visual characteristics of the image itself but must convey the same meaning as the image.
Example:- An image on a website provides a link to a free newsletter. The image contains the text "Free newsletter. Get free recipes, news, and more. Learn more". the alt text matches the text in the image.

125. What is the by default time in minutes when a page session ends and if user does not refresh or request page?
जब एक उपयोगकर्ता एक पेज को रिफ्रेश या रिक्वेस्ट नहीं करता है और पेज सेशन समाप्त हो जाता है, तो उस स्थिति में डिफॉल्ट के रूप में मिनट में कितना समय लगता है?
(a) 10 (b) 15
(c) 20 (d) 25

LT. Grade C.S. 2018

Ans. (c) : A session automatically ends if a user has not requested or refreshed a page in an application for a specified period of time. This value is 20 minutes by default.

126. What is related to application development strategies?
ऐप्लीकेशन डिवेलपमेंट स्ट्रैटजी से कौन-सा सम्बन्धित है?
(a) Brainstorming/ब्रेन्स्टॉर्मिंग
(b) Technical planning/तकनिकी योजना बनाना
(c) Writing the creative brief/रचनात्मक संक्षेप लिखना
(d) All of the above/उपर्युक्त सभी

LT. Grade C.S. 2018

Ans. (d) : Applications development strategies are plans or approaches used to guide the development process of a project, product, or system.
Brainstorming - often the initial step in development strategies to generate ideas and solutions collaboratively.
Technical planning - Involves defining technical requirements and resources for the development.
Writing the creative brief- Helps document objective, 5 key messages, and deliverables, guiding the project effectively.
All of these activities are interrelated and crucial to application development strategies.

127. A web page contains many small images. We can speed up the performance of the browser by
एक वेब पेज में अनेक छोटे इमेज होते हैं । हम निम्न द्वारा ब्राउसर की निष्पादन गति को बढ़ा सकते हैं .
(a) Having multiple processes executed simultaneously.
अनेक प्रक्रियाओं को एक साथ निष्पादित कर ।
(b) Having multiple threads./अनेक थ्रेड्स के साथ ।
(c) Increasing the speed of the transmission line.
प्रसारण लाईन की गति को बढ़ाकर ।
(d) Increasing the main memory of the system./प्रणाली की मुख्य मेमोरी को बढ़ाकर ।

KVS PGT C.S. 2013

Ans. (b) : Assign one thread for each image so that all images will load faster and performance of the browser will speed up.

128. What is the full form of DTD that is used to describe content in an XHTML document?
किसी XHTML दस्तावेज की विषयवस्तु का वर्णन करने में प्रयुक्त DTD का पूर्ण रूप क्या है?
(a) Document To Definition
(b) Document Type Definition
(c) Document Towards Definition
(d) Document Text Definition

KVS PGT C.S. 2017

Ans. (b) : DTD stands for Document Type Definition. A DTD defines the structure and the legal elements and attributes of an XHTML document. The XHTML Standard defines, 3 DTDs.
* Strict
* Transitional
* frameset.

129. Full form of MVC is:
MVC का पूर्ण रूप है–
(a) Memory Virtual Controller
(b) Model View Container
(c) Maximum View Controller
(d) Model View Controller

HTET PGT C.S. 30.07.2025

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Ans.(d): The full form of MVC is 'Model View Controller'.
MVC is a software design pattern that divides a large application into three inter-connected components the model (manages data), the view (handles user interface) and the controller (processes user input and updates the other two components).

130. A website's front page/main page is called
एक वेबसाइट के फ्रंट पेज या मुख्य पेज को कहा जाता है
(a) Browser Page (b) Search Page
(c) Home Page (d) Bookmark
Bihar STET C.S. 2020

Ans. (c) : Home page is that first page of website when user opens the websites.
It is the main Entry of any website.

(iv) Web Page Designing, Scripting, Server Site Programming

131. Which of the following is suitable tag for java script in HTML ?
निम्न में से कोन-सा एच.टी.एम.एल. में जावा स्क्रिप्ट के लिए उपयुक्त टैग है?
(a) <java> </java> (b) <script> </script>
(c) <jvm> </jvm> (d) <JS> </JS>
HTET PGT C.S. 2018

Ans. (b) : To write the java script code in HTML document, <script> tag is used within <script> tag we can write the programming codes of Java Script.
Syntax:– <Script>–JS codes–</Script>

132. In JavaScript a _____ function can be defined as a function passed into another function as a parameter.
JavaScript में एक _____ फ़ंक्शन को एक पैरामीटर के रूप में दूसरे फ़ंक्शन में पारित के रूप में परिभाषित किया जा सकता है।
(a) Call front/कॉल फ्रंट (b) Call back/क़ॉल बैक
(c) Roll back/रोल बैक (d) Roll front/रोल फ्रंट
DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (b) : In JavaScript a call back function is a function that is passed into another function as a parameter and is then invoked inside the outer function to complete some kind of action.
Callback functions are commonly used in asynchronous operations, event handling and higher order functions.

133. _____ is the person who runs our web servers.
_____वह व्यक्ति है जो हमारे वेब सर्वर चलाता है ।
(a) Data analyst /डेटा एनालिस्ट
(b) Web master /वेब मास्टर
(c) Database designer /डेटाबेस डिजाइनर
(d) Web Designer /वेब डिजाइनर
DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (b) : A webmaster is a person who creates, manages, and maintains a website. They handle various aspects of a website, including design development, content management, server administrator and web server.

134. In JavaScript, the _____ is the master container for all content you view in the web browser,
JavaScript में, _____ वेब ब्राउजर में दिखाई देने वाली सभी सामग्री के लिए मास्टर कंटेनर है।
(a) Document Object/डॉक्यूमेंट ऑब्जेक्ट
(b) Window Object/विंडो ऑब्जेक्ट
(c) Location Object/लोकेशन ऑब्जेक्ट
(d) Screen Object/स्क्रीन ऑब्जेक्ट
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : In JavaScript, the window object is the master container for all content you view in the web browser.
The window object represent the browser window itself and serves as the global object for all JavaScript code running in hat window.

135. Which window method generates a dialog box that displays whatever text you pass as parameter in JavaScript?
कौन सी विंडो विधि एक डायलॉग बॉक्स जनरेट करती है जो जावास्क्रिप्ट में पैरामीटर के भेजे जाने वाले किसी भी टेक्स्ट को प्रदर्शित करता है?
(a) Display() (b) Show()
(c) Alert() (d) Perform()
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (c) : Alert() method is used to generate a dialog box that displays whatever text you pass as parameter in JavaScript.

136. Which object holds the real content of the page in JavaScript?
जावास्क्रिप्ट में पृष्ठ की वास्तविक सामग्री कौन सा ऑब्जेक्ट रखती है?
(a) Document (b) Alert
(c) Location (d) Window
DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : In JavaScript, the Document Object Model (DOM) holds the real content of the page. More specifically, the 'document' object represents the entire HTML document and server as the entry point for interacting with the page's content, structure and style.

137. Which of the following commands will output "one two three"?
(a) For val; do echo-n $val; done < one two three
(b) For one two three; do echo-n-; done
(c) For n in one two three; do echo-n $n; done
(d) For n in one two three {echo -n $ n}
UGC NET C.S. June-2014 (Paper-II)

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14. Miscellaneous

(i) Languages

1. In Python_____operator is used to determine the element's presence or absence in list.
पायथन में_____ऑपरेटर का उपयोग लिस्ट में एलीमेंट की उपस्थिति एवं अनुपस्थिति को निर्धारित करने के लिए किया जाता है।
(a) find (b) in
(c) search (d) lookup
HTET PGT C.S. 30.07.2025

Ans.(b): In operator in Python is used to check if a value exists within a sequence like a list tuple, or string. It returns True if the value is found and false otherwise.
For example– To check if the number 5 is in the list my_list = [1, 2, 3, 4, 5], you would use the expression 5 in my_list.

2. In Python, _____is the type of sequence that resembles a list, except that, unlike a list, it is immutable.
पायथन में_____एक प्रकार का अनुक्रम है जो एक सूची जैसा दिखता है, सिवाय इसके कि, एक सूची के विपरीत, यह अपरिवर्तनीय है–
(a) Vector (b) Object
(c) Dictionary (d) Tuple
HTET PGT C.S. 30.07.2025

Ans.(d): A tuple is a type of sequence in python that is similar to a list, but it's immutable, meaning its elements can't be changed after creation.

3. In Python Dictionaries organize the information by_____, not position.
पायथन में Dictionaries जानकारी को स्थिति के अनुसार नहीं, _____द्वारा व्यवस्थित करता है।
(a) Sequencing (b) Ordering
(c) Inheritance (d) Association
HTET PGT C.S. 30.07.2025

Ans.(d): Python dictionaries organize information by association, not position. They store data as key value pairs where each key is uniquely associated with a specific value. This is different from data structures like lists or tuples, which store information in a sequential or ordered manner based on their index or position.

4. In Python_____function returns the number of characters in string argument.
पायथन में यह फंक्शन स्ट्रिंग में कैरेक्टर की संख्या बताता है–
(a) Length (b) Len
(c) L (d) Numchar
HTET PGT C.S. 30.07.2025

Ans.(b): len() is a built in method that returns the number of elements in a sequence, such as a string. It directly accesses the internal length attribute of the string making it the most efficient way to count characters.

5. What is output of the code in Python Language:
पायथन भाषा में दिए गए कोड का आउटपुट क्या है:
>>> str1 = 'All the Best'
>>> str1[–3]
(a) 'I' (b) 'e'
(c) 'h' (d) 'B'
DSSSB PGT C.S. M/F 11.07.2021

Ans. (b) : >>> str1 = 'All the best'
>>> 'str1[–3]' accesses the third character from the end of the string. The string "All the Best" is indexed as follows 'ALL the Best" from the end, the –3 index position is 'e'. The output of the code is 'e'.

6. Which one is NOT a feature of Python language?
निम्नलिखित में से कौन सी पायथन भाषा की विशेषता नहीं है?
(a) High level language/हाई लेवल लैंग्वेज
(b) Interpreted language/इंटरप्रेटेड लैंग्वेज
(c) Case insensitive/केस इंसेंसिटीव
(d) Portable/पोर्टेबल
DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Case insensitive is not a featured of python languages.
Python is a case-sensitive language, meaning that it distinguishes between upper case and lower case in identifiers.
Example- 'Variable' and 'variable' would be considered two different identifiers in python.

7. What is the meaning of >>> in Python Language?
पायथन भाषा में >>> का क्या अर्थ है?
(a) 3 left shift/लेफ्ट शिफ्ट
(b) 3 right shift/राइट शिफ्ट
(c) Interpreter is ready to take instruction
इंटरप्रेटर निर्देश लेने के लिए तैयार है
(d) Compiler is ready to take instruction
कम्पाइलर निर्देश लेने के लिए तैयार है
DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In Python, the '>>>' symbol is used the prompt in the interactive python interpreter, also known as the python shell when you open the python shell by typing 'python' in your terminal or command prompt. You will see the '>>>' prompt, indicating that the interpreter is ready to accept input.

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8. The given declaration statement in Python Language belongs to which data type?
पायथन लैंग्वेज में दिया गया डिक्लेरेशन स्टेटमेंट किस डेटा टाइप से संबंधित है?

>>> Student = {'SUBJECT' : 'Compiler', 'Marks' : '50', 'Grade' : 'C'}

(a) Numbers/नंबर्स (b) Sequence/सीक्वेंस
(c) Sets/सेट्स (d) Dictionary/डिक्शनरी

DSSSB PGT C.S. M/F 11.07.2021

Ans. (d) : The given declaration statement in python language belong to the Dictionary data type. In python, a dictionary is a built-in data type that stores collections of key-value pairs. Each key is unique and maps to a value.

9. Which statement(s) is/are NOT true about the variables in Python Language:
पायथन भाषा में वेरिएबल्स के बारे में कौन सा/से कथन सत्य नहीं है/हैं?

I. Values of every variable can be changed once a variable has been created and assigned value.
एक वेरिएबल बनाने और मान असाइन किए जाने के बाद प्रत्येक वेरिएबल के मान बदले जा सकते हैं।

II. Assignment statement is used to create a new variable and assign value.
असाइनमेंट स्टेटमेंट का उपयोग एक नया वेरिएबल बनाने और वैल्यू असाइन करने के लिए किया जाता है।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

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Ans. (a) : Statement (I)– is generally true for mutable objects (like list, dictionaries) etc. but its not true for immutable objects like integers strings, tupples, etc.
Statement (II)– Is true In python, an assignment statement is used to create a new variable and assign it a value.
So the statement (i) is not true about the variables in pythons.

10. What is the output of this code in Python Language:
पायथन भाषा में दिए गए कोड का आउटपुट क्या है:

```
>>> x = 125
>>> y = 13
>>> x // = y
>>> x
```

(a) 9 (b) 125/13
(c) 10 (d) 9.62

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Ans. (a) : The '//' operator performs floor division, it divides the numbers and rounds down to the nearest whole number '125 // '3' equals '9'. So the value of 'x' after the operation would be '9'. Thus, the output of code would be 9.

11. What minimum and maximum values will be generated for the following Python statement?
print (random.randrange (100)+1)
निम्नलिखित पायथन स्टेटमेंट में न्यूनतम और अधिकतम वैल्यू ज्ञात करें।

(a) 1 and 100 (b) 0 and 100
(c) 1 and 99 (d) 0 and 99

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Ans. (a) : The random.randrange (100) + 1 statement generates a random integer between 1 and 100 because random.randrange (100) generates a random integer from 0 to 99 and adding 1 to it shift the range to 1 to 100. So, the minimum value generated is 1 and the maximum value generated is 100.

12. Which of the following cannot be used as a valid identifier in Python?
निम्नलिखित में से किसे पायथन में एक वेलिड आइडेंटिफायर के रूप मे प्रयोग नहीं किया जा सकता है?

(a) 1st Name (b) Name1
(c) Name_1 (d) FirstName

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Ans. (a) : The first character of identifiers in python should always start with an alphabet or underscore, and then it can be followed by any of the digit, character or underscore, special characters such as '%', '#', '@', and '$' are not allowed as identifiers in python. A valid identifier can't start with a number, or contain any spaces.

13. Which of the following Function Headers in a Python code will result in error?
पायथन कोड में निम्नलिखित फंक्शन हेडर्स में से किसका परिणाम error होगा?

(a) def Compu (X = 100, Y = 10, Z):
(b) def Compu (X, Y, Z = 1):
(c) def Compu (X=100, Y =10, Z=1):
(d) def Compu (X, Y = 10, Z = 1):

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Ans. (a) : The function header that will result in an error is option (a). def Compu (x = 100, y = 10, z). In Python, when providing default parameter values, all parameters without default values. Therefore placing z before x and y without a default value will cause syntax error.

14. Consider the following dictionary:
निम्नलिखित डिक्शनरी पर विचार कीजिए:

Num = {10 : 'Ten', 100 : 'Hundred', 10: 'Decimal'}
What shall be the output of print (Num)?

(a) {10 : 'Ten', 100 : 'Hundred', 10 : 'Decimal'}
(b) {10 : 'Ten', 100 : 'Hundred'}
(c) {10 : 'Decimal', 100 : 'Hundred'}
(d) Error

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Ans. (c) : In Python a dictionary can't have duplicates keys. When you define a dictionary with duplicate keys like in example, only the last occurrence of the duplicate, key will be stored in the dictionary. Therefore, the output of point (Num)
{10 : 'Decimal', 100: 'Hundred'}

15. Identify the Logical Operator used in Python out of the following.
निम्नलिखित में से पायथन में प्रयुक्त लॉजिकल ऑपरेटर की पहचान कीजिए।

(a) And (b) In
(c) < = (d) + =

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Ans. (a) : In Python, logical operators are used an conditional statements (either true or false). They perform Logical AND, Logical OR and logical NOT operations.

16. Which of the following will not run the loop statement an infinite number of times in python?
निम्नलिखित पायथन कोड में कौन सा लूप स्टेट _____ अनगिनत बार नहीं चलेगा?

(a) While 21%2:
print ("ALWAYS")
(b) While 21/ /2:
print ("ALWAYS")
(c) While True :
Print ("ALWAYS")
(d) While 1 % 1:
print ("ALWAYS")

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Ans. (d) : The loop statement will not run an infinite number of times in Python if the condition inside the 'while' loop evaluates to 'False'.
While 1% 1: is 0, which is considered 'false' in python. Therefore, this loop will not run at all. So the correct answer is (d) while 1%1:

17. Which of the following is an invalid method/function in a Python dictionary?
पायथन डिक्शनरी में इनमें से कौन सा मैथड/फंक्शन वैध नहीं है?

(a) popitem () (b) remove ()
(c) get () (d) pop ()

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Ans. (b) : The remove () method is not valid in a Python dictionary, because remove () method related to lists. It is used to remove the first occurrence of a specified value from a list.

18. For the following Python statement
निम्नलिखित पायथन स्टेटमेंट को देखें

Msg = ('Good Day')
What shall be the data type of Msg?
Msg का डाटा टाइप क्या होगा?

(a) Tuple (b) List
(c) Error in statement (d) String

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Ans. (d) : Statement
Msg = ('Good Day')
This is a string data type if you print the msg the given output string.
Msg = ('Good Day')
print (type (msg))
output - <class 'str'>

19. For the following Python statement
निम्नलिखित पायथन स्टेटमेंट को देखें:

Txt=list ('PEACE')
What shall be the output of print (Txt)?
print (Txt) का आउटपुट क्या होगा?

(a) ['P', 'E', 'A', 'C', 'E'] (b) [P, E, A, C, E]
(c) Error in statement (d) ['PEACE']

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Ans. (a) : The given Python statement Txt = list ('PEACE') converts the string PEACE into a list of characters. So, print (Txt) would output-
['P', 'E', 'A', 'C', 'E']
The list () function converts the string PEACE into a list of characters. Each character in the string becomes an element in the list.

20. Select the correct output for the following Python code.
निम्नलिखित पायथन कोड का सही आउटपुट बताएं।

```
for I in range (6, 1, -2) :
    print (end = '*$$')
```

(a) *$$*$$
(b) *$$ *$$ *$$
(c) *$$*$$*$$
(d) *$$ *$$

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Ans. (c) : The given Python code snippet uses a for loop to iterate over a range of number from 6 to 1 with a step of –2. The print function is used with the end parameter set to '*$$', which means that after printing each item, it will not move to the next line and will instead append '*$$' to the output.
I start at 6 and goes down by 2 in each iteration until it reaches 2 for each iteration, the print (end = '*$$') statement prints '*$$', so the output would be - *$$*$$*$$.

21. Which of the following options is true in Python?
निम्नलिखित में से कौन सा विकल्प पायथन में सही है?

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(a) Execution/Runtime errors can be handled using try _____except
Execution/Runtime errors can be handled using try _____except द्वारा व्यवस्थित किा जा सकता है।

(b) No errors can be handled using try _____except
try _____except द्वारा किसी प्रकार की _____ को व्यवस्थित नहीं किया जा सकता है।

(c) Logical errors can be handled using try _____except
Logical errors को try _____except व्यवस्थित किया जा सकता है।

(d) Syntax errors can be handled using try _____ except
Syntax errors को try _____ except व्यवस्थित किया जा सकता है।

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Ans. (a) : In Python 'try .....except' blocks are used to handle runtime errors. also known as exceptions. syntax errors are caught by the python interpreter before the code is executed, so they can't be handled using 'try ---- except'. Logical errors. Which occur when the code does not produce the desired output due to flawed logic, are not typically handled using 'try ----- except' . They one debugged by inspecting the code and correcting the logic. So the correct option is (a).

22. Which out of the following module is required to be used to create a binary file in Python?
पायथन में एक बाइनरी फाइल _____ लिए निम्नलिखित में से कौन सा मॉड्यूल प्रयोग किया जाना आवश्यक है?

(a) Bin (b) Pickle
(c) Binfile (d) Binary

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Ans. (b) : To create a binary file in Python you would typically use the pickle module. The pickle module is used for serializing and desterilizing Python objects. which means you can use if to write Python objects to a binary file and later read them back into memory.

23. Consider the following Python code and answer the question given below:
निम्नलिखित पायथन कोड पर विचान करें और नीचे दिये गए प्रश्न का उत्तर दें:

```
with open ("NOTES. TXT","r") as F:
  Line=F.readlines()
  _____ # Statement 1
```

Which of the following Python statements will find and display the number of lines present in the file "NOTES.TXT"?
निम्नलिखित पायथन स्टेटमेंट में से कौन सी स्टेटमेंट "NOTES.TXT" के अंदर लाइनों की संख्या को खोज कर लगाएगी और उसे दर्शाएगी?

(a) print (len (LINE))
(b) print (LINE.count()
(c) print (LINE.count('\n'))
(d) print (LINE.len())

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Ans. (a) : The correct statement to find and display the number of lines present in the file 'NOTES.TxT' is-
print(len(LINE))
This statement uses the len() function to find the number of elements in the 'line' list, which contains all the lines of the file "NOTES.TXT".
Each element in the list represents a line in the file. So, the length of the list is equal to the number of lines in the file.

24. Which of the following strings do not belong the given regular expression? (a)*(a + cba)
निम्नलिखित में से कौन-सी स्ट्रिंग दी गई नियमित अभिव्यक्ति से सम्बन्धित नहीं है? (a)*(a + cba)

(a) acbacba
(b) aaa
(c) acba
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : The regular expression (a)*(a + cba) specifies a string that begins with zero or more 'a's followed by either a single 'a' or the string 'cba'.
Lets break down each option.
acbacba– This string starts with a but the remainder cbacba, does not match either a or cba. Therefore this string does not belong to the regular expression.
aaaa– This string matches the pattern. It can be interpreted as three 'a's from the (a)* part, folowed by one 'a' from the (a + cba) part.
acba– This string also matches the pattern. It can be interpreted as one 'a' from the (a)* part followed by the string cba from the (a + cba) part since only acbacba fails to match the regular expression, the correct answer is (a).

25. How can we describe an array in the best possible way?
हम किसी ऐरे का सर्वोत्तम संभव तरीके से वर्णन कैसे कर सकते हैं?

(a) Array are immutable
(b) Container that stores the elements of similar types
(c) The array is not a data structure
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : Array stores elements of the some type in a contiguous block of memory. So we can say that array is a container which stores elements of the same type.

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26. Which of the following gives the nth element of the Array?
निम्नलिखित में से कौन-सा Array का nवाँ तत्व देता है?

(a) array [a];
(b) array [n –1];
(c) array [n + 1];
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : n^{th} element of the array is array [n–1]. Where Array-name is the access name and n is the total number of elements in the array, indexing of array starts from 0 so to add n^{th} element we use n–1.

27. To use a reverse — iterator, you should
रिवर्स-इटरेटर (reverse — iterator) का उपयोग करने के लिए, आपको चाहिए

(a) Begin by initializing it to end ()
शुरू करें इसे प्रारम्भ करने के लिए end () से
(b) Begin by initializing it to rend()
शुरू करें इसे प्रारम्भ करने के लिए rend() से
(c) Increment it to move backward through the container
कंटेनर के माध्यम से पीछे जाने के लिए इसे बढ़ाना
(d) Decrement it to move backward through the container
कंटेनर के माध्यम से पीछे जाने के लिए इसे घटाना

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Ans (c) : Reverse iterates are designed to traverse a container from the end to the beginning rbegin() returns a reverse iterate pointing to the last element, which is the beginning of the reverse iteration. Incrementing a reverse iterate (++it) moves it to the previous element in the container, effectively moving backward through the sequence, conversely, decrementing a reverse iterate (--it) moves it to the next element.

28. In C++, a function contained within a class is called
C++, में, एक क्लास के अन्दर निहित फंक्शन को कहा जाता है

(a) A member function/एक सदस्य फंक्शन
(b) An operator/एक ऑपरेटर
(c) A vlass function/एक क्लास फंक्शन
(d) A method/एक विधि

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Ans (a) : C++, a class is a user defined data type that encapsulates data and functions that operate on that data. Function defined within a class are called member functions because they are members of that class. These functions are associated with an instance of the class and can access the class's data members.

29. A normal C++ operator that acts in special ways on newly defined data types is said to be
एक सामान्य C++ ऑपरेटर जो विशेष रूप से नये परिभाषित डाटा प्रकारों पर कार्य करता है, कहा जाता है

(a) Glorifie/ग्लोरिफाईड
(b) Encapsulated/इनकैप्सुलेटेड
(c) Classified/क्लासिफाईड
(d) Overloaded/ओवरलोडेड

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Ans (d) : In C++, operator overloading allows operators to have user defined meanings on user defined types (classes or strects). This means that operators like +, –, *, etc. can be given new behaviors when used with objects of these new data types.

30. To perform I/O operations in CPP, we must use……header file.
CPP में I/O ऑपरेशन करने के लिए हमें..... हेडर फाइल का उपयोग करना चाहिए।

(a) <iostream.h>
(b) <string.h>
(c) <math.h>
(d) <stdio.h>

Bihar STET C.S. 2020

Ans. (a) : To use input/output (I/O) operations like cin, cout in C^{++}, we need a special 'header file' that defines these operations.

Example :-
```
CPP
# include <iostream.h>
void main(){
cout <<"Hello World";
}
```

31. CPP keywords are also called
CPP कीवर्ड को..... भी कहा जाता है।

(a) Preprocessor
(b) Punctuation symbols
(c) Operators
(d) Reserve words

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Ans. (d) : "Keywords" are words that have a special meaning and are reserved for a specific purpose in programming . Therefore, they are also called "Reserved words".

32. Assume the C++ definition : class circle : public point
Which of the following is false?
मान लें C++ परिभाषाएं : क्लास सर्कल : पब्लिक प्वाइंट
निम्नलिखित में से कौनसा गलत है?

(a) 'Point' is the base class and 'circle' is the derived class./'प्वाइंट' आधार वर्ग है और 'सर्कल' व्युत्पन्न वर्ग है।

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(b) The colon (:) in the header of class definition indicates inheritance./क्लास डेफिनिशन के हेडर में कोलन (:) इनहेरिटेन्स को दर्शाता है।

(c) The keyword 'public' indicates type of inheritance./मुख्य शब्द 'पब्लिक' 'इनहरिटेन्स' के प्रकार को दर्शाता है।

(d) All the public and protected members of class 'circle' are inherited as public and protected members, respectively, into class 'point'./क्लास 'सर्कल' के सभी पब्लिक और प्रोटेक्टेड सदस्यों के क्रमशः पब्लिक एण्ड प्रोटेक्टेड मेम्बर्स के रूप 'प्वाइंट' वर्ग में इनहेरिट किए गए है।

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Ans. (d) : In this question option (d) is incorrect, because according to the given C++ definition, all the public and protected members of class 'point' are inherited as public and protected members, respectively, into class 'circle'. In public inheritance, public members of the base class, become public members of the derived class and protected members of the base class become protected members of the derived class.

33. What is the actual syntax of destructor in C++?
C++ में डिस्ट्रक्टर का वास्तविक सिन्टैक्स क्या है?

(a) !classname() (b) ~classname()
(c) $classname() (d) @classname()

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Ans (b) : A destructor is a special member function of a class that is automatically called when an object of that class goes out of scope or is explicitly deleted. Its primary purpose is to free up resources that the object might have acquired during, its lifetime, such as dynamically allocated memory, file handles, or network connections.

34. Which functions reduce call overload in C++?

(a) Friend functions
(b) Virtual functions
(c) Private functions
(d) Inline functions
(e) Not attempted

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Ans. (d) : Inline functions are a C++ feature that helps reduce call overload. When you declare a function as inline you're suggesting to the compiler that it should replace the function call with the functions' code itself at compile time this is known as inclining. The goal is to avoid the overhead associated with a normal function call, which includes saving registers, pushing arguments onto the stack and then jumping to the functions code.

35. In C++, it is possible to allow a non-member function, an access to private members of a class by declaring it as

(a) Public (b) Global
(c) Protected (d) Friend
(e) Not attempted

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Ans. (d) : A friend function is a non member function that is granted special permission by a class to access its private and protected members. This permission is given by declaring the function as a friend inside the class definition, using the friend keyword.

36. In C++ programming, by default, access to all data members and member functions of a struct is

(a) Public
(b) Private
(c) Protected
(d) Neither public nor private
(e) Not attempted

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Ans. (a) : In C++ programming access to all data member and member functions of a struct is public by default.
When you define a struct in C++ all its members both data members (variable) and member functions are accessible from outside the struct's definition without any specific access specified, This is in contrast to a class where members are private by default.

37. In C++, identify the name of a destructor for class X

(a) X ::X() (b) X :: ~X()
(c) X :: X~() (d) X ::*X()
(e) Not attempted

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Ans. (b) : In C++, a destructor is a special member function that is automatically called when an object is destroyed. Its name is the same as the class name, prefixed by a tilde (~). It is responsible for performing cleanup tasks, such as releasing dynamically allocated memory or closing files.

38. Assuming that ptr is a pointer to float, which of the following is a valid construct in C++?

(a) ++ptr (b) ptr&&
(c) 54& (d) &37
(e) Not attempted

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Ans. (a) : Pointer Arithmetic– In C++, pointer arithmetic is a valid operation. When ++ptr is executed, the pointer ptr is incremented to point to the next memory location of an object of its base type (float in this case). The increment depends on the size of the data type it points to for a float pointer, ptr will he incremented by size of (float) bytes.

39. Which of the following is not among manipulators in C++ ?
निम्न में से कौन-सा C++ मैनिप्युलेटर में से नहीं है?

(a) setw( ) (b) setfill( )
(c) resetiosflags( ) (d) unset( )

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Ans. (d) : Analyze each option to determine which one is not a C++ manipulator.
setw()– This is manipulator from the <iomanip> header. It sets the width of the next input/output field.
setfill()– This is also a manipulation from <iomanip>. It sets the fill character for output operations.
resetiosflags()– This is a manipulator from <iomanip>. It clears the format flags specified in its argument.
unset()– This is not a standard C++ manipulator. There is no such manipulator. There is no such manipulator in the C++ standard library.
Therefore, the correct answer is (d).

40. Polymorphism in C++ is implemented using
(a) Friend functions
(b) Absolute functions
(c) Virtual functions
(d) Inline functions
(e) Not attempted

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Ans. (c) : Polymorphism in C++ is primarily implemented using virtual functions, which enable dynamic (runtime) polymorphism. When a function is declared as virtual in a base class, it allows derived classes to override that function. Then, when a base class pointer or reference points to a derived class object the appropriate overridden function is called based on the actual object type at runtime.

41. In the context of while loop and do-while loop in C++, which of the following is not true?
C++ में while लूप तथा do-while लूप के परिप्रेक्ष्य में, निम्नलिखित में से कौन-सा कथन सही नहीं हैं?
(a) Both the loops are repetitive in nature/दोनों लूप आवृत्तिमूलक प्रकृति के है।
(b) Both are conditional loops/दोनों सशर्त (कंडीशनल) लूप है।
(c) Both will be executed at least once/दोनों को कम-से-कम एक बार निष्पादित किया जाएगा।
(d) Both are terminated when the condition becomes false./शर्त के असत्य होने पर दोनों समाप्त हो जाते हैं।

KVS PGT C.S. 2018

Ans. (c) : While loop and do-while loop, Both are conditional loops and repetitive in nature and both are terminated when the condition becomes false, whereas, while loop condition is checked first then statement is executed and in do-while loop statement is executed at -least once, thereafter condition is checked.

42. Which of the following is not true for overloaded functions C++?
C++ में अतिभारित फंक्शन के लिए निम्न में से क्या सही नहीं है ?
(a) Overloaded functions should have different argument lists/अतिभारित फंक्शन में विभिन्न तर्क सूची होनी चाहिए ।
(b) Overloaded functions having the same argument lists should have a different return type./समान तर्क सूची वाले अतिभारित फंक्शन में भिन्न रिटर्न टाईप होना चाहिए।
(c) Functions cannot be overloaded on the basis of one being static and the other non-static./किसी एक के स्थिर तथा अन्य के अस्थिर होने के आधार पर फंक्शन को अतिभारित नहीं किया जा सकता है।
(d) None of these/उपरोक्त कोई नहीं

KVS PGT C.S. 2013

Ans. (b) : Two or more functions having same name but different arguments are known as overload function.

43. Which constructor will be called by the following lines of code.?
निम्नलिखित कूट लाइनों से कौन सी रचना बुलायी जाएगी ?
(i) Student S1;
(ii) Student S2 = S1
(a) First copy constructor, then default constructor./पहला copy constructor, बाद में default constructor.
(b) First default constructor, then copy constructor./पहला default constructor, बाद में copy constructor.
(c) Default constructors for both lines of code./दोनों कूट लाइनों के लिए Default constructors.
(d) Copy constructors for both lines of code./दोनों कूट लाइनों के लिए Copy constructors.

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Ans. (b) : Constructor is a special method that is automatically called when an object of that class whose constructor is called created.
Default constructor:- A constructor that either has no parameters, or if it has parameters all the parameters have default values
Copy constructor:- A copy constructors is a member function that initializes an object using another object of the same class.

44. Which of the following commands in C++ will open a text file NOTES.TXT for reading the content from it ?
C++ में निम्नलिखित में से कौन सी कमांड से कन्टेन्ट को पढ़ने के लिए NOTES.TXT नामक टेक्स्ट फाइल खुलेगी?
(a) Ifstream F ("NOTES.TXT");
(b) Ofstream F ("NOTES.TXT");
(c) Fstream F ("NOTES.TXT");
(d) Fstream F ("NOTES.TXT", ios :: read);

NVS PGT C.S. 15-.12.2022 (Shift-II)

Sarkari Resource

Ans. (a) : If stream command in C^{++} will open a text file NOTES.TXT for reading the Content from it.
Ifstream ⇒ Reads from files.
Option (a) is correct if stream F ("NOTES.TXT")

45. Given below are the applications of "Queue" data structure.
नीचे ''क्यू'' डाटा स्ट्रक्चर के अनुप्रयोग दिए गए हैं।

A. Managing function calls in C++
C++ में फंक्शन कॉल मैनेज करना।

B. Managing calls in a call centre
कॉल सेन्टर में कॉल मैनेज करना।

C. Allocating terminals to the students in a computer lab, when the students sign up in a web-based request form.
किसी कंप्यूटर लैब में छात्रों को टर्मिनल आवंटित करना, जहाँ छात्र किसी वेब ब्राउज़र में साइन-अप करते है।

Choose the correct option :
सही विकल्प का चयन कीजिए:

(a) (A), (B) and (C) are correct applications
(A), (B) तथा (C) सही अनुप्रयोग है।
(b) Only (A) and (B) are the correct applications
केवल (A) तथा (B) सही अनुप्रयोग है।
(c) Only (B) and (C) are the correct applications
केवल (B) तथा (C) सही अनुप्रयोग है।
(d) Only (A) and (C) are the correct applications
केवल (A) तथा (C) सही अनुप्रयोग है।

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (c) : Queue:- A Queue is defined as a linear data structure that is open at both ends and operations are performed in first in first out (FIFO) order.
⇒ Option (B) and (C) follow above define Queue. and correct option is (B) and (C).

46. Name the type of Inheritance if Activity, Teacher and SUPW are C++ Classes and → denotes inheritance.
यदि Activity, Teacher तथा SUPW, C++ क्लास है और → इन्हेरिटेन्स को इंगित करता है तो इन्हेरिटेन्स का कौन सा प्रकार होगा?

Activity
↓
Teacher
↓
SUPW

(a) Multi Level Inheritance/मल्टी लेवल इन्हेरिटेन्स
(b) Multiple Inheritance/मल्टीपल इन्हेरिटेन्स
(c) Single Level Inheritance/सिंगल लेवल इन्हेरिटेन्स
(d) Hybrid Inheritance/हाइब्रिड इन्हेरिटेन्स

NVS PGT C.S. 15-.12.2022 (Shift-II)

Ans. (a) : In this Type of Inheritance, a derived class is created from another derived class.
⇒ Activity denotes → (Base class 2)
⇒ Teacher denotes → (Base class 1)
⇒ Supw denotes → (Derived class)

47. What will happen if in a C++ program, a class has no name?
यदि C++ प्रोग्राम में किसी क्लास का कोई नाम नहीं है, तो क्या होगा?

(a) It is not even allowed in C++
(b) It will not have the constructor
(c) It will not have the destructor
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c): In C++, if a class does not have a name, the code will be generated and will not compile. A class's name identifies it and is used to create it later. Therefore, code without a class name cannot be compiled.

48. Which of the following is the correct syntax to read the single character to console in the C++ language?
C++ भाषा में कंसोल के लिए एकल वर्ण को पढ़ने के लिए निम्नलिखित में से कौन-सा सही वाक्य विन्यास है?

(a) read ch()
(b) getlinech()
(c) get(ch)
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (e): The syntax for reading a single character from the console in c^{++} is typically done using cin.get (ch) where ch is a character variable. However, none of the given options correctly represents this syntax:-
• Option (a) read ch() is not a valid c^{++} function for reading a single character.
• Option (b) get line ch() is misspelled and invalid: get line is used for strings, not single characters.
• Option (C) get (ch) is incomplete because it does not specify a stream (like cin.get(ch)), and there are no independent get() functions for console input.
• Since none of the options are correct, the answer is (e) none of the above.

49. How many types of the array are there in the C++ programming langauge?
C++ प्रोग्रामिंग भाषा में ऐरे कितने प्रकार के होते हैं?

(a) There are three types of arrays
(b) There are four types of arrays

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(c) The are two types of arrays
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): There are two types of arrays in C^{++}
single-dimensional array- This is a collection of elements of the same data type, stored in a contiguous block of memory.
Multidimensional array- This is an array that contains one of more arrays as its elements.

50. In C++, for what purpose the ''rank()'' is used?
C++ में ''rank()'' का उपयोग किस उद्देश्य के लिए किया जाता है?

(a) It returns the size of each dimension
(b) It returns the maximum number of elements that can be stored in the array
(c) It returns the dimension of the specified array
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): The "rank()" function returns the rank of an array, i.e. the dimension of an array. For example, int arr [10][10] has rank of 2.

51. Which one of the following is the correct definition of the ''is_array();'' function in C++?
निम्नलिखित में से कौन-सी C++ में ''is_array();'' फंक्शन की सही परिभाषा है?

(a) It checks that the specified variable is of the array or not.
(b) It checks that the specified array is of single dimension or not.
(c) It checks that the array specified is of multi-dimension or not.
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (a) In C++, the "is _ array();" function is used to check whether the given value belongs to an array or not.

52. Which of the following is the original creator of the C++ language?
इनमें से कौन C++ भाषा का मूल निर्माता है?

(a) Dennis Ritchie/Brian Kernighan
(b) Ken Thompson
(c) Bjarne Stroustrup
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): The original creator of the C^{++} language is Bjarne stroustrup. He developed C^{++} in 1979. C^{++} is a high-level programming language that supports object-oriented programming.

53. What is abstract class in C++?
C++ में abstract class क्या है?

(a) Any class in C++ is an abstract class/Class from which any class is derived.
(b) Class specifically used as a base class with at least one virtual function.
(c) Class specifically used as a base class with at least one pure virtual function.
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(c): Abstract classes are classes that are defined only to be inherited by other classes. The purpose of abstract classes is to provide a draw structure to other classes that can be inherited. To make a class abstract, at least one pure virtual function must be defined in it. A pure virtual function is just a declaration of a method. A pure function is declared by O.

54. Which of the following constructors is provided by the C++ compiler if not defined in a class?
यदि किसी क्लास में परिभाषित नहीं किया गया है, तो निम्नलिखित में से कौन-सा कंस्ट्रक्टर C++ कंपाइलर द्वारा प्रदान किया जाता है?

(a) Copy constructor
(b) Default constructor
(c) Parameterized constructor
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b): If the constructor is not specified, a default contractor is specified, a default constructor is provided by the compiler.

55. How do structures and classes in C++ differ?
C++ में structures और classes किस प्रकार भिन्न हैं?

(a) In structures, members are private by default whereas in classes, they are public by default.
(b) In structures, members are public by default whereas in classes, they are private by default.
(c) Structures cannot have private members whereas classes can have.
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (b): The main difference between structure and classes in C++ is that the member of structures are public by default while the member of classes are private by default. Structure are generally used to organize data while classes provide functionality to organize data as well as work withit.

56. Pick the incorrect statement about inline functions in C++.
C++ में इनलाइन फंक्शंस के बारे में गलत कथन चुनें।

(a) They save overhead of a return call from a function.
(b) They are generally very large and complicated function.
(c) These functions are inserted/substituted at the point of call.
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b): Inlines are functions that are expanded when called. The entire code of on inline function is replaced at the point of call Inline help in reducing function call overheads Also, they avoid the return call overhead from a function. Inline functions are usually kept small.

57. What is the difference between delete and delete[ ] is C ++?
C ++ में delete और delete [ ] के बीच क्या अंतर है?

(a) Delete is syntactically correct but delete[] is wrong and hence will give an error if used in any case
(b) Delete is used to delete normal objects whereas delete[] is used to delete pointer objects
(c) Delete is used to delete single object whereas delete[] is used to delete multiple (array/pointer of) objects
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): Delete is used to delete single object whereas delete[] is used to delete multiple (array/pointer of) objects.

58. What happens if the following C++ statement is compiled and executed?
यदि निम्नलिखित C++ कथन संकलित और निष्पादित किया जाता है, तो क्या होता है?

```
int*ptr = NULL;
delete ptr;
```

(a) The program is not semantically correct
(b) The program is compiled and executed successfully
(c) The program is compiled successfully but throws an error during runtime
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (*): The above statement is correct because C++ allows the programmer to delete the NULL pointer, so the program compiles and executes successfully.
Note- BPSC has cancelled this question.

59. What is virtual inheritance in C++?
C++ में वर्चुअल इनहेरिटेन्स क्या होता है?

(a) C++ technique to enhance multiple inheritance.
(b) C++ technique to ensure that a private member of the base class can be accessed somehow
(c) C++ technique to avoid multiple copies of the base class into children derived class.
(d) More than one of the above/उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): In C++, virtual inheritance is a technique that ensures that only one copy (a single instance) of the member variables of a base class exists in the final derived class, specifically in diamond inheritance.

60. Which of the following approaches is used by C++?
निम्नलिखित में एक कौन-सा दृष्टिकोण C++ द्वारा उपयोग किया जाता है?

(a) Left-right
(b) Top-down
(c) Bottom-up
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (c): The bottom-up approach is a method of software development in which program development gebins with the implementation of small, low-level modules and then gradually builds up higher-level and more complex modules.

61. Let A be the base class in C++ and B be the derived class from A with protected inheritance. Which of the following statement is false for class B ?

(a) Member function of class B access protected data of class A
(b) Member function of Class can access public data of class A
(c) Member function of class B cannot access private data of class A
(d) Object of derived class B can access public base class data

UGC NET C.S. 4 December 2019 (Paper-II)

Ans. (d) : Object of Derived class cannot access any member of base class directly. So it cannot access public, protected as well as private member directly. Therefore, object of derived class B can access public base class data is false.

62. The control string in C++ consists of three important classifications of characters

(a) Escape sequence characters. Format specifies and White-space characters

(b) Special characters, White-space characters and Non-white space characters

(c) Format specifies, White-space characters and Non-white space characters

(d) Special characters, White-space characters and Format specifies

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Because, In programming language we use unique keywords to store data, just like that, the control string is use to access the different type of keyword, and control strings are the characters that use with the % (percent) and it is a classification of format specifier, white space and non-white space characters.

63. Which statements is correct regarding main () function in C++?

C++ में main () फंक्शन के संदर्भ में कौन सा कथन सही है ?

I. It is a user defined function.
यह यूजर डिफाइन्ड फंक्शन है।

II. It is a pre-defined function.
यह प्री-डिफाइन्ड फंक्शन है।

(a) Only I/केवल I

(b) Only II/केवल II

(c) Both I and II/ I तथा II दोनों

(d) Neither I nor II/ना ही I ना ही II

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : The main () function in C++ must be written by the programmer (user) and cannot be pre-defined. The implementation of main () is always user provided.

64. Which of the following are TRUE about constructors in C++ ?

(A) A constructor can be overloaded.

(B) A constructor does not have a return type.

(C) a constructor must be declared as a friend function.

(D) A constructor is called when an object is destroyed.

Choose the correct answer from the options given below :

(a) (B), (C), (D) Only

(b) (B), (C) Only

(c) (A), (B), (C) Only

(d) (A), (B) Only

UGC NET C.S. Re-Exam- June 2024

Ans. (d) : A constructor can be overloaded, true. In C++, you can have multiple constructors with different parameter lists in the same class (overloading).

A constructor does not have a return type, true. Constructors in C++ do not have a return type, not even void.

A constructor must be declared as a friend function, false. Constructors cannot be declared as friend functions. They are number functions of a class.

A constructor is called when an object is destroyed, false. A constructor is called when an object is created. A destructor is called object is created. A destructor is called when object is destroyed.

65. Which of the following is not correct for virtual function in C++?

(a) Must be declared in public section of class

(b) Virtual function can be static

(c) Virtual function should be accessed using pointers

(d) Virtual function is defined in base class

UGC NET C.S. November-2017 (Paper-II)

Ans : (b) Virtual function cannot be global or static because, a virtual function is a member function of a base class an relies on a specific object to determine which implementation of function is called. You can declare a virtual function to be a friend of another class.

66. Which of the following is not correct (in C++)?

(a) Class templates and function templates are instantiated in the same way

(b) Class templates differ from function templates in the way they are initiated

(c) Class template is initiated by defining an object using the template argument

(d) Class templates are generally used for storage classes

UGC NET C.S. November-2017 (Paper-II)

Ans : (a) **Template** are a feature of the C++ programming language that allow functions and classes to operate with generic types, which allow a function or class to work on many different data type without being rewritten for every one separately.

1. Function template: Behave like function except that the template can have argument of many different types.

(i) Function template does not occupy space in memory.

(ii) The function template does not save memory.

(iii) The function template can be instantiated with any copy constructible type for which the expression is valid.

2. Class template: Provides a specification for generating classes based on parameters.

(i) There are used commonly to implement containers.

(ii) A class template is instantiated by passing a given set of types to it as template arguments.

Sarkari Resource

67. The associativity of which of the following operators is left to right, in C++?

(a) Unary Operator
(b) Logical not
(c) Array element access
(d) Address of

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) Only array element access ([]) is left-to-right associative among the options listed. All others are unary and follow right-to-left.

68. A member function can always access the data in _____ , (in C++)

(a) the class of which it is member
(b) the object of which it is a member
(c) the public part of its class
(d) the private part of its class

UGC NET C.S. November-2017 (Paper-II)

Ans : (a) Member function can always access the data in the class of which is member in C++.

Code:-

```
class x {
Private String name;
Private int age;
x (string name, int age)
{
this . name = name;
this . age = age;
}
int get Age ()
{
return age;
}
}
```

69. Which of the following storage classes have global visibility in C/C++?

(a) Auto
(b) Extern
(c) Static
(d) Register

UGC NET C.S. June-2017 (Paper-II)

Ans : (b) Extern is used for declaring extern variables in C. This modifier is used for all data types like int, float, double etc. It is default storage class of all global variable as well all function.

70. Which of the following operators cannot be overloaded in C/C++?

(a) Bitwise right shift assignment
(b) Address of
(c) Indirection
(d) Structure reference

UGC NET C.S. June-2017 (Paper-II)

Ans : (d)

- Structure reference typically refers to the dot operator (.) which is one of the operators that cannot be overloaded in C^{++}.
- The other options involve operators that can generally be overloaded for custom types in C^{++}.

71. Which of the following cannot be passed to a function in C++?

(a) Constant (b) Structure
(c) Array (d) Header file

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Ans : (d) Header file cannot passes as to a function in C++. While array, constant and structure can be passed into a function.

72. Which one of the following is correct for overloaded functions in C++?

(a) Compiler sets up a separate function for every definition of function
(b) Compiler does not set up a separate function for every definition of function
(c) Overloaded functions cannot handle different types of objects
(d) Overloaded functions cannot have same number of arguments

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Ans : (a) When you call an overloaded function, the compiler determines the most appropriate definition to use by comparing the arguments types you used to call the function with the parameter types specified in the definitions. The process of selecting the most appropriate overloaded function is called overload resolution.

73. In which action of a shift-reduce parser, the parser ensures that the right end of handle is at the top of a stack, locates the left end of the handle within the stacks, and replaces the handle with an appropriate non-terminal?

किस क्रिया में, एक शिफ्ट-रिड्यूस पार्सर यह सुनिश्चित करता है कि हैंडल का दाँया सिरा स्टैक के शीर्ष पर हो, स्टैक के भीतर हैंडल के बायें सिरे का पता लगाता है, और हैंडल को उपयुक्त नॉन-टर्मिनल से प्रतिस्थापित करता है?

(a) Shift (b) Reduce
(c) Accept (d) Error
(e) Not attempted

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Ans. (b) : The "reduce" action in a shift reduce parser is responsible for indentifying a handle: on the top of the stack and replacing it with the corresponding non-terminal symbol on the left - hand side of that production rule. The description in the question perfectly matches the definition of a reduce action .

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74. When the inheritance is private, the private methods in base class are _____ in the derived class (in C++).

(a) Inaccessible (b) Accessible
(c) Protected (d) Public

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Ans.(a) : Inaccessible in the derived class, only public or protected members are inherited
So answer is (a)

75. Which of the following in C++, is inherited in a derived class from base class?

(a) Constructor (b) Destructor
(c) Data Members (d) Virtual Methods

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Ans.(c) : Data members in C++ is inherited in a derived class from base class.
Constructor is a member function of a class which initialize objects of a class. In C++, constructor is automatically called when object create. It is special member function of the class.
Destructor is a member function which destructs or deletes an object.
Virtual methods is a method which is redefined (over-ridden) in derived class.

76. Which of the following is incorrect in C++?

(a) When we write overloaded function we must code the function for each usage
(b) When we write function template we code the function only once
(c) It is difficult to debug macros
(d) Templates are more efficient than macros

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Ans.(d) : First 3 choice are correct, & so it's 4th Templates undoubtedly take larger.
However, templates are significantly more powerful and obey C++ syntactical rules, where as macros do not.
Template take more compilation time than macros. So, macros are more efficient in term of time.

77. In C++, if a function *f* is defined as a friend function, then which of the following category of data members can be accessed using the function *f* ?
C++ में, यदि कोई फंक्शन *f* फ्रेंड फंक्शन के रूप में परिभाषित किया गया है, तो फंक्शन *f* का उपयोग करके निम्नलिखित में से किस वर्ग डेटा मेम्बर तक पहुंचा जा सकता है?

(a) Only Private/केवल निजी
(b) Only Protected/केवल संरक्षित
(c) Only Public/केवल सार्वजनिक
(d) Public, Private and Protected/सार्वजनिक, निजी और संरक्षित

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Ans. (d) : Public, private and Protected
Explanations:-
In C++ if a mind that this is not a members function, but a non-member function, which has been given special permission.

78. In case of public inheritance in C++, which of the following correctly represents the accessibility of the data variables from derived class?
C++ में पब्लिक इनहेरिटेंस के मामले में, निम्नलिखित में से कौन सा व्युत्पन्न वर्ग से डेटा चर की पहुंच को सही ढंग से दर्शाता है?

(a) Private variables: Yes, Protected variables: Yes, Public variables: Yes/निजी चरः हाँ, संरक्षित चरः हाँ, सार्वजनिक चरः हाँ
(b) Private variables: Yes, Protected variables: No, Public variables: Yes/निजी चरः हाँ, संरक्षित चरः नहीं, सार्वजनिक चरः हाँ
(c) Private variables: No, Protected variables: Yes, Public variables: Yes/निजी चरः नहीं, संरक्षित चरः हाँ, सार्वजनिक चरः हाँ
(d) Private variables: No, Protected variables: No, Public variables: Yes/निजी चरः नहीं, संरक्षित चरः नहीं, सार्वजनिक चरः हाँ

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Ans. (c) : When a class is publicity inherited in C++:

Base class member	In derived class	Access from derived class
Public	remains public	Accessible
Protected	remain protected	Accessible
Private	not inherited as accessible	Not accessible

79. In case of protected inheritance in C++, which of the following correctly represents the accessibility of the data variables from second derived class?
C++ में संरक्षित इनहेरिटेंस के संबंध में, निम्नलिखित में से कौन सा दूसरे व्युत्पन्न वर्ग से डेटा चर की पहुंच को सही ढंग से दर्शाता है?

(a) Private variables: Yes, Protected variables: Yes, Public variables: Yes/निजी चरः हाँ, संरक्षित चरः हाँ, सार्वजनिक चरः हाँ
(b) Private variables: Yes, Protected variables: No, Public variables: Yes/निजी चरः हाँ, संरक्षित चरः नहीं, सार्वजनिक चरः हाँ
(c) Private variables: No, Protected variables: Yes, Public variables: Yes/निजी चरः नहीं, संरक्षित चरः हाँ, सार्वजनिक चरः हाँ

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(d) Private variables: No, Protected variables: No, Public variables: Yes/निजी चरः नहीं, संरक्षित चरः नहीं, सार्वजनिक चरः हाँ

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Ans. (c) : Private variables: No, Protected variables: Yes, Public variables : Yes

Explanation: In case of protected inheritance in C++, the members of the base class can be accessed in the derived class in the following way:

Private members: Private members of the base class cannot be accessed in the derived class. They are accessible only with in the base class.

Protected members: Protected members of the base class can be accessed as protected in the derived class.

Public members: The public members of the base class can be accessed as protected in the derived class. This means they will be accessible within the derived class and in any classes that are further derived from that class, but they will not be directly accessible outside the class.

80. Considering A as a one-dimensional array, which of the following C++ statements will print the third element of A?
A को वन-डायमेंशनल ऐरे के रूप में मानते हुए, निम्नलिखित में से कौन सा C++ स्टेटमेंट्स (विवरण) A के थर्ड एलीमेंट (तृतीय) को प्रिंट करेगा?

(a) cout << A[3];
(b) cout << 3[A];
(c) cout << 2[A];
(d) cout << A[2];

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Ans. (d): In one dimensional array, the third element can be printed usingh the statement cout<<A[2].
In C++, array indexing starts from 0.
The first element is at index 0 (e.g., A[0])
The second element is at index 1 (e.g., A[1])
The third element is at index 2 (e.g., A[2]).
In option (c) cout<<2[A]; similar to the above, this is also a valid but non-standard syntax. It is equivalent to *(2+A), which correctly accesses the third element.
Therefore, cout<<[2]; is the correct and conventional statement for printing the third element.
As per commission correct answer is option (c).

81. Consider the following declaration in C++; int C, *PTR;
Which of the following statements is TRUE?
C++ में निम्नलिखित डेक्लरेशन्स पर विचार करें:
int C, *PTR;
निम्नलिखित कथनों में से कौन सा कथन सत्य है?

(a) PTR = C;
(b) *PTR = &C;
(c) PTR = &C;
(d) C = &PTR;

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Ans. (c) : in c, PRT;
This declores c as an integer variable and PTR as a pointer to an integer.
option (a) PTR = c ;
This assings because the value of c (an integer) to PTR (a pointer). This is invalid because we cannot assign an integer to a pointer without a cast. This would generate a compiler error.
option (b): PTR = &c;
This dereferences PTR and assigns the address of c to that integer. This is invalid because we cannot assing a pointer to an integer. This would generate a compiler error.
option (c): PTR = &c:
This assings the address of c (which is an pointer to an integer) to PTR (which in a pointer to an integer). This is valid and correct. After this assignment, PTR to c. This is invalid. Therefore, the only true (valid) statement is (c).

82. What is the correct interpretation of the following declaration in C++?
int (*FPTR)(char*)
C++ में निम्नलिखित डेक्लरेशन की सही व्याख्या क्या है? int (*FPTR)(char*)

(a) FPTR is a pointer to character
FPTR कैरेक्टर (अक्षर) का पॉइंटर है
(b) FPTR is a pointer to a function, which takes character as an argument and returns integer value/FPTR एक फंक्शन का एक पॉइंटर है, जो कैरेक्टर (वर्ण) को तर्क के रूप में लेता है और पूर्णांक मान देता है
(c) FPTR is a pointer to a function, which takes character pointer as an argument and returns integer value/ FPTR एक फंक्शन का एक पॉइंटर है, जो कैरेक्टर (वर्ण) पॉइंटर को तर्क के रूप में लेता है और पूर्णांक मान देता है
(d) FPTR is a function, which takes character pointer as an argument and returns integer pointer/ FPTR एक फंक्शन है, जो वर्ण सूचक को तर्क के रूप में लेता है और पूर्णांक सूचक देता है

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Ans. (c) : In the statement int (* F PTR) (char*);
int represent that the function's return type is an integer.
(* FPRR) indicates that FPTR is a pointet (not a function returning a pointer).
(char *): The function takes a character pointer (char *) as its single parameter.
So, int (* FPTR) (char*) means FPTR is a pointer to a function, which takes character pointer as an argument and returns integer value.

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83. Which of the following is not a correct declaration statement in C++?
निम्न में से कौन सा C++ में करेक्ट डेक्लरेंश स्टेटमेंट (सही घोषणा विवरण) नहीं है?

(a) int *ptr = 0, p;
(b) int p, *ptr = 0;
(c) int p, double *fp = &P;
(d) int *p, *q, r;

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Ans. (c) : In c ++, a single declaration statement can list multiple variables only if they share the same base type. In option (c). the statements tries to declare p as an int and fp as a pointer to double with in the same declaration. C ++ does not allow mixing different base types (int and double *fp) in on declaration list. also, & P is invalid because C++ is
case- sensitive : if the variable is decleaned as p (lowercase), p (upper case) is a different identifier and is undeclared. Other option, are valid declaration.
Therefore, incorrect declaration statement in C++ is option (c) i.e. int p, double *fp = &P;

84. The programming language C++ was developed by:
प्रोग्रामिंग लैंग्वेज (भाषा) C++ किसने विकसित किया था?

(a) Blaise Pascal/ब्लेज पास्कल
(b) Dennis Ritchie/डेनिस रिची
(c) Alan Turing/एलन ट्यूरिंग
(d) Bjarne Stroustrup/बजेर्ने स्ट्रॉसट्रुप

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Ans. (d) : The C++ Programming language was developed by Bjarne Stroustrup at Bell Labs in 1979 as an enhanced, object-oriented extension of C. The name C++ comes from the increment operator (+ +) in C.
Other options ∴
Dennis Ritchie: Father of computer science and artificial intelligence, but not related to C++.
Blaise Pascal ∴ The language Pascal was named after him.

85. Which of the following characters is a terminal that terminates a statements in C++ ?
निम्नलिखित में से कौन सा वर्ण टर्मिनल है, जो C++ में कथन को समाप्त कर देता है?

(a) Semicolon (;)/ सेमीकॉलन (अर्धविराम) (;)
(b) Colon (:)/कॉलन (:)
(c) Ampersand (&)/डॉलर (&)
(d) Dollar ($)/डॉलर ($)

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Ans. (a) : In C++, every complete statement must end with a semicolon (;).
e.g. int x = 10 ; // semicolon terminates the statement
count << x; // semicolon terminates the statement
other option:
Collon(:) is used in other context (like lables, scope resolution::)
Ampersand (&) is used for reference or address of operator.

86. Which of the following is the insertion operator in C++, used to write formatted data into the stream?
निम्नलिखित में से कौन सा C++ में ऑपरेटर है, जो डिफॉर्मेटेड (स्वरूपित) डेटा को स्ट्रीम में लिखने के लिए इस्तेमाल किया जाता है?

(a) >> (b) <<
(c) && (d) <>

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Ans. (b) : In C++, the operator << is known as the insertion operator when used with output streams like cout. It is used to insert (write) formatted data into the output streams.
e.g. # include < iostream >
using namespace std;
int main ( ) {
cout<< "Hello World" ; // << "Hello World" into the cout stream
return 0 ;
?
>> is the extraction operator used with cin for input.
&& is the logical AND operator.

87. In C++, trying to access private data form outside of a class throws error. This feature is known as:
C++ में, क्लास के बाहर से निजी डेटा तक पहुँच प्राप्त करने में त्रुटि होती है। इस गुण को किस नाम से जाना जाता है?

(a) Data abstraction/डेटा ऐब्स्ट्रैक्शन
(b) Data encoding/डेटा एन्कोडिंग
(c) Data hiding/डेटा हाइडिंग
(d) Data mining/डेटा माइनिंग

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Ans. (c) : In C++, trying to access private data from outside of a class throws error. This feature is known as Data hiding. Data hiding means restricting direct access to a class's internal data from outside the class. In C++, members declared as private (or protected) can not be accessed directly from outside the class.
Access is allowed only through public member functions.
other options:
Data abstraction: showing only essential features while hiding details.
Data encoding: converting data into a specific format
Data mining: Discovering patterns from large data sets.

Sarkari Resource

88. **The operator C++, which is used to access a value referred by a pointer is known as _____**
C++ का ऑपरेटर, जिसका उपयोग किसी प्वॉइंटर द्वारा निर्दिष्ट मान तक पहुँचने के लिए किया जाता है, _____ के रूप में जाना जाता है।

(a) Indirect operator /इन्डाइरेक्ट ऑपरेटर
(b) Direct operator/डाइरेक्ट ऑपरेटर
(c) Indirection operator/इन्डायरेक्शन ऑपरेटर
(d) Inseartion operator/इन्शर्सन ऑपरेटर

DSSSB TGT C.S. 07.08.2021 (Shift-II)

Ans. (c) : In C++, the * (pointer) operator is used to acess the value pointed to by a pointer. This operator is known as the indirection operator (or deference operator).
e.g. int x = 10;
int *ptr = & x;
printf ("%d", *ptr); // Here,*ptr access the valve stored at address held by ptr.
So, the operator that accesses the value referred by a pointer is called the indirection operator.

89. **If a function is friend of a class, which one of the following is wrong?**

(a) A function can only be declared a friend by a class itself
(b) Friend functions are not members of a class, they are associated with it
(c) Friend functions are members of a class
(d) It can have access to all members of the class, even private ones

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Ans : (c) A friend function of a class is defined outside that class's scope and it has the right to access all private and protected members of the class. Even through the prototypes for friends functions appears in the class definition, friends are not members functions.

90. **Given the following statements :**

(1) To implement Abstract data type, a programming language require a syntactic unit to encapsulate type definition.
(2) To implement ADT, a programming language requires some primitive operations that are built in the language processor.
(3) C++, Ada, Java 5.0, C#2005 provide support for parameterized ADT

Which one of the following options is correct?

(a) (1), (2) and (3) are false
(b) (1) and (2) are true; (3) is false
(c) (1) is true; (2) and (3) are false
(d) (1), (2) and (3) are true

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Ans : (d) Language requirements for ADT'S
- A syntactic unit in which to encapsulate the type definition
- Ada provides package that simulate ADTs
- C++ data abstraction is provided by classes
- Java's data abstraction is similar to C++

So, 1, 2, 3 are true
Hence, answer is (d).

91. **Consider the code segment written below in C++ :**
C++ में नीचे लिखे गए कोड खंड पर विचार कीजिए:

```
if (count < 10)//if#1
  if ((count%4) ==2)//if#2
        cout<<" condition;white/n";
  else//(Indentation is wrong)
  cout<<" condition is:tan/n";
```

There are 2 if statements and one else.
To which if, the else statement belong?
दो if स्टेटमेंट हैं तथा एक else स्टेटमेंट है। कौन से if से else स्टेटमेंट संबंधित है?

(a) It belongs to if # 1/यह if # 1 से संबंधित है।
(b) It belongs to if # 2/यह if # 2 से संबंधित है।
(c) It belongs to both the if statements./यह दोनों if स्टेटमेंट से संबंधित है।
(d) It is independent./यह स्वतंत्र है।

KVS PGT C.S. 2018

Ans. (b) : There are two if statements (if # 1, if # 2) and one else, then else belongs to if # 2 because-
suppose, count = 8
then if # 1
if (8 < 10) is true
if # 2
if (8 % 4 = = 2) is false then it will related with the else statements.

92. **Which of the following is not correct in C++?**
C++ में निम्नलिखित में से कौन सा कथन सही नहीं है?

(a) x– =2; is the same as x = x–2;
x– =2; समान है x = x–2; के।
(b) x* = 2; is the same as x = x*2;
x* = 2; समान है x = x*2; के।
(c) x% = 2; is the same as x = x/2;
x% = 2; समान है x = x/2; के।
(d) x/=2; is the same as x = x/2;
x/=2; समान है x = x/2; के।

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Ans. (c) : The % (Modulus) operator given the remainder of an integer after division. In Assignment operator, % = (Modulus then assign) takes modulus using the values of the two operands and assigns the result to the left operand. So, x% = 2; is the same as x = x% 2.

Sarkari Resource

93. **Consider the following code segment in C++. How many times the for loop will be repeated?**
C++ में निम्नलिखित कोड खंड पर विचार करें। for लूप की कितनी बार आवृत्ति होती?

```
int main ( )
{
  int f = 1;
  for (; f ;)
  cout <<"f = " <<f++<<"\n";
  return 0;
}
```

(a) 10 times/10 बार
(b) Not even once/एक बार भी नहीं
(c) Repeated forever/हमेशा के लिए दोहराया जाएगा
(d) Only once/केवल एक बार

KVS PGT C.S. 2018

Ans. (c) : In given code segment there is no test condition in the for loop due to which loop will get executed infinite number.
So, the for loop will be repeated forever.

94. **How many characters does an escape sequence (\On, \Hn, \n, \f) in C++ consume?**
C++ में कोई पलायन अनुक्रम (\On, \Hn, \n, \f) कितने वर्णों की खपत करता है ?

(a) 1
(b) 3
(c) 2
(d) None of these/इनमें से कोई नहीं

KVS PGT C.S. 2013

Ans. (a) : Escape Sequences- They are primarily used to put non-printable characters in character and string literals.

All escape sequences consist of two or more characters. The first of which is the backslash \\ the remaining characters determine the interpretation of the escape sequence. If we treat it's a character constant means one character

An escape sequence is regarded as a single character and is therefore valid as a character constant.

95. **Consider the following two C++ statements and the accompanying contexts.**
निम्नलिखित दो C++ कथनों तथा संलग्न परिप्रेक्ष्यों पर विचार कीजिए।

(A) Return P→next = = NULL; // P is a pointer to an element in the link list L.
return P→next = = NULL; // P लिंक लिस्ट L में एक एलीमेंट का पॉइंटर है।

(B) Return S → next == NULL; // S is a stack implemented as a link list.
return S → next == NULL; // S एक स्टैक है जिसे लिंक लिस्ट के तौर पर क्रियान्वित किया गया है।

If the return value is TRUE in both cases, which of the following is correct ?
यदि दोनों मामलों में रिटर्न मान TRUE हो निम्नलिखित में से कौन सा कथन सत्य है?

(a) P points to the last element in L and S is empty./P, L में अंतिम एलीमेंट को सूचित करता है और S खाली है।
(b) P points to the last element in L and S is full./P, L में अंतिम एलीमेंट को सूचित करता है और S भरा हुआ है।
(c) L is empty and S is full.
L खाली है और S भरा हुआ है।
(d) Both L and S are empty.
L तथा S दोनों खाली है।

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Ans. (c) : Statement (A) – return P > next = NULL;
If the linked list L is empty, there is no node.
Assigning P – > next – NULL still returns True
Statement (B) : return S – > next = NULL;
If stack is full, it has nodes.
Setting the last node's next to NULL returns True.
So, both statements return True when L is empty and S is full.

96. **Consider the following Inheritance in C++ :**
C++ में निम्नलिखित इन्हेरिटेन्स पर विचार कीजिए:

```
class Person
{
char AadharNo[12];
protected :
char Name[20];
public :
_______
_______
void Readdata( );
};
class Student : public Person
{
int Grade;
public :
_____
_____
};
```

What will be the visibility mode of Name and Readdata () in the derived class student ?
डेराइव्ड क्लास student में Name and Readdata () का विजीबिलिटी मोड क्या होगा:

(a) Name → Protected, Readdata → Protected
(b) Name → Protected, Readdata → Public
(c) Name → Private, Readdata → Protected
(d) Name → Public, Readdata → Public

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Ans. (b) : Visibility mode inherited member
⇒ Public :- When the member is declared as public, it is accessible to all the functions of the program.
⇒ Private:- When the member is declared as private, it is accessible within the class only.
⇒ Protected :- When the member is declared as protected, it is accessible within its own class as well as the class immediately derived from it.
⇒ In Question - visibility mode of
Name → protected
Readdata → Public

97. This question is based upon the following class and code segment :
यह प्रश्न निम्नलिखित क्लास तथा कोड खंड पर आधारित हैं:

```
class STOCK
{ float price ;
public :
STOCK( ) ;                  // Function 1
STOCK(float P) ;            // Function 2
STOCK(STOCK &S) ;           // Function 3
void INIT( ) ;              // Function 4
void DISP( ) ;              // Function 5
~ STOCK( ) ;                // Function 6
};
void main( )
{
STOCK S2(10) ;              // Statement 1
_________
_________                   // Other Statements
_________
}
```

Which of the following declaration/statement will invoke the default constructor of the class - STOCK ?
निम्नलिखित में से कौन सी घोषणा/कथन class-STOCK के डिफॉल्ट कंस्ट्रक्टर को सक्रिय करेगी?

(a) STOCK S1(S2); (b) STOCK S2;
(c) STOCK S1(20); (d) S2.INIT();

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Ans. (c) : STOCK S1 (20): creates an object S1 of the class STOCK.
It calls the constructor STOCK (float P) to initialize S1.
It automatically runs to reate the object, even though technically it is a parmeterized constructor.
So, in simple words : "S1 is created and initialized using a constructor.
Hence, options (c) STOCK S1(20); is correct.

98. This question is based upon the following class and code segment :
यह प्रश्न निम्नलिखित क्लास तथा कोड खंड पर आधारित हैं:

```
class STOCK
{ float price ;
public :
STOCK( ) ;                  // Function 1
STOCK(float P) ;            // Function 2
STOCK(STOCK &S) ;           // Function 3
void INIT( ) ;              // Function 4
void DISP( ) ;              // Function 5
~ STOCK( ) ;                // Function 6
};
void main( )
{
STOCK S2(10) ;              // Statement 1
_________
_________                   // Other Statements
_________
}
```

Which of the following constructor will be invoked automatically because of the declaration given as Statement 1 in the given code ?
दिए गए कोड में Statement 1 के तौर पद दी गई घोषणा के कारण निम्नलिखित में से कौन सा कंस्ट्रक्टर स्वतः सक्रिय हो जाएगा?

(a) Function 2 (b) Copy Constructor
(c) Function 1 (d) Function 3

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Ans. (d) : When an object of the class is returned by value.
⇒ When an object of the class is passed by value as an argument
So, option is (d) correct.

99. This question is based upon the following class and code segment :
यह प्रश्न निम्नलिखित क्लास तथा कोड खंड पर आधारित हैं?

```
class STOCK
{ float price ;
public :
STOCK( ) ;                  // Function 1
STOCK(float P) ;            // Function 2
STOCK(STOCK &S) ;           // Function 3
void INIT( ) ;              // Function 4
void DISP( ) ;              // Function 5
~ STOCK( ) ;                // Function 6
};
void main( )
{
STOCK S2(10) ;              // Statement 1
_________
_________                   // Other Statements
_________
}
```

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Which of the following can be an example of copy constructor ?
निम्नलिखित में से कौन सा कॉपी कंस्ट्रक्टर का उदाहरण हो सकता है?

(a) Function 1 (b) Function 2
(c) Function 3 (d) Function 5

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Ans. (c) : A copy constructor is member function that Initializes an object using another object of the same class.
Hence option (c) is example of copy constructor.

100. Select the correct output of the following C++ program segment assuming all necessary header files are correctly included:
यह मानते हुए कि सभी आवश्यक हैडर फाइलें सही प्रकार से सम्मिलित हैं, निम्नलिखित C++ प्रोग्राम खंड के सही आउटपुट का चयन कीजिए:

```
void main( )
{
char *S = "MySQL";
S + = 2;
cout << S <<endl;
}
```

(a) SQL (b) QL
(c) My (d) MyS

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Ans. (a) : char * S = "MYSQL"
S+ = 2,
Cout << S << endl;
So ⇒ print answer is SQL.

101. Identify the correct output of the C++ code from the following options assuming all required header files are included:
यह मानते हुए कि सभी आवश्यक हैडर फाइलें शामिल हैं, निम्नलिखित विकल्पों में से C++ कोड के सही आउटपुट की पहचान कीजिए:

```
void main( )
{
int N[ ] = {1, 2, 3, 4};
int *P = N;
N[1] + = 10;
cout << *P << " : " ;
P ++;
cout << *P << endl;
}
```

(a) 1 : 12 (b) 1 : 2
(c) 11 : 2 (d) 11 : 12

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Ans. (a) : Output Analysis:
- First *P → 1
- After P + +, second *P → 12

Output = 1:12

102. If all required header files are already included, identify the possible output of the C++ code :
यदि सभी वांछित हैडर फाइलें पहले ही सम्मिलित कर ली गई हो तो C++ कोड के संभावित आउटपुट की पहचान कीजिए:

```
class Weather
{
int Humidity, Temp; public :
Weather (int H = 50, int T = 20)
{ Humidity = H; Temp = T;}
void HOT (int T) { Temp += T;}
void HUMID (int H) { Humidity = H; }
void DISP( ) { cout << Humidity <<" : "<<
Temp << endl;}
};
void main( )
{
Weather W(60);
W.HOT(5); W.HUMID(65);
W.DISP( );
}
```

(a) 65 : 25 (b) 60 : 25
(c) 60 : 20 (d) 65 : 5

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Ans. (a) : Execution step by step:
- **Object creation:** Weather W(60) → Humidity = 60, Temp = 20
- **HOT (5) call:** Temp = 20 + 5 = 25
- **HUMID (65) call:** Humidity = 65
- **DISP ( ) call:** prints → 65: 25

103. Write the output of the following program:
निम्नलिखित प्रोग्राम का आउटपुट लिखें :

```
int a[] = {1, 2, 3} *p;
```

(a) Junk value/जंक वैल्यू
(b) 3
(c) Runtime error/रनटाइम त्रुटि
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (e) : The following statement is syntactically different in C++: int a[] {1, 2, 3}* p; attempts to declare an array a and a pointer p, but it combines array a and a pointer p, but it combines array initialization with a pointer declaration in a different way. int a [] = {1, 2, 3} initializes an array with three elements 1,2, and 3* p is intended to declare a pointer, but in this context, the placement of *p after the array initialization is invalid.
Due to the syntactic problem, the program will not compile and a compile-time error will occur. if the correct option is to choose (e), then none of these will occur.

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104. **The C++ code which causes abnormal termination/behaviour of a program should be written under _____ block.**
C++ कोड, जो किसी प्रोग्राम के असामान्य समापन/व्यवहार का कारण बनता है, उसे ब्लॉक के अंतर्गत लिखा जाना चाहिए।

(a) Catch
(b) Throw
(c) Try
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

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Ans. (c): The code which leads to abnormal termination of the program should be written under the try block.

105. **Consider the following C++ function f() :**

```
usigned int f (unsigned int n){
Unsigned int b = 0
while (n) {
b + = n & 1;
n > > 1;
}
return b;
}
```

The function f() returns the int that represents the __p__ in the binary representation of positive integer n, where P is

(a) Number of 0' s
(b) Number of bits
(c) Number of consecutive 1's
(d) Number of 1's

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Ans. (d) : b + = n & 1;
binary AND operation and then add result with b consider n as 5;
So binary representation of n is 101
initially, b = 0
B + = n & 1; // It means b = 0 + 101
& 001; b = 0 + convert to decimal (001);
b = 1 n >> // n is right shifted to 1 so,
n becomes 010
b + = n & 1;
this gives b = 1 + 010 & 001; b = 1 + 0; b = 1
Again, n is right shifted; n becomes 001
b + = n & 1; // b = 1 + 001 & 001;
b = 1 + 1 = 2
Again, n is right shifted n becomes 000 while (n) //It becomes false
program will return the final value of b; which is 2.
Which is equal to the number of 1's in the binary representation of n.

106. **Consider the C/C++ function f() given below :**

```
void f(char w[])
{
    int x = strlen (w); //length of a string
    char c;
    for (int i = 0; i < x; i++)
    {
        c = w[i];
        w[i] = w[x–i–1];
        w[x–i–1] = c;
    }
}
```

Which of the following is the purpose of f( )?

(a) It outputs the contents of the array with the characters rearranged so they are no longer recognized as the words in the original phrase.
(b) It outputs the contents of the array in reverse order.
(c) It outputs the contents of the array in the original order.
(d) It outputs the contents of the array with the characters shifted ever by one position.

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Ans. (c) : Given an array arr[] and an integer c, we need to print k maximum elements of given array. The elements should printed in the order of the input.
Note : c is always less than or equal to **w**.
It outputs the contents of the array in the original order.

107. **Consider the following two C++ programs P_1 and P_2 and two statements S_1 and S_2 about these programs :**

P_1

```
void f(int a, int *b, int &c)
{   a = 1;
   *b = 2;
    c = 3;
}
int man( )
{   int i = 0;
    f(i, & i, i);
    cout << i;
}
```

P_2

```
double a = 1, b = 2;
double & f(double & d)
{   d = 4;
    return b;
}
int main ()
{
    f(a) = 5;
    cout << a << ":" << b;
}
```

S_1 : P_1 prints out 3
S_2 : P_2 prints out 4 : 2
What can you say about the statements S_1 and S_2?

(a) Only S_2 is true.
(b) Both S_1 and S_2 are true.
(c) Only S_1 is true.
(d) Neither S_1 and S_2 is true.

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Ans. (c) : in P_1: output is 3. But in P_2 Print() function is both overrided and overloaded in derived class, so option P_2 is wrong.

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P_1	P_2

void f(int a, int *b, int &c)
{ a = 1;
*b = 2;
c = 3;
}
int man( )
{ int i = 0;
f(i, & i, i);
	double a = 1, b = 2;
	double & f(double & d)
	{ d = 4;
	return b;
	}
	int main ()
	{
	f(a) = 5;

108. What is the output of code given below :

```
#include<iostream>
using namespace std;
class Base {
public :
Base ( ){
        cout <<"Base Const";
}
virtual ~Base ( )
{
        cout<<"Base    dest";
}
    };
class derived : public Base {
Public:
Derived ( )  {
    cout << "Derived Const" ;
~Derived ( ) {
        cout<<"Derived dest";
            }
            };
int main ( )
{
    Base *b = new Derived ( );
    delete b;
    return 0;
}
```

(a) Base Const Derived Const Derived dest Base dest
(b) Base Const Derived Const Base dest Derived dest
(c) Derived Const Base Const Base dest Derived dest
(d) Base Const Derived const Base dest

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Ans. (a) : The correct answer is–
Base const derived const derived dest base dest.
Explanation–
Base *b = new derived(); – A Derived Object is created but the pointer is of type base. The constructors are called in the order of inheritance–
The Base constructor (Base()) is called first, printing Base const. Then the derived constructor (derived()) is called, printing derived const.
Delete b; – Since the destructor in Bases is declared as virtual, the destructors are called in the reverse order of construction–
The derived destructor (~ Derived()) is called first, printing derived dest. Then the base destructor (~ Base()) is called, pointing base dest.
So the output is– Base const derived const Derived dest Base dest.

109. 'ptrdata' is a pointer to a data type. The expression *ptrdata++ is evaluated as (in C++):

(a) *(ptrdata++)
(b) (*ptrdata)++
(c) *(ptrdata)++
(d) Depends on compiler

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Ans : (a) Here ++ works first on the pointer (because it has higher priority than *)
It means –
(1) use the value at ptrdata (dereference).
(2) After using it, move the pointer to the next location.
So the option (a) is correct.
If it was written (*ptrdata)++, then it would increase the value stored at that location not the pointer.

110. Which option gives the correct interpretation of the following declaration in C++?
कौन सा विकल्प C++ में निम्नलिखित डेक्लेरेशन (घोषणा) की सही व्याख्या करता है?

int(*p[5])();

(a) p is a pointer to function
p फंक्शन का पॉइंटर है
(b) p is an array of pointers to function
p फंक्शन के पॉइंटर का एक ऐरे है
(c) p is a pointer to such function which return type is array
p ऐसे फंक्शन का पॉइंटर है जो रिटर्न टाइप ऐरे है
(d) p is a pointer to an array of functions
p ऐरे के फंक्शन का पॉइंटर है।

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : int (*p [5]) ( ); means:
• p[5] : p is an array of 5 elements.
• (*p[5] **:** each element is a pointer.
• ( ) **:** Each pointer points to a function that takes no arguments.
• Int: each function returns an int.
So, p is an array of 5 pointers to functions that take no arguments and return an int.

111. State whether True of False?
बताएं कि सही या गलत है?

(i) Pointer arithmetic can't be done on a void pointer./पॉइंटर अर्थमेटिक वॉइड पॉइंटर पर नहीं किया जा सकता है।

(ii) A void pointer cannot point to a class member in C++,/वॉइड पॉइंटर C++ में क्लास मेंबर को पॉइंट (इंगित) नहीं कर सकता है।

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(a) (i) True, (ii) True/ (i) सत्य, (ii) सत्य
(b) (i) True, (ii) False/ (i) सत्य, (ii) असत्य
(c) (i) False, (ii) True/ (i) असत्य, (ii) सत्य
(d) (i) False, (ii) False/ (i) असत्य, (ii) असत्य

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (a) : Explanation:
1. (i)True: Pointer arithmetic cannot be directly performed an a void* because void has no size, so the compiler cannot calculate how for to move the pointer.
2. (ii) True: A void* cannot directly point to a specific class member in C^{++} without casting to the correct member type.

112. What will be the output of the following C++ program?
निम्नलिखित C++ कार्यक्रम का आउटपुट क्या होगा?

```
#include <iostream>/#शामिल <iostream>
using namespace sted;/namespace sted का उपयोग करना;
int main (){
char array [10];
for(int i = 0; i < 5; i++)
*(arry + i) = 65 + 1;
*(arry + i) = '\0';
cout << array;
return(0);
}
```

(a) ABCDE\0 (b) ABCDE
(c) BCDEF (d) CDEFG

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (b) : # include < iostream >
Using name space std;
int main ()
{char array [10];
For (int = 0: i < 5; I ++)
* (array +i) = 65 + I ; // As}
* (array + 5) = '10'; // Add null
cout << array;
return (0) :}
step – by – step execution:
65 = 'A' in ASCll
Loop runs from = 0 to 4
- i = 0 → * (array [0] = 65 → 'A'
- i = 1 → * (array [1] = 66 → 'B'
- i = 2 → * (array [2] = 67 → 'C'
- i = 4 → array [3] = 68 → 'D'
- i = 4 → array [4] = 69 → 'E'

Now the loop ends → i = 5
→ *(array + 5) = '/0' (null terminator inserted)
('A', 'B', 'C','D', 'E', '/O',)
When cout << array: it prints string till null

113. Consider the following C++ code segment:
निम्नलिखित C++ कोड सेगमेंट पर विचार करें:

```
int a = 10, b = 20;
int *p = &a, *q = &b;
p = q;
```

With respect to the code segment mentioned above, which of the following statement is correct?
उपरोक्त वर्णित कोड सेगमेंट के संबंध में, निम्नलिखित में से कौन सा कथन सही है?

(a) Both a and b will contain 10
a और b दोनों में 10 होगा
(b) Both a and b will contain 20
a और b दोनों में 20 होगा
(c) Both p and q will contain a
p और q दोनों a की ओर इंगित करेंगे
(d) Both p and q will contain b
p और q दोनों b की ओर इंगित करेंगे

DSSSB PGT C.S. (Male) 27.07.2018 (Shift-II)

Ans. (d) : 1, int a = 10, b = 20: → There are two integer variables:
A = 10, b = 20
2. int p = & a * q = & b:
→ p pointer is pointing towards a → q pointer is pointing towards b.
3. p = q;
→ Now p has been assigned the value of
q → That means p will now point to the same address that q is pointing to i.e. b.
Final position :
a = 10
b = 20
p → will point to b
q → will point to b
The right option
[d] Both p and q will contain b

114. In C++ a wide – character constant begins with the letter _____ .
C++ में एक वाइड – कैरेक्टर कोंस्टंट _____ अक्षर से शुरु होता है।

(a) W (b) C
(c) L (d) S

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : A wide character constant in C++ is just a character that can store more symbols than a normal character. To tell the compiler it is wide, we start it with L.
e.g. wchar-t ch = L'A';
Here L'A' is a wide character constant of type wchar-t.

115. A function template in C++ provides _____ level of generalization.

(a) 4 (b) 3
(c) 2 (d) 1

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Sarkari Resource

Ans : (c) C++ provides two kinds of templates: class templates and function templates. A function template in C++ provides two level of generalization.

116. Which of the following statement is not True?

(a) A data member of a class can be declared as static and is normally used to maintain values common to the entire class.

(b) A friend function can be invoked like a normal function without the help of any object.

(c) The scope resolution operator (::) can be overloaded like normal operators.

(d) Multiple inheritances may lead to duplication of inherited members from a "Grandparent" base class. This may be avoided by making the common base class a virtual base class.

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Ans. (c) : Statement-a (True):- A static data member belongs to the class rather than any specific object and maintains common data for all objects.
Statement-b (True):- A friend function is not a member of the class, so it can be called like a normal function, without using an object of the class.
Statement-c (False):- The Scope resolution operator (: :) cannot be overload in C++, it is a fundamental part of the language used for accessing class member and namespaces.
Statement-d (True):- In multiple inheritance, it a class inherits from two classes that both inherit from a common base class, the base class members may get duplicate. This problem is avoided by using virtual base classes.
Statement -c is not True.

117. Consider the following C++ statements :
निम्नलिखित C++ कथनों पर विचार करें:

```
int x;
int& y = x;
The statement y = 25; x = 2*x + 30;
```

updates the value of x and y as :
कथन : y = 25; x = 2*x + 30; x और y के मान को अपडेट करता है:

(a) y = 25, x = 25 (b) y = 25, x = 80
(c) y = 25, x = 30 (d) y = 80, x = 80

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Ans. (d) : Let's analyze the code step by step.
1. int x; declares an integer variable x. Since it is not initialized, it contains an indeterminate value. However, for the purpose of this question. Actually we are going to assign to it through y first.
2. int& y = x; declares y as a reference to x. This means y is an alias for x; any operation on y is directly applied to x.
3. The next statement is y = 25;. Since y is a reference to x, this assigns 25 to x. so now both x and y are 25.
4. Then x = 2*x + 30; is executed. At this point, x is 25, so 2*x + 30 is 2*25 + 30 = 50 + 30 = 80. This value is assigned to x since y is a reference to x, y also becomes 80.

118. Which symbol should be used to pass the object by reference in C++ ?
C++ में रेफरेंस द्वारा ऑब्जेक्ट को पास करने के लिए किस चिन्ह का उपयोग किया जाना चाहिए?

(a) &
(b) @
(c) $
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans. (a) : The object to be passed by reference to the function should be preceded by & symbol in the argument list syntax of the function. This indicates the compiler not be use new object. The same object which is being passed have to be used.

119. In C++, << operator used for shifting bits can also be used for :
C++ में बिट खिसकाने वाले ऑपरेटर << को और किस कार्य में प्रयोग किया जा सकता है?

(a) Displaying values/मान दर्शाने में
(b) Addition/योग में
(c) Increment/वृद्धि में
(d) Decrement/ह्रास में

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Ans. (a) : The << operator is overloaded in C++ to work with streams, most notably std::cout, which is the standard output stream. When used with std::cout, it's called the stream insertion operator. This operator inserts data into the output stream effectively sending it to the console for display. For example- the statement std::cout << "Hello, world !"; uses this operator to send the string literal to the console.

120. In context of C++, consider the following statements :
C++ के सन्दर्भ में निम्न कथनों पर विचार करें:

I. Friend function of a class can directly access both private and protected data of the class./एक क्लास (कक्षा) का मित्र फलन (फ्रेंड फंक्शन) कक्षा के निजी (प्राइवेट) तथा संरक्षित (प्रोटेक्टेड) दोनों डाटा को सीधे ही अभिगम (एक्सेस) कर सकता है।

I.I Member function of derived class can directly access both private and protected data of the class./व्युत्पन्न (डिराइव्ड) कक्षा का सदस्य (मेम्बर) फलन कक्षा के निजी तथा संरक्षित दोनों डाटा को सीधे ही अभिगम कर सकता है।

The true statement(s) is/are :
सत्य कथन है।

(a) Only I /केवल I
(b) Only II /केवल II

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(c) Both I and II /I तथा II दोनों
(d) Neither I Nor II /न ही I और न ही II

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Ans. (a) : In C++, a friend function can be given a special grant to access private and protected members.
Member function of derived class cannot be directly access both private and protected data of the class.
Therefore only Ist statement is correct.

121. In C++, which of the following operator cannot be overloaded ?
C++ में, निम्न में से कौन-सा ऑपरेटर ओवरलोड नहीं किया जा सकता?

(a) ^ (b) ==
(c) .[dot] (d) !

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Ans. (c) : The dot operator (·) is used for member access and cannot be overloaded. This is to ensure that the behavior of accessing members of a class or struct remains unambiguous and safe.

122. In context C++, the process of individually copying each member of an object to the corresponding member of another object of the same class is known as _____ copying.
C++ के संदर्भ में, एक ऑब्जेक्ट के प्रत्येक सदस्य को एक-एक कर उसी क्लास के दूसरे ऑब्जेक्ट के संबंधित सदस्य में कॉपी (नकल) करने की प्रक्रिया को _____ कॉपी कहा जाता है।

(a) Mutable (b) Shallow
(c) Respective (d) Default

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Ans. (b) : In C++, the process of individually copying each member of an object to the corresponding member of another object of the same class is known as shallow copying.
This is because a shallow copy creates a new object and copies the values of the member variables from the original object. However, it any of these member variables are pointers or references to dynamically allocated memory, only the addresses of those resources are copied not the actual data they point to. This means both the original and the copied object will share the same underlying data for those members.

123. Which of the following is not supported in C++?
निम्न में कौन-सा C++ में आलम्बित (सपोर्टेड) नहीं है?

(a) Automatic garbage collection
ऑटोमेटिक गार्बेज कलेक्शन
(b) Multiple inheritance/मल्टीपल इनहेरिटेंस
(c) Smart pointer/स्मार्ट पॉइंटर
(d) Virtual destructor/वर्चुअल डेस्ट्रक्टर

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Ans. (a) : Automatic garbage collection– Not supported in C++, C++ relies on manual memory management (New/delete) or programmer controlled mechanisms, unlike java or C#, C++ does not have a built in garbage collector.
Multiple inheritance– fully supported in C++.

124. Which of the following is invalid C++ comment statement ?
निम्न में कौन-सा एक अनुचित C++ कमेंट कथन है?

(a) /*//Comment/*/ (b) //*Comment*/
(c) /*Comment* (d) //Comment

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Ans. (c) : C++ supports single line and multi-line comments. All characters available inside any comment are ignored by C++ compiler.
C++ comments start with /* and end with */.
For example- /* This is a comment */.
/*C++ comments can also * span multiple lines */.

125. In context of C++, consider the following statements :
C++ के सन्दर्भ में निम्न कथनों पर विचार करें:

I. Static member function of a class can access only static data members of the class./एक क्लास (कक्षा) का स्टेटिक सदस्य फंक्शन, क्लास के केवल स्टेटिक डाटा सदस्यों को ही अभिगम (एक्सेस) कर सकता है।

II. A static data member of class is shared by all the objects of the class./एक क्लास का स्टेटिक सदस्य डाटा, कक्षा के सभी ऑब्जेक्ट्स द्वारा साझा किया जाता है।

The true statement(s) is/are :
सत्य कथन है/हैं:

(a) Only I/केवल I
(b) Only II/केवल II
(c) Both I and II/I तथा II दोनों
(d) Neither I Nor II/न तो I और न ही II

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Ans. (c) : Static member function of a class can access only static data members of the class, static member functions belong to the class itself not to any specific object instance. They do not have a this pointer, which is used to refer to the current object's non static members. Therefore, a static member function can only access other static members of the class, as these also belong to the class and not to individual objects.
A static data member of a class is shared by all the objects of the class.
When a data member is declared as static, there is only one copy of that member for the entire class, regardless of how many objects of that class are created. All objects of the class share and access this single copy of the static data member. This is in contrast to non-static data members, where each object has its own separate copy.
Therefore, the correct answer is (c) Both I and II.

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126. Which of the following feature is not supported in C++ ?
निम्न में से कौन-सा फीचर C++ में अवलंबित (सपोर्टेड) नहीं है?

(a) Inheritance/इनहेरिटेंस
(b) Abstraction/एबस्ट्रैक्शन
(c) Reflection/रिफ्लेक्शन
(d) Dynamic Binding/डाइनेमिक बाइन्डिंग

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Ans. (c) : Inheritance is a fundamental feature of C++ that allows a new class to inherit properties and behaviors from an existing class (base class). This promotes code reusability and establishes an is a relationship between classes.
Abstraction– is a another core concept in C++. It involves hiding complex implementation details and showing only the essential features of an object.
Dynamic Binding– is the process of linking a function call to its corresponding function at runtime rather than at compile time.
Reflection– is the ability of a program to examine and modify its own structure and behavior at runtime. While some languages like java and C# have built-in reflection capabilities, C++ does not you can achieve some limited forms of reflection like behavior in C++ through various libraries.

127. _____ is used to declare and initialize an object from another object of same class in C++.
C++ में एक ऑब्जेक्ट को डिक्लेयर तथा उसी क्लास के दूसरे ऑब्जेक्ट से इनिशिएलाइज करने के लिए _____ प्रयुक्त होता है।

(a) Copy constructor
(b) Default constructor
(c) Overloaded constructor
(d) Non-parameterized constructor

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Ans. (a) : Copy constructor is used to declare and initialize an object from another object of same class in C++.

128. In context of C++, consider following loop-statement :
C++ के संदर्भ में, निम्न लूप स्टेटमेंट पर विचार करें:
while(5) {cout <<"Hello";}
Hello will be displayed _____ time(s) by the above loop.
इस लूप द्वारा_____बार Hello प्रदर्शित किया जायेगा।

(a) Only one/केवल एक
(b) Five/पाँच
(c) Zero/शून्य
(d) Infinite/अनंत

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Ans. (d) : In C++, any non-zero value used as a condition in a while loop is treated as true, The number 5 is a non-zero value, so the condition while (5) is always true. Because the condition never becomes false the loop will continue to execute indefinitely, printing "Hello" to the console over and over again.
To stop a while loop, its condition must eventually become false. For example, a loop like int i = 0; while (i < 5) {cout<<"Hello"; i++;} would print "Hello" five times because the value of i increases with each iteration, eventually making the condition i < 5 false.

129. The keywords private, protected and public may appear _____, in the declaration of a class in C++.
C++ में एक क्लास के डिक्लेरेशन में private protected तथा public कीवर्ड्स _____ आ सकते है।

(a) In any order and any number of times
किसी भी क्रम में तथा कितनी भी बार
(b) At the most one time and in any order
अधिकाधिक एक बार तथा किसी भी क्रम में
(c) At the most one time and strictly in the order
अधिकाधिक एक बार तथा ठीक इसी क्रम में
(d) Any number of times but strictly in the order
कितनी भी बार परन्तु ठीक इसी क्रम में

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Ans. (a) : In C++, the keywords private, protected and public are known as access specifies. They control the visibility and accessibility of a classes members.
Public members are accessible from anywhere.
Protected members are accessible within the class itself and by its derived classes.
Private members are accessible only within the class itself.

130. In C++, which of the following is not a valid access specifire ?
C++ में, निम्न में से कौन-सा एक उचित एक्सेस स्पेसिफायर नहीं है?

(a) Private
(b) Public
(c) Protected
(d) Mutable

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Ans. (d) : Mutable is a keyword in C++ but it is not an access specified. It is used to allow a data member of a const object to be modified.

131. Which one of the following is not a valid reserved keyword in C++ ?
निम्नलिखित में से कौन-सा C++ में मान्य आरक्षित कीवर्ड नहीं हैं?

(a) Explicit
(b) Public
(c) Implicit
(d) Private

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Ans. (c) : In C++, reserved keywords are words that have a predefined meaning and cannot be used as identifiers (such as variable names, function names, or class names). They are fundamental to the C++ language syntax and are used to declare or define various constructs. The keyword implicit is not part of the standard C++ language specification.

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132. In context of C++, which of the following is valid statement ?
C++ के संदर्भ में, निम्न में कौन-सा एक उचित कथन है?

(a) Friend function of class is always a member function of that class./एक क्लास का मित्र फलन सदैव उस क्लास का सदस्य फंक्शन होता है।

(b) Value of static data member cannot be manipulated once assigned./स्टेटिक डाटा सदस्य का मान एक बार असाइन करने के बाद परिवर्तित नहीं किया जा सकता।

(c) Static data member of a class can be accessed only by static member function of that class.
एक क्लास का स्टेटिक डाटा सदस्य केवल उस क्लास के स्टेटिक सदस्य फलन द्वारा ही एक्सेस किया जा सकता है।

(d) Keywords private, protected and public may appear in any order and any number of times in a class./एक क्लास में private, protected तथा public की वर्ड्स किसी भी क्रम में तथा कितनी भी बार आ सकते हैं।

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Ans. (d) : Valid statement (d)-
Keywords Private, Protected and Public may appear in any order and any number of times in a class.
The access specifies private, protected, and public can be used multiple times and in any sequence within a class definition. Each time an access specified appears, it changes the access level for all subsequent members until another specified is encountered.

133. Which of the following is the scope resolution operator in C++ language ?
C++ लैंग्वेज में निम्नलिखित में से कौन-सा स्कोप रिजॉल्यूशन ऑपरेटर है?

(a) · (b) : :
(c) * (d) ~

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Ans. (c) : In C++, the scope resolution operator (::) is used to access the identifiers such as variable names and function name defined inside some other scope in the current scope.

134. Inheritance allows the following in C++ program :
इनहेरिटेंस C++ प्रोग्राम में निम्नलिखित की अनुमति देता है:

(a) Class Re-usability/क्लास पुनः प्रयोज्य
(b) Creating a hierarchy of classes
कक्षाओं का एक पदानुक्रम बनाना
(c) Extendibility/विस्तारशीलता
(d) All options are correct/सभी विकल्प सही हैं

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Ans. (d) : Inheritance C++ allows–
Class Re-usability– Existing class members can be reused in derived classes.
Hierarchy of Classes– Classes can be organized in parent child relationship.
Extendibility– New functionality can be added without rewriting the base class.

135. The functions 'cin' and 'cout' resolve the necessary conversions at :
फंक्शन 'cin' और 'cout' आवश्यक रूपांतरणों को हल करते हैं:

(a) Compile time/संकलन के समय
(b) Execution time/निष्पादन के समय
(c) Linking time/लिंकिंग के समय
(d) None of these/इनमें से कोई नहीं

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Ans. (b) : The function cin and cout in C++ use operator overloading to handle tape conversions. The exact type to be used to is determined during Execution time when the stream extraction (>>) or insertion (<<) operators are applied to values.

136. What is a copy constructor ?
कॉपी कंस्ट्रक्टर क्या है?

(a) A constructor that allows a user to move data from one object to another./एक कंस्ट्रक्टर जो उपयोगकर्ता अथवा यूजर को डेटा को एक ऑब्जेक्ट से दूसरे ऑब्जेक्ट में स्थानांतरित करने की अनुमति देता है।

(b) A constructor to initialize an object with the values of another object./एक कंस्ट्रक्टर जो किसी ऑब्जेक्ट को किसी अन्य ऑब्जेक्ट के मानो या वैल्यू के साथ प्रारंभ करता है।

(c) A constructor to check whether objects are equal or not./एक कंस्ट्रक्टर यह जाँचने के लिए कि ऑब्जेक्टस बराबर हैं या नहीं।

(d) A constructor to delete other copies of a given object./एक कंस्ट्रक्टर जो किसी दिए ऑब्जेक्ट की अन्य प्रतियों को डिलीट करता है।

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Ans. (b) : A copy constructor is a special type of constructor in C++ that crates a new object by initializing it with the values of an existing object of the same class. It is used to initialize one object using another existing object of the same type.
Therefore, the correct option is (b).

137. In C++, namespace defines a _____ for the identifiers that are used in a program.
C++ में, नेमस्पेस प्रोग्राम में उपयोग किए जाने वाले आइडेंटिफायरस के लिए एक _____ को परिभाषित करता है।

(a) Size/साईज (b) Scope/स्कोप
(c) Syntax/सिंटेक्स (d) Standard/स्टैण्डर्ड

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Ans. (b) : In C++, namespace defines a scope for the identifiers that are used in a program.

138. In C++, operator '>>' is used along with object cin for reading input is known as _____.
C++ में, ऑपरेटर '>>' जिसे इनपुट रीड करने के लिए जाना जाता है उसे _____ कहते हैं।

(a) Extraction/एक्सट्रैक्शन
(b) Output/आउटपुट
(c) Get in/गेट इन
(d) Put to/पुट टू

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Ans. (a) : The extraction operator (>>), which is preprogrammed for all standard C++ data types, is the easiest way to get bytes from an input stream object.

139. Which of the following statement(s) is/are correct regarding void keyword in C++ ?
C++ में वोयड कीवर्ड के संदर्भ में निम्नलिखित में से कौन सा/से कथन सही है/हैं?

I. To specify the return type of a function when it is not returning any value.
किसी फंक्शन के रिटर्न टाइप को निर्दिष्ट करने के लिए जब वह कोई वैल्यू नहीं लौटा रहा हो।

II. To indicate an empty argument list to a function.
किसी फंक्शन के लिए एक खाली आर्गुमेंट सूची को इंगित करने के लिए।

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

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Ans. (c) : The void keyword serves two primary purposes in C++
1. Specifying a Non-Returning Function, A function declared with a return type of void indicates that it does not return a value. This is useful for functions that perform an action or a task like printing to the console or modifying a variable by reference, but don't need to pass a result back to the calling function.
2. Indicating an Empty Argument List- In c, the void keyword is explicitly used inside the parentheses of a function to indicate that the function takes no arguments, for example, into my function (void). While this is considered good practice in c for clarity in C++ an empty pair of parentheses() is sufficient and implies the something that the function takes no argument. However, using void inside the parenthesis is still void C++ code and performs the same function.
Both statements correctly describe valid uses of the void keyword in the context of C++.

140. In C++, bool data type is used to hold how many values?
C++ में, बूल डेटा टाइप का उपयोग कितनी वैल्यूज रखने के लिए किया जाता है?

(a) n^2
(b) 2
(c) 2n
(d) 2^n

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Ans. (b) : The bool data type, short for "Boolean" can represent only one of two possible states true or false. This is analogous to a light switch which can only be either on or off. It's a fundamental concept in computer science and digital logic, representing a binary state. Therefore, the correct answer is (b).

141. Which of the following is specified by function prototype in C++?
निम्नलिखित में से कौन सा फंक्शन प्रोटोटाइप द्वारा निर्दिष्ट किया जाता है?

I. Return Type/रिटर्न टाइप
II. Function name/फंक्शन नेम
III. Number and types of argument
आर्गुमेंट की संख्या तथा प्रकार

(a) I and II/ I तथा II
(b) I and III/ I तथा III
(c) II and III/ II तथा III
(d) I, II and III/ I, II तथा III

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Ans. (d) : A function prototypes (also known as a function declaration) provides the compiler with essential information about a function before it's actually defined. This is crucial for the compiler to correctly check for type errors and ensure that function calls are valid. The prototype doesn't include the function's body (the code within the curly braces).
Return Type– This specifies the data type of the value the fault will return.
Function Name– This is the unique identifier used to call the function.
Number and types of Arguments– This specifies the data type and order of the values (Parameters) the function expects to receive when it's called. The names of the parameters are optional in a function prototype.

142. What is the symbol and associativity of the assignment operator in C++?
C++ में असाइनमेंट ऑपरेटर का प्रतीक और संबद्धता क्या है?

(a) =, left to right/ =, बाएं से दायां
(b) = =, left to right/ = =, बाएं से दायां
(c) ®, right to left/ ®, दाएं से बाएं
(d) =, right to left/ ®, दाएं से बाएं

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Ans. (d) : In C++, the assignment operator is represented by the symbol '='.
The associatively of the assignment operator is right-to-left. This means that in an expression with multiple assignment operators, the assignments are performed from right to left.

143. Which statement is correct about static member function in C++?
C++ में स्टेटिक मेंबर फंक्शन के संदर्भ में कौन सा कथन सही है?

I. It can access to only other static members declared in the same class.
यह केवल उसी क्लास में घोषित अन्य स्टेटिक मेंबर्स तक पहुंच सकता है।

II. It cannot be called using the class name (instead of its objects)
इसे क्लास नेम का इस्तेमाल करके कॉल नहीं किया जा सकता है (इसके ऑब्जेक्ट्स के बजाय)

(a) Only I/ केवल I
(b) Only II/ केवल II
(c) Both I and II/ I तथा II दोनों
(d) Neither I nor II/ ना ही I ना ही II

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Ans. (a) : The correct statement about a static member function in C++ is.
(i) It can access only other static members declared in the same class because it does not operate on a specific instance of the class.
(ii) Statement (II) is false. A static member function can be called using the class name, without creating an object of the class.
So statement I is correct and statement II is incorrect.

144. By default in C++, the member of a class are _____, while, by default the member of structure are _____.
C++ में डिफॉल्ट रूप से, एक क्लास के मंबर्स_____ होते हैं, जबकि डिफॉल्ट रूप से स्ट्रक्चर के मेंसर्स _____ होते हैं।

(a) Private; private/प्राइवेटः प्राइवेट
(b) Public; public/पब्लिक, पब्लिक
(c) Private; public/प्राइवेटः पब्लिक
(d) Public; private/पब्लिकः प्राइवेट

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Ans. (c) : By default in C++, the member of a class are private while by default the member of structure are public.

145. Match the following file stream classes in C++ with its functions.
निम्नलिखित फाइल स्ट्रीम क्लासेस को C++ में इसके फंक्शन्स के साथ सुमेलित करें।

1	**Fstream base Fस्ट्रीम बेस**	**i**	**Provides support for simultaneous input and output operation**
2	**If stream If स्ट्रीम**	**ii**	**Provides output operation**
3	**Of stream Of स्ट्रीम**	**iii**	**Provides operations common to file streams**
4	**F stream F स्ट्रीम**	**iv**	**Provides input operation**

(a) 1 – i, 2 – ii, 3 – iv, 4 – iii
(b) 1 – iv, 2 – iii, 3 – i, 4 – ii
(c) 1 – i, 2 – iii, 3 – iv, 4 – ii
(d) 1 – iii, 2 – iv, 3 – ii, 4 – i

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Ans. (d) : The correct match

1	Fstream base	**iii**	Provides operations common to file streams
2	If stream	**iv**	Provides input operation
3	Of stream	**ii**	Provides output operation
4	F stream	**i**	Provides support for simultaneous input and output operation

146. In C++, "Out-of-range-index" belong to _____ type exception while keyboard interrupts are called _____ exceptions.
C++ में, ''आउट-ऑफ-रेंज-इंडेक्स''_____ प्रकार के एक्सेप्शन से संबंधित है।

(a) Synchronous; Synchronous
सिंक्रोनस; सिंक्रोनस
(b) Synchronous; Asynchronous
सिंक्रोनस; एसिंक्रोनस
(c) Asynchronous; Synchronous
एसिंक्रोनस; सिंक्रोनस
(d) Asynchronous; Asynchronous
एसिंक्रोनस; एसिंक्रोनस

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Ans. (b) : In C++, "Out-of-range-index" belong to synchronous; type exception while keyboard interrupts are called asynchronous exceptions.

147. What is output of the given code?
दिए गए कोड का आउटपुट क्या है?

```
void main ()
{
int a = 3, b, *p, **q;
p = & a;
q = & p;
b = 2 * (**q + *p);
cout <<b;
}
```

(a) 3 (b) 12
(c) 18 (d) 24

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Ans. (b) : Step by step execution-
int a = 3
p = & a; pointer p stores the address of a.
q = & b; pointer to pointer q stores the address of p.
**q → q points to p → *q is p → **q is value at p → value at a → 3.
*p → p points to a → *p = 3
(**q + *p) → 3 + 3 = 6
b = 2 * 6 = 12
Final output = 12

148. What is the output of code?
दिए गए कोड का आउटपुट क्या है?

```
void main ()
{
int x = 33, y, z;
z = x++ + 17;
y = ++z – ++x;
cout << "x" << x << "y" << y << "z" << z;
}
```

(a) x = 34 y = 17 z = 51
(b) x = 35 y = 16 z = 51
(c) x = 34 y = 16 z = 50
(d) x = 35 y = 17 z = 51

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Ans. (b) : Step by step execution
Step-1 initialization
x = 33
y = uninitialized
z = uninitialized
Step-2 z = x++ + 17;
x++ is post increment → use old value of x, then increment. old x = 33.
Expression becomes-
z = 33 + 17 = 50
After this, x is incremented to 34
Now-
x = 34
z = 50
Step-3 y = ++z – ++x;
++z → pre-increment z = 51
++x → pre-increment x = 35
Expression-
y = 51 – 35 = 16
Now-
x = 35
y = 16
z = 51.

149. How do structures and classes in C++ differ?
C++ में, संरचनाएँ और कक्षाएँ कैसे भिन्न होती हैं?

(a) Structures by default hide every member whereas classes do not/डिफ़ॉल्ट रूप से संचानाएँ प्रत्येक सदस्य को छुपाती हैं, जबकि कक्षाएँ नहीं होती हैं
(b) In structures, members are public by default, whereas, in classes, they are private by default/संरचाओं में, सदस्य डिफ़ॉल्ट रूप से सार्वजनिक होते हैं, जबकि कक्षाओं में व डिफ़ॉल्ट रूप से निजी होते हैं
(c) Structures cannot have private members, whereas classes can have/संरचनाओं में निजी सदस्य नहीं हो सकते हैं, जबकि कक्षाओं में हो सकते हैं
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (d) : In C++, structures and classes are two different ways of structuring data structures are used to group data, but they do not have special functions. In structures. members are public by default, while in classes they are private by default classes can hold data as well as special methods.

150. The matrix contains m rows and n columns. The matrix is called Sparse Matrix if
मैट्रिक्स में m पंक्तियाँ और n कॉलम होते हैं। मैट्रिक्स को विरल मैट्रिक्स कहा जाता है, यदि

(a) Total number of zero elements > (m* n)/2
शून्य तत्त्वों की कुल संख्या > (m* n)/2
(b) Total number of zero elements = m+ n
शून्य तत्त्वों की कुल संख्या = m+ n
(c) Total number of zero elements = m/n
शून्य तत्त्वों की कुल संख्या = m/n
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (a) : For matrix to be sparse matrix, it should contain zero elements more than the non zero elements. Total elements of the given matrix is m*n. So it total number of zero elements > (m*n)/2 then the matrix is called sparse matrix.

151. What will be the output of the following C++ code?
निम्नलिखित C++ कोड का आउटपुट क्या होगा?

```
#include<iostream>
#include<string>
using namespace std;
int main(int argc, char const *argv[ ])
{
char s1[6]=“Hello”;
char s2[6]=“World”;
char s3[12]=s1+“ ”+s2;
cout<<s3;
return 0;
}
```

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(a) Hello
(b) World
(c) Error
(d) More than one of the above
उपर्युक्त में से एक अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 1.0 2023 (11-12)

Ans. (c) : The given C++ code will output an error because the method of concatenating two character array S1 and S2 to form a new string S3 is incorrect. C++ std : : string is used to concatenate two character arrays.

(ii) Artificial Intelligence

152. Which of the following is a major cloud computing platform?
निम्न में से कौन एक प्रमुख क्लाउड कम्प्यूटिंग प्लेटफार्म है?

(a) Microsoft Azure/माइक्रोसाफ्ट अज्यूरे
(b) MS-Office/एम.एस.ऑफिस
(c) Java/जावा
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (a) : Microsoft Azure is a widely recognized cloud computing platform offered by Microsoft. It provides a range of services including computing, storage, networking, and artificial intelligence.

153. What is the frame?
फ्रेम क्या है?

(a) Data type
(b) Data Structure
(c) A way of representing knowledge
(d) None of the above

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : A frame is a data structure used in artificial intelligence to represent stereotypical situations or scenarios. It organize information about objects, concepts, and their relationship using attributes and values. This allow for efficient access and manipulation of complex knowledge frames are often structured hierarchically, enabling inheritance of properties from parent frames to child frames.

154. In constraint satisfaction problem, constraints can be stated as _____

(a) Arithmetic equations and inequalities that bind the values of variables
(b) Arithmetic equations and inequalities that doesn't bind any restriction over variables
(c) Arithmetic equations that impose restrictions over variables
(d) Arithmetic equations that discard constraints over the given variables

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) Remember map coloring problem, it is a constraint satisfaction problem we need mathematical equation and inequalities. Inequalities are must to check whether constraints are full field. Equation specify the constraint. In a nutshell 1 should be the proper choice.

155. Which of the following is not an application of AI ?
निम्नलिखित में से क्या ए आई (AI) की एक एप्लिकेशन नहीं है?

(a) Chat Bot/चैट बोट
(b) Financial Accounting System
फाइनेंशियल अकांउटिंग सिस्टम
(c) Driverless Car/ड्राइवरलेस कार
(d) Face Recognition System
फेस रिकगनिशन सिस्टम

KVS PGT 23.03.2023 (Shift-II)

Ans. (b) : Chat Bot, natural language processing, machine learning, face recognition system and driverless car, etc applications of AI, whereas database management system, financial accounting systems are not AI applications.

156. Overfitting is expected when we observe that?

(a) With training iterations error on training set as well as test set decreases
(b) With training iterations error on training set decrease but test set increases
(c) With training iterations error on training set as well as test set increases
(d) With training iterations training set as well as test error remains constant

UGC NET C.S. December-2022

Ans. (b) : Overfitting is expected–
- High training accuracy, but low test accuracy when a model achieves a high accuracy on the training data but performs poorly on unseen data, it suggests overfitting. The model has essentially "memorized" the training examples including the noise or irrelevant patterns and cannot generalize well.

157. What is the name of the form of machine learning in which the algorithm looks for patterns or groupings in data even when the labels are not explicit?

(a) Supervised learning
(b) Unsupervised learning
(c) Reinforcement learning
(d) Semi-supervised learning
(e) Not attempted

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Ans. (b) : Unsupervised learning is a type of machine learning where the algorithm is given unlabeled data and must find patterns or relationships within it without any explicit guidance from a human.

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158. The operating system and application stack is added in which level of cloud?
परिचालन तंत्र तथा एप्लीकेशन के समूहों को क्लाउड के किस स्तर पर रखा जाता है?
(a) PaaS
(b) SaaS
(c) IaaS
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (a) : PaaS (Platform as a Service)– PaaS provides developers with a complete platform including operating system, middleware, runtime and application stack. The user only focuses on building and deploying application without worrying about infrastructure or OS management. Example– Google App Engine, Microsoft Azure App Service.

159. What is the vanishing gradient problem in deep learning?
(a) Gradients become too small, slowing down learning
(b) Gradients become too large, causing instability
(c) Model over fits due to excessive training
(d) Model stops learning due to lack of data
(e) Not attempted

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Ans. (a) : The vanishing gradient problem occurs in deep neural networks particularly those with many layers. During back propagation, gradients are computed to update the weight. However, when gradients are propagated backwards through many layers they can become exponentially small due to repeated multiplication of small derivatives. This causes the weights in earlier layer to update very slowly or not at all, significantly slowing down, learning and making it difficult for the network to train effectively.

160. In supervised learning, the system learns from:
(a) Unlabeled data
(b) Metadata
(c) Labeled data
(d) Reinforcement singals
(e) Not attempted

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Ans. (c) : Supervised learning uses manually labeled data to produce predictions or classifications. Unsupervised learning aims to uncover and learn hidden patters from unlabeled data. In contrast to supervised learning, reinforcement learning does not use labeled examples of correct or incorrect behavior. But reinforcement learning also differs from unsupervised learning.

161. What is the main difference between a feedforward neural network and a recurrent neural network (RNN)?
(a) RNNs use feedback loops, while feedforward networks do not.
(b) Feedforward networks can handle sequential data, while RNNs cannot.
(c) Feedforward networks use feedback loops, while RNNs do not.
(d) Feedforward networks use backpropagation while RNNs do not.
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : The main difference between a feedforword neural network and a recurrent neural network (RNN) is that RNNs use feedback loops, while feedforward networks do not.
Feedforward Neural Networks– A feedforward neural network is the simplest type of artificial neural network. In this architecture, information moves in only one direction forward from the input layer through any hidden layers, to the output layer.
Recurrent Neural Networks– A recurrent neural network (RNN) is a type of neural network designed specifically for processing sequential data such as time series, speech or text. They have internal memory and feedback loops that allow them to remember information from previous step in a sequence.

162. Which of the following is NOT a key operation in genetic algorithms?
(a) Selection (b) Crossover
(c) Mutation (d) Backpropagation
(e) Not attempted

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Ans. (d) : Let's analyze the options–
Selection– This is a key operation in genetic algorithms where individuals are chosen from the population for reproduction based on their fitness.
Crossover– This is a key operation where two parent individuals are combined to produce off spring exchanging genetic material.
Mutation– This is a key operation that introduces random changes to individuals to maintain genetic diversity.
Backpropagation– This is not an operation in genetic algorithms. Instead it is a method used in training artificial neural networks to compute gradients and update weights, typically associated with gradient descent.

163. In constraint satisfaction problem, constraints can be stated as _____
(a) Arithmetic equations and inequalities that bind the values of variables
(b) Arithmetic equations and inequalities that doesn't bind any restriction over variables
(c) Arithmetic equations that impose restrictions over variables
(d) Arithmetic equations that discard constraints over the given variables

UGC NET C.S. December-2015 (Paper-III)

Ans : (a) Remember map coloring problem, it is a constraint satisfaction problem we need mathematical equation and inequalities.
Inequalities are must to check whether constraints are nullified. Equation specify the constraint. In a nutshell 1 should be the proper choice.

164. Logistic regression can binominal, _____ or multinominal.
लॉजिस्टिक रिग्रेशन बायोनोमिनल, या मल्टीनोमिनल हो सकता है।

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(a) Ordinal (b) Numbered
(c) Unimial (d) Binary

HTET PGT C.S. 30.07.2025

Ans.(a): Logistic regression is a statistical method used for predicting a categorical dependent variable from a set of independent variable. The key destination between the different types of logistic regression lies in the nature of the dependent variable.

165. What is the role of the inference engine in an expert system?
(a) To store knowledge in a structured format
(b) To apply logical rules and derive conclusions
(c) To interact with users through a GUI
(d) To collect data from the internet
(e) Not attempted

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Ans.(b): The inference engine is a core component of an expert system, acting as its brain. Its primary role is to manipulate and reason with the knowledge stored in the knowledge base to solve problems and draw conclusions. It does this by applying logical rules or "if then" statements to the facts and data provided by the user or from external sources.

166. In genetic algorithms, a potential solution is represented as
(a) A chromosome
(b) A neuron
(c) A rule-based system
(d) A deterministic model
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : In genetic algorithm, each potential solution to the optimization problem is represented as a chromosome, which is typically encoded as a string (binary, real valued, or other forms). These chromosomes undergo genetic operators like selection, crossover and mutation to evolve better solutions over generations.

167. Which of the following is a method of analogical problem solving that have been studied in AI?
AI में अध्ययन की जाने वाली एनालॉजिकल प्राब्लम साल्विंग विधि निम्न में से कौन सी है?
(a) Transformational analogy
ट्रांसफार्मेशनल एनालॉजी
(b) Derivational analogy/डेरिवेशनल एनालॉजी
(c) Both (a) and (b)/ (a) तथा (b) दोनों
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (c) : In Artificial Intelligence, analogical problem solving involves solving a new problem by recalling and adapting solutions from previously solved problems.
Transformational analogy– focuses on transforming the previous solution to fit the new problem.
Derivational analogy– focuses on reasoning the derivation process (the reasoning steps) from the old problem to solve the new one.
Both are well studied analogical reasoning methods in AI.

168. Which condition is used to influence a variable directly by all the others
अन्य के द्वारा सीधे चर को प्रभावित करने में किस दशा का प्रयोग होता है?
(a) Partially connected/आंशिक जुड़े
(b) Full connected/पूर्ण जुड़े
(c) Local connected/स्थानीय जुड़े
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (b) : A fully connected network is a type of neural network architecture where each neuron in one layer is connected to every neuron in the subsequent layer. This means that the output of any neuron in a layer is influenced by the outputs of all neurons in the preceding layer.

169. Supervised learning and unsupervised learning are two broad category of
सुपरवाइज्ड लर्निंग तथा अनसुपरवाइज्ड लर्निंग निम्न में से किसकी दो व्यापक श्रेणियाँ हैं?
(a) Data warehouse/डेटा वेयरहाउस
(b) DBMS/डी बी एम एस
(c) Neural Network/न्यूरल नेटवर्क
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (c) : Supervised learning and unsupervised learning are two broad category of machine learning. Neural networks are a type of model used in machine learning that can be trained using both supervised and unsupervised learning methods.

170. Which of the following deals with the precise model where accurate solutions are achieved quickly?
निम्न मे से कौन, जहाँ पर जल्दी से शुद्ध हल प्राप्त होते हैं, से सम्बन्ध रखने वाला उचित मॉडल है?
(a) Soft computing/सॉफ्ट कम्प्यूटिंग
(b) Mobile computing/मोबाइल कम्प्यूटिंग
(c) Hard computing/हार्ड कम्प्यूटिंग
(d) None of the above/उपरोक्त में से कोई नहीं

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Ans (c) : Hard computing uses traditional mathematical methods to solve problems such as algorithms and mathematical models. It is based on deterministic and precise calculations and is ideal for solving problems that have well-defined mathematical solution.

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171. In which of the following search agent operates by interleaving computation and action?
निम्न में से किस सर्च में एजेन्ट कम्प्यूटेशन तथा एक्शन इन्टरलीविंग द्वारा कार्य करता है?
(a) Offline search/ऑफलाइन सर्च
(b) Online search/ऑनलाइन सर्च
(c) Breadth first search/ब्रिड्थ फर्स्ट सर्च
(d) Depth first search/डेप्थ फर्स्ट सर्च

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Ans (b) : Online search agents interleave computation (Planning) and action (Execution). They perform actions in the real world while simultaneously searching for the next action to take. This is necessary in dynamic or unknown environments where the agent cannot precompute a complete plan offline.

172. What is Artificial Intelligence?
आर्टिफिशियल इंटेलीजेन्स क्या है?
(a) Putting your intelligence into computer
अपनी समझ को कम्प्यूटर में प्रेषित करना।
(b) Programming with your own intelligence
अपनी स्वयं की समझ के साथ प्रोग्रामिंग करना।
(c) Making a machine intelligent
मशीन को समझदार बनाना।
(d) Putting more memory into computer
कम्प्यूटर में और ज्यादा मेमोरी रखना।

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Ans (c) : Artificial intelligence is about developing systems that can perform tasks requiring human like intelligence such as learning, reasoning, problem solving, decision, making and understanding natural language.

173. In an Artificial neural network (ANN), a neuron is also referred to as a
(a) Filter (b) Perceptrton
(c) Genome (d) Data frame
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : In an Artificial Neural Network a single neuron is also referred to as a perceptron. A perceptron is the fundamental computational unit with an ANN, inspired by bio-logical neurons, and is responsible for receiving inputs, processing them and producing an output.

174. Which of the following can be an application of neural network?
निम्न में से कौन न्यूरल नेटवर्क का अनुप्रयोग हो सकता है?
(a) Sales forecasting/सेल्स फोरकास्टिंग
(b) Data validation/डेटा वैलिडेशन
(c) Risk management/रिस्क मैनेजमेंट
(d) All of the above/उपरोक्त सभी

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Ans (d) : Neural networks are a powerful type of machine learning model that can be applied to a wide range of tasks, including
Soles Forecasting– Predicting future sales trends from historical data.
Data Validation– Detecting anomalies, errors, or inconsistencies in datasets.
Risk management– Assessing and predicting financial, operational or credit risks.

175. Which of the following is an example of unsupervised neural network?
(a) Back-propagation network
(b) Hebb network
(c) Associative memory network
(d) Self-organizing feature map

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Ans. (d) : A Self-Organizing Feature Map– It is a type of artificial neural network (ANN) that is trained using unsupervised learning to produce a low-dimensional (typically two-dimensional), discretized representation of the input space of the training samples, called a map, and is therefore a method to do dimensionality reduction.
Self-organizing maps differ from other artificial neural network as opposed to error correction learning (such as back propagation with gradient descent), and in the sense that they use a neighborhood function to preserve the topological properties of the input space.

176. Reinforcement learning can be formalized in terms of ____ in which the agent initially only knows the set of possible _____ and the set of possible actions.
(a) Markov decision processes, objects
(b) Hidden states, objects
(c) Markov decision processes, states
(d) Objects, states

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : Reinforcement learning can be formalized in terms of Markov decision processes, in which the agent initially only knows the set of possible states and the set of possible actions.

177. Which of the following is NOT true in problem solving in artificial intelligence ?
(a) Implements heuristic search technique
(b) Solutions steps are not explicit
(c) Knowledge is imprecise
(d) It works on or implement repetitions mechanism

UGC NET C.S. June-2020 (Paper-II)

Ans. (d) : Every all program has to do the process of searching for the solution steps that are not explicit in nature.
- Knowledge plays a vital role in intelligence as well as creating artificial intelligence.
- Hence the incorrect statements. it work on or implements a repetition mechanism.

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178. In Artificial Intelligence (AI), an environment is uncertain if it is _____.

(a) Not fully observable and not deterministic
(b) Not fully observable or not deterministic
(c) Fully observable but not deterministic
(d) Not fully observable but deterministic

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : We say an environment is uncertain if it is not fully observable or not deterministic option (b).

179. In Artificial Intelligence (AI), a simple reflex agent selects actions on the basis of_____.

(a) Current percept, completely ignoring rest of the percept history.
(b) Rest of the percept history, completely ignoring current percept.
(c) Both current percept and complete percept history.
(d) Both current percept and just previous percept.

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : In Artificial Intelligence (AI), a simple reflex agent selects actions on the basis of current percept, completely ignoring rest of the percept history.

180. In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics h_1.....h_m is available for a problem and none of them dominates any of the others, which should we choose?

(a) $h(n) = \max\{h_1(n),....,h_m(n)\}$
(b) $h(n) = \min\{h_1(n),....,h_m(n)\}$
(c) $h(n) = \text{avg}\{h_1(n),....,h_m(n)\}$
(d) $h(n) = \text{sum}\{h_1(n),....,h_m(n)\}$

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics h_1.....h_m is available for a problem and none of them dominates any of the others then $h(n) = \max\{h_1(n), ... h_m(n)\}$.

181. Consider following sentences regarding A*, an informed search strategy in Artificial Intelligence (AI).

(a) A* expands all nodes with f(n) < C*.
(b) A* expands no nodes with f(n) /C*.
(c) Pruning is integral to A*.

Here, C* is the cost of the optimal solution path.
Which of the following is correct with respect to the above statements ?

(a) Both statement (a) and statement (b) are true.
(b) Both statement (a) and statement (c) are true.
(c) Both statement (b) and statement (c) are true.
(d) All the statements (a), (b) and (c) are true.

UGC NET C.S. June-2018 (Paper-II)

Ans. (a) : A* expands all nodes with f(n) < C*. **True**
A* expands no nodes with f(n) /C*. **True**
Pruning is integral to A*. **False**

182. Consider the following two sentences :

(a) The planning graph data structure can be used to give a better heuristic for a planning problem.
(b) Dropping negative effects from every action schema in a planning problem results in a relaxed problem.

Which of the following is correct with respect to the above sentences ?

(a) Both sentence (a) and sentence (b) are false.
(b) Both sentence (a) and sentence (b) are true.
(c) Sentence (a) is true but sentence (b) is false.
(d) Sentence (a) is false but sentence (b) is true.

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Ans. (b) : (a) The planning graph data structure can be used to give a better heuristic for a planning problem. **True**
(b) Dropping negative effects from every action schema in a planning problem results in a relaxed problem. **True**
So, both sentence (a) and sentence (b) are true.

183. Given below are two statements one is labelled as Assertion A and the other is labelled as Reason R

A. Artificial Neural Networks are the backbone of deep learning algorithms.
R. Back propagation is one of the techniques used to train an ANN.

In the light of the above statements, choose the most appropriate answer from the options given below :

(a) Both A and R are correct and R is correct explanation of A.
(b) Both A and R are correct but R is NOT the correct explanation of A
(c) A is correct but R is not correct
(d) A is not correct but R is correct

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Ans. (b) : The most appropriate answer is Both A and R are correct but R is Not the correct explanation of A.
Explanation
Artificial Neural Network are the backbone of deep learning algorithms.
Back propagation or backward propagation of error is an algorithm that is designed to test for errors working back from output nodes to input nodes. It is an important mathematical tool for improving the accuracy of predications in data mining and machine learning. Essentially, back propagation is an algorithm used to calculate derivation quickly.
So option (b) is correct.

184. Which of the following is not a property of a good system for representation of knowledge in a particular domain?

(a) Presentation adequac
(b) Inferential adequacy

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(c) Infertile efficiency
(d) Acquisitional efficiency

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Ans. (a) : Knowledge Representation is a necessary component of AI. It allows programmers to convert real-world data into a language that a computer can use to develop an information system that is critical in accomplishing tasks. The following properties should be possessed by a knowledge representation system.
(i) Representational Adequacy
(ii) Inferential Adequacy
(iii) Inferential Efficiency
(iv) Acquisitional adequacy
Presentation adequacy is not a property of a good system for representation of knowledge in a particular domain.

185. Which is not the component of the natural language understanding process?
(a) Morphological analysis
(b) Semantic analysis
(c) Pragmatic analysis
(d) Meaning analysis

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Ans. (d) : Natural language understanding (NLU) is a branch of Artificial Intelligence (AI) that uses computer software to understand input in the form of sentences using text or speech. NLU enables human-computer interaction by analyzing language versus just words. There are five components of the natural language understanding process.
(i) Morphological and lexical Analysis
(ii) Syntactic Analysis
(iii) Semantic Analysis
(iv) Discourse Integration
(v) Pragmatic Analysis

186. Which is not a basic approach to the problem of conflict resolution in a production system?
(a) Assigning a preference based on the rule that matched
(b) Assigning a preference based the object that matched
(c) Assigning a preference based on the action that the matched rule would perform
(d) Assigning a preference based on the action that the matched object would perform

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Ans. (d) : In a production system, conflicts are typically resolved by considering factors such as the rule that matched, the objects involved, or the actions that the matched rules would perform. Assigning preference based on the action that the matched object would perform does not directly address the conflict resolution within the system's rule-based framework. This approach might be less common because it's not directly tied to the rule-based logic of the system.

187. What is the generic structure of Multi Agent System (MAS)?
(a) Single agent with multiple objectives
(b) Multiagents with a single objectives
(c) Multiagents with diverse objectives and communication abilities
(d) Multiagent with two objectives

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Ans. (c) : The generic structure of a multi Agent system (MAS) typically involves multiagents with diverse objectives and communication, abilities. This structure allows for interaction between multiple agents with varying goals and the capability to communicate and collaborate to achieve those objectives.

188. What is the role of weights in an ANN ?
(a) To store data
(b) To adjust and improve network performance
(c) to control the speed
(d) To secure the network

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Ans. (b) : Weight in an Artificial neural network (ANN) determine the strength and importance of the input signal during the learning process. By adjusting the weights during training (using techniques like back propagation), the network improves its ability to predict or classify data accurately. This process helps optimize the network's performance.

189. Artificial Neural networks (ANNs) are inspired by :
(a) Quantum mechanics
(b) Human brain's neural network
(c) Computer Hardware architecture
(d) Genetic algorithm

UGC NET C.S. Re-Exam- June 2024

Ans. (b) : Artificial Neural network (ANNs) are inspired by Human brain's neural network. The design and functioning of ANNs are based on the structure and behavior of biological neural networks in the human brain. They aim to mimic the way humans learn and process information.

190. Which of the following is/are the application area(s) of ANN?
(A) Natural Language Processing
(B) Image Processing
(C) Pattern Recognition
(D) Speech Recognition
Choose the correct answer from the options given belw :
(a) (A) and (B) Only
(b) (B) and (C) Only
(c) (A), (B), and (C) Only
(d) (A), (B), (C) and (D)

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Ans. (d) : Artificial Neural Networks (ANNs) have a wide range of applications due to their ability to learn and generalize from data. The application area of ANNs include:
(i) Natural Language Processing (NLP)
(ii) Image Processing
(iii) Pattern Recognition
(iv) Speech Recognition
Therefore, the correct answer is (A), (B), (C) and (D).

191. Which of the following neural networks uses supervised learning?
(A) Multilayer perception
(B) Self organizing feature map
(C) Hopfield network
(a) (A) only (b) (B) only
(c) (A) and (B) only (d) (A) and (C) only
UGC NET C.S. June-2017 (Paper-III)

Ans : (a) Multilayer perceptron network and support vector machines are examples of **supervised learning,** while self organizing feature maps and Hopfield networks are examples of **unsupervised learning.**

192. Consider following two rules R1 and R2 in logical reasoning in Artificial Intelligence (AI):
R1 : from $\alpha \supset \beta$ and α inter β is known as Modus Tollens (MT)
R2 : Form $\alpha \supset \beta$ and $\neg\alpha$ inter $\neg\beta$ is known as Modus Ponens (MP)
(a) Only R1 is correct
(b) Only R2 is correct
(c) Both R1 and R2 are correct
(d) Neither R1 nor R2 is correct
UGC NET C.S. June-2017 (Paper-III)

Ans : (d) R1 holds the correct definition for modus Ponens & R2 hold the correct definition of modus tollens.

193. In Artificial Intelligence (AI), what is present in the planning graph?
(a) Sequence of levels (b) Literals
(c) Variables (d) Heuristic estimates
UGC NET C.S. June-2017 (Paper-III)

Ans : (a) A planning graph consists of sequence of levels correspond to time steps
- Used to achieve heuristic estimates
  - A solution can also directly be extracted using GRAPHPLAN
- Consist of a sequence of levels that correspond to steps in plan
  - Level 0 is the initial stage

Each level consist of a set of literals and a set of function
- **Literals :** All those that could be true at that time step, depending upon the actions executed at preceding time step.
- **Action :** All those action that could have their preconditions satisfied at that time step, depending on which of the literals actually hold.

194. What is the best method to go for the game playing problem?
(a) Optimal search (b) Random search
(c) Heuristic search (d) Stratified search
UGC NET C.S. June-2017 (Paper-III)

Ans : (c) We use a heuristic approach, as it will find out brute force computation, looking at hundreds of thousands of positions.

195. In Artificial Intelligence (AI), which agent deals with happy and unhappy state?
(a) Simple reflex agent (b) Model based agent
(c) Learning agent (d) Utility based agent
UGC NET C.S. November-2017 (Paper-III)

Ans : (b) Model based agent deals with happy & unhappy state in artificial intelligence (AI). So option (b) is correct

196. In Artificial Intelligence, a semantic network
(a) Is a graph-based method of knowledge representation where nodes represent concepts and arcs represent relations between concepts.
(b) Is a graph-based method of knowledge representation where nodes represent relations between concepts and arcs represent concepts.
(c) Represents an entity as a set of slots and associated rules.
(d) Is a subset of first-order logic.
UGC NET C.S. August-2016 (Paper-III)

Ans : (a) In Artificial Intelligence is a graph-based method of knowledge representation where nodes represent concepts and arcs represent relations between concepts. In other words Semantic Network representation nodes represents concepts, instances or events and link represents relationship between nodes, such as is-a, part-of, kind-of, interactive relations.

197. STRIPS addresses the problem of efficiently representing and implementation of a planner. It is not related to which one of the following?
(a) SHAKEY (b) SRI
(c) NLP (d) None of these
UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : STRIPs was devised **SRI** in the early 1970s to control a robot called **SHAKEY** strips is not related with NLP (Natural Language Processing).

198. Slots and facets are used in
(a) Semantic Networks (b) Frames
(c) Rules (d) All of these
UGC NET C.S. June-2014 (Paper-III)

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Ans. (b) :
Frames : Frames are structured objects that represent knowledge and contain labeled slots with their values.
Slots : Slots are almost similar to attributes as in ERD or object-oriented systems in the sense that they may contain declarative as well as procedural information.
Facets : Facets are almost similar to description to the slot values that may function as criteria or validity of the values inserted into a slop.

Name of frame		
Slots	Slot valves	Facets

199. Which of the following is the correct sequence in Pattern Recognition process?
निम्न में से कौन पैटर्न पहचान प्रक्रिया का सही क्रम है?

(a) Data Acquisition, Feature Extraction, Matching, Classification/डाटा एक्वीजीशन, फीचर एक्सट्रैक्शन, मैचिंग, क्लासिफिकेशन
(b) Feature Extraction, Data Acquisition, Matching, Classification/फीचर एक्सट्रैक्शन, डाटा एक्वीजीशन, मैचिंग, क्लासिफिकेशन
(c) Data Acquisition, Feature Extraction, Classification, Matching/डाटा एक्वीजीशन, फीचर एक्सट्रैक्शन, क्लासिफिकेशन, मैचिंग
(d) Matching, Data Acquisition, Feature Extraction, Classification/मैचिंग, डाटा एक्वीजीशन, फीचर एक्सट्रैक्शन, क्लासिफिकेशन

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : The correct sequence in the pattern Recognition process is.
Data Acquisition, Feature Extraction, Matching, Classification.
Data Acquisition– First, you collect raw data (Image, signals, text, etc.)
Feature Extraction– From the raw data, you extract meaningful features that are easier to process and analyze.
Matching– You compare the extracted features with known patterns or templates.
Classification– Finally, based on matching, you assign the data to a specific category or class.
Thus option (a) is correct.

200. What is a neuron in Neural Networks?
न्यूरल नेटवर्क में एक न्यूरॉन क्या है?

(a) A neuron is a small cell that receives electrochemical signals from various sources and transmit electrical impulses to other neurons./न्यूरॉन एक छोटी कोशिका होती है जो विभिन्न स्रोतों से विद्युत रासायनिक संकेत प्राप्त करती है और विद्युत आवेगों को अन्य न्यूरॉन्सं तक पहुंचाती है।
(b) A neuron is a cell that send electromagnetic signals to axons./एक न्यूरॉन एक कोशिका है जो अक्सॉन्स को विद्युत चुम्बकीय संकेत भेजती है।
(c) A neuron is a small cell that sends electric impulses to ions./न्यूरॉन एक छोटी कोशिका है जो आयनों को विद्युत आवेग भेजती है।
(d) A neuron is a cell which has the capability to predict the situation from past trends.
एक न्यूरॉन एक कोशिका है जो पिछले रुझानों से स्थिति की भविष्यवाणी करने की क्षमता रखती है।

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Ans. (a) : In a neural Network, a neural (Also called a node or unit) is the basic computational unit that mimics the function of a biological neuron. It receives one or more input values, processes them and produces an output.

201. What is axon in Neural Networks?
न्यूरल नेटवर्क में अक्सॉन क्या है?

(a) It is a long cylindrical connection that carries impulses from the neurons./यह एक लंबा बेलनाकार कनेक्श है जो न्यूरॉन्स से आवेगों को वहन करता है।
(b) It is a small cell used to perform basic logical operations./यह एक छोटा सेल है जिसका उपयोग बुनियादी तार्किक ऑपरेशन्स के लिए किया जाता है।
(c) It is a small cell which contain electric signal.
यह एक छोटा सेल है जिसमें विद्युत संकेत होता है।
(d) It is a collection of neurons.
यह न्यूरॉन्स का एक संग्रह है।

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Ans. (a) : In the context of neural networks, an axon is inspired by biological neurons. In biology, an axon is a long, cylindrical project of a neuron that conducts electrical impulses away from the neuron's cell body to other neurons, the term "axon" is sometimes used analogically to desirable the connections or pathways that transmit signals (data) between neuron in the network.

202. What is the full form of LISP Programming Language?
प्रोग्रामिंग लैंगुऐज में एल.आई.एस.पी. का असंक्षिप्त रूप क्या है?

(a) Language Processing/लैंग्वेज प्रोसेसिंग
(b) List Processing/लिस्ट प्रोसेसिंग
(c) Least Processing/लीस्ट प्रोसेसिंग
(d) Limited Processing/लीमिटेड प्रोसेसिंग

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Ans. (b) : LISP is one of the oldest high-level programming languages, created by John McCarthy in 1958. The full form of LISP is 'List Processing'.

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203. What is the full form of FOPL in artificial intelligence?
आर्टिफिशियल इंटेलिजेंस में एफ.ओ.पी.एल. का असंक्षिप्त रूप क्या है?

(a) First order predicate logic
फस्ट आर्डर प्रेडीकेट लॉजिक
(b) File open predicate logic
फाइल ओपन प्रेडीकेट लॉजिक
(c) First operator performance logic
फर्स्ट ऑपरेटर परफॉरमेंस लॉजिक
(d) File operator perforate logic
फाइल ऑपरेटर परफोरते लॉजिक

DSSSB PGT C.S. M/F 11.07.2021

Ans. (a) : FOPL stands for First-Order Predicate Logic, FOPL is a key technique in artificial intelligence (AI) that's used to represent objects and their relationships.

204. Which of the following are correct for the neural network?

A. The training time depends upon the size of network.
B. Neural network can be simulated on the conventional computer.
C. Neural network mimic the same way as that of the humans brain.
D. A neural network includes feedback.

Choose the correct answer from the options given below :

(a) A and B only (b) A, C, and D only
(c) A, B and C only (d) A and C only

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Ans. (b) : The correct answer for the Neural Network is-
(A) The training time depends upon the size of network.
(C) Neural Network mimic the same way as that of the humans brain.
(D) A Neural Network include feedback.

205. Which of the following is not a component of the classic Planning Definition?

(a) Init (b) Domain
(c) Action (d) Goal

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Ans. (b) : Init (initial state) : The starting condition of the system.
Action (operators): the set of possible actions that can be performed.
Goal: The desired end state that the planner aims to Achieve.
While the domain is important in defining the problem space, it is not typically considered a fundamental component of the classic planning definition instead the domain refers to the broader context in which planning problems are defined.

206. Machine that places the request to access the data, is generally called as _____.
वह मशीन जो कि डाटा तक पहुँच पाने का अनुरोध करती है, आमतौर पर _____ कहलाती है।

(a) Server machine/सर्वर मशीन
(b) Client machine/क्लाइंट मशीन
(c) Request machine/रिक्वेस्ट मशीन
(d) Intelligent machine/इंटेलीजेंट मशीन

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Ans. (b) : Machine that places the request to access the data, is generally called as client machine. A client is a computer or a program that, as part of its operation, relies on sending a request to another program or a computer hardware or software that accesses a service made available by a server.

207. Which of the following is an application of Artificial Intelligence?
निम्नलिखिम में से कौन-सा आर्टिफिशियल इंटेलिजेंस का अनुप्रयोग है?

(a) It helps to exploit vulnerabilities to secure the firm
(b) Language understanding and problem solving (Text analytics and NLP)
(c) It helps to deploy applications on the cloud
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (b) Artificial intelligence comprises a number of techniques, including natural language processing (NLP) and text analytics that are used to understand and process human language.

208. What does the Bayesian network provide?
बायेसियन नेटवर्क क्या प्रदान करता है?

(a) Partial description of the domain
(b) Complete description of the problem
(c) Complete description of the domain
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : Bayesian network provides a complete description of the domain. It represents the probability relationships between different variable using a specified acyclic graph, which gives a complete idea of the structure of the domain and the relationships between different elements.

209. The _____ input addresses the different styles of different users.
_____ इनपुट विभिन्न उपयोगकर्ताओं की विभिन्न शैलियों को संबोधित करता है।

(a) Multi-sense
(b) Multimedia
(c) Multi-sensory
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) : Multi-sensory– Refers to content or experiences that engage more than one of the user's senses simultaneously (e.g., sight, sound touch). Different users may prater to learn or interact using different combinations of these senses.
For example– Some people are visual learners, while others are auditory or kinesthetic.

210. Which of the following are the key feature(s) of cloud computing?
निम्नलिखित में से कौन से क्लाउड कंप्यूटिंग की प्रमुख विषेताएं हैं?
I. Scalability/स्केलेबिलिटी
II. Self service/सेल्फ सर्विस
III. Pay-per-use/पे-पर-यूज़
(a) I nad II
(b) II and III
(c) I and III
(d) I, II and III

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (d) : Cloud computing is a technology that allows you to store and access data and application over the internet instead of using your computer's hard drive or a local server.
Features of cloud computing–
1. On-demand self services
2. Broad network access
3. Rapid elasticity
4. Resource pooling
5. Measured service
6. Multi tenancy
7. Virtualization
8. Resilient computing
9. Flexible pricing models
10. Automation.

211. Match the List-I with List-II

List-I		List-II	
A.	**Decision Tree**	**I.**	**Delta Learning Rule**
B.	**Supervised Learning**	**II.**	**Self Organizing Map**
C.	**Artificial Neural**	**III.**	**C4.5 Algorithm**
D.	**Instance base Learning**	**IV.**	**Non-linear Regression Algorithm**

Choose the correct answer from the options given below :
(a) A-I, B-II, C-III, D-IV
(b) A-II, B-III, C-IV, D-I
(c) A-III, B-IV, C-I, D-II
(d) A-IV, B-I, C-II, D-III

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Ans. (c) : Correct match-

List-I			List-II
A.	Decision Tree	–	C4.5 Algorithm
B.	Supervised Learning	–	Non-linear Regression Algorithm
C.	Artificial Neural	–	Delta Learning Rule
D.	Instance base Learning	–	Self Organizing Map

212. Definition's of _____ organized into following four categories namely, Thinking Humanly, Thinking Rationally, Acting Humanly, Acting Rationally
(a) Machine Learning
(b) Deep Learning
(c) Artificial Intelligence
(d) Neural Network

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Ans. (c) : Artificial intelligence (AI) has become a major of four categories in topic:
Thinking humanly:- Definition : This approach relates to the internal Cognitive process of human. in other words, how humans think and how we can represent that process in computers.
Acting Humanly:- Definition: This focuses on the machines' ability to mimic human behavior to the point where it cannot be distinguished from a real human. The Turing Test is often associated with this approach.
Thinking rationally: Definition: This approach relates to thinking logically and making decision based on reasoning in other works' using logic in decision making.
Acting Rationally: Definition: This focuses on the end result of AI action, which is to act in the most optimal or efficient way to achieve a certain goal.

(iii) Internet of Thing

213. Which wireless communication technology is commonly used in IoT devices for short-range communication?
कौन-सी वायरलेस संचार तकनीक आमतौर पर छोटी दूरी के संचार के लिए IoT उपकरणों में उपयोग की जाती है?
(a) Bluetooth/ब्लूटूथ
(b) Wi-Fi/वाई-फाई
(c) 4G LTE/4जी एलटीई
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (a) : Bluetooth is a wireless technology specifically designed for short range communication between devices like smart phones, smart watches and other portable gadgets. It's low power and efficient making it ideal for use in Internet of Things (IoT) applications.

Sarkari Resource

214. Which ZigBee device is used to transfer data between ZigBee devices?
Zigbee उपकरणों के बीच डेटा स्थानांतरित करने के लिए किस Zigbee डिवाइस का उपयोग किया जाता है?

(a) Zigbee Coordinator device
(b) Zigbee Router
(c) Zigbee start devce
(d) Zigbee end devce

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (b) : Zigbee Router– This device acts as a relay station, forwarding data packets between other zegbee devices within the network. It can receive data from an end devices or another router and pass it on to the intended recipient.

215. The Internet of Things (IoT), access to real-time data, and the introduction of cyber-physical systems mark the big changes for this era.
इंटरनेट आफ थिंग्स (IoT), वास्तविक समय के डेटा तक पहुँच और साइबर भौतिक प्रणालियों की शुरूआत इस युग के लिए बड़े बदलावो को चिन्हित करती है।

(a) Industrial Revolution
(b) 3rd Industrial Revolution
(c) Industry 4.0
(d) Manufacturing automation

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (c) : The Internet of Things, real time data access, and cyber physical system are key character characteristics of the fourth industrial Revolution, Commonly referred to as industry 4.0. This era emphasizes smart technology, automation, and data exchange in manufacturing technologies, including, IoT, cyber physical systems, and real time data analytics.

216. _____ are used to overcome the challenges of managing the resources of the IoT.
संसाधनो के प्रबंधन की चुनौतियों पर काबू पाने के लिए किसका उपयोग किया जाता है।

(a) Clustering
(b) Software agents
(c) Synchronization techniques
(d) Cluster, Software agent, and Synchronization techniques

Bihar STET C.S. 12.09.2023 (Shift-II)

Ans. (d) : The use of Cluster, Software agent, and Synchronization techniques are used in order to overcome the challenges of managing the resources of the internal of things objects.

217. Which of the following is an example of an IoT protocol used for device communication?
निम्नलिखित में से कौन-सा डिवाइस संचार के लिए उपयोग किए जाने वाले IoT प्रोटोकॉल का एक उदाहरण है?

(a) HTTP
(b) FTP
(c) MQTT
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (c) : MQTT (Message Queuing Telemetry Transport) is a lightweight and publish subscribe based messaging protocol which is very popular for communication between IoT devices.
HTTP and FTP are also protocols but they are not IoT specific but are used for general web and file transfer.

218. Which of the given wireless technologies used in IoT, consumes the least amount of power?

(a) Zigbee (b) Bluetooth
(c) Wi-Fi (d) GSM/CDMA

UGC NET C.S. November-2017 (Paper-II)

Ans : (b) Bluetooth is a telecommunications industry specification that describes how mobile devices, computers and other device can easily communicate with each other using a short range wireless connection.

219. BLE stands for /BLE का पूर्ण रूप क्या है?

(a) Bluetooth large energy/ब्लूटूथ लार्ज एनर्जी
(b) Bluetooth low energy/ब्लूटूथ लो एनर्जी
(c) Bluetooth light energy/ब्लूटूथ लाइट एनर्जी
(d) Bluetooth long energy/लूटूथ लॉन्ग एनर्जी

Bihar STET C.S. 12.09.2023 (Shift-I)

Ans. (b) : BLE stands for Bluetooth Low Energy. It is a wireless technology that consumes less power and is used in IoT devices.

220. Which of the following is not an IoT platform?
निम्नलिखित में से कौन-सा IoT प्लेटफॉर्म नहीं है?

(a) Amazon Web Services
(b) Microsoft Azure
(c) Flipkart
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) Amazon Web Services, Microsoft Azure and Salesforce provide cloud computing IoT services. While Flipkart is an e-commerce website that provides marketing facilities.

221. Which layer is used for wireless connection in IoT devices?
IoT उपकरणों में वायरलेस कनेक्शन के लिए किस परत का उपयोग किया जाता है?

Sarkari Resource

(a) Application layer
(b) Network layer
(c) Data link layer
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (c) Data link layer is used for wireless connections in IoT devices. This layer handles the protocols and techniques of communication through physical wireless channels.

222. Which of the following is not a sensor in IoT?
निम्नलिखित में से कौन IoT में सेंसर नहीं है?

(a) BMP280
(b) DHT11
(c) Photo resistor
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (e) : BMP280– This is a pressure sensor often used in IoT applications to measure atmospheric pressure.
DHT11– This is a temperature and humidity sensor commonly used in IoT devices for environmental monitoring.
Photo resistor– This is a light sensitive resistor that changes its resistance depending on the amount of light it receives.

223. IoT gateway must provide _____
IoT गेटवे को _____ प्रदान करना होगा।

(a) Protocol abstraction
(b) Data storage / Security with hardware
(c) Simple and fast installation
(d) More than one of the above
(e) None of the above

BPSC TRE 3.0 22.07.2024 (6-10)

Ans. (a) : An IoT gateway acts as an intermediary between IoT devices and a control system or cloud. One of the key functions provided by an IoT gateway is protocol abstraction this means that the gateway should be able to translate and adapt the different communication protocols used by various IoT devices into a common format that can be understood by the central system.

(iv) Theory of Computation

224. For NP complete problems:

(a) Several polynomial time algorithms are available
(b) No polynomial time algorithm is discovered by
(c) Polynomial time algorithms exist but not discovered
(d) Polynomial time algorithms will not exist, hence cannot be discovered

CGPSC C.S. 2020 Set-A

Ans.(b): For NP-complete problems no polynomial time algorithm is known. If one were found for any NP-complete problem it would imply P = NP we also haven't proved such algorithm can't exist.

225. Which of the following is true?

(a) All NP complete problems are NP hard
(b) If an NP hard problem can be solved in polynomial time, then all NP complete problems can be solved in polynomial time
(c) NP hard problems are not known to be NP complete
(d) All of the above are true

CGPSC C.S. 2020 Set-A

Ans.(d): Let's analyze each option–
All NP complete problems are NP hard– This is true by definition. An NP-complete problem is one that is both in NP and NP-hand, so every NP-complete problem is necessarily NP-hard.
If an NP hard problem can be solved in polynomial time, then all NP complete problems can be solved in polynomial time– It's true. If an NP-hard problem can be solved in polynomial time then every problem in NP (including all NP complete problems) can be reduced to it and hence solved in polynomial time. This would imply P = NP.
NP hard problems are not known to be NP complete–
This is true. An NP-hard problem is at least as hard as the hardest problems in NP, but it might not be in NP itself. For example, the halting problem is NP complete unless it is also in NP.
Since (d) is correct.

226. The subset construction method is used to obtain a DFA from

(a) A regular expression
(b) An NFA
(c) A CFG
(d) A CSG
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : The subset construction method is specifically designed to convert a Non-Deterministic Finite Automaton (NFA) into a Deterministic Finite Automaton (DFA).

227. In which of the following transition diagrams, for each state S and input symbol a, there is at most one edge labeled a leaving S?

(a) NFA (b) SFA
(c) DFA (d) Recursive NFA
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (c) : A DFA (Deterministic Finite Automaton) is defined by having exactly one transition for each state and input symbol. This means for any given state, there's only one outgoing edge labeled with a particular input symbol.

Sarkari Resource

228. Which is used to generate membership function in fuzzy logic?
फजी लॉजिक में मेम्बरशिप फलन को बनाने के लिए किसका प्रयोग होता है?

(a) Approximate reasoning/अनुमानित तर्क
(b) Proportionality/प्रोपोर्शनलिटी
(c) Rationality/रेशनलिटी
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asist. Prof. 2017

Ans (d) : In fuzzy logic, membership functions and used to define the degree of membership of an element in a fuzzy set. The generation of membership functions is typically based on the context of the problem, expert knowledge or data driven methods.
(a) Approximate reasoning– This refers to the process of making inferences with fuzzy logic, but it is not directly used to generate membership functions.
(b) Proportionality– This is not a standard term for generating membership functions membership functions are not necessarily proportional.
(c) Rationality– This is a general concept and not specifically used for generating membership functions.

229. A turing machine crashes
एक ट्यूरिंग मशीन क्रैश होती है-

(a) If the machine reads all the input characters without traversing some state./यदि मशीन सभी इनपुट अक्षरों को बिना कुछ स्टेट पर पहुंचें पढ़ लें।
(b) If the machine traverses all its states till the input remains./यदि मशीन सभी स्टेट तक पहुंचती है और इनपुट बचा हो।
(c) If the transitional function is not defined./यदि परिवर्ती फलन परिभाषित न हो।
(d) None of the above/उपरोक्त में से कोई नहीं

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) : A Turing machine operates based on a transitional function that dictates its next move. This function takes the current state and the symbol under the head as input and determines the next state, the symbol to write, and the direction to move the head (left to right). If the machine reaches a state and reads a symbol for which no transition is defined in its program, it halts unexpectedly, which is what's meant by a crash. This is often referred to as an undefined transition.

230. Which of the following represents the output of the transition function (δ)

$\delta(q_0, a) = (q_1, x, R)$
$\delta(q_1, a) = (q_1, a, R)$
$\delta(q_1, y_1) = (q_1, y, R)$
$\delta(q_1, b) = (q_2, y, L)$
$\delta(q_2, y) = (q_2, y, L)$
$\delta(q_2, a) = (q_2, a, L)$
$\delta(q_2, x) = (q_0, x, R)$
$\delta(q_0, y) = (q_3, y, R)$
$\delta(q_3, y) = (q_3, y, R)$
$\delta(q_2, \square) = (q_2, \square, R)$

(a) $L = \{a^n b^n \mid n \geq 0\}$ (b) $L = \{a^n b^n \mid n \geq 1\}$
(c) $L = \{a^n b^n \mid n > 0\}$ (d) $L = \{a^n b^n \mid n > 1\}$

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Ans. (b) : The given transitions function defines how a Turing machine processes input symbol.
Each transition is of the form:

$$\delta(q, x) = (q', y, D)$$

Where:-
- q is the current state,
- x is the current symbol being read,
- q' is the next state,
- y is the symbol that replaces x,
- D is the direction of movement (left or right)

Language Recognized:-
- The transitions ensure that every 'a' is matched with a corresponding 'b'.
- The process starts with at least one 'a' and one 'b' (since q 0 starts by replacing 'a' and moves to 'b')
- This corresponds to the language:

$$L = \{a^n b^n \mid n \geq 1\}$$

Among the given options (b) is correct.

231. Given two languages :

$L1 = \{(ab)^n a^k \mid n > k, k \geq 0\}$
$L2 = \{a^n b^m \mid n \neq m\}$
Using pumping lemma for regular language, it can be shown that

(a) L_1 is regular and L_2 is not regular
(b) L_1 is not regular and L_2 is regular
(c) L_1 is regular and L_2 is regular
(d) L_1 is not regular and L_2 is not regular

UGC NET C.S. December-2014 (Paper-III)

Ans : (d) Whenever pumping lemma is there it can only be used to prove that language is not regular or not context free
So the answer is (d)

232. Context free grammar is not closed under :

(a) Concatenation (b) Complementation
(c) Kleene Star (d) Union

UGC NET C.S. November-2017 (Paper-III)

Ans : (b) Context free grammar are not closed under intersection context free language are accepted by pushdown automata but not by finite automata - context free grammar.

233. Which of the following problem is undividable?

(a) To determine if two finite automata are equivalent
(b) Membership problem for context free equivalent
(c) Finiteness problem for finite automata
(d) Ambiguity problem for context free grammar

UGC NET C.S. November-2017 (Paper-III)

Sarkari Resource

Ans : (d) Ambiguity problem for context free grammar is undecidable. The problem is undecidable if there is no turing machine will always half in finite amount of time.

234. Given a Turing Machine

$M = (\{q_0, q_1\}, \{0, 1\}, \{0, 1, B\}, \delta, B, \{q_1\})$
Where δ is a transition function defined as
$\delta(q_0, 0) = (q_0, 0, R)$
$\delta(q_0, B) = (q_1, B, R)$
The language L(M) accepted by turing machine is given as:

(a) 0* 1* (b) 00*
(c) 10* (d) 1*0*

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) This Turing Machine accept 0*.
00* is subset of 0*. So, Turing Machine accepts all strings of 00*. The only string except 00* is $\in$ accepted by this Turing Machine.

235. Let $G = (V, T, S, P)$ be a context-free grammar such that every one of its productions is of the form $A \rightarrow v$, with $|v| = k > 1$. The derivation tree for any string $W \in L(G)$ has a height such that

(a) $h < \frac{(|W|-1)}{K-1}$

(b) $\log_k |W| \le h$

(c) $\log_k |W| < h < \frac{(|W|-1)}{K-1}$

(d) $\log_k |W| \le h \le \frac{(|W|-1)}{K-1}$

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) The derivation tree for any string w $\in L(G)$ has a height such that

$$\log_k |w| \le h \le \frac{(|w|-1)}{k-1}$$

236. Which of the following pairs have different expressive power?

(a) Single-tape-turning machine and multi-dimensional turning machine
(b) Multi-tape turning machine and multi-dimensional turning machine
(c) Deterministic push down automata and non-deterministic pushdown automata.
(d) Deterministic finite automata and Non-deterministic finite automata

UGC NET C.S. June-2017 (Paper-III)

Ans : (c) Deterministic push down automata & non deterministic push down automata.

237. The sum of minimum and maximum number of final states for a Deterministic Finite Automata (DFA) having 'P' state is equal to:

(a) p (b) p – 1
(c) p + 1 (d) p + 2

UGC NET C.S. December-2023

Ans. (c) : The maximum number of final states for a DFA can be total number of states itself and minimum would always be 1, as no DFA exists without a final state. Therefore, Then sum of minimum and maximum number

238. Consider the following statements:

A. For every regular language, we can design Turing Machine.
B. For every context free language, we can design Turing Machine.

Which of the following is correct?

(a) Statement-A is false and Statement-B is true.
(b) Statement-A and Statement-B both are false.
(c) Statement-A is true and Statement-B is false
(d) Statement-A and Statement-B both are true

UGC NET C.S. December-2022

Ans. (d) : Both statement (A) and (B) are correct.
A. For every regular language, we can design a Turing Machine. Regular language are equivalent in power to the languages that can be recognized by a Turing Machine, So there exists a Turing Machine for every regular language.
B. For every context free language, we can design a Turing Machine. Similarly, context free language are also Turing -recognizable meaning there exists a Turing Machine that can recognize every context-free language.

239. A pushdown automata behaves like a Turing machine when the number of auxiliary memory is :

(a) 0 (b) 1
(c) 1 or more (d) 2 or more

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : Pushdown automata have only one stack with 2 stacks we can implement queue = $\Rightarrow$ T(n + 2) is sufficient form Turing Machine.
Moreover expressive power of T(n + 2) = T(n + 3) = T (n + 4) etc.
Therefore we need one more extra auxiliary stack memory for PDA or we need at least 2 auxiliary memory to implement TM
Option (d) is correct.

240. Pushdown automata can recognize language generated by_____.

(a) Only context free grammar
(b) Only regular grammar
(c) Context free grammar or regular grammar
(d) Only context sensitive grammar

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : Pushdown automata recognize regular as well as context free grammar.

241. Context sensitive language can be recognized by a :

(a) Finite state machine
(b) Deterministic finite automata
(c) Non-deterministic finite automata
(d) Linear bounded automata

UGC NET C.S. June-2018 (Paper-II)

Ans. (d) : Context sensitive language can be recognized by a linear bounded automata for context sensitive grammar we need linear bounded automata.

Sarkari Resource

242. How can the decision algorithm be constructed for deciding whether context-free language L is finite?

(A) By constructing redundant CFG G in CNF generating language L

(B) By constructing non-redundant CFG G in CNF generating language L

(C) By constructing non-redundant CFG G in CNF generating language L – {^}(^ stands for null)

Which of the following is correct?

(a) (A) only
(b) (B) only
(c) (C) only
(d) None of (A), (B) and (C)

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : We can check that whether a context free language L is finite or not by contracting non redundant context free grammar in which there are no null productions no unit productions and no useless productions. Steps that are used in deciding whether a context free language is finite or not?

1) Construct the context free grammar corresponding to that language.
2) Reduce the grammar by removing null, unit and useless productions.
3) Contract the directed graph of that grammar
4) If that graph contains a cycle, then that language is infinite.

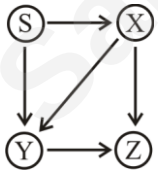
If graph does not contain cycle, it means language is finite.

$S \rightarrow XY/X$

$X \rightarrow YZ/Y$

$Y \rightarrow ZZ/Z$

This grammar is already reduced. Graph for this is



This graph does not contain any cycle, so Language is finite for this.

The decision algorithm be constructed for deciding whether context free language L is finite by constructing non redundant CFG G in CNF generating language L – {^} (^ stands for null)

243. Pumping lemma for regular sets is used to:-

(a) Prove that certain sets are regular
(b) Prove that certain sets are not regular
(c) Generate context free language
(d) Generate context sensitive language

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : The Pumping Lemma states that if a language is regular, then any sufficiently long string in the language can be "pumped" (i.e., a middle section of the string can be repeated an arbitrary number of times) and the resulting strings will still be in the language.

Therefore, to prove that a set is not regular, one uses a proof by contradiction-

1. Assume the language is regular
2. Apply the Pumping Lemma to a carefully.
3. Show the pumping this string repeating the middle section lends to a string that is not in the language, which contradicts the Pumping Lemma.

244. The logic of pumping lemma is a good example of

(a) Pigeon hole principle
(b) Recursion
(c) Divide and conquer technique
(d) Iteration

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : The logic of the pumping lemma is an example of the pigeon hole principle.

Pigeon hole principle if there are total p pigeons seating into q holes then what pigeon hole principle says that at least one hole must have more than one pigeon.

Pumping lemma– There is an infinite number of input sequences possible and for every finite automation there is a finite number of states.

Therefore, the logic pumping lemma is a good example of the pigeon hole principle.

245. In data mining k-means is used for the purpose of:-

(a) Data cleaning
(b) Data Reduction
(c) Clustering by portioning
(d) Outlier detection

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : In data mining, the d-means method is primarily used for "clustering by partitioning".

K-means is a clustering algorithm that divides a dataset into a specified number of clusters, where each cluster contains data points that are similar to each other.

246. Which of the following is not a measure of dispersion of data:-

(a) Median (b) Range
(c) Standard deviation (d) Variance

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : Measures of dispersion describe how spread out or varied the data is. Common measures of dispersion include

Range (max-min)

Standard Deviation (Average distance from the mean)

variance (square of standard deviation)

However, the median is a measure of control tendency, not dispersion, It represent the middle value of a dataset and does not indicate how data is spread.

247. For sampling of a large dataset D, n-tuples of D were divided into multiple disjoint subsets, then Random sampling was applied an each subset to obtain a sample of D. Sampling of D is:-

(a) Cluster sampling (b) Partial sampling
(c) Stratified sampling (d) Random sampling

RPSC C.S. 2016 (PAPER-II)

Sarkari Resource

Ans. (c) : In stratified sampling, the dataset D is divided into disjoint subsets based on some criteria and then random sampling is applied within each stratum. This ensures that each subgroup is properly represented in the final sample.

In description– Dataset D was divided into disjoint subsets.

Random sampling was applied on each subset.

So, this matches the definition of stratified sampling.

248. In mobile computing which of the following is not a dynamic update scheme of location management:-

(a) Periodic updates
(b) Movement-based updates
(c) Cost-based-updates
(d) distance-based updates

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : In mobile computing, location management involves tracking the location of mobile devices to ensure efficient communication. Dynamic update schemes are strategies used to update the location of a mobile device based on certain criteria.

Cost based update is not a standard dynamic update scheme in traditional location management. The commonly recognized dynamic update schemes are periodic, movement based, and distance-based updates, cost considerations may influence the choice of update scheme but are not a standalone update mechanism.

249. One of the purposes of using intermediate code in compilers is to

(a) Make parsing and semantic analysis simpler
(b) Improve error recovery and error reporting
(c) Increase the chances of reusing the machine-independent code optimizer in other compliers
(d) improve the register allocation

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : Intermediate code is a machine independent representation of a program, meaning it's not tied to a specific hardware architecture. This allows compilers to reuse the same optimization routines for different target machines. The front end of the compiler translates the source code into intermediate code, and the back end then translates the intermediate code into machine code for the specific target architecture. This modular design enables the reuse of the optimization stage across different compilers.

250. Data marts that incorporate data mining tools to extract sets of data are called_____.

(a) Independent data mart
(b) Dependent data marts
(c) Intra-entry data mart
(d) Inter-entry data mart

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : Dependent Data Mart is created by extracting the data from central repository. Data warehouse, First data warehouse is created by extracting data from external sources and then data mart is created from data warehouse. Dependent data mart is created in top-down approach of data warehouse architecture. This model of data mart is used by big organizations.

251. The most common source of change data in refreshing a data warehouse is

(a) Query able change data
(b) Cooperative change data
(c) Logged change data
(d) Snapshot change data

RPSC C.S. 2016 (PAPER-II)

Ans. (d) : Snapshot change data involves taking periodic full or partial snapshots of source systems and comparing them to identify what has changed since the last snapshot. This is the most widely used approach in practice.

252. The term that is not associate with data cleaning process is_____.

(a) Domain consistence (b) De-duplication
(c) Disambiguation (d) Segmentation

RPSC C.S. 2016 (PAPER-II)

Ans. (d) : Data cleaning focuses on fixing errors removing duplicates, and clarifying ambiguous data. Segmentation, on the other hand, is a process of dividing a dataset into distinct groups based on specific characteristics, which is typically used for analysis not data cleaning.

253. Eclat (Equivalence Class Transformation) algorithm Mines using

(a) Horizontal data format
(b) Vertical data format
(c) Mixed data format
(d) Class Data format

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : The Eclat (Equivalence Class Transformation) algorithm mines frequent item sets using the vertical data format.

Vertical Data Format each item is associated with a list of transaction IDs where it appears, for example, instead of a row for each transaction showing what items were bought, you have a row for each item showing which transactions it was part of.

254. OLE, A Microsoft's component document technology, means :
ओ एल ई, माइक्रोसॉफ्ट की कम्पोनेंट डाक्यूमेंट टेक्नोलॉजी, का अभिप्राय है:

(a) Overlay Linking and Exchange
ओवरले लिंकिंग एण्ड एक्सचेंज
(b) Online Linking and Embedding
ऑनलाइन लिंकिंग एण्ड एम्बेडिंग
(c) Open Learning Exchange
ओपन लर्निंग एक्सचेंज

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(d) Object Linking and Embedding
ऑब्जेक्ट लिंकिंग एण्ड एम्बेडिंग

KVS PGT C.S. 2017

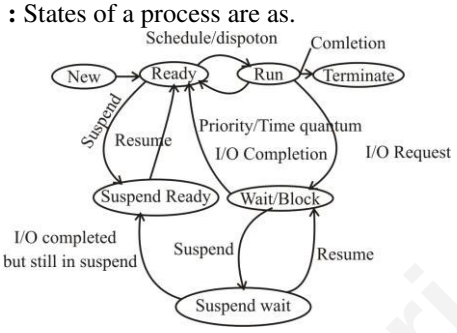
Ans. (d) : Object Linking & Embedding (OLE) is a proprietary technology developed by Microsoft that allows embedding and linking to documents and other objects. OLE allows an editing application to export part of a document to another editing application and then import it with additional content.

255. If a process is runnable but is temporarily stopped to let another process run, in which state is the process said to be ?
यदि प्रक्रिया अनुगमनीय है परंतु अस्थायी रूप से अन्य प्राक्रिया के अनुगमन के लिए रूकी हुई है, यह प्रक्रिया किस अवस्था में कही जाएगी ?

(a) Running/अनुगमित (b) Ready/तैयार
(c) Interrupted/बाधित (d) Blocked/अवरूद्ध

KVS PGT C.S. 2013

Ans. (d) : States of a process are as.



Process state Diagram

1. New 2. Ready
2. Run 4. Blocked or wait
5. Terminated or Completed
6. Suspend Ready state
7. Suspend wait state.

Blocked or wait – Whenever the process requests access to I|O or needs input from the user or needs access to a critical region it enters the blocked or wait state. The process continues to wait in the main memory and does not require CPU.

256. When a key is pressed on keyboard, which standard is used for converting the keystroke into the corresponding bits?
जब कीबोर्ड पर एक कुंजी दबाई जाती है, तो कीस्ट्रोक को संबंधित बिट्स में परिवर्तित करने के लिए किस मानक का उपयोग किया जाता है?

(a) ANSI
(b) ASCII
(c) EBCDIC
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans.(a): ANSI (American National Standards Institute) has provided standards for converting keystrokes into corresponding bits.

257. A Web application and its support environment has not been fully fortified against attack. Web engineers estimate that the likelihood of repelling an attack is only 30 percent. The application does not contain sensitive or controversial information, so the threat probability is 25 percent. What is the integrity of the web application?

(a) 0.625 (b) 0.725
(c) 0.775 (d) 0.825

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (d) : Integrity of web application :
Integrity of web application can be found out using the security and threat in that.
Formula :
Integrity = [1 – threat (1 – security)]
Calculation -
It is given that, probability or Likelihood of repelling an attack = 30%
i.e probability of security = 0.30
Threat probability is given as 25% or 0.25
So integrity = [1 – 0.25 (1 – 0.30)]
integrity = [1 – 0.25 (0.70)]
Integrity = 1 – 0.175 = 0.825

258. The data node and name node in HADOOP are

(a) Worker Node and Master Node respectively
(b) Master Node and Worker Node respectively
(c) Both Worker Nodes
(d) Both Master Nodes

UGC NET C.S. June-2020 (Paper-II)

Ans. (a) : The main difference between Namenode and Datanode in HADOOP is that the Namenode is master node in HADDOP distributed file system (HDFS) that manages the file system meta data distributed file system that stores the actual data as instructed by the Namenode.

259. Match the following with reference to Functional programming history:

List-I		List-II	
A.	Lambda calculus	**1.**	Church, 1932
B.	Lambda calculus as programming language	**2.**	Wordsworth, 1970
C.	Lazy evaluation	**3.**	Haskel, 1990
D.	Type classes	**4.**	Mecarthy, 1960

Codes:

	A	**B**	**C**	**D**
(a)	4	1	3	2
(b)	1	4	2	3
(c)	3	2	4	1
(d)	2	1	4	3

UGC NET C.S. August-2016 (Paper-III)

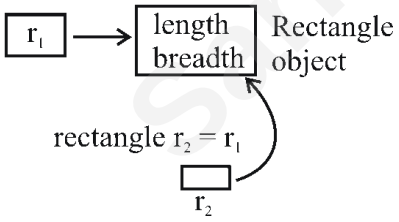
Sarkari Resource

Ans : (b) (i) Lambda calculus: is a formal system in mathematical logic for expressing computation based on function abstraction and application using variable binding and substitution. The lambda calculus was introduced by mathematician Alonzo church in the 1930's as part of an investigation into the "foundation of mathematics (1932)".
Until the 1960's when its relation to programming languages was clarified, the λ-calculus was only a formalism. To LISP by MeCarthy.
(ii) Lazy evaluation: Was introduced in 1970 by wordsworth.
(iii) Type classes: Was introduced in 1990 by Haskel.

260. When one object reference variable is assigned to another object reference variable then
(a) A copy of the object is created
(b) A copy of the reference is created
(c) A copy of the reference is not created
(d) It is illegal to assign one object reference variable to another object reference variable

UGC NET C.S. June-2016 (Paper-III)

Ans : (b)
(1) We can assign value of reference variable to another reference variable.
(2) Reference variable is used to store the address of the variable
(3) Assigning Reference will not create distinct copies of object
(4) All reference variables are referring to some object
(5) Assigning object reference variable does not create distinct object
rectangle r1 = now rectangle ( ) :



261. The region of feasible solution of a linear programming problem has a _____ property in geometry, provided the feasible solution of the problem exists.
(a) Concavity (b) Convexity
(c) Quadratic (d) Polyhedron

UGC NET C.S. June-2016 (Paper-III)

Ans : (b) Geometric interpretation : The set of point in IRn at which any single constraint holds with equality is a hyper plane in IRn. Thus each constraint is satisfied by the points of a closed half space of IRn, and the set of feasible solution is the intersection of all these half spaces. Convex polyhedron p.
So answer is (b)

262. Consider the following statements :
(1) Revised simplex method requires lesser computations than the Simplex method.
(2) Revised simplex method automatically generates the inverse of the current basis matrix.
(3) Less number of entries are needed in each table of revised simplex method than usual simplex method.
Which of these statements are correct?
(a) (1) and (2) only (b) (1) and (3) only
(c) (2) and (3) only (d) (1), (2) and (3)

UGC NET C.S. June-2016 (Paper-III)

Ans : (d) Revised simplex method is an improvement over simplex method. Revised simplex method is computationally more efficient and accurate. Benefit of revised simplex method is comprehended in case of large LP problems. In simplex method the entire simplex tableau is updated while a small part of it is used. The revised simplex method uses exactly the same steps as those in simplex method.
So answer is option (d)

263. What is the full form of GSM is mobile computing?
मोबाइल कंप्यूटिंग में GSM क्रा असंक्षिप्त रूप क्या है?
(a) Global Systems for mobile communication
ग्लोबल सिस्टम फॉर मोबाइल कम्यूनिकेशन
(b) Global Scientific mobile computing
ग्लोबल साइंटिफिक मोबाइल कम्पयूटिंग
(c) Global Services for mobile computing
ग्लोबल सर्विस फॉर मोबाइल कम्पयूटिंग
(d) Global Signalling for mobile computing
ग्लोबल सिगंनलिगं फॉर मोबाइल कम्पयूटिंग

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : GSM stands for Global System for Mobile Communications. It is a family of standards to describe the protocols for second generation digital cellular networks, as used by mobile devices such as mobile phones and mobile broadband modems.

264. _____is used to reduce the size of multimedia files.
_____ का उपयोग मल्टीमीडिया फाइलों के आकार को कम करने के लिया जाता है।
(a) Data compression/डेटा कंप्रेशन
(b) Data coupling/डेटा कपलिंग
(c) Data handing/डेटा हैंडलिंग
(d) Data dependency/डेटा डिपेंडेंसी

DSSSB TGT C.S. 07.08.2021 (Shift-I)

Ans. (a) : Data compression is the process of reducing the size of a file by eliminating redundant or unnecessary information. This is done by using algorithms to represent the data more efficiently thereby taking up less space.

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265. Given the following process with their times and priority (lower number indicates higher priority)

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

Which of the following is a value of average waiting time using priority scheduling algorithm?

(a) 9.2 (b) 6.8
(c) 7.6 (d) 8.2

UGC NET C.S. December-2024

Ans. (d) : Priority scheduling (Non-primitive) we will schedule the processes based on their priority, lower number indicates higher priority.

(1) P2 (Priority 1): Starts immediately (waiting time = 0): runs for 1 unit
(2) P5 (Priority 2) : Waits for P2 to finish (1 unit), runs for 5 units.
(3) P1 (Priority 3): Waits for P2 and P5 to finish (its = 6 units); runs for 10 units
(4) P3 (Priority 4) : waits for P_2, P5 and P1 to finish (1 + 5 + 10 = 16 units), runs for 2 units
(5) P4(Priority 5): Waits for P2, P5 P1 and P3 to finish (1 + 5 + 10 + 2 = 18 units), runs for 1 units

Waiting times **Average waiting time:**

- P2 = 0 (0 + 1 + 6 + 16 + 18)/5
- P5 = 1 = 41/5 = 8.2
- P1 = 6
- P3 = 16
- P4 = 18

266. Regarding the code optimization, choose the correct sequence

A. Algebraic Simplification
B. Use of machine idioms
C. Redundant – instruction elimination
D. Flow of control optimization
E. Improved target code

Choose the correct answer from the options below :

(a) A, C, B, D, E (b) B, C, D, A, E
(c) C, D, A, B, E (d) D, B, A, C, E

UGC NET C.S. December-2024

Ans. (a) :

(A) Algebraic simplification– Simplifying expression to reduce complexity and improve efficiency.
(C) Redundant instruction elimination– Removing unnecessary instructions that do not contribute to the final output.
(B) Use of machine idioms– Replacing high-level contracts with optimized machine level instructions or operations specific to the target architecture.
(D) Flow of control optimization– Optimizing the flow of control to enhance performance branching overhead.
(E) Improved target code– Generating the most efficient code for the target machine or environment.

267. Which of the following is TRUE about the Pumping Lemma for regular language?

(a) It applies to all regular language
(b) It applies only to infinite regular languages
(c) It applies to all context – free languages
(d) It applies to all recursively enumerable language

UGC NET C.S. December-2024

Ans. (a) : The Pumping Lemma for regular languages states that every regular language has a property where sufficiently long strings can be pumped (repeated a certain part) to generate new strings that are also in the language. This applies to all regular languages including both finite and infinite ones.

268. Match the List-I with List-II

	List-I		**List-II**
A.	**Service Level Agreement**	**I.**	**The technology that connects and enables communication between physical devices over the Internet.**
B.	**Virtualization**	**II.**	**A develop environment provides over the internet to create and deploy applications.**
C.	**Internet of Things**	**III.**	**Contracts that specify performance standards and services availability.**
D.	**Platform as a service**	**IV.**	**A method allowing multiple operating system to run on a single physical server.**

Choose the correct answer from the options given below:

(a) A-III, B-II, C-I, D-IV
(b) A-III, B-IV, C-I, D-II
(c) A-III, B-II, C-IV, D-I
(d) A-I, B-IV, C-III, D-II

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Ans. (b) : The correct match is-

	List-I		**List-II**
A.	Service Level Agreement	–	Contracts that specify performance standards and services availability.
B.	Virtualization	–	A method allowing multiple operating system to run on a single physical server.

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C.	Internet of Things	–	The technology that connects and enables communication between physical devices over the Internet.
D.	Platform as a service	–	A develop environment provides over the internet to create and deploy applications.

269. Which of the following table contains the primary information in the data warehouse?
(a) Dimension table (b) Fact table
(c) Lookup table (d) Primary table

UGC NET C.S. December-2024

Ans. (b) : The table that contains the primary information in a data warehouse is the fact table-
A fact table is the central table in a data warehouse, storing the quantitative measurement and metrics related to a business process, making it the primary source of information for analysis.

270. The ability to input data directly into a computer system by speaking to it is knows as _____.
कंप्यूटर सिस्टम में सीधे बोलकर डेटा इनपुट करने की क्षमता को_____के रूप में जाना ज़ात है।
(a) Speech recognition/स्पीच रेकग्निशन
(b) Speaker recognition/स्पीकर रेकग्निशन
(c) Speaker identification/स्पीकर आइडेंटिफिकेशन
(d) Speech enhancement/स्पीच एनहांसमेंट

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Speech recognition is the technology that allows a computer system to convert spoken words into text or commands, enabling direct data input by speaking.

271. _____is a controlled electronic switch fabricated using a semiconductor.
_____एक सेमीकंडक्टर का उपयोग करके निर्मित एक नियंत्रित इलेक्ट्रॉनिक स्विच है।
(a) Transistor/ट्रांजिस्टर (b) Hard-disk/हार्ड डिस्क
(c) Monitor/मॉनिटर (d) Coupler/कप्लर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : A transistor is a controlled electronic switch mode from semiconductor materials. It can amplify signals or act as an electronic switch in circuits.

272. _____is a system software package that converts a high language program to machine language.
_____एक सिस्टम सॉफ्टवेयर पैकेज है जो एक हाइ लेवल भाषा को मशीनी भाषा में परिवर्तित करता है।
(a) Compiler/कम्पाइलर (b) Emulator/एमुलेटर
(c) Editor/एडिटर (d) Debugger/डीबअगर

DSSSB TGT C.S. 01.08.2021 (Shift-II)

Ans. (a) : Compiler is a system software package that converts a high level language program to machine language. It translates the entire source code into machine code before execution.
Emulator– An emulator is a software or hardware that mimics another system's environment. It allows programs made for one type of computer or device to run an a different one.
Editor– An editor is a tool used to write and modify source code.
Debugger– A debugger is a program that helps in finding and fixing errors in your code. It allows you to run your program step by step, inspect variables, and understand where and why a program is not working as expected.

273. A compiler is a translating program which:
कंपाइलर एक अनुवाद करने वाला प्रोग्राम है, जो–
(a) Translates instruction of a high-level language into machine language
(b) Translates entire source program into machine language program
(c) It is not involved in program's execution
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d) : A compiler is a program that translates a program from high-level programming language to low-level machine language. It compiles the source code written by the user so that it can be understood by the computer.

274. Match the description of several parts of a classic optimizing compiler in List-I, with the names of those parts in List - II :

	List-I		**List-II**
A.	**A part of a compiler that is responsible for recognizing syntax**	**i.**	**Optimizer**
B.	**A part of a complier that takes as input a stream of characters and produces as output a stream of words along with their associated syntactic categories**	**ii.**	**Semantic Analysis**
C.	**A part of a compiler that understand the meanings of variable names and other symbols and checks that they are used in ways consistent with their definitions.**	**iii.**	**Parser**
D.	**An IR-to-IR transformer that tries to improve the IR program in some way (Intermediate Representation)**	**iv.**	**Scanner**

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Code :

	A	B	C	D
(a)	iii	iv	ii	i
(b)	iv	iii	ii	i
(c)	ii	iv	i	iii
(d)	ii	iv	iii	i

UGC NET C.S. November-2017 (Paper-II)

Ans : (a)

(i) **Parser** which is also called **syntax analyzer** who is responsible for recognizing syntax.

(ii) **Scanner** which is also called **lexical analyzer** who is responsible for tokenizing input stream of characters and produce tokens.

(iii) **Semantic analyzer** is part of compiler that understand the meaning of variable and check that they are used in same way consistent with their definitions.

(iv) **Optimizer** is responsible for **IR-IR transformation** that tries to improve the IR program in same way.

275. A compiler is a translating program which:
कंपाइलर एक अनुवाद करने वाला प्रोग्राम है, जो–

(a) Translates instruction of a high-level language into machine language

(b) Translates entire source program into machine language program

(c) It is not involved in program's execution

(d) More than one of the above
उपर्युक्त में से एक से अधिक

(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (d) : A compiler is a program that translates a program from high-level programming language to low-level machine language. It compiles the source code written by the user so that it can be understood by the computer.

276. What is the name of the virtual machine component of Microsoft .NET Framework which manages the execution of .NET program?
माइक्रोसॉफ्ट .NET फ्रेमवर्क के आभासी मशीन घटक, जोकि .NET प्रोग्राम के निष्पादन का प्रबंधन करता है, का नाम क्या है

(a) Code level security /कोड लेवल सिक्योरिटी

(b) Cross language class inheritance
क्रोस लैंग्वेज क्लास इन्हैरिटैंस

(c) Common language runtime
कॉमन लैंग्वेज रनटाइम

(d) Common type system/कोमन टाइप सिस्टम

DSSSB TGT C.S. 01.08.2021 (Shift-I)

Ans. (c) : The Common Language Runtime is the virtual machine component of the Microsoft .NET framework. It manages the execution of .NET programs by providing services like.

1. Memory Management
2. Garbage Collection
3. Exception Handling
4. Security
5. Just-In-Time (JIT) Compilation

277. _____ is an imitation of the dynamics of a real world process or system over time.
_____ समय के साथ वास्तविक दुनिया की प्रक्रिय या प्रणाली की गतिशीलता की नकल है।

(a) Designing/डिजाइनिंग

(b) Cryptography/क्रिप्टोग्राफी

(c) Simulation/सिमुलेशन

(d) Copying/कॉपिंग

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Simulation is the imitation of the dynamics of a real-world process or system over time often used for analysis, training, and prediction.

278. _____act as a interface between a domain expert and an expert system.
_____एक डोमेन विशेषज्ञ और एक विशेषज्ञ प्रणाली के बीच एक इंटरफेस के रूप में कार्य करता है।

(a) Language Editor/लैंग्वेज एडिटर

(b) K editor/ K एडिटर

(c) Intelligent editor/इंटेलीजेंट एडिटर

(d) Knowledge Editor/नॉलेज एडिटर

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Intelligent Editor– act as a interface between a domain expert and an expert system. An intelligent editor is a software tool or feature that uses artificial intelligence (AI) and machine learning (ML) to assist with editing and improving written content.

279. Match the following with respect to Event Driven Programming.
इवेंट ड्रिवेन प्रोग्रामिंग के संबंध में निम्नलिखित का मिलान करें।

I	**Event** **इवेंट**	**A**	**Perform said event** **उक्त इवेंट का अंजाम दें**
II	**Event listener** **इवेंट लिस्टनेर**	**B**	**User interaction** **यूजर इंटरेक्शन**
III	**Event handler** **इवेंट हैंडलर**	**C**	**Event monitored by code/कोड द्वारा निगरानी की गई इवेंट**

(a) I-A, II-B, III-C (b) I-C, II-A, III-B

(c) I-B, II-A, III-C (d) I-B, II-C, III-A

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : Correct matching with respect to Event Driven Programming.

I	Event	**B**	User interaction
II	Event listener	**A**	Perform said event
III	Event handler	**C**	Event monitored by code

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280. What is responsibility of event queue in event driven programming?
इवेंट ड्रिवेन प्रोग्रामिंग में इंवेट क्यू की क्या जिम्मेदारी होती है?

(a) Create new event/नया ईवेंट बनाएं
(b) Delete the event/इवेंट को हटायें
(c) Make a sequence in which event is executed एक क्रम बनाएं जिसमें ईवेंट एक्सेक्यूटे हो
(d) Change the event/इवेंट को बदले

DSSSB PGT C.S. M/F 11.07.2021

Ans. (c) : In event-driven programming, the event queue is responsible for managing and maintaining the order of events that need to be processed. It ensures that events are executed in the sequence they are received or based on priority.

281. _____files are spreadsheet files in a text format of data when cell contents are separated by a character for example a comma or semi-colon.

(a) Comma-separated-values
(b) Semi-colon Separated Values
(c) Column and Row Headers
(d) CVS

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (a) : Spreadsheet files that store data in a text format where cell contents are separated by a specific character are commonly known as Comma-separated-values (CSV) files. In a CSV, each line represents a row in the spreadsheet and the values within that row are separated by a comma. For example– Name, age, city.

282. Which of the following suffices to convert an arbitrary CFG to an LL(1) grammar?

(a) Removing left recursion alone
(b) Removing the grammar alone
(c) Removing left recursion and factoring the grammar
(d) None of the above

UGC NET C.S. June-2014 (Paper-II)

Ans : (d) Removing left recursion and factoring from the grammar do not suffice to convert an arbitrary CFG to LL(1) grammar.
Since, LL(1) parser is top down parser. For top-down parsers, the grammar should be unambiguous, deterministic and from left recursion.
All the conditions must be satisfied.

283. Match the following :

	List–I		**List–II**
A.	Chomsky Normal form	i.	$S \rightarrow b\ S\ S \mid a\ S \mid c$
B.	Greibach Normal form	ii.	$S \rightarrow a\ S\ b \mid ab$
C.	S-grammar	iii.	$S \rightarrow AS \mid a$ $A \rightarrow SA \mid b$
D.	LL grammar	iv.	$S \rightarrow a\ B\ S\ B$ $B \rightarrow b$

Codes :

	A	**B**	**C**	**D**
(a)	iv	iii	i	ii
(b)	iv	iii	ii	i
(c)	iii	iv	i	ii
(d)	iii	iv	ii	i

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) :
(a) Chomsky Normal Form : Its production rules should be of the form
$A \rightarrow BC$, or
$A \rightarrow a$
$S \rightarrow \varepsilon$
(b) Greiback Normal Form : Its production rules should be of form
$A \rightarrow aA_1A_2 — An$ or
$A \rightarrow a$
(c) S-Grammar : If every production's right hand side begins with a terminal signal and this terminal is different for any two productions with the left hand side
(d) LL Grammar : It two production rules can begin with same terminal symbols if they have same select sets.

284. Which one of the following describes the syntax of prolog program?

I. Rules and facts are terminated by full stop(.)
II. Rules and facts are terminated by semicolon (;)
III. Variables names must start with upper case alphabets
IV. Variables names must start with lower case alphabets.

Codes :
(a) I, II (b) III, IV
(c) I, III (d) II, IV

UGC NET C.S. June-2014 (Paper-III)

Ans. (c) : Prolog variable begin with an upper case letter, predicate names, function names and the names for objects must begin with a lowercase letter and rules and facts are terminated by full stop (·) So the right answer is option (c).

285. Which of the following is false for the programming language PROLOG?

(a) A PROLOG variable can only be assigned to a value once
(b) PROLOG is a strongly typed language
(c) The scope of a variable in PROLOG is a single clause or rule
(d) The scope of a variable in PROLOG is a single query

UGC NET C.S. June-2015 (Paper-III)

Ans : (b) Since prolog is a dynamically typed language. Which is to say that programs are not checked for type errors until run time.

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286. Criticism free idea generation is a factor of _____.

(a) Decision Support System
(b) Group Decision Support System
(c) Enterprise Resource Support System
(d) Artificial Intelligence

UGC NET C.S. August-2016 (Paper-III)

Ans : (b) Criticism free idea generation is a factor of group decision support system.

287. Which of the following is a source coding technique?

(a) Huffman coding (b) Arithmetic coding
(c) Run-length coding (d) DPCM

UGC NET C.S. August-2016 (Paper-III)

Ans : (d) DPCM is differential pulse code modulation which convert source analog into digital signal (i.e. source coding).

288. Which of the following Super Computers in the fastest Super Computer?

(a) Sun-way Taihu Light (b) Titan
(c) Piz Daint (d) Sequoria

UGC NET C.S. November-2017 (Paper-II)

Ans : (a) Sun-way Taihu Light has been named as the world's fastest supercomputer title, as per recently released semiannual top 500 list of super computers. It is developed by chain's National research centre of parallel computer engineering & technology (NRCPL).

289. Which of the following statements about ERP system is true?

(a) Most ERP software implementations fully achieve seamless integration.
(b) ERP software package are themselves combinations of separate applications for manufacturing, materials, resource planning, general ledger, human resources, procurement and order entry
(c) Integration of ERP systems can be achieved in only one way
(d) An ERP package implemented uniformly throughout an enterprise is likely to contain very flexible connections to allow charges and software variations.

UGC NET C.S. November-2017 (Paper-II)

Ans : (b) Some ERP software packages are themselves combinations for manufacturing, material resource planning, general ledger, human resources procurement & order entry.

290. Which of the following is not a clustering method?

(a) K–Mean method
(b) Self Organizing feature map method
(c) K – nearest neighbor method
(d) Agglomerative method

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) In pattern recognition, the k-nearest neighbors algorithm (K-NN) is a non-parametric method used for classification and regression.

291. Which of the following regular expressions, each describing a language of binary numbers (MsB to LSB) that represents non-negative decimal values, does not include even values?

(a) 0*1+0*1*
(b) 0*10+1*
(c) 0*1*0*1+
(d) 0*1*0*1*

Where {+, *} are quantification characters.

UGC NET C.S. November-2017 (Paper-II)

Ans : (c) When a number is represented in binary form, for add numbers, we have a 1 in the LSB while a 0 in LSB for even numbers
So, only the language generated by option (c) 0*1*0*1+ will, for sure, here 1 in the LSB, So this language won't contain even number.

292. What is the role of Back Propagation Algorithm?

(a) To reduce error
(b) To secure network
(c) To control speed of data
(d) To add different layers

UGC NET C.S. Re-Exam- June 2024

Ans. (a) : The Back Propagation Algorithm is primarily used in training neural networks. Its role is to minimize the error by adjusting the weights of the network through gradient decent, ensuring the output becomes closer to the desired result.

293. Match List-I with List-II.

List-I		List-II	
A.	**The activation function**	**I.**	**Is called the delta rule**
B.	**The learning method of perceptron**	**II.**	**Is one of the key components of the penseptron as in the most common neural network architecture**
C.	**Areas of application of artificial neural network include**	**III.**	**Is always boolean like a switch.**
D.	**The output of the perceptron**	**IV.**	**System identification and control**

Choose the correct answer from the options given below :

(a) A-II, B-IV, C-III, D-I
(b) A-IV, B-III, C-II, D-I
(c) A-II, B-I, C-IV, D-III
(d) A-III, B-IV, C-II, D-I

UGC NET C.S. Re-Exam- June 2024

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Ans. (c) : The correct matching is–

List-I		List-II	
A.	The activation function	**ii**	Is one of the key components of the penseptron as in the most common neural network architecture
B.	The learning method of perceptron	**i**	Is called the delta rule
C.	Areas of application of artificial neural network include	**iv**	System identification and control
D.	The output of the perceptron	**iii**	Is always Boolean like a switch.

294. Match List-I with List-II.

List-I		List-II	
A	**National language processing**	**i**	**A method of training algorithm by rewarding desired behavior and/or punishing undesired one.**
B	**Reinforcement learning**	**ii**	**System designed to emulate the decision making abilities of a human expert.**
C	**Support vector machine**	**iii**	**A branch of AI focused on understanding and generating human language.**
D	**Expert system**	**iv**	**A machine learning technique that finds the hyper plane that best separates different class in a feature space.**

Choose the correct answer from the options given below :

(a) A-i, B-ii, C-iv, D-iii
(b) A-iii, B-ii, C-i, D-iv
(c) A-iii, B-i, C-iv, D-ii
(d) A-ii, B-iv, C-iii, D-i

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Ans. (c) : Correct match-

List-I		List-II	
A.	National language processing	**iii**	A branch of AI focused onunderstanding and generating human language.
B.	Reinforcement learning	**i**	A method of training algorithm by rewarding desired behavior and/or punishing undesired one.
C.	Support vector machine	**iv**	A machine learning technique that finds the hyper plane that best separates different class in a feature space.
D.	Expert system	**ii**	System designed to emulate the decision making abilities of a human expert.

295. Arrange the following phases of a compiler as per their order of execution (start to end)

(A) Target code generation
(B) Syntax Analysis
(C) Code optimization
(D) Semantic Analysis
(E) Lexical Analysis

Choose the correct answer from the options given below

(a) (B), (E), (D), (A), (C)
(b) (E), (D), (B), (A), (C)
(c) (E), (B), (D), (C), (A)
(d) (B), (D), (E), (A), (C)

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Ans. (c) : Compiler operates in various phases each phase transforms the source program from one representation to another. Every phase takes inputs from its previous stage and feeds its output to the next phase of the compiler.
The phases of a compiler are-
1. Lexical analysis
2. Syntax analysis
3. Semantic analysis
4. Intermediate code generator
5. Code optimizer
6. Target code generation

296. Which of the following are commonly used parsing techniques in NLP (Natural Language Processing) for syntactic analysis.

(A) Top down parsing
(B) Bottom Up parsing
(C) Dependency parsing
(D) Statistical machine translation
(E) Early parsing

Choose the correct answer from the options given below :

(a) (A), (C), (D), (E) Only
(b) (B), (C), (D), (E) Only
(c) (A), (B), (C), (E) Only
(d) (A) and (B) Only

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Ans. (c) : The commonly used parsing techniques in NLP. for syntactic analysis are-
(1) Top-down parsing
(2) Bottom-up parsing
(3) Dependency parsing
(5) Early parsing
(4) Statistical machine translation is not a parsing technique. It is a method used for translation form on language to another statistical models.

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297. One of the purposes of using intermediate code in compilers is to:

(a) Make parsing and semantic analysis simpler
(b) Improve error recovery and error reporting
(c) Increase the chances of reusing the machine independent code optimizer in other compilers
(d) Improve the register allocation

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Ans. (c) : After semantic Analysis, the code is converted into intermediate code which is platform (os + hardware) independent, the advantage of converting into intermediate code is to improve the performance of code generation and to increase the chances of reusing the machine- independent code optimizer in other compilers. So, option (c) is correct.

298. The compiler for high level language that runs on one machine and produces code for other machine is called-

(a) Cross compiler
(b) Multipass compiler
(c) Optimizing Compiler
(d) One pass Compiler

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Ans. (a) : A cross compiler is a compiler capable of creating executable code for a platform other than the one on which the complier is running, for example, a compiler that runs on a PC but generates code that runs or Android devices is a cross compiler.

299. Which of the following parser is most powerful parser?

(a) Operator precedence
(b) SLR
(c) Canonical LR
(d) LALR

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Ans. (c) : The 'canonical LR' parser is more powerful parser because it utilizes the full LR (1) parsing technique, which means it considers one look ahead symbol when making parsing decisions. This additional look ahead allows it to handle more complex grammars and resolve parsing conflicts more effectively than parsers with limited lookahead, such as SLR and LALR parsers. canonical LR parsers construct a coronial collection of LR (1) sets, providing a more comprehensive analysis of the grammar's structure.

300. Match List-I with List-II

LIST-I		**LIST-II**	
A.	**Expert system**	**I.**	**Decision tree**
B.	**Fuzzy system**	**II.**	**Scramble**
C.	**Operator in genetic algorithm**	**III.**	**Inference engine**
D.	**Supervised technique**	**IV.**	**Mycin**

Choose the correct from the option given below.

(a) A-IV B-I C-III D-II
(b) A-III B-IV C-II D-I
(c) A-IV B-III C-II D-I
(d) A-I B-II C-III D-IV

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Ans. (c) : Correct match-

LIST I		**LIST II**	
A.	Expert system	IV.	Mycin
B.	Fuzzy system	III.	Inference engine
C.	Operator in genetic algorithm	II.	Scramble
D.	Supervised technique	I.	Decision tree

301. The maximum yield length of the following CNF, CFG is.

S→AB
A→CD
B→e
C→a
D→b

(a) 8 (b) 7
(c) 4 (d) 5

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Ans. (c) : To find the maximum yield length of a CNF (Chomsky Normal Form) CFG (Context Free Grammar), we need to look at the rule and find the longest path from the start symbol to a terminal symbol.

$S \rightarrow AB$

$S \rightarrow CDe$ $\quad \because A \rightarrow CD$, $B \rightarrow e$

$S \rightarrow abcd$ $\quad \because C \rightarrow a$, $D \rightarrow b$

We get the longest string-

$S \rightarrow abcd$

Therefore, the maximum yield length is 4.

302. Let (Z, +) denote the group of all integers under addition. Then the number of all automorphisms of (Z, +) is

(a) 1 (b) 2
(c) 3 (d) 4

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Ans. (b) : The number of automorphisms of the group of integers under addition is 2. One is identity automorphism and other is the negation automorphism.

303. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
The most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using _____

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(a) Divide and Conquer
(b) Dynamic Programming
(c) Greedy Algorithm
(d) Backtracking

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Ans. (c) : A knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
The most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using Greedy Algorithm.
So option (c) is correct.

304. FFTs are techniques used to encode digital video and audio information and helps analyse the signal in _____ domain.

(a) Encryption, time
(b) Encryption, frequency
(c) Compression, time
(d) Compression, frequency

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Ans. (d) : Fourier Transform is a mathematical concept that can covert a continuous signal from time - domain to frequency - domain. FFTs are compression techniques used to analyze digital video and audio information and help analyze the signal in the frequency domain. So the correct answer is compression, frequency.

305. Match List I with List II :

List I		List II	
A.	**Type 0**	**i.**	**Finite automata**
B.	**Type 1**	**ii.**	**Tuning machine**
C.	**Type 2**	**iii.**	**Linear bound automata**
D.	**Type 3**	**iv.**	**Pushdown automata**

Choose the correct answer from the options given below :

(a) A-iii, B-iv, C-ii, D-i
(b) A-ii, B-iii, C-iv, D-i
(c) A-iii, B-iv, C-i, D-ii
(d) A-ii, B-iii, C-i, D-iv

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Ans. (b) : The correct match-

List I		List II	
A.	Type 0	**ii.**	Tuning machine
B.	Type 1	**iii.**	Linear bound automata
C.	Type 2	**iv.**	Pushdown automata
D.	Type 3	**i.**	Finite automata

306. Which mode as a block cipher implementation as a self synchronizing stream cipher?

(a) Cipher Block Chaining Mode
(b) Cipher Feedback Mode
(c) Electronic Codebook Mode
(d) Output Feedback Mode

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Ans. (b) : Cipher Block Chaining Mode (CBC) is a mode of operation for a block Cipher one in which a sequence of bits are encrypted as a single unit, or block, with a Cipher key applied to the entire block. Cipher block chaining uses what is known as an initialization vector (IV) of a certain length.

Cipher Feedback Mode

In this mode the Cipher is given as feedback to the next block of encryption with some new specifications. First, an Initial Vector (IV) is used for first encryption and output bits are divided as a set of 5 and b-s bits. The left hand side 5 bits are selected along with plaintext bit to which an XOR operation is applied.

The result is given as input to a shift register having b-s bits to rhs, s bits to rhs and the process continues. The encryption and decryption process for the same is show below, both of them encryption algorithms.

Electronic code book mode– Electronic code book is the easiest block Cipher mode of functioning. It is easier because of direct encryption of each block of input plaintext and output is in form of blocks of encrypted Cipher text. Generally, if a massage is larger than b bits in size, it can be broken down into a branch of block and the procedure is repeated.

Output Feedback Mode– The output feedback it involves feeding the successive output block from the underlying block Cipher back to it.

So option (b) is correct.

307. In reference to Big data, consider the following database:

(A) Memcached
(B) Couch DB
(C) Infinite graph

Choose the most appropriate answer from the options given below.

(a) (A) and (B) only
(b) (A) and (C) only
(c) (A) and (B) only
(d) (A) (B) and (C) only

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Ans. (d) : Big data – The definition of big data is data that contains greater variety, arriving in increasing volume and with more velocity.

This is also known as the three vs.

The three V_3 means volume, velocity, variety.

Memcached is a free and open source high-performance memory caching system. It's typically used to cache data base data, API call or page rendering chunks in RAM to increase the application performance.

It can store data as little as a number or as big, as a finished HTML page.

Couch DB is an open source database developed by Apache software foundation. The focus is on the ease of use embracing the web. It is a No SQL document store database.

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The prominent big data analytics tools that use non-relation databases are Mongo DB Cassandra, oracle No-SQL and Apache Couch DB.
Couch DB is a NoSQL database for document storage. Documents are the fundamental unit of data in Couch DB, and they also contain metadata.
Infinite graph is a distributed graph database implementated in Java and C^{++} and is from a class of NOSQL database technologies that focus on graph data structures. Developers use infinite graph to find useful and often hidden relationship in highly connected, complex big data set.
So option (a), (b) and (c) is correct answer.

308. A bottom-up parser generates :
(a) Left-most derivation in reverse
(b) Right-most derivation in reverse
(c) Left-most derivation
(d) Right-most derivation

UGC NET C.S. June-2018 (Paper-II)

Ans. (b) : A bottom-up parser generates is right-most derivation in reverse.

309. Which of the following statements is false about convex minimization problem?
(a) If a local minimum exists, then it is a global minimum
(b) The set of all global minima is convex set
(c) The set of all global minima is concave set
(d) For each strictly convex function, if the function has a minimum, then the minimum is unique

UGC NET C.S. June-2018 (Paper-II)

Ans. (c) : The following statements are true about the convex minimization problem:
- If a local minimum exists, then it is a global minimum.
- The set of all (global) minima is convex.
- For each strictly convex function, if the function has a minimum, then the minimum is unique.

310. Data scrubbing is
(a) A process to upgrade the quality of data after it is moved into a data warehouse.
(b) A process to load the data in the data warehouse and to create the necessary indexes.
(c) A process to upgrade the quality of data before it is moved into a data warehouse.
(d) A process to reject data from the data warehouse and to create the necessary indexes.

UGC NET C.S. December-2018 (Paper-II)

Ans. (c) : Data scrubbing, also called data cleansing, is the process of amending or removing data in a database that is incorrect, incomplete, improperly formatted, or duplicated before it is moved into a data warehouse.

311. Which of the following statement/s is/are true?
(i) Facebook has the world's largest Hadoop Cluster.
(ii) Hadoop 2.0 allows live stream processing of Real time data.
Choose the correct answer from the code given below :
Code :
(a) (ii) only (b) Neither (i) nor (ii)
(c) (i) only (d) Both (i) and (ii)

UGC NET C.S. December-2018 (Paper-II)

Ans. (d) : The Data warehouse Hadoop cluster at Facebook has become the largest known Hadoop storage cluster in the world. Here are some of the details about this single HDFS cluster:
21 PB of storage in a single HDFS cluster 2000 machines
12 TB per machine (a few machines have 24 TB each)
1200 machines with 8 cores each + 800 machines with 16 cores each
32 GB of RAM per machine
15 map-reduce tasks per machine
That's a total of more than 21 PB of configured storage capacity! This is larger than the previously known Yahoo!'s cluster of 14 PB. Here are the cluster statistics from the HDFS cluster at Facebook. So, option (i) is correct.
Hadoop batch processes data distributed over a number of computers ranging in 100s and 1000s. So, option (ii) is correct.
So, Both Statements are correct.

312. Hadoop (a big data tool) works with number of related tools, Choose from the following, the common tools included into Hadoop :
(a) MySQL, Google API and Map reduce
(b) Map reduce, Scala and Hummer
(c) Map reduce, H Base and Hive
(d) Map reduce, Hummer and Heron

UGC NET C.S. Exam-24 June-2019 (Paper-II)

Ans. (c) : Hadoop is an open source platform providing highly reliable, scalable, distributed processing of large data sets using simple programming models.
Hadoop provides a reliable shared storage and analysis system. In this, storage, is provide by HDFS (Hadoop distributed file system) and analysis by map reduce.
HDFS breaks a file into chunks and distributed them across the nodes of the cluster. It stores large amount of data on local disk across a distributed cluster of computers. It is written in Java. Hadoop work with the tools : Map Reduce, H base, hive.
Map Reduce A distributed data processing model and execution environment that runs on large clusters of commodity machines.
Hbase → It is a distributed, column-oriented database. HBase uses HDFS for its underlying storage and

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supports both batch style computations using map reduce and point queries.
HIVE → It is a distributed data ware house. Hive manages data stored in HDFS and provides a query language based on SQL.

313. Which of the following is an example of a digital Electronic?
निम्नलिखित में से कौन डिजिटल इलेक्ट्रॉनिक का उदाहरण है?

(a) Computer/कम्प्यूटर
(b) Mobile phone/मोबाइल फोन
(c) Digital cameras/डिजिटल कैमरे
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (d) : Computers, information appliance, digital cameras, digital television, flash memory, USB memory, mobile phones, die disks, and computer memory devices are example of digital electronics. After converting analog signals to digital form, digital signal processing is applied on them.

314. Which of the following is NOT a component of a Turing machine?
निम्नलिखित में से कौन ट्यूरिंग मशीन का घटक नहीं है?

(a) Input tape/इनपुट टेप
(b) Output tape/आउटपुट टेप
(c) Control unit/नियंत्रण इकाई
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 3.0 Exam-22.07.2024 (11-12)

Ans. (b) **:** Turing machine generally has the following components-
■ Input tape in which the input is stored.
■ The tape head that reads and writes on the tape.
■ The control unit that determines the behavior of the machine based on the current state and input symbols.
A Turing machine has no separate output tape:
The output is usually thought of as the contents of the tape when the machine halts.

315. UNIVAC is
UNIVAC है

(a) Universal Automatic Computer
(b) Universal Array Computer
(c) Unique Automatic Computer
(d) Unvalued Automatic Computer

Bihar STET C.S. 2019 (Shift-II)

Ans. (a) : UNIVAC Stands for Universal Automatic Computer. The first general purpose electronic digital computer design for robot business applications was the UNIVAC 1. This computer was produced in the unite states. It was designed by John Mauchly and J. Presper Eckert.

316. What is the term used when you press and hold the left mouse key and move the mouse around the slide?
जब आप बाई माउस कुँजी दबाते हैं और माउस को स्लाइड के चारों ओर दबाते हैं, तो क्या शब्द का उपयोग किया जाता है?

(a) Moving (b) Dragging
(c) Monitoring (d) Highlighting

Bihar STET C.S. 2019 (Shift-II)

Ans. (b) : When you hold down the left mouse button and move the mouse around on the screen or slides, this action is called dragging. It is used when you select an area move an object from one place to another.

317. Computer is free from tiresome and boredom. We call it
कम्प्यूटर थकान और विश्सता से मुक्त होता है, जिसे हम कहते हैं

(a) Accuracy (b) Reliability
(c) Diligence (d) Versatility

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : Diligence– It means that the computer does not get tired, does not get bored, and can work continuously, without any breaks, so this characteristic shows the computer's freedom from fatigue and relaxation.

318. A hybrid computer
एक hybrid computer

(a) Resembles digital computer
डिजिटल कंप्यूटर जैसा है
(b) Resembles analogue computer
एनालॉग कंप्यूटर जैसा है
(c) Resembles both as digital and analogue computer
दोनों डिजिटल तथा एनालॉग कंप्यूटर जैसा है
(d) None of these/इनमें से कोई नहीं

Bihar STET C.S. 2019 (Shift-II)

Ans. (c) : Hybrid Computer is a computer that works by combining the features of both Analog and Digital computers. These computers are used for tasks where some part of the data is in continuous and some part in quantitative form such as in medical equipment, scientific experiments, etc.

319. Which of the following is not a method of obtaining hash function:

(a) Random Probing (b) Folding
(c) Truncation (d) Modular arithmetic

RPSC C.S. 2016 (PAPER-II)

Sarkari Resource

Ans. (a) : The methods used for obtaining hash function typically include-
Folding– Breaking the key into parts and combining them (e.g., adding them).
Truncation– Using only a part of the key to compute the hash.
Modular Arithmetic– Using the module operation (key % table size) to map keys to hash values.
Random probing is not a method for creating a hash function. Instead, it is a collision resolution technique used in open addressing hash tables, where a random sequence is used to find an empty slot when a collision occurs.
Thus, the correct option is (a).

320. In context of compilers, a lexical analyser generates:-
(a) Syntax tree (b) Token stream
(c) Regular expression (d) Intermediate code

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : In the context of compilers, the lexical analyzer (also known as the scanner or tokenizes) is responsible for breaking the source code into a series of tokens.
Token stream– The output of the lexical analyzer is a stream of tokens, where each token represents a basic unit of the source code such as keywords, identifiers, literals or symbols.

321. The following function is not performed by a loader:-
(a) Allocation (b) Translation
(c) Linking (d) Relocation

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : Loader functions– A loader is responsible for loading the program into memory for execution. Its primary functions include.
Allocation– Assigning memory locations to the program.
Relocation– Adjusting addresses in the program to match the assigned memory locations.
Linking– Resolving external references between object modules.
Translation– Translation, which involves converting source code into machine code, is typically performed by a compiler or assembler, not a loader.

322. Which of the following is not a machine independent phase of compiler?
(a) Code generation
(b) Lexical Analysis
(c) Syntax Analysis
(d) Intermediate code generation

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : Compiler Phases– The phases of a compiler can be categorized into machine dependent and machine independent phases.
Machine Independent Phases– The following phases are typically machine independent.
1. Lexical Analysis- Breaking the source code into tokens
2. Syntax Analysis- Parsing the tokens into a syntax tree.
3. Intermediate Code Generation- Generating platform- agnostic intermediate code.

Machine Dependent Phases Code Generation– This phase generates machine specific code from the intermediate code, making it machine dependent.
So, the correct option is (a).

323. An assembler is a system software that:-
(a) Assembles modules of high level language program
(b) Translates assembly language program to machine language equivalent
(c) Translate HLL Program into object file
(d) Converts a high level language program to assembly language

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : An Assembler is a system software that takes assembly language code (Which is human readable and low-level) and converts it into machine code (Which is directly executed by the CPU). This machine code is also known as the object code.

324. An assembly language is a/an-
एक एसेंबली भाषा है–
(a) Low-level programming language
(b) Middle-level programming language
(c) Internet-based programming language
(d) More than one of the above
उपर्युक्त में से एक से अधिक
(e) None of the above/उपर्युक्त में से कोई नहीं

BPSC TRE 2.0 22.02.2024 (9-10)

Ans. (b) : Assembly language is a low-level language help communicate directly with computer hardware. It uses (mnemonics) to represent the actions performed by the processor. It is an Intermediate language between high-level language such as C++ and binary language. It uses hexadecimal and binary values, which can be read by the user.

325. Choose a false statement about MACRO in context of an assembly language source code:-
(a) It is single line abbreviation for group of instructions
(b) Macro-name is used as an operation mnemonic
(c) Macro calls are replaced by Macro definition
(d) Macro definition also appears in expanded source code

RPSC C.S. 2016 (PAPER-II)

Ans. (d) : Macro in Assembly Language– A Macro in assembly language is a way to define a block of code that can be reused throughout a program.

Sarkari Resource

Macro characteristics–

1. Macro Definition– A macro definition is a block of code that is defined once and can be invoked multiple times in a program.

2. Macro Call– When a macro is called, the assembler replaces the macro call with the actual code defined in the macro definition.

3. Expended Source Code– After macro expansion, the macro calls are replaced by the macro definition, so the macro definition itself does not appear in the expanded source code as a separate entity, instead, its content is inserted where the macro is called.

So, the false statement is (d) macro definition also appears in expanded source code.

326. Resolving symbolic references when a program is loaded is main memory for execution is called:-

(a) Allocation (b) Linking
(c) Relocation (d) Loading

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : When a Program is loaded into main memory for execution, the process of resolving symbolic references is called linking.

This is the process of combining various code and data modules (Which might have symbolic references to each other, like function calls or global variable access) into a single executable file or a loadable program. It resolves these symbolic references by finding the actual memory addresses or offsets of the referenced symbols.

327. YACC stands for:-

(a) Yet Accepted compiler construction
(b) Yet another compiler- compiler
(c) Yet Available compiler- complier
(d) Yet Another compiler construction

RPSC C.S. 2016 (PAPER-II)

Ans. (b) : YACC stands for Yet Another Compiler, Compiler. It is a tool used for generating parsers based on a given grammar, It is commonly used in compiler construction to parse the syntax of programming language.

328. Consider various forms of a program code for code optimisation. It is possible to optimise:-

(a) Source code and intermediate code only
(b) Intermediate code and Target code only
(c) Source code, Intermediate code and Target code
(d) Source code and Target code only

RPSC C.S. 2016 (PAPER-II)

Ans. (c) : Code optimization can be applied at various stages of the compilation process.

1. Source code optimization– This involves improving the source code before it's compiled (e.g., removing redundant expressions, using better algorithm).

2. Intermediate code optimization– Most modern compilers use an intermediate representation of code, which is platform independent.

3. Target code optimization– After code generation, further optimizations specific to the target machine (like instruction scheduling register allocation) can be applied.

329. Consider task T1: control execution speed, T2: search a word, T3: delete a paragraph and T4: Track flow of execution. Choose appropriate match:-

(a) Debugger : T1, T4 and editor : T2, T3
(b) Debugger : T2, T3 and editor : T1, T4
(c) Debugger : T1, T3 and editor : T2, T4
(d) Debugger : T2, T4 and editor : T1, T3

RPSC C.S. 2016 (PAPER-II)

Ans. (a) : Analyze the tasks-

T1 control execution speed– This task is typically associated with a debugger, which allows you to step through code, set breakpoints, and control the execution flow.

T2 Search a word– This task is commonly performed in an editor, where you can search for specific text within a document.

T3 Delete a paragraph– This task is also typically performed in an editor, where you can modify and edit text.

T4 Track Flow of execution– This task is usually done with a debugger, which helps you understand the flow of your program's execution.

Matching Tasks–

Debugger- T1 and T4
Editor- T2 and T3

So, the best option is (a).

330. The E-waste (management) rules 2016 of India, recognizes two broad categories: IT and telecommunication equipment and consumer electrical and electronics including which of the following material?

I. Fluorescent
II. Mercury

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Ans. (c) : According to the E-waste Management Rules 2016 of India, "Consumer electrical and electronics" includes fluorescent and mercury containing lamps, among other items. Therefore the correct answer is (c) Both I and II.

331. In E-waste management, from which factors, there comes the exposure variability in electronic waste?

(a) Type of E-waste
(b) Quantity of the pollutants in air
(c) Management processes of the air pollution
(d) Medical waste

DSSSB TGT C.S. 07.02.2024 (Shift-II)

Sarkari Resource

Ans. (a) : The primary factor contributing to exposure variability in electronic waste (e-waste) is the type of e-waste. The variability in exposure to hazardous substances in electronic waste primarily comes from the type of e-waste being handled. Different electronic devices contain different materials and a varying composition of hazardous substances.

332. A compound statement which is neither a tautology nor a contradiction is called_____
एक कम्पाउण्ड स्टेटमेन्ट जो न तो एक टोटोलॉजी है और न ही विरोधाभास है, कहलाता है

(a) Contingency (b) Equivalence
(c) Condition (d) Inference

UKPSC PGT C.S. Asist. Prof. 2017

Ans (a) : A compound statement that is neither a tautology (always true) nor a contradiction (always false) is called a contingency. This type of statement has truth values that can vary being sometimes true and sometimes false depending on the truth values of its components.

333. Which of the following is not a PaaS in cloud environment?
निम्न में से कौन क्लाउड परिवेश में PaaS नहीं है?

(a) OS and application software
ओ.एस. तथा एप्लीकेशन सॉफ्टवेयर
(b) Database Server/डेटाबेस सर्वर
(c) Virtual Machine (with specified CPU, memory and disk)/वर्चुअल मशीन (निर्दिष्ट सी.पी.यू., स्मृति तथा डिस्क के साथ)
(d) Web Server/वेब सर्वर

UKPSC PGT C.S. Asist. Prof. 2017

Ans (c) : In Cloud Computing Service Models-
IaaS (Infrastructure as a Service) provides virtual machines, storage, CPU, memory etc.
PaaS (Platform as a Service) provides OS, application software, databases, web servers, development tools so that developers can build and deploy applications.
SaaS (Software as a Service) provides ready to use applications over the internet.
Database server and web server are both example of PaaS components because the are managed platforms provided to the user.
OS and application software can be a part of a PaaS offering, as the platform often includes pre-configured operating system and software to run applications.

334. In which category of SaaS services does CRM software fall?
SaaS सेवाओं की किस श्रेणी में सी.आर.एम. सॉफ्टवेयर आता है?

(a) Consumer Services/उपभोक्ता सेवाओं में
(b) Communication Services/संचार सेवाओं में
(c) Infrastructure Services/अवसंरचना सेवाओं में
(d) Business Services/व्यापारिक सेवाओं में

UKPSC PGT C.S. Asist. Prof. 2017

Ans (d) : CRM (Customer Relationship Management) software is a type of SaaS that businesses use to manage and analyze customer interactions and data throughout the customer lifecycle. It helps with improving business relationships with customers, assisting in customer relation and driving sales growth. Because it's a tool designed to support the internal operations and functions of a business it is categorized as a business service.
1. Consumer Services
2. Communication Services
3. Infrastructure Services.

335. _____analyzes the business situation, identities opportunities for improvement and designs an information system to implement the improvement.
_____व्यवसायिक स्थिति का विश्लेषण करता है, सुधार के अवसरों की पहचान करता है और सुधारों को लागू करने के लिए एक सूचना प्रणाली डिजाइन करता है।

(a) System Analyst (b) Developer
(c) Programmer (d) Debugger

HTET PGT C.S. 30.07.2025

Ans.(a): A system analyst is responsible for assessing a business's current processes, identifying areas for improvement and designing information system to address those needs.

336. Which of the following is/are example of cloud computing?
निम्नलिखित में से कौन क्लाउड कम्प्यूटिंग का/के उदाहरण है/हैं?

(a) Web applications/वेब एप्लीकेशन्स
(b) Saas/सास
(c) Software Plus Services/सॉफ्टवेयर प्लस सर्विसेस
(d) All options are correct/सभी विकल्प सही हैं

HTET PGT C.S. 30.07.2025

Ans.(d): All of the listed options are examples of or directly related to cloud computing. Cloud computing is the delivery of computing services including servers, storage, databases, networking, software analytics and intelligence over the internet.
Web Application– Applications hosted and accessed through the internet are part of cloud computing.
SaaS– A core service model of cloud computing where software is delivered over the internet.
Software Plus Services– A hybrid model combining locally installed software with services delivered from the cloud.

337. FOPL stands for:
FOPL का मतलब है–

(a) First Order Predicate Logic
(b) Full Order Predicate Logic
(c) First Out Predicate Logic
(d) First Order Prepositional Logic

HTET PGT C.S. 30.07.2025

Ans.(a): FOPL stands for First Order Predicate Logic. It is an extension of propositional logic that allows for quantification over objects, properties, and relations, FOPL includes constants, variables, predicates, functions and quantifiers.

338. Which of the following terms indicates a record structure used to implement three-address statement in compilers?

(a) Triple (b) Handle
(c) Ripple (d) Tuple
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : Triple is a data structure used in intermediate code generation to represent there address code. In consists of three fields an operator and two operands (which can be references to other triples or symbols). Triples are a compact way to represent three address statements without storing temporary variable names explicitly.

339. 'E' within the ETL tools used for data warehouses stands for _____.

(a) Extract (b) Execute
(c) Explore (d) Expect
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (a) : In data transformation, extract, transform and load tools transform and consolidate the raw data in the staging area to prepare it for the target data warehouse. The data transformation phase can involve the following types of data changes.

340. Which phase of a compiler has the main role of generating a sequence of tokens from the input source program?

(a) Semantic analyzer
(b) Lexical analyzer
(c) Parser
(d) Code optimizer
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : The phase of a compiler that has the main role of generating a sequence of tokens from the input source program is the lexical analyzer.
The lexical analyzer, also known as the scanner or laxer, is the first phase of a compiler. It reads the source code character by character and groups them into meaningful units called tokens. These tokens represent keywords, identifier, operators, constants, and other elements of the programming language.

341. A _____ is a data structure, used by a compiler that contains a record for each identifier, with fields for the attributes of the identifier.

(a) Parsing table (b) Symbol table
(c) Transition table (d) Transition table
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : A symbol table is a crucial data structure used by compilers and interpreters to store information about the identifiers in a program.
A Symbol table is a data structure, typically a hash table or a tree, that maps identifiers to their attributes. The compiler uses this information during the various phases of compilation, particularly during semantic analysis and code generation.

342. The input for the code generator is a:

(a) Tree at lexical level
(b) Tree at semantic level
(c) Sequence of assembly language instructions
(d) Sequence of machine idioms

CGPSC C.S. 2020 Set-A

Ans. (b) : A code generator is a component of a compiler that translates an intermediate representation of source code into a target machine code. This process happens after the compiler has already performed lexical and syntactic analysis, which cheeks for error in the code's structure.
The semantic level representation is a parse Tree or an Abstract Syntax Tree (AST) that has been checked for semantic correctness. This means the tree not only represents the grammatical structure of the program, but also contains information about the meaning and relationships between the program's elements.

343. Problems that can be solved in polynomial time are called:

(a) Tractable (b) Decidable
(c) Solvable (d) Computable

CGPSC C.S. 2020 Set-A

Ans.(a): Problems that can be solved in polynomial time are known as tractable. Problems that cannot be solved in polynomial time are intractable.

344. Consider Column - 1 and Column - 2 :
स्तम्भ-I तथा स्तम्भ-II पर विचार करें:

Column-I/स्तम्भ-I		Column-II/स्तम्भ-II	
I.	Bing/बिंग	A.	Social Networking सोशल नेटवर्किंग
II.	Flipkart/फ्लिपकार्ट	B.	e-mail/ई-मेल
III.	Facebook/फेसबुक	C.	Search Engine सर्च इंजन
IV.	Google Chrome गूगल क्रोम	D.	Online Shopping ऑनलाइन शॉपिंग
V.	gmail/जीमेल	E.	Web Browser वेब ब्राउजर

Match of Column-1 and Column-2 is :
स्तम्भ-I तथा स्तम्भ-II का सुमेल है:

(a) I- D, II-A, III-C, IV-B V-E
(b) I- C, II-D, III-A, IV-E V-B
(c) I- E, II-A, III-B, IV-C V-D
(d) I- C, II-E, III-B, IV-D V-A

HTET PGT C.S. 2018

Sarkari Resource

Ans. (b) : The correct match–

Column-I		Column-II	
I.	Bing	**C.**	Search Engine
II.	Flip kart	**D.**	Online Shopping
III.	Facebook	**A.**	Social Networking
IV.	Google Chrome	**E.**	Web Browser
V.	Gmail	**B.**	e-mail

345. Consider Column - 1 and Column - 2 :
स्तम्भ-I तथा स्तम्भ-II पर विचार करें:

Column-I/स्तम्भ-I		Column-II/स्तम्भ-II	
I.	**Microsoft Office माइक्रोसॉफ्ट आफिस**	**A.**	**File Archiver and Compressor/फाइल अकाइवर तथा कंप्रेशर**
II.	**Windows-10 विंडोज-10**	**B.**	**Digital Library डिजिटल लाइब्रेरी**
III.	**WinZip/विनजिप**	**C.**	**Application Software एप्लीकेशन सॉफ्टवेयर**
IV.	**A. V. G/ए.वी.जी.**	**D.**	**Operating System ऑपरेटिंग सिस्टम**
V.	**N. D. L. I. एन.डी.एल.आई.**	**E.**	**Antivirus/एंटीवायरस**

Match of Column-1 and Column-2 is :
स्तम्भ-I तथा स्तम्भ-II का सुमेल है:

(a) I- E, II-A, III-B, IV-C V-D
(b) I- C, II-E, III-B, IV-D V-A
(c) I- D, II-A, III-C, IV-B V-E
(d) I- C, II-D, III-A, IV-E V-B

HTET PGT C.S. 2018

Ans. (d) : The correct match–

Column-I		Column-II	
I	Microsoft Office	**C.**	Application Software
II	Windows-10	**D.**	Operating System
III	WinZip	**A.**	File Archiver and Compressor
IV	A. V. G	**E.**	Antivirus
V	N. D. L. I.	**B.**	Digital Library

346. Which of the following is not a type of computer port ?
निम्न में से कौन-सा कम्प्यूटर पोर्ट का एक प्रकार नहीं है?

(a) SCSI (b) POP3
(c) PS/2 (d) USB

HTET PGT C.S. 2019

Ans. (b) : Let's analyze each option-
SCSI– It's stands for Small Computer System Interface, which is a type of hardware port used for connecting peripherals like hard drives and scanners.
POP3– It is stands for post-office protocol version 3. It is a protocol used for receiving emails. It is not a physical computer port, it is a network protocol.
PS/2– It is a type of port used for connecting keyboards and mice.
USB– It is stands for Universal Serial Bus, which is a common port for connecting a wide variety of devices.

347. Which of the following is a top-down parser?

(a) LR parser
(b) Operator precedence parser
(c) Shift reduce parser
(d) Recursive-descent parser
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (d) : A top-down parser starts at the root of the parse tree and works its way down, attempting to match the input string against the grammar rules. Recursive descent parsing is a top-down parsing technique that uses recursive functions to parse the input. It constructs the parse tree from the root node.

348. Which of the following statements is FALSE in view of the process of constructing a composite NFA N(st) for the regular expression st using thompson's construction algorithm?

(a) The start state of N(s) becomes the start of N(st).
(b) The accepting state of N(t) becomes the accepting state of N(st).
(c) The accepting state of N(s) is merged with the start state of N(t).
(d) The accepting state of N(s) becomes the accepting state of N(st).
(e) Not attempted

Haryana PGT C.S. Asstt. Prof. 2024

Ans. (d) : statement (d) is False because, in Thompson's construction for the composite NFA of st the accepting state of the NFA for s is merged with the start state of the NFA for t through an epsilon transition while the final accepting state of st comes from the new accepting state created to accommodate this merge, not the accepting state of **s** alone.

349. Which of the following can be viewed as an attempt to construct a parse tree for the input starting from the root and creating the nodes of the parse tree in preorder?

(a) Bottom-up parsing
(b) Top-down parsing
(c) Left-right parsing
(d) Right-left parsing
(e) Not attempted

Haryana PGT C.S. Asist. Prof. 2024

Ans. (b) : Top down parsing starts from the start symbol (root of the parse tree).
It expands productions recursively to try to match the input string.
This corresponds to constructing the parse tree in preorder travel (root → left → right).

Sarkari Resource

350. Consider Column-I and Column-II in context of tools to support UML diagram for analysis modeling :
एनालिसिस मॉडलिंग के लिए UML डायग्राम को सपोर्ट करने वाले टूल्स के संदर्भ में स्तम्भ-I तथा स्तम्भ-II पर विचार कीजिए:

Column-I/स्तम्भ-I		Column-II/स्तम्भ-I	
I.	**Power Designer**	**A.**	**IBM**
II.	**Rational Rose**	**B.**	**Microsoft**
III.	**Visio**	**C.**	**Sybase**
IV.	**Argo UML**	**D.**	**Open source**

Match of Column-I and Column-II is :
स्तम्भ-I तथा स्तम्भ-II का सुमेल है:

(a) I-B, II-A, III-D, IV-C
(b) I-C, II-A, III-B, IV-D
(c) I-D, II-B, III-C, IV-A
(d) I-C, II-D, III-A, IV-B

HTET PGT C.S. 2022

Ans. (b) : The correct match–

Column-I		Column-II	
I.	Power Designer	**C.**	Sybase
II.	Rational Rose	**A.**	IBM
III.	Visio	**B.**	Microsoft
IV.	Argo UML	**D.**	Open source

114. Consider Column-I and Column-II :
स्तम्भ-I तथा स्तम्भ-II पर विचार कीजिए:

Column-I/स्तम्भ-I		Column-II/स्तम्भ-II	
I.	**3D Studio** **3D स्टूडियो**	**A.**	**Search Engine** **सर्च इंजन**
II.	**Ubuntu** **उबुन्तु**	**B.**	**Anti virus** **एन्टी वायरस**
III.	**Duck Duck Go** **डक डक गो**	**C.**	**Animation** **एनीमेशन**
IV.	**Maxthon** **मैक्सथोन**	**D.**	**Operating System** **ऑपरेटिंग सिस्टम**
V.	**Vipre Rescue** **वाइप्रे रेस्क्यु**	**E.**	**Web Browser** **वेब ब्राउजर**

Match the Column-I and Column-II is :
स्तम्भ-I तथा स्तम्भ-II का सुमेल है :

(a) I-D, II-A, III-C, IV-B, V-E
(b) I-C, II-D, III-A, IV-E, V-B
(c) I-E, II-A, III-B, IV-C, V-D
(d) I-C, II-E, III-B, IV-D, V-A

HTET PGT C.S. 2022

Ans. (b) : The correct match–

Column-I		Column-II	
I.	3D Studio	**C.**	Animation
II.	Ubuntu	**D.**	Operating System
III.	Duck Duck Go	**A.**	Search Engine
IV.	Maxthon	**E.**	Web Browser
V.	Vipre Rescue	**B.**	Anti virus

351. Choose valid set of output devices :
आउटपुट डिवाइसेज का उचित सेट चुनिए:

(a) Printer, MICR, Bar Code Reader
(b) Speaker, OMR, VDU
(c) Plotter, VDU, Headphone
(d) Headphone, OCR, Joystick

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Ans. (c) : Plotter– An output device that creates graphical output, often used for large scale prints.
VDU (Visual Display Unit)– A monitor which displays visual information from a computer.
Headphone– An output device that allows users to listen audio privately.

352. The producer and Consumer processes share the following Variables:
प्रोड्यूसर तथा कंज्यूमर प्रोसेस निम्नलिखित चरों को साझा करते हैं:

int n,
Semaphore M = 1
Semaphore E = n
Semaphore F = 0
The consumer process must execute _____ and _____ before removing an item from buffer.
कंज्यूमर प्रोसेस को किसी मद को बफर से हटाने से पहले _____ तथा _____ को निष्पादित करना होगा।

(a) Signal (M), signal (F)
(b) Signal (M), wait (F)
(c) Signal (F), wait (M)
(d) Wait (F), wait (M)

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Ans. (d) : We assume that the pool consists of n buffers, each capable of holding one item, The mutex (M) semaphore provides mutual exclusion for accesses the buffer pool and a initialized to the value 1. The empty (E) and full (F) semaphores count the number of empty and full buffers. The semaphore empty is initialized to the value n, the semaphore full is initialized to the value 0.
The code for the consumer process is

```
do {
        wait (full);
        wait (mutex);
        /*remove an item from buffer to next-consumed*/
        signal (mutex);
        Signal (empty);
/* consume the item in next-consumed */
        } while (true);
```

So, the consumer process must execute wait (F) and wait (M) before removing an item from buffer.